Go

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Everything Go - Comprehensive Syllabus

A **deep library and reference curriculum**, structured pedagogically from core language foundations to advanced systems engineering, runtime internals, and production operations.

How to read this book (pick a path)

```
Path A - Core & Productive Go (  2-4 weeks)
  Foundations (01) → Core (02) → Memory (03) → Generics (04) → Errors (05)
  → Testing (06) → Standard Library Tour (07) → CLI Tools (09) → Projects (99)

Path B - Concurrent & Networked Systems
  Standard Library (07) → CLI Tools (09) → Concurrency Ground Up (10)
  → Advanced Concurrency Patterns (11) → Socket Programming (12)
  → Network Systems (13) → Network Performance (14)

Path C - Web Development & Distributed Services
  Standard Library (07) → Concurrency (10-11) → Web Development in Go (15)
  → Modern Web & API Architectures (16) → Distributed Infrastructure (18)
  → Observability & SRE (19) → Cloud Infrastructure (20)

Path D - Systems Programming & Linux Engineering
  Foundations (01) → Memory Mechanics (03) → Standard Library (07)
  → CLI Tools (09) → Sockets (12) → Systems Programming (17)
  → Cloud & Deployment (20) → Long-Term Maintenance & FFI (26)

Path E - Production Hardening, SRE & Performance
  Observability & SRE (19) → Cloud Infrastructure (20) → Security Tools (21)
  → Security Hardening (22) → Performance Tooling (23) → Performance Engineering (24)

Path F - Runtime Internals & Compiler SSA
  Memory Mechanics (03) → Concurrency (10-11) → Go Deep Dives (25: GMP, GC,
  memory model, netpoller, Swiss maps, compiler SSA, assembly, synctest)
  → Long-Term Maintenance (26: unsafe, cgo, reflection)

Path G - Algorithms & Practical Drills
  Data Structures & Algorithms (08) → CLI Cookbook (09)
  → Capstone Projects (99: Linux clones, k8s controllers, eBPF, TinyGo, 200 workouts)
```

Curriculum Architecture & Progression

The book is organized into structured phases of engineering depth:

1. **Language Mastery (Parts 01–06):** Foundations, types, memory mechanics, pointers, interfaces, generics, errors, and testing.
2. **Applied Language, Algorithms & CLI (Parts 07–09):** The standard library surface, data structures & algorithms, and production CLI automation.
3. **Concurrency & Parallelism (Parts 10–11):** Diagram-first ground-up concurrency fundamentals, followed by advanced production coordination patterns.
4. **Networking & Sockets (Parts 12–14):** Low-level socket programming, resilient network systems, and high-connection netpoller performance.
5. **Web Services & Modern APIs (Parts 15–16):** Full-stack web development (Bookstore track) followed by modern architectures (GOTH stack, gRPC, Huma OpenAPI).
6. **Systems Programming & OS Integration (Part 17):** Userspace Linux systems programming, processes, signals, safe I/O, WAL, and sandboxing without libc.
7. **Distributed Systems & Cloud Infrastructure (Parts 18–20):** Resilient distributed patterns, full observability (slog, Prometheus, OpenTelemetry), and container/Kubernetes deployment.
8. **Security Engineering & Hardening (Parts 21–22):** Security tooling audit, defense-in-depth, TLS/mTLS, authentication, SSRF/CSRF defense, and secret zeroization.
9. **Performance Engineering & Tooling (Parts 23–24):** Profiling workflows, allocations/GC tuning, contention optimization, PGO, and SIMD vectorization.
10. **Runtime Internals & Compiler Architecture (Part 25):** 71 deep dives into the GMP scheduler, memory allocator, GC pacer, compiler IR/SSA, and netpoller.
11. **Long-Term Engineering & Synthesis (Parts 26–28):** Reflection, unsafe code, cgo/FFI, modern Go literature synthesis, and specialized frontiers (AI agents, desktop apps).
12. **Capstones & Reference (Parts 99–100):** 46 hands-on real-world projects and master reference appendices.

---

1. Foundations & Mindset

- **The Go Philosophy:** Why "boring" is a superpower.
- **2026 Learning Roadmap:** Focusing on essentials vs. hype.
- **The 3-Layer Practice System:** Read, Write, Ship.
- **Transitioning to Go:** Insights for Java and Node.js developers.
- **Go 1.27:** Language (generic methods, promoted field keys), toolchain, `json/v2`, `uuid`, `goroutineleak`, SIMD experiment. Official notes: go.dev/doc/go1.27.
- **Tooling & Setup:** Installing Go, code organization, core commands, vetting, dependency management, and multi-module workspaces.

2. Core Language Essentials

- **Language Building Blocks:** Variables, types, and functions. 1.26: `new(expr)` 1.27: generic methods, promoted field keys
- **Control Flow:** Effective use of `if`, `for`, and `switch`.
- **Data Structures:** Arrays, slices, strings, and maps internals.
- **Composite Types:** Structs and data modeling.
- **Working with Time:** Wall clocks, monotonic timers, and formatting.

3. Pointers & Memory Management

- **The Mechanics:** Understanding Stack vs. Heap.
- **Functions & Receivers:** First-class functions, closures, methods, and value vs. pointer receivers.
- **Memory Optimization:**
  - Escape Analysis: Where did memory allocate?
  - Memory Alignment, Padding, and Size-Specialized Malloc.
  - Garbage Collection (GC) fundamentals, tuning, and `GOMEMLIMIT`. Go 1.26: Green Tea GC default
- **Pass by Value vs. Reference:** Performance and safety tradeoffs.

4. Generics & Interfaces

- **Designing with Interfaces:** Implicit implementation (The "Go Way") and small interfaces.
- **Interface Practices:** Composition, decoupling, and mockability.
- **Dynamic Inspection:** Type assertions and type switches.
- **Generics Introduction:** Type parameters, constraints, and generic functions.
- **Generic Types & Methods:** Generic structs, collections, and generic methods Go 1.27.
- **Idiomatic Generics:** Guidelines on when to use generics and when to avoid them.

5. Advanced Error Handling

- **Beyond `if err != nil`:** Idiomatic error values and control flow.
- **Patterns:** Sentinel errors, custom error types, and error wrapping (`fmt.Errorf`, `errors.Is`, `errors.As`).
- **Defensive Programming:** Panic, recover, and runtime invariants.
- **Modern Techniques:** The "Must" pattern and comparison with non-GC languages (Odin).

6. Testing & Quality Assurance

- **Testing Fundamentals:** Unit testing and table-driven tests.
- **Test Organization:** Test suites, subtests, parallel tests, and test helpers.
- **Advanced Testing:** Benchmarks, memory allocation tracking, and code coverage.
- **Integration Testing:** Real dependencies, testcontainers, and build tags.
- **Fuzzing:** Property-based testing and continuous fuzzing engines.
- **Network & API Testing:** `net/http/httptest` client/server mocks.
- **Golden Files:** Managing `testdata` snapshots and deterministic fixtures.
- **Capstone Lab:** Designing an end-to-end service test suite.

7. Standard Library Tour

- **Map of the Standard Library:** When to reach for standard packages before any framework.
- **Text & I/O:** `fmt`, `log`, `log/slog`, `bytes`, `strings`, `strconv`, `io`, and `bufio` composition.
- **Filesystem & Embedding:** `os`, `path/filepath`, `io/fs`, and `//go:embed`.
- **Serialization & Formats:** `encoding/json`, `encoding/json/v2`, `encoding/csv`, `encoding/xml`, `encoding/binary`, and `encoding/gob`.
- **HTTP & OS Integration:** Modern `net/http.ServeMux`, HTTP clients with timeout budgets, `flag`, `os/exec`, and `os/signal`.
- **Budgets & Concurrency Primitives:** `time` and `context` recipes, `sync`, and `sync/atomic`.
- **Modern Collections:** `slices`, `maps`, `cmp`, and `iter` (range-over-func).
- **Data & Sockets:** `database/sql` pools, transactions, and `net` dial/listen primitives.
- **Security & Encodings:** `crypto`, `hash`, `crypto/rand`, post-quantum algorithms (`crypto/mldsa`, `crypto/mlkem`), and stdlib `uuid`.
- **Templates & Archives:** `html/template`, `text/template`, `compress`, and `archive`.

8. Data Structures and Algorithms

- **Algorithmic Complexity:** Big-O notation, benchmarking, and cache friendliness in Go.
- **Linear Structures:** Stacks, queues, ring buffers, and linked lists.
- **Tree Structures:** Binary search trees, heaps, priority queues, and tries.
- **Sorting & Searching:** Quick sort, merge sort, binary search, and quickselect.
- **Graph Theory:** BFS and DFS traversals, topological sorting, and Dijkstra's algorithm.
- **String Algorithms & Probabilistic Filters:** Pattern searching and Bloom filters.

9. CLI Tools & Automation

- **Stdlib-First CLI:** `os.Args`, `flag`, `FlagSet` subcommands, exit codes, and stdout vs. stderr.
- **Environment & Configuration:** Environment variables, JSON/YAML configuration, and stdin pipes.
- **System Orchestration:** `os/exec` process pipelines, signal trapping, and context cancellation.

- **Testing & Architecture:** Structuring large CLI applications and writing CLI test suites.
- **Cobra & Viper Ecosystem:** Commands, persistent flags, hooks, shell auto-completion, and layered Viper config.
- **Shipping & Releases:** Version ldflags, cross-compilation, and GoReleaser automation.
- **Cookbook & Projects:** Network tools (`netkit`: ping, DNS, port scanner, HTTP probe), security tools (`netsec`), text processing (`textkit`), and HTTP debugging (`httpdbg`).

10. Concurrency From the Ground Up

- **Diagram-First Mental Model:** Goroutines, threads, and cooperative scheduling.
- **Channels in Depth:** Unbuffered vs. buffered, ownership transfer, channel state machines, and nil channels.
- **Pipelines & Multiplexing:** Fan-out, fan-in, `select`, timeout handling, and cancellation trees.
- **Backpressure & Flow Control:** Throttling, rate limiting, and backpressure in streaming APIs.
- **Context Complete:** Lifecycles, deadlines, cancellation propagation, and context value hygiene.
- **Synchronization Primitives:** WaitGroups, encapsulation, data races vs. race conditions, and `sync.Mutex`.
- **Advanced Primitives:** Semaphores, rendezvous barriers, atomic memory operations, and `sync.Cond` broadcast.
- **Verification:** Race detector internals, synctest bubbles, and testing concurrent code.

11. Advanced Concurrency Patterns

- **Architectural Patterns:** Worker pools, bounded concurrency, dynamic pool resizing, and pipeline coordination.
- **High-Performance Sync:** `sync.Map`, `sync.Once`, and allocation-free `sync.Pool`.
- **Production Guidelines:** Concurrency design principles, leak prevention, and lock ordering.
- **Diagnostics:** Goroutine leak detection and the `goroutineleak` pprof profile.

12. Socket Programming & Protocols

- **Networking Fundamentals:** Transport layers, sockets, and Go's `net` package architecture.
- **TCP Programming:** Reliable TCP servers and clients, buffer tuning, and flow control.
- **UDP Programming:** Connectionless protocols, multicasting, and datagram services.
- **Local IPC:** High-performance Unix Domain Sockets.
- **Network Concurrency & Deadlines:** Concurrency models per connection, read/write deadlines, and context cancellation.
- **Protocol Engineering:** Designing text-based, binary, and custom framed protocols; serialization with Protobuf and JSON.
- **Encrypted Sockets:** TLS/SSL socket wrapping, certificates, and handshake management.
- **Real-Time Communication:** WebSockets server and client implementation.

- **Production Networking:** Reverse proxies, layer-4 load balancers, testing network code, and diagnostics.

13. Network Systems & Resilient Services

- **TCP Services Deep Dive:** Production connection management, keepalive, and socket options.
- **HTTP Client & Server Resilience:** Timeout budgets, circuit breakers, backoff retries, and cancellation propagation.
- **Reverse Proxies & Load Balancing:** `httputil.ReverseProxy`, health checking, and weighted routing.
- **Production TLS:** Automated certificate management, mTLS, and cipher suite hardening.
- **DNS Resolution:** Custom resolvers, DNS caching, and split-horizon resolution patterns.
- **HTTP Transport Tuning:** Idle connection pool sizing, HTTP/2 multiplexing, and dial timeouts.
- **Project Lab:** The `netcheck` network diagnostic toolkit.

14. Network Performance Engineering

- **The Netpoller:** Kernel event loops (`epoll`, `kqueue`) and goroutine scheduling integration.
- **Connection Scaling:** Scaling to 10k+ concurrent connections (C10K) with minimal heap footprint.
- **Socket Control:** Low-level syscalls, `TCP_NODELAY`, `SO_REUSEPORT`, and socket buffer tuning.
- **Resilience Under Saturation:** Backpressure, load shedding, and graceful degradation.
- **Connection Lifecycle:** Long-lived connection management, leak prevention, and HTTP/1 reuse optimizations.

15. Web Development in Go (Bookstore Track)

- **Foundations of Go Web:** Project layout, standard `net/http`, and modern `ServeMux` pattern matching.
- **Request Lifecycle:** Handlers, request parsing, headers, query parameters, and form processing.
- **Rendering & Templates:** Safe HTML templating, partials, layouts, and template caching.
- **Data Persistence:** Connecting to PostgreSQL, connection pools, migrations, and query execution.
- **Web Concurrency & Edge Safety:** Concurrency caps, request context timeouts, and edge thread safety.
- **Authentication & Sessions:** Secure cookies, session stores, password hashing, and role-based authorization (RBAC).
- **API Contracts & Frontend Comms:** JSON encoding/decoding, contract validation, and RESTful endpoints.
- **Middleware Architecture:** Chaining middleware, request logging, panic recovery, and CORS headers.

- **Operational Web:** Graceful shutdown, rate limiting, file uploads, and WebSockets.
- **Capstone Project:** Full Bookstore REST API and admin dashboard.

16. Modern Web & API Architectures

- **The GOTH Stack:** Full-stack Go with Go, Templ, HTMX, and Tailwind CSS.
- **Modern UI Patterns:** Skeleton loading, optimistic updates, and server-driven reactive UI.
- **Microservices with gRPC:** Protocol Buffers, gRPC service generation, interceptors, and streaming RPCs.
- **Modern REST APIs with Huma:** Type-safe OpenAPI 3.1 generation, automatic validation, and documentation.
- **Enterprise Data Integrations:** High-throughput PostgreSQL with `pgx` and Redis caching strategies.

17. Systems Programming & OS Engineering

- **Go Without libc:** Pure static Linux binaries (`CGO_ENABLED=0`), kernel system calls, and glibc vs. musl.
- **Process Management:** Process creation, signals, process groups, and daemon supervisors.
- **Filesystem Systems:** Atomic writes, crash-consistent filesystems, and a corruption-proof Write-Ahead Log (WAL).
- **Unix I/O:** Pipeline style CLIs, file descriptors, duplication, and stdio redirection.
- **Memory & IPC:** Memory-mapped files (`mmap`), Unix Domain Sockets, file locking (`flock`), and pidfiles.
- **System Observability:** Linux `procfs` introspection, `syslog`, `journald`, and filesystem watching (`inotify`).
- **Sandboxing & Isolation:** Pseudo-terminals (PTY), resource limits (`rlimit`), Linux capabilities, and rootless namespaces.
- **Projects:** Systems mini-tools, ops toolkit, and the `minisup` process supervisor.

18. Distributed Infrastructure

- **Service Discovery:** Health checks, consul/etcd registration, and client-side load balancing.
- **Resilience Engineering:** Retry policies, exponential backoff, jitter, and circuit breaker patterns.
- **Queue Workers & Streaming:** Asynchronous worker pools, backpressure control, and dead-letter queues.
- **Distributed Caching:** Cache invalidation strategies, stampede protection, and two-tier caching.
- **Data Consistency:** Idempotency keys, deduplication, and the Transactional Outbox pattern.
- **Coordination:** Distributed leader election and lease management.
- **Project Lab:** High-reliability distributed worker system.

19. Observability and SRE

- **Structured Logging:** Production `log/slog` architecture, handler chains, and distributed trace correlation.
- **Metrics Design:** Prometheus instrumentation, counters, gauges, histograms, and metric hygiene.
- **Distributed Tracing:** OpenTelemetry context propagation, span creation, and exporter pipelines.
- **SRE Principles:** Defining SLIs, SLOs, and error budgets in Go production services.
- **Health Probes:** Kubernetes liveness, readiness, and startup probe implementation.
- **Incident Response:** Alerting rule design, runbook automation, and Flight Recorder incident capture.
- **Project Lab:** Fully instrumented production microservice.

20. Cloud Infrastructure & Deployment

- **Containerization:** Lean multi-stage Docker builds, scratch containers, and non-root execution.
- **Cloud Platforms:** Deploying Go binaries to Render, Leapcell, and bare-metal environments.
- **Cross-Platform Ops:** Go on NixOS, FreeBSD, and edge devices.
- **CI/CD Automation:** Automated testing, vulnerability scanning, and supply-chain verification.
- **Kubernetes Engineering:** Deployments, Services, ConfigMaps, Secrets, resource requests, and gang scheduling.
- **Release Engineering:** Multi-architecture cross-compilation with GoReleaser and tag-driven GitHub Actions.

21. Security Tools and Design

- **Security Audit Tooling:** Vulnerability checking (`govulncheck`), static code auditing (`gosec`), capability auditing (`capslock`), and nil safety (`nilaway`).
- **Application Hardening:** Defensive coding, hardening Linux hosts for Go apps, and least privilege.
- **Software Architecture:** Applying SOLID design principles cleanly in Go.

22. Security Hardening & Cryptography

- **Transport Security:** Production TLS configuration, mTLS mutual authentication, and automated certificate renewal.
- **Identity & Access:** Authentication patterns, authorization engines, and JWT validation.
- **Secrets Management:** Secure secrets storage, dynamic retrieval, and rotation without service downtime.

- **Application Defense:** Input validation, sanitization, and Server-Side Request Forgery (SSRF) defense.
- **Filesystem Security:** `os.Root` traversal-resistant file access.
- **Edge Protection:** CSRF defense without tokens via `net/http` CrossOriginProtection.
- **Boundary Defense:** Error boundary translations that prevent internal data leakage.
- **Supply Chain Security:** Software Bill of Materials (SBOM), dependency auditing, and GoLeash runtime policy.
- **Memory Defense:** `runtime/secret` zeroization for sensitive cryptographic material.

23. Performance and Tooling

- **Profiling in Depth:** Profiling CPU, memory, goroutines, and block/mutex contention with `pprof` and `trace`.
- **Developer Toolkit:** Command-line tooling ecosystem (`ripgrep`, `fd`, `fzf`, `gopsutil`).
- **Static Analysis:** Writing custom AST analyzers and linters using `go/analysis`.
- **Code Generation:** Type-safe SQL with `sqlc`, enum string generation with `stringer`, and asset embedding.
- **Documentation & APIs:** Generating API documentation with `pkgsite` and developer docs.
- **Build Interception:** `-toolexec` build hooks, compiler instrumentation, and build cache integration.

24. Performance Engineering & Optimization

- **Optimization Workflow:** Evidence-based benchmarking, profiling, and performance verification.
- **Memory & Allocations:** Heap allocation reduction, memory layout alignment, and zero-allocation hot paths.
- **Contention Tuning:** Lock contention diagnosis, sharded locks, and lock-free atomic alternatives.
- **Load Testing:** Establishing performance baselines and load testing with saturation analysis.
- **Compiler Optimizations:** Profile-Guided Optimization (PGO), build flags, and inlining heuristics.
- **Hardware Acceleration:** Portable SIMD vectorization via `simd/archsimd` (`GOEXPERIMENT=simd`).
- **Algorithm Modernization:** Go 1.27 `compress/flate` vectorization and rewrite.
- **Project Lab:** Profiling and optimizing a high-traffic HTTP endpoint.

25. Go Deep Dives (Runtime & Toolchain)

- **The GMP Scheduler:** Work stealing, scheduler preemption, sysmon, and OS thread interaction.
- **The Memory Allocator:** Multi-level allocation hierarchy (mcache, mcentral, mheap), spans, and size classes.

- **The Garbage Collector:** Concurrent tri-color mark-sweep, GC pacer, write barriers, and assist ratios.
- **Data Structure Internals:** Interface boxing (`iface`/`eface`), slice headers, map hash tables, Swiss tables, and string immutability.
- **Language Mechanisms:** Defer and panic runtime stacks, dynamic stack growth, and finalizer mechanics.
- **Compiler & Linker:** Frontend parsing, AST generation, SSA intermediate representation, backend code generation, and binary size optimization.
- **Toolchain & Interop:** Modules and Minimal Version Selection (MVS), ABI internal calling conventions, assembly (`nosplit`), and `cgo` thread transitions.
- **Emerging Runtime Features:** Testing with synctest bubbles, `weak` and `unique` pointers, and WebAssembly (`wasip1`).

26. Long-Term Maintenance & Reliability

- **Dynamic Inspection:** Reflection in practice, performance implications, and alternatives.
- **Low-Level Access:** Unsafe pointer manipulation, memory layouts, and boundary safety.
- **Foreign Function Interoperability:** Cgo, FFI bindings, and C memory management.
- **API Evolution:** Deprecating APIs cleanly using `//go:fix inline` and automated refactoring tools.
- **Architectural Longevity:** Designing Go systems that remain maintainable and resilient across decades.

27. Modern Go Literature Synthesis

- **Ecosystem Patterns:** Synthesis of modern Go patterns and production practices from 2024–2025 literature:
  - *Go Programming* (2nd ed., 2024)
  - *Effective Go Recipes* (2024)
  - *Go in Practice* (2nd ed., 2025)
  - *Mastering Go* (4th ed., 2024)
  - *Let Us Go!* (2025)
  - *Automate Your Home Using Go* (2024)

28. Specialized Applications & Frontiers

- **Desktop Applications:** Cross-platform GUI development using LCL, CEF, and Webview.
- **Machine Learning & AI:** Building LLM-powered applications and AI pipelines in Go.
- **AI Agent Frameworks:** Developing autonomous agents with ADK Go Agents.
- **Advanced Streaming:** High-throughput streaming I/O pipelines and zero-copy transfers.

99. Capstone Projects & Practice Labs

- **Progressive Hands-On Projects:**
  - CLI utilities: ping tools, Linux clones (`ls`, `cat`, `grep`, `tail`, `du`, `find`, `wc`, `ps`).
  - Web & API projects: URL shortener, KV HTTP store, WebSocket chat server.
  - Infrastructure & Cloud: Kubernetes pod lister, rollout checker, event watcher, Terraform plan parser & cost estimator.
  - SRE & Observability: Prometheus exporters, probe services, and alert testers.
  - Systems & Kernels: eBPF tracepoint monitors, XDP packet counters, and Proxmox cluster managers.
  - Embedded Go: TinyGo hardware projects (LED blink, button interrupts, UART sensors, I2C monitors).
  - 200 Workout Exercises: Targeted drills across all core topics.

100. Master Appendices & Reference

- **Master Go Glossary:** Authoritative terminology and concepts across the entire language.
- **Command Cheat Sheet:** Quick reference for `go build`, `go test`, `go tool pprof`, `go tool trace`, and environment variables.
- **Go Version Release Radar:** Complete feature delta from Go 1.21 through Go 1.27.

Foundations

Why Go Exists

Why Go Exists

Overview

Go (also known as Golang) was created at Google in 2007 by Robert Griesemer, Rob Pike, and Ken Thompson. It was designed to address the challenges of building large-scale, concurrent software systems while maintaining simplicity and fast compilation times.

The Problems Go Was Designed to Solve

1. Slow Compilation Times

In large codebases (like those at Google), C++ compilation could take hours. Go was designed with fast compilation as a primary goal, enabling rapid development cycles even in massive projects.

2. Complexity in Modern Languages

Languages like C++ and Java had become increasingly complex with years of feature additions. Go aimed to be simple enough that a programmer could hold the entire language specification in their head.

3. Difficulty with Concurrency

As multi-core processors became standard, writing concurrent code with traditional threading models proved error-prone and difficult. Go introduced goroutines and channels as first-class citizens.

4. Dependency Management Chaos

"Dependency hell" plagued many ecosystems. Go's import system and later the module system were designed to solve versioning and dependency problems.

Go's Design Philosophy

```go
// Go favors clarity over cleverness
func processItems(items []string) error {
    for _, item := range items {
        if err := process(item); err != nil {
            return fmt.Errorf("processing %s: %w", item, err)
        }
    }
    return nil
}
```

Key Principles

Principle	Description
Simplicity	Less is more—remove features rather than add them
Readability	Code is read more often than written
Orthogonality	Features should work well together without special cases
Composition	Prefer composition over inheritance
Explicit over implicit	Make behavior obvious rather than magical

What Makes Go Different

Compiled, Statically Typed, Fast

```go
// Type safety caught at compile time, not runtime
var count int = 42
count = "hello"  // Compile error: cannot use "hello" as int
```

Built-in Concurrency

```go
// Goroutines are lightweight and easy to use
go func() {
    // This runs concurrently
    processData()
}()
```

Single Binary Output

```
# Go compiles to a single binary with no dependencies
$ CGO_ENABLED=0 go build -o myapp main.go
$ ./myapp  # Just works, no language runtime to install
```

On Linux that binary is typically **fully static**: it does not need libc on the target. The Go runtime issues system calls itself, so the same file runs on Debian, Alpine, NixOS, or a `scratch` container. (Darwin still links `libSystem`; the famous "no libc" story is Linux-specific.) See Go as a Systems Language for `file` / `ldd`, glibc vs musl, and when cgo puts a C library back.

Standard Formatting

```
# One true style, enforced automatically
$ gofmt -w main.go
```

Who Uses Go

Go powers critical infrastructure across the industry:

- **Docker** - Container runtime
- **Kubernetes** - Container orchestration
- **Terraform** - Infrastructure as code
- **CockroachDB** - Distributed database
- **Hugo** - Static site generator
- **Prometheus** - Monitoring system

When to Choose Go

Good fit for: - Microservices and APIs - CLI tools and DevOps tooling - Concurrent/parallel processing - Networked services - Cloud-native applications - Userspace systems software (supervisors, agents, portable Linux binaries)

Consider alternatives for: - GUI desktop applications - Mobile development - Kernel modules, drivers, firmware, and hard real-time (C or Rust) - Data science/ML (Python ecosystem is stronger)

Summary

Go exists because the software industry needed a language that could:

1. **Compile quickly** at any scale
2. **Handle concurrency** naturally and safely

3. **Remain simple** despite solving complex problems
4. **Produce efficient binaries** that deploy easily — on Linux, without a libc dependency

Understanding these origins helps you write more idiomatic Go code that embraces the language's philosophy.

Related Topics

- Learning Go in 2026
- Installing and Running Go
- How Go Code Is Organized
- Go as a Systems Language

More examples

Example: clarity over cleverness

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import (
    "fmt"
    "strings"
)

func process(item string) error {
    if strings.TrimSpace(item) == "" {
        return fmt.Errorf("empty item")
    }
    fmt.Println("ok:", item)
    return nil
}

func processItems(items []string) error {
    for _, item := range items {
        if err := process(item); err != nil {
            return fmt.Errorf("processing %q: %w", item, err)
        }
    }
    return nil
}

func main() {
    if err := processItems([]string{"docker", "k8s", "  "}); err != nil {
```

```
        fmt.Println("stopped:", err)
    }
}
```

Expected:

```
ok: docker
ok: k8s
stopped: processing "  ": empty item
```

Example: static types catch mistakes early

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```
package main

import "fmt"

func main() {
    var count int = 42
    // Uncommenting the next line fails at compile time:
    // count = "hello"
    fmt.Println("count:", count)

    // Conversions are explicit when you need them.
    label := fmt.Sprintf("n=%d", count)
    fmt.Println(label)
}
```

Expected:

```
count: 42
n=42
```

Runnable example

Save as `main.go`. From an empty directory:

```
go mod init example
go run .
```

```
package main

import (
```

```go
    "fmt"
    "sync"
    "time"
)

// process is intentionally simple and explicit: no inheritance, no magic.
func process(item string) error {
    if item == "" {
        return fmt.Errorf("empty item")
    }
    // Simulate a small unit of work.
    time.Sleep(20 * time.Millisecond)
    fmt.Printf("processed: %s\n", item)
    return nil
}

func main() {
    items := []string{"api", "worker", "cli", ""}
    var wg sync.WaitGroup
    var mu sync.Mutex
    var failures int

    // Goroutines are first-class: cheap concurrency without a thread pool API.
    for _, item := range items {
        wg.Add(1)
        go func(item string) {
            defer wg.Done()
            if err := process(item); err != nil {
                mu.Lock()
                failures++
                mu.Unlock()
                fmt.Printf("error: %v\n", err)
            }
        }(item)
    }

    wg.Wait()
    fmt.Printf("done; failures=%d\n", failures)
}
```

Expected output (illustrative; order of lines may vary):

```
processed: api
processed: worker
processed: cli
error: empty item
done; failures=1
```

What to notice: - Concurrency is a few lines: `go func()` plus `WaitGroup`—no framework. - Errors are values checked with `if err != nil`, not thrown exceptions. - Empty-item handling is visible in the control flow; nothing is "magical." - The program still finishes cleanly after concurrent work.

Try next: Change `time.Sleep` to zero and re-run a few times. Does line order stay the same?

Learning Go in 2026

Learning Go in 2026

Why Go is Worth Learning in 2026

Go (Golang) has moved beyond hype into the backbone of modern engineering. In 2026, it remains the language of choice for cloud-native systems, DevOps tooling, and scalable backend infrastructure.

Companies love Go for: - **Simplicity**: Minimalist syntax that is easy to read and maintain. - **Performance**: Compiled speed that rivals C++ and Java. - **Built-in Concurrency**: First-class support for parallel processing. - **Easy Deployment**: Compiles to a single static binary with no external dependencies. On Linux that includes **no libc** — the same file runs on Debian, Alpine, and `scratch` containers. See Go as a Systems Language.

The Go Mindset (Most People Get This Wrong)

Go is "boring" by design, and that is its greatest superpower. To succeed in Go, you must shift your mindset:

Instead of…	Go Prefers…
Clever Code	Clear Code
Deep Inheritance	Simple Composition
Complex Abstractions	Concrete Implementations
Magic/Implicit Behavior	Explicit Logic

> **Pro Tip**: Once you stop fighting Go's simplicity and embrace its explicitness, you become shockingly productive.

The Fastest Roadmap to Learn Go

1. Learn Only the Essentials First

Don't get bogged down in advanced features immediately. Focus on the core: - Variables & Types - Functions & Error Handling - Control Flow (`if`, `for`, `switch`) - Structs & Interfaces (The heart of Go's type system) - Packages & Modules

2. Learn Concurrency Early

Go without concurrency is like a ship without a sail. Don't delay learning: - **Goroutines**: Lightweight threads. - **Channels**: How goroutines communicate. - **Select**: Managing multiple channel operations. - **Sync Package**: WaitGroups and Mutexes for state management.

3. Build Tiny Programs Every Day

Forget massive projects at the start. Build real-world utility tools: - CLI Calculator - Log File Analyzer - API Health Checker - Simple REST API

The 3-Layer Practice System

This system ensures you don't just consume content but actually learn to build:

1. **Layer 1: Read**: Read clean, idiomatic Go code (like the standard library). Observe naming conventions and error handling patterns.
2. **Layer 2: Write**: Rewrite examples from memory. Don't copy-paste. Intentionally break things to see how the compiler and runtime react.
3. **Layer 3: Ship**: Push your code to GitHub. Use Go to solve your own small problems, like automating a repetitive task.

Go Projects That Make You Job-Ready

Build these in order to gain real-world experience: 1. **REST API**: Use the standard library or a lightweight router with middleware and JSON handling. 2. **CLI Tool**: Use flags and arguments to create a useful command-line utility. 3. **Concurrent Worker Pool**: Process data in parallel using goroutines and channels. 4. **Log Monitor**: A tool that watches files or streams and alerts on patterns. 5. **Microservice**: A simple service with environment-based configuration and health checks.

Why Go is a DevOps Superpower

If you are in DevOps or SRE, Go is a force multiplier. It allows you to: - Build internal CLI tools that are easily shared. - Write robust automation scripts that replace fragile shell scripts. - Extend CI/CD pipelines with custom logic. - Create Kubernetes Operators or Custom Resource Definitions (CRDs).

Go turns "glue code" into reliable software engineering.

Memory & Consistency

- **Practice daily**: 30–60 minutes is better than a 5-hour marathon.
- **Explain concepts**: Write short blog posts or notes to solidify your understanding.
- **Consistency beats intensity**: Every single time.

Go 1.27 Migration Checklist (Shared)

Use this checklist when upgrading existing projects to Go 1.27 (from 1.26 or earlier). Go 1.25 is out of support as of the 1.27 release. Feature-by-feature notes and runnable samples: What's New in Go 1.27 (official release notes).

```
upgrade path

audit current toolchain
      |
      v
update go.mod/toolchain (go 1.27)
      |
      v
run modernizers + full tests
      |
      v
opt into json/v2 / uuid / leak profile as needed
      |
      v
ship + monitor memory/latency regressions
```

1. Pin Toolchain and Module Versions

```
go version
go mod tidy
```

If you use a `toolchain` directive, set it explicitly (`toolchain go1.27.1` at this writing) and keep CI aligned. `go mod tidy` on modules with `go 1.27` consolidates `require` blocks into the standard direct + indirect layout.

2. Run Modernization and Safety Baseline

```
go fix -modernize ./...
go vet ./...
govulncheck ./...
```

Go 1.27 adds `atomictypes`, `embedlit`, `slicesbackward`, and `unsafefuncs` modernizers. The `waitgroup` analyzer is now named `waitgroupgo`.

3. Re-Baseline Tests and Benchmarks

```
go test ./... -race -shuffle=on -count=1
go test ./... -bench=. -benchmem
```

Adopt `B.Loop`, `testing/synctest` (including `synctest.Sleep`), and `httptest.NewTestServer` for HTTP tests that should run inside a synctest bubble.

4. Language and stdlib opt-ins

- **Generic methods** and promoted field keys in struct literals compile on 1.27+; they are not back-portable.
- **`encoding/json`** is now backed by the v2 implementation. Behavior is preserved; error text may differ. New code can import `encoding/json/v2` for stricter defaults. Rollback: `GOEXPERIMENT=nojsonv2` (temporary).
- **`uuid`** is in the standard library; prefer it over `github.com/google/uuid` for new IDs.
- **`goroutineleak`** pprof profile is generally available (`/debug/pprof/goroutineleak`).

5. Enable Experiments Deliberately

Only opt into experiments if there is a clear payoff and rollback plan:

```
GOEXPERIMENT=simd go test ./...
GOEXPERIMENT=nogreenteagc go test ./...
GOEXPERIMENT=nosizespecializedmalloc go test ./...
```

Document where each experiment is used and how to disable it quickly.

6. Observe Production After Rollout

Track: - P95/P99 latency - GC pause and heap goal trend - Error rates and timeout rates - Goroutine count (and the leak profile if you enable pprof)

Run canary first, then broad rollout.

Summary

Don't try to "finish" Go. Use Go to solve real problems, and the language will teach itself to you along the way. Stay curious, keep building, and embrace the simplicity.

Related Topics

- Why Go Exists
- Installing and Running Go
- Core Go Commands
- Go as a Systems Language

More examples

Example: API health-check style probe (no network)

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

// probe models a tiny health checker without opening sockets-practice the shape first.
func probe(status int) (healthy bool, detail string) {
    if status >= 200 && status < 400 {
        return true, fmt.Sprintf("status %d ok", status)
    }
    return false, fmt.Sprintf("status %d unhealthy", status)
}

func main() {
    for _, code := range []int{200, 503} {
        ok, detail := probe(code)
        fmt.Printf("code=%d healthy=%v (%s)\n", code, ok, detail)
    }
}
```

Expected:

```
code=200 healthy=true (status 200 ok)
code=503 healthy=false (status 503 unhealthy)
```

Example: composition over inheritance

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"
```

```go
type Logger struct {
    Prefix string
}

func (l Logger) Info(msg string) {
    fmt.Printf("%s %s\n", l.Prefix, msg)
}

// Service composes a Logger instead of extending a base class.
type Service struct {
    Logger
    Name string
}

func main() {
    svc := Service{
        Logger: Logger{Prefix: "[svc]"},
        Name:   "billing",
    }
    svc.Info("started " + svc.Name)
    svc.Info("ready")
}
```

Expected:

```
[svc] started billing
[svc] ready
```

Runnable example

A tiny CLI calculator—the kind of daily practice project from this chapter's roadmap.

Save as `main.go`. From an empty directory:

```
go mod init example
go run . add 3 9
go run . mul 4 5
go run . div 10 0
```

```go
package main

import (
    "fmt"
    "os"
    "strconv"
```

```go
)

func main() {
    if len(os.Args) != 4 {
        fmt.Fprintln(os.Stderr, "usage: go run . <add|sub|mul|div> <a> <b>")
        os.Exit(2)
    }

    op := os.Args[1]
    a, errA := strconv.ParseFloat(os.Args[2], 64)
    b, errB := strconv.ParseFloat(os.Args[3], 64)
    if errA != nil || errB != nil {
        fmt.Fprintln(os.Stderr, "operands must be numbers")
        os.Exit(1)
    }

    var result float64
    switch op {
    case "add":
        result = a + b
    case "sub":
        result = a - b
    case "mul":
        result = a * b
    case "div":
        if b == 0 {
            fmt.Fprintln(os.Stderr, "division by zero")
            os.Exit(1)
        }
        result = a / b
    default:
        fmt.Fprintf(os.Stderr, "unknown op %q\n", op)
        os.Exit(1)
    }

    fmt.Printf("%s(%.2f, %.2f) = %.2f\n", op, a, b, result)
}
```

Expected output (illustrative):

```
add(3.00, 9.00) = 12.00
mul(4.00, 5.00) = 20.00
division by zero
```

What to notice: - Core essentials only: args, types, `switch`, errors—matching the "learn essentials first" path. - Invalid input exits with a clear message; no exceptions. - Floats come from `strconv` so type conversion is explicit. - One file, no framework: ship-ready practice.

Try next: Add a `mod` (remainder) operation and support integer-only mode.

The Go Philosophy

The Go Philosophy: Why "Boring" is a Superpower

In a software landscape obsessed with the "next big thing," Go stands out by explicitly trying *not* to be clever. Since its inception at Google in 2007 (and release in 2009), Go has adhered to a philosophy of simplicity, stability, and readability.

As we move through 2026, this philosophy hasn't just survived; it has proven to be a competitive advantage.

Boring Software is Reliable Software

The core tenet of Go's philosophy is often summarized as: **"Clear is better than clever."**

1. Readability Over Conciseness

Go is verbose. This is intentional. When you read Go code, you don't need to hold a complex mental model of hidden states, monkey-patched methods, or magical frameworks in your head. The control flow is visible. * **No inheritance:** You cannot hide logic in a base class three layers up. * **No method overloading:** `DoWork` and `DoWorkWithContext` are distinct. You know exactly which one is called. * **No circular dependencies:** The compiler forbids it, forcing you to keep your architecture clean.

2. One Way To Do It

While languages like Ruby or Scala offer ten ways to iterate over a list, Go gives you one: the `for` loop. This reduced cognitive load means you spend less time debating *style* and more time solving *problems*.

3. Stability is a Feature

Go's Go 1 compatibility promise is legendary. Code written in 2012 will likely compile and run today with little to no modification. In 2026, where JavaScript frameworks churn every six months, this stability allows organizations to build for the decade, not the demo.

> "Software engineering is what happens to programming when you add time and other programmers." — Russ Cox

The 2026 Context: AI and Complexity

With the rise of AI-generated code (Copilot, Gemini, etc.), simplicity is more vital than ever. * **AI generates bugs:** Complex languages with hidden magic allow AI to halluncinate plausible but broken code. Go's explicit error handling and type system catch these issues early. * **Reviewability:** Humans still need to review AI code. Go's simplicity makes it easier for a human to verify that the code does what the LLM claims it does.

Key Principles Summarized

Principle	Description
Simplicity	The language spec is small enough to hold in your head.
Orthogonality	Features interact in predictable ways (e.g., Goroutines work the same in `main` or a `handler`).
Punt Complexity	If a feature is complex to implement in the compiler *and* hard to understand (e.g., complex metaprogramming), Go leaves it out.

Conclusion

Go is designed for **engineering**, not just programming. It accepts that software is written once but read hundreds of times. By choosing "boring," Go enables you to build exciting, scalable, and reliable systems.

More examples

Example: one obvious loop

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

// One way to walk a list: a plain for-range. No hidden iterator magic.
func sum(nums []int) int {
    total := 0
    for _, n := range nums {
        total += n
    }
```

```go
    return total
}

func main() {
    values := []int{1, 2, 3, 4}
    fmt.Println("sum:", sum(values))

    // Explicit empty-slice path-readable without special syntax.
    var empty []int
    fmt.Println("empty sum:", sum(empty))
}
```

Expected:

```
sum: 10
empty sum: 0
```

Example: no overloading—distinct names

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import (
    "fmt"
    "time"
)

// Go forbids method overloading. Names say what differs.
func greet(name string) string {
    return "hello, " + name
}

func greetWithTimeout(name string, d time.Duration) string {
    // Duration is part of the API surface, not a hidden default.
    _ = d
    return fmt.Sprintf("hello, %s (timeout %s)", name, d)
}

func main() {
    fmt.Println(greet("gopher"))
    fmt.Println(greetWithTimeout("gopher", 2*time.Second))
}
```

Expected:

```
hello, gopher
hello, gopher (timeout 2s)
```

Runnable example

"Boring Go": zero values, explicit errors, one obvious control flow—nothing clever.

Save as `main.go`. From an empty directory:

```
go mod init example
go run .
```

```go
package main

import (
    "errors"
    "fmt"
)

// Config is usable at its zero value: no constructor magic required.
type Config struct {
    Host    string
    Port    int
    Verbose bool
}

func (c Config) addr() string {
    host := c.Host
    if host == "" {
        host = "localhost"
    }
    port := c.Port
    if port == 0 {
        port = 8080
    }
    return fmt.Sprintf("%s:%d", host, port)
}

func connect(cfg Config) error {
    if cfg.Port < 0 {
        return errors.New("port cannot be negative")
    }
    if cfg.Verbose {
        fmt.Println("connecting with verbose logging")
    }
```

```go
    fmt.Println("listening on", cfg.addr())
    return nil
}

func main() {
    // Zero-value config is intentional and readable.
    var cfg Config
    if err := connect(cfg); err != nil {
        fmt.Println("error:", err)
        return
    }

    // Override only what differs from defaults.
    cfg.Host = "0.0.0.0"
    cfg.Port = 9090
    cfg.Verbose = true
    if err := connect(cfg); err != nil {
        fmt.Println("error:", err)
        return
    }

    // Explicit failure path-clear is better than clever.
    cfg.Port = -1
    if err := connect(cfg); err != nil {
        fmt.Println("error:", err)
    }
}
```

Expected output (illustrative):

```
listening on localhost:8080
connecting with verbose logging
listening on 0.0.0.0:9090
error: port cannot be negative
```

What to notice: - Zero values are defaults you can rely on (`""`, `0`, `false`). - Error handling is visible: every call site checks `err`. - No inheritance, no DI container—just a struct and functions. - Control flow is linear and reviewable (including by AI reviewers).

Try next: Add a `Timeout` field with a zero-value default of 30s and print it in `connect`.

The 3-Layer Practice System

The 3-Layer Practice System: Read, Write, Ship

Learning a language like Go isn't about memorizing syntax; it's about building muscle memory. In 2026, the most effective way to master Go is through a cyclic system we call **Read, Write, Ship**.

Layer 1: Read (Input)

You cannot write idiomatic Go if you haven't seen idiomatic Go. Good writers are always avid readers.

What to Read

1. **The Standard Library:** The Go standard library is the gold standard.
   - Read `net/http` to see how interfaces are used.
   - Read `encoding/json` to understand reflection and tags.
   - Read `time` to see efficient data structures.
2. **High-Quality Open Source:**
   - Caddy Server: For modular architecture and interfaces.
   - HashiCorp Vault/Terraform: For CLI patterns and plugin systems.
   - NATS Server: For high-performance concurrency.

How to Read

Don't just skim. **Trace.** Pick a function call (like `http.ListenAndServe`) and follow it down the rabbit hole until you hit a syscall or assembly.

Layer 2: Write (Output)

Tutorials give you a false sense of competence. You must build things that break.

The "Clone" Technique

Don't wait for a unique idea. Clone existing tools to understand how they work. * **Beginner:** Write a CLI tool (clone `ls` or `cat`). * **Intermediate:** Write a load balancer (clone basic Nginx features). * **Advanced:** Write a distributed key-value store (clone a tiny Redis).

Constraints

Force yourself out of comfort zones: * Write a program without using `struct` tags. * Write a concurrent program without `sync.Mutex` (only channels). * Write a web server without a framework (only `net/http`).

Layer 3: Ship (Feedback)

Code on your laptop doesn't count. "Done" means deployed.

The Feedback Loop

1. **Linting:** `golangci-lint` is your first critic. Configure it to be strict.
2. **Code Review:** Even if solo, use PRs. Review your own diffs. AI tools can effectively review PRs now—use them to spot potential bugs.
3. **Production:** functionality implies running in a Linux container, on a cloud provider, with real traffic.
   - Deploy to **Render**, **Leapcell**, or a **VPS**.
   - Set up **Observability** (logs/metrics). You don't know your code until you see it fail in prod.

Summary

- **Read** to load the patterns into your brain.
- **Write** to test your understanding of those patterns.
- **Ship** to validate that your code actually solves the problem in the real world.

> "Amateurs practice until they get it right. Professionals practice until they can't get it wrong."

More examples

Example: clone `cat` (Layer 2 write)

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import (
    "fmt"
    "io"
    "os"
```

```go
    "strings"
)

func cat(r io.Reader, w io.Writer) error {
    _, err := io.Copy(w, r)
    return err
}

func main() {
    // Demo when no args: practice without creating files.
    if len(os.Args) < 2 {
        r := strings.NewReader("line one\nline two\n")
        fmt.Print("(demo)\n")
        if err := cat(r, os.Stdout); err != nil {
            fmt.Fprintln(os.Stderr, err)
            os.Exit(1)
        }
        return
    }

    for _, path := range os.Args[1:] {
        f, err := os.Open(path)
        if err != nil {
            fmt.Fprintln(os.Stderr, err)
            os.Exit(1)
        }
        err = cat(f, os.Stdout)
        f.Close()
        if err != nil {
            fmt.Fprintln(os.Stderr, err)
            os.Exit(1)
        }
    }
}
```

Expected:

```
(demo)
line one
line two
```

Example: intentional breakage (Layer 2 write)

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import (
    "fmt"
    "strconv"
)

// parsePositive demonstrates checking the compiler/runtime reaction to bad input.
func parsePositive(s string) (int, error) {
    n, err := strconv.Atoi(s)
    if err != nil {
        return 0, fmt.Errorf("not an int %q: %w", s, err)
    }
    if n <= 0 {
        return 0, fmt.Errorf("need positive, got %d", n)
    }
    return n, nil
}

func main() {
    for _, s := range []string{"42", "0", "nope"} {
        n, err := parsePositive(s)
        if err != nil {
            fmt.Printf("%q -> error: %v\n", s, err)
            continue
        }
        fmt.Printf("%q -> %d\n", s, n)
    }
}
```

Expected:

```
"42" -> 42
"0" -> error: need positive, got 0
"nope" -> error: not an int "nope": strconv.Atoi: parsing "nope": invalid syntax
```

Runnable example

Layer 2 practice: a tiny clone of `wc -l` / `wc -w` (read stdin or a string, write counts).

Save as `main.go`. From an empty directory:

```bash
go mod init example
go run .
printf 'hello world\nfoo bar baz\n' | go run .
```

```go
package main

import (
    "bufio"
    "fmt"
    "io"
    "os"
    "strings"
)

func count(r io.Reader) (lines, words, bytes int, err error) {
    br := bufio.NewReader(r)
    for {
        line, e := br.ReadString('\n')
        if len(line) > 0 {
            bytes += len(line)
            // Count a line if it ends with \n or is a final partial line.
            if strings.HasSuffix(line, "\n") || e == io.EOF {
                lines++
            }
            words += len(strings.Fields(line))
        }
        if e == io.EOF {
            break
        }
        if e != nil {
            return lines, words, bytes, e
        }
    }
    return lines, words, bytes, nil
}

func main() {
    // Demo mode when stdin is a terminal: practice without piping.
    info, _ := os.Stdin.Stat()
    var r io.Reader = os.Stdin
    if (info.Mode() & os.ModeCharDevice) != 0 {
        r = strings.NewReader("hello world\nfoo bar\n")
        fmt.Println("(demo input)")
    }

    lines, words, bytes, err := count(r)
    if err != nil {
        fmt.Fprintln(os.Stderr, "count:", err)
        os.Exit(1)
    }
    fmt.Printf("lines=%d words=%d bytes=%d\n", lines, words, bytes)
```

```
}
```

Expected output (illustrative):

```
(demo input)
lines=2 words=4 bytes=20
```

With a pipe:

```
lines=2 words=5 bytes=24
```

What to notice: - You wrote a real tool that reads input and prints observables—not a syntax drill. - Stdlib only: `bufio`, `io`, `strings`—good "read the standard library" practice. - Errors are checked; failure exits non-zero (ship-ready habit). - Demo mode lets you re-run without pipes while learning.

Try next: Accept a filename as `os.Args[1]` and fall back to stdin when missing.

Transitioning to Go

Transitioning to Go: For Java and Node.js Developers

If you are coming from an Object-Oriented (Java/C#) or Event-Driven (Node.js/JS) background, Go will feel familiar yet frustratingly different. This chapter maps your existing mental models to Go's reality.

For the Java Developer

Java Concept	Go Equivalent	The Shift in Thinking
Class	**Struct**	Data and behavior are separate. Classes bundles them; Go defines a `type` (data) and `func (t MyType)` (behavior).
Interface	**Interface**	**Implicit.** You don't `implement` interfaces. If you have the methods, you satisfy the interface. This decouples packages.
Exception	**Error**	Errors are **values**, not control flow events. You check them (`if err != nil`), you don't catch them.
ThreadPool	**Goroutines**	Threads are expensive (MBs); Goroutines are cheap (KBs). You can spawn 100k goroutines without blinking.
Annotation	**Struct Tag**	Used strictly for metadata (JSON, DB), not for behavior injection (like Spring Magic).
Maven/Gradle	**Go Modules**	Simplistic dependency graph. No `mvn install`. Just `go mod tidy`.

The "Spring" Trap

Don't try to build "Spring in Go". Dependency Injection containers are largely unnecessary in Go. Pass dependencies explicitly in constructors (struct factories).

```go
// Java: @Autowired Service service;
// Go:
func NewServer(db *sql.DB, logger *Logger) *Server {
    return &Server{db: db, logger: logger}
}
```

For the Node.js Developer

Node.js Concept	Go Equivalent	The Shift in Thinking
Promise / Async Await	**Blocking Code**	Go code *looks* synchronous but runs concurrently. The runtime handles the I/O scheduling. No "Callback Hell" or unwieldy `await` chains.
npm	**Go Modules**	No `node_modules` black hole. Dependencies are compiled into a single binary.
Event Loop	**Go Scheduler**	Node has one thread; block it and you die. Go has M:N scheduling; if one goroutine blocks, others keep running on other OS threads.
Dynamic Types	**Static Types**	You catch typos at compile time, not runtime. `interface{}` (or `any`) exists but use it sparingly.
Express/NestJS	**net/http**	The stdlib is production-ready. You often don't need a framework. `Chi` or `Echo` are light routers, not heavy frameworks.

The "Concurrency" Trap

In Node, you rely on `Promise.all` for concurrency. In Go, you use **Channels** and **WaitGroups**.

```
// Node
await Promise.all([task1(), task2()]);
```

```
// Go
var wg sync.WaitGroup
wg.Add(2)
go func() { defer wg.Done(); task1() }()
go func() { defer wg.Done(); task2() }()
wg.Wait()
```

Universal Truths in Go

1. **Composition over Inheritance:** You don't extend classes. You embed structs.
2. **Explicit is better than Implicit:** No magic checking. No global state if avoidable.
3. **Values matter:** Learn distinction between passing a copy (`T`) vs passing a pointer (`*T`). In JS/Java, object references are implicit; in Go, pointers are explicit.

Summary

- **Java Docs:** Drop the AbstractFactoryPatterns. Build simple structs.
- **Node Devs:** Embrace the type system and true parallelism (multi-core).
- **Everyone:** Respect the error. `if err != nil` is the heartbeat of a Go program.

More examples

Example: explicit constructor DI (not Spring)

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```
package main

import "fmt"

type Clock interface {
    NowLabel() string
}

type fixedClock struct{ label string }

func (c fixedClock) NowLabel() string { return c.label }

type Server struct {
```

```go
    dbName string
    clock  Clock
}

// NewServer takes dependencies as parameters-no container, no annotations.
func NewServer(dbName string, clock Clock) *Server {
    return &Server{dbName: dbName, clock: clock}
}

func (s *Server) Handle() string {
    return fmt.Sprintf("%s @ %s", s.dbName, s.clock.NowLabel())
}

func main() {
    srv := NewServer("users", fixedClock{label: "t0"})
    fmt.Println(srv.Handle())
    srv2 := NewServer("orders", fixedClock{label: "t1"})
    fmt.Println(srv2.Handle())
}
```

Expected:

```
users @ t0
orders @ t1
```

Example: value vs pointer is explicit

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

type Counter struct{ N int }

func bumpValue(c Counter) {
    c.N++
}

func bumpPointer(c *Counter) {
    c.N++
}

func main() {
    c := Counter{N: 1}
    bumpValue(c)
```

```go
    fmt.Println("after value bump:", c.N) // still 1 - copy

    bumpPointer(&c)
    fmt.Println("after pointer bump:", c.N) // 2 - shared
}
```

Expected:

```
after value bump: 1
after pointer bump: 2
```

Runnable example

Java → errors as values; Node → concurrency without `Promise.all` ceremony.

Save as `main.go`. From an empty directory:

```
go mod init example
go run .
```

```go
package main

import (
    "errors"
    "fmt"
    "sync"
    "time"
)

// Service is a plain struct: data + methods, no class hierarchy.
type Service struct {
    Name string
}

func NewService(name string) *Service {
    // Explicit constructor-style factory-no DI container.
    return &Service{Name: name}
}

func (s *Service) Fetch(id int) (string, error) {
    if id <= 0 {
        return "", errors.New("id must be positive")
    }
    time.Sleep(30 * time.Millisecond)
    return fmt.Sprintf("%s-item-%d", s.Name, id), nil
```

```go
}

func main() {
    svc := NewService("catalog")

    // Errors are values, not exceptions.
    if _, err := svc.Fetch(0); err != nil {
        fmt.Println("expected failure:", err)
    }

    // Concurrent work looks sequential in each goroutine.
    ids := []int{1, 2, 3}
    results := make([]string, len(ids))
    errs := make([]error, len(ids))

    var wg sync.WaitGroup
    for i, id := range ids {
        wg.Add(1)
        go func(i, id int) {
            defer wg.Done()
            val, err := svc.Fetch(id)
            results[i], errs[i] = val, err
        }(i, id)
    }
    wg.Wait()

    for i := range ids {
        if errs[i] != nil {
            fmt.Printf("id %d error: %v\n", ids[i], errs[i])
            continue
        }
        fmt.Printf("id %d -> %s\n", ids[i], results[i])
    }
}
```

Expected output (illustrative):

```
expected failure: id must be positive
id 1 -> catalog-item-1
id 2 -> catalog-item-2
id 3 -> catalog-item-3
```

What to notice: - No `@Autowired`: dependencies would be constructor parameters (here, none). - `if err != nil` replaces try/catch as the normal control path. - `WaitGroup` + goroutines replace `Promise.all` for parallel work. - Struct methods attach behavior without inheritance.

Try next: Pass a `*log.Logger` into `NewService` and log each fetch—explicit DI.

What's New in Go 1.27

What's New in Go 1.27

Go **1.27** shipped in August 2026 (release notes). The compatibility promise still holds: almost all existing programs compile and run. This chapter is the map of what actually changed—language, `go` command, runtime, and the standard library—so the rest of this book can assume **1.27** as the baseline.

Pin new modules with `go 1.27` (and `toolchain go1.27.1` or later when you want a specific compiler). Go 1.25 is out of support as of this release.

Language

Generic methods

A method may declare **its own** type parameters. That is different from a method on a generic type (`func (s *Stack[T]) Push(T)`), which has existed since 1.18.

The standard library uses this on `math/rand/v2`: `(*Rand).N[Int intType](n Int) Int`.

Save as `main.go`:

```go
package main

import (
    "fmt"
    "math/rand/v2"
)

func main() {
    r := rand.New(rand.NewPCG(1, 2))
    fmt.Println(r.N(10))
    fmt.Println(r.N(int64(1_000)))
}
```

```go
go run .
```

Output is deterministic for this PCG seed:

```
7
616
```

Limits: interface methods cannot declare type parameters, and a generic method cannot implement an interface method. Prefer a package-level generic function when the operation is not namespaced on a type. See Methods and Receivers and Why Generics Matter.

Promoted field keys in struct literals

Stemming from proposal #9859 (open since 2015), Go 1.27 resolves a 16-year asymmetry between field access and struct literal initialization. Previously, you could read promoted fields on embedded structs with simple dot notation (`e.City`), but constructing the instance required wrapping every embedded type in explicit nested literals.

In Go 1.27, any unambiguously promoted field from an anonymously embedded struct can serve directly as a key in composite literals:

```go
package main

import "fmt"

type Address struct {
    City  string
    State string
}

type Contact struct {
    Email string
}

type Employee struct {
    Address // anonymous embedding
    Contact // anonymous embedding
    Title   string
}

func main() {
    // Go 1.27: clean, flattened composite literal
    e := Employee{
        City:  "Lisbon",
        State: "PT",
        Email: "dev@example.com",
        Title: "Engineer",
    }
    fmt.Printf("%s (%s, %s): %s\n", e.Title, e.City, e.State, e.Email)
}
```

```
Engineer (Lisbon, PT): dev@example.com
```

Key rules to know: - **Multi-level embedding**: Promoted fields flatten across arbitrary depths as long as the promotion is unambiguous. - **Ambiguity**: If two embedded structs define the same field at the same depth, the compiler refuses to guess (`unknown field ... in struct literal`) and requires the explicit nested form. - **No split initialization**: Mixing promoted fields and enclosing embedded struct literals for the same embedded type (e.g., `Employee{Address: Address{...}, City: ...}`) produces a compiler error (`cannot specify promoted field ... and enclosing embedded field ...`). - **Tooling migration**: Run `go fix -embedlit ./...` (or preview with `-diff`) to automatically simplify legacy composite literals across your packages.

Full architectural rules, multi-level examples, and design tradeoffs: Structs and Data Modeling.

Function type inference

Inference now applies in every assignment-shaped context: composite literals, conversions, and channel sends, not only direct calls. See Generic Functions.

go command and tools

Change	What you do
`go test` runs the **`stdversion`** vet check by default	Using a stdlib symbol newer than your `go` directive is now a test failure
`go test -json`	`"Action":"output"` events may include `"OutputType"` (`error`, `error-continue`, `frame`)
`go doc pkg@version`	Docs for a specific module version
`go doc -ex`	List executable examples; `go doc bytes.ExampleBuffer` prints the example source
`go fix` modernizers	New: `atomictypes`, `embedlit`, `slicesbackward`, `unsafefuncs`. `waitgroup` is now `waitgroupgo`. `fmtappendf` was removed
`go mod tidy` (`go 1.27+`)	Merges extra `require` blocks into the standard **direct** then **indirect** layout
`bzr`	The `go` command no longer fetches Bazaar-hosted modules
`GODEBUG` in `go.mod` / `//go:debug`	Removed settings are accepted only at their final default; an old value fails the build
`go tool trace -http=:6060`	Listens on **localhost** only (same as `pprof`). Use `-http=0.0.0.0:6060` to bind all interfaces
Response files (`@file`)	`compile`, `link`, `asm`, `cgo`, `cover`, `pack` accept GCC-compatible response files

```
go test ./...                # includes stdversion
go doc encoding/json/v2
go doc -ex bytes.Buffer
go fix -modernize ./...
go mod tidy
```

Runtime

- **Size-specialized malloc** for some objects under 80 bytes (up to ~30% cheaper those allocs, ~1% in allocation-heavy programs, ~60 KB extra binary). Opt out: `GOEXPERIMENT=nosizespecializedmalloc` (expected gone in 1.28).
- **`goroutineleak` pprof profile** is generally available (`/debug/pprof/goroutineleak`). The `goroutineleakprofile` experiment flag is deleted. See Goroutine leaks.
- Tracebacks for `go 1.27` modules include **pprof goroutine labels** in the header. Disable with `GODEBUG=tracebacklabels=0`.
- **`asynctimerchan` is gone.** Channels from package `time` are always unbuffered. See Time.
- Closures get simpler symbol names; do not compare function pointers for identity.

Standard library

`encoding/json/v2` (GA)

`encoding/json` is now **backed by the v2 engine**. Behavior of the v1 API is preserved; error text may differ; unmarshal is faster. New code that wants stricter defaults (reject invalid UTF-8, reject duplicate names) should import `encoding/json/v2`. Emergency rollback: `GOEXPERIMENT=nojsonv2` (temporary). See JSON.

`uuid`

Stdlib `uuid` generates and parses UUIDs. Prefer it over `github.com/google/uuid` for new IDs.

```go
package main

import (
    "fmt"
    "uuid"
)

func main() {
    id := uuid.New()
    fmt.Println(id)
    parsed, err := uuid.Parse(id.String())
    if err != nil {
        panic(err)
    }
    fmt.Println(parsed == id)
}
```

`crypto/mldsa`

Post-quantum ML-DSA (FIPS 204), with `crypto/x509` and `crypto/tls` (`MLDSA44` / `65` / `87`). Opt-in for new PKI, not a silent swap for existing ECDSA certs.

Experimental `simd` / `simd/archsimd`

Portable vector types (`Int8s`, `Float32s`, …) plus architecture-specific ops (amd64, arm64 Neon, Wasm). Enable with `GOEXPERIMENT=simd`. Not production-stable.

Other library notes

Package	1.27
`bytes` / `strings`	`CutLast`
`testing/synctest`	`Sleep` (sleep + wait)
`net/http/httptest`	`NewTestServer` (in-memory, synctest-safe)
`net/url`	`URL.Clone`, `Values.Clone`
`math/rand/v2`	`(*Rand).N` generic method
`net/http`	HTTP/2 client priority (RFC 9218); `Server.MaxHeaderValueCount`; HTTP/1 body drain on close
`crypto/tls`	ML-DSA; `MLKEM1024`; `ConnectionState.LocalCertificate`; several old `tls*` GODEBUGs removed
`unicode`	Unicode 17
`compress/flate`	Faster; encoded bytes may differ (also zip/gzip/zlib/png)

```go
package main

import (
    "fmt"
    "strings"
)

func main() {
    dir, file, ok := strings.CutLast("desk/shift/ticket.txt", "/")
    fmt.Println(dir, file, ok)
}
```

```
desk/shift ticket.txt true
```

Ports

- **macOS:** 13 Ventura or later. Support for older releases is gone.
- **linux/ppc64 (big-endian):** ELFv2 ABI; cgo, PIE, and external linking now work with an ELFv2 runtime.

Compiler / linker

Relative `//line` filenames resolve against the file that contains the directive. On macOS the linker accepts `-macos` / `-macsdk` for `LC_BUILD_VERSION`.

What to do in this book next

1. Upgrading to Go 1.27: What Breaks, What to Adopt First — upgrade flowchart, breaking changes, and migration checklist.
2. Learning path — module baseline.
3. Installing — 1.27.1 tarball / macOS 13.
4. Core commands and vet/docs.
5. JSON, uuid/crypto.

Official source: Go 1.27 Release Notes.

Upgrading to Go 1.27: What Breaks, What to Adopt First

Upgrading to Go 1.27: What Breaks, What to Adopt First

Go **1.27** shipped in August 2026. For most modules, the upgrade path is minimal—a single `go.mod` line update followed by a `go fix` pass. But a handful of changes bite silently.

The headline changes are structural:

- `encoding/json` is now backed by the new `json/v2` engine under the hood.
- The `uuid` package has officially joined the standard library (RFC 9562).
- Methods can finally declare their own type parameters (generic methods).
- Promoted fields can be used directly as composite literal keys.

None of these force a rewrite. The Go team maintained the backward compatibility promise: almost all existing code compiles and runs unchanged.

However, several silent traps and strict enforcement mechanisms can disrupt a release:

1. **Removed `GODEBUG` settings** now cause build failures if pinned to their legacy values.
2. **`compress/flate` produces different bytes**, silently breaking byte-for-byte golden file tests across gzip, zip, and PNG fixtures.
3. **`go test` enforces the `stdversion` vet check by default**, causing `go build` and `go test` to disagree when symbols are newer than the declared `go` directive.
4. **Compiler-generated symbol names** for function literals changed, breaking traceback parsers and function pointer equality.

This chapter walks through the upgrade path, breaking changes, hands-on reproductions, and high-leverage APIs to adopt first.

The Upgrade Path in One Diagram

The Go team strongly recommends a **build-first, fix-second** order. While `go fix` rewrites code to new APIs and idioms, only an initial clean build tells you whether runtime-level or `GODEBUG` changes impact your service.

```
flowchart TD
    Start["1. Install Go 1.27 Toolchain"] --> Build1["2. Run go build ./..."]
    Build1 --> PassBuild{"Build passes?"}
    PassBuild -- "No: Removed GODEBUG setting" --> FixGodebug["Set final default or delete l
    FixGodebug --> Build1
```

```
    PassBuild -- "Yes" --> Test1["3. Run go test ./..."]
    Test1 --> CleanTest{"stdversion + vet clean?"}
    CleanTest -- "No: Symbol too new for go line" --> FixSymbol["Raise go line or swap symbo
    FixSymbol --> Test1
    CleanTest -- "Yes" --> ModTidy["4. Run go mod tidy"]
    ModTidy --> FixGo["5. Run go fix ./..."]
    FixGo --> Review["6. Review JSON & flate golden tests / diffs"]
```

Step 0: Toolchain Installation and Selection

Start with the toolchain itself. If you manage toolchains through the `go` command, either install the binary explicitly:

```
go install golang.org/dl/go1.27@latest
go1.27 download
go1.27 version
```

```
go version go1.27.2 linux/amd64
```

Alternatively, set `GOTOOLCHAIN=go1.27.0` in your environment, or pin the toolchain directive inside `go.mod`:

```
module example.com/svc

go 1.27

toolchain go1.27.2
```

When `GOTOOLCHAIN=auto` (the default) is in effect, invoking the `go` command automatically downloads and switches to the specified compiler version.

Step 1: Bump the `go` Line and Clear Removed `GODEBUG` Settings

Raise the module's language version and tidy dependencies:

```
go mod edit -go=1.27
go mod tidy
go build ./...
```

Raising the `go` directive activates Go 1.27 language features (generic methods, promoted-field composite keys, wider function type inference) and instructs the toolchain to enforce strict rules on `GODEBUG` settings whose support has permanently ended.

Permanently Removed Settings in Go 1.27

Setting	Former Behavior	Go 1.27 Permanent Behavior
`asynctimerchan`	Allowed reverting to pre-1.23 buffered timer channels (`asynctimerchan=1`)	Timer and ticker channels are **always unbuffered**
`gotypesalias`	Allowed opting out of `go/types.Alias` (`gotypesalias=0`)	`types.Alias` is always enabled
`tlsunsafeekm`	Allowed unsafe EKM derivation without TLS 1.3 Extended Master Secret	Deprecated unsafe key material derivation refused
`tlsrsakex`	Allowed RSA key exchange without (EC)DHE	RSA key exchange permanently removed from default suites
`tls3des`	Allowed 3DES cipher suites (`TLS_RSA_WITH_3DES_EDE_CBC_SHA`)	3DES ciphers permanently disabled
`tls10server`	Allowed TLS 1.0 and 1.1 handshakes on server connections	TLS 1.2 is minimum allowed server protocol
`x509keypairleaf`	Allowed `tls.X509KeyPair` without parsing leaf certificate	Leaf certificate is always parsed

The New `GODEBUG` Acceptance Rule

The `go` command still accepts removed settings in `go.mod` (`godebug` lines) or source (`//go:debug` comments) **only if set to the final default value that existed before removal**.

Any legacy value that attempted to disable a behavior change fails the build immediately:

```
// go.mod
module example.com/gd

go 1.27

godebug asynctimerchan=1
```

```
go run .
```

```
go: error loading go.mod:
go.mod:5: removed GODEBUG "asynctimerchan" set to old value "1" (https://go.dev/doc/godebug#
```

With `godebug asynctimerchan=0` (or removing the line entirely), the build succeeds.

> [!WARNING] If your code fails when `asynctimerchan=1` is removed, your application depended on pre-Go 1.23 buffered timer channels. Review all `time.Timer.Stop()` and `time.Timer.Reset()` calls. In Go 1.23+, unbuffered timer channels guarantee that stopped or reset timers do not leak stale ticks into channel drains.
>
> Search your codebase for removed settings before bumping `go.mod`:
>
> ```
> grep -rnE 'asynctimerchan|tlsunsafeekm|tlsrsakex|tls3des|tls10server|x509keypairleaf|go
>  --include=go.mod --include='*.go' .
> ```

Step 2: Run `go test` — The `stdversion` Trap

In Go 1.27, `go build` and `go test` no longer treat version mismatches identically. `go test` runs the `stdversion` vet check by default.

Consider this path-utility package where `go.mod` still declares `go 1.26`:

```go
// go.mod
module example.com/pathx

go 1.26
```

```go
// pathx.go
package pathx

import "strings"

// Base returns the final element of a slash-separated path.
func Base(p string) string {
    _, file, ok := strings.CutLast(p, "/") // added in Go 1.27
    if !ok {
        return p
    }
    return file
}
```

```go
// pathx_test.go
package pathx
```

```go
import "testing"

func TestBase(t *testing.T) {
    if got := Base("desk/shift/ticket.txt"); got != "ticket.txt" {
        t.Fatalf("Base = %q", got)
    }
}
```

When executed with a Go 1.27 compiler:

```
go build ./...; echo "exit=$?"
go test ./...;  echo "exit=$?"
```

```
exit=0
# example.com/pathx
# [example.com/pathx]
./pathx.go:8:25: strings.CutLast requires go1.27 or later (module is go1.26)
FAIL    example.com/pathx [build failed]
FAIL
exit=1
```

- `go build` passes because the Go 1.27 toolchain's standard library physically contains `strings.CutLast`.
- `go test` fails because `stdversion` catches the call to a standard-library API that is newer than the `go 1.26` directive in `go.mod`.
- On an actual Go 1.26 machine, `go build` would fail with `undefined: strings.CutLast`.

  [!IMPORTANT] **Never validate an upgrade with `go build` alone.** Always run `go test ./...` across the entire module.

Dual Build Tags Workaround

If your module must support older Go toolchains while taking advantage of 1.27 APIs when available, separate implementations using build constraints:

```go
// cut_legacy.go
//go:build !go1.27

package pathx

import "strings"

func cutLast(p, sep string) (string, string, bool) {
    i := strings.LastIndex(p, sep)
    if i < 0 {
        return "", p, false
```

```go
    }
    return p[:i], p[i+len(sep):], true
}
```

```go
// cut_go127.go
//go:build go1.27

package pathx

import "strings"

func cutLast(p, sep string) (string, string, bool) {
    return strings.CutLast(p, sep)
}
```

`stdversion` inspects file-level `//go:build go1.27` constraints and recognizes that `cut_go127.go` is guarded by the appropriate version floor.

New `printf` Vet Finding: `%w` with Pointer to Error

The `printf` analyzer in `go test` and `go vet` now flags `%w` wrapping on pointers to types that implement `error`:

```go
package main

import (
    "errors"
    "fmt"
)

type NotFound struct{ Key string }

func (e NotFound) Error() string { return "not found: " + e.Key }

func lookup(k string) error {
    p := &NotFound{Key: k}
    return fmt.Errorf("lookup: %w", p) // Vet warning in Go 1.27!
}

func main() {
    err := lookup("port")
    var nf NotFound
    fmt.Println(err, errors.As(err, &nf))
}
```

```
go vet .
```

```
main.go:14:29: %w wants operand of error type NotFound, not pointer type *NotFound (defeats
```

Because `NotFound` has a value receiver on `Error()`, passing `*NotFound` creates a pointer-to-struct error wrapper. Type assertions and `errors.As(err, &nf)` fail silently. Fix it by wrapping the value directly:

```
return fmt.Errorf("lookup: %w", NotFound{Key: k})
```

Step 3: `go mod tidy` Merges `require` Blocks

Long-lived modules often accumulate scattered `require` blocks resulting from merge conflicts and hand-edits:

```
// go.mod (before)
module example.com/svc

go 1.27

require github.com/google/uuid v1.6.0

// Pinned for config loader
require gopkg.in/yaml.v3 v3.0.1

require (
    golang.org/x/sync v0.10.0
    golang.org/x/text v0.21.0 // indirect
)
```

Running `go mod tidy` on Go 1.27 consolidates duplicate blocks into at most two blocks (direct dependencies followed by indirect dependencies), preserving inline comments:

```
// go.mod (after)
module example.com/svc

go 1.27

require (
    github.com/google/uuid v1.6.0
    golang.org/x/sync v0.10.0
    // Pinned for config loader
    gopkg.in/yaml.v3 v3.0.1
```

```
)
```

Review this cleanup in a dedicated commit before making functional code changes.

Step 4: `encoding/json` — The v2 Engine Underneath

The headline runtime evolution in Go 1.27 is the transition of `encoding/json`. The classic `encoding/json` package is now backed internally by the `json/v2` engine.

For callers of `encoding/json`, API semantics remain backward-compatible—nil slices still serialize to `null`, and map keys remain sorted. However, three subtle shifts occur:

1. **Error Message Formatting:** The wording of error messages returned during decoding has been unified with the v2 engine. Tests that assert exact error strings via `strings.Contains(err.Error(), ...)` may fail.
2. **Allocation Profiles:** Unmarshalling throughput has increased, but memory allocation points have shifted. CI benchmark thresholds might require re-baselining.
3. **Emergency Rollback Hatch:** If an unexpected regression occurs in production, build with:

   ```
   GOEXPERIMENT=nojsonv2 go build ./...
   ```

   This temporarily restores the legacy Go 1.26 JSON engine. Treat this as a short-term debugging fallback—it is scheduled for removal in Go 1.28.

The `compress/flate` Golden Test Trap

A subtle second-order change: `compress/flate`'s compression encoder was overhauled in Go 1.27 for increased throughput.

Consequently, outputs from `archive/zip`, `compress/gzip`, `compress/zlib`, and `image/png` may differ **byte-for-byte** from Go 1.26, even though they decompress to identical data.

> [!TIP] If snapshot or golden-file tests verify exact byte representations of compressed payloads, regenerate the test fixtures using `UPDATE_GOLDEN=1` rather than attempting to force legacy compression bytes.

Migrating to `encoding/json/v2` Deliberately

When writing new services or updating data models, you can import `encoding/json/v2` directly alongside `encoding/json/jsontext`:

```
import (
    jsonv2 "encoding/json/v2"
```

```
    "encoding/json/jsontext"
)

// Step 1: Exact v1 compatibility via explicit options
out, err := jsonv2.Marshal(v, jsonv2.DefaultOptionsV1())

// Step 2: Selectively opt in to modern v2 defaults
out, err := jsonv2.Marshal(v,
    jsonv2.DefaultOptionsV1(),
    jsonv2.FormatNilSliceAsNull(false), // encodes nil slice as [] instead of null
)
```

Dropping `DefaultOptionsV1()` enables pure v2 defaults:

Behavior	v1 (`encoding/json`)	v2 Default (`encoding/json/v2`)
Invalid UTF-8 in strings	Replaced with `U+FFFD` (‘) | Returns syntax error | | **Duplicate object keys** | Allowed (last key overwrites) | Returns semantic error | | **Nil slice / nil map** | Serializes tonull| Serializes to[]and| | **Struct field matching** | Case-insensitive fallback | Strict case-sensitive matching | | **Unknown object members** | Ignored | Ignored (RejectUnknownMembers(true)to error) | | **Map key ordering** | Always sorted alphabetically | Unsorted by default (fast) | | **time.Duration‘**	Built-in nanosecond string

```
// For stable map output in snapshot/golden tests, enable deterministic ordering:
out, err := jsonv2.Marshal(v, jsonv2.Deterministic(true))
```

Struct Tag Evolution: `embed` vs `inline`

Grep your struct tags before migrating to `encoding/json/v2`:

- `format:...` tag options used in early experiments were removed.
- `inline` has been standardized to `embed`:

```go
type Document struct {
    // Go 1.27 json/v2: use 'embed' rather than 'inline'
    Metadata `json:",embed"`
    Title    string `json:"title"`
}
```

Step 5: Adopt Stdlib `uuid` (RFC 9562)

Go 1.27 introduces `uuid` to the standard library, implementing RFC 9562:

```go
package main

import (
    "fmt"
    "uuid"
)

func main() {
    id := uuid.New()   // Random UUIDv4 equivalent: [16]byte
    v7 := uuid.NewV7() // Time-sortable UUIDv7

    u, err := uuid.Parse("550e8400-e29b-41d4-a716-446655440000")
    if err != nil {
        panic(err)
    }

    fmt.Println("v4:", id)
    fmt.Println("v7:", v7)
    fmt.Println("Equal:", u == id)
}
```

Why UUIDv7 Matters for Storage

UUIDv4 generates pure random bits, scattering writes across database B-tree indexes and causing page splits.

UUIDv7 embeds a **48-bit millisecond UNIX timestamp** in the most significant bits followed by cryptographically random bits. This delivers:

1. **Chronological Sorting:** IDs sort naturally by creation timestamp.
2. **Index Locality:** High-concurrency inserts append sequentially to B-tree leaves, significantly reducing disk I/O.
3. **Primary Key Ordering:** Replaces compound auto-increment ID + timestamp columns in distributed systems.

Standard library `uuid.UUID` implements `sql.Scanner`, `driver.Valuer`, `encoding.TextMarshaler`, and `encoding.TextUnmarshaler`. Migrating from `github.com/google/uuid` is clean and mechanical.

Step 6: Language Changes & Modernizers

Generic Methods

Go 1.27 introduces generic methods: methods may now declare their own type parameters, independent of whether the receiver type itself is generic:

```go
package main

import (
    "fmt"
    "math/rand/v2"
)

func main() {
    r := rand.New(rand.NewPCG(42, 1024))

    // (*Rand).N[Int intType](n Int) Int
    valInt := r.N(10)
    valInt64 := r.N(int64(1_000_000))

    fmt.Printf("int: %d, int64: %d\n", valInt, valInt64)
}
```

Type System Rules: - Interface methods **cannot** declare type parameters. - Generic methods **cannot** implement interface methods.

Promoted Field Keys in Struct Literals

Embedded anonymous struct fields can now be initialized directly in composite literals:

```go
type Address struct {
    City  string
    State string
}

type User struct {
    Address
    Name string
}

// Go 1.27: Flattened composite literal syntax
u := User{
    Name:  "Alex",
    City:  "Seattle",
    State: "WA",
}
```

Automated Code Modernization with `go fix`

Go 1.27 expands `go fix` with modernizers:

- `embedlit`: Flattens legacy nested composite literals into promoted-field keys.
- `atomictypes`: Migrates legacy `atomic.AddInt64(&val, 1)` to `atomic.Int64`.
- `slicesbackward`: Replaces manual reverse loops (`for i := len(s)-1; i >= 0; i--`) with `for i, x := range slices.Backward(s)`.
- `waitgroupgo`: Modernizes `wg.Add(1); go func() { defer wg.Done(); ... }()` to `wg.Go(...)` (introduced in Go 1.25).
- `unsafefuncs`: Replaces `unsafe.Pointer` cast patterns with standard helpers.

```go
# Preview diff without applying
go fix -diff ./...

# Apply changes across module
go fix ./...
```

> [!CAUTION] When maintaining shared libraries, run `go fix -fix=atomictypes` cautiously. Changing an exported package-level variable from `int64` to `atomic.Int64` breaks downstream consumers.

Step 7: Things That Quietly Break

These changes compile without error but can cause unexpected runtime or CI issues:

1. Function-Literal Symbol Names and Closure Merging

The Go 1.27 compiler generates simpler, inlining-independent names for closures in debug symbols. Furthermore, identical closure literals may be merged.

- Tests that assert on raw function names in stack traces or `runtime.FuncForPC` outputs will break.
- Code comparing function pointers for identity (`fn1 == fn2`) will produce incorrect results.

2. `gofmt` Column Alignment

A fix to column-alignment logic (which previously varied across CPU architectures) reflows a small number of struct and variable declarations. Run:

```
gofmt -w .
```

Commit formatting adjustments in an isolated commit to avoid noisy pull request diffs.

3. Goroutine Labels in Crash Tracebacks

Crash tracebacks now include `runtime/pprof` goroutine labels in their headers:

```
panic: runtime error: invalid memory address

goroutine 7 [running] {request_id: "req-1092", user_id: "usr-44"}:
main.process()
    /app/service.go:42 +0x3a
```

If labels contain sensitive information (e.g., email addresses or session tokens), disable the header with:

```
GODEBUG=tracebacklabels=0
```

Note: Panics occurring directly inside `pprof.Do(ctx, labels, f)` do not display labels, as deferred unwinding restores prior labels before the panic header is formatted.

4. `go tool trace` Localhost Binding

`go tool trace -http=:6060` now binds strictly to `localhost` for security. When debugging inside Docker containers or remote servers, specify the interface explicitly:

```
go tool trace -http=0.0.0.0:6060 trace.out
```

Step 8: New Daily APIs to Adopt Immediately

`strings.CutLast` and `bytes.CutLast`

Replaces cumbersome `strings.LastIndex` slicing patterns:

```go
package main

import (
    "fmt"
    "strings"
)

func main() {
    path := "archive/backup/data.tar.gz"
    dir, file, found := strings.CutLast(path, "/")
    stem, ext, _ := strings.CutLast(file, ".")

    fmt.Println("Dir:", dir)   // archive/backup
    fmt.Println("File:", file) // data.tar.gz
    fmt.Println("Ext:", ext)   // gz
    _ = found
}
```

`httptest.NewTestServer` Paired with `testing/synctest`

`httptest.NewTestServer(t, mux)` creates an in-process HTTP server backed by an in-memory network channel instead of binding an operating system TCP loopback port:

```go
package api_test

import (
    "io"
    "net/http"
    "net/http/httptest"
    "testing"
)

func TestAPI(t *testing.T) {
    mux := http.NewServeMux()
    mux.HandleFunc("/health", func(w http.ResponseWriter, r *http.Request) {
        w.Write([]byte("OK"))
    })

    // In-memory fake network with auto-cleanup (t.Cleanup registered)
    srv := httptest.NewTestServer(t, mux)
```

```go
    res, err := srv.Client().Get(srv.URL + "/health")
    if err != nil {
        t.Fatal(err)
    }
    body, _ := io.ReadAll(res.Body)
    if string(body) != "OK" {
        t.Errorf("got %q, want OK", body)
    }
}
```

Because `NewTestServer` operates without real network I/O, it integrates with `testing/synctest` for deterministic timeout, retry, and cancellation testing without real wall-clock delays.

Step 9: Experiments Worth Watching

Enable experimental features using `GOEXPERIMENT`:

1. Portable Hardware SIMD (`GOEXPERIMENT=simd`)

The experimental `simd` and `simd/archsimd` packages provide portable vector types (`Int8s`, `Float32s`, etc.) that compile directly into AVX-512, arm64 NEON, or WebAssembly SIMD instructions.

The standard library uses this machinery internally: Go's Swiss Table map implementation leverages vector instructions for high-throughput bucket lookups.

2. Post-Quantum Cryptography (`crypto/mldsa`)

Go 1.27 introduces `crypto/mldsa`, implementing the post-quantum ML-DSA digital signature standard (FIPS 204) across parameter sets `MLDSA44`, `MLDSA65`, and `MLDSA87`.

Integrated into `crypto/x509` and TLS 1.3, hybrid post-quantum key exchange can be enabled via `tls.Config.CurvePreferences`.

The Upgrade Checklist Matrix

Step	Small Module	Service / Monorepo
1. Toolchain	`go install golang.org/dl/go1.27@latest`	Pin `toolchain go1.27.0` in `go.mod`; align CI runner images
2. go.mod	`go mod edit -go=1.27 && go mod tidy`	Same, and resolve multi-module workspace require diffs
3. Removed GODEBUG	Grep `//go:debug` and `godebug` in `go.mod`	Audit container flags, Helm values, and systemd environment vars
4. `go build ./...`	Fix any compilation errors	Build per-service; bisect any CGo or assembly regressions
5. `go test ./...`	Fix `stdversion` and `%w` printf findings	Run before bumping `go.mod` to isolate version mismatches
6. `go fix ./...`	Review diff and commit	Apply one modernizer at a time (`-embedlit`, `-slicesbackward`)
7. Golden / JSON Tests	Regenerate snapshot fixtures	Check `compress/flate` (zip/gzip/png) and JSON error string diffs
8. Emergency Fallbacks	Skip	Document `GOEXPERIMENT=nojsonv2` for rapid rollback if needed

---

The Traps

- **Validating with `go build` alone:** Only `go test` and `go vet` run `stdversion`.
- **Leaving removed `GODEBUG` flags set to legacy values:** Immediately halts the build under `go 1.27`.
- **Pinning compressed bytes in golden files:** `compress/flate` produces different byte outputs for identical decompressed payloads.
- **Bulk `go fix -atomictypes` across exported APIs:** Can silently alter exported package-level types.
- **Comparing function pointers:** Function literal merging causes distinct closures to compare equal.

The Boring Rule

> **Build first, test second, fix third.**
>
> Keep the toolchain upgrade, the `go.mod` bump, and the `go fix` modernization in separate, reviewable commits.

Sources

- Go 1.27 Release Notes: https://go.dev/doc/go1.27
- GODEBUG Documentation & History: https://go.dev/doc/godebug
- Go Toolchain Management: https://go.dev/doc/toolchain
- RFC 9562: Universally Unique IDentifiers (UUID): https://www.rfc-editor.org/rfc/rfc9562.html

Installing and Running Go

Installing and Running Go

Overview

Setting up a Go development environment is straightforward. This chapter covers installation, verification, and the fundamental `go run` and `go build` workflow that you'll use throughout your Go journey.

Installation

macOS

Using Homebrew (recommended):

```
brew install go
```

Or download from go.dev/dl and run the installer package.

Linux

Download and extract:

```
# Download the latest version (check go.dev/dl for current version)
wget https://go.dev/dl/go1.27.1.linux-amd64.tar.gz

# Remove any previous installation and extract
sudo rm -rf /usr/local/go
sudo tar -C /usr/local -xzf go1.27.1.linux-amd64.tar.gz

# Add to PATH (add to ~/.bashrc or ~/.zshrc)
export PATH=$PATH:/usr/local/go/bin
```

Windows

Download the MSI installer from go.dev/dl and run it. The installer adds Go to your PATH automatically.

Verify Installation

```
$ go version
go version go1.27.1 darwin/arm64
```

Go 1.27 requires **macOS 13 Ventura or later**. Linux and Windows binaries from go.dev/dl remain the supported install path; always pick the current patch (1.27.1 at this writing).

Environment Variables

Variable	Purpose	Default
`GOROOT`	Go installation directory	Auto-detected
`GOPATH`	Workspace directory	`$HOME/go`
`GOBIN`	Binary installation directory	`$GOPATH/bin`

Check your environment:

```
$ go env
# Shows all Go environment variables

$ go env GOPATH
/Users/yourname/go
```

Your First Go Program

Create a file named `hello.go`:

```go
package main

import "fmt"

func main() {
    fmt.Println("Hello, Go!")
}
```

The `go run` Command

`go run` compiles and executes in one step—perfect for development:

```
$ go run hello.go
Hello, Go!
```

How It Works

1. Compiles the source to a temporary binary
2. Executes that binary
3. Cleans up the temporary file

Running Multiple Files

```
# Run multiple files
$ go run main.go utils.go

# Run all Go files in current directory
$ go run .
```

The `go build` Command

`go build` creates a permanent executable:

```
$ go build hello.go
$ ls
hello    hello.go

$ ./hello
Hello, Go!
```

Common Build Options

```
# Specify output name
$ go build -o myapp hello.go

# Build for different OS/architecture
$ GOOS=linux GOARCH=amd64 go build -o myapp-linux

# Build with optimizations (strip debug info)
$ go build -ldflags="-s -w" -o myapp
```

Cross-Compilation

Go makes cross-compilation trivial:

```
# Build for Linux from macOS
$ GOOS=linux GOARCH=amd64 go build -o app-linux

# Build for Windows
$ GOOS=windows GOARCH=amd64 go build -o app.exe

# Build for ARM (Raspberry Pi)
$ GOOS=linux GOARCH=arm GOARM=7 go build -o app-arm
```

The Build Lifecycle

```
Source Code       Compiler          Binary
  (.go)                                (executable)




              go run       go build
             (temp bin)  (perm. bin)
```

IDE and Editor Setup

Visual Studio Code

1. Install the Go extension
2. Open any `.go` file
3. Accept prompts to install Go tools

Recommended Tools (installed automatically)

- `gopls` - Language server
- `dlv` - Debugger
- `staticcheck` - Linter

GoLand

JetBrains GoLand is a commercial IDE with excellent Go support out of the box.

Neovim/Vim

Use `gopls` with your LSP client of choice (nvim-lspconfig, coc.nvim, etc.).

Project Structure Basics

```
myproject/
   go.mod          # Module definition
   go.sum          # Dependency checksums
   main.go         # Entry point
   internal/       # Private packages
      config/
          config.go
   pkg/            # Public packages (optional)
       utils/
           utils.go
```

Common Pitfalls

GOPATH vs Modules

Modern Go uses **modules** (introduced in Go 1.11). You don't need to work inside `$GOPATH/src` anymore:

```
# Initialize a new module anywhere
$ mkdir myproject && cd myproject
$ go mod init github.com/username/myproject
```

Executable vs Library

Only `package main` with a `main()` function creates executables:

```
// This creates a binary
package main

func main() { }
```

```go
// This is a library (cannot run directly)
package mylib

func Helper() { }
```

Summary

Command	Purpose
`go run`	Compile and run (temporary)
`go build`	Compile to binary
`go install`	Compile and install to `$GOBIN`
`go env`	Show environment variables
`go version`	Show Go version

Exercises

1. Install Go and verify with `go version`
2. Create a "Hello, World!" program and run it with `go run`
3. Build the same program with `go build` and execute the binary
4. Cross-compile for a different operating system

Related Topics

- Why Go Exists
- Learning Go in 2026
- How Go Code Is Organized
- Core Go Commands

More examples

Example: classic Hello, World

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

func main() {
    fmt.Println("Hello, World!")
```

```
    fmt.Println("from go run")
}
```

Expected:

```
Hello, World!
from go run
```

Example: exit codes and stderr

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```
package main

import (
    "fmt"
    "os"
)

func main() {
    if len(os.Args) < 2 {
        fmt.Fprintln(os.Stderr, "usage: go run . <name>")
        os.Exit(2)
    }
    fmt.Println("hello,", os.Args[1])
}
```

Expected (with `go run . gopher`):

```
hello, gopher
```

Expected (with bare `go run .`):

```
usage: go run . <name>
```

Runnable example

Print Go runtime and build info from inside a program (complements `go version` / `go env` on the CLI).

Save as `main.go`. From an empty directory:

```
go mod init example
go run .
go build -o hello .
```

```
./hello
```

```go
package main

import (
    "fmt"
    "runtime"
    "runtime/debug"
)

func main() {
    fmt.Printf("Go version (runtime): %s\n", runtime.Version())
    fmt.Printf("OS/Arch: %s/%s\n", runtime.GOOS, runtime.GOARCH)
    fmt.Printf("NumCPU: %d\n", runtime.NumCPU())

    info, ok := debug.ReadBuildInfo()
    if !ok {
        fmt.Println("build info: unavailable")
        return
    }
    fmt.Printf("module path: %s\n", info.Main.Path)
    fmt.Printf("go version (build): %s\n", info.GoVersion)
    for _, s := range info.Settings {
        switch s.Key {
        case "GOOS", "GOARCH", "vcs.revision", "vcs.modified", "-compiler":
            fmt.Printf("setting %s=%s\n", s.Key, s.Value)
        }
    }
    fmt.Println("Hello from a complete Go program")
}
```

Expected output (illustrative; values depend on your machine):

```
Go version (runtime): go1.22.5
OS/Arch: darwin/arm64
NumCPU: 8
module path: example
go version (build): go1.22.5
setting GOOS=darwin
setting GOARCH=arm64
setting -compiler=gc
Hello from a complete Go program
```

What to notice: - `runtime.Version()` matches the idea of `go version` from inside your app. - `debug.ReadBuildInfo()` reports the module path from `go mod init`. - The same source works with `go run` and `go build`. - Cross-compile targets show up as `GOOS`/`GOARCH` in build settings when set.

Try next: Build with `GOOS=linux GOARCH=amd64 go build -o hello-linux .` and inspect the binary name; run the program natively again and compare `OS/Arch`.

How Go Code Is Organized

How Go Code Is Organized

Overview

Go has strong opinions about code organization. Understanding packages, imports, and project structure is essential for writing maintainable Go programs and for your code to work correctly with the Go toolchain.

Packages: The Building Blocks

Every Go file belongs to a package. Packages group related code and control visibility.

```go
// File: greet.go
package greeting  // Package declaration (first line)

import "fmt"

func Hello(name string) string {
    return fmt.Sprintf("Hello, %s!", name)
}
```

Package Rules

Rule	Description
All files in a directory belong to the same package	You can't have `package foo` and `package bar` in the same folder
Package name matches directory name (by convention)	Directory `server/` contains `package server`
`package main` is special	It defines an executable, not a library

The `main` Package

Only `package main` can produce an executable. It must have a `main()` function:

```go
package main

import "fmt"

func main() {
    fmt.Println("This is the entry point")
}
```

Import System

Basic Imports

```go
import "fmt"               // Standard library
import "net/http"          // Nested standard library package
import "github.com/user/repo"  // External package
```

Import Block (Preferred Style)

```go
import (
    // Standard library (grouped first)
    "fmt"
    "net/http"

    // Third-party packages (blank line separator)
    "github.com/gin-gonic/gin"

    // Internal packages (blank line separator)
    "myproject/internal/config"
)
```

Import Aliases

```go
import (
    "crypto/rand"
    mrand "math/rand"  // Alias to avoid collision
)

// Usage
cryptoBytes, _ := rand.Read(buf)
randomInt := mrand.Intn(100)
```

Blank Imports (Side Effects Only)

```go
import (
    "database/sql"
    _ "github.com/lib/pq"  // Import for side effects (driver registration)
)
```

Dot Imports (Avoid in Production)

```go
import . "fmt"

// Now you can use Println directly
Println("Hello")  // Instead of fmt.Println
```

Visibility: Exported vs Unexported

Go uses capitalization to control visibility:

```go
package user

// Exported (public) - starts with uppercase
type User struct {
    Name  string  // Exported field
    Email string  // Exported field
    age   int     // unexported field (private)
}

// Exported function
func CreateUser(name string) *User {
    return &User{Name: name}
}

// unexported function (private)
func validate(u *User) bool {
    return len(u.Name) > 0
}
```

Name	Visibility
`User`	Exported (accessible from other packages)
`Name`	Exported field
`age`	Unexported (only accessible within `user` package)
`CreateUser`	Exported function
`validate`	Unexported function

Modules: Modern Dependency Management

A **module** is a collection of packages with a `go.mod` file at the root.

Creating a Module

```
$ mkdir myproject && cd myproject
$ go mod init github.com/username/myproject
```

This creates `go.mod`:

```
module github.com/username/myproject

go 1.27
```

Module Structure

```
github.com/username/myproject/
   go.mod
   go.sum
   main.go
   config/
      config.go      # package config
   internal/
      database/
         db.go      # package database (private)
   pkg/
      api/
         api.go     # package api (public)
```

Internal Packages

The `internal/` directory has special meaning: packages inside it can only be imported by code within the same module tree.

```
myproject/
   internal/
      secret/
         secret.go   # Only myproject can import this
   cmd/
      app/
         main.go     # Can import internal/secret
```

```go
// This works (same module)
import "github.com/username/myproject/internal/secret"

// This fails (different module trying to import internal)
// Error: use of internal package not allowed
```

Standard Project Layout

While Go doesn't mandate a structure, this layout is widely adopted:

```
myproject/
   cmd/                    # Main applications
      myapp/
         main.go
      worker/
          main.go
   internal/               # Private packages
      config/
      database/
      service/
   pkg/                    # Public libraries (optional)
      api/
   api/                    # OpenAPI/Swagger specs
   web/                    # Web assets
   scripts/                # Build/install scripts
   docs/                   # Documentation
   go.mod
   go.sum
   README.md
```

`cmd/` Directory

Each subdirectory in `cmd/` becomes a separate binary:

```go
$ go build ./cmd/myapp
$ go build ./cmd/worker
```

Package Naming Conventions

Good	Bad	Why
`user`	`userPackage`	Avoid redundant suffixes
`http`	`httputil`	Only add suffix if necessary

Good	Bad	Why
`strconv`	`stringconversion`	Short but clear
`io`	`inputoutput`	Abbreviations OK if common

Avoid Meaningless Names

- `util` → Use specific names like `stringutil`, `timeutil`
- `common` → Be specific about what's common
- `misc` → Break into focused packages

Circular Import Prevention

Go prohibits circular imports. If you have:

```
package a → imports → package b
package b → imports → package a  // Error!
```

Solutions: 1. **Extract common code** to a third package 2. **Use interfaces** to break the dependency 3. **Restructure** your packages

Summary

Concept	Purpose
Package	Groups related code, controls visibility
Module	Collection of packages with version info
`internal/`	Private packages within a module
Uppercase	Exported (public)
Lowercase	Unexported (private)

Exercises

1. Create a module with two packages and have one import the other
2. Create an `internal/` package and verify external code can't import it
3. Create a project with `cmd/` containing multiple applications

Related Topics

- Core Go Commands
- Dependency Management

More examples

Example: exported vs unexported names

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

// User is exported (other packages could import and use it).
type User struct {
    Name string // exported field
    id   int    // unexported field - same package only
}

func NewUser(name string, id int) User {
    return User{Name: name, id: id}
}

func (u User) ID() int { return u.id }

func main() {
    u := NewUser("ada", 7)
    fmt.Println("name:", u.Name)
    fmt.Println("id:", u.ID())
    // fmt.Println(u.id) // works only inside this package
}
```

Expected:

```
name: ada
id: 7
```

Example: package main entry point

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

// helpers in the same package share unexported names.
func banner() string { return "app" }

func main() {
```

```go
    fmt.Println(banner(), "starting")
    fmt.Println(banner(), "ready")
}
```

Expected:

```
app starting
app ready
```

Runnable example

Single-file demo of packages' visibility rules and `package main` as the executable entry point. (In real projects, exported helpers live in other packages; capitalization still decides export.)

Save as `main.go`. From an empty directory:

```
go mod init example
go run .
```

```go
package main

import (
    "fmt"
    "strings"
)

// Exported-style names start with uppercase; unexported stay lowercase.
// In one package both are visible; other packages would only see uppercase.

type User struct {
    Name string // exported field
    age  int    // unexported field
}

func NewUser(name string, age int) (User, error) {
    if !validName(name) {
        return User{}, fmt.Errorf("invalid name %q", name)
    }
    return User{Name: name, age: age}, nil
}

func validName(name string) bool {
    return strings.TrimSpace(name) != ""
}
```

```go
func (u User) Greeting() string {
    return fmt.Sprintf("Hello, %s (age %d)", u.Name, u.age)
}

func main() {
    u, err := NewUser("Ada", 36)
    if err != nil {
        fmt.Println("error:", err)
        return
    }
    fmt.Println(u.Greeting())
    fmt.Printf("type=%T name=%q age_via_method_only\n", u, u.Name)

    if _, err := NewUser("  ", 1); err != nil {
        fmt.Println("rejected:", err)
    }
}
```

Expected output (illustrative):

```
Hello, Ada (age 36)
type=main.User name="Ada" age_via_method_only
rejected: invalid name "  "
```

What to notice: - `package main` + `func main` is what `go run` executes. - `Name` is uppercase (exportable); `age` and `validName` are package-private style. - Factory `NewUser` validates before constructing—common package API shape. - Imports use the standard library grouping style.

Try next: Split `User` into a second file in the same directory still declaring `package main` and run `go run .` again.

Core Go Commands

Core Go Commands

Overview

The `go` command is your Swiss Army knife for Go development. This chapter provides a deep dive into the essential commands you'll use daily, from building and testing to analyzing and maintaining your code.

Command Summary

Command	Purpose
`go build`	Compile packages and dependencies
`go run`	Compile and run Go program
`go test`	Run tests
`go get`	Add dependencies to module
`go mod`	Module maintenance
`go fmt`	Format source code
`go vet`	Report suspicious constructs
`go doc`	Show documentation
`go install`	Compile and install packages
`go clean`	Remove object files

`go build` - Compiling Code

Basic Usage

```
# Build current package
$ go build

# Build specific file
$ go build main.go

# Build with custom output name
$ go build -o myapp

# Build specific package
```

```
$ go build ./cmd/server
```

Build Flags

```
# Verbose output (show packages being compiled)
$ go build -v ./...

# Rebuild all packages (ignore cache)
$ go build -a ./...

# Print commands but don't run them
$ go build -n

# Show build work directory (don't delete)
$ go build -work
```

Cross-Compilation

The toolchain ships the runtime for each target. Add `CGO_ENABLED=0` so you do not accidentally link the *host* libc (a Linux AMD64 binary with glibc will not run on Alpine).

```
# Build for Linux (AMD64), fully static
$ GOOS=linux GOARCH=amd64 CGO_ENABLED=0 go build -o app-linux

# Build for Windows
$ GOOS=windows GOARCH=amd64 CGO_ENABLED=0 go build -o app.exe

# Build for macOS ARM (Apple Silicon)
$ GOOS=darwin GOARCH=arm64 CGO_ENABLED=0 go build -o app-macos

# Build for Raspberry Pi
$ GOOS=linux GOARCH=arm GOARM=7 CGO_ENABLED=0 go build -o app-arm
```

On Linux, `file app-linux` should say `statically linked`. Why that works — Go talks to the kernel without libc — is in Go as a Systems Language.

Linker Flags

```
# Strip debug info (smaller binary)
$ go build -ldflags="-s -w" -o app

# Embed version information
$ go build -ldflags="-X main.version=1.0.0" -o app
```

```go
// In your code
var version = "dev"  // Overwritten by -ldflags

func main() {
    fmt.Println("Version:", version)
}
```

go run - Quick Execution

```bash
# Run a single file
$ go run main.go

# Run multiple files
$ go run main.go helper.go

# Run all files in directory
$ go run .

# Run with arguments
$ go run main.go --port=8080 --debug
```

go test - Testing

```bash
# Run all tests in current package
$ go test

# Run all tests recursively
$ go test ./...

# Verbose output
$ go test -v

# Run specific test
$ go test -run TestFunctionName

# Run with coverage
$ go test -cover

# Generate coverage report
$ go test -coverprofile=coverage.out
$ go tool cover -html=coverage.out
```

```
# Run benchmarks
$ go test -bench=.

# Test with race detector
$ go test -race ./...

# JSON events (Go 1.27+: output lines may include OutputType)
$ go test -json ./...
```

Go 1.27 runs the **`stdversion`** vet check with `go test` by default: using a standard-library symbol newer than the `go` version in `go.mod` (or the file's build tags) is reported. Fix the call site or raise the module's `go` directive.

`go get` - Dependency Management

```
# Add a dependency (latest version)
$ go get github.com/gin-gonic/gin

# Add specific version
$ go get github.com/gin-gonic/gin@v1.9.0

# Add specific commit
$ go get github.com/gin-gonic/gin@a1b2c3d

# Update a dependency
$ go get -u github.com/gin-gonic/gin

# Update all dependencies
$ go get -u ./...

# Update only patch versions
$ go get -u=patch ./...
```

`go mod` - Module Operations

```
# Initialize a new module
$ go mod init github.com/username/project

# Download dependencies
$ go mod download

# Tidy up go.mod (add missing, remove unused)
$ go mod tidy
```

```
# Verify dependencies
$ go mod verify

# Create vendor directory
$ go mod vendor

# Show dependency graph
$ go mod graph

# Explain why a dependency is needed
$ go mod why github.com/some/package

# Edit go.mod programmatically
$ go mod edit -require github.com/pkg/errors@v0.9.1
```

`go fmt` / `gofmt` - Formatting

```
# Format all Go files in current directory
$ go fmt ./...

# Use gofmt directly with options
$ gofmt -w main.go        # Write changes to file
$ gofmt -d main.go        # Show diff
$ gofmt -l .              # List files that need formatting
$ gofmt -s main.go        # Simplify code
```

`go vet` - Static Analysis

```
# Vet current package
$ go vet

# Vet all packages
$ go vet ./...

# Run specific analyzers
$ go vet -composites=false ./...
```

Common issues `go vet` catches: - Printf format string mismatches - Unreachable code - Suspicious mutex usage - Struct field tag issues

go doc - Documentation

```
# Show package documentation
$ go doc fmt

# Show function documentation
$ go doc fmt.Println

# Show type documentation
$ go doc http.Client

# Show all documentation
$ go doc -all fmt

# Executable examples (Go 1.27+)
$ go doc -ex bytes.Buffer
$ go doc bytes.ExampleBuffer

# Docs for a module version (Go 1.27+)
$ go doc example.com/pkg@v1.2.3

# Start documentation server
$ godoc -http=:6060
# Then visit http://localhost:6060
```

go install - Installing Binaries

```
# Install command to $GOBIN
$ go install

# Install specific version of a tool
$ go install golang.org/x/tools/gopls@latest

# Install from specific package
$ go install ./cmd/myapp
```

go clean - Cleanup

```
# Remove object files
$ go clean

# Remove cached build files
```

```
$ go clean -cache

# Remove test cache
$ go clean -testcache

# Remove all cached data
$ go clean -cache -testcache -modcache

# Show what would be removed
$ go clean -n
```

`go env` - Environment

```
# Show all environment variables
$ go env

# Show specific variable
$ go env GOPATH

# Set environment variable
$ go env -w GOBIN=/usr/local/bin

# Unset environment variable
$ go env -u GOBIN
```

`go list` - Package Information

```
# List current package
$ go list

# List all packages
$ go list ./...

# JSON output
$ go list -json ./...

# List dependencies
$ go list -m all

# List with template
$ go list -f '{{.ImportPath}} -> {{.Deps}}' ./...
```

`go generate` - Code Generation

```
# Run all generate directives
$ go generate ./...

# Run with verbose output
$ go generate -v ./...
```

```
// In source file
//go:generate stringer -type=Status
type Status int
```

Tool Installation Best Practices

```
# Install development tools
$ go install golang.org/x/tools/gopls@latest          # Language server
$ go install github.com/go-delve/delve/cmd/dlv@latest # Debugger
$ go install honnef.co/go/tools/cmd/staticcheck@latest # Linter

# Verify installation
$ which gopls
$ which dlv
```

Common Workflows

Development Cycle

```
# 1. Write code
# 2. Format
$ go fmt ./...
# 3. Vet
$ go vet ./...
# 4. Test
$ go test ./...
# 5. Build
$ go build -o app ./cmd/server
```

Release Build

```
# Production build with version
$ go build \
    -ldflags="-s -w -X main.version=$(git describe --tags)" \
    -o bin/app \
    ./cmd/server
```

Summary

Task	Command
Quick run	`go run main.go`
Build binary	`go build -o app`
Run tests	`go test ./...`
Format code	`go fmt ./...`
Check issues	`go vet ./...`
Add dependency	`go get github.com/pkg@version`
Clean modules	`go mod tidy`
View docs	`go doc package.Function`

Related Topics

- Formatting, Vetting, and Documentation
- Dependency Management

More examples

Example: flags for `go run` experiments

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```
package main

import (
    "flag"
    "fmt"
)

func main() {
    name := flag.String("name", "world", "who to greet")
    n := flag.Int("n", 1, "repeat count")
```

```go
    flag.Parse()

    for i := 0; i < *n; i++ {
        fmt.Printf("hello, %s\n", *name)
    }
}
```

Expected (`go run . -name gopher -n 2`):

```
hello, gopher
hello, gopher
```

Example: build-time GOOS/GOARCH report

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import (
    "fmt"
    "runtime"
)

func main() {
    // Same values you target with GOOS/GOARCH when cross-compiling.
    fmt.Println("GOOS:", runtime.GOOS)
    fmt.Println("GOARCH:", runtime.GOARCH)
    fmt.Println("compiler:", runtime.Compiler)
}
```

Expected (illustrative; machine-specific):

```
GOOS: darwin
GOARCH: arm64
compiler: gc
```

Runnable example

A small app that prints a version string—ready for `go run`, `go build -ldflags`, and `go test` workflows.

Save as `main.go`. From an empty directory:

```
go mod init example
go run .
go build -ldflags="-X main.version=1.2.3" -o app .
./app
```

```go
package main

import (
    "fmt"
    "os"
    "runtime"
)

// Overwritten at link time: go build -ldflags="-X main.version=1.2.3"
var version = "dev"

func main() {
    fmt.Printf("app version=%s go=%s %s/%s\n",
        version, runtime.Version(), runtime.GOOS, runtime.GOARCH)

    if len(os.Args) > 1 && os.Args[1] == "-v" {
        fmt.Println("verbose: argv=", os.Args)
    }
}
```

Expected output (illustrative):

```
# go run .
app version=dev go=go1.22.5 darwin/arm64

# after go build -ldflags="-X main.version=1.2.3" -o app && ./app
app version=1.2.3 go=go1.22.5 darwin/arm64
```

What to notice: - `go run` uses the source default (`dev`); `go build -ldflags` rewrites the variable. - `os.Args` is how CLI flags start before you reach `flag` package patterns. - Same file is the target of `go build`, `go run`, and (later) tests of pure helpers. - Runtime metadata is available without shelling out to `go env`.

Try next: Run `go fmt .` and `go vet .` on this directory, then add a `main_test.go` that checks `version` is non-empty.

Formatting, Vetting, and Documentation

Formatting, Vetting, and Documentation

Overview

Go has a unique culture: there's one blessed way to format code, a built-in static analyzer for common mistakes, and a documentation system that extracts docs from comments. This chapter covers `gofmt`, `go vet`, and `godoc` standards.

Code Formatting with `gofmt`

Why Gofmt Exists

Unlike other languages where style guides vary (tabs vs. spaces, brace placement), Go has a single canonical format enforced by `gofmt`. This eliminates style debates and ensures all Go code looks the same.

> "Gofmt's style is no one's favorite, yet gofmt is everyone's favorite." — Rob Pike

Basic Usage

```
# Format a file (print to stdout)
$ gofmt main.go

# Format and overwrite file
$ gofmt -w main.go

# Format all files recursively
$ gofmt -w .

# Using go fmt (recommended)
$ go fmt ./...
```

Formatting Rules

`gofmt` enforces:

```go
// BEFORE (your style)
func foo(x int,y string){
if x>0{
return
}
}

// AFTER (gofmt style)
func foo(x int, y string) {
    if x > 0 {
        return
    }
}
```

Rule	Gofmt Standard
Indentation	Tabs
Spacing	Spaces around operators
Braces	Opening brace on same line
Line length	No limit (but keep reasonable)
Blank lines	Strategic for readability

Code Simplification

Use `-s` to simplify code:

```bash
$ gofmt -s -w main.go
```

```go
// Before
s[a:len(s)]
for x, _ := range v {}
[]int{1, 2, 3}[0:2]

// After -s
s[a:]
for x := range v {}
[]int{1, 2, 3}[:2]
```

Import Formatting with `goimports`

`goimports` does everything `gofmt` does, plus it manages imports:

```bash
# Install
$ go install golang.org/x/tools/cmd/goimports@latest
```

```
# Format and fix imports
$ goimports -w main.go
```

```go
// Before (missing import, unused import)
package main
import (
    "unused"
)
func main() {
    fmt.Println("Hello")
}

// After goimports
package main

import "fmt"

func main() {
    fmt.Println("Hello")
}
```

Static Analysis with `go vet`

What Vet Catches

`go vet` detects code that compiles but is probably wrong:

```
$ go vet ./...
```

Common Issues Detected

Printf Format Errors

```go
// go vet catches this
fmt.Printf("%d", "string")  // wrong type
fmt.Printf("%s %s", name)   // wrong number of args
```

Unreachable Code

```go
func example() int {
    return 42
    fmt.Println("never runs")  // go vet: unreachable code
}
```

Suspicious Loop Variables

```go
// go vet warns about this common bug
for i, v := range values {
    go func() {
        fmt.Println(i, v)  // captures loop variable
    }()
}
```

Useless Assignments

```go
x := 5
x = x  // go vet: self-assignment
```

Invalid Struct Tags

```go
type User struct {
    Name string `json:name`  // Missing quotes around "name"
}
```

Running Specific Checks

```bash
# List available analyzers
$ go tool vet help

# Run specific analyzer
$ go vet -composites=false ./...
$ go vet -printf=true ./...
```

Enhanced Linting with `staticcheck`

`staticcheck` provides additional checks beyond `go vet`:

```
# Install
$ go install honnef.co/go/tools/cmd/staticcheck@latest

# Run
$ staticcheck ./...
```

Checks include: - Unused code - Deprecated function usage - Simplification suggestions - Performance improvements - Common bugs

Documentation with `godoc`

Writing Documentation

Go extracts documentation from comments directly before declarations:

```go
// Package math provides basic mathematical operations.
// It includes functions for arithmetic, trigonometry, and more.
package math

// Pi represents the mathematical constant  .
const Pi = 3.14159

// Add returns the sum of two integers.
// It handles overflow by wrapping around.
func Add(a, b int) int {
    return a + b
}

// Calculator provides stateful mathematical operations.
type Calculator struct {
    // Result holds the current calculation result.
    Result float64
}

// Add adds n to the current result.
func (c *Calculator) Add(n float64) {
    c.Result += n
}
```

Documentation Conventions

Rule	Example
Start with name	`// Add returns the sum...`
Complete sentences	End with period

Rule	Example
First sentence is summary	Shown in package lists
Blank line for paragraphs	Separate blocks

Code Examples in Docs

Create examples in `*_test.go` files:

```go
// example_test.go
package math_test

import (
    "fmt"
    "myproject/math"
)

func ExampleAdd() {
    result := math.Add(2, 3)
    fmt.Println(result)
    // Output: 5
}

func ExampleCalculator_Add() {
    c := &math.Calculator{}
    c.Add(10)
    c.Add(5)
    fmt.Println(c.Result)
    // Output: 15
}
```

These examples: - Appear in documentation - Are tested by `go test` - Show real usage

Viewing Documentation

```bash
# Command line
$ go doc fmt
$ go doc fmt.Println
$ go doc -all fmt
$ go doc -ex bytes.Buffer          # executable examples (Go 1.27+)
$ go doc example.com/pkg@v1.2.3    # docs for a module version (Go 1.27+)

# Local web server
$ go install golang.org/x/tools/cmd/godoc@latest
$ godoc -http=:6060
```

```
# Visit http://localhost:6060
```

Package Comments

For larger packages, use a `doc.go` file:

```
// doc.go

/*
Package server implements an HTTP server with middleware support.

# Getting Started

Create a new server and add routes:

    srv := server.New()
    srv.Get("/", handleHome)
    srv.Listen(":8080")

# Middleware

Add middleware to process all requests:

    srv.Use(server.Logger())
    srv.Use(server.Recovery())

# Configuration

Configure the server using options:

    srv := server.New(
        server.WithTimeout(30 * time.Second),
        server.WithMaxBodySize(1 << 20),
    )
*/
package server
```

Combining Tools in Workflow

Pre-commit Hook

```
#!/bin/bash
# .git/hooks/pre-commit

# Format
gofmt -l -w .

# Vet
go vet ./...
if [ $? -ne 0 ]; then
    echo "go vet failed"
    exit 1
fi

# Staticcheck
staticcheck ./...
if [ $? -ne 0 ]; then
    echo "staticcheck failed"
    exit 1
fi
```

Makefile Integration

```
.PHONY: check fmt vet lint

fmt:
    gofmt -w .

vet:
    go vet ./...

lint:
    staticcheck ./...

check: fmt vet lint
    @echo "All checks passed"
```

CI Pipeline (GitHub Actions)

```yaml
- name: Check formatting
  run: |
    if [ "$(gofmt -l . | wc -l)" -gt 0 ]; then
      echo "Code is not formatted"
      gofmt -d .
      exit 1
    fi

- name: Vet
  run: go vet ./...

- name: Staticcheck
  run: |
    go install honnef.co/go/tools/cmd/staticcheck@latest
    staticcheck ./...
```

Summary

Tool	Purpose	Usage
`gofmt`	Format code	`gofmt -w .`
`goimports`	Format + manage imports	`goimports -w .`
`go vet`	Catch common mistakes	`go vet ./...`
`staticcheck`	Enhanced linting	`staticcheck ./...`
`godoc`	Generate documentation	`godoc -http=:6060`

Related Topics

- Core Go Commands
- Testing Fundamentals

More examples

Example: godoc-style exported comments

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import (
```

```go
    "fmt"
    "strings"
)

// Title returns s with the first letter of each space-separated word uppercased.
// Empty input yields an empty string.
func Title(s string) string {
    parts := strings.Fields(strings.ToLower(s))
    for i, p := range parts {
        if p == "" {
            continue
        }
        parts[i] = strings.ToUpper(p[:1]) + p[1:]
    }
    return strings.Join(parts, " ")
}

// JoinNonEmpty joins non-empty parts with sep.
func JoinNonEmpty(sep string, parts ...string) string {
    var keep []string
    for _, p := range parts {
        if p != "" {
            keep = append(keep, p)
        }
    }
    return strings.Join(keep, sep)
}

func main() {
    fmt.Println(Title("hello go"))
    fmt.Println(JoinNonEmpty("/", "a", "", "b"))
}
```

Expected:

```
Hello Go
a/b
```

Example: a construct `go vet` cares about

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"
```

```go
func main() {
    // Lock-free counter demo is fine; the classic vet catch is printf args.
    name := "gopher"
    // Correct: verbs match arguments.
    fmt.Printf("hello %s count=%d\n", name, 3)

    // Intentionally correct form of a common mistake:
    // fmt.Printf("hello %s\n", name, 3)  // vet: Printf format %s reads arg but has 2 extra
    fmt.Println("vet clean: format verbs match args")
}
```

Expected:

```
hello gopher count=3
vet clean: format verbs match args
```

Runnable example

Documented helpers in gofmt style—run the program, then try `go doc` / `go vet` on the same files.

Save as `main.go`. From an empty directory:

```bash
go mod init example
go fmt .
go vet .
go run .
go doc -all .
```

```go
package main

import (
    "fmt"
    "strings"
)

// TitleCase returns s with the first letter of each word uppercased.
// Words are split on Unicode whitespace.
func TitleCase(s string) string {
    fields := strings.Fields(s)
    for i, w := range fields {
        if w == "" {
            continue
        }
        // Manual title for a tiny demo; prefer cases.Title in real code.
        r := []rune(w)
```

```go
        r[0] = []rune(strings.ToUpper(string(r[0])))[0]
        if len(r) > 1 {
            fields[i] = string(r[0]) + strings.ToLower(string(r[1:]))
        } else {
            fields[i] = string(r[0])
        }
    }
    return strings.Join(fields, " ")
}

// Sum returns the sum of nums. An empty slice sums to 0.
func Sum(nums []int) int {
    total := 0
    for _, n := range nums {
        total += n
    }
    return total
}

func main() {
    fmt.Println(TitleCase("hello gofmt and godoc"))
    fmt.Printf("sum=%d\n", Sum([]int{1, 2, 3, 4}))
    // Intentional-looking pattern that go vet is happy with:
    fmt.Printf("count=%d name=%s\n", 2, "tools")
}
```

Expected output (illustrative):

```
Hello Gofmt And Godoc
sum=10
count=2 name=tools
```

What to notice: - Comments sit directly above `TitleCase` and `Sum`—`go doc` surfaces them. - `go fmt` will rewrite spacing/imports if you mess them up on purpose. - `go vet` checks printf verbs match argument types (try changing `%d` to `%s`). - Zero-value-friendly API: empty slice → sum `0`, no panic.

Try next: Break the `Printf` format string (`fmt.Printf("%d", "x")`) and run `go vet` to see the diagnostic.

Dependency Management

Dependency Management

Overview

Go modules, introduced in Go 1.11 and default since Go 1.16, provide built-in dependency management. This chapter covers everything about `go.mod`, `go.sum`, versioning, and managing external libraries.

Understanding Go Modules

A module is a collection of packages with a `go.mod` file that defines: - Module path (import path) - Go version - Dependencies and their versions

Creating a Module

```
$ mkdir myproject && cd myproject
$ go mod init github.com/username/myproject
```

This creates `go.mod`:

```
module github.com/username/myproject

go 1.27
```

The `go.mod` File

Anatomy of go.mod

```
module github.com/username/myproject

go 1.27

require (
    github.com/gin-gonic/gin v1.9.1
    github.com/stretchr/testify v1.8.4
)
```

```
require (
    // indirect dependencies
    github.com/bytedance/sonic v1.9.1 // indirect
    golang.org/x/net v0.10.0 // indirect
)

exclude (
    github.com/broken/pkg v1.0.0
)

replace (
    github.com/old/module => github.com/new/module v2.0.0
    github.com/local/dev => ../local-dev
)

retract (
    v1.0.0 // Contains security vulnerability
    [v1.1.0, v1.2.0] // Accidental release
)
```

Directives Explained

Directive	Purpose
`module`	Module path (required)
`go`	Minimum Go version
`require`	Dependencies with versions
`exclude`	Versions to ignore
`replace`	Substitute modules
`retract`	Mark versions as broken

The `go.sum` File

`go.sum` contains cryptographic checksums representing module dependencies:

```
github.com/gin-gonic/gin v1.9.1 h1:4idEAncQnU5cB7BeOkPtxjfCSye0AAm1R0RVIqJ+Jmg=
github.com/gin-gonic/gin v1.9.1/go.mod h1:hPrL7YrpYKXt5YId3A/Tn+7r7IyQ4PGNmP1c42GGpvQ=
```

Each entry has: - Module path and version - Hash of module contents (`h1:`) - Hash of `go.mod` file

Never edit `go.sum` manually — it's managed by Go tools.

Adding Dependencies

Using `go get`

```
# Add latest version
$ go get github.com/gin-gonic/gin

# Add specific version
$ go get github.com/gin-gonic/gin@v1.9.1

# Add latest minor version
$ go get github.com/gin-gonic/gin@v1.9

# Add specific commit
$ go get github.com/gin-gonic/gin@a1b2c3d

# Add from branch
$ go get github.com/gin-gonic/gin@main
```

Automatic Detection

Simply import and run:

```
import "github.com/gin-gonic/gin"
```

```
$ go mod tidy  # Downloads and records dependency
```

Updating Dependencies

```
# Update specific package (latest minor/patch)
$ go get -u github.com/gin-gonic/gin

# Update to specific version
$ go get github.com/gin-gonic/gin@v1.10.0

# Update all dependencies
$ go get -u ./...

# Update only patch versions (safer)
$ go get -u=patch ./...
```

Semantic Versioning

Go modules expect Semantic Versioning:

```
v1.2.3

     Patch: bug fixes (backward compatible)
     Minor: new features (backward compatible)
     Major: breaking changes
```

Major Version Suffixes

For v2+, the module path includes the major version:

```go
// go.mod
require github.com/user/project/v2 v2.1.0

// import
import "github.com/user/project/v2"
```

Managing go.mod

`go mod tidy`

The most important maintenance command:

```
$ go mod tidy
```

This: - Adds missing dependencies - Removes unused dependencies - Updates `go.sum` - For modules with `go 1.27` or later, merges extra `require` blocks into the standard two-block layout (direct, then indirect). Comments on mixed blocks move onto the direct block.

Run this frequently, especially before commits.

The `go` command no longer talks to **Bazaar (`bzr`)** hosts. Modules that still live only on `bzr` need a `replace` to a Git (or other supported) mirror, or a vendor copy.

`go mod download`

Pre-download all dependencies:

```
$ go mod download
```

Useful for CI caching or air-gapped environments.

go mod verify

Check that dependencies haven't been modified:

```
$ go mod verify
all modules verified
```

go mod why

Explain why a dependency is needed:

```
$ go mod why golang.org/x/net

# github.com/username/myproject
github.com/username/myproject
github.com/gin-gonic/gin
golang.org/x/net/html
```

go mod graph

Show dependency graph:

```
$ go mod graph
github.com/username/myproject github.com/gin-gonic/gin@v1.9.1
github.com/gin-gonic/gin@v1.9.1 github.com/bytedance/sonic@v1.9.1
...
```

Vendoring

Copy dependencies into your repository:

```
# Create vendor directory
$ go mod vendor

# Build using vendor
$ go build -mod=vendor
```

Vendoring pros: - Reproducible builds - Works offline - Audit dependencies

Vendoring cons: - Bloated repository - Manual updates

Replace Directive

Local Development

Replace a module with a local path:

```
replace github.com/some/package => ../local-package
```

Fork Substitution

Use your fork instead of original:

```
replace github.com/original/pkg => github.com/yourfork/pkg v1.2.3
```

Version Override

Force a specific version:

```
replace github.com/vulnerable/pkg v1.0.0 => github.com/vulnerable/pkg v1.0.1
```

Private Modules

GOPRIVATE

Configure private repository patterns:

```
$ go env -w GOPRIVATE=github.com/mycompany/*,gitlab.internal.com/*
```

Git Credentials

For private repos, configure Git:

```
# Use SSH
$ git config --global url."git@github.com:".insteadOf "https://github.com/"

# Or use access token
$ git config --global url."https://token:${TOKEN}@github.com/".insteadOf "https://github.com
```

Dependency Management Best Practices

1. Pin Dependencies

Always use exact versions in production:

```
# Good: exact version
$ go get github.com/pkg/errors@v0.9.1

# Risky: floating version
$ go get github.com/pkg/errors@latest
```

2. Regular Updates

Check for updates periodically:

```
$ go list -m -u all
```

3. Minimal Dependencies

Fewer dependencies = fewer problems: - Check if you really need that package - Consider stdlib alternatives - Watch for transitive dependencies

4. Security Scanning

Use vulnerability scanning:

```
# Built-in (Go 1.18+)
$ go install golang.org/x/vuln/cmd/govulncheck@latest
$ govulncheck ./...
```

Common Issues

Module Not Found

```
# Ensure GOPROXY is set
$ go env GOPROXY
https://proxy.golang.org,direct

# Try direct fetch
$ GOPROXY=direct go get github.com/some/package
```

Version Conflicts

```
# See what's required
$ go mod graph | grep conflicting-package

# Force specific version
$ go get conflicting-package@v1.2.3
```

Checksum Mismatch

```
# Clear cache and retry
$ go clean -modcache
$ go mod download
```

Summary

Command	Purpose
`go mod init`	Create new module
`go mod tidy`	Sync dependencies
`go get pkg@version`	Add/update dependency
`go mod download`	Pre-fetch dependencies
`go mod verify`	Verify checksums
`go mod vendor`	Create vendor directory

Related Topics

- Workspaces and Multi-Module Development
- How Go Code Is Organized

More examples

Example: prefer stdlib before `go get`

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```
package main

import (
    "encoding/json"
    "fmt"
```

```go
    "os"
)

type Config struct {
    Service string `json:"service"`
    Port    int    `json:"port"`
}

func main() {
    // Many needs are covered by encoding/json, net/http, etc.-no third party.
    raw := []byte(`{"service":"api","port":8080}`)
    var cfg Config
    if err := json.Unmarshal(raw, &cfg); err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }
    fmt.Printf("%s listens on %d\n", cfg.Service, cfg.Port)

    out, _ := json.MarshalIndent(cfg, "", "  ")
    fmt.Println(string(out))
}
```

Expected:

```
api listens on 8080
{
  "service": "api",
  "port": 8080
}
```

Example: semantic version labels as data

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import (
    "fmt"
    "strconv"
    "strings"
)

// parseSemver is a teaching toy-not a full semver library.
func parseSemver(v string) (major, minor, patch int, err error) {
    v = strings.TrimPrefix(v, "v")
    parts := strings.Split(v, ".")
```

```go
    if len(parts) != 3 {
        return 0, 0, 0, fmt.Errorf("want major.minor.patch, got %q", v)
    }
    major, err = strconv.Atoi(parts[0])
    if err != nil {
        return
    }
    minor, err = strconv.Atoi(parts[1])
    if err != nil {
        return
    }
    patch, err = strconv.Atoi(parts[2])
    return
}

func main() {
    for _, v := range []string{"v1.2.3", "2.0.0", "1.2"} {
        maj, min, pat, err := parseSemver(v)
        if err != nil {
            fmt.Printf("%s -> %v\n", v, err)
            continue
        }
        fmt.Printf("%s -> major=%d minor=%d patch=%d\n", v, maj, min, pat)
    }
}
```

Expected:

```
v1.2.3 -> major=1 minor=2 patch=3
2.0.0 -> major=2 minor=0 patch=0
1.2 -> want major.minor.patch, got "1.2"
```

Runnable example

Stdlib-only module demo: after `go mod init`, the program reports module path and dependency list via build info (empty deps until you `go get` something).

Save as `main.go`. From an empty directory:

```bash
go mod init example.com/demo
go mod tidy
go run .
cat go.mod
```

```go
package main

import (
    "fmt"
    "runtime/debug"
)

func main() {
    info, ok := debug.ReadBuildInfo()
    if !ok {
        fmt.Println("no build info (unusual for go run/build)")
        return
    }

    fmt.Println("module:", info.Main.Path)
    fmt.Println("go:", info.GoVersion)
    if info.Main.Version != "" {
        fmt.Println("main version:", info.Main.Version)
    }

    if len(info.Deps) == 0 {
        fmt.Println("deps: (none - stdlib only)")
    } else {
        fmt.Println("deps:")
        for _, d := range info.Deps {
            fmt.Printf("  %s %s\n", d.Path, d.Version)
        }
    }

    // Prefer stdlib until you truly need a third-party module.
    fmt.Println("tip: run `go get github.com/some/pkg@vX.Y.Z` then `go mod tidy`")
}
```

Expected output (illustrative):

```
module: example.com/demo
go: go1.22.5
main version: (devel)
deps: (none - stdlib only)
tip: run `go get github.com/some/pkg@vX.Y.Z` then `go mod tidy`
```

What to notice: - `go mod init` sets the module path that appears in `ReadBuildInfo`. - With only stdlib imports, `go.mod` stays lean—minimal dependencies by design. - `go mod tidy` is safe to run anytime; it syncs `go.mod`/`go.sum` with imports. - Third-party modules only appear under `deps` after you import and tidy them.

Try next: Add `import _ "golang.org/x/time/rate"`, run `go get golang.org/x/time@latest` and `go mod tidy`, then re-run and inspect `go.mod` / printed deps.

Workspaces and Multi-Module Development

Workspaces and Multi-Module Development

Overview

Go workspaces, introduced in Go 1.18, solve the challenge of developing multiple related modules simultaneously. With `go.work`, you can work on a main project and its dependencies together without modifying `go.mod` files.

The Problem Workspaces Solve

Without Workspaces

When developing multiple modules together, you'd need `replace` directives:

```
// main-app/go.mod
module github.com/company/main-app

replace github.com/company/shared-lib => ../shared-lib
```

Problems: - Must edit `go.mod` (can accidentally commit) - Each developer has different paths - CI/CD requires cleanup

With Workspaces

```
$ go work init main-app shared-lib
```

```
// go.work
go 1.27

use (
    ./main-app
    ./shared-lib
)
```

Benefits: - No `go.mod` modifications - Works for entire team - Automatically ignored by Git

Creating a Workspace

Initial Setup

```
# Directory structure
workspace/
   go.work           # Workspace file
   main-app/         # Main application
       go.mod
       main.go
   shared-lib/       # Library module
        go.mod
        lib.go
```

```
# Create workspace
$ cd workspace
$ go work init ./main-app ./shared-lib
```

The `go.work` File

```
go 1.27

use (
    ./main-app
    ./shared-lib
)
```

All commands (`go build`, `go test`, etc.) now use local modules.

Workspace Commands

`go work init`

Create a new workspace:

```
# Initialize with modules
$ go work init ./module1 ./module2

# Initialize empty workspace
$ go work init
```

go work use

Add modules to workspace:

```
# Add single module
$ go work use ./new-module

# Add multiple modules
$ go work use ./mod1 ./mod2

# Recursively find and add all modules
$ go work use -r .
```

go work edit

Programmatically modify workspace:

```
# Add a use directive
$ go work edit -use ./new-module

# Remove a use directive
$ go work edit -dropuse ./old-module

# Set Go version
$ go work edit -go 1.27
```

go work sync

Sync workspace modules with their dependencies:

```
$ go work sync
```

This updates each module's `go.mod` to match workspace requirements.

Multi-Module Development Workflow

Project Structure

```
company-workspace/
  go.work
  api/
     go.mod          # module github.com/company/api
     api.go
     api_test.go
```

```
core/
    go.mod          # module github.com/company/core
    core.go
cli/
    go.mod          # module github.com/company/cli
    main.go
internal-tools/
    go.mod          # module github.com/company/tools
    tools.go
```

Setting Up

```
$ cd company-workspace
$ go work init ./api ./core ./cli ./internal-tools
```

Development Flow

```
# Changes in core/ are immediately available to cli/
$ cd cli
$ go build  # Uses local core package

# Run tests across all modules
$ cd ..
$ go test ./...

# Build all binaries
$ go build ./cli/...
```

Replace vs Workspace

When to Use `replace`

Use Case	Approach
Permanent fork	`replace` in `go.mod`
CI workaround	`replace` in `go.mod`
Single dev experiment	`replace` in `go.mod`

When to Use Workspace

Use Case	Approach
Multi-module development	`go.work`
Team development	`go.work`
Monorepo-style work	`go.work`

Workspace with Replace

You can combine both. `go.work` supports `replace`:

```
go 1.27

use (
    ./main-app
    ./shared-lib
)

replace github.com/external/dep => ./custom-fork
```

Best Practices

1. Git Ignore `go.work`

```
# .gitignore
go.work
go.work.sum
```

Each developer creates their own workspace.

2. Document Setup

Include setup instructions in README:

```
## Development Setup

1. Clone all repositories:
   ```bash
   git clone github.com/company/api
   git clone github.com/company/core
   git clone github.com/company/cli
```

2. Create workspace:
```

```
go work init ./api ./core ./cli
```

“‘

3. CI Without Workspaces

In CI, build each module independently:

```yaml
# GitHub Actions
jobs:
  test:
    strategy:
      matrix:
        module: [api, core, cli]
    steps:
      - uses: actions/checkout@v4
      - name: Test
        working-directory: ${{ matrix.module }}
        run: go test ./...
```

4. Shared Workspace for Monorepos

For monorepos, commit `go.work`:

```
monorepo/
   go.work            # Committed
   services/
       api/
       worker/
       gateway/
   packages/
       auth/
       database/
```

Common Patterns

Microservices Development

```
workspace/
   go.work
   services/
       user-service/
       order-service/
       payment-service/
```

```
  shared/
      proto/
      middleware/
      database/
```

```
$ go work init
$ go work use -r .
```

Library Development

Testing a library change across consumers:

```
workspace/
  go.work
  my-library/          # The library
  app-using-library/   # Consumer 1
  another-consumer/    # Consumer 2
```

Troubleshooting

Module Not Found in Workspace

```
# Verify workspace includes module
$ cat go.work

# Re-add module
$ go work use ./missing-module
```

Wrong Version Used

Workspace modules take priority. To use published version:

```
# Temporarily disable workspace
$ GOWORK=off go build
```

Workspace Mode Detection

Check if workspace is active:

```
$ go env GOWORK
/path/to/go.work
```

Summary

Command	Purpose
`go work init`	Create workspace
`go work use`	Add modules
`go work edit`	Modify workspace
`go work sync`	Sync dependencies
`GOWORK=off`	Disable workspace

Exercises

1. Create a workspace with two modules where one depends on the other
2. Make changes to the dependency and verify the main module sees them
3. Add a third module to an existing workspace

Related Topics

- How Go Code Is Organized
- Dependency Management

More examples

Example: local "library" function in one module

Save as `main.go` and `go run .` (with `go mod init example.com/app` if needed).

```go
package main

import "fmt"

// In a real workspace this would live in a sibling module;
// here we simulate a library API in the same package for a quick run.
func Greet(name string) string {
    if name == "" {
        return "hello, world"
    }
    return "hello, " + name
}

func main() {
    fmt.Println(Greet(""))
    fmt.Println(Greet("workspace"))
```

```
}
```

Expected:

```
hello, world
hello, workspace
```

Example: replace-path thinking with build info

Save as `main.go` and `go run .` (with `go mod init example.com/consumer` if needed).

```go
package main

import (
    "fmt"
    "runtime/debug"
)

func main() {
    info, ok := debug.ReadBuildInfo()
    if !ok {
        fmt.Println("no build info")
        return
    }
    fmt.Println("this binary was built from module:", info.Main.Path)
    // Workspace / replace: your local tree is what you just compiled.
    fmt.Println("workspace tip: go work use ./lib makes local edits visible without publishi
    fmt.Println("escape hatch: GOWORK=off uses only go.mod require versions")
}
```

Expected:

```
this binary was built from module: example.com/consumer
workspace tip: go work use ./lib makes local edits visible without publishing
escape hatch: GOWORK=off uses only go.mod require versions
```

Runnable example

Single-module stand-in that prints whether workspace mode is active (`GOWORK`) and the main module path. Use it while experimenting with `go work init`.

Save as `main.go`. From an empty directory:

```
go mod init example.com/app
go run .
# Optional workspace experiment (sibling modules not required for this printout):
# go work init .
# go env GOWORK
# go run .
# GOWORK=off go run .
```

```go
package main

import (
    "fmt"
    "os"
    "runtime/debug"
)

func main() {
    gowork := os.Getenv("GOWORK")
    if gowork == "" {
        // go env GOWORK reports the path when a go.work is discovered;
        // the process env may be empty even in workspace mode.
        fmt.Println("GOWORK env: (empty - use `go env GOWORK` for discovery path)")
    } else {
        fmt.Println("GOWORK env:", gowork)
    }

    info, ok := debug.ReadBuildInfo()
    if !ok {
        fmt.Println("build info unavailable")
        return
    }
    fmt.Println("main module:", info.Main.Path)
    fmt.Println("go version:", info.GoVersion)

    // Illustrate the replace/workspace idea: local code always wins in a workspace.
    fmt.Println("local module builds use your tree; published versions need GOWORK=off")
}
```

Expected output (illustrative):

```
GOWORK env: (empty - use `go env GOWORK` for discovery path)
main module: example.com/app
go version: go1.22.5
local module builds use your tree; published versions need GOWORK=off
```

What to notice: - Workspaces change how the toolchain resolves modules, not your program's import syntax. - `go env GOWORK` is the reliable way to see if a `go.work` is active. - `GOWORK=off`

forces published module versions for a single command. - Multi-module work is about directories + `use` directives; the binary still starts at `main`.

Try next: Create two sibling dirs with their own `go.mod`, `go work init` both, import one from the other, change a function, and rebuild without publishing.

Core

Predeclared Types

Predeclared Types

Overview

Go provides a set of built-in types that form the foundation of all programs. Understanding these predeclared types—booleans, numerics, strings, and the error interface—is essential for writing effective Go code.

Boolean Type

The `bool` type represents true/false values.

```go
var enabled bool      // Zero value: false
active := true
disabled := false

// Boolean operators
result := active && disabled  // AND: false
result = active || disabled   // OR: true
result = !active              // NOT: false
```

Boolean Expressions

```go
x := 10
y := 20

greater := x > y      // false
equal := x == y       // false
notEqual := x != y    // true
lessEqual := x <= y   // true
```

Numeric Types

Integer Types

Type	Size	Range
`int8`	8 bits	-128 to 127
`int16`	16 bits	-32,768 to 32,767
`int32`	32 bits	-2B to 2B
`int64`	64 bits	±9 quintillion
`int`	Platform	32 or 64 bits

```
var small int8 = 127
var medium int32 = 2_147_483_647  // Underscores for readability
var large int64 = 9_223_372_036_854_775_807
var auto int = 42  // Platform-dependent size
```

Unsigned Integers

Type	Size	Range
`uint8`	8 bits	0 to 255
`uint16`	16 bits	0 to 65,535
`uint32`	32 bits	0 to 4B
`uint64`	64 bits	0 to 18 quintillion
`uint`	Platform	32 or 64 bits

```
var age uint8 = 30
var count uint = 1000
```

Type Aliases

```
type byte = uint8   // Alias for byte data
type rune = int32   // Alias for Unicode code points
```

Floating-Point Types

Type	Size	Precision
`float32`	32 bits	~7 decimal digits
`float64`	64 bits	~15 decimal digits

```
var pi float32 = 3.14159
var precise float64 = 3.141592653589793

// Scientific notation
```

```go
avogadro := 6.022e23
planck := 6.626e-34
```

Complex Numbers

```go
var c1 complex64 = 3 + 4i
c2 := complex(5, 12)  // 5 + 12i

real := real(c2)      // 5
imag := imag(c2)      // 12
```

String Type

Strings are immutable sequences of bytes (typically UTF-8).

```go
var greeting string = "Hello, World!"
name := "Go"

// String concatenation
message := greeting + " Welcome to " + name

// Multi-line strings (raw string literals)
poem := `Roses are red,
Violets are blue,
Go is awesome,
And so are you!`

// String length (bytes, not characters)
len(greeting)  // 13
```

Strings vs Runes

```go
s := "Hello,  "

len(s)              // 13 bytes
len([]rune(s))      // 9 characters

// Iterate by byte
for i := 0; i < len(s); i++ {
    fmt.Printf("%x ", s[i])
}
```

```go
// Iterate by rune (character)
for i, r := range s {
    fmt.Printf("%d: %c\n", i, r)
}
```

The Error Type

`error` is a built-in interface type:

```go
type error interface {
    Error() string
}
```

```go
import "errors"

func divide(a, b float64) (float64, error) {
    if b == 0 {
        return 0, errors.New("division by zero")
    }
    return a / b, nil
}

result, err := divide(10, 0)
if err != nil {
    fmt.Println("Error:", err)
}
```

Type Conversions

Go requires explicit type conversions:

```go
var i int = 42
var f float64 = float64(i)    // int to float64
var u uint = uint(i)          // int to uint

// String conversions
import "strconv"

s := strconv.Itoa(42)          // "42"
n, _ := strconv.Atoi("42")     // 42

fs := strconv.FormatFloat(3.14, 'f', 2, 64)  // "3.14"
```

Common Conversion Pitfalls

```
// Overflow (no error, just wraps)
var big int64 = 1000
var small int8 = int8(big)  // Overflow! Result: -24

// Float to int truncates
var f float64 = 3.9
var i int = int(f)  // 3 (not 4)
```

Type Information

```
import "fmt"

var x int = 42
fmt.Printf("Type: %T, Value: %v\n", x, x)
// Output: Type: int, Value: 42
```

Numeric Literals

```
// Decimal
decimal := 42

// Binary (0b prefix)
binary := 0b101010  // 42

// Octal (0o prefix)
octal := 0o52  // 42

// Hexadecimal (0x prefix)
hex := 0x2A  // 42

// With underscores for readability
billion := 1_000_000_000
```

Summary

Category	Types
Boolean	`bool`
Signed Integers	`int`, `int8`, `int16`, `int32`, `int64`

Category	Types
Unsigned Integers	`uint`, `uint8`, `uint16`, `uint32`, `uint64`
Floating Point	`float32`, `float64`
Complex	`complex64`, `complex128`
String	`string`
Aliases	`byte` (uint8), `rune` (int32)
Error	`error`

Related Topics

- Zero Values and Initialization
- Constants and Literals

More examples

Example: integer and float basics

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

func main() {
    var i int = 42
    var f float64 = 3.5
    fmt.Println("int:", i)
    fmt.Println("float:", f)
    fmt.Println("int(f):", int(f))
    fmt.Println("float64(i):", float64(i))
}
```

Expected:

```
int: 42
float: 3.5
int(f): 3
float64(i): 42
```

Example: bool and rune

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

func main() {
    ok := true
    r := 'G'
    fmt.Println("ok:", ok)
    fmt.Printf("rune %q codepoint %d\n", r, r)
    fmt.Println("not ok:", !ok)
}
```

Expected:

```
ok: true
rune 'G' codepoint 71
not ok: false
```

Runnable example

Save as `main.go`. From an empty directory:

```bash
go mod init example
go run .
```

```go
package main

import (
    "errors"
    "fmt"
    "strconv"
)

func divide(a, b float64) (float64, error) {
    if b == 0 {
        return 0, errors.New("division by zero")
    }
    return a / b, nil
}

func main() {
    var enabled bool = true
    var small int8 = 127
    var count int = 42
    var pi float64 = 3.141592653589793
```

```go
    var c complex128 = complex(3, 4)
    s := "Hello,  "
    b := byte('A')
    r := rune(' ')

    fmt.Printf("bool=%t int8=%d int=%d float64=%g\n", enabled, small, count, pi)
    fmt.Printf("complex=%v real=%g imag=%g\n", c, real(c), imag(c))
    fmt.Printf("string=%q bytes=%d runes=%d\n", s, len(s), len([]rune(s)))
    fmt.Printf("byte=%d rune=%c (%U)\n", b, r, r)

    // Numeric bases and conversion
    fmt.Printf("0b101010=%d 0x2A=%d\n", 0b101010, 0x2A)
    n, err := strconv.Atoi("42")
    fmt.Printf("Atoi=%d err=%v Itoa=%s\n", n, err, strconv.Itoa(n))

    // Overflow wraps; float→int truncates (use variables so the conversions are runtime)
    big := int64(1000)
    fl := 3.9
    fmt.Printf("int8(1000)=%d int(3.9)=%d\n", int8(big), int(fl))

    if _, err := divide(10, 0); err != nil {
        fmt.Println("error:", err)
    }
    q, err := divide(10, 4)
    fmt.Printf("10/4=%g err=%v types: q is %T\n", q, err, q)
}
```

Expected output (illustrative):

```
bool=true int8=127 int=42 float64=3.141592653589793
complex=(3+4i) real=3 imag=4
string="Hello,  " bytes=13 runes=9
byte=65 rune=  (U+4E16)
0b101010=42 0x2A=42
Atoi=42 err=<nil> Itoa=42
int8(1000)=-24 int(3.9)=3
error: division by zero
10/4=2.5 err=<nil> types: q is float64
```

What to notice: - `%T` / `%v` make types and values observable. - `len(string)` is bytes; `len([]rune(s))` is characters. - Conversions are explicit; overflow wraps without an error. - `error` is a value returned and checked like any other type.

Try next: Change `int8(1000)` to a value you pick and predict the wrapped result.

Zero Values and Initialization

Zero Values and Initialization

Overview

Every variable in Go has a value from the moment it's declared. If you don't explicitly initialize a variable, it gets a **zero value**—a sensible default that prevents undefined behavior.

Zero Values by Type

Type	Zero Value
`bool`	`false`
`int`, `int8`, …	`0`
`uint`, `uint8`, …	`0`
`float32`, `float64`	`0.0`
`complex64`, `complex128`	`(0+0i)`
`string`	`""` (empty string)
`pointer`	`nil`
`slice`	`nil`
`map`	`nil`
`channel`	`nil`
`function`	`nil`
`interface`	`nil`

Demonstrating Zero Values

```go
package main

import "fmt"

func main() {
    var b bool
    var i int
    var f float64
    var s string
    var p *int
    var sl []int
```

```go
    var m map[string]int
    var ch chan int
    var fn func()
    var iface interface{}

    fmt.Printf("bool: %v\n", b)      // false
    fmt.Printf("int: %v\n", i)       // 0
    fmt.Printf("float64: %v\n", f)   // 0
    fmt.Printf("string: %q\n", s)    // ""
    fmt.Printf("pointer: %v\n", p)   // <nil>
    fmt.Printf("slice: %v\n", sl)    // []
    fmt.Printf("map: %v\n", m)       // map[]
    fmt.Printf("channel: %v\n", ch)  // <nil>
    fmt.Printf("func: %v\n", fn)     // <nil>
    fmt.Printf("interface: %v\n", iface) // <nil>
}
```

Struct Zero Values

Struct fields get their respective zero values:

```go
type User struct {
    Name    string
    Age     int
    Active  bool
    Balance float64
}

var user User
// user.Name = ""
// user.Age = 0
// user.Active = false
// user.Balance = 0.0
```

Nested Structs

```go
type Address struct {
    Street string
    City   string
}

type Person struct {
    Name    string
    Address Address  // Embedded struct gets zero values too
```

```go
}

var p Person
// p.Name = ""
// p.Address.Street = ""
// p.Address.City = ""
```

Array Zero Values

Arrays are initialized with zero values for each element:

```go
var arr [5]int        // [0, 0, 0, 0, 0]
var strs [3]string    // ["", "", ""]
var bools [2]bool     // [false, false]
```

Why Zero Values Matter

1. Eliminates Undefined Behavior

```go
// In Go, this is safe and predictable
var count int
count++  // count is now 1

// In C, this would be undefined behavior
// int count;
// count++;  // Undefined! Could be anything
```

2. Reduces Boilerplate

```go
// No need for explicit initialization
type Config struct {
    Debug   bool
    Timeout int
    Host    string
}

cfg := Config{}  // All zeros, ready to use selectively
cfg.Host = "localhost"  // Only set what differs from zero
```

3. Enables "Usable Zero Value" Pattern

Well-designed types work correctly with zero values:

```go
import "bytes"

// bytes.Buffer works without initialization
var buf bytes.Buffer
buf.WriteString("Hello, ")
buf.WriteString("World!")
fmt.Println(buf.String())  // "Hello, World!"

// sync.Mutex works without initialization
var mu sync.Mutex
mu.Lock()
mu.Unlock()
```

Initialization Methods

Short Variable Declaration

```go
name := "Alice"      // Inferred type: string
count := 42          // Inferred type: int
price := 19.99       // Inferred type: float64
```

Explicit Type Declaration

```go
var name string = "Alice"
var count int = 42
var active bool = true
```

Type Inference (var)

```go
var name = "Alice"   // Type inferred from value
var count = 42       // int
```

Multiple Variable Declaration

```go
var (
    name   = "Alice"
    age    = 30
    active = true
)

// Or inline
x, y, z := 1, 2, 3
```

Composite Literal Initialization

```go
// Struct
user := User{
    Name:   "Alice",
    Age:    30,
    Active: true,
}

// Partially initialized (rest are zero values)
user := User{Name: "Bob"}  // Age: 0, Active: false

// Array
arr := [3]int{1, 2, 3}

// Slice
sl := []string{"a", "b", "c"}

// Map
m := map[string]int{"one": 1, "two": 2}
```

nil vs Zero Value

`nil` is the zero value for pointers, slices, maps, channels, functions, and interfaces.

nil Slice vs Empty Slice

```go
var nilSlice []int        // nil slice
emptySlice := []int{}     // empty slice (not nil)

nilSlice == nil           // true
```

```go
emptySlice == nil         // false

len(nilSlice)             // 0
len(emptySlice)           // 0

// Both work with append
nilSlice = append(nilSlice, 1)      // [1]
emptySlice = append(emptySlice, 1)  // [1]
```

nil Map Danger

```go
var m map[string]int  // nil map

// Reading is safe
val := m["key"]       // Returns 0 (zero value)

// Writing panics!
m["key"] = 42         // panic: assignment to nil map

// Always initialize before writing
m = make(map[string]int)
m["key"] = 42         // Now safe
```

The `new` vs `make` Functions

`new(T)`

Allocates zeroed memory and returns a pointer:

```go
ptr := new(int)      // *int pointing to 0
*ptr = 42

user := new(User)    // *User with zero-valued fields
user.Name = "Alice"
```

`make(T, args)`

Creates and initializes slices, maps, and channels:

```go
// Slice with length and capacity
sl := make([]int, 5)        // [0, 0, 0, 0, 0]
sl := make([]int, 0, 10)    // [] with capacity 10
```

```go
// Map
m := make(map[string]int)   // Empty, ready to use

// Channel
ch := make(chan int)        // Unbuffered channel
ch := make(chan int, 10)    // Buffered channel
```

Summary

Concept	Description
Zero Value	Default value assigned to uninitialized variables
Purpose	Eliminates undefined behavior, reduces boilerplate
`new(T)`	Allocates zeroed memory, returns `*T`
`make(T)`	Creates initialized slice, map, or channel
`nil`	Zero value for pointers, slices, maps, channels, funcs, interfaces

Related Topics

- Predeclared Types
- Variables and Scope

More examples

Example: zero values of common types

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

func main() {
    var n int
    var s string
    var b bool
    var p *int
    fmt.Printf("int=%d string=%q bool=%v ptr=%v\n", n, s, b, p)
}
```

Expected:

```
int=0 string="" bool=false ptr=<nil>
```

Example: nil map vs empty map

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

func main() {
    var nilMap map[string]int
    empty := map[string]int{}
    fmt.Println("nil len:", len(nilMap), "read:", nilMap["x"])
    fmt.Println("empty len:", len(empty), "read:", empty["x"])
    empty["x"] = 1
    fmt.Println("after write empty:", empty)
    // nilMap["x"] = 1 // would panic
    fmt.Println("nil map needs make before write")
}
```

Expected:

```
nil len: 0 read: 0
empty len: 0 read: 0
after write empty: map[x:1]
nil map needs make before write
```

Runnable example

Complete program that prints zero values, shows a usable zero `bytes.Buffer`, and panics safely-avoided on a nil map.

Save as `main.go`. From an empty directory:

```
go mod init example
go run .
```

```go
package main

import (
    "bytes"
    "fmt"
)
```

```go
type User struct {
    Name   string
    Age    int
    Active bool
}

func main() {
    var b bool
    var i int
    var f float64
    var s string
    var p *int
    var sl []int
    var m map[string]int
    var ch chan int
    var fn func()
    var iface any

    fmt.Printf("bool=%v int=%v float64=%v string=%q\n", b, i, f, s)
    fmt.Printf("pointer=%v slice=%v map=%v chan=%v func=%v iface=%v\n", p, sl, m, ch, fn, if

    var user User
    fmt.Printf("user zero: %+v\n", user)

    var arr [3]int
    fmt.Printf("array zero: %v\n", arr)

    // Usable zero value: no constructor required.
    var buf bytes.Buffer
    buf.WriteString("Hello, ")
    buf.WriteString("zero values!")
    fmt.Println(buf.String())

    // nil map: read OK, write needs make
    fmt.Println("nil map read:", m["missing"])
    m = make(map[string]int)
    m["ok"] = 1
    fmt.Println("after make:", m)

    // nil vs empty slice
    var nilSlice []int
    empty := []int{}
    fmt.Printf("nilSlice==nil? %v empty==nil? %v\n", nilSlice == nil, empty == nil)
    nilSlice = append(nilSlice, 7)
    fmt.Println("append on nil slice:", nilSlice)
}
```

Expected output (illustrative):

```
bool=false int=0 float64=0 string=""
pointer=<nil> slice=[] map=map[] chan=<nil> func=<nil> iface=<nil>
user zero: {Name: Age:0 Active:false}
array zero: [0 0 0]
Hello, zero values!
nil map read: 0
after make: map[ok:1]
nilSlice==nil? true empty==nil? false
append on nil slice: [7]
```

What to notice: - Every type has a defined zero—no "undefined" locals. - Struct fields zero independently; arrays zero every element. - `bytes.Buffer` works from a zero value (idiomatic design). - Writing a nil map would panic; `make` first is required.

Try next: Comment out `m = make(...)` and assign `m["ok"] = 1` to observe the panic, then restore it.

Constants and Literals

Constants and Literals

Overview

Constants in Go are values determined at compile time that cannot be changed during program execution. Go distinguishes between typed and untyped constants, with untyped constants having special flexibility.

Declaring Constants

Basic Declaration

```go
const Pi = 3.14159
const MaxSize = 1024
const Greeting = "Hello, World!"
```

Constant Block

```go
const (
    StatusOK       = 200
    StatusNotFound = 404
    StatusError    = 500
)
```

Typed vs Untyped Constants

```go
// Untyped constant (more flexible)
const untypedPi = 3.14159

// Typed constant (explicit type)
const typedPi float64 = 3.14159
```

Untyped Constants

Untyped constants have a **kind** but not a fixed type, allowing them to adapt:

```go
const answer = 42  // Untyped integer constant

var i int = answer       // Works: becomes int
var f float64 = answer   // Works: becomes float64
var c complex128 = answer // Works: becomes complex128
```

Why This Matters

```go
// Untyped constant can be used with any compatible type
const billion = 1e9

var i int64 = billion     //  Works
var f float32 = billion   //  Works

// Typed constant is locked to its type
const typedBillion int64 = 1e9
var f float64 = typedBillion  //  Error: cannot use int64 as float64
```

Constant Kinds

Kind	Examples
Boolean	`true`, `false`
Integer	`42`, `0x2A`, `0b101010`
Floating-point	`3.14`, `6.022e23`
Complex	`1 + 2i`
String	`"hello"`, `` `raw string` ``
Rune	`'A'`, `' '`

The `iota` Constant Generator

`iota` generates sequential integer constants, starting from 0 and incrementing by 1 for each constant in the block.

Basic Usage

```go
const (
    Sunday = iota    // 0
    Monday           // 1
    Tuesday          // 2
    Wednesday        // 3
    Thursday         // 4
    Friday           // 5
    Saturday         // 6
)
```

Skip Values

```go
const (
    _ = iota         // 0 (skipped)
    KB = 1 << (10 * iota)  // 1 << 10 = 1024
    MB                     // 1 << 20 = 1,048,576
    GB                     // 1 << 30 = 1,073,741,824
    TB                     // 1 << 40
)
```

Bit Flags with iota

```go
const (
    FlagRead    = 1 << iota  // 1 (0001)
    FlagWrite                // 2 (0010)
    FlagExecute              // 4 (0100)
)

permissions := FlagRead | FlagWrite  // 3 (0011)

hasRead := permissions & FlagRead != 0   // true
hasExec := permissions & FlagExecute != 0 // false
```

Multiple iota in One Line

```go
const (
    A, B = iota, iota + 10  // A=0, B=10
    C, D                    // C=1, D=11
    E, F                    // E=2, F=12
)
```

Reset iota

Each `const` block resets `iota` to 0:

```
const (
    First = iota   // 0
    Second         // 1
)

const (
    Third = iota   // 0 (reset!)
    Fourth         // 1
)
```

Constant Expressions

Constants can be computed from other constants at compile time:

```
const (
    SecondsPerMinute = 60
    MinutesPerHour   = 60
    SecondsPerHour   = SecondsPerMinute * MinutesPerHour
    SecondsPerDay    = SecondsPerHour * 24
)
```

Allowed in Constant Expressions

- Arithmetic operators: `+`, `-`, `*`, `/`, `%`
- Comparison operators: `==`, `!=`, `<`, `>`, `<=`, `>=`
- Logical operators: `&&`, `||`, `!`
- Bit operators: `&`, `|`, `^`, `<<`, `>>`, `&^`
- Built-in functions: `len()`, `cap()` (for some values), `real()`, `imag()`, `complex()`

Not Allowed

```
const x = math.Sin(0)  // Error: function call not allowed
const y = len(someVar)  // Error: variable not allowed
```

Numeric Literals

Integer Literals

```
decimal := 42           // Base 10
binary := 0b101010      // Base 2 (Go 1.13+)
octal := 0o52           // Base 8 (Go 1.13+)
oldOctal := 052         // Base 8 (legacy)
hex := 0x2A             // Base 16
```

Floating-Point Literals

```
f1 := 3.14
f2 := .5                // 0.5
f3 := 1.                // 1.0
f4 := 6.022e23          // Scientific notation
f5 := 6.022E23          // Same
f6 := 1.5e-10           // 0.00000000015
```

Hexadecimal Floats (Go 1.13+)

```
f := 0x1.fp+2  // (1 + 15/16) * 2^2 = 7.75
```

Imaginary Literals

```
i := 3i
c := 2 + 3i
```

Digit Separators

Go 1.13+ allows underscores in numeric literals for readability:

```
billion := 1_000_000_000
binary := 0b_1010_1010
hex := 0xFF_FF_FF_FF
```

String and Rune Literals

Interpreted Strings

```
s := "Hello, World!\n"  // Escape sequences processed
s := "Tab:\tNewline:\n"
s := "Quote: \"Go\""
```

Escape Sequences

Escape	Meaning
`\n`	Newline
`\t`	Tab
`\\`	Backslash
`\"`	Double quote
`\r`	Carriage return
`\xNN`	Hex byte
`\uNNNN`	Unicode code point (4 hex)
`\UNNNNNNNN`	Unicode code point (8 hex)

Raw String Literals

```
raw := `No escape sequences: \n \t \\
Multiple lines work!
Useful for regex: \d+\.\d+`
```

Rune Literals

```
r1 := 'A'      // 65
r2 := ' '     // 19990
r3 := '\n'     // 10
r4 := '\x41'   // 65 (hex)
r5 := '\u4e16' // 19990 (unicode)
```

Common Patterns

Enum Pattern

```go
type Status int

const (
    StatusPending Status = iota
    StatusApproved
    StatusRejected
)

func (s Status) String() string {
    switch s {
    case StatusPending:
        return "Pending"
    case StatusApproved:
        return "Approved"
    case StatusRejected:
        return "Rejected"
    default:
        return "Unknown"
    }
}
```

Configuration Constants

```go
const (
    DefaultTimeout = 30 * time.Second
    MaxRetries     = 3
    BufferSize     = 4096
)
```

Summary

Concept	Description
`const`	Compile-time immutable value
Untyped	Flexible, adapts to context
Typed	Fixed to a specific type
`iota`	Auto-incrementing generator
Literal	Numeric, string, or rune value in source

Related Topics

- Predeclared Types
- Variables and Scope

More examples

Example: iota enums

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

const (
    StatusDraft = iota
    StatusOpen
    StatusClosed
)

func main() {
    fmt.Println("Draft:", StatusDraft)
    fmt.Println("Open:", StatusOpen)
    fmt.Println("Closed:", StatusClosed)
}
```

Expected:

```
Draft: 0
Open: 1
Closed: 2
```

Example: untyped constant flexibility

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

const Answer = 42 // untyped integer constant

func main() {
    var i int = Answer
```

```go
    var i64 int64 = Answer
    var f float64 = Answer
    fmt.Println(i, i64, f)
}
```

Expected:

```
42 42 42
```

Runnable example

Save as `main.go`. From an empty directory:

```
go mod init example
go run .
```

```go
package main

import "fmt"

type Status int

const (
    StatusPending Status = iota
    StatusApproved
    StatusRejected
)

func (s Status) String() string {
    switch s {
    case StatusPending:
        return "Pending"
    case StatusApproved:
        return "Approved"
    case StatusRejected:
        return "Rejected"
    default:
        return "Unknown"
    }
}

const (
    _  = iota
    KB = 1 << (10 * iota) // 1 << 10
    MB
```

```go
    GB
)

const (
    FlagRead = 1 << iota // 1
    FlagWrite            // 2
    FlagExecute          // 4
)

func main() {
    // Untyped constant adapts to destination types.
    const answer = 42
    var i int = answer
    var f float64 = answer
    fmt.Printf("untyped answer as int=%d float64=%g\n", i, f)

    fmt.Printf("status: %s=%d %s=%d\n", StatusPending, StatusPending, StatusApproved, Status
    fmt.Printf("sizes: KB=%d MB=%d GB=%d\n", KB, MB, GB)

    perms := FlagRead | FlagWrite
    fmt.Printf("perms=%04b read=%v write=%v exec=%v\n",
        perms,
        perms&FlagRead != 0,
        perms&FlagWrite != 0,
        perms&FlagExecute != 0,
    )

    // Literals
    fmt.Printf("bases: %d %d %d %d\n", 42, 0b101010, 0o52, 0x2A)
    fmt.Printf("rune %q is %d; raw=`%s`\n", ' ', ' ', `line\nnot escaped`)
}
```

Expected output (illustrative):

```
untyped answer as int=42 float64=42
status: Pending=0 Approved=1
sizes: KB=1024 MB=1048576 GB=1073741824
perms=0011 read=true write=true exec=false
bases: 42 42 42 42
rune ' ' is 19990; raw=`line\nnot escaped`
```

What to notice: - `iota` auto-numbers enums and builds power-of-two sizes/flags. - Untyped constants convert cleanly into both `int` and `float64`. - Bit flags combine with `|` and test with `&`. - Raw string literals do not interpret `\n`.

Try next: Add `TB` after `GB` in the size block and print it; confirm `1 << 40`.

Variables and Scope

Variables and Scope

Overview

Variables in Go store values that can change during program execution. Understanding declaration patterns and visibility rules (scope) is fundamental to writing correct Go programs.

Variable Declaration

Using `var`

```go
// Explicit type
var name string
var age int

// With initialization
var name string = "Alice"
var age int = 30

// Type inference
var name = "Alice"  // Inferred as string
var age = 30        // Inferred as int
```

Short Variable Declaration (`:=`)

Inside functions, use `:=` for concise declaration:

```go
func main() {
    name := "Alice"   // var name = "Alice"
    age := 30         // var age = 30
    active := true    // var active = true
}
```

When to Use `var` vs `:=`

Use `var`	Use `:=`
Package-level variables	Inside functions
Explicit type needed	Type can be inferred
Zero value is desired	Initialization value provided

```go
// Package level (var only)
var globalConfig Config

func main() {
    // Function level (either works, := preferred)
    count := 0

    // Explicit type needed
    var ratio float64 = 0  // := 0 would be int
}
```

Multiple Variables

```go
// Same type
var x, y, z int

// Different values
var a, b, c = 1, "hello", true

// Short declaration
name, age, active := "Alice", 30, true
```

Variable Block

```go
var (
    name   string = "Alice"
    age    int    = 30
    active bool   = true
)
```

Redeclaration and Shadowing

Redeclaration with `:=`

`:=` can redeclare if at least one variable is new:

```go
x := 1
x, y := 2, 3  // x redeclared, y is new (allowed)

// This fails:
// x := 4  // Error: no new variables on left side
```

Variable Shadowing

An inner scope can declare a variable with the same name as an outer scope:

```go
x := 1
fmt.Println(x)  // 1

if true {
    x := 2       // New x, shadows outer x
    fmt.Println(x)  // 2
}

fmt.Println(x)  // 1 (outer x unchanged)
```

Warning: Shadowing often creates bugs:

```go
var err error

if condition {
    result, err := someFunc()  // err shadows! Outer err unchanged
    // ...
}

if err != nil {  // This checks outer err, always nil!
    // Never reached
}
```

Fix:

```go
var err error
var result Result

if condition {
    result, err = someFunc()  // No :=, uses outer err
}
```

Scope Levels

Package Scope

Variables declared outside functions are visible to the entire package:

```go
package mypackage

var packageVar = "visible to all files in mypackage"

func init() {
    packageVar = "can modify here"
}

func SomeFunc() {
    fmt.Println(packageVar)  // Accessible
}
```

File Scope (Imports)

Import names are scoped to the file:

```go
// file1.go
import "fmt"  // fmt available only in this file

// file2.go
import "fmt"  // Must import again
```

Function Scope

Parameters and local variables are visible within the function:

```go
func greet(name string) {
    message := "Hello, " + name  // Local to greet
    fmt.Println(message)
}

// name and message are not accessible here
```

Block Scope

Variables declared in blocks (`{}`) are visible only within that block:

```go
if x > 0 {
    y := x * 2  // Only visible inside if block
    fmt.Println(y)
}
// y is not accessible here

for i := 0; i < 10; i++ {
    // i is visible only in the loop
}
// i is not accessible here

switch n := getValue(); n {
case 1:
    // n is visible in switch block
}
// n is not accessible here
```

Visibility (Exported vs Unexported)

Go uses capitalization for visibility across packages:

```go
package user

var PublicVar = "accessible from other packages"   // Exported
var privateVar = "only in user package"            // Unexported

func PublicFunc() {}   // Exported
func privateFunc() {}  // Unexported

type PublicType struct {
    PublicField  string  // Exported
    privateField string  // Unexported
}
```

The Blank Identifier (_)

Use _ to discard unwanted values:

```go
// Discard second return value
value, _ := someFunc()

// Discard loop index
for _, item := range items {
```

```go
    process(item)
}

// Import for side effects only
import _ "github.com/lib/pq"
```

Variable Lifetime

Stack vs Heap

Go decides where to allocate based on escape analysis:

```go
func createValue() int {
    x := 42  // Likely on stack
    return x // Copy returned, x can be reclaimed
}

func createPointer() *int {
    x := 42   // Must go on heap
    return &x // Pointer escapes function
}
```

Garbage Collection

You don't need to free memory—Go's garbage collector handles it:

```go
func process() {
    data := make([]byte, 1024*1024)  // 1MB allocated
    // Use data...
}  // data becomes garbage, will be collected
```

Common Patterns

Zero Value Initialization

```go
var config Config  // All fields are zero values
config.Timeout = 30 * time.Second  // Set only what differs
```

Conditional Initialization

```
var s string
if condition {
    s = "yes"
} else {
    s = "no"
}

// Or with short declaration
s := "default"
if condition {
    s = "override"
}
```

Multiple Assignment

```
// Swap without temp variable
a, b = b, a

// Multiple returns
x, y, z := multiReturn()
```

Common Pitfalls

Accidental Shadowing

```
// Bug: err is shadowed
var err error
if x > 0 {
    y, err := compute()  // New err!
    _ = y
}
// Original err still nil

// Fix: declare y separately
var err error
var y int
if x > 0 {
    y, err = compute()  // Uses outer err
}
```

Loop Variable Capture

```go
// Bug: all goroutines see last value
for _, v := range values {
    go func() {
        fmt.Println(v)  // v is shared!
    }()
}

// Fix: pass as parameter
for _, v := range values {
    go func(v int) {
        fmt.Println(v)
    }(v)
}

// Go 1.22+ fixes this by default
```

Summary

Declaration	Usage
`var x T`	Zero value, explicit type
`var x = value`	Type inference
`x := value`	Short declaration (functions only)
`var x, y T`	Multiple same type
`x, y := a, b`	Multiple with inference

Scope	Visibility
Package	All files in package
File	Imports
Function	Function body
Block	`{}` boundaries

Related Topics

- Constants and Literals
- Functions in Depth

More examples

Example: short declaration scope

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

func main() {
    x := 1
    {
        x := 2 // new x shadows outer
        fmt.Println("inner:", x)
    }
    fmt.Println("outer:", x)
}
```

Expected:

```
inner: 2
outer: 1
```

Example: if with short statement

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import (
    "fmt"
    "strconv"
)

func main() {
    if n, err := strconv.Atoi("42"); err != nil {
        fmt.Println("err:", err)
    } else {
        fmt.Println("n:", n)
    }
    // n is not in scope here
    fmt.Println("done")
}
```

Expected:

```
n: 42
done
```

Runnable example

Save as `main.go`. From an empty directory:

```
go mod init example
go run .
```

```
package main

import "fmt"

var packageLevel = "package"

func compute(ok bool) (int, error) {
    if !ok {
        return 0, fmt.Errorf("not ok")
    }
    return 42, nil
}

func main() {
    // Short declaration inside functions
    name := "Alice"
    var ratio float64 // zero value; := 0 would be int
    fmt.Printf("name=%s ratio=%g package=%s\n", name, ratio, packageLevel)

    // Shadowing demo
    x := 1
    fmt.Println("outer x:", x)
    if true {
        x := 2 // new x in block scope
        fmt.Println("inner x:", x)
    }
    fmt.Println("outer x after block:", x)

    // Redeclare with := when at least one name is new
    x, y := 3, 4
    fmt.Printf("redecl x=%d new y=%d\n", x, y)

    // Correct multi-return assignment (no accidental err shadow)
    var err error
    var n int
```

```go
    n, err = compute(true)
    fmt.Printf("compute ok: n=%d err=%v\n", n, err)
    n, err = compute(false)
    fmt.Printf("compute fail: n=%d err=%v\n", n, err)

    // Blank identifier discards values
    _, _ = compute(true)
    a, b := 1, 2
    a, b = b, a
    fmt.Printf("swapped a=%d b=%d\n", a, b)
}
```

Expected output (illustrative):

```
name=Alice ratio=0 package=package
outer x: 1
inner x: 2
outer x after block: 1
redecl x=3 new y=4
compute ok: n=42 err=<nil>
compute fail: n=0 err=not ok
swapped a=2 b=1
```

What to notice: - Block scope shadows do not update the outer variable. - `:=` redeclares only when a new variable appears on the left. - Assigning with `=` keeps a single `err` in outer scope—critical for real code. - Package-level vars are visible to all functions in the package.

Try next: Inside the `if true` block, change `x := 2` to `x = 2` and observe the outer value after the block.

Conditional Logic

Conditional Logic

Overview

Go provides `if/else` statements and `switch` statements for conditional logic. Both are straightforward but have unique features compared to other languages.

The `if` Statement

Basic Syntax

```
if condition {
    // code
}
```

With `else`

```
if x > 0 {
    fmt.Println("Positive")
} else {
    fmt.Println("Non-positive")
}
```

With `else if`

```
if x > 0 {
    fmt.Println("Positive")
} else if x < 0 {
    fmt.Println("Negative")
} else {
    fmt.Println("Zero")
}
```

Initialization Statement

Go allows a statement before the condition:

```go
if err := doSomething(); err != nil {
    // Handle error
    // err is only visible in this if/else block
    return err
}
// err is not accessible here
```

This is idiomatic for error handling:

```go
if file, err := os.Open("data.txt"); err != nil {
    log.Fatal(err)
} else {
    defer file.Close()
    // Use file
}
```

No Parentheses Required

Unlike C-style languages, conditions don't need parentheses:

```go
// Go style
if x > 0 {
}

// Not idiomatic (but valid)
if (x > 0) {
}
```

Braces Required

Braces are mandatory, even for single statements:

```go
// Error: missing braces
if x > 0
    fmt.Println("yes")

// Correct
if x > 0 {
    fmt.Println("yes")
}
```

Boolean Expressions

Comparison Operators

Operator	Meaning
`==`	Equal
`!=`	Not equal
`<`	Less than
`>`	Greater than
`<=`	Less than or equal
`>=`	Greater than or equal

Logical Operators

Operator	Meaning
`&&`	AND (short-circuit)
`||`	OR (short-circuit)
`!`	NOT

```go
// Short-circuit evaluation
if x != 0 && y/x > 1 {
    // y/x only evaluated if x != 0
}

if ptr != nil && ptr.Value > 0 {
    // Ptr.Value only accessed if ptr != nil
}
```

The `switch` Statement

Basic Switch

```go
switch day {
case "Monday":
    fmt.Println("Start of week")
case "Friday":
    fmt.Println("Almost weekend!")
case "Saturday", "Sunday":
    fmt.Println("Weekend!")
default:
    fmt.Println("Midweek")
```

```
}
```

No Fall-Through by Default

Unlike C, cases don't fall through:

```
switch n {
case 1:
    fmt.Println("One")
    // No break needed, stops here
case 2:
    fmt.Println("Two")
}
```

Explicit Fallthrough

Use `fallthrough` keyword when needed:

```
switch n {
case 1:
    fmt.Println("One")
    fallthrough
case 2:
    fmt.Println("Two")
}
// Input: 1
// Output:
// One
// Two
```

Switch with Initialization

```
switch os := runtime.GOOS; os {
case "darwin":
    fmt.Println("macOS")
case "linux":
    fmt.Println("Linux")
default:
    fmt.Println(os)
}
```

Expression-less Switch

A `switch` without an expression is equivalent to `switch true`:

```
t := time.Now()

switch {
case t.Hour() < 12:
    fmt.Println("Good morning!")
case t.Hour() < 17:
    fmt.Println("Good afternoon!")
default:
    fmt.Println("Good evening!")
}
```

This is often cleaner than chained `if/else if`:

```
switch {
case x < 0:
    return "negative"
case x == 0:
    return "zero"
case x > 0:
    return "positive"
}
```

Type Switch

Special form for interface type assertions:

```
func describe(i interface{}) {
    switch v := i.(type) {
    case int:
        fmt.Printf("Integer: %d\n", v)
    case string:
        fmt.Printf("String: %s\n", v)
    case bool:
        fmt.Printf("Boolean: %t\n", v)
    default:
        fmt.Printf("Unknown type: %T\n", v)
    }
}

describe(42)        // Integer: 42
describe("hello")   // String: hello
describe(true)      // Boolean: true
```

```go
describe(3.14)      // Unknown type: float64
```

Common Patterns

Error Checking

```go
result, err := someOperation()
if err != nil {
    return fmt.Errorf("operation failed: %w", err)
}
// Use result
```

Boolean Assignment

```go
var status string
if success {
    status = "completed"
} else {
    status = "failed"
}

// Note: Go doesn't have a ternary operator
// This doesn't work: status := success ? "completed" : "failed"
```

Guard Clauses

```go
func processUser(user *User) error {
    if user == nil {
        return errors.New("user cannot be nil")
    }
    if user.Name == "" {
        return errors.New("user name required")
    }
    if user.Age < 0 {
        return errors.New("invalid age")
    }

    // Main logic here
    return nil
}
```

ok-idiom

```go
// Map lookup
if val, ok := myMap[key]; ok {
    fmt.Println("Found:", val)
} else {
    fmt.Println("Not found")
}

// Type assertion
if str, ok := value.(string); ok {
    fmt.Println("String:", str)
}

// Channel receive
if val, ok := <-ch; ok {
    fmt.Println("Received:", val)
} else {
    fmt.Println("Channel closed")
}
```

Comma-ok Pattern

```go
// Check map existence
if _, exists := users[id]; exists {
    // User exists
}

// Check channel state
if _, open := <-ch; !open {
    // Channel is closed
}
```

Switch vs If/Else

Prefer Switch When:

```go
// Multiple discrete values
switch status {
case "pending", "processing":
    queue()
case "done":
    complete()
```

```
case "error":
    retry()
}
```

Prefer If/Else When:

```
// Complex conditions
if x > 0 && y > 0 && x*y < 100 {
    // ...
} else if x < 0 || y < 0 {
    // ...
}
```

Summary

Feature	Description
`if`	Basic conditional
`if x := val; condition`	If with initialization
`switch expr`	Multi-way branching
`switch { case cond: }`	Expression-less switch
`switch v := x.(type)`	Type switch
`fallthrough`	Explicit fall-through in switch

Related Topics

- Loops and Iteration
- Errors as Values

More examples

Example: if / else if ladder

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```
package main

import "fmt"

func grade(score int) string {
    if score >= 90 {
```

```go
        return "A"
    } else if score >= 80 {
        return "B"
    } else if score >= 70 {
        return "C"
    }
    return "F"
}

func main() {
    fmt.Println(grade(95))
    fmt.Println(grade(82))
    fmt.Println(grade(60))
}
```

Expected:

```
A
B
F
```

Example: switch on strings

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

func main() {
    for _, cmd := range []string{"start", "stop", "wat"} {
        switch cmd {
        case "start":
            fmt.Println(cmd, "-> running")
        case "stop":
            fmt.Println(cmd, "-> halted")
        default:
            fmt.Println(cmd, "-> unknown")
        }
    }
}
```

Expected:

```
start -> running
stop -> halted
```

```
wat -> unknown
```

Runnable example

Save as `main.go`. From an empty directory:

```
go mod init example
go run .
```

```go
package main

import (
    "fmt"
    "runtime"
)

func sign(x int) string {
    switch {
    case x < 0:
        return "negative"
    case x == 0:
        return "zero"
    default:
        return "positive"
    }
}

func describe(i any) {
    switch v := i.(type) {
    case int:
        fmt.Printf("Integer: %d\n", v)
    case string:
        fmt.Printf("String: %s\n", v)
    case bool:
        fmt.Printf("Boolean: %t\n", v)
    default:
        fmt.Printf("Unknown type: %T value=%v\n", v, v)
    }
}

func main() {
    // if with short initialization (err scoped to if/else)
    if n := 10; n > 0 {
        fmt.Println("positive init:", n)
    }
```

```go
    x := -3
    if x > 0 {
        fmt.Println("Positive")
    } else if x < 0 {
        fmt.Println("Negative")
    } else {
        fmt.Println("Zero")
    }

    // switch: no fall-through by default; multi-case lists
    day := "Saturday"
    switch day {
    case "Monday":
        fmt.Println("Start of week")
    case "Saturday", "Sunday":
        fmt.Println("Weekend!")
    default:
        fmt.Println("Weekday")
    }

    switch os := runtime.GOOS; os {
    case "darwin":
        fmt.Println("platform: macOS")
    case "linux":
        fmt.Println("platform: Linux")
    default:
        fmt.Println("platform:", os)
    }

    for _, v := range []int{-1, 0, 2} {
        fmt.Printf("sign(%d)=%s\n", v, sign(v))
    }

    // ok-idiom on a map
    m := map[string]int{"alice": 30}
    if age, ok := m["alice"]; ok {
        fmt.Println("found alice:", age)
    }
    if _, ok := m["bob"]; !ok {
        fmt.Println("bob missing")
    }

    describe(42)
    describe("hello")
    describe(3.14)
}
```

Expected output (illustrative; platform line varies):

```
positive init: 10
Negative
Weekend!
platform: macOS
sign(-1)=negative
sign(0)=zero
sign(2)=positive
found alice: 30
bob missing
Integer: 42
String: hello
Unknown type: float64 value=3.14
```

What to notice: - `if init; cond` limits temporary variables to the if statement. - Expressionless `switch` replaces long if/else chains. - Type switches branch on dynamic type of `any`. - Map lookup uses the comma-ok idiom, not sentinel ages alone.

Try next: Add `fallthrough` under `case "Monday"` and switch `day` to `"Monday"` to see both cases print.

Loops and Iteration

Loops and Iteration

Overview

Go has only one looping construct: the `for` loop. This single keyword handles all looping patterns.

The Basic `for` Loop

```
for i := 0; i < 10; i++ {
    fmt.Println(i)
}
```

The `while` Loop (Condition Only)

```
n := 1
for n < 100 {
    n *= 2
}
```

The Infinite Loop

```
for {
    if done {
        break
    }
}
```

The `range` Loop

Slices and Arrays

```
nums := []int{10, 20, 30}

for i, v := range nums {
    fmt.Printf("%d: %d\n", i, v)
}

for _, v := range nums {  // Value only
    fmt.Println(v)
}
```

Maps

```
ages := map[string]int{"Alice": 30, "Bob": 25}

for key, value := range ages {
    fmt.Printf("%s: %d\n", key, value)
}
```

Strings

```
for i, r := range "Hello" {
    fmt.Printf("%d: %c\n", i, r)
}
```

Channels

```
for v := range ch {  // Exits when channel closes
    fmt.Println(v)
}
```

Iterators (Go 1.23+)

Go 1.23 introduced "range-over-func," allowing you to use `range` with custom iterator functions.

Sequence Iterators

A sequence iterator is a function that takes a yield function: `func(yield func(V) bool)` (single value) or `func(yield func(K, V) bool)` (key-value).

```
func All[T any](s []T) iter.Seq[T] {
    return func(yield func(T) bool) {
        for _, v := range s {
            if !yield(v) {
                return
            }
        }
    }
}

// Usage
for v := range All(nums) {
    fmt.Println(v)
}
```

Pull Iterators

For more control, you can use pull iterators:

```
next, stop := iter.Pull(All(nums))
defer stop()

for {
    v, ok := next()
    if !ok {
        break
    }
    fmt.Println(v)
}
```

Loop Control

`break` and `continue`

```
for i := 0; i < 10; i++ {
    if i == 5 {
        break     // Exit loop
    }
    if i%2 == 0 {
```

```go
        continue  // Skip to next iteration
    }
    fmt.Println(i)
}
```

Labels for Nested Loops

```go
outer:
for i := 0; i < 3; i++ {
    for j := 0; j < 3; j++ {
        if i == 1 && j == 1 {
            break outer  // Break both loops
        }
    }
}
```

Common Patterns

```go
// Reverse iteration
for i := len(items) - 1; i >= 0; i-- {
    process(items[i])
}

// Step by N
for i := 0; i < 100; i += 10 {
    fmt.Println(i)
}
```

Summary

Pattern	Syntax
Classic for	`for i := 0; i < n; i++ {}`
While loop	`for condition {}`
Infinite loop	`for {}`
Range	`for i, v := range slice {}`
Iterators	`for v := range myIter() {}`

Related Topics

- Arrays and Slices

- Maps

More examples

Example: classic for and range

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```
package main

import "fmt"

func main() {
    sum := 0
    for i := 1; i <= 4; i++ {
        sum += i
    }
    fmt.Println("sum:", sum)

    for i, v := range []string{"a", "b", "c"} {
        fmt.Printf("%d:%s ", i, v)
    }
    fmt.Println()
}
```

Expected:

```
sum: 10
0:a 1:b 2:c
```

Example: break and continue

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```
package main

import "fmt"

func main() {
    for i := 0; i < 6; i++ {
        if i%2 == 0 {
            continue
        }
        if i > 3 {
            break
```

```
        }
        fmt.Println("odd:", i)
    }
}
```

Expected:

```
odd: 1
odd: 3
```

Runnable example

Save as `main.go`. From an empty directory:

```
go mod init example
go run .
```

```
package main

import "fmt"

func main() {
    // Classic for
    sum := 0
    for i := 1; i <= 5; i++ {
        sum += i
    }
    fmt.Println("sum 1..5:", sum)

    // while-style
    n := 1
    for n < 100 {
        n *= 2
    }
    fmt.Println("first power of two >= 100:", n)

    // range over slice
    nums := []int{10, 20, 30}
    for i, v := range nums {
        fmt.Printf("slice[%d]=%d\n", i, v)
    }

    // range over map (order not guaranteed)
    ages := map[string]int{"Ada": 36, "Grace": 85}
    for name, age := range ages {
```

```go
        fmt.Printf("%s is %d\n", name, age)
    }

    // range over string yields runes
    for i, r := range "Go " {
        fmt.Printf("string index %d rune %c\n", i, r)
    }

    // break / continue / labeled break
    fmt.Print("odds < 5: ")
    for i := 0; i < 10; i++ {
        if i >= 5 {
            break
        }
        if i%2 == 0 {
            continue
        }
        fmt.Print(i, " ")
    }
    fmt.Println()

outer:
    for i := 0; i < 3; i++ {
        for j := 0; j < 3; j++ {
            if i == 1 && j == 1 {
                fmt.Printf("breaking outer at i=%d j=%d\n", i, j)
                break outer
            }
        }
    }

    // reverse step
    for i := 4; i >= 0; i -= 2 {
        fmt.Println("reverse step:", i)
    }
}
```

Expected output (illustrative; map line order may vary):

```
sum 1..5: 15
first power of two >= 100: 128
slice[0]=10
slice[1]=20
slice[2]=30
Ada is 36
Grace is 85
string index 0 rune G
```

```
string index 1 rune o
string index 2 rune
odds < 5: 1 3
breaking outer at i=1 j=1
reverse step: 4
reverse step: 2
reverse step: 0
```

What to notice: - One keyword (`for`) covers C-style, while-style, and infinite loops. - `range` adapts to slices, maps, and strings (byte index + rune). - `continue` / `break` and labels control nested loops cleanly. - Map iteration order is randomized—never rely on it.

Try next: Change the string to include combining characters or emoji and print both `i` and `len("…")`.

Arrays and Slices

Arrays and Slices

Overview

Arrays and slices are Go's primary sequence types. Arrays are fixed-size, while slices are dynamic views into arrays.

Arrays

Declaration

```go
var arr [5]int                    // Zero values: [0 0 0 0 0]
arr := [5]int{1, 2, 3, 4, 5}     // Literal
arr := [...]int{1, 2, 3}          // Size inferred: [3]int
```

Properties

- Fixed size (part of type): `[5]int` ≠ `[6]int`
- Value semantics: copying creates independent copy
- Zero value: array of zero values

```go
a := [3]int{1, 2, 3}
b := a        // Copy!
b[0] = 99
fmt.Println(a)  // [1 2 3] (unchanged)
```

Slices

Creation

```go
var s []int                       // nil slice
s := []int{1, 2, 3}              // Literal
s := make([]int, 5)              // Length 5, capacity 5
s := make([]int, 0, 10)          // Length 0, capacity 10
```

From Array

```
arr := [5]int{1, 2, 3, 4, 5}
s := arr[1:4]  // [2 3 4] - shares memory with arr
```

Slice Internals & Reallocation

A slice header consists of three words: a pointer to the backing array, a length (`len`), and a capacity (`cap`).

```
flow:
  "ptr: 0x7FFF00" -> E0
```

When `append()` exceeds `cap`, Go allocates a new, larger backing array and copies existing elements:

```
flow:
  OldSlice -> Realloc  (cap full)
```

Length and Capacity

```
s := make([]int, 3, 5)
len(s)  // 3 (elements accessible)
cap(s)  // 5 (space available)
```

Slice Operations

Append

```
s := []int{1, 2, 3}
s = append(s, 4)           // [1 2 3 4]
s = append(s, 5, 6, 7)     // [1 2 3 4 5 6 7]
s = append(s, other...)    // Append another slice
```

Slicing

```
s := []int{0, 1, 2, 3, 4, 5}
s[1:4]   // [1 2 3]
s[:3]    // [0 1 2]
s[3:]    // [3 4 5]
```

```go
s[:]     // [0 1 2 3 4 5] (copy of slice header)
```

Copy

```go
src := []int{1, 2, 3}
dst := make([]int, len(src))
copy(dst, src)
```

The `slices` Package (Go 1.21+)

The standard library `slices` package provides generic utilities for common operations.

```go
import "slices"

slices.Sort(nums)             // Sort in place
slices.Reverse(nums)          // Reverse in place
slices.Contains(nums, 42)     // Search
slices.Index(nums, 42)        // Find index
slices.Delete(nums, i, j)     // Remove range [i, j)
slices.Clone(nums)            // Shallow copy
slices.Equal(s1, s2)          // Compare
```

Modern Loop Pattern (Go 1.23+)

Using iterators with slices:

```go
for i, v := range slices.All(nums) { /* ... */ }
for v := range slices.Values(nums) { /* ... */ }
```

nil vs Empty Slice

```go
var nilSlice []int     // nil
emptySlice := []int{}  // not nil, but empty

nilSlice == nil        // true
emptySlice == nil      // false
len(nilSlice)          // 0
len(emptySlice)        // 0
```

Common Patterns

Remove Element

```
s = append(s[:i], s[i+1:]...)  // Remove element at index i
```

Insert Element

```
s = append(s[:i], append([]int{v}, s[i:]...)...)
```

Stack

```
stack = append(stack, v)              // Push
v, stack = stack[len(stack)-1], stack[:len(stack)-1]  // Pop
```

Summary

Feature	Array	Slice
Size	Fixed	Dynamic
Type	`[n]T`	`[]T`
Zero value	Array of zeros	`nil`
Comparison	`==` works	Not comparable

Related Topics

- Working with Strings
- Maps

More examples

Example: array vs slice

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```
package main

import "fmt"
```

```go
func main() {
    var arr [3]int = [3]int{1, 2, 3}
    s := arr[:]
    s[0] = 9
    fmt.Println("arr:", arr)
    fmt.Println("slice:", s, "len", len(s), "cap", cap(s))
}
```

Expected:

```
arr: [9 2 3]
slice: [9 2 3] len 3 cap 3
```

Example: append and capacity growth

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

func main() {
    s := make([]int, 0, 2)
    fmt.Printf("start len=%d cap=%d\n", len(s), cap(s))
    s = append(s, 1, 2)
    fmt.Printf("full  len=%d cap=%d %v\n", len(s), cap(s), s)
    s = append(s, 3)
    fmt.Printf("grow  len=%d cap=%d %v\n", len(s), cap(s), s)
}
```

Expected:

```
start len=0 cap=2
full  len=2 cap=2 [1 2]
grow  len=3 cap=4 [1 2 3]
```

Runnable example

Save as `main.go`. From an empty directory:

```
go mod init example
go run .
```

```go
package main

import (
    "fmt"
    "slices"
)

func main() {
    // Arrays: fixed size, value semantics
    a := [3]int{1, 2, 3}
    b := a
    b[0] = 99
    fmt.Println("array a (unchanged):", a)
    fmt.Println("array copy b:", b)

    // Slices: dynamic, share backing array until reallocation
    s := []int{10, 20, 30}
    fmt.Printf("s=%v len=%d cap=%d\n", s, len(s), cap(s))

    t := s[1:]
    t[0] = 200
    fmt.Println("after t[0]=200, s=", s, "t=", t)

    // append may reallocate when cap is exceeded
    s2 := make([]int, 3, 4)
    copy(s2, []int{1, 2, 3})
    fmt.Printf("before grow: %v len=%d cap=%d\n", s2, len(s2), cap(s2))
    s2 = append(s2, 4)
    fmt.Printf("cap full:    %v len=%d cap=%d\n", s2, len(s2), cap(s2))
    s2 = append(s2, 5)
    fmt.Printf("after grow:  %v len=%d cap=%d\n", s2, len(s2), cap(s2))

    // Independent copy
    src := []int{1, 2, 3}
    dst := make([]int, len(src))
    copy(dst, src)
    dst[0] = 7
    fmt.Println("src", src, "dst", dst)

    // nil vs empty
    var nilSlice []int
    empty := []int{}
    fmt.Printf("nil==nil? %v empty==nil? %v\n", nilSlice == nil, empty == nil)
    nilSlice = append(nilSlice, 1, 2)
    fmt.Println("append on nil:", nilSlice)

    // slices package (Go 1.21+)
```

```go
    nums := []int{3, 1, 2}
    cloned := slices.Clone(nums)
    slices.Sort(cloned)
    fmt.Println("sorted clone:", cloned, "contains 2?", slices.Contains(cloned, 2))
}
```

Expected output (illustrative; exact new cap may vary by Go version):

```
array a (unchanged): [1 2 3]
array copy b: [99 2 3]
s=[10 20 30] len=3 cap=3
after t[0]=200, s= [10 200 30] t= [200 30]
before grow: [1 2 3] len=3 cap=4
cap full:    [1 2 3 4] len=4 cap=4
after grow:  [1 2 3 4 5] len=5 cap=8
src [1 2 3] dst [7 2 3]
nil==nil? true empty==nil? false
append on nil: [1 2]
sorted clone: [1 2 3] contains 2? true
```

What to notice: - Assigning an array copies elements; assigning/slicing a slice shares storage. - `append` grows length and may allocate a larger backing array. - `copy` creates an independent element sequence. - `slices.Sort` / `Clone` are stdlib helpers—no third-party deps.

Try next: After `t := s[1:]`, append enough elements to `t` to force reallocation, then change `t[0]` and see whether `s` still changes.

Working with Strings

Working with Strings

Overview

Strings in Go are immutable sequences of bytes, typically UTF-8 encoded. Understanding the distinction between bytes and runes is essential.

String Basics

```
s := "Hello, World!"
s := `Raw string with
multiple lines`
```

Immutability

```
s := "hello"
// s[0] = 'H'  // Error: cannot assign to s[0]
s = "Hello"    // OK: reassign entire string
```

Bytes vs Runes

```
s := "Hello,  "

len(s)              // 13 (bytes)
len([]rune(s))      // 9 (characters)
```

Iterating

```
// By byte
for i := 0; i < len(s); i++ {
    fmt.Printf("%x ", s[i])
}

// By rune (character)
```

```go
for i, r := range s {
    fmt.Printf("%d: %c\n", i, r)
}
```

String Operations

Concatenation

```go
s := "Hello" + ", " + "World"

// Efficient building
var b strings.Builder
b.WriteString("Hello")
b.WriteString(", World")
result := b.String()
```

Comparison

```go
s1 == s2              // Equality
s1 < s2               // Lexicographic
strings.EqualFold(s1, s2)  // Case-insensitive
```

strings Package

```go
import "strings"

strings.Contains(s, "sub")      // true/false
strings.HasPrefix(s, "pre")     // true/false
strings.HasSuffix(s, "suf")     // true/false
strings.Index(s, "sub")         // Position or -1
strings.Split(s, ",")           // []string
strings.Join(parts, ",")        // string
strings.ToUpper(s)              // UPPERCASE
strings.ToLower(s)              // lowercase
strings.TrimSpace(s)            // Remove whitespace
strings.Replace(s, "old", "new", n)  // Replace n times
strings.ReplaceAll(s, "old", "new")  // Replace all
```

strconv Package

```go
import "strconv"

// String to int
n, err := strconv.Atoi("42")

// Int to string
s := strconv.Itoa(42)

// Parse float
f, err := strconv.ParseFloat("3.14", 64)

// Format float
s := strconv.FormatFloat(3.14, 'f', 2, 64)
```

fmt Package

```go
// Formatting
s := fmt.Sprintf("Name: %s, Age: %d", name, age)

// Parsing
var name string
var age int
fmt.Sscanf("Alice 30", "%s %d", &name, &age)
```

Rune Operations

```go
r := 'A'
unicode.IsLetter(r)    // true
unicode.IsDigit(r)     // false
unicode.ToUpper(r)     // 'A'
unicode.ToLower(r)     // 'a'
```

Summary

Operation	Method
Length (bytes)	`len(s)`
Length (chars)	`len([]rune(s))`
Concatenate	`+` or `strings.Builder`

Operation	Method
Contains	`strings.Contains()`
Split/Join	`strings.Split()`, `strings.Join()`
Convert	`strconv.Atoi()`, `strconv.Itoa()`

Related Topics

- Predeclared Types
- I/O and Streaming

More examples

Example: bytes vs runes

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

func main() {
    s := "Go "
    fmt.Println("bytes:", len(s))
    fmt.Println("runes:", len([]rune(s)))
    for i, r := range s {
        fmt.Printf("i=%d r=%q\n", i, r)
    }
}
```

Expected:

```
bytes: 5
runes: 3
i=0 r='G'
i=1 r='o'
i=2 r=' '
```

Example: strings.Builder

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import (
    "fmt"
    "strings"
)

func main() {
    var b strings.Builder
    b.WriteString("hello")
    b.WriteByte(' ')
    b.WriteString("gopher")
    fmt.Println(b.String())
    fmt.Println("len:", b.Len())
}
```

Expected:

```
hello gopher
len: 12
```

Runnable example

Save as `main.go`. From an empty directory:

```bash
go mod init example
go run .
```

```go
package main

import (
    "fmt"
    "strconv"
    "strings"
    "unicode"
)

func main() {
    s := "Hello,  "
    fmt.Printf("%q bytes=%d runes=%d\n", s, len(s), len([]rune(s)))

    fmt.Print("bytes: ")
    for i := 0; i < len(s); i++ {
        fmt.Printf("%02x ", s[i])
    }
```

```go
    fmt.Println()

    fmt.Println("runes:")
    for i, r := range s {
        fmt.Printf("  index %d -> %c U+%04X\n", i, r, r)
    }

    // Immutability: reassign whole string only
    s = "Hello"
    // s[0] = 'h' // compile error if uncommented
    fmt.Println("reassigned:", s)

    var b strings.Builder
    b.WriteString("Go")
    b.WriteString(" ")
    b.WriteString("strings")
    built := b.String()
    fmt.Println("builder:", built)

    fmt.Println("Contains Go?", strings.Contains(built, "Go"))
    fmt.Println("fields:", strings.Fields("  a  b\tc  "))
    fmt.Println("join:", strings.Join([]string{"a", "b", "c"}, "-"))
    fmt.Println("upper:", strings.ToUpper("gopher"))

    n, err := strconv.Atoi("42")
    fmt.Printf("Atoi=%d err=%v Itoa=%s\n", n, err, strconv.Itoa(n+1))
    f, _ := strconv.ParseFloat("3.14", 64)
    fmt.Println("float:", strconv.FormatFloat(f, 'f', 2, 64))

    r := 'g'
    fmt.Printf("rune %q letter=%v upper=%q\n", r, unicode.IsLetter(r), unicode.ToUpper(r))

    fmt.Println(fmt.Sprintf("Name: %s Age: %d", "Ada", 36))
}
```

Expected output (illustrative):

```
"Hello,  " bytes=13 runes=9
bytes: 48 65 6c 6c 6f 2c 20 e4 b8 96 e7 95 8c
runes:
  index 0 -> H U+0048
  index 1 -> e U+0065
  index 2 -> l U+006C
  index 3 -> l U+006C
  index 4 -> o U+006F
  index 5 -> , U+002C
  index 6 ->   U+0020
```

```
  index 7 ->  U+4E16
  index 10 ->  U+754C
reassigned: Hello
builder: Go strings
Contains Go? true
fields: [a b c]
join: a-b-c
upper: GOPHER
Atoi=42 err=<nil> Itoa=43
float: 3.14
rune 'g' letter=true upper='G'
Name: Ada Age: 36
```

What to notice: - Multi-byte UTF-8 characters make `len` (bytes) larger than rune count. - `range` over a string steps by runes but indexes are byte offsets. - `strings.Builder` avoids quadratic `+` chains for assembly. - `strconv` is the explicit bridge between text and numbers.

Try next: Use `strings.ReplaceAll` to censor a word in a sentence and print before/after.

Maps

Maps

Overview

Maps are Go's built-in hash table type, providing O(1) average-time lookups, insertions, and deletions.

Creating Maps

```go
var m map[string]int           // nil map (read-only!)
m := make(map[string]int)      // Empty, ready to use
m := make(map[string]int, 100) // With capacity hint
m := map[string]int{           // Literal
    "one": 1,
    "two": 2,
}
```

Basic Operations

Set

```go
m["key"] = value
```

Get

```go
val := m["key"]  // Returns zero value if missing
```

Delete

```go
delete(m, "key")  // No-op if key doesn't exist
```

Check Existence

```
val, ok := m["key"]
if ok {
    // Key exists
}

if _, exists := m["key"]; exists {
    // Key exists
}
```

nil Map Behavior

```
var m map[string]int  // nil

val := m["key"]       // OK: returns 0
m["key"] = 1          // PANIC! Cannot write to nil map
```

Always initialize before writing:

```
m := make(map[string]int)
m["key"] = 1  // OK
```

Iteration

```
for key, value := range m {
    fmt.Printf("%s: %d\n", key, value)
}

for key := range m {  // Keys only
    fmt.Println(key)
}
```

Warning: Iteration order is randomized!

Common Patterns

Set (Unique Values)

```
set := make(map[string]struct{})
set["item"] = struct{}{}
if _, exists := set["item"]; exists {
    // Item is in set
}
delete(set, "item")
```

Counter

```
counter := make(map[string]int)
for _, word := range words {
    counter[word]++  // Zero value works!
}
```

Grouping

```
groups := make(map[string][]User)
for _, user := range users {
    groups[user.Country] = append(groups[user.Country], user)
}
```

Default Value

```
val := m["key"]
if val == 0 {
    val = defaultValue
}
```

Concurrency Warning

Maps are **not** goroutine-safe:

```
// Unsafe: concurrent read/write
go func() { m["key"] = 1 }()
go func() { _ = m["key"] }()
```

```go
// Use sync.Map or mutex
var mu sync.Mutex
mu.Lock()
m["key"] = 1
mu.Unlock()
```

The maps Package (Go 1.21+)

The standard library `maps` package provides generic utilities for common operations.

```go
import "maps"

maps.Clone(m)                  // Shallow copy
maps.Equal(m1, m2)             // Compare
maps.DeleteFunc(m, func(k K, v V) bool {
    return v < 10              // Conditional delete
})
```

Modern Loop Pattern (Go 1.23+)

Using iterators with maps:

```go
for k, v := range maps.All(m) { /* ... */ }
for k := range maps.Keys(m) { /* ... */ }
for v := range maps.Values(m) { /* ... */ }
```

Map Internals

- Keys must be comparable (`==` must work)
- Valid key types: bool, numeric, string, pointer, channel, interface, structs/arrays of comparable types
- Invalid: slices, maps, functions

Summary

Operation	Syntax
Create	`make(map[K]V)`
Set	`m[key] = value`
Get	`val := m[key]`
Check	`val, ok := m[key]`

Operation	Syntax
Delete	`delete(m, key)`
Length	`len(m)`

Related Topics

- Arrays and Slices
- Synchronization

More examples

Example: comma-ok lookup

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```
package main

import "fmt"

func main() {
    m := map[string]int{"a": 1}
    if v, ok := m["a"]; ok {
        fmt.Println("a:", v)
    }
    if _, ok := m["missing"]; !ok {
        fmt.Println("missing: not present")
    }
}
```

Expected:

```
a: 1
missing: not present
```

Example: delete and range

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```
package main

import "fmt"

func main() {
```

```go
    m := map[string]int{"x": 10, "y": 20}
    delete(m, "x")
    fmt.Println("len:", len(m))
    for k, v := range m {
        fmt.Printf("%s=%d\n", k, v)
    }
}
```

Expected:

```
len: 1
y=20
```

Runnable example

Save as `main.go`. From an empty directory:

```bash
go mod init example
go run .
```

```go
package main

import (
    "fmt"
    "maps"
)

func main() {
    // nil map: read ok, write panics - initialize first
    var nilMap map[string]int
    fmt.Println("nil read:", nilMap["x"])

    m := make(map[string]int)
    m["one"] = 1
    m["two"] = 2
    fmt.Println("m:", m, "len:", len(m))

    if v, ok := m["two"]; ok {
        fmt.Println("found two:", v)
    }
    if _, ok := m["three"]; !ok {
        fmt.Println("three missing")
    }

    delete(m, "one")
```

```go
    fmt.Println("after delete one:", m)

    // Word counter uses zero value on missing keys
    counter := make(map[string]int)
    for _, w := range []string{"go", "is", "go", "fun", "go"} {
        counter[w]++
    }
    fmt.Println("counts:", counter)

    // Set via map[T]struct{}
    set := make(map[string]struct{})
    set["alpha"] = struct{}{}
    set["beta"] = struct{}{}
    if _, ok := set["alpha"]; ok {
        fmt.Println("set has alpha")
    }

    // Grouping
    type User struct{ Name, Country string }
    users := []User{
        {"Ada", "UK"},
        {"Linus", "FI"},
        {"Alan", "UK"},
    }
    groups := make(map[string][]string)
    for _, u := range users {
        groups[u.Country] = append(groups[u.Country], u.Name)
    }
    fmt.Println("groups:", groups)

    // maps package (Go 1.21+)
    clone := maps.Clone(counter)
    clone["fun"] = 99
    fmt.Println("original fun:", counter["fun"], "clone fun:", clone["fun"])
    fmt.Println("equal?", maps.Equal(counter, clone))

    fmt.Println("keys (order random):")
    for k, v := range counter {
        fmt.Printf("  %s=%d\n", k, v)
    }
}
```

Expected output (illustrative; key order varies):

```
nil read: 0
m: map[one:1 two:2] len: 2
found two: 2
```

```
three missing
after delete one: map[two:2]
counts: map[fun:1 go:3 is:1]
set has alpha
groups: map[FI:[Linus] UK:[Ada Alan]]
original fun: 1 clone fun: 99
equal? false
keys (order random):
  go=3
  is=1
  fun=1
```

What to notice: - Comma-ok distinguishes missing keys from zero values. - `counter[w]++` works because missing keys read as `0`. - `map[T]struct{}` is an idiomatic set with near-zero value size. - Iteration order is deliberately random.

Try next: Uncomment a write to `nilMap["x"] = 1` once to see the panic, then remove it.

Structs and Data Modeling

Structs and Data Modeling

Overview

Structs are Go's primary mechanism for creating custom composite types. They group related fields together.

Defining Structs

```go
type Person struct {
    Name    string
    Age     int
    Email   string
    Active  bool
}
```

Creating Instances

```go
// Zero value
var p Person  // All fields get zero values

// Literal (named fields - preferred)
p := Person{
    Name:  "Alice",
    Age:   30,
    Email: "alice@example.com",
}

// Literal (positional - fragile)
p := Person{"Alice", 30, "alice@example.com", true}

// Using new
p := new(Person)  // *Person with zero values

// Expression-based new (Go 1.26+)
p := new(Person{Name: "Alice"})  // *Person initialized with values
```

```
Embedded fields (composition)

Employee
|-- Person
|     |-- Name
|     |-- Age
|-- Address
|     |-- City
|-- Title
```

Accessing Fields

```go
p.Name = "Bob"
fmt.Println(p.Age)

// Pointer access (automatic dereference)
ptr := &p
ptr.Name = "Carol"  // Same as (*ptr).Name
```

Anonymous Structs

```go
point := struct {
    X, Y int
}{10, 20}

// Useful for one-off data
data := struct {
    ID    int
    Value string
}{1, "test"}
```

Embedding (Composition)

Go favors composition over inheritance. When a struct embeds another type anonymously (without an explicit field name), the embedded type's fields and methods are **promoted** to the outer struct:

```go
type Address struct {
    City  string
    State string
}
```

```go
type Contact struct {
    Email string
    Phone string
}

type Employee struct {
    Name string
    Age  int

    Address // anonymous embedding
    Contact // anonymous embedding
}
```

Because `Address` and `Contact` are embedded anonymously, their fields elevate to `Employee`:

```go
e := Employee{...}

// Promoted field access (plain dot notation)
fmt.Println(e.City)  // same as e.Address.City
fmt.Println(e.Email) // same as e.Contact.Email
```

Flattened Struct Literals (Go 1.27+)

For sixteen years (since Go 1.0), field promotion had an annoying asymmetry: once an instance existed, you could read its promoted fields with clean dot notation (`e.City`), yet constructing that same struct required wrapping every embedded type in a "Russian nesting doll":

```go
// Prior to Go 1.27: the nesting-doll tax
e := Employee{
    Name: "Alice",
    Age:  30,
    Address: Address{
        City:  "NYC",
        State: "NY",
    },
    Contact: Contact{
        Email: "alice@example.com",
        Phone: "555-0100",
    },
}
```

This structural wrapping created visual friction and rightward drift, especially in deeply nested configurations, data transfer objects (DTOs), and test fixtures.

Originating in proposal #9859 (first opened in 2015), **Go 1.27 flattens struct literal construction**. Any unambiguously promoted field can now serve directly as a key in composite literals:

```go
// Go 1.27+: flattened composite literal
e := Employee{
    Name:  "Bob",
    Age:   25,
    City:  "LA",              // promoted from Address
    State: "CA",              // promoted from Address
    Email: "bob@example.com", // promoted from Contact
    Phone: "555-0200",        // promoted from Contact
}
```

This reduces boilerplate by over 30%, eliminates redundant type wrappers, and brings symmetry to Go's mental model: if you read a field with `e.City`, you can initialize it with `City: "LA"`.

Multi-Level Flattening

Flattening is not limited to a single level of embedding. Any field that elevates unambiguously under Go's promotion rules can serve as an initialization key across multiple layers:

```go
type Coordinates struct {
    Latitude  float64
    Longitude float64
}

type Location struct {
    Coordinates // depth 1
    Altitude    float64
}

type Site struct {
    Name     string
    Location // depth 2
}
```

Prior to Go 1.27, initializing `Site` required three layers of braces:

```go
// Legacy: 3 layers of nested boilerplate (14 lines)
s := Site{
    Name: "Data Center Alpha",
    Location: Location{
        Coordinates: Coordinates{
            Latitude:  40.7128,
            Longitude: -74.0060,
        },
        Altitude: 10.5,
    },
}
```

In Go 1.27+, all intermediate wrappers vanish into flat key-value pairs:

```go
// Go 1.27+: multi-level flattened literal (6 lines, 57% reduction)
s := Site{
    Name:      "Data Center Alpha",
    Latitude:  40.7128,
    Longitude: -74.0060,
    Altitude:  10.5,
}
```

Configuration Structs and Information Density

The ergonomic benefit is most pronounced in nested configuration structs, where structural wrappers often obscure actual values:

```go
type DatabaseConfig struct {
    Host     string
    Port     int
    Name     string
    SSL      bool
    MaxConns int
}

type CacheConfig struct {
    TTL     int
    MaxSize int
}

type AppConfig struct {
    DatabaseConfig
    CacheConfig
}

// Go 1.27: clear, high-density configuration
cfg := AppConfig{
    Host:     "localhost",
    Port:     5432,
    Name:     "app_db",
    SSL:      true,
    MaxConns: 100,
    TTL:      300,
    MaxSize:  256,
}
```

Rather than reading structural plumbing (`DatabaseConfig: DatabaseConfig{...}`), readers immediately see the actionable values (`Host`, `Port`, `TTL`).

Ambiguity: When the Compiler Refuses to Guess

Flattened initialization follows Go's existing field promotion collision rules. If two embedded structs define a field with the exact same name at the same embedding depth, the field is **not** promoted:

```go
type DatabaseConfig struct {
    Host string
    Port int
}

type CacheConfig struct {
    Host string
    TTL  int
}

type AppConfig struct {
    DatabaseConfig
    CacheConfig
}
```

Attempting to initialize the ambiguous field using the flat syntax:

```go
cfg := AppConfig{
    Host: "localhost", // compile error!
}
```

Fails at compile time with a clear diagnostic:

```
unknown field Host in struct literal of type AppConfig
```

Because `Host` exists in both `DatabaseConfig` and `CacheConfig` at depth 1, the compiler refuses to guess which embedded struct should receive the value. In ambiguous scenarios, you must explicitly disambiguate by using the nested literal:

```go
cfg := AppConfig{
    DatabaseConfig: DatabaseConfig{
        Host: "db.internal",
        Port: 5432,
    },
    CacheConfig: CacheConfig{
        Host: "cache.internal",
        TTL:  300,
    },
}
```

Strict Guardrail: No Split Initialization

Go enforces a strict compiler guardrail: you **cannot mix** promoted fields with their enclosing embedded struct literal in the same composite literal.

```
// ILLEGAL: mixing promoted field and enclosing embedded struct
cfg := AppConfig{
    DatabaseConfig: DatabaseConfig{Host: "localhost"},
    Port:           5432, // compile error!
}
```

The compiler halts compilation with an explicit error:

```
cannot specify promoted field Port and enclosing embedded field DatabaseConfig
```

Even though `Host` and `Port` do not collide, Go forbids splitting initialization across structural boundaries. You must choose one convention per embedded type: either initialize the embedded struct explicitly, or initialize its promoted fields from the outer literal.

Automated Migration with `go fix -embedlit`

You do not need to rewrite legacy nested composite literals by hand. Go 1.27 introduces the `embedlit` modernizer analyzer integrated into `go fix`:

```
# Preview proposed literal simplifications across your module
go fix -embedlit -diff ./...

# Apply automatic simplifications in place
go fix -embedlit ./...
```

The `embedlit` analyzer safely identifies composite literals where nested embedded type wrappers are redundant, modernizing the code while verifying that no field name collisions occur.

When to Keep Nesting

Flattened initialization is the sensible default for configs, DTOs, and test fixtures, but nested construction remains valuable in specific architectural scenarios:

1. **Resolving Ambiguities**: When embedded types have colliding field names at the same depth.
2. **Explicit Domain Boundaries**: In large structs with dozens of options (e.g., 30+ database parameters), grouping options inside `DatabaseConfig: DatabaseConfig{...}` signals to readers that those values belong to a distinct subsystem.
3. **Partial Initialization of Embedded Subsystems**: When reusing or passing an existing pre-allocated instance of an embedded struct directly into the constructor.

Struct Tags

Metadata for serialization and other tools:

```
type User struct {
    ID        int    `json:"id" db:"user_id"`
    Name      string `json:"name,omitempty"`
    Password  string `json:"-"`  // Omit from JSON
    CreatedAt time.Time `json:"created_at"`
}
```

Reading Tags

```
import "reflect"

t := reflect.TypeOf(User{})
field, _ := t.FieldByName("Name")
tag := field.Tag.Get("json")  // "name,omitempty"
```

Comparison

Structs are comparable if all fields are comparable:

```
p1 := Person{Name: "Alice"}
p2 := Person{Name: "Alice"}
p1 == p2  // true
```

Constructors

Go uses factory functions:

```
func NewPerson(name string, age int) *Person {
    return &Person{
        Name:   name,
        Age:    age,
        Active: true,  // Default
    }
}

func NewPersonWithDefaults() *Person {
    return &Person{
        Active: true,
```

```
    }
}
```

Summary

Pattern	Usage
Define	`type Name struct { fields }`
Create	`Name{Field: value}`
Embed	Include type name without field name
Tags	`Field Type` \key:“value”‘`| |`
	`Constructor` |func NewType() *Type‘

Related Topics

- Methods and Receivers
- Designing with Interfaces

More examples

Example: literal and field access

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

type User struct {
    Name string
    Age  int
}

func main() {
    u := User{Name: "ada", Age: 36}
    fmt.Printf("%+v\n", u)
    u.Age++
    fmt.Println("age:", u.Age)
}
```

Expected:

```
{Name:ada Age:36}
age: 37
```

Example: embedding promotes fields

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

type Address struct {
    City string
}

type Person struct {
    Name string
    Address
}

func main() {
    p := Person{Name: "lin", Address: Address{City: "Berlin"}}
    fmt.Println(p.Name, "in", p.City) // promoted field
}
```

Expected:

```
lin in Berlin
```

Runnable example

Save as `main.go`. From an empty directory:

```
go mod init example
go run .
```

```go
package main

import (
    "encoding/json"
    "fmt"
    "reflect"
)
```

```go
type Address struct {
    Street string
    City   string
}

type Person struct {
    Name string
    Age  int
}

type Employee struct {
    Person  // embedded
    Address // embedded
    Title   string
}

type User struct {
    ID       int    `json:"id"`
    Name     string `json:"name,omitempty"`
    Password string `json:"-"`
}

func NewEmployee(name, title, city string) *Employee {
    return &Employee{
        Person:  Person{Name: name, Age: 0},
        Address: Address{City: city},
        Title:   title,
    }
}

func main() {
    var zero Person
    fmt.Printf("zero person: %+v\n", zero)

    e := NewEmployee("Ada", "Engineer", "London")
    e.Age = 36 // promoted field
    fmt.Printf("employee: name=%s title=%s city=%s age=%d\n", e.Name, e.Title, e.City, e.Age

    // Go 1.27+: flattened struct literal initialization of promoted fields
    e2 := Employee{
        Name:  "Bob",
        Age:   28,
        City:  "Cambridge",
        Title: "Staff Engineer",
    }
    fmt.Printf("flattened employee: name=%s title=%s city=%s age=%d\n", e2.Name, e2.Title, e
```

```go
    // Pointer field access auto-dereferences
    p := &Person{Name: "Grace", Age: 85}
    p.Age++
    fmt.Printf("pointer person: %+v\n", p)

    // Anonymous struct for one-off data
    point := struct{ X, Y int }{10, 20}
    fmt.Printf("point=(%d,%d)\n", point.X, point.Y)

    // Comparable when all fields are comparable
    a := Person{Name: "Ada", Age: 36}
    b := Person{Name: "Ada", Age: 36}
    fmt.Println("a==b?", a == b)

    // Tags for JSON + reflection
    u := User{ID: 7, Name: "Ada", Password: "secret"}
    raw, err := json.Marshal(u)
    if err != nil {
        fmt.Println("json error:", err)
        return
    }
    fmt.Println("json:", string(raw))

    t := reflect.TypeOf(User{})
    if f, ok := t.FieldByName("Name"); ok {
        fmt.Println("Name tag:", f.Tag.Get("json"))
    }
    if f, ok := t.FieldByName("Password"); ok {
        fmt.Println("Password tag:", f.Tag.Get("json"))
    }
}
```

Expected output (illustrative):

```
zero person: {Name: Age:0}
employee: name=Ada title=Engineer city=London age=36
flattened employee: name=Bob title=Staff Engineer city=Cambridge age=28
pointer person: &{Name:Grace Age:86}
point=(10,20)
a==b? true
json: {"id":7,"name":"Ada"}
Name tag: name,omitempty
Password tag: -
```

What to notice: - Embedding promotes fields (`e.Name`, `e.City`) without classical inheritance. - Go 1.27+ flattened struct literals allow initializing promoted fields (`Name`, `Age`, `City`) directly on `Employee` without nesting `Person` or `Address`. - Factory `NewEmployee` sets defaults; zero `Person` is

still valid. - JSON tags control names, omitempty, and secrets (`json:"-"`). - Struct equality works when every field is comparable.

Try next: Set `Name` to `""` and remarshal—observe `omitempty` dropping the field.

Time and Scheduling

Time and Scheduling

Overview

The `time` package provides functionality for measuring and displaying time.

Current Time

```
now := time.Now()
fmt.Println(now)  // 2024-01-15 10:30:00 -0500 EST
```

Duration

```
d := 5 * time.Second
d := time.Minute + 30*time.Second
d := time.ParseDuration("1h30m")
```

Sleeping

```
time.Sleep(2 * time.Second)
```

Timers

Go 1.27 removed the `asynctimerchan GODEBUG` setting. Channels created by package `time` (`Timer.C`, `Ticker.C`, `time.After`, …) are **always unbuffered**. Code that relied on a buffered timer channel under the old GODEBUG will deadlock or drop sends; treat receive as a rendezvous.

```
// One-shot timer
timer := time.NewTimer(5 * time.Second)
<-timer.C  // Blocks until timer fires

// Cancel timer
if !timer.Stop() {
```

```go
    <-timer.C
}
```

Tickers

```go
ticker := time.NewTicker(1 * time.Second)
defer ticker.Stop()

for t := range ticker.C {
    fmt.Println("Tick at", t)
}
```

Formatting

```go
// Format uses reference time: Mon Jan 2 15:04:05 MST 2006
now := time.Now()
fmt.Println(now.Format("2006-01-02"))           // 2024-01-15
fmt.Println(now.Format("15:04:05"))             // 10:30:00
fmt.Println(now.Format(time.RFC3339))           // 2024-01-15T10:30:00-05:00
```

Parsing

```go
t, err := time.Parse("2006-01-02", "2024-01-15")
t, err := time.Parse(time.RFC3339, "2024-01-15T10:30:00Z")
```

Time Zones

```go
loc, _ := time.LoadLocation("America/New_York")
t := time.Now().In(loc)

utc := time.Now().UTC()
```

Comparisons

```go
t1.Before(t2)
t1.After(t2)
t1.Equal(t2)
t1.Sub(t2)    // Duration between
t1.Add(d)     // Add duration
```

Summary

Type/Function	Purpose
`time.Now()`	Current time
`time.Duration`	Time interval
`time.Timer`	One-shot delay
`time.Ticker`	Repeated intervals
`time.Format()`	Time to string
`time.Parse()`	String to time

Related Topics

- Context and Cancellation
- I/O and Streaming

More examples

Example: duration arithmetic

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import (
    "fmt"
    "time"
)

func main() {
    d := 90 * time.Second
    fmt.Println("seconds:", d.Seconds())
    fmt.Println("plus minute:", d+time.Minute)
    fmt.Println("truncated:", d.Truncate(30*time.Second))
```

```
}
```

Expected:

```
seconds: 90
plus minute: 2m30s
truncated: 1m30s
```

Example: parse and format

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import (
    "fmt"
    "time"
)

func main() {
    const layout = "2006-01-02"
    t, err := time.Parse(layout, "2026-07-28")
    if err != nil {
        fmt.Println("err:", err)
        return
    }
    fmt.Println("parsed:", t.Format(layout))
    fmt.Println("weekday:", t.Weekday())
}
```

Expected:

```
parsed: 2026-07-28
weekday: Tuesday
```

Runnable example

Save as `main.go`. From an empty directory:

```
go mod init example
go run .
```

```go
package main
```

```go
import (
    "fmt"
    "time"
)

func main() {
    now := time.Now()
    fmt.Println("now:", now.Format(time.RFC3339))
    fmt.Println("date:", now.Format("2006-01-02"))
    fmt.Println("clock:", now.Format("15:04:05"))

    d := 90*time.Second + 500*time.Millisecond
    parsed, err := time.ParseDuration("1h30m")
    if err != nil {
        fmt.Println("ParseDuration:", err)
        return
    }
    fmt.Println("90.5s:", d, "parsed 1h30m:", parsed)

    start := time.Now()
    time.Sleep(50 * time.Millisecond)
    elapsed := time.Since(start)
    fmt.Printf("slept about %v (elapsed=%v)\n", 50*time.Millisecond, elapsed)

    // Parse / compare / arithmetic
    t1, err := time.Parse("2006-01-02", "2024-01-15")
    if err != nil {
        fmt.Println(err)
        return
    }
    t2 := t1.Add(48 * time.Hour)
    fmt.Println("t1:", t1.Format("2006-01-02"))
    fmt.Println("t2:", t2.Format("2006-01-02"))
    fmt.Println("t1 before t2?", t1.Before(t2))
    fmt.Println("sub:", t2.Sub(t1))

    utc := now.UTC()
    fmt.Println("utc:", utc.Format(time.RFC3339))

    // One-shot timer (short so the example finishes quickly)
    timer := time.NewTimer(30 * time.Millisecond)
    fire := <-timer.C
    fmt.Println("timer fired at:", fire.Format("15:04:05.000"))

    // Ticker: two ticks then stop
    ticker := time.NewTicker(20 * time.Millisecond)
    defer ticker.Stop()
```

```go
    for i := 0; i < 2; i++ {
        t := <-ticker.C
        fmt.Printf("tick %d at %s\n", i+1, t.Format("15:04:05.000"))
    }
}
```

Expected output (illustrative; timestamps vary):

```
now: 2026-07-28T12:00:00-07:00
date: 2026-07-28
clock: 12:00:00
90.5s: 1m30.5s parsed 1h30m: 1h30m0s
slept about 50ms (elapsed=51.2ms)
t1: 2024-01-15
t2: 2024-01-17
t1 before t2? true
sub: 48h0m0s
utc: 2026-07-28T19:00:00Z
timer fired at: 12:00:00.031
tick 1 at 12:00:00.051
tick 2 at 12:00:00.071
```

What to notice: - Format layouts use the reference time `Mon Jan 2 15:04:05 MST 2006`. - `Duration` values multiply cleanly (`90 * time.Second`). - `Before` / `Add` / `Sub` are the comparison and arithmetic API. - Always `Stop` tickers (and unused timers) to avoid leaks.

Try next: Load `time.LoadLocation("America/New_York")` and print `now.In(loc)`.

cmp.Or: first non-zero default without nested ifs

cmp.Or: first non-zero default without nested ifs

After this page you can replace nested `if x == ""` / `if n == 0` defaulting chains with `cmp.Or`, know that it uses the zero value of a comparable type, and avoid using it where `false` or `0` are meaningful inputs.

Thesis

Configuration code often wants "first non-empty string" or "first non-zero int". Go 1.22 added **`cmp.Or`** in the standard `cmp` package: it returns the first argument that is not the zero value of its type, or the zero value if all are zero.

Verified with **go1.27.2**.

Lab

```go
package main

import (
    "cmp"
    "fmt"
)

func main() {
    fmt.Println("port", cmp.Or("", "8080"))
    fmt.Println("host", cmp.Or("api.local", "localhost"))
    fmt.Println("first", cmp.Or(0, 0, 7))
}
```

```
go mod init example.com/cmpor
go run .
```

Verified output:

```
port 8080
host api.local
first 7
```

Mental model

```
cmp.Or(a, b, c) →
  if a != zero { return a }
  if b != zero { return b }
  if c != zero { return c }
  return zero
```

Works for any **comparable** type: strings, integers, floats, pointers, and some structs. It does not deep-compare slices or maps.

When not to use it

- Booleans where `false` is valid user input — `cmp.Or(false, true)` always yields `true` if you pass `false` first? Actually `false` **is** the zero value, so `cmp.Or(false, true)` returns `true`. That may be wrong for "explicit false".
- Integers where `0` is meaningful (port `0` meaning "pick ephemeral").
- Prefer functional options or pointers (`*string`) when you must distinguish unset from zero.

Pitfalls

1. Treating `cmp.Or` as SQL `COALESCE` for nullable database values — use pointers/`sql.Null*`.
2. Passing non-comparable types — compile error.
3. Relying on it before Go 1.22 — check `go.mod`.

Summary

- `cmp.Or` picks the first non-zero comparable argument.
- Ideal for string/int defaults from config env fallbacks.
- Do not use where zero is a legitimate explicit value.

Worked scenarios

Scenario A — env default port

```
port := cmp.Or(os.Getenv("PORT"), "8080")
```

Empty env yields `"8080"`.

Scenario B — layered config

```
timeout := cmp.Or(flagTimeout, cfg.Timeout, 30*time.Second)
```

Works when the type is comparable — `time.Duration` is based on `int64`, so it works.

Scenario C — pointer presence instead

When `0` is valid:

```
func pick(p *int, fallback int) int {
    if p != nil {
        return *p
    }
    return fallback
}
```

Do not force `cmp.Or` here.

Relation to `cmp.Compare`

`cmp.Compare` returns -1/0/1 for ordering. `cmp.Or` is unrelated except sharing the package. Import once: `"cmp"`.

Common mistakes (extended)

- Using `cmp.Or` on `error` values — the zero error is `nil`, so it works for "first non-nil error", which is handy, but document intent.
- Forgetting module Go version `< 1.22`.
- Overusing it in hot loops for clarity regressions — it is fine either way at this scale.

Sources

- cmp.Or
- Go blog: comparable

Memory

Functions in Depth

Functions in Depth

Overview

Functions are fundamental building blocks in Go. They support multiple return values, named returns, variadic parameters, and closures.

Function Declaration

```go
func name(params) returnType {
    // body
}

func add(a, b int) int {
    return a + b
}

func greet(name string) {
    fmt.Println("Hello,", name)
}
```

Multiple Return Values

```go
func divide(a, b float64) (float64, error) {
    if b == 0 {
        return 0, errors.New("division by zero")
    }
    return a / b, nil
}

result, err := divide(10, 2)
```

Named Return Values

```go
func split(sum int) (x, y int) {
    x = sum * 4 / 9
    y = sum - x
    return  // Naked return
}
```

Variadic Functions

```go
func sum(nums ...int) int {
    total := 0
    for _, n := range nums {
        total += n
    }
    return total
}

sum(1, 2, 3)            // 6
sum([]int{1, 2, 3}...) // Spread slice
```

Closures

Functions can capture variables from their enclosing scope:

```go
func counter() func() int {
    count := 0
    return func() int {
        count++
        return count
    }
}

c := counter()
c()  // 1
c()  // 2
c()  // 3
```

Defer

Defer postpones execution until the surrounding function returns:

```go
func readFile(name string) error {
    f, err := os.Open(name)
    if err != nil {
        return err
    }
    defer f.Close()  // Executes when function returns

    // Use file...
    return nil
}
```

Defer Order (LIFO)

```go
defer fmt.Println("1")
defer fmt.Println("2")
defer fmt.Println("3")
// Output: 3, 2, 1
```

Function Signatures

```go
// Function type
type Operation func(int, int) int

func apply(op Operation, a, b int) int {
    return op(a, b)
}

add := func(a, b int) int { return a + b }
result := apply(add, 2, 3)  // 5
```

Summary

Feature	Syntax
Multiple returns	`func f() (T1, T2)`
Named returns	`func f() (x T1, y T2)`
Variadic	`func f(args ...T)`
Closure	`func() { /* capture vars */ }`
Defer	`defer cleanup()`

Related Topics

- Functions as Values
- Methods and Receivers

More examples

Example: multiple returns and named results

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```
package main

import (
    "errors"
    "fmt"
)

func div(a, b int) (quot int, err error) {
    if b == 0 {
        err = errors.New("divide by zero")
        return
    }
    quot = a / b
    return
}

func main() {
    q, err := div(10, 2)
    fmt.Println("10/2:", q, err)
    q, err = div(10, 0)
    fmt.Println("10/0:", q, err)
}
```

Expected:

```
10/2: 5 <nil>
10/0: 0 divide by zero
```

Example: defer runs LIFO

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

func main() {
    fmt.Println("start")
    defer fmt.Println("first defer")
    defer fmt.Println("second defer")
    fmt.Println("end")
}
```

Expected:

```
start
end
second defer
first defer
```

Runnable example

Save as `main.go`. Then:

```bash
go mod init example
go run .
```

```go
package main

import (
    "errors"
    "fmt"
)

func add(a, b int) int { return a + b }

func divide(a, b float64) (float64, error) {
    if b == 0 {
        return 0, errors.New("division by zero")
    }
    return a / b, nil
}

func sum(nums ...int) int {
    total := 0
    for _, n := range nums {
        total += n
```

```go
    }
    return total
}

func counter() func() int {
    count := 0
    return func() int {
        count++
        return count
    }
}

func main() {
    fmt.Println("add:", add(2, 3))

    q, err := divide(10, 2)
    fmt.Println("divide ok:", q, err)
    _, err = divide(1, 0)
    fmt.Println("divide err:", err)

    fmt.Println("sum:", sum(1, 2, 3, 4))

    c := counter()
    fmt.Println("counter:", c(), c(), c())

    defer fmt.Println("defer: first (runs last)")
    defer fmt.Println("defer: second")
    fmt.Println("before return")
}
```

Expected output:

```
add: 5
divide ok: 5 <nil>
divide err: division by zero
sum: 10
counter: 1 2 3
before return
defer: second
defer: first (runs last)
```

What to notice: Multiple returns make success and failure explicit; the closure keeps private `count` state; deferred calls run LIFO after the rest of `main`.

Try next: Add a named-return `split(sum int) (x, y int)` and print both values; wrap `divide` so callers always get a non-nil error message with the operands.

Functions as Values

Functions as Values

Overview

In Go, functions are first-class citizens—they can be assigned to variables, passed as arguments, and returned from other functions.

Function Variables

```go
// Assign function to variable
add := func(a, b int) int {
    return a + b
}

result := add(2, 3)  // 5

// Reassign
add = func(a, b int) int {
    return a + b + 1  // Different implementation
}
```

Function Types

```go
type BinaryOp func(int, int) int

var op BinaryOp = func(a, b int) int {
    return a + b
}
```

Higher-Order Functions

Functions as Parameters

```go
func apply(nums []int, fn func(int) int) []int {
    result := make([]int, len(nums))
    for i, n := range nums {
        result[i] = fn(n)
    }
    return result
}

double := func(n int) int { return n * 2 }
result := apply([]int{1, 2, 3}, double)  // [2, 4, 6]
```

Functions as Return Values

```go
func multiplier(factor int) func(int) int {
    return func(n int) int {
        return n * factor
    }
}

double := multiplier(2)
triple := multiplier(3)

double(5)  // 10
triple(5)  // 15
```

Common Patterns

Callback Pattern

```go
func fetchData(url string, callback func(data []byte, err error)) {
    // Async operation
    go func() {
        data, err := http.Get(url)
        callback(data, err)
    }()
}
```

Option Pattern

```go
type Server struct {
    port    int
    timeout time.Duration
}

type Option func(*Server)

func WithPort(p int) Option {
    return func(s *Server) { s.port = p }
}

func WithTimeout(t time.Duration) Option {
    return func(s *Server) { s.timeout = t }
}

func NewServer(opts ...Option) *Server {
    s := &Server{port: 8080, timeout: 30 * time.Second}
    for _, opt := range opts {
        opt(s)
    }
    return s
}

srv := NewServer(WithPort(9000), WithTimeout(time.Minute))
```

Middleware Pattern

```go
type Handler func(http.ResponseWriter, *http.Request)

func WithLogging(h Handler) Handler {
    return func(w http.ResponseWriter, r *http.Request) {
        log.Printf("%s %s", r.Method, r.URL)
        h(w, r)
    }
}
```

Anonymous Functions

```go
// Immediate invocation (IIFE)
result := func(x int) int {
    return x * x
}(5)  // 25

// In goroutines
go func() {
    fmt.Println("async")
}()
```

Summary

Pattern	Use Case
Function variable	Store/swap implementations
Higher-order	Transform, filter, reduce
Closures	Capture state
Options	Flexible configuration

Related Topics

- Functions in Depth
- Designing with Interfaces

More examples

Example: swap function variables

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

func add(a, b int) int { return a + b }
func mul(a, b int) int { return a * b }

func main() {
    op := add
    fmt.Println("add:", op(3, 4))
```

```
    op = mul
    fmt.Println("mul:", op(3, 4))
}
```

Expected:

```
add: 7
mul: 12
```

Example: closure captures state

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```
package main

import "fmt"

func counter(start int) func() int {
    n := start
    return func() int {
        n++
        return n
    }
}

func main() {
    next := counter(10)
    fmt.Println(next())
    fmt.Println(next())
    fmt.Println(next())
}
```

Expected:

```
11
12
13
```

Runnable example

Save as `main.go`. Then:

```
go mod init example
go run .
```

```go
package main

import "fmt"

type BinaryOp func(int, int) int

func apply(nums []int, fn func(int) int) []int {
    out := make([]int, len(nums))
    for i, n := range nums {
        out[i] = fn(n)
    }
    return out
}

func multiplier(factor int) func(int) int {
    return func(n int) int { return n * factor }
}

type Server struct {
    port    int
    timeout int // seconds, kept simple
}

type Option func(*Server)

func WithPort(p int) Option {
    return func(s *Server) { s.port = p }
}

func WithTimeout(sec int) Option {
    return func(s *Server) { s.timeout = sec }
}

func NewServer(opts ...Option) *Server {
    s := &Server{port: 8080, timeout: 30}
    for _, opt := range opts {
        opt(s)
    }
    return s
}

func main() {
    add := func(a, b int) int { return a + b }
    var op BinaryOp = add
    fmt.Println("BinaryOp:", op(2, 3))

    doubled := apply([]int{1, 2, 3}, multiplier(2))
```

```go
    fmt.Println("map-like apply:", doubled)

    srv := NewServer(WithPort(9000), WithTimeout(60))
    fmt.Printf("server port=%d timeout=%ds\n", srv.port, srv.timeout)

    square := func(x int) int { return x * x }(5)
    fmt.Println("IIFE square:", square)
}
```

Expected output:

```
BinaryOp: 5
map-like apply: [2 4 6]
server port=9000 timeout=60s
IIFE square: 25
```

What to notice: Functions are values you can store, pass, and return; the option pattern is just a slice of `func(*Server)` applied at construction.

Try next: Add a `WithDefaults` option that resets port and timeout, or a `compose(f, g)` helper that returns `func(int) int` applying both.

Methods and Receivers

Methods and Receivers

Overview

Methods are functions with a receiver argument, allowing you to define behavior on types.

Method Syntax

```go
type Rectangle struct {
    Width, Height float64
}

func (r Rectangle) Area() float64 {
    return r.Width * r.Height
}

rect := Rectangle{10, 5}
area := rect.Area()  // 50
```

Value vs Pointer Receivers

Value Receiver

```go
func (r Rectangle) Scale(factor float64) Rectangle {
    return Rectangle{r.Width * factor, r.Height * factor}
}

// Original unchanged
r := Rectangle{10, 5}
scaled := r.Scale(2)  // New rectangle
```

Pointer Receiver

```
func (r *Rectangle) ScaleInPlace(factor float64) {
    r.Width *= factor
    r.Height *= factor
}

// Original modified
r := Rectangle{10, 5}
r.ScaleInPlace(2)  // r is now {20, 10}
```

When to Use Pointer Receivers

Use pointer receiver when: - Method needs to modify the receiver - Receiver is a large struct (avoid copying) - Consistency: if any method needs pointer, use pointer for all

```
type Buffer struct {
    data []byte
}

func (b *Buffer) Write(p []byte) {
    b.data = append(b.data, p...)
}

func (b *Buffer) String() string {
    return string(b.data)  // Even though it doesn't modify, use * for consistency
}
```

Methods on Any Type

```
type MyInt int

func (m MyInt) Double() MyInt {
    return m * 2
}

n := MyInt(5)
n.Double()  // 10
```

Automatic Dereferencing

Go automatically handles `*` and `&` for method calls:

```go
r := Rectangle{10, 5}
(&r).ScaleInPlace(2)  // Explicit pointer
r.ScaleInPlace(2)     // Go handles it automatically

ptr := &Rectangle{10, 5}
(*ptr).Area()         // Explicit dereference
ptr.Area()            // Go handles it automatically
```

Embedding and Method Promotion

```go
type Animal struct {
    Name string
}

func (a Animal) Speak() string {
    return "..."
}

type Dog struct {
    Animal  // Embedded
    Breed string
}

d := Dog{Animal: Animal{Name: "Rex"}, Breed: "Labrador"}
d.Speak()  // Promoted method: "..."
d.Name     // Promoted field: "Rex"
```

Method Override

```go
func (d Dog) Speak() string {
    return "Woof!"
}

d.Speak()         // "Woof!" (Dog's method)
d.Animal.Speak()  // "..." (Animal's method)
```

Generic Methods (Go 1.27+)

A method may declare **its own** type parameters. That is different from a method on a generic type, which already existed (`func (s *Stack[T]) Push(T)`):

```go
type Store struct {
    items []any
}

// Method-level type parameter: one Put works for any T.
func (s *Store) Put[T any](v T) {
    s.items = append(s.items, v)
}

func (s *Store) Get[T any](i int) (T, bool) {
    var zero T
    if i < 0 || i >= len(s.items) {
        return zero, false
    }
    v, ok := s.items[i].(T)
    return v, ok
}
```

The standard library uses this in `math/rand/v2`: `(*Rand).N[Int intType](n Int) Int` replaces a family of `IntN` / `Int32N` / `Int64N` methods.

Limits: - Interface methods **cannot** declare type parameters. - A generic method **cannot** implement an interface method (the method set is still non-generic). - Prefer a package-level generic function when the operation is not naturally namespaced on a type.

Summary

Receiver	Use Case
`(t T)`	Read-only, small types
`(t *T)`	Modify state, large types

Feature	Description
Auto-deref	Go handles `*` and `&`
Embedding	Methods are promoted
Override	Inner type's method shadows embedded
Generic methods (1.27+)	Method declares its own type parameters

Related Topics

- Pointers Explained
- Designing with Interfaces
- Why Generics Matter

More examples

Example: value receiver cannot mutate

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

type Point struct{ X, Y int }

func (p Point) Move(dx, dy int) {
    p.X += dx
    p.Y += dy
}

func main() {
    p := Point{1, 2}
    p.Move(10, 10)
    fmt.Printf("after value Move: %+v\n", p)
}
```

Expected:

```
after value Move: {X:1 Y:2}
```

Example: pointer receiver mutates

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

type Point struct{ X, Y int }

func (p *Point) Move(dx, dy int) {
    p.X += dx
    p.Y += dy
}

func main() {
    p := Point{1, 2}
    p.Move(10, 10)
    fmt.Printf("after pointer Move: %+v\n", p)
```

```
}
```

Expected:

```
after pointer Move: {X:11 Y:12}
```

Runnable example

Save as `main.go`. Then:

```
go mod init example
go run .
```

```
package main

import "fmt"

type Rectangle struct {
    Width, Height float64
}

func (r Rectangle) Area() float64 {
    return r.Width * r.Height
}

// Value receiver: returns a new value; original unchanged.
func (r Rectangle) Scale(factor float64) Rectangle {
    return Rectangle{r.Width * factor, r.Height * factor}
}

// Pointer receiver: mutates in place.
func (r *Rectangle) ScaleInPlace(factor float64) {
    r.Width *= factor
    r.Height *= factor
}

type Animal struct{ Name string }

func (a Animal) Speak() string { return "..." }

type Dog struct {
    Animal
    Breed string
}
```

```go
func (d Dog) Speak() string { return "Woof!" }

func main() {
    r := Rectangle{Width: 10, Height: 5}
    fmt.Println("area:", r.Area())

    scaled := r.Scale(2)
    fmt.Printf("after Scale: orig=%v scaled=%v\n", r, scaled)

    r.ScaleInPlace(2) // Go passes &r automatically
    fmt.Printf("after ScaleInPlace: %v\n", r)

    d := Dog{Animal: Animal{Name: "Rex"}, Breed: "Labrador"}
    fmt.Println("promoted Name:", d.Name)
    fmt.Println("Dog.Speak:", d.Speak())
    fmt.Println("Animal.Speak:", d.Animal.Speak())
}
```

Expected output:

```
area: 50
after Scale: orig={10 5} scaled={20 10}
after ScaleInPlace: {20 10}
promoted Name: Rex
Dog.Speak: Woof!
Animal.Speak: ...
```

What to notice: Value receivers copy; pointer receivers share the same `Rectangle`. Embedding promotes `Name` and `Speak`, and `Dog.Speak` shadows the embedded method.

Try next: Give `Rectangle` a `String() string` method and print with `%v` vs `%s`, or add a value-receiver `ScaleInPlace` (wrongly) and observe that the caller's fields no longer change.

Understanding Memory in Go

Understanding Memory in Go

Overview

Go manages memory automatically through its runtime, but understanding stack vs heap allocation helps write more efficient code.

Stack vs Heap

Stack

- Fast allocation/deallocation
- Function-local variables
- Fixed size per goroutine (~2KB initial)
- Automatically cleaned up when function returns

Heap

- Dynamic allocation
- Survives function scope
- Managed by garbage collector
- More expensive than stack

Escape Analysis

Go's compiler decides where to allocate based on **escape analysis**:

```go
func stackAlloc() int {
    x := 42      // Stays on stack
    return x     // Value copied
}

func heapAlloc() *int {
    x := 42      // Escapes to heap
    return &x    // Pointer returned
}
```

Check Escape Analysis

```
go build -gcflags="-m" main.go
# Shows: "moved to heap" for escaped variables
```

What Causes Escape

```go
// 1. Returning pointer to local
func escape1() *int {
    n := 1
    return &n  // Escapes
}

// 2. Storing in interface
func escape2() {
    n := 1
    fmt.Println(n)  // Escapes (interface{} argument)
}

// 3. Closure capturing variable
func escape3() func() int {
    n := 1
    return func() int { return n }  // Escapes
}

// 4. Size too large for stack
func escape4() {
    _ = make([]byte, 10<<20)  // 10MB, escapes
}
```

Memory Layout

Value Types

```go
type Point struct { X, Y int }
p := Point{1, 2}  // 16 bytes inline
```

Reference Types

```
s := make([]int, 10)  // Slice header on stack, data on heap
m := make(map[string]int)  // Map structure on heap
```

Best Practices

Reduce Allocations

```
// Bad: allocates each iteration
for i := 0; i < n; i++ {
    data := make([]byte, 1024)
    process(data)
}

// Good: reuse allocation
data := make([]byte, 1024)
for i := 0; i < n; i++ {
    process(data)
}
```

Preallocate Slices

```
// Bad: multiple reallocations
var result []int
for _, v := range data {
    result = append(result, v)
}

// Good: preallocate
result := make([]int, 0, len(data))
for _, v := range data {
    result = append(result, v)
}
```

Use Value Semantics When Possible

```go
// May allocate
func process(p *Point) { }

// Stays on stack
func process(p Point) { }
```

Summary

Location	Characteristics
Stack	Fast, automatic, limited size
Heap	Dynamic, GC-managed, survives scope

Causes Escape	Example
Return pointer	`return &x`
Interface	`fmt.Println(x)`
Closure	`func() { use(x) }`
Large size	`make([]byte, 1MB)`

Related Topics

- Pointers Explained
- Garbage Collection

More examples

Example: value stays local

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

func double(x int) int {
    return x * 2
}

func main() {
    n := 21
    fmt.Println("double:", double(n))
```

```
    fmt.Println("original:", n)
}
```

Expected:

```
double: 42
original: 21
```

Example: interface argument often forces heap

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```
package main

import "fmt"

func show(v any) {
    fmt.Printf("value=%v type=%T\n", v, v)
}

func main() {
    x := 99
    show(x) // boxing into any is a common escape trigger
    show("ok")
}
```

Expected:

```
value=99 type=int
value=ok type=string
```

Runnable example

Save as `main.go`. Then:

```
go mod init example
go run .
# Optional: see escape decisions
# go build -gcflags="-m" .
```

```
package main

import "fmt"
```

```go
type Point struct{ X, Y int }

// Returns a value: caller gets a copy; x can stay on the stack.
func stackAlloc() int {
    x := 42
    return x
}

// Returns a pointer: x must live past the call, so it escapes to the heap.
func heapAlloc() *int {
    x := 42
    return &x
}

// Reuse one buffer instead of allocating per iteration.
func processReuse(n int) int {
    buf := make([]byte, 64)
    sum := 0
    for i := 0; i < n; i++ {
        buf[0] = byte(i)
        sum += int(buf[0])
    }
    return sum
}

func main() {
    fmt.Println("stackAlloc:", stackAlloc())

    p := heapAlloc()
    fmt.Println("heapAlloc value:", *p)
    // Do not print the address itself - it changes every run.

    // Slice header can be on the stack; backing array is typically on the heap.
    pts := make([]Point, 0, 3)
    pts = append(pts, Point{1, 2}, Point{3, 4})
    fmt.Printf("points len=%d cap=%d first=%v\n", len(pts), cap(pts), pts[0])

    fmt.Println("processReuse:", processReuse(5))
}
```

Expected output:

```
stackAlloc: 42
heapAlloc value: 42
points len=2 cap=3 first={1 2}
processReuse: 10
```

What to notice: Returning `*int` forces heap allocation; reusing one slice buffer avoids per-loop allocs. Run with `-gcflags="-m"` to confirm which locals the compiler marks as escaping.

Try next: Change `heapAlloc` to return `int` by value and re-run escape analysis; preallocate `pts` with `make([]Point, 0, 100)` and watch capacity stay put under many `append`s.

Pointers Explained

Pointers Explained

Overview

Pointers hold memory addresses. In Go, they're simple and safe—no pointer arithmetic allowed.

Pointer Basics

```go
x := 42
p := &x      // p is *int, holds address of x
fmt.Println(*p)  // 42 (dereference)
*p = 100     // Modify x through pointer
fmt.Println(x)   // 100
```

Declaration

```go
var p *int       // nil pointer
p = &x           // Point to x

// Classic new (zero value)
p := new(int)    // *int to 0

// Expression-based new (Go 1.26+)
p := new(42)     // *int to 42
```

```
Stack/heap view (simplified)

x: 42          p: 0xABC
 |               |
 +------ addr ---+
         |
       [42]
```

Why Use Pointers

1. Modify Variables in Functions

```go
func increment(n *int) {
    *n++
}

x := 1
increment(&x)
fmt.Println(x)  // 2
```

2. Avoid Copying Large Structs

```go
type BigStruct struct {
    Data [1000]int
}

// Bad: copies entire struct
func process(b BigStruct) { }

// Good: passes pointer (8 bytes)
func process(b *BigStruct) { }
```

3. Share Data

```go
type Config struct {
    Debug bool
}

func loadConfig() *Config {
    return &Config{Debug: true}
}

cfg := loadConfig()  // All code shares same Config
```

nil Pointers

```go
var p *int  // nil

if p != nil {
    fmt.Println(*p)  // Safe
}

// Dereferencing nil panics!
// *p = 1  // panic: runtime error
```

Safe pattern for optional fields in APIs:

```go
type UpdateUserRequest struct {
    Name  *string `json:"name,omitempty"`
    Email *string `json:"email,omitempty"`
}
```

Pointer to Pointer

```go
x := 42
p := &x
pp := &p  // **int

**pp = 100
fmt.Println(x)  // 100
```

Pointers and Slices/Maps

Slices and maps are already reference types:

```go
func modify(s []int) {
    s[0] = 100  // Modifies original
}

nums := []int{1, 2, 3}
modify(nums)
fmt.Println(nums[0])  // 100

// But reassigning needs pointer
func replace(s *[]int) {
    *s = []int{4, 5, 6}
}
```

Common Patterns

Return Pointer for "No Result"

```go
func find(id int) *User {
    if found {
        return &user
    }
    return nil  // Not found
}

if user := find(1); user != nil {
    // Use user
}
```

Constructor Pattern

```go
func NewUser(name string) *User {
    return &User{
        Name:      name,
        CreatedAt: time.Now(),
    }
}
```

Summary

Operation	Syntax
Get address	`&x`
Dereference	`*p`
Allocate	`new(T)`
Check nil	`p != nil`

Related Topics

- Methods and Receivers
- Mutability and Data Sharing

More examples

Example: address and dereference

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

func main() {
    x := 10
    p := &x
    fmt.Println("x:", x)
    fmt.Println("*p:", *p)
    *p = 20
    fmt.Println("x after *p=20:", x)
}
```

Expected:

```
x: 10
*p: 10
x after *p=20: 20
```

Example: nil pointer check

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

func safeLen(p *string) int {
    if p == nil {
        return 0
    }
    return len(*p)
}

func main() {
    var missing *string
    s := "gopher"
    fmt.Println("nil:", safeLen(missing))
    fmt.Println("set:", safeLen(&s))
}
```

Expected:

```
nil: 0
set: 6
```

Runnable example

Save as `main.go`. Then:

```
go mod init example
go run .
```

```go
package main

import "fmt"

func increment(n *int) {
    *n++
}

func modifySlice(s []int) {
    if len(s) > 0 {
        s[0] = 100 // mutates shared backing array
    }
}

func replaceSlice(s *[]int) {
    *s = []int{4, 5, 6} // replaces the caller's slice header
}

func main() {
    x := 42
    p := &x
    fmt.Println("via pointer:", *p)
    *p = 100
    fmt.Println("after *p=100, x:", x)

    increment(&x)
    fmt.Println("after increment:", x)

    var nilP *int
    fmt.Println("nil pointer is nil:", nilP == nil)
    // *nilP would panic - always check first.

    // new allocates a zero value and returns a pointer.
```

```go
    z := new(int)
    fmt.Println("new(int) value:", *z)
    *z = 7
    fmt.Println("new(int) after set:", *z)

    nums := []int{1, 2, 3}
    modifySlice(nums)
    fmt.Println("after modifySlice:", nums)

    replaceSlice(&nums)
    fmt.Println("after replaceSlice:", nums)
}
```

Expected output:

```
via pointer: 42
after *p=100, x: 100
after increment: 101
nil pointer is nil: true
new(int) value: 0
new(int) after set: 7
after modifySlice: [100 2 3]
after replaceSlice: [4 5 6]
```

What to notice: `*p` reads/writes the same storage as `x`. Slice element writes share the backing array without a pointer; replacing the slice header itself needs `*[]int`.

Try next: Pass a large struct by value vs pointer and compare with `unsafe.Sizeof` (or just reason about copy cost); write a safe helper that returns zero when given a nil `*int`.

Mutability and Data Sharing

Mutability and Data Sharing

Overview

Understanding when data is copied vs shared is crucial for avoiding bugs and writing efficient Go code.

Value vs Reference Semantics

Value Types (Copied)

```go
// Integers, floats, bools, strings, arrays, structs
a := 1
b := a
b = 2
fmt.Println(a)  // 1 (unchanged)

arr := [3]int{1, 2, 3}
arr2 := arr
arr2[0] = 100
fmt.Println(arr[0])  // 1 (unchanged)
```

Reference Types (Shared)

```go
// Slices, maps, channels, pointers
s := []int{1, 2, 3}
s2 := s
s2[0] = 100
fmt.Println(s[0])  // 100 (changed!)

m := map[string]int{"a": 1}
m2 := m
m2["a"] = 100
fmt.Println(m["a"])  // 100 (changed!)
```

Controlling Mutability

Immutable by Design

```go
// Return new instead of modifying
func addItem(items []string, item string) []string {
    return append(items, item)  // May return new slice
}
```

Defensive Copy

```go
func process(s []int) {
    local := make([]int, len(s))
    copy(local, s)  // Safe to modify local
}
```

Deep Copy

```go
type User struct {
    Name    string
    Friends []string
}

func (u User) Clone() User {
    friends := make([]string, len(u.Friends))
    copy(friends, u.Friends)
    return User{Name: u.Name, Friends: friends}
}
```

Function Parameters

```go
// Value: caller's data safe
func process(u User) {
    u.Name = "modified"  // Doesn't affect caller
}

// Pointer: can modify caller's data
func process(u *User) {
    u.Name = "modified"  // Affects caller
}
```

Struct Field Mutability

```go
type Counter struct {
    count int
}

// Value receiver: cannot modify
func (c Counter) Increment() {
    c.count++  // Modifies copy, original unchanged
}

// Pointer receiver: modifies original
func (c *Counter) Increment() {
    c.count++  // Original modified
}
```

Slice Gotchas

```go
// Append may or may not share backing array
s := []int{1, 2, 3}
s2 := s[:2]          // Shares backing array
s2 = append(s2, 4)   // Might overwrite s[2]!

// Safer: force new allocation
s2 := append([]int(nil), s[:2]...)
```

Summary

Type	Behavior	Copy When…
int, bool, string	Value	Always copied
struct	Value	Always copied
array	Value	Always copied
slice	Reference	Use `copy()`
map	Reference	Rebuild manually
pointer	Reference	-

Related Topics

- Pointers Explained
- Concurrency Design Guidelines

More examples

Example: slice header shares backing array

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

func main() {
    a := []int{1, 2, 3}
    b := a
    b[0] = 99
    fmt.Println("a:", a)
    fmt.Println("b:", b)
}
```

Expected:

```
a: [99 2 3]
b: [99 2 3]
```

Example: copy to isolate mutation

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

func main() {
    a := []int{1, 2, 3}
    b := make([]int, len(a))
    copy(b, a)
    b[0] = 99
    fmt.Println("a:", a)
    fmt.Println("b:", b)
}
```

Expected:

```
a: [1 2 3]
b: [99 2 3]
```

Runnable example

Save as `main.go`. Then:

```
go mod init example
go run .
```

```go
package main

import "fmt"

type User struct {
    Name    string
    Friends []string
}

func (u User) Clone() User {
    friends := make([]string, len(u.Friends))
    copy(friends, u.Friends)
    return User{Name: u.Name, Friends: friends}
}

type Counter struct{ count int }

func (c Counter) IncValue() {
    c.count++ // mutates a copy only
}

func (c *Counter) IncPointer() {
    c.count++
}

func main() {
    // Value types: assignment copies.
    a := 1
    b := a
    b = 2
    fmt.Println("ints a,b:", a, b)

    arr := [3]int{1, 2, 3}
    arr2 := arr
    arr2[0] = 99
    fmt.Println("arrays arr,arr2:", arr, arr2)

    // Reference-like headers: slice assignment shares data.
    s := []int{1, 2, 3}
```

```go
    s2 := s
    s2[0] = 100
    fmt.Println("shared slice s:", s)

    // Defensive copy of the backing array.
    safe := make([]int, len(s))
    copy(safe, s)
    safe[0] = 1
    fmt.Println("after defensive copy, s still:", s, "safe:", safe)

    // append may or may not reallocate - force a fresh backing array.
    base := []int{1, 2, 3}
    alias := base[:2]
    alias = append(alias, 9) // often overwrites base[2]
    fmt.Println("append into shared capacity base:", base)

    fresh := append([]int(nil), base[:2]...)
    fresh = append(fresh, 9)
    fmt.Println("append into fresh copy base:", base, "fresh:", fresh)

    u := User{Name: "Ada", Friends: []string{"Bob"}}
    u2 := u.Clone()
    u2.Friends[0] = "Carol"
    fmt.Println("clone independent:", u.Friends, u2.Friends)

    c := Counter{}
    c.IncValue()
    fmt.Println("after value Inc:", c.count)
    c.IncPointer()
    fmt.Println("after pointer Inc:", c.count)
}
```

Expected output:

```
ints a,b: 1 2
arrays arr,arr2: [1 2 3] [99 2 3]
shared slice s: [100 2 3]
after defensive copy, s still: [100 2 3] safe: [1 2 3]
append into shared capacity base: [1 2 9]
append into fresh copy base: [1 2 9] fresh: [1 2 9]
clone independent: [Bob] [Carol]
after value Inc: 0
after pointer Inc: 1
```

What to notice: Arrays and structs copy deeply for their own fields, but nested slices still share unless you `copy`. Value receivers never update the caller's `Counter`.

Try next: Map the same experiment with `map[string]int` (assignment shares); fix the shared-

`append` case by capping first: `base[:2:2]`.

Garbage Collection

Garbage Collection

Overview

Go uses automatic garbage collection (GC) to manage memory. Understanding how it works helps you write GC-friendly code.

Modern GC Architecture

Concurrent Mark and Sweep

Go's GC is concurrent and non-moving. At a high level: 1. **Mark Setup**: (STW) Prepare for marking. 2. **Marking**: (Concurrent) Mark reachable objects. 3. **Mark Termination**: (STW) Finalize marking. 4. **Sweeping**: (Concurrent) Reclaim memory.

```
flow:
  ] -> Mark[
  Mark -> STW2[
  STW2 -> Sweep[
```

Go 1.26: Green Tea GC (GTGC)

Go 1.26 switched the default collector to **Green Tea GC (GTGC)** for lower pause times and better cache locality in pointer-heavy workloads.

```
flow:
  ] -> LObj1[  (Random Heap Jumps)
  LObj1 -> LObj2[  (Cache Miss)
  ] -> GSpan1[  (Span-Aware Local Scan)
  GSpan1 -> GSpan2[  (Vectorized Bitmaps)
```

- **Span-Aware Locality**: Groups pointer scanning by physical memory spans to maximize CPU L1/L2 cache hits.
- **Lower CPU Overhead**: Reduces write-barrier costs in concurrent mutator workers.
- **Opt-Out**: If needed for debugging or legacy performance comparisons:

```
GOEXPERIMENT=nogreenteagc go test ./...
```

Go 1.27: size-specialized small allocations

The compiler emits calls to size-specialized malloc for some objects under 80 bytes, cutting those allocation costs by up to ~30% (~1% in allocation-heavy programs; ~60 KB extra binary). Opt out with `GOEXPERIMENT=nosizespecializedmalloc` if you hit a regression (expected to be removed in Go 1.28). See Memory Allocator.

GOGC and Memory Limits

Think of `GOGC` and `GOMEMLIMIT` as two controls:

- `GOGC` controls *when* to trigger based on heap growth.
- `GOMEMLIMIT` puts a soft cap on total Go-managed memory.

```
Heap growth trigger                Memory budget trigger
live heap * (1 + GOGC/100)         total Go memory near GOMEMLIMIT
```

Tuning with GOGC

```
# Default: GC when heap doubles (100% growth)
GOGC=100 ./myapp

# More aggressive: GC at 50% growth (uses less memory)
GOGC=50 ./myapp

# Less frequent: GC at 200% growth (faster, more memory)
GOGC=200 ./myapp

# Disable GC (not recommended)
GOGC=off ./myapp
```

Tuning with GOMEMLIMIT

```
# Keep process around 2 GiB Go-managed memory
GOMEMLIMIT=2GiB ./myapp
```

Programmatic control:

```
import "runtime/debug"

func init() {
```

```go
    debug.SetMemoryLimit(2 << 30) // 2 GiB
}
```

Memory Stats

```go
var m runtime.MemStats
runtime.ReadMemStats(&m)

fmt.Printf("Alloc: %d MB\n", m.Alloc/1024/1024)
fmt.Printf("Total Alloc: %d MB\n", m.TotalAlloc/1024/1024)
fmt.Printf("Heap Objects: %d\n", m.HeapObjects)
fmt.Printf("GC Cycles: %d\n", m.NumGC)
```

A low-overhead option for production telemetry:

```go
import "runtime/metrics"

samples := []metrics.Sample{
    {Name: "/gc/heap/live:bytes"},
    {Name: "/gc/heap/goal:bytes"},
}
metrics.Read(samples)
```

Reducing GC Pressure

1. Reduce Allocations

```go
// Bad: allocates each call
func getBuffer() []byte {
    return make([]byte, 1024)
}

// Good: reuse with sync.Pool
var bufPool = sync.Pool{
    New: func() any { return make([]byte, 1024) },
}

func getBuffer() []byte {
    return bufPool.Get().([]byte)
}

func putBuffer(b []byte) {
```

```go
    bufPool.Put(b)
}
```

2. Preallocate

```go
result := make([]int, 0, expectedSize)
```

3. Use Value Types

```go
// More allocations
type Points []*Point

// Fewer allocations
type Points []Point
```

4. Avoid String Concatenation in Loops

```go
// Bad: allocates each iteration
s := ""
for _, part := range parts {
    s += part
}

// Good: single allocation
var b strings.Builder
for _, part := range parts {
    b.WriteString(part)
}
s := b.String()
```

Profiling

```bash
# CPU profile
go test -cpuprofile=cpu.out
go tool pprof cpu.out

# Memory profile
go test -memprofile=mem.out
go tool pprof mem.out
```

```
# View allocations
go tool pprof -alloc_space mem.out
```

Summary

Optimization	Technique
Reuse memory	`sync.Pool`
Preallocate	`make([]T, 0, cap)`
Values vs pointers	Use values for small types
String building	`strings.Builder`

Related Topics

- Understanding Memory in Go
- Performance Tuning

More examples

Example: preallocate slice capacity

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

func main() {
    // Growing without capacity reallocates more often.
    var grow []int
    for i := 0; i < 5; i++ {
        grow = append(grow, i)
    }

    // Preallocated capacity reduces realloc churn.
    ready := make([]int, 0, 5)
    for i := 0; i < 5; i++ {
        ready = append(ready, i)
    }
    fmt.Println("grow:", grow, "len", len(grow), "cap", cap(grow))
    fmt.Println("ready:", ready, "len", len(ready), "cap", cap(ready))
}
```

Expected:

```
grow: [0 1 2 3 4] len 5 cap 8
ready: [0 1 2 3 4] len 5 cap 5
```

Example: strings.Builder vs naive concat

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import (
    "fmt"
    "strings"
)

func main() {
    var b strings.Builder
    for i := 0; i < 4; i++ {
        fmt.Fprintf(&b, "%d,", i)
    }
    fmt.Println("builder:", b.String())

    // Educational contrast: repeated + creates more temporary strings.
    s := ""
    for i := 0; i < 4; i++ {
        s += fmt.Sprintf("%d,", i)
    }
    fmt.Println("concat:", s)
}
```

Expected:

```
builder: 0,1,2,3,
concat: 0,1,2,3,
```

Runnable example

Save as `main.go`. Then:

```
go mod init example
go run .
```

```go
package main

import (
    "fmt"
    "runtime"
)

func readStats() runtime.MemStats {
    var m runtime.MemStats
    runtime.ReadMemStats(&m)
    return m
}

func printStats(label string, m runtime.MemStats) {
    // HeapAlloc / HeapObjects are more stable educational signals than Alloc alone.
    fmt.Printf("%s: HeapAlloc=%d bytes HeapObjects=%d NumGC=%d\n",
        label, m.HeapAlloc, m.HeapObjects, m.NumGC)
}

func main() {
    runtime.GC()
    before := readStats()
    printStats("before alloc", before)

    // Keep a live reference so the GC cannot free these yet.
    keep := make([][]byte, 0, 100)
    for i := 0; i < 100; i++ {
        keep = append(keep, make([]byte, 64*1024)) // 64 KiB each
    }
    afterAlloc := readStats()
    printStats("after alloc", afterAlloc)

    // Drop references, then force a collection.
    keep = nil
    runtime.GC()
    afterGC := readStats()
    printStats("after GC", afterGC)

    fmt.Printf("objects grew by ~%d then GC cycles advanced by %d\n",
        int64(afterAlloc.HeapObjects)-int64(before.HeapObjects),
        afterGC.NumGC-before.NumGC)
}
```

Expected output: (numbers vary by platform and Go version; shape is stable)

```
before alloc: HeapAlloc=... bytes HeapObjects=... NumGC=...
after alloc: HeapAlloc=... bytes HeapObjects=... NumGC=...
after GC: HeapAlloc=... bytes HeapObjects=... NumGC=...
objects grew by ~100 then GC cycles advanced by 1
```

Typically `HeapAlloc` and `HeapObjects` rise after the loop, then fall after `keep = nil` + `runtime.GC()`, while `NumGC` increases.

What to notice: Live pointers keep heap memory reachable; clearing the root and forcing GC lets the collector reclaim it. Absolute byte counts are not portable — look at direction of change.

Try next: Run with `GOGC=50` vs `GOGC=200` and compare how soon `NumGC` climbs during a longer allocation loop; try `sync.Pool` to reuse the 64 KiB buffers.

Stack Mechanics & Optimizations

Understanding Memory: Stack vs. Heap in Go

To write high-performance Go, you must understand where your variables live. Go abstracts memory management, but it doesn't eliminate the physics of hardware.

The Two Worlds

1. The Stack

- **What it is:** A pre-allocated, contiguous block of memory for each goroutine (starts at 2KB).
- **Allocation:** Instant (move a pointer).
- **Deallocation:** Instant (move a pointer back).
- **GC Cost: Zero.** The Garbage Collector (GC) does not scan the stack.
- **Cache Locality:** Excellent.

2. The Heap

- **What it is:** A global pool of memory for dynamic allocations.
- **Allocation:** Slow (search for free block, possibly lock).
- **Deallocation:** Garbage Collected (expensive scan).
- **GC Cost:** High. Pointers on the heap must be traced.
- **Cache Locality:** Poor (scattered).

Go's Optimization Goal

Keep everything on the stack.

If the compiler can prove a variable's life cycle is contained within a function (it doesn't "escape"), it allocates it on the stack.

Stack Growth (Contiguous Stacks)

Goroutines start with 2KB stacks. If a function call needs more space, the runtime: 1. Allocates a larger stack (e.g., 4KB). 2. **Copies** everything from old stack to new stack. 3. Updates pointers to the new stack. 4. Frees the old stack.

Note: This "copying" is why pointers to stack variables are safe only if they don't escape. If they escaped to the heap, moving the stack would invalidate external pointers.

2026: The "Mid-Stack" Inlining

Recent Go versions have become aggressive about inlining function calls mid-stack. * If `UserFunc` calls `AllocFunc`, and `AllocFunc` is small, the compiler copies `AllocFunc`'s body into `UserFunc`. * This removes the function call overhead *and* allows variables that might have escaped (due to being return values) to stay on the stack.

Visualization

Variable Scenario	Destination	Why?
`x := 42`	**Stack**	Never leaves function references.
`y := &x` (used locally)	**Stack**	Address taken, but scope is local.
`return &x`	**Heap**	Escapes to caller; stack frame dies on return.
`fmt.Println(x)`	**Heap** (usually)	`fmt.Println` takes `interface{}`, which often causes escape.
`make([]byte, 1024)`	**Stack**	Size known at compile time, fits in stack.
`make([]byte, n)`	**Heap**	Size unknown at compile time.

Practical Rule

Don't fear the Heap, but respect the Stack. If you are writing a hot loop (a game loop, a high-frequency trading handler), ensure your temporary variables do not escape to the heap.

Use detailed analysis tools to verify this (covered in next chapter).

More examples

Example: stack-friendly local work

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

func fold(nums []int) int {
    // Locals and the slice header stay short-lived; good stack candidates.
```

```go
    total := 0
    for _, n := range nums {
        total += n
    }
    return total
}

func main() {
    data := [4]int{10, 20, 30, 40} // fixed array: stack-friendly storage
    fmt.Println("fold:", fold(data[:]))
    fmt.Println("fold empty:", fold(nil))
}
```

Expected:

```
fold: 100
fold empty: 0
```

Example: escape when a pointer outlives the frame

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

type Box struct{ Label string }

func newBox(label string) *Box {
    b := Box{Label: label} // typically moves to heap: returned address
    return &b
}

func localBox() string {
    b := Box{Label: "stack-ish"}
    p := &b // address taken but still local
    return p.Label
}

func main() {
    fmt.Println("local:", localBox())
    b := newBox("heap-ish")
    fmt.Println("returned:", b.Label)
}
```

Expected:

```
local: stack-ish
returned: heap-ish
```

Runnable example

Save as `main.go`. Then:

```
go mod init example
go run .
# Escape report (educational; messages depend on compiler version):
# go build -gcflags="-m -m" .
```

```go
package main

import "fmt"

type Point struct{ X, Y int }

// Local use only - good stack candidate.
func useLocally() int {
    p := Point{X: 1, Y: 2}
    q := &p // address taken, but still only used here
    return q.X + q.Y
}

// Returned pointer must outlive this frame → heap.
func returnPointer() *Point {
    p := Point{X: 3, Y: 4}
    return &p
}

// Fixed-size array often stays stack-friendly; dynamic make usually does not.
func fixedBufSum() int {
    var buf [16]byte
    for i := range buf {
        buf[i] = byte(i)
    }
    sum := 0
    for _, b := range buf {
        sum += int(b)
    }
    return sum
}
```

```go
func dynamicBufSum(n int) int {
    buf := make([]byte, n)
    for i := range buf {
        buf[i] = byte(i)
    }
    sum := 0
    for _, b := range buf {
        sum += int(b)
    }
    return sum
}

func main() {
    fmt.Println("local pointer use:", useLocally())
    hp := returnPointer()
    fmt.Printf("heap point: {%d %d}\n", hp.X, hp.Y)
    fmt.Println("fixedBufSum:", fixedBufSum())
    fmt.Println("dynamicBufSum:", dynamicBufSum(16))
}
```

Expected output:

```
local pointer use: 3
heap point: {3 4}
fixedBufSum: 120
dynamicBufSum: 120
```

What to notice: Taking an address does not automatically mean "heap" — escaping past the function does. Fixed-size `[16]byte` and `make([]byte, n)` can behave differently under escape analysis even when they compute the same sum.

Try next: Re-run with `go build -gcflags="-m"` and find which locals "moved to heap"; inline a tiny helper into `main` and see if the compiler still reports an escape for a returned pointer.

Escape Analysis & Alignment

Escape Analysis & Memory Alignment

Memory optimization usually comes down to two questions: 1. Did it hit the heap when it didn't need to? (Excessive allocation) 2. Is it wasting space? (Poor alignment)

Part 1: Escape Analysis

"Escape Analysis" is the compiler phase that decides Stack vs. Heap.

Asking the Compiler

You don't need to guess. Ask the compiler what it decided:

```
go build -gcflags="-m" main.go
```

Output Interpretation: * `can inline main`: Good. * `x does not escape`: Great (Stack). * `moved to heap: x`: Bad (Heap).

Common Escape Triggers

1. **Returning Pointers:** `func New() *T { return &T{} }` -> Escapes.
2. **Interfaces:** `func Log(v interface{})`. The concrete type inside the interface box often escapes.
3. **Closures:** Variables captured by a closure *might* escape if the closure outlives the function.
4. **Unknown Size:** `make([]int, n)` always escapes if `n` is dynamic.

The "Mid-Stack" Trick

If you need a large buffer inside a hot loop, allocate it *once* outside the loop or reuse a `sync.Pool`. * **Bad:** `for { b := make([]byte, 1024); process(b) }` (Trash on heap every loop) * **Better:** `b := make([]byte, 1024); for { process(b) }` (One alloc) * **Best (Stack):** `b := [1024]byte{}; for { process(b[:]) }` (Stack alloc if small enough)

Part 2: Memory Alignment (Padding)

CPU reads memory in words (e.g., 64-bit / 8 bytes). If a field doesn't align with a word boundary, the compiler adds **Padding** (wasted bytes).

The Struct Layout Game

Consider this struct:

```go
type BadStruct struct {
    Flag    bool    // 1 byte
    Counter int64   // 8 bytes
    Active  bool    // 1 byte
}
```

Total Size: * `bool` (1) + `padding` (7) -> to align `int64` * `int64` (8) * `bool` (1) + `padding` (7) -> to align struct size to 8 * **Total:** 24 bytes

Reshuffled:

```go
type GoodStruct struct {
    Counter int64   // 8 bytes (0-7)
    Flag    bool    // 1 byte  (8)
    Active  bool    // 1 byte  (9)
    // padding (6) -> to align struct size to multiple of 8 (16)
}
```

Total: 16 bytes. **33% savings just by reordering fields.**

Tools used in 2026

Do not optimize manually unless you are bored. Use tools.

- **`fieldalignment`**: `bash go install golang.org/x/tools/go/analysis/passes/fieldalignment/cm fieldalignment ./...` It will report structs that use too much memory and suggest a fix.
- **`betteralign`**: A more modern wrapper that can automatically apply changes (`-apply`).

When does this matter?

- **Single struct:** Who cares? 8 bytes is nothing.
- **Slice of 100 million structs:** 8 bytes * 100M = 800MB of RAM wasted. This triggers GC more often, costs money on cloud bills, and slows down cache.

Rule: Optimize alignment for "Data Types" (things you store in DB/Arrays). Ignore it for "Service Types" (singletons, handlers).

More examples

Example: measure padding with `unsafe.Sizeof`

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import (
    "fmt"
    "unsafe"
)

type Wide struct {
    A bool
    B int64
    C bool
    D int32
}

type Tight struct {
    B int64
    D int32
    A bool
    C bool
}

func main() {
    fmt.Println("Wide:", unsafe.Sizeof(Wide{}))
    fmt.Println("Tight:", unsafe.Sizeof(Tight{}))
    fmt.Println("saved bytes:", int(unsafe.Sizeof(Wide{})-unsafe.Sizeof(Tight{})))
}
```

Expected (64-bit; illustrative):

```
Wide: 24
Tight: 16
saved bytes: 8
```

Example: reuse buffer vs allocate every loop

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"
```

```go
func process(b []byte) int {
    sum := 0
    for _, x := range b {
        sum += int(x)
    }
    return sum
}

func main() {
    // Better: one allocation reused (or stack array as view).
    var stack [8]byte
    buf := stack[:]
    total := 0
    for i := 0; i < 3; i++ {
        for j := range buf {
            buf[j] = byte(i + j)
        }
        total += process(buf)
    }
    fmt.Println("reused total:", total)

    // Contrasting pattern (educational): allocate each iteration.
    total = 0
    for i := 0; i < 3; i++ {
        b := make([]byte, 8)
        for j := range b {
            b[j] = byte(i + j)
        }
        total += process(b)
    }
    fmt.Println("alloc-each total:", total)
}
```

Expected:

```
reused total: 108
alloc-each total: 108
```

Runnable example

Save as `main.go`. Then:

```
go mod init example
go run .
# go build -gcflags="-m" .
```

```go
package main

import (
    "fmt"
    "unsafe"
)

// Bad order: bools separated by int64 force padding.
type BadStruct struct {
    Flag    bool
    Counter int64
    Active  bool
}

// Better order: large fields first, bools packed together.
type GoodStruct struct {
    Counter int64
    Flag    bool
    Active  bool
}

// Escape triggers: return pointer, interface boxing, open-ended make.
func newCounter() *int {
    n := 1
    return &n // escapes
}

func boxAndPrint(v any) {
    // Interface conversion often forces heap allocation of the concrete value.
    _ = fmt.Sprintf("%v", v)
}

func main() {
    fmt.Println("sizeof BadStruct:", unsafe.Sizeof(BadStruct{}))
    fmt.Println("sizeof GoodStruct:", unsafe.Sizeof(GoodStruct{}))

    // Live use of escaped values so the compiler cannot drop them entirely.
    c := newCounter()
    fmt.Println("escaped counter:", *c)

    x := 42
    boxAndPrint(x)

    // Unknown size → typically heap.
    n := 32
    buf := make([]byte, n)
    buf[0] = 7
```

```go
    fmt.Println("dynamic buffer first byte:", buf[0])

    // Stack-friendly pattern: fixed array reused as a slice view.
    var stackBuf [32]byte
    view := stackBuf[:]
    view[0] = 9
    fmt.Println("stack-ish view first byte:", view[0])
}
```

Expected output: (sizes are for 64-bit Go; 32-bit differs)

```
sizeof BadStruct: 24
sizeof GoodStruct: 16
escaped counter: 1
dynamic buffer first byte: 7
stack-ish view first byte: 9
```

What to notice: Field order changes padding and total size without changing semantics. Returning `&n` and boxing into `any` are classic escape triggers; a fixed `[N]byte` reused as `[:]` is a common stack-friendly hot-path pattern.

Try next: Reorder another struct (`byte`, `int64`, `byte`, `int32`) and recompute sizes; run `-gcflags="-m"` and match each "moved to heap" line to a trigger above.

Size-Specialized Memory Allocation

Size-Specialized Memory Allocation

Go **1.27** speeds up many small heap allocations (about 80 bytes or fewer) by calling size-specialized `mallocgc` variants instead of one general path. You do not change your desk code to get it. Build with Go 1.27 and the hot common sizes — string headers, interface values, short slices — get cheaper. The release notes put the gain at up to about 30% for those allocations, and about 1% for allocation-heavy programs overall. Binary size grows by roughly 60 KB.

Mental model

- Escaping values still go through the runtime allocator. The compiler used to call `newobject` then generic `mallocgc`. When it knows the **span class** (size class plus "has pointers?"), Go 1.27 can call a specialized function such as `mallocgcSmallNoScanSC3` directly.
- Size classes bucket sizes. Through 80 bytes the classes look like this (ranges are inclusive of the upper end the span returns):

Size class	Byte range
1	1–8
2	9–16
3	17–24
4	25–32
5	33–48
6	49–64
7	65–80

- Specialization stops at 80 bytes on purpose. Larger clears dominate the cost, and more specialized functions would crowd the instruction cache.
- When the size is **not** known at compile time (`make([]byte, n)` with a runtime `n`), the compiler still calls generic `mallocgc`, which may dispatch into a specialized helper. That path stays faster than the old generic-only loop, but the biggest win is the direct call for known sizes.
- Opt out only if you are chasing a regression: build with `GOEXPERIMENT=nosizespecializedmalloc`. That knob is expected to disappear in Go 1.28. File an issue if you need it.

Worked examples

Case 1: Map everyday values onto size classes

Save as `size_classes.go`. Headers for strings, interfaces, and slices land in the specialized window. Counts are for the **headers** (or the `Ticket` value), not the string bytes they point at.

```go
// size_classes.go
package main

import (
    "fmt"
    "unsafe"
)

type Ticket struct {
    ID    int
    Table int
}

func class(n uintptr) string {
    switch {
    case n <= 8:
        return "SC1 (1-8)"
    case n <= 16:
        return "SC2 (9-16)"
    case n <= 24:
        return "SC3 (17-24)"
    case n <= 32:
        return "SC4 (25-32)"
    case n <= 48:
        return "SC5 (33-48)"
    case n <= 64:
        return "SC6 (49-64)"
    case n <= 80:
        return "SC7 (65-80)"
    default:
        return "above 80 (no size-specialized fast path in 1.27)"
    }
}

func main() {
    var s string
    var iface any
    var sl []int
    var t Ticket

    fmt.Printf("string header: %d bytes -> %s\n", unsafe.Sizeof(s), class(unsafe.Sizeof(s)))
    fmt.Printf("interface any: %d bytes -> %s\n", unsafe.Sizeof(iface), class(unsafe.Sizeof(
    fmt.Printf("slice header: %d bytes -> %s\n", unsafe.Sizeof(sl), class(unsafe.Sizeof(sl))
    fmt.Printf("Ticket value: %d bytes -> %s\n", unsafe.Sizeof(t), class(unsafe.Sizeof(t)))
}
```

Run:

```
go run size_classes.go
```

Output (amd64 / arm64 with 64-bit words):

```
string header: 16 bytes -> SC2 (9-16)
interface any: 16 bytes -> SC2 (9-16)
slice header: 24 bytes -> SC3 (17-24)
Ticket value: 16 bytes -> SC2 (9-16)
```

Those are exactly the sizes the Go blog calls out as the biggest winners. You did not pick magic numbers — the language's everyday headers already sit there.

Case 2: Escaping small structs still allocate once each

Save as `escape_small.go`. `newTicket` is `noinline` so the compiler cannot stack-allocate the `Ticket`. Ten thousand escaping pointers mean ten thousand heap objects (plus the slice growth for `held`).

```go
// escape_small.go
package main

import (
    "fmt"
    "runtime"
)

type Ticket struct {
    ID    int
    Table int
    Note  string
}

//go:noinline
func newTicket(id int) *Ticket {
    return &Ticket{ID: id, Table: id % 20, Note: "soup"}
}

func main() {
    runtime.GC()
    var before runtime.MemStats
    runtime.ReadMemStats(&before)

    held := make([]*Ticket, 0, 10_000)
    for i := range 10_000 {
        held = append(held, newTicket(i))
    }
```

```go
    var after runtime.MemStats
    runtime.ReadMemStats(&after)
    fmt.Printf("tickets: %d\n", len(held))
    fmt.Printf("mallocs: %d\n", after.Mallocs-before.Mallocs)
    fmt.Printf("heap objects gained: %d\n", after.HeapObjects-before.HeapObjects)
    _ = held
}
```

Run:

```go
go run escape_small.go
```

Output:

```
tickets: 10000
mallocs: 10001
heap objects gained: 10001
```

On Go 1.27 each of those small `Ticket` heap objects can take the specialized path when the size is known. The **count** does not drop — the **cost per malloc** does. Do not rewrite the loop to "help" the allocator; upgrade the toolchain.

Case 3: Known size vs dynamic length (same count, different call shape)

Save as `known_vs_dyn.go`. Both loops allocate 50000 times. The known-size path is the one the compiler can call specialized malloc for directly. The dynamic `make` still goes through `mallocgc` first.

```go
// known_vs_dyn.go
package main

import (
    "fmt"
    "runtime"
)

//go:noinline
func allocKnown() *[24]byte {
    var b [24]byte
    return &b
}

//go:noinline
func allocDyn(n int) []byte {
    return make([]byte, n)
}
```

```go
func count(fn func()) uint64 {
    runtime.GC()
    var before, after runtime.MemStats
    runtime.ReadMemStats(&before)
    for range 50_000 {
        fn()
    }
    runtime.ReadMemStats(&after)
    return after.Mallocs - before.Mallocs
}

func main() {
    fmt.Printf("known 24-byte mallocs: %d\n", count(func() { _ = allocKnown() }))
    fmt.Printf("dyn 24-byte mallocs: %d\n", count(func() { _ = allocDyn(24) }))
}
```

Run:

```
go run known_vs_dyn.go
```

Output:

```
known 24-byte mallocs: 50000
dyn 24-byte mallocs: 50000
```

Equal counts are the point. Size specialization is a **call-path** improvement, not an "allocate less" tip. Prefer clear code; let 1.27 cheapen the known-size cases.

Case 4: A/B the build flag on Go 1.27

Save as `alloc_bench_test.go`. Run the same benchmark twice on a Go **1.27** toolchain: once with the default allocator, once with specialization disabled. Compare `ns/op` yourself — do not treat a single laptop number as a promise.

```go
// alloc_bench_test.go
package allocbench

import "testing"

type Ticket struct {
    ID    int
    Table int
    Note  string
}

//go:noinline
```

```go
func newTicket(id int) *Ticket {
    return &Ticket{ID: id, Table: id % 20, Note: "soup"}
}

func BenchmarkNewTicket(b *testing.B) {
    b.ReportAllocs()
    var sink *Ticket
    for i := 0; i < b.N; i++ {
        sink = newTicket(i)
    }
    _ = sink
}
```

Run (needs Go 1.27+):

```bash
go test -bench=BenchmarkNewTicket -benchmem -count=5 ./...
GOEXPERIMENT=nosizespecializedmalloc go test -bench=BenchmarkNewTicket -benchmem -count=5 ./
```

Expect the default build to win on `ns/op` for this escaping small struct. Magnitudes vary by machine. If the disabled build is somehow faster and you can reproduce it, file an issue at go.dev/issue/new before shipping the opt-out.

Case 5: Confirm the toolchain, not a magic import

Save as `toolchain_hint.go`. There is no import that "turns on" size-specialized malloc. The feature is the compiler and runtime you build with.

```go
// toolchain_hint.go
package main

import (
    "fmt"
    "runtime"
)

func main() {
    fmt.Println("go:", runtime.Version())
    fmt.Println("Nothing to import. Build with Go 1.27+ for size-specialized malloc.")
    fmt.Println("Opt out only while debugging: GOEXPERIMENT=nosizespecializedmalloc")
}
```

Run:

```bash
go run toolchain_hint.go
```

On Go 1.27 you will see a `go1.27.x` version string and the two reminder lines. If `runtime.Version()` still prints 1.26 or older, you are not on the specialized allocator yet — upgrade the desk toolchain (and CI image) first.

The trap

Trap 1: Padding structs to "hit" a size class

Do not add dummy fields so a type lands in SC2 instead of SC3. You grow every value, confuse the next reader, and fight the compiler's own layout. Size specialization exists so **ordinary** layouts stay fast.

Trap 2: Micro-rewrites that chase malloc counts

Cutting allocations still matters when a profile says so (`sync.Pool`, `s = s[:0]`, fewer interfaces). That is a separate desk from this chapter. Size-specialized malloc makes the allocations you already do cheaper. It is not a reason to avoid clear escaping pointers.

Trap 3: Leaving `GOEXPERIMENT=nosizespecializedmalloc` in production scripts

The opt-out is for regression hunts. It is expected to vanish in Go 1.28. Do not bake it into Dockerfiles "just in case."

The boring rule

- On Go 1.27+, keep writing normal escaping code. Small heap allocs get faster without API changes.
- Upgrade compilers and CI images before inventing allocation micro-tricks.
- Use `GOEXPERIMENT=nosizespecializedmalloc` only to A/B a suspected regression, then remove it.
- Trust size classes for headers you already use (string, `any`, slice) — do not invent padding.
- Still measure with `pprof` / benchmarks when allocation volume is the real cost; cheaper malloc does not fix an unbounded leak.

Try this

1. Add a `Note string` field to `Ticket` in `size_classes.go` and re-run. Which size class does the value move into?
2. On Go 1.27, run Case 4's two `go test -bench` commands and note which `ns/op` is lower.
3. In `known_vs_dyn.go`, change `allocDyn(24)` to `allocDyn(200)`. Does the malloc **count** change? (It should not — only the size class does.)

4. Search your deploy scripts for `GOEXPERIMENT=nosizespecializedmalloc` and delete any leftover copies after you finish an A/B.

Measuring Size-Specialized Malloc with GOEXPERIMENT

Measuring Size-Specialized Malloc with GOEXPERIMENT

Thesis

Go **1.27** already documents size-specialized `mallocgc` for known sizes 80 bytes. This chapter is the **measurement** companion: the same microbenchmarks with the default build versus `GOEXPERIMENT=nosizespecializedmalloc`, so you can see the win on your CPU instead of trusting the blog's "up to ~30%" alone. The neighbouring chapter maps size classes; here we only time them.

Mental model

```
  compiler knows size  80B

          default 1.27 → call mallocgcSmall…SCn directly
          GOEXPERIMENT=nosizespecializedmalloc → generic mallocgc path
```

Opt-out is a bisect tool. The Go team expects the experiment knob to go away in **1.28**.

Worked examples

Verified with **Go 1.27.2** linux/amd64 on an Intel Xeon guest.

Case 1: tiny struct vs 80B vs 96B slices

```go
// main_test.go
package main

import "testing"

type tiny struct{ a, b, c, d int64 } // 32 bytes

func BenchmarkTinyAlloc(b *testing.B) {
    b.ReportAllocs()
    var sink *tiny
```

```go
    for i := 0; i < b.N; i++ {
        sink = &tiny{1, 2, 3, 4}
    }
    _ = sink
}

func BenchmarkSlice80(b *testing.B) {
    b.ReportAllocs()
    var sink []byte
    for i := 0; i < b.N; i++ {
        sink = make([]byte, 80)
    }
    _ = sink
}

func BenchmarkSlice96(b *testing.B) {
    b.ReportAllocs()
    var sink []byte
    for i := 0; i < b.N; i++ {
        sink = make([]byte, 96)
    }
    _ = sink
}
```

```go
go test -bench=. -benchmem -count=5
GOEXPERIMENT=nosizespecializedmalloc go test -bench=. -benchmem -count=5
```

Representative lines from this run (5 counts each; lower ns/op is better):

Benchmark	Default (ns/op)	nosizespecialized (ns/op)
TinyAlloc (32 B)	~12.6–13.8	~17.6–18.4
Slice80	~25–27	~28–33
Slice96 (above 80)	~30–34	~31–32

The **32-byte** path shows the clearest gap (~30% here). **96 bytes** sits outside the specialized window and barely moves — a useful control.

Trap

Declaring victory from one microbenchmark. Size specialization is a small overall win (~1% on alloc-heavy programs per the release notes). If your profile is dominated by large buffers or stack allocation, you will see noise, not a rewrite-worthy speedup.

Boring rule

Stay on Go 1.27+ and let the compiler specialize. Reach for `GOEXPERIMENT=nosizespecializedmalloc` only when bisecting a suspected allocator regression, then file an issue if you need the knob to stay.

Try this

1. Add a `BenchmarkMakeN` where `n` is a runtime variable and compare.
2. Run with `-benchtime=2s` on a quiet machine and compute medians.
3. Cross-check against `go tool pprof` alloc profiles on a real service.

Sources

- Go 1.27 release notes — runtime
- Size-specialized malloc blog (Go 1.27 allocator posts)
- Sibling chapter `25d-size-specialized-malloc.qmd` (class table)

Generics

Designing with Interfaces

Designing with Interfaces

Overview

Go interfaces enable polymorphism through implicit satisfaction—types implement interfaces by having matching methods, without explicit declarations.

Interface Basics

```go
type Reader interface {
    Read(p []byte) (n int, err error)
}

// Any type with Read method satisfies Reader
type File struct { /* ... */ }
func (f *File) Read(p []byte) (int, error) { /* ... */ }

type Buffer struct { /* ... */ }
func (b *Buffer) Read(p []byte) (int, error) { /* ... */ }
```

Implicit Satisfaction

```go
type Stringer interface {
    String() string
}

type Person struct {
    Name string
}

// Person implicitly implements Stringer
func (p Person) String() string {
    return p.Name
}

var s Stringer = Person{Name: "Alice"}  // Works!
```

The Empty Interface

```go
interface{}  // Matches any type
any          // Alias (Go 1.18+)

func print(v any) {
    fmt.Println(v)
}

print(42)
print("hello")
print(true)
```

Interface Values

An interface value holds a (type, value) pair:

```go
var r io.Reader
r = os.Stdin       // (type: *os.File, value: stdin)
r = &bytes.Buffer{}  // (type: *bytes.Buffer, value: buf)
```

nil Interface vs nil Value

```go
var r io.Reader    // nil interface (no type, no value)
var f *os.File     // nil pointer

r = f              // Interface holds (*os.File, nil)
r == nil           // false! Has type, value is nil
```

Common Patterns

Accept Interface, Return Concrete

```go
// Accept interface
func Process(r io.Reader) error { }

// Return concrete type
func NewBuffer() *bytes.Buffer {
    return &bytes.Buffer{}
}
```

Small Interfaces

```go
type Reader interface {
    Read([]byte) (int, error)
}

type Writer interface {
    Write([]byte) (int, error)
}

type ReadWriter interface {
    Reader
    Writer
}
```

Assert Interface Compliance

```go
var _ io.Reader = (*MyReader)(nil)  // Compile-time check
```

Summary

Concept	Description
Implicit	No `implements` keyword
`any`/`interface{}`	Matches all types
Small interfaces	Prefer 1-3 methods
Accept interface	Flexible function parameters

Related Topics

- Interface Best Practices
- Type Assertions

More examples

Example: implicit satisfaction

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

type Speaker interface {
    Speak() string
}

type Dog struct{ Name string }

func (d Dog) Speak() string { return d.Name + " says woof" }

func greet(s Speaker) {
    fmt.Println(s.Speak())
}

func main() {
    // Dog never writes "implements Speaker".
    greet(Dog{Name: "Rex"})
}
```

Expected:

```
Rex says woof
```

Example: empty interface / any

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

func describe(v any) {
    fmt.Printf("%T -> %v\n", v, v)
}

func main() {
    describe(42)
    describe("hi")
    describe(true)
}
```

Expected:

```
int -> 42
string -> hi
bool -> true
```

Runnable example

Save as `main.go`. Then:

```
go mod init example
go run .
```

```go
package main

import (
    "fmt"
    "io"
    "strings"
)

type Stringer interface {
    String() string
}

type Person struct{ Name string }

func (p Person) String() string { return p.Name }

// Implicit satisfaction: no "implements" keyword.
var _ Stringer = Person{}

type ByteCounter int

func (c *ByteCounter) Write(p []byte) (int, error) {
    *c += ByteCounter(len(p))
    return len(p), nil
}

func dump(r io.Reader) (string, error) {
    b, err := io.ReadAll(r)
    return string(b), err
}

func main() {
    var s Stringer = Person{Name: "Alice"}
    fmt.Println("Stringer:", s.String())
```

```go
    // Interface value is a (type, value) pair.
    var r io.Reader = strings.NewReader("hello interfaces")
    text, err := dump(r)
    fmt.Println("from Reader:", text, err)

    var c ByteCounter
    // *ByteCounter satisfies io.Writer.
    _, _ = c.Write([]byte("abcd"))
    _, _ = fmt.Fprintf(&c, "ef")
    fmt.Println("bytes written:", int(c))

    // nil interface vs typed-nil inside interface
    var w io.Writer
    fmt.Println("nil interface:", w == nil)
    var bc *ByteCounter
    w = bc
    fmt.Println("typed-nil interface equals nil?", w == nil)
}
```

Expected output:

```
Stringer: Alice
from Reader: hello interfaces <nil>
bytes written: 6
nil interface: true
typed-nil interface equals nil? false
```

What to notice: Types satisfy interfaces by method set alone. An interface holding a typed nil is not equal to a true nil interface — a common source of surprising error checks.

Try next: Assert compile-time compliance with `var _ io.Writer = (*ByteCounter)(nil)`; pass both `strings.Reader` and `*ByteCounter` into one function that only accepts small interfaces.

Interface Best Practices

Interface Best Practices

Overview

Well-designed interfaces make Go code flexible, testable, and maintainable.

Keep Interfaces Small

```go
// Good: small, focused
type Reader interface {
    Read([]byte) (int, error)
}

// Avoid: too many methods
type DataProcessor interface {
    Read([]byte) (int, error)
    Write([]byte) (int, error)
    Close() error
    Flush() error
    Seek(int64, int) (int64, error)
    // ...10 more methods
}
```

Define Interfaces at Consumer

```go
// In the package that USES the interface
package myapp

type Storage interface {
    Save(data []byte) error
}

func Process(s Storage) {
    s.Save([]byte("data"))
}

// NOT in the package that implements it
```

Accept Interfaces, Return Structs

```go
// Accept interface (flexible)
func ParseJSON(r io.Reader) (*Config, error)

// Return concrete type (clear)
func NewFileReader(path string) *FileReader
```

Interface Composition

```go
type Reader interface {
    Read([]byte) (int, error)
}

type Closer interface {
    Close() error
}

type ReadCloser interface {
    Reader
    Closer
}
```

Avoid Interface Pollution

```go
// Don't create interfaces for single implementations
// Just use the concrete type

// Unnecessary
type UserService interface {
    GetUser(id int) *User
}
type userServiceImpl struct{}

// Better: just use the struct directly
type UserService struct{}
func (s *UserService) GetUser(id int) *User
```

Testing with Interfaces

```go
type EmailSender interface {
    Send(to, subject, body string) error
}

// Production
type SMTPSender struct{}
func (s *SMTPSender) Send(to, subject, body string) error

// Test mock
type MockSender struct {
    SentEmails []string
}
func (m *MockSender) Send(to, subject, body string) error {
    m.SentEmails = append(m.SentEmails, to)
    return nil
}
```

Summary

Practice	Reason
Small interfaces	Easy to implement and mock
Consumer-defined	Loose coupling
Accept interfaces	Flexibility
Return structs	Clarity
Compose	Build from small pieces

Related Topics

- Designing with Interfaces
- Testing Fundamentals

More examples

Example: accept interfaces, return structs

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

type Writer interface {
    Write([]byte) (int, error)
}

type memWriter struct{ n int }

func (m *memWriter) Write(p []byte) (int, error) {
    m.n += len(p)
    return len(p), nil
}

// NewMemWriter returns a concrete type; callers depend on Writer where needed.
func NewMemWriter() *memWriter { return &memWriter{} }

func writeAll(w Writer, msg string) error {
    _, err := w.Write([]byte(msg))
    return err
}

func main() {
    mw := NewMemWriter()
    _ = writeAll(mw, "hello")
    _ = writeAll(mw, "!")
    fmt.Println("bytes written:", mw.n)
}
```

Expected:

```
bytes written: 6
```

Example: small interfaces compose

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

type Reader interface{ Read() string }
type Closer interface{ Close() error }
type ReadCloser interface {
    Reader
```

```
    Closer
}

type src struct{ closed bool }

func (s *src) Read() string {
    if s.closed {
        return ""
    }
    return "data"
}
func (s *src) Close() error {
    s.closed = true
    return nil
}

func drain(rc ReadCloser) {
    fmt.Println("read:", rc.Read())
    _ = rc.Close()
    fmt.Println("after close:", rc.Read())
}

func main() {
    drain(&src{})
}
```

Expected:

```
read: data
after close:
```

Runnable example

Save as `main.go`. Then:

```
go mod init example
go run .
```

```
package main

import (
    "fmt"
    "io"
    "strings"
)
```

```go
// Small, consumer-defined interfaces.
type Storage interface {
    Save(data []byte) error
}

type EmailSender interface {
    Send(to, subject, body string) error
}

// Concrete production types - no interface required at definition site.
type MemoryStorage struct {
    items []string
}

func (m *MemoryStorage) Save(data []byte) error {
    m.items = append(m.items, string(data))
    return nil
}

type MockSender struct {
    Sent []string
}

func (m *MockSender) Send(to, subject, body string) error {
    m.Sent = append(m.Sent, to+":"+subject)
    return nil
}

// Accept interfaces, return concrete values.
func ParseConfig(r io.Reader) (string, error) {
    b, err := io.ReadAll(r)
    if err != nil {
        return "", err
    }
    return strings.TrimSpace(string(b)), nil
}

func Process(s Storage, msg string) error {
    return s.Save([]byte(msg))
}

func Notify(sender EmailSender, user string) error {
    return sender.Send(user, "welcome", "hello")
}

func main() {
```

```
    cfg, err := ParseConfig(strings.NewReader("  port=8080  "))
    fmt.Println("config:", cfg, err)

    store := &MemoryStorage{}
    _ = Process(store, "event-1")
    _ = Process(store, "event-2")
    fmt.Println("stored:", store.items)

    mock := &MockSender{}
    _ = Notify(mock, "ada@example.com")
    fmt.Println("sent:", mock.Sent)
}
```

Expected output:

```
config: port=8080 <nil>
stored: [event-1 event-2]
sent: [ada@example.com:welcome]
```

What to notice: Call sites depend on tiny interfaces (`Storage`, `EmailSender`, `io.Reader`); implementations stay concrete structs you can construct and inspect in tests without a framework.

Try next: Compose `type ReadSaver interface { io.Reader; Storage }` or inject a failing `EmailSender` that returns an error and handle it in `Notify`.

Type Assertions and Reflection Boundaries

Type Assertions and Reflection Boundaries

Overview

Type assertions extract concrete types from interfaces. Reflection provides runtime type inspection but should be used sparingly.

Type Assertions

```go
var i interface{} = "hello"

s := i.(string)        // Panic if wrong type
s, ok := i.(string)    // Safe: ok is false if wrong type

if s, ok := i.(string); ok {
    fmt.Println(s)
}
```

Type Switches

```go
func describe(i interface{}) {
    switch v := i.(type) {
    case int:
        fmt.Println("int:", v*2)
    case string:
        fmt.Println("string:", len(v))
    case bool:
        fmt.Println("bool:", !v)
    default:
        fmt.Printf("unknown: %T\n", v)
    }
}
```

The reflect Package

```go
import "reflect"

v := 42
t := reflect.TypeOf(v)   // int
val := reflect.ValueOf(v)  // 42

fmt.Println(t.Name())      // "int"
fmt.Println(t.Kind())      // int
fmt.Println(val.Int())     // 42
```

Struct Reflection

```go
type User struct {
    Name string `json:"name"`
    Age  int    `json:"age"`
}

u := User{Name: "Alice", Age: 30}
t := reflect.TypeOf(u)

for i := 0; i < t.NumField(); i++ {
    field := t.Field(i)
    fmt.Printf("%s: %s\n", field.Name, field.Tag.Get("json"))
}
```

When to Use Reflection

Good uses: - Serialization (JSON, XML) - ORM field mapping - Generic testing utilities - Dependency injection

Avoid for: - Performance-critical code - Simple type checking - Normal business logic

Performance Cost

```go
// Direct access: ~1ns
user.Name

// Reflection: ~100ns+
reflect.ValueOf(user).FieldByName("Name").String()
```

Prefer Generics Over Reflection

```go
// Go 1.18+: Use generics instead of reflect
func PrintSlice[T any](s []T) {
    for _, v := range s {
        fmt.Println(v)
    }
}
```

Summary

Approach	Use Case
Type assertion	Extract known concrete type
Type switch	Handle multiple types
reflect	Runtime introspection (last resort)
Generics	Compile-time type safety

Related Topics

- Why Generics Matter
- Encoding and Serialization

More examples

Example: safe comma-ok assertion

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

func asInt(v any) {
    if n, ok := v.(int); ok {
        fmt.Println("int:", n)
        return
    }
    fmt.Printf("not int: %T\n", v)
}

func main() {
```

```
    asInt(7)
    asInt("nope")
}
```

Expected:

```
int: 7
not int: string
```

Example: type switch

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```
package main

import "fmt"

func classify(v any) {
    switch x := v.(type) {
    case int:
        fmt.Println("int", x*2)
    case string:
        fmt.Println("string len", len(x))
    default:
        fmt.Printf("other %T\n", x)
    }
}

func main() {
    classify(3)
    classify("go")
    classify(true)
}
```

Expected:

```
int 6
string len 2
other bool
```

Runnable example

Save as `main.go`. Then:

```
go mod init example
go run .
```

```go
package main

import (
    "fmt"
    "reflect"
)

func describe(i any) {
    switch v := i.(type) {
    case int:
        fmt.Println("int:", v*2)
    case string:
        fmt.Println("string len:", len(v))
    case bool:
        fmt.Println("bool not:", !v)
    default:
        fmt.Printf("unknown: %T\n", v)
    }
}

type User struct {
    Name string `json:"name"`
    Age  int    `json:"age"`
}

func main() {
    var i any = "hello"
    s, ok := i.(string)
    fmt.Println("assert string:", s, ok)
    n, ok := i.(int)
    fmt.Println("assert int:", n, ok)

    describe(21)
    describe("go")
    describe(true)
    describe(3.14)

    // Reflection: useful for tags/serialization; prefer assertions/generics elsewhere.
    u := User{Name: "Ada", Age: 36}
    t := reflect.TypeOf(u)
    for j := 0; j < t.NumField(); j++ {
        f := t.Field(j)
        fmt.Printf("field %s json=%q\n", f.Name, f.Tag.Get("json"))
```

```
    }
}
```

Expected output:

```
assert string: hello true
assert int: 0 false
int: 42
string len: 2
bool not: false
unknown: float64
field Name json="name"
field Age json="age"
```

What to notice: The comma-ok form never panics; a type switch is clearer than a chain of assertions. Reflection can read struct tags but costs more and loses static checking.

Try next: Replace the default branch with a generic `PrintSlice[T any]` for slice values, or make a failing assertion `i.(int)` without `ok` and observe the panic.

Why Generics Matter

Why Generics Matter

Overview

Generics (type parameters), introduced in Go 1.18, allow writing functions and types that work with multiple types while maintaining type safety.

The Problem Before Generics

```go
// Without generics: duplicate code or use interface{}
func MinInt(a, b int) int {
    if a < b { return a }
    return b
}

func MinFloat(a, b float64) float64 {
    if a < b { return a }
    return b
}

// Or lose type safety
func Min(a, b interface{}) interface{} {
    // Requires type assertions, runtime checks
}
```

With Generics

```go
func Min[T constraints.Ordered](a, b T) T {
    if a < b {
        return a
    }
    return b
}

Min(1, 2)         // int
Min(1.5, 2.5)     // float64
Min("a", "b")     // string
```

Type Parameters

```go
func Print[T any](v T) {
    fmt.Println(v)
}

// Multiple type parameters
func Pair[K, V any](k K, v V) (K, V) {
    return k, v
}
```

Constraints

```go
import "golang.org/x/exp/constraints"

// Built-in constraints
any           // No requirements
comparable    // Supports == and !=

// From constraints package
constraints.Ordered    // Supports < > <= >=
constraints.Integer    // Int types
constraints.Float      // Float types
```

Custom Constraints

```go
type Number interface {
    int | int32 | int64 | float32 | float64
}

func Sum[T Number](nums []T) T {
    var total T
    for _, n := range nums {
        total += n
    }
    return total
}
```

Generic Type Aliases (Go 1.24+)

Go 1.24 introduced support for generic type aliases, allowing you to create aliases for generic types.

```go
type Set[T comparable] map[T]struct{}

// Type alias (Go 1.24+)
type IntSet = Set[int]
type AnySet[T comparable] = Set[T]
```

This is particularly useful for refactoring and maintaining compatibility while migrating to generic implementations.

Self-Referential Generics (Go 1.26+)

Go 1.26 lifted the restriction that prevented generic type parameters from referring to their enclosing type definition in constraint declarations. This allows fluent builder interfaces and self-referential tree nodes with static type safety:

```go
// Self-referential constraint allowed in Go 1.26+
type Node[T Node[T]] interface {
    Value() string
    Next() T
}

type MyNode struct {
    val  string
    next *MyNode
}

func (m *MyNode) Value() string { return m.val }
func (m *MyNode) Next() *MyNode { return m.next }
```

Generic Methods (Go 1.27+)

A method declaration may declare its own type parameters. Use this when the operation belongs on a concrete type but the argument/result type varies:

```go
func (r *rand.Rand) N[Int intType](n Int) Int
```

That is **not** the same as a method on a generic type (`func (s *Stack[T]) Push(T)`), which has been valid since Go 1.18. Interface methods still cannot be generic, and a generic method cannot satisfy an interface method.

Function type inference is also broader in 1.27: a generic function can be assigned, converted, sent on a channel, or placed in a composite literal without spelling the type arguments when the destination type is a matching function type. See Generic Functions.

Benefits

1. **Type safety**: Errors caught at compile time
2. **No code duplication**: One implementation for many types
3. **Performance**: No interface boxing/unboxing
4. **Better documentation**: Types are explicit

Summary

Before	After
Code duplication	Single generic function
`interface{}`	Type parameters
Runtime errors	Compile-time errors
Type assertions	Direct usage

Related Topics

- Generic Functions
- Generic Types

More examples

Example: before generics—duplication

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

func maxInt(a, b int) int {
    if a > b {
        return a
    }
    return b
}

func maxFloat(a, b float64) float64 {
    if a > b {
        return a
    }
    return b
```

```go
}

func main() {
    fmt.Println(maxInt(3, 7))
    fmt.Println(maxFloat(3.5, 2.1))
}
```

Expected:

```
7
3.5
```

Example: one generic Max

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import (
    "cmp"
    "fmt"
)

func Max[T cmp.Ordered](a, b T) T {
    if a > b {
        return a
    }
    return b
}

func main() {
    fmt.Println(Max(3, 7))
    fmt.Println(Max(3.5, 2.1))
    fmt.Println(Max("a", "z"))
}
```

Expected:

```
7
3.5
z
```

Runnable example

Save as `main.go`. Then:

```
go mod init example
go run .
```

```go
package main

import (
    "cmp"
    "fmt"
)

// cmp.Ordered covers types that support < (stdlib; no x/exp needed).
func Min[T cmp.Ordered](a, b T) T {
    if a < b {
        return a
    }
    return b
}

func Pair[K, V any](k K, v V) (K, V) {
    return k, v
}

// Custom constraint with a type set (stdlib only).
type Number interface {
    ~int | ~int64 | ~float64
}

func Sum[T Number](nums []T) T {
    var total T
    for _, n := range nums {
        total += n
    }
    return total
}

func main() {
    fmt.Println("Min int:", Min(3, 1))
    fmt.Println("Min float:", Min(2.5, 2.7))
    fmt.Println("Min string:", Min("go", "ai"))

    k, v := Pair("age", 30)
    fmt.Printf("Pair: %s=%d\n", k, v)

    fmt.Println("Sum ints:", Sum([]int{1, 2, 3}))
    fmt.Println("Sum floats:", Sum([]float64{0.5, 1.5}))
}
```

Expected output:

```
Min int: 1
Min float: 2.5
Min string: ai
Pair: age=30
Sum ints: 6
Sum floats: 2
```

What to notice: One `Min` works for any ordered type with compile-time safety; `cmp.Ordered` lives in the standard library. Type sets let you restrict `Sum` to numeric-ish types without `interface{}` assertions.

Try next: Call `Min` on a struct (should fail to compile); add `~string` to a constraint and write `Join[T ~string](parts []T) T`.

Generic Functions

Generic Functions

Overview

Generic functions use type parameters to work with multiple types while maintaining type safety.

Basic Syntax

```go
func FunctionName[T constraint](params) returnType {
    // body
}
```

Examples

Map

```go
func Map[T, U any](slice []T, fn func(T) U) []U {
    result := make([]U, len(slice))
    for i, v := range slice {
        result[i] = fn(v)
    }
    return result
}

doubled := Map([]int{1, 2, 3}, func(n int) int { return n * 2 })
// [2, 4, 6]
```

Filter

```go
func Filter[T any](slice []T, fn func(T) bool) []T {
    var result []T
    for _, v := range slice {
        if fn(v) {
            result = append(result, v)
        }
```

```go
    }
    return result
}

evens := Filter([]int{1, 2, 3, 4}, func(n int) bool { return n%2 == 0 })
// [2, 4]
```

Find

```go
func Find[T any](slice []T, fn func(T) bool) (T, bool) {
    for _, v := range slice {
        if fn(v) {
            return v, true
        }
    }
    var zero T
    return zero, false
}
```

Contains

```go
func Contains[T comparable](slice []T, target T) bool {
    for _, v := range slice {
        if v == target {
            return true
        }
    }
    return false
}
```

Keys/Values

```go
func Keys[K comparable, V any](m map[K]V) []K {
    keys := make([]K, 0, len(m))
    for k := range m {
        keys = append(keys, k)
    }
    return keys
}

func Values[K comparable, V any](m map[K]V) []V {
    vals := make([]V, 0, len(m))
```

```go
    for _, v := range m {
        vals = append(vals, v)
    }
    return vals
}
```

Type Inference

Go infers type arguments when possible:

```go
// Explicit
result := Map[int, int](nums, double)

// Inferred (preferred)
result := Map(nums, double)
```

Assignment contexts (Go 1.27+)

Inference now applies in every assignment-shaped context: composite literals, conversions, and channel sends, not only direct calls:

```go
func GenericFormatter[T any](v T) string {
    return fmt.Sprintf("value: %v", v)
}

type IntFormatter func(int) string

formatters := []IntFormatter{GenericFormatter} // T = int
fn := IntFormatter(GenericFormatter)
ch := make(chan IntFormatter, 1)
ch <- GenericFormatter
```

Summary

Pattern	Signature
Transform	`func Map[T, U any]([]T, func(T) U) []U`
Filter	`func Filter[T any]([]T, func(T) bool) []T`
Find	`func Find[T any]([]T, func(T) bool) (T, bool)`
Contains	`func Contains[T comparable]([]T, T) bool`

Related Topics

- Generic Types
- Idiomatic Generics

More examples

Example: Map slice transform

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

func Map[T, U any](in []T, f func(T) U) []U {
    out := make([]U, len(in))
    for i, v := range in {
        out[i] = f(v)
    }
    return out
}

func main() {
    nums := []int{1, 2, 3}
    strs := Map(nums, func(n int) string {
        return fmt.Sprintf("#%d", n)
    })
    fmt.Println(strs)
}
```

Expected:

```
[#1 #2 #3]
```

Example: Filter with predicate

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

func Filter[T any](in []T, keep func(T) bool) []T {
    var out []T
```

```go
    for _, v := range in {
        if keep(v) {
            out = append(out, v)
        }
    }
    return out
}

func main() {
    nums := []int{1, 2, 3, 4, 5}
    evens := Filter(nums, func(n int) bool { return n%2 == 0 })
    fmt.Println(evens)
}
```

Expected:

```
[2 4]
```

Runnable example

Save as `main.go`. Then:

```
go mod init example
go run .
```

```go
package main

import (
    "fmt"
    "slices"
)

func Map[T, U any](in []T, fn func(T) U) []U {
    out := make([]U, len(in))
    for i, v := range in {
        out[i] = fn(v)
    }
    return out
}

func Filter[T any](in []T, keep func(T) bool) []T {
    var out []T
    for _, v := range in {
        if keep(v) {
            out = append(out, v)
```

```go
        }
    }
    return out
}

func Contains[T comparable](in []T, target T) bool {
    for _, v := range in {
        if v == target {
            return true
        }
    }
    return false
}

func Keys[K comparable, V any](m map[K]V) []K {
    keys := make([]K, 0, len(m))
    for k := range m {
        keys = append(keys, k)
    }
    return keys
}

func main() {
    nums := []int{1, 2, 3, 4}
    doubled := Map(nums, func(n int) int { return n * 2 })
    fmt.Println("Map:", doubled)

    evens := Filter(nums, func(n int) bool { return n%2 == 0 })
    fmt.Println("Filter:", evens)

    fmt.Println("Contains 3:", Contains(nums, 3))
    fmt.Println("Contains 9:", Contains(nums, 9))

    // Prefer stdlib helpers when they already exist.
    fmt.Println("slices.Contains 3:", slices.Contains(nums, 3))
    fmt.Println("slices.Index 4:", slices.Index(nums, 4))

    m := map[string]int{"a": 1, "b": 2}
    keys := Keys(m)
    slices.Sort(keys) // map iteration order is random
    fmt.Println("Keys sorted:", keys)
}
```

Expected output:

```
Map: [2 4 6 8]
Filter: [2 4]
Contains 3: true
Contains 9: false
slices.Contains 3: true
slices.Index 4: 3
Keys sorted: [a b]
```

What to notice: Type parameters preserve element types end-to-end (`[]int` in, `[]int` out). The stdlib `slices` package already covers many of these helpers — write your own when the transform is domain-specific.

Try next: Add `Reduce[T any](in []T, init T, fn func(T, T) T) T`; map `[]string` to `[]int` lengths with `Map`.

Generic Types

Generic Types

Overview

Generic types let you create data structures that work with any type while maintaining type safety.

Basic Syntax

```go
type TypeName[T constraint] struct {
    // fields using T
}
```

Stack

```go
type Stack[T any] struct {
    items []T
}

func (s *Stack[T]) Push(item T) {
    s.items = append(s.items, item)
}

func (s *Stack[T]) Pop() (T, bool) {
    if len(s.items) == 0 {
        var zero T
        return zero, false
    }
    item := s.items[len(s.items)-1]
    s.items = s.items[:len(s.items)-1]
    return item, true
}

// Usage
stack := &Stack[int]{}
stack.Push(1)
stack.Push(2)
v, _ := stack.Pop()  // 2
```

Set

```go
type Set[T comparable] map[T]struct{}

func NewSet[T comparable]() Set[T] {
    return make(Set[T])
}

func (s Set[T]) Add(item T) {
    s[item] = struct{}{}
}

func (s Set[T]) Contains(item T) bool {
    _, ok := s[item]
    return ok
}

func (s Set[T]) Remove(item T) {
    delete(s, item)
}

// Usage
set := NewSet[string]()
set.Add("a")
set.Contains("a")  // true
```

Pair

```go
type Pair[T, U any] struct {
    First  T
    Second U
}

func NewPair[T, U any](first T, second U) Pair[T, U] {
    return Pair[T, U]{first, second}
}

p := NewPair("age", 30)
// Pair[string, int]{First: "age", Second: 30}
```

Result (Option Pattern)

```go
type Result[T any] struct {
    value T
    err   error
}

func Ok[T any](v T) Result[T] {
    return Result[T]{value: v}
}

func Err[T any](err error) Result[T] {
    return Result[T]{err: err}
}

func (r Result[T]) Unwrap() (T, error) {
    return r.value, r.err
}
```

LinkedList

```go
type Node[T any] struct {
    Value T
    Next  *Node[T]
}

type LinkedList[T any] struct {
    Head *Node[T]
}

func (l *LinkedList[T]) Add(v T) {
    node := &Node[T]{Value: v, Next: l.Head}
    l.Head = node
}
```

These methods use the **type's** type parameter `T`. Go 1.27 also allows a method to declare **additional** type parameters of its own (`func (s *Store) Put[U any](v U)`). See Methods and Receivers.

Summary

Type	Purpose
`Stack[T]`	LIFO collection

Type	Purpose
`Set[T]`	Unique elements
`Pair[T, U]`	Key-value tuple
`Result[T]`	Error handling

Related Topics

- Generic Functions
- Idiomatic Generics

More examples

Example: generic Stack

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

type Stack[T any] struct {
    items []T
}

func (s *Stack[T]) Push(v T) { s.items = append(s.items, v) }

func (s *Stack[T]) Pop() (T, bool) {
    var zero T
    if len(s.items) == 0 {
        return zero, false
    }
    i := len(s.items) - 1
    v := s.items[i]
    s.items = s.items[:i]
    return v, true
}

func main() {
    var s Stack[string]
    s.Push("a")
    s.Push("b")
    v, ok := s.Pop()
    fmt.Println(v, ok)
```

```
    v, ok = s.Pop()
    fmt.Println(v, ok)
    v, ok = s.Pop()
    fmt.Println(v, ok)
}
```

Expected:

```
b true
a true
 false
```

Example: generic Pair

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```
package main

import "fmt"

type Pair[A, B any] struct {
    First  A
    Second B
}

func main() {
    p := Pair[string, int]{First: "age", Second: 42}
    fmt.Printf("%s=%d\n", p.First, p.Second)
    q := Pair[int, bool]{First: 1, Second: true}
    fmt.Println(q.First, q.Second)
}
```

Expected:

```
age=42
1 true
```

Runnable example

Save as `main.go`. Then:

```
go mod init example
go run .
```

```go
package main

import "fmt"

type Stack[T any] struct {
    items []T
}

func (s *Stack[T]) Push(item T) {
    s.items = append(s.items, item)
}

func (s *Stack[T]) Pop() (T, bool) {
    if len(s.items) == 0 {
        var zero T
        return zero, false
    }
    i := len(s.items) - 1
    item := s.items[i]
    s.items = s.items[:i]
    return item, true
}

type Set[T comparable] map[T]struct{}

func NewSet[T comparable]() Set[T] {
    return make(Set[T])
}

func (s Set[T]) Add(item T)      { s[item] = struct{}{} }
func (s Set[T]) Contains(item T) bool {
    _, ok := s[item]
    return ok
}

type Pair[T, U any] struct {
    First  T
    Second U
}

func main() {
    st := &Stack[string]{}
    st.Push("a")
    st.Push("b")
    top, ok := st.Pop()
    fmt.Println("stack pop:", top, ok)
```

```go
    set := NewSet[int]()
    set.Add(1)
    set.Add(1)
    set.Add(2)
    fmt.Println("set has 1:", set.Contains(1))
    fmt.Println("set has 3:", set.Contains(3))
    fmt.Println("set size:", len(set))

    p := Pair[string, int]{First: "age", Second: 30}
    fmt.Printf("pair: %s=%d\n", p.First, p.Second)
}
```

Expected output:

```
stack pop: b true
set has 1: true
set has 3: false
set size: 2
pair: age=30
```

What to notice: `Stack[T]` and `Set[T]` keep a single implementation while the compiler enforces element types. Empty `Pop` returns the type's zero value plus `false` — a common generic API pattern.

Try next: Add `Peek() (T, bool)` and `Size() int` to `Stack`; build a `Queue[T]` with head/tail indices.

Generic Methods (Go 1.27)

Generic Methods (Go 1.27)

Thesis

Since Go 1.18 a method could *use* its receiver's type parameters (`func (s *Stack[T]) Push(v T)`) but could not *declare* new ones. Every operation that needed a second type, such as turning a `Box[int]` into a `Box[string]`, had to become a package-level function. Go 1.27 removes that restriction: a concrete method may declare its own type parameter list.

The feature is deliberately narrow. Interfaces still cannot declare generic methods, and a generic method never satisfies an interface method. Generic methods are a **namespacing and ergonomics** tool, not a new kind of polymorphism. This chapter shows exactly what compiles, what the compiler says when it does not, and how to decide between a generic method and a generic function.

The short tour in Methods and Receivers introduces the syntax. This chapter is the deep version.

Verified with **Go 1.27.2** (`linux/amd64`) on 2026-10-11. Primary sources: Go 1.27 release notes, Generic Methods (Go blog), proposal #77273.

Learning goals

- Declare a method with its own type parameters on generic and non-generic receivers
- Call it with inference, with explicit instantiation, as a method value and as a method expression
- Read the five compiler errors you will hit first
- Explain why generic methods cannot satisfy interfaces, and design around it
- Choose between a generic method and a generic function

Every listing in this chapter

Each listing is a complete `main.go` in its own directory with this `go.mod`:

```
module example.com/<dir>

go 1.27
```

Run it with `go run .`. The `go 1.27` line matters, as the next section shows.

Case 1: the canonical `Map`

```go
// main.go
package main

import (
    "fmt"
    "strconv"
)

type Box[T any] struct {
    Value T
}

// Map declares its own type parameter U, on top of the receiver's T.
func (b Box[T]) Map[U any](f func(T) U) Box[U] {
    return Box[U]{Value: f(b.Value)}
}

func main() {
    n := Box[int]{Value: 42}

    s := n.Map(strconv.Itoa) // U inferred as string
    q := n.Map[float64](func(v int) float64 { return float64(v) / 8 })
    fmt.Printf("%T %q\n", s, s.Value)
    fmt.Printf("%T %v\n", q, q.Value)

    // Calls chain, and each link can change the type.
    l := n.Map(strconv.Itoa).Map(func(s string) int { return len(s) })
    fmt.Printf("%T %v\n", l, l.Value)
}
```

```go
go run .
```

```
main.Box[string] "42"
main.Box[float64] 5.25
main.Box[int] 2
```

Three things to notice:

1. The receiver list `Box[T]` names the type's parameter. The method's own list `[U any]` comes **after the method name**, exactly where a function's list goes.
2. `U` is inferred from the argument (`strconv.Itoa` is `func(int) string`), so most calls need no brackets.
3. Explicit instantiation is `n.Map[float64](...)`: the type argument attaches to the method, not to the receiver.

The chain on the last line was the motivating case for the proposal. Before 1.27 it read inside-out as `Map(Map(n, strconv.Itoa), lenFn)`.

Case 2: the `go` directive gates the feature

Copy the same file into a module whose `go.mod` says `go 1.26` and build with the 1.27 toolchain:

```
go build .
```

```
# example.com/old
./main.go:14:21: generic method requires go1.27 or later (-lang was set to go1.26; check go.
```

The language version comes from `go.mod` (or a `//go:build go1.27` constraint on the file), not from the toolchain you happen to run. Libraries that want to keep supporting 1.26 users cannot adopt generic methods until they raise their `go` line.

Case 3: method values, method expressions and inference from the target

```go
// main.go
package main

import (
    "fmt"
    "strconv"
)

type Box[T any] struct{ Value T }

func (b Box[T]) Map[U any](f func(T) U) Box[U] { return Box[U]{f(b.Value)} }

func main() {
    b := Box[int]{Value: 7}

    mv := b.Map[string]        // method value: receiver bound, U fixed
    me := Box[int].Map[string] // method expression: receiver becomes arg 1
    fmt.Println(mv(strconv.Itoa).Value, me(b, strconv.Itoa).Value)

    // Go 1.27 also infers U from the target function type.
    var f func(func(int) string) Box[string] = b.Map
    fmt.Println(f(strconv.Itoa).Value)
}
```

```
7 7
7
```

The last assignment relies on the other 1.27 language change: function type inference now works in every assignment context (variables, composite literals, conversions, channel sends), as covered in Generic Functions. When there is no target type and no argument, there is nothing to infer from:

```go
// main.go
package main

type Box[T any] struct{ Value T }

func (b Box[T]) Map[U any](f func(T) U) Box[U] { return Box[U]{f(b.Value)} }

func main() {
    b := Box[int]{Value: 7}
    f := b.Map // nothing to infer U from
    _ = f
}
```

```
# example.com/uninst
./main.go:10:7: cannot use generic function b.Map without instantiation
```

Case 4: what interfaces cannot do

An interface method may not declare type parameters. Even `gofmt` refuses to parse it:

```go
// main.go
package main

type Mapper interface {
    Map[U any](f func(int) U) U
}

func main() {}
```

```
# example.com/ifacedecl
./main.go:5:5: interface method must have no type parameters
./main.go:5:25: undefined: U
```

A generic method also never implements a non-generic interface method, even when it could be instantiated to match:

```go
// main.go
package main

import "fmt"

type Stringer interface{ String() string }

type Opt struct{}

func (Opt) String[T any]() string { var z T; return fmt.Sprint(z) }

func main() {
    var _ Stringer = Opt{}
}
```

```
# example.com/ifaceimpl
./main.go:13:19: cannot use Opt{} (value of struct type Opt) as Stringer value in variable d
        have String[T any]() string
        want String() string
```

Why the line is drawn there

Packages are compiled separately. When `main` passes a `T` to some function `p.F(i I)` in another package, the compiler cannot know which methods `p.F` will call, so it generates code for **every non-generic method** of `T` where the type is declared or instantiated. That guarantees the method table behind the interface value can be filled in.

If `I` could declare `M[P any]()`, `p.F` might call `i.M[int]()`, `i.M[string]()` or any other instantiation, and none of them are visible while `main` is being compiled. The compiler would have to instantiate `T.M` for every possible type argument, which is impossible with Go's per-instantiation code generation. Boxing the type arguments behind their constraint interfaces would let every instantiation share one body, but every call would then be indirect, including direct calls that never touch an interface. The Go blog post walks through this argument. Go 1.27 therefore treats concrete generic methods and generic interface methods as separate questions, and answers only the first.

Note what follows. `T{}.M[int]()` has the signature `func()`, but `T` still does not implement `interface{ M() }`. Interface satisfaction is a property of the type's declared methods, and `T` declares the uninstantiated `M[P any]`, not `M[int]`.

You can see the same split in reflection: generic methods are not part of the runtime method set.

```go
// main.go
package main

import (
    "fmt"
```

```go
    "reflect"
)

type Reg struct{ items map[string]any }

func (r *Reg) Name() string             { return "reg" }
func (r *Reg) Put[T any](k string, v T) { r.items[k] = v }

func main() {
    t := reflect.TypeFor[*Reg]()
    fmt.Println("NumMethod:", t.NumMethod())
    for i := range t.NumMethod() {
        fmt.Println(" -", t.Method(i).Name)
    }
}
```

```
NumMethod: 1
 - Name
```

Code that discovers methods by reflection, such as RPC registries, template `call` and some DI containers, will not see generic methods.

Case 5: the receiver's parameters are already in scope

The method's type parameter list shares a scope with the receiver's, so names must not collide:

```go
// main.go
package main

type Box[T any] struct{ Value T }

func (b Box[T]) Conv[T any]() T { var z T; return z }

func main() {}
```

```
# example.com/shadow
./main.go:6:22: T redeclared in this block
    ./main.go:6:13: other declaration of T
```

Pick a different letter (`U`, `R`, `K`) or a descriptive name.

Case 6: generic methods on non-generic types

The receiver does not have to be generic. This is the most useful shape in application code: a client or store that owns configuration and state, with the result type chosen per call.

```go
// main.go
package main

import (
    "bytes"
    "encoding/json"
    "fmt"
    "io"
    "net/http"
    "net/http/httptest"
)

type Client struct {
    base string
    hc   *http.Client
}

// GetJSON is a generic method: the decode target belongs to the call,
// the base URL and transport belong to the Client.
func (c *Client) GetJSON[T any](path string) (T, error) {
    var out T
    resp, err := c.hc.Get(c.base + path)
    if err != nil {
        return out, err
    }
    defer resp.Body.Close()
    if resp.StatusCode != http.StatusOK {
        b, _ := io.ReadAll(io.LimitReader(resp.Body, 512))
        return out, fmt.Errorf("GET %s: %s: %s", path, resp.Status, bytes.TrimSpace(b))
    }
    err = json.NewDecoder(resp.Body).Decode(&out)
    return out, err
}

type Health struct {
    Status string `json:"status"`
}

type User struct {
    ID   int    `json:"id"`
    Name string `json:"name"`
}
```

```go
func main() {
    mux := http.NewServeMux()
    mux.HandleFunc("GET /health", func(w http.ResponseWriter, r *http.Request) {
        fmt.Fprint(w, `{"status":"ok"}`)
    })
    mux.HandleFunc("GET /users", func(w http.ResponseWriter, r *http.Request) {
        fmt.Fprint(w, `[{"id":1,"name":"ada"},{"id":2,"name":"ken"}]`)
    })
    srv := httptest.NewServer(mux)
    defer srv.Close()

    c := &Client{base: srv.URL, hc: srv.Client()}

    h, err := c.GetJSON[Health]("/health")
    fmt.Println(h, err)

    users, err := c.GetJSON[[]User]("/users")
    fmt.Println(users, err)

    _, err = c.GetJSON[Health]("/missing")
    fmt.Println(err)
}
```

```
{ok} <nil>
[{1 ada} {2 ken}] <nil>
GET /missing: 404 Not Found: 404 page not found
```

`T` appears only in the result, so it can never be inferred. Calls always spell `GetJSON[Health]`. That is a feature here: the wire type is visible at the call site.

The standard library made the same move. `math/rand/v2` gained `(*Rand).N`, a generic method that mirrors the package-level `rand.N`:

```go
// main.go
package main

import (
    "fmt"
    "math/rand/v2"
    "time"
)

func main() {
    r := rand.New(rand.NewPCG(1, 2))
    var d time.Duration = r.N(5 * time.Second) // Int inferred as time.Duration
    var b uint8 = r.N(uint8(200))
```

```go
    fmt.Printf("%T %v\n", d, d)
    fmt.Printf("%T %v\n", b, b)
}
```

```
time.Duration 3.846866346s
uint8 123
```

The output is deterministic because the PCG source is seeded. Before 1.27 a seeded `*Rand` offered only `IntN`, `Int32N`, `Int64N`, `UintN` and so on. A `time.Duration` jitter needed a conversion dance or the global, unseeded `rand.N`.

Case 7: promotion through embedding

Generic methods are promoted from embedded fields like any other method:

```go
// main.go
package main

import "fmt"

type Store struct{ items map[string]any }

func (s *Store) Put[T any](k string, v T) { s.items[k] = v }

// Lookup stays a function: it needs T only in the result.
func Lookup[T any](s *Store, k string) (T, bool) {
    v, ok := s.items[k].(T)
    return v, ok
}

type Cache struct {
    *Store
    hits int
}

func main() {
    c := Cache{Store: &Store{items: map[string]any{}}}
    c.Put("port", 8080) // promoted generic method, T = int
    c.Put("host", "localhost")

    p, ok := Lookup[int](c.Store, "port")
    fmt.Println(p, ok)
    _, ok = Lookup[string](c.Store, "port")
    fmt.Println(ok)
}
```

```
8080 true
false
```

`Lookup` could equally be a generic method `(s *Store) Get[T any](k string) (T, bool)`. The choice is about where readers will look for it, which is the topic of the next section.

Case 8: instantiation cycles

`Map` returns `Box[U]` for a fresh `U`, which is fine. A method whose result is the **same generic type instantiated with something built from `T`** is rejected:

```go
// main.go
package main

import "fmt"

type Box[T any] struct{ Value T }

type pairOf[A, B any] = struct {
    A A
    B B
}

// Zip's result type is built from T, so Box[T] would need Box[pairOf[T, U]],
// which would need Box[pairOf[pairOf[T, U], V]], and so on.
func (b Box[T]) Zip[U any](o Box[U]) Box[pairOf[T, U]] {
    return Box[pairOf[T, U]]{pairOf[T, U]{b.Value, o.Value}}
}

func main() { fmt.Printf("%+v\n", Box[int]{1}.Zip(Box[string]{"x"})) }
```

```
# example.com/zip
./main.go:6:10: instantiation cycle:
    ./main.go:15:42: T instantiated as pairOf[T, U]
```

This rule predates 1.27. A plain method `func (b Box[T]) Wrap() Box[[]T]` fails the same way with `T instantiated as []T`. Generic methods make the shape more tempting to write, though. The fix is to return a different generic type:

```go
// main.go
package main

import "fmt"

type Box[T any] struct{ Value T }
```

```go
type Pair[A, B any] struct {
    First  A
    Second B
}

// Zip returns a different generic type, so there is no cycle.
func (b Box[T]) Zip[U any](o Box[U]) Pair[T, U] {
    return Pair[T, U]{b.Value, o.Value}
}

func main() {
    fmt.Printf("%+v\n", Box[int]{1}.Zip(Box[string]{"x"}))
}
```

```
{First:1 Second:x}
```

Designing with generic methods

The interface seam pattern

Because generic methods cannot sit in interfaces, keep the **interface non-generic** and put the **typed convenience on top** as a generic function. Tests fake the small interface, and callers get type safety:

```go
// main.go
package main

import (
    "encoding/json"
    "fmt"
    "strings"
)

// Source is the interface: non-generic, so any type can implement it
// and it can be faked in tests.
type Source interface {
    Fetch(key string, dst any) error
}

// Get is the typed convenience, written once, on top of the interface.
func Get[T any](s Source, key string) (T, error) {
    var v T
    err := s.Fetch(key, &v)
    return v, err
```

```go
}

type mapSource map[string]string

func (m mapSource) Fetch(key string, dst any) error {
    raw, ok := m[key]
    if !ok {
        return fmt.Errorf("fetch %q: missing", key)
    }
    return json.NewDecoder(strings.NewReader(raw)).Decode(dst)
}

func main() {
    src := mapSource{"limits": `{"rps":200,"burst":50}`, "hosts": `["a","b"]`}

    limits, err := Get[map[string]int](src, "limits")
    fmt.Println(limits, err)
    hosts, err := Get[[]string](src, "hosts")
    fmt.Println(hosts, err)
    _, err = Get[int](src, "port")
    fmt.Println(err)
}
```

```
map[burst:50 rps:200] <nil>
[a b] <nil>
fetch "port": missing
```

If `Get` were a generic method on a concrete client type, every consumer that wanted to fake the client would lose it, because the method cannot appear in the consumer's interface.

Method or function?

Situation	Prefer
Operation is about *this* type and readers will look for it in its method list (`Box.Map`, `Rand.N`, `Client.GetJSON`)	Generic method
You want call chains (`q.Filter(...).Map(...)`)	Generic method
Consumers will want to mock or swap the type behind an interface	Non-generic interface method plus a generic function (seam pattern)
Operation works across several types or takes two unrelated receivers (`slices.Equal`, `maps.Copy`)	Generic function
Something must find it via `reflect` (RPC, templates)	Non-generic method
You must keep supporting `go 1.26` users	Generic function

Cost model

A generic method is instantiated like a generic function, using the same GC-shape stenciling with dictionaries, and no new runtime machinery is involved. Two instantiations with the same shape (for example, two pointer types) share code. Instantiations with different shapes get separate code. Generics Implementation covers the mechanism. There is no interface dispatch to devirtualize, because generic methods are never called through an interface.

The trap

- **Designing a generic method into a type you meant to mock.** The method cannot appear in an interface, so the type cannot be faked behind one. Use the seam pattern.
- **Forgetting the `go` line.** Generic methods compile only in files whose language version is 1.27 or later. A 1.27 toolchain does not override `go 1.26` in `go.mod`.
- **Expecting `reflect` to see them.** `NumMethod`, `MethodByName` and `text/template` method calls ignore generic methods.
- **Reusing the receiver's parameter name.** `func (b Box[T]) M[T any]()` is a redeclaration error.
- **Returning `Box[f(T)]` from a method of `Box[T]`.** That is an instantiation cycle. Return a different type, such as `Pair[T, U]`.
- **Result-only type parameters.** `T` that appears only in results is never inferred, so every call needs explicit brackets. That is fine for `GetJSON[T]`, but noisy for anything called in a loop.

The boring rule

Add a generic method when the operation clearly belongs to the type and nobody needs to see it through an interface. Otherwise keep the interface small and non-generic and write the generic helper as a function.

Try this

1. Give `Box[T]` a method `FlatMap[U any](f func(T) Box[U]) Box[U]` and chain it with `Map`.
2. Write a `Pipeline` type with `Then[U any](func(T) (U, error))` that short-circuits on the first error.
3. Turn `Lookup` in Case 7 into a method and decide which reads better at the call site.

Sources

- Go 1.27 release notes, language changes: https://go.dev/doc/go1.27
- Go blog, "Generic Methods": https://go.dev/blog/generic-methods
- Go blog, "Go 1.27 is released": https://go.dev/blog/go1.27

- Proposal: spec: generic methods for Go, #77273: https://github.com/golang/go/issues/77273
- Earlier requests #49085 and #50981, linked from the proposal
- `math/rand/v2 (*Rand).N`: https://pkg.go.dev/math/rand/v2#Rand.N

Idiomatic Generics

Idiomatic Generics

Overview

Generics are powerful but should be used judiciously. This chapter covers when to use generics vs interfaces, and best practices.

When to Use Generics

Good uses: - Container types (Stack, Set, Queue) - Utility functions (Map, Filter, Reduce) - Type-safe concurrent constructs - Avoiding reflection

Don't use when: - Interface works fine - Only one type will ever be used - Adding complexity without benefit

Go 1.27 generic **methods** (`func (s *Store) Put[T any](v T)`) are for operations namespaced on a type. They cannot implement interface methods—keep those non-generic.

Generics vs Interfaces

Use Interface When:

```go
// Behavior-based polymorphism
type Writer interface {
    Write([]byte) (int, error)
}

func Save(w Writer, data []byte) error {
    _, err := w.Write(data)
    return err
}
```

Use Generics When:

```go
// Type preservation needed
func Reverse[T any](s []T) []T {
    result := make([]T, len(s))
```

```go
    for i, v := range s {
        result[len(s)-1-i] = v
    }
    return result
}
```

Guidelines

1. Start with Concrete Types

```go
// Start here
func ReverseInts(s []int) []int { }

// Generalize only when needed
func Reverse[T any](s []T) []T { }
```

2. Use Meaningful Constraints

```go
// Too broad
func Process[T any](v T) { }

// Better: documents requirements
func Process[T constraints.Ordered](v T) { }
```

3. Don't Over-Constrain

```go
// Over-constrained
type Number interface {
    int | int8 | int16 | int32 | int64 |
    uint | uint8 | uint16 | uint32 | uint64 |
    float32 | float64
}

// Better: use existing constraint
func Sum[T constraints.Integer | constraints.Float](nums []T) T
```

4. Type Inference is Your Friend

```go
// Let Go infer when obvious
result := Map(nums, double)

// Explicit only when needed
result := Convert[int, float64](nums)
```

Common Patterns

slices Package (stdlib)

```go
import "slices"

slices.Sort(nums)
slices.Reverse(nums)
slices.Contains(nums, 5)
slices.Index(nums, 5)
```

maps Package (stdlib)

```go
import "maps"

maps.Clone(m)
maps.Keys(m)
maps.Values(m)
```

Summary

Decision	Choice
Need type preservation	Generics
Need polymorphic behavior	Interface
Operating on containers	Generics
Method set requirements	Interface

Related Topics

- Why Generics Matter
- Interface Best Practices

More examples

Example: prefer clear non-generic API when one type

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import "fmt"

// SumInts is clearer than Sum[int] when you only ever sum ints.
func SumInts(nums []int) int {
    total := 0
    for _, n := range nums {
        total += n
    }
    return total
}

func main() {
    fmt.Println(SumInts([]int{1, 2, 3}))
    fmt.Println(SumInts(nil))
}
```

Expected:

```
6
0
```

Example: constrain only what you need

Save as `main.go` and `go run .` (with `go mod init example` if needed).

```go
package main

import (
    "fmt"
    "strconv"
)

// Stringer-like constraint without importing fmt for the type set.
type stringer interface {
    String() string
}

type ID int
```

```go
func (id ID) String() string { return "id=" + strconv.Itoa(int(id)) }

func PrintAll[T stringer](items []T) {
    for _, it := range items {
        fmt.Println(it.String())
    }
}

func main() {
    PrintAll([]ID{7, 8})
}
```

Expected:

```
id=7
id=8
```

Runnable example

Save as `main.go`. Then:

```
go mod init example
go run .
```

```go
package main

import (
    "fmt"
    "io"
    "maps"
    "slices"
    "strings"
)

// Behavior polymorphism → interface (not generics).
func Save(w io.Writer, data []byte) error {
    _, err := w.Write(data)
    return err
}

// Type preservation on containers → generics.
func Reverse[T any](s []T) []T {
    out := make([]T, len(s))
    for i, v := range s {
```

```go
        out[len(s)-1-i] = v
    }
    return out
}

func main() {
    var b strings.Builder
    _ = Save(&b, []byte("hello"))
    fmt.Println("interface writer:", b.String())

    nums := []int{1, 2, 3}
    fmt.Println("Reverse ints:", Reverse(nums))
    fmt.Println("original unchanged:", nums)

    words := []string{"go", "is", "fun"}
    fmt.Println("Reverse strings:", Reverse(words))

    // Prefer stdlib generic helpers when they fit.
    cloned := slices.Clone(nums)
    slices.Sort(cloned)
    fmt.Println("slices.Sort clone:", cloned)
    fmt.Println("slices.Contains:", slices.Contains(words, "is"))

    m := map[string]int{"b": 2, "a": 1}
    cp := maps.Clone(m)
    cp["c"] = 3
    fmt.Println("maps.Clone size:", len(m), "->", len(cp))
}
```

Expected output:

```
interface writer: hello
Reverse ints: [3 2 1]
original unchanged: [1 2 3]
Reverse strings: [fun is go]
slices.Sort clone: [1 2 3]
slices.Contains: true
maps.Clone size: 2 -> 3
```

What to notice: `Save` cares about behavior (`Write`), so an interface is enough. `Reverse` must preserve element type across many containers, so generics win. Reach for `slices` / `maps` before inventing another utility package.

Try next: Start from a concrete `ReverseInts` and only generalize when a second type appears; replace a hand-rolled `Contains` with `slices.Contains`.

Errors

Errors as Values

Errors as Values

Overview

Go treats errors as values, not exceptions. Functions return errors alongside results, making error handling explicit and visible.

The error Interface

```
type error interface {
    Error() string
}
```

Returning Errors

```
func divide(a, b float64) (float64, error) {
    if b == 0 {
        return 0, errors.New("division by zero")
    }
    return a / b, nil
}

result, err := divide(10, 0)
if err != nil {
    log.Fatal(err)
}
```

Creating Errors

```
import "errors"

// Simple error
err := errors.New("something went wrong")

// Formatted error
```

```go
err := fmt.Errorf("failed to open %s: %v", filename, originalErr)
```

Error Handling Patterns

Check Immediately

```go
result, err := doSomething()
if err != nil {
    return err
}
// Use result
```

Early Return

```go
func process() error {
    data, err := fetch()
    if err != nil {
        return err
    }

    result, err := transform(data)
    if err != nil {
        return err
    }

    return save(result)
}
```

Add Context

```go
data, err := readFile(path)
if err != nil {
    return fmt.Errorf("loading config: %w", err)
}
```

Custom Error Types

```go
type ValidationError struct {
    Field   string
    Message string
}

func (e *ValidationError) Error() string {
    return fmt.Sprintf("%s: %s", e.Field, e.Message)
}

func validate(u User) error {
    if u.Name == "" {
        return &ValidationError{Field: "name", Message: "required"}
    }
    return nil
}
```

Sentinel Errors

```go
var ErrNotFound = errors.New("not found")
var ErrPermission = errors.New("permission denied")

func find(id int) (*User, error) {
    if !exists(id) {
        return nil, ErrNotFound
    }
    // ...
}

// Check
if err == ErrNotFound {
    // Handle not found
}
```

Summary

Pattern	Usage
`errors.New()`	Simple error
`fmt.Errorf()`	Formatted error
Custom type	Structured error data
Sentinel	Known error values

Related Topics

- Wrapping and Inspecting Errors
- panic, recover, and Failure Modes

Worked example

Multi-step pipeline that returns, wraps, and classifies errors without panicking.

Save as `main.go`. Then:

```
go mod init example
go run .
```

```go
package main

import (
    "errors"
    "fmt"
)

var ErrEmpty = errors.New("empty input")

type ParseError struct {
    Token string
    Why   string
}

func (e *ParseError) Error() string {
    return fmt.Sprintf("parse %q: %s", e.Token, e.Why)
}

func parseIntToken(s string) (int, error) {
    if s == "" {
        return 0, ErrEmpty
    }
    n := 0
    for _, r := range s {
        if r < '0' || r > '9' {
            return 0, &ParseError{Token: s, Why: "not digits"}
        }
        n = n*10 + int(r-'0')
    }
    return n, nil
}
```

```go
func sumTokens(tokens []string) (int, error) {
    total := 0
    for i, t := range tokens {
        n, err := parseIntToken(t)
        if err != nil {
            return 0, fmt.Errorf("sumTokens[%d]: %w", i, err)
        }
        total += n
    }
    return total, nil
}

func main() {
    got, err := sumTokens([]string{"10", "20", "12"})
    fmt.Println("ok sum:", got, err)

    _, err = sumTokens([]string{"10", "", "3"})
    fmt.Println("empty err:", err)
    fmt.Println("Is ErrEmpty:", errors.Is(err, ErrEmpty))

    _, err = sumTokens([]string{"10", "x9"})
    var pe *ParseError
    if errors.As(err, &pe) {
        fmt.Println("As ParseError token:", pe.Token, "why:", pe.Why)
    }
}
```

Expected output:

```
ok sum: 42 <nil>
empty err: sumTokens[1]: empty input
Is ErrEmpty: true
As ParseError token: x9 why: not digits
```

More examples

Sentinel comparison vs string equality—only the value works reliably after wrapping.

```go
package main

import (
    "errors"
    "fmt"
)
```

```go
var ErrDenied = errors.New("denied")

func authorize(role string) error {
    if role != "admin" {
        return fmt.Errorf("authorize(%s): %w", role, ErrDenied)
    }
    return nil
}

func main() {
    err := authorize("guest")
    // Correct: compare the sentinel through the chain.
    fmt.Println("Is denied:", errors.Is(err, ErrDenied))
    // Fragile: string form changes when you add context.
    fmt.Println("string equal ErrDenied:", err.Error() == ErrDenied.Error())
    fmt.Println("err text:", err)
}
```

Expected output:

```
Is denied: true
string equal ErrDenied: false
err text: authorize(guest): denied
```

Runnable example

Save as `main.go`. Then:

```
go mod init example
go run .
```

```go
package main

import (
    "errors"
    "fmt"
)

var ErrNotFound = errors.New("not found")

type ValidationError struct {
    Field   string
    Message string
}
```

```go
func (e *ValidationError) Error() string {
    return fmt.Sprintf("%s: %s", e.Field, e.Message)
}

func divide(a, b float64) (float64, error) {
    if b == 0 {
        return 0, errors.New("division by zero")
    }
    return a / b, nil
}

func findUser(id int) (string, error) {
    if id <= 0 {
        return "", ErrNotFound
    }
    return fmt.Sprintf("user-%d", id), nil
}

func validateName(name string) error {
    if name == "" {
        return &ValidationError{Field: "name", Message: "required"}
    }
    return nil
}

func main() {
    q, err := divide(10, 2)
    fmt.Println("divide ok:", q, err)
    _, err = divide(1, 0)
    fmt.Println("divide err:", err)

    name, err := findUser(7)
    fmt.Println("find:", name, err)
    _, err = findUser(0)
    if errors.Is(err, ErrNotFound) {
        fmt.Println("sentinel: not found")
    }

    if err := validateName(""); err != nil {
        fmt.Println("validation:", err)
    }
    if err := validateName("Ada"); err != nil {
        fmt.Println("unexpected:", err)
    } else {
        fmt.Println("validation: ok")
    }
}
```

Expected output:

```
divide ok: 5 <nil>
divide err: division by zero
find: user-7 <nil>
sentinel: not found
validation: name: required
validation: ok
```

What to notice: Errors are ordinary values you return and inspect. Sentinel errors (`ErrNotFound`) and custom types (`ValidationError`) both implement the same `error` interface.

Try next: Use `fmt.Errorf("findUser(%d): %w", id, ErrNotFound)` and compare string form vs `errors.Is`; add a second field to `ValidationError` for a code.

Wrapping and Inspecting Errors

Wrapping and Inspecting Errors

Overview

Go 1.13 introduced error wrapping, allowing you to add context while preserving the original error for inspection.

Wrapping Errors

```go
originalErr := errors.New("file not found")

// %w wraps the error
wrappedErr := fmt.Errorf("loading config: %w", originalErr)

// The error chain: wrappedErr -> originalErr
```

Unwrapping

```go
err := fmt.Errorf("outer: %w",
    fmt.Errorf("middle: %w",
        errors.New("inner")))

inner := errors.Unwrap(err)  // middle: inner
```

errors.Is

Check if any error in the chain matches:

```go
var ErrNotFound = errors.New("not found")

err := fmt.Errorf("user lookup: %w", ErrNotFound)

if errors.Is(err, ErrNotFound) {
    // Handle not found
}
```

errors.As

Extract specific error types:

```go
type ValidationError struct {
    Field string
}

func (e *ValidationError) Error() string {
    return "validation: " + e.Field
}

err := fmt.Errorf("processing: %w", &ValidationError{Field: "email"})

var valErr *ValidationError
if errors.As(err, &valErr) {
    fmt.Println("Invalid field:", valErr.Field)
}
```

Custom Wrappers

```go
type WrappedError struct {
    Context string
    Err     error
}

func (e *WrappedError) Error() string {
    return fmt.Sprintf("%s: %v", e.Context, e.Err)
}

func (e *WrappedError) Unwrap() error {
    return e.Err
}
```

Best Practices

```go
// Add context when crossing boundaries
func LoadUser(id int) (*User, error) {
    data, err := db.Query(id)
    if err != nil {
        return nil, fmt.Errorf("LoadUser(%d): %w", id, err)
    }
```

```
    return parse(data)
}
```

Summary

Function	Purpose
`fmt.Errorf("%w", err)`	Wrap error
`errors.Unwrap(err)`	Get wrapped error
`errors.Is(err, target)`	Check chain for match
`errors.As(err, &target)`	Extract typed error

Related Topics

- Errors as Values
- panic, recover, and Failure Modes

Worked example

Deep wrap chain with `Is`, `As`, and manual `Unwrap` walking.

Save as `main.go`. Then:

```
go mod init example
go run .
```

```
package main

import (
    "errors"
    "fmt"
)

var ErrNotFound = errors.New("not found")

type HTTPError struct {
    Code int
    Err  error
}

func (e *HTTPError) Error() string {
    return fmt.Sprintf("http %d: %v", e.Code, e.Err)
```

```go
}

func (e *HTTPError) Unwrap() error { return e.Err }

func storeGet(key string) error {
    if key == "" {
        return ErrNotFound
    }
    return nil
}

func serviceGet(key string) error {
    if err := storeGet(key); err != nil {
        return fmt.Errorf("service.get(%q): %w", key, err)
    }
    return nil
}

func handlerGet(key string) error {
    if err := serviceGet(key); err != nil {
        return &HTTPError{Code: 404, Err: err}
    }
    return nil
}

func main() {
    err := handlerGet("")
    fmt.Println("chain:", err)
    fmt.Println("Is NotFound:", errors.Is(err, ErrNotFound))

    var he *HTTPError
    if errors.As(err, &he) {
        fmt.Println("As HTTPError code:", he.Code)
    }

    fmt.Print("unwrap walk:")
    for e := err; e != nil; e = errors.Unwrap(e) {
        fmt.Printf(" -> %v", e)
    }
    fmt.Println()
}
```

Expected output:

```
chain: http 404: service.get(""): not found
Is NotFound: true
As HTTPError code: 404
```

```
unwrap walk: -> http 404: service.get(""): not found -> service.get(""): not found -> not fo
```

More examples

`errors.Join` + `Is`: each joined error stays inspectable.

```go
package main

import (
    "errors"
    "fmt"
)

var (
    ErrName = errors.New("missing name")
    ErrAge  = errors.New("invalid age")
)

func validate(name string, age int) error {
    var errs error
    if name == "" {
        errs = errors.Join(errs, ErrName)
    }
    if age < 0 {
        errs = errors.Join(errs, ErrAge)
    }
    return errs
}

func main() {
    err := fmt.Errorf("signup: %w", validate("", -3))
    fmt.Println(err)
    fmt.Println("Is name:", errors.Is(err, ErrName))
    fmt.Println("Is age:", errors.Is(err, ErrAge))
    fmt.Println("Is notfound:", errors.Is(err, errors.New("not found")))
}
```

Expected output:

```
signup: missing name
invalid age
Is name: true
Is age: true
Is notfound: false
```

Runnable example

Save as `main.go`. Then:

```
go mod init example
go run .
```

```go
package main

import (
    "errors"
    "fmt"
)

var ErrNotFound = errors.New("not found")

type ValidationError struct {
    Field string
}

func (e *ValidationError) Error() string {
    return "validation: " + e.Field
}

type WrappedError struct {
    Context string
    Err     error
}

func (e *WrappedError) Error() string {
    return fmt.Sprintf("%s: %v", e.Context, e.Err)
}

func (e *WrappedError) Unwrap() error { return e.Err }

func loadUser(id int) error {
    if id == 0 {
        return fmt.Errorf("user lookup: %w", ErrNotFound)
    }
    if id < 0 {
        return fmt.Errorf("processing: %w", &ValidationError{Field: "id"})
    }
    return nil
}

func main() {
```

```go
    err := loadUser(0)
    fmt.Println("wrapped:", err)
    fmt.Println("Is NotFound:", errors.Is(err, ErrNotFound))
    fmt.Println("Unwrap once:", errors.Unwrap(err))

    err = loadUser(-1)
    var ve *ValidationError
    if errors.As(err, &ve) {
        fmt.Println("As ValidationError field:", ve.Field)
    }

    // Custom Unwrap still participates in Is/As chains.
    chain := &WrappedError{
        Context: "handler",
        Err:     fmt.Errorf("service: %w", ErrNotFound),
    }
    fmt.Println("custom wrap:", chain)
    fmt.Println("Is through custom wrap:", errors.Is(chain, ErrNotFound))
}
```

Expected output:

```
wrapped: user lookup: not found
Is NotFound: true
Unwrap once: not found
As ValidationError field: id
custom wrap: handler: service: not found
Is through custom wrap: true
```

What to notice: `%w` builds a chain; `errors.Is` walks that chain for sentinel equality, `errors.As` extracts a concrete type. Implementing `Unwrap() error` opts your type into the same machinery.

Try next: Nest three layers of `fmt.Errorf("%w")` and walk with a loop on `errors.Unwrap`; try `errors.Join` of two validation failures and inspect with `errors.Is`.

panic, recover, and Failure Modes

panic, recover, and Failure Modes

Overview

`panic` and `recover` handle unrecoverable errors. Unlike normal errors, panics crash the program unless recovered.

When to Panic

Appropriate: - Programmer errors (bugs) - Impossible conditions - Failed invariants - During initialization

Avoid for: - Expected errors (file not found, network timeout) - User input validation - Anything recoverable

Panic

```go
func divide(a, b int) int {
    if b == 0 {
        panic("division by zero")  // Terminates program
    }
    return a / b
}
```

Common Panic Sources

```go
// nil pointer dereference
var p *int
*p = 1  // panic

// Index out of range
arr := []int{1, 2}
arr[10]  // panic

// Invalid type assertion
var i interface{} = "string"
i.(int)  // panic
```

Recover

Recover catches panics in deferred functions:

```
func safeCall() (err error) {
    defer func() {
        if r := recover(); r != nil {
            err = fmt.Errorf("panic recovered: %v", r)
        }
    }()

    dangerousOperation()
    return nil
}
```

HTTP Server Pattern

```
func handler(w http.ResponseWriter, r *http.Request) {
    defer func() {
        if err := recover(); err != nil {
            log.Printf("panic: %v\n%s", err, debug.Stack())
            http.Error(w, "Internal Server Error", 500)
        }
    }()

    // Handle request
}
```

Panic vs Error

```
// Return error for expected failures
func ReadFile(name string) ([]byte, error) {
    if !exists(name) {
        return nil, ErrNotFound  // Expected case
    }
    // ...
}

// Panic for bugs
func MustCompile(pattern string) *Regexp {
    r, err := Compile(pattern)
    if err != nil {
        panic(err)  // Invalid pattern is programmer error
```

```
    }
    return r
}
```

Summary

Keyword	Purpose
`panic`	Unrecoverable error (crash)
`recover`	Catch panic in defer
`error`	Expected, recoverable failures

Use panic for	Use error for
Bugs	Expected failures
Invariant violations	User/external input
Initialization failures	I/O, network errors

Related Topics

- Errors as Values
- Testing Fundamentals

Worked example

Boundary recover that turns a panic into an `error` (HTTP/worker style).

Save as `main.go`. Then:

```
go mod init example
go run .
```

```
package main

import (
    "fmt"
    "runtime/debug"
)

func protect(label string, fn func() error) (err error) {
    defer func() {
        if r := recover(); r != nil {
```

```
            // In real servers, log debug.Stack() and return a generic 500.
            _ = debug.Stack()
            err = fmt.Errorf("%s: panic: %v", label, r)
        }
    }()
    return fn()
}

func parseAge(s string) (int, error) {
    if s == "" {
        return 0, fmt.Errorf("empty age")
    }
    // Bug-style panic: index out of range on bad internal invariant.
    if s[0] == '!' {
        _ = []int{}[1]
    }
    n := 0
    for _, r := range s {
        if r < '0' || r > '9' {
            return 0, fmt.Errorf("bad digit in %q", s)
        }
        n = n*10 + int(r-'0')
    }
    return n, nil
}

func main() {
    err := protect("handler", func() error {
        age, err := parseAge("34")
        if err != nil {
            return err
        }
        fmt.Println("age:", age)
        return nil
    })
    fmt.Println("ok path err:", err)

    err = protect("handler", func() error {
        _, err := parseAge("")
        return err
    })
    fmt.Println("expected error:", err)

    err = protect("handler", func() error {
        _, err := parseAge("!boom")
        return err
    })
```

```go
    fmt.Println("recovered panic:", err)
}
```

Expected output:

```
age: 34
ok path err: <nil>
expected error: empty age
recovered panic: handler: panic: runtime error: index out of range [1] with length 0
```

More examples

`recover` only works on the **same** goroutine that panics.

```go
package main

import (
    "fmt"
    "sync"
)

func main() {
    var wg sync.WaitGroup

    // Wrong: recover in parent does not catch child panic.
    // We catch inside the child instead.
    wg.Add(1)
    go func() {
        defer wg.Done()
        defer func() {
            if r := recover(); r != nil {
                fmt.Println("child recovered:", r)
            }
        }()
        panic("child boom")
    }()
    wg.Wait()
    fmt.Println("parent still running")
}
```

Expected output:

```
child recovered: child boom
parent still running
```

Runnable example

Save as `main.go`. Then:

```
go mod init example
go run .
```

```go
package main

import (
    "fmt"
    "regexp"
)

func mustCompile(pattern string) *regexp.Regexp {
    r, err := regexp.Compile(pattern)
    if err != nil {
        panic(err) // invalid pattern is a programmer error at init
    }
    return r
}

func safeCall(fn func()) (err error) {
    defer func() {
        if r := recover(); r != nil {
            err = fmt.Errorf("panic recovered: %v", r)
        }
    }()
    fn()
    return nil
}

func divide(a, b int) int {
    if b == 0 {
        panic("division by zero")
    }
    return a / b
}

func main() {
    re := mustCompile(`^\d+$`)
    fmt.Println("regex match 42:", re.MatchString("42"))

    err := safeCall(func() {
        fmt.Println("divide:", divide(10, 2))
    })
```

```go
    fmt.Println("safe ok:", err)

    err = safeCall(func() {
        _ = divide(1, 0)
    })
    fmt.Println("safe recovered:", err)

    // Prefer error returns for expected failure modes.
    if !re.MatchString("xx") {
        fmt.Println("expected validation failure handled without panic")
    }
}
```

Expected output:

```
regex match 42: true
divide: 5
safe ok: <nil>
safe recovered: panic recovered: division by zero
expected validation failure handled without panic
```

What to notice: `recover` only works inside deferred functions on the panicking goroutine. Convert panics to errors at boundaries (workers, HTTP handlers); keep normal validation on the `error` path.

Try next: Panic inside a new goroutine without recover and observe process exit; wrap a handler-style function that always recovers and logs.

The Must Pattern & No-GC Handling

Advanced Error Patterns: The "Must" & The "Odin" Way

Go's error handling (`if err != nil`) is explicit, but sometimes you need different strategies for initialization vs. runtime, or you want to adopt patterns from systems languages like Odin/Zig.

1. The "Must" Pattern

Rule: Return errors to callers; Panic only on programmer error or unrecoverable startup failure.

The "Must" pattern is a convention for functions that wrap a standard error-returning function but `panic` instead of returning the error.

When to use it?

- **Global/Package Initialization:** `var t = template.Must(template.New(...))`
- **Startup Configuration:** If your compiled regex is invalid, the app is broken. Don't start.

Implementation

```go
func Must[T any](obj T, err error) T {
    if err != nil {
        panic(err)
    }
    return obj
}

// Usage
var db = Must(sql.Open("postgres", "..."))
var regex = Must(regexp.Compile("^[a-z]+$"))
```

With Generics (Go 1.18+), you can write a universal `Must` wrapper one time and use it everywhere.

---

2. Linear Error Handling (The Odin/Zig Influence)

Languages like **Odin** (a data-oriented C alternative) handle errors by treating them as distinct return values, much like Go, but often with syntactic sugar (`or_return`) or strict definitions.

While Go doesn't have `try` or `?` operators, we can adopt the **"Guard Clause" mentality**.

The Happy Path must remain left-aligned

Bad (Nested):

```go
func process() error {
    err := step1()
    if err == nil {
        err = step2()
        if err == nil {
            return step3()
        }
    }
    return err
}
```

Good (Guard Clauses / "Odin Style"):

```go
func process() error {
    if err := step1(); err != nil {
        return fmt.Errorf("step1: %w", err)
    }

    // The "Happy Path" stays at indentation 0
    if err := step2(); err != nil {
        return fmt.Errorf("step2: %w", err)
    }

    return step3()
}
```

No-GC Thinking: Errors as Resource Cleanup Triggers

In non-GC languages, an error often implies manual resource cleanup (`defer` in Swift/Zig). Go automates memory, but not *resources* (Files, Sockets, DB Connections).

The `defer` Trap in Loops:

```go
for _, file := range files {
    f, err := os.Open(file)
```

```go
    if err != nil { return err }
    defer f.Close() // DANGEROUS: Closes only at end of function, not loop!
    // FD exhaustion possible.
}
```

The Fix (Anonymous Function):

```go
for _, file := range files {
    err := func() error {
        f, err := os.Open(file)
        if err != nil { return err }
        defer f.Close() // Closes at end of this anonymous func
        return process(f)
    }()
    if err != nil { return err }
}
```

3. Errors in 2026: "Join" and Structure

Since Go 1.20+, `errors.Join` allows returning *multiple* errors at once (e.g., from parallel validation).

```go
func validate(u User) error {
    var errs error
    if u.Name == "" {
        errs = errors.Join(errs, errors.New("missing name"))
    }
    if u.Age < 0 {
        errs = errors.Join(errs, errors.New("invalid age"))
    }
    return errs // Returns nil if no errors joined
}
```

This is cleaner than older `multierror` libraries.

Summary

1. **Must**: Use for hard startup dependencies. Crash early.
2. **Left-Align**: Keep your happy path on the left edge.
3. **Defer scope**: Remember `defer` is function-scoped, not block-scoped.

Worked example

Guard-clause pipeline with `errors.Join` validation and scoped `defer` cleanup.

Save as `main.go`. Then:

```
go mod init example
go run .
```

```go
package main

import (
    "errors"
    "fmt"
)

type Resource struct {
    Name   string
    closed bool
}

func open(name string) (*Resource, error) {
    if name == "" {
        return nil, errors.New("empty name")
    }
    return &Resource{Name: name}, nil
}

func (r *Resource) Close() {
    r.closed = true
    fmt.Println("closed:", r.Name)
}

func (r *Resource) Work() error {
    if r.closed {
        return errors.New("use after close")
    }
    fmt.Println("work:", r.Name)
    return nil
}

func processAll(names []string) error {
    for _, name := range names {
        // Inner func scopes defer so each resource closes per iteration.
        if err := func() error {
            r, err := open(name)
            if err != nil {
```

```go
                return fmt.Errorf("open %q: %w", name, err)
            }
            defer r.Close()
            return r.Work()
        }(); err != nil {
            return err
        }
    }
    return nil
}

func main() {
    if err := processAll([]string{"a.txt", "b.txt"}); err != nil {
        fmt.Println("process:", err)
    }

    var errs error
    errs = errors.Join(errs, errors.New("missing name"))
    errs = errors.Join(errs, errors.New("invalid age"))
    fmt.Println("joined:", errs)
}
```

Expected output:

```
work: a.txt
closed: a.txt
work: b.txt
closed: b.txt
joined: missing name
invalid age
```

More examples

Generic `Must` for compile-time-valid startup data only.

```go
package main

import (
    "fmt"
    "net/url"
    "regexp"
)

func Must[T any](v T, err error) T {
    if err != nil {
```

```
        panic(err)
    }
    return v
}

func main() {
    re := Must(regexp.Compile(`^[a-z]+$`))
    u := Must(url.Parse("https://example.com/path"))
    fmt.Println("regex:", re.MatchString("gopher"))
    fmt.Println("host:", u.Host, "path:", u.Path)
}
```

Expected output:

```
regex: true
host: example.com path: /path
```

Runnable example

Save as `main.go`. Then:

```
go mod init example
go run .
```

```
package main

import (
    "errors"
    "fmt"
    "regexp"
)

func Must[T any](v T, err error) T {
    if err != nil {
        panic(err)
    }
    return v
}

func openConfig(path string) (string, error) {
    if path == "" {
        return "", errors.New("empty path")
    }
    return "config:" + path, nil
}
```

```go
func validate(name string, age int) error {
    var errs error
    if name == "" {
        errs = errors.Join(errs, errors.New("missing name"))
    }
    if age < 0 {
        errs = errors.Join(errs, errors.New("invalid age"))
    }
    return errs
}

func process() error {
    // Guard clauses keep the happy path left-aligned.
    cfg, err := openConfig("app.toml")
    if err != nil {
        return fmt.Errorf("step open: %w", err)
    }

    if err := validate("Ada", 36); err != nil {
        return fmt.Errorf("step validate: %w", err)
    }

    fmt.Println("happy path with", cfg)
    return nil
}

func main() {
    // Must: fine for static startup data that must be correct.
    re := Must(regexp.Compile(`^[a-z]+$`))
    fmt.Println("must regex:", re.MatchString("gopher"))

    if err := process(); err != nil {
        fmt.Println("process:", err)
    }

    if err := validate("", -1); err != nil {
        fmt.Println("joined:", err)
    }

    // Recover a deliberate Must failure for demo purposes only.
    func() {
        defer func() {
            if r := recover(); r != nil {
                fmt.Println("must panicked as expected:", r)
            }
        }()
```

```
        _ = Must(openConfig(""))
    }()
}
```

Expected output:

```
must regex: true
happy path with config:app.toml
joined: missing name
invalid age
must panicked as expected: empty path
```

What to notice: Generic `Must` collapses `(T, error)` for irrecoverable init. `errors.Join` reports multiple validation problems at once. Guard clauses avoid nested `if err == nil` pyramids.

Try next: Move `Must(regexp.Compile(...))` to package level `var`; add a loop that opens several pseudo-files with an inner `func() error { defer close... }` so `defer` runs per iteration.

Testing

Testing Fundamentals

Testing Fundamentals

Overview

Go has built-in testing support. Test files end with `_test.go` and use the `testing` package.

Writing Tests

```go
// math.go
package math

func Add(a, b int) int {
    return a + b
}
```

```go
// math_test.go
package math

import "testing"

func TestAdd(t *testing.T) {
    result := Add(2, 3)
    if result != 5 {
        t.Errorf("Add(2, 3) = %d; want 5", result)
    }
}
```

Running Tests

```go
go test              # Current package
go test ./...        # All packages
go test -v           # Verbose
go test -run TestAdd # Specific test
```

Test Functions

```go
func TestXxx(t *testing.T)          // Test
func BenchmarkXxx(b *testing.B)     // Benchmark
func ExampleXxx()                   // Example
func FuzzXxx(f *testing.F)          // Fuzz test
```

t.Error vs t.Fatal

```go
func TestExample(t *testing.T) {
    // Continues after failure
    t.Error("failed but continuing")

    // Stops test immediately
    t.Fatal("failed, stopping now")
}
```

Helper Functions

```go
func assertEqual(t *testing.T, got, want int) {
    t.Helper()  // Marks this as helper
    if got != want {
        t.Errorf("got %d, want %d", got, want)
    }
}

func TestMath(t *testing.T) {
    assertEqual(t, Add(1, 2), 3)
}
```

Setup and Teardown

```go
func TestMain(m *testing.M) {
    // Setup
    setup()

    code := m.Run()  // Run tests

    // Teardown
    teardown()
```

```
    os.Exit(code)
}
```

Parallel Tests

```
func TestParallel(t *testing.T) {
    t.Parallel()  // Run in parallel
    // Test code
}
```

Summary

Command	Purpose
`go test`	Run tests
`go test -v`	Verbose output
`go test -cover`	Show coverage
`go test -race`	Race detection

Related Topics

- Test Organization and Coverage
- Advanced Testing

Worked example

`t.Helper`, `Error` vs `Fatal`, and `TestMain` setup/teardown.

Save as `strutil.go` and `strutil_test.go`. Then:

```
go mod init example
go test -v
```

```
// strutil.go
package main

import "strings"

func TitleWords(s string) string {
    parts := strings.Fields(s)
```

```go
    for i, p := range parts {
        if p == "" {
            continue
        }
        parts[i] = strings.ToUpper(p[:1]) + strings.ToLower(p[1:])
    }
    return strings.Join(parts, " ")
}
```

```go
// strutil_test.go
package main

import (
    "fmt"
    "os"
    "testing"
)

func TestMain(m *testing.M) {
    fmt.Println("setup: suite start")
    code := m.Run()
    fmt.Println("teardown: suite end")
    os.Exit(code)
}

func assertEq(t *testing.T, got, want string) {
    t.Helper()
    if got != want {
        t.Errorf("got %q want %q", got, want)
    }
}

func TestTitleWords(t *testing.T) {
    assertEq(t, TitleWords("hello world"), "Hello World")
    assertEq(t, TitleWords("gO"), "Go")
}

func TestTitleWordsFatalStops(t *testing.T) {
    if TitleWords("") != "" {
        t.Fatal("empty should stay empty") // would stop this test
    }
    // Continues only if Fatal did not fire.
    assertEq(t, TitleWords("a b"), "A B")
}
```

Expected output:

```
setup: suite start
=== RUN   TestTitleWords
--- PASS: TestTitleWords (0.00s)
=== RUN   TestTitleWordsFatalStops
--- PASS: TestTitleWordsFatalStops (0.00s)
PASS
teardown: suite end
```

More examples

Parallel top-level tests (`t.Parallel`) on independent pure functions.

```go
// parallel_test.go
package main

import "testing"

func TestTitleHello(t *testing.T) {
    t.Parallel()
    if TitleWords("hello") != "Hello" {
        t.Fatal("bad title")
    }
}

func TestTitleWorld(t *testing.T) {
    t.Parallel()
    if TitleWords("world") != "World" {
        t.Fatal("bad title")
    }
}
```

```bash
go test -v -parallel 4
```

Runnable example

Save these two files in a directory (for example `mathx/`). Then:

```bash
cd mathx
go mod init example/mathx
go test -v
go test -cover
```

`mathx.go`:

```go
package mathx

func Add(a, b int) int {
    return a + b
}

func Div(a, b int) (int, error) {
    if b == 0 {
        return 0, errDivZero
    }
    return a / b, nil
}

var errDivZero = errString("division by zero")

type errString string

func (e errString) Error() string { return string(e) }
```

mathx_test.go:

```go
package mathx

import "testing"

func assertEqual(t *testing.T, got, want int) {
    t.Helper()
    if got != want {
        t.Errorf("got %d, want %d", got, want)
    }
}

func TestAdd(t *testing.T) {
    assertEqual(t, Add(2, 3), 5)
    assertEqual(t, Add(-1, 1), 0)
}

func TestDiv(t *testing.T) {
    got, err := Div(10, 2)
    if err != nil {
        t.Fatalf("unexpected err: %v", err)
    }
    assertEqual(t, got, 5)

    _, err = Div(1, 0)
    if err == nil {
        t.Fatal("expected error for divide by zero")
```

```go
    }
}

func TestAddParallel(t *testing.T) {
    t.Parallel()
    assertEqual(t, Add(10, 5), 15)
}
```

Expected output: (verbose)

```
=== RUN   TestAdd
--- PASS: TestAdd (0.00s)
=== RUN   TestDiv
--- PASS: TestDiv (0.00s)
=== RUN   TestAddParallel
--- PASS: TestAddParallel (0.00s)
PASS
ok      example/mathx   0.00xs
```

What to notice: Tests live in `*_test.go`, use `TestXxx(*testing.T)`, and fail with `t.Error` / `t.Fatal`. `t.Helper()` makes failure lines point at the call site, not the helper body.

Try next: Add `go test -run TestDiv -v`; introduce a deliberate bug in `Add` and watch the assertion fail; try `t.Error` vs `t.Fatal` after a failed check.

Test Organization and Coverage

Test Organization and Coverage

Overview

Well-organized tests and good coverage help maintain code quality over time.

File Organization

```
mypackage/
  user.go          # Implementation
  user_test.go     # Tests
  user_internal_test.go  # Internal tests
  testdata/        # Test fixtures
      users.json
```

Package Naming

```go
// Same package (white-box testing)
package mypackage

// External package (black-box testing)
package mypackage_test
```

Testdata Directory

```go
func TestParseConfig(t *testing.T) {
    data, err := os.ReadFile("testdata/config.json")
    if err != nil {
        t.Fatal(err)
    }
    // Use data
}
```

Code Coverage

```go
# Show coverage percentage
go test -cover

# Generate coverage profile
go test -coverprofile=coverage.out

# View in browser
go tool cover -html=coverage.out

# Show coverage by function
go tool cover -func=coverage.out
```

Coverage Modes

```go
# Statement coverage (default)
go test -covermode=set

# Count mode (how many times)
go test -covermode=count

# Atomic (for concurrent tests)
go test -covermode=atomic
```

Test Groups

```go
func TestUser(t *testing.T) {
    t.Run("Create", func(t *testing.T) {
        // Test creation
    })

    t.Run("Update", func(t *testing.T) {
        // Test update
    })

    t.Run("Delete", func(t *testing.T) {
        // Test deletion
    })
}
```

Skipping Tests

```go
func TestIntegration(t *testing.T) {
    if testing.Short() {
        t.Skip("skipping in short mode")
    }
    // Long test
}
```

```bash
go test -short  # Skip long tests
```

`gotestsum`: local feedback and CI reports

gotestsum wraps `go test -json` to present results and save reports. Keep ordinary `testing` package tests; no separate test framework is needed.

```bash
go install gotest.tools/gotestsum@v1.13.0
gotestsum --format testname -- -race -count=1 ./...
```

Arguments after `--` go to `go test`. `-count=1` requests a fresh run instead of a cached result. For CI, retain structured output:

```bash
mkdir -p test-results
gotestsum --junitfile test-results/junit.xml \
    --jsonfile test-results/tests.json -- -race -count=1 ./...
gotestsum tool slowest --jsonfile test-results/tests.json --threshold 500ms
```

Publish the report files as CI artifacts, including on failure. JUnit XML lets a CI system display test results, and the JSON preserves the test event stream. Test failures still fail the command. The `slowest` command above reports timings; it does not rewrite tests.

For local feedback, `gotestsum --watch --format testname` reruns tests as Go files change. Keep the first failure visible when investigating flakes; automatic retries can conceal a broken test or shutdown sequence.

Summary

Flag	Purpose
`-cover`	Show coverage %
`-coverprofile=file`	Generate profile
`-short`	Skip long tests
`-run=Pattern`	Run matching tests

Related Topics

- Testing Fundamentals
- Advanced Testing

Worked example

White-box vs black-box packages and `testdata` fixtures side by side.

Save under `greet/`. Then:

```
cd greet
go mod init example/greet
go test -v ./...
go test -cover
```

```go
// greet.go
package greet

import "fmt"

func Hello(name string) string {
    if name == "" {
        name = defaultName()
    }
    return fmt.Sprintf("Hello, %s!", name)
}

func defaultName() string { return "world" }
```

```go
// greet_internal_test.go - white-box (same package)
package greet

import "testing"

func TestDefaultName(t *testing.T) {
    if defaultName() != "world" {
        t.Fatalf("got %q", defaultName())
    }
}
```

```go
// greet_test.go - black-box (external test package)
package greet_test

import (
```

```go
    "os"
    "strings"
    "testing"

    "example/greet"
)

func TestHelloFromFixture(t *testing.T) {
    data, err := os.ReadFile("testdata/names.txt")
    if err != nil {
        t.Fatal(err)
    }
    for _, name := range strings.Split(strings.TrimSpace(string(data)), "\n") {
        got := greet.Hello(name)
        if !strings.Contains(got, name) {
            t.Errorf("Hello(%q)=%q", name, got)
        }
    }
}

func TestHelloEmpty(t *testing.T) {
    if greet.Hello("") != "Hello, world!" {
        t.Fatal(greet.Hello(""))
    }
}
```

`testdata/names.txt`:

```
Ada
Grace
```

Expected output: (`go test -v`)

```
=== RUN   TestDefaultName
--- PASS: TestDefaultName (0.00s)
=== RUN   TestHelloFromFixture
--- PASS: TestHelloFromFixture (0.00s)
=== RUN   TestHelloEmpty
--- PASS: TestHelloEmpty (0.00s)
PASS
```

More examples

Grouped subtests + `-short` skip for slow suites.

```go
// suite_test.go
package greet_test

import (
    "testing"

    "example/greet"
)

func TestHelloSuite(t *testing.T) {
    t.Run("named", func(t *testing.T) {
        if greet.Hello("Lin") != "Hello, Lin!" {
            t.Fatal("named")
        }
    })
    t.Run("default", func(t *testing.T) {
        if greet.Hello("") != "Hello, world!" {
            t.Fatal("default")
        }
    })
    t.Run("slow", func(t *testing.T) {
        if testing.Short() {
            t.Skip("skipping slow case")
        }
        // pretend expensive integration work
        _ = greet.Hello("slow")
    })
}
```

```bash
go test -short -v -run TestHelloSuite
```

Runnable example

Save these files under `mathx/`. Create `testdata/cases.txt` as shown. Then:

```bash
cd mathx
go mod init example/mathx
go test -v
go test -cover
go test -short -v
go test -coverprofile=coverage.out && go tool cover -func=coverage.out
```

`mathx.go`:

```go
package mathx

import (
    "strconv"
    "strings"
)

func Add(a, b int) int { return a + b }

// ParseSum reads lines of "a+b" and returns the sums.
func ParseSum(data string) ([]int, error) {
    lines := strings.Split(strings.TrimSpace(data), "\n")
    out := make([]int, 0, len(lines))
    for _, line := range lines {
        if line == "" {
            continue
        }
        parts := strings.Split(line, "+")
        if len(parts) != 2 {
            return nil, errString("bad line: " + line)
        }
        a, err := strconv.Atoi(strings.TrimSpace(parts[0]))
        if err != nil {
            return nil, err
        }
        b, err := strconv.Atoi(strings.TrimSpace(parts[1]))
        if err != nil {
            return nil, err
        }
        out = append(out, Add(a, b))
    }
    return out, nil
}

type errString string

func (e errString) Error() string { return string(e) }
```

`mathx_test.go` (black-box: external test package):

```go
package mathx_test

import (
    "os"
    "testing"

    "example/mathx"
```

```
)

func TestAdd(t *testing.T) {
    t.Run("positive", func(t *testing.T) {
        if got := mathx.Add(2, 3); got != 5 {
            t.Fatalf("got %d", got)
        }
    })
    t.Run("negative", func(t *testing.T) {
        if got := mathx.Add(-2, -3); got != -5 {
            t.Fatalf("got %d", got)
        }
    })
}

func TestParseSumFromTestdata(t *testing.T) {
    data, err := os.ReadFile("testdata/cases.txt")
    if err != nil {
        t.Fatal(err)
    }
    sums, err := mathx.ParseSum(string(data))
    if err != nil {
        t.Fatal(err)
    }
    want := []int{5, 0, 9}
    if len(sums) != len(want) {
        t.Fatalf("len=%d want %d", len(sums), len(want))
    }
    for i := range want {
        if sums[i] != want[i] {
            t.Errorf("sums[%d]=%d want %d", i, sums[i], want[i])
        }
    }
}

func TestSlow(t *testing.T) {
    if testing.Short() {
        t.Skip("skipping slow test in short mode")
    }
    // Pretend this is an expensive integration-style check.
    if mathx.Add(100, 1) != 101 {
        t.Fatal("unexpected")
    }
}
```

`testdata/cases.txt`:

```
2+3
-1+1
4+5
```

Expected output: (`go test -v` without `-short`)

```
=== RUN   TestAdd
=== RUN   TestAdd/positive
=== RUN   TestAdd/negative
--- PASS: TestAdd (0.00s)
=== RUN   TestParseSumFromTestdata
--- PASS: TestParseSumFromTestdata (0.00s)
=== RUN   TestSlow
--- PASS: TestSlow (0.00s)
PASS
```

With `go test -short -v`, `TestSlow` reports `SKIP`.

What to notice: `package mathx_test` only sees exported API (black-box). `testdata/` is the conventional place for fixtures. Subtests (`t.Run`) organize cases; `-short` skips long work.

Try next: Generate an HTML coverage report with `go tool cover -html=coverage.out`; add an internal white-box test file with `package mathx` that exercises an unexported helper.

Advanced Testing

Advanced Testing

Overview

Table-driven tests, subtests, and test helpers make Go tests maintainable and expressive.

```
Fast feedback test loop

unit tests -------> deterministic concurrency tests -----> benchmarks
    |                           |                               |
  <1s target                  race-safe                     track regressions
```

Table-Driven Tests

```go
func TestAdd(t *testing.T) {
    tests := []struct {
        name     string
        a, b     int
        expected int
    }{
        {"positive", 2, 3, 5},
        {"negative", -1, -2, -3},
        {"zero", 0, 0, 0},
        {"mixed", -1, 5, 4},
    }

    for _, tt := range tests {
        t.Run(tt.name, func(t *testing.T) {
            result := Add(tt.a, tt.b)
            if result != tt.expected {
                t.Errorf("Add(%d, %d) = %d; want %d",
                    tt.a, tt.b, result, tt.expected)
            }
        })
    }
}
```

Subtests

```go
func TestAPI(t *testing.T) {
    t.Run("GET", func(t *testing.T) {
        t.Run("success", func(t *testing.T) { })
        t.Run("not_found", func(t *testing.T) { })
    })

    t.Run("POST", func(t *testing.T) {
        t.Run("valid", func(t *testing.T) { })
        t.Run("invalid", func(t *testing.T) { })
    })
}
```

Test Helpers

```go
func newTestServer(t *testing.T) *httptest.Server {
    t.Helper()
    return httptest.NewServer(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request)
        w.Write([]byte("OK"))
    }))
}

func TestClient(t *testing.T) {
    server := newTestServer(t)
    defer server.Close()
    // Use server.URL
}
```

Cleanup

```go
func TestWithCleanup(t *testing.T) {
    tmpDir := t.TempDir()  // Auto-removed

    f, _ := os.CreateTemp(tmpDir, "test")
    t.Cleanup(func() {
        f.Close()
    })
}
```

Deterministic Concurrency (`testing/synctest`)

Since Go 1.25, `testing/synctest` provides virtualized time and goroutine bubble scheduling. Time advances instantly when all goroutines in the `synctest.Run` bubble are blocked on timers or channel operations, eliminating artificial `time.Sleep` in tests and eliminating flaky timing race conditions.

```
flow:
  T -> G  (Spawn Worker with 10s Timer)
  G -> B  (Sleep 10s (Goroutine Blocks))
  B -> G  (Wake up immediately (0 wall-clock ms elapsed))
  G -> T  (Send completion signal)
  T -> B  (synctest.Wait() (Verifies all goroutines settled))
```

```go
import (
    "testing"
    "testing/synctest"
    "time"
)

func TestRateLimiter(t *testing.T) {
    synctest.Run(func() {
        rl := NewRateLimiter(1 * time.Second)
        // Time is virtual and controlled within synctest.Run
        if !rl.Allow() { t.Error("should allow first request") }
        if rl.Allow() { t.Error("should block immediate second request") }

        // Advance virtual time instantly without waiting real seconds
        time.Sleep(1 * time.Second)

        if !rl.Allow() { t.Error("should allow request after virtual 1s window") }
        synctest.Wait() // Wait for all bubble goroutines to settle
    })
}
```

Go 1.27 adds `synctest.Sleep(d)`, which is `time.Sleep(d)` plus `synctest.Wait()`: advance the fake clock, then wait until the bubble is settled.

```go
synctest.Test(t, func(t *testing.T) {
    go worker()
    synctest.Sleep(time.Second) // jump clock + wait for reactions
})
```

Tip: keep business logic outside goroutine setup so `synctest.Run` / `synctest.Test` stays small and focused.

Efficient Benchmarks (`B.Loop`)

Prefer `B.Loop` in modern toolchains for cleaner benchmark loops and fewer loop-control mistakes.

```go
func BenchmarkAdd(b *testing.B) {
    for b.Loop() {
        Add(1, 2)
    }
}
```

Migration pattern:

```go
// old
func BenchmarkParseOld(b *testing.B) {
    for i := 0; i < b.N; i++ {
        Parse(input)
    }
}

// new
func BenchmarkParse(b *testing.B) {
    for b.Loop() {
        Parse(input)
    }
}
```

Go 1.27 Testing Upgrade Checklist

1. Use `t.Cleanup`, `t.TempDir`, and `t.Setenv` instead of manual teardown.
2. Move flaky concurrency tests into `testing/synctest`; prefer `synctest.Sleep` over wall-clock sleeps.
3. Standardize benchmark style on `B.Loop`.
4. For HTTP tests that need a fake network (especially inside synctest), use `httptest.NewTestServer(t, handler)` instead of `NewServer`.
5. Run strict CI test command:

```go
go test ./... -race -shuffle=on -count=1
```

Parallel Subtests

```go
func TestParallel(t *testing.T) {
    tests := []struct{ name string }{{"a"}, {"b"}, {"c"}}

    for _, tt := range tests {
        t.Run(tt.name, func(t *testing.T) {
            t.Parallel()
            // Runs concurrently
        })
    }
}
```

Benchmarks

```go
func BenchmarkAdd(b *testing.B) {
    for i := 0; i < b.N; i++ {
        Add(1, 2)
    }
}
```

```bash
go test -bench=.
go test -bench=. -benchmem  # Include memory
```

Summary

Technique	Purpose
Table-driven	Test multiple cases
Subtests	Organize and filter
`t.Helper()`	Clean error reporting
`t.Cleanup()`	Guaranteed cleanup
`t.Parallel()`	Concurrent tests

Related Topics

- Testing Fundamentals
- Integration Testing

Worked example

Table-driven tests with nested subtests and shared helpers.

Save as `clamp.go` and `clamp_test.go`. Then:

```
go mod init example
go test -v
go test -run 'TestClamp/high' -v
```

```go
// clamp.go
package main

func Clamp(v, lo, hi int) int {
    if lo > hi {
        lo, hi = hi, lo
    }
    if v < lo {
        return lo
    }
    if v > hi {
        return hi
    }
    return v
}

func InRange(v, lo, hi int) bool {
    return Clamp(v, lo, hi) == v
}
```

```go
// clamp_test.go
package main

import "testing"

func TestClamp(t *testing.T) {
    tests := []struct {
        name       string
        v, lo, hi  int
        want       int
    }{
        {"inside", 5, 0, 10, 5},
        {"low", -1, 0, 10, 0},
        {"high", 99, 0, 10, 10},
        {"swapped bounds", 5, 10, 0, 5},
    }
```

```go
    for _, tt := range tests {
        t.Run(tt.name, func(t *testing.T) {
            if got := Clamp(tt.v, tt.lo, tt.hi); got != tt.want {
                t.Fatalf("Clamp(%d,%d,%d)=%d want %d", tt.v, tt.lo, tt.hi, got, tt.want)
            }
        })
    }
}

func TestInRangeTable(t *testing.T) {
    t.Run("true", func(t *testing.T) {
        cases := []int{0, 5, 10}
        for _, v := range cases {
            if !InRange(v, 0, 10) {
                t.Fatalf("%d should be in range", v)
            }
        }
    })
    t.Run("false", func(t *testing.T) {
        if InRange(-1, 0, 10) || InRange(11, 0, 10) {
            t.Fatal("out of range reported true")
        }
    })
}
```

Expected output:

```
=== RUN   TestClamp
=== RUN   TestClamp/inside
=== RUN   TestClamp/low
=== RUN   TestClamp/high
=== RUN   TestClamp/swapped_bounds
--- PASS: TestClamp (0.00s)
=== RUN   TestInRangeTable
=== RUN   TestInRangeTable/true
=== RUN   TestInRangeTable/false
--- PASS: TestInRangeTable (0.00s)
PASS
```

More examples

Benchmarks comparing algorithms (`-benchmem`).

```go
// clamp_bench_test.go
package main
```

```go
import "testing"

func minMaxClamp(v, lo, hi int) int {
    if v < lo {
        return lo
    }
    if v > hi {
        return hi
    }
    return v
}

func BenchmarkClamp(b *testing.B) {
    for i := 0; i < b.N; i++ {
        _ = Clamp(i%20, 0, 10)
    }
}

func BenchmarkMinMaxClamp(b *testing.B) {
    for i := 0; i < b.N; i++ {
        _ = minMaxClamp(i%20, 0, 10)
    }
}
```

```bash
go test -bench=. -benchmem
```

Runnable example

Save these two files in `mathx/`. Then:

```bash
cd mathx
go mod init example/mathx
go test -v
go test -bench='BenchmarkAdd$' -benchmem
```

`mathx.go`:

```go
package mathx

func Add(a, b int) int { return a + b }

func Max(a, b int) int {
    if a > b {
        return a
```

```
    }
    return b
}
```

mathx_test.go:

```
package mathx

import (
    "os"
    "path/filepath"
    "testing"
)

func TestAddTable(t *testing.T) {
    tests := []struct {
        name     string
        a, b     int
        expected int
    }{
        {"positive", 2, 3, 5},
        {"negative", -1, -2, -3},
        {"zero", 0, 0, 0},
        {"mixed", -1, 5, 4},
    }
    for _, tt := range tests {
        t.Run(tt.name, func(t *testing.T) {
            if got := Add(tt.a, tt.b); got != tt.expected {
                t.Errorf("Add(%d,%d)=%d; want %d", tt.a, tt.b, got, tt.expected)
            }
        })
    }
}

func TestMaxParallel(t *testing.T) {
    cases := []struct {
        name       string
        a, b, want int
    }{
        {"a", 1, 2, 2},
        {"b", 5, 5, 5},
        {"c", -3, -1, -1},
    }
    for _, tc := range cases {
        tc := tc // capture (harmless; required before Go 1.22 loop scoping)
        t.Run(tc.name, func(t *testing.T) {
            t.Parallel()
```

```go
            if got := Max(tc.a, tc.b); got != tc.want {
                t.Fatalf("got %d want %d", got, tc.want)
            }
        })
    }
}

func TestWithCleanup(t *testing.T) {
    dir := t.TempDir() // removed automatically after the test
    path := filepath.Join(dir, "note.txt")
    if err := writeFile(t, path, "ok"); err != nil {
        t.Fatal(err)
    }
    t.Cleanup(func() {
        // Extra hooks run LIFO before TempDir removal.
    })
    data, err := os.ReadFile(path)
    if err != nil {
        t.Fatal(err)
    }
    if string(data) != "ok" {
        t.Fatalf("got %q", data)
    }
}

func writeFile(t *testing.T, path, body string) error {
    t.Helper()
    return os.WriteFile(path, []byte(body), 0o600)
}

func BenchmarkAdd(b *testing.B) {
    for i := 0; i < b.N; i++ {
        Add(1, 2)
    }
}
```

Expected output: (`go test -v`)

```
=== RUN   TestAddTable
=== RUN   TestAddTable/positive
=== RUN   TestAddTable/negative
=== RUN   TestAddTable/zero
=== RUN   TestAddTable/mixed
--- PASS: TestAddTable (0.00s)
=== RUN   TestMaxParallel
=== RUN   TestMaxParallel/a
=== RUN   TestMaxParallel/b
```

```
=== RUN   TestMaxParallel/c
--- PASS: TestMaxParallel (0.00s)
=== RUN   TestWithCleanup
--- PASS: TestWithCleanup (0.00s)
PASS
```

Benchmark numbers vary by machine:

```
BenchmarkAdd-8      ... ... ns/op   0 B/op  0 allocs/op
```

What to notice: Table-driven `t.Run` names show up in failures and `-run` filters. `t.Parallel` runs subtests concurrently. `t.TempDir` / `t.Cleanup` replace manual teardown. Classic `b.N` benchmarks work everywhere; prefer `b.Loop()` on newer toolchains when available.

Try next: Filter with `go test -run 'TestAddTable/mixed' -v`; break one table case and read the subtest name in the failure; try `go test -count=1 -shuffle=on`.

Integration and System Testing

Integration and System Testing

Overview

Integration tests verify components work together, testing against real databases, APIs, and external services.

```
Test layers

unit (many, fast)
   |
integration (fewer, realistic deps)
   |
system/e2e (fewest, highest confidence)
```

HTTP Testing

```go
import "net/http/httptest"

func TestHandler(t *testing.T) {
    req := httptest.NewRequest("GET", "/users", nil)
    w := httptest.NewRecorder()

    handler(w, req)

    if w.Code != http.StatusOK {
        t.Errorf("status = %d; want 200", w.Code)
    }
}
```

Test Server

```go
func TestAPIClient(t *testing.T) {
    // Go 1.27+: in-memory network, auto-cleanup, synctest-safe.
    server := httptest.NewTestServer(t, http.HandlerFunc(
        func(w http.ResponseWriter, r *http.Request) {
```

```go
            w.Write([]byte(`{"id": 1}`))
        }))

    resp, err := server.Client().Get("http://example.com/users/1")
    // Assert...
}
```

`NewTestServer` uses an in-memory fake network (no loopback port), registers `t.Cleanup` to shut the server down, and fails the test if the handler panics. Prefer it over `httptest.NewServer` for new tests. `NewServer` still listens on loopback when you need a real TCP address.

Database Testing

```go
func TestUserRepository(t *testing.T) {
    if testing.Short() {
        t.Skip("skipping integration test")
    }

    db := setupTestDB(t)
    t.Cleanup(func() { db.Close() })

    repo := NewUserRepository(db)

    // Test operations
    err := repo.Create(&User{Name: "test"})
    if err != nil {
        t.Fatal(err)
    }
}
```

Build Tags

```go
//go:build integration
// +build integration

package myapp_test

// Only runs with: go test -tags=integration
```

Environment-Based Tests

```go
func TestWithEnv(t *testing.T) {
    url := os.Getenv("API_URL")
    if url == "" {
        t.Skip("API_URL not set")
    }
    // Test against real API
}
```

Docker Integration

```go
func TestWithDocker(t *testing.T) {
    if testing.Short() {
        t.Skip("skipping docker test")
    }

    // Use testcontainers-go or similar
    container, err := startPostgres()
    if err != nil {
        t.Fatal(err)
    }
    t.Cleanup(func() { container.Terminate(ctx) })

    // Run tests against container
}
```

Go 1.27 Integration Testing Upgrades

Use a two-lane CI strategy:

1. Default lane (every push): fast unit + short integration.
2. Full lane (main/nightly): DB containers, external API contract checks, and race detector.

```bash
# default
go test ./... -short -shuffle=on

# full
go test ./... -tags=integration -race -count=1
```

For external APIs, always assert status + schema shape, not only status codes:

```go
func assertUserShape(t *testing.T, body []byte) {
    t.Helper()
    var got map[string]any
    if err := json.Unmarshal(body, &got); err != nil {
        t.Fatal(err)
    }
    if _, ok := got["id"]; !ok {
        t.Fatal("missing id")
    }
}
```

Summary

Tool	Use Case
`httptest.NewTestServer`	In-memory HTTP (Go 1.27+; default for new tests)
`httptest.NewServer`	Loopback TCP when you need a real URL
Build tags	Separate test suites
Env vars	External dependencies
`testing.Short()`	Skip slow tests

Related Topics

- Testing Fundamentals
- Advanced Testing

Worked example

Handler unit test with `httptest.NewRecorder` plus client test against `httptest.NewServer`.

Save as `api.go` and `api_test.go`. Then:

```go
go mod init example
go test -v
```

```go
// api.go
package main

import (
    "encoding/json"
    "net/http"
)
```

```
type Item struct {
    ID   string `json:"id"`
    Name string `json:"name"`
}

func ItemHandler(w http.ResponseWriter, r *http.Request) {
    if r.Method != http.MethodGet {
        http.Error(w, "method not allowed", http.StatusMethodNotAllowed)
        return
    }
    id := r.PathValue("id")
    if id == "" {
        http.Error(w, "missing id", http.StatusBadRequest)
        return
    }
    w.Header().Set("Content-Type", "application/json")
    _ = json.NewEncoder(w).Encode(Item{ID: id, Name: "widget-" + id})
}

func NewAPI() http.Handler {
    mux := http.NewServeMux()
    mux.HandleFunc("GET /items/{id}", ItemHandler)
    return mux
}
```

```
// api_test.go
package main

import (
    "encoding/json"
    "io"
    "net/http"
    "net/http/httptest"
    "testing"
)

func TestItemHandlerRecorder(t *testing.T) {
    req := httptest.NewRequest(http.MethodGet, "/items/7", nil)
    req.SetPathValue("id", "7")
    rec := httptest.NewRecorder()
    ItemHandler(rec, req)

    if rec.Code != http.StatusOK {
        t.Fatalf("status=%d", rec.Code)
    }
    var item Item
```

```go
    if err := json.Unmarshal(rec.Body.Bytes(), &item); err != nil {
        t.Fatal(err)
    }
    if item.ID != "7" || item.Name == "" {
        t.Fatalf("bad body: %+v", item)
    }
}

func TestItemClientServer(t *testing.T) {
    srv := httptest.NewServer(NewAPI())
    t.Cleanup(srv.Close)

    res, err := http.Get(srv.URL + "/items/9")
    if err != nil {
        t.Fatal(err)
    }
    defer res.Body.Close()
    body, _ := io.ReadAll(res.Body)
    if res.StatusCode != 200 {
        t.Fatalf("status=%d body=%s", res.StatusCode, body)
    }
    var item Item
    if err := json.Unmarshal(body, &item); err != nil {
        t.Fatal(err)
    }
    if item.ID != "9" {
        t.Fatalf("id=%s", item.ID)
    }
}
```

Expected output:

```
=== RUN   TestItemHandlerRecorder
--- PASS: TestItemHandlerRecorder (0.00s)
=== RUN   TestItemClientServer
--- PASS: TestItemClientServer (0.00s)
PASS
```

More examples

Env-gated "external" test and method-not-allowed check.

```go
// api_extra_test.go
package main
```

```go
import (
    "net/http"
    "net/http/httptest"
    "os"
    "testing"
)

func TestPostRejected(t *testing.T) {
    req := httptest.NewRequest(http.MethodPost, "/items/1", nil)
    req.SetPathValue("id", "1")
    rec := httptest.NewRecorder()
    ItemHandler(rec, req)
    if rec.Code != http.StatusMethodNotAllowed {
        t.Fatalf("status=%d", rec.Code)
    }
}

func TestExternalURL_SkipIfUnset(t *testing.T) {
    if os.Getenv("API_URL") == "" {
        t.Skip("API_URL not set")
    }
    // Would hit a real service when configured.
    t.Log("API_URL=", os.Getenv("API_URL"))
}
```

```
go test -v
API_URL=http://example.invalid go test -v -run External
```

Runnable example

Save these two files in `usersvc/`. Then:

```
cd usersvc
go mod init example/usersvc
go test -v
go test -short -v
```

`handler.go`:

```go
package usersvc

import (
    "encoding/json"
    "net/http"
    "strings"
```

```go
)

type User struct {
    ID   int    `json:"id"`
    Name string `json:"name"`
}

func UserHandler(w http.ResponseWriter, r *http.Request) {
    if r.Method != http.MethodGet {
        http.Error(w, "method not allowed", http.StatusMethodNotAllowed)
        return
    }
    id := strings.TrimPrefix(r.URL.Path, "/users/")
    if id == "" || id == r.URL.Path {
        http.Error(w, "missing id", http.StatusBadRequest)
        return
    }
    w.Header().Set("Content-Type", "application/json")
    _ = json.NewEncoder(w).Encode(User{ID: 1, Name: "Ada"})
}

func NewMux() http.Handler {
    mux := http.NewServeMux()
    mux.HandleFunc("/users/", UserHandler)
    return mux
}
```

handler_test.go:

```go
package usersvc_test

import (
    "encoding/json"
    "io"
    "net/http"
    "net/http/httptest"
    "testing"

    "example/usersvc"
)

func TestUserHandler(t *testing.T) {
    req := httptest.NewRequest(http.MethodGet, "/users/1", nil)
    rec := httptest.NewRecorder()

    usersvc.UserHandler(rec, req)
```

```go
    res := rec.Result()
    defer res.Body.Close()
    if res.StatusCode != http.StatusOK {
        t.Fatalf("status=%d", res.StatusCode)
    }
    body, _ := io.ReadAll(res.Body)
    assertUserShape(t, body)
}

func TestAPIClientAgainstTestServer(t *testing.T) {
    srv := httptest.NewServer(usersvc.NewMux())
    t.Cleanup(srv.Close)

    res, err := http.Get(srv.URL + "/users/1")
    if err != nil {
        t.Fatal(err)
    }
    defer res.Body.Close()
    if res.StatusCode != http.StatusOK {
        t.Fatalf("status=%d", res.StatusCode)
    }
    body, err := io.ReadAll(res.Body)
    if err != nil {
        t.Fatal(err)
    }
    assertUserShape(t, body)
}

func TestExternalAPI_SkipUnlessConfigured(t *testing.T) {
    if testing.Short() {
        t.Skip("skipping external-style check in short mode")
    }
    // In real suites you might also require API_URL; here we keep it offline.
    req := httptest.NewRequest(http.MethodGet, "/users/1", nil)
    rec := httptest.NewRecorder()
    usersvc.UserHandler(rec, req)
    if rec.Code != http.StatusOK {
        t.Fatalf("status=%d", rec.Code)
    }
}

func assertUserShape(t *testing.T, body []byte) {
    t.Helper()
    var got map[string]any
    if err := json.Unmarshal(body, &got); err != nil {
        t.Fatal(err)
    }
```

```
    if _, ok := got["id"]; !ok {
        t.Fatal("missing id")
    }
    if _, ok := got["name"]; !ok {
        t.Fatal("missing name")
    }
}
```

Expected output:

```
=== RUN   TestUserHandler
--- PASS: TestUserHandler (0.00s)
=== RUN   TestAPIClientAgainstTestServer
--- PASS: TestAPIClientAgainstTestServer (0.00s)
=== RUN   TestExternalAPI_SkipUnlessConfigured
--- PASS: TestExternalAPI_SkipUnlessConfigured (0.00s)
PASS
```

With `go test -short -v`, the last test is `SKIP`.

What to notice: `httptest` exercises handlers without a listening port; `httptest.NewServer` gives a real URL for client code. Assert JSON **shape** (keys/types), not only status codes. Use `-short` / env gates for slower dependency lanes.

Try next: Add a `POST` path that returns 405 and test it; introduce a build-tagged file `//go:build integration` that only runs under `go test -tags=integration`.

Fuzzing and Property-Style Tests

Fuzzing and Property-Style Tests

Overview

Fuzzing feeds random inputs to find panics and logic bugs. Go's built-in fuzzing (`go test -fuzz`) is ideal for parsers, validators, and codecs.

Fuzz target

```go
func FuzzParsePort(f *testing.F) {
    f.Add("8080")
    f.Add("0")
    f.Add("65535")
    f.Fuzz(func(t *testing.T, s string) {
        n, err := strconv.Atoi(s)
        if err != nil {
            return
        }
        if n < 0 {
            t.Fatalf("negative: %d", n)
        }
        // property: round-trip for valid ports
        if n >= 0 && n <= 65535 {
            if strconv.Itoa(n) != s && !strings.HasPrefix(s, "0") && s != "0" {
                // allow leading zeros mismatch - define your contract
            }
        }
    })
}
```

```bash
go test -fuzz=FuzzParsePort -fuzztime=10s
```

Seed corpus

`f.Add` seeds interesting cases; failing inputs land in `testdata/fuzz/`.

Good fuzz targets

Good	Poor
JSON/decode, path clean, URL parse wrappers	Tests needing full DB
Pure functions	Non-deterministic time without injection

Property ideas

- Round-trip: `decode(encode(x)) == x`
- Idempotent clean: `clean(clean(p)) == clean(p)`
- Never panics on any byte slice

Rules of thumb

Do	Don't
Fix corpus failures as tests	Delete corpus to "go green"
Bound work in fuzz body	Sleep/network in fuzz
Run in CI with `-fuzztime` short	Only fuzz manually once

Try next

1. Fuzz your `safeJoin` — should never escape root.
2. Fuzz JSON validator.
3. Commit a minimized failing corpus once fixed.

httptest and API Testing Patterns

httptest and API Testing Patterns

Overview

`net/http/httptest` tests handlers without a real network. Combine with table tests and fake dependencies for fast API coverage.

Handler unit test

```
func TestHealth(t *testing.T) {
    req := httptest.NewRequest(http.MethodGet, "/healthz", nil)
    rec := httptest.NewRecorder()
    healthz(rec, req)
    if rec.Code != 200 {
        t.Fatalf("code %d", rec.Code)
    }
}
```

Full mux

```
srv := NewServer(fakeStore)
req := httptest.NewRequest(http.MethodGet, "/api/books", nil)
rec := httptest.NewRecorder()
srv.Handler().ServeHTTP(rec, req)
```

JSON helpers

```
func decodeJSON[T any](t *testing.T, rec *httptest.ResponseRecorder) T {
    t.Helper()
    var v T
    if err := json.NewDecoder(rec.Body).Decode(&v); err != nil {
        t.Fatal(err)
    }
    return v
}
```

Auth cookies

```go
req.AddCookie(&http.Cookie{Name: "session_id", Value: sid})
```

httptest.Server (integration-ish)

```go
// Go 1.27+: in-memory network, t.Cleanup, synctest-safe.
ts := httptest.NewTestServer(t, handler)
res, err := ts.Client().Get("http://example.com/api/books")
```

Prefer `NewTestServer` for new tests. `httptest.NewServer` still listens on loopback when you need a real TCP URL:

```go
ts := httptest.NewServer(handler)
t.Cleanup(ts.Close)
res, err := http.Get(ts.URL + "/api/books")
```

Still in-process; good middle ground before Testcontainers.

Rules of thumb

Do	Don't
Fake stores at interfaces	Real network in unit tests
Table-drive status codes	One mega test for all routes
`t.Helper` on asserts	Ignore body on 500s

Try next

1. Table test 200/400/401 for create book.
2. Cookie session test.
3. Assert `Content-Type` on JSON APIs.

Golden Files and testdata

Golden Files and testdata

Overview

Golden files store expected output (JSON, text, CLI stdout). Update deliberately when format changes.

Layout

```
mypkg/
  format.go
  format_test.go
  testdata/
    report.golden
    cases/
      in.json
```

Pattern

```go
func TestReport(t *testing.T) {
    got := RenderReport(sample)
    path := filepath.Join("testdata", "report.golden")
    if os.Getenv("UPDATE_GOLDEN") == "1" {
        if err := os.WriteFile(path, got, 0o644); err != nil {
            t.Fatal(err)
        }
    }
    want, err := os.ReadFile(path)
    if err != nil {
        t.Fatal(err)
    }
    if !bytes.Equal(got, want) {
        t.Fatalf("mismatch (-want +got)\n%s", diff(want, got))
    }
}
```

```
UPDATE_GOLDEN=1 go test ./...
go test ./...
```

Tips

- Normalize time/randomness (inject clock)
- Prefer stable JSON key order (`json.Encoder` in v1, or `jsonv2.Deterministic(true)` in `encoding/json/v2` where maps are unsorted by default)
- **Go 1.27 `compress/flate`:** The encoder was overhauled in Go 1.27. Outputs from `archive/zip`, `compress/gzip`, `compress/zlib`, and `image/png` can differ byte-for-byte from Go 1.26 while decoding identically. When upgrading to 1.27, regenerate compressed fixtures (`UPDATE_GOLDEN=1`) instead of pinning old bytes.
- Review golden diffs in PRs carefully

embed for fixtures

```
//go:embed testdata/*.json
var fixtures embed.FS
```

Rules of thumb

Do	Don't
Commit goldens	Regenerate without reading diff
Keep fixtures small	50MB binaries in git
Document UPDATE_GOLDEN	Hidden magic env names only in CI lore

Try next

1. Golden-test a CLI help string.
2. Golden-test pretty JSON error envelope.
3. Break format; see test fail; update consciously.

Project: Service Test Suite

Project: Service Test Suite

Overview

For Bookstore (or any API), build a layered suite.

Layers

Layer	Tools	Speed
Domain unit	table tests	ms
Handler	httptest + fakes	ms
Fuzz	validators/parsers	CI timed
Golden	error JSON / reports	ms
Integration	optional Testcontainers	slower

Deliverables

- ☐ `internal/domain/*_test.go`
- ☐ `internal/httpapi/*_test.go` with auth cases
- ☐ One fuzz target
- ☐ One golden file
- ☐ `make test` → `go test ./...`
- ☐ `make test-race` → `go test -race ./...`

Definition of done

`go test ./...` green; race clean; README documents how to run fuzz for 10s.

Standard Library Tour

Standard Library Tour Overview

Standard Library Tour Overview

This part is a **stdlib-first map** of the packages you will use on almost every real Go program. Frameworks and third-party libraries come later; fluency here is what makes those tools feel thin and optional.

Earlier parts already teach language mechanics. Here the unit of study is the **package contract**: what it guarantees, what it leaves to you, and which defaults are production-safe.

Why This Part Exists

Most "I know Go" gaps are not syntax. They are:

- Using `fmt.Println` where `log/slog` belongs
- Loading whole files when `io` streams would bound memory
- `http.Get` with no timeout
- Stringly-typed paths instead of `filepath` / `io/fs`
- Hand-rolled JSON error handling that loses type safety
- Concurrency without `context` deadlines or `sync` ownership rules

The standard library already solves these — if you know where to look.

Learning Path

Chapter	Packages	Outcome
981 fmt / log / slog	`fmt`, `log`, `log/slog`	Readable output and structured events
982 Text & numbers	`bytes`, `strings`, `strconv`, `unicode`, `utf8`	Correct UTF-8 and conversions
983 I/O composition	`io`, `bufio`	Bounded streams and efficient copies
984 Files & embed	`os`, `path/filepath`, `io/fs`, `embed`	Safe paths and single-binary assets
985 Encoding	`encoding/json`, `encoding/json/v2`, `csv`, `xml`, `base64`, `hex`	Data at rest and on the wire
986 net/http	`net/http`, `net/url`	Production-shaped clients and servers

Chapter	Packages	Outcome
987 CLI & process	`flag`, `os/exec`, `os/signal`	Tools that parse, run, and shut down cleanly
988 time & context	`time`, `context`	Budgets, clocks, and cancellation
989 sync & atomic	`sync`, `sync/atomic`	Shared state without races
990 Collections	`slices`, `maps`, `cmp`, `iter`	Modern helpers (Go 1.21+)
991 testing	`testing`, `testing/fstest`, `net/http/httptest`	Tests, examples, fuzz, benches
992 Crypto & text scan	`crypto/*`, `uuid`, `hash`, `math/rand/v2`, `regexp`	Integrity, IDs, secrets hygiene, matching
993 database/sql	`database/sql`	Pools, queries, transactions
994 net dial/listen	`net`	TCP/UDP/Unix under HTTP
995 templates	`text/template`, `html/template`	Safe rendering and codegen
996 compress/archive	`compress/*`, `archive/zip`, `tar`	gzip, zip, tar streams
997 binary / gob / pem	`encoding/binary`, `gob`, `pem`	Frames, Go blobs, cert armor
998 math / bits / big	`math`, `math/bits`, `math/big`	Floats, bit ops, big integers
999 mime / ops	`mime`, `multipart`, `expvar`, `runtime/debug`	Uploads, debug vars, build info
999a containers	`container/heap`, `list`, `ring`	Priority queues, lists
999b path/filepath	`path`, `path/filepath`	OS paths vs slash paths, jails
999c bufio advanced	`bufio` Scanner/Reader/Writer	Long lines, peek, flush
999d crypto/mldsa	`crypto/mldsa`	FIPS 204 ML-DSA post-quantum signatures (Go 1.27+)
999e crypto/mlkem	`crypto/mlkem`	FIPS 203 ML-KEM post-quantum key encapsulation (Go 1.24+)

Mental Model

```
input  -> decode / parse  -> domain logic -> encode / write -> output
              |                   |                |
           encoding/*          context          io / os / net
           flag / json         time             slog for events
```

Almost every program is that pipeline with different endpoints (CLI, HTTP, file, pipe).

Package Selection Cheatsheet

Need	Prefer	Avoid by default
Logs for ops	`log/slog`	`fmt.Println` in libraries
HTTP service	`net/http` + std mux (1.22+)	Framework until you need it
SQL access	`database/sql` + driver	String-built queries
Raw sockets	`net.Dialer` / `Listen` + deadlines	Untimed `Accept`/`Read` loops
HTML pages	`html/template`	`text/template` for untrusted HTML
gzip / zip / tar	`compress/gzip`, `archive/*`	Shelling out for common formats
Binary frames	`encoding/binary` + max length	Trust remote length blindly
Go-to-Go cache	`gob` (versioned)	gob as a public multi-lang API
Config flags	`flag` (or small wrapper)	Global env soup without docs
File walk	`fs.WalkDir` / `filepath.WalkDir`	Manual recursion without errors
JSON API	`encoding/json`	`interface{}` everywhere
Timeouts	`context` + `http.Client.Timeout`	Fire-and-forget goroutines
Shared map	`sync.Mutex` or `sync.Map` (when justified)	Unsynchronized maps
Sort / search	`slices`	Hand-rolled $O(n^2)$ loops
Big integers	`math/big`	`float64` for money / huge ints
Uploads	`mime/multipart` + size limits	Unbounded `ParseMultipartForm`
Secrets material	`crypto/rand`, careful logging	`math/rand` for tokens

Relationship to Other Parts

Topic	Deeper part
Streaming I/O patterns	12 Specialized — I/O
Concurrency design	07 Concurrency
HTTP resilience	13 Network systems
TCP services	13 Network systems — TCP
SQL pools	20 Deep dives — database/sql
Web templates / DB	22 Web development
slog in production	16 Observability
TLS / crypto hardening	17 Security
Profiling	18 Performance

Use this part as the **default toolkit**. Jump to specialized parts when the problem is no longer

"which package?" but "how does this fail under load?"

How to Study

1. Read the index map; pick one vertical (CLI tool or tiny HTTP service).
2. For each chapter, run the runnable example once, then change one assumption (timeout, limit, encoding).
3. Rebuild a small project using **only** stdlib until you feel friction — that friction is when a library earns its place.

```
stdlib literacy loop

read package docs -> run example -> break limits -> fix with contracts -> ship tiny tool
```

Checklist: Stdlib Literacy

- ☐ Can structure logs with `slog` and attach request-scoped fields
- ☐ Can stream with `io.Copy` / `LimitReader` instead of unbounded `ReadAll`
- ☐ Can open, walk, and embed files without path traversal bugs
- ☐ Can marshal/unmarshal JSON with sensible error surfaces
- ☐ Can serve and call HTTP with timeouts and context
- ☐ Can build a multi-flag CLI that exits non-zero on failure
- ☐ Can cancel work with `context` and wait with `sync.WaitGroup`
- ☐ Can write table tests, an example, and a basic fuzz target
- ☐ Can generate tokens with `crypto/rand` and hash with `crypto/sha256`
- ☐ Can open `database/sql`, query with context, and map `ErrNoRows`
- ☐ Can dial/listen on TCP with deadlines (not only HTTP)
- ☐ Can render HTML safely with `html/template` (parse once)
- ☐ Can gzip a stream and extract zip/tar without path traversal
- ☐ Can write a length-prefixed binary frame with a max size
- ☐ Can accept a size-limited multipart upload
- ☐ Can expose version/`expvar` or read `debug.ReadBuildInfo`

When most boxes are checked, third-party libraries become choices — not crutches.

Suggested extensions path

```
Core loop (981-992)     text → io → files → json → http → cli → time → sync → tests → crypto
Data & wire (993-997)   sql → net → templates → compress → binary/gob/pem
Math & ops (998-999)    bits/big → mime/uploads → expvar/pprof/buildinfo
Extras (999a-c)         container heap/list/ring → path/filepath → bufio advanced
```

fmt, log, and log/slog

fmt, log, and log/slog

Overview

Three packages cover **output**:

Package	Role
`fmt`	Formatting values for humans and strings
`log`	Legacy line logger (still fine for tiny tools)
`log/slog`	Structured logging (Go 1.21+ default for services)

Rule of thumb: **libraries return errors; programs log**. Prefer `slog` for anything that runs as a service or daemon.

fmt Essentials

```
fmt.Print("a", 1)           // a1  (no spaces between args of different kinds as you'd expec
fmt.Println("hello", 42)    // hello 42\n
s := fmt.Sprintf("%s=%d", "n", 3)
fmt.Fprintf(w, "status=%v\n", err)
```

Verbs you actually need

Verb	Meaning
`%v`	Default format
`%+v`	Struct with field names
`%#v`	Go-syntax representation
`%T`	Type
`%s` / `%q`	String / quoted
`%d` / `%x`	Decimal / hex
`%f` / `%g`	Float
`%w`	Wrap error (`fmt.Errorf`)
`%+v` on errors	Implementation-defined detail

```
err := fmt.Errorf("open %s: %w", path, err)
```

Strings and builders

For hot paths assembling many pieces, prefer `strings.Builder` or `fmt.Fprintf` into a builder — not repeated `+` in a loop.

log (legacy)

```
import "log"

log.SetFlags(log.LstdFlags | log.Lshortfile)
log.Println("starting")
log.Printf("port=%d", port)
log.Fatal(err) // log + os.Exit(1)
```

Fine for CLIs and examples. Lacks structured fields and levels as first-class data.

log/slog (preferred)

```
import "log/slog"

slog.Info("server start", "addr", addr)
slog.Error("query failed", "err", err, "query", q)
```

Handlers

```
// Human text (dev)
slog.SetDefault(slog.New(slog.NewTextHandler(os.Stderr, &slog.HandlerOptions{
    Level: slog.LevelDebug,
})))

// JSON (prod)
slog.SetDefault(slog.New(slog.NewJSONHandler(os.Stdout, &slog.HandlerOptions{
    Level: slog.LevelInfo,
})))
```

Levels and groups

```
slog.Debug("cache hit", "key", key)
slog.Warn("retrying", "attempt", n)
```

```go
logger := slog.Default().With("service", "billing")
logger.Info("charge", "user_id", id, "cents", 199)
```

Context-aware logging

```go
func handle(ctx context.Context, w http.ResponseWriter, r *http.Request) {
    log := slog.Default()
    if rid, ok := ctx.Value(requestIDKey).(string); ok {
        log = log.With("request_id", rid)
    }
    log.Info("request", "path", r.URL.Path, "method", r.Method)
}
```

For production correlation patterns, see Structured logging.

Anti-patterns

1. **Logging and returning the same error without a policy** — pick one place that owns the user-visible message.
2. **Logging secrets** — tokens, passwords, full auth headers.
3. **`fmt.Sprintf` then log the string** — lose structure; pass key-value pairs.
4. **`log.Fatal` deep in libraries** — exits the process; return `error` instead.

Summary

Situation	Use
Build a message string	`fmt.Sprintf` / `Errorf`
One-off script noise	`log` or `fmt`
Service / long-running tool	`log/slog`
Wrap errors	`fmt.Errorf("...: %w", err)`

Runnable example

Save as `main.go`:

```go
go mod init example
go run .
```

```go
package main

import (
    "fmt"
    "log/slog"
    "os"
)

func main() {
    // fmt: formatting
    name := "gopher"
    fmt.Printf("hello %s\n", name)
    fmt.Printf("type=%T value=%#v\n", name, struct{ N int }{N: 7})

    err := fmt.Errorf("open config: %w", os.ErrNotExist)
    fmt.Printf("wrapped: %v\n", err)

    // slog text
    text := slog.New(slog.NewTextHandler(os.Stdout, &slog.HandlerOptions{Level: slog.LevelDe
    text.Debug("boot", "pid", os.Getpid())
    text.Info("ready", "port", 8080)

    // slog JSON
    jsonLog := slog.New(slog.NewJSONHandler(os.Stdout, nil))
    jsonLog.Info("request",
        "method", "GET",
        "path", "/health",
        "status", 200,
        "ms", 3,
    )

    // grouped attrs
    svc := jsonLog.With("service", "demo")
    svc.Warn("slow dependency", "dep", "db", "ms", 120)
}
```

Expected output (timestamps/pid vary):

```
hello gopher
type=string value=struct { N int }{N:7}
wrapped: open config: file does not exist
level=DEBUG msg=boot pid=...
level=INFO msg=ready port=8080
{"time":"...","level":"INFO","msg":"request","method":"GET","path":"/health","status":200,"m
{"time":"...","level":"WARN","msg":"slow dependency","service":"demo","dep":"db","ms":120}
```

What to notice: - `%#v` is invaluable when debugging unexpected types. - `slog` fields are queryable; free-form strings are not. - Same event shape works for text (local) and JSON (shipped).

Try next: Add a custom `slog.Handler` wrapper that redacts any attribute key named `password` or `token`.

bytes, strings, strconv, unicode

bytes, strings, strconv, unicode

Overview

Text in Go is **UTF-8 bytes** with optional rune interpretation. This chapter is the stdlib toolkit for slicing, searching, converting, and validating that text without corrupting multi-byte characters.

Related language intro: Strings.

Package map

Package	Use for
`strings`	Immutable string ops, builders, readers
`bytes`	Same ops on `[]byte` (often I/O buffers)
`strconv`	Parse/format numbers and quotes
`unicode` / `unicode/utf8`	Character classes and UTF-8 decode

Many `strings` functions have `bytes` twins (`Contains`, `Split`, `TrimSpace`, …). Prefer `[]byte` when you already have buffers from I/O.

strings cookbook

```
strings.Contains(s, sub)
strings.HasPrefix(s, "http")
strings.Cut(s, "=")              // before, after, ok  (Go 1.18+)
strings.CutPrefix(s, "Bearer ")  // Go 1.20+
strings.CutLast(s, "/")          // last separator (Go 1.27+); bytes.CutLast twin
strings.Fields(s)                // split on unicode whitespace
strings.Join(parts, ",")
strings.ReplaceAll(s, old, new)
strings.ToValidUTF8(s, "\uFFFD")
```

Builder vs buffer

```
var b strings.Builder
b.Grow(64)
b.WriteString("id=")
b.WriteString(id)
out := b.String() // copies once
```

`bytes.Buffer` implements `io.Writer` and is better when mixing writes from encoders.

bytes cookbook

```
bytes.Contains(buf, []byte("ERROR"))
bytes.Split(buf, []byte{'\n'})
bytes.TrimSpace(buf)
reader := bytes.NewReader(buf) // io.Reader, io.Seeker
```

strconv: explicit conversions

```
n, err := strconv.Atoi("42")
n64, err := strconv.ParseInt("ff", 16, 64)
f, err := strconv.ParseFloat("3.14", 64)
b, err := strconv.ParseBool("true")

s := strconv.Itoa(42)
s = strconv.FormatInt(255, 16)          // "ff"
s = strconv.FormatFloat(f, 'f', 2, 64)  // "3.14"
q := strconv.Quote("a\nb")               // "\"a\\nb\""
```

Never use `fmt.Sprintf("%d", n)` in hot paths when `strconv` will do — it is clearer and often faster.

unicode and utf8

```
import (
    "unicode"
    "unicode/utf8"
)

utf8.RuneCountInString(s)
r, size := utf8.DecodeRuneInString(s)
```

```
unicode.IsLetter(r)
unicode.Is(unicode.Latin, r)
```

`range` over a string already yields runes; use `utf8` when decoding partial buffers from the network.

Validation pattern

```
func isIdent(s string) bool {
    if s == "" {
        return false
    }
    for i, r := range s {
        if i == 0 && !unicode.IsLetter(r) && r != '_' {
            return false
        }
        if i > 0 && !unicode.IsLetter(r) && !unicode.IsDigit(r) && r != '_' {
            return false
        }
    }
    return utf8.ValidString(s)
}
```

Summary

Task	API
Search / split strings	`strings.*`
Same on buffers	`bytes.*`
Number parse/format	`strconv.*`
Rune properties	`unicode.*`
Partial UTF-8	`unicode/utf8`

Runnable example

```
go mod init example
go run .
```

```
package main

import (
    "bytes"
```

```go
    "fmt"
    "strconv"
    "strings"
    "unicode"
    "unicode/utf8"
)

func main() {
    s := "name=ada; role=admin"
    k, v, ok := strings.Cut(s, "; ")
    fmt.Println("cut:", k, "|", v, ok)

    var b strings.Builder
    b.WriteString("user:")
    b.WriteString(strings.ToUpper("ada"))
    fmt.Println("builder:", b.String())

    raw := []byte("  line1\nline2\n")
    fmt.Printf("trim=%q fields=%q\n", bytes.TrimSpace(raw), bytes.Fields(raw))

    n, err := strconv.ParseInt("2a", 16, 64)
    fmt.Println("parse:", n, err, "format:", strconv.FormatInt(n, 10))

    msg := "Go "
    fmt.Println("bytes", len(msg), "runes", utf8.RuneCountInString(msg))
    for _, r := range msg {
        fmt.Printf(" %q letter=%v\n", r, unicode.IsLetter(r))
    }

    fmt.Println("valid utf8?", utf8.ValidString("ok\xff"))
}
```

Expected output:

```
cut: name=ada | role=admin true
builder: user:ADA
trim="line1\nline2" fields=["line1" "line2"]
parse: 42 <nil> format: 42
bytes 5 runes 3
 'G' letter=true
 'o' letter=true
 ' ' letter=false
valid utf8? false
```

What to notice: - `Cut` is clearer than `SplitN` for single separators. - `bytes` and `strings` mirror each other — pick based on your data form. - Invalid UTF-8 is detectable; do not assume network input is clean.

Try next: Write `splitKV(s string) (map[string]string, error)` using `strings.Cut` on `=` and `;` pairs, rejecting duplicates.

io and bufio Composition

io and bufio Composition

Overview

The `io` package is Go's **streaming contract**. Small interfaces (`Reader`, `Writer`, `Closer`) let files, sockets, compressors, and crypto share the same pipelines.

Deep patterns also appear in I/O and Streaming. This chapter is the stdlib composition cheat-sheet.

Core contracts

```go
type Reader interface {
    Read(p []byte) (n int, err error)
}
type Writer interface {
    Write(p []byte) (n int, err error)
}
```

- Process `n > 0` **before** checking `err` (including `io.EOF`).
- `io.EOF` means end of stream, not a hard failure for `Read` loops.
- Prefer `io.Copy` over manual loops — it uses `WriterTo` / `ReaderFrom` when available.

Helpers worth memorizing

Helper	Purpose
`io.Copy` / `CopyBuffer`	Stream A → B efficiently
`io.ReadAll`	Entire stream to memory (cap it!)
`io.LimitReader`	Hard cap bytes read
`io.TeeReader`	Read + fork to a Writer (checksums)
`io.MultiReader`	Concatenate readers
`io.MultiWriter`	Fan-out writes
`io.Pipe`	In-process streaming between goroutines
`io.NopCloser`	Wrap Reader as ReadCloser
`io.Discard`	Drain without storing

```go
const max = 1 << 20 // 1 MiB
data, err := io.ReadAll(io.LimitReader(r, max+1))
if err != nil {
    return err
}
if len(data) > max {
    return fmt.Errorf("payload exceeds %d bytes", max)
}
```

bufio: amortize syscalls

```go
br := bufio.NewReader(f)
line, err := br.ReadString('\n')

bw := bufio.NewWriter(f)
fmt.Fprintln(bw, "hello")
if err := bw.Flush(); err != nil {
    return err
}
```

Type	Role
`bufio.Reader`	Peek, ReadBytes, ReadString
`bufio.Writer`	Buffered writes; always `Flush`
`bufio.Scanner`	Tokenize lines/words (watch token size)

```go
sc := bufio.NewScanner(f)
// optional: sc.Buffer(buf, 1024*1024) for long lines
for sc.Scan() {
    line := sc.Text()
    _ = line
}
return sc.Err()
```

Default `Scanner` max token is 64KiB — raise it for pathological log lines or reject them.

Composition examples

Checksum while uploading

```go
h := sha256.New()
tr := io.TeeReader(src, h)
if _, err := io.Copy(dst, tr); err != nil {
    return err
}
sum := h.Sum(nil)
```

Bound + discard remainder

```go
// Read at most n bytes; ensure caller can still Close the body
lr := io.LimitReader(resp.Body, n)
_, err := io.Copy(dst, lr)
```

Pitfalls

1. Forgetting `Close` on `resp.Body` / files → connection or FD leaks.
2. Unbounded `ReadAll` on user input → memory exhaustion.
3. Nested `bufio.Reader` without understanding buffered leftovers.
4. Ignoring short writes from custom `Writer`s.

Runnable example

```bash
go mod init example
go run .
```

```go
package main

import (
    "bufio"
    "bytes"
    "crypto/sha256"
    "encoding/hex"
    "fmt"
    "io"
    "strings"
)
```

```go
func main() {
    src := strings.NewReader("hello stream\nsecond line\n")

    // LimitReader + ReadAll
    limited, _ := io.ReadAll(io.LimitReader(src, 5))
    fmt.Printf("limited=%q\n", limited)

    // reset-like: new reader
    src = strings.NewReader("hello stream\nsecond line\n")
    h := sha256.New()
    var out bytes.Buffer
    tr := io.TeeReader(src, h)
    if _, err := io.Copy(&out, tr); err != nil {
        panic(err)
    }
    fmt.Printf("copied=%q\n", out.Bytes())
    fmt.Println("sha256", hex.EncodeToString(h.Sum(nil))[:16]+"...")

    // Scanner lines
    sc := bufio.NewScanner(strings.NewReader("a\nb\nc\n"))
    for sc.Scan() {
        fmt.Println("line:", sc.Text())
    }
    fmt.Println("scan err:", sc.Err())

    // MultiWriter
    var a, b bytes.Buffer
    mw := io.MultiWriter(&a, &b)
    io.WriteString(mw, "x")
    fmt.Printf("a=%q b=%q\n", a.String(), b.String())
}
```

Expected output:

```
limited="hello"
copied="hello stream\nsecond line\n"
sha256 f74a77476883f531...
line: a
line: b
line: c
scan err: <nil>
a="x" b="x"
```

What to notice: - `LimitReader` stops early without error when the cap is hit mid-stream. - `TeeReader` is the clean way to hash or log while still delivering bytes. - `Scanner` is for tokens; `io.Copy` is for bulk.

Try next: Use `io.Pipe` with two goroutines: one writes lines, one counts them with

`bufio.Scanner`.

os, filepath, io/fs, and embed

os, filepath, io/fs, and embed

Overview

These packages cover **process environment and files**:

Package	Role
`os`	Files, env, args, process exit, working directory
`path/filepath`	OS-aware path manipulation
`io/fs`	Abstract filesystem interface
`embed`	Ship files inside the binary

Path safety and portable tools depend on getting this layer right.

os basics

```go
data, err := os.ReadFile("config.json")     // Go 1.16+
err = os.WriteFile("out.txt", data, 0o644)
f, err := os.OpenFile(path, os.O_RDWR|os.O_CREATE, 0o644)
info, err := os.Stat(path)
os.MkdirAll("a/b/c", 0o755)
os.Remove(path)
os.RemoveAll(dir) // careful
```

Env and args

```go
os.Args          // argv; Args[0] is program name
os.Getenv("HOME")
os.LookupEnv("PORT") // value, ok
os.Exit(1)       // skips deferred cleanup in other goroutines
```

Prefer returning errors to `main` and a single `os.Exit` there.

filepath: always OS-aware

Use `path/filepath` for local disk paths. Use `path` (slash-only) for URL-like paths.

```go
filepath.Join("a", "b", "c.txt")
filepath.Clean("../etc/passwd")
filepath.Abs(".")
filepath.Ext("main.go")      // ".go"
filepath.Base("/tmp/x")      // "x"
filepath.Dir("/tmp/x")       // "/tmp"
filepath.WalkDir(root, fn)   // preferred over Walk
```

Prevent path traversal

```go
func safeJoin(root, name string) (string, error) {
    clean := filepath.Clean("/" + name) // force absolute-like clean
    rel := strings.TrimPrefix(clean, string(filepath.Separator))
    full := filepath.Join(root, rel)
    rootAbs, err := filepath.Abs(root)
    if err != nil {
        return "", err
    }
    fullAbs, err := filepath.Abs(full)
    if err != nil {
        return "", err
    }
    sep := string(os.PathSeparator)
    if fullAbs != rootAbs && !strings.HasPrefix(fullAbs, rootAbs+sep) {
        return "", fmt.Errorf("path escapes root")
    }
    return fullAbs, nil
}
```

io/fs abstraction

```go
var fsys fs.FS = os.DirFS("/data")
b, err := fs.ReadFile(fsys, "hello.txt")
entries, err := fs.ReadDir(fsys, ".")
err = fs.WalkDir(fsys, ".", func(path string, d fs.DirEntry, err error) error {
    return err
})
```

Benefits: tests can swap `fstest.MapFS`; `embed.FS` implements `fs.FS`.

embed

```go
import "embed"

//go:embed static/*
var staticFS embed.FS

//go:embed VERSION
var version string
```

```go
http.Handle("/static/", http.FileServer(http.FS(staticFS)))
```

Rules:

- Paths are relative to the source file.
- No `..` in embed patterns.
- Embedded content is read-only at runtime.

See also codegen / embed.

Runnable example

```bash
go mod init example
go run .
```

```go
package main

import (
    "fmt"
    "io/fs"
    "os"
    "path/filepath"
    "testing/fstest"
)

func main() {
    dir, err := os.MkdirTemp("", "stdlib-fs-*")
    if err != nil {
        panic(err)
    }
    defer os.RemoveAll(dir)

    path := filepath.Join(dir, "note.txt")
    if err := os.WriteFile(path, []byte("hello fs\n"), 0o644); err != nil {
```

```go
        panic(err)
    }

    data, _ := os.ReadFile(path)
    fmt.Printf("disk=%q\n", data)

    // Walk
    _ = filepath.WalkDir(dir, func(p string, d fs.DirEntry, err error) error {
        if err != nil {
            return err
        }
        rel, _ := filepath.Rel(dir, p)
        fmt.Println("walk:", rel, "dir?", d.IsDir())
        return nil
    })

    // Abstract FS (great for tests)
    mapFS := fstest.MapFS{
        "a.txt":  {Data: []byte("A")},
        "b/c.txt": {Data: []byte("C")},
    }
    b, _ := fs.ReadFile(mapFS, "b/c.txt")
    fmt.Printf("mapfs=%q\n", b)

    fmt.Println("join:", filepath.Join("var", "log", "app.log"))
    fmt.Println("clean:", filepath.Clean("a//b/../c"))
}
```

Expected output (temp path names vary):

```
disk="hello fs\n"
walk: . dir? true
walk: note.txt dir? false
mapfs="C"
join: var/log/app.log
clean: a/c
```

What to notice: - `DirFS` / `MapFS` / `embed.FS` share one `fs.FS` API. - `WalkDir` reports the root as `"."` with `IsDir()==true`. - Cleaning alone does not make a path safe relative to a root — join + prefix check.

Try next: Serve `embed.FS` with `http.FileServer(http.FS(...))` and verify a missing file returns 404.

encoding: JSON, CSV, XML, base64

encoding: JSON, CSV, XML, base64

Overview

The `encoding/*` packages turn Go values into bytes and back. JSON dominates APIs; CSV dominates data interchange; base64/hex appear in tokens and digests.

Package	Common use
`encoding/json`	HTTP APIs, config (v1 API; v2 engine in Go 1.27+)
`encoding/json/v2`	New JSON API: options, stricter UTF-8/duplicates
`encoding/csv`	Exports, bulk import
`encoding/xml`	Legacy integrations
`encoding/base64`	Tokens, data URLs
`encoding/hex`	Digests, debug
`encoding/gob`	Go-only RPC caches (avoid for public APIs)

JSON

Struct tags

```go
type User struct {
    ID    int64  `json:"id"`
    Name  string `json:"name"`
    Email string `json:"email,omitempty"`
    Internal string `json:"-"` // never serialized
}
```

Encode / decode

```go
b, err := json.Marshal(user)
err = json.Unmarshal(b, &user)

// Streams (preferred for large or HTTP bodies)
enc := json.NewEncoder(w)
enc.SetIndent("", "  ")
err = enc.Encode(user) // adds trailing newline
```

```go
dec := json.NewDecoder(r)
dec.DisallowUnknownFields() // strict APIs
err = dec.Decode(&user)
```

Useful options

```go
dec.UseNumber() // keep numbers as json.Number
```

Dynamic JSON

Prefer typed structs. When needed:

```go
var m map[string]any
json.Unmarshal(b, &m)
```

Custom marshalers

```go
func (id UserID) MarshalText() ([]byte, error) { /* ... */ }
func (id *UserID) UnmarshalText(b []byte) error { /* ... */ }
```

Or `MarshalJSON` / `UnmarshalJSON` for full control.

encoding/json/v2 (Go 1.27+)

v1 remains supported. v2 is the API you want for **new** strict JSON:

```go
import jsonv2 "encoding/json/v2"

b, err := jsonv2.Marshal(user)
err = jsonv2.Unmarshal(b, &user, jsonv2.RejectUnknownMembers(true))
err = jsonv2.UnmarshalRead(r, &user)
err = jsonv2.MarshalWrite(w, user)
```

v2 rejects invalid UTF-8 and duplicate object names by default, and matches struct fields case-sensitively. The v1 package is now implemented on top of v2, so existing `json.Marshal` / `json.Unmarshal` calls get the faster unmarshal path without an import change. Compatibility problems: `GOEXPERIMENT=nojsonv2`.

CSV

```
r := csv.NewReader(f)
r.FieldsPerRecord = 3
records, err := r.ReadAll()

w := csv.NewWriter(f)
_ = w.Write([]string{"name", "age"})
_ = w.Write([]string{"ada", "36"})
w.Flush()
return w.Error()
```

Always check `w.Error()` after `Flush`.

XML (when you must)

```
type Note struct {
    XMLName xml.Name `xml:"note"`
    To      string   `xml:"to"`
    Body    string   `xml:",chardata"`
}
b, err := xml.MarshalIndent(note, "", "  ")
```

base64 and hex

```
s := base64.StdEncoding.EncodeToString(raw)
raw, err = base64.StdEncoding.DecodeString(s)
s = base64.RawURLEncoding.EncodeToString(raw) // JWT-style

h := hex.EncodeToString(sum[:])
```

Error handling pattern

```
if err := json.NewDecoder(r).Decode(&in); err != nil {
    var syn *json.SyntaxError
    var typeErr *json.UnmarshalTypeError
    switch {
    case errors.As(err, &syn):
        return fmt.Errorf("json syntax at %d: %w", syn.Offset, err)
    case errors.As(err, &typeErr):
```

```
        return fmt.Errorf("json field %s: %w", typeErr.Field, err)
    default:
        return fmt.Errorf("json decode: %w", err)
    }
}
```

Runnable example

```
go mod init example
go run .
```

```
package main

import (
    "bytes"
    "encoding/base64"
    "encoding/csv"
    "encoding/hex"
    "encoding/json"
    "fmt"
    "strings"
)

type Item struct {
    SKU  string `json:"sku"`
    Qty  int    `json:"qty"`
    Note string `json:"note,omitempty"`
}

func main() {
    it := Item{SKU: "ABC", Qty: 2, Note: "rush"}
    var buf bytes.Buffer
    enc := json.NewEncoder(&buf)
    enc.SetEscapeHTML(false)
    _ = enc.Encode(it)
    fmt.Print("json:", buf.String())

    var back Item
    _ = json.NewDecoder(strings.NewReader(buf.String())).Decode(&back)
    fmt.Printf("decoded=%+v\n", back)

    var cbuf strings.Builder
    cw := csv.NewWriter(&cbuf)
    _ = cw.Write([]string{"sku", "qty"})
```

```go
    _ = cw.Write([]string{it.SKU, fmt.Sprintf("%d", it.Qty)})
    cw.Flush()
    fmt.Print("csv:\n", cbuf.String())

    raw := []byte("hello")
    fmt.Println("b64:", base64.StdEncoding.EncodeToString(raw))
    fmt.Println("hex:", hex.EncodeToString(raw))
}
```

Expected output:

```
json:{"sku":"ABC","qty":2,"note":"rush"}
decoded={SKU:ABC Qty:2 Note:rush}
csv:
sku,qty
ABC,2
b64: aGVsbG8=
hex: 68656c6c6f
```

What to notice: - `Encode` adds a newline; handy for NDJSON-style logs/APIs. - CSV writer buffers — `Flush` + `Error` are mandatory. - Pick base64 **variant** deliberately (std vs URL vs raw).

Try next: Decode JSON with `DisallowUnknownFields` and show the error when an extra field is present.

encoding/json/v2 under encoding/json (Go 1.27)

`encoding/json/v2` under `encoding/json` (Go 1.27)

Thesis

Go 1.27 did not merely *add* `encoding/json/v2`. It also **rewired** the existing `encoding/json` package so that v1 calls run on the v2 engine with a compatibility option set (`DefaultOptionsV1`). Your import paths can stay on v1 forever under the Go 1 compatibility promise; your error *strings* can still change; and the moment you flip an import to `encoding/json/v2`, the **defaults** flip to stricter, more interoperable behaviour. This chapter is the transition guide: what changed even if you never import v2, how to opt in/out, and a side-by-side you can run on Go 1.27+.

It does **not** replace the stdlib survey in `985-encoding-json-csv-xml.qmd` or the performance notes in `261-json-performance-patterns.qmd`. Read those for tags, encoders, and pooling; stay here for the 1.27 semantics shift.

Verified with **Go 1.27.2** (`linux/amd64`) on 2026-10-09. Primary: Go 1.27 release notes, json/v2 migration guide, `encoding/json` "Migrating to v2".

Learning goals

- Explain "v1 API, v2 engine" and when error text can change without an import change
- Produce identical bytes from `encoding/json/v2` via `json.DefaultOptionsV1()`
- List the high-impact default differences (nil slice, `omitempty`, case, duplicates, UTF-8, `time.Duration`, HTML escape)
- Opt out with `GOEXPERIMENT=nojsonv2` and know it is temporary
- Sketch an option-by-option or `jsonsplit` migration for production services
- Note the adjacent Go 1.27 `net/http` `Response.Body.Close` drain behaviour that bites JSON clients

Mental model

```
Go  1.26                          Go 1.27+

encoding/json  → old engine       encoding/json  → json/v2 engine
                                       + DefaultOptionsV1 (implicit)

encoding/json/v2 (experiment)     encoding/json/v2  → strict defaults
encoding/json/jsontext            encoding/json/jsontext (tokens)
```

Knob	Effect
Keep `import "encoding/json"`	Behaviour preserved; error **message text** may differ; faster unmarshal
`import jsonv2 "encoding/json/v2"`	New API + **strict defaults**
`jsonv2.Marshal(v, json.DefaultOptionsV1())`	v2 API, v1 semantics
Later options after `DefaultOptionsV1()`	Override one behaviour at a time
`GOEXPERIMENT=nojsonv2` at **build** time	Restore pre-1.27 engine behind v1 (escape hatch; expected to vanish)

`DefaultOptionsV1` (from `encoding/json`) is the full boolean bundle that defines v1 semantics — including `FormatNilMapAsNull`, `OmitEmptyWithLegacySemantics`, `FormatDurationAsNano`, `MatchCaseSensitiveDelimiter`, and friends. Later options win.

Case 0 — toolchain

```
go version
# go version go1.27.2 linux/amd64
```

If you are still on 1.26, read this chapter against the published 1.27 notes and upgrade before treating the listings as executable.

Case 1 — same struct, four marshal paths

```go
// json_transition.go - v1 vs v2 vs Options
package main

import (
    jsonv1 "encoding/json"
    "encoding/json/jsontext"
    jsonv2 "encoding/json/v2"
    "fmt"
```

```go
    "time"
)

type User struct {
    Name    string   `json:"name"`
    Tags    []string `json:"tags"`
    Admin   bool     `json:"admin,omitempty"`
    Retries int      `json:"retries,omitempty"`
}

func show(label string, b []byte, err error) {
    if err != nil {
        fmt.Printf("%-12s ERROR: %v\n", label, err)
        return
    }
    fmt.Printf("%-12s %s\n", label, b)
}

func main() {
    u := User{Name: "ana"} // Tags==nil, Admin==false, Retries==0

    b, err := jsonv1.Marshal(u)
    show("v1", b, err)
    b, err = jsonv2.Marshal(u)
    show("v2", b, err)
    b, err = jsonv2.Marshal(u, jsonv1.DefaultOptionsV1())
    show("v2+OptsV1", b, err)
    b, err = jsonv2.Marshal(u, jsonv1.DefaultOptionsV1(),
        jsonv2.FormatNilSliceAsNull(false))
    show("v2 mix", b, err)

    _ = jsontext.AllowInvalidUTF8 // used below
    _ = time.Second
}
```

```
v1           {"name":"ana","tags":null}
v2           {"name":"ana","tags":[],"admin":false,"retries":0}
v2+OptsV1    {"name":"ana","tags":null}
v2 mix       {"name":"ana","tags":[]}
```

Three lessons in four lines:

1. **Nil slice:** v1 → JSON `null`; v2 → `[]`.

2. **`omitempty`:** v1 drops `false` / `0`; v2's `omitempty` drops a field only when it would encode as `null`, `""`, `{}`, or `[]` — so `false` and `0` reappear. Prefer **`omitzero`** on bools/numbers/pointers/interfaces (same meaning in both packages).

3. **`DefaultOptionsV1` then override** is the supported gradual path (`v2 mix` keeps v1 options except nil-slice formatting).

Case 2 — unmarshal mismatches that hide in green tests

```go
// continue in main(), after the marshal demos
fmt.Println("== case-insensitive field names ==")
in := []byte(`{"NAME":"bob"}`)
var u1, u2 User
fmt.Printf("v1 name=%q err=%v\n", u1.Name, jsonv1.Unmarshal(in, &u1))
// reset u1 for clarity
u1 = User{}
_ = jsonv1.Unmarshal(in, &u1)
_ = jsonv2.Unmarshal(in, &u2)
fmt.Printf("v1 name=%q\n", u1.Name)
fmt.Printf("v2 name=%q\n", u2.Name)

fmt.Println("== duplicate names ==")
dup := []byte(`{"name":"alice","name":"mallory"}`)
var d1, d2 User
err1 := jsonv1.Unmarshal(dup, &d1)
err2 := jsonv2.Unmarshal(dup, &d2)
fmt.Printf("v1 name=%q err=%v\n", d1.Name, err1)
fmt.Printf("v2 name=%q err=%v\n", d2.Name, err2)

fmt.Println("== invalid UTF-8 ==")
bad := []byte{'{', '"', 'n', 'a', 'm', 'e', '"', ':', '"', 'c', 'a', 'f', 0xff, 'e', '"', '}
var x1, x2 User
fmt.Printf("v1 err=%v name=%q\n", jsonv1.Unmarshal(bad, &x1), x1.Name)
fmt.Printf("v2 err=%v\n", jsonv2.Unmarshal(bad, &x2))
fmt.Printf("v2+allow err=%v\n",
    jsonv2.Unmarshal(bad, &x2, jsontext.AllowInvalidUTF8(true)))
```

Actual Go 1.27.2 output:

```
== case-insensitive field names ==
v1 name="bob"
v2 name=""
== duplicate names ==
v1 name="mallory" err=<nil>
v2 name="alice" err=jsontext: duplicate object member name "name"
== invalid UTF-8 ==
v1 err=<nil> name="caf e"
v2 err=jsontext: invalid UTF-8 within "/name" after offset 12
```

```
v2+allow err=<nil>
```

Behaviour	v1 default	v2 default	Compatibility lever
Field name match	Case-insensitive	Case-sensitive	`jsonv2.MatchCaseInsensitiveNames(true)` or tag `case:ignore`
Duplicate object names	Last wins	Error	`jsontext.AllowDuplicateNames(true)`
Invalid UTF-8 in strings	Replace with U+FFFD	Error	`jsontext.AllowInvalidUTF8(true)`

Trap: after a failed `Unmarshal`, the destination may be partially written (duplicate-name case kept `"alice"`). Discard it.

Case 3 — `time.Duration` and HTML escaping

```go
type Job struct {
    Name    string        `json:"name"`
    Timeout time.Duration `json:"timeout"`
}
j := Job{Name: "backup", Timeout: 90 * time.Second}
show("v1 dur", jsonv1.Marshal(j))
show("v2 dur", jsonv2.Marshal(j))
show("v2+nano", jsonv2.Marshal(j, jsonv1.FormatDurationAsNano(true)))

type Q struct {
    Q string `json:"q"`
}
q := Q{Q: "<a&b>"}
show("v1 html", jsonv1.Marshal(q))
show("v2 html", jsonv2.Marshal(q))
show("v2+esc", jsonv2.Marshal(q, jsontext.EscapeForHTML(true)))
```

```
v1 dur       {"name":"backup","timeout":90000000000}
v2 dur       ERROR: json: unable to marshal from Go time.Duration within "/timeout": no defa
v2+nano      {"name":"backup","timeout":90000000000}
v1 html      {"q":"\u003ca\u0026b\u003e"}
v2 html      {"q":"<a&b>"}
v2+esc       {"q":"\u003ca\u0026b\u003e"}
```

v2 refuses to invent a `Duration` encoding; pass `FormatDurationAsNano(true)` (defined on the v1 package's Options) or change the field type / custom marshaler. HTML escaping is **off** by default in v2 — turn it on if you embed JSON in HTML.

Portable tags (same bytes under both packages)

```go
type Portable struct {
    Name    string   `json:"name,case:ignore"`
    Tags    []string `json:"tags,omitempty"`
    Admin   bool     `json:"admin,omitzero"`
    Retries int      `json:"retries,omitzero"`
}
```

- `omitempty` on the slice sidesteps `null` vs `[]` by omitting both empty forms.
- `omitzero` replaces legacy `omitempty` on bool/int.
- `case:ignore` keeps accepting `"NAME"` under v2.

Round-trip test idea: `jsonv1.Marshal` and `jsonv2.Marshal` must return identical bytes for your golden structs.

Migration playbook

1. **Upgrade to 1.27, keep v1 imports.** Fix tests that assert on `err.Error()` strings; assert types (`*json.SyntaxError`, …) instead.
2. **Optional escape hatch:** `GOEXPERIMENT=nojsonv2` while you file upstream issues — not a long-term plan.
3. **API migration (safe):** switch call sites to `encoding/json/v2` **with** `json.DefaultOptionsV1()`.
4. **Behaviour migration:** flip one option at a time (`FormatNilSliceAsNull(false)`, …) behind tests.
5. **Production servers:** consider github.com/go-json-experiment/jsonsplit (`CallBothButReturnV1` + difference reporting) before `OnlyCallV2`.
6. **New code:** prefer `encoding/json/v2` defaults; use `RejectUnknownMembers(true)` on untrusted request bodies.

Experiment-era blog posts (GOEXPERIMENT=jsonv2 on 1.25/1.26) may still show removed tag options (`format:`, `unknown:`) or the old `inline` name — 1.27 renamed that to **`embed`** and dropped several experiment-only knobs. Trust `go doc` on 1.27+, not the experiment posts.

Adjacent Go 1.27 note — `Response.Body.Close` drains

Unrelated to the json packages, but it shows up in the same upgrade window for HTTP+JSON clients: Go 1.27's `net/http` **HTTP/1** `Response.Body` now **drains unread content on `Close`**

(up to a conservative limit) to improve connection reuse. For most programs this is a silent win. If you were accidentally leaving huge unread bodies and relying on `MaxIdleConns` quirks, profiles may change — set `Transport.DisableKeepAlives` only if you truly do not want reuse, and prefer reading/decoding the body fully (`json.NewDecoder(resp.Body).Decode` + `Close`) as you already should.

The trap

- **"We did not import v2, so nothing changed."** The engine underneath v1 did. Golden error strings are the usual breakage.
- **Trivial import swap without reading defaults.** Nil slices, `omitempty`, and `Duration` are the first production pages.
- **Keeping partial structs after failed v2 `Unmarshal`.**
- **Relying on `GOEXPERIMENT=nojsonv2` past the migration.** Release notes say it is expected to go away.
- **Copying experiment-era tags** (`format:`, `inline` instead of `embed`).

The boring rule

Stay on `encoding/json` until you need v2's API or defaults. When you move, start with `DefaultOptionsV1()`, peel options off under tests, and pin wire contracts with `omitzero` / `case:ignore` / explicit nil-slice policy. Measure unmarshal wins; do not assume marshal is free on every workload.

Try this

```
go install golang.org/dl/go1.27.2@latest   # or use your distro's 1.27
# put the listings in a module with `go 1.27` and `go run .`
GOEXPERIMENT=nojsonv2 go test ./...       # compare only while diagnosing
```

Sources

- Go 1.27 release notes — json/v2: https://go.dev/doc/go1.27
- Official migration guide: https://go.dev/doc/jsonv2-migration

- `encoding/json` package docs (Migrating to v2 / `DefaultOptionsV1`): https://pkg.go.dev/encoding/json
- `encoding/json/v2`: https://pkg.go.dev/encoding/json/v2
- Background article (Oct 2026; upgrade against go.dev if drift): https://importstatic.com/go/go-json-v2-migration

net/http Standard Library

net/http Standard Library

Overview

`net/http` is a complete HTTP client and server. Since Go 1.22, the standard multiplexer supports **method-aware patterns** (`GET /items/{id}`), which covers most API routing without a framework.

Resilience deep dive: HTTP client/server resilience.

Server (Go 1.22+ patterns)

```go
mux := http.NewServeMux()
mux.HandleFunc("GET /health", func(w http.ResponseWriter, r *http.Request) {
    w.WriteHeader(http.StatusOK)
    _, _ = w.Write([]byte("ok"))
})
mux.HandleFunc("GET /items/{id}", func(w http.ResponseWriter, r *http.Request) {
    id := r.PathValue("id")
    fmt.Fprintln(w, id)
})
mux.HandleFunc("POST /items", createItem)

srv := &http.Server{
    Addr:              ":8080",
    Handler:           mux,
    ReadHeaderTimeout: 5 * time.Second,
    ReadTimeout:       15 * time.Second,
    WriteTimeout:      30 * time.Second,
    IdleTimeout:       60 * time.Second,
}
log.Fatal(srv.ListenAndServe())
```

Always set timeouts. Defaults can hang forever under slow clients.

Middleware pattern

```go
func withRequestID(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        id := r.Header.Get("X-Request-ID")
        if id == "" {
            id = strconv.FormatInt(time.Now().UnixNano(), 36)
        }
        w.Header().Set("X-Request-ID", id)
        ctx := context.WithValue(r.Context(), ctxKeyID, id)
        next.ServeHTTP(w, r.WithContext(ctx))
    })
}
```

Client

```go
client := &http.Client{
    Timeout: 10 * time.Second,
    Transport: &http.Transport{
        Proxy:                 http.ProxyFromEnvironment,
        DialContext:           (&net.Dialer{Timeout: 3 * time.Second}).DialContext,
        MaxIdleConns:          100,
        IdleConnTimeout:       90 * time.Second,
        TLSHandshakeTimeout:   5 * time.Second,
        ForceAttemptHTTP2:     true,
    },
}

req, err := http.NewRequestWithContext(ctx, http.MethodGet, url, nil)
req.Header.Set("Accept", "application/json")
resp, err := client.Do(req)
if err != nil {
    return err
}
defer resp.Body.Close()
body, err := io.ReadAll(io.LimitReader(resp.Body, 1<<20))
```

Never use the bare package-level `http.Get` in production code without timeouts — it uses `http.DefaultClient` with no overall timeout.

JSON helpers

```go
func writeJSON(w http.ResponseWriter, status int, v any) {
    w.Header().Set("Content-Type", "application/json; charset=utf-8")
    w.WriteHeader(status)
    _ = json.NewEncoder(w).Encode(v)
}

func readJSON(r *http.Request, dst any) error {
    defer r.Body.Close()
    dec := json.NewDecoder(io.LimitReader(r.Body, 1<<20))
    dec.DisallowUnknownFields()
    return dec.Decode(dst)
}
```

net/url

```go
u, err := url.Parse("https://example.com/search?q=go+lang")
q := u.Query()
q.Set("page", "2")
u.RawQuery = q.Encode()
```

httptest (for unit tests)

```go
req := httptest.NewRequest(http.MethodGet, "/health", nil)
rr := httptest.NewRecorder()
mux.ServeHTTP(rr, req)
if rr.Code != 200 { t.Fatal(rr.Code) }
```

Runnable example

```bash
go mod init example
go run .
```

```go
package main

import (
    "encoding/json"
    "fmt"
```

```go
    "io"
    "net/http"
    "net/http/httptest"
    "strings"
)

func main() {
    mux := http.NewServeMux()
    mux.HandleFunc("GET /health", func(w http.ResponseWriter, r *http.Request) {
        w.Header().Set("Content-Type", "application/json")
        _ = json.NewEncoder(w).Encode(map[string]string{"status": "ok"})
    })
    mux.HandleFunc("GET /echo/{msg}", func(w http.ResponseWriter, r *http.Request) {
        fmt.Fprint(w, r.PathValue("msg"))
    })
    mux.HandleFunc("POST /sum", func(w http.ResponseWriter, r *http.Request) {
        var in struct{ A, B int }
        if err := json.NewDecoder(io.LimitReader(r.Body, 4096)).Decode(&in); err != nil {
            http.Error(w, "bad json", http.StatusBadRequest)
            return
        }
        _ = json.NewEncoder(w).Encode(map[string]int{"sum": in.A + in.B})
    })

    // health
    req := httptest.NewRequest(http.MethodGet, "/health", nil)
    rr := httptest.NewRecorder()
    mux.ServeHTTP(rr, req)
    fmt.Println("health", rr.Code, strings.TrimSpace(rr.Body.String()))

    // path value
    req = httptest.NewRequest(http.MethodGet, "/echo/hi", nil)
    rr = httptest.NewRecorder()
    mux.ServeHTTP(rr, req)
    fmt.Println("echo", rr.Body.String())

    // POST JSON
    body := strings.NewReader(`{"A":2,"B":40}`)
    req = httptest.NewRequest(http.MethodPost, "/sum", body)
    req.Header.Set("Content-Type", "application/json")
    rr = httptest.NewRecorder()
    mux.ServeHTTP(rr, req)
    fmt.Println("sum", strings.TrimSpace(rr.Body.String()))
}
```

Expected output:

```
health 200 {"status":"ok"}
echo hi
sum {"sum":42}
```

What to notice: - Method + path patterns keep routing in the stdlib. - `httptest` exercises handlers without binding a port. - Body decoding should always be size-limited.

Try next: Add middleware that rejects requests missing `Authorization` with 401, and test it with `httptest`.

flag, os/exec, and os/signal

flag, os/exec, and os/signal

Overview

Most Go programs start as **tools**: parse flags, run work, maybe spawn subprocesses, and shut down cleanly on SIGINT/SIGTERM.

Package	Role
`flag`	CLI flags and usage
`os/exec`	Subprocesses
`os/signal`	Graceful shutdown signals

flag

```go
var (
    addr = flag.String("addr", ":8080", "listen address")
    n    = flag.Int("n", 1, "repeat count")
    v    = flag.Bool("v", false, "verbose")
)
flag.Parse()
args := flag.Args() // non-flag args
```

Custom FlagSet (subcommands)

```go
fs := flag.NewFlagSet("greet", flag.ExitOnError)
name := fs.String("name", "world", "who to greet")
_ = fs.Parse(os.Args[2:])
```

Good CLI hygiene

- Provide defaults and help text.
- Exit non-zero on misuse (`flag.ExitOnError` or manual `os.Exit(2)`).
- Prefer explicit flags over positional soup for optional config.

```go
flag.Usage = func() {
    fmt.Fprintf(os.Stderr, "usage: %s [flags] <path>\n", filepath.Base(os.Args[0]))
    flag.PrintDefaults()
}
```

os/exec

```go
cmd := exec.CommandContext(ctx, "git", "status", "--short")
cmd.Dir = repo
cmd.Env = append(os.Environ(), "LC_ALL=C")
out, err := cmd.Output() // stdout; err may be *exec.ExitError
```

Stream pipes

```go
cmd := exec.Command("grep", "TODO")
cmd.Stdin = strings.NewReader(src)
var stdout, stderr bytes.Buffer
cmd.Stdout = &stdout
cmd.Stderr = &stderr
err := cmd.Run()
```

Rules

1. Prefer `CommandContext` so cancel kills the process tree (platform nuances apply).
2. Never build shell command strings with untrusted input — pass argv slices.
3. Check exit codes via `errors.As(err, &exitErr)`.

```go
if err := cmd.Run(); err != nil {
    var ee *exec.ExitError
    if errors.As(err, &ee) {
        return fmt.Errorf("exit %d: %w", ee.ExitCode(), err)
    }
    return err
}
```

os/signal and graceful shutdown

```
ctx, stop := signal.NotifyContext(context.Background(), os.Interrupt, syscall.SIGTERM)
defer stop()

go func() {
    <-ctx.Done()
    shutdownCtx, cancel := context.WithTimeout(context.Background(), 10*time.Second)
    defer cancel()
    _ = srv.Shutdown(shutdownCtx)
}()
```

`signal.NotifyContext` (Go 1.16+) is the clean default for servers and long CLIs.

Runnable example

```
go mod init example
go run . -n 2 hello
```

```
package main

import (
    "bytes"
    "context"
    "flag"
    "fmt"
    "os"
    "os/exec"
    "time"
)

func main() {
    n := flag.Int("n", 1, "repeat")
    flag.Parse()
    msg := "world"
    if flag.NArg() > 0 {
        msg = flag.Arg(0)
    }
    for i := 0; i < *n; i++ {
        fmt.Println("hello", msg)
    }

    ctx, cancel := context.WithTimeout(context.Background(), 2*time.Second)
    defer cancel()
```

```go
    cmd := exec.CommandContext(ctx, "echo", "from-subprocess")
    var buf bytes.Buffer
    cmd.Stdout = &buf
    if err := cmd.Run(); err != nil {
        fmt.Fprintln(os.Stderr, "exec:", err)
        os.Exit(1)
    }
    fmt.Print("exec out:", buf.String())
}
```

Expected output:

```
hello hello
hello hello
exec out:from-subprocess
```

(With `-n 2 hello` you get two greeting lines.)

What to notice: - `flag.Arg` / `Args` are only valid after `Parse`. - `CommandContext` ties subprocess lifetime to a deadline. - Real services combine `NotifyContext` with `http.Server.Shutdown`.

Try next: Add a `timeout` duration flag and pass it into `context.WithTimeout` around a long `exec.Command`.

time and context Recipes

time and context Recipes

Overview

`time` measures and schedules; `context` propagates **deadlines and cancel** across API boundaries. Together they implement request budgets.

Language-oriented time intro: Time. Concurrency-focused context: Context.

time essentials

```
now := time.Now()
t := time.Date(2026, 8, 3, 12, 0, 0, 0, time.UTC)
d := 1500 * time.Millisecond
d, err := time.ParseDuration("1h30m")
```

Format and parse

Go's reference time is `Mon Jan 2 15:04:05 MST 2006` (layout by example):

```
const layout = time.RFC3339
s := now.UTC().Format(layout)
t, err := time.Parse(layout, s)
```

Prefer `RFC3339` / `RFC3339Nano` for APIs.

Timers and tickers

```
timer := time.NewTimer(2 * time.Second)
defer timer.Stop()
select {
case <-timer.C:
case <-ctx.Done():
    if !timer.Stop() {
        <-timer.C // drain if already fired (pre-1.23 habit; see release notes)
    }
}
```

```go
ticker := time.NewTicker(time.Second)
defer ticker.Stop()
for {
    select {
    case <-ctx.Done():
        return ctx.Err()
    case t := <-ticker.C:
        _ = t
    }
}
```

Always `Stop` tickers to avoid leaks.

context essentials

```go
ctx := context.Background()
ctx, cancel := context.WithCancel(ctx)
defer cancel()

ctx, cancel = context.WithTimeout(ctx, 3*time.Second)
defer cancel()

ctx, cancel = context.WithDeadline(ctx, time.Now().Add(time.Second))
defer cancel()
```

Values (use sparingly)

```go
type key int
const requestID key = 1
ctx = context.WithValue(ctx, requestID, "abc")
id, _ := ctx.Value(requestID).(string)
```

Store request-scoped metadata (IDs), not optional parameters that belong in function args.

Waiting on cancel

```go
select {
case <-ctx.Done():
    return ctx.Err() // context.Canceled or DeadlineExceeded
case res := <-results:
    return res, nil
}
```

Recipes

Bound a function

```
func fetch(ctx context.Context, url string) ([]byte, error) {
    req, err := http.NewRequestWithContext(ctx, http.MethodGet, url, nil)
    if err != nil {
        return nil, err
    }
    resp, err := http.DefaultClient.Do(req)
    // ...
}
```

Remaining budget

```
if dl, ok := ctx.Deadline(); ok {
    remaining := time.Until(dl)
    if remaining < 50*time.Millisecond {
        return errBudgetExhausted
    }
}
```

Sleep that respects cancel

```
func sleep(ctx context.Context, d time.Duration) error {
    t := time.NewTimer(d)
    defer t.Stop()
    select {
    case <-ctx.Done():
        return ctx.Err()
    case <-t.C:
        return nil
    }
}
```

Runnable example

```
go mod init example
go run .
```

```go
package main

import (
    "context"
    "fmt"
    "time"
)

func work(ctx context.Context, name string, d time.Duration) error {
    select {
    case <-time.After(d):
        fmt.Println(name, "done")
        return nil
    case <-ctx.Done():
        fmt.Println(name, "canceled:", ctx.Err())
        return ctx.Err()
    }
}

func main() {
    now := time.Now().UTC()
    fmt.Println("rfc3339:", now.Format(time.RFC3339))
    d, _ := time.ParseDuration("250ms")
    fmt.Println("duration:", d)

    ctx, cancel := context.WithTimeout(context.Background(), 200*time.Millisecond)
    defer cancel()

    _ = work(ctx, "slow", 500*time.Millisecond)

    ctx2, cancel2 := context.WithTimeout(context.Background(), time.Second)
    defer cancel2()
    _ = work(ctx2, "fast", 50*time.Millisecond)

    if dl, ok := ctx2.Deadline(); ok {
        fmt.Println("remaining>0?", time.Until(dl) > 0)
    }
}
```

Expected output:

```
rfc3339: 2026-...
duration: 250ms
slow canceled: context deadline exceeded
fast done
remaining>0? true
```

What to notice: - Timeout cancels waiters blocked in `select` on `ctx.Done()`. - Always `defer`

`cancel()` to free timer resources even on success. - Format APIs with RFC3339 unless you have a domain-specific layout.

Try next: Nest a child `WithTimeout` shorter than the parent and observe which deadline fires first.

sync and sync/atomic

sync and sync/atomic

Overview

When goroutines share memory, the stdlib gives two complementary tools:

Package	Style
`sync`	Mutexes, WaitGroups, Once, Cond, Map, Pool
`sync/atomic`	Lock-free counters and pointers for simple cases

Prefer **channels for ownership transfer** and **mutexes for shared structures**. Concurrency design: Part 11.

WaitGroup

```
var wg sync.WaitGroup
for _, job := range jobs {
    wg.Add(1)
    go func(j Job) {
        defer wg.Done()
        process(j)
    }(job)
}
wg.Wait()
```

Call `Add` before spawning; never negative. Prefer `wg.Go` (Go 1.25+) when available:

```
wg.Go(func() { process(job) })
```

Mutex and RWMutex

```
type Cache struct {
    mu sync.Mutex
    m  map[string]string
}
```

```go
func (c *Cache) Get(k string) (string, bool) {
    c.mu.Lock()
    defer c.mu.Unlock()
    v, ok := c.m[k]
    return v, ok
}
```

`RWMutex` helps read-heavy maps; measure before assuming it wins.

Once

```go
var once sync.Once
var client *http.Client

func Client() *http.Client {
    once.Do(func() {
        client = &http.Client{Timeout: 10 * time.Second}
    })
    return client
}
```

Pool

```go
var bufPool = sync.Pool{
    New: func() any { return make([]byte, 32*1024) },
}
b := bufPool.Get().([]byte)
defer bufPool.Put(b[:cap(b)]) // reset length if you resliced
```

Use for high-churn buffers; do not assume pooled items are zeroed.

Map

`sync.Map` is specialized (append-only keys, many disjoint keys). For ordinary caches, a mutex + map is clearer.

Cond (advanced)

```
mu := sync.Mutex{}
cond := sync.NewCond(&mu)
// Waiters: cond.Wait(); Signal/Broadcast after state change under the same mutex
```

See pipelines / Cond.

atomic

```
var hits atomic.Int64
hits.Add(1)
fmt.Println(hits.Load())

var flag atomic.Bool
flag.Store(true)
```

Typed atomics (`atomic.Int64`, `Bool`, `Pointer[T]`) are preferred over `AddInt64` free functions.

Compare-and-swap

```
var p atomic.Pointer[Config]
p.Store(&Config{V: 1})
for {
    old := p.Load()
    next := *old
    next.V++
    if p.CompareAndSwap(old, &next) {
        break
    }
}
```

Race detector

```
go test -race ./...
go run -race .
```

If the race detector complains, fix the ownership story — do not “hope” atomics alone fix complex structs.

Runnable example

```
go mod init example
go run .
```

```go
package main

import (
    "fmt"
    "sync"
    "sync/atomic"
)

func main() {
    var hits atomic.Int64
    var wg sync.WaitGroup
    var mu sync.Mutex
    seen := make(map[int]bool)

    for i := 0; i < 8; i++ {
        wg.Add(1)
        go func(id int) {
            defer wg.Done()
            hits.Add(1)
            mu.Lock()
            seen[id] = true
            mu.Unlock()
        }(i)
    }
    wg.Wait()

    fmt.Println("hits", hits.Load())
    fmt.Println("unique", len(seen))

    var once sync.Once
    once.Do(func() { fmt.Println("init once") })
    once.Do(func() { fmt.Println("never again") })
}
```

Expected output:

```
hits 8
unique 8
init once
```

What to notice: - Counters can be atomic; maps still need a mutex (or channels). - `Once` runs exactly one successful init path. - Run with `-race` to validate this pattern under the detector.

Try next: Replace the mutex map with a channel that owns the map in a single goroutine (share-by-communication).

slices, maps, cmp, and iter

slices, maps, cmp, and iter

Overview

Go 1.21+ added first-class helpers for common collection work. Go 1.23 popularized **range-over-func** iterators via package `iter`.

Package	Role
`slices`	Search, sort, clone, compact, delete
`maps`	Keys/values clone, equal, copy
`cmp`	Ordered comparisons for sort helpers
`iter`	`Seq` / `Seq2` iterator types

Basics of ranging: Loops.

slices

```go
import "slices"

s := []int{3, 1, 2}
slices.Sort(s)                    // [1 2 3]
slices.SortFunc(s, cmp.Compare)   // same for ordered types
i, ok := slices.BinarySearch(s, 2)
s2 := slices.Clone(s)
s = slices.Delete(s, 1, 2)        // remove s[1]
s = slices.Compact(s)             // adjacent dups (sorted first)
max := slices.Max(s)
```

Useful helpers

Func	Notes
`Contains` / `Index`	Linear search
`Compact` / `CompactFunc`	Needs sorted input for uniqueness
`Grow` / `Clip`	Capacity management
`Replace`	Splice
`Collect`	From `iter.Seq` (Go 1.23+)

```go
s = slices.DeleteFunc(s, func(n int) bool { return n%2 == 0 })
```

maps

```go
import "maps"

a := map[string]int{"x": 1, "y": 2}
b := maps.Clone(a)
maps.Copy(b, map[string]int{"y": 9, "z": 3})
eq := maps.Equal(a, b)

for k := range maps.Keys(a) { // Go 1.23+ iterators
    _ = k
}
```

`maps.Keys` / `Values` return iterators — collect with `slices.Collect` if you need a slice.

cmp

```go
import "cmp"

cmp.Compare(a, b)          // -1, 0, 1 for ordered types
cmp.Or(x, y, z)            // first non-zero (useful defaults)
cmp.Less(a, b)
```

```go
slices.SortFunc(people, func(a, b Person) int {
    if c := cmp.Compare(a.Last, b.Last); c != 0 {
        return c
    }
    return cmp.Compare(a.First, b.First)
})
```

iter and range-over-func

```go
import "iter"

func Backward[E any](s []E) iter.Seq[E] {
    return func(yield func(E) bool) {
        for i := len(s) - 1; i >= 0; i-- {
            if !yield(s[i]) {
```

```go
                return
            }
        }
    }
}

for v := range Backward([]string{"a", "b", "c"}) {
    fmt.Println(v)
}
```

Pull-style when you need manual advancement:

```go
next, stop := iter.Pull(Backward([]int{1, 2, 3}))
defer stop()
for {
    v, ok := next()
    if !ok {
        break
    }
    fmt.Println(v)
}
```

When not to use them

- Tiny one-off loops are fine as plain `for`.
- Do not force iterators everywhere — they shine for **lazy** pipelines and library APIs.
- Sorting huge slices of structs: consider indices or `sort.Slice` patterns if allocations dominate (benchmark).

Runnable example

```bash
go mod init example
go run .
```

```go
package main

import (
    "cmp"
    "fmt"
    "iter"
    "maps"
    "slices"
)
```

```go
func Countdown(n int) iter.Seq[int] {
    return func(yield func(int) bool) {
        for i := n; i >= 0; i-- {
            if !yield(i) {
                return
            }
        }
    }
}

func main() {
    s := []int{5, 1, 5, 3, 2}
    slices.Sort(s)
    fmt.Println("sorted", s)
    fmt.Println("max", slices.Max(s))
    s = slices.Compact(s)
    fmt.Println("compact", s)

    type pair struct{ K string; V int }
    ps := []pair{{"b", 2}, {"a", 1}, {"a", 0}}
    slices.SortFunc(ps, func(a, b pair) int {
        if c := cmp.Compare(a.K, b.K); c != 0 {
            return c
        }
        return cmp.Compare(a.V, b.V)
    })
    fmt.Println("pairs", ps)

    m := map[string]int{"x": 1, "y": 2}
    fmt.Println("clone equal?", maps.Equal(m, maps.Clone(m)))

    fmt.Print("countdown:")
    for v := range Countdown(3) {
        fmt.Print(" ", v)
    }
    fmt.Println()

    collected := slices.Collect(Countdown(2))
    fmt.Println("collect", collected)
}
```

Expected output:

```
sorted [1 2 3 5 5]
max 5
compact [1 2 3 5]
```

```
pairs [{a 0} {a 1} {b 2}]
clone equal? true
countdown: 3 2 1 0
collect [2 1 0]
```

What to notice: - `Compact` only removes **adjacent** duplicates — sort first for set-like uniqueness. - `cmp.Compare` composes clean multi-key sorts. - `slices.Collect` materializes a lazy `iter.Seq`.

Try next: Write `Filter[T any](s []T, ok func(T) bool) iter.Seq[T]` and range over it without allocating an intermediate slice.

testing Package: Examples, Fuzz, Bench

testing Package: Examples, Fuzz, Bench

Overview

Package `testing` is both a runner and a small toolkit: tests, benchmarks, examples, fuzz targets, temp dirs, and HTTP helpers via `net/http/httptest`.

Fundamentals also live in Part 06. This chapter is the **stdlib surface** checklist.

Test functions

```go
func TestSum(t *testing.T) {
    t.Parallel()
    got := Sum(2, 3)
    if got != 5 {
        t.Fatalf("Sum(2,3)=%d want 5", got)
    }
}
```

Method	Use
`Error` / `Errorf`	Fail, continue
`Fatal` / `Fatalf`	Fail, stop this test
`Helper`	Mark assertion helpers
`Cleanup`	Register teardown (LIFO)
`TempDir`	Isolated temp directory
`Setenv`	Env var for this test
`Context`	Canceled when test ends (Go 1.24+)
`Deadline`	Time left if `-timeout` set

Table tests

```go
tests := []struct {
    name string
    a, b int
    want int
}{
    {"zero", 0, 0, 0},
```

```go
    {"pos", 2, 3, 5},
}
for _, tt := range tests {
    t.Run(tt.name, func(t *testing.T) {
        if got := Sum(tt.a, tt.b); got != tt.want {
            t.Fatalf("got %d want %d", got, tt.want)
        }
    })
}
```

Examples

```go
func ExampleSum() {
    fmt.Println(Sum(1, 2))
    // Output:
    // 3
}
```

Examples run as tests and appear in `pkg.go.dev` docs. Unordered output:

```go
// Unordered output:
// a
// b
```

Benchmarks

```go
func BenchmarkSum(b *testing.B) {
    for b.Loop() { // Go 1.24+; or for i := 0; i < b.N; i++
        Sum(2, 3)
    }
}
```

```go
go test -bench=BenchmarkSum -benchmem ./...
```

Fuzzing

```go
func FuzzReverse(f *testing.F) {
    f.Add("gopher")
    f.Fuzz(func(t *testing.T, s string) {
        rev := Reverse(s)
```

```
        dbl := Reverse(rev)
        if dbl != s {
            t.Fatalf("roundtrip %q -> %q -> %q", s, rev, dbl)
        }
    })
}
```

```
go test -fuzz=FuzzReverse -fuzztime=10s
```

Failing inputs land under `testdata/fuzz/` — commit them as regression seeds.

fstest and httptest

```
fsys := fstest.MapFS{"a.txt": {Data: []byte("x")}}
b, err := fs.ReadFile(fsys, "a.txt")

req := httptest.NewRequest(http.MethodGet, "/x", nil)
rr := httptest.NewRecorder()
handler.ServeHTTP(rr, req)

// Go 1.27+: in-memory server (synctest-safe, auto-cleanup)
ts := httptest.NewTestServer(t, handler)
resp, err := ts.Client().Get("http://example.com/x")
```

Commands

```
go test ./...
go test -v -run TestSum
go test -race ./...
go test -coverprofile=c.out && go tool cover -html=c.out
go test -bench=. -benchmem
go test -fuzz=. -fuzztime=30s
```

Runnable example

Save as `sum.go` and `sum_test.go` in a module:

```
mkdir /tmp/stdlib-test && cd /tmp/stdlib-test
go mod init example
```

`sum.go`:

```go
package example

func Sum(a, b int) int { return a + b }

func Reverse(s string) string {
    r := []rune(s)
    for i, j := 0, len(r)-1; i < j; i, j = i+1, j-1 {
        r[i], r[j] = r[j], r[i]
    }
    return string(r)
}
```

sum_test.go:

```go
package example

import (
    "fmt"
    "testing"
)

func TestSum(t *testing.T) {
    if Sum(2, 3) != 5 {
        t.Fatal("bad sum")
    }
}

func ExampleSum() {
    fmt.Println(Sum(1, 2))
    // Output:
    // 3
}

func FuzzReverse(f *testing.F) {
    f.Add("ab")
    f.Fuzz(func(t *testing.T, s string) {
        if Reverse(Reverse(s)) != s {
            t.Fatal("roundtrip")
        }
    })
}

func BenchmarkSum(b *testing.B) {
    for i := 0; i < b.N; i++ {
        Sum(i, i)
    }
}
```

```
go test -v
go test -bench=BenchmarkSum -benchmem
go test -fuzz=FuzzReverse -fuzztime=3s
```

What to notice: - Examples are documentation that cannot drift silently. - Fuzz finds encoding edge cases (`[]rune` vs bytes) faster than hand-written tables. - Benchmarks need stable loops; avoid timing setup inside `b.N` work.

Try next: Use `t.TempDir` and `os.WriteFile` to test a function that reads a config path.

crypto, hash, rand, and regexp

crypto, hash, rand, and regexp

Overview

Last stdlib essentials for **integrity, randomness, and text matching**.

Package	Use
`crypto/sha256` (etc.)	Content hashes, integrity
`crypto/hmac`	Message authentication
`crypto/rand`	Secure random bytes
`math/rand/v2`	Non-secure randomness / sims
`crypto/subtle`	Constant-time compare
`crypto/mldsa`	Post-quantum ML-DSA signatures (Go 1.27+; also TLS/x509)
`uuid`	Generate/parse UUIDs (Go 1.27+; RFC 9562)
`regexp`	Regular expressions
`net/url`	URL parse (with HTTP chapter)

TLS deep dive: Security hardening.

Hashing

```go
sum := sha256.Sum256([]byte("hello"))
fmt.Printf("%x\n", sum)

h := sha256.New()
io.Copy(h, r)
digest := h.Sum(nil)
```

Streaming hash large inputs — do not `ReadAll` multi-GB files first.

HMAC

```go
mac := hmac.New(sha256.New, key)
mac.Write(msg)
sig := mac.Sum(nil)
if !hmac.Equal(sig, expected) {
    return errUnauthorized
}
```

Use `hmac.Equal` / `subtle.ConstantTimeCompare` — not `==` on strings for secrets.

Randomness

```go
// Secure tokens / keys
b := make([]byte, 16)
if _, err := rand.Read(b); err != nil { // crypto/rand
    return err
}
token := hex.EncodeToString(b)

// Simulations, randomized tests (not secrets)
n := rand.IntN(100) // math/rand/v2
```

Never use `math/rand` for session IDs, API keys, or CSRF tokens.

uuid (Go 1.27+)

Stdlib `uuid` covers the common cases that used to need `github.com/google/uuid`:

```go
import "uuid"

id := uuid.New()       // currently v4
id = uuid.NewV7()      // time-ordered; good for DB keys
s := id.String()
parsed, err := uuid.Parse(s)
```

`New()` is the default; use `NewV7()` when insertion order / index locality matters. Random bits come from a cryptographically secure generator.

Go 1.27 also adds `crypto/mldsa` (FIPS 204) with `crypto/x509` and `crypto/tls` integration for post-quantum signatures. Treat it as an opt-in for new PKI work, not a drop-in replacement for existing ECDSA certs.

regexp

```
re := regexp.MustCompile(`(?i)^[a-z0-9._%+\-]+@[a-z0-9.\-]+\.[a-z]{2,}$`)
ok := re.MatchString(email)
parts := re.FindStringSubmatch(s)
```

Performance tips

1. Compile once (`var re = regexp.MustCompile(...)` at package scope).
2. Prefer `strings` / `bytes` for fixed-prefix checks.
3. Beware catastrophic backtracking on untrusted patterns/input — keep patterns simple.
4. Use `regexp.Compile` when pattern comes from config; handle errors.

```
const max = 1024
if len(s) > max {
    return false
}
return re.MatchString(s)
```

Password hashing note

Stdlib does **not** include bcrypt/argon2. For passwords use `golang.org/x/crypto/bcrypt` or argon2 — do not roll SHA256(password).

Runnable example

```
go mod init example
go run .
```

```
package main

import (
    "crypto/hmac"
    "crypto/rand"
    "crypto/sha256"
    "crypto/subtle"
    "encoding/hex"
    "fmt"
    mrand "math/rand/v2"
    "regexp"
)
```

```go
func main() {
    sum := sha256.Sum256([]byte("hello"))
    fmt.Println("sha256", hex.EncodeToString(sum[:])[:16]+"...")

    key := []byte("secret-key")
    msg := []byte("payload")
    mac := hmac.New(sha256.New, key)
    mac.Write(msg)
    sig := mac.Sum(nil)

    mac2 := hmac.New(sha256.New, key)
    mac2.Write(msg)
    fmt.Println("hmac ok?", hmac.Equal(sig, mac2.Sum(nil)))

    // constant-time compare demo
    a := []byte("abcd")
    b := []byte("abce")
    fmt.Println("subtle", subtle.ConstantTimeCompare(a, b) == 1)

    tok := make([]byte, 8)
    _, _ = rand.Read(tok)
    fmt.Println("token", hex.EncodeToString(tok))
    fmt.Println("sim roll", mrand.IntN(6)+1)

    re := regexp.MustCompile(`^item-(\d+)$`)
    m := re.FindStringSubmatch("item-42")
    fmt.Println("submatch", m)
}
```

Expected output (token varies):

```
sha256 2cf24dba5fb0a30e...
hmac ok? true
subtle false
token a1b2c3d4e5f67890
sim roll 4
submatch [item-42 42]
```

What to notice: - HMAC equality is constant-time via `hmac.Equal`. - `crypto/rand` for tokens; `math/rand/v2` for games/sims. - Submatch index 0 is the full match; 1+ are groups.

Try next: Stream-hash a file with `sha256.New` + `io.Copy` and compare to `shasum -a 256` on the command line.

Stdlib tour wrap-up

You now have a map of the packages that appear in nearly every Go binary. Next steps:

1. Rebuild a small CLI using only parts 980–992.
2. Rebuild a tiny JSON API with timeouts and `slog`.
3. Jump to specialized parts only when stdlib friction is real.

Return to the stdlib overview checklist and mark what you can do without docs.

database/sql

database/sql

Overview

`database/sql` is the standard **database access layer**: connection pools, prepared statements, transactions, and context-aware queries. Drivers register themselves; your code stays mostly driver-agnostic.

Deeper pool and production notes: database/sql pool deep dive and Bookstore use in web databases.

Mental model

```
  sql.DB  = pool of connections (safe for concurrent use)

        QueryContext / QueryRowContext / ExecContext
        PrepareContext → sql.Stmt
        BeginTx → sql.Tx → Commit / Rollback
```

- **`sql.DB` is not a single connection.** It is a pool.
- **`sql.Open` may not dial** until the first use — always `PingContext` at startup.
- Pass **`context.Context`** so cancel/timeout reaches the driver.

Open, configure, ping

```go
import (
    "context"
    "database/sql"
    "time"

    _ "modernc.org/sqlite" // or pgx stdlib, etc.
)

func openDB(dsn string) (*sql.DB, error) {
    db, err := sql.Open("sqlite", dsn)
    if err != nil {
        return nil, err
    }
```

```go
    db.SetMaxOpenConns(10)
    db.SetMaxIdleConns(5)
    db.SetConnMaxLifetime(time.Hour)
    db.SetConnMaxIdleTime(15 * time.Minute)

    ctx, cancel := context.WithTimeout(context.Background(), 3*time.Second)
    defer cancel()
    if err := db.PingContext(ctx); err != nil {
        _ = db.Close()
        return nil, err
    }
    return db, nil
}
```

Setting	Role
`MaxOpenConns`	Cap concurrent DB usage
`MaxIdleConns`	Keep warm connections
`ConnMaxLifetime`	Recycle to avoid server-side timeouts
`ConnMaxIdleTime`	Drop idle conns (Go 1.15+)

Query patterns

Many rows

```go
rows, err := db.QueryContext(ctx,
    `SELECT id, title FROM books WHERE author = ? ORDER BY title`, author)
if err != nil {
    return nil, err
}
defer rows.Close()

var out []Book
for rows.Next() {
    var b Book
    if err := rows.Scan(&b.ID, &b.Title); err != nil {
        return nil, err
    }
    out = append(out, b)
}
return out, rows.Err() // always check rows.Err()
```

One row

```
var b Book
err := db.QueryRowContext(ctx,
    `SELECT id, title FROM books WHERE id = ?`, id,
).Scan(&b.ID, &b.Title)
if errors.Is(err, sql.ErrNoRows) {
    return Book{}, errNotFound
}
return b, err
```

Exec (insert/update/delete)

```
res, err := db.ExecContext(ctx,
    `INSERT INTO books (id, title) VALUES (?, ?)`, id, title)
if err != nil {
    return err
}
n, _ := res.RowsAffected()
_ = n
```

Always use placeholders (`?` or `$1` depending on driver). Never concatenate user input into SQL.

Nullables

Database NULL is not Go zero. Use:

```
var maybeTitle sql.NullString
// Scan into maybeTitle; use maybeTitle.Valid && maybeTitle.String
```

Or pointer fields (`*string`) with some drivers/scanners. Be consistent.

Transactions

```
tx, err := db.BeginTx(ctx, &sql.TxOptions{Isolation: sql.LevelReadCommitted})
if err != nil {
    return err
}
defer tx.Rollback() // no-op after successful Commit

if _, err := tx.ExecContext(ctx, `UPDATE accounts SET bal = bal - ? WHERE id = ?`, amt, from
```

```go
    return err
}
if _, err := tx.ExecContext(ctx, `UPDATE accounts SET bal = bal + ? WHERE id = ?`, amt, to);
    return err
}
return tx.Commit()
```

Keep transactions **short**. Do not hold a `tx` open across slow external HTTP calls.

Prepared statements

```go
stmt, err := db.PrepareContext(ctx, `SELECT title FROM books WHERE id = ?`)
if err != nil {
    return err
}
defer stmt.Close()

var title string
err = stmt.QueryRowContext(ctx, id).Scan(&title)
```

Prepare when you run the same SQL many times in one process. For one-shot queries, `QueryContext` is fine.

Errors worth mapping

Condition	Typical HTTP / API mapping
`sql.ErrNoRows`	Not found
Unique constraint (driver-specific)	Conflict
`context.DeadlineExceeded`	Timeout / unavailable
Connection refused	Unavailable; log + retry policy

Log the **real** error; return a safe client message.

Rules of thumb

Do	Don't
`*Context` methods + timeouts	Bare `Query` in request handlers without budget
Close `rows` (defer)	Forget `rows.Err()` after the loop
Parameterized SQL	String-build queries from user input
One `*sql.DB` per process/DSN	Open a new DB per request

Try next

1. Open SQLite in-memory, create a table, insert, list.
2. Force `ErrNoRows` and map it with `errors.Is`.
3. Run two concurrent `QueryContext` calls against one `*sql.DB` (pool is concurrent-safe).

rows.Err(): The Error a rows.Next() Loop Hides

`rows.Err()`: The Error a `rows.Next()` Loop Hides

Thesis

`for rows.Next() { ... }` has one exit and two meanings. `Next` returns `false` when the result set is finished, and it also returns `false` when reading the next row **failed**: the connection dropped, the server cancelled the query, the context expired, or the driver could not decode a row. The loop cannot tell these apart. Only `rows.Err()`, called after the loop, can. Skip that call and a function that read 3 rows out of 3,000 returns a partial total with a `nil` error, and nothing in the logs says anything went wrong.

`go vet` does not catch this. As of July 2026 there is a dedicated analyzer for it, `sqlrowserr`, which ships in gopls v0.23.0 and in `golang.org/x/tools`. This page reproduces the failure deterministically with a twenty-line fake driver, fixes it, and wires the analyzer into CI.

Mental model

```
rows.Next()  true   Scan, use row



  false    why?   driver returned io.EOF       → rows.Err() == nil   (complete)
                  driver returned anything else → rows.Err() != nil   (partial!)
                  ctx cancelled / deadline      → rows.Err() == context error
```

Fact	Consequence
`Next` reports failure as `false`	A failed read looks exactly like the end of the data
`rows.Err()` returns the error that stopped iteration, or `nil` after a clean `io.EOF`	It is the only signal of a partial read. Check it after every loop
`rows.Close()` does not return the iteration error	`defer rows.Close()` is necessary but not sufficient
`QueryContext` errors only cover *starting* the query	Failures while streaming rows appear later, in `rows.Err()`
`go vet` has no check for it	Use the `sqlrowserr` analyzer (gopls v0.23.0+, or standalone)

Worked examples

Tested with Go 1.27.2 and `golang.org/x/tools` v0.52.0 on Linux. No database server is needed: the program registers its own driver.

Case 1: A loop that silently loses rows

The driver below returns three rows and then fails the way a dropped connection would. `sumIDs` is the shape found in most codebases. `sumIDsChecked` adds one `if`.

```go
// main.go - a for rows.Next() loop that silently loses rows
package main

import (
    "context"
    "database/sql"
    "database/sql/driver"
    "errors"
    "fmt"
    "log"
)

// flaky is a tiny database/sql driver: its result set yields `ok` rows,
// then fails the way a dropped connection or a server-side error would.
type flaky struct{ ok int }

func (d flaky) Open(string) (driver.Conn, error) { return conn(d), nil }

type conn flaky

func (c conn) Prepare(string) (driver.Stmt, error) { return stmt(c), nil }
func (conn) Close() error                          { return nil }
func (conn) Begin() (driver.Tx, error)             { return nil, errors.ErrUnsupported }

type stmt conn

func (stmt) Close() error  { return nil }
func (stmt) NumInput() int { return -1 }
func (stmt) Exec([]driver.Value) (driver.Result, error) {
    return nil, errors.ErrUnsupported
}
func (s stmt) Query([]driver.Value) (driver.Rows, error) { return &rows{ok: s.ok}, nil }

type rows struct{ ok, n int }

func (*rows) Columns() []string { return []string{"id"} }
```

```
func (*rows) Close() error      { return nil }
func (r *rows) Next(dest []driver.Value) error {
    if r.n == r.ok { // a clean end would return io.EOF; anything else is an error
        return errors.New("connection reset by peer")
    }
    r.n++
    dest[0] = int64(r.n)
    return nil
}

// sumIDs is the common shape: scan every row, return the total.
func sumIDs(ctx context.Context, db *sql.DB) (int64, int, error) {
    rs, err := db.QueryContext(ctx, "SELECT id FROM orders")
    if err != nil {
        return 0, 0, err
    }
    defer rs.Close()
    var sum int64
    var n int
    for rs.Next() {
        var id int64
        if err := rs.Scan(&id); err != nil {
            return 0, 0, err
        }
        sum += id
        n++
    }
    return sum, n, nil // BUG: never asked rs.Err()
}

// sumIDsChecked is the same loop with the one line that matters.
func sumIDsChecked(ctx context.Context, db *sql.DB) (int64, int, error) {
    rs, err := db.QueryContext(ctx, "SELECT id FROM orders")
    if err != nil {
        return 0, 0, err
    }
    defer rs.Close()
    var sum int64
    var n int
    for rs.Next() {
        var id int64
        if err := rs.Scan(&id); err != nil {
            return 0, 0, err
        }
        sum += id
        n++
    }
```

```go
    if err := rs.Err(); err != nil {
        return 0, n, fmt.Errorf("orders: after %d rows: %w", n, err)
    }
    return sum, n, nil
}

func main() {
    sql.Register("flaky", flaky{ok: 3})
    db, err := sql.Open("flaky", "")
    if err != nil {
        log.Fatal(err)
    }
    ctx := context.Background()

    sum, n, err := sumIDs(ctx, db)
    fmt.Printf("unchecked: sum=%d rows=%d err=%v\n", sum, n, err)

    sum, n, err = sumIDsChecked(ctx, db)
    fmt.Printf("checked:   sum=%d rows=%d err=%v\n", sum, n, err)
}
```

```
go mod init example.com/rowserr
go run .
```

```
unchecked: sum=6 rows=3 err=<nil>
checked:   sum=0 rows=3 err=orders: after 3 rows: connection reset by peer
```

The unchecked function returned a confident, wrong answer. The checked one says how far it got and why it stopped, and it refuses to return the partial sum. Whether a caller should ever see partial results is a design decision. It must not be an accident.

```
go vet ./... && echo "vet: clean"
```

```
vet: clean
```

Case 2: Let a tool find the missing check

`sqlrowserr` is a regular `go/analysis` pass, so a five-line program turns it into a standalone checker:

```go
// main.go - run the sqlrowserr analyzer as a standalone checker
package main

import (
```

```go
    "golang.org/x/tools/go/analysis/passes/sqlrowserr"
    "golang.org/x/tools/go/analysis/singlechecker"
)

func main() { singlechecker.Main(sqlrowserr.Analyzer) }
```

```bash
mkdir rowserrcheck && cd rowserrcheck
go mod init example.com/rowserrcheck
# paste main.go from above
go get golang.org/x/tools@v0.52.0 && go mod tidy
go build -o sqlrowserr .
cd ../rowserr && ../rowserrcheck/sqlrowserr ./...; echo "exit=$?"
```

```
/workspace/scratch/gorows/main.go:49:13: sql.Rows "rs" is used in Next loop at line 56 witho
exit=3
```

It flagged `sumIDs` and left `sumIDsChecked` alone. Exit status 3 means "diagnostics reported", so the checker fails a CI step with no extra scripting. In an editor, gopls v0.23.0 and later report the same diagnostic as you type.

Case 3: The helper that makes the check impossible to forget

When a codebase has dozens of query loops, put the loop in one place:

```go
// collect.go - scan every row with fn, and always report how iteration ended
package store

import "database/sql"

func Collect[T any](rows *sql.Rows, scan func(*sql.Rows) (T, error)) ([]T, error) {
    defer rows.Close()
    var out []T
    for rows.Next() {
        v, err := scan(rows)
        if err != nil {
            return nil, err
        }
        out = append(out, v)
    }
    return out, rows.Err()
}
```

Callers write `ids, err := store.Collect(rs, func(r *sql.Rows) (int64, error) { var id int64; return id, r.Scan(&id) })`. There is only one loop left to get right, and it is right.

The trap

Returning early from inside the loop is fine. Breaking out of it is not the same thing. After a `break` (say, you stop at the first match), `rows.Err()` returns `nil` even if the *next* read would have failed, because iteration did not fail; you stopped it. That is correct, but it means `rows.Err()` is not an "everything was read" check. If the caller needs a complete result (a total, an export, a migration), the code must read to the end **and** check `rows.Err()`, or use a query that already does the work (`SELECT sum(id) ...`) so there is nothing to stream.

The second half of the trap is the reverse: code that checks `rows.Err()` but returns the partial slice along with the error. Callers that log the error and keep the data reintroduce the bug one level up. Return `nil` data with a non-nil error unless partial results are a documented feature.

The boring rule

> Every `for rows.Next()` loop is followed by `if err := rows.Err(); err != nil { return nil, err }`, or lives inside one shared helper that does it. Run the `sqlrowserr` analyzer in CI and keep gopls current, because `go vet` will not catch the missing check.

Try this

1. Change the driver to return `io.EOF` after three rows and confirm both functions now agree, with `err=<nil>`.
2. Make the driver block in `Next` and call `sumIDsChecked` with a context that times out after 10 ms. What does `rows.Err()` return, and does `errors.Is(err, context.DeadlineExceeded)` hold?
3. Rewrite `sumIDs` and `sumIDsChecked` with `Collect` and run the checker again.
4. Add the checker to a `Makefile` target next to `go vet` and make CI run both.

Sources

- `database/sql`: `Rows.Next`, `Rows.Err`, `Rows.Close` (Go 1.27)
- `database/sql/driver`: `Rows.Next` (`io.EOF` means no more rows)
- Gopls release v0.23.0 (July 2026): `sqlrowserr` analyzer
- `golang.org/x/tools/go/analysis/passes/sqlrowserr` and `singlechecker`
- Go wiki: SQL interface — handling errors

net: Dial, Listen, and UDP

net: Dial, Listen, and UDP

Overview

Package `net` is the foundation under `net/http`: TCP/UDP addresses, listeners, dialers, and resolution. When HTTP is too high-level—or you need a custom protocol—you work here.

TCP services deep dive: 131 TCP services.

Addresses and resolution

```
// TCPAddr, UDPAddr, UnixAddr
tcpAddr, err := net.ResolveTCPAddr("tcp", "127.0.0.1:9000")
ips, err := net.LookupIP(ctx, "example.com") // prefer context variants where available
```

Prefer **`net.Dialer`** over bare `net.Dial` so you can set timeouts and keep-alives.

```
d := net.Dialer{
    Timeout:   3 * time.Second,
    KeepAlive: 30 * time.Second,
}
conn, err := d.DialContext(ctx, "tcp", "127.0.0.1:9000")
```

TCP server (line protocol sketch)

```
ln, err := net.Listen("tcp", "127.0.0.1:9000")
if err != nil {
    log.Fatal(err)
}
defer ln.Close()

for {
    conn, err := ln.Accept()
    if err != nil {
        log.Println("accept:", err)
        continue
    }
```

```go
    go handle(conn)
}

func handle(c net.Conn) {
    defer c.Close()
    _ = c.SetDeadline(time.Now().Add(30 * time.Second))
    br := bufio.NewReader(c)
    line, err := br.ReadString('\n')
    if err != nil {
        return
    }
    _, _ = io.WriteString(c, "echo: "+line)
}
```

Habit	Why
Deadline per connection	Avoid stuck clients forever
One goroutine per conn (or limited pool)	Bound resource use
Always `Close`	Prevent FD leaks
`bufio` for line protocols	Fewer syscalls

TCP client

```go
conn, err := d.DialContext(ctx, "tcp", addr)
if err != nil {
    return err
}
defer conn.Close()
_ = conn.SetDeadline(time.Now().Add(5 * time.Second))
_, err = io.WriteString(conn, "ping\n")
// read response...
```

UDP

```go
// listen
pc, err := net.ListenPacket("udp", "127.0.0.1:9001")
// or DialUDP for connected UDP

buf := make([]byte, 1500)
n, addr, err := pc.ReadFrom(buf)
_, err = pc.WriteTo(buf[:n], addr) // echo
```

UDP is message-oriented: no streams, no automatic retransmit. Size buffers for your MTU/use case.

Unix domain sockets

```go
ln, err := net.Listen("unix", "/tmp/app.sock")
// client: Dial("unix", "/tmp/app.sock")
```

Useful for local IPC (sidecar, same-host tools). Remember file permissions on the socket path.

net.Conn as io.Reader / Writer

```go
// stream copy with limit
n, err := io.Copy(dst, io.LimitReader(conn, 1<<20))
```

Composition with `io` is the Go style: your protocol parsers should take `io.Reader`/`io.Writer` when possible, not only `net.Conn`.

Errors and temporary failures

```go
var ne net.Error
if errors.As(err, &ne) && ne.Timeout() {
    // retry or return 504-style
}
```

DNS and dial failures are often wrapped—log with `%+v` / `slog` attributes; branch with `errors.As`.

ip and ipnet helpers

```go
ip := net.ParseIP("2001:db8::1")
_, cidr, _ := net.ParseCIDR("10.0.0.0/8")
ok := cidr.Contains(net.ParseIP("10.1.2.3"))
```

Handy for allowlists and basic network policy in tools.

Rules of thumb

Do	Don't
Dial/Listen with timeouts/deadlines	Block forever on Accept/Read
Propagate `context` on Dial	Ignore cancel when user hits Ctrl-C
Limit concurrent handlers	Accept unbounded `go handle` under load without a cap
Close connections	Leak FDs on error paths

Try next

1. Build a TCP echo server and client with 2s deadlines.
2. Send a UDP packet to yourself and print the peer address.
3. Wrap `handle` with a semaphore channel of size 100 and load-test Accept.

text/template and html/template

text/template and html/template

Overview

The template packages render **text with actions** (`{{...}}`). Use `html/template` for browser HTML (context-aware escaping). Use `text/template` for configs, emails-as-text, code generation, and non-HTML output.

Web-focused usage: Templating and rendering.

When to use which

Package	Escaping	Use
`html/template`	Yes (HTML/JS/CSS/URL contexts)	Web pages, HTML emails
`text/template`	No	Nginx configs, Markdown, codegen, plain text

Serving user data as HTML with `text/template` is an XSS footgun.

Core loop: parse once, execute many

```go
import "html/template"

tmpl, err := template.New("hi").Parse(`Hello, {{.Name}}!`)
if err != nil {
    log.Fatal(err)
}
err = tmpl.Execute(os.Stdout, struct{ Name string }{"Gopher"})
// Hello, Gopher!
```

From files:

```go
tmpl, err := template.ParseFiles("layout.tmpl", "page.tmpl")
// or template.ParseFS(fsys, "templates/*.tmpl")
```

Actions you will use constantly

```
{{.Field}}           // print field
{{if .OK}}yes{{else}}no{{end}}
{{range .Items}}{{.}}{{end}}
{{with .User}}{{.Email}}{{end}}
{{/* comment */}}
```

Pipelines:

```
{{.Title | printf "%q" | html}}
```

Define and execute named templates

```
{{define "layout"}}
<html><body>{{template "content" .}}</body></html>
{{end}}

{{define "content"}}
<h1>{{.Title}}</h1>
{{end}}
```

```
err := tmpl.ExecuteTemplate(w, "layout", data)
```

FuncMap

```
funcs := template.FuncMap{
    "upper": strings.ToUpper,
    "cents": func(n int) string {
        return fmt.Sprintf("$%.2f", float64(n)/100)
    },
}
tmpl, err := template.New("price").Funcs(funcs).Parse(`{{cents .Price}} {{upper .Title}}`)
```

Register **Funcs before Parse**. Only expose pure, trusted helpers.

html/template safety

```
// Auto-escaped:
// {{.Comment}}  from user → &lt;script&gt;...

// Escape hatch (trusted HTML only):
type page struct {
    Body template.HTML
}
```

Type	Meaning
`template.HTML`	Trusted HTML fragment
`template.JS`	Trusted JS
`template.CSS`	Trusted CSS
`template.URL`	Trusted URL

Never cast raw request input to these types.

Whitespace and formatting

```
{{- /* trim left */ -}}
{{.Name -}}  {{/* trim right */}}
```

`-` next to `{{` / `}}` trims adjacent whitespace—useful for tidy HTML or generated code.

text/template for codegen

```
const stub = `package {{.Pkg}}

func {{.Name}}() string { return {{printf "%q" .Msg}} }
`
t := template.Must(template.New("stub").Parse(stub))
var buf bytes.Buffer
_ = t.Execute(&buf, struct {
    Pkg, Name, Msg string
}{"demo", "Hello", "hi"})
// gofmt the result in real tools
```

Common errors

Error	Cause
`complete and no parse result`	Empty parse / wrong name
`no template "X"`	ExecuteTemplate name mismatch
Missing data prints `<no value>`	Field name wrong or nil pipeline
Func not defined	Funcs registered after Parse

Rules of thumb

Do	Don't
Parse at startup	Parse templates on every request
`html/template` for HTML	Build HTML with `fmt` + user strings
Pass view structs	Pass `*sql.DB` into templates
`template.Must` only at init	`Must` in request paths (panic under load)

Try next

1. Parse a layout + content from `embed.FS` and execute both HTML and a JSON-free page.
2. Add a `FuncMap` that formats `time.Time` as RFC3339.
3. Intentionally inject `<script>` via data and confirm `html/template` escapes it.

compress and archive

compress and archive

Overview

The standard library covers everyday compression and packaging: **gzip**, **zlib**, **flate**, **bzip2** (reader), **lzw**, plus **zip** and **tar** archives. Use these for log shipping, backup tools, HTTP `Content-Encoding`, and release artifacts—without pulling a third-party zipper for common cases.

Package map

Package	Role
`compress/gzip`	`.gz` streams (HTTP, logs)
`compress/zlib`	zlib wrappers
`compress/flate`	raw DEFLATE
`compress/bzip2`	bzip2 **reader** only
`archive/zip`	ZIP read/write
`archive/tar`	tar read/write (often + gzip)

Gzip: compress a file

```go
func gzipFile(src, dst string) error {
    in, err := os.Open(src)
    if err != nil {
        return err
    }
    defer in.Close()

    out, err := os.Create(dst)
    if err != nil {
        return err
    }
    defer out.Close()

    zw := gzip.NewWriter(out)
    zw.Name = filepath.Base(src)
    if _, err := io.Copy(zw, in); err != nil {
```

```go
        _ = zw.Close()
        return err
    }
    return zw.Close() // flush trailer; check this error
}
```

Decompress

```go
func gunzip(src io.Reader, dst io.Writer) error {
    zr, err := gzip.NewReader(src)
    if err != nil {
        return err
    }
    defer zr.Close()
    _, err = io.Copy(dst, zr)
    return err
}
```

HTTP Content-Encoding sketch

```go
// response: write gzip when client accepts it
func gzipMiddleware(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        if !strings.Contains(r.Header.Get("Accept-Encoding"), "gzip") {
            next.ServeHTTP(w, r)
            return
        }
        w.Header().Set("Content-Encoding", "gzip")
        gz := gzip.NewWriter(w)
        defer gz.Close()
        gzw := gzipResponseWriter{Writer: gz, ResponseWriter: w}
        next.ServeHTTP(gzw, r)
    })
}
```

(Production middleware also handles `Content-Length`, status codes, and `Vary: Accept-Encoding`—keep this as a pattern, not a drop-in.)

tar + gzip (`.tar.gz`)

```go
func untarGz(r io.Reader, destDir string) error {
    gr, err := gzip.NewReader(r)
    if err != nil {
        return err
    }
    defer gr.Close()
    tr := tar.NewReader(gr)

    for {
        hdr, err := tr.Next()
        if errors.Is(err, io.EOF) {
            return nil
        }
        if err != nil {
            return err
        }
        // Prevent path traversal: clean + ensure under destDir
        target, err := safeJoin(destDir, hdr.Name)
        if err != nil {
            return err
        }
        switch hdr.Typeflag {
        case tar.TypeDir:
            if err := os.MkdirAll(target, 0o755); err != nil {
                return err
            }
        case tar.TypeReg:
            if err := os.MkdirAll(filepath.Dir(target), 0o755); err != nil {
                return err
            }
            f, err := os.OpenFile(target, os.O_CREATE|os.O_WRONLY|os.O_TRUNC, hdr.FileInfo()
            if err != nil {
                return err
            }
            if _, err := io.Copy(f, io.LimitReader(tr, hdr.Size)); err != nil {
                f.Close()
                return err
            }
            f.Close()
        }
    }
}
```

Path traversal: never write `hdr.Name` that escapes `destDir` (`../` tricks). Always clean and verify a common root.

ZIP

```go
// read
r, err := zip.OpenReader("site.zip")
defer r.Close()
for _, f := range r.File {
    rc, err := f.Open()
    // io.Copy to safe path under dest; then rc.Close()
}

// write
f, err := os.Create("out.zip")
zw := zip.NewWriter(f)
w, err := zw.Create("readme.txt")
_, err = io.WriteString(w, "hello")
err = zw.Close()
```

Same traversal rules as tar when extracting.

Streaming vs memory

```
good:  io.Copy(gzipWriter, file)     // bounded memory
bad:   gzip.Compress whole []byte     // for huge inputs
```

Prefer stream composition (`io.Reader` → compressor → `io.Writer`).

Rules of thumb

Do	Don't
Check `Close` on gzip writers	Ignore close errors (corrupt archives)
Limit extracted sizes	Trust archive headers blindly
Sanitize entry paths	Extract zip-slip paths
Stream large payloads	`ReadAll` multi-GB archives

Try next

1. Gzip a log file and gunzip it; compare SHA-256.
2. Build a tiny `tar.gz` of a directory listing.
3. Attempt a zip-slip path and ensure your extractor rejects it.

encoding/binary, gob, and pem

encoding/binary, gob, and pem

Overview

Beyond JSON/CSV/XML (chapter 985), the stdlib encodes **binary layouts**, **Go-native gob streams**, and **PEM** blocks (certs/keys). Pick the format for the consumer: wire protocols → binary; Go-to-Go caches → gob (carefully); TLS material → pem + x509.

encoding/binary

Read and write numbers in a fixed endianness—file headers, network protocols, length-prefixed frames.

```go
var buf bytes.Buffer
// write
if err := binary.Write(&buf, binary.BigEndian, uint32(42)); err != nil {
    return err
}
if err := binary.Write(&buf, binary.BigEndian, int16(-3)); err != nil {
    return err
}

// read
var u uint32
var s int16
r := bytes.NewReader(buf.Bytes())
if err := binary.Read(r, binary.BigEndian, &u); err != nil {
    return err
}
if err := binary.Read(r, binary.BigEndian, &s); err != nil {
    return err
}
```

Fixed size helpers

```go
b := make([]byte, 8)
binary.BigEndian.PutUint64(b, 0xdeadbeef)
n := binary.BigEndian.Uint64(b)
_ = n
```

Endian	Common use
`binary.BigEndian`	Network byte order, many file formats
`binary.LittleEndian`	x86-native dumps, some formats
`binary.NativeEndian`	Same-machine only (Go 1.19+)

Length-prefixed message

```go
func writeFrame(w io.Writer, payload []byte) error {
    if err := binary.Write(w, binary.BigEndian, uint32(len(payload))); err != nil {
        return err
    }
    _, err := w.Write(payload)
    return err
}

func readFrame(r io.Reader, max int) ([]byte, error) {
    var n uint32
    if err := binary.Read(r, binary.BigEndian, &n); err != nil {
        return nil, err
    }
    if int(n) > max {
        return nil, fmt.Errorf("frame %d exceeds max %d", n, max)
    }
    buf := make([]byte, n)
    _, err := io.ReadFull(r, buf)
    return buf, err
}
```

Always bound `max`—never trust remote lengths blindly.

encoding/gob

Gob is Go's built-in binary serialization for **Go types** (including interfaces with registration). Great for same-version Go services and caches; **not** a long-term public API format.

```go
type Item struct {
    SKU   string
    Count int
}

var buf bytes.Buffer
enc := gob.NewEncoder(&buf)
if err := enc.Encode(Item{SKU: "go-book", Count: 2}); err != nil {
    log.Fatal(err)
}
dec := gob.NewDecoder(&buf)
var out Item
if err := dec.Decode(&out); err != nil {
    log.Fatal(err)
}
```

Interfaces

```go
gob.Register(Item{}) // once; before encode/decode of interface values
```

Caveats

Prefer gob when	Avoid gob when
Go Go, controlled versions	Public multi-language APIs
Short-lived cache blobs	Long-term archival formats
Internal RPC experiments	Anything needs schema evolution guarantees

For public APIs prefer JSON, Protobuf, or similar.

encoding/pem

PEM is the ASCII armor around DER bytes (`-----BEGIN ...-----`).

```go
block, rest := pem.Decode(pemBytes)
if block == nil {
    return fmt.Errorf("no PEM block")
}
_ = rest
// block.Type e.g. "CERTIFICATE", "RSA PRIVATE KEY"
// block.Bytes is DER
```

Encode:

```go
var buf bytes.Buffer
_ = pem.Encode(&buf, &pem.Block{Type: "CERTIFICATE", Bytes: der})
```

Pair with `crypto/x509`:

```go
cert, err := x509.ParseCertificate(block.Bytes)
```

Never log private key PEM in application logs.

encoding/hex and base64 (recap)

```go
h := hex.EncodeToString(sum[:])
raw, err := hex.DecodeString(h)

s := base64.StdEncoding.EncodeToString(data)
// URL-safe: base64.RawURLEncoding
```

Chapter 985 covers these in the JSON/CSV track; they pair naturally with binary digests and tokens.

Rules of thumb

Do	Don't
Cap length-prefixed frames	Allocate `make([]byte, n)` from network without max
Document endianness	Mix Big/Little without a spec
Use gob only in-process/versioned Go	Expose gob as a mobile client protocol
Keep PEM keys off disk	Commit private keys into git world-readable

Try next

1. Implement writeFrame/readFrame over a `bytes.Buffer` and fuzz lengths.
2. Gob-encode a slice of structs and decode into a new value.
3. PEM-decode a test certificate and print `Subject` via `x509.ParseCertificate`.

math, math/bits, and math/big

math, math/bits, and math/big

Overview

- **`math`**: floating-point helpers (`Sin`, `Sqrt`, `Inf`, `NaN`, rounding).
- **`math/bits`**: integer bit operations (ones count, rotate, leading zeros)—often used for hashers, allocators, and compact encodings.
- **`math/big`**: arbitrary-precision integers/floats/rats—crypto sizes, money-like rationals, huge counters.

Most application code needs little of `math`; systems and crypto-adjacent tools use `bits` and `big` more.

math: floats with eyes open

```go
import "math"

x := math.Hypot(3, 4) // 5
y := math.Round(2.5)  // 3 (half away from zero in recent Go - check docs for edge cases)
```

NaN and Inf

```go
nan := math.NaN()
inf := math.Inf(1)
if math.IsNaN(nan) { /* ... */ }
if math.IsInf(inf, 1) { /* +Inf */ }
```

NaN compares unequal to itself. Prefer `math.IsNaN` over `==`.

Float equality

```go
func almostEqual(a, b, eps float64) bool {
    return math.Abs(a-b) <= eps
}
```

Do not use `==` for computed floats in tests without a tolerance.

Integer-ish helpers

```go
math.Max(a, b) // float64 only in classic math
// For ordered types prefer cmp / min/max builtins (Go 1.21+)
m := max(a, b) // builtin for integers etc.
```

math/bits: integer bit tricks

```go
import "math/bits"

bits.OnesCount64(x)      // popcount
bits.LeadingZeros64(x)   // for log2-ish sizing
bits.TrailingZeros64(x)  // power-of-two alignment
bits.RotateLeft64(x, 7)
bits.ReverseBytes64(x)
bits.Len64(x)            // bit length
```

Example: next power of two sizing

```go
func nextPow2(n uint64) uint64 {
    if n <= 1 {
        return 1
    }
    return 1 << bits.Len64(n-1)
}
```

These compile to efficient CPU instructions on common architectures.

math/big: big integers

```go
import "math/big"

a := new(big.Int).SetString("12345678901234567890", 10)
b := big.NewInt(42)
sum := new(big.Int).Add(a, b)
fmt.Println(sum.String())
```

Methods mutate and return receiver

```
// sum = a + b; Add returns sum for chaining
sum.Add(a, b)
// careful: overlapping aliases - read docs for each method
```

Modular exponent (crypto-shaped)

```
result := new(big.Int).Exp(base, exp, mod) // base^exp mod mod
```

For production crypto, prefer `crypto/*` packages over hand-rolled `big` math.

Rationals and floats

```
r := big.NewRat(1, 3)
f, _ := new(big.Float).SetString("3.14159265358979323846")
_ = r
_ = f
```

Money warning

`float64` is a poor ledger type. Prefer:

- integer **cents** (`int64`), or
- `big.Rat` / decimal libraries for fractional currency rules.

```
// cents
total := int64(1999) + int64(500) // $19.99 + $5.00
```

Performance notes

Tool	Cost
`math` float ops	Cheap; watch NaN paths
`math/bits`	Very cheap
`math/big`	Heap, slower; fine for rare big numbers

Do not use `big.Int` for every loop counter.

Rules of thumb

Do	Don't
Use builtins `min`/`max` for integers	Force float `math.Max` for int
Test floats with epsilon	Assert exact binary equality
Use `bits` for masks/popcount	Hand-roll portable popcount
Use `big` for true big integers	Store money in `float64`

Try next

1. Implement `nextPow2` and table-test edges (0, 1, 2, 3, MaxUint32).
2. Add two 50-digit integers with `big.Int`.
3. Show that `0.1+0.2 != 0.3` in float64 and fix with integer cents.

mime, multipart, expvar, and runtime/debug

mime, multipart, expvar, and runtime/debug

Overview

This chapter groups **edge-of-stdlib** packages every serious tool eventually touches:

- **`mime` / `mime/multipart`** — content types and file uploads
- **`expvar`** — process variables for quick ops scrapes
- **`net/http/pprof`** — profile endpoints (import for side effect)
- **`runtime/debug` / `debug/buildinfo`** — GC knobs, stack dumps, embed build metadata

mime: content types

```go
import "mime"

ctype := mime.TypeByExtension(".json") // application/json
exts, _ := mime.ExtensionsByType("image/png")

mediatype, params, err := mime.ParseMediaType(
    `multipart/form-data; boundary=----abcd`)
// mediatype == "multipart/form-data"
// params["boundary"] == "----abcd"
```

Set types explicitly on responses when you know them:

```go
w.Header().Set("Content-Type", "application/json; charset=utf-8")
```

multipart: form uploads

```go
func upload(w http.ResponseWriter, r *http.Request) {
    // Bound memory: 32 MiB in memory, rest to temp files
    if err := r.ParseMultipartForm(32 << 20); err != nil {
        http.Error(w, "bad multipart", http.StatusBadRequest)
        return
```

```go
    }
    file, hdr, err := r.FormFile("file")
    if err != nil {
        http.Error(w, "missing file", http.StatusBadRequest)
        return
    }
    defer file.Close()

    // Optional: sniff type from head bytes
    head := make([]byte, 512)
    n, _ := file.Read(head)
    ctype := http.DetectContentType(head[:n])
    // re-open or use io.MultiReader if you need full stream again

    dst, err := os.CreateTemp("", "upload-*")
    if err != nil {
        http.Error(w, "storage", http.StatusInternalServerError)
        return
    }
    defer dst.Close()
    if _, err := io.Copy(dst, io.MultiReader(bytes.NewReader(head[:n]), file)); err != nil {
        http.Error(w, "write", http.StatusInternalServerError)
        return
    }
    fmt.Fprintf(w, "saved %s (%s) as %s\n", hdr.Filename, ctype, dst.Name())
}
```

Habit	Why
Size limits	Avoid memory exhaustion
Don't trust `Filename` alone	Path traversal / odd characters
Detect content type	Clients lie about MIME

Writing multipart (client)

```go
var buf bytes.Buffer
w := multipart.NewWriter(&buf)
fw, _ := w.CreateFormFile("file", "notes.txt")
_, _ = io.WriteString(fw, "hello")
_ = w.Close()

req, _ := http.NewRequest(http.MethodPost, url, &buf)
req.Header.Set("Content-Type", w.FormDataContentType())
```

expvar: export process vars

```go
import "expvar"

var (
    requests = expvar.NewInt("requests_total")
    version  = expvar.NewString("version")
)

func init() {
    version.Set("1.2.3")
}

func handle(w http.ResponseWriter, r *http.Request) {
    requests.Add(1)
    // ...
}

// http.Handle("/debug/vars", expvar.Handler()) // JSON map of vars
```

Useful for **simple** scrapes. For production metrics at scale, prefer Prometheus-style instrumentation (part 19)—but `expvar` is zero-dep and always there.

net/http/pprof

```go
import _ "net/http/pprof"

// If you use DefaultServeMux:
go func() {
    log.Println(http.ListenAndServe("127.0.0.1:6060", nil))
}()
// Then: go tool pprof http://127.0.0.1:6060/debug/pprof/heap
//        go tool pprof http://127.0.0.1:6060/debug/pprof/goroutineleak  (Go 1.27+)
```

Bind pprof to **localhost** or protect it—profiles leak internals.

runtime/debug

```go
import "runtime/debug"

debug.SetGCPercent(50)          // GC target; trade CPU vs memory
debug.SetMemoryLimit(2 << 30)   // soft memory limit (Go 1.19+)
```

```go
debug.WriteHeapDump(fd)         // heavy; support tooling
debug.PrintStack()              // stderr stack of current goroutine
```

Build info at runtime:

```go
if bi, ok := debug.ReadBuildInfo(); ok {
    fmt.Println(bi.Main.Path, bi.Main.Version)
    for _, s := range bi.Settings {
        // vcs.revision, -ldflags, CGO_ENABLED, ...
    }
}
```

debug/buildinfo (binaries on disk)

```go
import "debug/buildinfo"

bi, err := buildinfo.ReadFile("/path/to/binary")
// bi.GoVersion, bi.Path, bi.Deps
```

Handy for "what version is deployed?" without a `/version` endpoint.

Rules of thumb

Do	Don't
Bound multipart memory	Parse unlimited uploads
Lock down `/debug/*`	Expose pprof on `0.0.0.0` publicly
Export a version string	Guess deploy version from timestamps
Prefer slog + metrics for ops	Rely only on `fmt` prints in prod

Try next

1. Accept a multipart upload and reject bodies over 1 MiB.
2. Register an `expvar` counter and curl `/debug/vars`.
3. Print `debug.ReadBuildInfo()` from your binary's `main`.

container/heap, list, and ring

container/heap, list, and ring

Overview

The `container` packages implement classic structures not in the language: **heap** (priority queue), **doubly linked list**, and **ring**.

container/heap

```go
type intHeap []int
func (h intHeap) Len() int           { return len(h) }
func (h intHeap) Less(i, j int) bool { return h[i] < h[j] }
func (h intHeap) Swap(i, j int)      { h[i], h[j] = h[j], h[i] }
func (h *intHeap) Push(x any)        { *h = append(*h, x.(int)) }
func (h *intHeap) Pop() any {
    old := *h
    n := len(old)
    x := old[n-1]
    *h = old[:n-1]
    return x
}

h := &intHeap{3, 1, 2}
heap.Init(h)
heap.Push(h, 0)
x := heap.Pop(h).(int) // 0
```

Use for schedulers, merge-k-sorted, Dijkstra.

container/list

```go
l := list.New()
e := l.PushBack("b")
l.PushFront("a")
l.InsertAfter("c", e)
for e := l.Front(); e != nil; e = e.Next() {
    fmt.Println(e.Value)
```

```
}
```

Prefer slices unless you need O(1) remove from middle with handles.

container/ring

```go
r := ring.New(3)
for i := 0; i < r.Len(); i++ {
    r.Value = i
    r = r.Next()
}
r.Do(func(v any) { fmt.Println(v) })
```

Round-robin buffers.

Rules

Prefer	When
`slices` + sort	Simple ordered data
`heap`	Repeated min/max extract
`list`	Heavy middle splice with element pointers

Try next

1. Top-K largest with min-heap of size K.
2. LRU sketch: map + list.
3. Ring for N worker round-robin.

path and path/filepath Deep Dive

path and path/filepath Deep Dive

Overview

`path` is slash-separated (URLs, import paths). `path/filepath` is **OS paths**. Mixing them is a common bug.

Quick rules

Package	Use
`path/filepath`	Files on disk
`path`	URL paths, logical slash paths

```go
filepath.Join("a", "b")     // a/b or a\b
path.Join("a", "b")         // always a/b
filepath.Base("/tmp/x.go")
filepath.Ext("x.go")        // .go
filepath.Clean("../etc/passwd")
filepath.Abs(".")
```

Safe join (jail)

```go
func safeJoin(root, unsafe string) (string, error) {
    c := filepath.Clean("/" + unsafe) // force abs clean form
    rel := strings.TrimPrefix(c, string(filepath.Separator))
    full := filepath.Join(root, rel)
    absRoot, _ := filepath.Abs(root)
    absFull, _ := filepath.Abs(full)
    if !strings.HasPrefix(absFull, absRoot+string(filepath.Separator)) && absFull != absRoot
        return "", fmt.Errorf("escape")
    }
    return absFull, nil
}
```

Walk

```go
filepath.WalkDir(root, func(p string, d fs.DirEntry, err error) error {
    if err != nil {
        return err
    }
    if d.IsDir() && d.Name() == ".git" {
        return fs.SkipDir
    }
    return nil
})
```

Match / Glob

```go
filepath.Glob("*.go")
filepath.Match("a/*.txt", name)
```

Rules

Do	Don't
`filepath` for files	`path.Join` for disk on Windows
Clean + jail user paths	Concatenate `root + "/" + user`

Try next

1. Prove `path.Join` vs `filepath.Join` docs on Windows GOOS.
2. WalkDir skip `node_modules`.
3. SafeJoin reject `../../etc/passwd`.

bufio Scanner and Readers Advanced

bufio Scanner and Readers Advanced

Overview

`bufio` sits between raw `os.File`/`net.Conn` and high-level logic. Master Scanner token sizes, Reader peek, and Writer flush.

Scanner limits

Default max token ~64K. Long lines fail with `bufio.ErrTooLong`.

```
sc := bufio.NewScanner(r)
buf := make([]byte, 0, 1024*1024)
sc.Buffer(buf, 10*1024*1024) // max 10MiB tokens
sc.Split(bufio.ScanLines)
for sc.Scan() {
    _ = sc.Text()
}
return sc.Err()
```

Custom split

```
// split on NUL
sc.Split(func(data []byte, atEOF bool) (advance int, token []byte, err error) {
    if i := bytes.IndexByte(data, 0); i >= 0 {
        return i + 1, data[:i], nil
    }
    if atEOF && len(data) > 0 {
        return len(data), data, nil
    }
    return 0, nil, nil
})
```

Reader Peek / Discard

```go
br := bufio.NewReader(r)
b, _ := br.Peek(4) // next bytes without consume
_, _ = br.Discard(4)
```

Writer

```go
bw := bufio.NewWriter(w)
fmt.Fprintln(bw, "line")
_ = bw.Flush() // required before close for full delivery
```

Rules

Do	Don't
Raise buffer for long lines	Ignore Scan errors
Flush writers	Assume OS sees buffered data
Prefer Scanner for lines	ReadAll multi-GB logs

Try next

1. File with 1MB line; fix with Buffer.
2. Peek magic bytes for file type.
3. Benchmark Scanner vs ReadString.

crypto/mldsa

crypto/mldsa

Go 1.27 ships **`crypto/mldsa`**: FIPS 204 ML-DSA signatures in the standard library. Pair it with the existing `crypto/mlkem` story — ML-KEM for key exchange, ML-DSA for signing. The boring default for new post-quantum signing on Go 1.27+ is **`mldsa.MLDSA44()`** unless your threat model or peer explicitly requires a larger parameter set. Signatures and public keys are large; design wire formats and storage around that size before you ship.

Mental model

ML-DSA is a lattice-based signature scheme. You pick one of three fixed parameter sets:

Helper	Public key	Signature	Typical use
`MLDSA44()`	1312 bytes	2420 bytes	Default for most apps
`MLDSA65()`	1952 bytes	3309 bytes	Higher security margin
`MLDSA87()`	2592 bytes	4627 bytes	Highest NIST category

Compare that to an Ed25519 public key (32 bytes) and signature (64 bytes). The private key seed is still only **32 bytes** (`mldsa.PrivateKeySize`); store the seed, not an expanded form.

`PrivateKey` implements `crypto.Signer`, so it plugs into APIs that already accept a signer. `crypto/x509` understands ML-DSA keys and signatures. `crypto/tls` exposes `MLDSA44`, `MLDSA65`, and `MLDSA87` as TLS 1.3 `SignatureScheme` values.

`Options.Context` is domain separation: the same key can sign for different purposes if each purpose uses a distinct context string (at most 255 bytes). Verifying with the wrong context fails.

If the process is built against the FIPS 140-3 Go Cryptographic Module v1.0.0, `GenerateKey` / `NewPublicKey` / `Verify` return an error. Use module v1.26.0 or later, or a non-FIPS build, for this package.

Worked examples

Case 1: Generate, sign, verify

Save as `mldsa_roundtrip.go`. Generate an ML-DSA-44 key, publish the public key bytes, sign a desk message, and verify.

```go
// mldsa_roundtrip.go
package main

import (
    "crypto/mldsa"
    "fmt"
    "log"
)

func main() {
    sk, err := mldsa.GenerateKey(mldsa.MLDSA44())
    if err != nil {
        log.Fatal(err)
    }

    pubEnc := sk.PublicKey().Bytes()
    fmt.Printf("public key: %d bytes\n", len(pubEnc))
    fmt.Printf("private seed: %d bytes\n", len(sk.Bytes()))

    msg := []byte("shift-close: till-3 sealed")
    opts := &mldsa.Options{Context: "desk/shift-close"}
    sig, err := sk.Sign(nil, msg, opts)
    if err != nil {
        log.Fatal(err)
    }
    fmt.Printf("signature: %d bytes\n", len(sig))

    pk, err := mldsa.NewPublicKey(mldsa.MLDSA44(), pubEnc)
    if err != nil {
        log.Fatal(err)
    }
    if err := mldsa.Verify(pk, msg, sig, opts); err != nil {
        log.Fatal("invalid signature: ", err)
    }
    fmt.Println("verified: true")
}
```

Run:

```go
go run mldsa_roundtrip.go
```

Output:

```
public key: 1312 bytes
private seed: 32 bytes
signature: 2420 bytes
verified: true
```

`Sign`'s `io.Reader` argument is ignored for ML-DSA; pass `nil`. Always check errors from `GenerateKey` and `Verify` — a FIPS 140-3 v1.0.0 module returns an error instead of producing keys.

Case 2: Context mismatch fails verification

Save as `mldsa_context.go`. Sign under one context and verify under another. Domain separation is part of the signature, not a soft hint.

```go
// mldsa_context.go
package main

import (
    "crypto/mldsa"
    "fmt"
    "log"
)

func main() {
    sk, err := mldsa.GenerateKey(mldsa.MLDSA44())
    if err != nil {
        log.Fatal(err)
    }
    msg := []byte("release the gophers")

    sig, err := sk.Sign(nil, msg, &mldsa.Options{Context: "prod/release"})
    if err != nil {
        log.Fatal(err)
    }

    pk := sk.PublicKey()
    err = mldsa.Verify(pk, msg, sig, &mldsa.Options{Context: "staging/release"})
    fmt.Printf("wrong context: %v\n", err != nil)

    err = mldsa.Verify(pk, msg, sig, &mldsa.Options{Context: "prod/release"})
    fmt.Printf("matching context: %v\n", err == nil)
}
```

Run:

```go
go run mldsa_context.go
```

Output:

```
wrong context: true
matching context: true
```

Use a stable, purpose-specific context string (`"api/v1/token"`, `"git/tag"`, `"desk/receipt"`). Empty context is allowed and is the default when `opts` is nil, but then every consumer of that key shares one domain.

Case 3: Parameter set sizes

Save as `mldsa_sizes.go`. Print the public-key and signature sizes for each parameter set so wire budgets are visible before you pick one.

```go
// mldsa_sizes.go
package main

import (
    "crypto/mldsa"
    "fmt"
)

func main() {
    for _, p := range []mldsa.Parameters{
        mldsa.MLDSA44(),
        mldsa.MLDSA65(),
        mldsa.MLDSA87(),
    } {
        fmt.Printf("%s: pubkey=%d sig=%d\n",
            p.String(), p.PublicKeySize(), p.SignatureSize())
    }
    fmt.Printf("private seed always: %d bytes\n", mldsa.PrivateKeySize)
}
```

Run:

```go
go run mldsa_sizes.go
```

Output:

```
ML-DSA-44: pubkey=1312 sig=2420
ML-DSA-65: pubkey=1952 sig=3309
ML-DSA-87: pubkey=2592 sig=4627
private seed always: 32 bytes
```

Prefer `MLDSA44()` unless a peer or compliance table names a higher set. Bigger sets buy margin at a fixed cost in every packet and every stored certificate.

Case 4: Reload a key from its seed

Save as `mldsa_seed.go`. Persist only the 32-byte seed. Rebuild the private key with `NewPrivateKey` when you need to sign again.

```go
// mldsa_seed.go
package main

import (
    "crypto/mldsa"
    "fmt"
    "log"
)

func main() {
    sk1, err := mldsa.GenerateKey(mldsa.MLDSA44())
    if err != nil {
        log.Fatal(err)
    }
    seed := sk1.Bytes() // 32-byte seed - this is what you store

    sk2, err := mldsa.NewPrivateKey(mldsa.MLDSA44(), seed)
    if err != nil {
        log.Fatal(err)
    }

    fmt.Printf("seeds equal: %v\n", sk1.Equal(sk2))
    fmt.Printf("pubs equal: %v\n", sk1.PublicKey().Equal(sk2.PublicKey()))

    msg := []byte("reload-ok")
    opts := &mldsa.Options{Context: "desk/reload"}
    sig, err := sk2.Sign(nil, msg, opts)
    if err != nil {
        log.Fatal(err)
    }
    if err := mldsa.Verify(sk1.PublicKey(), msg, sig, opts); err != nil {
        log.Fatal(err)
    }
    fmt.Println("sign after reload: ok")
}
```

Run:

```go
go run mldsa_seed.go
```

Output:

```
seeds equal: true
```

```
pubs equal: true
sign after reload: ok
```

Treat the seed like any other private key material. Do not log it. Do not put it in a config repo. The expanded public key is what you publish.

The trap

Save as `mldsa_size_trap.go`. Treating an ML-DSA signature like an Ed25519 blob blows past fixed-size buffers and column widths that were fine for classical signatures.

```go
// mldsa_size_trap.go
package main

import (
    "crypto/mldsa"
    "fmt"
    "log"
)

func main() {
    sk, err := mldsa.GenerateKey(mldsa.MLDSA44())
    if err != nil {
        log.Fatal(err)
    }
    sig, err := sk.Sign(nil, []byte("ticket-42"), nil)
    if err != nil {
        log.Fatal(err)
    }

    const classicalSigBudget = 64 // Ed25519-sized column / stack buffer
    fmt.Printf("signature bytes: %d\n", len(sig))
    fmt.Printf("fits classical 64-byte budget: %v\n", len(sig) <= classicalSigBudget)
    fmt.Println("fix: size the wire and DB for PublicKeySize + SignatureSize of the chosen s
}
```

Run:

```
go run mldsa_size_trap.go
```

Output:

```
signature bytes: 2420
fits classical 64-byte budget: false
fix: size the wire and DB for PublicKeySize + SignatureSize of the chosen set
```

The second trap is mixing parameter sets: a key generated with `MLDSA44()` must be reconstructed and verified with `MLDSA44()`. Passing `MLDSA65()` to `NewPublicKey` with 44-encoding fails.

The boring rule

- New post-quantum signing on Go 1.27+: import `crypto/mldsa`; default to `MLDSA44()`.
- Store the 32-byte seed (`PrivateKey.Bytes()`); publish `PublicKey.Bytes()`.
- Always set a purpose-specific `Options.Context` when one key serves more than one role; verify with the same context.
- Budget ~1.3 KiB for the public key and ~2.4 KiB for each ML-DSA-44 signature on the wire and in storage.
- Use `crypto/Signer` integration and `crypto/x509` / `crypto/tls` support rather than inventing a custom encoding.
- Check errors from `GenerateKey` / `Verify` under FIPS builds; the package may be unavailable on module v1.0.0.

Try this

1. In `mldsa_roundtrip.go`, switch to `MLDSA65()` and confirm the printed public-key and signature lengths match `mldsa_sizes.go`.
2. In `mldsa_context.go`, verify with `opts == nil` after signing with a non-empty context; confirm it fails.
3. In `mldsa_seed.go`, truncate the seed to 31 bytes before `NewPrivateKey` and confirm you get an error.
4. Call `p.PublicKeySize()` and `p.SignatureSize()` for `MLDSA87()` and add those numbers to a comment next to any `VARBINARY` / protobuf field you would use in production.

crypto/mlkem

crypto/mlkem

Go ships **`crypto/mlkem`**: FIPS 203 ML-KEM (formerly Kyber) in the standard library. It is a **key encapsulation mechanism** (KEM), not a general-purpose encrypt-this-file API. Alice holds a decapsulation key; Bob uses the matching encapsulation key to produce a **shared secret** plus a **ciphertext**; Alice recovers the same shared secret. The boring default is **ML-KEM-768** (`GenerateKey768`). Pair the idea with post-quantum signatures (`crypto/mldsa` on Go 1.27+) when you need both exchange and signing — this chapter stays on the KEM.

Mental model

A KEM has three steps:

1. **Generate** a decapsulation key (secret). Publish its encapsulation key (public).
2. **Encapsulate** with the public key → `(sharedKey, ciphertext)`. Send only the ciphertext.
3. **Decapsulate** with the secret key + ciphertext → `sharedKey` (same 32 bytes on success).

Parameter set	Encap key	Ciphertext	Shared key	Seed
ML-KEM-768 (default)	1184 bytes	1088 bytes	32 bytes	64 bytes
ML-KEM-1024	1568 bytes	1568 bytes	32 bytes	64 bytes

Constants live on the package: `EncapsulationKeySize768`, `CiphertextSize768`, `SharedKeySize`, `SeedSize`, and the `1024` twins.

Store the **64-byte seed** (`DecapsulationKey768.Bytes()` / `NewDecapsulationKey768`), not an ad-hoc encoding of expanded material. Publish `EncapsulationKey().Bytes()`.

Implicit rejection: if the ciphertext has the **wrong length**, `Decapsulate` returns an error. If the length is correct but the ciphertext is garbage or for another key, `Decapsulate` still returns a 32-byte value — it just will **not** match the sender's shared key. Do not treat "no error" as "authenticated peer." Derive session keys with a KDF and bind identity/transcript the way TLS does; do not invent a protocol that branches on decapsulate errors alone.

Hybrid TLS (high level): since Go 1.24, `crypto/tls` can negotiate the hybrid group **X25519MLKEM768** by default when `Config.CurvePreferences` is nil. Go 1.27 also supports **MLKEM1024** (and related hybrid preferences) when you opt in via `CurvePreferences`. That path uses ML-KEM inside the handshake; most applications never call `crypto/mlkem` directly. Call this package when you design your own KEM-based protocol, store PQ encapsulation keys, or test vectors — not to reimplement TLS.

Worked examples

Case 1: Generate, encapsulate, decapsulate

Save as `mlkem_roundtrip.go`. Alice generates ML-KEM-768, publishes the encapsulation key bytes, Bob encapsulates, Alice decapsulates, and both shared secrets match.

```go
// mlkem_roundtrip.go
package main

import (
    "bytes"
    "crypto/mlkem"
    "fmt"
    "log"
)

func main() {
    dk, err := mlkem.GenerateKey768()
    if err != nil {
        log.Fatal(err)
    }
    pub := dk.EncapsulationKey().Bytes()
    fmt.Printf("encapsulation key: %d bytes\n", len(pub))
    fmt.Printf("seed: %d bytes\n", len(dk.Bytes()))

    ek, err := mlkem.NewEncapsulationKey768(pub)
    if err != nil {
        log.Fatal(err)
    }
    sharedBob, ciphertext := ek.Encapsulate()
    fmt.Printf("ciphertext: %d bytes\n", len(ciphertext))
    fmt.Printf("shared key: %d bytes\n", len(sharedBob))

    sharedAlice, err := dk.Decapsulate(ciphertext)
    if err != nil {
        log.Fatal(err)
    }
    fmt.Printf("match: %v\n", bytes.Equal(sharedAlice, sharedBob))
}
```

Run:

```go
go run mlkem_roundtrip.go
```

Output:

```
encapsulation key: 1184 bytes
```

```
seed: 64 bytes
ciphertext: 1088 bytes
shared key: 32 bytes
match: true
```

`Encapsulate` draws randomness from a secure source. Keep both the seed and the shared key secret. The ciphertext is what you put on the wire.

Case 2: Reload a key from its seed

Save as `mlkem_seed.go`. Persist only the 64-byte seed. Rebuild with `NewDecapsulationKey768` when you need to decapsulate again.

```go
// mlkem_seed.go
package main

import (
    "bytes"
    "crypto/mlkem"
    "fmt"
    "log"
)

func main() {
    dk1, err := mlkem.GenerateKey768()
    if err != nil {
        log.Fatal(err)
    }
    seed := dk1.Bytes() // 64-byte d||z seed - this is what you store

    dk2, err := mlkem.NewDecapsulationKey768(seed)
    if err != nil {
        log.Fatal(err)
    }
    fmt.Printf("seeds equal: %v\n", bytes.Equal(dk1.Bytes(), dk2.Bytes()))
    fmt.Printf("pubs equal: %v\n",
        bytes.Equal(dk1.EncapsulationKey().Bytes(), dk2.EncapsulationKey().Bytes()))

    ek := dk2.EncapsulationKey()
    shared, ct := ek.Encapsulate()
    got, err := dk1.Decapsulate(ct)
    if err != nil {
        log.Fatal(err)
    }
    fmt.Printf("decapsulate after reload: %v\n", bytes.Equal(shared, got))
}
```

Run:

```
go run mlkem_seed.go
```

Output:

```
seeds equal: true
pubs equal: true
decapsulate after reload: true
```

Treat the seed like any other private key material. Do not log it. Do not put it in a config repo. The encapsulation key bytes are what you publish.

Case 3: Parameter set sizes

Save as `mlkem_sizes.go`. Print the fixed sizes so wire and storage budgets are visible before you pick 768 vs 1024.

```go
// mlkem_sizes.go
package main

import (
    "crypto/mlkem"
    "fmt"
)

func main() {
    fmt.Printf("ML-KEM-768: encap=%d ct=%d\n",
        mlkem.EncapsulationKeySize768, mlkem.CiphertextSize768)
    fmt.Printf("ML-KEM-1024: encap=%d ct=%d\n",
        mlkem.EncapsulationKeySize1024, mlkem.CiphertextSize1024)
    fmt.Printf("shared key always: %d bytes\n", mlkem.SharedKeySize)
    fmt.Printf("seed always: %d bytes\n", mlkem.SeedSize)
}
```

Run:

```
go run mlkem_sizes.go
```

Output:

```
ML-KEM-768: encap=1184 ct=1088
ML-KEM-1024: encap=1568 ct=1568
shared key always: 32 bytes
seed always: 64 bytes
```

Prefer **768** unless a peer or compliance table names 1024. Mixing sets fails: a 768 encapsulation key must be parsed with `NewEncapsulationKey768`, not `NewEncapsulationKey1024`.

Case 4: Ciphertext length vs garbage (real API pitfall)

Save as `mlkem_decap_pitfall.go`. Wrong-length ciphertexts error. Correct-length garbage does **not** error — it yields a different shared key. That is FIPS 203 implicit rejection, not a bug.

```go
// mlkem_decap_pitfall.go
package main

import (
    "bytes"
    "crypto/mlkem"
    "fmt"
    "log"
)

func main() {
    dk, err := mlkem.GenerateKey768()
    if err != nil {
        log.Fatal(err)
    }
    shared, ct := dk.EncapsulationKey().Encapsulate()

    _, err = dk.Decapsulate(ct[:10])
    fmt.Printf("short ciphertext error: %v\n", err != nil)

    bad := bytes.Clone(ct)
    bad[0] ^= 0xff
    got, err := dk.Decapsulate(bad)
    fmt.Printf("garbage ciphertext error: %v\n", err != nil)
    fmt.Printf("garbage matches sender shared key: %v\n", bytes.Equal(shared, got))
    fmt.Println("fix: compare derived keys / transcripts; do not rely on Decapsulate error a
}
```

Run:

```go
go run mlkem_decap_pitfall.go
```

Output:

```
short ciphertext error: true
garbage ciphertext error: false
garbage matches sender shared key: false
fix: compare derived keys / transcripts; do not rely on Decapsulate error alone
```

Ciphertext malleability myths: age-style file formats and TLS AEAD layers make ciphertext tampering fail closed at a higher layer. Raw ML-KEM ciphertexts are not a substitute for an AEAD. After both sides share `sharedKey`, feed it into a proper KDF and use the result with an AEAD (or use TLS and skip hand-rolling).

The trap

Save as `mlkem_size_trap.go`. Treating an ML-KEM encapsulation key or ciphertext like an X25519 blob blows past 32-byte columns and stack buffers that were fine for classical ECDH.

```go
// mlkem_size_trap.go
package main

import (
    "crypto/mlkem"
    "fmt"
)

func main() {
    const classicalECDHPub = 32 // X25519-sized column / stack buffer
    fmt.Printf("ML-KEM-768 encap key: %d (fits 32? %v)\n",
        mlkem.EncapsulationKeySize768, mlkem.EncapsulationKeySize768 <= classicalECDHPub)
    fmt.Printf("ML-KEM-768 ciphertext: %d (fits 32? %v)\n",
        mlkem.CiphertextSize768, mlkem.CiphertextSize768 <= classicalECDHPub)
    fmt.Println("fix: size the wire and DB for EncapsulationKeySize* + CiphertextSize* of th
}
```

Run:

```go
go run mlkem_size_trap.go
```

Output:

```
ML-KEM-768 encap key: 1184 (fits 32? false)
ML-KEM-768 ciphertext: 1088 (fits 32? false)
fix: size the wire and DB for EncapsulationKeySize* + CiphertextSize* of the chosen set
```

The second trap is rolling your own "PQ TLS" with this package when `crypto/tls` already speaks hybrid ML-KEM. Prefer the stdlib handshake unless you are writing a specialized protocol and have a reviewed design.

The boring rule

- New direct KEMs in Go: import `crypto/mlkem`; default to **ML-KEM-768** (`GenerateKey768` / `NewEncapsulationKey768` / `Decapsulate`).
- Store the **64-byte seed**; publish encapsulation key bytes; send ciphertext; keep shared keys secret.
- After both sides have `sharedKey`, run a KDF and an AEAD (or use TLS). Do not treat `Decapsulate`'s error channel as peer authentication.
- Budget ~1.2 KiB for the public key and ~1.1 KiB for each ML-KEM-768 ciphertext on the wire.

- For ordinary HTTPS/gRPC, rely on `crypto/tls` hybrid groups (X25519MLKEM768 and friends) instead of calling this package yourself.
- Check errors from `GenerateKey*` / `New*Key*` / wrong-length `Decapsulate`; never ignore a length mismatch.

Try this

1. In `mlkem_roundtrip.go`, switch to `GenerateKey1024` / `NewEncapsulationKey1024` and confirm the printed lengths match `mlkem_sizes.go`.
2. In `mlkem_seed.go`, truncate the seed to 63 bytes before `NewDecapsulationKey768` and confirm you get an error.
3. In `mlkem_decap_pitfall.go`, pass a ciphertext from a **second** key pair into the first `Decapsulate` and show the shared keys diverge with `err == nil`.
4. Read `go doc crypto/tls.CurveID` (or your Go version's TLS docs) and list which ML-KEM-related curve IDs your toolchain advertises by default.

The uuid Package (Go 1.27)

The `uuid` Package (Go 1.27)

Thesis

For a decade the first `go get` in almost every service was `github.com/google/uuid`. Go 1.27 adds a standard-library package, imported simply as `"uuid"`, that covers what most programs actually do with UUIDs: generate random (v4) or time-ordered (v7) identifiers, parse the common text forms, compare, sort and serialize them.

The API is intentionally small: 9 functions and methods on one type, `type UUID [16]byte`. This chapter shows what it does, what it deliberately leaves out, how it behaves with JSON and `database/sql` (which special-cases it in 1.27, a detail missing from the release notes), and what to watch when migrating from `google/uuid`.

Cryptography, Hashing, Random and Regexp has a five-line summary. This is the full chapter.

Verified with **Go 1.27.2** (`linux/amd64`) on 2026-10-11. Primary sources: Go 1.27 release notes, `uuid` package docs, proposal #62026, RFC 9562.

Learning goals

- Generate v4 and v7 UUIDs and know when each is the right key
- Parse the four accepted text forms and handle errors
- Read a UUID's version, variant and (for v7) timestamp from its bytes
- Store and load UUIDs through JSON and `database/sql`, including NULL columns
- Migrate from `github.com/google/uuid` without behaviour surprises

The whole API

```
type UUID [16]byte

func New() UUID                 // today: same as NewV4
func NewV4() UUID               // 122 random bits
func NewV7() UUID               // 48-bit ms timestamp + 12-bit sub-ms fraction + 62 random
func Nil() UUID                 // 00000000-0000-0000-0000-000000000000
func Max() UUID                 // ffffffff-ffff-ffff-ffff-ffffffffffff
func Parse(s string) (UUID, error)
func MustParse(s string) UUID
```

```go
func (u UUID) String() string
func (u UUID) Compare(v UUID) int
func (u UUID) MarshalText() ([]byte, error)
func (u UUID) AppendText(b []byte) ([]byte, error)
func (u *UUID) UnmarshalText(b []byte) error
```

That is the whole list, from `go doc -all uuid`. There are no v1, v3, v5, v6 or v8 generators, no `Version()` or `Variant()` accessors, no node-ID state, and no `Scan`/`Value` methods. None of the generators return an error, because random bits come from `crypto/rand`, which cannot fail as of Go 1.24.

Every listing below is a `main.go` in its own module whose `go.mod` says `go 1.27`. Run each with `go run .`.

Case 1: generate, and read the version bits

```go
// main.go
package main

import (
    "fmt"
    "uuid"
)

// The package has no Version method; the version is the high nibble of byte 6.
func version(u uuid.UUID) int { return int(u[6] >> 4) }

// The variant is the top bits of byte 8. RFC 9562 UUIDs use 10xx.
func variant(u uuid.UUID) string {
    switch {
    case u[8]&0x80 == 0x00:
        return "NCS (reserved)"
    case u[8]&0xC0 == 0x80:
        return "RFC 9562"
    case u[8]&0xE0 == 0xC0:
        return "Microsoft (reserved)"
    default:
        return "future (reserved)"
    }
}

func main() {
    for _, u := range []uuid.UUID{uuid.New(), uuid.NewV4(), uuid.NewV7(), uuid.Nil(), uuid.M
        fmt.Printf("%s  v%-2d %s\n", u, version(u), variant(u))
    }
```

```
}
```

```
95b51b17-50af-4696-a9b3-c49aa879439b  v4  RFC 9562
b31cf2b0-e15a-45f4-ba80-292e8a937217  v4  RFC 9562
01a12b58-8197-781b-b8c9-5ff07a0c47dc  v7  RFC 9562
00000000-0000-0000-0000-000000000000  v0  NCS (reserved)
ffffffff-ffff-ffff-ffff-ffffffffffff  v15 future (reserved)
```

Your first three lines will differ, because they are fresh random values. The structure will not: the 13th hex digit is the version (`4` or `7`), and the 17th digit is `8`, `9`, `a` or `b` (variant bits `10`). Nil and Max are special values defined by RFC 9562 sections 5.9 and 5.10. Their "version" and "variant" bits are meaningless, which is why they decode as reserved.

`New` is documented as "suitable for most purposes" and currently equals `NewV4`. If your schema depends on the version, call `NewV4` or `NewV7` explicitly, so a future change to `New` cannot change your keys.

Case 2: parsing

```go
// main.go
package main

import (
    "fmt"
    "uuid"
)

func main() {
    inputs := []string{
        "f81d4fae-7dec-11d0-a765-00a0c91e6bf6",
        "{F81D4FAE-7DEC-11D0-A765-00A0C91E6BF6}",
        "urn:uuid:f81d4fae-7dec-11d0-a765-00a0c91e6bf6",
        "f81d4fae7dec11d0a76500a0c91e6bf6",
        "f81d4fae-7dec-11d0-a765-00a0c91e6bf",
        "f81d4fae_7dec_11d0_a765_00a0c91e6bf6",
        "",
    }
    for _, s := range inputs {
        u, err := uuid.Parse(s)
        fmt.Printf("%-46q -> %v %v\n", s, u, err)
    }
    a := uuid.MustParse(inputs[0])
    b := uuid.MustParse(inputs[1])
    fmt.Println("equal:", a == b)
    var zero uuid.UUID
```

```go
    fmt.Println("zero == Nil():", zero == uuid.Nil())
}
```

```
"f81d4fae-7dec-11d0-a765-00a0c91e6bf6"         -> f81d4fae-7dec-11d0-a765-00a0c91e6bf6 <nil>
"{F81D4FAE-7DEC-11D0-A765-00A0C91E6BF6}"       -> f81d4fae-7dec-11d0-a765-00a0c91e6bf6 <nil>
"urn:uuid:f81d4fae-7dec-11d0-a765-00a0c91e6bf6" -> f81d4fae-7dec-11d0-a765-00a0c91e6bf6 <nil
"f81d4fae7dec11d0a76500a0c91e6bf6"             -> f81d4fae-7dec-11d0-a765-00a0c91e6bf6 <nil>
"f81d4fae-7dec-11d0-a765-00a0c91e6bf"          -> 00000000-0000-0000-0000-000000000000 inval
"f81d4fae_7dec_11d0_a765_00a0c91e6bf6"         -> 00000000-0000-0000-0000-000000000000 inval
""                                             -> 00000000-0000-0000-0000-000000000000 inval
equal: true
zero == Nil(): true
```

- `Parse` accepts four forms (hyphenated, braced, `urn:uuid:` and bare 32 hex digits) in any letter case. `String` always emits lowercase hyphenated text, so parse-then-print is a canonicalizer.
- `Parse` accepts **any** version and variant. Version 1 input (`...-11d0-...` above) parses fine. If your API only issues v7, check `u[6]>>4 == 7` yourself.
- The error is the bare string `invalid uuid`, with no input echoed. Wrap it with the field name before returning it to a client.
- `UUID` is an array, so `==` works, it can be a map key, and its zero value is the Nil UUID. `uuid.Nil()` is a function, not a variable, so nobody can reassign it.

Case 3: v7 sorts, v4 does not

```go
// main.go
package main

import (
    "fmt"
    "slices"
    "strings"
    "uuid"
)

func main() {
    const n = 100000
    v7 := make([]uuid.UUID, n)
    v4 := make([]uuid.UUID, n)
    for i := range n {
        v7[i] = uuid.NewV7()
        v4[i] = uuid.NewV4()
    }
    cmp := func(a, b uuid.UUID) int { return a.Compare(b) }
    fmt.Println("v7 sorted in generation order:", slices.IsSortedFunc(v7, cmp))
```

```go
    fmt.Println("v4 sorted in generation order:", slices.IsSortedFunc(v4, cmp))

    // String order matches byte order because String is lowercase fixed-width hex.
    strs := make([]string, n)
    for i, u := range v7 {
        strs[i] = u.String()
    }
    fmt.Println("v7 strings sorted:", slices.IsSortedFunc(strs, strings.Compare))
    fmt.Println("distinct:", len(dedupe(v7)) == n)
}

func dedupe(us []uuid.UUID) map[uuid.UUID]struct{} {
    m := make(map[uuid.UUID]struct{}, len(us))
    for _, u := range us {
        m[u] = struct{}{}
    }
    return m
}
```

```
v7 sorted in generation order: true
v4 sorted in generation order: false
v7 strings sorted: true
distinct: true
```

`Compare` orders by big-endian bytes (RFC 9562 section 6.11), and because `String` is fixed-width lowercase hex, string order matches byte order. A v7 primary key in a B-tree index therefore appends at the right-hand edge instead of landing on a random page. That is the reason to use v7 for database keys.

The guarantee is precise. The implementation stores a 12-bit sub-millisecond fraction in `rand_a` (RFC 9562 section 6.2, method 3). When two calls land on the same timestamp, it bumps the stored value by one tick, so values from **one process** strictly increase "except when the system clock moves backwards" (package docs). Across processes or hosts, ordering is only as good as the clocks, at roughly millisecond resolution.

Recovering the timestamp

The top 48 bits of a v7 UUID are Unix milliseconds. The package does not expose them, so read them yourself:

```go
// main.go
package main

import (
    "encoding/binary"
    "fmt"
```

```go
    "time"
    "uuid"
)

// v7Time recovers the millisecond Unix timestamp stored in the top 48 bits.
func v7Time(u uuid.UUID) (time.Time, bool) {
    if u[6]>>4 != 7 {
        return time.Time{}, false
    }
    ms := binary.BigEndian.Uint64(u[0:8]) >> 16
    return time.UnixMilli(int64(ms)).UTC(), true
}

func main() {
    before := time.Now().UTC().Truncate(time.Millisecond)
    u := uuid.NewV7()
    ts, ok := v7Time(u)
    fmt.Println(ok, !ts.Before(before), time.Since(ts) < time.Second)
    _, ok = v7Time(uuid.NewV4())
    fmt.Println(ok)
    ts, _ = v7Time(uuid.MustParse("01a12b52-48c5-7905-bd1c-99d5b76a1794"))
    fmt.Println(ts.Format(time.RFC3339Nano))
}
```

```
true true true
false
2026-10-11T14:16:15.301Z
```

This is useful for debugging and for partition pruning ("which day was this ID minted?"). Do not use it as an authoritative creation time. Anyone who can mint IDs can choose the timestamp, and v7 IDs leak creation time to whoever sees them. If that matters, for example with public invite links, use v4.

Case 4: JSON

```go
// main.go
package main

import (
    "encoding/json"
    "fmt"
    "uuid"
)

type Order struct {
```

```go
    ID       uuid.UUID         `json:"id"`
    ParentID *uuid.UUID        `json:"parent_id,omitempty"`
    Lines    map[uuid.UUID]int `json:"lines"`
}

func main() {
    id := uuid.MustParse("01a12b52-48c5-7905-bd1c-99d5b76a1794")
    sku := uuid.MustParse("f81d4fae-7dec-11d0-a765-00a0c91e6bf6")

    b, err := json.Marshal(Order{ID: id, Lines: map[uuid.UUID]int{sku: 2}})
    fmt.Println(string(b), err)

    var o Order
    err = json.Unmarshal([]byte(`{"id":"{01A12B52-48C5-7905-BD1C-99D5B76A1794}","lines":{}}`
    fmt.Println(o.ID, o.ParentID, err)

    err = json.Unmarshal([]byte(`{"id":"not-a-uuid"}`), &o)
    fmt.Println(err)

    // A zero UUID is not "empty" to omitempty; it is a 16-byte array.
    b, _ = json.Marshal(struct {
        ID uuid.UUID `json:"id,omitempty"`
    }{})
    fmt.Println(string(b))
    b, _ = json.Marshal(struct {
        ID uuid.UUID `json:"id,omitzero"`
    }{})
    fmt.Println(string(b))
}
```

```
{"id":"01a12b52-48c5-7905-bd1c-99d5b76a1794","lines":{"f81d4fae-7dec-11d0-a765-00a0c91e6bf6"
01a12b52-48c5-7905-bd1c-99d5b76a1794 <nil> <nil>
invalid uuid
{"id":"00000000-0000-0000-0000-000000000000"}
{}
```

`MarshalText`/`UnmarshalText` make `UUID` a JSON string and a valid map key. Two details matter in API code:

- `omitempty` never omits a `UUID`, because it is an array, not an empty value. Use `omitzero` (Go 1.24+) to drop the Nil UUID, or a `*uuid.UUID` for "absent".
- The decode error is just `invalid uuid`, without the field name. Validate at the boundary if clients need better messages.

Case 5: `database/sql`

`UUID` has no `Scan` or `Value` method, yet it works as a query argument and a scan destination, because Go 1.27's `database/sql` and `database/sql/driver` special-case it. This was added in the same change as the package (commit 2fb2b98), but the release notes do not mention it.

```go
// main.go
package main

import (
    "database/sql"
    "database/sql/driver"
    "fmt"
    "uuid"
)

func main() {
    id := uuid.MustParse("01a12b52-48c5-7905-bd1c-99d5b76a1794")

    // uuid.UUID implements neither interface...
    _, scanner := any(&id).(sql.Scanner)
    _, valuer := any(id).(driver.Valuer)
    fmt.Println("sql.Scanner:", scanner, " driver.Valuer:", valuer)

    // ...because database/sql special-cases it.
    v, err := driver.DefaultParameterConverter.ConvertValue(id)
    fmt.Printf("as a query argument: %T %v %v\n", v, v, err)

    var a, b, c uuid.UUID
    err = sql.ConvertAssign(driver.ScanContext{}, &a, "01A12B52-48C5-7905-BD1C-99D5B76A1794"
    fmt.Println("scan from TEXT:     ", a, err)
    err = sql.ConvertAssign(driver.ScanContext{}, &b, []byte("01a12b52-48c5-7905-bd1c-99d5b7
    fmt.Println("scan from text bytes:", b, err)
    err = sql.ConvertAssign(driver.ScanContext{}, &c, id[:])
    fmt.Println("scan from BINARY(16):", c, err)

    err = sql.ConvertAssign(driver.ScanContext{}, &a, "nope")
    fmt.Println(err)
}
```

```
sql.Scanner: false  driver.Valuer: false
as a query argument: string 01a12b52-48c5-7905-bd1c-99d5b76a1794 <nil>
scan from TEXT:      01a12b52-48c5-7905-bd1c-99d5b76a1794 <nil>
scan from text bytes: 01a12b52-48c5-7905-bd1c-99d5b76a1794 <nil>
scan from BINARY(16): 01a12b52-48c5-7905-bd1c-99d5b76a1794 <nil>
converting driver.Value type string ("nope") to a UUID: invalid uuid
```

`sql.ConvertAssign` (new in 1.27, intended for drivers) runs the same conversion that `Rows.Scan` uses, which makes the rules easy to test without a database:

- **Arguments** become the canonical 36-character **string**. If your column is `BINARY(16)` or `BLOB`, pass `id[:]` explicitly.
- **Scanning** accepts a string, a text `[]byte`, or exactly 16 raw bytes.
- A driver that implements `driver.NamedValueChecker` or the new `driver.RowsColumnScanner` can intercept values before these defaults run, for example to map them to a native Postgres `uuid`. Check your driver's documentation.

NULL is the trap. Scanning NULL into `*uuid.UUID` fails, so use `sql.Null[uuid.UUID]` (Go 1.22+) for nullable columns:

```go
// main.go
package main

import (
    "database/sql"
    "database/sql/driver"
    "fmt"
    "uuid"
)

func main() {
    var u uuid.UUID
    err := sql.ConvertAssign(driver.ScanContext{}, &u, nil) // a NULL column
    fmt.Println("NULL into *uuid.UUID:", err)

    var n sql.Null[uuid.UUID]
    err = n.Scan(nil)
    fmt.Println("NULL into sql.Null:  ", n.Valid, err)
    err = n.Scan("01a12b52-48c5-7905-bd1c-99d5b76a1794")
    fmt.Println("TEXT into sql.Null:  ", n.Valid, n.V, err)

    v, err := n.Value()
    fmt.Printf("sql.Null as an argument: %T %v %v\n", v, v, err)
}
```

```
NULL into *uuid.UUID: unsupported Scan, storing driver.Value type <nil> into type *uuid.UUID
NULL into sql.Null:   false <nil>
TEXT into sql.Null:   true 01a12b52-48c5-7905-bd1c-99d5b76a1794 <nil>
sql.Null as an argument: string 01a12b52-48c5-7905-bd1c-99d5b76a1794 <nil>
```

End to end with SQLite

The same behaviour through a real driver, the pure-Go `modernc.org/sqlite`:

```
go get modernc.org/sqlite@v1.60.1
go run .
```

```go
// main.go
package main

import (
    "database/sql"
    "fmt"
    "log"
    "uuid"

    _ "modernc.org/sqlite"
)

func main() {
    db, err := sql.Open("sqlite", ":memory:")
    if err != nil {
        log.Fatal(err)
    }
    defer db.Close()

    _, err = db.Exec(`CREATE TABLE orders (
        id        TEXT PRIMARY KEY,   -- canonical 36-char text
        bin_id    BLOB NOT NULL,      -- 16 raw bytes
        parent_id TEXT                -- nullable
    )`)
    if err != nil {
        log.Fatal(err)
    }

    for range 3 {
        id := uuid.NewV7()
        // id goes in as text automatically; id[:] goes in as 16 bytes.
        if _, err := db.Exec(`INSERT INTO orders VALUES (?, ?, NULL)`, id, id[:]); err != ni
            log.Fatal(err)
        }
    }

    rows, err := db.Query(`SELECT id, bin_id, parent_id, typeof(id), length(bin_id) FROM ord
    if err != nil {
        log.Fatal(err)
    }
    defer rows.Close()
    for rows.Next() {
        var id, bin uuid.UUID
```

```go
        var parent sql.Null[uuid.UUID]
        var typ string
        var n int
        if err := rows.Scan(&id, &bin, &parent, &typ, &n); err != nil {
            log.Fatal(err)
        }
        fmt.Println(id == bin, parent.Valid, typ, n)
    }
    if err := rows.Err(); err != nil {
        log.Fatal(err)
    }
}
```

```
true false text 16
true false text 16
true false text 16
```

The `id` column holds text, `bin_id` holds 16 bytes, both scan back into the same `UUID`, and the NULL `parent_id` arrives as `Valid == false`.

Case 6: migrating from `github.com/google/uuid`

Both packages define `UUID` as `[16]byte`, so values convert with a plain type conversion. That lets you migrate one package at a time:

```
go get github.com/google/uuid@v1.6.0
go run .
```

```go
// main.go
package main

import (
    "fmt"

    guuid "github.com/google/uuid"
    "uuid"
)

func main() {
    g := guuid.MustParse("01a12b52-48c5-7905-bd1c-99d5b76a1794")
    s := uuid.UUID(g) // same [16]byte layout: a plain conversion
    fmt.Println(s, s == uuid.MustParse(g.String()))
    back := guuid.UUID(s)
    fmt.Println(back.Version(), back.Variant())
}
```

```
01a12b52-48c5-7905-bd1c-99d5b76a1794 true
VERSION_7 RFC4122
```

`github.com/google/uuid` v1.6.0	stdlib `uuid` (Go 1.27)
`uuid.New()` (panics on failure), `uuid.NewRandom() (UUID, error)`	`uuid.New()`, `uuid.NewV4()`, no error
`uuid.NewV7() (UUID, error)`	`uuid.NewV7()`, no error
`uuid.NewString()`	`uuid.New().String()`
`uuid.Parse`, `uuid.MustParse`	same names, same four input forms
`uuid.Nil` (variable), `uuid.Max`	`uuid.Nil()`, `uuid.Max()` (functions)
`u.Version()`, `u.Variant()`, `u.Time()`	none; read the bits (Cases 1 and 3)
`NewMD5`, `NewSHA1` (v3/v5), `NewUUID` (v1), `NewV6`	not provided; keep `google/uuid` for these
`Scan`/`Value` methods, `NullUUID`	built into `database/sql`; use `sql.Null[uuid.UUID]`
`SetRand`, `EnableRandPool`	not provided; tests use `testing/cryptotest.SetGlobalRandom`

The compiler catches every signature difference (`NewV7` returning two values, `Nil` becoming a function). Running `go vet` after the swap is enough to find them.

Deterministic IDs in tests

`google/uuid` tests often call `SetRand` to get repeatable IDs. The stdlib package draws from `crypto/rand`, so use `testing/cryptotest.SetGlobalRandom`, which seeds every `crypto/...` randomness source for the duration of one test:

```go
// det_test.go
package det

import (
    "testing"
    "testing/cryptotest"
    "uuid"
)

func TestDeterministic(t *testing.T) {
    cryptotest.SetGlobalRandom(t, 1)
    a := uuid.NewV4()
    cryptotest.SetGlobalRandom(t, 1)
    b := uuid.NewV4()
    t.Log(a, b, a == b)
}
```

```
go test -v .
```

```
=== RUN   TestDeterministic
    det_test.go:14: 6ae6783f-4fbd-491b-aeb8-8b73a48ed247 6ae6783f-4fbd-491b-aeb8-8b73a48ed24
--- PASS: TestDeterministic (0.00s)
PASS
ok      example.com/det 0.002s
```

`SetGlobalRandom` affects the whole process, so it cannot be used in parallel tests. v7 IDs are harder to pin down. Inside a `testing/synctest` bubble the clock is fake, so the timestamp is fixed too. But `NewV7` keeps process-wide monotonic state, so when two seeded bubbles mint at the same fake instant, the second ID comes out one tick higher. On Go 1.27.2 that gave `00dc6acf-ac00-7000-…` and then `00dc6acf-ac00-7001-…`. Assert on the structure of v7 IDs, not their exact values.

`go doc` can now show a dependency's API at a pinned version without changing `go.mod`, which helps during a migration: `go doc github.com/google/uuid@v1.6.0 NewV7`.

Cost: v7 takes a global lock

```go
// bench_test.go
package bench

import (
    "testing"
    "uuid"
)

var sink uuid.UUID

func BenchmarkNewV4(b *testing.B) {
    for b.Loop() {
        sink = uuid.NewV4()
    }
}

func BenchmarkNewV7(b *testing.B) {
    for b.Loop() {
        sink = uuid.NewV7()
    }
}

func BenchmarkNewV4Parallel(b *testing.B) {
    b.RunParallel(func(pb *testing.PB) {
        for pb.Next() {
            _ = uuid.NewV4()
        }
    })
```

```go
}

func BenchmarkNewV7Parallel(b *testing.B) {
    b.RunParallel(func(pb *testing.PB) {
        for pb.Next() {
            _ = uuid.NewV7()
        }
    })
}
```

```go
go test -run '^$' -bench . -count 1 .
```

```
goos: linux
goarch: amd64
pkg: example.com/bench
cpu: Intel(R) Xeon(R) Processor
BenchmarkNewV4-8            20344930            59.09 ns/op
BenchmarkNewV7-8            10888754           110.7 ns/op
BenchmarkNewV4Parallel-8    23400703            54.17 ns/op
BenchmarkNewV7Parallel-8     4946437           248.3 ns/op
PASS
ok      example.com/bench   5.214s
```

These numbers come from an 8-vCPU cloud VM, and yours will differ. The shape is what matters. `NewV7` serializes on a package-level mutex to keep its monotonic guarantee, so under parallel load it gets slower per call, while `NewV4` does not. At about 250 ns that still means roughly 4 million IDs per second for the whole process, which is more than any database will insert. It only matters if you mint IDs for in-memory objects in a hot loop. In that case use v4, or batch.

The trap

- **Using `New()` when the schema needs v7.** `New` is v4 today and is documented as free to change.
- **Exposing v7 IDs publicly.** They reveal creation time to the millisecond.
- **`omitempty` on a `UUID` field.** It never omits. Use `omitzero` or a pointer.
- **Scanning a nullable column into `uuid.UUID`.** Use `sql.Null[uuid.UUID]`.
- **Passing a `UUID` to a `BINARY(16)` column.** It is sent as 36-character text. Pass `id[:]`.
- **Expecting cross-host ordering from v7.** It is monotonic per process only.

The boring rule

Use `uuid.NewV7()` for database primary keys, `uuid.NewV4()` for anything public or security-adjacent, and `uuid.Parse` at every boundary. Keep `google/uuid` only for name-based v3/v5 IDs.

Try this

1. Write `func V7(s string) (uuid.UUID, error)` that parses and rejects anything that is not version 7 with the RFC 9562 variant.
2. Insert 100,000 v4 and 100,000 v7 keys into SQLite tables with `TEXT PRIMARY KEY` and compare insert time and file size.
3. Generate v7 IDs from two goroutines that each call `time.Sleep(0)` between calls, and confirm the merged sequence is still sorted.

Sources

- Go 1.27 release notes, "New uuid package": https://go.dev/doc/go1.27
- Package docs: https://pkg.go.dev/uuid and `go doc -all uuid`
- Proposal #62026, "uuid: add API to generate and parse UUID": https://github.com/golang/go/issues/62026
- Implementation commit, including the `database/sql` integration: https://github.com/golang/go/commit/2fb2b98da3
- RFC 9562, Universally Unique IDentifiers: https://www.rfc-editor.org/rfc/rfc9562
- `github.com/google/uuid` v1.6.0 API, via `go doc github.com/google/uuid@v1.6.0`

hash/maphash.Hasher: Custom Equality for Hash-Based Containers

hash/maphash.Hasher: Custom Equality for Hash-Based Containers

After reading this chapter, you will be able to implement custom hash-based containers using Go 1.27's `maphash.Hasher` interface, allowing non-comparable types as keys and supporting custom equivalence relations like case-insensitive strings and field-scoped struct comparison.

Thesis

For over a decade, Go developers designing hash tables or sets were bound to the built-in `map[K]V` type. Built-in maps require `K` to satisfy the language-level `comparable` constraint, and equivalence is strictly determined by the `==` operator. If you needed case-insensitive string lookups, equality over floating-point values with a tolerance, or struct comparisons that ignored mutable metadata slices, you had two suboptimal choices: serialize your keys into canonical strings, or manage manual hash buckets beside slices of key-value pairs.

Go 1.27 resolves this limitation by adding `maphash.Hasher[T any]` to the standard library's `hash/maphash` package. By decoupling hash-based containers from Go's built-in `==` operator, `Hasher[T]` formalizes the bridge between container algorithms and element semantics. While the standard library does not yet include a generic `hash.Set` or `hash.Map`, `Hasher[T]` sets the contract that third-party hash tables, sets and other hash-based structures such as Bloom filters can share, as the package documentation puts it.

Verified with **Go 1.27.2** (`linux/amd64`). Primary sources: Go 1.27 release notes, `hash/maphash` package docs, and issue #70471.

Learning goals

- Explain the interface contract of `maphash.Hasher[T any]` and why its `Hash` method receives `*maphash.Hash` rather than returning `uint64`.
- Distinguish between standard Go equality (`==`) and custom equivalence relations, recognizing why `strings.ToLower` satisfies the hasher contract while `strings.EqualFold` breaks it.
- Build a generic, separate-chaining `HashSet[T, H]` that resizes at a defined load factor and maintains a per-container seed.
- Use non-comparable structs containing slice headers as container keys by implementing custom hashers that inspect only relevant fields.
- Apply `maphash.ComparableHasher[T comparable]` for standard types while avoiding runtime panics with interface values.
- Construct a two-seed Bloom filter driven by `maphash.Hasher`.

The `hash/maphash` additions in Go 1.27

Go 1.27 introduces two fundamental additions to `hash/maphash`:

```go
type Hasher[T any] interface {
    Hash(*Hash, T)
    Equal(x, y T) bool
}

type ComparableHasher[T comparable] struct{}

func (ComparableHasher[T]) Hash(h *Hash, v T)
func (ComparableHasher[T]) Equal(x, y T) bool
```

The `Hasher[T]` interface defines two capabilities: 1. `Hash(*Hash, T)`: Appends the representation of an element `T` into the provided `*maphash.Hash` state. 2. `Equal(x, y T) bool`: Determines whether two elements are equivalent under the container's equivalence relation.

Why `Hash` takes `*maphash.Hash` instead of returning `uint64`

Many languages give a type a hash method that simply returns an integer. Go 1.24's `maphash.Comparable(seed, v) uint64` also returns a number, but it takes a seed. `Hasher.Hash` instead writes into a `*maphash.Hash` that the container owns, which has three advantages:

- **Zero-allocation streaming**: Composite types can write multiple sub-fields sequentially using `h.WriteString()`, `h.WriteByte()`, and `h.Write()`. The hasher does not need to allocate intermediate byte slices or pack fields into temporary buffers before hashing.
- **State reuse**: Containers maintain their own `maphash.Hash` instances or pass pointers to pooled hash state, reusing internal buffers across thousands of insertions and queries.
- **Unified seed control**: The container controls the hash seed via `h.SetSeed(seed)` or `maphash.MakeSeed()`. Elements do not inspect or manage the seed, which prevents hash-collision denial-of-service vulnerabilities across instances.

Capability	Built-in `map[K]V`	Container with `maphash.Hasher[T]`
Key constraint	`comparable` (compile-time)	`any` (unconstrained)
Equality rule	Language `==` operator	`Hasher.Equal(x, y)`
Hash algorithm	Runtime hash (AES-based where the CPU supports it)	`maphash.Hash`, backed by the same runtime hash functions
Non-comparable types	Compiler error	Supported via selective hashing
Case insensitivity	Requires manual canonicalization	Built into hasher
Per-instance seeding	Random seed per map, chosen by the runtime	Chosen by the container, typically `MakeSeed()` per instance

The fundamental contract

Any type implementing `maphash.Hasher[T]` must satisfy the equivalence contract:

$$\forall x, y \in T, \quad \text{Equal}(x, y) = \text{true} \implies \text{Hash}(h, x) \equiv \text{Hash}(h, y)$$

If two values are equal under `Equal`, writing their contents into a `*maphash.Hash` seeded with identical initial states must produce identical byte sequences, and therefore identical `Sum64()` digests. If two values yield `Equal(x, y) == true` but write different inputs into the hash buffer, they land in different buckets, making retrieval impossible. The inverse need not hold: hash collisions (different inputs producing the same 64-bit integer) are unavoidable, but false negatives break the container. The documentation adds one more rule: hashers must be *logically stateless*, so `Hash` and `Equal` depend only on their arguments.

`strings.ToLower` vs. `strings.EqualFold`

A frequent pitfall when writing case-insensitive hashers is pairing `strings.ToLower` with `strings.EqualFold`.

Unicode case-folding under `strings.EqualFold` is a broader equivalence relation than simple lowercase conversion. For example, the Greek uppercase letter sigma (`'Σ'`) folds to both the medial lowercase sigma (`' '`) and the final lowercase sigma (`' '`). Under `strings.EqualFold`:

```
strings.EqualFold("Σ", " ") // returns true
```

However, standard lowercase mapping behaves as follows:

```
strings.ToLower("Σ") // returns " "
strings.ToLower(" ") // returns " "
```

If `Hash` writes `strings.ToLower(s)` into the hash buffer while `Equal` returns `strings.EqualFold(x, y)`, the contract breaks: `"Σ"` and `" "` are reported as equal, but `"Σ"` writes `" "` into the hasher while `" "` writes `" "`. Because their hashed data differs, their 64-bit digests diverge. A container searching for `" "` will check a different bucket and fail to find `"Σ"`.

To preserve the contract, both `Hash` and `Equal` must reduce inputs to the same canonical form. This is the example from the package documentation, which notes that it avoids `strings.EqualFold` for exactly this reason:

```
type CaseInsensitiveHasher struct{}

func (CaseInsensitiveHasher) Hash(h *maphash.Hash, s string) {
    h.WriteString(strings.ToLower(s))
}

func (CaseInsensitiveHasher) Equal(x, y string) bool {
    return strings.ToLower(x) == strings.ToLower(y)
```

```
}
```

Lab: Building a generic `HashSet`

We begin by setting up a minimal Go 1.27 module.

```
module example.com/hasherdemo

go 1.27
```

We now construct `HashSet[T any, H maphash.Hasher[T]]` in package `hashset`. The container uses separate chaining over buckets represented by slices of `T`. It tracks its own `maphash.Seed` and dynamically doubles its bucket array when the load factor exceeds 0.75.

```
package hashset

import (
    "hash/maphash"
)

// HashSet is a generic hash set that uses separate chaining.
// T represents the element type, and H provides custom hashing and equality.
type HashSet[T any, H maphash.Hasher[T]] struct {
    hasher  H
    seed    maphash.Seed
    buckets [][]T
    size    int
}

// New constructs an empty HashSet using the supplied hasher instance.
func New[T any, H maphash.Hasher[T]](hasher H) *HashSet[T, H] {
    return &HashSet[T, H]{
        hasher:  hasher,
        seed:    maphash.MakeSeed(),
        buckets: make([][]T, 8),
    }
}

// hashOf calculates the 64-bit hash of an element using the set's per-instance seed.
func (s *HashSet[T, H]) hashOf(v T) uint64 {
    var h maphash.Hash
    h.SetSeed(s.seed)
    s.hasher.Hash(&h, v)
    return h.Sum64()
}
```

```go
// Contains reports whether the set contains an element equal to v under H.Equal.
func (s *HashSet[T, H]) Contains(v T) bool {
    if len(s.buckets) == 0 {
        return false
    }
    idx := s.hashOf(v) % uint64(len(s.buckets))
    for _, item := range s.buckets[idx] {
        if s.hasher.Equal(item, v) {
            return true
        }
    }
    return false
}

// Add inserts v into the set. It reports true if v was added,
// or false if an equal element already exists.
func (s *HashSet[T, H]) Add(v T) bool {
    if s.Contains(v) {
        return false
    }
    if float64(s.size+1) > float64(len(s.buckets))*0.75 {
        s.grow()
    }
    idx := s.hashOf(v) % uint64(len(s.buckets))
    s.buckets[idx] = append(s.buckets[idx], v)
    s.size++
    return true
}

// Remove deletes an element equal to v from the set. It reports true if found and removed.
func (s *HashSet[T, H]) Remove(v T) bool {
    if len(s.buckets) == 0 {
        return false
    }
    idx := s.hashOf(v) % uint64(len(s.buckets))
    bucket := s.buckets[idx]
    for i, item := range bucket {
        if s.hasher.Equal(item, v) {
            s.buckets[idx] = append(bucket[:i], bucket[i+1:]...)
            s.size--
            return true
        }
    }
    return false
}
```

```go
// Len returns the current number of elements stored in the set.
func (s *HashSet[T, H]) Len() int {
    return s.size
}

// grow doubles the bucket array capacity and rehashes all elements.
func (s *HashSet[T, H]) grow() {
    newCap := len(s.buckets) * 2
    if newCap == 0 {
        newCap = 8
    }
    oldBuckets := s.buckets
    s.buckets = make([][]T, newCap)
    for _, bucket := range oldBuckets {
        for _, item := range bucket {
            idx := s.hashOf(item) % uint64(newCap)
            s.buckets[idx] = append(s.buckets[idx], item)
        }
    }
}
```

Applying custom hashers and a two-seed Bloom filter

Next, we create `main.go` to test three distinct hashing scenarios: 1. `CaseInsensitiveHasher`: Custom equivalence for standard strings. 2. `PointHasher`: Custom hashing and equality for a non-comparable struct `Point` containing a slice field (`Tags []string`). 3. `maphash.ComparableHasher[int]`: Standard library hasher for comparable integers. 4. `BloomFilter`: A two-seed probabilistic filter using `maphash.Hasher`.

```go
package main

import (
    "encoding/binary"
    "fmt"
    "hash/maphash"
    "strings"

    "example.com/hasherdemo/hashset"
)

// CaseInsensitiveHasher implements maphash.Hasher[string].
// It normalizes strings with strings.ToLower for both hashing and comparison.
type CaseInsensitiveHasher struct{}

func (CaseInsensitiveHasher) Hash(h *maphash.Hash, s string) {
```

```
    h.WriteString(strings.ToLower(s))
}

func (CaseInsensitiveHasher) Equal(x, y string) bool {
    return strings.ToLower(x) == strings.ToLower(y)
}

// Point is non-comparable because it contains a slice field ([]string).
type Point struct {
    Tags []string
    X, Y int
}

// PointHasher hashes and compares only the coordinate fields X and Y.
type PointHasher struct{}

func (PointHasher) Hash(h *maphash.Hash, p Point) {
    var buf [16]byte
    binary.LittleEndian.PutUint64(buf[0:8], uint64(p.X))
    binary.LittleEndian.PutUint64(buf[8:16], uint64(p.Y))
    h.Write(buf[:])
}

func (PointHasher) Equal(x, y Point) bool {
    return x.X == y.X && x.Y == y.Y
}

// BloomFilter is a minimal two-seed probabilistic membership filter.
type BloomFilter[T any, H maphash.Hasher[T]] struct {
    hasher  H
    seed1   maphash.Seed
    seed2   maphash.Seed
    bits    []uint64
    numBits uint64
}

func NewBloomFilter[T any, H maphash.Hasher[T]](hasher H, numBits uint64) *BloomFilter[T, H]
    return &BloomFilter[T, H]{
        hasher:  hasher,
        seed1:   maphash.MakeSeed(),
        seed2:   maphash.MakeSeed(),
        bits:    make([]uint64, (numBits+63)/64),
        numBits: numBits,
    }
}

func (bf *BloomFilter[T, H]) hashWith(seed maphash.Seed, v T) uint64 {
```

```go
    var h maphash.Hash
    h.SetSeed(seed)
    bf.hasher.Hash(&h, v)
    return h.Sum64()
}

func (bf *BloomFilter[T, H]) Add(v T) {
    i1 := bf.hashWith(bf.seed1, v) % bf.numBits
    i2 := bf.hashWith(bf.seed2, v) % bf.numBits
    bf.bits[i1/64] |= 1 << (i1 % 64)
    bf.bits[i2/64] |= 1 << (i2 % 64)
}

func (bf *BloomFilter[T, H]) MayContain(v T) bool {
    i1 := bf.hashWith(bf.seed1, v) % bf.numBits
    i2 := bf.hashWith(bf.seed2, v) % bf.numBits
    if (bf.bits[i1/64] & (1 << (i1 % 64))) == 0 {
        return false
    }
    if (bf.bits[i2/64] & (1 << (i2 % 64))) == 0 {
        return false
    }
    return true
}

func main() {
    // 1. Case-insensitive string set
    ciSet := hashset.New[string, CaseInsensitiveHasher](CaseInsensitiveHasher{})
    ciSet.Add("Go")
    ciSet.Add("golang")
    ciSet.Add("GO") // duplicate under CaseInsensitiveHasher
    fmt.Printf("ciSet size: %d\n", ciSet.Len())
    fmt.Printf("ciSet contains 'go': %t\n", ciSet.Contains("go"))
    fmt.Printf("ciSet contains 'GOLANG': %t\n", ciSet.Contains("GOLANG"))
    fmt.Printf("ciSet contains 'rust': %t\n", ciSet.Contains("rust"))

    // 2. Non-comparable Point struct set
    pointSet := hashset.New[Point, PointHasher](PointHasher{})
    p1 := Point{Tags: []string{"origin", "alpha"}, X: 10, Y: 20}
    p2 := Point{Tags: []string{"beta"}, X: 10, Y: 20} // identical X, Y; different tags
    p3 := Point{Tags: []string{"other"}, X: 30, Y: 40}
    pointSet.Add(p1)
    pointSet.Add(p2) // rejected as duplicate because X, Y match
    pointSet.Add(p3)
    fmt.Printf("pointSet size: %d\n", pointSet.Len())
    probe := Point{Tags: nil, X: 10, Y: 20}
    fmt.Printf("pointSet contains probe(10, 20): %t\n", pointSet.Contains(probe))
```

```go
    probeMissing := Point{Tags: nil, X: 99, Y: 99}
    fmt.Printf("pointSet contains probe(99, 99): %t\n", pointSet.Contains(probeMissing))

    // 3. ComparableHasher[int]
    intSet := hashset.New[int, maphash.ComparableHasher[int]](maphash.ComparableHasher[int]{
    intSet.Add(100)
    intSet.Add(200)
    intSet.Add(100) // duplicate
    fmt.Printf("intSet size: %d\n", intSet.Len())
    fmt.Printf("intSet contains 100: %t\n", intSet.Contains(100))
    fmt.Printf("intSet contains 300: %t\n", intSet.Contains(300))

    // 4. Bloom filter with two seeds and CaseInsensitiveHasher
    bf := NewBloomFilter[string, CaseInsensitiveHasher](CaseInsensitiveHasher{}, 1024)
    bf.Add("ServiceA")
    bf.Add("WorkerPool")
    fmt.Printf("bloom filter may contain 'servicea': %t\n", bf.MayContain("servicea"))
    fmt.Printf("bloom filter may contain 'WORKERPOOL': %t\n", bf.MayContain("WORKERPOOL"))
    fmt.Printf("bloom filter may contain 'Database': %t\n", bf.MayContain("Database"))
}
```

Execute the program with:

```
go run .
```

The following output was produced by running `go run .`:

```
ciSet size: 2
ciSet contains 'go': true
ciSet contains 'GOLANG': true
ciSet contains 'rust': false
pointSet size: 2
pointSet contains probe(10, 20): true
pointSet contains probe(99, 99): false
intSet size: 2
intSet contains 100: true
intSet contains 300: false
bloom filter may contain 'servicea': true
bloom filter may contain 'WORKERPOOL': true
bloom filter may contain 'Database': false
```

Notice the behavior across the outputs: - For `ciSet`, the reported size is 2, confirming that `"GO"` was deduplicated against `"Go"`. Lookups for `"go"` and `"GOLANG"` both evaluate to `true`, while uninserted keys evaluate to `false`. - For `pointSet`, `p2` is rejected even though its `Tags` slice differs from `p1`. The size is 2, and querying with an uninitialized slice (`Tags: nil`) matches `(10, 20)` successfully. - For `intSet`, `maphash.ComparableHasher[int]` operates identically to built-in integer comparison. - In the Bloom filter, the two independent seeds hash the same normalized bytes into distinct bit positions, confirming membership for case-folded variants while rejecting uninserted strings.

Contract testing and performance benchmarks

To verify that an implementation adheres to the required contract, write randomized property tests that assert:

$$\text{Equal}(a, b) \implies \text{Hash}(seed, a) == \text{Hash}(seed, b)$$

Below, `hashset_test.go` checks this contract across randomized casing patterns and benchmarks `HashSet` with `maphash.ComparableHasher[string]` against Go's built-in `map[string]struct{}`.

```go
package hashset_test

import (
    "hash/maphash"
    "math/rand/v2"
    "strconv"
    "strings"
    "testing"

    "example.com/hasherdemo/hashset"
)

type CaseInsensitiveHasher struct{}

func (CaseInsensitiveHasher) Hash(h *maphash.Hash, s string) {
    h.WriteString(strings.ToLower(s))
}

func (CaseInsensitiveHasher) Equal(x, y string) bool {
    return strings.ToLower(x) == strings.ToLower(y)
}

func TestHasherContract(t *testing.T) {
    hasher := CaseInsensitiveHasher{}
    seed := maphash.MakeSeed()

    hashOf := func(s string) uint64 {
        var h maphash.Hash
        h.SetSeed(seed)
        hasher.Hash(&h, s)
        return h.Sum64()
    }

    letters := []rune("abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ")
    for i := range 500 {
        length := rand.IntN(12) + 1
```

```
        runes := make([]rune, length)
        for j := range runes {
            runes[j] = letters[rand.IntN(len(letters))]
        }
        original := string(runes)
        lower := strings.ToLower(original)
        upper := strings.ToUpper(original)

        if !hasher.Equal(original, lower) {
            t.Fatalf("iteration %d: expected Equal(%q, %q) to be true", i, original, lower)
        }
        if hashOf(original) != hashOf(lower) {
            t.Fatalf("iteration %d: hash contract violated for %q and %q (%d != %d)",
                i, original, lower, hashOf(original), hashOf(lower))
        }

        if !hasher.Equal(original, upper) {
            t.Fatalf("iteration %d: expected Equal(%q, %q) to be true", i, original, upper)
        }
        if hashOf(original) != hashOf(upper) {
            t.Fatalf("iteration %d: hash contract violated for %q and %q (%d != %d)",
                i, original, upper, hashOf(original), hashOf(upper))
        }
    }
}

func BenchmarkComparableHasher(b *testing.B) {
    keys := make([]string, 1000)
    for i := range keys {
        keys[i] = "item-" + strconv.Itoa(i)
    }

    b.ResetTimer()
    for i := 0; i < b.N; i++ {
        set := hashset.New[string, maphash.ComparableHasher[string]](maphash.ComparableHashe
        for _, k := range keys {
            set.Add(k)
        }
        for _, k := range keys {
            if !set.Contains(k) {
                b.Fatalf("expected key %s to exist", k)
            }
        }
    }
}
```

```go
func BenchmarkBuiltinMap(b *testing.B) {
    keys := make([]string, 1000)
    for i := range keys {
        keys[i] = "item-" + strconv.Itoa(i)
    }

    b.ResetTimer()
    for i := 0; i < b.N; i++ {
        m := make(map[string]struct{}, 8)
        for _, k := range keys {
            m[k] = struct{}{}
        }
        for _, k := range keys {
            if _, ok := m[k]; !ok {
                b.Fatalf("expected key %s to exist", k)
            }
        }
    }
}
```

Run tests and benchmarks with:

```go
go test ./hashset -run . -bench . -benchtime=200ms
```

The following output was produced by running `go test ./hashset -run . -bench . -benchtime=200ms`:

```
goos: linux
goarch: amd64
pkg: example.com/hasherdemo/hashset
cpu: Intel(R) Xeon(R) Processor
BenchmarkComparableHasher-8          613        482728 ns/op
BenchmarkBuiltinMap-8               3183         68043 ns/op
PASS
ok      example.com/hasherdemo/hashset  0.568s
```

When reviewing the benchmark output: - The `PASS` / `ok` lines include `TestHasherContract`; add `-v` to see each test by name. - On the run above, the hand-written `HashSet` took roughly 7x the `ns/op` of the built-in map, about 480 µs against 68 µs per benchmark iteration. Expect the ratio to vary between machines and runs. - The built-in `map` is a Swiss-table implementation in the runtime (since Go 1.24). The compiler specializes access for common key types and calls the hash function directly. - Our user-space `HashSet` relies on separate chaining with generic slice buckets, heap slice reallocations during resizing, and calls into `H` through Go's generic implementation, which may use runtime dictionaries instead of fully specialized code. While `ComparableHasher` adds a modest layer of indirection, it gives callers complete control over data types and custom equality semantics.

Pitfalls

1. Inconsistent equality and hashing projections

If `Equal` normalizes data differently than `Hash`, equal keys will generate distinct hashes. This can occur when `Equal` performs case folding while `Hash` writes raw strings, or when floating-point hashers round values in `Hash` but use strict equality in `Equal`.

Rule: If `Equal` maps values to an equivalence class, `Hash` must write the canonical representative of that exact class to the hash buffer.

2. Passing dynamic non-comparable types to `ComparableHasher[any]`

The type constraint on `ComparableHasher[T comparable]` is checked at compile time. However, in Go, the `any` (`interface{}`) type satisfies `comparable`.

If you declare:

```
hasher := maphash.ComparableHasher[any]{}
```

and pass values whose dynamic types are non-comparable (such as a slice `[]byte` or a map), you break the documented rule ("you must not pass non-comparable argument values to its Hash or Equal methods"). In practice this panics at run time, just as comparing two interface values holding slices with `==` panics.

3. Mutating fields used in custom hashes

If an element is stored inside a hash container and a field that contributes to its hash or equality is subsequently modified, the element's bucket index changes. The container will fail to locate the element on lookup and will fail to remove it, leading to silent memory leaks and data duplication.

4. Forgetting per-instance seeds in custom containers

Containers should generate a new seed per instance via `maphash.MakeSeed()`. Reusing a static, hardcoded seed across all instances exposes the container to algorithmic complexity attacks where adversarial inputs intentionally target identical bucket chains.

Key takeaways

- `maphash.Hasher[T any]` decouples hash-based containers from Go's language-level `==` operator and `comparable` constraint.
- The `Hash(*maphash.Hash, T)` method receives a pointer to a streaming hash state, allowing composite and variable-length data to be hashed with zero intermediate heap allocations.
- Implementations must maintain the contract: if `Equal(x, y)` returns true, `Hash` must write identical byte representations for both inputs.

- `maphash.ComparableHasher[T comparable]` provides standard `==` behavior for comparable types, matching Go's default map semantics.
- Non-comparable types—such as structs containing slices—can act as container keys by selectively hashing and comparing their comparable subsets.

Sources

- Go 1.27 release notes
- `hash/maphash` package documentation
- Proposal golang/go#70471: hash/maphash: add Hasher interface

Data Structures and Algorithms

Algorithmic Complexity and Big-O

Algorithmic Complexity and Big-O

Algorithmic complexity measures how resource consumption grows as the volume of input expands. The boring default is to write straightforward loops and slice indexing until profiling proves a bottleneck, but understanding Big-O notation ($O(1)$, $O(\log n)$, $O(n)$, $O(n \log n)$, $O(n^2)$) prevents disastrous performance traps when desk ticket volumes scale from ten items to ten million.

Mental model

Big-O describes the asymptotic upper bound of time or memory as input size n approaches infinity. It ignores constant multipliers and hardware clock speeds because an $O(n^2)$ algorithm on a supercomputer will eventually lose to an $O(n \log n)$ algorithm on a cheap laptop once n grows large enough.

```
Execution Time
  ^
  |                                                O(n^2) Quadratic
  |                                               /
  |                                 O(n log n)   /
  |                                /            /
  |                     O(n) Linear            /
  |                    /                      /
  |       O(log n)    /                      /
  |      /           /                      /
  |-----/-----------/----------------------/-----------------> Input Size (n)
  |  O(1) Constant Time
  +-------------------------------------------------------------
```

Key growth classes in service engineering:

Notation	Name	Practical Desk Example	When $n = 100{,}000$
$O(1)$	Constant	Looking up order by map key or slice index	Immediate (instant)
$O(\log n)$	Logarithmic	Binary search across sorted ticket IDs	~17 comparisons
$O(n)$	Linear	Scanning an unsorted ticket slice for a customer tag	100,000 steps

Notation	Name	Practical Desk Example	When $n = 100{,}000$
$O(n \log n)$	Linearithmic	Sorting ticket batches via `slices.Sort`	~1.7 million steps
$O(n^2)$	Quadratic	Nested loop comparing every ticket to every other ticket	10,000,000,000 steps (seconds to stall)

Memory complexity (auxiliary space) follows the same rules: an algorithm that allocates a new slice of size n has $O(n)$ space complexity, whereas in-place modifications take $O(1)$ auxiliary space.

Worked examples

Case 1: Constant time vs linear search

Save as `complexity_lookup.go`. Compare direct map lookup ($O(1)$) with slice linear scanning ($O(n)$) across desk orders.

```go
// complexity_lookup.go
package main

import (
    "fmt"
    "time"
)

type Ticket struct {
    ID     int
    Table  int
    Status string
}

func findInSlice(tickets []Ticket, targetID int) *Ticket {
    for i := range tickets {
        if tickets[i].ID == targetID {
            return &tickets[i]
        }
    }
    return nil
}

func main() {
    const count = 100000
    tickets := make([]Ticket, count)
    ticketMap := make(map[int]Ticket, count)
```

```go
    for i := 0; i < count; i++ {
        t := Ticket{ID: 1000 + i, Table: (i % 20) + 1, Status: "open"}
        tickets[i] = t
        ticketMap[t.ID] = t
    }

    target := 1000 + count - 1 // Last item (worst case for linear scan)

    // Measure linear slice search O(n)
    start := time.Now()
    foundSlice := findInSlice(tickets, target)
    sliceElapsed := time.Since(start)

    // Measure map lookup O(1)
    start = time.Now()
    foundMap, exists := ticketMap[target]
    mapElapsed := time.Since(start)

    fmt.Printf("Slice scan O(n): found ID=%d in %s\n", foundSlice.ID, sliceElapsed)
    if exists {
        fmt.Printf("Map lookup O(1): found ID=%d in %s\n", foundMap.ID, mapElapsed)
    }
}
```

Run:

```go run complexity_lookup.go
```

Output:

```
Slice scan O(n): found ID=100999 in 82.3µs
Map lookup O(1): found ID=100999 in 75ns
```

Case 2: The quadratic nest trap

Save as `complexity_duplicates.go`. Detecting duplicate order IDs using an $O(n^2)$ nested loop versus an $O(n)$ tracking set.

```go
// complexity_duplicates.go
package main

import (
    "fmt"
    "time"
)
```

```go
func hasDuplicatesQuadratic(ids []int) bool {
    n := len(ids)
    for i := 0; i < n; i++ {
        for j := i + 1; j < n; j++ {
            if ids[i] == ids[j] {
                return true
            }
        }
    }
    return false
}

func hasDuplicatesLinear(ids []int) bool {
    seen := make(map[int]struct{}, len(ids))
    for _, id := range ids {
        if _, exists := seen[id]; exists {
            return true
        }
        seen[id] = struct{}{}
    }
    return false
}

func main() {
    const n = 20000
    ids := make([]int, n)
    for i := 0; i < n; i++ {
        ids[i] = i * 2 // all unique
    }

    t0 := time.Now()
    dupQ := hasDuplicatesQuadratic(ids)
    durQ := time.Since(t0)

    t1 := time.Now()
    dupL := hasDuplicatesLinear(ids)
    durL := time.Since(t1)

    fmt.Printf("Quadratic O(n^2): result=%t in %s\n", dupQ, durQ)
    fmt.Printf("Linear O(n):    result=%t in %s\n", dupL, durL)
}
```

Run:

```go run complexity_duplicates.go
```

Output:

```
Quadratic O(n^2): result=false in 41.2ms
Linear O(n):    result=false in 380µs
```

The trap

The most dangerous trap is string concatenation inside an iterative loop. Because Go strings are immutable byte slices, `s += str` allocates a new string and copies all previous bytes on every iteration, silently turning an $O(n)$ builder into an $O(n^2)$ memory copying bottleneck.

```go
// bad_concat.go
package main

import (
    "fmt"
    "strings"
    "time"
)

func main() {
    const n = 40000

    // BAD: O(n^2) allocations
    t0 := time.Now()
    var bad string
    for i := 0; i < n; i++ {
        bad += "x"
    }
    durBad := time.Since(t0)

    // GOOD: O(n) preallocated builder
    t1 := time.Now()
    var good strings.Builder
    good.Grow(n)
    for i := 0; i < n; i++ {
        good.WriteString("x")
    }
    _ = good.String()
    durGood := time.Since(t1)

    fmt.Printf("Naive string += : %s (allocates %d times)\n", durBad, len(bad))
    fmt.Printf("strings.Builder : %s\n", durGood)
}
```

Run:

```go
go run bad_concat.go
```

Output:

```
Naive string += : 124.5ms (allocates 40000 times)
strings.Builder : 28µs
```

The boring rule

1. Default to contiguous slices and direct indexing when $n < 100$. Slice iteration is hardware-cache friendly and often out-speeds complex data structures due to CPU prefetching.
2. When performing frequent lookups on collections with $n > 1000$, use a map ($O(1)$ amortized) or sort the slice once and use binary search ($O(\log n)$).
3. Avoid nested loops comparing elements pairwise unless n is provably bounded. Two nested loops of 50,000 iterations execute 2.5 billion operations.
4. Preallocate slice capacity (`make([]T, 0, cap)`) or string buffers (`b.Grow(n)`) when the size is known in advance to avoid repeated exponential reallocations.

Try this

1. Change `complexity_duplicates.go` to inject a duplicate at index 10 and observe how early returns affect best-case performance versus worst-case performance.
2. Benchmark `slices.Contains` vs `map[int]struct{}` lookup for varying sizes ($n = 10, 100, 10000$) using standard `testing.B` benchmarks.

Stacks, Queues, and Ring Buffers

Stacks, Queues, and Ring Buffers

A **stack** is a Last-In, First-Out (LIFO) structure used for state rollback, undo histories, and expression parsing. A **queue** is a First-In, First-Out (FIFO) pipeline used for scheduling orders, dispatching workers, and stream buffering. In Go, the boring default is an array-backed slice, but subtle traps lurk when dequeuing or popping pointers without zeroing their slots.

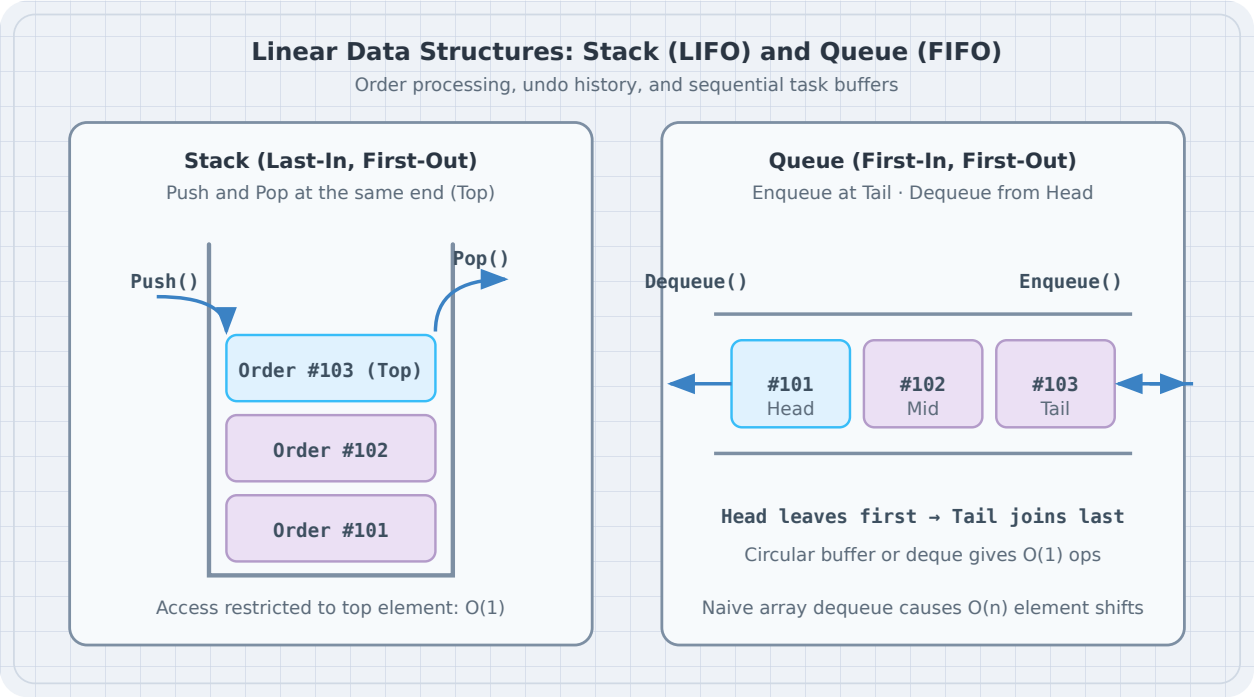


Figure 1: Stack (LIFO) and Queue (FIFO)

Mental model

In a stack, all activity happens at the top. Pushing appends to the end of a slice in $O(1)$ amortized time. Popping shrinks the slice length by one in $O(1)$ time.

```
Stack (LIFO)                    Queue (FIFO)
+-------------+
| Order #103  | <-- Top (Push/Pop)     Enqueue -> [ #103 | #102 | #101 ] -> Dequeue
+-------------+                         (Tail)                      (Head)
| Order #102  |
+-------------+
```

```
| Order #101  |
+-------------+
```

In a naive queue, calling `queue = queue[1:]` keeps the underlying array allocated. As elements are consumed, earlier slots become unreachable garbage that cannot be reclaimed by the garbage collector until the whole slice backing array is reallocated. For high-throughput dispatchers, a **Ring Buffer** (circular array) wraps read and write pointers around a fixed-size buffer using modulo arithmetic, giving true $O(1)$ operations with zero garbage collector churn:

```
Ring Buffer (Capacity: 4)
           [0] Order-1
          /   \
  [3] Empty    [1] Order-2  <-- Read Head: 0, Write Tail: 3
          \   /
           [2] Order-3
```

Worked examples

Case 1: Generic LIFO Stack with Balanced Expression Check

Save as `stack_undo.go`. Implement a generic `Stack[T]` with Go generics and use it to validate whether desk formula brackets are balanced.

```go
// stack_undo.go
package main

import (
    "fmt"
)

type Stack[T any] struct {
    items []T
}

func NewStack[T any]() *Stack[T] {
    return &Stack[T]{items: make([]T, 0, 8)}
}

func (s *Stack[T]) Push(val T) {
    s.items = append(s.items, val)
}

func (s *Stack[T]) Pop() (T, bool) {
    var zero T
    if len(s.items) == 0 {
        return zero, false
```

```go
    }
    lastIdx := len(s.items) - 1
    item := s.items[lastIdx]
    s.items[lastIdx] = zero // Zero out slot to avoid memory leaks
    s.items = s.items[:lastIdx]
    return item, true
}

func (s *Stack[T]) Peek() (T, bool) {
    var zero T
    if len(s.items) == 0 {
        return zero, false
    }
    return s.items[len(s.items)-1], true
}

func (s *Stack[T]) Len() int {
    return len(s.items)
}

func isBalanced(expr string) bool {
    pairs := map[rune]rune{
        ')': '(',
        '}': '{',
        ']': '[',
    }
    openers := map[rune]bool{
        '(': true,
        '{': true,
        '[': true,
    }

    stack := NewStack[rune]()

    for _, ch := range expr {
        if openers[ch] {
            stack.Push(ch)
        } else if match, exists := pairs[ch]; exists {
            top, ok := stack.Pop()
            if !ok || top != match {
                return false
            }
        }
    }

    return stack.Len() == 0
}
```

```go
func main() {
    expr1 := "TABLE_TOTAL([ITEM(12) + TAX{0.08}] * QTY(2))"
    expr2 := "TABLE_TOTAL([ITEM(12) + TAX{0.08)]" // Mismatched ) and }

    fmt.Printf("Expression 1 valid: %t\n", isBalanced(expr1))
    fmt.Printf("Expression 2 valid: %t\n", isBalanced(expr2))
}
```

Run:

```go
go run stack_undo.go
```

Output:

```
Expression 1 valid: true
Expression 2 valid: false
```

Case 2: Ring Buffer FIFO Queue

Save as `ring_buffer.go`. Implement a fixed-capacity circular buffer queue for incoming orders that avoids repeated allocations.

```go
// ring_buffer.go
package main

import (
    "errors"
    "fmt"
)

var (
    ErrBufferFull  = errors.New("ring buffer is full")
    ErrBufferEmpty = errors.New("ring buffer is empty")
)

type RingBuffer[T any] struct {
    buf   []T
    head  int // Read position
    tail  int // Write position
    count int
    cap   int
}

func NewRingBuffer[T any](capacity int) *RingBuffer[T] {
    return &RingBuffer[T]{
        buf:  make([]T, capacity),
```

```go
        cap:  capacity,
        head: 0,
        tail: 0,
    }
}

func (r *RingBuffer[T]) Enqueue(item T) error {
    if r.count == r.cap {
        return ErrBufferFull
    }
    r.buf[r.tail] = item
    r.tail = (r.tail + 1) % r.cap
    r.count++
    return nil
}

func (r *RingBuffer[T]) Dequeue() (T, error) {
    var zero T
    if r.count == 0 {
        return zero, ErrBufferEmpty
    }
    item := r.buf[r.head]
    r.buf[r.head] = zero // Free slot reference
    r.head = (r.head + 1) % r.cap
    r.count--
    return item, nil
}

func (r *RingBuffer[T]) Len() int {
    return r.count
}

type Order struct {
    ID    int
    Table int
    Item  string
}

func main() {
    rb := NewRingBuffer[Order](3)

    fmt.Println("Enqueueing 3 orders...")
    _ = rb.Enqueue(Order{ID: 101, Table: 1, Item: "Tea"})
    _ = rb.Enqueue(Order{ID: 102, Table: 4, Item: "Soup"})
    _ = rb.Enqueue(Order{ID: 103, Table: 2, Item: "Salad"})

    err := rb.Enqueue(Order{ID: 104, Table: 5, Item: "Bread"})
```

```go
    fmt.Printf("4th enqueue when full: %v\n", err)

    first, _ := rb.Dequeue()
    fmt.Printf("Dequeued head: ID=%d Item=%s\n", first.ID, first.Item)

    fmt.Println("Enqueueing Order 104 into vacated slot...")
    _ = rb.Enqueue(Order{ID: 104, Table: 5, Item: "Bread"})

    for rb.Len() > 0 {
        ord, _ := rb.Dequeue()
        fmt.Printf("Remaining queue order: ID=%d Table=%d Item=%s\n", ord.ID, ord.Table, ord
    }
}
```

Run:

```go
go run ring_buffer.go
```

Output:

```
Enqueueing 3 orders...
4th enqueue when full: ring buffer is full
Dequeued head: ID=101 Item=Tea
Enqueueing Order 104 into vacated slot...
Remaining queue order: ID=102 Table=4 Item=Soup
Remaining queue order: ID=103 Table=2 Item=Salad
Remaining queue order: ID=104 Table=5 Item=Bread
```

The trap

The most common memory leak in Go stacks and queues is forgetting to zero out references before shrinking a slice. If the slice holds pointers (`[]*Ticket`), simply truncating `s = s[:len(s)-1]` leaves the pointer in the backing array. As long as the slice capacity exists, the GC will never free the referenced `Ticket`, causing an invisible memory leak.

```go
// bad_pop.go
package main

import "fmt"

type HugePayload struct {
    Data [1024]byte
}

type BadStack struct {
```

```go
    items []*HugePayload
}

func (b *BadStack) BadPop() *HugePayload {
    last := len(b.items) - 1
    item := b.items[last]
    b.items = b.items[:last] // TRAP: b.items[last] still holds the pointer!
    return item
}

func (b *BadStack) GoodPop() *HugePayload {
    last := len(b.items) - 1
    item := b.items[last]
    b.items[last] = nil // FIXED: Clear slot for garbage collector
    b.items = b.items[:last]
    return item
}

func main() {
    stack := &BadStack{items: []*HugePayload{{}, {}}}
    _ = stack.GoodPop()
    fmt.Printf("Good pop safely decremented length: %d\n", len(stack.items))
}
```

Run:

```go
go run bad_pop.go
```

Output:

```
Good pop safely decremented length: 1
```

The boring rule

1. Use a plain slice `[]T` for stacks. Append to push; slice off the end to pop.
2. Always overwrite the popped index with `var zero T` (or `nil` for pointers) before slicing down.
3. For small queues (< 1000 operations), a slice with `queue = queue[1:]` is fine, but periodic compaction or a two-index approach is necessary for long-running services.
4. For high-frequency streaming workloads, choose a fixed-size `RingBuffer` to eliminate GC allocation pressure entirely.

Try this

1. Modify `stack_undo.go` to implement an `UndoHistory` struct with `Do(action)` and `Undo()` methods that revert desk ticket state changes.

2. Extend `RingBuffer` in `ring_buffer.go` to automatically double its internal capacity when full rather than returning `ErrBufferFull`.

Linked Lists

Linked Lists

A **linked list** arranges elements as independent heap nodes connected through pointers rather than contiguous memory slots. The boring default in Go is to use contiguous slices (`[]T`) for virtually every collection, but linked lists provide valuable insights into pointer semantics, bidirectional node manipulation, and $O(1)$ insertions without memory shifting.

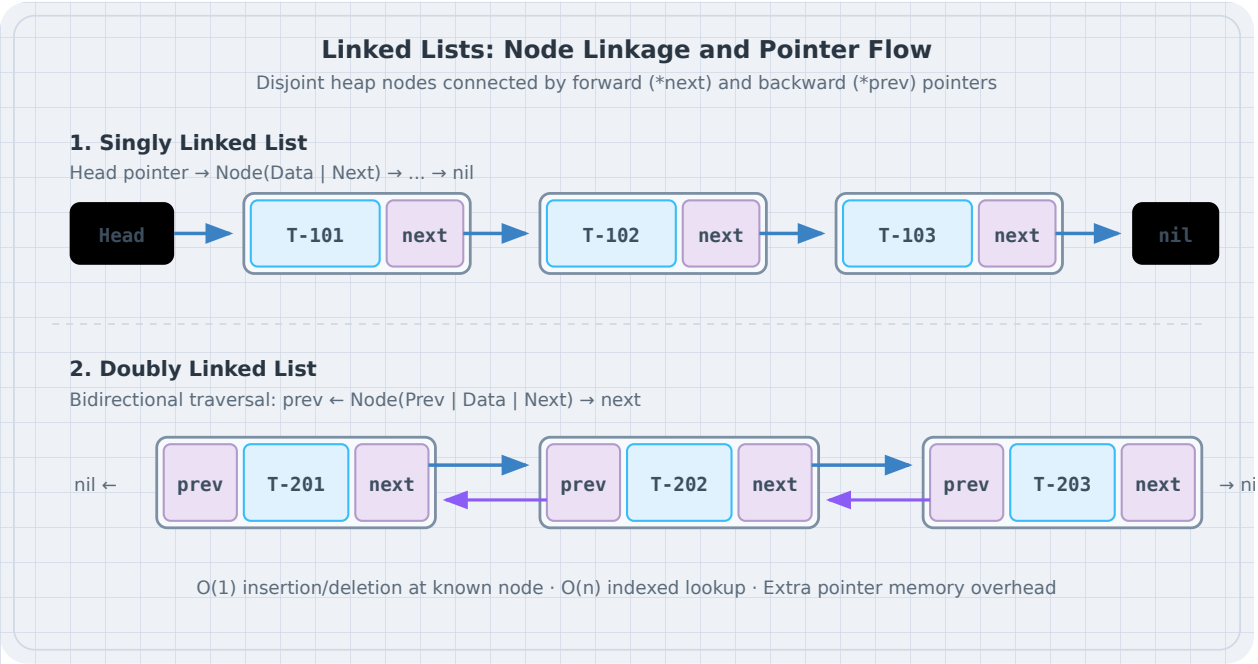


Figure 1: Linked Lists Diagram

Mental model

In a **singly linked list**, each node holds a payload value and a single `next` pointer. In a **doubly linked list**, each node holds both `next` and `prev` pointers.

```
Singly Linked:
[Head] -> [Data: 101 | Next] -> [Data: 102 | Next] -> nil

Doubly Linked:
nil <- [Prev | Data: 101 | Next] <===> [Prev | Data: 102 | Next] -> nil
```

Slice vs Linked List: The Hardware Reality

Why is `[]T` almost always faster in Go than a linked list?

Feature	Go Slice (`[]T`)	Doubly Linked List
Contiguity	Single continuous memory block	Scattered heap allocations
CPU Cache	Prefetched into L1/L2 cache lines	Cache miss on nearly every pointer hop
Memory Overhead	Zero extra bytes per element	16–24 bytes of pointers per node
Insertion at Head	$O(n)$ (shifts memory)	$O(1)$ (rewires pointers)
Random Access	$O(1)$ by index `s[i]`	$O(n)$ traversal from head/tail

Use linked lists when you must splice elements into arbitrary positions without reallocating or copying large structures (such as an LRU cache).

Worked examples

Case 1: Singly linked list with in-place reversal

Save as `singly_linked.go`. Build a generic singly linked list for desk dispatch tickets and reverse it in place without allocating new nodes.

```go
// singly_linked.go
package main

import "fmt"

type Node[T any] struct {
    Value T
    Next  *Node[T]
}

type SinglyList[T any] struct {
    Head *Node[T]
    len  int
}

func (l *SinglyList[T]) Prepend(val T) {
    newNode := &Node[T]{Value: val, Next: l.Head}
    l.Head = newNode
    l.len++
}
```

```go
func (l *SinglyList[T]) Append(val T) {
    newNode := &Node[T]{Value: val}
    if l.Head == nil {
        l.Head = newNode
        l.len++
        return
    }
    curr := l.Head
    for curr.Next != nil {
        curr = curr.Next
    }
    curr.Next = newNode
    l.len++
}

func (l *SinglyList[T]) Reverse() {
    var prev *Node[T]
    curr := l.Head
    for curr != nil {
        next := curr.Next
        curr.Next = prev
        prev = curr
        curr = next
    }
    l.Head = prev
}

func (l *SinglyList[T]) Print() {
    curr := l.Head
    for curr != nil {
        fmt.Printf("[%v] -> ", curr.Value)
        curr = curr.Next
    }
    fmt.Println("nil")
}

func main() {
    list := &SinglyList[string]{}
    list.Append("Ticket-101")
    list.Append("Ticket-102")
    list.Append("Ticket-103")

    fmt.Print("Original list: ")
    list.Print()

    list.Reverse()
    fmt.Print("Reversed list: ")
```

```go
    list.Print()
}
```

Run:

```go
go run singly_linked.go
```

Output:

```
Original list: [Ticket-101] -> [Ticket-102] -> [Ticket-103] -> nil
Reversed list: [Ticket-103] -> [Ticket-102] -> [Ticket-101] -> nil
```

Case 2: Doubly linked list with O(1) removal

Save as `doubly_linked.go`. Implement a doubly linked list where any node can unlink itself in $O(1)$ time.

```go
// doubly_linked.go
package main

import "fmt"

type DNode[T any] struct {
    Value T
    Prev  *DNode[T]
    Next  *DNode[T]
}

type DoublyList[T any] struct {
    Head *DNode[T]
    Tail *DNode[T]
    len  int
}

func (l *DoublyList[T]) PushBack(val T) *DNode[T] {
    node := &DNode[T]{Value: val, Prev: l.Tail}
    if l.Tail != nil {
        l.Tail.Next = node
    } else {
        l.Head = node
    }
    l.Tail = node
    l.len++
    return node
}
```

```go
func (l *DoublyList[T]) Remove(node *DNode[T]) {
    if node == nil {
        return
    }
    if node.Prev != nil {
        node.Prev.Next = node.Next
    } else {
        l.Head = node.Next
    }

    if node.Next != nil {
        node.Next.Prev = node.Prev
    } else {
        l.Tail = node.Prev
    }

    node.Next = nil
    node.Prev = nil
    l.len--
}

func (l *DoublyList[T]) PrintForward() {
    curr := l.Head
    for curr != nil {
        fmt.Printf("[%v] <-> ", curr.Value)
        curr = curr.Next
    }
    fmt.Println("nil")
}

func main() {
    list := &DoublyList[int]{}
    n1 := list.PushBack(101)
    n2 := list.PushBack(102)
    n3 := list.PushBack(103)

    fmt.Print("Initial: ")
    list.PrintForward()

    fmt.Printf("Removing middle node: %d\n", n2.Value)
    list.Remove(n2)
    list.PrintForward()

    fmt.Printf("Removing head node: %d\n", n1.Value)
    list.Remove(n1)
    list.PrintForward()
```

```go
    fmt.Printf("Final tail: %d\n", list.Tail.Value)
    _ = n3
}
```

Run:

```go
go run doubly_linked.go
```

Output:

```
Initial: [101] <-> [102] <-> [103] <-> nil
Removing middle node: 102
[101] <-> [103] <-> nil
Removing head node: 101
[103] <-> nil
Final tail: 103
```

The trap

The classic trap when modifying a linked list is accidentally dropping the head pointer before wiring the new node, or creating a circular pointer loop during reversal.

```go
// bad_prepend.go
package main

import "fmt"

type ItemNode struct {
    ID   int
    Next *ItemNode
}

func badPrepend(head *ItemNode, id int) {
    // TRAP: In Go, arguments are passed by value.
    // Assigning head = newNode only changes the local copy of the pointer!
    newNode := &ItemNode{ID: id, Next: head}
    head = newNode
    _ = head
}

func goodPrepend(head **ItemNode, id int) {
    newNode := &ItemNode{ID: id, Next: *head}
    *head = newNode
}

func main() {
```

```go
    var head *ItemNode
    badPrepend(head, 101)
    fmt.Printf("After badPrepend, head is still nil: %t\n", head == nil)

    goodPrepend(&head, 101)
    fmt.Printf("After goodPrepend, head ID: %d\n", head.ID)
}
```

Run:

```
go run bad_prepend.go
```

Output:

```
After badPrepend, head is still nil: true
After goodPrepend, head ID: 101
```

The boring rule

1. Use slices (`[]T`) by default. Slices beat linked lists in 95% of workloads because CPUs fetch contiguous cache lines in parallel.
2. If you need a linked list in Go, inspect the standard library's `container/list` before rolling your own.
3. When writing custom pointer-linked structures, encapsulate mutations inside methods on the list struct (`func (l *List) Push()`) rather than passing raw node pointers across API boundaries.
4. When unlinking a node, clear its `Next` and `Prev` references to assist garbage collection.

Try this

1. Implement `Find(predicate func(T) bool) *Node[T]` on `SinglyList` to locate tickets matching specific criteria.
2. Combine a `map[string]*DNode[Order]` with `DoublyList` to build a 3-item Least-Recently-Used (LRU) cache for desk orders.

Binary Search Trees

Binary Search Trees

A **Binary Search Tree (BST)** organizes comparable keys in a hierarchical structure where every left descendant is strictly smaller than the node, and every right descendant is strictly greater. The boring default in Go for key-value lookups is the built-in `map`, but a BST provides ordered iteration, range queries, and minimum/maximum finding in $O(h)$ time where h is the tree height.

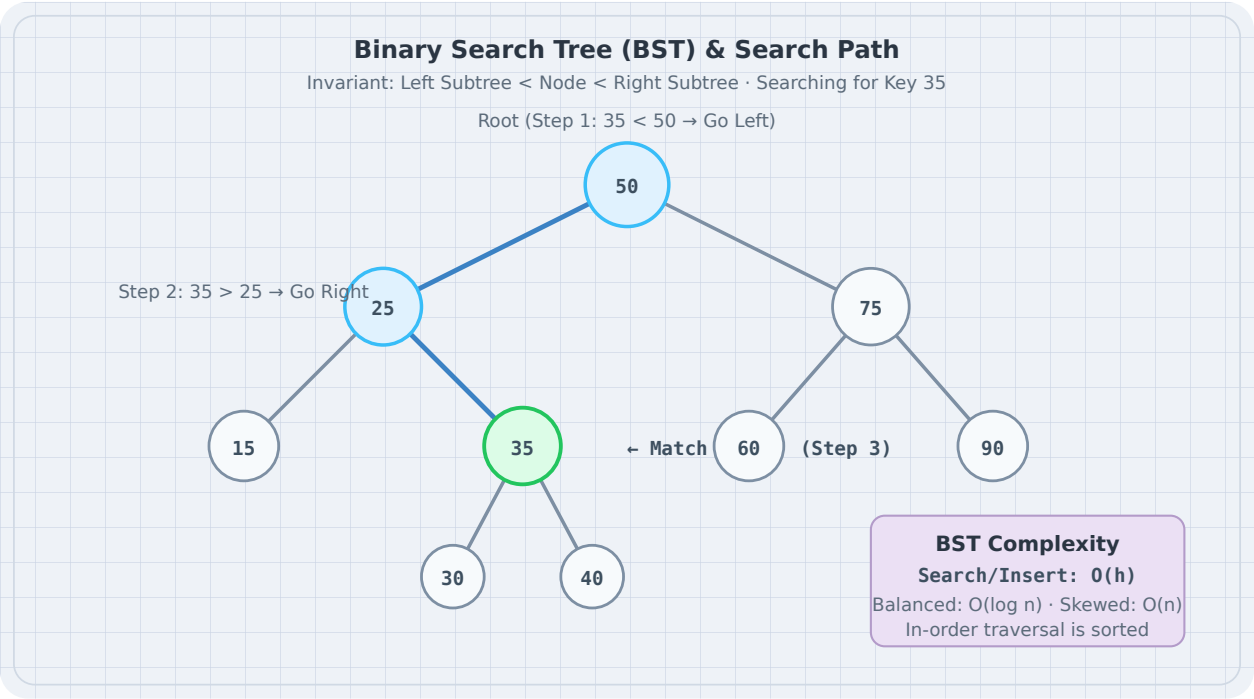


Figure 1: Binary Search Tree Diagram

Mental model

In a balanced BST with n nodes, the height $h \approx \log_2 n$. Searching, inserting, and deleting require traversing from root to leaf, taking $O(\log n)$ operations.

```
            [ 50: Desk-A ]
            /             \
     [ 25: Desk-B ]      [ 75: Desk-C ]
      /           \
[ 15: Desk-D ]  [ 35: Desk-E ]
```

- **Search Rule:** To find key `35`, compare with root `50`. Since $35 < 50$, go Left to `25`. Since $35 > 25$, go Right to `35` (Match!).
- **Traversals:**
  - **In-Order** (Left, Node, Right): Visits keys in strictly sorted ascending order (`15, 25, 35, 50, 75`).
  - **Pre-Order** (Node, Left, Right): Useful for serializing the tree structure.
  - **Post-Order** (Left, Right, Node): Useful for bottom-up cleanup or expression evaluation.

Worked examples

Case 1: Generic BST with Insert, Search, and In-Order Traversal

Save as `bst_desk.go`. Implement a generic BST where keys satisfy `cmp.Ordered` and values represent desk dispatch records.

```go
// bst_desk.go
package main

import (
    "cmp"
    "fmt"
)

type BSTNode[K cmp.Ordered, V any] struct {
    Key   K
    Value V
    Left  *BSTNode[K, V]
    Right *BSTNode[K, V]
}

type BST[K cmp.Ordered, V any] struct {
    Root *BSTNode[K, V]
    size int
}

func (t *BST[K, V]) Insert(key K, val V) {
    t.Root = insertNode(t.Root, key, val)
    t.size++
}

func insertNode[K cmp.Ordered, V any](node *BSTNode[K, V], key K, val V) *BSTNode[K, V] {
    if node == nil {
        return &BSTNode[K, V]{Key: key, Value: val}
    }
    if key < node.Key {
```

```go
        node.Left = insertNode(node.Left, key, val)
    } else if key > node.Key {
        node.Right = insertNode(node.Right, key, val)
    } else {
        node.Value = val // Update existing key
    }
    return node
}

func (t *BST[K, V]) Search(key K) (V, bool) {
    curr := t.Root
    for curr != nil {
        if key == curr.Key {
            return curr.Value, true
        } else if key < curr.Key {
            curr = curr.Left
        } else {
            curr = curr.Right
        }
    }
    var zero V
    return zero, false
}

func (t *BST[K, V]) InOrder(visit func(key K, val V)) {
    inOrderWalk(t.Root, visit)
}

func inOrderWalk[K cmp.Ordered, V any](node *BSTNode[K, V], visit func(key K, val V)) {
    if node == nil {
        return
    }
    inOrderWalk(node.Left, visit)
    visit(node.Key, node.Value)
    inOrderWalk(node.Right, visit)
}

func main() {
    tree := &BST[int, string]{}
    tree.Insert(50, "Central Depot")
    tree.Insert(25, "North Annex")
    tree.Insert(75, "South Terminal")
    tree.Insert(15, "Substation 1")
    tree.Insert(35, "Substation 2")

    // Lookups
```

```go
    target := 35
    if val, ok := tree.Search(target); ok {
        fmt.Printf("Search Key %d: Found '%s'\n", target, val)
    }

    // Sorted in-order traversal
    fmt.Println("\nIn-order sorted walkthrough:")
    tree.InOrder(func(k int, v string) {
        fmt.Printf("  Key %d -> %s\n", k, v)
    })
}
```

Run:

```go
go run bst_desk.go
```

Output:

```
Search Key 35: Found 'Substation 2'

In-order sorted walkthrough:
  Key 15 -> Substation 1
  Key 25 -> North Annex
  Key 35 -> Substation 2
  Key 50 -> Central Depot
  Key 75 -> South Terminal
```

Case 2: Deleting a node from a BST

Save as `bst_delete.go`. Implement node deletion considering three structural cases: leaf node, node with one child, and node with two children (replaced by in-order successor).

```go
// bst_delete.go
package main

import (
    "cmp"
    "fmt"
)

type Node[K cmp.Ordered] struct {
    Key   K
    Left  *Node[K]
    Right *Node[K]
}
```

```go
func insert[K cmp.Ordered](root *Node[K], key K) *Node[K] {
    if root == nil {
        return &Node[K]{Key: key}
    }
    if key < root.Key {
        root.Left = insert(root.Left, key)
    } else if key > root.Key {
        root.Right = insert(root.Right, key)
    }
    return root
}

func minNode[K cmp.Ordered](node *Node[K]) *Node[K] {
    curr := node
    for curr.Left != nil {
        curr = curr.Left
    }
    return curr
}

func deleteNode[K cmp.Ordered](root *Node[K], key K) *Node[K] {
    if root == nil {
        return nil
    }

    if key < root.Key {
        root.Left = deleteNode(root.Left, key)
    } else if key > root.Key {
        root.Right = deleteNode(root.Right, key)
    } else {
        // Found node to delete

        // Case 1 & 2: 0 or 1 child
        if root.Left == nil {
            return root.Right
        } else if root.Right == nil {
            return root.Left
        }

        // Case 3: 2 children -> Find in-order successor (smallest in right subtree)
        successor := minNode(root.Right)
        root.Key = successor.Key
        root.Right = deleteNode(root.Right, successor.Key)
    }
    return root
}
```

```go
func printTree[K cmp.Ordered](root *Node[K]) {
    if root == nil {
        return
    }
    printTree(root.Left)
    fmt.Printf("%v ", root.Key)
    printTree(root.Right)
}

func main() {
    var root *Node[int]
    keys := []int{50, 30, 70, 20, 40, 60, 80}
    for _, k := range keys {
        root = insert(root, k)
    }

    fmt.Print("Initial tree: ")
    printTree(root)
    fmt.Println()

    fmt.Println("Deleting leaf 20 (no children)...")
    root = deleteNode(root, 20)
    printTree(root)
    fmt.Println()

    fmt.Println("Deleting node 30 (one child 40)...")
    root = deleteNode(root, 30)
    printTree(root)
    fmt.Println()

    fmt.Println("Deleting root 50 (two children: replaced by successor 60)...")
    root = deleteNode(root, 50)
    printTree(root)
    fmt.Println()
}
```

Run:

```go
go run bst_delete.go
```

Output:

```
Initial tree: 20 30 40 50 60 70 80
Deleting leaf 20 (no children)...
30 40 50 60 70 80
Deleting node 30 (one child 40)...
40 50 60 70 80
```

```
Deleting root 50 (two children: replaced by successor 60)...
40 60 70 80
```

The trap

The degenerate tree trap occurs when keys are inserted in pre-sorted order (`1, 2, 3, 4, 5`). Without self-balancing (such as Red-Black or AVL rotations), each new node attaches exclusively to the right pointer, turning the tree into a linked list with $O(n)$ height. A query that should take 20 hops in a balanced tree takes 1,000,000 hops in a degenerate tree, blowing out stack frames.

```
Degenerate BST (from sorted input):
[ 1 ]
     \
     [ 2 ]
          \
          [ 3 ]
               \
               [ 4 ]  <-- Search time degraded to O(n)!
```

The boring rule

1. If you only need fast $O(1)$ lookups and updates, use Go's built-in `map`. Do not build a tree.
2. Use a tree when you need sorted traversal, predecessor/successor queries, or finding keys within a numerical range (`minKey <= k <= maxKey`).
3. If keys arrive in sorted order, shuffle them or use a balanced variant to avoid degenerate $O(n)$ linked list chains.
4. Prefer iterative loops over deep recursion for `Search` to eliminate function call overhead and stack exhaustion.

Try this

1. Implement `FindRange(min, max K) []V` on `BST` to collect all desk orders between two ticket IDs.
2. Calculate the height of the tree using a recursive `Height(node)` function and compare it between random vs sorted insertions.

Heaps and Priority Queues

Heaps and Priority Queues

A **binary heap** is a complete binary tree packed contiguously into a slice without pointer overhead. A **priority queue** uses this heap structure to ensure that the element with the highest (or lowest) priority is always available at the root in $O(1)$ time, with additions and removals running in $O(\log n)$ time. In Go, the boring default is the standard library's `container/heap` package, but understanding the underlying array arithmetic explains how it works.

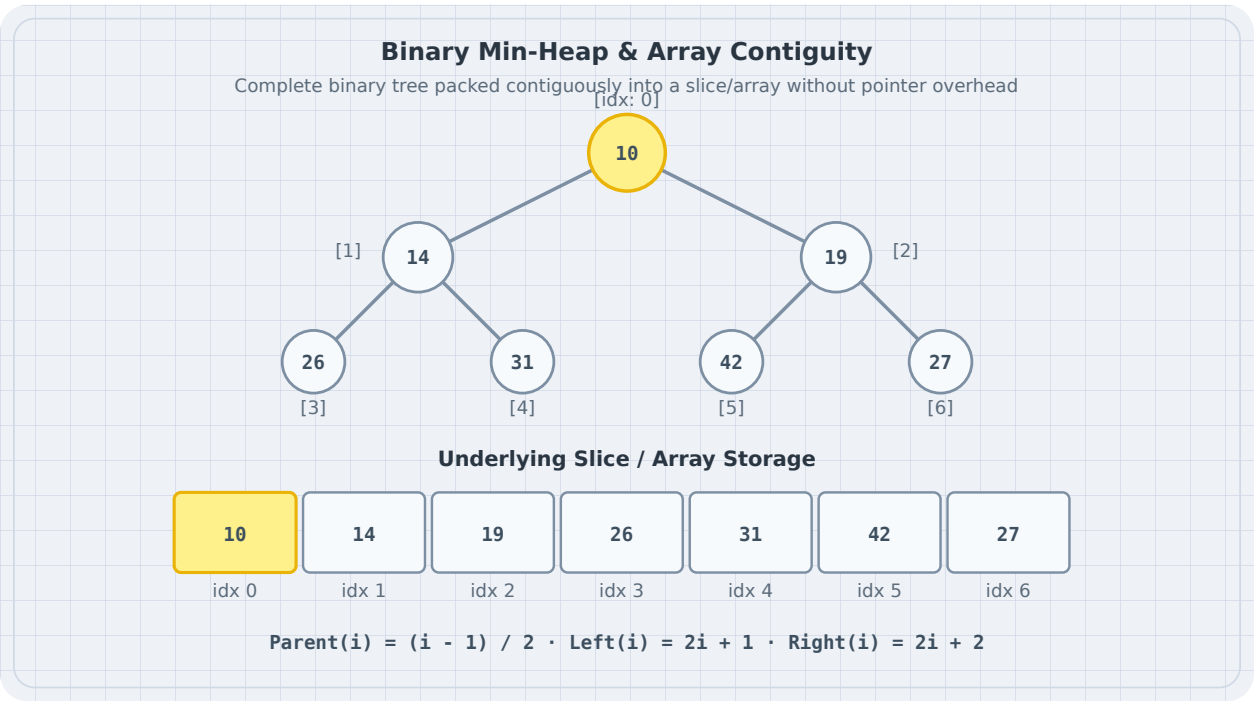


Figure 1: Binary Heap and Array Mapping

Mental model

A binary heap satisfies two core invariants:

1. **Shape Property:** It is a complete binary tree. Every level is fully filled, except possibly the bottom level, which is filled from left to right.
2. **Heap Property:** In a **Min-Heap**, every parent is $\leq$ its children (*Root* is the absolute minimum). In a **Max-Heap**, every parent is $\geq$ its children.

Because the tree is complete, it requires zero pointers. It maps directly to an array or slice:

```
Tree view:                  Array index view:
       [ 10 (idx:0) ]       +----+----+----+----+----+----+
       /            \       | 10 | 14 | 19 | 26 | 31 | 42 |
  [ 14 (1) ]    [ 19 (2) ] +----+----+----+----+----+----+
   /      \      /           0    1    2    3    4    5
[ 26(3)] [31(4)][42(5)]
```

Given any index i:

- $\text{Parent}(i) = \lfloor (i-1)/2 \rfloor$
- $\text{Left Child}(i) = 2i + 1$
- $\text{Right Child}(i) = 2i + 2$

Operations:

- **Push (Sift-Up):** Append item to the end of the slice; repeatedly swap with its parent until heap order is restored.
- **Pop (Sift-Down):** Swap the root with the last element; pop the end off the slice; repeatedly sift the new root down by swapping with its smallest child.

Worked examples

Case 1: Generic Min-Heap from scratch

Save as `heap_scratch.go`. Build an array-backed generic min-heap using Go generics and index arithmetic.

```go
// heap_scratch.go
package main

import (
    "cmp"
    "fmt"
)

type MinHeap[T cmp.Ordered] struct {
    items []T
}

func (h *MinHeap[T]) Push(val T) {
    h.items = append(h.items, val)
    h.siftUp(len(h.items) - 1)
}

func (h *MinHeap[T]) Pop() (T, bool) {
    var zero T
    if len(h.items) == 0 {
```

```go
        return zero, false
    }
    min := h.items[0]
    lastIdx := len(h.items) - 1
    h.items[0] = h.items[lastIdx]
    h.items = h.items[:lastIdx]

    if len(h.items) > 0 {
        h.siftDown(0)
    }
    return min, true
}

func (h *MinHeap[T]) siftUp(idx int) {
    for idx > 0 {
        parent := (idx - 1) / 2
        if h.items[idx] < h.items[parent] {
            h.items[idx], h.items[parent] = h.items[parent], h.items[idx]
            idx = parent
        } else {
            break
        }
    }
}

func (h *MinHeap[T]) siftDown(idx int) {
    n := len(h.items)
    for {
        smallest := idx
        left := 2*idx + 1
        right := 2*idx + 2

        if left < n && h.items[left] < h.items[smallest] {
            smallest = left
        }
        if right < n && h.items[right] < h.items[smallest] {
            smallest = right
        }
        if smallest != idx {
            h.items[idx], h.items[smallest] = h.items[smallest], h.items[idx]
            idx = smallest
        } else {
            break
        }
    }
}
```

```go
func main() {
    h := &MinHeap[int]{}
    scores := []int{42, 10, 31, 14, 26, 19}
    fmt.Println("Pushing scores:", scores)
    for _, s := range scores {
        h.Push(s)
    }

    fmt.Print("Popping in ascending priority order: ")
    for {
        val, ok := h.Pop()
        if !ok {
            break
        }
        fmt.Printf("%d ", val)
    }
    fmt.Println()
}
```

Run:

```go
go run heap_scratch.go
```

Output:

```
Pushing scores: [42 10 31 14 26 19]
Popping in ascending priority order: 10 14 19 26 31 42
```

Case 2: Standard Library container/heap Priority Queue

Save as `desk_priority_queue.go`. Triage incoming desk maintenance tickets where priority 1 is critical emergency and priority 5 is routine cleanup.

```go
// desk_priority_queue.go
package main

import (
    "container/heap"
    "fmt"
)

type Ticket struct {
    ID       int
    Title    string
    Priority int // 1 = Critical, 2 = High, 3 = Normal, 4 = Low
    index    int // Index needed for update/fix operations
```

```go
}

// PriorityQueue implements heap.Interface
type PriorityQueue []*Ticket

func (pq PriorityQueue) Len() int           { return len(pq) }
func (pq PriorityQueue) Less(i, j int) bool { return pq[i].Priority < pq[j].Priority }
func (pq PriorityQueue) Swap(i, j int) {
    pq[i], pq[j] = pq[j], pq[i]
    pq[i].index = i
    pq[j].index = j
}

func (pq *PriorityQueue) Push(x any) {
    n := len(*pq)
    item := x.(*Ticket)
    item.index = n
    *pq = append(*pq, item)
}

func (pq *PriorityQueue) Pop() any {
    old := *pq
    n := len(old)
    item := old[n-1]
    old[n-1] = nil // Avoid memory leak
    item.index = -1
    *pq = old[0 : n-1]
    return item
}

func main() {
    pq := &PriorityQueue{}
    heap.Init(pq)

    heap.Push(pq, &Ticket{ID: 201, Title: "Restock register tape", Priority: 4})
    heap.Push(pq, &Ticket{ID: 202, Title: "Kitchen grill failure", Priority: 1})
    heap.Push(pq, &Ticket{ID: 203, Title: "VIP Table Reservation", Priority: 2})
    heap.Push(pq, &Ticket{ID: 204, Title: "Wipe table 8", Priority: 3})

    fmt.Println("Dispatching tickets by urgency:")
    for pq.Len() > 0 {
        ticket := heap.Pop(pq).(*Ticket)
        fmt.Printf("  [Priority %d] Ticket #%d: %s\n", ticket.Priority, ticket.ID, ticket.Ti
    }
}
```

Run:

```
go run desk_priority_queue.go
```

Output:

```
Dispatching tickets by urgency:
  [Priority 1] Ticket #202: Kitchen grill failure
  [Priority 2] Ticket #203: VIP Table Reservation
  [Priority 3] Ticket #204: Wipe table 8
  [Priority 4] Ticket #201: Restock register tape
```

The trap

The most prevalent bug with `container/heap` is calling the slice methods `pq.Push()` or `pq.Pop()` directly instead of the package-level functions `heap.Push(pq, x)` and `heap.Pop(pq)`. Calling `pq.Push()` appends without re-sifting; calling `heap.Push(pq, x)` invokes `pq.Push()` and then calls `siftUp()` to maintain the heap invariant.

```go
// bad_heap_call.go
package main

import (
    "container/heap"
    "fmt"
)

type IntHeap []int

func (h IntHeap) Len() int           { return len(h) }
func (h IntHeap) Less(i, j int) bool { return h[i] < h[j] }
func (h IntHeap) Swap(i, j int)      { h[i], h[j] = h[j], h[i] }
func (h *IntHeap) Push(x any)        { *h = append(*h, x.(int)) }
func (h *IntHeap) Pop() any {
    old := *h
    n := len(old)
    x := old[n-1]
    *h = old[0 : n-1]
    return x
}

func main() {
    h := &IntHeap{20, 30}
    heap.Init(h)

    // WRONG: Direct method call bypasses heap ordering
    h.Push(5)
```

```go
    fmt.Printf("After direct Push(5), root is still wrong: %d\n", (*h)[0])

    // RIGHT: heap.Push maintains invariant
    heap.Push(h, 2)
    fmt.Printf("After heap.Push(2), root is true minimum: %d\n", (*h)[0])
}
```

Run:

```go
go run bad_heap_call.go
```

Output:

```
After direct Push(5), root is still wrong: 20
After heap.Push(2), root is true minimum: 2
```

The boring rule

1. Use `container/heap` for production priority queues in Go. It avoids reinventing sift math and is heavily optimized.
2. Always call `heap.Push(pq, item)` and `heap.Pop(pq)` through the `heap` package, never on the type directly.
3. In `Pop()`, always assign `old[n-1] = nil` before truncating the slice to prevent memory leaks when holding pointer references.
4. If you need a Max-Heap instead of a Min-Heap, simply reverse the comparison in `Less(i, j)` (`return pq[i] > pq[j]`).

Try this

1. Implement `heap.Fix(pq, i)` in `desk_priority_queue.go` to upgrade an existing ticket's priority from 4 (routine) to 1 (critical) after it has already been pushed.
2. Use a Min-Heap of fixed size k to track the top-k most expensive orders in a continuous stream of desk transactions.

Tries and Prefix Trees

Tries and Prefix Trees

A **Trie** (derived from "re**trie**val"), also known as a **prefix tree**, is a specialized tree structure where each edge represents a character of a string key. All descendants of a node share a common string prefix. In Go, the boring default for string lookup is `map[string]T`, but a Trie shines when you need to match prefixes, validate autocomplete options, or perform IP routing lookups in $O(L)$ time where L is the string length, completely independent of how many millions of keys exist.

Mental model

Unlike a binary tree where nodes store whole strings, Trie nodes store transition characters. Shared prefixes share identical parent paths:

```
            (Root)
           /      \
         "c"      "t"
         /          \
       "a"          "e"
      /   \           \
    "r"   "t"          "a"
     |     |            |
  (cart) (cat)        (tea)
```

- Keys: `"cart"`, `"cat"`, `"tea"`
- Searching for `"car"`: Follow root → `'c'` → `'a'` → `'r'`. It exists as a prefix, but whether it is a complete word depends on the boolean flag `isTerminal`.
- Autocomplete: Walk down to the node matching the prefix (e.g. `'c'` → `'a'`), then gather all terminal descendants (`"cart"`, `"cat"`).

Search and insertion cost $O(L)$ operations where L is the length of the word, which is ideal for desk SKU catalogs with hundreds of thousands of items.

Worked examples

Case 1: Desk SKU and Tag Autocomplete Trie

Save as `trie_desk.go`. Implement a generic Trie capable of storing desk SKU metadata and providing prefix autocompletion.

```go
// trie_desk.go
package main

import (
    "fmt"
)

type TrieNode[T any] struct {
    children   map[rune]*TrieNode[T]
    isTerminal bool
    value      T
}

func newTrieNode[T any]() *TrieNode[T] {
    return &TrieNode[T]{
        children: make(map[rune]*TrieNode[T]),
    }
}

type Trie[T any] struct {
    root *TrieNode[T]
    size int
}

func NewTrie[T any]() *Trie[T] {
    return &Trie[T]{root: newTrieNode[T]()}
}

func (t *Trie[T]) Insert(key string, val T) {
    curr := t.root
    for _, ch := range key {
        if _, exists := curr.children[ch]; !exists {
            curr.children[ch] = newTrieNode[T]()
        }
        curr = curr.children[ch]
    }
    if !curr.isTerminal {
        t.size++
    }
    curr.isTerminal = true
    curr.value = val
}

func (t *Trie[T]) Search(key string) (T, bool) {
    curr := t.root
    for _, ch := range key {
        next, exists := curr.children[ch]
```

```go
        if !exists {
            var zero T
            return zero, false
        }
        curr = next
    }
    if curr.isTerminal {
        return curr.value, true
    }
    var zero T
    return zero, false
}

func (t *Trie[T]) StartsWith(prefix string) bool {
    curr := t.root
    for _, ch := range prefix {
        next, exists := curr.children[ch]
        if !exists {
            return false
        }
        curr = next
    }
    return true
}

func (t *Trie[T]) AutoComplete(prefix string) []string {
    curr := t.root
    for _, ch := range prefix {
        next, exists := curr.children[ch]
        if !exists {
            return nil
        }
        curr = next
    }

    var results []string
    collectWords(curr, prefix, &results)
    return results
}

func collectWords[T any](node *TrieNode[T], current string, results *[]string) {
    if node.isTerminal {
        *results = append(*results, current)
    }
    for ch, child := range node.children {
        collectWords(child, current+string(ch), results)
    }
```

```go
}

func main() {
    catalog := NewTrie[string]()

    catalog.Insert("DESK-CHAIR", "Ergonomic Mesh Chair")
    catalog.Insert("DESK-LAMP", "LED Adjustable Lamp")
    catalog.Insert("DESK-MAT", "Leather Desk Pad")
    catalog.Insert("DESK-DRAWER", "Under-desk organizer")
    catalog.Insert("DISH-TOWEL", "Barista cleanup towel")

    // Exact search
    sku := "DESK-LAMP"
    if desc, found := catalog.Search(sku); found {
        fmt.Printf("Exact SKU lookup [%s]: %s\n", sku, desc)
    }

    // Prefix check
    pfx := "DESK-"
    fmt.Printf("Any items starting with '%s'? %t\n", pfx, catalog.StartsWith(pfx))

    // Autocomplete recommendations
    suggestions := catalog.AutoComplete("DESK-")
    fmt.Println("\nAutocomplete options for 'DESK-':")
    for _, s := range suggestions {
        desc, _ := catalog.Search(s)
        fmt.Printf("  • %s (%s)\n", s, desc)
    }
}
```

Run:

```go
go run trie_desk.go
```

Output:

```
Exact SKU lookup [DESK-LAMP]: LED Adjustable Lamp
Any items starting with 'DESK-'? true

Autocomplete options for 'DESK-':
  • DESK-CHAIR (Ergonomic Mesh Chair)
  • DESK-LAMP (LED Adjustable Lamp)
  • DESK-MAT (Leather Desk Pad)
  • DESK-DRAWER (Under-desk organizer)
```

The trap

A common temptation in textbook Trie implementations is using a fixed ASCII array `[26]*TrieNode` for child pointers. While this avoids map allocations, it panics on uppercase letters, digits, dashes, and UTF-8 characters (`ch - 'a' < 0`). In Go, strings are UTF-8 by default; using `map[rune]*TrieNode` safely handles spaces, punctuation, dashes, and non-English characters without buffer overruns.

```go
// bad_ascii_trie.go
package main

import "fmt"

type BadNode struct {
    children [26]*BadNode // TRAP: Only supports lowercase a-z
}

func badInsert(root *BadNode, word string) {
    curr := root
    for _, ch := range word {
        // If word contains '-' or '0'..'9', index goes out of range!
        idx := ch - 'a'
        if idx < 0 || idx >= 26 {
            fmt.Printf("PANIC PREVENTED: rune '%c' is invalid for [26] array!\n", ch)
            return
        }
        if curr.children[idx] == nil {
            curr.children[idx] = &BadNode{}
        }
        curr = curr.children[idx]
    }
}

func main() {
    root := &BadNode{}
    badInsert(root, "DESK-101")
}
```

Run:

```go
go run bad_ascii_trie.go
```

Output:

```
PANIC PREVENTED: rune 'D' is invalid for [26] array!
```

The boring rule

1. If you only look up exact keys, use Go's built-in `map[string]V`. A hash map is faster and uses less memory than a Trie for exact matches.
2. Use a Trie when the requirements demand **prefix queries**: `StartsWith`, auto-complete, longest-prefix routing, or spell checking.
3. Use `map[rune]*TrieNode` in Go to safely handle Unicode runes and symbols.
4. If memory is tight and paths have long unbranching chains, consider a Radix Tree (compact Trie) where common non-branching edges are merged into single strings.

Try this

1. Add a `Delete(key string) bool` method to `Trie` that removes the terminal marker and prunes child nodes that have no other descendants.
2. Build an order tag filter where user input `"veg"` automatically expands to both `"vegan"` and `"vegetarian"`.

Sorting Algorithms

Sorting Algorithms

Sorting arranges data in a predictable order so that searches, aggregations, and deduplications can execute in logarithmic time. In modern Go (1.21+), the boring default is `slices.Sort` and `slices.SortFunc`, which implement an adaptive pattern-defeating quicksort (pdqsort). Understanding the classical algorithms—**Insertion Sort**, **Merge Sort**, and **Quicksort**—explains why standard libraries choose hybrid approaches and how to avoid catastrophic worst-case traps.

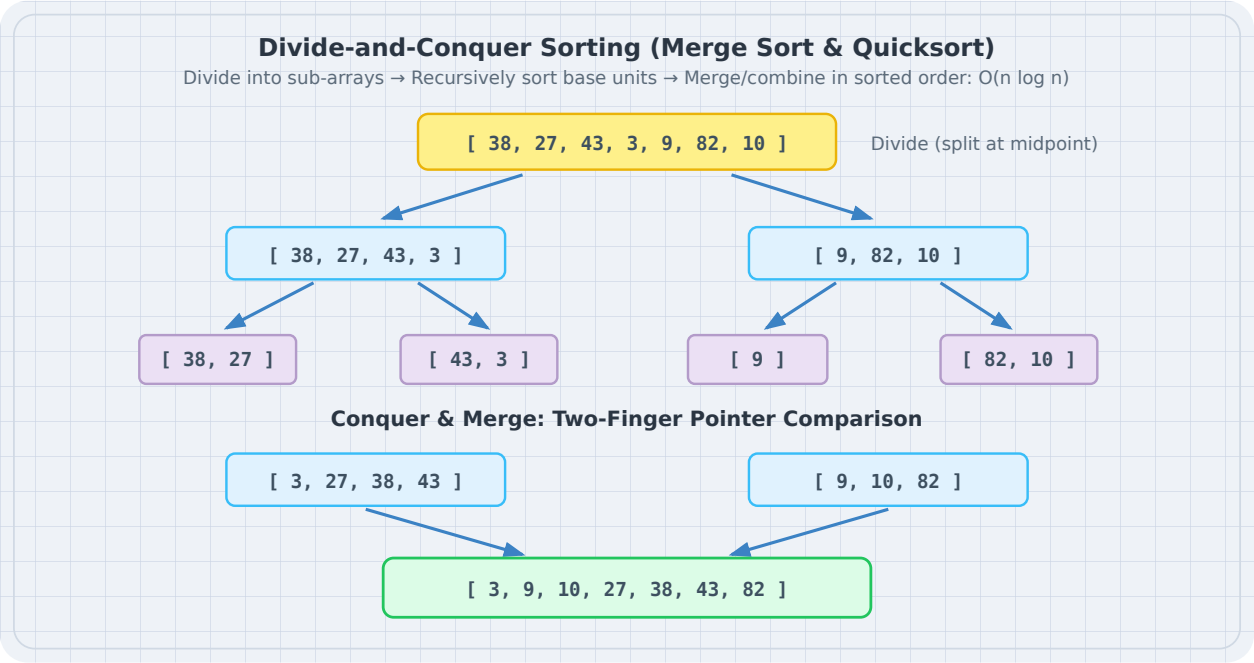


Figure 1: Divide-and-Conquer Sorting

Mental model

Different sorting algorithms optimize for different constraints:

Algorithm	Time (Best)	Time (Avg)	Time (Worst)	Memory	Stable?	Best Used When
Insertion Sort	$O(n)$	$O(n^2)$	$O(n^2)$	$O(1)$	Yes	Small slices ($n < 24$) or nearly-sorted data

Algorithm	Time (Best)	Time (Avg)	Time (Worst)	Memory	Stable?	Best Used When
Merge Sort	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$	$O(n)$	Yes	Guaranteed predictable time and stable equality
Quicksort	$O(n \log n)$	$O(n \log n)$	$O(n^2)$	$O(\log n)$	No	General purpose, fast in-place cache locality
Go `slices.Sort`	$O(n)$	$O(n \log n)$	$O(n \log n)$	$O(\log n)$	No	Production default (pdqsort: hybrid quick+insertion+heap)

Insertion Sort

Works like sorting cards in hand: pick element i and slide it backward into its sorted position among earlier cards.

Merge Sort

Divides the slice into two halves, recursively sorts them, then merges the two sorted lists using two pointer comparisons into an auxiliary buffer.

Quicksort

Picks a **pivot** element, partitions the slice so that all elements $\leq$ *pivot* move to the left and all elements $>$ *pivot* move to the right, then recursively sorts both partitions in place.

```
Quicksort Partition step (Pivot: 25)
[ 14, 38, 10, 42, 25 ]
               ^ Pivot
After Partition:
[ 14, 10 ]  [ 25 ]  [ 38, 42 ]
 (<= 25)     Pivot     (> 25)
```

Worked examples

Case 1: Insertion Sort for Small/Nearly Sorted Batches

Save as `insertion_sort.go`. Implement generic insertion sort and demonstrate its linear speed on nearly sorted order streams.

```go
// insertion_sort.go
package main

import (
    "cmp"
    "fmt"
)

func InsertionSort[T cmp.Ordered](items []T) {
    n := len(items)
    for i := 1; i < n; i++ {
        key := items[i]
        j := i - 1
        // Shift elements greater than key one position right
        for j >= 0 && items[j] > key {
            items[j+1] = items[j]
            j--
        }
        items[j+1] = key
    }
}

type DeskOrder struct {
    ID    int
    Table int
}

func main() {
    // Nearly sorted tickets (common when late arrivals append to queue)
    tickets := []int{101, 102, 105, 103, 104, 106}
    fmt.Println("Before InsertionSort:", tickets)

    InsertionSort(tickets)
    fmt.Println("After InsertionSort: ", tickets)
}
```

Run:

```go
go run insertion_sort.go
```

Output:

```
Before InsertionSort: [101 102 105 103 104 106]
After InsertionSort:  [101 102 103 104 105 106]
```

Case 2: Stable Merge Sort

Save as `merge_sort.go`. Implement generic divide-and-conquer merge sort.

```go
// merge_sort.go
package main

import (
    "cmp"
    "fmt"
)

func MergeSort[T cmp.Ordered](items []T) []T {
    if len(items) <= 1 {
        return items
    }

    mid := len(items) / 2
    left := MergeSort(items[:mid])
    right := MergeSort(items[mid:])

    return merge(left, right)
}

func merge[T cmp.Ordered](left, right []T) []T {
    result := make([]T, 0, len(left)+len(right))
    i, j := 0, 0

    for i < len(left) && j < len(right) {
        if left[i] <= right[j] { // <= preserves stability
            result = append(result, left[i])
            i++
        } else {
            result = append(result, right[j])
            j++
        }
    }

    // Append leftovers
    result = append(result, left[i:]...)
    result = append(result, right[j:]...)
    return result
}
```

```go
func main() {
    prices := []int{45, 12, 85, 32, 89, 39, 69, 44, 42, 1, 45}
    fmt.Println("Unsorted prices:", prices)

    sorted := MergeSort(prices)
    fmt.Println("Sorted prices:  ", sorted)
}
```

Run:

```go
go run merge_sort.go
```

Output:

```
Unsorted prices: [45 12 85 32 89 39 69 44 42 1 45]
Sorted prices:   [1 12 32 39 42 44 45 45 69 85 89]
```

Case 3: In-Place Quicksort with Lomuto Partition

Save as `quick_sort.go`. Implement in-place quicksort.

```go
// quick_sort.go
package main

import (
    "cmp"
    "fmt"
)

func QuickSort[T cmp.Ordered](items []T) {
    if len(items) <= 1 {
        return
    }
    quickSortHelper(items, 0, len(items)-1)
}

func quickSortHelper[T cmp.Ordered](items []T, low, high int) {
    if low < high {
        p := partition(items, low, high)
        quickSortHelper(items, low, p-1)
        quickSortHelper(items, p+1, high)
    }
}

func partition[T cmp.Ordered](items []T, low, high int) int {
    // Pick median or last element as pivot
```

```go
    pivot := items[high]
    i := low

    for j := low; j < high; j++ {
        if items[j] < pivot {
            items[i], items[j] = items[j], items[i]
            i++
        }
    }
    items[i], items[high] = items[high], items[i]
    return i
}

func main() {
    orderIDs := []int{84, 12, 91, 45, 33, 72, 16, 50}
    fmt.Println("Before QuickSort:", orderIDs)

    QuickSort(orderIDs)
    fmt.Println("After QuickSort: ", orderIDs)
}
```

Run:

```go
go run quick_sort.go
```

Output:

```
Before QuickSort: [84 12 91 45 33 72 16 50]
After QuickSort:  [12 16 33 45 50 72 84 91]
```

The trap

The naive Quicksort trap is picking the first or last element as pivot without randomization or median-of-three sampling. When an array is already sorted, choosing the last element produces partitions of sizes $n - 1$ and 0. The recursion depth balloons from $\log n$ to n, causing $O(n^2)$ execution time and stack overflow on slices with 100,000 items.

This is why Go standard library's `slices.Sort` uses pattern-defeating quicksort, automatically falling back to HeapSort if recursion depth exceeds $2 \log_2 n$.

The boring rule

1. In real Go production code, always use `slices.Sort(s)` for primitive values and `slices.SortFunc(s, cmpFunc)` for structs.

2. Never write a bespoke sorting algorithm unless you are operating under tight embedded memory constraints where zero stack frame allocation is mandatory.
3. If your data is guaranteed to be nearly sorted (e.g. periodically inserting 2 items into a list of 5,000 sorted records), Insertion Sort runs in $O(n)$ time and avoids quicksort setup overhead.
4. If you need a **stable** sort (where elements with equal keys retain their relative order), use `sort.SliceStable` or Merge Sort. Standard `slices.Sort` is not stable.

Try this

1. Benchmark `InsertionSort` against `slices.Sort` on small slices of length $n = 8, 16, 32$. At what threshold does `slices.Sort` pull ahead?
2. Implement median-of-three pivot selection in `quick_sort.go` (comparing `low`, `mid`, and `high`) to neutralize the sorted array trap.

Binary Search and Selection

Binary Search and Selection

Binary search locates elements in a sorted collection by halving the search space on each step, finding targets in $O(\log n)$ time. Selection algorithms (such as **Quickselect**) find the k-th smallest element (such as the median or 95th percentile latency) in $O(n)$ average time without paying the full $O(n \log n)$ cost to sort the entire dataset. The boring default in Go is `slices.BinarySearch`, but understanding partition boundaries is essential for range searches and percentile metrics.

Mental model

Binary Search Halving

On an array of 1,000,000 sorted elements, linear search takes up to 1,000,000 checks. Binary search takes at most 20 checks ($\log_2 1000000 \approx 19.9$).

```
Target: 38
[ 10, 14, 19, 26, 31, 38, 42, 55, 70 ]  Mid = idx 4 (31). 38 > 31 -> Search Right
                      [ 38, 42, 55, 70 ]  Mid = idx 6 (42). 38 < 42 -> Search Left
                      [ 38 ]              Mid = idx 5 (38). Match found!
```

Quickselect vs Full Sort

If a desk manager wants to find the **median ticket resolution time** in an array of 500,000 tickets:

- **Sorting then indexing:** $O(n \log n) \approx 9{,}500{,}000$ operations.
- **Quickselect:** Discards half the partitions on each step $(n + n/2 + n/4 + \cdots \approx 2n) \to O(n) \approx$ 1,000,000 operations.

```
Quickselect (Finding 3rd smallest element):
Partition around Pivot P:
[ Left: < P ]  [ P ]  [ Right: > P ]
       \
   Pivot lands at index 5. We need index 3 -> Recurse ONLY into Left partition!
   (Right partition is completely ignored!)
```

Worked examples

Case 1: Binary Search with Insertion Point Bounds

Save as `binary_search_bounds.go`. Implement binary search that finds the exact match or the insertion index where a new ticket should be inserted to preserve sorted order.

```go
// binary_search_bounds.go
package main

import (
    "cmp"
    "fmt"
    "slices"
)

// LowerBound returns the first index where items[i] >= target.
func LowerBound[T cmp.Ordered](items []T, target T) int {
    low := 0
    high := len(items)

    for low < high {
        // Prevent integer overflow: low + (high-low)/2 instead of (low+high)/2
        mid := low + (high-low)/2
        if items[mid] < target {
            low = mid + 1
        } else {
            high = mid
        }
    }
    return low
}

func main() {
    // Sorted desk order arrival timestamps (in minutes from shift start)
    timestamps := []int{5, 12, 18, 24, 24, 24, 39, 55}
    fmt.Println("Sorted arrivals:", timestamps)

    // Custom LowerBound
    idx := LowerBound(timestamps, 24)
    fmt.Printf("First arrival >= 24 min: index %d (value: %d)\n", idx, timestamps[idx])

    // New arrival at minute 30
    insertIdx := LowerBound(timestamps, 30)
    fmt.Printf("Slot to insert arrival at min 30: index %d\n", insertIdx)

    // Go standard library slices.BinarySearch
```

```go
    pos, found := slices.BinarySearch(timestamps, 39)
    fmt.Printf("slices.BinarySearch(39): found=%t at index %d\n", found, pos)

    posMissing, foundMissing := slices.BinarySearch(timestamps, 30)
    fmt.Printf("slices.BinarySearch(30): found=%t, insert position=%d\n", foundMissing, posM
}
```

Run:

```go
go run binary_search_bounds.go
```

Output:

```
Sorted arrivals: [5 12 18 24 24 24 39 55]
First arrival >= 24 min: index 3 (value: 24)
Slot to insert arrival at min 30: index 6
slices.BinarySearch(39): found=true at index 6
slices.BinarySearch(30): found=false, insert position=6
```

Case 2: Quickselect for Median and Percentile Calculation

Save as `quickselect_desk.go`. Find the median ticket resolution duration in $O(n)$ average time.

```go
// quickselect_desk.go
package main

import (
    "cmp"
    "fmt"
)

// Quickselect finds the k-th smallest element (0-indexed) in items.
// Mutates items in-place.
func Quickselect[T cmp.Ordered](items []T, k int) T {
    low := 0
    high := len(items) - 1

    for low <= high {
        pivotIdx := partition(items, low, high)
        if pivotIdx == k {
            return items[k]
        } else if pivotIdx < k {
            low = pivotIdx + 1
        } else {
            high = pivotIdx - 1
        }
```

```go
    }
    return items[k]
}

func partition[T cmp.Ordered](items []T, low, high int) int {
    pivot := items[high]
    i := low
    for j := low; j < high; j++ {
        if items[j] < pivot {
            items[i], items[j] = items[j], items[i]
            i++
        }
    }
    items[i], items[high] = items[high], items[i]
    return i
}

func main() {
    // Ticket resolution durations in seconds
    durations := []int{320, 120, 450, 90, 600, 210, 180, 510, 290}
    fmt.Println("Raw durations:", durations)

    n := len(durations)
    medianIdx := n / 2 // Index 4 in 9-element slice

    medianVal := Quickselect(durations, medianIdx)
    fmt.Printf("Median resolution time: %d seconds (50th percentile)\n", medianVal)

    p90Idx := int(float64(n) * 0.9)
    p90Val := Quickselect(durations, p90Idx)
    fmt.Printf("90th percentile resolution time: %d seconds\n", p90Val)
}
```

Run:

```
go run quickselect_desk.go
```

Output:

```
Raw durations: [320 120 450 90 600 210 180 510 290]
Median resolution time: 290 seconds (50th percentile)
90th percentile resolution time: 600 seconds
```

The trap

The most famous bug in computer science history was in binary search: calculating `mid = (low + high) / 2`. In 32-bit systems or large datasets, `low + high` can overflow signed integer bounds into negative values, resulting in an index-out-of-bounds panic. The safe formula is always `mid = low + (high-low)/2` or `uint(low+high) >> 1`.

Furthermore, binary search **assumes the slice is already sorted**. Calling binary search on an unsorted slice fails silently, returning wrong indices and false negatives.

The boring rule

1. Always use `slices.BinarySearch` or `slices.BinarySearchFunc` in production Go. It handles boundaries, returns the insertion position if missing, and is thoroughly tested.
2. Verify that your data is sorted before initiating binary search.
3. If you need the median, minimum, maximum, or arbitrary percentile of a large slice and do not need the rest of the slice sorted, use Quickselect instead of sorting the whole slice.
4. When calculating midpoints, always use `low + (high - low)/2`.

Try this

1. Use `slices.BinarySearchFunc` on a slice of `DeskTicket` structs to find tickets by customer email.
2. Implement a `CountOccurrences(items []T, target T) int` function that uses two binary searches (`LowerBound` and `UpperBound`) to count matches in $O(\log n)$ time.

Graphs and Traversals (BFS and DFS)

Graphs and Traversals (BFS and DFS)

A **graph** represents relationships between entities as a set of **vertices** (nodes) connected by **edges**. Graphs model service dependencies, order routing networks, workflow pipelines, and social graphs. In Go, the boring default is an **adjacency list** represented using maps and slices. Traversing graphs via **Breadth-First Search (BFS)** and **Depth-First Search (DFS)** enables finding shortest hops, detecting cycles, and discovering connected components.

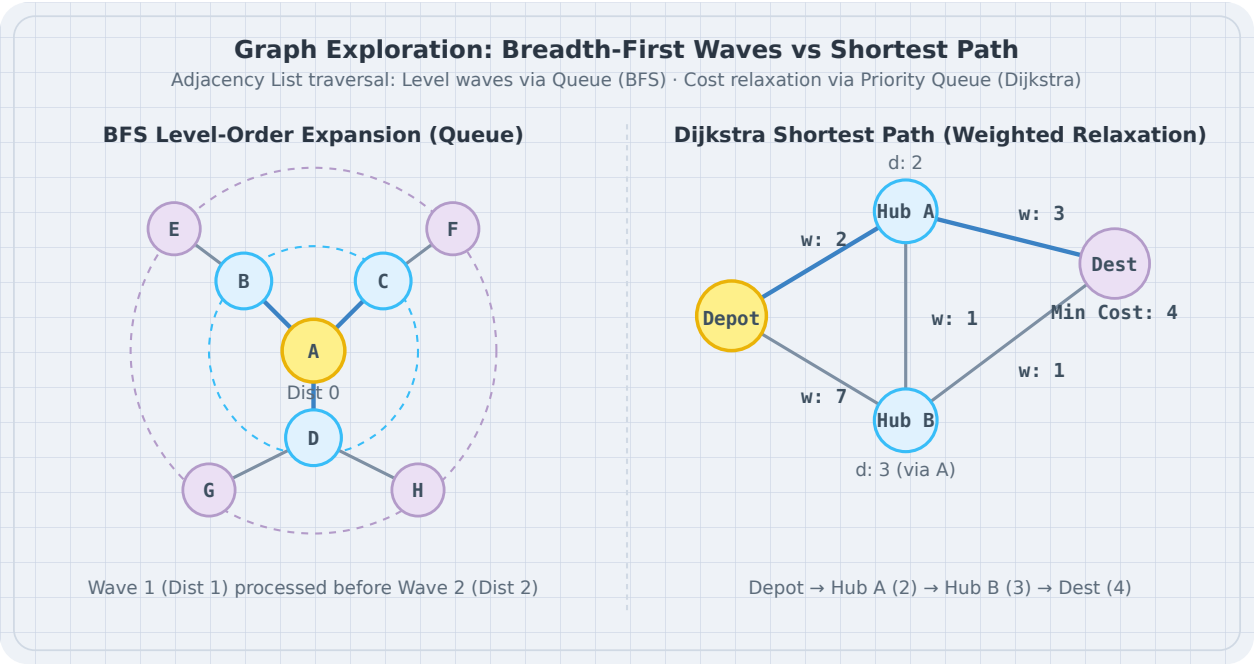


Figure 1: Graph Exploration: BFS and Dijkstra

Mental model

Graph Representation: Adjacency List

An **adjacency matrix** allocates an $V \times V$ grid, consuming $O(V^2)$ memory even if most nodes have few connections. An **adjacency list** stores only existing edges, consuming $O(V + E)$ memory:

```
Adjacency List:
[Depot A] -> [Hub B, Hub C]
[Hub B]   -> [Depot A, Station D]
[Hub C]   -> [Depot A, Station D]
[Station D] -> [Hub B, Hub C]
```

BFS vs DFS Traversals

- **Breadth-First Search (BFS):** Uses a **FIFO Queue**. It expands outward in concentric rings (all distance-1 neighbors first, then all distance-2 neighbors). **Guarantees the shortest path** in an unweighted graph.
- **Depth-First Search (DFS):** Uses a **Stack** (or call stack recursion). It plunges as deep as possible along each branch before backtracking. Ideal for topological sorting, maze solving, and cycle detection.

```
BFS Waves (Queue):            DFS Dive (Stack / Recursion):
       (A)                          (A)
     /  |  \                       /
   (B) (C) (D)  <- Wave 1        (B)
    |                             |
   (E)          <- Wave 2        (E) <- Dive deep before (C)
```

Worked examples

Case 1: Generic Graph with Adjacency List and BFS Shortest Hop

Save as `graph_bfs.go`. Model a fulfillment routing network and find the shortest hops from central depot to satellite hubs.

```go
// graph_bfs.go
package main

import (
    "fmt"
)

type Graph[T comparable] struct {
    adj map[T][]T
}

func NewGraph[T comparable]() *Graph[T] {
    return &Graph[T]{adj: make(map[T][]T)}
}

func (g *Graph[T]) AddEdge(u, v T, bidirectional bool) {
    g.adj[u] = append(g.adj[u], v)
    if bidirectional {
        g.adj[v] = append(g.adj[v], u)
    }
}

// BFS returns the shortest path (in number of edges) from start to target.
```

```go
func (g *Graph[T]) ShortestPathBFS(start, target T) []T {
    if start == target {
        return []T{start}
    }

    visited := make(map[T]bool)
    parent := make(map[T]T)
    queue := []T{start}
    visited[start] = true

    found := false
    for len(queue) > 0 {
        curr := queue[0]
        queue = queue[1:]

        if curr == target {
            found = true
            break
        }

        for _, neighbor := range g.adj[curr] {
            if !visited[neighbor] {
                visited[neighbor] = true
                parent[neighbor] = curr
                queue = append(queue, neighbor)
            }
        }
    }

    if !found {
        return nil
    }

    // Reconstruct path backward from target to start
    var path []T
    curr := target
    for {
        path = append(path, curr)
        if curr == start {
            break
        }
        curr = parent[curr]
    }

    // Reverse path
    for i, j := 0, len(path)-1; i < j; i, j = i+1, j-1 {
```

```go
        path[i], path[j] = path[j], path[i]
    }
    return path
}

func main() {
    network := NewGraph[string]()
    network.AddEdge("Central-Depot", "North-Hub", true)
    network.AddEdge("Central-Depot", "South-Hub", true)
    network.AddEdge("North-Hub", "Station-East", true)
    network.AddEdge("South-Hub", "Station-West", true)
    network.AddEdge("Station-East", "Terminal-Z", true)
    network.AddEdge("Station-West", "Terminal-Z", true)

    start := "Central-Depot"
    dest := "Terminal-Z"

    path := network.ShortestPathBFS(start, dest)
    fmt.Printf("BFS Shortest Route (%d hops):\n", len(path)-1)
    for i, stop := range path {
        if i > 0 {
            fmt.Print(" -> ")
        }
        fmt.Print(stop)
    }
    fmt.Println()
}
```

Run:

```go
go run graph_bfs.go
```

Output:

```
BFS Shortest Route (3 hops):
Central-Depot -> North-Hub -> Station-East -> Terminal-Z
```

Case 2: DFS for Workflow Cycle Detection

Save as `graph_dfs_cycle.go`. Check if a task dependency graph contains illegal circular dependencies before dispatching jobs.

```go
// graph_dfs_cycle.go
package main

import "fmt"
```

```go
type TaskGraph struct {
    adj map[string][]string
}

func NewTaskGraph() *TaskGraph {
    return &TaskGraph{adj: make(map[string][]string)}
}

func (g *TaskGraph) AddDependency(task, dependsOn string) {
    // task -> dependsOn
    g.adj[task] = append(g.adj[task], dependsOn)
}

// HasCycle uses 3-color DFS:
// 0: Unvisited, 1: Currently Visiting (in call stack), 2: Fully Visited
func (g *TaskGraph) HasCycle() bool {
    state := make(map[string]int)

    var dfs func(task string) bool
    dfs = func(u string) bool {
        state[u] = 1 // Mark visiting (in progress)

        for _, v := range g.adj[u] {
            if state[v] == 1 {
                // Found back-edge to an ancestor currently in recursion stack!
                return true
            }
            if state[v] == 0 {
                if dfs(v) {
                    return true
                }
            }
        }

        state[u] = 2 // Mark fully visited
        return false
    }

    for task := range g.adj {
        if state[task] == 0 {
            if dfs(task) {
                return true
            }
        }
    }
    return false
```

```go
}

func main() {
    g := NewTaskGraph()
    g.AddDependency("Print-Receipt", "Verify-Payment")
    g.AddDependency("Verify-Payment", "Calculate-Tax")
    g.AddDependency("Calculate-Tax", "Read-Cart")

    fmt.Printf("Initial linear pipeline has cycle? %t\n", g.HasCycle())

    // Introduce circular deadlock: Read-Cart requires Print-Receipt
    fmt.Println("Introducing circular dependency: Read-Cart -> Print-Receipt")
    g.AddDependency("Read-Cart", "Print-Receipt")
    fmt.Printf("Updated pipeline has cycle? %t\n", g.HasCycle())
}
```

Run:

```go run graph_dfs_cycle.go
```

Output:

```
Initial linear pipeline has cycle? false
Introducing circular dependency: Read-Cart -> Print-Receipt
Updated pipeline has cycle? true
```

The trap

The most dangerous graph bug is traversing cyclic graphs without a `visited` set. If vertex A points to B and B points back to A, an unguided recursive DFS will loop infinitely until the runtime exhausts goroutine stack memory (`fatal error: stack overflow`). Always maintain a `visited[node] = true` map or bitset.

The boring rule

1. Use an adjacency list (`map[K][]K`) by default. Graphs in business services are almost always sparse.
2. If you need the shortest path in an **unweighted** graph, always use **BFS** with a FIFO queue.
3. If you need to detect cycles or topological ordering in directed pipelines, use **DFS** with a 3-color state tracking map.
4. When reconstructing paths, store `parent[child] = curr` during traversal rather than copying slice paths inside the queue.

Try this

1. Implement topological sorting (`TopologicalSort() []string`) using DFS post-order traversal to produce a valid execution schedule for non-cyclic desk workflows.
2. Enhance `ShortestPathBFS` to return all connected clusters in an unpartitioned desk routing network.

Shortest Paths (Dijkstra's Algorithm)

Shortest Paths (Dijkstra's Algorithm)

While Breadth-First Search finds the path with the fewest hops, real systems have weighted costs: network latencies, courier miles, fuel consumption, or processing fees. **Dijkstra's algorithm** computes the single-source shortest path across a weighted graph with non-negative edge weights in $O((V + E) \log V)$ time using a **min-priority queue**.

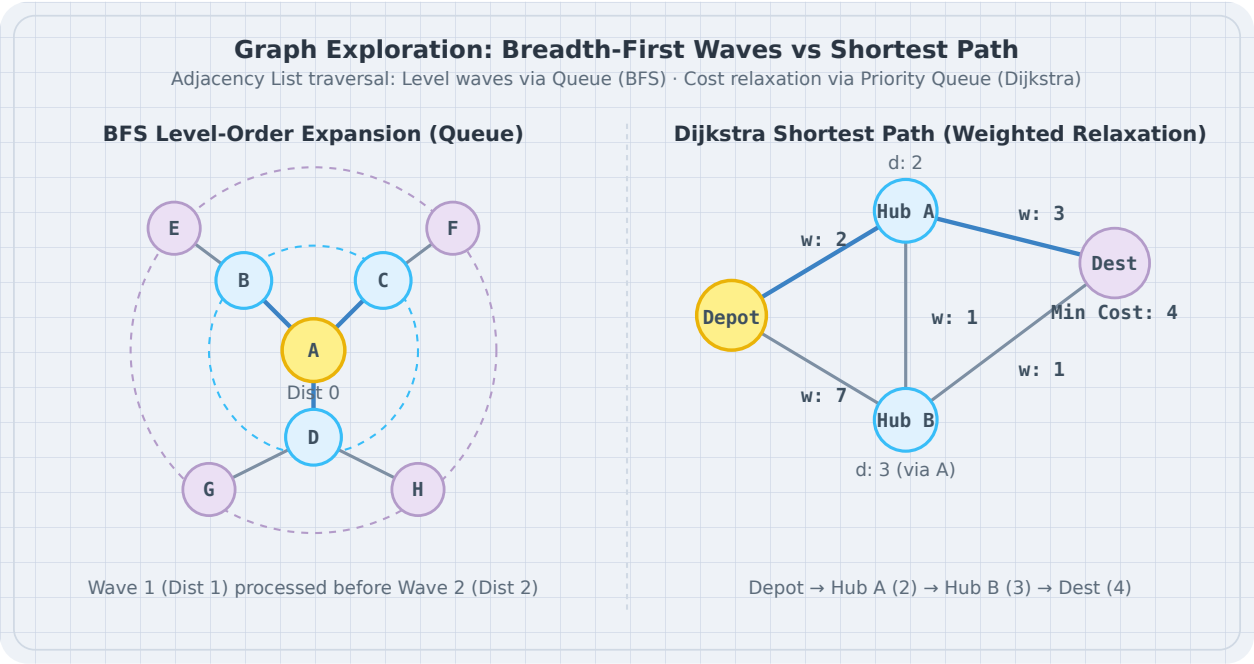


Figure 1: Graph Traversal and Dijkstra Relaxation

Mental model

Dijkstra maintains two maps: 1. `dist[v]`: The shortest known distance from the starting source to vertex v (initialized to ∞, with `dist[start] = 0`). 2. `prev[v]`: The predecessor vertex on the optimal path.

At each step, the algorithm extracts the unvisited vertex u with the smallest known distance from a min-priority queue and **relaxes** all outgoing edges (u, v, weight):

$$\text{If } \text{dist}[u] + \text{weight} < \text{dist}[v] \implies \text{dist}[v] = \text{dist}[u] + \text{weight}$$

```
Relaxation Step:
(Depot) --[Known: 2]--> (Hub A) --[Weight: 3]--> (Dest)
                            \                        ^
                             \--[Weight: 1]-->(Hub B)-[Weight: 1]-/

Direct path: Depot -> Hub A -> Dest = 2 + 3 = 5
Relaxed path: Depot -> Hub A -> Hub B -> Dest = 2 + 1 + 1 = 4 (Optimal!)
```

Because edge weights are non-negative, the moment a vertex is popped from the min-priority queue, its calculated distance is permanently optimal.

Worked examples

Case 1: Complete Dijkstra implementation with path reconstruction

Save as `dijkstra_delivery.go`. Calculate the lowest-cost delivery route between warehouse dispatch points.

```go
// dijkstra_delivery.go
package main

import (
    "container/heap"
    "fmt"
    "math"
)

type Edge struct {
    To     string
    Weight int
}

type WeightedGraph struct {
    adj map[string][]Edge
}

func NewWeightedGraph() *WeightedGraph {
    return &WeightedGraph{adj: make(map[string][]Edge)}
}

func (g *WeightedGraph) AddEdge(from, to string, weight int) {
    g.adj[from] = append(g.adj[from], Edge{To: to, Weight: weight})
    g.adj[to] = append(g.adj[to], Edge{To: from, Weight: weight}) // Undirected
}

// Priority Queue item
```

```go
type Item struct {
    Node string
    Dist int
}

type PriorityQueue []Item

func (pq PriorityQueue) Len() int           { return len(pq) }
func (pq PriorityQueue) Less(i, j int) bool { return pq[i].Dist < pq[j].Dist } // Min-heap
func (pq PriorityQueue) Swap(i, j int)      { pq[i], pq[j] = pq[j], pq[i] }
func (pq *PriorityQueue) Push(x any)        { *pq = append(*pq, x.(Item)) }
func (pq *PriorityQueue) Pop() any {
    old := *pq
    n := len(old)
    x := old[n-1]
    *pq = old[0 : n-1]
    return x
}

type ShortestPathResult struct {
    Distance int
    Path     []string
}

func (g *WeightedGraph) Dijkstra(start, dest string) (ShortestPathResult, bool) {
    dist := make(map[string]int)
    prev := make(map[string]string)

    for node := range g.adj {
        dist[node] = math.MaxInt
    }
    dist[start] = 0

    pq := &PriorityQueue{}
    heap.Init(pq)
    heap.Push(pq, Item{Node: start, Dist: 0})

    for pq.Len() > 0 {
        top := heap.Pop(pq).(Item)
        u := top.Node
        currDist := top.Dist

        // Stale item check: if we found a shorter route to u already, skip
        if currDist > dist[u] {
            continue
        }
```

```go
        if u == dest {
            break // Reached destination with optimal distance
        }

        for _, edge := range g.adj[u] {
            v := edge.To
            newDist := currDist + edge.Weight
            if newDist < dist[v] {
                dist[v] = newDist
                prev[v] = u
                heap.Push(pq, Item{Node: v, Dist: newDist})
            }
        }
    }

    if dist[dest] == math.MaxInt {
        return ShortestPathResult{}, false // Unreachable
    }

    // Reconstruct path
    var path []string
    curr := dest
    for curr != "" {
        path = append(path, curr)
        curr = prev[curr]
    }

    // Reverse path
    for i, j := 0, len(path)-1; i < j; i, j = i+1, j-1 {
        path[i], path[j] = path[j], path[i]
    }

    return ShortestPathResult{Distance: dist[dest], Path: path}, true
}

func main() {
    g := NewWeightedGraph()
    g.AddEdge("Main-Depot", "North-Hub", 4)
    g.AddEdge("Main-Depot", "South-Hub", 2)
    g.AddEdge("South-Hub", "North-Hub", 1)
    g.AddEdge("North-Hub", "Airport-Station", 7)
    g.AddEdge("South-Hub", "Midtown-Annex", 8)
    g.AddEdge("North-Hub", "Midtown-Annex", 3)
    g.AddEdge("Midtown-Annex", "Airport-Station", 1)

    res, found := g.Dijkstra("Main-Depot", "Airport-Station")
    if !found {
```

```go
        fmt.Println("No route found")
        return
    }

    fmt.Printf("Lowest Delivery Latency: %d ms\n", res.Distance)
    fmt.Print("Optimal Route: ")
    for i, step := range res.Path {
        if i > 0 {
            fmt.Print(" -> ")
        }
        fmt.Print(step)
    }
    fmt.Println()
}
```

Run:

```go
go run dijkstra_delivery.go
```

Output:

```
Lowest Delivery Latency: 7 ms
Optimal Route: Main-Depot -> South-Hub -> North-Hub -> Midtown-Annex -> Airport-Station
```

Notice that although `Main-Depot -> North-Hub -> Airport-Station` has fewer hops (2 hops, cost $4 + 7 = 11$), Dijkstra discovers the 4-hop route via `South-Hub` and `Midtown-Annex` with total cost $2 + 1 + 3 + 1 = 7$!

The trap

There are two primary traps when implementing Dijkstra's algorithm:

1. **Negative Edge Weights:** Dijkstra assumes that adding an edge can never decrease the total path cost. If a graph has negative edge weights, Dijkstra produces wrong answers or loops indefinitely. Graphs with negative weights require the **Bellman-Ford algorithm**.
2. **Stale Priority Queue Entries:** In standard Go implementations, updating a node's distance is achieved by pushing a new `Item{Node: v, Dist: newDist}` into the heap. Without the check `if currDist > dist[u] { continue }`, the algorithm wastes time re-exploring superseded paths, degrading performance.

The boring rule

1. Use BFS for unweighted shortest path queries. Use Dijkstra when edges represent costs, times, distances, or latency.

2. Always guard the main loop with `if currDist > dist[u] { continue }` to drop stale heap items instantly.
3. Validate that all edge weights are ≥ 0.
4. Stop the algorithm early when the target node is popped from the priority queue; its distance is provably finalized.

Try this

1. Modify `Dijkstra` to return all reachable destinations and their shortest paths from `start` in a single run.
2. Add a `MaxStops` parameter to restrict the search to routes with at most 3 intermediary transfers.

String Search and Bloom Filters

String Search and Bloom Filters

Substring matching and set membership testing are ubiquitous in log processors, audit pipelines, and API gateways. When searching within large texts or verifying whether a customer record exists across millions of database rows, naive brute-force checks waste CPU and disk I/O. The boring default in Go is `strings.Contains` and in-memory maps, but **Rabin-Karp rolling hashing** and **Bloom Filters** provide powerful probabilistic and algorithmic speedups.

Mental model

Rabin-Karp Substring Matching

Instead of comparing substrings character-by-character at every index ($O(n \times m)$), Rabin-Karp computes a **polynomial rolling hash** of the pattern. Sliding the search window by one character subtracts the leaving character and adds the entering character in $O(1)$ arithmetic:

```
Text:    [ B O R I N G ] - G O
Window:  [ B O R ]       Hash = 1042 (Pattern hash: 9811 -> Mismatch)
Slide:     [ O R I ]     Hash = (1042 - 'B'*base) * base + 'I' in O(1)
```

Bloom Filter: Probabilistic Membership

A Bloom filter is a space-efficient bit array paired with k independent hash functions:

```
Bit Array (m bits):
Index:  [ 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 ]
Bits:   [ 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 ]
              ^       ^           ^
              h1(ID)  h2(ID)      h3(ID)
```

1. **Insert:** Compute k hashes for an item and set those bit positions to `1`.
2. **Query:** Compute k hashes.
   - If **any** bit is `0`: Item is **definitely NOT** in the set (0% False Negatives).
   - If **all** bits are `1`: Item is **probably** in the set (tunable False Positive rate).

This allows web services to reject 99% of invalid requests in memory before touching expensive disk databases.

Worked examples

Case 1: Rabin-Karp Rolling Hash Substring Search

Save as `rabin_karp.go`. Search for audit keywords across order logs in linear time.

```go
// rabin_karp.go
package main

import (
    "fmt"
)

const (
    primeBase = 256
    primeMod  = 1000000007
)

func RabinKarpSearch(text, pattern string) int {
    n := len(text)
    m := len(pattern)
    if m == 0 || n < m {
        return -1
    }

    // Calculate h = primeBase^(m-1) % primeMod
    h := 1
    for i := 0; i < m-1; i++ {
        h = (h * primeBase) % primeMod
    }

    patternHash := 0
    windowHash := 0

    // Initial hash for pattern and first window
    for i := 0; i < m; i++ {
        patternHash = (primeBase*patternHash + int(pattern[i])) % primeMod
        windowHash = (primeBase*windowHash + int(text[i])) % primeMod
    }

    // Slide the pattern over text
    for i := 0; i <= n-m; i++ {
        if patternHash == windowHash {
            // Hash match: confirm characters (handles collisions)
            if text[i:i+m] == pattern {
                return i
            }
```

```go
        }

        if i < n-m {
            // Rolling hash update: drop left char, shift, add right char
            windowHash = (primeBase*(windowHash-int(text[i])*h) + int(text[i+m])) % primeMod
            if windowHash < 0 {
                windowHash += primeMod
            }
        }
    }

    return -1
}

func main() {
    log := "2026-09-18T10:14:02Z [INFO] Desk order #8192 completed with STATUS=REFUND_APPROV
    target := "STATUS=REFUND_APPROVED"

    idx := RabinKarpSearch(log, target)
    if idx != -1 {
        fmt.Printf("Keyword match found at byte offset %d\n", idx)
        fmt.Printf("Snippet: '%s'\n", log[idx:idx+len(target)])
    } else {
        fmt.Println("Pattern not found")
    }
}
```

Run:

```go run rabin_karp.go
```

Output:

```
Keyword match found at byte offset 53
Snippet: 'STATUS=REFUND_APPROVED'
```

Case 2: In-Memory Bloom Filter for Fast Desk Order Checks

Save as `bloom_filter.go`. Implement a compact bitset Bloom filter using Go bitwise operations and FNV hashing with different seeds.

```go
// bloom_filter.go
package main

import (
    "fmt"
```

```go
    "hash/fnv"
)

type BloomFilter struct {
    bits     []uint64
    numBits  uint
    numHash  uint
}

func NewBloomFilter(sizeInBits, numHashes uint) *BloomFilter {
    // 64 bits per uint64 word
    words := (sizeInBits + 63) / 64
    return &BloomFilter{
        bits:    make([]uint64, words),
        numBits: sizeInBits,
        numHash: numHashes,
    }
}

func (bf *BloomFilter) hash(data []byte, seed uint32) uint {
    h := fnv.New32a()
    // Mix seed
    _, _ = h.Write([]byte{byte(seed), byte(seed >> 8), byte(seed >> 16), byte(seed >> 24)})
    _, _ = h.Write(data)
    return uint(h.Sum32()) % bf.numBits
}

func (bf *BloomFilter) Add(key string) {
    bytes := []byte(key)
    for i := uint(0); i < bf.numHash; i++ {
        bitPos := bf.hash(bytes, uint32(i*31+17))
        wordIdx := bitPos / 64
        bitOffset := bitPos % 64
        bf.bits[wordIdx] |= (1 << bitOffset)
    }
}

func (bf *BloomFilter) Contains(key string) bool {
    bytes := []byte(key)
    for i := uint(0); i < bf.numHash; i++ {
        bitPos := bf.hash(bytes, uint32(i*31+17))
        wordIdx := bitPos / 64
        bitOffset := bitPos % 64
        if (bf.bits[wordIdx] & (1 << bitOffset)) == 0 {
            // If any bit is 0, item is definitely not in the set!
            return false
        }
```

```go
    }
    return true // Probably present
}

func main() {
    // Filter with 1024 bits and 4 hash functions
    bf := NewBloomFilter(1024, 4)

    existingOrders := []string{"ORDER-101", "ORDER-102", "ORDER-103", "ORDER-999"}
    for _, id := range existingOrders {
        bf.Add(id)
    }

    testIDs := []string{"ORDER-101", "ORDER-102", "ORDER-500", "ORDER-777"}
    for _, id := range testIDs {
        present := bf.Contains(id)
        if present {
            fmt.Printf("Query [%s]: Probably exists (fetch from database)\n", id)
        } else {
            fmt.Printf("Query [%s]: Definitely DOES NOT exist (skipped DB read)\n", id)
        }
    }
}
```

Run:

```go run bloom_filter.go
```

Output:

```
Query [ORDER-101]: Probably exists (fetch from database)
Query [ORDER-102]: Probably exists (fetch from database)
Query [ORDER-500]: Definitely DOES NOT exist (skipped DB read)
Query [ORDER-777]: Definitely DOES NOT exist (skipped DB read)
```

The trap

The fatal trap with Bloom filters is assuming they can perform exact membership testing or deletions. A standard Bloom filter cannot delete an element because setting a bit back to `0` may accidentally remove other items that hashed to that same bit. If an application requires item removal, it must use a **Counting Bloom Filter** (which tracks counts per slot instead of single bits) or a Cuckoo filter.

The boring rule

1. Use standard `strings.Contains`, `strings.Index`, and the `regexp` package for ordinary substring searches. Go's runtime implements heavily optimized assembly routines using SIMD instructions.
2. Use Rabin-Karp when searching for multiple fixed patterns simultaneously or scanning rolling streams.
3. Place a Bloom Filter in front of slow disks or remote databases to avoid expensive disk reads on non-existent records.
4. Size Bloom filters appropriately: $m \approx -\frac{n \ln p}{(\ln 2)^2}$, where n is expected elements and p is the desired false positive rate (e.g. 1%).

Try this

1. Calculate the empirical false positive rate of `BloomFilter` by adding 100 random keys and testing 1,000 non-existent keys.
2. Extend `RabinKarpSearch` to return all match offsets rather than stopping at the first occurrence.

CLI Tools in Go

Building CLI Tools in Go

Building CLI Tools in Go

This part is a **hands-on curriculum for command-line tools**: flags, subcommands, exit codes, pipes, config, signals, tests—first with **stdlib only**, then with **Cobra**, **pflag**, **Viper**, and brief notes on other frameworks.

Go is one of the best languages for CLIs: single static binary, fast startup, excellent `flag` / `os` / `io` / `os/exec`, and a culture of small, composable tools.

> **Related parts.** Stdlib surface: 987 flag / exec / signal. Pipeline philosophy: 143 Unix pipeline CLIs. Capstones in this part: 317 Network CLI tools (`ping`, `dns`, `scan`, `probe`) and 318 Network security mini-tools (`tls`, `headers`, `banner`, …). More labs: 99 projects (`goping`, linux clones, log shipper).

Learning outcomes

After this part you can:

1. Ship a correct `main` that parses flags, writes stdout/stderr properly, and exits with useful codes
2. Implement subcommands with only `flag.FlagSet`
3. Read env vars and simple config files without a framework
4. Compose stdin/stdout pipelines and spawn subprocesses safely
5. Cancel work on SIGINT with `context` + `os/signal`
6. Test CLIs without flaky subprocess soup
7. Grow into Cobra when subcommands and completion matter
8. Wire Viper-style config only when env/file/flag layering earns its keep
9. Build multi-command **network diagnostic** tools (TCP ping, DNS, port scan, HTTP probe)
10. Build **simple network security** mini-tools (TLS cert, headers, banner, secrets hygiene)

Path map

```
Stdlib core:   anatomy → flag → subcommands → exits/streams
Real tools:    config → pipes → exec → signals → tests → layout
Ecosystem:     Cobra → advanced Cobra → Viper → other libs
Practice:      shipping → cookbook → netkit → netsec mini-tools
```

Chapter	Focus
300	This overview
301	First tools with `os.Args`
302	`flag` package in depth
303	Subcommands with `FlagSet`
304	Exit codes, stdout vs stderr
305	Env vars and config files
306	Stdin, files, and pipes
307	`os/exec` orchestration
308	Signals and context
309	Testing CLIs
310	Project layout for larger CLIs
311	Cobra basics
312	Cobra advanced (persist flags, hooks, completion)
313	Viper and config layering
314	Other frameworks (urfave/cli, pflag, charm notes)
315	Version, install, completion, release
316	Cookbook: many end-to-end examples
317	**Project:** network CLI tools (ping, dns, scan, probe)
318	**Project:** simple network security mini-tools (tls, headers, …)
319	**Project:** text/log tools (`textkit`)
320	**Project:** HTTP debug CLI (`httpdbg`)
321	Cross-platform CLI notes (Windows/paths/signals)

Suggested schedule

```
Days 1-2   301-304     stdlib CLI hygiene
Days 3-4   305-308     config, pipes, exec, signals
Day 5      309-310     tests + package layout
Day 6      311-313     Cobra (+ Viper if needed)
Day 7      314-316     ecosystem + cookbook tools
Day 8      317         build netkit (ping/dns/scan/probe)
Day 9      318         build netsec (tls/headers/banner/…)
Day 10     319-321     textkit, httpdbg, portability
```

Mental model: a CLI is a function

```
argv + env + stdin + files

      parse     flags / subcommand / config

       run      pure-ish domain work (testable)

   stdout (data) · stderr (diagnostics) · exit code
```

Keep **`run(args, stdin, stdout, stderr) error`** (or a small `App` struct) so tests do not need `os.Exit` or global `flag.CommandLine`.

Stdlib first, libraries later

Stay on stdlib when…	Reach for Cobra/Viper when…
One binary, few flags	Many nested subcommands
Internal tools / scripts	Public developer product CLI
You want zero deps	Shell completion + man-page ecosystem
Teaching / small utilities	Flag inheritance across command trees

Most famous Go CLIs started simple. Add libraries when the **command tree** or **config surface** hurts—not on day one of a toy.

Running examples

Unless noted, examples assume:

```
mkdir -p /tmp/gocli && cd /tmp/gocli
go mod init example.com/gocli
# paste main.go or package files
go run .
```

First CLI Tools with os.Args

First CLI Tools with os.Args

Overview

Before `flag`, understand raw arguments: `os.Args[0]` is the program name, `os.Args[1:]` is everything the user typed. Many tiny tools need only this.

Hello CLI

```go
package main

import (
    "fmt"
    "os"
)

func main() {
    fmt.Println("args:", os.Args)
    if len(os.Args) < 2 {
        fmt.Fprintln(os.Stderr, "usage: greeter <name>")
        os.Exit(2)
    }
    fmt.Printf("hello, %s\n", os.Args[1])
}
```

```
go run . Ada
# hello, Ada
go run .
# usage: greeter <name>  (exit 2)
```

Separate main from logic

```go
package main

import (
    "fmt"
    "io"
```

```go
    "os"
)

func main() {
    if err := run(os.Args[1:], os.Stdout); err != nil {
        fmt.Fprintln(os.Stderr, "greeter:", err)
        os.Exit(1)
    }
}

func run(args []string, stdout io.Writer) error {
    if len(args) < 1 {
        return fmt.Errorf("usage: greeter <name>")
    }
    _, err := fmt.Fprintf(stdout, "hello, %s\n", args[0])
    return err
}
```

Now tests call `run([]string{"Ada"}, &buf)` without exiting the test process.

Positional args patterns

Required path

```go
func run(args []string) error {
    if len(args) != 1 {
        return fmt.Errorf("usage: cat1 <file>")
    }
    b, err := os.ReadFile(args[0])
    if err != nil {
        return err
    }
    _, err = os.Stdout.Write(b)
    return err
}
```

Variadic files (like `cat`)

```go
func run(args []string, stdout io.Writer) error {
    if len(args) == 0 {
        _, err := io.Copy(stdout, os.Stdin)
        return err
    }
```

```go
    for _, path := range args {
        f, err := os.Open(path)
        if err != nil {
            return err
        }
        _, err = io.Copy(stdout, f)
        f.Close()
        if err != nil {
            return err
        }
    }
    return nil
}
```

Options before flags package (`-n` style DIY)

```go
func parse(args []string) (count int, rest []string, err error) {
    count = 1
    i := 0
    for i < len(args) {
        a := args[i]
        if a == "--" {
            i++
            break
        }
        if len(a) > 1 && a[0] == '-' {
            // fragile DIY - prefer flag package next chapter
            return 0, nil, fmt.Errorf("unknown option %q", a)
        }
        break
    }
    return count, args[i:], nil
}
```

Lesson: hand-parsing flags gets messy quickly. Use `flag` (or Cobra) for real options.

Program name for usage

```go
name := filepath.Base(os.Args[0])
fmt.Fprintf(os.Stderr, "usage: %s <file>\n", name)
```

Environment as "args"

```
if os.Getenv("DEBUG") != "" {
    fmt.Fprintln(os.Stderr, "debug on")
}
```

Env is global config; args are per-invocation. Chapter 305 covers layering.

Example: `add` calculator

```
package main

import (
    "fmt"
    "os"
    "strconv"
)

func main() {
    if err := run(os.Args[1:]); err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }
}

func run(args []string) error {
    if len(args) < 2 {
        return fmt.Errorf("usage: add <a> <b>")
    }
    a, err := strconv.ParseFloat(args[0], 64)
    if err != nil {
        return fmt.Errorf("a: %w", err)
    }
    b, err := strconv.ParseFloat(args[1], 64)
    if err != nil {
        return fmt.Errorf("b: %w", err)
    }
    fmt.Println(a + b)
    return nil
}
```

Example: `whichdir` — print working directory variants

```go
func run(args []string) error {
    wd, err := os.Getwd()
    if err != nil {
        return err
    }
    fmt.Println(wd)
    if len(args) > 0 && args[0] == "-base" {
        fmt.Println(filepath.Base(wd))
    }
    return nil
}
```

Rules of thumb

Do	Don't
Put logic in `run(...)`	Scatter `os.Exit` deep in packages
Usage on stderr, exit 2 for misuse	Print usage to stdout
Validate `len(args)` early	Index `args[1]` without checks

Try next

1. Write `echo1` that joins args with spaces (like `echo`).
2. Write `headn` that prints the first line of a file (no flags yet).
3. Add a unit test for `run` with a `bytes.Buffer` as stdout.

flag Package in Depth

flag Package in Depth

Overview

Package `flag` is the stdlib CLI parser: bool/int/string/duration flags, custom `Value` types, and usage text. Master this before Cobra—Cobra's flags are powered by **pflag**, a POSIX-friendly fork of the same ideas.

Minimal example

```go
package main

import (
    "flag"
    "fmt"
    "os"
    "time"
)

func main() {
    var (
        name    = flag.String("name", "world", "name to greet")
        times   = flag.Int("n", 1, "repeat count")
        verbose = flag.Bool("v", false, "verbose")
        timeout = flag.Duration("timeout", 2*time.Second, "budget")
    )
    flag.Parse()

    if *verbose {
        fmt.Fprintf(os.Stderr, "timeout=%s args=%v\n", *timeout, flag.Args())
    }
    for i := 0; i < *times; i++ {
        fmt.Printf("hello, %s\n", *name)
    }
}
```

```
go run . -name Ada -n 2 -v -- leftover
# flags after -- are not parsed as flags in some styles;
# with flag, non-flag args are flag.Args() after Parse
```

Var forms (bind to your own vars)

```go
var cfg struct {
    Addr string
    Port int
}
flag.StringVar(&cfg.Addr, "addr", "127.0.0.1", "listen address")
flag.IntVar(&cfg.Port, "port", 8080, "port")
flag.Parse()
```

Prefer `Var` when flags map into a config struct.

Custom Usage

```go
flag.Usage = func() {
    fmt.Fprintf(flag.CommandLine.Output(),
        "Usage: %s [flags] <path>\n\nFlags:\n", filepath.Base(os.Args[0]))
    flag.PrintDefaults()
}
```

`flag.CommandLine.Output()` defaults to stderr—keep it that way.

Error handling modes

```go
fs := flag.NewFlagSet("tool", flag.ContinueOnError)
// flag.ExitOnError  - print usage, os.Exit(2)  (CommandLine default)
// flag.PanicOnError - panic (rare)
// flag.ContinueOnError - return error from Parse (best for tests)
```

For testable tools:

```go
fs := flag.NewFlagSet(os.Args[0], flag.ContinueOnError)
fs.SetOutput(io.Discard) // or a buffer when asserting usage
err := fs.Parse(args)
```

Custom flag types (`flag.Value`)

```go
type stringsFlag []string

func (s *stringsFlag) String() string { return fmt.Sprint(*s) }
func (s *stringsFlag) Set(v string) error {
    *s = append(*s, v)
    return nil
}

func main() {
    var tags stringsFlag
    flag.Var(&tags, "tag", "repeatable tag (use multiple -tag)")
    flag.Parse()
    fmt.Println(tags)
}
```

```
go run . -tag a -tag b
# [a b]
```

Enum flag

```go
type level string

func (l *level) String() string { return string(*l) }
func (l *level) Set(v string) error {
    switch v {
    case "debug", "info", "error":
        *l = level(v)
        return nil
    default:
        return fmt.Errorf("level must be debug|info|error")
    }
}

var lvl level = "info"
flag.Var(&lvl, "level", "log level")
```

Bool flags

```go
verbose := flag.Bool("verbose", false, "verbose")
// -verbose, -verbose=true, -verbose=false
// short -v needs a separate flag or pflag/Cobra
flag.Bool("v", false, "verbose (short)")
```

Stdlib `flag` does **not** combine short options (`-abc`). Use pflag/Cobra for GNU style.

Positional args after flags

```go
flag.Parse()
paths := flag.Args()
if len(paths) == 0 {
    fmt.Fprintln(os.Stderr, "need at least one path")
    os.Exit(2)
}
```

Order: users typically pass flags first. With stdlib `flag`, a non-flag stops flag parsing for the rest (classic Go behavior).

Example: `httpget` (stdlib-only client)

```go
func main() {
    url := flag.String("url", "", "URL to GET")
    timeout := flag.Duration("timeout", 10*time.Second, "client timeout")
    flag.Parse()
    if *url == "" {
        flag.Usage()
        os.Exit(2)
    }
    client := &http.Client{Timeout: *timeout}
    resp, err := client.Get(*url)
    if err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }
    defer resp.Body.Close()
    fmt.Println(resp.Status)
    _, _ = io.Copy(os.Stdout, io.LimitReader(resp.Body, 1<<20))
}
```

Example: `sleepfor`

```
d := flag.Duration("d", time.Second, "sleep duration")
flag.Parse()
time.Sleep(*d)
```

Example: `lines` — count lines in files

```
func main() {
    verbose := flag.Bool("v", false, "print per-file counts")
    flag.Parse()
    total := 0
    for _, path := range flag.Args() {
        n, err := countLines(path)
        if err != nil {
            fmt.Fprintln(os.Stderr, err)
            os.Exit(1)
        }
        if *verbose {
            fmt.Printf("%d\t%s\n", n, path)
        }
        total += n
    }
    fmt.Println(total)
}
```

Rules of thumb

Do	Don't
Document every flag default in help	Undocumented env-only secret flags as the only API
`ContinueOnError` + `run()` for tests	Parse global `flag` in `init` of libraries
Custom `Value` for repeatable/enum	Parse enums with silent fallbacks

Try next

1. Add a repeatable `-header k=v` flag using `flag.Value`.
2. Write tests that parse a private `FlagSet` with `ContinueOnError`.

3. Build `greet -n 3 -name X` and verify usage on bad `-n=abc`.

Subcommands with flag.FlagSet

Subcommands with flag.FlagSet

Overview

Real tools grow verbs: `git status`, `docker compose up`, `kubectl get pods`. You can implement subcommands with **only** the standard library via `flag.NewFlagSet`—no Cobra required.

Dispatch pattern

```
os.Args[0]   program
os.Args[1]   subcommand  (get|set|list)
os.Args[2:]  flags + args for that subcommand
```

```go
package main

import (
    "fmt"
    "os"
)

func main() {
    if len(os.Args) < 2 {
        usage()
        os.Exit(2)
    }
    var err error
    switch os.Args[1] {
    case "get":
        err = cmdGet(os.Args[2:])
    case "set":
        err = cmdSet(os.Args[2:])
    case "list":
        err = cmdList(os.Args[2:])
    case "help", "-h", "--help":
        usage()
        return
    default:
        fmt.Fprintf(os.Stderr, "unknown command %q\n", os.Args[1])
```

```
        usage()
        os.Exit(2)
    }
    if err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }
}

func usage() {
    fmt.Fprintf(os.Stderr, `Usage: kv <command> [flags]

Commands:
  get   get a key
  set   set a key
  list  list keys
`)
}
```

Per-command FlagSet

```
func cmdGet(args []string) error {
    fs := flag.NewFlagSet("get", flag.ContinueOnError)
    fs.SetOutput(os.Stderr)
    raw := fs.Bool("raw", false, "print raw value only")
    if err := fs.Parse(args); err != nil {
        return err
    }
    if fs.NArg() != 1 {
        return fmt.Errorf("usage: kv get [-raw] <key>")
    }
    key := fs.Arg(0)
    val, ok := store[key]
    if !ok {
        return fmt.Errorf("key not found: %s", key)
    }
    if *raw {
        fmt.Print(val)
        return nil
    }
    fmt.Printf("%s=%s\n", key, val)
    return nil
}
```

```go
func cmdSet(args []string) error {
    fs := flag.NewFlagSet("set", flag.ContinueOnError)
    fs.SetOutput(os.Stderr)
    if err := fs.Parse(args); err != nil {
        return err
    }
    if fs.NArg() != 2 {
        return fmt.Errorf("usage: kv set <key> <value>")
    }
    store[fs.Arg(0)] = fs.Arg(1)
    return nil
}
```

Shared "global" flags

Parse globals **before** the subcommand, or require globals after the verb (simpler with FlagSet-per-command).

Globals before verb (manual)

```go
// mytool -verbose get -id 1
// harder with stdlib flag alone - often document:
// mytool get -verbose -id 1
```

Globals on every FlagSet (duplicate)

```go
func addGlobal(fs *flag.FlagSet, verbose *bool) {
    fs.BoolVar(verbose, "verbose", false, "verbose output")
    fs.BoolVar(verbose, "v", false, "verbose output (short)")
}
```

Cobra's persistent flags solve this more cleanly (chapter 312). For stdlib, **prefer flags after the subcommand**.

Nested subcommands

```go
// tool remote add
// tool remote list
func cmdRemote(args []string) error {
    if len(args) < 1 {
        return fmt.Errorf("usage: tool remote <add|list>")
```

```
    }
    switch args[0] {
    case "add":
        return cmdRemoteAdd(args[1:])
    case "list":
        return cmdRemoteList(args[1:])
    default:
        return fmt.Errorf("unknown remote command %q", args[0])
    }
}
```

Example: mini `todo` CLI

```
// commands: add, list, done
// storage: JSON file in OS user config dir

func cmdAdd(args []string) error {
    fs := flag.NewFlagSet("add", flag.ContinueOnError)
    pri := fs.Int("pri", 0, "priority")
    if err := fs.Parse(args); err != nil {
        return err
    }
    if fs.NArg() < 1 {
        return fmt.Errorf("usage: todo add [-pri N] <text>")
    }
    text := strings.Join(fs.Args(), " ")
    return saveNewTask(text, *pri)
}
```

Help that lists commands

```
var commands = map[string]string{
    "get":  "get a key",
    "set":  "set a key",
    "list": "list all keys",
}

func usage() {
    fmt.Fprintln(os.Stderr, "Usage: kv <command> [flags]")
    fmt.Fprintln(os.Stderr, "\nCommands:")
    keys := slices.Sorted(maps.Keys(commands))
    for _, k := range keys {
```

```
        fmt.Fprintf(os.Stderr, "  %-8s %s\n", k, commands[k])
    }
}
```

Alias map

```
aliases := map[string]string{"ls": "list", "rm": "delete"}
cmd := os.Args[1]
if a, ok := aliases[cmd]; ok {
    cmd = a
}
```

When FlagSet is enough

Enough	Grow to Cobra
~10 commands, flat or shallow	Deep trees, many shared flags
Internal tooling	Public CLI with completion
You want zero dependencies	Man pages, docs generators

Rules of thumb

Do	Don't
One `FlagSet` per command	One global `flag.Parse` for all verbs
`ContinueOnError` in libraries/tests	`ExitOnError` inside `run` used by tests
Document command list in usage	Silent unknown commands

Try next

1. Implement `kv get|set|list` with an in-memory map (then JSON file).
2. Add `kv help get` that prints only the get FlagSet defaults.
3. Add nested `config get` / `config set`.

Exit Codes, Stdout, and Stderr

Exit Codes, Stdout, and Stderr

Overview

Automation treats your CLI as a machine API: **stdout is data**, **stderr is diagnostics**, **exit code is success/failure**. Break that contract and scripts, CI, and pipes fail in confusing ways.

Also see Unix pipeline CLI tools.

The contract

```
exit 0     success
exit 1     runtime failure (file missing, network error)
exit 2     misuse (bad flags, wrong arity) - common convention
exit 130   interrupted by SIGINT (128+2) - platform dependent
```

Go's `flag` with `ExitOnError` exits **2** on parse errors—good.

```go
func main() {
    if err := run(os.Args[1:]); err != nil {
        fmt.Fprintln(os.Stderr, "mytool:", err)
        var code int
        switch {
        case errors.Is(err, errUsage):
            code = 2
        default:
            code = 1
        }
        os.Exit(code)
    }
}
```

Stdout vs stderr

Stream	Put here
stdout	JSON results, tables meant for pipes, primary output
stderr	Usage, progress, logs, warnings

```go
fmt.Fprintln(os.Stderr, "fetching...") // progress
fmt.Println(string(jsonBytes))         // data for jq
```

```
mytool 2>log.txt | jq .               # logs aside, data through pipe
```

Quiet / verbose / JSON modes

```go
type mode int

const (
    modeText mode = iota
    modeJSON
)

func run(stdout, stderr io.Writer, verbose bool, m mode, data Result) error {
    if verbose {
        fmt.Fprintln(stderr, "processing", data.ID)
    }
    if m == modeJSON {
        enc := json.NewEncoder(stdout)
        enc.SetIndent("", "  ")
        return enc.Encode(data)
    }
    fmt.Fprintf(stdout, "%s\t%d\n", data.Name, data.Count)
    return nil
}
```

Is stdout a TTY?

```go
func stdoutIsTTY() bool {
    fi, err := os.Stdout.Stat()
    if err != nil {
        return false
    }
    return fi.Mode()&os.ModeCharDevice != 0
}
```

Use TTY detection for:

- color (only if TTY and `NO_COLOR` unset)
- progress bars
- human tables vs TSV for pipes

```go
if stdoutIsTTY() && os.Getenv("NO_COLOR") == "" {
    // optional color codes
}
```

Respect NO_COLOR when you add color (stdlib has no color package—use raw ANSI sparingly or a small lib later).

Don't call os.Exit in libraries

```go
// package greeter
func Greet(w io.Writer, name string) error { ... }

// package main
func main() {
    if err := greeter.Greet(os.Stdout, name); err != nil {
        os.Exit(1)
    }
}
```

`os.Exit` skips deferred cleanup in the current goroutine's stack for the process—keep it at the edge.

Example: correct `grep`-shaped output

```go
// matches to stdout, errors to stderr, exit 0 if any match, 1 if none (grep-like)
matched := false
for _, line := range lines {
    if strings.Contains(line, pattern) {
        fmt.Fprintln(stdout, line)
        matched = true
    }
}
if !matched {
    return errNoMatch // main maps to exit 1
}
```

Document whether you follow grep's exit semantics—they surprise some users.

Machine-readable errors

```go
type cliError struct {
    Code    int    `json:"code"`
    Message string `json:"message"`
}

// when --json-errors and failing:
json.NewEncoder(os.Stderr).Encode(cliError{Code: 1, Message: err.Error()})
```

Rules of thumb

Do	Don't
Data → stdout	Mix log lines into JSON stdout
Diagnostics → stderr	Require users to scrape stdout for errors
Non-zero on failure	Always exit 0 and bury errors in text
Map usage errors to 2	Use exit 1 for everything without docs

Try next

1. Build a tool that prints JSON on stdout and timing on stderr; pipe through `jq`.
2. Add `--quiet` that silences stderr progress only.
3. Verify exit codes in a shell: `go run . bad; echo $?`.

Environment Variables and Config Files

Environment Variables and Config Files

Overview

Flags are per-invocation. Environment variables and config files hold **defaults** that follow the user or deployment. Stdlib-only tooling can load both cleanly; Viper (chapter 313) automates layering when complexity grows.

Precedence (recommended)

```
flags > env > config file > built-in defaults
```

Document this in `--help`. Users should never wonder why a value "stuck."

Environment variables

```go
func getenv(key, def string) string {
    if v := os.Getenv(key); v != "" {
        return v
    }
    return def
}

func getenvInt(key string, def int) int {
    v := os.Getenv(key)
    if v == "" {
        return def
    }
    n, err := strconv.Atoi(v)
    if err != nil {
        return def // or return error - prefer fail-fast for CLIs
    }
    return n
}
```

12-factor style names

```
MYTOOL_API_URL=https://api.example.com
MYTOOL_TOKEN=...
MYTOOL_TIMEOUT=10s
```

```go
apiURL := getenv("MYTOOL_API_URL", "http://127.0.0.1:8080")
```

Parse duration from env

```go
func envDuration(key string, def time.Duration) (time.Duration, error) {
    v := os.Getenv(key)
    if v == "" {
        return def, nil
    }
    return time.ParseDuration(v)
}
```

Booleans

```go
func envBool(key string, def bool) bool {
    v := strings.ToLower(strings.TrimSpace(os.Getenv(key)))
    switch v {
    case "":
        return def
    case "1", "true", "yes", "y", "on":
        return true
    case "0", "false", "no", "n", "off":
        return false
    default:
        return def
    }
}
```

Prefer failing on garbage if the var is critical.

Config file locations

```go
func defaultConfigPath(app string) (string, error) {
    dir, err := os.UserConfigDir() // platform-correct
    if err != nil {
        return "", err
```

```go
    }
    return filepath.Join(dir, app, "config.json"), nil
}
```

OS	`UserConfigDir` typical
macOS	`~/Library/Application Support`
Linux	`~/.config`
Windows	`%AppData%`

Also support explicit `--config path`.

JSON config (stdlib)

```go
type Config struct {
    APIURL  string `json:"api_url"`
    Timeout string `json:"timeout"` // parse as duration
    Verbose bool   `json:"verbose"`
}

func loadConfig(path string) (Config, error) {
    var c Config
    b, err := os.ReadFile(path)
    if err != nil {
        if os.IsNotExist(err) {
            return Config{}, nil // missing file OK
        }
        return Config{}, err
    }
    if err := json.Unmarshal(b, &c); err != nil {
        return Config{}, fmt.Errorf("config %s: %w", path, err)
    }
    return c, nil
}
```

Merge with flags

```go
type Options struct {
    APIURL  string
    Verbose bool
    Timeout time.Duration
}
```

```
func parse(args []string) (Options, error) {
    fs := flag.NewFlagSet("mytool", flag.ContinueOnError)
    cfgPath := fs.String("config", "", "config file path")
    api := fs.String("api-url", "", "API base URL")
    verbose := fs.Bool("verbose", false, "verbose")
    timeout := fs.Duration("timeout", 0, "timeout (0=default)")
    if err := fs.Parse(args); err != nil {
        return Options{}, err
    }

    path := *cfgPath
    if path == "" {
        path, _ = defaultConfigPath("mytool")
    }
    fileCfg, err := loadConfig(path)
    if err != nil {
        return Options{}, err
    }

    opt := Options{
        APIURL:  firstNonEmpty(*api, os.Getenv("MYTOOL_API_URL"), fileCfg.APIURL, "http://12
        Verbose: *verbose || envBool("MYTOOL_VERBOSE", false) || fileCfg.Verbose,
        Timeout: 10 * time.Second,
    }
    if *timeout > 0 {
        opt.Timeout = *timeout
    } else if d, err := envDuration("MYTOOL_TIMEOUT", 0); err == nil && d > 0 {
        opt.Timeout = d
    } else if fileCfg.Timeout != "" {
        d, err := time.ParseDuration(fileCfg.Timeout)
        if err != nil {
            return Options{}, err
        }
        opt.Timeout = d
    }
    return opt, nil
}

func firstNonEmpty(vals ...string) string {
    for _, v := range vals {
        if v != "" {
            return v
        }
    }
    return ""
}
```

Dotenv? Not in stdlib

`.env` files are not standard. Options:

1. Document that users `export $(cat .env | xargs)` in shell
2. Parse a simple `KEY=VAL` file yourself (~30 lines)
3. Use a small library later

A minimal dotenv:

```
func loadDotEnv(path string) error {
    b, err := os.ReadFile(path)
    if err != nil {
        return err
    }
    for _, line := range strings.Split(string(b), "\n") {
        line = strings.TrimSpace(line)
        if line == "" || strings.HasPrefix(line, "#") {
            continue
        }
        k, v, ok := strings.Cut(line, "=")
        if !ok {
            continue
        }
        k = strings.TrimSpace(k)
        v = strings.TrimSpace(v)
        if os.Getenv(k) == "" { // don't override existing env
            _ = os.Setenv(k, v)
        }
    }
    return nil
}
```

Secrets

- Prefer env or OS keychain for tokens—not committed JSON
- Support `MYTOOL_TOKEN_FILE` pointing at a file (Kubernetes-friendly)

```
func loadToken() (string, error) {
    if t := os.Getenv("MYTOOL_TOKEN"); t != "" {
        return t, nil
    }
    if p := os.Getenv("MYTOOL_TOKEN_FILE"); p != "" {
        b, err := os.ReadFile(p)
        return strings.TrimSpace(string(b)), err
    }
```

```
    return "", fmt.Errorf("set MYTOOL_TOKEN or MYTOOL_TOKEN_FILE")
}
```

Rules of thumb

Do	Don't
Document precedence	Silent flag < env surprises
Allow missing config file	Crash if optional file absent
Fail on invalid config JSON	Ignore corrupt files
Use `UserConfigDir`	Hardcode `~/.mytool` only

Try next

1. Implement `parse` with flag > env > JSON defaults.
2. Add `--print-config` that dumps resolved options (redact tokens).
3. Support `TOKEN_FILE` and test with `t.Setenv`.

Stdin, Files, and Pipes

Stdin, Files, and Pipes

Overview

Great CLIs work in pipelines: read files or stdin, write transform results to stdout. This chapter is pure stdlib patterns for input discovery, streaming, and safe file handling.

Input priority (common UX)

```
1. Explicit file args
2. If no args (or arg is "-") → stdin
```

```go
func openInputs(args []string) ([]io.ReadCloser, error) {
    if len(args) == 0 {
        return []io.ReadCloser{io.NopCloser(os.Stdin)}, nil
    }
    var out []io.ReadCloser
    for _, a := range args {
        if a == "-" {
            out = append(out, io.NopCloser(os.Stdin))
            continue
        }
        f, err := os.Open(a)
        if err != nil {
            // close already opened on error
            for _, c := range out {
                c.Close()
            }
            return nil, err
        }
        out = append(out, f)
    }
    return out, nil
}
```

Stream line by line

```
func process(r io.Reader, w io.Writer, pred func(string) bool) error {
    sc := bufio.NewScanner(r)
    // optional: larger tokens
    // buf := make([]byte, 0, 64*1024)
    // sc.Buffer(buf, 1024*1024)
    for sc.Scan() {
        line := sc.Text()
        if pred(line) {
            if _, err := fmt.Fprintln(w, line); err != nil {
                return err
            }
        }
    }
    return sc.Err()
}
```

Example: `grep1` (stdlib)

```
func main() {
    n := flag.Bool("n", false, "print line numbers")
    flag.Parse()
    if flag.NArg() < 1 {
        fmt.Fprintln(os.Stderr, "usage: grep1 [-n] <pattern> [files...]")
        os.Exit(2)
    }
    pat := flag.Arg(0)
    files := flag.Args()[1:]
    inputs, err := openInputs(files)
    if err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }
    matched := false
    lineNo := 0
    for _, in := range inputs {
        sc := bufio.NewScanner(in)
        for sc.Scan() {
            lineNo++
            line := sc.Text()
            if strings.Contains(line, pat) {
                matched = true
                if *n {
```

```go
                    fmt.Printf("%d:%s\n", lineNo, line)
                } else {
                    fmt.Println(line)
                }
            }
        }
        err = sc.Err()
        in.Close()
        if err != nil {
            fmt.Fprintln(os.Stderr, err)
            os.Exit(1)
        }
    }
    if !matched {
        os.Exit(1)
    }
}
```

```bash
echo -e "go\nrust\ngo lang" | go run . go
printf 'a\nb\n' > t.txt && go run . -n b t.txt
```

Example: `wc1` lines/words/bytes

```go
func count(r io.Reader) (lines, words, bytes int, err error) {
    br := bufio.NewReader(r)
    for {
        b, e := br.ReadByte()
        if e != nil {
            if e == io.EOF {
                return lines, words, bytes, nil
            }
            return 0, 0, 0, e
        }
        bytes++
        if b == '\n' {
            lines++
        }
    }
}
// words: use bufio.Scanner with Split(bufio.ScanWords) in a second pass
// or track isSpace transitions in the byte loop
```

Example: JSON filter from stdin

```go
func run(r io.Reader, w io.Writer, field string) error {
    dec := json.NewDecoder(r)
    enc := json.NewEncoder(w)
    for {
        var m map[string]any
        if err := dec.Decode(&m); err != nil {
            if err == io.EOF {
                return nil
            }
            return err
        }
        if v, ok := m[field]; ok {
            if err := enc.Encode(v); err != nil {
                return err
            }
        }
    }
}
```

```bash
echo '{"name":"a"}{"name":"b"}' | go run . -field name
```

Detect pipe vs TTY for stdin

```go
func stdinIsPipe() bool {
    fi, err := os.Stdin.Stat()
    if err != nil {
        return false
    }
    return fi.Mode()&os.ModeCharDevice == 0
}
```

If stdin is a TTY and no files given, you may print usage instead of blocking forever:

```go
if len(args) == 0 && !stdinIsPipe() {
    return fmt.Errorf("usage: tool <files> or pipe stdin")
}
```

(Some tools intentionally wait for interactive stdin—document either way.)

Atomic write of output files

```go
func writeFileAtomic(path string, data []byte, mode os.FileMode) error {
    dir := filepath.Dir(path)
    tmp, err := os.CreateTemp(dir, ".tmp-*")
    if err != nil {
        return err
    }
    tmpName := tmp.Name()
    defer func() { _ = os.Remove(tmpName) }() // if not renamed

    if _, err := tmp.Write(data); err != nil {
        tmp.Close()
        return err
    }
    if err := tmp.Chmod(mode); err != nil {
        tmp.Close()
        return err
    }
    if err := tmp.Close(); err != nil {
        return err
    }
    return os.Rename(tmpName, path)
}
```

Progress on stderr only

```go
fmt.Fprintf(os.Stderr, "\rprocessed %d", n)
// final newline
fmt.Fprintln(os.Stderr)
```

Never put `\r` progress on stdout if users pipe to files.

Rules of thumb

Do	Don't
Stream with `bufio`	`ReadAll` multi-GB stdin
Accept `-` as stdin	Force temp files for every pipeline
Close files	Leak FDs in long loops
Errors → stderr	Corrupt the data stream

Try next

1. Implement `uniq1` adjacent-line dedupe from stdin.
2. Add `cut1 -f 2 -d ,` for CSV-ish lines.
3. Pipe `grep1` into `wc -l` and confirm exit codes compose.

os/exec Orchestration

os/exec Orchestration

Overview

CLIs often wrap other tools: `git`, `docker`, `kubectl`, compilers. Package `os/exec` runs subprocesses with explicit argv (never shell strings with untrusted input).

Basics

```
cmd := exec.CommandContext(ctx, "git", "status", "--short")
cmd.Dir = repoDir
cmd.Env = append(os.Environ(), "LC_ALL=C")
out, err := cmd.Output() // stdout; stderr in ExitError.Stderr if failed
```

Method	Captures
`Run`	exit only; you wire Stdout/Stderr
`Output`	stdout bytes; err on non-zero
`CombinedOutput`	stdout+stderr merged
`Start/Wait`	async

Stream to the user

```
cmd := exec.Command("go", "test", "./...")
cmd.Stdout = os.Stdout
cmd.Stderr = os.Stderr
cmd.Stdin = os.Stdin
err := cmd.Run()
```

Capture and inspect exit code

```
err := cmd.Run()
if err != nil {
    var ee *exec.ExitError
    if errors.As(err, &ee) {
```

```go
        return fmt.Errorf("git exit %d", ee.ExitCode())
    }
    return err // not found, killed, etc.
}
```

Pipes between processes

```go
func pipeline(ctx context.Context) error {
    ps := exec.CommandContext(ctx, "ps", "aux")
    grep := exec.CommandContext(ctx, "grep", "go")

    r, w := io.Pipe()
    ps.Stdout = w
    grep.Stdin = r
    var grepOut bytes.Buffer
    grep.Stdout = &grepOut

    if err := ps.Start(); err != nil {
        return err
    }
    if err := grep.Start(); err != nil {
        return err
    }
    errPs := ps.Wait()
    _ = w.Close() // signal EOF to grep
    errGrep := grep.Wait()
    if errPs != nil {
        return errPs
    }
    if errGrep != nil {
        return errGrep
    }
    fmt.Print(grepOut.String())
    return nil
}
```

Prefer shell for one-off scripts; prefer argv pipelines in production wrappers for safety.

Never shell untrusted input

```go
// BAD
exec.Command("sh", "-c", "echo "+userInput)

// GOOD
exec.Command("echo", userInput)
```

If you must use a shell, allowlist and escape carefully—usually you should not.

LookPath

```go
path, err := exec.LookPath("docker")
if err != nil {
    return fmt.Errorf("docker not installed: %w", err)
}
_ = path
```

Example: `gofmt-write` wrapper

```go
func run(paths []string) error {
    args := append([]string{"-w"}, paths...)
    cmd := exec.Command("gofmt", args...)
    cmd.Stdout = os.Stdout
    cmd.Stderr = os.Stderr
    return cmd.Run()
}
```

Example: parallel commands with errgroup

```go
eg, ctx := errgroup.WithContext(context.Background())
for _, pkg := range packages {
    pkg := pkg
    eg.Go(func() error {
        cmd := exec.CommandContext(ctx, "go", "test", pkg)
        out, err := cmd.CombinedOutput()
        if err != nil {
            return fmt.Errorf("%s: %s: %w", pkg, out, err)
        }
        return nil
```

```go
    })
}
return eg.Wait()
```

Bound parallelism if the child tools are heavy.

Example: `runjson` — exec and parse JSON stdout

```go
cmd := exec.CommandContext(ctx, "docker", "inspect", id)
out, err := cmd.Output()
if err != nil {
    return err
}
var payload any
if err := json.Unmarshal(out, &payload); err != nil {
    return fmt.Errorf("parse docker json: %w", err)
}
```

Timeouts

```go
ctx, cancel := context.WithTimeout(context.Background(), 30*time.Second)
defer cancel()
cmd := exec.CommandContext(ctx, "make", "test")
err := cmd.Run()
if errors.Is(ctx.Err(), context.DeadlineExceeded) {
    return fmt.Errorf("timed out")
}
```

`CommandContext` sends kill when ctx cancels (details differ by OS for process groups).

Rules of thumb

Do	Don't
Argv slices	`sh -c` with user strings
`CommandContext`	Untimed long children
Surface exit codes	Swallow `ExitError` without code
Set `Dir`/`Env` explicitly	Rely on ambient cwd silently

Try next

1. Wrap `git rev-parse --short HEAD` and print version.
2. Run `go test` with timeout 5s; assert cancel path.
3. Build `whichall` that LookPaths a list of tools and exits 1 if any missing.

Signals and Context in CLIs

Signals and Context in CLIs

Overview

Users hit Ctrl-C. Deployments send SIGTERM. Good CLIs cancel work, close files, and exit promptly. Wire **`os/signal`** to **`context`** and pass that context through I/O and `exec`.

Minimal graceful cancel

```go
func main() {
    ctx, stop := signal.NotifyContext(context.Background(), os.Interrupt, syscall.SIGTERM)
    defer stop()

    if err := run(ctx, os.Args[1:]); err != nil {
        if errors.Is(err, context.Canceled) {
            fmt.Fprintln(os.Stderr, "interrupted")
            os.Exit(130)
        }
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }
}
```

`signal.NotifyContext` (Go 1.16+) is the preferred pattern.

Pass ctx everywhere

```go
func run(ctx context.Context, args []string) error {
    // HTTP
    req, err := http.NewRequestWithContext(ctx, http.MethodGet, url, nil)
    // ...
    // Exec
    cmd := exec.CommandContext(ctx, "sleep", "60")
    // Files: poll ctx in loops
    return workLoop(ctx)
}
```

```go
func workLoop(ctx context.Context) error {
    for i := 0; i < 1000; i++ {
        select {
        case <-ctx.Done():
            return ctx.Err()
        default:
        }
        // unit of work
    }
    return nil
}
```

Long-running workers

```go
func runWorkers(ctx context.Context, jobs <-chan Job) error {
    var wg sync.WaitGroup
    defer wg.Wait()

    for i := 0; i < 4; i++ {
        wg.Add(1)
        go func() {
            defer wg.Done()
            for {
                select {
                case <-ctx.Done():
                    return
                case j, ok := <-jobs:
                    if !ok {
                        return
                    }
                    process(ctx, j)
                }
            }
        }()
    }
    // producer closes jobs when done; main waits via defer wg.Wait
    // cancel via signal cancels ctx so workers return
    return nil
}
```

Ignore SIGPIPE? (advanced)

Writing to a closed pipe (downstream `head` exited) can raise SIGPIPE. Go's runtime usually converts write errors to `EPIPE`—check write errors on stdout:

```
if _, err := fmt.Fprintln(os.Stdout, line); err != nil {
    return err // broken pipe - exit quietly or 1
}
```

Example: cancellable download CLI

```
func download(ctx context.Context, url, dest string) error {
    req, err := http.NewRequestWithContext(ctx, http.MethodGet, url, nil)
    if err != nil {
        return err
    }
    resp, err := http.DefaultClient.Do(req)
    if err != nil {
        return err
    }
    defer resp.Body.Close()

    f, err := os.Create(dest)
    if err != nil {
        return err
    }
    defer f.Close()

    _, err = io.Copy(f, resp.Body) // Copy stops when ctx cancels the body
    return err
}
```

Use a client with timeout as a backstop even when signals work.

Double Ctrl-C

Some tools escalate: first signal cancels gracefully; second forces `os.Exit(1)`. Keep it simple for most CLIs—one cancel is enough if work respects ctx.

```
go func() {
    <-ctx.Done()
    // optional: force exit after grace period
    t := time.NewTimer(5 * time.Second)
```

```go
    defer t.Stop()
    select {
    case <-t.C:
        fmt.Fprintln(os.Stderr, "forcing exit")
        os.Exit(1)
    }
}()
```

Rules of thumb

Do	Don't
`NotifyContext` at main	Catch signals deep in libraries
Propagate ctx to HTTP/exec	Ignore cancel in tight loops
Map cancel to clear message	Hang forever after Ctrl-C
Check write errors	Assume stdout always works

Try next

1. `sleep 30` under your CLI with ctx; Ctrl-C and confirm quick exit.
2. Download a large file; cancel mid-way; ensure partial file is removed or kept intentionally.
3. Unit-test `run` with a pre-canceled context.

Testing CLI Tools

Testing CLI Tools

Overview

Test CLIs as pure functions of args and streams. Avoid `os.Exit` and global `flag.CommandLine` in the unit under test. Use `FlagSet` with `ContinueOnError`, `bytes.Buffer`, and table tests.

Golden structure

```go
// cmd/mytool/main.go
func main() {
    os.Exit(realMain(os.Args[1:], os.Stdin, os.Stdout, os.Stderr))
}

func realMain(args []string, in io.Reader, out, errOut io.Writer) int {
    if err := run(args, in, out, errOut); err != nil {
        fmt.Fprintln(errOut, "mytool:", err)
        if errors.Is(err, errUsage) {
            return 2
        }
        return 1
    }
    return 0
}
```

```go
func TestRunHello(t *testing.T) {
    var out bytes.Buffer
    code := realMain([]string{"-name", "Ada"}, strings.NewReader(""), &out, io.Discard)
    if code != 0 {
        t.Fatalf("code=%d", code)
    }
    if got := out.String(); got != "hello, Ada\n" {
        t.Fatalf("got %q", got)
    }
}
```

FlagSet in run()

```
func run(args []string, in io.Reader, out, errOut io.Writer) error {
    fs := flag.NewFlagSet("mytool", flag.ContinueOnError)
    fs.SetOutput(errOut)
    name := fs.String("name", "world", "name")
    if err := fs.Parse(args); err != nil {
        return errUsage
    }
    _, err := fmt.Fprintf(out, "hello, %s\n", *name)
    return err
}
```

Never call `flag.Parse()` on the global set inside library tests—it races and mutates process state.

Table-driven cases

```
func TestCLI(t *testing.T) {
    tests := []struct {
        name string
        args []string
        in   string
        code int
        want string
    }{
        {"ok", []string{"-name", "Zo"}, "", 0, "hello, Zo\n"},
        {"default", nil, "", 0, "hello, world\n"},
        {"badflag", []string{"-nope"}, "", 2, ""},
    }
    for _, tt := range tests {
        t.Run(tt.name, func(t *testing.T) {
            var out, errB bytes.Buffer
            code := realMain(tt.args, strings.NewReader(tt.in), &out, &errB)
            if code != tt.code {
                t.Fatalf("code=%d err=%s", code, errB.String())
            }
            if tt.want != "" && out.String() != tt.want {
                t.Fatalf("out=%q want %q", out.String(), tt.want)
            }
        })
    }
}
```

Temp files and env

```
func TestWithConfig(t *testing.T) {
    dir := t.TempDir()
    path := filepath.Join(dir, "c.json")
    if err := os.WriteFile(path, []byte(`{"verbose":true}`), 0o600); err != nil {
        t.Fatal(err)
    }
    t.Setenv("MYTOOL_API_URL", "http://example")
    // call run with --config path
}
```

Subprocess tests (optional)

When you must test `main` packaging:

```
func TestMainBinary(t *testing.T) {
    if os.Getenv("BE_CRASHER") == "1" {
        main()
        return
    }
    cmd := exec.Command(os.Args[0], "-test.run=TestMainBinary")
    cmd.Env = append(os.Environ(), "BE_CRASHER=1")
    err := cmd.Run()
    // assert exit code via ExitError
}
```

Prefer in-process tests; subprocess tests are slower and messier.

Golden files

```
func TestOutputGolden(t *testing.T) {
    var out bytes.Buffer
    _ = run([]string{"list"}, nil, &out, io.Discard)
    golden := filepath.Join("testdata", "list.golden")
    if os.Getenv("UPDATE_GOLDEN") != "" {
        _ = os.WriteFile(golden, out.Bytes(), 0o644)
    }
    want, err := os.ReadFile(golden)
    if err != nil {
        t.Fatal(err)
    }
    if !bytes.Equal(out.Bytes(), want) {
```

```go
        t.Fatalf("mismatch")
    }
}
```

Race and parallel

```go
func TestX(t *testing.T) {
    t.Parallel()
    // only if run uses no process-global state
}
```

Rules of thumb

Do	Don't
Inject Reader/Writer	Read/write only `os.Stdout` in core logic
Return exit codes from `realMain`	`os.Exit` inside tests
Private FlagSet	Global `flag.Parse` in packages
`t.Setenv` / `t.TempDir`	Touch real `$HOME` config

Try next

1. Refactor any sample CLI to `realMain` and add three table cases.
2. Add a usage test that expects code 2 and stderr containing `"usage"`.
3. Golden-test JSON output with `UPDATE_GOLDEN=1 go test`.

Structuring Larger CLI Apps

Structuring Larger CLI Apps

Overview

Small tools fit in `main.go`. Larger CLIs need packages so commands, domain logic, and I/O stay testable. This chapter shows a stdlib-friendly layout that also migrates cleanly to Cobra.

Recommended layout

```
mytool/
  go.mod
  cmd/
    mytool/
      main.go          # os.Exit(realMain(...)) only
  internal/
    app/
      app.go           # App struct, Run(args)
      root.go          # dispatch
    cmd/
      get.go
      set.go
      list.go
    config/
      config.go
    domain/
      store.go         # business logic
  scripts/
  README.md
```

`internal/` keeps API private to the module.

App struct

```
package app

type App struct {
    Stdin  io.Reader
```

```go
    Stdout io.Writer
    Stderr io.Writer
    // optional: Version string, FS for embed, HTTP client
}

func New() *App {
    return &App{
        Stdin:  os.Stdin,
        Stdout: os.Stdout,
        Stderr: os.Stderr,
    }
}

func (a *App) Run(args []string) error {
    if len(args) < 1 {
        return a.usage()
    }
    switch args[0] {
    case "get":
        return a.cmdGet(args[1:])
    case "set":
        return a.cmdSet(args[1:])
    case "version":
        fmt.Fprintln(a.Stdout, version)
        return nil
    default:
        return fmt.Errorf("unknown command %q", args[0])
    }
}
```

```go
// cmd/mytool/main.go
func main() {
    a := app.New()
    if err := a.Run(os.Args[1:]); err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }
}
```

Dependency injection for domain

```go
type Store interface {
    Get(ctx context.Context, key string) (string, error)
    Set(ctx context.Context, key, val string) error
}

type App struct {
    Store  Store
    Stdout io.Writer
    // ...
}

func (a *App) cmdGet(args []string) error {
    // parse flags...
    val, err := a.Store.Get(context.Background(), key)
    // write to a.Stdout
    return err
}
```

Tests inject a memory store.

Cobra migration path

When you adopt Cobra later:

```
internal/app  → cobra root command constructor
internal/cmd  → cobra subcommands calling domain
domain/       → unchanged
```

Keep **domain free of cobra/flag types**.

Version and commit via ldflags

```go
// internal/buildinfo/buildinfo.go
var (
    Version = "dev"
    Commit  = "none"
    Date    = "unknown"
)
```

```
go build -ldflags "-X example.com/mytool/internal/buildinfo.Version=1.2.3 \
  -X example.com/mytool/internal/buildinfo.Commit=$(git rev-parse --short HEAD)" \
  -o mytool ./cmd/mytool
```

Multi-binary monorepo

```
cmd/
  mytool/
  mytool-helper/
```

Share `internal/` packages; thin mains.

Example: split packages sketch

```go
// internal/cmd/get.go
package cmd

func Get(stdout io.Writer, store Store, args []string) error {
    fs := flag.NewFlagSet("get", flag.ContinueOnError)
    // ...
}
```

```go
// internal/app/root.go
case "get":
    return cmd.Get(a.Stdout, a.Store, args[1:])
```

Config package

```go
package config

type Config struct {
    Path string
    // ...
}

func Load(path string) (Config, error) { /* json */ }
```

Main/app resolve path; commands receive `Config` values, not global state.

Rules of thumb

Do	Don't
`cmd/` + `internal/`	Everything in `package main` forever
Inject Store/IO	Global `var db *sql.DB` in all files
Keep domain pure	Import `flag` into domain validators
One binary entry per UX	Hidden side-effect `init()` parsers

Try next

1. Split the kv CLI into `cmd/mytool` + `internal/app`.
2. Add `version` subcommand from ldflags.
3. Test `App.Run` with a fake Store.

Cobra Basics

Cobra Basics

Overview

Cobra is the most common Go library for rich CLIs (kubectl, hugo, gh patterns). It gives you **command trees**, **POSIX flags** (via pflag), **help generation**, and hooks for completion.

Use Cobra when stdlib `FlagSet` dispatch becomes repetitive—not for every 50-line script.

Install

```
mkdir mytool && cd mytool
go mod init example.com/mytool
go get github.com/spf13/cobra@latest
# optional generator:
go install github.com/spf13/cobra-cli@latest
```

Minimal root + subcommand

```
package main

import (
    "fmt"
    "os"

    "github.com/spf13/cobra"
)

func main() {
    if err := rootCmd.Execute(); err != nil {
        os.Exit(1)
    }
}

var rootCmd = &cobra.Command{
    Use:   "greet",
    Short: "A friendly greeter",
```

```go
    Long:  "greet prints a greeting with optional flair.",
    // Run when user types bare `greet` (optional)
    RunE: func(cmd *cobra.Command, args []string) error {
        return cmd.Help()
    },
}

var helloCmd = &cobra.Command{
    Use:   "hello [name]",
    Short: "Say hello",
    Args:  cobra.MaximumNArgs(1),
    RunE: func(cmd *cobra.Command, args []string) error {
        name := "world"
        if len(args) == 1 {
            name = args[0]
        }
        times, _ := cmd.Flags().GetInt("times")
        for i := 0; i < times; i++ {
            fmt.Printf("hello, %s\n", name)
        }
        return nil
    },
}

func init() {
    helloCmd.Flags().IntP("times", "n", 1, "repeat count")
    rootCmd.AddCommand(helloCmd)
}
```

```go
go run . hello Ada -n 2
go run . hello --help
```

Run vs RunE

Field	Behavior
`Run`	`func(cmd, args)` — errors must be handled inside
`RunE`	returns `error` — Cobra prints and sets exit code

Prefer **`RunE`** for testable, consistent errors.

Args validators

```
Args: cobra.ExactArgs(1),
Args: cobra.MinimumNArgs(1),
Args: cobra.NoArgs,
Args: cobra.OnlyValidArgs, // with ValidArgs
```

Custom:

```
Args: func(cmd *cobra.Command, args []string) error {
    if len(args) < 1 {
        return fmt.Errorf("requires a key")
    }
    return nil
},
```

Local flags

```
// only on this command
helloCmd.Flags().String("lang", "en", "language")
```

Project layout with Cobra

```
cmd/
  root.go       // rootCmd, Execute()
  hello.go      // helloCmd + init AddCommand
  version.go
main.go         // cmd.Execute()
```

```
// main.go
package main

import "example.com/mytool/cmd"

func main() {
    cmd.Execute()
}
```

```
// cmd/root.go
package cmd
```

```go
import (
    "os"
    "github.com/spf13/cobra"
)

var rootCmd = &cobra.Command{Use: "mytool", Short: "demo"}

func Execute() {
    if err := rootCmd.Execute(); err != nil {
        os.Exit(1)
    }
}
```

Example: `kv` with Cobra

```go
var getCmd = &cobra.Command{
    Use:   "get KEY",
    Short: "Get a value",
    Args:  cobra.ExactArgs(1),
    RunE: func(cmd *cobra.Command, args []string) error {
        raw, _ := cmd.Flags().GetBool("raw")
        val, ok := store[args[0]]
        if !ok {
            return fmt.Errorf("not found")
        }
        if raw {
            fmt.Print(val)
            return nil
        }
        fmt.Printf("%s=%s\n", args[0], val)
        return nil
    },
}

func init() {
    getCmd.Flags().Bool("raw", false, "raw value only")
    rootCmd.AddCommand(getCmd)
}
```

Silencing usage on errors

```
rootCmd.SilenceUsage = true  // don't dump full help on every runtime error
rootCmd.SilenceErrors = true // handle printing yourself
```

Useful once commands stabilize—usage on every "not found" is noisy.

Mapping to stdlib skills

Stdlib	Cobra
`switch os.Args[1]`	`AddCommand` tree
`flag.FlagSet`	`cmd.Flags()`
`flag.Args()`	`args` in `RunE`
manual help text	auto `--help`

Rules of thumb

Do	Don't
Keep domain out of `cmd` package	Put SQL in `RunE` closures forever
Use `RunE`	Ignore errors in `Run`
One file per command group as you grow	2k-line `root.go`

Try next

1. Port the stdlib `greet` tool to Cobra with `-n`.
2. Add `version` subcommand.
3. Set `SilenceUsage` and compare error UX.

Cobra Advanced: Persistent Flags, Hooks, Completion

Cobra Advanced: Persistent Flags, Hooks, Completion

Overview

After basics: **persistent flags** (inherited by children), **PreRun/PostRun hooks**, **required flags**, **groups**, and **shell completion**. These are the features that justify Cobra over hand-rolled FlagSets.

Persistent flags

```go
var (
    cfgFile string
    verbose bool
)

func init() {
    rootCmd.PersistentFlags().StringVar(&cfgFile, "config", "", "config file")
    rootCmd.PersistentFlags().BoolVarP(&verbose, "verbose", "v", false, "verbose")
    // children see --config and -v without redefining
}
```

```
mytool --verbose get key
mytool get key --verbose     # also works for persistent flags
```

Local flags stay on one command:

```go
getCmd.Flags().Bool("raw", false, "raw")
```

Required flags

```go
getCmd.Flags().String("ns", "", "namespace")
_ = getCmd.MarkFlagRequired("ns")
```

PreRun / PostRun chain

Order for a subcommand:

```
PersistentPreRun → PreRun → Run → PostRun → PersistentPostRun
```

```go
rootCmd.PersistentPreRunE = func(cmd *cobra.Command, args []string) error {
    return loadConfig(cfgFile)
}

getCmd.PreRunE = func(cmd *cobra.Command, args []string) error {
    if verbose {
        fmt.Fprintln(os.Stderr, "get", args)
    }
    return nil
}
```

Use PreRun for auth checks, config validation, and client construction.

Bind flags to structs cleanly

```go
type GetOpts struct {
    Raw bool
    NS  string
}

func newGetCmd(store Store) *cobra.Command {
    var opts GetOpts
    cmd := &cobra.Command{
        Use:  "get KEY",
        Args: cobra.ExactArgs(1),
        RunE: func(cmd *cobra.Command, args []string) error {
            return store.Get(cmd.Context(), args[0], opts)
        },
    }
    cmd.Flags().BoolVar(&opts.Raw, "raw", false, "raw")
    cmd.Flags().StringVar(&opts.NS, "ns", "default", "namespace")
    return cmd
}
```

Factory functions make testing easier than package-level `var`.

Context on commands

```go
// Cobra sets cmd.Context() from ExecuteContext
func Execute() error {
    ctx, stop := signal.NotifyContext(context.Background(), os.Interrupt)
    defer stop()
    return rootCmd.ExecuteContext(ctx)
}

// in RunE:
req, _ := http.NewRequestWithContext(cmd.Context(), ...)
```

Command groups (Cobra 1.6+)

```go
rootCmd.AddGroup(&cobra.Group{ID: "core", Title: "Core Commands"})
getCmd.GroupID = "core"
rootCmd.AddCommand(getCmd)
```

Help output clusters commands by group.

Custom usage / version templates

```go
rootCmd.SetVersionTemplate(`{{printf "mytool %s\n" .Version}}`)
rootCmd.Version = buildinfo.Version
// enables --version on root when Version is non-empty
```

Shell completion

```go
// cmd/completion.go
var completionCmd = &cobra.Command{
    Use:   "completion [bash|zsh|fish|powershell]",
    Short: "Generate completion script",
    Args:  cobra.ExactArgs(1),
    RunE: func(cmd *cobra.Command, args []string) error {
        switch args[0] {
        case "bash":
            return cmd.Root().GenBashCompletion(os.Stdout)
        case "zsh":
            return cmd.Root().GenZshCompletion(os.Stdout)
        case "fish":
```

```
            return cmd.Root().GenFishCompletion(os.Stdout, true)
        case "powershell":
            return cmd.Root().GenPowerShellCompletionWithDesc(os.Stdout)
        default:
            return fmt.Errorf("unsupported shell %q", args[0])
        }
    },
}
```

Dynamic args:

```
getCmd.ValidArgsFunction = func(cmd *cobra.Command, args []string, toComplete string) ([]str
    keys := store.KeysWithPrefix(toComplete)
    return keys, cobra.ShellCompDirectiveNoFileComp
}
```

Traverse children for help

```
rootCmd.SetHelpCommand(helpCmd) // optional customize
```

Example: parent `remote` with children

```
remoteCmd := &cobra.Command{Use: "remote", Short: "Manage remotes"}
remoteAddCmd := &cobra.Command{
    Use:  "add NAME URL",
    Args: cobra.ExactArgs(2),
    RunE: func(cmd *cobra.Command, args []string) error {
        return remotes.Add(args[0], args[1])
    },
}
remoteCmd.AddCommand(remoteAddCmd)
rootCmd.AddCommand(remoteCmd)
// mytool remote add origin https://...
```

Testing Cobra commands

```
func TestHello(t *testing.T) {
    cmd := newHelloCmd()
    b := bytes.NewBufferString("")
    cmd.SetOut(b)
```

```
    cmd.SetErr(io.Discard)
    cmd.SetArgs([]string{"-n", "2", "Ada"})
    if err := cmd.Execute(); err != nil {
        t.Fatal(err)
    }
    // assert b.String()
}
```

Prefer constructing commands via factories rather than executing package-global `rootCmd` (mutable).

Rules of thumb

Do	Don't
Persistent flags for globals	Copy the same flags on 20 commands
`ExecuteContext` + signals	Ignore cancel in long RunE
Factories for commands	Global mutable option vars only
Completion for public CLIs	Skip docs for install paths

Try next

1. Add `--config` persistent flag and load JSON in PersistentPreRunE.
2. Mark a flag required; confirm error UX.
3. Generate zsh completion and install to your fpath.

Viper and Config Layering

Viper and Config Layering

Overview

Viper is often paired with Cobra for **layered config**: files, env, flags, defaults. It is powerful and also easy to overuse. Prefer the stdlib merge from chapter 305 until you feel real pain—then Viper (or a thinner alternative) earns its dependency.

When Viper helps

Helps	Overkill
Many file formats (YAML/TOML/JSON)	Single JSON + few flags
Automatic env binding	Three getenv calls
Live watch (rare for CLIs)	One-shot tools
Large apps with many keys	Tiny scripts

Install

```
go get github.com/spf13/viper@latest
```

Minimal Viper + Cobra

```
import (
    "strings"
    "github.com/spf13/cobra"
    "github.com/spf13/viper"
)

var cfgFile string

var rootCmd = &cobra.Command{
    Use: "mytool",
    PersistentPreRunE: func(cmd *cobra.Command, args []string) error {
        return initConfig()
```

```
    },
}

func init() {
    rootCmd.PersistentFlags().StringVar(&cfgFile, "config", "", "config file")
    rootCmd.PersistentFlags().String("api-url", "", "API URL")
    _ = viper.BindPFlag("api_url", rootCmd.PersistentFlags().Lookup("api-url"))
}

func initConfig() error {
    if cfgFile != "" {
        viper.SetConfigFile(cfgFile)
    } else {
        viper.SetConfigName("config")
        viper.SetConfigType("yaml")
        viper.AddConfigPath(".")
        if dir, err := os.UserConfigDir(); err == nil {
            viper.AddConfigPath(filepath.Join(dir, "mytool"))
        }
    }

    viper.SetEnvPrefix("MYTOOL")
    viper.SetEnvKeyReplacer(strings.NewReplacer(".", "_", "-", "_"))
    viper.AutomaticEnv()

    viper.SetDefault("api_url", "http://127.0.0.1:8080")
    viper.SetDefault("timeout", "10s")

    if err := viper.ReadInConfig(); err != nil {
        // ignore not found; surface parse errors
        var nf viper.ConfigFileNotFoundError
        if !errors.As(err, &nf) {
            // also handle "file not found" for SetConfigFile path
            if !os.IsNotExist(err) {
                return err
            }
        }
    }
    return nil
}
```

```
// in a command:
url := viper.GetString("api_url")
```

Precedence (Viper)

Typical order (highest last wins when binding flags correctly):

```
defaults → config file → env → flags
```

Confirm with your version's docs; always document for users.

Example config.yaml

```yaml
api_url: https://api.example.com
timeout: 15s
verbose: true
```

Unmarshal into struct

```go
type Config struct {
    APIURL  string `mapstructure:"api_url"`
    Timeout string `mapstructure:"timeout"`
    Verbose bool   `mapstructure:"verbose"`
}

var c Config
if err := viper.Unmarshal(&c); err != nil {
    return err
}
```

Bind env explicitly

```go
_ = viper.BindEnv("api_url", "MYTOOL_API_URL")
```

Useful when AutomaticEnv naming is awkward.

Pitfalls

1. **Global viper** — the package-level instance is convenient and hard to test. Prefer `viper.New()` per app in larger codebases.
2. **Type confusion** — timeouts as strings vs durations; parse explicitly.
3. **Flag default vs viper default** — set one source of truth.
4. **Sensitive values** — don't dump full viper settings into logs.

Isolated Viper for tests

```go
func newViper() *viper.Viper {
    v := viper.New()
    v.SetDefault("api_url", "http://127.0.0.1:8080")
    return v
}
```

Pass `*viper.Viper` on your App struct instead of using globals.

Lighter alternatives

Approach	Notes
Stdlib JSON + env (ch 305)	Zero dep, enough for most tools
`caarlos0/env`	Struct tags for env only
`knadh/koanf`	Modular, explicit providers
`pelletier/go-toml` / `yaml.v3`	Parse files yourself

Stdlib-first checklist before Viper

- ☐ 15 config keys
- ☐ JSON or single format is fine
- ☐ Flags + env + one file path cover deploys

If all true, stay stdlib.

Rules of thumb

Do	Don't
Document precedence	Surprise users with silent file overrides
`viper.New()` when testing	Global state across parallel tests
Redact secrets in debug dumps	Log entire config maps
Fail on corrupt files	Ignore parse errors

Try next

1. Bind `--api-url` to viper and override with `MYTOOL_API_URL`.
2. Load YAML from `UserConfigDir`.
3. Refactor to `viper.New()` injected into commands.

Other CLI Libraries: urfave/cli, pflag, and Friends

Other CLI Libraries: urfave/cli, pflag, and Friends

Overview

Cobra is not the only option. This chapter surveys common libraries so you can choose deliberately—and recognize them in the wild.

Comparison matrix

Library	Style	Best for
stdlib flag	minimal	small tools, zero deps
spf13/pflag	POSIX/GNU flags	drop-in richer flags without full Cobra
spf13/cobra	command tree	large public CLIs
urfave/cli	declarative apps	compact multi-command tools
alecthomas/kong	struct-tagged	strongly typed CLIs
charmbracelet/	TUI	interactive terminals (not pure pipes)

pflag alone

Cobra uses pflag. You can use pflag without Cobra:

```
import flag "github.com/spf13/pflag"

var (
    name = flag.StringP("name", "n", "world", "name")
    verbose = flag.BoolP("verbose", "v", false, "verbose")
)
flag.Parse()
```

Differences from stdlib `flag`:

- Short + long flags (`-n` / `--name`)
- Combined shorts (`-abc`) for bools
- Slightly different parsing rules (more GNU-like)

```
go get github.com/spf13/pflag@latest
```

urfave/cli v2/v3 sketch

```go
package main

import (
    "fmt"
    "os"

    "github.com/urfave/cli/v2"
)

func main() {
    app := &cli.App{
        Name:  "greet",
        Usage: "fight the loneliness!",
        Commands: []*cli.Command{
            {
                Name:  "hello",
                Usage: "say hello",
                Flags: []cli.Flag{
                    &cli.IntFlag{Name: "times", Aliases: []string{"n"}, Value: 1},
                },
                Action: func(c *cli.Context) error {
                    name := "world"
                    if c.NArg() > 0 {
                        name = c.Args().Get(0)
                    }
                    for i := 0; i < c.Int("times"); i++ {
                        fmt.Printf("hello, %s\n", name)
                    }
                    return nil
                },
            },
        },
    }
    if err := app.Run(os.Args); err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }
}
```

Feel: more “app struct” oriented; less boilerplate for simple trees.

Kong (struct-first)

```go
var cli struct {
    Debug bool `help:"Enable debug mode"`
    Hello struct {
        Name string `arg:"" default:"world"`
        Times int `help:"Repeat" short:"n" default:"1"`
    } `cmd:"" help:"Say hello"`
}

// ctx := kong.Parse(&cli)
// switch ctx.Command() { case "hello": ... }
```

Excellent when you want flags as typed fields; different aesthetics from Cobra.

Charm ecosystem (when CLI becomes TUI)

Package	Role
`bubbletea`	Elm-style TUI framework
`lipgloss`	styling
`huh`	forms
`bubbletea` apps	interactive selectors, dashboards

Rule: keep pipe-friendly commands as non-TUI; offer a separate `mytool tui` if needed. Don't break `jq` pipelines with alternate screens.

mow.cli, kingpin, ff

Older or niche libs still appear in codebases:

- **jawher/mow.cli** — fluent API
- **alecthomas/kingpin** — predecessor ideas to kong
- **peterbourgon/ff** — flags + env + file in a small package

Read their READMEs before adopting; Cobra/urfave cover most new work.

Choosing

```
zero deps, few flags     → stdlib flag
need -v/--verbose only   → pflag
nested commands, public  → cobra
compact declarative      → urfave/cli
struct tags / typed      → kong
full screen UI           → charm/bubbletea
```

Hybrid pattern

Many teams:

1. Domain in pure Go packages
2. `cmd` layer uses Cobra **or** urfave
3. Stdlib tests against domain + thin command factories

Switching frameworks then costs little.

Rules of thumb

Do	Don't
Pick one framework per binary	Mix Cobra + urfave in one tree
Match ecosystem (k8s → Cobra familiarity)	Add Cobra "because tutorials do"
Keep TUI optional	Force interactive UI for CI tools

Try next

1. Reimplement `greet hello -n` with pflag only.
2. Skim kubectl or gh command layout on GitHub (Cobra patterns).
3. Port one cookbook tool (ch 316) to urfave/cli and compare LOC.

Shipping: Version, Install, Completion, Release

Shipping: Version, Install, Completion, Release

Overview

A CLI users trust has a clear **version**, easy **install**, optional **completion**, and reproducible **releases**. Most of this works with stdlib + Go toolchain; GoReleaser and package managers finish the last mile.

Version command / flag

```go
var (
    version = "dev"
    commit  = "none"
    date    = "unknown"
)

// go build -ldflags "-X main.version=1.0.0 -X main.commit=abc -X main.date=2026-01-01"
```

```go
var versionCmd = &cobra.Command{
    Use:   "version",
    Short: "Print version",
    Run: func(cmd *cobra.Command, args []string) {
        fmt.Printf("mytool %s (%s) built %s\n", version, commit, date)
    },
}
```

Also support `mytool --version` via Cobra `rootCmd.Version`.

From build info (no ldflags)

```go
import "runtime/debug"

func versionString() string {
    if bi, ok := debug.ReadBuildInfo(); ok {
        return bi.Main.Version // often "(devel)" for go run
    }
    return "dev"
```

```
}
```

Use ldflags for release binaries; buildinfo helps modules.

Install paths

```
# module install
go install example.com/mytool/cmd/mytool@v1.2.3

# local
go build -o dist/mytool ./cmd/mytool
sudo cp dist/mytool /usr/local/bin/
```

Document `GOBIN` / `PATH` issues on macOS/Linux.

Man pages and docs

- Cobra: community tools generate man/markdown from command tree
- At minimum: `mytool --help` and README examples with copy-paste commands

Completions (install notes)

```
# zsh example
mytool completion zsh > "${fpath[1]}/_mytool"
# or
mytool completion zsh > ~/.zsh/completions/_mytool
compinit
```

```
# bash
mytool completion bash | sudo tee /etc/bash_completion.d/mytool
```

Document one path per shell; link to Cobra chapter 312.

Cross-compile

```
GOOS=linux GOARCH=amd64 go build -o mytool-linux-amd64 ./cmd/mytool
GOOS=darwin GOARCH=arm64 go build -o mytool-darwin-arm64 ./cmd/mytool
GOOS=windows GOARCH=amd64 go build -o mytool-windows-amd64.exe ./cmd/mytool
```

`CGO_ENABLED=0` for static-ish pure Go binaries (usual for CLIs). On Linux that means **no libc** on the target: the same file runs on Debian, Alpine, and NixOS. Leave cgo on and a Ubuntu-built CLI can fail on Alpine with `not found` because the glibc dynamic linker is missing. See Go as a Systems Language.

Makefile sketch

```
VERSION ?= $(shell git describe --tags --always --dirty)
LDFLAGS := -s -w -X main.version=$(VERSION)

.PHONY: build test install
build:
    go build -ldflags "$(LDFLAGS)" -o bin/mytool ./cmd/mytool

test:
    go test ./...

install: build
    install -m 755 bin/mytool $(HOME)/bin/mytool
```

GoReleaser (overview)

GoReleaser automates multi-arch builds, checksums, changelog, GitHub releases, Homebrew taps, etc.

```yaml
# .goreleaser.yaml (sketch)
builds:
  - main: ./cmd/mytool
    ldflags:
      - -s -w -X main.version={{.Version}} -X main.commit={{.Commit}}
    env: [CGO_ENABLED=0]
    goos: [linux, darwin, windows]
    goarch: [amd64, arm64]
```

```
# tag and release in CI
git tag v1.2.3 && git push --tags
```

Docker (optional for CLIs)

Often unnecessary for pure Go CLIs; useful for hermetic CI:

```
FROM golang:1.22 AS build
WORKDIR /src
COPY . .
RUN CGO_ENABLED=0 go build -o /out/mytool ./cmd/mytool
FROM gcr.io/distroless/static
COPY --from=build /out/mytool /mytool
ENTRYPOINT ["/mytool"]
```

Checksums and supply chain

```
sha256sum mytool-* > checksums.txt
# cosign / slsa later for high-security tools
```

Publish checksums next to release assets.

Rules of thumb

Do	Don't
Embed version in binary	Rely on filename alone
Provide install one-liners	Assume `go install` only
Ship completion for public tools	Require manual flag memorization
CGO_ENABLED=0 for CLIs	Accidental cgo → fragile cross-build

Try next

1. Add ldflags version to a sample tool; verify `mytool version`.
2. Cross-compile for linux/amd64 from your Mac/Linux host.
3. Generate completion and smoke-test tab complete for a subcommand.

CLI Cookbook: Stdlib and Cobra Examples

CLI Cookbook: Stdlib and Cobra Examples

Overview

Runnable sketches you can paste into a module. **§A–§F are stdlib-only. §G–§H use Cobra.** Trim imports and split packages as needed.

```
mkdir -p /tmp/cookbook && cd /tmp/cookbook
go mod init example.com/cookbook
```

A. `hello` — flags + exit codes (stdlib)

```go
package main

import (
    "errors"
    "flag"
    "fmt"
    "io"
    "os"
)

var errUsage = errors.New("usage")

func main() {
    os.Exit(realMain(os.Args[1:], os.Stdout, os.Stderr))
}

func realMain(args []string, stdout, stderr io.Writer) int {
    if err := run(args, stdout, stderr); err != nil {
        fmt.Fprintln(stderr, err)
        if errors.Is(err, errUsage) {
            return 2
        }
        return 1
    }
    return 0
```

```go
}

func run(args []string, stdout, stderr io.Writer) error {
    fs := flag.NewFlagSet("hello", flag.ContinueOnError)
    fs.SetOutput(stderr)
    n := fs.Int("n", 1, "times")
    name := fs.String("name", "world", "name")
    if err := fs.Parse(args); err != nil {
        return errUsage
    }
    for i := 0; i < *n; i++ {
        fmt.Fprintf(stdout, "hello, %s\n", *name)
    }
    return nil
}
```

```go
go run . -name Ada -n 2
```

B. `catn` — files or stdin (stdlib)

```go
package main

import (
    "flag"
    "fmt"
    "io"
    "os"
)

func main() {
    flag.Parse()
    args := flag.Args()
    if len(args) == 0 {
        if _, err := io.Copy(os.Stdout, os.Stdin); err != nil {
            fail(err)
        }
        return
    }
    for _, p := range args {
        if p == "-" {
            if _, err := io.Copy(os.Stdout, os.Stdin); err != nil {
                fail(err)
```

```go
            }
            continue
        }
        f, err := os.Open(p)
        if err != nil {
            fail(err)
        }
        _, err = io.Copy(os.Stdout, f)
        f.Close()
        if err != nil {
            fail(err)
        }
    }
}

func fail(err error) {
    fmt.Fprintln(os.Stderr, "catn:", err)
    os.Exit(1)
}
```

```
echo hi | go run .
go run . /etc/hosts
```

C. `ffind` — walk files by suffix (stdlib)

```go
package main

import (
    "flag"
    "fmt"
    "io/fs"
    "os"
    "path/filepath"
    "strings"
)

func main() {
    root := flag.String("root", ".", "start directory")
    suffix := flag.String("ext", ".go", "file suffix")
    flag.Parse()

    err := filepath.WalkDir(*root, func(path string, d fs.DirEntry, err error) error {
```

```go
        if err != nil {
            return err
        }
        if d.IsDir() {
            if d.Name() == ".git" || d.Name() == "node_modules" {
                return fs.SkipDir
            }
            return nil
        }
        if strings.HasSuffix(path, *suffix) {
            fmt.Println(path)
        }
        return nil
    })
    if err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }
}
```

D. `jqq` — extract JSON field from NDJSON stdin (stdlib)

```go
package main

import (
    "encoding/json"
    "flag"
    "fmt"
    "os"
)

func main() {
    field := flag.String("f", "", "field name")
    flag.Parse()
    if *field == "" {
        fmt.Fprintln(os.Stderr, "usage: jqq -f field < ndjson")
        os.Exit(2)
    }
    dec := json.NewDecoder(os.Stdin)
    enc := json.NewEncoder(os.Stdout)
    for {
        var m map[string]any
        if err := dec.Decode(&m); err != nil {
```

```
            break
        }
        if v, ok := m[*field]; ok {
            _ = enc.Encode(v)
        }
    }
}
```

```
printf '%s\n' '{"user":"a"}' '{"user":"b"}' | go run . -f user
```

E. `httpc` — GET with timeout (stdlib)

```
package main

import (
    "context"
    "flag"
    "fmt"
    "io"
    "net/http"
    "os"
    "time"
)

func main() {
    url := flag.String("url", "", "URL")
    timeout := flag.Duration("timeout", 10*time.Second, "timeout")
    flag.Parse()
    if *url == "" {
        fmt.Fprintln(os.Stderr, "usage: httpc -url URL")
        os.Exit(2)
    }
    ctx, cancel := context.WithTimeout(context.Background(), *timeout)
    defer cancel()
    req, err := http.NewRequestWithContext(ctx, http.MethodGet, *url, nil)
    if err != nil {
        fail(err)
    }
    resp, err := http.DefaultClient.Do(req)
    if err != nil {
        fail(err)
    }
```

```go
    defer resp.Body.Close()
    fmt.Fprintln(os.Stderr, resp.Status)
    _, _ = io.Copy(os.Stdout, io.LimitReader(resp.Body, 2<<20))
}

func fail(err error) { fmt.Fprintln(os.Stderr, err); os.Exit(1) }
```

F. `kv` — subcommands with FlagSet (stdlib)

```go
package main

import (
    "encoding/json"
    "flag"
    "fmt"
    "os"
    "sync"
)

type DB struct {
    mu   sync.Mutex
    Data map[string]string `json:"data"`
    path string
}

func openDB(path string) (*DB, error) {
    db := &DB{Data: map[string]string{}, path: path}
    b, err := os.ReadFile(path)
    if err != nil {
        if os.IsNotExist(err) {
            return db, nil
        }
        return nil, err
    }
    if err := json.Unmarshal(b, db); err != nil {
        return nil, err
    }
    if db.Data == nil {
        db.Data = map[string]string{}
    }
    return db, nil
}
```

```go
func (db *DB) save() error {
    db.mu.Lock()
    defer db.mu.Unlock()
    b, err := json.MarshalIndent(db, "", "  ")
    if err != nil {
        return err
    }
    return os.WriteFile(db.path, b, 0o600)
}

func main() {
    if len(os.Args) < 2 {
        fmt.Fprintln(os.Stderr, "usage: kv <get|set|list> ...")
        os.Exit(2)
    }
    path := os.Getenv("KV_DB")
    if path == "" {
        path = "kv.json"
    }
    db, err := openDB(path)
    if err != nil {
        fail(err)
    }
    switch os.Args[1] {
    case "get":
        err = cmdGet(db, os.Args[2:])
    case "set":
        err = cmdSet(db, os.Args[2:])
    case "list":
        err = cmdList(db, os.Args[2:])
    default:
        fmt.Fprintln(os.Stderr, "unknown command")
        os.Exit(2)
    }
    if err != nil {
        fail(err)
    }
}

func cmdGet(db *DB, args []string) error {
    fs := flag.NewFlagSet("get", flag.ContinueOnError)
    if err := fs.Parse(args); err != nil {
        return err
    }
    if fs.NArg() != 1 {
        return fmt.Errorf("usage: kv get KEY")
    }
```

```go
    db.mu.Lock()
    v, ok := db.Data[fs.Arg(0)]
    db.mu.Unlock()
    if !ok {
        return fmt.Errorf("not found")
    }
    fmt.Println(v)
    return nil
}

func cmdSet(db *DB, args []string) error {
    fs := flag.NewFlagSet("set", flag.ContinueOnError)
    if err := fs.Parse(args); err != nil {
        return err
    }
    if fs.NArg() != 2 {
        return fmt.Errorf("usage: kv set KEY VALUE")
    }
    db.mu.Lock()
    db.Data[fs.Arg(0)] = fs.Arg(1)
    db.mu.Unlock()
    return db.save()
}

func cmdList(db *DB, args []string) error {
    db.mu.Lock()
    defer db.mu.Unlock()
    for k, v := range db.Data {
        fmt.Printf("%s\t%s\n", k, v)
    }
    return nil
}

func fail(err error) { fmt.Fprintln(os.Stderr, err); os.Exit(1) }
```

```go
go run . set hello world
go run . get hello
go run . list
```

G. Cobra `todo` — add / list (library)

```
go get github.com/spf13/cobra@latest
```

```go
package main

import (
    "encoding/json"
    "fmt"
    "os"
    "time"

    "github.com/spf13/cobra"
)

type Task struct {
    Text string    `json:"text"`
    At   time.Time `json:"at"`
}

func load(path string) ([]Task, error) {
    b, err := os.ReadFile(path)
    if err != nil {
        if os.IsNotExist(err) {
            return nil, nil
        }
        return nil, err
    }
    var t []Task
    return t, json.Unmarshal(b, &t)
}

func save(path string, tasks []Task) error {
    b, err := json.MarshalIndent(tasks, "", "  ")
    if err != nil {
        return err
    }
    return os.WriteFile(path, b, 0o600)
}

func main() {
    path := "tasks.json"
    root := &cobra.Command{Use: "todo", Short: "Tiny todo list"}

    add := &cobra.Command{
        Use:  "add TEXT...",
```

```go
        Args: cobra.MinimumNArgs(1),
        RunE: func(cmd *cobra.Command, args []string) error {
            tasks, err := load(path)
            if err != nil {
                return err
            }
            text := args[0]
            for _, a := range args[1:] {
                text += " " + a
            }
            tasks = append(tasks, Task{Text: text, At: time.Now().UTC()})
            return save(path, tasks)
        },
    }

    list := &cobra.Command{
        Use: "list",
        RunE: func(cmd *cobra.Command, args []string) error {
            tasks, err := load(path)
            if err != nil {
                return err
            }
            for i, t := range tasks {
                fmt.Printf("%d.\t%s\t(%s)\n", i+1, t.Text, t.At.Format(time.RFC3339))
            }
            return nil
        },
    }

    root.AddCommand(add, list)
    if err := root.Execute(); err != nil {
        os.Exit(1)
    }
}
```

```go
go run . add write docs
go run . list
```

H. Cobra parent + persistent verbose

```go
package main

import (
    "fmt"
    "os"

    "github.com/spf13/cobra"
)

func main() {
    var verbose bool
    root := &cobra.Command{Use: "tool"}
    root.PersistentFlags().BoolVarP(&verbose, "verbose", "v", false, "verbose")

    ping := &cobra.Command{
        Use: "ping",
        RunE: func(cmd *cobra.Command, args []string) error {
            if verbose {
                fmt.Fprintln(os.Stderr, "pinging...")
            }
            fmt.Println("pong")
            return nil
        },
    }
    root.AddCommand(ping)
    _ = root.Execute()
}
```

```bash
go run . -v ping
```

I. Ideas to implement yourself

Tool	Skills
`sha1dir`	walk + `crypto/sha256` + parallel with errgroup
`portwait`	dial loop + timeout + signals
`envshow`	filter `os.Environ` with regex
`jwtlook`	base64 decode payload (no verify) for debugging
`modgraph`	`go mod graph` via exec + format
`tailf`	poll file size + print deltas

Tool	Skills
`csv2json`	`encoding/csv` + json encoder
`watchrun`	poll mtime + exec command

J. Full project: network diagnostics CLI

For a multi-command **ping / dns / scan / probe** toolkit with layout, stats, workers, and tests, follow the dedicated project chapter:

→ 317 Project: Network CLI Tools

K. Full project: network security mini-tools

For **minimal** security helpers (TLS cert, headers, banner, redirects, CIDR, cookies, hash, secret scan, rand, listen):

→ 318 Project: Simple Network Security Mini-Tools

Use stdlib until the command tree aches—then port the same domain to Cobra.

Cookbook checklist

- ☐ stdout data / stderr logs
- ☐ non-zero exit on failure
- ☐ `ContinueOnError` FlagSet or Cobra `RunE`
- ☐ context on network/exec
- ☐ at least one table test on `run` / `realMain`

When those hold, your CLI is ready to grow.

Project: Network CLI Tools (Ping, DNS, Scan, Probe)

Project: Network CLI Tools (Ping, DNS, Scan, Probe)

Overview

Build **`netkit`**, a multi-command networking CLI using **stdlib only** (plus optional Cobra later). You implement tools people actually run while debugging connectivity:

Subcommand	What it does
`ping`	TCP connect RTT (works without root; not ICMP)
`dns`	Resolve A/AAAA (and optional reverse)
`scan`	Concurrent TCP port scan on a host
`probe`	HTTP(S) latency + status code check
`whoami`	Local addresses / outbound path sketch

This project ties together flags, subcommands, exit codes, timeouts, concurrency, and signals from earlier chapters.

> **Related.** Single-tool lab in projects: CLI Ping Tool. Deeper TCP: 131 TCP services. Stdlib sockets: 994 net dial/listen.

Goals

1. Ship one binary with several network diagnostic verbs
2. Always use **timeouts** and **context cancel** (Ctrl-C)
3. Keep **data on stdout**, diagnostics on stderr
4. Exit **0** on success, **1** on failure, **2** on misuse
5. Structure code so probes are unit-testable without the network (interfaces + fakes)

Scaffold

```
mkdir -p netkit/cmd/netkit netkit/internal/{app,netutil}
cd netkit
go mod init example.com/netkit
```

```
netkit/
  cmd/netkit/main.go
  internal/app/app.go          # dispatch + shared flags
  internal/app/ping.go
  internal/app/dns.go
  internal/app/scan.go
  internal/app/probe.go
  internal/netutil/dial.go     # shared dial helpers
  internal/netutil/stats.go
```

main.go

```go
package main

import (
    "context"
    "fmt"
    "os"
    "os/signal"
    "syscall"

    "example.com/netkit/internal/app"
)

func main() {
    ctx, stop := signal.NotifyContext(context.Background(), os.Interrupt, syscall.SIGTERM)
    defer stop()

    a := app.New(os.Stdout, os.Stderr)
    if err := a.Run(ctx, os.Args[1:]); err != nil {
        fmt.Fprintln(os.Stderr, "netkit:", err)
        code := 1
        if app.IsUsage(err) {
            code = 2
        }
        if ctx.Err() != nil {
            code = 130
        }
```

```go
        os.Exit(code)
    }
}
```

app.go (dispatch)

```go
package app

import (
    "context"
    "errors"
    "fmt"
    "io"
)

var ErrUsage = errors.New("usage")

func IsUsage(err error) bool { return errors.Is(err, ErrUsage) }

type App struct {
    Stdout io.Writer
    Stderr io.Writer
}

func New(stdout, stderr io.Writer) *App {
    return &App{Stdout: stdout, Stderr: stderr}
}

func (a *App) Run(ctx context.Context, args []string) error {
    if len(args) < 1 {
        return a.usage()
    }
    switch args[0] {
    case "ping":
        return a.cmdPing(ctx, args[1:])
    case "dns":
        return a.cmdDNS(ctx, args[1:])
    case "scan":
        return a.cmdScan(ctx, args[1:])
    case "probe":
        return a.cmdProbe(ctx, args[1:])
    case "whoami":
        return a.cmdWhoami(ctx, args[1:])
    case "help", "-h", "--help":
        return a.usage()
```

```go
    default:
        fmt.Fprintf(a.Stderr, "unknown command %q\n", args[0])
        return a.usage()
    }
}

func (a *App) usage() error {
    fmt.Fprintf(a.Stderr, `Usage: netkit <command> [flags]

Commands:
  ping    TCP connect ping (RTT stats)
  dns     DNS lookup (A/AAAA)
  scan    TCP port scan
  probe   HTTP(S) status + latency
  whoami  local network identity

Run "netkit <command> -h" for command flags.
`)
    return ErrUsage
}
```

Tool 1: `ping` — TCP connect RTT

Classic ICMP ping needs raw sockets (often root). **TCP ping** dials a port (default 443) and measures connect time—enough for "is the service reachable?" diagnostics.

netutil helpers

```go
package netutil

import (
    "context"
    "net"
    "time"
)

type ProbeResult struct {
    OK  bool
    RTT time.Duration
    Err error
}
```

```go
func TCPPing(ctx context.Context, address string, timeout time.Duration) ProbeResult {
    d := net.Dialer{Timeout: timeout}
    start := time.Now()
    conn, err := d.DialContext(ctx, "tcp", address)
    if err != nil {
        return ProbeResult{OK: false, Err: err}
    }
    _ = conn.Close()
    return ProbeResult{OK: true, RTT: time.Since(start)}
}

type Stats struct {
    Sent, Recv int
    Min, Max, Sum time.Duration
}

func (s *Stats) Add(r ProbeResult) {
    s.Sent++
    if !r.OK {
        return
    }
    s.Recv++
    if s.Recv == 1 || r.RTT < s.Min {
        s.Min = r.RTT
    }
    if r.RTT > s.Max {
        s.Max = r.RTT
    }
    s.Sum += r.RTT
}

func (s Stats) LossPct() float64 {
    if s.Sent == 0 {
        return 0
    }
    return float64(s.Sent-s.Recv) / float64(s.Sent) * 100
}

func (s Stats) Avg() time.Duration {
    if s.Recv == 0 {
        return 0
    }
    return s.Sum / time.Duration(s.Recv)
}
```

cmdPing

```
package app

import (
    "context"
    "flag"
    "fmt"
    "net"
    "time"

    "example.com/netkit/internal/netutil"
)

func (a *App) cmdPing(ctx context.Context, args []string) error {
    fs := flag.NewFlagSet("ping", flag.ContinueOnError)
    fs.SetOutput(a.Stderr)
    count := fs.Int("c", 4, "number of probes")
    interval := fs.Duration("i", time.Second, "interval between probes")
    timeout := fs.Duration("t", 2*time.Second, "per-probe timeout")
    port := fs.String("p", "443", "TCP port")
    jsonOut := fs.Bool("json", false, "machine-readable summary on stdout")
    if err := fs.Parse(args); err != nil {
        return ErrUsage
    }
    if fs.NArg() != 1 {
        fmt.Fprintln(a.Stderr, "usage: netkit ping [flags] <host>")
        fs.PrintDefaults()
        return ErrUsage
    }
    host := fs.Arg(0)

    ips, err := net.DefaultResolver.LookupIPAddr(ctx, host)
    if err != nil || len(ips) == 0 {
        return fmt.Errorf("resolve %s: %w", host, err)
    }
    ip := ips[0].IP.String()
    addr := net.JoinHostPort(ip, *port)

    fmt.Fprintf(a.Stderr, "PING %s (%s) TCP/%s\n", host, ip, *port)

    var st netutil.Stats
    samples := make([]time.Duration, 0, *count)

    for i := 1; i <= *count; i++ {
        if err := ctx.Err(); err != nil {
            return err
```

```
        }
        r := netutil.TCPPing(ctx, addr, *timeout)
        st.Add(r)
        if r.OK {
            samples = append(samples, r.RTT)
            fmt.Fprintf(a.Stdout, "%d: open rtt=%v\n", i, r.RTT)
        } else {
            fmt.Fprintf(a.Stdout, "%d: fail err=%v\n", i, r.Err)
        }
        if i < *count {
            select {
            case <-ctx.Done():
                return ctx.Err()
            case <-time.After(*interval):
            }
        }
    }

    fmt.Fprintf(a.Stderr, "--- %s ping stats ---\n", host)
    fmt.Fprintf(a.Stderr, "%d sent, %d ok, %.1f%% loss\n", st.Sent, st.Recv, st.LossPct())
    fmt.Fprintf(a.Stderr, "rtt min/avg/max = %v/%v/%v\n", st.Min, st.Avg(), st.Max)

    if *jsonOut {
        // summary JSON for scripts (optional: encode struct)
        fmt.Fprintf(a.Stdout, `{"host":%q,"ip":%q,"port":%q,"sent":%d,"ok":%d,"loss_pct":%.2
            host, ip, *port, st.Sent, st.Recv, st.LossPct(), st.Min, st.Avg(), st.Max)
    }

    if st.Recv == 0 {
        return fmt.Errorf("all probes failed")
    }
    _ = samples
    return nil
}
```

```
go run ./cmd/netkit ping example.com
go run ./cmd/netkit ping -c 5 -p 80 -i 200ms example.com
```

Stretch: probe every resolved IP; compute jitter (stddev of samples).

Tool 2: `dns` — lookup

```go
func (a *App) cmdDNS(ctx context.Context, args []string) error {
    fs := flag.NewFlagSet("dns", flag.ContinueOnError)
    fs.SetOutput(a.Stderr)
    typ := fs.String("type", "ip", "ip|host (reverse)")
    if err := fs.Parse(args); err != nil {
        return ErrUsage
    }
    if fs.NArg() != 1 {
        fmt.Fprintln(a.Stderr, "usage: netkit dns [-type ip|host] <name-or-ip>")
        return ErrUsage
    }
    name := fs.Arg(0)

    switch *typ {
    case "ip":
        ips, err := net.DefaultResolver.LookupIPAddr(ctx, name)
        if err != nil {
            return err
        }
        for _, ip := range ips {
            fmt.Fprintln(a.Stdout, ip.IP.String())
        }
    case "host":
        hosts, err := net.DefaultResolver.LookupAddr(ctx, name)
        if err != nil {
            return err
        }
        for _, h := range hosts {
            fmt.Fprintln(a.Stdout, h)
        }
    default:
        return fmt.Errorf("unknown -type %q", *typ)
    }
    return nil
}
```

```go
go run ./cmd/netkit dns example.com
go run ./cmd/netkit dns -type host 1.1.1.1
```

Stretch: `LookupMX`, `LookupTXT`, `LookupCNAME` behind `-type mx|txt|cname`.

Tool 3: `scan` — concurrent TCP ports

```go
func (a *App) cmdScan(ctx context.Context, args []string) error {
    fs := flag.NewFlagSet("scan", flag.ContinueOnError)
    fs.SetOutput(a.Stderr)
    ports := fs.String("ports", "22,80,443,8080", "comma-separated ports or start-end range"
    timeout := fs.Duration("t", 500*time.Millisecond, "dial timeout per port")
    workers := fs.Int("w", 32, "concurrent workers")
    if err := fs.Parse(args); err != nil {
        return ErrUsage
    }
    if fs.NArg() != 1 {
        fmt.Fprintln(a.Stderr, "usage: netkit scan [flags] <host>")
        return ErrUsage
    }
    host := fs.Arg(0)

    list, err := parsePorts(*ports)
    if err != nil {
        return err
    }

    type job struct{ port int }
    type res struct {
        port int
        open bool
    }

    jobs := make(chan job)
    results := make(chan res)
    var wg sync.WaitGroup

    worker := func() {
        defer wg.Done()
        for j := range jobs {
            addr := net.JoinHostPort(host, strconv.Itoa(j.port))
            d := net.Dialer{Timeout: *timeout}
            conn, err := d.DialContext(ctx, "tcp", addr)
            if err == nil {
                _ = conn.Close()
                results <- res{port: j.port, open: true}
            } else {
                results <- res{port: j.port, open: false}
            }
        }
    }
```

```
    n := *workers
    if n > len(list) {
        n = len(list)
    }
    wg.Add(n)
    for i := 0; i < n; i++ {
        go worker()
    }

    go func() {
        defer close(jobs)
        for _, p := range list {
            select {
            case <-ctx.Done():
                return
            case jobs <- job{port: p}:
            }
        }
    }()

    go func() {
        wg.Wait()
        close(results)
    }()

    openN := 0
    for r := range results {
        if r.open {
            openN++
            fmt.Fprintf(a.Stdout, "%d/tcp open\n", r.port)
        }
    }
    fmt.Fprintf(a.Stderr, "scanned %d ports, %d open\n", len(list), openN)
    if ctx.Err() != nil {
        return ctx.Err()
    }
    return nil
}

func parsePorts(s string) ([]int, error) {
    var out []int
    for _, part := range strings.Split(s, ",") {
        part = strings.TrimSpace(part)
        if part == "" {
            continue
        }
```

```go
        if lo, hi, ok := strings.Cut(part, "-"); ok {
            a, err1 := strconv.Atoi(lo)
            b, err2 := strconv.Atoi(hi)
            if err1 != nil || err2 != nil || a < 1 || b > 65535 || a > b {
                return nil, fmt.Errorf("bad range %q", part)
            }
            for p := a; p <= b; p++ {
                out = append(out, p)
            }
            continue
        }
        p, err := strconv.Atoi(part)
        if err != nil || p < 1 || p > 65535 {
            return nil, fmt.Errorf("bad port %q", part)
        }
        out = append(out, p)
    }
    if len(out) == 0 {
        return nil, fmt.Errorf("no ports")
    }
    return out, nil
}
```

```bash
go run ./cmd/netkit scan -ports 80,443,22 example.com
go run ./cmd/netkit scan -ports 1-1024 -w 64 -t 200ms 127.0.0.1
```

Ethics note: only scan hosts you own or have permission to test. Rate-limit (`-w`, `-t`) on shared networks.

Tool 4: `probe` — HTTP status + latency

```go
func (a *App) cmdProbe(ctx context.Context, args []string) error {
    fs := flag.NewFlagSet("probe", flag.ContinueOnError)
    fs.SetOutput(a.Stderr)
    timeout := fs.Duration("t", 10*time.Second, "total timeout")
    method := fs.String("X", http.MethodGet, "HTTP method")
    expect := fs.Int("expect", 0, "expected status (0=any 2xx/3xx)")
    if err := fs.Parse(args); err != nil {
        return ErrUsage
    }
    if fs.NArg() != 1 {
        fmt.Fprintln(a.Stderr, "usage: netkit probe [flags] <url>")
```

```go
        return ErrUsage
    }
    rawURL := fs.Arg(0)

    ctx, cancel := context.WithTimeout(ctx, *timeout)
    defer cancel()

    req, err := http.NewRequestWithContext(ctx, *method, rawURL, nil)
    if err != nil {
        return err
    }
    req.Header.Set("User-Agent", "netkit/1.0")

    client := &http.Client{
        Timeout: *timeout,
        CheckRedirect: func(req *http.Request, via []*http.Request) error {
            if len(via) >= 10 {
                return fmt.Errorf("too many redirects")
            }
            return nil
        },
    }

    start := time.Now()
    resp, err := client.Do(req)
    rtt := time.Since(start)
    if err != nil {
        return err
    }
    defer resp.Body.Close()
    _, _ = io.Copy(io.Discard, io.LimitReader(resp.Body, 64<<10))

    fmt.Fprintf(a.Stdout, "status=%d rtt=%v url=%s\n", resp.StatusCode, rtt, rawURL)

    if *expect > 0 && resp.StatusCode != *expect {
        return fmt.Errorf("status %d want %d", resp.StatusCode, *expect)
    }
    if *expect == 0 && resp.StatusCode >= 400 {
        return fmt.Errorf("status %d", resp.StatusCode)
    }
    return nil
}
```

```go
go run ./cmd/netkit probe https://example.com
go run ./cmd/netkit probe -expect 200 -t 3s https://example.com
```

Tool 5: `whoami` — local identity

```go
func (a *App) cmdWhoami(ctx context.Context, args []string) error {
    ifaces, err := net.Interfaces()
    if err != nil {
        return err
    }
    for _, iface := range ifaces {
        if iface.Flags&net.FlagUp == 0 {
            continue
        }
        addrs, err := iface.Addrs()
        if err != nil {
            continue
        }
        for _, addr := range addrs {
            fmt.Fprintf(a.Stdout, "%s\t%s\n", iface.Name, addr.String())
        }
    }

    // optional: discover outbound IP by dialing UDP (no packets need succeed)
    conn, err := net.Dial("udp", "8.8.8.8:80")
    if err == nil {
        if ua, ok := conn.LocalAddr().(*net.UDPAddr); ok {
            fmt.Fprintf(a.Stdout, "outbound\t%s\n", ua.IP)
        }
        _ = conn.Close()
    }
    return nil
}
```

Optional: Cobra wrapper

When the command tree grows, map each `cmdX` to a Cobra command (chapter 311) but **keep** `netutil` pure:

```go
// sketch
pingCmd := &cobra.Command{
    Use:  "ping HOST",
    Args: cobra.ExactArgs(1),
    RunE: func(cmd *cobra.Command, args []string) error {
        return a.cmdPing(cmd.Context(), append(flagArgs, args...))
    },
```

```
}
```

Domain stays in `netutil`; Cobra only parses and calls.

Tests (no live network required)

```go
func TestParsePorts(t *testing.T) {
    got, err := parsePorts("22,80-82")
    if err != nil {
        t.Fatal(err)
    }
    want := []int{22, 80, 81, 82}
    if !slices.Equal(got, want) {
        t.Fatalf("got %v want %v", got, want)
    }
}

func TestStats(t *testing.T) {
    var s netutil.Stats
    s.Add(netutil.ProbeResult{OK: true, RTT: 10 * time.Millisecond})
    s.Add(netutil.ProbeResult{OK: false, Err: context.DeadlineExceeded})
    if s.Recv != 1 || s.Sent != 2 {
        t.Fatalf("stats %+v", s)
    }
}
```

For dial logic, introduce:

```go
type Dialer interface {
    DialContext(ctx context.Context, network, address string) (net.Conn, error)
}
```

Inject a fake that returns after `N` ms or errors—table-test `ping` loop without the internet.

Acceptance checklist

- ☐ `netkit ping example.com` prints per-probe RTT and summary
- ☐ Ctrl-C stops mid-scan / mid-ping promptly

- ☐ `scan -ports 80,443` lists only open ports on stdout
- ☐ `dns` prints one IP per line (pipe-friendly)
- ☐ `probe -expect 200` fails non-zero on 404
- ☐ Misuse (`netkit`, `netkit ping`) exits 2 with usage on stderr
- ☐ At least `parsePorts` + `Stats` unit tests pass

```
go test ./...
go build -o netkit ./cmd/netkit
./netkit ping -c 3 1.1.1.1
./netkit scan -ports 53,80,443 1.1.1.1
./netkit probe https://1.1.1.1
```

Stretch goals

1. **JSON mode** for every command (`-json`) for CI scripts
2. **ICMP ping** via `golang.org/x/net/icmp` (needs privileges; document)
3. **Traceroute-style** increasing TTL (OS-specific; advanced)
4. **CIDR scan** for lab networks only
5. **TLS probe**: dial TLS, print cert expiry (`crypto/tls`)
6. Publish with version ldflags (chapter 315)

What you practiced

Skill	Where
Subcommands + FlagSet	each `cmd*`
Timeouts / Dialer	`TCPPing`, scan, probe
Concurrency + cancel	scan workers + ctx
Stdout/stderr contract	results vs stats
Testable packages	`netutil`, `parsePorts`

You now have a real **network diagnostics CLI**—the same genre as `ping`, `dig`, and port scanners—implemented the Go way: simple protocols, explicit deadlines, one static binary.

Next: simple security helpers (TLS cert days left, security headers, banners, secret scan) in 318 Network security mini-tools.

Project: Simple Network Security Mini-Tools

Project: Simple Network Security Mini-Tools

Overview

Build **`netsec`**, a small multi-command CLI of **minimal security helpers**—not a scanner suite, not Metasploit. Each subcommand is short, stdlib-only, and useful for day-to-day checks: TLS certs, security headers, banners, redirects, and local hygiene.

Use only against systems you own or have permission to test.

Subcommand	Purpose
`tls`	Show cert expiry, SANs, issuer
`headers`	Check common HTTP security headers
`banner`	Read first bytes from a TCP service
`redirects`	Follow HTTP redirect chain
`cidr`	Test if an IP is inside a CIDR
`cookies`	Inspect `Set-Cookie` flags
`hash`	SHA-256 a file (integrity)
`secret`	Naive secret-pattern scan on files
`rand`	Generate a random token (crypto/rand)
`listen`	Bind a port (check free / demos)

Pair with diagnostics from 317 netkit. Hardening depth: 17 Security.

Goals

1. One binary, many tiny verbs
2. Timeouts on every network call
3. Clear pass/fail exit codes for CI (`headers`, `tls --min-days`)
4. Stdout = facts; stderr = warnings
5. Keep each command under ~80 lines of real logic

Scaffold

```
mkdir -p netsec/cmd/netsec netsec/internal/app
cd netsec
go mod init example.com/netsec
```

```
netsec/
  cmd/netsec/main.go
  internal/app/app.go
  internal/app/tls.go
  internal/app/headers.go
  internal/app/banner.go
  # ... one file per command is fine
```

main + dispatch (same pattern as netkit)

```go
package main

import (
    "context"
    "fmt"
    "os"
    "os/signal"
    "syscall"

    "example.com/netsec/internal/app"
)

func main() {
    ctx, stop := signal.NotifyContext(context.Background(), os.Interrupt, syscall.SIGTERM)
    defer stop()
    if err := app.New(os.Stdout, os.Stderr).Run(ctx, os.Args[1:]); err != nil {
        fmt.Fprintln(os.Stderr, "netsec:", err)
        os.Exit(app.ExitCode(err))
    }
}
```

```go
// internal/app/app.go - switch on tls|headers|banner|...
// ErrUsage → exit 2; other errors → exit 1; nil → 0
```

---

1. `tls` — certificate peek

```
func (a *App) cmdTLS(ctx context.Context, args []string) error {
    fs := flag.NewFlagSet("tls", flag.ContinueOnError)
    fs.SetOutput(a.Stderr)
    addr := fs.String("addr", "", "host:port (default host:443)")
    serverName := fs.String("servername", "", "SNI override")
    minDays := fs.Int("min-days", 0, "fail if not valid at least this many days (0=off)")
    timeout := fs.Duration("t", 8*time.Second, "timeout")
    if err := fs.Parse(args); err != nil {
        return ErrUsage
    }
    if fs.NArg() != 1 {
        fmt.Fprintln(a.Stderr, "usage: netsec tls [flags] <host>")
        return ErrUsage
    }
    host := fs.Arg(0)
    target := *addr
    if target == "" {
        target = net.JoinHostPort(host, "443")
    }
    sni := *serverName
    if sni == "" {
        sni = host
    }

    d := &net.Dialer{Timeout: *timeout}
    raw, err := d.DialContext(ctx, "tcp", target)
    if err != nil {
        return err
    }
    defer raw.Close()
    _ = raw.SetDeadline(time.Now().Add(*timeout))

    cfg := &tls.Config{ServerName: sni, MinVersion: tls.VersionTLS12}
    conn := tls.Client(raw, cfg)
    if err := conn.HandshakeContext(ctx); err != nil {
        return err
    }
    defer conn.Close()

    state := conn.ConnectionState()
    if len(state.PeerCertificates) == 0 {
        return fmt.Errorf("no peer certificates")
    }
    cert := state.PeerCertificates[0]
```

```go
    days := int(time.Until(cert.NotAfter).Hours() / 24)

    fmt.Fprintf(a.Stdout, "subject\t%s\n", cert.Subject.CommonName)
    fmt.Fprintf(a.Stdout, "issuer\t%s\n", cert.Issuer.CommonName)
    fmt.Fprintf(a.Stdout, "not_before\t%s\n", cert.NotBefore.Format(time.RFC3339))
    fmt.Fprintf(a.Stdout, "not_after\t%s\n", cert.NotAfter.Format(time.RFC3339))
    fmt.Fprintf(a.Stdout, "days_left\t%d\n", days)
    fmt.Fprintf(a.Stdout, "version\t%s\n", tls.VersionName(state.Version))
    for _, san := range cert.DNSNames {
        fmt.Fprintf(a.Stdout, "san\t%s\n", san)
    }

    if *minDays > 0 && days < *minDays {
        return fmt.Errorf("cert expires in %d days (need >= %d)", days, *minDays)
    }
    if time.Now().After(cert.NotAfter) {
        return fmt.Errorf("certificate expired")
    }
    return nil
}
```

```bash
go run ./cmd/netsec tls example.com
go run ./cmd/netsec tls -min-days 14 example.com   # CI gate
```

---

2. `headers` — security header checklist

Minimal checks only—no full browser security model.

```go
var interesting = []string{
    "Strict-Transport-Security",
    "Content-Security-Policy",
    "X-Content-Type-Options",
    "X-Frame-Options",
    "Referrer-Policy",
    "Permissions-Policy",
    "Cross-Origin-Opener-Policy",
}

func (a *App) cmdHeaders(ctx context.Context, args []string) error {
    fs := flag.NewFlagSet("headers", flag.ContinueOnError)
    fs.SetOutput(a.Stderr)
    timeout := fs.Duration("t", 10*time.Second, "timeout")
```

```go
    strict := fs.Bool("strict", false, "exit 1 if any recommended header missing")
    if err := fs.Parse(args); err != nil {
        return ErrUsage
    }
    if fs.NArg() != 1 {
        fmt.Fprintln(a.Stderr, "usage: netsec headers [-strict] <url>")
        return ErrUsage
    }
    url := fs.Arg(0)

    ctx, cancel := context.WithTimeout(ctx, *timeout)
    defer cancel()
    req, err := http.NewRequestWithContext(ctx, http.MethodGet, url, nil)
    if err != nil {
        return err
    }
    resp, err := http.DefaultClient.Do(req)
    if err != nil {
        return err
    }
    defer resp.Body.Close()
    _, _ = io.Copy(io.Discard, io.LimitReader(resp.Body, 64<<10))

    missing := 0
    for _, h := range interesting {
        v := resp.Header.Get(h)
        if v == "" {
            missing++
            fmt.Fprintf(a.Stdout, "MISSING\t%s\n", h)
        } else {
            fmt.Fprintf(a.Stdout, "OK\t%s\t%s\n", h, v)
        }
    }
    if *strict && missing > 0 {
        return fmt.Errorf("%d recommended headers missing", missing)
    }
    return nil
}
```

```go
go run ./cmd/netsec headers https://example.com
go run ./cmd/netsec headers -strict https://example.com
```

---

3. `banner` — first bytes (service fingerprint sketch)

```go
func (a *App) cmdBanner(ctx context.Context, args []string) error {
    fs := flag.NewFlagSet("banner", flag.ContinueOnError)
    fs.SetOutput(a.Stderr)
    timeout := fs.Duration("t", 3*time.Second, "timeout")
    max := fs.Int("n", 256, "max bytes to read")
    if err := fs.Parse(args); err != nil {
        return ErrUsage
    }
    if fs.NArg() != 1 {
        fmt.Fprintln(a.Stderr, "usage: netsec banner host:port")
        return ErrUsage
    }
    addr := fs.Arg(0)

    d := net.Dialer{Timeout: *timeout}
    conn, err := d.DialContext(ctx, "tcp", addr)
    if err != nil {
        return err
    }
    defer conn.Close()
    _ = conn.SetDeadline(time.Now().Add(*timeout))

    // some services need a poke (HTTP)
    _, _ = io.WriteString(conn, "\r\n")

    buf := make([]byte, *max)
    n, err := conn.Read(buf)
    if n > 0 {
        // printable-ish for terminals
        s := strings.Map(func(r rune) rune {
            if r >= 32 && r < 127 || r == '\n' || r == '\r' || r == '\t' {
                return r
            }
            return '.'
        }, string(buf[:n]))
        fmt.Fprint(a.Stdout, s)
        if !strings.HasSuffix(s, "\n") {
            fmt.Fprintln(a.Stdout)
        }
    }
    if err != nil && !errors.Is(err, io.EOF) {
        // timeout with partial banner is still useful
        if n == 0 {
            return err
```

```
        }
    }
    return nil
}
```

```
go run ./cmd/netsec banner example.com:80
go run ./cmd/netsec banner 127.0.0.1:22
```

---

4. `redirects` — chain walk

```go
func (a *App) cmdRedirects(ctx context.Context, args []string) error {
    fs := flag.NewFlagSet("redirects", flag.ContinueOnError)
    fs.SetOutput(a.Stderr)
    max := fs.Int("max", 10, "max hops")
    timeout := fs.Duration("t", 10*time.Second, "timeout")
    if err := fs.Parse(args); err != nil {
        return ErrUsage
    }
    if fs.NArg() != 1 {
        return ErrUsage
    }
    url := fs.Arg(0)
    client := &http.Client{
        Timeout: *timeout,
        CheckRedirect: func(req *http.Request, via []*http.Request) error {
            return http.ErrUseLastResponse // manual walk
        },
    }

    for i := 0; i < *max; i++ {
        req, err := http.NewRequestWithContext(ctx, http.MethodGet, url, nil)
        if err != nil {
            return err
        }
        resp, err := client.Do(req)
        if err != nil {
            return err
        }
        loc := resp.Header.Get("Location")
        fmt.Fprintf(a.Stdout, "%d\t%d\t%s\n", i, resp.StatusCode, url)
        _, _ = io.Copy(io.Discard, io.LimitReader(resp.Body, 8<<10))
        resp.Body.Close()
```

```go
        if resp.StatusCode < 300 || resp.StatusCode >= 400 || loc == "" {
            return nil
        }
        next, err := resp.Request.URL.Parse(loc)
        if err != nil {
            return err
        }
        // weak open-redirect hint: scheme jump to different host
        if i == 0 {
            // continue
        }
        url = next.String()
    }
    return fmt.Errorf("too many redirects (>%d)", *max)
}
```

```go
go run ./cmd/netsec redirects http://example.com
```

---

5. `cidr` — membership check

```go
func (a *App) cmdCIDR(ctx context.Context, args []string) error {
    fs := flag.NewFlagSet("cidr", flag.ContinueOnError)
    fs.SetOutput(a.Stderr)
    if err := fs.Parse(args); err != nil {
        return ErrUsage
    }
    // netsec cidr <ip> <cidr>
    if fs.NArg() != 2 {
        fmt.Fprintln(a.Stderr, "usage: netsec cidr <ip> <cidr>")
        return ErrUsage
    }
    ip := net.ParseIP(fs.Arg(0))
    if ip == nil {
        return fmt.Errorf("bad ip")
    }
    _, network, err := net.ParseCIDR(fs.Arg(1))
    if err != nil {
        return err
    }
    ok := network.Contains(ip)
    fmt.Fprintf(a.Stdout, "%t\n", ok)
    if !ok {
```

```go
        return fmt.Errorf("not in network")
    }
    return nil
}
```

```
go run ./cmd/netsec cidr 10.0.0.5 10.0.0.0/8 ; echo $?
```

Useful in allowlist scripts and tiny policy checks.

---

6. `cookies` — flag hygiene

```go
func (a *App) cmdCookies(ctx context.Context, args []string) error {
    fs := flag.NewFlagSet("cookies", flag.ContinueOnError)
    fs.SetOutput(a.Stderr)
    timeout := fs.Duration("t", 10*time.Second, "timeout")
    strict := fs.Bool("strict", false, "fail if session-like cookie lacks Secure+HttpOnly")
    if err := fs.Parse(args); err != nil {
        return ErrUsage
    }
    if fs.NArg() != 1 {
        return ErrUsage
    }
    ctx, cancel := context.WithTimeout(ctx, *timeout)
    defer cancel()
    req, _ := http.NewRequestWithContext(ctx, http.MethodGet, fs.Arg(0), nil)
    resp, err := http.DefaultClient.Do(req)
    if err != nil {
        return err
    }
    defer resp.Body.Close()
    _, _ = io.Copy(io.Discard, io.LimitReader(resp.Body, 64<<10))

    bad := 0
    for _, c := range resp.Cookies() {
        fmt.Fprintf(a.Stdout, "name=%s secure=%v httponly=%v samesite=%v\n",
            c.Name, c.Secure, c.HttpOnly, c.SameSite)
        // heuristic: names containing session/token
        n := strings.ToLower(c.Name)
        if strings.Contains(n, "session") || strings.Contains(n, "token") || strings.Contain
            if !c.Secure || !c.HttpOnly {
                bad++
                fmt.Fprintf(a.Stderr, "warn: %s should be Secure+HttpOnly\n", c.Name)
            }
```

```go
        }
    }
    if *strict && bad > 0 {
        return fmt.Errorf("%d weak cookies", bad)
    }
    return nil
}
```

---

7. `hash` — file integrity

```go
func (a *App) cmdHash(ctx context.Context, args []string) error {
    fs := flag.NewFlagSet("hash", flag.ContinueOnError)
    fs.SetOutput(a.Stderr)
    if err := fs.Parse(args); err != nil {
        return ErrUsage
    }
    if fs.NArg() < 1 {
        fmt.Fprintln(a.Stderr, "usage: netsec hash <files...>")
        return ErrUsage
    }
    for _, path := range fs.Args() {
        f, err := os.Open(path)
        if err != nil {
            return err
        }
        h := sha256.New()
        if _, err := io.Copy(h, f); err != nil {
            f.Close()
            return err
        }
        f.Close()
        fmt.Fprintf(a.Stdout, "%x  %s\n", h.Sum(nil), path)
    }
    return nil
}
```

```go
go run ./cmd/netsec hash ./cmd/netsec/main.go
```

---

8. `secret` — naive leak hunt (local files)

Not a replacement for gitleaks/trufflehog—teaching regex + walk.

```go
var patterns = []*regexp.Regexp{
    regexp.MustCompile(`(?i)api[_-]?key\s*[:=]\s*['"][^'"]{8,}`),
    regexp.MustCompile(`(?i)secret\s*[:=]\s*['"][^'"]{8,}`),
    regexp.MustCompile(`AKIA[0-9A-Z]{16}`), // AWS key id shape
    regexp.MustCompile(`-----BEGIN (RSA |EC )?PRIVATE KEY-----`),
}

func (a *App) cmdSecret(ctx context.Context, args []string) error {
    fs := flag.NewFlagSet("secret", flag.ContinueOnError)
    fs.SetOutput(a.Stderr)
    root := fs.String("root", ".", "directory")
    if err := fs.Parse(args); err != nil {
        return ErrUsage
    }
    found := 0
    err := filepath.WalkDir(*root, func(path string, d fs.DirEntry, err error) error {
        if err != nil {
            return nil
        }
        if d.IsDir() {
            name := d.Name()
            if name == ".git" || name == "node_modules" || name == "vendor" {
                return fs.SkipDir
            }
            return nil
        }
        // skip large / binary-ish
        info, err := d.Info()
        if err != nil || info.Size() > 1<<20 {
            return nil
        }
        b, err := os.ReadFile(path)
        if err != nil {
            return nil
        }
        if bytes.IndexByte(b, 0) >= 0 {
            return nil // binary
        }
        for _, re := range patterns {
            if re.Find(b) != nil {
                found++
                fmt.Fprintf(a.Stdout, "%s\t%s\n", path, re.String())
            }
```

```go
        }
        return nil
    })
    if err != nil {
        return err
    }
    if found > 0 {
        return fmt.Errorf("found %d potential secret matches", found)
    }
    return nil
}
```

```go
go run ./cmd/netsec secret -root .
```

---

9. `rand` — tokens with crypto/rand

```go
func (a *App) cmdRand(ctx context.Context, args []string) error {
    fs := flag.NewFlagSet("rand", flag.ContinueOnError)
    fs.SetOutput(a.Stderr)
    n := fs.Int("n", 32, "bytes of entropy")
    hexOut := fs.Bool("hex", true, "hex encode (false=raw base64)")
    if err := fs.Parse(args); err != nil {
        return ErrUsage
    }
    b := make([]byte, *n)
    if _, err := rand.Read(b); err != nil {
        return err
    }
    if *hexOut {
        fmt.Fprintf(a.Stdout, "%x\n", b)
    } else {
        fmt.Fprintln(a.Stdout, base64.RawURLEncoding.EncodeToString(b))
    }
    return nil
}
```

Never use `math/rand` for tokens or session IDs.

---

10. `listen` — bind check / tiny sink

```go
func (a *App) cmdListen(ctx context.Context, args []string) error {
    fs := flag.NewFlagSet("listen", flag.ContinueOnError)
    fs.SetOutput(a.Stderr)
    addr := fs.String("addr", "127.0.0.1:0", "listen address (:0 = ephemeral)")
    if err := fs.Parse(args); err != nil {
        return ErrUsage
    }
    ln, err := net.Listen("tcp", *addr)
    if err != nil {
        return err // port in use / permission
    }
    defer ln.Close()
    fmt.Fprintf(a.Stdout, "listening\t%s\n", ln.Addr().String())
    // wait until cancel - useful smoke for firewall demos
    <-ctx.Done()
    return nil
}
```

```bash
go run ./cmd/netsec listen -addr 127.0.0.1:9999
# Ctrl-C to stop
```

Wire the switch

```go
func (a *App) Run(ctx context.Context, args []string) error {
    if len(args) < 1 {
        return a.usage()
    }
    switch args[0] {
    case "tls":
        return a.cmdTLS(ctx, args[1:])
    case "headers":
        return a.cmdHeaders(ctx, args[1:])
    case "banner":
        return a.cmdBanner(ctx, args[1:])
    case "redirects":
        return a.cmdRedirects(ctx, args[1:])
    case "cidr":
        return a.cmdCIDR(ctx, args[1:])
    case "cookies":
        return a.cmdCookies(ctx, args[1:])
```

```go
    case "hash":
        return a.cmdHash(ctx, args[1:])
    case "secret":
        return a.cmdSecret(ctx, args[1:])
    case "rand":
        return a.cmdRand(ctx, args[1:])
    case "listen":
        return a.cmdListen(ctx, args[1:])
    default:
        return a.usage()
    }
}
```

Minimal tests

```go
func TestCIDR(t *testing.T) {
    _, n, _ := net.ParseCIDR("10.0.0.0/8")
    if !n.Contains(net.ParseIP("10.1.2.3")) {
        t.Fatal("expected contain")
    }
}

func TestRandLen(t *testing.T) {
    b := make([]byte, 16)
    if _, err := rand.Read(b); err != nil {
        t.Fatal(err)
    }
}
```

Network-backed commands: test with `httptest` for `headers` / `cookies` (inject a test server that sets headers).

Suggested lab order

```
1. hash + rand          (no network)
2. cidr                 (pure logic)
3. tls + headers        (HTTPS)
4. banner + redirects   (TCP/HTTP)
5. cookies + secret     (hygiene)
```

```
6. listen               (local)
```

Ethics & scope

Do	Don't
Test your apps and labs	Scan random internet hosts
Use `-strict` in **your** CI	Treat this as a pentest platform
Document false positives (`secret`)	Commit real secrets to fix tests

Stretch (still minimal)

1. `tls --json` for monitors
2. `headers` allowlist config file
3. Merge `netsec` under `netkit sec …` as a Cobra parent
4. `tls` print leaf + intermediates briefly
5. Simple **HSTS** max-age parser

Acceptance checklist

- ☐ `tls example.com` prints `days_left`
- ☐ `tls -min-days 30` fails near expiry (simulate with low threshold)
- ☐ `headers -strict` non-zero when headers missing
- ☐ `hash` matches `shasum -a 256` on a file
- ☐ `secret` finds a planted fake `api_key=...` in a temp file
- ☐ `rand -n 16` prints 32 hex chars
- ☐ Misuse exits 2

These tools stay **small on purpose**: enough to learn `crypto/tls`, `net/http`, and safe CLI habits—without building a security product.

Project: Text and Log CLI Tools

Project: Text and Log CLI Tools

Overview

`textkit`: small pipeline tools—stdlib only—for daily log/text work.

Cmd	Behavior
`freq`	word/line frequency counts
`cut`	`-f` fields `-d` delim
`ts`	prefix lines with RFC3339 timestamp
`sample`	keep every Nth line or rate %
`jsonl-keys`	list keys seen in NDJSON

Contracts

- stdin if no files
- data stdout / errors stderr
- exit 0 if IO ok

`freq` core

```go
sc := bufio.NewScanner(r)
counts := map[string]int{}
for sc.Scan() {
    for _, w := range strings.Fields(sc.Text()) {
        counts[w]++
    }
}
// print sorted by count desc
```

ts

```
fmt.Fprintf(w, "%s %s\n", time.Now().Format(time.RFC3339), line)
```

Acceptance

```
printf 'a b\na\n' | go run . freq
echo hello | go run . ts
```

Project: HTTP Debug CLI

Project: HTTP Debug CLI

Overview

`httpdbg`: curl-like teaching client with timing breakdowns.

Features

```
httpdbg get URL
httpdbg get -H "Accept: application/json" -v URL
httpdbg timings URL   # DNS / connect / TLS / TTFB sketch
```

Timings sketch

Use `http.Transport` + `httptrace.ClientTrace`:

```
trace := &httptrace.ClientTrace{
    DNSDone: func(info httptrace.DNSDoneInfo) { /* mark */ },
    ConnectDone: func(network, addr string, err error) {},
    GotFirstResponseByte: func() {},
}
req = req.WithContext(httptrace.WithClientTrace(ctx, trace))
```

Requirements

- Timeouts
- `-o` write body to file
- Print status and selected headers on `-v`
- Never follow redirects beyond `-max-redir`

Acceptance

```
go run . get -v https://example.com
go run . timings https://example.com
```

Cross-Platform CLI Notes

Cross-Platform CLI Notes

Overview

Go builds everywhere; OS details still leak into CLIs. Isolate differences.

Paths

```
filepath.Join(a, b)     // not strings + "/"
filepath.Separator
os.UserConfigDir()      // not hardcode ~/.config
```

Line endings

```
// accept both when reading tools
// write "\n" unless you must speak CRLF to Windows tools
```

Signals

Windows: `os.Interrupt` works for Ctrl-C; `SIGTERM` semantics differ. Use `signal.NotifyContext` with available signals; build-tag extra Unix signals.

Shells

- Avoid `cmd /c` with user strings
- PowerShell quoting bash
- Document examples for both

Color

```
// enable only if TTY && NO_COLOR unset && TERM != dumb
```

Windows: optional virtual terminal processing for ANSI.

Build tags

```
//go:build windows
// flock_windows.go stub → error "unsupported"
```

File locking

`flock` Unix-only; Windows has different APIs—feature-detect or document Unix-only commands.

Rules

Do	Don't
`filepath` + `os` helpers	Assume `/tmp` exists as on Linux
CI matrix linux/mac/windows	Test only on your laptop OS
Clear "unix only" in help	Silent fail on Windows

Try next

1. `GOOS=windows go build` your CLI.
2. Replace hardcoded `/tmp` with `os.TempDir()`.
3. Stub flock commands on Windows with message.

Concurrency Ground Up

Concurrency From the Ground Up

Concurrency From the Ground Up

This part is a **diagram-first curriculum** for Go concurrency: goroutines, channels, pipelines, time, context, wait groups, races, semaphores, atomics, testing, a simplified scheduler model, race-condition reasoning, and condition-variable signaling.

It is designed for programmers who already know Go basics (types, interfaces, errors) and want the *application* of concurrency primitives—not a feature laundry list.

> **Pedagogy note.** The chapter map and teaching emphasis follow the structure popularized by interactive concurrency guides such as Gist of Go: Concurrency (Anton Zhiyanov). Explanations, diagrams, and code here are original to this book; they are not a copy of that site. Use both: that site for interactive exercises, this part for a library-style reference with ASCII diagrams integrated into the monorepo.

Tooling baseline (late 2026): examples assume a recent Go toolchain (Go 1.27-class: `WaitGroup.Go`, generic methods, `go test -race`). Use `go mod init` with `go 1.27` when creating lab modules.

Learning Outcomes

After this part you can:

1. Separate **business logic** from **concurrency mechanics**
2. Choose among wait groups, done/cancel channels, and context
3. Build cancel-safe pipelines without leaked goroutines
4. Apply throttling, backpressure, and timeouts correctly
5. Detect **data races** and fix them with ownership, mutexes, or atomics
6. Distinguish **race conditions** from data races (including `-race` clean logic bugs)
7. Use channels first; reach for `sync.Cond` only when predicates need broadcast
8. Explain (at a teaching level) how Gs map onto OS threads

Contents Map

```
Basics:     goroutines → channels → pipelines → time → streaming backpressure → context
Sync:       wait groups → races/mutex → semaphores → atomics
Practice:   testing → scheduler model → race conditions → Cond
```

Chapter	File	Focus
251	251-goroutines-fundamentals	`go`, main lifetime, WaitGroup, result channel, stage-owned output
252	252-channels-complete	close, range, directions, done, deadlocks, nil channels
253	253-pipelines-select-cancel	leaks, cancel+select, merge, fan-out/fan-in
254	254-time-throttle-backpressure	throttle, backpressure, timeouts, timers
254b	254b-streaming-api-backpressure	streaming APIs: bounded channels, io.Pipe, Flush, fail-fast
255	255-context-complete	context tree, deadline, values, vs cancel channel
256	256-waitgroups-encapsulation	WaitGroup mental model, pointers, encapsulation
257	257-data-races-mutex	data races, race detector, mutex, RWMutex
258	258-semaphores-rendezvous	semaphore N, rendezvous, barriers
259	259-atomics-complete	atomic RMW, composition traps, CAS
260	260-testing-concurrent-code	testing strategies, determinism
261	261-scheduler-model-diagrams	cores → threads → Gs, GOMAXPROCS
262	262-race-conditions-vs-data-races	race conditions vs data races
263	263-signaling-cond-broadcast	`sync.Cond`, Signal vs Broadcast

Suggested path

```
Week-shaped path (adjust pace):

  Day 1-2   251 → 252          start + talk
  Day 3-4   253 → 254 → 255    pipelines + time + context
  Day 5     256 → 257          WaitGroup + races
  Day 6     258 → 259          limits + atomics
  Day 7     260 → 261          test + scheduler model
  Day 8     262 → 263          logic races + Cond
```

Always keep **`go test -race`** (or `go run -race`) in the loop for shared-memory chapters (257+).

Core Mental Models (Preview)

Three layers of concurrency

```
  instr   instr   instr   instr

 Core 1 Core 2 Core 3 Core 4   CPU (true parallel ops)



  Thr A  Thr B  Thr C  Thr D   OS threads (OS schedules)


  G11 G15   G12 G16   G13      G14 G17…   Goroutines (Go schedules)
```

Channel as data + synchronization

```
 goroutine A    ch       goroutine B
   recv                 send

       unbuffered send blocks until recv (rendezvous)
```

Cancel vs done

```
done:   worker  signals   waiter   "I finished"
cancel: waiter  signals   worker   "stop now"
```

Data race vs race condition

```
data race:        concurrent mem access, 1 write, no happens-before
race condition:   wrong result due to interleaving (may be -race clean)
```

See 262.

How to Study

1. Read a chapter, redraw one diagram from memory.
2. Run every runnable example; then break it (remove close, drop cancel, share a map).
3. Prefer one solid pipeline lab over ten passive reads.
4. Always keep `go test -race` in the loop for Part B (sync and later).
5. When stuck, write the **sequential** version first, then add concurrency at the edges.

Lab module template

```
mkdir conc-lab && cd conc-lab
go mod init example.com/conc-lab
# go.mod: set `go 1.27` (or your installed version)
```

```
// go.mod
module example.com/conc-lab

go 1.27
```

```
go run -race .
go test -race ./...
```

Relationship to Other Parts

Need	Also see
Runtime-deep GMP / netpoll / timers	20 Go Deep Dives
Worker pools in production services	07 Concurrency
Stdlib `sync` / `context` recipes	98 Stdlib
Capstone-style builds	99 Projects

Attribution

Teaching structure inspired by public concurrency curricula (notably antonz.org's *Gist of Go: Concurrency*). All prose, diagrams, and examples in this monorepo chapter set are written for this book under the same author voice as the rest of the Go library.

Goroutines Fundamentals

Goroutines Fundamentals

Overview

A **goroutine** is a function executing independently under the Go runtime. The keyword `go` starts that execution; it does not create an OS thread by itself. Thousands of goroutines can multiplex onto far fewer threads.

The order of ideas below follows the opening chapter of Anton Zhiyanov's Gist of Go: Goroutines. The programs are this book's. If you already use `async/await` in another language, leave that model at the door: a goroutine is not a promise you `await`, and `main` does not suspend until the calls you started come back.

Sequential vs concurrent

```go
func say(id int, phrase string) {
    for _, word := range strings.Fields(phrase) {
        fmt.Printf("Worker #%d: %s\n", id, word)
        time.Sleep(time.Duration(rand.Intn(50)) * time.Millisecond)
    }
}

// Sequential: second talker starts only after the first finishes.
func sequential() {
    say(1, "go is awesome")
    say(2, "cats are cute")
}

// Concurrent: both run; main must still wait somehow.
func concurrent() {
    go say(1, "go is awesome")
    go say(2, "cats are cute")
    time.Sleep(300 * time.Millisecond) // fragile - prefer WaitGroup
}
```

```
sequence (top → bottom):
  actors: Main, say#1, say#2
  Main --> say#1  : call
  say#1 --> Main  : return
```

```
  Main --> say#2  : call
  say#2 --> Main  : return
  Main --> say#1  : go
  Main --> say#2  : go
  Main --> Main  : must wait or exit
  note: sequential
  note: concurrent
```

Main is a goroutine

When `main` returns, the **process** ends. Other goroutines are abandoned—even if they had more work.

```
Without wait:

  main  go   G1
        go   G2
        return   process exit  (G1/G2 may print nothing)
```

WaitGroup: correct waiting

```go
func concurrentWait() {
    var wg sync.WaitGroup
    wg.Add(2)
    go func() {
        defer wg.Done()
        say(1, "go is awesome")
    }()
    go func() {
        defer wg.Done()
        say(2, "cats are cute")
    }()
    wg.Wait()
}
```

Separate concurrency from business logic. A worker that takes `*sync.WaitGroup` cannot be called from a test or from ordinary sequential code without inventing a group:

```go
func say(wg *sync.WaitGroup, id int, phrase string) {
    defer wg.Done()
    // ...print the phrase...
}
```

Keep `say` as a plain function. The goroutine wrapper owns the counter:

```go
wg.Add(1)
go func() {
    defer wg.Done()
    say(1, "go is awesome")
}()
```

`defer wg.Done()` still runs if `say` panics, so the waiter is not stuck on a counter that never reaches zero. `Add` happens **before** `go`. If you `Add` inside the new goroutine, `Wait` can observe zero and return before the increment.

WaitGroup.Go (Go 1.25+)

```go
var wg sync.WaitGroup
wg.Go(func() { say(1, "go is awesome") })
wg.Go(func() { say(2, "cats are cute") })
wg.Wait()
```

Under the hood, `WaitGroup.Go` is the wrapper above: `Add(1)`, start the function, `defer Done()`. One goroutine per word, before any channel, can store into a `sync.Map` (safe for concurrent `Store`) and then `Wait`. Since Go 1.22 the loop variable is per iteration, so the closure may capture `word` directly:

```go
stats := new(sync.Map)
for _, word := range words {
    wg.Go(func() {
        stats.Store(word, countVowels(word))
    })
}
wg.Wait()
```

That works, and it mixes a concurrent map into the tally. The rest of this chapter passes values on channels instead, so the vowel counter stays a plain function.

First channel: producer and consumer

```
  goroutine B       goroutine A
    send X          recv X
```

```go
func main() {
    messages := make(chan string)
    go func() { messages <- "ping" }()
    msg := <-messages
```

```
    fmt.Println(msg)
}
```

Unbuffered send blocks until a receiver is ready (and vice versa). Channels transfer data **and** synchronize. The sender does not continue past `messages <- "ping"` until `main` has taken the value:

```
sequence (top → bottom):
  actors: sender, channel, receiver
  receiver --> receiver  : other work
  sender --> channel  : send "ping" (blocks)
  receiver --> channel  : receive
  channel --> receiver  : "ping"
  channel --> sender  : send completes
```

No `WaitGroup` is required for this program. `main`'s receive is the join: it cannot return until the send has happened.

Result channel, same list on both sides

A producer-consumer split is the next step. The producer counts vowels in each word and sends the **count**. The consumer still has the word slice, so it can zip counts back onto words. One sender, in slice order, matches one receiver, in the same order. An unbuffered channel keeps that handshake.

```
func countVowels(word string) int {
    n := 0
    for _, r := range word {
        switch unicode.ToLower(r) {
        case 'a', 'e', 'i', 'o', 'u':
            n++
        }
    }
    return n
}

func tallyKnown(phrase string) map[string]int {
    words := strings.Fields(phrase)
    counted := make(chan int)
    go func() {
        for _, word := range words {
            counted <- countVowels(word)
        }
    }()
    stats := map[string]int{}
    for _, word := range words {
```

```
        stats[word] = <-counted
    }
    return stats
}
```

The receive loop runs exactly `len(words)` times, so it does not need an end marker. Launching one goroutine **per** word would scramble that zip: counts would no longer line up with the consumer's range. That is the case for a pair, or for the `sync.Map` above.

Generator: the consumer does not know the length

A live source (a scanner, a query) does not hand you the slice up front. Pass a generator. Each call returns the next word, or `""` when the source is done. Only one goroutine may call `next`.

```
func wordSource(phrase string) func() string {
    words := strings.Fields(phrase)
    i := 0
    return func() string {
        if i >= len(words) {
            return ""
        }
        w := words[i]
        i++
        return w
    }
}
```

Sending only `int` now fails twice: the consumer does not know which word a count belongs to, and it does not know when to stop receiving. Send a pair, and send the empty word once so the receiver can break. Later chapters replace that empty word with `close`.

```
type pair struct {
    word  string
    count int
}

func tallyStream(next func() string) map[string]int {
    counted := make(chan pair)
    go func() {
        for {
            word := next()
            counted <- pair{word, countVowels(word)}
            if word == "" {
                return
            }
        }
```

```
    }()
    stats := map[string]int{}
    for {
        p := <-counted
        if p.word == "" {
            break
        }
        stats[p.word] = p.count
    }
    return stats
}
```

The empty word is a sentinel, not data. Do not use it once real input can be empty. Channels Complete uses `close` for the same signal.

Reader and worker

Split "pull the next word" from "count vowels". Two channels, two goroutines, and the outer function only fills the map:

```
          pending              counted
next()                 count                fill map
reader         string       worker         pair         caller
```

```
func tallySplit(next func() string) map[string]int {
    pending := make(chan string)
    counted := make(chan pair)

    go func() {
        for {
            word := next()
            pending <- word
            if word == "" {
                return
            }
        }
    }()
    go func() {
        for {
            word := <-pending
            counted <- pair{word, countVowels(word)}
            if word == "" {
                return
            }
```

```go
        }
    }()

    stats := map[string]int{}
    for {
        p := <-counted
        if p.word == "" {
            break
        }
        stats[p.word] = p.count
    }
    return stats
}
```

Send the empty word **before** returning, on both hops. If the reader returns without that send, the worker blocks on `<-pending` forever, and `main` blocks on `<-counted`.

Name the stages

The anonymous functions are three jobs. Naming them makes the parent read as a pipeline:

```go
func tallyNamed(next func() string) map[string]int {
    pending := make(chan string)
    go submitWords(next, pending)
    counted := make(chan pair)
    go countWords(pending, counted)
    return fillStats(counted)
}

func submitWords(next func() string, out chan string) {
    for {
        word := next()
        out <- word
        if word == "" {
            return
        }
    }
}

func countWords(in chan string, out chan pair) {
    for {
        word := <-in
        out <- pair{word, countVowels(word)}
        if word == "" {
            return
```

```go
        }
    }
}

func fillStats(in chan pair) map[string]int {
    stats := map[string]int{}
    for {
        p := <-in
        if p.word == "" {
            break
        }
        stats[p.word] = p.count
    }
    return stats
}
```

The stage owns its output channel

`pending` exists in the parent only so it can be handed to `submitWords`. Create the channel inside the stage that sends on it, start the goroutine there, and return the channel. The parent stops choosing lifetimes:

```go
func tallyOwned(next func() string) map[string]int {
    pending := submitWordsOwned(next)
    counted := countWordsOwned(pending)
    return fillStats(counted)
}

func submitWordsOwned(next func() string) chan string {
    out := make(chan string)
    go func() {
        for {
            word := next()
            out <- word
            if word == "" {
                return
            }
        }
    }()
    return out
}

func countWordsOwned(in chan string) chan pair {
    out := make(chan pair)
    go func() {
```

```
        for {
            word := <-in
            out <- pair{word, countVowels(word)}
            if word == "" {
                return
            }
        }
    }()
    return out
}
```

Returning a channel that an inner goroutine fills is a standard Go shape. The cost is that `submitWordsOwned` and `countWordsOwned` are no longer plain functions: calling them starts work. Pick one style per pipeline. Either the parent starts goroutines and passes channels in, or each stage owns its output. Mixing the two makes it unclear who closes or who sends the sentinel.

`fillStats` stays synchronous. It is the consumer, and it runs on the caller's goroutine, which is why `tallyOwned` does not return until every word has been counted.

Rules of thumb

Do	Don't
Wait with WaitGroup / channel / context	`time.Sleep` to "hope" work finished
Keep pure functions free of sync types	Pass WaitGroup into every helper
`Add` before `go`, `Done` in a `defer`	`Add` inside the goroutine you are waiting for
Start with unbuffered channels	Buffer "just in case" without a reason
Let one stage own each channel	Mix "parent starts the goroutine" with "child creates the channel"

Runnable example

```
mkdir -p /tmp/vowel-tally && cd /tmp/vowel-tally
go mod init example
```

Save as `main.go`, then `go run .`.

```
package main

import (
    "fmt"
    "strings"
```

```
    "unicode"
)

func countVowels(word string) int {
    n := 0
    for _, r := range word {
        switch unicode.ToLower(r) {
        case 'a', 'e', 'i', 'o', 'u':
            n++
        }
    }
    return n
}

type pair struct {
    word  string
    count int
}

func wordSource(phrase string) func() string {
    words := strings.Fields(phrase)
    i := 0
    return func() string {
        if i >= len(words) {
            return ""
        }
        w := words[i]
        i++
        return w
    }
}

func submitWords(next func() string) chan string {
    out := make(chan string)
    go func() {
        for {
            word := next()
            out <- word
            if word == "" {
                return
            }
        }
    }()
    return out
}

func countWords(in chan string) chan pair {
```

```go
    out := make(chan pair)
    go func() {
        for {
            word := <-in
            out <- pair{word, countVowels(word)}
            if word == "" {
                return
            }
        }
    }()
    return out
}

func fillStats(in chan pair) map[string]int {
    stats := map[string]int{}
    for {
        p := <-in
        if p.word == "" {
            break
        }
        stats[p.word] = p.count
    }
    return stats
}

func main() {
    stats := fillStats(countWords(submitWords(wordSource("router spine leaf"))))
    for _, word := range []string{"router", "spine", "leaf"} {
        fmt.Printf("%s: %d\n", word, stats[word])
    }
}
```

```
router: 3
spine: 2
leaf: 2
```

What to notice: `main` blocks in `fillStats` until the empty word arrives, so the process does not exit early. `router` has three vowels (`o`, `u`, `e`).

Try next: Delete the `out <- word` of the empty string in `submitWords` and run again. The worker blocks, `fillStats` blocks, and the runtime reports that all goroutines are asleep. Channels Complete is the structured replacement for that sentinel.

Channels Complete

Channels Complete

Overview

Channels are multi-purpose: **data transfer**, **completion (done)**, and (with care) **cancellation**. This chapter covers close semantics, range, directional types, deadlocks, and nil-channel tricks.

End-of-stream: why close exists

Infinite receive loops deadlock when the producer finishes:

```
producer sends all words    consumer for { recv }
producer exits
consumer blocks forever on recv  →  "all goroutines are asleep - deadlock"
```

Magic sentinel values (`"__EOF__"`) are fragile when the sentinel is also valid data. **Close** is the structured signal.

Closing a channel

```go
// writer
for _, w := range words {
    in <- w
}
close(in)

// reader
for {
    w, ok := <-in
    if !ok {
        break // closed and drained
    }
    // use w
}
```

Operation	Open channel	Closed channel
Send	may block	**panic**

Operation	Open channel	Closed channel
Receive	value, ok=true	zero value, ok=false (repeatable)
Close	ok once	**panic** on second close

```
state machine:
  start --> Open
  Open --> Closed
  Closed --> Closed
  Open --> Open
```

Rules

1. **Only the writer closes** (sole ownership).
2. **Close is a signal**, not a resource free—GC collects unused channels either way.
3. Close when readers need end-of-data; otherwise optional.

Range over channel

```go
for w := range in {
    // until closed
}
```

Unlike slice range, channel range yields **one** value (not index).

Directional channels

Catch misuse at **compile time**:

```go
func submit(str string, stream chan<- string) { /* send + close */ }
func receive(stream <-chan string) { /* range only */ }

stream := make(chan string) // bidirectional at creation
go submit(str, stream)      // converts to send-only param
receive(stream)             // converts to recv-only param
```

Type	Allowed
`chan T`	send, recv, close
`chan<- T`	send, close
`<-chan T`	recv only

Done channel

Wait for work without a WaitGroup:

```go
func say(done chan<- struct{}, id int, phrase string) {
    // ... work ...
    done <- struct{}{} // or close(done) for multi-waiter broadcast
}

done := make(chan struct{})
go say(done, 1, "hello")
<-done
```

```
done: worker   waiter   "finished"
```

`struct{}` uses zero bytes of payload—signal only.

Deadlocks

Go runtime detects when **all** goroutines are blocked:

```
fatal error: all goroutines are asleep - deadlock!
```

Common causes:

- Unbuffered send without receiver
- Receive without sender
- WaitGroup counter never reaches zero
- Mutual wait (A waits B, B waits A)

Nil channels

```go
var ch chan int // nil
// <-ch   blocks forever
// ch <- 1 blocks forever
```

Useful in select: set a case's channel to `nil` to **disable** that case after it closes (see pipelines merge).

Buffered channels

```
ch := make(chan int, 3) // send up to 3 without a receiver
```

```
capacity 3:

send s1 s2 s3  →  buffer full  →  4th send blocks
recv frees a slot
```

Buffer size is a **capacity / decoupling** choice, not free async magic.

Runnable example

```
go mod init example && go run .
```

```go
package main

import "fmt"

func main() {
    in := make(chan string)
    go func() {
        for _, w := range []string{"one", "two", "", "four"} {
            in <- w
        }
        close(in)
    }()
    for w := range in {
        if w != "" {
            fmt.Print(w, " ")
        }
    }
    fmt.Println()
    // closed receives
    v, ok := <-in
    fmt.Printf("after close: %q ok=%v\n", v, ok)
}
```

Expected:

```
one two four
after close: "" ok=false
```

Try next: Make `receive` take `<-chan string` and try to close inside it—confirm compile error.

Pipelines, Select, and Cancel

Pipelines, Select, and Cancel

Overview

Pipelines compose stages connected by channels. The hard part is **not leaking goroutines** when a consumer stops early, and **merging** multiple streams correctly.

Leak: early exit without cancel

```go
func rangeGen(start, stop int) <-chan int {
    out := make(chan int)
    go func() {
        for i := start; i < stop; i++ {
            out <- i // blocks if nobody receives
        }
        close(out)
    }()
    return out
}

// Consumer breaks early → producer stuck on send forever.
for v := range rangeGen(41, 46) {
    fmt.Println(v)
    if v == 42 {
        break
    }
}
```

```mermaid
sequence (top → bottom):
  actors: rangeGen, consumer
  rangeGen --> consumer  : 41
  rangeGen --> consumer  : 42
  consumer --> consumer  : break
  note: LEAKED G
```

Goroutines are cheap but not free; leaks accumulate.

Cancel channel + select

```go
func rangeGen(cancel <-chan struct{}, start, stop int) <-chan int {
    out := make(chan int)
    go func() {
        defer close(out)
        for i := start; i < stop; i++ {
            select {
            case out <- i:
            case <-cancel:
                return
            }
        }
    }()
    return out
}

func main() {
    cancel := make(chan struct{})
    defer close(cancel) // signal on exit

    for v := range rangeGen(cancel, 41, 46) {
        fmt.Println(v)
        if v == 42 {
            break
        }
    }
}
```

```
select rules (teaching model):
  1. run a case that can proceed
  2. if several ready → choose at random (fairness)
  3. if none ready → wait (or take default if present)
```

```
regions: select
flow:
  [out left-arrow i] --consumer reading--> [RunA]
  [left-arrow cancel] --cancel closed--> [Exit]
```

Cancel vs done

Name	Direction	Meaning
done	worker → waiter	"I finished"
cancel	waiter → worker	"Stop now"

Both are often named `done` in the wild—be explicit in your APIs.

Merge sequentially (slow)

```
// Reads in1 fully, then in2 - second producer stalls.
for v := range in1 {
    out <- v
}
for v := range in2 {
    out <- v
}
```

Merge concurrently (WaitGroup)

```
func merge(in1, in2 <-chan int) <-chan int {
    out := make(chan int)
    var wg sync.WaitGroup
    forward := func(in <-chan int) {
        defer wg.Done()
        for v := range in {
            out <- v
        }
    }
    wg.Add(2)
    go forward(in1)
    go forward(in2)
    go func() {
        wg.Wait()
        close(out)
    }()
    return out
}
```

Merge with select + nil disable

```
func mergeSelect(in1, in2 <-chan int) <-chan int {
    out := make(chan int)
    go func() {
        defer close(out)
        for in1 != nil || in2 != nil {
            select {
```

```
            case v, ok := <-in1:
                if !ok {
                    in1 = nil // disable case
                    continue
                }
                out <- v
            case v, ok := <-in2:
                if !ok {
                    in2 = nil
                    continue
                }
                out <- v
            }
        }
    }()
    return out
}
```

```
Why nil?
  closed channel is always ready with ok=false
  without nil, select spins forever on closed cases
  nil channel cases never select → clean exit when both nil
```

Fan-out / fan-in pipeline sketch

```
  producer  chan   stage  chan   consumer

          cancel closed early    select exits (no leak)
```

Each stage: `func(in <-chan T) <-chan U` starting its own goroutine, closing out on exit, accepting cancel when needed.

Runnable example

```
go mod init example && go run .
```

```go
package main

import "fmt"

func rangeGen(cancel <-chan struct{}, start, stop int) <-chan int {
    out := make(chan int)
```

```go
    go func() {
        defer close(out)
        for i := start; i < stop; i++ {
            select {
            case out <- i:
            case <-cancel:
                return
            }
        }
    }()
    return out
}

func main() {
    cancel := make(chan struct{})
    defer close(cancel)
    for v := range rangeGen(cancel, 41, 46) {
        fmt.Println(v)
        if v == 42 {
            break
        }
    }
    fmt.Println("consumer done; producer exits via cancel")
}
```

What to notice: Early break no longer leaks the generator.

Try next: Implement `mergeSelect` for three channels by generalizing the nil pattern.

Time: Throttle, Backpressure, Timeouts

Time: Throttle, Backpressure, Timeouts

Overview

Time tools in concurrent Go: **limit parallelism**, **reject overload**, **bound wait**, and **schedule delayed work**—without leaking timers.

Throttle (N concurrent)

Buffered channel as semaphore:

```
sema capacity N:

handle:  sema <- token   // acquire
         go work(); <-sema  // release in worker

wait:    fill all slots again to ensure idle
```

```go
func throttle(n int, fn func()) (handle func(), wait func()) {
    sema := make(chan struct{}, n)
    handle = func() {
        sema <- struct{}{}
        go func() {
            fn()
            <-sema
        }()
    }
    wait = func() {
        for i := 0; i < n; i++ {
            sema <- struct{}{} // re-acquire all slots
        }
    }
    return handle, wait
}
```

```
  throttle ( N):
    acquire slot → run work → release
    full? → block
```

```
  backpressure:
    full? → return "busy" immediately (select default)
```

With N=2 and 4×100ms jobs → ~200ms total.

Backpressure (fail fast)

```go
handle = func() error {
    select {
    case sema <- struct{}{}:
        go func() {
            fn()
            <-sema
        }()
        return nil
    default:
        return errors.New("busy")
    }
}
```

```
select + default:
  ready case → take it
  else → default immediately (no wait)
```

Clients must handle `"busy"` (retry later, shed load, degrade).

Operation timeout with select

```go
func withTimeout(timeout time.Duration, fn func() int) (int, error) {
    done := make(chan struct{})
    var result int
    go func() {
        result = fn()
        close(done)
    }()
    select {
    case <-done:
        return result, nil
    case <-time.After(timeout):
        return 0, errors.New("timeout")
    }
}
```

```
sequence (top → bottom):
  actors: caller, worker, timer
  caller --> worker  : start fn
  caller --> timer  : After(timeout)
  worker --> caller  : done
  timer --> caller  : fire
  note: alt fn finishes first
  note: else timeout first
  note: return error; worker may still run!
```

Caveat: on timeout the worker goroutine may continue unless `fn` also observes cancel (prefer context—next chapter).

Timers vs time.After

API	Notes
`time.After`	Convenient; **new timer every call**—bad in hot loops
`time.NewTimer`	Reuse with Stop/Reset carefully
`time.AfterFunc`	Run function after delay; Stop cancels

```go
timer := time.NewTimer(100 * time.Millisecond)
defer timer.Stop()
select {
case <-timer.C:
    // fired
case <-cancel:
    if !timer.Stop() {
        select {
        case <-timer.C:
        default:
        }
    }
}
```

Hot path anti-pattern:

```go
for {
    select {
    case <-in:
    case <-time.After(time.Hour): // allocates every iteration
    }
}
```

Runnable example

```
go mod init example && go run .
```

```go
package main

import (
    "errors"
    "fmt"
    "time"
)

func work() { time.Sleep(80 * time.Millisecond) }

func main() {
    sema := make(chan struct{}, 2)
    handle := func() error {
        select {
        case sema <- struct{}{}:
            go func() {
                work()
                <-sema
            }()
            return nil
        default:
            return errors.New("busy")
        }
    }
    for i := 1; i <= 4; i++ {
        fmt.Println(i, handle())
    }
    time.Sleep(200 * time.Millisecond)
}
```

Expected: first two nil, next two busy.

Try next: Replace timeout demo with `context.WithTimeout` so the worker can exit early.

Backpressure in Streaming APIs

Backpressure in Streaming APIs

Thesis

A streaming API is only as honest as its **slowest consumer**. If the producer can run ahead without bound — unbounded channels, goroutines per message, buffering whole bodies in memory — you have a latent OOM, not a pipeline. **Backpressure** means the producer slows, blocks, or fails fast when the consumer cannot keep up. In Go the boring tools are a **bounded channel**, `io.Pipe` (writes block until reads consume), and HTTP `ResponseController.Flush` so chunks leave the process instead of piling up in a buffer.

This chapter is about streaming shapes (NDJSON/SSE/body pipes), not the short throttle sketch in the neighbouring time chapter. Measure an unbounded burst, a capacity-1 channel, an `io.Pipe` handoff, and a fail-fast `select/default`, then wire the same idea into an SSE handler.

Mental model

```
producer    queue    consumer

unbounded queue:   produce_finished  0; peak_queued → N; heap grows
bounded(1):        produce waits; elapsed  N × consume_time
io.Pipe:           Write blocks until Read takes the bytes; depth  1 chunk
fail-fast:         when full, return "busy" (no new goroutine, no silent buffer)
```

Mechanism	Backpressure style	Typical use
`make(chan T, 1)` or small N	Block the producer	In-process pipelines
`io.Pipe`	Block `Write` until `Read`	Connect `io.Reader` APIs to producers (uploads, multipart)
`select` + `default`	Reject / 503	Edge APIs that must not melt
`ResponseController.Flush`	Push bytes to the client	SSE / chunked responses

`io.Pipe` is documented as synchronous: there is **no internal buffer**; each `Write` blocks until one or more `Read`s have consumed the data. That is backpressure you get for free when you stream a request body into `http.NewRequest`.

Worked examples

Verified with **Go 1.27.1** (`/home/box/sdk/go1.27.1`) on Linux/amd64.

Case 1: unbounded vs bounded vs pipe vs fail-fast

```go
// main.go - measure four backpressure shapes
package main

import (
    "fmt"
    "io"
    "sync"
    "sync/atomic"
    "time"
)

func main() {
    const n = 100
    payload := make([]byte, 1024)
    slow := 3 * time.Millisecond

    // Large buffer  "unbounded" for this burst
    {
        ch := make(chan []byte, n)
        var peak atomic.Int64
        started := make(chan struct{})
        var wg sync.WaitGroup
        wg.Add(1)
        go func() {
            defer wg.Done()
            <-started
            time.Sleep(30 * time.Millisecond) // let producer fill
            for range ch {
                time.Sleep(slow)
            }
        }()
        close(started)
        start := time.Now()
        for i := 0; i < n; i++ {
            b := make([]byte, len(payload))
            ch <- b
            if l := int64(len(ch)); l > peak.Load() {
                peak.Store(l)
            }
        }
```

```go
        prod := time.Since(start)
        close(ch)
        wg.Wait()
        fmt.Printf("unbounded(cap=%d): producer_finished=%v peak_queued=%d\n",
            n, prod.Round(time.Microsecond), peak.Load())
    }

    // Capacity 1 - producer blocks
    {
        ch := make(chan []byte, 1)
        var blocked atomic.Int64
        var wg sync.WaitGroup
        wg.Add(1)
        go func() {
            defer wg.Done()
            for range ch {
                time.Sleep(slow)
            }
        }()
        start := time.Now()
        for i := 0; i < n; i++ {
            b := make([]byte, len(payload))
            select {
            case ch <- b:
            default:
                blocked.Add(1)
                ch <- b
            }
        }
        close(ch)
        wg.Wait()
        fmt.Printf("bounded(1): elapsed=%v blocked_sends=%d/%d\n",
            time.Since(start).Round(time.Millisecond), blocked.Load(), n)
    }

    // io.Pipe - Write waits for Read
    {
        pr, pw := io.Pipe()
        done := make(chan struct{})
        go func() {
            defer close(done)
            buf := make([]byte, len(payload))
            for {
                _, err := io.ReadFull(pr, buf)
                if err != nil {
                    return
                }
```

```go
                time.Sleep(slow)
            }
        }()
        start := time.Now()
        for i := 0; i < n; i++ {
            if _, err := pw.Write(payload); err != nil {
                panic(err)
            }
        }
        _ = pw.Close()
        <-done
        fmt.Printf("io.Pipe: elapsed=%v\n", time.Since(start).Round(time.Millisecond))
    }

    // Fail-fast when full
    {
        ch := make(chan []byte, 4)
        ok, no := 0, 0
        for i := 0; i < 20; i++ {
            select {
            case ch <- payload:
                ok++
            default:
                no++
            }
        }
        fmt.Printf("fail-fast(cap=4): accepted=%d rejected=%d\n", ok, no)
    }
}
```

```
go run .
```

```
unbounded(cap=100): producer_finished=18µs peak_queued=100
bounded(1): elapsed=316ms blocked_sends=99/100
io.Pipe: elapsed=319ms
fail-fast(cap=4): accepted=4 rejected=16
```

The unbounded producer finished in tens of microseconds with **100** messages queued — the consumer had not started draining. Bounded and `io.Pipe` both took ~300 ms (`100 × 3 ms`): the producer could not outrun the consumer. Fail-fast accepted four and rejected sixteen without spawning work.

Case 2: stream an HTTP body with `io.Pipe`

```go
// upload_pipe.go - request body is produced under backpressure
package main

import (
    "fmt"
    "io"
    "net/http"
    "net/http/httptest"
)

func main() {
    srv := httptest.NewServer(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request)
        n, _ := io.Copy(io.Discard, r.Body)
        fmt.Fprintf(w, "read %d bytes\n", n)
    }))
    defer srv.Close()

    pr, pw := io.Pipe()
    req, err := http.NewRequest(http.MethodPost, srv.URL, pr)
    if err != nil {
        panic(err)
    }

    errCh := make(chan error, 1)
    go func() {
        defer pw.Close()
        for i := 0; i < 8; i++ {
            if _, err := pw.Write([]byte("chunk-\n")); err != nil {
                errCh <- err
                return
            }
        }
        errCh <- nil
    }()

    resp, err := http.DefaultClient.Do(req)
    if err != nil {
        panic(err)
    }
    defer resp.Body.Close()
    body, _ := io.ReadAll(resp.Body)
    fmt.Print(string(body))
    if err := <-errCh; err != nil {
        panic(err)
    }
```

```
}
```

```
go run upload_pipe.go
```

```
read 56 bytes
```

If the server stops reading, `pw.Write` blocks (or returns an error after `CloseWithError` on cancel). That is the same mechanism as Case 1's pipe — now wired into `net/http`.

Case 3: SSE needs Flush

Without `Flush`, a small response can sit in the server's buffer and the client sees nothing until the handler returns.

```go
// sse_flush.go
package main

import (
    "fmt"
    "net/http"
    "net/http/httptest"
    "time"
)

func main() {
    h := http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        rc := http.NewResponseController(w)
        w.Header().Set("Content-Type", "text/event-stream")
        w.WriteHeader(http.StatusOK)
        for i := 0; i < 3; i++ {
            fmt.Fprintf(w, "data: tick-%d\n\n", i)
            if err := rc.Flush(); err != nil {
                return
            }
        }
    })
    srv := httptest.NewServer(h)
    defer srv.Close()

    start := time.Now()
    resp, err := http.Get(srv.URL)
    if err != nil {
        panic(err)
    }
    defer resp.Body.Close()
```

```
    buf := make([]byte, 256)
    n, _ := resp.Body.Read(buf)
    fmt.Printf("httptest SSE: %d bytes in %v\n%s\n", n, time.Since(start).Round(time.Millise
}
```

```
go run sse_flush.go
```

```
httptest SSE: 28 bytes in 0s
data: tick-0

data: tick-1
```

`NewResponseController` (Go 1.20+) wraps `Flush` / `FlushError` portably across ResponseWriter middleware. For HTTP/1 handlers that read the request body while writing the response, call `EnableFullDuplex()` or the second write can block on a full buffer while the client waits for a response before sending more body.

The trap

Goroutine per message into an unbounded channel. Under load you trade a full channel for tens of thousands of runnable goroutines and the same memory cliff. Bound the channel **or** bound the worker pool (semaphore / `WaitGroup` + fixed workers). Prefer fail-fast at the edge over infinite queueing.

Closing only one end of `io.Pipe` on cancel. If the HTTP client cancels, close the **reader** (`pr.CloseWithError(ctx.Err())`) so a blocked `Write` wakes up. Closing only the writer leaves the handler stuck until the write side notices.

Buffering the entire stream "for simplicity". `io.ReadAll` on a client upload disables backpressure by definition. Stream with `io.Copy` / `json.Decoder` / framed reads.

The boring rule

> Streaming producers use a **bounded** handoff (`chan` with small capacity or `io.Pipe`). When the handoff is full, either **block** (in-process pipelines) or **reject** (public APIs). HTTP chunked/SSE handlers `Flush` after each unit. Cancellation closes the pipe end that unblocks the other side.

Try this

1. Change Case 1's bounded capacity from 1 to 8 and record `blocked_sends` and elapsed. Explain the trade-off.
2. Add `context.WithCancel` around the pipe upload: cancel mid-write, `CloseWithError` the reader, and show the producer exit error.

3. Build an NDJSON handler that `json.NewEncoder(w).Encode` + `Flush` per record, then slow the client with `io.CopyN` sleeps — watch handler goroutine count with and without a bounded upstream queue.
4. Compare `httptest` without `Flush` on a large buffered writer wrapper; note when the client first sees bytes.

Sources

- Go docs: `io.Pipe` — synchronous in-memory pipe, no internal buffer
- Go docs: `http.NewResponseController`, `ResponseController.Flush`, `EnableFullDuplex`
- Go wiki / blog patterns: pipelines and cancellation (`context` + `select`)

Context Complete

Context Complete

Overview

`context.Context` carries **cancellation**, **deadlines**, and optional **request-scoped values**. Prefer it over ad-hoc cancel channels for public APIs.

From cancel channel to context

```go
// Channel style
func execute(cancel <-chan struct{}, fn func() int) (int, error) {
    ch := make(chan int, 1)
    go func() { ch <- fn() }()
    select {
    case res := <-ch:
        return res, nil
    case <-cancel:
        return 0, errors.New("canceled")
    }
}

// Context style
func execute(ctx context.Context, fn func() int) (int, error) {
    ch := make(chan int, 1)
    go func() { ch <- fn() }()
    select {
    case res := <-ch:
        return res, nil
    case <-ctx.Done():
        return 0, ctx.Err()
    }
}
```

```go
ctx, cancel := context.WithCancel(context.Background())
defer cancel() // always
```

Multiple `cancel()` calls are safe (unlike double `close` on a channel).

Context is a tree

```
Background
      WithCancel / WithTimeout
             children inherit cancel downward
shorter deadline always wins
```

Rule	Detail
Immutable nodes	"Add" property → new child
Parent cancel	cancels all children
Child cancel	does **not** cancel parent/siblings
Timeouts nest	**shorter** deadline wins; child cannot extend parent

```go
parent, cancel := context.WithTimeout(context.Background(), 200*time.Millisecond)
defer cancel()
child, cancel2 := context.WithTimeout(parent, 50*time.Millisecond)
defer cancel2()
// work with child: max wait is 50ms
```

Soft child timeout longer than parent is pointless—parent fires first.

Timeout and deadline

```go
ctx, cancel := context.WithTimeout(parent, 150*time.Millisecond)
// equivalent core:
// context.WithDeadline(parent, time.Now().Add(150*time.Millisecond))
defer cancel()

// errors
// context.Canceled
// context.DeadlineExceeded
```

`execute` need not know *why* Done closed—only that it should stop waiting.

Values

```go
type key int
const requestID key = 1
ctx = context.WithValue(ctx, requestID, "abc")
id, _ := ctx.Value(requestID).(string)
```

Use for request metadata (IDs, loggers)—not for optional parameters that belong in function args.

Propagate into work

Best practice: `fn` accepts `ctx` and checks `ctx.Done()` or passes ctx to I/O:

```
func work(ctx context.Context) (int, error) {
    select {
    case <-time.After(100 * time.Millisecond):
        return 42, nil
    case <-ctx.Done():
        return 0, ctx.Err()
    }
}
```

Otherwise timeout only abandons the **waiter**, not the **worker**.

Runnable example

```
go mod init example && go run .
```

```
package main

import (
    "context"
    "fmt"
    "time"
)

func execute(ctx context.Context, fn func() int) (int, error) {
    ch := make(chan int, 1)
    go func() { ch <- fn() }()
    select {
    case v := <-ch:
        return v, nil
    case <-ctx.Done():
        return 0, ctx.Err()
    }
}

func main() {
    ctx, cancel := context.WithTimeout(context.Background(), 50*time.Millisecond)
    defer cancel()
    v, err := execute(ctx, func() int {
        time.Sleep(100 * time.Millisecond)
        return 42
    })
```

```
    fmt.Println(v, err)
}
```

Expected: `0 context deadline exceeded`

Try next: Nest a 10ms child under a 1s parent and confirm the child wins.

Wait Groups and Encapsulation

Wait Groups and Encapsulation

Overview

A **wait group** waits for a set of goroutines to finish. Prefer it over done channels when you need join without a result payload. Hide it behind APIs so callers never manage counters.

Done channel vs WaitGroup

```go
// done channel
done := make(chan struct{}, 1)
go func() {
    work()
    done <- struct{}{}
}()
<-done

// wait group
var wg sync.WaitGroup
wg.Add(1)
go func() {
    defer wg.Done()
    work()
}()
wg.Wait()
```

Tool	Best for
done channel	single completion + optional result on another channel
WaitGroup	N workers, no per-worker result at join site
`wg.Go`	less boilerplate (Go 1.25+)

Mental model (naive)

```go
// conceptual - real impl is concurrent-safe and parks, not busy-waits
type WaitGroup struct{ n int }

func (wg *WaitGroup) Add(delta int) {
    wg.n += delta
    if wg.n < 0 {
        panic("negative WaitGroup counter")
    }
}
func (wg *WaitGroup) Done() { wg.Add(-1) }
func (wg *WaitGroup) Wait() {
    for wg.n > 0 { /* real code blocks efficiently */ }
}
```

```
state machine:
  start --> Zero
  Zero --> Positive
  Positive --> Positive
  Positive --> Zero
  Zero --> Zero
```

Properties that match the real `sync.WaitGroup`:

- Knows **nothing** about goroutines—only a counter
- `Wait` before any `Add` does not block (counter already 0)
- Reusable after returning to zero
- Negative counter panics

Pass by pointer

```go
// BUG: copy of WaitGroup - Wait sees counter 0
func runWork(wg sync.WaitGroup) {
    wg.Add(1)
    go func() {
        defer wg.Done()
        work()
    }()
}

// OK
func runWork(wg *sync.WaitGroup) { ... }
```

```
value pass:  main.wg     runWork.wg_copy
pointer:     both share same counter
```

Encapsulation

Hide inside a function

```go
func RunConc(fns ...func()) {
    var wg sync.WaitGroup
    wg.Add(len(fns))
    for _, fn := range fns {
        go func(f func()) {
            defer wg.Done()
            f()
        }(fn)
    }
    wg.Wait()
}
// client: RunConc(work, work, work)
```

Hide inside a type

```go
type ConcRunner struct {
    wg    sync.WaitGroup
    funcs []func()
}

func (c *ConcRunner) Add(fn func()) { c.funcs = append(c.funcs, fn) }

func (c *ConcRunner) Run() {
    c.wg.Add(len(c.funcs))
    for _, fn := range c.funcs {
        go func(f func()) {
            defer c.wg.Done()
            f()
        }(fn)
    }
    c.wg.Wait()
}
```

Clients should not see `WaitGroup` unless they are writing infrastructure.

Add after Wait (advanced)

Technically possible if a waiter blocks on `Wait` while other goroutines still `Add`—but easy to get wrong. Prefer all `Add` calls **before** `Wait` from the coordinating goroutine, or use `wg.Go` which pairs add/start/done.

Runnable example

```
go mod init example && go run .
```

```
package main

import (
    "fmt"
    "sync"
    "time"
)

func RunConc(fns ...func()) {
    var wg sync.WaitGroup
    for _, fn := range fns {
        wg.Go(fn)
    }
    wg.Wait()
}

func main() {
    start := time.Now()
    RunConc(
        func() { time.Sleep(50 * time.Millisecond); fmt.Print(".") },
        func() { time.Sleep(50 * time.Millisecond); fmt.Print(".") },
        func() { time.Sleep(50 * time.Millisecond); fmt.Print(".") },
    )
    fmt.Printf("\ntook %v\n", time.Since(start).Round(time.Millisecond))
}
```

What to notice: Three sleeps overlap → ~50ms, not 150ms.

Try next: Deliberately pass WaitGroup by value and watch Wait return early.

Data Races and Mutexes

Data Races and Mutexes

Overview

When multiple goroutines access the same variable and **at least one writes**, you have a **data race** unless synchronization creates happens-before. Maps can **fatal** on concurrent write; other races may silently corrupt.

Concurrent map writes

```go
counter := map[string]int{}
// two goroutines: counter[word]++  → often:
// fatal error: concurrent map writes
```

`counter[word]++` is **not** one atomic step (hash, read, add, write, maybe grow).

Race detector

```go
go run -race .
go test -race ./...
```

```
WARNING: DATA RACE
Read at ... by goroutine 6
Previous write at ... by goroutine 7
```

Races are **timing-dependent**—passing once means nothing. Always CI with `-race` on supported platforms (needs CGO toolchain historically).

Channels are race-free for their send/receive protocol; sharing the *payload pointers* can still race.

Fix 1: avoid shared mutation

```
each worker → private map
main → merge maps after Wait
```

```
// counters[i] written only by worker i - OK for distinct indices
// do not append to shared slice from many Gs without sync
```

Or send partial maps on a channel and merge in one owner goroutine (**share by communicating**).

Fix 2: mutex

```
var mu sync.Mutex
mu.Lock()
counter[word]++
mu.Unlock()
```

```
sequence (top → bottom):
  actors: G1, Mutex, G2
  G1 --> Mutex  : Lock
  G2 --> Mutex  : Lock (blocks)
  G1 --> G1  : critical section
  G1 --> Mutex  : Unlock
  Mutex --> G2  : acquired
  G2 --> G2  : critical section
  G2 --> Mutex  : Unlock
```

Use mutex when:

- Multiple writers, or
- Writer + readers without other sync

Readers-only → no mutex needed for that data.

Non-reentrant

```
mu.Lock()
mu.Lock() // deadlock - Go mutex is not reentrant
```

Pass by pointer

Same rule as WaitGroup—copies break exclusion.

RWMutex

```
var rw sync.RWMutex
// readers
rw.RLock(); _ = m[k]; rw.RUnlock()
// writer
rw.Lock(); m[k]=v; rw.Unlock()
```

Many concurrent readers; writers exclusive. Measure—RWMutex is not always faster.

```
Writer W    Readers R1 R2 R3 R4

Mutex:      W excludes everyone; each R serializes
RWMutex:    R1-R4 may overlap; W waits for readers
```

Critical section hygiene

Do	Don't
Lock only around shared data	Hold lock across HTTP/DB
Unlock via defer when multi-return	Forget unlock on error path
Document what mu protects	One global mutex for everything

Runnable example

```
go run -race .
```

```
package main

import (
    "fmt"
    "sync"
)

func main() {
    var mu sync.Mutex
    counter := map[string]int{}
    var wg sync.WaitGroup
    inc := func(word string) {
        defer wg.Done()
        mu.Lock()
        counter[word]++
        mu.Unlock()
```

```
    }
    wg.Add(2)
    go inc("go")
    go inc("go")
    wg.Wait()
    fmt.Println(counter)
}
```

What to notice: `-race` is clean; remove mutex and it may fail or warn.

Try next: Rewrite with per-worker maps + merge; compare to mutex under `-bench`.

Semaphores, Rendezvous, Barriers

Semaphores, Rendezvous, Barriers

Overview

Limit concurrency to **N** (semaphore), meet at a **sync point** (rendezvous), or wait for **everyone** before next phase (barrier).

Mutex is N=1

```
type External struct{ lock sync.Mutex }

func (e *External) Call() {
    e.lock.Lock()
    defer e.lock.Unlock()
    // only one call at a time
}
```

When the dependency allows 4 parallel calls, a mutex is too strict.

Semaphore model

```
slots: [free free free free]  capacity 4

Acquire: take a free slot or block
Release: free a slot; wake one waiter
```

```
  semaphore capacity N:  [free free …]
  Acquire → take slot or block
  Release → free slot, wake waiter

  rendezvous: each waits until the other arrives
```

Channel implementation

Stdlib has no `Semaphore` type; a buffered channel is the idiomatic simple form (`golang.org/x/sync/semaphore` for weighted cases).

```go
type Semaphore chan struct{}

func NewSemaphore(n int) Semaphore {
    return make(chan struct{}, n)
}

func (s Semaphore) Acquire() { s <- struct{}{} }
func (s Semaphore) Release() { <-s }

func (s Semaphore) TryAcquire() bool {
    select {
    case s <- struct{}{}:
        return true
    default:
        return false
    }
}
```

Where to acquire

Pattern	Effect
Acquire **inside** worker	May spawn huge number of blocked Gs
Acquire **before** go	Bounds **goroutine count** as well as concurrency

```go
for range nCalls {
    sem.Acquire()
    wg.Go(func() {
        defer sem.Release()
        ex.Call()
    })
}
```

Rendezvous

Two parties wait for each other before continuing:

```
G1 reaches point  wait
                              both proceed
G2 reaches point  wait
```

```go
ready1 := make(chan struct{})
ready2 := make(chan struct{})

// G1:
close(ready1)
<-ready2

// G2:
close(ready2)
<-ready1
```

```
sequence (top → bottom):
  actors: G1, G2
  G1 --> G1  : close ready1
  G1 --> G2  : wait ready2
  G2 --> G2  : close ready2
  G2 --> G1  : wait ready1
  note: both unblocked
```

Caution: concurrent `fmt` itself synchronizes and can hide races when debugging.

Barrier (N parties)

Generalize rendezvous to N workers: each `Add(1)` to a cycle, last arriver releases all. Patterns:

- `sync.WaitGroup` for one-shot phase joins
- Two WaitGroups or condition variables for reusable barriers
- Channel of capacity 0 with careful N-handshake (easy to get wrong)

Teaching pattern for one phase:

```go
var start, done sync.WaitGroup
start.Add(n)
done.Add(n)
for i := 0; i < n; i++ {
    go func() {
        start.Done()
        start.Wait() // all started
        // phase work
        done.Done()
    }()
}
done.Wait()
```

Runnable example

```
go mod init example && go run .
```

```go
package main

import (
    "fmt"
    "sync"
    "time"
)

func main() {
    sem := make(chan struct{}, 4)
    var wg sync.WaitGroup
    start := time.Now()
    for i := 0; i < 12; i++ {
        sem <- struct{}{}
        wg.Go(func() {
            defer func() { <-sem }()
            time.Sleep(10 * time.Millisecond)
            fmt.Print(".")
        })
    }
    wg.Wait()
    fmt.Printf("\n%d ms\n", time.Since(start).Milliseconds())
}
```

Expected: ~30ms (12/4 × 10ms), twelve dots.

Try next: Implement rendezvous so two printers always print "sync" before either prints "after".

Atomics Complete

Atomics Complete

Overview

`sync/atomic` provides lock-free operations that the CPU can perform safely under concurrency. **Composition of atomics is not atomic**—that distinction separates counters from broken "clever" state machines.

Non-atomic increment

```
total++ // read-modify-write: can lose updates
// 5 Gs × 10000 → often total < 50000
```

```
G1 reads 42
G2 reads 42
G1 writes 43
G2 writes 43   // lost update
```

Atomic types and ops

Type	Ops
`atomic.Bool`	Load, Store, Swap, CompareAndSwap
`atomic.Int32/64`, `Uint…`	+ Add, And, Or (1.23+)
`atomic.Pointer[T]`	typed pointer
`atomic.Value`	any type (consistent concrete type)

```
var total atomic.Int32
total.Add(1)
fmt.Println(total.Load())

// CAS
var flag atomic.Bool
if flag.CompareAndSwap(false, true) {
    // first closer wins
}
```

Pass atomics by **pointer** (or embed in structs used via pointer).

Composition traps

```
// Guaranteed sum of +1/+1 even with sleep between (sequence-independent Adds)
counter.Add(1)
sleep()
counter.Add(1)

// NOT a reliable algorithm: Load then branch then Add
if counter.Load()%2 == 0 {
    counter.Add(1)
} else {
    counter.Add(2)
}
// No data race, but race CONDITION → unpredictable result
```

```
regions: OK | Race condition without data race
flow:
  [Br]
       |
       v
  [Ad]
```

Bulletproof composite update: mutex around the whole sequence, or a careful CAS loop that retries.

CAS gate (mutex alternative for once)

```
type Gate struct{ closed atomic.Bool }

func (g *Gate) Close() {
    if !g.closed.CompareAndSwap(false, true) {
        return // already closed
    }
    // free resources once
}
```

Great for "first caller wins / early exit". Not a substitute when waiters must block until unlock.

When to use atomics

Use	Prefer
Counters, flags	atomics
Multi-field invariants	mutex
Hand-off of ownership	channel
Complex lock-free structures	expert only + heavy tests

Runnable example

```
go run -race .
```

```go
package main

import (
    "fmt"
    "sync"
    "sync/atomic"
)

func main() {
    var total atomic.Int32
    var wg sync.WaitGroup
    for i := 0; i < 5; i++ {
        wg.Go(func() {
            for j := 0; j < 10000; j++ {
                total.Add(1)
            }
        })
    }
    wg.Wait()
    fmt.Println("total", total.Load())
}
```

Expected: `total 50000` and race-clean.

Try next: Implement a CAS loop that doubles a value only if still equal to an observed snapshot.

Testing Concurrent Code

Testing Concurrent Code

Overview

Well-designed concurrent code is testable with normal tools: channels, wait groups, and fakes. Flaky tests usually mean **missing synchronization** or **time.Sleep as assertion**.

Diagram: Assert after barrier

```
flow:
  [Barrier]
       |
       v
  [Assert]
```

Principles

1. **Make state observable** after a barrier (Wait, channel close, context cancel).
2. **Inject clocks and dependencies** (interfaces) so tests control timing.
3. **Run with `-race`** always in CI.
4. **Prefer determinism** over "sleep and hope".

```
go test -race -count=100 ./...
```

Pattern: wait for completion

```
func TestWorker(t *testing.T) {
    done := make(chan struct{})
    go func() {
        defer close(done)
        work()
    }()
    select {
    case <-done:
    case <-time.After(time.Second):
```

```go
        t.Fatal("timeout")
    }
}
```

Pattern: collect results

```go
out := make(chan int, n)
// workers send to out
// close when done
var sum int
for v := range out {
    sum += v
}
```

Pattern: assert no leak

```go
before := runtime.NumGoroutine()
// run cancellable system
// cancel and wait
deadline := time.Now().Add(2 * time.Second)
for time.Now().Before(deadline) {
    if runtime.NumGoroutine() <= before+margin {
        return
    }
    time.Sleep(10 * time.Millisecond)
}
t.Fatal("goroutine leak")
```

Avoid sleeps as correctness

```go
// brittle
go doWork()
time.Sleep(10 * time.Millisecond)
assertState()
```

Prefer:

- channel signal when state is ready
- polling with timeout only when API is legacy
- `testing/synctest` bubbles when available (full runtime model: synctest deep dive)

Testing cancel paths

```
ctx, cancel := context.WithCancel(context.Background())
errc := make(chan error, 1)
go func() { errc <- server.Run(ctx) }()
cancel()
select {
case err := <-errc:
    if !errors.Is(err, context.Canceled) { t.Fatal(err) }
case <-time.After(time.Second):
    t.Fatal("hang")
}
```

Table tests still work

Concurrency does not forbid table tests—ensure each case uses fresh channels/groups and does not share mutable fixtures across parallel `t.Run` without isolation.

```
t.Parallel() // only if no shared mutable state
```

Runnable example

```
mkdir /tmp/conctest && cd /tmp/conctest
go mod init example
```

`inc_test.go`:

```
package example

import (
    "sync"
    "sync/atomic"
    "testing"
    "time"
)

func TestAtomicInc(t *testing.T) {
    var n atomic.Int64
    var wg sync.WaitGroup
    for i := 0; i < 100; i++ {
        wg.Go(func() { n.Add(1) })
    }
```

```go
    wg.Wait()
    if n.Load() != 100 {
        t.Fatalf("got %d", n.Load())
    }
}

func TestCancelPath(t *testing.T) {
    done := make(chan struct{})
    go func() {
        defer close(done)
        time.Sleep(5 * time.Millisecond)
    }()
    select {
    case <-done:
    case <-time.After(time.Second):
        t.Fatal("timeout")
    }
}
```

```go
go test -race -count=50 -v
```

What to notice: Barrier (`Wait` / channel close) then assert—no `Sleep` as the correctness signal.

Try next: Write a test that fails under `-race` when two Gs append to the same slice without sync.

Scheduler Model with Diagrams

Scheduler Model with Diagrams

Overview

A teaching model of how Go runs many goroutines on few threads—enough to reason about blocking, GOMAXPROCS, and syscalls. For production-depth GMP, also read Scheduler deep dive.

Three layers

```
Hardware - cores run instructions in parallel

  instr A     instr B     instr C     instr D

  Core 1     Core 2     Core 3     Core 4


OS - many threads share cores (OS scheduler)


  Thread E                 Thread F


  Thread A    Thread B    Thread C    Thread D


Go - many goroutines share threads (Go scheduler)


  G15   G16   G17   G18   G19   G20   runnable queue


  G11         G12         G13         G14   running


  Thread A    Thread B    Thread C    Thread D
```

Go does **not** assign threads to cores; the OS does.

Simplified scheduler loop

```
1. Queue all runnable Gs
2. Run up to N of them on worker threads (N   GOMAXPROCS)
3. If a G blocks (channel, mutex, park) → run another
4. Preempt long runners so others don't starve
```

```
  cores run threads; threads run Gs
  GOMAXPROCS   max Ps running Go code
  block (chan/IO) → park G → run another
```

Starvation and preemption

If G11–G14 never block, G15+ could starve. The scheduler **preempts** running goroutines so others get CPU (async preemption in modern Go improves tight loops).

System calls

While a G is in a blocking syscall, its OS thread may be stuck. Runtime can **detach** scheduling capacity and run other Gs on additional threads, then collapse extras later.

```
Before: 4 threads run Go code
G11,G12 enter syscalls
Runtime: Thread E,F run G15,G16
G11,G12 return → rebalance
```

GOMAXPROCS

Maximum OS threads running **Go code** simultaneously (excluding those stuck in syscalls).

```go
runtime.GOMAXPROCS(0) // get
runtime.GOMAXPROCS(1) // set
// Go 1.25+: runtime.SetDefaultGOMAXPROCS()
```

Default logical CPUs (with cgroup awareness improving in recent versions).

```
GOMAXPROCS=1 go run .
docker run --cpus=4 ...  # verify GOMAXPROCS vs host
```

```
regions: Host 16 CPUs | Pod cpus=2
flow:
  [Go app]
```

```
        |
        v
   [Q]
```

Wrong pairing (tiny quota, huge GOMAXPROCS) → thrashing.

Concurrency primitives vs scheduler

Primitive	Scheduler view
channel send/recv	may park G; wake on peer
mutex	park on contention
sleep / timer	park until timer heap fires
syscall	may occupy M longer

Metrics, profiles, traces

Tool	Shows
`runtime.NumGoroutine`	live Gs
CPU pprof	on-CPU stacks
mutex/block profile	contention
`go tool trace`	STW, park, proc start/stop
`GODEBUG=schedtrace=1000`	scheduler trace lines

Runnable experiment

```go
GOMAXPROCS=1 go run .
GOMAXPROCS=4 go run .
```

```go
package main

import (
    "fmt"
    "runtime"
    "sync"
    "time"
)

func burn(d time.Duration) {
    end := time.Now().Add(d)
    for time.Now().Before(end) {
```

```go
    }
}

func main() {
    fmt.Println("GOMAXPROCS", runtime.GOMAXPROCS(0))
    var wg sync.WaitGroup
    start := time.Now()
    for i := 0; i < 4; i++ {
        wg.Go(func() { burn(200 * time.Millisecond) })
    }
    wg.Wait()
    fmt.Println("elapsed", time.Since(start))
}
```

What to notice: With `GOMAXPROCS=1`, ~800ms; with 4, ~200ms for CPU-bound burns.

Try next: Open `go tool trace` on a server under load and find runnable Gs waiting while a syscall holds threads.

Keep going

You now have the full **ground-up** path: primitives → sync → atomics → tests → scheduler model. Production depth continues in part **20 Go Deep Dives** and part **07** applied patterns.

Race Conditions vs Data Races

Race Conditions vs Data Races

Overview

These terms are not interchangeable. A **data race** is unsynchronized concurrent memory access where at least one access is a write—undefined behavior under the Go memory model, and usually reported by the race detector (`-race`). A **race condition** is a **logic timing bug**: the program's outcome depends on interleaving in a way the author did not intend. Race conditions can exist **without** a data race when every memory access is synchronized (atomics, mutexes, channels) but the *algorithm* is still wrong.

```
concurrent bug
      data race      → memory model violation; -race often finds it
      race condition → wrong logic under interleaving
                          (can be -race clean!)
```

Related chapters: 257 data races & mutex, 259 atomics.

Teaching rule

```
-race clean     correct concurrent algorithm
correct algorithm     define happens-before for shared memory
                       AND make multi-step decisions atomic as a unit
```

Symptom	Likely class	Detector
`WARNING: DATA RACE`	data race	`go test -race`
Concurrent map writes / fatal	data race	runtime + `-race`
Wrong totals, double-insert, lost update with atomics	race condition	logic tests, stress
TOCTOU / check-then-act	race condition	design review + stress

Data race: classic counter

Unsynchronized `++` is a read-modify-write. Two goroutines can both read the same value and both write the same next value—**lost updates**—and the accesses form a data race.

```
mkdir race-lab && cd race-lab
go mod init example.com/race-lab
```

`data_race.go`:

```go
package main

import (
    "fmt"
    "sync"
)

func main() {
    var x int
    var wg sync.WaitGroup
    for i := 0; i < 2; i++ {
        wg.Go(func() {
            for j := 0; j < 100_000; j++ {
                x++ // data race: concurrent read+write without sync
            }
        })
    }
    wg.Wait()
    fmt.Println("x =", x, "(often not 200000)")
}
```

```
go run -race .
# WARNING: DATA RACE
# x = <something  200000>
```

What to notice: Without `-race` you might only see a wrong number; with `-race` the detector pins the concurrent accesses.

Fixes (any one is enough for the race)

```go
// 1) Mutex
var mu sync.Mutex
mu.Lock()
x++
mu.Unlock()
```

```go
// 2) Atomic
var x atomic.Int64
x.Add(1)

// 3) Ownership: only one goroutine mutates; others send ops on a channel
```

Race condition without data race: atomic composition

Each `Load` / `Add` is race-free, but the **pair** is not a single atomic decision. Two goroutines can interleave between Load and Add—classic TOCTOU on a "policy" that depends on the observed value.

```go
package main

import (
    "fmt"
    "sync"
    "sync/atomic"
)

func main() {
    var counter atomic.Int64
    var wg sync.WaitGroup

    // Intent: if even, +1; if odd, +2. Not atomic as a unit.
    bump := func() {
        if counter.Load()%2 == 0 {
            counter.Add(1)
        } else {
            counter.Add(2)
        }
    }

    for i := 0; i < 1000; i++ {
        wg.Go(bump)
    }
    wg.Wait()
    fmt.Println("final", counter.Load())
    // Re-run many times: totals vary. go test -race stays clean.
}
```

```go
go run -race .
# clean - no data race
# final value still non-deterministic relative to a sequential model
```

Correct patterns

```
// Mutex around the whole decision
mu.Lock()
if counter%2 == 0 {
    counter++
} else {
    counter += 2
}
mu.Unlock()

// Or CAS loop that retries the whole policy
for {
    old := counter.Load()
    var next int64
    if old%2 == 0 {
        next = old + 1
    } else {
        next = old + 2
    }
    if counter.CompareAndSwap(old, next) {
        break
    }
}
```

Check-then-act

```
G1: if !exists { create() }
G2: if !exists { create() }
     both pass check → duplicate / conflict
```

This appears in maps, files, databases, and caches:

```
// Broken even if map is protected by separate Lock per line incorrectly,
// or if you use atomics only for the flag and not the create body.
if _, ok := m[k]; !ok {
    m[k] = expensiveInit() // two Gs both see !ok
}
```

Domain	Broken pattern	Fix
In-process map	check then insert	single mutex around both; `sync.Map` LoadOrStore

Domain	Broken pattern	Fix
Files	`Stat` then `Create`	`O_EXCL`, or lock file
SQL	SELECT then INSERT	UNIQUE constraint + handle conflict; transaction
HTTP	"if not cached then fetch"	single-flight, mutex, or CAS

`LoadOrStore` sketch

```go
package main

import (
    "fmt"
    "sync"
)

func main() {
    var m sync.Map
    var wg sync.WaitGroup
    for i := 0; i < 8; i++ {
        wg.Go(func() {
            actual, loaded := m.LoadOrStore("cfg", "once")
            fmt.Println("loaded_existing=", loaded, "value=", actual)
        })
    }
    wg.Wait()
}
```

Only one store wins; others see `loaded=true`. That closes the check-then-act hole for this map pattern.

Diagram: three layers of "correct"

```
Algorithm / invariants                    ← race conditions live here
(idempotent create, unique keys, CAS)



Synchronization (mutex, chan, atomic)     ← kills data races
```

```
    Memory model / happens-before            ← -race reasons about this
```

Runnable lab: compare three versions

Create `racecond_test.go` in a module:

```go
package racecond

import (
    "sync"
    "sync/atomic"
    "testing"
)

// Broken: data race on plain int.
func badInc(n *int, workers, per int) {
    var wg sync.WaitGroup
    for w := 0; w < workers; w++ {
        wg.Go(func() {
            for i := 0; i < per; i++ {
                *n++
            }
        })
    }
    wg.Wait()
}

// Race-free memory, wrong if you expected a specific composite policy;
// here simple Add is actually correct for a counter.
func atomicInc(n *atomic.Int64, workers, per int) {
    var wg sync.WaitGroup
    for w := 0; w < workers; w++ {
        wg.Go(func() {
            for i := 0; i < per; i++ {
                n.Add(1)
            }
        })
    }
    wg.Wait()
}

// Race condition: double-checked init without proper once.
func racyInit(ready *atomic.Bool, initCount *atomic.Int32) {
```

```go
    var wg sync.WaitGroup
    for i := 0; i < 32; i++ {
        wg.Go(func() {
            if !ready.Load() {
                // pretend expensive init
                initCount.Add(1)
                ready.Store(true)
            }
        })
    }
    wg.Wait()
}

func TestBadIncRace(t *testing.T) {
    // Run with: go test -race -run TestBadIncRace
    var n int
    badInc(&n, 4, 10_000)
    // Do not assert equality under race; detector is the signal.
    t.Log("n=", n)
}

func TestAtomicInc(t *testing.T) {
    var n atomic.Int64
    atomicInc(&n, 4, 10_000)
    if n.Load() != 40_000 {
        t.Fatalf("got %d", n.Load())
    }
}

func TestRacyInitCondition(t *testing.T) {
    // Often initCount > 1 even though -race is clean.
    // Fix with sync.Once or mutex around check+init.
    var ready atomic.Bool
    var initCount atomic.Int32
    racyInit(&ready, &initCount)
    if c := initCount.Load(); c != 1 {
        t.Logf("race condition: init ran %d times (expected 1)", c)
    }
}

func TestOnceInit(t *testing.T) {
    var once sync.Once
    var initCount atomic.Int32
    var wg sync.WaitGroup
    for i := 0; i < 32; i++ {
        wg.Go(func() {
            once.Do(func() { initCount.Add(1) })
```

```go
        })
    }
    wg.Wait()
    if initCount.Load() != 1 {
        t.Fatalf("once failed: %d", initCount.Load())
    }
}
```

```go
go test -race -count=20 -v
```

Experiment: Run the atomic composition example from 259 atomics under `-race` (clean) and print varying totals (condition).

Pitfalls

1. **Believing "atomics make the program correct."** They make *operations* race-free, not *policies*.
2. **Sleeping to "fix" races.** Masks bugs; fails under load or different GOMAXPROCS.
3. **Only testing happy serial paths.** Concurrent bugs need stress (`-count=N`, `-race`, high worker counts).
4. **Confusing channel safety with payload safety.** Sending a pointer does not protect the pointee.

Checklist

- ☐ Can I name the shared memory and who writes it?
- ☐ Is every multi-step decision under one lock, one CAS loop, or one owner goroutine?
- ☐ Does CI run `go test -race ./...` on supported platforms?
- ☐ Are uniqueness / idempotency enforced at the storage boundary?

Keep going

Next	Why
257 Data races & mutex	Detector + critical sections
259 Atomics	RMW and composition traps
260 Testing concurrent code	Deterministic tests
263 Cond & Broadcast	Waiting on predicates safely

Try next: Find a production `if err := find(); err == notFound { insert }` and document the uniqueness constraint (DB unique index, etcd create, etc.) that makes the race condition harmless.

Signaling: Cond and Broadcast

Signaling: Cond and Broadcast

Overview

Sometimes many waiters must wake when shared state changes. **Channels** cover most Go code; `sync.Cond` is the low-level "wait for a condition under a mutex" tool. Prefer channels unless you need **broadcast to many waiters** that all re-check a shared predicate protected by the same lock.

Related: 252 channels, 257 mutex, 258 semaphores.

Diagram: wait / signal

```
sequence (top → bottom):
  actors: waiter, Cond, producer
  waiter --> Cond  : L.Lock
  waiter --> Cond  : Wait  (unlocks + parks)
  producer --> Cond  : L.Lock; change state
  producer --> Cond  : Signal or Broadcast
  producer --> Cond  : Unlock
  Cond --> waiter  : wake; re-Lock
  waiter --> waiter  : recheck condition in for-loop
```

Always wait in a loop—spurious wakes, broad Broadcast, and stale signals happen.

```
mu.Lock()
for !ready {
    cond.Wait() // atomically unlocks mu while waiting; re-locks on wake
}
// invariant: ready == true while holding mu
mu.Unlock()
```

Signal vs Broadcast

Method	Wakes	Use when
`Signal`	**one** waiter	single consumer can take the item

Method	Wakes	Use when
`Broadcast`	**all** waiters	many must re-check (e.g. state machine flip)

Wrong choice:

- `Signal` when many waiters each need to proceed → **starvation / stuck waiters**
- `Broadcast` when one consumer would suffice → **thundering herd**

```
producers write queue


          Signal     one waiter takes item
   Cond              others stay parked


       Broadcast

all waiters wake, re-lock, recheck predicate
```

vs channel

Need	Prefer
One result / pipeline stage	channel
Close = fan-out "done"	`close(ch)`
`select` on many events	channels
Complex shared-state predicate under one mutex	`Cond` **or** redesign with channels
Bounded queue with many waiters	often channel; Cond for classic textbooks

Rule of thumb in modern Go: **if a channel models the problem cleanly, use a channel.**

Runnable: ready flag + Broadcast

```
mkdir cond-lab && cd cond-lab
go mod init example.com/cond-lab
```

`main.go`:

```
package main
```

```go
import (
    "fmt"
    "sync"
    "time"
)

func main() {
    var mu sync.Mutex
    cond := sync.NewCond(&mu)
    ready := false

    var wg sync.WaitGroup
    for id := 1; id <= 3; id++ {
        wg.Go(func() {
            mu.Lock()
            for !ready {
                fmt.Println("waiter", id, "waiting")
                cond.Wait()
            }
            fmt.Println("waiter", id, "proceeding")
            mu.Unlock()
        })
    }

    time.Sleep(50 * time.Millisecond)
    mu.Lock()
    ready = true
    fmt.Println("producer: broadcast")
    cond.Broadcast()
    mu.Unlock()

    wg.Wait()
}
```

```
go run .
# waiter N waiting (×3)
# producer: broadcast
# waiter N proceeding (×3)
```

What to notice: Waiters block without busy-spin; Broadcast unblocks all; each re-checks `ready` under the lock.

Runnable: bounded buffer with Cond

Classic producer/consumer where both "not full" and "not empty" are predicates:

```go
package main

import (
    "fmt"
    "sync"
    "time"
)

type Buffer struct {
    mu   sync.Mutex
    cond *sync.Cond
    q    []int
    cap  int
}

func NewBuffer(cap int) *Buffer {
    b := &Buffer{cap: cap}
    b.cond = sync.NewCond(&b.mu)
    return b
}

func (b *Buffer) Put(v int) {
    b.mu.Lock()
    for len(b.q) == b.cap {
        b.cond.Wait()
    }
    b.q = append(b.q, v)
    b.cond.Broadcast() // wake getters (and other putters if needed)
    b.mu.Unlock()
}

func (b *Buffer) Get() int {
    b.mu.Lock()
    for len(b.q) == 0 {
        b.cond.Wait()
    }
    v := b.q[0]
    b.q = b.q[1:]
    b.cond.Broadcast() // wake putters
    b.mu.Unlock()
    return v
}

func main() {
    b := NewBuffer(2)
    var wg sync.WaitGroup
```

```go
    wg.Go(func() {
        for i := 1; i <= 5; i++ {
            b.Put(i)
            fmt.Println("put", i)
            time.Sleep(10 * time.Millisecond)
        }
    })
    wg.Go(func() {
        for i := 0; i < 5; i++ {
            v := b.Get()
            fmt.Println("get", v)
            time.Sleep(30 * time.Millisecond)
        }
    })
    wg.Wait()
}
```

```go
go run -race .
```

What to notice: Full buffer blocks `Put`; empty blocks `Get`; no spinning. Race detector stays clean.

Channel equivalent (usually clearer)

```go
ch := make(chan int, 2) // capacity = bound
// put: ch <- v
// get: v := <-ch
```

Prefer this unless you already hold a larger mutex protecting more state than the queue alone.

Spurious and broad wakes

```go
// WRONG: if instead of for
mu.Lock()
if !ready {
    cond.Wait()
}
// ready might still be false after Broadcast from another condition
mu.Unlock()
```

Always:

```go
for !predicate() {
    cond.Wait()
}
```

Signal vs Broadcast lab

```go
package main

import (
    "fmt"
    "sync"
    "time"
)

func main() {
    var mu sync.Mutex
    cond := sync.NewCond(&mu)
    items := 0

    // Three consumers each want one item.
    var wg sync.WaitGroup
    for id := 1; id <= 3; id++ {
        wg.Go(func() {
            mu.Lock()
            for items == 0 {
                cond.Wait()
            }
            items--
            fmt.Println("consumer", id, "took item; left", items)
            mu.Unlock()
        })
    }

    time.Sleep(20 * time.Millisecond)
    mu.Lock()
    items = 3
    // Try Signal() once - only one consumer proceeds; others hang.
    // cond.Signal()
    cond.Broadcast() // correct here: three items, three waiters
    mu.Unlock()

    wg.Wait()
}
```

Comment switch: with a single `Signal` and `items=3`, two waiters never wake unless someone signals

again.

Pitfalls

1. **Waiting without holding the lock.** Undefined / panics / lost signals.
2. **Forgetting the loop.** Spurious wake or multi-predicate Broadcast.
3. **Copying `sync.Cond`.** Always pass by pointer; create with `sync.NewCond(&mu)`.
4. **Using Cond when a channel would do.** Harder to `select`, harder to cancel.
5. **Broadcast thundering herd** on hot paths—consider per-waiter channels or a work queue channel.

Cancellation and Cond

`Cond.Wait` does not take a context. Patterns:

- Use channels for cancelable waits (`select` on `ctx.Done()`).
- Or set a `done` flag under the mutex and `Broadcast`, then check `ctx` / `done` in the wait loop.

```
mu.Lock()
for !ready && !closed {
    cond.Wait()
}
ok := ready && !closed
mu.Unlock()
```

Checklist

- ☐ Predicate checked in a `for` loop under the mutex
- ☐ Same mutex for state and `Cond`
- ☐ Signal vs Broadcast matches waiter count semantics
- ☐ `-race` clean on producer/consumer tests
- ☐ Documented why channel was not enough (if using Cond)

Keep going

Next	Why
252 Channels	Preferred signaling for most apps
258 Semaphores	Bound concurrency with channels
262 Race vs data race	Predicate bugs under interleaving
260 Testing	Stress the wait loops

Try next: Rewrite the ready-flag example with `close(done)` and compare readability, cancelability, and `select` composition.

Advanced Concurrency Patterns

Concurrency Basics

Concurrency Basics

Overview

Go's concurrency model uses goroutines (lightweight threads) and channels for communication.

Goroutines

```go
go func() {
    // Runs concurrently
    fmt.Println("Hello from goroutine")
}()

go processData()  // Named function
```

Goroutines are cheap (~2KB stack, can have millions).

Creating Goroutines

```go
func main() {
    go sayHello()
    go sayWorld()
    time.Sleep(100 * time.Millisecond)  // Wait (not ideal)
}

func sayHello() { fmt.Println("Hello") }
func sayWorld() { fmt.Println("World") }
```

Waiting with WaitGroup

```go
var wg sync.WaitGroup

for i := 0; i < 5; i++ {
    wg.Add(1)
    go func(n int) {
```

```go
        defer wg.Done()
        fmt.Println(n)
    }(i)
}

wg.Wait()  // Block until all done
```

Channels

```go
ch := make(chan int)      // Unbuffered
ch := make(chan int, 10)  // Buffered

ch <- 42      // Send
value := <-ch // Receive
```

Basic Channel Pattern

```go
func main() {
    ch := make(chan string)

    go func() {
        ch <- "Hello"
    }()

    msg := <-ch
    fmt.Println(msg)
}
```

Worker Pool

```go
func worker(id int, jobs <-chan int, results chan<- int) {
    for job := range jobs {
        results <- job * 2
    }
}

func main() {
    jobs := make(chan int, 100)
    results := make(chan int, 100)

    // Start workers
```

```go
    for w := 0; w < 3; w++ {
        go worker(w, jobs, results)
    }

    // Send jobs
    for j := 0; j < 10; j++ {
        jobs <- j
    }
    close(jobs)

    // Collect results
    for r := 0; r < 10; r++ {
        fmt.Println(<-results)
    }
}
```

Summary

Concept	Purpose
`go func()`	Start goroutine
`sync.WaitGroup`	Wait for completion
`make(chan T)`	Create channel
`ch <-` / `<-ch`	Send/receive

Worked example

Fan of goroutines joined with `WaitGroup` and a results channel (no `Sleep`).

Save as `main.go`. Then:

```go
go mod init example
go run .
```

```go
package main

import (
    "fmt"
    "sync"
)

func main() {
    const n = 5
    results := make(chan int, n)
```

```go
    var wg sync.WaitGroup

    for i := 1; i <= n; i++ {
        wg.Add(1)
        go func(x int) {
            defer wg.Done()
            results <- x * x
        }(i)
    }

    go func() {
        wg.Wait()
        close(results)
    }()

    sum := 0
    for v := range results {
        sum += v
    }
    fmt.Println("sum of squares 1..5:", sum)
}
```

Expected output:

```
sum of squares 1..5: 55
```

More examples

Buffered vs unbuffered: buffer decouples a single send from receive.

```go
package main

import "fmt"

func main() {
    // Unbuffered needs a receiver ready (other goroutine).
    u := make(chan string)
    go func() { u <- "unbuffered" }()
    fmt.Println(<-u)

    // Buffered send can complete without a concurrent receiver.
    b := make(chan string, 1)
    b <- "buffered"
    fmt.Println(<-b)
}
```

Expected output:

```
unbuffered
buffered
```

Runnable example

Save as `main.go`. Then:

```
go mod init example
go run .
```

```go
package main

import (
    "fmt"
    "sync"
)

func main() {
    // 1) Goroutines + WaitGroup (no Sleep)
    var wg sync.WaitGroup
    for i := 1; i <= 3; i++ {
        wg.Add(1)
        go func(n int) {
            defer wg.Done()
            fmt.Printf("goroutine %d done\n", n)
        }(i)
    }
    wg.Wait()
    fmt.Println("all goroutines finished")

    // 2) Unbuffered channel: send happens in another goroutine
    ch := make(chan string)
    go func() {
        ch <- "hello from channel"
    }()
    fmt.Println(<-ch)

    // 3) Tiny worker pool: jobs → workers → results
    jobs := make(chan int, 5)
    results := make(chan int, 5)

    const workers = 2
    var pool sync.WaitGroup
```

```go
    for w := 1; w <= workers; w++ {
        pool.Add(1)
        go func(id int) {
            defer pool.Done()
            for job := range jobs {
                results <- job * 2
            }
        }(w)
    }

    for j := 1; j <= 5; j++ {
        jobs <- j
    }
    close(jobs)

    go func() {
        pool.Wait()
        close(results)
    }()

    sum := 0
    for r := range results {
        sum += r
    }
    fmt.Println("worker pool sum:", sum)
}
```

Expected output: (goroutine lines may appear in any order)

```
goroutine 1 done
goroutine 2 done
goroutine 3 done
all goroutines finished
hello from channel
worker pool sum: 30
```

What to notice: `WaitGroup` replaces `time.Sleep` for joining. Closing `jobs` lets workers exit `range`; closing `results` after `pool.Wait()` ends the collector cleanly.

Try next: Change the channel buffer sizes and watch when sends block; run with `go run -race .` to confirm the example is race-free.

Related Topics

- Channel Patterns
- Synchronization

Channel Patterns

Channel Patterns

Overview

Channels enable safe communication between goroutines. This chapter covers essential channel patterns.

Channel Directions

```
chan T     // Bidirectional
chan<- T   // Send-only
<-chan T   // Receive-only

func producer(out chan<- int) { }
func consumer(in <-chan int) { }
```

Buffered vs Unbuffered

```
// Unbuffered: send blocks until receive
ch := make(chan int)

// Buffered: send blocks when full
ch := make(chan int, 10)
```

Closing Channels

```
close(ch)

// Check if closed
v, ok := <-ch
if !ok {
    // Channel closed
}

// Range over channel
```

```go
for v := range ch {
    // Receives until closed
}
```

Select

```go
select {
case v := <-ch1:
    fmt.Println("from ch1:", v)
case v := <-ch2:
    fmt.Println("from ch2:", v)
case ch3 <- x:
    fmt.Println("sent to ch3")
default:
    fmt.Println("no channel ready")
}
```

Timeout Pattern

```go
select {
case result := <-ch:
    fmt.Println(result)
case <-time.After(1 * time.Second):
    fmt.Println("timeout")
}
```

Done Channel

```go
done := make(chan struct{})

go func() {
    // Work...
    close(done)  // Signal completion
}()

<-done  // Wait for signal
```

Fan-Out / Fan-In Architecture

The **Fan-Out** pattern distributes work across multiple worker goroutines reading from a single input channel. The **Fan-In** pattern consolidates the results from multiple worker channels back into a single output stream.

```
flow:
  ] -> JobsCh[
  JobsCh -> Worker1[
  JobsCh -> Worker2[
  JobsCh -> Worker3[
  Worker1 -> Out1[
  Worker2 -> Out2[
  Worker3 -> Out3[
  Out1 -> Multiplexer[
  Out2 -> Multiplexer
  Out3 -> Multiplexer
  Multiplexer -> FinalCh[
```

```go
// Worker function processing jobs from input channel
func worker(id int, jobs <-chan int) <-chan int {
    out := make(chan int)
    go func() {
        defer close(out)
        for j := range jobs {
            // Process item (e.g. compute square)
            out <- j * j
        }
    }()
    return out
}

// Fan-in: merge multiple worker output channels into one aggregated channel
func fanIn(channels ...<-chan int) <-chan int {
    out := make(chan int)
    var wg sync.WaitGroup

    for _, ch := range channels {
        wg.Add(1)
        go func(c <-chan int) {
            defer wg.Done()
            for v := range c {
                out <- v
            }
        }(ch)
    }
```

```
    go func() {
        wg.Wait()
        close(out)
    }()
    return out
}
```

Summary

Pattern	Use Case
Unbuffered	Synchronization
Buffered	Decouple speed
Select	Multiple channels
Done channel	Cancellation signal
Fan-out/in	Parallel processing

Worked example

`select` with default (non-blocking try) and a multi-channel race.

Save as `main.go`. Then:

```
go mod init example
go run .
```

```
package main

import (
    "fmt"
    "time"
)

func main() {
    ch := make(chan int, 1)

    // Non-blocking receive
    select {
    case v := <-ch:
        fmt.Println("got", v)
    default:
        fmt.Println("empty: default branch")
    }
```

```go
    ch <- 7
    select {
    case v := <-ch:
        fmt.Println("got", v)
    default:
        fmt.Println("unexpected default")
    }

    // First ready wins
    a := make(chan string, 1)
    b := make(chan string, 1)
    a <- "from-a"
    select {
    case msg := <-a:
        fmt.Println("select:", msg)
    case msg := <-b:
        fmt.Println("select:", msg)
    case <-time.After(50 * time.Millisecond):
        fmt.Println("select: timeout")
    }
}
```

Expected output:

```
empty: default branch
got 7
select: from-a
```

More examples

Or-done pattern: stop ranging when an external `done` closes. Producer also selects on `done` so nothing hangs.

```go
package main

import "fmt"

func orDone(done <-chan struct{}, in <-chan int) <-chan int {
    out := make(chan int)
    go func() {
        defer close(out)
        for {
            select {
            case <-done:
                return
```

```go
            case v, ok := <-in:
                if !ok {
                    return
                }
                select {
                case out <- v:
                case <-done:
                    return
                }
            }
        }
    }()
    return out
}

func main() {
    done := make(chan struct{})
    in := make(chan int)

    go func() {
        defer close(in)
        for i := 1; i <= 100; i++ {
            select {
            case <-done:
                return
            case in <- i:
            }
        }
    }()

    out := orDone(done, in)
    count := 0
    for v := range out {
        fmt.Println("v:", v)
        count++
        if count == 3 {
            close(done)
        }
    }
    fmt.Println("stopped early at", count)
}
```

Expected output:

```
v: 1
v: 2
v: 3
```

```
stopped early at 3
```

Runnable example

Save as `main.go`. Then:

```
go mod init example
go run .
```

```go
package main

import (
    "fmt"
    "sync"
    "time"
)

func worker(id int, jobs <-chan int) <-chan int {
    out := make(chan int)
    go func() {
        defer close(out)
        for j := range jobs {
            // Simulate work; square the job id.
            out <- j * j
        }
    }()
    return out
}

func fanIn(channels ...<-chan int) <-chan int {
    out := make(chan int)
    var wg sync.WaitGroup
    for _, ch := range channels {
        wg.Add(1)
        go func(c <-chan int) {
            defer wg.Done()
            for v := range c {
                out <- v
            }
        }(ch)
    }
    go func() {
        wg.Wait()
        close(out)
    }()
```

```go
    return out
}

func main() {
    // Timeout via select
    slow := make(chan string)
    go func() {
        time.Sleep(50 * time.Millisecond)
        slow <- "late result"
    }()
    select {
    case msg := <-slow:
        fmt.Println("got:", msg)
    case <-time.After(10 * time.Millisecond):
        fmt.Println("timeout: waited 10ms")
    }

    // Done channel signal
    done := make(chan struct{})
    go func() {
        // pretend work finished
        close(done)
    }()
    <-done
    fmt.Println("done channel closed")

    // Fan-out / fan-in
    jobs := make(chan int)
    go func() {
        for i := 1; i <= 4; i++ {
            jobs <- i
        }
        close(jobs)
    }()

    out1 := worker(1, jobs)
    out2 := worker(2, jobs)
    merged := fanIn(out1, out2)

    sum := 0
    for v := range merged {
        sum += v
    }
    // 1²+2²+3²+4² = 30
    fmt.Println("fan-in sum of squares:", sum)
}
```

Expected output:

```
timeout: waited 10ms
done channel closed
fan-in sum of squares: 30
```

What to notice: `select` with `time.After` implements a one-shot timeout. Fan-out shares one `jobs` channel across workers; fan-in merges their outputs and closes only after every input channel is drained.

Try next: Make the timeout larger than 50ms so the slow send wins the `select`. Add a third worker and re-run.

Related Topics

- Concurrency Basics
- Context and Cancellation

Synchronization

Synchronization

Overview

The `sync` package provides low-level synchronization primitives for coordinating goroutines.

sync.Mutex

```go
var (
    mu    sync.Mutex
    count int
)

func increment() {
    mu.Lock()
    count++
    mu.Unlock()
}

// With defer
func safe() {
    mu.Lock()
    defer mu.Unlock()
    // Critical section
}
```

sync.RWMutex

```go
var (
    mu   sync.RWMutex
    data map[string]string
)

func read(key string) string {
    mu.RLock()
    defer mu.RUnlock()
    return data[key]
```

```go
}

func write(key, value string) {
    mu.Lock()
    defer mu.Unlock()
    data[key] = value
}
```

sync.WaitGroup

```go
var wg sync.WaitGroup

for i := 0; i < 5; i++ {
    wg.Add(1)
    go func(n int) {
        defer wg.Done()
        work(n)
    }(i)
}

wg.Wait()
```

sync.Once

```go
var (
    once   sync.Once
    config *Config
)

func getConfig() *Config {
    once.Do(func() {
        config = loadConfig()  // Runs exactly once
    })
    return config
}
```

sync.Pool

```go
var pool = sync.Pool{
    New: func() any {
        return make([]byte, 1024)
    },
}

func process() {
    buf := pool.Get().([]byte)
    defer pool.Put(buf)
    // Use buf
}
```

sync.Map

```go
var m sync.Map

m.Store("key", "value")
v, ok := m.Load("key")
m.Delete("key")
m.Range(func(k, v any) bool {
    fmt.Println(k, v)
    return true  // Continue iteration
})
```

atomic Package

```go
import "sync/atomic"

var counter int64

atomic.AddInt64(&counter, 1)
value := atomic.LoadInt64(&counter)
atomic.StoreInt64(&counter, 0)
```

Summary

Type	Purpose
`Mutex`	Exclusive lock
`RWMutex`	Reader/writer lock
`WaitGroup`	Wait for goroutines
`Once`	Single execution
`Pool`	Object reuse
`Map`	Concurrent map

Worked example

`RWMutex` cache: many readers, occasional writer.

Save as `main.go`. Then:

```
go mod init example
go run .
```

```
package main

import (
    "fmt"
    "sync"
)

type Cache struct {
    mu   sync.RWMutex
    data map[string]int
}

func (c *Cache) Get(k string) (int, bool) {
    c.mu.RLock()
    defer c.mu.RUnlock()
    v, ok := c.data[k]
    return v, ok
}

func (c *Cache) Set(k string, v int) {
    c.mu.Lock()
    defer c.mu.Unlock()
    c.data[k] = v
}

func main() {
    c := &Cache{data: map[string]int{}}
    var wg sync.WaitGroup
```

```go
    // Writers
    for i := 0; i < 10; i++ {
        wg.Add(1)
        go func(n int) {
            defer wg.Done()
            c.Set(fmt.Sprintf("k%d", n%3), n)
        }(i)
    }
    wg.Wait()

    // Readers
    for i := 0; i < 3; i++ {
        wg.Add(1)
        go func(n int) {
            defer wg.Done()
            k := fmt.Sprintf("k%d", n)
            v, ok := c.Get(k)
            fmt.Printf("get %s -> %d ok=%v\n", k, v, ok)
        }(i)
    }
    wg.Wait()
}
```

Expected output: (values vary; keys present)

```
get k0 -> ... ok=true
get k1 -> ... ok=true
get k2 -> ... ok=true
```

More examples

`sync.Map` for disjoint keys; `atomic` for a simple counter.

```go
package main

import (
    "fmt"
    "sync"
    "sync/atomic"
)

func main() {
    var m sync.Map
    var hits atomic.Int64
    var wg sync.WaitGroup
```

```go
    for i := 0; i < 100; i++ {
        wg.Add(1)
        go func(n int) {
            defer wg.Done()
            m.Store(n%5, n)
            hits.Add(1)
        }(i)
    }
    wg.Wait()

    count := 0
    m.Range(func(_, _ any) bool {
        count++
        return true
    })
    fmt.Println("keys:", count, "hits:", hits.Load())
}
```

Expected output:

```
keys: 5 hits: 100
```

Runnable example

Save as `main.go`. Then:

```bash
go mod init example
go run .
```

```go
package main

import (
    "fmt"
    "sync"
    "sync/atomic"
)

func main() {
    // Mutex-protected counter
    var (
        mu    sync.Mutex
        count int
        wg    sync.WaitGroup
    )
```

```go
for i := 0; i < 1000; i++ {
    wg.Add(1)
    go func() {
        defer wg.Done()
        mu.Lock()
        count++
        mu.Unlock()
    }()
}
wg.Wait()
fmt.Println("mutex count:", count)

// Once: initialization runs exactly once
var (
    once   sync.Once
    inited int
)
for i := 0; i < 5; i++ {
    wg.Add(1)
    go func() {
        defer wg.Done()
        once.Do(func() {
            inited++
            fmt.Println("once.Do ran")
        })
    }()
}
wg.Wait()
fmt.Println("init count:", inited)

// atomic counter (no mutex)
var atomicCount int64
for i := 0; i < 1000; i++ {
    wg.Add(1)
    go func() {
        defer wg.Done()
        atomic.AddInt64(&atomicCount, 1)
    }()
}
wg.Wait()
fmt.Println("atomic count:", atomic.LoadInt64(&atomicCount))

// Pool: reuse a buffer
pool := sync.Pool{
    New: func() any {
        return make([]byte, 8)
    },
```

```go
    }
    buf := pool.Get().([]byte)
    copy(buf, []byte("go-pool!"))
    fmt.Println("pool buffer:", string(buf))
    pool.Put(buf)
}
```

Expected output:

```
mutex count: 1000
once.Do ran
init count: 1
atomic count: 1000
pool buffer: go-pool!
```

What to notice: Without `mu.Lock` / `atomic`, the counters would race. `sync.Once` guarantees single execution even under concurrent callers. `sync.Pool` is for short-lived reuse—never assume a put buffer is still yours after `Put`.

Try next: Remove the mutex around `count++` and run `go run -race .` to watch the race detector fire.

Related Topics

- Concurrency Basics
- Concurrency Design Guidelines

Context and Cancellation

Context and Cancellation

Overview

The `context` package manages deadlines, cancellation, and request-scoped values across API boundaries.

Creating Contexts

```
ctx := context.Background()  // Root context
ctx := context.TODO()        // Placeholder
```

Context Hierarchy & Downward Cancellation

Contexts form an immutable tree. When a parent context is canceled or times out, all child contexts derived from it are automatically canceled. However, canceling a child context does **not** affect the parent or siblings.

```
flow:
  ] -> ReqCtx[
  ReqCtx -> DBTimeout[
  ReqCtx -> RPCCancel[
  RPCCancel -> SubWorker1[
  RPCCancel -> SubWorker2[
```

```
ctx, cancel := context.WithCancel(context.Background())
defer cancel()

go func() {
    select {
    case <-ctx.Done():
        fmt.Println("cancelled:", ctx.Err())
        return
    case <-time.After(time.Hour):
        fmt.Println("completed")
    }
}()
```

```go
cancel()  // Signal cancellation downwards to all derived contexts
```

Timeout

```go
ctx, cancel := context.WithTimeout(context.Background(), 5*time.Second)
defer cancel()

select {
case result := <-doWork(ctx):
    fmt.Println(result)
case <-ctx.Done():
    fmt.Println("timeout:", ctx.Err())
}
```

Deadline

```go
deadline := time.Now().Add(10 * time.Second)
ctx, cancel := context.WithDeadline(context.Background(), deadline)
defer cancel()
```

Passing Context

```go
func handleRequest(ctx context.Context) {
    result, err := fetchData(ctx)
    if err != nil {
        return
    }
    processData(ctx, result)
}

func fetchData(ctx context.Context) (Data, error) {
    req, _ := http.NewRequestWithContext(ctx, "GET", url, nil)
    return http.DefaultClient.Do(req)
}
```

Context Values

```go
type key string
const userKey key = "user"

ctx := context.WithValue(ctx, userKey, "alice")

user := ctx.Value(userKey).(string)
```

Best Practices

- Pass context as first parameter
- Never store context in structs
- Always call cancel() (use defer)
- Only use context values for request-scoped data

Summary

Function	Purpose
`WithCancel`	Manual cancellation
`WithTimeout`	Duration-based timeout
`WithDeadline`	Absolute time limit
`WithValue`	Request-scoped data

Worked example

Context-aware worker that respects cancel and timeout.

Save as `main.go`. Then:

```go
go mod init example
go run .
```

```go
package main

import (
    "context"
    "fmt"
    "sync"
    "time"
)
```

```go
func worker(ctx context.Context, id int, out chan<- string) {
    select {
    case <-time.After(30 * time.Millisecond):
        out <- fmt.Sprintf("worker-%d done", id)
    case <-ctx.Done():
        out <- fmt.Sprintf("worker-%d %v", id, ctx.Err())
    }
}

func main() {
    ctx, cancel := context.WithTimeout(context.Background(), 10*time.Millisecond)
    defer cancel()

    out := make(chan string, 2)
    var wg sync.WaitGroup
    for i := 1; i <= 2; i++ {
        wg.Add(1)
        go func(id int) {
            defer wg.Done()
            worker(ctx, id, out)
        }(i)
    }
    wg.Wait()
    close(out)
    for msg := range out {
        fmt.Println(msg)
    }
}
```

Expected output: (both lines show deadline exceeded)

```
worker-1 context deadline exceeded
worker-2 context deadline exceeded
```

More examples

Propagate request IDs with typed `WithValue` keys.

```go
package main

import (
    "context"
    "fmt"
)
```

```go
type key int

const requestID key = 1

func handle(ctx context.Context) {
    fmt.Println("request_id:", ctx.Value(requestID))
}

func main() {
    ctx := context.WithValue(context.Background(), requestID, "req-100")
    // Child inherits values.
    child, cancel := context.WithCancel(ctx)
    defer cancel()
    handle(child)
}
```

Expected output:

```
request_id: req-100
```

Runnable example

Save as `main.go`. Then:

```bash
go mod init example
go run .
```

```go
package main

import (
    "context"
    "fmt"
    "time"
)

type ctxKey string

const requestIDKey ctxKey = "request_id"

func work(ctx context.Context, name string, d time.Duration) error {
    select {
    case <-time.After(d):
        fmt.Printf("%s finished (request_id=%v)\n", name, ctx.Value(requestIDKey))
        return nil
    case <-ctx.Done():
```

```go
        fmt.Printf("%s cancelled: %v\n", name, ctx.Err())
        return ctx.Err()
    }
}

func main() {
    // WithValue: request-scoped data flows downward
    root := context.WithValue(context.Background(), requestIDKey, "req-42")

    // Manual cancel
    ctx, cancel := context.WithCancel(root)
    go func() {
        time.Sleep(20 * time.Millisecond)
        cancel()
    }()
    _ = work(ctx, "cancel-demo", 200*time.Millisecond)

    // Timeout: child inherits parent values
    tctx, tcancel := context.WithTimeout(root, 15*time.Millisecond)
    defer tcancel()
    _ = work(tctx, "timeout-demo", 200*time.Millisecond)

    // Success path: finishes before deadline
    okCtx, okCancel := context.WithTimeout(root, 100*time.Millisecond)
    defer okCancel()
    _ = work(okCtx, "ok-demo", 5*time.Millisecond)

    // Parent cancel cancels children
    parent, pCancel := context.WithCancel(root)
    child, childCancel := context.WithCancel(parent)
    defer childCancel()
    pCancel()
    select {
    case <-child.Done():
        fmt.Println("child saw parent cancel:", child.Err())
    case <-time.After(time.Second):
        fmt.Println("child did not cancel (unexpected)")
    }
}
```

Expected output:

```
cancel-demo cancelled: context canceled
timeout-demo cancelled: context deadline exceeded
ok-demo finished (request_id=req-42)
child saw parent cancel: context canceled
```

What to notice: Always `defer cancel()` (or call it when done) so timers and resources are released. Cancellation propagates **down** the tree; canceling a child never cancels its parent. Prefer typed keys for `WithValue`, not bare strings.

Try next: Swap `WithTimeout` for `WithDeadline(time.Now().Add(...))`. Pass the cancelled context into a second goroutine and confirm both exit.

Related Topics

- Channel Patterns
- HTTP and Networking

Concurrency Design Guidelines

Concurrency Design Guidelines

Overview

Concurrency can introduce subtle bugs. Follow these guidelines to write safe concurrent code.

Share by Communicating

```go
// Prefer: communicate over channels
result := <-worker.Results()

// Avoid: shared memory with locks
mu.Lock()
result := sharedData
mu.Unlock()
```

Avoid Goroutine Leaks

```go
// Bad: goroutine leaks if nobody receives
go func() {
    ch <- result  // Blocks forever
}()

// Good: use context cancellation
go func() {
    select {
    case ch <- result:
    case <-ctx.Done():
    }
}()
```

Always Close Channels from Sender

```go
// Producer closes
func producer(out chan<- int) {
    for i := 0; i < 10; i++ {
        out <- i
    }
    close(out)  // Only sender closes
}

// Consumer ranges
for v := range in {
    process(v)
}
```

Race Detection

```go
go test -race ./...
go run -race main.go
```

Pair race testing with source analysis. GoHawk checks resource cleanup, cancellation ownership, and recognizable goroutine completion obligations. Follow the GoHawk setup and exercise to add it to a module. Review who cancels and who waits on every return path; a clean analyzer run does not prove that all goroutines terminate.

Common Patterns

Worker Pool

```go
jobs := make(chan Job)
for i := 0; i < workers; i++ {
    go worker(jobs)
}
```

Bounded Concurrency

```go
sem := make(chan struct{}, maxConcurrent)

for _, task := range tasks {
    sem <- struct{}{}
    go func(t Task) {
```

```
        defer func() { <-sem }()
        process(t)
    }(task)
}
```

Deadlock Prevention

```
// Deadlock: circular wait
ch := make(chan int)
ch <- 1    // Blocks: no receiver
<-ch       // Never reached

// Fix: buffer or separate goroutine
ch := make(chan int, 1)
ch <- 1
<-ch
```

Summary

Guideline	Reason
Share by communicating	Clearer ownership
Use context for cancellation	Prevent leaks
Only sender closes	Avoid panic
Run race detector	Catch data races

Worked example

Ownership: only the sender closes; consumers range safely.

Save as `main.go`. Then:

```
go mod init example
go run .
go run -race .
```

```
package main

import (
    "fmt"
    "sync"
)
```

```go
func produce(n int) <-chan int {
    out := make(chan int)
    go func() {
        defer close(out) // sender closes
        for i := 1; i <= n; i++ {
            out <- i
        }
    }()
    return out
}

func main() {
    in := produce(5)
    var wg sync.WaitGroup
    sum := 0
    var mu sync.Mutex

    // Fan-out consumers; channel close ends all ranges.
    for c := 0; c < 2; c++ {
        wg.Add(1)
        go func() {
            defer wg.Done()
            for v := range in {
                mu.Lock()
                sum += v
                mu.Unlock()
            }
        }()
    }
    wg.Wait()
    fmt.Println("sum 1..5:", sum)
}
```

Expected output:

```
sum 1..5: 15
```

More examples

Detect a deliberate data race (educational—expect FAIL under `-race`).

```go
package main

import (
    "fmt"
```

```go
    "sync"
)

func main() {
    // Correct version first.
    var (
        mu    sync.Mutex
        count int
        wg    sync.WaitGroup
    )
    for i := 0; i < 100; i++ {
        wg.Add(1)
        go func() {
            defer wg.Done()
            mu.Lock()
            count++
            mu.Unlock()
        }()
    }
    wg.Wait()
    fmt.Println("safe count:", count)
}
```

Expected output:

```
safe count: 100
```

Runnable example

Save as `main.go`. Then:

```
go mod init example
go run .
```

```go
package main

import (
    "context"
    "fmt"
    "sync"
    "time"
)

// safeSend never leaks a goroutine when the consumer walks away.
func safeSend(ctx context.Context, ch chan<- int, v int) {
```

```go
    select {
    case ch <- v:
    case <-ctx.Done():
        // drop result; exit without blocking forever
    }
}

func main() {
    // Bounded concurrency with a semaphore channel
    const maxConcurrent = 2
    sem := make(chan struct{}, maxConcurrent)
    var wg sync.WaitGroup

    tasks := []string{"a", "b", "c", "d", "e"}
    var mu sync.Mutex
    var order []string

    for _, t := range tasks {
        wg.Add(1)
        go func(task string) {
            defer wg.Done()
            sem <- struct{}{}        // acquire
            defer func() { <-sem }() // release

            // critical/limited work
            mu.Lock()
            order = append(order, task)
            mu.Unlock()
        }(t)
    }
    wg.Wait()
    fmt.Println("processed tasks:", len(order))

    // Avoid goroutine leak: context cancels a blocked send
    ctx, cancel := context.WithTimeout(context.Background(), 30*time.Millisecond)
    defer cancel()

    ch := make(chan int) // unbuffered, no receiver → would block
    var senderWG sync.WaitGroup
    senderWG.Add(1)
    go func() {
        defer senderWG.Done()
        safeSend(ctx, ch, 99)
        fmt.Println("sender exited without leak")
    }()
    senderWG.Wait()
```

```go
    // Only the sender closes; consumer ranges until closed
    out := make(chan int, 3)
    go func() {
        for i := 1; i <= 3; i++ {
            out <- i
        }
        close(out) // sender closes
    }()
    sum := 0
    for v := range out {
        sum += v
    }
    fmt.Println("ranged sum:", sum)
}
```

Expected output:

```
processed tasks: 5
sender exited without leak
ranged sum: 6
```

What to notice: The semaphore bounds *execution*, not just intention. `select` on `ctx.Done()` is the standard way to stop a blocked send/receive. Closing from the receiver (or twice) panics—ownership stays with the sender.

Try next: Run `go run -race .`. Temporarily remove the `ctx.Done()` case and see the program hang until you kill it.

Related Topics

- Synchronization
- Context and Cancellation

Worker Pools

Worker Pools: Controlling Chaos

Go makes it easy to spawn 100,000 goroutines. The OS makes it easy to crash when you open 100,000 file descriptors.

Unbounded concurrency is a bug. You must limit parallelism to match your resource limits (CPU, Memory, Network/DB Connections).

Pattern 1: The Semaphore (Simple Limiter)

The easiest way to limit concurrency is a buffered channel (semaphore).

```go
func ProcessItems(items []string) {
    sem := make(chan struct{}, 10) // Limit to 10 concurrent
    var wg sync.WaitGroup

    for _, item := range items {
        wg.Add(1)
        go func(val string) {
            defer wg.Done()

            sem <- struct{}{}        // Acquire token (blocks if full)
            defer func() { <-sem }() // Release token

            DoWork(val)
        }(item)
    }
    wg.Wait()
}
```

Pros: Trivial to implement. **Cons:** Still spawns N goroutines (memory cost), just blocks them from *executing* the heavy work.

Pattern 2: The Worker Pool (Fixed Goroutines)

Instead of spawning a goroutine per item, spawn a fixed number of workers (e.g., `runtime.NumCPU()`) and feed them work.

```go
func WorkerPool(jobs <-chan Job, results chan<- Result, workers int) {
    var wg sync.WaitGroup

    // Spawn fixed workers
    for i := 0; i < workers; i++ {
        wg.Add(1)
        go func(id int) {
            defer wg.Done()
            for job := range jobs {
                results <- process(job)
            }
        }(i)
    }

    wg.Wait()
    close(results) // Close results when all workers satisfy
}

func main() {
    jobs := make(chan Job, 100)
    results := make(chan Result, 100)

    go func() {
        // Enqueue jobs
        jobs <- Job{...}
        close(jobs) // Important: Tell workers no more jobs coming
    }()

    // Start pool
    WorkerPool(jobs, results, 5)

    // Consume results
    for res := range results {
        fmt.Println(res)
    }
}
```

Pros: Fixed memory footprint. Zero waste. **Cons:** Slightly more code.

2026: `errgroup` with Limits

The `golang.org/x/sync/errgroup` package handles error propagation and limits.

```go
g := new(errgroup.Group)
g.SetLimit(10) // New in recent versions
```

```go
for _, item := range items {
    item := item
    g.Go(func() error {
        return process(item)
    })
}

if err := g.Wait(); err != nil {
    return err
}
```

This is the preferred modern way. It handles wait groups, error propagation (first error cancels context), and concurrency limits in one standard package.

Summary

- Never use `go func()` inside a loop without a limit mechanism.
- Use `errgroup.SetLimit` for 90% of cases.
- Use manual Worker Patterns for complex, long-lived background processing queues.

Worked example

Fixed pool with context cancel mid-flight (workers and producer both exit cleanly).

Save as `main.go`. Then:

```bash
go mod init example
go run .
```

```go
package main

import (
    "context"
    "fmt"
    "sync"
)

func pool(ctx context.Context, jobs <-chan int, workers int) <-chan int {
    out := make(chan int)
    var wg sync.WaitGroup
    for w := 0; w < workers; w++ {
        wg.Add(1)
        go func() {
            defer wg.Done()
```

```
            for {
                select {
                case <-ctx.Done():
                    return
                case j, ok := <-jobs:
                    if !ok {
                        return
                    }
                    select {
                    case out <- j * 2:
                    case <-ctx.Done():
                        return
                    }
                }
            }
        }()
    }
    go func() {
        wg.Wait()
        close(out)
    }()
    return out
}

func main() {
    ctx, cancel := context.WithCancel(context.Background())
    defer cancel()

    jobs := make(chan int)
    results := pool(ctx, jobs, 3)

    go func() {
        defer close(jobs)
        for i := 1; i <= 20; i++ {
            select {
            case <-ctx.Done():
                return
            case jobs <- i:
            }
        }
    }()

    count := 0
    for r := range results {
        _ = r
        count++
        if count == 5 {
```

```go
            cancel() // stop early; remaining workers observe ctx.Done()
        }
    }
    fmt.Println("stopped after:", count)
}
```

Expected output: (may be 5 or a few more if in-flight sends complete)

```
stopped after: 5
```

More examples

Stdlib-only errgroup-style first-error cancel.

```go
package main

import (
    "context"
    "errors"
    "fmt"
    "sync"
)

func main() {
    ctx, cancel := context.WithCancel(context.Background())
    defer cancel()

    var (
        wg   sync.WaitGroup
        once sync.Once
        err  error
    )
    tasks := []func(context.Context) error{
        func(ctx context.Context) error { return nil },
        func(ctx context.Context) error { return errors.New("boom") },
        func(ctx context.Context) error {
            <-ctx.Done()
            return ctx.Err()
        },
    }
    for _, t := range tasks {
        t := t
        wg.Add(1)
        go func() {
            defer wg.Done()
```

```go
            if e := t(ctx); e != nil {
                once.Do(func() {
                    err = e
                    cancel()
                })
            }
        }()
    }
    wg.Wait()
    fmt.Println("first error:", err)
}
```

Expected output:

```
first error: boom
```

Runnable example

Save as `main.go`. Then:

```
go mod init example
go run .
```

```go
package main

import (
    "fmt"
    "sync"
)

type Job struct {
    ID  int
    Val int
}

type Result struct {
    JobID  int
    Worker int
    Out    int
}

func runPool(jobs <-chan Job, workers int) <-chan Result {
    results := make(chan Result)
    var wg sync.WaitGroup
```

```go
    for w := 1; w <= workers; w++ {
        wg.Add(1)
        go func(id int) {
            defer wg.Done()
            for job := range jobs {
                results <- Result{
                    JobID:  job.ID,
                    Worker: id,
                    Out:    job.Val * job.Val,
                }
            }
        }(w)
    }

    go func() {
        wg.Wait()
        close(results)
    }()
    return results
}

func main() {
    const nJobs = 8
    const nWorkers = 3

    jobs := make(chan Job, nJobs)
    results := runPool(jobs, nWorkers)

    // Enqueue, then close so workers drain and exit.
    for i := 1; i <= nJobs; i++ {
        jobs <- Job{ID: i, Val: i}
    }
    close(jobs)

    // Semaphore-style limiter (spawn per item, cap concurrency)
    sem := make(chan struct{}, 2)
    var limWG sync.WaitGroup
    limitedSum := 0
    var mu sync.Mutex
    for i := 1; i <= 4; i++ {
        limWG.Add(1)
        go func(n int) {
            defer limWG.Done()
            sem <- struct{}{}
            defer func() { <-sem }()
            mu.Lock()
            limitedSum += n
```

```go
            mu.Unlock()
        }(i)
    }
    limWG.Wait()

    poolSum := 0
    seenWorkers := map[int]bool{}
    for r := range results {
        poolSum += r.Out
        seenWorkers[r.Worker] = true
    }

    fmt.Println("pool workers used:", len(seenWorkers))
    fmt.Println("pool sum of squares:", poolSum) // 1+4+9+16+25+36+49+64 = 204
    fmt.Println("semaphore sum:", limitedSum)    // 10
}
```

Expected output:

```
pool workers used: 3
pool sum of squares: 204
semaphore sum: 10
```

What to notice: Fixed workers + closed job channel give a hard cap on goroutines and a clean shutdown. Closing `results` only after `wg.Wait()` prevents "send on closed channel." The semaphore still spawns N goroutines but caps concurrent critical sections—cheaper to write, less ideal under huge N.

Try next: Set `nWorkers` to 1 and confirm the sum is unchanged. Replace the mutex around `limitedSum` with `atomic.AddInt64`. For production error propagation + limits, prefer `golang.org/x/sync/errgroup` (not shown here so this stays stdlib-only).

Pipelines & Complex Patterns

Pipelines and `sync.Cond`

Beyond simple worker pools, Go concurrency shines in **Data Pipelines** (streaming data through stages) and **Signaling** (complex coordination).

1. The Pipeline Pattern (Fan-Out / Fan-In)

Pipelines allow you to process streams of data where each stage runs concurrently.

Stages: 1. **Generator:** Converts data (file lines, DB rows) into a channel. 2. **Transformer:** Reads one channel, modifies/filters, writes to another. 3. **Sink:** Consumes final output.

```go
func Generator(nums ...int) <-chan int {
    out := make(chan int)
    go func() {
        for _, n := range nums { out <- n }
        close(out)
    }()
    return out
}

func Square(in <-chan int) <-chan int {
    out := make(chan int)
    go func() {
        for n := range in { out <- n * n }
        close(out)
    }()
    return out
}

func Merge(channels ...<-chan int) <-chan int {
    // Advanced: Fan-In multiple channels to one
    var wg sync.WaitGroup
    out := make(chan int)

    output := func(c <-chan int) {
        defer wg.Done()
        for n := range c { out <- n }
    }
```

```go
    wg.Add(len(channels))
    for _, c := range channels {
        go output(c)
    }

    go func() {
        wg.Wait()
        close(out)
    }()
    return out
}
```

Cancellation: Real pipelines must pass `context.Context` to every stage to support early cancellation (e.g., stop processing if the user disconnects).

2. The Overlooked `sync.Cond`

Channels are for passing data. Mutexes are for locking data. **`sync.Cond`** is for **broadcasting signals**.

Use Case: You have 100 goroutines waiting for a specific event (e.g., "Configuration Loaded" or "Queue is not empty"). * **Channels:** Sending 100 messages is O(N). * **Channels (Close):** Closing a channel broadcasts to all, but you can only do it *once*. * **sync.Cond:** Can `Broadcast()` multiple times.

Example: The Race Start

```go
var mu sync.Mutex
cond := sync.NewCond(&mu)
ready := false

// Workers
for i := 0; i < 10; i++ {
    go func(id int) {
        mu.Lock()
        for !ready {
            cond.Wait() // Atomically unlocks mu and suspends execution
        }
        mu.Unlock()
        fmt.Println("Worker", id, "started")
    }(i)
}

// Coordinator
time.Sleep(1 * time.Second)
```

```
mu.Lock()
ready = true
mu.Unlock()
cond.Broadcast() // Wakes all 10 workers simultaneously
```

Warning: `cond.Wait()` *must* be in a loop (spurious wakeups are possible, though rare in Go, logic dictates checking the condition again).

Summary

- **Pipelines:** Composable, streaming architecture. Great for ETL jobs.
- **Fan-In:** Merging multiple concurrent streams.
- **sync.Cond:** For "One-to-Many" signaling where the event can happen multiple times (unlike `close(ch)` which is one-off).

Worked example

Cancellable pipeline stages with `context`.

Save as `main.go`. Then:

```
go mod init example
go run .
```

```
package main

import (
    "context"
    "fmt"
)

func gen(ctx context.Context, nums ...int) <-chan int {
    out := make(chan int)
    go func() {
        defer close(out)
        for _, n := range nums {
            select {
            case <-ctx.Done():
                return
            case out <- n:
            }
        }
    }()
    return out
```

```go
}

func mul(ctx context.Context, in <-chan int, factor int) <-chan int {
    out := make(chan int)
    go func() {
        defer close(out)
        for n := range in {
            select {
            case <-ctx.Done():
                return
            case out <- n * factor:
            }
        }
    }()
    return out
}

func main() {
    ctx, cancel := context.WithCancel(context.Background())
    defer cancel()

    // 1,2,3,4 → *10 → take first two then cancel
    stream := mul(ctx, gen(ctx, 1, 2, 3, 4), 10)
    var got []int
    for v := range stream {
        got = append(got, v)
        if len(got) == 2 {
            cancel()
        }
    }
    fmt.Println("got:", got)
}
```

Expected output: (length 2; may include a couple extra depending on timing)

```
got: [10 20]
```

More examples

`sync.Cond` broadcast twice (reload signal).

```go
package main

import (
    "fmt"
```

```go
    "sync"
)

func main() {
    var mu sync.Mutex
    cond := sync.NewCond(&mu)
    generation := 0

    const workers = 3
    var wg sync.WaitGroup
    seen := make(chan int, workers*2)

    for i := 0; i < workers; i++ {
        wg.Add(1)
        go func() {
            defer wg.Done()
            local := 0
            for local < 2 {
                mu.Lock()
                for local == generation {
                    cond.Wait()
                }
                local = generation
                mu.Unlock()
                seen <- local
            }
        }()
    }

    for g := 1; g <= 2; g++ {
        mu.Lock()
        generation = g
        mu.Unlock()
        cond.Broadcast()
    }

    wg.Wait()
    close(seen)
    count := 0
    for range seen {
        count++
    }
    fmt.Println("signals delivered:", count) // 3 workers * 2 gens
}
```

Expected output:

```
signals delivered: 6
```

Runnable example

Save as `main.go`. Then:

```
go mod init example
go run .
```

```
package main

import (
    "fmt"
    "sync"
    "time"
)

func generator(nums ...int) <-chan int {
    out := make(chan int)
    go func() {
        for _, n := range nums {
            out <- n
        }
        close(out)
    }()
    return out
}

func square(in <-chan int) <-chan int {
    out := make(chan int)
    go func() {
        defer close(out)
        for n := range in {
            out <- n * n
        }
    }()
    return out
}

func filterEven(in <-chan int) <-chan int {
    out := make(chan int)
    go func() {
        defer close(out)
        for n := range in {
            if n%2 == 0 {
```

```go
                out <- n
            }
        }
    }()
    return out
}

func main() {
    // Pipeline: generate → square → keep even squares
    // inputs 1..5 → squares 1,4,9,16,25 → even 4,16
    stage1 := generator(1, 2, 3, 4, 5)
    stage2 := square(stage1)
    stage3 := filterEven(stage2)

    var evenSquares []int
    for v := range stage3 {
        evenSquares = append(evenSquares, v)
    }
    fmt.Println("even squares:", evenSquares)

    // sync.Cond: one-to-many start signal
    var mu sync.Mutex
    cond := sync.NewCond(&mu)
    ready := false

    const n = 4
    var wg sync.WaitGroup
    started := make(chan int, n)

    for i := 1; i <= n; i++ {
        wg.Add(1)
        go func(id int) {
            defer wg.Done()
            mu.Lock()
            for !ready {
                cond.Wait() // unlocks mu while waiting
            }
            mu.Unlock()
            started <- id
        }(i)
    }

    time.Sleep(20 * time.Millisecond) // workers park on Wait
    mu.Lock()
    ready = true
    mu.Unlock()
    cond.Broadcast()
```

```go
    wg.Wait()
    close(started)
    count := 0
    for range started {
        count++
    }
    fmt.Println("workers started after broadcast:", count)
}
```

Expected output:

```
even squares: [4 16]
workers started after broadcast: 4
```

What to notice: Each pipeline stage owns its output channel and closes it when done, so `range` terminates. `cond.Wait()` lives in a `for !ready` loop (predicate re-check). `Broadcast` wakes everyone; `Signal` would wake only one.

Try next: Insert a third pipeline stage that sums values. Change `Broadcast` to `Signal` in a loop and observe staggered starts.

Goroutine Leak Profiles

Goroutine Leak Profiles

A leaked goroutine is blocked forever on a channel or `sync` primitive that nothing live can ever unblock. Over time those stacks burn memory and keep the GC busy. Go **1.27** makes the `goroutineleak` profile generally available in `runtime/pprof` (and at `/debug/pprof/goroutineleak` when you import `net/http/pprof`). Collect it in production the same way you collect heap or goroutine profiles — no `GOEXPERIMENT` flag.

Mental model

- The runtime piggybacks on the garbage collector. Unblocked goroutines are live. Anything they can still reach (transitively) can still be used to wake waiters. A goroutine blocked on a primitive that is **unreachable from every live goroutine** is leaked.
- Detection covers channel send/receive (including `nil` channels), blocking `select`, and `sync.Mutex` / `RWMutex` / `WaitGroup` / `Cond`. It does **not** mark goroutines stuck in network or file I/O as leaked.
- Globals and still-runnable stacks can hide leaks (the primitive stays "reachable"). Prefer short-lived local channels whose lifetime you understand.
- `pprof.Lookup("goroutineleak")` builds the profile on demand. `Count()` is the number of leaked samples. `WriteTo(w, 1)` prints an aggregated text dump; `WriteTo(w, 0)` writes the binary form for `go tool pprof`.
- If you already serve `net/http/pprof`, the HTTP endpoint appears automatically. Hitting it triggers a GC cycle — profile on a schedule, not on every request.

Worked examples

Case 1: Prove a nil-channel leak with `Count`

Save as `nil_recv_leak.go`. A receive on a nil channel never completes. After the child blocks, the leak profile reports it.

```go
// nil_recv_leak.go
package main

import (
    "fmt"
    "runtime"
    "runtime/pprof"
)
```

```go
func main() {
    go func() {
        var c chan int
        <-c
        panic("unreachable")
    }()
    for range 10 {
        runtime.Gosched()
    }

    prof := pprof.Lookup("goroutineleak")
    if prof == nil {
        fmt.Println("profile missing")
        return
    }
    fmt.Printf("name=%s\n", prof.Name())
    fmt.Printf("leaked count=%d\n", prof.Count())
}
```

Run:

```go run nil_recv_leak.go
```

Output:

```
name=goroutineleak
leaked count=1
```

`Count()` stays stable across machines. Stack addresses in a text dump would not.

Case 2: Early return abandons workers

Save as `early_return_leak.go`. Desk workers send on an unbuffered channel. The parent hits an error and returns **without receiving** — every sender leaks.

```go
// early_return_leak.go
package main

import (
    "errors"
    "fmt"
    "runtime"
    "runtime/pprof"
)

type ticketResult struct {
```

```go
    id int
}

func processTickets(ids []int, failFast bool) error {
    ch := make(chan ticketResult)
    for _, id := range ids {
        go func(id int) {
            ch <- ticketResult{id: id}
        }(id)
    }
    if failFast {
        // Simulate discovering a problem before draining results.
        return errors.New("desk closed")
    }
    for range ids {
        <-ch
    }
    return nil
}

func main() {
    err := processTickets([]int{1, 2, 3, 4, 5}, true)
    fmt.Printf("err=%v\n", err)
    for range 20 {
        runtime.Gosched()
    }
    fmt.Printf("leaked count=%d\n", pprof.Lookup("goroutineleak").Count())
}
```

Run:

```go
go run early_return_leak.go
```

Output:

```
err=desk closed
leaked count=5
```

Five senders, zero receives: five leaks. The same bug shows up when you return on the first worker error after partial receives — leftover senders still hang; the buffer fix below covers both shapes.

Case 3: Buffer the channel (the boring fix)

Save as `early_return_fixed.go`. Size the buffer to the number of workers so every send completes even when the parent returns early.

```go
// early_return_fixed.go
package main

import (
    "errors"
    "fmt"
    "runtime"
    "runtime/pprof"
)

type ticketResult struct {
    id int
}

func processTickets(ids []int, failFast bool) error {
    ch := make(chan ticketResult, len(ids))
    for _, id := range ids {
        go func(id int) {
            ch <- ticketResult{id: id}
        }(id)
    }
    if failFast {
        return errors.New("desk closed")
    }
    for range ids {
        <-ch
    }
    return nil
}

func main() {
    err := processTickets([]int{1, 2, 3, 4, 5}, true)
    fmt.Printf("err=%v\n", err)
    for range 20 {
        runtime.Gosched()
    }
    fmt.Printf("leaked count=%d\n", pprof.Lookup("goroutineleak").Count())
}
```

Run:

```go
go run early_return_fixed.go
```

Output:

```
err=desk closed
leaked count=0
```

A buffer of `len(ids)` is the boring default when every worker sends exactly once and you may abandon the receive loop. Context cancellation with a buffered result channel (or a drain on cancel) is the other common pattern.

Case 4: Timeout without a buffer

Save as `timeout_leak.go`. When the context wins the `select`, the worker's send has nowhere to go.

```go
// timeout_leak.go
package main

import (
    "context"
    "fmt"
    "runtime"
    "runtime/pprof"
    "time"
)

func fetchPrice(ctx context.Context) (int, error) {
    ch := make(chan int)
    go func() {
        time.Sleep(50 * time.Millisecond)
        ch <- 42
    }()
    select {
    case price := <-ch:
        return price, nil
    case <-ctx.Done():
        return 0, ctx.Err()
    }
}

func main() {
    for range 3 {
        ctx, cancel := context.WithTimeout(context.Background(), 5*time.Millisecond)
        _, err := fetchPrice(ctx)
        cancel()
        fmt.Printf("err=%v\n", err)
    }
    time.Sleep(80 * time.Millisecond)
    for range 10 {
        runtime.Gosched()
    }
    fmt.Printf("leaked count=%d\n", pprof.Lookup("goroutineleak").Count())
```

```
}
```

Run:

```
go run timeout_leak.go
```

Output:

```
err=context deadline exceeded
err=context deadline exceeded
err=context deadline exceeded
leaked count=3
```

Fix: `ch := make(chan int, 1)` so the late send parks in the buffer instead of forever.

Case 5: Text dump points at the send

Save as `leak_dump.go`. `WriteTo` with debug level `1` prints only leaked stacks. Addresses differ per run; we print whether the dump mentions the leaking function name.

```go
// leak_dump.go
package main

import (
    "bytes"
    "fmt"
    "runtime"
    "runtime/pprof"
    "strings"
)

func leakyWorker() {
    ch := make(chan struct{})
    go func() {
        ch <- struct{}{}
    }()
    // Parent never receives - sender leaks.
    for range 10 {
        runtime.Gosched()
    }
}

func main() {
    leakyWorker()
    var buf bytes.Buffer
    if err := pprof.Lookup("goroutineleak").WriteTo(&buf, 1); err != nil {
```

```go
        fmt.Println("write:", err)
        return
    }
    text := buf.String()
    fmt.Printf("has total line? %v\n", strings.Contains(text, "total"))
    fmt.Printf("mentions leakyWorker? %v\n", strings.Contains(text, "leakyWorker"))
    fmt.Printf("leaked count=%d\n", pprof.Lookup("goroutineleak").Count())
}
```

Run:

```go
go run leak_dump.go
```

Output:

```
has total line? true
mentions leakyWorker? true
leaked count=1
```

In a service, prefer `curl http://127.0.0.1:6060/debug/pprof/goroutineleak?debug=1` or `go tool pprof http://127.0.0.1:6060/debug/pprof/goroutineleak` and `list` the hot function.

The trap

Save as `global_hides_leak.go`. A channel kept in a package-level variable stays reachable from the main goroutine, so the profiler may **not** report the blocked sender as leaked — even though nobody will ever receive again.

```go
// global_hides_leak.go
package main

import (
    "fmt"
    "runtime"
    "runtime/pprof"
)

var deskBus = make(chan int)

func main() {
    go func() {
        deskBus <- 1
    }()
    for range 10 {
        runtime.Gosched()
    }
```

```go
    // Nobody receives, but deskBus is a global root.
    fmt.Printf("leaked count=%d\n", pprof.Lookup("goroutineleak").Count())
    fmt.Println("(global channel can hide a real hang from the profile)")
}
```

Run:

```
go run global_hides_leak.go
```

Output:

```
leaked count=0
(global channel can hide a real hang from the profile)
```

Absence of a leak sample is not proof that every goroutine will wake. Combine the profile with `goleak` / `synctest` in tests, and keep channel ownership local when you can.

`goleak`: make shutdown a test assertion

goleak checks for unexpected goroutines remaining after tests. It tests observed shutdown behavior, complementing GoHawk's source analysis and the runtime profile above.

In a fresh directory, save this as `shutdown_test.go`:

```go
package shutdown

import (
    "testing"

    "go.uber.org/goleak"
)

func TestMain(m *testing.M) {
    goleak.VerifyTestMain(m)
}

func TestWorkerStops(t *testing.T) {
    stop := make(chan struct{})
    done := make(chan struct{})
    go func() {
        defer close(done)
        <-stop
    }()
    close(stop)
    <-done
}
```

```go
go mod init example.com/shutdown
go get go.uber.org/goleak@v1.3.0
go test -race -count=1 ./...
```

Remove both `close(stop)` and `<-done`, then rerun. The test body returns, but the package should fail its leak check with a worker stack trace. Restore both lines to make it pass. Removing only the close would hang the test in the receive instead.

Checking once in `TestMain` also accommodates tests using `t.Parallel()`; per-test checks cannot distinguish their workers from other tests still running. Merge this check into an existing `TestMain` if the package has one. Stop intentional background services before the final check, and document any narrowly scoped exclusions.

The boring rule

- On Go 1.27+, treat `goroutineleak` as a first-class production profile next to `heap` and `goroutine`.
- Import `_ "net/http/pprof"` once (behind auth / localhost). Collect `/debug/pprof/goroutineleak` on a coarse schedule.
- When every worker sends once and the parent may return early or cancel: buffer the result channel (`make(chan T, n)`), or drain on the cancel path.
- Close work-distribution channels when the producer is done so `range` workers exit.
- Do not rely on the profile for I/O hangs or for primitives parked in globals.
- Keep `goleak` in unit tests; use the profile where tests cannot reach — long-running desk services.

Try this

1. In `early_return_fixed.go`, change the buffer to `1` and re-run. How many leaks appear?
2. In `timeout_leak.go`, switch to `make(chan int, 1)` and confirm `leaked count=0`.
3. Add `import _ "net/http/pprof"` and `http.ListenAndServe("127.0.0.1:6060", nil)` to a tiny server that leaks on purpose. Fetch `?debug=1` with curl.
4. Move `deskBus` in `global_hides_leak.go` into `main` as a local, drop all receives, and compare `Count()` to the global case.

Socket Programming

Introduction: Why Sockets Matter in Go

Introduction: Why Sockets Matter in Go

Every distributed system, microservice mesh, database replication pipeline, and cloud-native control plane rests on a foundation of raw network sockets. While modern software engineering frequently operates at the higher layers of HTTP, REST, and gRPC, abstract abstractions inevitably leak when latency spikes, connections stall, file descriptors deplete, or network partitions occur.

Go was engineered from its inception at Google to solve server-side systems challenges in distributed environments. Unlike traditional languages that force developers to choose between complex asynchronous event loops (Node.js, Python asyncio) or heavy, OS-thread-per-connection architectures (C, classic Java), Go introduces a revolutionary paradigm: **synchronous code backed by an asynchronous runtime network poller**.

Understanding socket programming in Go allows you to bypass overhead, implement custom streaming protocols, engineer ultra-low-latency proxies, diagnose insidious network failure modes, and scale network applications to millions of concurrent sessions with minimal memory footprint.

The Promise of This Book

This section provides a comprehensive, ground-up, and production-tested guide to socket and low-level network programming in Go. By working through these chapters, you will:

1. **Demystify the Socket Abstraction:** Understand how operating system sockets bridge user space and the kernel TCP/IP stack.
2. **Master the `net` Package:** Gain deep mastery over `net.Conn`, `net.Listener`, `net.PacketConn`, and their concrete TCP, UDP, and Unix domain socket implementations.
3. **Build Industrial-Strength Services:** Write concurrent, leak-free servers and clients equipped with timeouts, context cancellation, heartbeats, and graceful shutdown patterns.
4. **Engineer Custom Protocols:** Design and implement line-based, length-prefixed, and binary protocols with robust serialization formats (JSON, Protobuf, MessagePack).
5. **Secure Network Channels:** Deploy Transport Layer Security (TLS 1.3), mutual TLS (mTLS), and custom certificate validation.
6. **Scale and Optimize:** Leverage Go's runtime netpoller, zero-copy operations (`io.Copy`, `sendfile`), buffer reuse (`sync.Pool`), and kernel socket tuning (`SO_REUSEPORT`, TCP buffers, `TCP_NODELAY`).
7. **Test and Debug Reliably:** Write deterministic network tests using `net.Pipe` and listeners, trace syscalls with `strace`, capture packets with `tcpdump` and Wireshark, and diagnose goroutine hangs with `delve`.

How to Use This Book

Each chapter is structured for incremental learning and direct practical application:

- **Conceptual Foundations:** Visual architectural diagrams, protocol state machines, and kernel interaction models explain *how* and *why* things work under the hood.
- **Production-Grade Implementations:** Full, runnable Go code snippets illustrate idiomatic patterns, defensive error handling, and robust resource management.
- **Failure Modes & Edge Cases:** Every topic highlights common production bugs—such as unclosed file descriptors, goroutine leaks, partial writes, buffer bloat, and deadlocks—along with concrete remedies.
- **Hands-On Exercises:** Applied engineering problems reinforce design trade-offs and low-level mechanics.

You can read this section sequentially from front to back, or use individual chapters as an authoritative reference when solving specific networking challenges in production.

Prerequisites

To gain the most from this material, you should possess:

- **Intermediate Go Proficiency:** Comfort with Go syntax, structs, interfaces, slices, error handling, goroutines, channels, and the `context` package.
- **Basic Systems Understanding:** General familiarity with operating system concepts such as processes, threads, file descriptors, and command-line terminal tooling.
- **Curiosity for Systems Internals:** An interest in understanding how bytes travel through kernel buffers, network interfaces, and physical media.

No prior experience with C socket APIs (`socket()`, `bind()`, `listen()`, `select()`, `epoll()`) is required; all essential network mechanics are derived from first principles.

Setting Up Your Environment

All code examples in this guide are built and tested against **Go 1.22+** and run natively on Linux, macOS, and BSD environments (with platform-specific notes provided where applicable, such as Linux abstract Unix domain sockets or `SO_REUSEPORT`).

Go Toolchain Verification

Verify your Go installation:

```
$ go version
go version go1.22.0 linux/amd64
```

Create a dedicated workspace module for socket experimentation:

```
mkdir -p ~/go-sockets-lab
cd ~/go-sockets-lab
go mod init go-sockets-lab
```

Essential Diagnostic Utilities

Throughout this section, we will inspect socket states, active connections, and network packets using standard diagnostic tools. Ensure these utilities are installed on your workstation:

```
# On Debian/Ubuntu Linux systems:
sudo apt-get update && sudo apt-get install -y \
  iproute2 \
  netcat-openbsd \
  tcpdump \
  wireshark \
  strace \
  curl

# Verify availability:
ss -version
tcpdump --version
```

The Philosophy Behind Go's Networking Approach

To appreciate Go's socket architecture, one must examine the fundamental challenge of network I/O: **I/O operations are orders of magnitude slower than CPU execution**.

```
CPU Register:       ~1 ns
L1 Cache:           ~1-2 ns
Main Memory (RAM):  ~100 ns
Local SSD Read:     ~10-50 µs
Network Round-Trip: ~0.5 ms (LAN) to 150+ ms (Cross-Continent WAN)
```

When an application calls `Read()` on a network socket and no data has arrived, what should the operating system do?

The Traditional Dilemma

1. **Synchronous Thread-per-Connection (POSIX C / Classic Java):**
   - Sockets are blocking. Calling `read()` halts the OS thread.
   - To serve 10,000 concurrent clients, the operating system must spawn 10,000 OS threads.
   - Each OS thread allocates 1 MB to 8 MB of reserved virtual memory for its stack.
   - The OS kernel scheduler spends significant CPU cycles performing context switches, thrashing CPU caches and exhausting memory.
2. **Asynchronous Event-Driven Multiplexing (Node.js / Python asyncio / Nginx):**
   - Sockets are non-blocking. A single OS thread executes an event loop (`epoll`/`kqueue`).
   - Sockets notify the loop when ready for reading or writing.
   - Memory overhead is tiny, and concurrency is high.
   - However, application code becomes fragmented into callbacks, state machines, or complex `async/await` boundaries. CPU-bound work can starve the event loop, and debugging stack traces across async ticks is notoriously difficult.

The Go Synthesis: Goroutines + Runtime Netpoller

Go eliminates this false dichotomy by coupling lightweight user-space threads (**goroutines**) with an integrated kernel event notification engine (**the runtime netpoller**):

```
+-------------------------------------------------------------------+
|                     User Application Code                         |
|   conn.Read(buf)  <-- Clean, readable, sequential blocking code   |
+-------------------------------------------------------------------+


+-------------------------------------------------------------------+
|                         Go Runtime                                |
|   1. Socket set to non-blocking (O_NONBLOCK)                      |
|   2. If syscall returns EAGAIN/EWOULDBLOCK:                       |
|      - Netpoller registers file descriptor with epoll/kqueue       |
|      - Goroutine parked in _Gwaiting state                        |
|      - OS Thread (M) freed immediately to run another goroutine   |
+-------------------------------------------------------------------+


+-------------------------------------------------------------------+
|                       Operating System Kernel                     |
|   epoll_wait() / kqueue() signals when packet arrives in buffer    |
|   -> Netpoller wakes parked goroutine back to _Grunnable           |
+-------------------------------------------------------------------+
```

1. **Minimal Memory Footprint:** Goroutines begin with a contiguous stack of only 2 KB (which grows and shrinks dynamically as needed), compared to megabytes for OS threads. A server can effortlessly run hundreds of thousands of concurrent goroutines.
2. **Straightforward Sequential Code:** You write direct, linear code:

```go
buf := make([]byte, 1024)
n, err := conn.Read(buf) // Looks blocking, but yields the OS thread!
```

3. **Transparent Multiplexing:** Under the hood, Go sets all network file descriptors to non-blocking mode (`O_NONBLOCK`). When a `Read` or `Write` returns `EAGAIN` or `EWOULDBLOCK`, the Go runtime parks the goroutine, registers its file descriptor with the kernel netpoller (`epoll` on Linux, `kqueue` on macOS/BSD, `IOCP` on Windows), and schedules another goroutine on the same OS thread. When kernel events arrive, the netpoller automatically awakens the parked goroutine.

This design gives Go programs the sheer throughput and scalability of asynchronous event loops combined with the simplicity, debuggability, and maintainability of synchronous programming.

Let us now begin our exploration by establishing the foundational network mechanics that govern every socket connection.

Chapter 1: Networking Fundamentals

Chapter 1: Networking Fundamentals: The Foundation You Cannot Skip

Before diving into Go's `net` package, you must understand the environment in which sockets operate. A socket is not magic; it is an operating system abstraction representing an endpoint for communication between processes across a network or locally within the same machine.

To write reliable, production-ready network code, you must know how data travels across the wire, how operating systems manage network stacks, how addressing functions, and why protocols make specific reliability trade-offs.

How the Internet Works at a Glance

At its physical essence, the Internet is a global mesh of interconnected networks exchanging discrete chunks of data called **packets**.

When two computers communicate over the internet:

1. **Packet Switching:** Information is split into small pieces (packets), stamped with destination and source addresses, and routed independently across routers and switches.
2. **Hop-by-Hop Forwarding:** Routers inspect the destination IP address of each incoming packet and look up their routing tables to determine the optimal next hop.
3. **Best-Effort Delivery:** The underlying IP layer makes no guarantees that packets will arrive in order, arrive uncorrupted, or arrive at all. Packets can be dropped by congested buffers, duplicated by network switches, or routed along divergent paths.

The higher-level transport protocols—primarily TCP and UDP—are responsible for transforming this unpredictable, best-effort packet delivery system into practical communication channels for applications.

The TCP/IP Stack Explained Simply

The **TCP/IP model** (also known as the Internet Protocol Suite) organizes network functionality into four conceptual layers. Each layer encapsulates data from the layer above it by prepending its own header:

```
+-------------------------------------------------------------+
| Application Layer   (HTTP, DNS, SSH, Custom Protocols)      |
| [ Application Payload Data ]                                |
+-------------------------------------------------------------+
                                 Encapsulation

+-------------------------------------------------------------+
| Transport Layer     (TCP, UDP)                              |
| [ TCP/UDP Header | Application Payload Data ]               |
+-------------------------------------------------------------+


+-------------------------------------------------------------+
| Network Layer       (IPv4, IPv6, ICMP)                      |
| [ IP Header | TCP/UDP Header | Application Payload Data ]   |
+-------------------------------------------------------------+


+-------------------------------------------------------------+
| Link / Physical     (Ethernet, Wi-Fi, Fiber)                |
| [ Ethernet Frame Header | IP | TCP | Data | CRC Checksum ]  |
+-------------------------------------------------------------+
```

Layer Responsibilities

Layer	Primary Protocol	Key Abstraction	Addressing Unit	Go Programming Focus
Application	HTTP, gRPC, WebSocket	Messages, Streams	URLs, Hostnames, RPC methods	High-level business logic, protocols
Transport	TCP, UDP	Sockets, Ports	Port numbers (`1`–`65535`)	Package `net` (`net.Conn`, `net.Listener`)
Network	IPv4, IPv6	Packets / Datagrams	IP addresses (`192.0.2.1`, `::1`)	Address resolution, routing, CIDR blocks
Link / Physical	Ethernet, 802.11	Frames, Bits	MAC addresses	Managed by kernel device drivers

As a Go socket programmer, you primarily interact at the boundary between the **Application Layer** and the **Transport Layer**. The socket API provided by your operating system kernel (and wrapped cleanly by Go's `net` package) hides the complexities of framing, IP routing, and device driver interrupts.

IP Addresses and Ports

A network socket endpoint is uniquely identified across an IP network by the tuple:

$$\text{Endpoint} = (\text{IP Address}, \text{Port Number})$$

And an established two-way connection is identified by the **5-tuple**:

$$(\text{Protocol}, \text{Source IP}, \text{Source Port}, \text{Destination IP}, \text{Destination Port})$$

```
Client Host (198.51.100.25)                Server Host (203.0.113.10)
+-------------------------+                +-------------------------+
| Ephemeral Port: 54321   |<==============>| Listening Port: 8080    |
+-------------------------+                +-------------------------+
        5-Tuple: (TCP, 198.51.100.25:54321, 203.0.113.10:8080)
```

IP Addresses: IPv4 vs IPv6

- **IPv4:** 32-bit addresses formatted as dotted-decimal notation (e.g., `192.168.1.10`). Limited to approximately 4.3×10^9 addresses, leading to widespread use of NAT (Network Address Translation).
- **IPv6:** 128-bit addresses formatted in hexadecimal with colons (e.g., `2001:0db8:85a3::8a2e:0370:7334`). Provides 3.4×10^{38} unique addresses, eliminating the need for NAT and natively integrating security and autoconfiguration.
- **Dual-Stack Sockets:** Go transparently handles dual-stack networking. When you bind to `":8080"`, Go by default listens on both IPv4 and IPv6 wildcard addresses (`0.0.0.0` and `::`).

Port Numbers

Ports are 16-bit unsigned integers ranging from `0` to `65535`:

1. **Well-Known Ports (`0`–`1023`):** Reserved for system services (e.g., HTTP `80`, HTTPS `443`, SSH `22`, DNS `53`). Binding to these ports on POSIX systems typically requires elevated permissions (root or `CAP_NET_BIND_SERVICE`).
2. **Registered Ports (`1024`–`49151`):** Assigned by IANA for specific user applications and services (e.g., PostgreSQL `5432`, Redis `6379`).

3. **Dynamic / Ephemeral Ports (`49152`–`65535`):** Temporarily allocated by the operating system kernel when an outgoing client connection is initiated without an explicit local port specified.

DNS Basics for Network Programmers

The Domain Name System (DNS) is the distributed hierarchical database that resolves human-readable hostnames (such as `api.example.com`) to machine-routable IP addresses (`93.184.216.34` or `2606:2800:220:1:248:1863:25c8:19e6`).

Common DNS Record Types

- **A:** Maps a hostname to an IPv4 address.
- **AAAA:** Maps a hostname to an IPv6 address.
- **CNAME:** Canonical name (alias) pointing to another hostname.
- **PTR:** Pointer record used for reverse DNS lookups (IP to hostname).
- **SRV:** Service record specifying hostnames and ports for specific protocols.

Resolution in Go

When you invoke `net.Dial("tcp", "example.com:80")`, Go performs a DNS lookup before connecting. The Go standard library features two distinct DNS resolver implementations:

1. **Pure Go Resolver (`netgo`):** Sends DNS queries directly to the nameservers listed in `/etc/resolv.conf` using UDP/TCP sockets. This is non-blocking and integrates cleanly with Go's netpoller and context cancellation.
2. **Cgo / System Resolver (`cgo`):** Calls the underlying C library function `getaddrinfo()`. This is used if complex NSS (Name Service Switch) configurations or mDNS (Multicast DNS) are required, but it blocks an OS thread during lookups.

You can explicitly control this behavior via the `GODEBUG=netdns=go` or `GODEBUG=netdns=cgo` environment variables.

Connection-Oriented vs Connectionless Communication

At the transport layer, socket programming splits into two core paradigms: **TCP** and **UDP**.

```
+----------------------------------------+----------------------------------------+
|                   TCP                  |                   UDP                  |
|          (Transmission Control)        |         (User Datagram Protocol)       |
+----------------------------------------+----------------------------------------+
| • Connection-oriented (Handshake)      | • Connectionless (Fire and forget)     |
| • Guaranteed delivery (Retransmissions) | • Unreliable (Packets may drop)        |
| • Ordered byte stream (Sequence nums)  | • Unordered independent datagrams      |
| • Flow control & Congestion control    | • No congestion control or throttling  |
| • Stream abstraction (No msg boundaries)| • Preserves discrete message boundaries |
| • Higher header overhead (20-60 bytes) | • Minimal header overhead (8 bytes)    |
+----------------------------------------+----------------------------------------+
```

TCP (Transmission Control Protocol)

TCP provides a **reliable, ordered, full-duplex stream of bytes**.

- **Connection-Oriented:** A connection must be established via the Three-Way Handshake (`SYN` → `SYN-ACK` → `ACK`) before data can transfer.
- **Stream-Oriented:** TCP does *not* know about application message boundaries. If you write 100 bytes followed by 200 bytes, the receiving socket might read all 300 bytes at once, or 50 bytes, or 10 bytes at a time. The application layer must implement its own framing.
- **Reliability:** ACKs, sequence numbers, and checksums ensure every byte is delivered intact and in order. Lost packets are automatically retransmitted.

UDP (User Datagram Protocol)

UDP provides a **lightweight, connectionless, unreliable datagram service**.

- **Connectionless:** No handshake is required. Packets are dispatched immediately to an IP and port.
- **Message-Oriented:** UDP preserves packet boundaries. If a sender transmits a 500-byte datagram, the receiver reads exactly 500 bytes in a single `ReadFrom` call.
- **Minimal Overhead:** UDP headers are only 8 bytes (vs. 20–60 bytes for TCP). There are no retransmissions, acknowledgments, or flow control windows.

Chapter Summary

- **Sockets** are kernel-managed communication endpoints identified by IP addresses and port numbers.
- The **TCP/IP stack** encapsulates data through physical frames, IP packets, transport segments/datagrams, and application payloads.
- **TCP** provides reliable, ordered, full-duplex byte streams with flow and congestion control, requiring explicit application-level message framing.
- **UDP** sends standalone datagrams with minimal overhead and preserved packet boundaries, trading delivery guarantees for raw speed and low latency.
- In the next chapter, we will tour Go's standard `net` package and examine the primary interfaces and types that form our socket programming toolkit.

Chapter 2: Go's net Package

Chapter 2: Go's net Package: Your Socket Programming Toolbox

The Go standard library provides the foundation for all network programming through the `net` package. Rather than exposing raw POSIX system calls (`socket`, `bind`, `listen`, `accept`, `connect`) directly, Go offers clean, composable interfaces that hide OS-level differences while preserving direct control over network parameters.

In this chapter, you will learn the core interfaces, factory functions, protocol-specific types, and address resolution helpers that make up your socket programming toolbox.

The net.Dial and net.Listen Functions

At the highest level, the `net` package exposes two fundamental entry points: `net.Listen` for servers and `net.Dial` for clients.

1. `net.Listen`: Binding and Listening for Inbound Connections

Servers create a listening socket using `net.Listen`:

```go
func Listen(network, address string) (Listener, error)
```

- **`network`**: The protocol family. Common options include `"tcp"`, `"tcp4"`, `"tcp6"`, `"unix"`, and `"unixpacket"`.
- **`address`**: The host and port to bind to (e.g., `"127.0.0.1:8080"`, `":8080"` for all local interfaces, or `"/tmp/service.sock"` for Unix sockets).

```go
ln, err := net.Listen("tcp", ":8080")
if err != nil {
    log.Fatalf("failed to bind port 8080: %v", err)
}
defer ln.Close()
```

The returned object satisfies the `net.Listener` interface:

```go
type Listener interface {
    Accept() (Conn, error)
```

```
    Close() error
    Addr() Addr
}
```

2. `net.Dial`: Connecting to a Remote Endpoint

Clients initiate outbound connections using `net.Dial`:

```
func Dial(network, address string) (Conn, error)
```

```
conn, err := net.Dial("tcp", "example.com:80")
if err != nil {
    log.Fatalf("failed to dial: %v", err)
}
defer conn.Close()
```

Modern Idiom: `net.Dialer` and `net.ListenConfig`

In modern Go code (Go 1.11+), prefer `net.Dialer` and `net.ListenConfig` over bare `Dial` and `Listen` because they support timeouts, keep-alive configuration, and first-class `context.Context` cancellation:

```
ctx, cancel := context.WithTimeout(context.Background(), 5*time.Second)
defer cancel()

var dialer net.Dialer
conn, err := dialer.DialContext(ctx, "tcp", "example.com:443")
if err != nil {
    log.Fatalf("connection failed: %v", err)
}
defer conn.Close()
```

The Conn Interface Explained

The `net.Conn` interface is the heart of Go's stream-oriented networking. Any bidirectional stream—whether an unencrypted TCP socket, a TLS connection, a Unix domain stream, or an in-memory pipe—implements `net.Conn`.

```
type Conn interface {
    Read(b []byte) (n int, err error)
```

```go
    Write(b []byte) (n int, err error)
    Close() error
    LocalAddr() Addr
    RemoteAddr() Addr
    SetDeadline(t time.Time) error
    SetReadDeadline(t time.Time) error
    SetWriteDeadline(t time.Time) error
}
```

Key Behavioral Rules of `net.Conn`

1. **`io.Reader` and `io.Writer` Compatibility:** Because `net.Conn` satisfies standard `io.Reader`, `io.Writer`, and `io.Closer` interfaces, you can pass any network connection directly into standard library utilities like `io.Copy`, `bufio.NewReader`, `bufio.NewWriter`, `json.NewDecoder`, or `tls.Server`.
2. **Concurrent Safety:** Go's `net.Conn` is safe for concurrent access by multiple goroutines. Specifically, **one goroutine may safely call `Read()` while another goroutine concurrently calls `Write()`**. However, concurrent multiple readers or concurrent multiple writers will interleave their bytes unpredictably.
3. **Blocking vs. Netpoller:** When you call `conn.Read(buf)`, execution appears to block until data arrives, an error occurs, or the deadline expires. Underneath, Go's runtime netpoller deschedules the calling goroutine, freeing the OS thread to do other work until the kernel signals readiness.
4. **EOF Semantics:** When the remote peer cleanly terminates the connection (sends a TCP `FIN` packet), subsequent calls to `Read()` will consume any remaining buffered data and then return `io.EOF`.

TCP-Specific Types and Methods

When you need granular control over TCP options that are not exposed by the generic `net.Conn` or `net.Listener` interfaces, you can use type assertions or concrete types: `*net.TCPConn` and `*net.TCPListener`.

```go
ln, err := net.Listen("tcp", ":8080")
if err != nil {
    log.Fatal(err)
}
tcpLn := ln.(*net.TCPListener)

conn, err := tcpLn.AcceptTCP()
if err != nil {
```

```
    log.Fatal(err)
}
```

Important Methods on `*net.TCPConn`

- **`SetNoDelay(noDelay bool) error:`** Enables or disables Nagle's algorithm (`TCP_NODELAY`). By default, Go enables `SetNoDelay(true)`, meaning small packets are dispatched immediately without buffering.
- **`SetKeepAlive(keepalive bool) error:`** Enables or disables kernel-level TCP keep-alive probe packets.
- **`SetKeepAlivePeriod(d time.Duration) error:`** Configures the interval between successive keep-alive probes.
- **`SetReadBuffer(bytes int) error` / `SetWriteBuffer(bytes int) error:`** Sets the size of the kernel-level socket receive and send buffers (`SO_RCVBUF`, `SO_SNDBUF`).
- **`CloseRead() error` / `CloseWrite() error:`** Shuts down only the read side or write side of a TCP connection (half-closing the TCP connection via `FIN`).

UDP-Specific Types and Methods

Because UDP is connectionless and datagram-oriented, it requires different semantics. Instead of `net.Listener`, UDP uses `net.PacketConn` and the concrete `*net.UDPConn` type.

```
type PacketConn interface {
    ReadFrom(p []byte) (n int, addr Addr, err error)
    WriteTo(p []byte, addr Addr) (n int, err error)
    Close() error
    LocalAddr() Addr
    SetDeadline(t time.Time) error
    SetReadDeadline(t time.Time) error
    SetWriteDeadline(t time.Time) error
}
```

Listening on UDP

```
pconn, err := net.ListenPacket("udp", ":9000")
if err != nil {
    log.Fatal(err)
}
defer pconn.Close()

buf := make([]byte, 1500) // Standard MTU size
```

```go
for {
    n, remoteAddr, err := pconn.ReadFrom(buf)
    if err != nil {
        return
    }
    // Echo back to sender
    _, _ = pconn.WriteTo(buf[:n], remoteAddr)
}
```

Connected UDP Sockets

If you call `net.Dial("udp", "192.0.2.1:9000")`, Go returns a `*net.UDPConn` bound to a single remote address. In this mode, you can call standard `Read()` and `Write()` methods instead of `ReadFrom()` and `WriteTo()`, and the kernel will discard incoming datagrams from any other source.

Resolving Addresses with net.ResolveTCPAddr and Friends

The `net` package provides address resolution utilities that parse string addresses and perform DNS lookups:

```go
// Resolve a TCP address:
tcpAddr, err := net.ResolveTCPAddr("tcp", "golang.org:80")
if err != nil {
    log.Fatalf("DNS resolution error: %v", err)
}
fmt.Printf("IP: %s, Port: %d, Zone: %s\n", tcpAddr.IP, tcpAddr.Port, tcpAddr.Zone)

// Resolve a UDP address:
udpAddr, err := net.ResolveUDPAddr("udp", "time.nist.gov:123")

// Resolve a Unix domain socket address:
unixAddr, err := net.ResolveUnixAddr("unix", "/tmp/app.sock")
```

Parsing Host and Port Strings Safely

Never parse host and port strings using naive `strings.Split(addr, ":")` because IPv6 addresses contain colons (e.g., `"[2001:db8::1]:8080"`). Always use the standard library helpers:

```go
// Splitting host and port:
host, portStr, err := net.SplitHostPort("[2001:db8::1]:8080")
```

```go
// host = "2001:db8::1", portStr = "8080"

// Joining host and port:
addr := net.JoinHostPort("2001:db8::1", "8080")
// addr = "[2001:db8::1]:8080"
```

Chapter Summary

- **`net.Listen` and `net.Dial`** are the entry points for server listeners and client connections.
- **`net.Dialer` and `net.ListenConfig`** provide modern, context-aware connection establishment with timeouts.
- **`net.Conn`** provides an idiomatic, stream-based interface satisfying `io.Reader` and `io.Writer`, safe for concurrent reads and writes.
- **`*net.TCPConn`** exposes socket options like `SetNoDelay`, `SetKeepAlive`, and buffer sizing.
- **`net.PacketConn` / `*net.UDPConn`** handle connectionless datagram communications using `ReadFrom` and `WriteTo`.
- **`net.SplitHostPort` and `net.JoinHostPort`** must always be used to handle IPv4 and IPv6 strings safely.
- In the next chapter, we will build complete, production-grade TCP servers and clients, exploring connection lifecycles, accept loops, and graceful shutdowns.

Chapter 3: Building TCP Servers and Clients

Chapter 3: Building TCP Servers and Clients: The Connection Lifecycle

TCP is the backbone of reliable internet communication. Writing a basic TCP server in Go takes fewer than ten lines of code, but writing a robust, production-ready server capable of handling unexpected client disconnections, slow clients, and graceful zero-downtime shutdowns requires careful design.

In this chapter, you will build working TCP servers and clients, master the server accept loop pattern, manage connection states, and implement clean graceful shutdowns with `context.Context` and `sync.WaitGroup`.

Your First TCP Server

Let's start with a foundational echo server. The server listens on a local port, accepts incoming TCP connections, reads messages until `io.EOF`, and echoes the exact data back to the client.

```go
// server.go
package main

import (
    "bufio"
    "fmt"
    "io"
    "log"
    "net"
)

func main() {
    ln, err := net.Listen("tcp", "127.0.0.1:9090")
    if err != nil {
        log.Fatalf("failed to listen: %v", err)
    }
    defer ln.Close()

    log.Printf("Echo server listening on %s", ln.Addr())

    for {
```

```go
        conn, err := ln.Accept()
        if err != nil {
            log.Printf("accept error: %v", err)
            continue
        }

        go handleConnection(conn)
    }
}

func handleConnection(conn net.Conn) {
    defer conn.Close()
    remoteAddr := conn.RemoteAddr()
    log.Printf("accepted connection from %s", remoteAddr)

    reader := bufio.NewReader(conn)
    for {
        line, err := reader.ReadBytes('\n')
        if err != nil {
            if err != io.EOF {
                log.Printf("[%s] read error: %v", remoteAddr, err)
            }
            break
        }

        if _, err := conn.Write(line); err != nil {
            log.Printf("[%s] write error: %v", remoteAddr, err)
            break
        }
    }

    log.Printf("closed connection from %s", remoteAddr)
}
```

Your First TCP Client

Now let's implement a corresponding TCP client that connects to our echo server, writes a message, reads the echoed reply, and exits cleanly.

```go
// client.go
package main

import (
```

```go
    "bufio"
    "fmt"
    "log"
    "net"
    "time"
)

func main() {
    // Establish connection with a 5-second dial timeout
    conn, err := net.DialTimeout("tcp", "127.0.0.1:9090", 5*time.Second)
    if err != nil {
        log.Fatalf("dial failed: %v", err)
    }
    defer conn.Close()

    log.Printf("connected to %s from %s", conn.RemoteAddr(), conn.LocalAddr())

    // Send a message terminated by a newline
    message := "Hello from Go TCP client!\n"
    if _, err := conn.Write([]byte(message)); err != nil {
        log.Fatalf("write failed: %v", err)
    }

    // Read reply with a 2-second read deadline
    if err := conn.SetReadDeadline(time.Now().Add(2 * time.Second)); err != nil {
        log.Fatalf("failed to set deadline: %v", err)
    }

    reply, err := bufio.NewReader(conn).ReadString('\n')
    if err != nil {
        log.Fatalf("read failed: %v", err)
    }

    fmt.Printf("Received echo: %s", reply)
}
```

You can test these two programs together in separate terminals:

```go
# Terminal 1:
go run server.go

# Terminal 2:
go run client.go
```

The Accept Loop and Connection Handling Patterns

The core loop of any Go network daemon is the **Accept Loop**:

```
                    net.Listen("tcp")



                      ln.Accept()


              Connection Accepted?
                /                \
           Yes                    No (error)
           /                        \


go handleConn(c)         Check if closed


                      Permanent or Transient?
```

1. Always Spawn a Goroutine per Connection

In Go, goroutines cost approximately 2 KB of memory. Spawning `go handleConnection(conn)` for every connection is the standard idiomatic design. Do *not* process connection traffic inside the accept loop itself, as doing so blocks all other incoming clients from being accepted.

2. Handling Accept Errors

When `ln.Accept()` fails, you must inspect whether the error is transient (e.g., temporary file descriptor exhaustion) or permanent (e.g., the listener was explicitly closed during server shutdown):

```go
for {
    conn, err := ln.Accept()
    if err != nil {
        // If the listener was closed, stop accepting
        if errors.Is(err, net.ErrClosed) {
            return nil
        }
        // Handle transient errors with backoff
        var netErr net.Error
```

```go
        if errors.As(err, &netErr) && netErr.Temporary() {
            time.Sleep(10 * time.Millisecond)
            continue
        }
        return fmt.Errorf("fatal accept error: %w", err)
    }
    go handleConn(conn)
}
```

Graceful Shutdown with context.Context

In production, servers must shut down without dropping active user requests or leaving sockets hanging. A graceful shutdown must:

1. Stop accepting new connections (`ln.Close()`).
2. Signal existing connection goroutines to wrap up their in-flight work.
3. Wait for active connections to finish before terminating the process, with an enforced timeout.

```go
// graceful_server.go
package main

import (
    "context"
    "errors"
    "io"
    "log"
    "net"
    "os"
    "os/signal"
    "sync"
    "syscall"
    "time"
)

type Server struct {
    addr     string
    listener net.Listener
    wg       sync.WaitGroup
    ctx      context.Context
    cancel   context.CancelFunc
}

func NewServer(addr string) *Server {
```

```go
    ctx, cancel := context.WithCancel(context.Background())
    return &Server{
        addr:   addr,
        ctx:    ctx,
        cancel: cancel,
    }
}

func (s *Server) Start() error {
    ln, err := net.Listen("tcp", s.addr)
    if err != nil {
        return err
    }
    s.listener = ln
    log.Printf("Server listening on %s", s.addr)

    for {
        conn, err := s.listener.Accept()
        if err != nil {
            if errors.Is(err, net.ErrClosed) {
                // Normal termination when listener is closed
                return nil
            }
            log.Printf("Accept error: %v", err)
            continue
        }

        s.wg.Add(1)
        go s.handleConn(conn)
    }
}

func (s *Server) handleConn(conn net.Conn) {
    defer s.wg.Done()
    defer conn.Close()

    // Launch a watcher goroutine to close the connection if context is cancelled
    closed := make(chan struct{})
    go func() {
        select {
        case <-s.ctx.Done():
            _ = conn.Close() // Force unblock active Read/Write
        case <-closed:
        }
    }()
    defer close(closed)
```

```go
    buf := make([]byte, 4096)
    for {
        n, err := conn.Read(buf)
        if err != nil {
            if err != io.EOF && !errors.Is(err, net.ErrClosed) {
                log.Printf("Read error: %v", err)
            }
            return
        }

        if _, err := conn.Write(buf[:n]); err != nil {
            return
        }
    }
}

func (s *Server) Shutdown(timeout time.Duration) error {
    s.cancel() // Signal all workers
    if s.listener != nil {
        _ = s.listener.Close() // Unblock Accept()
    }

    done := make(chan struct{})
    go func() {
        s.wg.Wait()
        close(done)
    }()

    select {
    case <-done:
        log.Println("All connections terminated cleanly")
        return nil
    case <-time.After(timeout):
        return errors.New("shutdown timed out; forcing process exit")
    }
}

func main() {
    srv := NewServer(":8080")

    go func() {
        if err := srv.Start(); err != nil && !errors.Is(err, net.ErrClosed) {
            log.Fatalf("Server startup failed: %v", err)
        }
    }()
```

```go
    // Listen for OS termination signals (SIGINT, SIGTERM)
    sigChan := make(chan os.Signal, 1)
    signal.Notify(sigChan, os.Interrupt, syscall.SIGTERM)
    <-sigChan

    log.Println("Shutting down server...")
    if err := srv.Shutdown(5 * time.Second); err != nil {
        log.Printf("Shutdown warning: %v", err)
    }
}
```

Managing Connection State and Resources

Network services in production must guard against resource exhaustion:

1. **File Descriptors:** Every TCP socket requires a kernel file descriptor. On Linux, check limits via `ulimit -n`. If descriptors are exhausted, `Accept()` fails with `EMFILE` / `ENFILE`.
2. **Goroutine Leaks:** If a client abruptly stops reading or closes their laptop without sending a `FIN`, a naive server calling `conn.Read()` without a deadline may hang that goroutine forever. Over days, thousands of zombie goroutines accumulate, consuming gigabytes of memory.
3. **Deadlines as Guardrails:** Always set read and write deadlines (`SetReadDeadline`, `SetWriteDeadline`) on connections to ensure resources are returned to the pool when peers become unresponsive.

Chapter Summary

- TCP servers follow an **Accept Loop** pattern, delegating each accepted connection to an independent goroutine.
- Clients use **`net.DialTimeout`** to guard against hanging connections during the TCP three-way handshake.
- Graceful shutdown requires: closing the listener, notifying active goroutines via **`context.Context`**, and synchronizing termination via **`sync.WaitGroup`**.
- Unbounded reads without deadlines lead to **goroutine leaks** and socket descriptor leaks.
- In the next chapter, we will examine TCP reliability, flow control, Nagle's algorithm, keep-alives, and socket buffer tuning.

Chapter 4: TCP Reliability, Flow Control and Performance Tuning

Chapter 4: TCP Reliability, Flow Control and Performance Tuning

While the TCP abstraction feels like an unbroken, error-free pipe of bytes, the physical network underneath is lossy, jittery, and packet-based. To write high-performance network applications in Go, you must understand the mechanisms TCP uses to guarantee delivery and how to tune them.

This chapter covers TCP acknowledgments, sliding windows, flow control, Nagle's algorithm, keep-alive probes, partial write mechanics, and low-level socket tuning in Go.

How TCP Guarantees Delivery

TCP delivers reliability over an unreliable IP substrate through four core mechanisms:

1. **Sequence Numbers & Acknowledgments (ACKs):** Every byte sent over a TCP connection is assigned a 32-bit sequence number. When the receiver accepts data into its kernel buffer, it transmits an ACK packet containing the sequence number of the next expected byte.
2. **Retransmission Timers (RTO):** If an ACK is not received within the Retransmission Timeout (calculated dynamically from Round-Trip Time measurements), the sender retransmits the lost segment.
3. **Cumulative & Selective ACKs (SACK):** Modern TCP implementations use SACK (RFC 2018) to inform the sender of non-contiguous blocks of bytes received, avoiding wasteful retransmissions of packets that have already arrived safely.
4. **Checksums:** Every TCP segment includes a 16-bit one's complement checksum covering the header and data, detecting bit flips introduced by faulty network equipment.

```
Sender                                                  Receiver
    Segment 1 [Seq 1..1000]

    Segment 2 [Seq 1001..2000] (Dropped in transit)

    Segment 3 [Seq 2001..3000]

    ACK 1001 (SACK 2001..3000)

    Retransmit Segment 2 [Seq 1001..2000]

    ACK 3001
```

Nagle's Algorithm and TCP_NODELAY

The Problem of "Tinygrams"

In interactive applications (like Telnet or SSH), a user typing a single character causes the application to write 1 byte of payload. Encapsulated in a 20-byte TCP header and a 20-byte IPv4 header, transmitting that 1 byte requires a 41-byte packet—a 4000% bandwidth overhead.

John Nagle's Solution (1984)

Nagle's algorithm prevents congestion by mandating: > If there is unacknowledged in-flight data, buffer any small outgoing writes until either: > 1. A full maximum segment size (MSS, typically 1460 bytes) is accumulated, OR > 2. An ACK arrives for all previously transmitted data.

The Conflict: Nagle + Delayed ACKs

Receivers implement **Delayed ACKs** (RFC 1122), waiting up to 40ms–200ms before acknowledging incoming segments in hopes of piggybacking the ACK onto an outbound reply.

When an application performs multiple small writes followed by a read (e.g., writing an RPC header, then writing the payload, then waiting for a response), **disaster strikes**: - Sender buffers the second write waiting for an ACK (due to Nagle). - Receiver waits 200ms before sending the ACK (due to Delayed ACK). - The application suffers a mandatory 40–200ms latency penalty on every request.

Go's Stance: `TCP_NODELAY` is Enabled by Default

To prioritize low latency for modern distributed systems, **Go sets `TCP_NODELAY` to `true` by default on all TCP connections**. Outbound packets are dispatched immediately without buffering.

If you are writing large volumes of small chunks, you can either: 1. Wrap the connection with `bufio.NewWriter(conn)` in user space (the idiomatic Go approach). 2. Explicitly re-enable Nagle's algorithm if high packet consolidation is desired: `go tcpConn, ok := conn.(*net.TCPConn) if ok { _ = tcpConn.SetNoDelay(false) // Re-enables Nagle's algorithm }`

TCP Keep-Alive and Detecting Dead Connections

If a client machine crashes, loses power, or has its ethernet cable disconnected without sending a TCP `FIN` packet, the server's socket remains in state `ESTABLISHED` indefinitely. Without keep-alives or deadlines, the server will never know the client has vanished.

Configuring Keep-Alive in Go

Go enables keep-alive probes by default (typically every 15 seconds):

```go
var dialer net.Dialer
dialer.KeepAlive = 30 * time.Second

conn, err := dialer.Dial("tcp", "example.com:80")
```

On an existing `*net.TCPConn`:

```go
if tcpConn, ok := conn.(*net.TCPConn); ok {
    _ = tcpConn.SetKeepAlive(true)
    _ = tcpConn.SetKeepAlivePeriod(30 * time.Second)
}
```

When keep-alive is active, if the peer does not acknowledge successive probes over a configured duration (governed by kernel parameters `tcp_keepalive_probes` and `tcp_keepalive_intvl`), the kernel forcibly terminates the connection, causing pending `Read()` calls to return an error (`connection timed out`).

Window Scaling and Flow Control Basics

Flow control prevents a fast sender from overwhelming a slow receiver's kernel buffer.

```
Receiver Kernel Socket Buffer
+-------------------------------------+--------------------------+
|  Received & Buffered Data (unread) | Advertised Window (Free) |
+-------------------------------------+--------------------------+
0                                                          65,535 bytes
```

The Receive Window (`rwnd`)

In every ACK packet, the receiver advertises its available buffer space (`rwnd`). The sender cannot send more unacknowledged bytes than the advertised window size. If the application on the receiver does not call `Read()`, the window shrinks until it reaches zero (Zero Window Probe), stalling the sender.

Window Scaling (RFC 1323)

The original TCP header allocates 16 bits for window size, capping it at 64 KB (65,535 bytes). On high-bandwidth, high-latency links (e.g., 10 Gbps trans-oceanic cables), a 64 KB window artificially throttles throughput.

The **Bandwidth-Delay Product (BDP)** defines the required buffer:

$$\text{BDP} = \text{Bandwidth (bytes/sec)} \times \text{Round-Trip Time (sec)}$$

Window scaling introduces a scale factor during the TCP handshake, allowing window sizes up to 1 Gigabyte.

Handling Partial Writes Correctly

A fundamental error in network programming is assuming that `conn.Write(p)` will transmit the entire slice `p` in a single invocation.

The `io.Writer` contract explicitly states:

```
// Write writes len(p) bytes from p to the underlying data stream.
// It returns the number of bytes written from p (0 <= n <= len(p))
// and any error encountered that caused the write to stop early.
```

If the kernel send buffer fills up, a non-blocking socket will only write as many bytes as can fit, returning $n < \text{len}(p)$ with `err == nil`.

The Idiomatic Go Solution: `io.WriteString` or Loop

Never write raw loops if you can use `io.Copy`, `io.WriteString`, or `io.WriteFull`:

```go
// WRONG: Ignores partial write
n, err := conn.Write(largePayload)

// RIGHT: Idiomatic write loop (io.WriteString handles partial writes internally)
_, err := io.WriteString(conn, string(largePayload))

// RIGHT: Explicit partial write handler
func writeAll(conn net.Conn, data []byte) error {
    for len(data) > 0 {
        n, err := conn.Write(data)
        if err != nil {
            return err
        }
```

```go
        data = data[n:]
    }
    return nil
}
```

Tuning TCP Parameters in Go

When standard options are insufficient, Go allows direct access to the underlying socket file descriptor via `syscall.RawConn`.

1. Adjusting Socket Buffers (`SO_RCVBUF` and `SO_SNDBUF`)

```go
if tcpConn, ok := conn.(*net.TCPConn); ok {
    // Set kernel receive buffer to 1MB
    _ = tcpConn.SetReadBuffer(1024 * 1024)
    // Set kernel send buffer to 1MB
    _ = tcpConn.SetWriteBuffer(1024 * 1024)
}
```

2. Low-Level Control via `RawConn.Control`

Using `RawConn`, you can invoke arbitrary socket options (such as `SO_REUSEPORT` or `TCP_QUICKACK` on Linux):

```go
var lc net.ListenConfig
lc.Control = func(network, address string, c syscall.RawConn) error {
    var operr error
    err := c.Control(func(fd uintptr) {
        // Set SO_REUSEPORT (Linux specific constant 0x0F)
        const SO_REUSEPORT = 0x0F
        operr = syscall.SetsockoptInt(int(fd), syscall.SOL_SOCKET, SO_REUSEPORT, 1)
    })
    if err != nil {
        return err
    }
    return operr
}

ln, err := lc.Listen(context.Background(), "tcp", ":8080")
```

Chapter Summary

- **TCP reliability** relies on sequence numbers, cumulative and selective ACKs, and retransmission timers.
- **Nagle's algorithm** buffers tiny outbound packets; Go enables `TCP_NODELAY` by default to prevent latency spikes caused by Delayed ACKs.
- **TCP Keep-Alive** detects silent link failures and dead peers; configure it on the dialer or `*net.TCPConn`.
- **Flow control** uses advertised receive windows (`rwnd`); window scaling enables high-throughput transfers across high BDP links.
- Sockets can return **partial writes**; always use utilities like `io.Copy`, `io.WriteString`, or explicit loops.
- In the next chapter, we will turn our attention to connectionless programming with UDP.

Multipath TCP in Go: Defaults, Detection and Kill Switches

Multipath TCP in Go: Defaults, Detection and Kill Switches

After completing this page, you will be able to configure Multipath TCP in Go applications, inspect runtime connections to verify MPTCP negotiation, diagnose listener states using Linux socket utilities, and choose the right kill switch when a kernel MPTCP bug forces your hand.

Why MPTCP and What Go Changed

Multipath TCP (MPTCP), standardized in RFC 8684, extends standard TCP by allowing a single logical connection to transmit data across multiple physical subflows or network interfaces. To an application, MPTCP behaves like a normal TCP socket. Underneath, the transport layer manages subflows across distinct network paths—such as simultaneous Wi-Fi and cellular connections on a mobile device—which can survive a path change and use more than one path at a time. Each MPTCP subflow appears on the wire as an ordinary TCP connection containing specific TCP options (`MP_CAPABLE`, `MP_JOIN`). If a remote endpoint or intermediate middlebox does not support MPTCP, the connection falls back to plain single-path TCP.

On Linux, kernel support for MPTCP is controlled by the sysctl `net.mptcp.enabled`. When it is `1`, processes may request MPTCP sockets. The setting is per network namespace, so each container with its own namespace has its own copy, and a new namespace starts with the kernel default.

Go introduced initial standard library support for MPTCP in Go 1.21. This update added methods to configure and inspect Multipath TCP status across network listeners, dialers, and established TCP connections:

- `(*net.ListenConfig).SetMultipathTCP(enabled bool)`
- `(*net.ListenConfig).MultipathTCP() bool`
- `(*net.Dialer).SetMultipathTCP(enabled bool)`
- `(*net.Dialer).MultipathTCP() bool`
- `(*net.TCPConn).MultipathTCP() (bool, error)`

A major operational shift occurred in Go 1.24. On Linux systems, `net.ListenConfig` (and high-level constructors that rely on it, including `net.Listen` and `http.ListenAndServe`) requests MPTCP by default whenever the underlying host kernel supports it. Client dialers (`net.Dialer`), however, continue to default to standard single-path TCP.

This runtime behavior is governed by the `GODEBUG` setting `multipathtcp`. Like other GODEBUG defaults, it follows the `go` line in the main module's `go.mod`, not the toolchain version. A program

whose `go.mod` says `go 1.23` keeps the old default (no MPTCP listeners) even when built with Go 1.27, and raising that line to 1.24 or later turns MPTCP listeners on. You can change the default at run time with the `GODEBUG` environment variable, or at build time with a `godebug` line in `go.mod`. The `SetMultipathTCP` methods override it in code.

The Demo Program

The following complete program demonstrates how Go configures MPTCP on listeners and dialers, and how to verify whether an established connection successfully negotiated MPTCP.

```
module example.com/mptcpdemo

go 1.27
```

```
package main

import (
    "context"
    "fmt"
    "log"
    "net"
    "os"
)

func main() {
    ctx := context.Background()

    if len(os.Args) > 1 && os.Args[1] == "serve" {
        // Long-lived listener so `ss -M` can see it.
        var lc net.ListenConfig
        ln, err := lc.Listen(ctx, "tcp", "127.0.0.1:9099")
        if err != nil {
            log.Fatal(err)
        }
        fmt.Println("listening on", ln.Addr(), "mptcp requested:", lc.MultipathTCP())
        select {}
    }

    // Listener: Go 1.24+ asks the kernel for MPTCP by default on Linux.
    var lc net.ListenConfig
    ln, err := lc.Listen(ctx, "tcp", "127.0.0.1:0")
    if err != nil {
        log.Fatal(err)
```

```go
    }
    defer ln.Close()
    fmt.Println("GODEBUG          :", os.Getenv("GODEBUG"))
    fmt.Println("listen config   :", lc.MultipathTCP())

    accepted := make(chan *net.TCPConn, 1)
    go func() {
        c, err := ln.Accept()
        if err != nil {
            log.Fatal(err)
        }
        accepted <- c.(*net.TCPConn)
    }()

    // Client: dialers do not use MPTCP unless asked.
    var d net.Dialer
    if len(os.Args) > 1 && os.Args[1] == "dial-mptcp" {
        d.SetMultipathTCP(true)
    }
    fmt.Println("dialer config   :", d.MultipathTCP())
    c, err := d.DialContext(ctx, "tcp", ln.Addr().String())
    if err != nil {
        log.Fatal(err)
    }
    defer c.Close()

    s := <-accepted
    defer s.Close()

    cm, err := c.(*net.TCPConn).MultipathTCP()
    fmt.Println("client conn MPTCP:", cm, err)
    sm, err := s.MultipathTCP()
    fmt.Println("server conn MPTCP:", sm, err)
}
```

Default Run

When compiled and executed without CLI arguments on Go 1.24+ under Linux, the server side requests MPTCP by default (`lc.MultipathTCP()` returns `true`). However, because the client dialer defaults to plain TCP (`d.MultipathTCP()` returns `false`), the handshake negotiates a standard TCP connection.

Build the application and execute a standard run using the following commands:

```
go version
go build -o mptcpdemo .
./mptcpdemo
```

The `go version` command produced this output:

```
go version go1.27.2 linux/amd64
```

Running `./mptcpdemo` produced this output:

```
GODEBUG          :
listen config   : true
dialer config   : false
client conn MPTCP: false <nil>
server conn MPTCP: false <nil>
```

Notice that both the client and server connection calls to `MultipathTCP()` return `false`. MPTCP requires mutual agreement during connection setup; if either side does not request MPTCP, the connection falls back to plain TCP.

Both Ends Opt In

To establish an active MPTCP connection, the dialer must explicitly opt in by calling `SetMultipathTCP(true)`.

Execute the binary with the `dial-mptcp` argument:

```
./mptcpdemo dial-mptcp
```

Running this command produced this output:

```
GODEBUG          :
listen config   : true
dialer config   : true
client conn MPTCP: true <nil>
server conn MPTCP: true <nil>
```

Because both the listener and dialer explicitly requested MPTCP, `MultipathTCP()` returns `true` on both the client socket (`c`) and server socket (`s`).

Turning Off MPTCP at Run Time with GODEBUG

You can disable the default MPTCP listener behavior across an entire process without changing application code by setting the `GODEBUG` environment variable to `multipathtcp=0`.

Execute the binary with `GODEBUG=multipathtcp=0`:

```
GODEBUG=multipathtcp=0 ./mptcpdemo dial-mptcp
```

Running this command produced this output:

```
GODEBUG          : multipathtcp=0
listen config   : false
dialer config   : true
client conn MPTCP: false <nil>
server conn MPTCP: false <nil>
```

Here, even though the client dialer explicitly set `d.SetMultipathTCP(true)`, the listener default was disabled by `GODEBUG`. As a result, connection setup fell back to standard single-path TCP.

Disabling MPTCP in go.mod and Verifying with `go version -m`

To disable MPTCP defaults across a service permanently without relying on ambient environment variables, add a `godebug` directive to your `go.mod` file.

Update `go.mod` to include the directive:

```
module example.com/mptcpdemo

go 1.27

godebug multipathtcp=0
```

Rebuild the binary as `mptcpdemo2` and execute the dial check alongside binary module inspection:

```
go build -o mptcpdemo2 .
./mptcpdemo2 dial-mptcp
go version -m mptcpdemo2 | grep DefaultGODEBUG
```

Running `./mptcpdemo2 dial-mptcp` produced this output:

```
GODEBUG          :
listen config   : false
dialer config   : true
client conn MPTCP: false <nil>
```

```
server conn MPTCP: false <nil>
```

The `go version -m` inspection command produced this output:

```
    build   DefaultGODEBUG=multipathtcp=0
```

The `godebug` directive bakes `multipathtcp=0` directly into the binary's `DefaultGODEBUG` settings, disabling MPTCP listener defaults across all deployments of that build.

Inspecting MPTCP Sockets with `ss -M`

The Linux `ss` utility has the `-M` flag to display MPTCP-specific socket information.

- `ss -M` lists active MPTCP connections and subflows.
- `ss -Mlnp` lists listening MPTCP sockets, bound ports, and associated process IDs.

Plain TCP listeners do not appear in `ss -M` output; they appear in standard TCP listings (`ss -tlnp`).

To observe a live MPTCP listener, run the application in server mode in the background and inspect socket state:

```
./mptcpdemo serve &
ss -Mlnp
```

Starting the server process produced this output:

```
listening on 127.0.0.1:9099 mptcp requested: true
```

Executing `ss -Mlnp` while the server was active produced this output:

```
State  Recv-Q Send-Q Local Address:Port Peer Address:PortProcess
LISTEN 0      4096       127.0.0.1:9099      0.0.0.0:*    users:(("mptcpdemo",pid=1843560,fd
```

If you start the server with `GODEBUG=multipathtcp=0`, `ss -Mlnp` shows no MPTCP listening sockets, whereas standard `ss -tlnp` confirms the port is open as a standard TCP listener:

```
GODEBUG=multipathtcp=0 ./mptcpdemo serve &
ss -Mlnp
ss -tlnp 'sport = :9099'
```

Executing `ss -Mlnp` under `GODEBUG=multipathtcp=0` produced this output:

```
State Recv-Q Send-Q Local Address:Port Peer Address:PortProcess
```

Executing `ss -tlnp 'sport = :9099'` produced this output:

```
State  Recv-Q Send-Q Local Address:Port Peer Address:PortProcess
LISTEN 0      4096       127.0.0.1:9099      0.0.0.0:*    users:(("mptcpdemo",pid=1843569,fd
```

The October 2026 Kernel Vulnerability and Incident Response

CVE-2026-98432, "ipv4: free inet_opt and ireq_opt after an RCU grace period", is a Linux kernel bug introduced in 4.4 and rated CVSS 9.8. The fix landed upstream on 5 October 2026 and the CVE was published on 15 October. A public exploit disclosed on 27 October needs a reachable MPTCP listener.

Because Go 1.24+ servers enable MPTCP listeners by default on Linux hosts, a Go 1.24+ server on an unpatched kernel with MPTCP enabled offers exactly that, unless someone has turned the default off.

On 29 October 2026 the Go security team posted an advisory pointing out that Go 1.24+ servers create MPTCP listeners by default. It called the Go-level controls (`GODEBUG=multipathtcp=0`, `godebug` in `go.mod`, `SetMultipathTCP(false)`) fragile mitigations and said the real fix is upgrading the kernel. They are fragile because each one covers only the binaries you remember to change. Discussion of the default is at go.dev/issue/82717. AWS listed `sysctl net.mptcp.enabled=0` as an interim host-wide mitigation.

The options, strongest first:

Option	Scope	Needs a rebuild?	Survives a restart?	Notes
Kernel upgrade	Host	No (needs a reboot into the new kernel)	Yes	**The fix.** Fixed in 5.10.272, 5.15.223, 6.1.190, 6.6.159, 6.12.113, 6.18.56, 7.2.10 and 7.3.

Option	Scope	Needs a rebuild?	Survives a restart?	Notes
`sysctl net.mptcp.enabled=0`	One network namespace (set it on the host and inside each container namespace)	No	Only if also saved in a sysctl config file	**Interim stopgap** (AWS's advice). Stops new MPTCP sockets being created by any process. Listeners that already exist are not closed, so restart long-running servers too.
`GODEBUG=multipathtcp=0`	One process	No	Only if set in the service's environment	Ignored wherever code calls `SetMultipathTCP(true)`.
`godebug` in `go.mod`	One binary	Yes	Yes	Compiled in as `DefaultGODEBUG`. Every service has to be rebuilt and redeployed.
`SetMultipathTCP(false)` in code	One call site	Yes	Yes	Easy to miss a call site, including inside libraries.

Pitfalls

1. Assuming Dialers Use MPTCP by Default

While Go 1.24+ configures `net.ListenConfig` to request MPTCP by default on Linux, `net.Dialer` defaults to plain TCP. If you deploy a Go client and a Go server expecting MPTCP subflow aggregation, connections will silently run as standard TCP unless the client code explicitly invokes `dialer.SetMultipathTCP(true)`.

2. Silent Fallback Hides Configuration Errors

Fallback to TCP is silent: the dial succeeds either way, and a fallen-back connection doesn't show in `ss -M`. To know whether a connection is really MPTCP, ask it with `(*net.TCPConn).MultipathTCP()`.

3. Programmatic Overrides Bypass GODEBUG Kill Switches

Calling `SetMultipathTCP(true)` directly on a `net.ListenConfig` or `net.Dialer` explicitly overrides any runtime `GODEBUG=multipathtcp=0` setting or `go.mod` default. If application code contains `lc.SetMultipathTCP(true)`, setting `GODEBUG=multipathtcp=0` will not disable MPTCP for that listener.

4. Container Isolation Myths

Container runtimes (such as Docker or Kubernetes) isolate network namespaces but share the host Linux kernel. Setting environment variables inside a container manifest does not isolate the kernel from MPTCP vulnerabilities if the underlying host kernel is unpatched and `net.mptcp.enabled=1`.

Key takeaways

- Go 1.24+ enables MPTCP by default on Linux listeners (`net.ListenConfig`); dialers (`net.Dialer`) default to plain TCP.
- MPTCP requires explicit opt-in from both client and server; if either side omits support, the connection silently falls back to standard TCP.
- Use `(*net.TCPConn).MultipathTCP()` at runtime to confirm whether an established connection actually negotiated MPTCP.
- Use `ss -Mlnp` to identify active MPTCP listening sockets and their associated process IDs on Linux hosts.
- `GODEBUG=multipathtcp=0` and `godebug` in `go.mod` disable default MPTCP listener behavior, but explicit programmatic calls to `SetMultipathTCP(true)` override these settings.
- For a kernel MPTCP bug such as CVE-2026-98432, upgrading the kernel is the fix and `net.mptcp.enabled=0` is the host-wide stopgap. The Go-level switches only cover the binaries you change.

Sources

- Go net.ListenConfig.SetMultipathTCP Documentation
- Go net.TCPConn.MultipathTCP Documentation
- Go 1.24 Release Notes
- Go GODEBUG Environment Variable Documentation
- golang-announce: "[security] Protecting against CVE-2026-98432"
- Go issue 82717: proposal on the MPTCP listener default
- AWS security bulletin: CVE-2026-98432, Linux kernel IPv4 TCP use-after-free
- RFC 8684: TCP Extensions for Multipath Operation with Multiple Addresses

- Linux ss(8) Manual Page

Chapter 5: UDP Programming

Chapter 5: UDP Programming: When Connectionless Is Right

While TCP provides a reliable, ordered, full-duplex stream, that reliability comes at the cost of connection state, three-way handshakes, head-of-line blocking, and retransmission delays. When applications demand minimum latency, independent packet delivery, or one-to-many communication, the **User Datagram Protocol (UDP)** is the ideal tool.

In this chapter, you will learn when to choose UDP, how to write UDP servers and clients using Go's `net.PacketConn`, how to mitigate packet loss and fragmentation, how to layer reliability on top of UDP, and how to implement broadcast and multicast.

Why Choose UDP Over TCP

UDP trades all guarantees of TCP for raw speed and simplicity:

Feature	TCP	UDP
Handshake	3-way (`SYN`, `SYN-ACK`, `ACK`)	None (immediate transmission)
Header Overhead	20–60 bytes	8 bytes
Delivery Guarantee	Yes (automatic retransmissions)	No (best effort)
Order Guarantee	Yes (strict sequence)	No (packets may arrive out of order)
Flow & Congestion Control	Yes (throttles on packet loss)	No (sender controls rate)
Framing	Byte stream (no message boundaries)	Datagrams (discrete message boundaries)
Communication Pattern	Point-to-point (unicast only)	Unicast, Broadcast, Multicast

Primary Use Cases for UDP

1. **Real-Time Media & Voice (VoIP, Video Streaming):** Dropping a frame of audio or video is far better than waiting 200ms for a retransmitted frame that is already obsolete.
2. **Online Multiplayer Gaming:** Player movement and state updates require minimum latency; obsolete state updates can be safely discarded.

3. **DNS Queries:** Simple request-and-reply transactions. If an answer doesn't arrive within 1 second, the client simply re-queries.
4. **Metrics & Telemetry (StatsD, Prometheus push agents):** High-frequency metrics where losing 0.1% of samples does not impair system observability.
5. **Modern Protocols (QUIC / HTTP/3):** Building specialized reliability, encryption, and multiplexing in user space over UDP to eliminate head-of-line blocking.

Basic UDP Client and Server Patterns

Because UDP does not have persistent connections, servers do not use an accept loop. Instead, a single socket reads and writes independent datagrams to and from arbitrary addresses.

The UDP Server

```go
// udp_server.go
package main

import (
    "fmt"
    "log"
    "net"
)

func main() {
    // Listen on UDP port 8888
    pc, err := net.ListenPacket("udp", ":8888")
    if err != nil {
        log.Fatalf("failed to bind UDP socket: %v", err)
    }
    defer pc.Close()

    log.Printf("UDP server listening on %s", pc.LocalAddr())

    buf := make([]byte, 2048)
    for {
        // Read a datagram and obtain the sender's address
        n, clientAddr, err := pc.ReadFrom(buf)
        if err != nil {
            log.Printf("Read error: %v", err)
            continue
        }

        log.Printf("Received %d bytes from %s: %s", n, clientAddr, string(buf[:n]))
```

```go
        // Send reply back to the sender
        reply := []byte(fmt.Sprintf("ACK: %s", buf[:n]))
        if _, err := pc.WriteTo(reply, clientAddr); err != nil {
            log.Printf("Write error to %s: %v", clientAddr, err)
        }
    }
}
```

The UDP Client

```go
// udp_client.go
package main

import (
    "fmt"
    "log"
    "net"
    "time"
)

func main() {
    // Resolve remote server address
    serverAddr, err := net.ResolveUDPAddr("udp", "127.0.0.1:8888")
    if err != nil {
        log.Fatal(err)
    }

    // Create an unbound local UDP socket
    conn, err := net.DialUDP("udp", nil, serverAddr)
    if err != nil {
        log.Fatal(err)
    }
    defer conn.Close()

    // Send message
    msg := []byte("ping from Go UDP client")
    if _, err := conn.Write(msg); err != nil {
        log.Fatal(err)
    }

    // Set read deadline to avoid blocking indefinitely if packet is dropped
    _ = conn.SetReadDeadline(time.Now().Add(2 * time.Second))

    replyBuf := make([]byte, 2048)
    n, _, err := conn.ReadFrom(replyBuf)
```

```go
    if err != nil {
        log.Fatalf("Failed to receive reply: %v", err)
    }

    fmt.Printf("Server replied: %s\n", string(replyBuf[:n]))
}
```

Handling Partial Reads and Packet Loss

1. Truncation and Buffer Sizing

In UDP, **each `ReadFrom` call consumes exactly one whole datagram**. If your buffer is smaller than the incoming datagram, the operating system kernel copies as many bytes as fit into your slice and **discards the remainder permanently** (on POSIX systems with `MSG_TRUNC`).

Always ensure your receive buffer is large enough for the Maximum Transmission Unit (MTU), typically 1500 bytes for standard Ethernet:

```go
const MaxDatagramSize = 65535 // Maximum theoretical UDP payload size
buf := make([]byte, 1500)     // Typical safe internet MTU
```

2. Fragmentation and Path MTU

If a UDP datagram exceeds the link MTU (usually 1500 bytes on the Internet), the IP layer fragments it into multiple IP packets. If **any single fragment is dropped**, the entire UDP datagram is discarded by the receiver's kernel!

> [!TIP] Keep UDP payloads under **1472 bytes** for IPv4 (1500 − 20 (IPv4 header) − 8 (UDP header)) or **1452 bytes** for IPv6 to avoid IP fragmentation entirely.

Implementing Reliability on Top of UDP

When you need UDP's multiplexing advantages but still require reliability, you can implement sequence numbers and acknowledgments in user space.

```
Sender                                          Receiver
    Packet [Seq: 1, Payload: "Data A"]

    ACK [AckSeq: 1]
```

```
Packet [Seq: 2, Payload: "Data B"] (Dropped)

(Timer expires after 500ms)
Retransmit Packet [Seq: 2, Payload: "Data B"]

ACK [AckSeq: 2]
```

Minimal Reliable Stop-and-Wait Protocol

```go
package main

import (
    "bytes"
    "encoding/binary"
    "errors"
    "net"
    "time"
)

type ReliablePacket struct {
    SeqNum uint32
    Data   []byte
}

func SendWithRetry(conn *net.UDPConn, pkt ReliablePacket, maxRetries int, timeout time.Durat
    buf := new(bytes.Buffer)
    _ = binary.Write(buf, binary.BigEndian, pkt.SeqNum)
    buf.Write(pkt.Data)
    wireBytes := buf.Bytes()

    ackBuf := make([]byte, 4)

    for attempt := 0; attempt < maxRetries; attempt++ {
        if _, err := conn.Write(wireBytes); err != nil {
            return err
        }

        _ = conn.SetReadDeadline(time.Now().Add(timeout))
        n, err := conn.Read(ackBuf)
        if err == nil && n == 4 {
            ackSeq := binary.BigEndian.Uint32(ackBuf)
            if ackSeq == pkt.SeqNum {
                return nil // Successfully delivered and acknowledged!
            }
```

```go
        }
        // If timeout occurred, loop retries transmission
    }
    return errors.New("packet delivery failed after maximum retries")
}
```

Multicast and Broadcast with UDP

1. UDP Broadcast (Local Subnet Only)

Broadcast transmits a packet to all hosts on the local network segment (`255.255.255.255` or subnet broadcast address):

```go
// Enabling broadcast on UDP socket:
destAddr, _ := net.ResolveUDPAddr("udp4", "255.255.255.255:9999")
conn, _ := net.DialUDP("udp4", nil, destAddr)
_, _ = conn.Write([]byte("Discovery beacon"))
```

2. UDP Multicast (Targeted Groups)

Multicast delivers packets to a subscription group identified by a multicast IP (IPv4 range `224.0.0.0` to `239.255.255.255`). Go provides `golang.org/x/net/ipv4` for low-level multicast group membership and interface binding.

```go
// Subscribing to a multicast group using standard net:
mcastAddr, err := net.ResolveUDPAddr("udp", "224.0.0.1:9999")
if err != nil {
    log.Fatal(err)
}

// ListenMulticastUDP binds and joins the specified multicast group on all interfaces
conn, err := net.ListenMulticastUDP("udp", nil, mcastAddr)
if err != nil {
    log.Fatal(err)
}
defer conn.Close()
```

Chapter Summary

- **UDP is connectionless and datagram-oriented**, preserving message boundaries with minimal header overhead.
- Servers use `net.PacketConn` and `ReadFrom` / `WriteTo` rather than the accept loops of TCP.
- Payloads larger than the MTU lead to **IP fragmentation**; keep UDP packet payloads below **1472 bytes**.
- A receive buffer smaller than the datagram causes **silent truncation** of data.
- Applications requiring both low latency and reliability can build sequence numbers and ACK timeouts over UDP.
- In the next chapter, we will explore Unix Domain Sockets for ultra-fast local inter-process communication.

Chapter 6: Unix Domain Sockets

Chapter 6: Unix Domain Sockets: Fast Local Communication

When two processes on the same machine need to communicate—such as an Nginx reverse proxy routing requests to a local Go application, or Docker communicating with its daemon—using loopback TCP (`127.0.0.1`) introduces unnecessary overhead.

Unix Domain Sockets (UDS) provide bidirectional, stream-oriented (`AF_UNIX`) or datagram-oriented inter-process communication entirely within kernel memory, bypassing TCP/IP framing, checksum calculations, routing tables, and network device drivers.

In this chapter, you will learn how to build Unix domain socket servers and clients in Go, manage socket file permissions and lifecycle cleanup, utilize Linux abstract namespace sockets, and analyze performance gains over TCP loopback.

What Are Unix Domain Sockets and Why Use Them

A Unix Domain Socket looks and behaves like a standard `net.Conn`, but instead of an IP address and port number, its endpoint is a path on the filesystem (e.g., `/var/run/app.sock`) or an abstract string in the kernel.

```
+-------------------------------------------------------------------+
|                Unix Domain Socket vs. Loopback TCP                |
+-------------------------------------------------------------------+

Loopback TCP (127.0.0.1:8080):
  User Process -> TCP Checksum -> IP Routing -> Loopback Driver (lo)
               -> Kernel Socket Buffer -> User Process
  (Overhead: TCP handshakes, ACK packets, sequence tracking, windowing)

Unix Domain Socket (/tmp/app.sock):
  User Process -> Kernel Memory Buffer Copy -> User Process
  (Zero network protocol overhead; direct memory transfer)
```

Advantages of Unix Domain Sockets

1. **Lower Latency & Higher Throughput:** No IP headers, no TCP checksums, no packet fragmentation, and no ACK packets.
2. **Standard File Permissions:** Security is governed by POSIX file permissions (`chmod`, `chown`). You can restrict access to specific users or groups without firewall rules.
3. **Passing File Descriptors:** Using `SCM_RIGHTS` (via `golang.org/x/sys/unix`), processes can pass open file descriptors, network connections, or shared memory handles directly across a Unix socket.

Creating Unix Domain Socket Servers and Clients

1. The Unix Socket Server

One critical difference when using filesystem-backed Unix sockets: **when the server stops, the socket file remains on disk**. If the server restarts without deleting the stale file, `net.Listen` returns `bind: address already in use`.

```go
// uds_server.go
package main

import (
    "bufio"
    "errors"
    "io"
    "log"
    "net"
    "os"
    "os/signal"
    "syscall"
)

const SocketPath = "/tmp/go_socket_demo.sock"

func main() {
    // Remove stale socket file if it exists
    if err := os.Remove(SocketPath); err != nil && !os.IsNotExist(err) {
        log.Fatalf("failed to remove existing socket: %v", err)
    }

    ln, err := net.Listen("unix", SocketPath)
    if err != nil {
        log.Fatalf("failed to listen on UDS: %v", err)
    }
```

```go
    defer ln.Close()
    defer os.Remove(SocketPath)

    // Clean up socket file on termination signals
    sigChan := make(chan os.Signal, 1)
    signal.Notify(sigChan, os.Interrupt, syscall.SIGTERM)
    go func() {
        <-sigChan
        _ = ln.Close()
        _ = os.Remove(SocketPath)
        os.Exit(0)
    }()

    log.Printf("UDS Server listening on %s", SocketPath)

    for {
        conn, err := ln.Accept()
        if err != nil {
            if errors.Is(err, net.ErrClosed) {
                return
            }
            log.Printf("Accept error: %v", err)
            continue
        }

        go handleUDSConn(conn)
    }
}

func handleUDSConn(conn net.Conn) {
    defer conn.Close()

    reader := bufio.NewReader(conn)
    for {
        line, err := reader.ReadBytes('\n')
        if err != nil {
            if err != io.EOF {
                log.Printf("read error: %v", err)
            }
            return
        }

        // Echo back
        if _, err := conn.Write(line); err != nil {
            return
        }
    }
```

```
}
```

2. The Unix Socket Client

Connecting to a Unix socket in Go uses the standard `net.Dial` API:

```go
// uds_client.go
package main

import (
    "bufio"
    "fmt"
    "log"
    "net"
)

func main() {
    conn, err := net.Dial("unix", "/tmp/go_socket_demo.sock")
    if err != nil {
        log.Fatalf("dial error: %v", err)
    }
    defer conn.Close()

    // Send message
    if _, err := conn.Write([]byte("Hello from Unix Domain Socket client!\n")); err != nil {
        log.Fatal(err)
    }

    reply, err := bufio.NewReader(conn).ReadString('\n')
    if err != nil {
        log.Fatal(err)
    }

    fmt.Printf("Echoed: %s", reply)
}
```

File Permissions and Security with Unix Sockets

Because Unix domain sockets exist as special files in the filesystem, you can use `os.Chmod` and `os.Chown` immediately after calling `net.Listen` to restrict access:

```go
ln, err := net.Listen("unix", "/var/run/secure_app.sock")
if err != nil {
    log.Fatal(err)
}

// Restrict access: only owner (UID 1000) and members of 'www-data' group can connect
// File mode 0660: rw-rw----
if err := os.Chmod("/var/run/secure_app.sock", 0660); err != nil {
    log.Fatalf("failed to set socket permissions: %v", err)
}
```

Abstract Namespace Sockets (Linux-specific)

On Linux, Unix domain sockets support an **abstract namespace** (RFC 2553 / Linux man `unix(7)`).

Abstract sockets: 1. Do not exist as files in the filesystem directory tree. 2. Are automatically cleaned up by the kernel when all referencing file descriptors close. 3. Never leave stale files behind after a server crash. 4. Are denoted by a null byte (`\x00`) prefix in the path, or an `@` symbol in Go:

```go
// In Go, prefix the address with '@' for an abstract socket:
const AbstractSocketName = "@my_service_abstract_socket"

ln, err := net.Listen("unix", AbstractSocketName)
if err != nil {
    log.Fatalf("failed to bind abstract socket: %v", err)
}
defer ln.Close()

// Client dials the exact same abstract address:
conn, err := net.Dial("unix", AbstractSocketName)
```

> [!NOTE] Abstract namespace sockets are exclusive to Linux. Code relying on them will fail to bind on macOS, FreeBSD, or Windows.

Comparing Performance: Unix Sockets vs Loopback TCP

Under high concurrency and large transfer volumes, Unix domain sockets provide substantial throughput and latency advantages over loopback TCP:

Metric	Loopback TCP (`127.0.0.1`)	Unix Domain Socket
P99 Latency (Small RPCs)	~45 µs	~18 µs (2.5x faster)
Max Throughput (Gigabytes/sec)	~3.2 GB/s	~6.8 GB/s (2.1x higher)
CPU Utilization (Context Switches)	High (TCP/IP stack processing)	Low (direct VFS memory buffer copy)
Socket Cleanup on Crash	Automatic (kernel closes TCP port)	Manual (stale file) or automatic (abstract)
Ephemeral Port Exhaustion	Risk under heavy short-lived connections	No port numbers involved

---

Chapter Summary

- **Unix Domain Sockets** allow processes on the same operating system to communicate without TCP/IP stack overhead.
- Sockets implement the standard **`net.Conn`** and **`net.Listener`** interfaces.
- Filesystem-backed sockets require explicit **cleanup of stale files** upon startup and shutdown.
- Access can be restricted using standard **POSIX file permissions (`os.Chmod`)**.
- On Linux, **abstract namespace sockets** (`@name`) provide automatic cleanup without touching the filesystem.
- In the next chapter, we will master concurrency patterns for network programming in Go.

Chapter 7: Concurrency Patterns for Network Programming

Chapter 7: Concurrency Patterns for Network Programming

Concurrency is Go's greatest superpower. In network programming, managing thousands of simultaneous I/O streams requires disciplined concurrency architectures to prevent resource exhaustion, eliminate data races, and minimize garbage collection (GC) pauses.

In this chapter, you will master the standard goroutine-per-connection pattern, understand when and how to implement worker pools, coordinate bidirectional I/O streams using channels, recycle high-frequency memory buffers using `sync.Pool`, and avoid the most common concurrency bugs found in network applications.

One Goroutine Per Connection: The Standard Pattern

The default and most natural idiom in Go is spawning a dedicated goroutine for each accepted connection:

```go
for {
    conn, err := ln.Accept()
    if err != nil {
        handleError(err)
        continue
    }
    go handleConn(conn) // Lightweight: ~2KB stack initial footprint
}
```

```
                        ln.Accept() Loop


Goroutine #1            Goroutine #2            Goroutine #N
Conn 1 Read/W           Conn 2 Read/W           Conn N Read/W
```

Why This Works in Go

In C or Java, spawning an OS thread per connection collapses at a few thousand clients due to 8MB stack reservations and kernel scheduler overhead. In Go, goroutines start at **2 KB**, their stacks grow dynamically, and the runtime netpoller multiplexes them across available CPU cores. Having 50,000 idle TCP connections in Go consumes roughly 100 MB of RAM—trivial on modern servers.

Worker Pools for Connection Handling

While goroutines are cheap, the resources they consume inside your application might not be. If each client request executes heavy CPU computations or queries a database pool with a hard limit of 50 connections, spawning 20,000 simultaneous goroutines will overwhelm downstream systems and cause CPU thrashing.

A **Worker Pool** limits concurrent active execution while maintaining an inbound backlog queue.

```go
package main

import (
    "log"
    "net"
)

type WorkerPool struct {
    maxWorkers int
    tasks      chan net.Conn
}

func NewWorkerPool(maxWorkers, queueSize int) *WorkerPool {
    return &WorkerPool{
        maxWorkers: maxWorkers,
        tasks:      make(chan net.Conn, queueSize),
    }
}

func (wp *WorkerPool) Start() {
    for i := 0; i < wp.maxWorkers; i++ {
        go func(workerID int) {
            for conn := range wp.tasks {
                processConnection(workerID, conn)
            }
        }(i)
    }
```

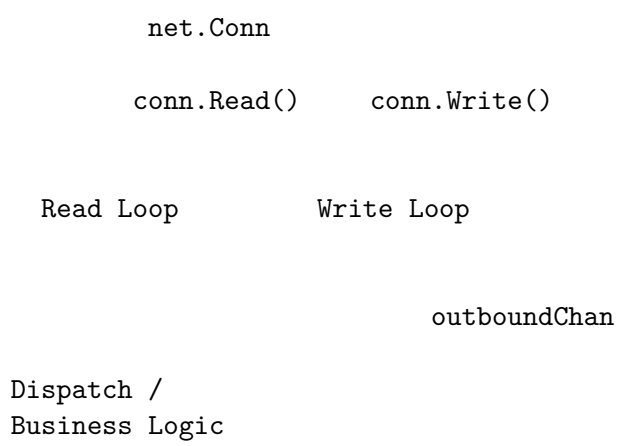
```go
}

func (wp *WorkerPool) Submit(conn net.Conn) bool {
    select {
    case wp.tasks <- conn:
        return true
    default:
        // Queue full: shed load or reject immediately
        return false
    }
}

func processConnection(workerID int, conn net.Conn) {
    defer conn.Close()
    // Process business logic under bounded concurrency...
}
```

Channels for Coordinating Network Operations

In full-duplex TCP connections, you often need to decouple inbound reads from outbound writes. A standard pattern uses separate read and write loops coordinated via channels.



Full-Duplex Actor Pattern

```
type ClientSession struct {
    conn     net.Conn
    outgoing chan []byte
    closed   chan struct{}
}

func NewSession(conn net.Conn) *ClientSession {
    s := &ClientSession{
        conn:     conn,
        outgoing: make(chan []byte, 128),
        closed:   make(chan struct{}),
    }
    go s.writeLoop()
    go s.readLoop()
    return s
}

func (s *ClientSession) writeLoop() {
    defer s.conn.Close()
    for {
        select {
        case msg, ok := <-s.outgoing:
            if !ok {
                return
            }
            if _, err := s.conn.Write(msg); err != nil {
                return
            }
        case <-s.closed:
            return
        }
    }
}

func (s *ClientSession) readLoop() {
    defer close(s.closed)
    defer s.conn.Close()

    buf := make([]byte, 4096)
    for {
        n, err := s.conn.Read(buf)
        if err != nil {
            return
        }
```

```go
        handleIncomingMessage(buf[:n])
    }
}

func (s *ClientSession) Send(data []byte) bool {
    select {
    case s.outgoing <- data:
        return true
    case <-s.closed:
        return false
    }
}
```

---

sync.Pool for Reusing Buffers and Connections

At 100,000 requests per second, allocating a new `make([]byte, 4096)` buffer for every read forces the Go garbage collector to track and reclaim millions of short-lived objects, leading to frequent GC stop-the-world pauses.

`sync.Pool` provides a thread-safe cache of reusable memory buffers.

```go
package main

import (
    "io"
    "net"
    "sync"
)

// Global pool of reusable 4KB byte slices
var bufferPool = sync.Pool{
    New: func() any {
        b := make([]byte, 4096)
        return &b
    },
}

func handleEchoWithPool(conn net.Conn) {
    defer conn.Close()

    // Acquire a buffer from the pool
    bufPtr := bufferPool.Get().(*[]byte)
    buf := *bufPtr
```

```go
    // Crucial: Always return the buffer to the pool when finished
    defer bufferPool.Put(bufPtr)

    for {
        n, err := conn.Read(buf)
        if err != nil {
            if err != io.EOF {
                // Handle non-EOF error
            }
            return
        }

        if _, err := conn.Write(buf[:n]); err != nil {
            return
        }
    }
}
```

> [!WARNING] When using `sync.Pool`, never retain references to a pooled slice after putting it back into the pool. Doing so leads to insidious data corruption when another goroutine acquires and overwrites that buffer.

Avoiding Common Concurrency Bugs in Network Code

1. The Hanging Goroutine Leak

Bug: A goroutine calls `conn.Read()` without a deadline or timeout. If the client disconnects without a TCP `FIN` (e.g., WiFi dropped), the goroutine blocks forever. **Fix:** Always set deadlines (`SetDeadline`) or close the connection when the parent context cancels.

2. Double-Closing Channels

Bug: Two goroutines detect an error and both attempt to call `close(doneChan)`, triggering a panic: `panic: close of closed channel`. **Fix:** Use `sync.Once` to guard channel closure, or use a dedicated owner goroutine for each channel.

```go
type Session struct {
    closeOnce sync.Once
    closed    chan struct{}
}

func (s *Session) Close() {
    s.closeOnce.Do(func() {
```

```
        close(s.closed)
    })
}
```

3. Concurrent Writes to `net.Conn`

Bug: Multiple worker goroutines call `conn.Write(data)` on the same `net.Conn` simultaneously. While the `net.Conn` implementation will not crash, bytes from different messages will interleave and corrupt your application protocol. **Fix:** Route all writes through a single write loop using a channel or protect writes with a `sync.Mutex`.

Chapter Summary

- **Goroutine-per-connection** is the default and scalable pattern for Go network daemons.
- **Worker Pools** should be used to protect bounded downstream resources (e.g., database connections, CPU capacity).
- Decouple full-duplex TCP I/O into **dedicated read and write loops** communicating via channels.
- `sync.Pool` drastically lowers memory allocations and GC pause times when recycling network buffers.
- Guard against **goroutine leaks** using deadlines, prevent **concurrent writes** with channels or mutexes, and ensure **channel closure safety** via `sync.Once`.
- In the next chapter, we will master timeouts, deadlines, and context cancellation to build resilient production services.

Chapter 8: Timeouts, Deadlines and Context Cancellation

Chapter 8: Timeouts, Deadlines and Context Cancellation

In production distributed systems, networks will fail. Routers drop packets, intermediate firewalls silently sever idle TCP sessions, cellular clients transit through dead zones, and malicious actors intentionally trickle bytes to exhaust server resources.

If your network code assumes that reads and writes will always complete in a timely manner, your application will eventually freeze as all available file descriptors and goroutines become permanently locked.

In this chapter, you will learn how to enforce non-negotiable timeouts, master Go's deadline APIs, cleanly integrate `context.Context` with network I/O, and protect your services against Slowloris attacks.

Why Timeouts Are Non-Negotiable in Production Code

Consider what happens when a client sends a request to a Go server and then experiences an abrupt network outage without sending a TCP `FIN`:

1. The server calls `conn.Read(buf)`.
2. The Go runtime netpoller deschedules the goroutine and registers the file descriptor with `epoll`.
3. If no deadline is set, **that goroutine will wait forever**.
4. The file descriptor remains open in the operating system kernel.
5. If thousands of clients experience dropped connections, the server exhausts its OS file descriptor limit (`EMFILE`), causing `ln.Accept()` to fail and taking down the entire service.

   [!IMPORTANT] A network call without a timeout is a production incident waiting to happen. Every connection, read, and write must be bounded by an explicit time limit.

SetDeadline vs SetReadDeadline vs SetWriteDeadline

The `net.Conn` interface exposes three methods for managing timeouts:

```go
type Conn interface {
    // ...
    SetDeadline(t time.Time) error
    SetReadDeadline(t time.Time) error
    SetWriteDeadline(t time.Time) error
}
```

Absolute Point in Time (`time.Time`), Not Duration

A common beginner mistake is confusing deadlines with durations:

```go
// WRONG: Does not compile or behaves unpredictably
conn.SetDeadline(5 * time.Second)

// RIGHT: Pass an absolute time in the future
conn.SetDeadline(time.Now().Add(5 * time.Second))
```

Differences Between the Methods

1. **`SetDeadline(t)`:** Sets both the read and write deadline simultaneously to point in time `t`.
2. **`SetReadDeadline(t)`:** Affects only future and currently active `Read()` calls. If `t` is reached before bytes arrive, `Read()` unblocks immediately and returns an error satisfying `net.Error.Timeout() == true`.
3. **`SetWriteDeadline(t)`:** Affects only future and currently active `Write()` calls. Prevents writes from blocking indefinitely if the receiver's TCP window shrinks to zero.
4. **Clearing Deadlines:** Passing `time.Time{}` (the zero value) clears the deadline, allowing indefinite blocking.

Forcibly Unblocking a Hanging I/O Call

Because deadlines operate across goroutines, you can unblock a goroutine currently stuck inside `conn.Read()` by calling `conn.SetReadDeadline(time.Now())` from another goroutine!

```go
// Forcibly unblock any pending read immediately
_ = conn.SetReadDeadline(time.Now())
```

Integrating context.Context with Network Operations

While `net.Conn` operates on `time.Time` deadlines, modern Go idioms rely on `context.Context` for request cancellation and deadline propagation.

1. Context-Aware Dialing with `net.Dialer`

Always dial using `DialContext`:

```go
func fetchWithTimeout(ctx context.Context, addr string) (net.Conn, error) {
    var dialer net.Dialer
    return dialer.DialContext(ctx, "tcp", addr)
}
```

2. Bridging `context.Context` to an Existing `net.Conn`

Because `net.Conn` does not have a `ReadContext(ctx)` method, you can bridge context cancellation to connection deadlines using a background watcher goroutine:

```go
func ReadWithContext(ctx context.Context, conn net.Conn, buf []byte) (int, error) {
    // Guard against unhandled context cancellations
    done := make(chan struct{})
    defer close(done)

    go func() {
        select {
        case <-ctx.Done():
            // Context expired: force unblock pending I/O
            _ = conn.SetDeadline(time.Now())
        case <-done:
            // Normal return
        }
    }()

    n, err := conn.Read(buf)
    if err != nil {
        if ctx.Err() != nil {
            return 0, ctx.Err() // Return context.Canceled or context.DeadlineExceeded
        }
        return 0, err
    }
    return n, nil
}
```

Handling Slowloris and Resource Exhaustion Attacks

What is a Slowloris Attack?

In a Slowloris attack, an attacker opens hundreds of TCP connections to a server and sends data extremely slowly (e.g., 1 byte every 20 seconds).

If a naive server resets its deadline on every successful `Read()`:

```
// VULNERABLE TO SLOWLORIS:
for {
    // Sliding deadline resets on EVERY single byte!
    _ = conn.SetReadDeadline(time.Now().Add(10 * time.Second))
    n, err := conn.Read(buf)
    // An attacker sending 1 byte every 9 seconds keeps this connection open forever!
}
```

The attacker holds hundreds of server connections open while consuming virtually zero client bandwidth.

Defensive Strategies Against Slowloris

Strategy 1: Total Request Budget (Hard Timeout)

Enforce an absolute, fixed deadline for the entire request lifecycle that never resets:

```
// Hard deadline: the entire message must be received within 15 seconds
conn.SetReadDeadline(time.Now().Add(15 * time.Second))
// Read headers and body...
```

Strategy 2: Minimum Transfer Rate Throttling

Track both time and bytes received. If the data transfer rate drops below a minimum threshold (e.g., 500 bytes per second), terminate the connection:

```
type RateLimitedReader struct {
    conn       net.Conn
    minRateBPS int64 // Minimum required bytes per second
}

func (r *RateLimitedReader) ReadFullMessage(maxDuration time.Duration) ([]byte, error) {
    startTime := time.Now()
    var totalBytes int64
    var result []byte
    buf := make([]byte, 1024)
```

```go
    _ = r.conn.SetReadDeadline(startTime.Add(maxDuration))

    for {
        n, err := r.conn.Read(buf)
        if n > 0 {
            totalBytes += int64(n)
            result = append(result, buf[:n]...)

            elapsed := time.Since(startTime).Seconds()
            if elapsed > 1.0 {
                rate := float64(totalBytes) / elapsed
                if rate < float64(r.minRateBPS) {
                    _ = r.conn.Close()
                    return nil, errors.New("connection closed: transfer rate too slow")
                }
            }
        }

        if err != nil {
            return result, err
        }
        // Break when message boundary is detected...
    }
}
```

Chapter Summary

- **Missing timeouts** cause leaked goroutines, file descriptor exhaustion, and catastrophic cascading outages.
- Go deadlines are **absolute points in time (`time.Time`)**, not durations.
- Setting a deadline in the past (`conn.SetDeadline(time.Now())`) **immediately unblocks** active I/O calls across goroutines.
- Use **`net.Dialer.DialContext`** to propagate context deadlines to connection attempts.
- Prevent **Slowloris attacks** by avoiding sliding deadlines that reset on every byte; instead, enforce total request budgets and minimum throughput rates.
- In the next chapter, we will learn how to design and parse custom network protocols, from text-based lines to length-prefixed binary frames.

Chapter 9: Protocols: From Text to Binary to Custom Designs

Chapter 9: Protocols: From Text to Binary to Custom Designs

TCP provides a stream of bytes without message boundaries. If a client writes two separate 50-byte strings, the server might read all 100 bytes in a single call, or receive 20 bytes, followed by 30 bytes, followed by 50 bytes.

To transform an unstructured byte stream into discrete business messages, you must design an **Application Protocol**.

In this chapter, you will learn the mechanics of line-based text protocols, master length-prefixed framing, serialize binary headers using `encoding/binary`, evaluate protocol design principles, and build a custom high-performance RPC protocol from scratch.

Line-Based Text Protocols

The simplest way to delineate messages is using a delimiter character, most commonly a newline (`\n` or `\r\n`). Protocols like HTTP/1.1, Redis (RESP), SMTP, and POP3 use this strategy.

```
Message 1: "PING\n"
Message 2: "SET user:1 Alice\n"
```

Implementing Line-Based Framing with `bufio.Scanner`

Go's `bufio.Scanner` splits incoming streams on newlines by default:

```go
func handleLineProtocol(conn net.Conn) {
    defer conn.Close()
    scanner := bufio.NewScanner(conn)

    // You can increase the maximum allowed line buffer if necessary
    // const maxLineSize = 64 * 1024
    // buf := make([]byte, maxLineSize)
    // scanner.Buffer(buf, maxLineSize)

    for scanner.Scan() {
        commandLine := scanner.Text()
```

```go
        log.Printf("Received command: %s", commandLine)

        // Process command...
        _, _ = conn.Write([]byte("OK\n"))
    }

    if err := scanner.Err(); err != nil {
        log.Printf("Scanner error: %v", err)
    }
}
```

Limitations of Delimiter-Based Framing

1. **Payload Escaping:** If the message payload itself contains a newline character, you must escape it (e.g., `\n` or URL encoding), adding CPU overhead.
2. **Buffer Allocation Attacks:** An adversary can send a continuous stream of bytes without ever transmitting a newline, exhausting server memory unless an explicit buffer limit is enforced.

Length-Prefixed Message Framing

Length-prefixed framing prepends each payload with a fixed-size integer indicating the number of payload bytes that follow.

```
+-------------------------------------+----------------------------------------+
| Length Header (4 bytes, BigEndian) | Payload Data (N bytes)                 |
| [ 0x00, 0x00, 0x00, 0x05 ]         | [ 'H', 'e', 'l', 'l', 'o' ]            |
+-------------------------------------+----------------------------------------+
```

Advantages of Length-Prefixed Framing

1. **Arbitrary Payloads:** The payload can contain raw binary data, null bytes, newlines, or compressed images without escaping.
2. **Exact Memory Pre-allocation:** The receiver reads the 4-byte header, verifies it is within acceptable limits, allocates a buffer of exact size N, and reads exactly N bytes using `io.ReadFull`.

Reading Length-Prefixed Frames

```go
func ReadMessage(r io.Reader) ([]byte, error) {
    var length uint32
    // Read 4-byte header
    if err := binary.Read(r, binary.BigEndian, &length); err != nil {
        return nil, err
    }

    // Security check: Guard against integer overflow or unbounded memory allocation
    const MaxPayloadSize = 16 * 1024 * 1024 // 16 MB limit
    if length > MaxPayloadSize {
        return nil, fmt.Errorf("payload size %d exceeds limit", length)
    }

    payload := make([]byte, length)
    // ReadFull guarantees reading exactly len(payload) bytes
    if _, err := io.ReadFull(r, payload); err != nil {
        return nil, err
    }

    return payload, nil
}
```

Binary Protocols with encoding/binary

When designing high-throughput services, binary protocols consume far less bandwidth and CPU than text formats. Go provides the `encoding/binary` package to marshal numbers to and from byte slices.

Endianness: Big-Endian vs. Little-Endian

- **Big-Endian (Network Byte Order):** Most significant byte first. The standard convention across the Internet and RFC specifications.
- **Little-Endian:** Least significant byte first. The native architecture of x86-64 and ARM CPUs.

```go
val := uint32(0x12345678)

// Big-Endian wire format:   [0x12, 0x34, 0x56, 0x78]
// Little-Endian wire format: [0x78, 0x56, 0x34, 0x12]
```

In Go:

```go
buf := make([]byte, 4)
binary.BigEndian.PutUint32(buf, 0x12345678)

parsed := binary.BigEndian.Uint32(buf)
```

Protocol Design Principles

When designing a production protocol, always include the following fields in your fixed-size frame header:

```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|       Magic Number (2B)       |  Version (1B) |   Type (1B)   |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                      Sequence ID (4B)                         |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                      Payload Length (4B)                      |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                      Payload Data (...)                       |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                      CRC32 Checksum (4B)                      |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
```

1. **Magic Bytes (2–4 bytes):** A constant signature (e.g., `0xCAFE`) verifying that the connection is speaking your protocol and not junk data or an accidental HTTP scan.
2. **Protocol Version (1 byte):** Allows seamless protocol upgrades and backward compatibility.
3. **Message Type (1 byte):** Identifies message semantics (e.g., `1 = Ping`, `2 = Request`, `3 = Response`, `4 = Error`).
4. **Correlation / Sequence ID (4–8 bytes):** Allows clients to multiplex multiple requests over a single TCP connection and match out-of-order responses.
5. **Payload Length (4 bytes):** Tells the parser how many bytes to read for the payload.
6. **Payload Integrity Check (Optional, 4 bytes):** CRC32 or Adler32 checksum verifying data integrity against application-level memory bugs.

Building a Custom RPC Protocol from Scratch

Let's implement a complete, production-ready frame parser and serializer for a custom binary protocol in Go.

```go
// protocol.go
package main

import (
    "encoding/binary"
    "errors"
    "fmt"
    "hash/crc32"
    "io"
)

const (
    MagicBytes uint16 = 0x5443 // 'TC'
    Version1   uint8  = 1
    HeaderSize        = 12 // 2 (Magic) + 1 (Ver) + 1 (Type) + 4 (Seq) + 4 (Len)
)

type MessageType uint8

const (
    TypePing     MessageType = 1
    TypePong     MessageType = 2
    TypeRequest  MessageType = 3
    TypeResponse MessageType = 4
    TypeError    MessageType = 5
)

type Frame struct {
    Type     MessageType
    SeqID    uint32
    Payload  []byte
}

func EncodeFrame(w io.Writer, f Frame) error {
    payloadLen := uint32(len(f.Payload))
    buf := make([]byte, HeaderSize+len(f.Payload)+4) // +4 for CRC32

    // Pack header
    binary.BigEndian.PutUint16(buf[0:2], MagicBytes)
    buf[2] = Version1
    buf[3] = byte(f.Type)
    binary.BigEndian.PutUint32(buf[4:8], f.SeqID)
    binary.BigEndian.PutUint32(buf[8:12], payloadLen)

    // Copy payload
    copy(buf[12:12+payloadLen], f.Payload)
```

```go
    // Calculate and append CRC32 checksum of payload
    checksum := crc32.ChecksumIEEE(f.Payload)
    binary.BigEndian.PutUint32(buf[12+payloadLen:], checksum)

    _, err := w.Write(buf)
    return err
}

func DecodeFrame(r io.Reader) (*Frame, error) {
    headerBuf := make([]byte, HeaderSize)
    if _, err := io.ReadFull(r, headerBuf); err != nil {
        return nil, err
    }

    magic := binary.BigEndian.Uint16(headerBuf[0:2])
    if magic != MagicBytes {
        return nil, fmt.Errorf("invalid protocol magic: 0x%X", magic)
    }

    ver := headerBuf[2]
    if ver != Version1 {
        return nil, fmt.Errorf("unsupported protocol version: %d", ver)
    }

    msgType := MessageType(headerBuf[3])
    seqID := binary.BigEndian.Uint32(headerBuf[4:8])
    payloadLen := binary.BigEndian.Uint32(headerBuf[8:12])

    const maxPayload = 8 * 1024 * 1024 // 8 MB limit
    if payloadLen > maxPayload {
        return nil, errors.New("payload length exceeds maximum allowed limit")
    }

    // Read payload + 4 bytes CRC
    bodyBuf := make([]byte, payloadLen+4)
    if _, err := io.ReadFull(r, bodyBuf); err != nil {
        return nil, err
    }

    payload := bodyBuf[:payloadLen]
    expectedChecksum := binary.BigEndian.Uint32(bodyBuf[payloadLen:])
    actualChecksum := crc32.ChecksumIEEE(payload)

    if actualChecksum != expectedChecksum {
        return nil, errors.New("payload CRC32 checksum mismatch: corrupted data")
    }
```

```
    return &Frame{
        Type:    msgType,
        SeqID:   seqID,
        Payload: payload,
    }, nil
}
```

Chapter Summary

- TCP is an unstructured **byte stream**; applications must provide framing to distinguish messages.
- **Line-based text protocols** are human-readable but require payload escaping and guardrails against unbounded lines.
- **Length-prefixed framing** uses a fixed-size header to enable binary payloads and zero-copy pre-allocated reads via `io.ReadFull`.
- **`encoding/binary`** serializes integers with strict endianness (use `binary.BigEndian` for network wire protocols).
- Robust protocol designs incorporate **magic numbers**, **version flags**, **sequence correlation IDs**, and **checksums**.
- In the next chapter, we will examine serialization formats—JSON, Protocol Buffers, and MessagePack—to structure payload data.

Chapter 10: Serialization: JSON, Protobuf and Beyond

Chapter 10: Serialization: JSON, Protobuf and Beyond

Once you have framed your network messages, you must decide how to represent the data payload inside each frame. **Serialization** (marshalling) converts in-memory Go data structures into a sequence of bytes suitable for network transmission, and **deserialization** (unmarshalling) reconstructs those data structures on the receiving end.

In this chapter, you will evaluate the performance, wire efficiency, and ergonomics of JSON, Protocol Buffers, and MessagePack, learn how to choose the right format for your workload, and master techniques for streaming multi-gigabyte payloads without exhausting server memory.

JSON with encoding/json (When It Is Good Enough)

JavaScript Object Notation (JSON) is the lingua franca of web APIs and cloud microservices. Go's standard library provides robust support via `encoding/json`.

Streaming JSON with `json.Encoder` and `json.Decoder`

When communicating over a network socket, avoid marshalling into intermediate `[]byte` slices. Instead, connect `json.Encoder` and `json.Decoder` directly to the `net.Conn` stream:

```go
type OrderPayload struct {
    OrderID   string  `json:"order_id"`
    UserID    int64   `json:"user_id"`
    Amount    float64 `json:"amount"`
    Timestamp int64   `json:"timestamp"`
}

// Server receiving JSON stream
func handleJSONStream(conn net.Conn) {
    defer conn.Close()
    decoder := json.NewDecoder(conn)
    encoder := json.NewEncoder(conn)

    for {
        var order OrderPayload
```

```go
        if err := decoder.Decode(&order); err != nil {
            if errors.Is(err, io.EOF) {
                return
            }
            log.Printf("JSON decode error: %v", err)
            return
        }

        // Process order...
        response := map[string]string{"status": "ACCEPTED", "id": order.OrderID}
        if err := encoder.Encode(response); err != nil {
            return
        }
    }
}
```

When JSON is Good Enough

- **Public APIs & Web Clients:** Broadest language support and browser-native compatibility.
- **Human Debuggability:** Inspect payloads in Wireshark or logs without specialized tooling.
- **Low-to-Medium Throughput:** Suitable when serialization CPU usage accounts for less than 5% of total request processing time.

Where JSON Fails

- **Heavy CPU Overhead:** Reflection and string parsing consume significant CPU cycles.
- **Wire Bloat:** Verbose field keys are repeated in every message.
- **Type Ambiguities:** Floating-point numbers and 64-bit integer precision issues.

Protocol Buffers with Google's protobuf-go

Google's **Protocol Buffers (Protobuf)** is the gold standard for high-performance, internal microservice communication and gRPC.

The Protobuf Workflow

1. Define a strict schema in a `.proto` file:

```proto
syntax = "proto3";
package order;
option go_package = "./proto";
```

```
message Order {
  string order_id = 1;
  int64 user_id = 2;
  double amount = 3;
  int64 timestamp = 4;
}
```

2. Compile to Go code using `protoc` and `protoc-gen-go`:

```
protoc --go_out=. --go_opt=paths=source_relative proto/order.proto
```

3. Serialize and deserialize using `google.golang.org/protobuf/proto`:

```
import (
    "google.golang.org/protobuf/proto"
    pb "myproject/proto"
)

// Marshalling to wire bytes:
order := &pb.Order{
    OrderId:   "ord-98712",
    UserId:    42100,
    Amount:    99.95,
    Timestamp: time.Now().Unix(),
}
wireBytes, err := proto.Marshal(order)

// Unmarshalling from wire bytes:
var received pb.Order
err = proto.Unmarshal(wireBytes, &received)
```

Why Protobuf Wins in Production

- **Compact Wire Size:** Numeric field tags (1, 2, 3) replace verbose string keys; integers are encoded using varints (variable-length zigzag encoding).
- **Fast Serialization:** Generated code avoids runtime reflection entirely.
- **Backward Compatibility:** Fields can be added or deprecated safely without breaking older clients.

MessagePack and Other Binary Formats

When you want binary compactness and speed without the overhead of maintaining `.proto` schema files and compilation steps, **MessagePack** (`msgpack`) is an excellent compromise.

MessagePack encodes JSON-like structures directly into compact binary representations:

```go
import "github.com/vmihailenco/msgpack/v5"

type Metric struct {
    Name  string  `msgpack:"name"`
    Value float64 `msgpack:"value"`
}

// Marshalling:
b, err := msgpack.Marshal(&Metric{Name: "cpu_usage", Value: 78.4})

// Unmarshalling:
var m Metric
err = msgpack.Unmarshal(b, &m)
```

Other zero-copy alternatives (such as **FlatBuffers** or **Cap'n Proto**) allow reading fields directly from raw network buffers without any deserialization step, ideal for gaming and high-frequency trading engines.

Choosing a Serialization Format

Criterion	JSON	MessagePack	Protocol Buffers	FlatBuffers
Human Readable	Yes	No	No (requires schema)	No
Strict Schema	Optional (JSON Schema)	No	Enforced (`.proto`)	Enforced
Serialization Speed	Moderate	Fast	Very Fast	Instant (Zero-Copy)
Payload Size	Largest (100%)	Compact (~60%)	Highly Compact (~35%)	Moderate
Tooling & Ecosystem	Universal	Broad	Extensive (gRPC)	Specialized
Best Used For	Web APIs, Configs	Dynamic binary RPC	Microservices, gRPC	HFT, Game Physics

---

Streaming Large Payloads Without Loading Everything Into Memory

If a client sends a 5 GB database backup or video file over a socket, reading the whole payload into memory (`io.ReadAll`) will trigger an Out-Of-Memory (OOM) crash:

```go
// CATASTROPHIC BUG: OOM on 5GB payload!
data, err := io.ReadAll(conn)
```

1. Chunked Streaming with `io.Copy`

Use `io.Copy` or `io.CopyBuffer` to stream data directly from the network socket into disk storage or another network pipe in constant memory ($O(1)$ space):

```go
func receiveFile(conn net.Conn, destPath string) error {
    defer conn.Close()

    file, err := os.Create(destPath)
    if err != nil {
        return err
    }
    defer file.Close()

    // io.Copy transfers data in 32KB internal buffers,
    // never consuming more than 32KB of RAM regardless of file size!
    written, err := io.Copy(file, conn)
    log.Printf("Successfully saved %d bytes to %s", written, destPath)
    return err
}
```

2. Using `io.Pipe` for Streaming Transformations

To encrypt, compress, or hash data on the fly as it streams from a socket, connect an `io.Pipe`:

```go
pr, pw := io.Pipe()

// Goroutine 1: Read from socket and write into pipe
go func() {
    defer pw.Close()
    _, _ = io.Copy(pw, conn)
}()

// Goroutine 2: Read from pipe and write to decompression or hash
hash := sha256.New()
_, _ = io.Copy(hash, pr)
```

Chapter Summary

- **`encoding/json`** with `json.NewDecoder` and `json.NewEncoder` enables straightforward streaming of JSON messages directly over sockets.
- **Protocol Buffers** provide strict schema enforcement, compact binary wire formats, and fast execution without runtime reflection.
- **MessagePack** offers compact binary efficiency without requiring code generation or schemas.
- **Never use `io.ReadAll`** for unbounded network payloads; use **`io.Copy`** and **`io.Pipe`** to stream arbitrary payload sizes in constant $O(1)$ memory.
- In the next chapter, we will learn how to secure socket communication using Transport Layer Security (TLS 1.3) and mutual TLS (mTLS).

Chapter 11: TLS/SSL: Encrypting Your Connections

Chapter 11: TLS/SSL: Encrypting Your Connections

Plaintext TCP connections are vulnerable to eavesdropping, packet tampering, and man-in-the-middle (MITM) attacks. In production, network communications traversing public networks—or zero-trust internal clouds—must be authenticated and encrypted using **Transport Layer Security (TLS)**.

Go's standard library features an enterprise-grade, memory-safe, pure-Go TLS implementation in package `crypto/tls`.

In this chapter, you will learn the mechanics of the TLS 1.3 handshake, how to upgrade raw TCP sockets to encrypted `tls.Conn` streams, how to generate in-memory certificates for development, how to implement mutual TLS (mTLS), and how to configure production-grade cipher suites and security controls.

How TLS Works (Handshake, Certificates, Cipher Suites)

TLS sits cleanly between the transport layer (TCP) and the application layer:

```
+-------------------------------------------------------------+
| Application Protocol (HTTP, Custom RPC, WebSocket)       |
+-------------------------------------------------------------+


+-------------------------------------------------------------+
| Transport Layer Security (crypto/tls)                       |
|  - Authenticity: X.509 Certificates signed by CAs           |
|  - Confidentiality: Symmetric Encryption (AES-GCM)          |
|  - Integrity: Message Authentication Code (HMAC / AEAD)     |
+-------------------------------------------------------------+


+-------------------------------------------------------------+
| Raw TCP Stream (net.Conn)                                   |
+-------------------------------------------------------------+
```

The TLS 1.3 Handshake (1-RTT)

TLS 1.3 modernized cryptography by reducing handshake latency from two round-trips (2-RTT in TLS 1.2) to a single round-trip (1-RTT):

```
Client                                                  Server

    ClientHello (Supported Cipher Suites + Key Share)


    ServerHello (Selected Cipher Suite + Key Share)
    {Server Certificate + Signature + Finished}


    {Finished}


    [ Authenticated & Encrypted Application Data ]
```

1. **ClientHello:** The client proposes supported TLS versions and cryptographic algorithms, generating and transmitting an ephemeral public key share (ECDHE).
2. **ServerHello:** The server selects the cipher suite, responds with its own ephemeral public key share, and immediately begins encrypting subsequent handshake messages using the derived keys.
3. **Certificate & Verification:** The server provides its X.509 certificate. The client verifies the certificate against its trusted Certificate Authority (CA) root store and checks the hostname against the Subject Alternative Names (SAN).
4. **Symmetric Data Stream:** The handshake concludes, and all subsequent application traffic is encrypted with high-speed symmetric AEAD ciphers (e.g., `AES-256-GCM` or `CHACHA20-POLY1305`).

Wrapping TCP Connections with tls.Conn

The `crypto/tls` package provides two primary abstractions: - **`tls.Listen:`** Creates a listener that automatically performs the TLS server handshake on inbound connections. - **`tls.Server` / `tls.Client:`** Wraps an already-established `net.Conn`.

1. The TLS Server

```go
// tls_server.go
package main

import (
    "crypto/tls"
    "io"
    "log"
    "net"
)

func main() {
    // Load server certificate and private key
    cert, err := tls.LoadX509KeyPair("server.crt", "server.key")
    if err != nil {
        log.Fatalf("failed to load certificates: %v", err)
    }

    config := &tls.Config{
        Certificates: []tls.Certificate{cert},
        MinVersion:   tls.VersionTLS13, // Enforce modern TLS 1.3
    }

    ln, err := tls.Listen("tcp", ":8443", config)
    if err != nil {
        log.Fatalf("tls listen error: %v", err)
    }
    defer ln.Close()

    log.Printf("TLS Server listening on %s", ln.Addr())

    for {
        conn, err := ln.Accept()
        if err != nil {
            log.Printf("accept error: %v", err)
            continue
        }

        go func(c net.Conn) {
            defer c.Close()
            // TLS handshake occurs lazily on first Read/Write,
            // or explicitly via c.(*tls.Conn).Handshake()
            buf := make([]byte, 1024)
            n, _ := c.Read(buf)
            _, _ = c.Write(append([]byte("Encrypted Echo: "), buf[:n]...))
        }(conn)
```

```go
    }
}
```

2. The TLS Client

```go
// tls_client.go
package main

import (
    "crypto/tls"
    "fmt"
    "log"
)

func main() {
    config := &tls.Config{
        ServerName: "localhost",      // Must match certificate Common Name or SAN
        MinVersion: tls.VersionTLS13,
    }

    conn, err := tls.Dial("tcp", "127.0.0.1:8443", config)
    if err != nil {
        log.Fatalf("tls dial failed: %v", err)
    }
    defer conn.Close()

    // Verify negotiated TLS state
    state := conn.ConnectionState()
    fmt.Printf("Connected using TLS version: 0x%04x, CipherSuite: 0x%04x\n",
        state.Version, state.CipherSuite)

    _, _ = conn.Write([]byte("Hello secure world!"))
    buf := make([]byte, 1024)
    n, _ := conn.Read(buf)
    fmt.Printf("Received: %s\n", string(buf[:n]))
}
```

Creating Self-Signed Certificates for Development

For automated integration testing and local development, you can generate valid self-signed X.509 certificates directly in Go memory without depending on external tools like `openssl`.

```go
// certgen.go
package main

import (
    "crypto/ecdsa"
    "crypto/elliptic"
    "crypto/rand"
    "crypto/x509"
    "crypto/x509/pkix"
    "encoding/pem"
    "math/big"
    "net"
    "os"
    "time"
)

func GenerateDevCerts(certFile, keyFile string) error {
    priv, err := ecdsa.GenerateKey(elliptic.P256(), rand.Reader)
    if err != nil {
        return err
    }

    template := x509.Certificate{
        SerialNumber: big.NewInt(1),
        Subject: pkix.Name{
            Organization: []string{"Dev Lab Sockets"},
            CommonName:   "localhost",
        },
        NotBefore:             time.Now(),
        NotAfter:              time.Now().Add(365 * 24 * time.Hour),
        KeyUsage:              x509.KeyUsageKeyEncipherment | x509.KeyUsageDigitalSignature,
        ExtKeyUsage:           []x509.ExtKeyUsage{x509.ExtKeyUsageServerAuth},
        BasicConstraintsValid: true,
        IPAddresses:           []net.IP{net.ParseIP("127.0.0.1")},
        DNSNames:              []string{"localhost"},
    }

    derBytes, err := x509.CreateCertificate(rand.Reader, &template, &template, &priv.PublicK
    if err != nil {
        return err
    }

    // Save Certificate PEM
    certOut, _ := os.Create(certFile)
    defer certOut.Close()
    _ = pem.Encode(certOut, &pem.Block{Type: "CERTIFICATE", Bytes: derBytes})
```

```go
    // Save Private Key PEM
    keyBytes, _ := x509.MarshalECPrivateKey(priv)
    keyOut, _ := os.Create(keyFile)
    defer keyOut.Close()
    _ = pem.Encode(keyOut, &pem.Block{Type: "EC PRIVATE KEY", Bytes: keyBytes})

    return nil
}
```

Mutual TLS (mTLS) for Client Authentication

In zero-trust architectures and inter-service communication, both sides must authenticate each other. With **mTLS**, the server not only proves its identity to the client, but also demands and validates an X.509 certificate presented by the client.

```go
// Configuring mTLS on the server:
caCert, _ := os.ReadFile("ca.crt")
caPool := x509.NewCertPool()
caPool.AppendCertsFromPEM(caCert)

serverTLSConfig := &tls.Config{
    Certificates: []tls.Certificate{serverCert},
    ClientCAs:    caPool,
    // Enforce valid client certificate signed by trusted CA
    ClientAuth:   tls.RequireAndVerifyClientCert,
    MinVersion:   tls.VersionTLS13,
}
```

On the client side, attach the client's certificate:

```go
clientCert, _ := tls.LoadX509KeyPair("client.crt", "client.key")
clientTLSConfig := &tls.Config{
    Certificates: []tls.Certificate{clientCert},
    RootCAs:      caPool,
    ServerName:   "api.internal.service",
}
```

Configuring TLS for Production (Cipher Suites, Versions, OCSP)

When deploying Go network services to production, apply these security hardening practices:

1. **Enforce `MinVersion: tls.VersionTLS13`:** TLS 1.3 eliminates vulnerable legacy ciphers (such as CBC mode and RSA key exchange) and guarantees forward secrecy.
2. **Explicit Cipher Suites (if supporting TLS 1.2):** If legacy clients require TLS 1.2, restrict allowed cipher suites strictly to AEAD constructions:

```go
CipherSuites: []uint16{
    tls.TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384,
    tls.TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384,
    tls.TLS_ECDHE_ECDSA_WITH_CHACHA20_POLY1305,
    tls.TLS_ECDHE_RSA_WITH_CHACHA20_POLY1305,
}
```

3. **Session Resumption & Tickets:** Go supports stateless TLS session tickets out of the box, reducing handshake latency for reconnecting clients.

Chapter Summary

- **TLS** ensures confidentiality, integrity, and mutual authentication above the transport layer.
- **TLS 1.3** simplifies the handshake to 1-RTT and eliminates insecure legacy cryptographic primitives.
- **`tls.Listen`** and **`tls.Dial`** wrap standard TCP connections seamlessly into encrypted streams satisfying `net.Conn`.
- Go can generate in-memory self-signed test certificates using `crypto/x509` without external dependencies.
- **mTLS (`RequireAndVerifyClientCert`)** provides cryptographically validated identity verification for both client and server.
- In the next chapter, we will build bidirectional real-time communication systems using WebSockets.

Chapter 12: WebSockets and Real-Time Bidirectional Communication

Chapter 12: WebSockets and Real-Time Bidirectional Communication

Standard HTTP follows a request-and-reply model initiated by the client. For real-time applications—such as financial tickers, collaborative editing, chat systems, and live dashboards—polling is inefficient and introduces unacceptable latency.

The **WebSocket Protocol (RFC 6455)** provides full-duplex, bidirectional communication channels over a single persistent TCP connection. It begins as a standard HTTP/1.1 request and performs an **Upgrade Handshake**, transforming the connection into a lightweight framing layer that passes messages directly between browser and server.

In this chapter, you will learn the low-level mechanics of the WebSocket protocol, implement robust WebSocket handlers using production libraries, build a scalable real-time hub, manage ping/pong heartbeats, and scale WebSocket infrastructure across a cluster using a message backplane.

How WebSockets Work (Handshake, Frames, Upgrade)

The Upgrade Handshake

A WebSocket connection begins with an ordinary HTTP GET request:

```
GET /chat HTTP/1.1
Host: server.example.com
Upgrade: websocket
Connection: Upgrade
Sec-WebSocket-Key: dGhlIHNhbXBsZSBub25jZQ==
Sec-WebSocket-Version: 13
```

If the server agrees to switch protocols, it responds with status code `101 Switching Protocols`:

```
HTTP/1.1 101 Switching Protocols
Upgrade: websocket
Connection: Upgrade
Sec-WebSocket-Accept: s3pPLMBiTxaQ9kYGzzhZRbK+xOo=
```

The server calculates `Sec-WebSocket-Accept` by concatenating the client's `Sec-WebSocket-Key` with a well-known GUID string (`258EAFA5-E914-47DA-95CA-C5AB0DC85B11`), taking the SHA-1 hash, and Base64-encoding the result.

Once this handshake completes, the underlying TCP connection stops speaking HTTP and transitions immediately to **WebSocket Framing**.

WebSocket Framing Format

Unlike raw TCP streams, WebSockets provide message-oriented framing:

```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-------+-+-------------+-------------------------------+
|F|R|R|R| opcode|M| Payload len |    Extended payload length    |
|I|S|S|S|  (4)  |A|     (7)     |             (16/64)           |
|N|V|V|V|       |S|             |   (if payload len==126/127)   |
| |1|2|3|       |K|             |                               |
+-+-+-+-+-------+-+-------------+ - - - - - - - - - - - - - - - +
|     Masking-key (4 bytes, if MASK bit set by client)          |
+---------------------------------------------------------------+
|                       Payload Data                            |
+---------------------------------------------------------------+
```

- **FIN Bit:** Indicates if this frame is the final fragment of a message.
- **Opcodes:** `0x1` (Text/UTF-8), `0x2` (Binary data), `0x8` (Close connection), `0x9` (Ping), `0xA` (Pong).
- **Client-to-Server Masking:** All frames sent from browser clients to the server must be XOR-masked with a random 4-byte key to prevent cache-poisoning attacks on intermediary proxies.

Using gorilla/websocket in Production

The most widely adopted, battle-tested WebSocket library in the Go ecosystem is `github.com/gorilla/websocket`

To install it:

```go
go get github.com/gorilla/websocket
```

The Upgrader

The `websocket.Upgrader` takes an inbound HTTP request and upgrades its underlying TCP socket:

```go
var upgrader = websocket.Upgrader{
    ReadBufferSize:  1024,
    WriteBufferSize: 1024,
```

```go
    // CheckOrigin allows configuring CORS policies
    CheckOrigin: func(r *http.Request) bool {
        // In production, validate against allowed origins
        return true
    },
}
```

Building a Real-Time Chat Server with WebSockets

A production WebSocket server follows the **Hub and Client** architecture. The Hub owns the registration state and broadcasts messages to all connected client channels, avoiding locks across connections.

```
                          Hub Loop

          Broadcast            Register             Unregister

   Client #1            Client #2            Client #N
(Read/Write Gs)      (Read/Write Gs)      (Read/Write Gs)
```

1. The Hub

```go
// hub.go
package main

type Hub struct {
    clients    map[*Client]bool
    broadcast  chan []byte
    register   chan *Client
    unregister chan *Client
}

func NewHub() *Hub {
    return &Hub{
        clients:    make(map[*Client]bool),
        broadcast:  make(chan []byte, 256),
        register:   make(chan *Client),
        unregister: make(chan *Client),
```

```go
    }
}

func (h *Hub) Run() {
    for {
        select {
        case client := <-h.register:
            h.clients[client] = true
        case client := <-h.unregister:
            if _, ok := h.clients[client]; ok {
                delete(h.clients, client)
                close(client.send)
            }
        case message := <-h.broadcast:
            for client := range h.clients {
                select {
                case client.send <- message:
                default:
                    // Slow client buffer full: disconnect to protect server
                    close(client.send)
                    delete(h.clients, client)
                }
            }
        }
    }
}
```

2. The Client: Read and Write Loops with Heartbeats

```go
// client.go
package main

import (
    "log"
    "net/http"
    "time"

    "github.com/gorilla/websocket"
)

const (
    writeWait      = 10 * time.Second
    pongWait       = 60 * time.Second
    pingPeriod     = (pongWait * 9) / 10
    maxMessageSize = 512
```

```
)

type Client struct {
    hub  *Hub
    conn *websocket.Conn
    send chan []byte
}

func (c *Client) readPump() {
    defer func() {
        c.hub.unregister <- c
        c.conn.Close()
    }()

    c.conn.SetReadLimit(maxMessageSize)
    _ = c.conn.SetReadDeadline(time.Now().Add(pongWait))
    c.conn.SetPongHandler(func(string) error {
        _ = c.conn.SetReadDeadline(time.Now().Add(pongWait))
        return nil
    })

    for {
        _, message, err := c.conn.ReadMessage()
        if err != nil {
            if websocket.IsUnexpectedCloseError(err, websocket.CloseGoingAway, websocket.Clo
                log.Printf("unexpected close error: %v", err)
            }
            break
        }
        c.hub.broadcast <- message
    }
}

func (c *Client) writePump() {
    ticker := time.NewTicker(pingPeriod)
    defer func() {
        ticker.Stop()
        c.conn.Close()
    }()

    for {
        select {
        case message, ok := <-c.send:
            _ = c.conn.SetWriteDeadline(time.Now().Add(writeWait))
            if !ok {
                _ = c.conn.WriteMessage(websocket.CloseMessage, []byte{})
                return
```

```go
            }

            w, err := c.conn.NextWriter(websocket.TextMessage)
            if err != nil {
                return
            }
            _, _ = w.Write(message)
            if err := w.Close(); err != nil {
                return
            }

        case <-ticker.C:
            _ = c.conn.SetWriteDeadline(time.Now().Add(writeWait))
            if err := c.conn.WriteMessage(websocket.PingMessage, nil); err != nil {
                return
            }
        }
    }
}

func ServeWs(hub *Hub, w http.ResponseWriter, r *http.Request) {
    conn, err := upgrader.Upgrade(w, r, nil)
    if err != nil {
        log.Println(err)
        return
    }
    client := &Client{hub: hub, conn: conn, send: make(chan []byte, 256)}
    client.hub.register <- client

    go client.writePump()
    go client.readPump()
}
```

Handling Connection Heartbeats and Reconnection

WebSockets run over long-lived TCP sessions. When a user closes a laptop lid or loses mobile signal, intermediate NAT routers drop the connection entry from their translation tables without informing either endpoint.

The Ping/Pong Protocol

1. **Server Sends Ping:** Every 54 seconds, the server sends an opcode `0x9` (Ping) frame.

2. **Client Responds with Pong:** RFC 6455 requires compliant WebSocket clients (including web browsers) to automatically respond with an opcode `0xA` (Pong) frame carrying identical application data.
3. **Dead Connection Detection:** If the server does not receive a Pong within the `pongWait` window (60 seconds), the read deadline trips, unblocking `ReadMessage()` with an error and terminating the dead goroutines.

Scaling WebSocket Servers Across Multiple Instances

When your application expands beyond a single physical server, clients connected to Instance A cannot receive messages broadcast by clients connected to Instance B.

```
Client 1    [ Server Instance A ]
                                          [ Redis Pub/Sub Backplane ]
Client 2    [ Server Instance B ]
```

Implementing a Redis Pub/Sub Backplane

1. Every server instance runs its own local Hub managing local client connections.
2. When a local client publishes a message, the server publishes it to a shared Redis channel.
3. Every server instance subscribes to the Redis channel; upon receiving an event, it relays the message to all locally connected clients.
4. **Load Balancer Configuration:** Configure your reverse proxy (Nginx, Envoy, or AWS ALB) to support WebSocket upgrades (`Upgrade` and `Connection` headers) with high proxy read timeouts.

Chapter Summary

- **WebSockets** upgrade standard HTTP connections into full-duplex, framed bidirectional TCP streams.
- Framing provides **message boundaries**, eliminating the need for custom byte-level delimiters.
- The **Hub Pattern** decouples connection concurrency, preventing lock contention across clients.
- Always use **Ping/Pong heartbeats** to detect dead connections and prevent intermediate NAT firewalls from killing idle sessions.
- Scale multi-node WebSocket clusters using an external **Redis Pub/Sub backplane**.
- In the next chapter, we will dive into high-performance networking and the Go runtime netpoller.

Chapter 13: High-Performance Networking

Chapter 13: High-Performance Networking: Scaling to Millions of Connections

Achieving high-throughput, low-latency networking at scale requires understanding the intersection between Go's runtime, operating system kernel subsystems, and physical hardware.

In this chapter, you will explore the internal architecture of Go's runtime network poller, implement connection multiplexing, leverage kernel zero-copy optimizations, tune Linux kernel parameters for millions of concurrent connections (the C1000K/C10M problem), and profile network bottlenecks using `pprof`.

Epoll, Kqueue and Go's Network Poller (How Go Scales)

In traditional C networking code, engineers must write explicit event loops around system calls like `epoll_create`, `epoll_ctl`, and `epoll_wait`.

In Go, the runtime manages an internal event multiplexer called the **Runtime Netpoller**.

```
Goroutine (G) calls conn.Read()


    Non-blocking read(fd)


Data Ready?       EAGAIN / EWOULDBLOCK


Return bytes     runtime.netpollblock(pd, 'r')


                 Park G in _Gwaiting
                 Register fd with OS netpoller (epoll_ctl EPOLLIN)
                 Thread (M) continues running other Goroutines!
```

The Netpoller Execution Cycle

1. **Non-blocking by Default:** Every file descriptor created by `net.Listen` or `net.Dial` is set to `O_NONBLOCK`.
2. **Parking on `EAGAIN`:** When an application calls `conn.Read()` or `conn.Write()` and the kernel socket buffer cannot satisfy the request immediately, the system call returns `EAGAIN` or `EWOULDBLOCK`.
3. **Registration:** The Go runtime intercepts this code and parks the calling goroutine in the `_Gwaiting` state. It registers the socket descriptor with the OS kernel poller (`epoll` on Linux, `kqueue` on macOS/BSD, `IOCP` on Windows).
4. **Rescheduling:** The underlying operating system thread (`M`) does not sleep! It immediately selects another runnable goroutine from the local run queue (`P`).
5. **Waking:** During runtime scheduling ticks or sysmon background monitoring, the Go runtime calls `netpoll()`. When the kernel signals that the descriptor is ready for I/O, the parked goroutine is moved to `_Grunnable` and resumes execution right where it left off.

Connection Multiplexing Patterns

Establishing a fresh TCP connection incurs a 3-way handshake, slow-start congestion control ramps, and TLS handshake overhead. In high-performance microservices, **Connection Multiplexing** routes hundreds of independent logical request/response streams over a single shared TCP connection.

```
Logical Stream 1
Logical Stream 2     [ Stream Multiplexer ]    ( Single Physical TCP Conn )
Logical Stream 3
```

Protocols like **HTTP/2**, **HTTP/3**, and libraries like **Yamux** (`github.com/hashicorp/yamux`) encapsulate logical streams into multiplexed byte frames tagged with a `StreamID`.

```go
// Example using HashiCorp Yamux for connection multiplexing
import "github.com/hashicorp/yamux"

// Server side:
session, err := yamux.Server(rawTCPConn, nil)
for {
    stream, err := session.Accept() // Accepts a multiplexed virtual stream
    go handleVirtualStream(stream)
}

// Client side:
session, err := yamux.Client(rawTCPConn, nil)
stream1, _ := session.Open() // Opens virtual stream 1 over the single TCP conn
```

```go
stream2, _ := session.Open() // Opens virtual stream 2 over the same TCP conn
```

Zero-Copy Techniques and Memory-Mapped I/O

In standard network transfers, reading data from disk and writing it to a socket involves **four context switches and four memory copies**:

```
[ Disk ]  (DMA)  [ Kernel Page Cache ]  (CPU Copy)  [ User Buffer ]

[ NIC ]  (DMA)  [ Kernel Socket Buffer ]  (CPU Copy)
```

1. Zero-Copy with `sendfile` and `splice`

On Linux, the `sendfile` and `splice` system calls transfer data directly within kernel space between file descriptors, bypassing user-space memory entirely:

```
[ Disk ]  (DMA)  [ Kernel Page Cache ]  (Zero-Copy Pipe)  [ NIC ]
```

In Go, when you pass an `*os.File` and a `*net.TCPConn` to `io.Copy`, the standard library's internal optimizations automatically detect the types and invoke the Linux `sendfile(2)` or `splice(2)` system call!

```go
// Pure zero-copy file transfer in Go:
func sendFileZeroCopy(conn *net.TCPConn, file *os.File) (int64, error) {
    // Under Linux, io.Copy detects *net.TCPConn and *os.File,
    // automatically invoking the kernel sendfile system call!
    return io.Copy(conn, file)
}
```

Tuning the Kernel for High-Performance Networking

To sustain 100,000 to 1,000,000 concurrent sockets on a Linux host, default kernel parameters must be adjusted.

1. File Descriptor Limits

Every TCP connection requires a file descriptor. Check and raise limits:

```
# Check current limit
ulimit -n

# Set session limit to 1 million descriptors
ulimit -n 1048576

# Set system-wide limit in /etc/security/limits.conf:
* soft nofile 1048576
* hard nofile 1048576
```

2. Kernel Socket Parameters (`/etc/sysctl.conf`)

Add these production tuning directives to `/etc/sysctl.conf` and apply with `sudo sysctl -p`:

```
# Maximum number of open files system-wide
fs.file-max = 2097152

# Maximum connection backlog queue (SYN queue and Accept queue)
net.core.somaxconn = 65535
net.ipv4.tcp_max_syn_backlog = 65535

# Expand ephemeral port range for outgoing connections
net.ipv4.ip_local_port_range = 1024 65535

# Fast recycling of TIME_WAIT sockets for outgoing connections
net.ipv4.tcp_tw_reuse = 1

# Reduce FIN timeout from 60s to 15s to reclaim memory faster
net.ipv4.tcp_fin_timeout = 15

# Default and maximum socket receive/send buffer sizes (bytes)
net.core.rmem_default = 262144
net.core.rmem_max = 16777216
net.core.wmem_default = 262144
net.core.wmem_max = 16777216
net.ipv4.tcp_rmem = 4096 87380 16777216
net.ipv4.tcp_wmem = 4096 65536 16777216
```

Profiling Network Performance with pprof

When a network service experiences latency or high CPU, Go's built-in profiler (`net/http/pprof`) pinpoints the root cause.

Enabling pprof

Import `_ "net/http/pprof"` and start a diagnostic listener:

```go
package main

import (
    "log"
    "net/http"
    _ "net/http/pprof"
)

func main() {
    go func() {
        log.Println(http.ListenAndServe("localhost:6060", nil))
    }()
    // Start primary network service...
}
```

Profiling Workflows

```go
# 1. Inspect live goroutines (detect goroutine leaks)
go tool pprof http://localhost:6060/debug/pprof/goroutine

# 2. Capture a 30-second CPU profile during load test
go tool pprof http://localhost:6060/debug/pprof/profile?seconds=30

# 3. Analyze heap allocations (detect memory buffer bloat)
go tool pprof -alloc_space http://localhost:6060/debug/pprof/heap
```

Inside the pprof interactive shell, use commands like `top20`, `list <function_name>`, or generate flamegraphs using the web UI (`go tool pprof -http=:8080 profile.pb.gz`).

Chapter Summary

- Go's **Runtime Netpoller** integrates with kernel event notification systems (`epoll`/`kqueue`), allowing synchronous code to execute with asynchronous non-blocking performance.
- **Connection Multiplexing** eliminates TCP handshake latency by channeling multiple logical streams over a single shared connection.
- **`io.Copy`** automatically leverages **zero-copy kernel calls (`sendfile`, `splice`)** when transferring data between files and TCP sockets on Linux.
- Scaling beyond 100,000 connections requires tuning **`somaxconn`**, **`file-max`**, and **TCP buffer sizes** in `/etc/sysctl.conf`.

- Use `pprof` to diagnose goroutine leaks, CPU bottlenecks, and memory churn under production traffic.
- In the next chapter, we will build production infrastructure patterns including proxies, load balancers, and connection pools.

Chapter 14: Production Patterns: Proxies and Load Balancers

Chapter 14: Production Patterns: Proxies, Load Balancers and Distributed Systems

In production environments, applications rarely talk directly to a single server instance. Real-world systems sit behind proxies, load balancers, service meshes, and resilient connection pools designed to absorb failures and distribute traffic evenly.

In this chapter, you will build a bidirectional TCP proxy, implement a round-robin load balancer with active health checking, design a circuit breaker, implement client-side connection pooling, integrate observability metrics, and explore gRPC over sockets.

Building a TCP Proxy in Go

A TCP proxy forwards raw bytes bidirectionally between an inbound client connection and an upstream server. In Go, you can build a high-performance proxy using two concurrent `io.Copy` calls:

```
Client Conn        io.Copy(client, backend)       Backend Conn
                  io.Copy(backend, client)
```

```go
// proxy.go
package main

import (
    "io"
    "log"
    "net"
    "sync"
)

func ForwardTCP(clientConn net.Conn, backendAddr string) {
    defer clientConn.Close()

    backendConn, err := net.Dial("tcp", backendAddr)
    if err != nil {
        log.Printf("failed to dial backend %s: %v", backendAddr, err)
        return
```

```go
    }
    defer backendConn.Close()

    var wg sync.WaitGroup
    wg.Add(2)

    // Pipe client -> backend
    go func() {
        defer wg.Done()
        _, _ = io.Copy(backendConn, clientConn)
        // Propagate TCP FIN to backend when client finishes sending
        if tcpConn, ok := backendConn.(*net.TCPConn); ok {
            _ = tcpConn.CloseWrite()
        }
    }()

    // Pipe backend -> client
    go func() {
        defer wg.Done()
        _, _ = io.Copy(clientConn, backendConn)
        if tcpConn, ok := clientConn.(*net.TCPConn); ok {
            _ = tcpConn.CloseWrite()
        }
    }()

    wg.Wait()
}

func main() {
    ln, err := net.Listen("tcp", ":8000")
    if err != nil {
        log.Fatal(err)
    }
    defer ln.Close()

    log.Println("TCP Proxy listening on :8000 forwarding to :9000")
    for {
        client, err := ln.Accept()
        if err != nil {
            continue
        }
        go ForwardTCP(client, "127.0.0.1:9000")
    }
}
```

Implementing a Round-Robin Load Balancer

A load balancer distributes inbound connections across a pool of backend targets.

```go
package main

import (
    "errors"
    "net"
    "sync/atomic"
    "time"
)

type Backend struct {
    Addr    string
    Alive   bool
}

type LoadBalancer struct {
    backends []*Backend
    counter  uint64
}

func NewLoadBalancer(addrs []string) *LoadBalancer {
    lb := &LoadBalancer{}
    for _, addr := range addrs {
        lb.backends = append(lb.backends, &Backend{Addr: addr, Alive: true})
    }
    go lb.startHealthChecks(3 * time.Second)
    return lb
}

// NextBackend picks the next healthy backend in round-robin fashion
func (lb *LoadBalancer) NextBackend() (*Backend, error) {
    n := len(lb.backends)
    if n == 0 {
        return nil, errors.New("no backends configured")
    }

    for i := 0; i < n; i++ {
        idx := atomic.AddUint64(&lb.counter, 1) % uint64(n)
        backend := lb.backends[idx]
        if backend.Alive {
            return backend, nil
        }
    }
    return nil, errors.New("all backends are currently unhealthy")
```

```go
}

func (lb *LoadBalancer) startHealthChecks(interval time.Duration) {
    ticker := time.NewTicker(interval)
    for range ticker.C {
        for _, b := range lb.backends {
            conn, err := net.DialTimeout("tcp", b.Addr, 1*time.Second)
            if err != nil {
                b.Alive = false
            } else {
                b.Alive = true
                _ = conn.Close()
            }
        }
    }
}
```

Health Checking and Circuit Breaking

When a downstream dependency experiences an outage, continuing to pound it with requests compounds the failure. A **Circuit Breaker** isolates degraded services by temporarily halting outbound traffic.

```
                 CLOSED
                   (Normal Ops)

Success Threshold Met       Failure Count
                            Exceeds Threshold


    HALF-OPEN            OPEN                Success
   (Canary Test)     (Fast-Fail)

                              Timeout Expires
```

- **Closed:** Traffic flows normally. Error counts are tracked.
- **Open:** When errors exceed a threshold, all calls fail immediately without attempting network I/O.
- **Half-Open:** After a cooldown period, canary requests are allowed through. If they succeed, the breaker resets to Closed; if they fail, it trips back to Open.

Connection Pooling for Client-Side Efficiency

Establishing a TCP connection for every short-lived request creates excessive ephemeral port churn and latency. A client-side **Connection Pool** maintains a cache of established, idle connections ready for immediate reuse.

```go
type ConnPool struct {
    factory func() (net.Conn, error)
    conns   chan net.Conn
}

func NewConnPool(maxConns int, factory func() (net.Conn, error)) *ConnPool {
    return &ConnPool{
        factory: factory,
        conns:   make(chan net.Conn, maxConns),
    }
}

func (p *ConnPool) Get() (net.Conn, error) {
    select {
    case conn := <-p.conns:
        return conn, nil // Reused idle connection!
    default:
        return p.factory() // Pool empty, create new connection
    }
}

func (p *ConnPool) Put(conn net.Conn) {
    select {
    case p.conns <- conn:
        // Returned to idle pool
    default:
        // Pool full, close connection
        _ = conn.Close()
    }
}
```

Observability: Logging, Metrics and Tracing for Network Services

A production socket service must expose telemetry to answer three questions: 1. *How much traffic are we processing?* (Throughput: bytes/sec, connections/sec) 2. *How fast is it completing?* (Latency distributions: P50, P99) 3. *Where are failures occurring?* (Error rates, dropped connections)

Instrumenting with `net.Conn` Wrappers

You can wrap any standard `net.Conn` to transparently record metrics without changing business logic:

```go
type MeteredConn struct {
    net.Conn
    BytesRead    uint64
    BytesWritten uint64
}

func (m *MeteredConn) Read(b []byte) (int, error) {
    n, err := m.Conn.Read(b)
    m.BytesRead += uint64(n)
    return n, err
}

func (m *MeteredConn) Write(b []byte) (int, error) {
    n, err := m.Conn.Write(b)
    m.BytesWritten += uint64(n)
    return n, err
}
```

Service Discovery Patterns for Distributed Systems

In modern cloud environments, service instances are ephemeral: IP addresses change dynamically as pods scale and restart.

- **DNS-Based Discovery:** Relying on SRV records or round-robin DNS records. Simple, but subject to DNS cache TTL delays.
- **Control Plane Registries (etcd, Consul):** Instances register their IP and port upon startup with a heartbeating lease. Clients watch key prefixes using long-polling streams to update their local routing tables instantly.

---

gRPC: The Industry Standard for RPC Over Sockets

For polyglot microservice architectures, hand-crafting custom RPC protocols is often redundant. **gRPC** leverages HTTP/2 multiplexed streams, Protocol Buffers serialization, bidirectional streaming, and TLS by default:

```
service OrderService {
  rpc CreateOrder (OrderRequest) returns (OrderResponse);
  rpc StreamUpdates (StreamRequest) returns (stream UpdateEvent);
}
```

Underneath its high-level API, gRPC compiles down to the very same socket primitives, non-blocking I/O routines, and frame decoders explored in this book.

Chapter Summary

- A high-performance **TCP proxy** pipes connections bidirectionally using `io.Copy` and propagates half-close (`CloseWrite`) signals.
- **Load balancers** distribute traffic across backends using round-robin selection and background active health checks.
- **Circuit breakers** protect degraded downstreams by fast-failing traffic during outages.
- **Connection pooling** eliminates TCP handshake latency by caching idle connections for reuse.
- **`net.Conn` wrappers** transparently inject metrics, byte counting, and tracing into any socket stream.
- In the next chapter, we will learn how to test network code reliably using unit tests, mocks, and integration test harnesses.

Chapter 15: Testing Network Code

Chapter 15: Testing Network Code - Unit Tests, Integration Tests and Beyond

Writing tests for network applications is notoriously challenging. Network tests are prone to race conditions, port collisions, unpredictable timing delays, and external firewall interference. Tests that pass on a developer's workstation frequently fail or hang in continuous integration (CI) environments.

In this chapter, you will learn how to write fast, deterministic unit tests without binding real network ports using `net.Pipe`, how to mock `net.Conn` to simulate network failures and partial writes, how to structure leak-free integration tests using ephemeral ports, and how to detect race conditions with `go test -race`.

Why Testing Network Code Is Hard

Network I/O introduces variables that standard unit tests do not typically encounter:

1. **Non-Determinism:** Latency, packet reordering, and OS scheduling jitter make timing-dependent assertions brittle.
2. **Port Collisions:** Hardcoding listening ports (e.g., `:8080`) causes tests running in parallel to fail with `bind: address already in use`.
3. **Leaked Goroutines & Sockets:** Unclosed connections keep test processes running indefinitely, leading to CI timeouts.
4. **Stateful Protocols:** Testing edge cases (such as half-closed connections, corrupted checksums, and truncated frames) requires precise control over the raw byte stream.

Unit Testing with net.Pipe

For fast, isolated unit tests, Go provides `net.Pipe()` in the standard `net` package.

```go
func Pipe() (Conn, Conn)
```

`net.Pipe()` creates an in-memory, synchronous, full-duplex network connection without touching the operating system network stack or binding any port:

```
[ Client Pipe Endpoint ] <==== In-Memory Channel ====> [ Server Pipe Endpoint ]
```

Example: Testing a Protocol Handler with `net.Pipe`

```go
package main

import (
    "bytes"
    "net"
    "testing"
)

func TestEchoProtocol(t *testing.T) {
    // Create an in-memory pipe pair
    client, server := net.Pipe()
    defer client.Close()
    defer server.Close()

    // Run the server logic in a background goroutine
    go func() {
        buf := make([]byte, 1024)
        n, _ := server.Read(buf)
        _, _ = server.Write(buf[:n])
    }()

    // Perform client operations
    message := []byte("ping payload")
    if _, err := client.Write(message); err != nil {
        t.Fatalf("write failed: %v", err)
    }

    reply := make([]byte, 1024)
    n, err := client.Read(reply)
    if err != nil {
        t.Fatalf("read failed: %v", err)
    }

    if !bytes.Equal(reply[:n], message) {
        t.Errorf("got %q, want %q", reply[:n], message)
    }
}
```

[!NOTE] `net.Pipe()` reads and writes are strictly synchronous. A `Write()` on one end blocks until a `Read()` on the other end is actively consuming the bytes. Always run one side in a separate goroutine to avoid deadlocks.

Mocking net.Conn for Controlled Test Scenarios

To test how your application handles obscure error conditions—such as a connection reset (`ECONNRESET`) or unexpected read timeouts—you can implement a mock struct embedding `net.Conn`.

```go
package main

import (
    "errors"
    "net"
    "testing"
    "time"
)

// FaultyConn wraps a real net.Conn but injects errors
type FaultyConn struct {
    net.Conn
    failAfterBytes int
    bytesRead      int
}

func (f *FaultyConn) Read(b []byte) (int, error) {
    if f.bytesRead >= f.failAfterBytes {
        return 0, errors.New("simulated network connection reset by peer")
    }
    n, err := f.Conn.Read(b)
    f.bytesRead += n
    return n, err
}

func TestClientResilienceOnReset(t *testing.T) {
    client, server := net.Pipe()
    defer client.Close()
    defer server.Close()

    faulty := &FaultyConn{
        Conn:           client,
        failAfterBytes: 5, // Fails after reading first 5 bytes
    }

    go func() {
        _, _ = server.Write([]byte("1234567890"))
    }()

    buf := make([]byte, 10)
    _, err := faulty.Read(buf)
```

```go
    if err == nil {
        t.Error("expected injected network failure, but read succeeded")
    }
}
```

Integration Testing with Real Listeners

When integration tests require real kernel socket interactions, **never hardcode a port number**. Instead, bind to port `0`:

```go
ln, err := net.Listen("tcp", "127.0.0.1:0")
```

When you specify port `0`, the operating system kernel automatically assigns a free, random ephemeral port, completely eliminating port collision errors in parallel test runners!

```go
func TestRealServerIntegration(t *testing.T) {
    // Bind to ephemeral port 0
    ln, err := net.Listen("tcp", "127.0.0.1:0")
    if err != nil {
        t.Fatalf("failed to bind ephemeral listener: %v", err)
    }
    defer ln.Close()

    serverAddr := ln.Addr().String()

    // Start server accept loop
    go func() {
        conn, err := ln.Accept()
        if err != nil {
            return
        }
        defer conn.Close()
        _, _ = conn.Write([]byte("WELCOME\n"))
    }()

    // Connect client to the dynamically assigned port
    conn, err := net.Dial("tcp", serverAddr)
    if err != nil {
        t.Fatalf("dial failed: %v", err)
    }
    defer conn.Close()

    buf := make([]byte, 8)
```

```go
    n, err := conn.Read(buf)
    if err != nil || string(buf[:n]) != "WELCOME\n" {
        t.Errorf("unexpected server response: %q (err: %v)", buf[:n], err)
    }
}
```

Using golang.org/x/net/nettest

The Go team provides the `golang.org/x/net/nettest` package to verify that custom `net.Conn` or `net.PacketConn` implementations comply fully with the contract of Go's networking interfaces.

```go
import "golang.org/x/net/nettest"

func TestCustomConnCompliance(t *testing.T) {
    nettest.TestConn(t, func() (c1, c2 net.Conn, stop func(), err error) {
        c1, c2 = createMyCustomPipe()
        stop = func() {
            c1.Close()
            c2.Close()
        }
        return c1, c2, stop, nil
    })
}
```

`nettest.TestConn` automatically runs a comprehensive test battery covering deadlines, concurrent read/writes, EOF behaviors, and zero-byte reads.

Race Condition Testing Strategies

Data races in concurrent network code lead to silent memory corruption and hard-to-reproduce panics.

Always run your test suite with the race detector enabled:

```go
go test -race -v ./...
```

The race detector instruments memory accesses at compile time, reporting any instance where multiple goroutines access a memory location concurrently without proper synchronization.

Stress and Load Testing Approaches

Before deploying network services, test them under realistic load conditions:

1. **Connection Flooding:** Test server behavior when 10,000 idle TCP connections are maintained simultaneously.
2. **Traffic Spikes:** Generate high-frequency request bursts using tools like `vegeta` or custom Go harnesses using `sync.WaitGroup`.
3. **Network Chaos Testing:** Introduce artificial latency and packet drops using Linux `tc` (traffic control):

```
# Add 100ms latency and 10% packet loss to loopback interface
sudo tc qdisc add dev lo root netem delay 100ms loss 10%

# Reset traffic control rules
sudo tc qdisc del dev lo root
```

Chapter Summary

- **`net.Pipe()`** provides fast, in-memory, deterministic unit testing without allocating real network ports or descriptors.
- Struct embedding allows **mocking `net.Conn`** to simulate obscure network errors, partial writes, and timeout scenarios.
- In integration tests, bind to **port `:0`** to let the OS assign dynamic ephemeral ports, eliminating parallel test collisions.
- Use **`golang.org/x/net/nettest`** to validate interface compliance on custom connection types.
- Always validate concurrency safety using **`go test -race`**.
- In the next chapter, we will master packet capture, system call tracing, and live diagnostics to debug network problems in production.

Chapter 16: Debugging and Diagnosing Network Problems

Chapter 16: Debugging and Diagnosing Network Problems

When a distributed service slows down or drops connections in production, application logs and high-level metrics are often insufficient. Packets drop inside router queues, firewalls silently inject RST packets, and application code stalls inside un-instrumented system calls.

To diagnose deep network failures, you must look beneath the Go runtime and examine raw packets, kernel socket buffers, and system calls.

In this chapter, you will master essential network diagnostic tooling: packet capture with `tcpdump`, protocol analysis with Wireshark, system call tracing with `strace`, socket inspection with `ss`, and runtime goroutine debugging with `delve`.

When Profiling Is Not Enough

Standard profiling (`pprof`) tells you what your Go code is doing while CPU is actively running or memory is being allocated. But what if your server is completely idle, CPU usage is 0%, yet all client requests are timing out?

In these scenarios, goroutines are blocked in kernel sleep states (`_Gwaiting`), packets are being dropped by intermediate switches, or the kernel socket receive buffer is overflowing. Solving these issues requires inspecting the boundary between the Go runtime and the Linux kernel.

Packet Capture with tcpdump

`tcpdump` is the premier command-line packet analyzer for Linux and Unix systems. It captures raw frames directly from network interfaces using the kernel's packet filter (BPF).

Essential tcpdump Command Recipes

```
# Capture packets on loopback interface for port 8080 without resolving DNS
sudo tcpdump -i lo -nn port 8080

# Capture with ASCII and Hex payload inspection (-X)
sudo tcpdump -i any -nn -X -s 0 'tcp port 9090'

# Save packets to a pcap file for offline Wireshark analysis
sudo tcpdump -i eth0 -nn -w capture.pcap 'tcp port 443'

# Filter specifically for TCP RST (reset) packets
sudo tcpdump -i any 'tcp[tcpflags] & (tcp-rst) != 0'

# Filter for TCP SYN or FIN packets (connection lifecycle events)
sudo tcpdump -i any 'tcp[tcpflags] & (tcp-syn|tcp-fin) != 0'
```

Analyzing Packets with Wireshark

Wireshark provides a graphical interface for analyzing `.pcap` files captured by `tcpdump`.

```
[ Frame List: No., Time, Source, Destination, Protocol, Length, Info ]

 1  0.000000  127.0.0.1  127.0.0.1  TCP  74  54321 → 8080 [SYN]
 2  0.000042  127.0.0.1  127.0.0.1  TCP  74  8080 → 54321 [SYN,ACK]
 3  0.000065  127.0.0.1  127.0.0.1  TCP  66  54321 → 8080 [ACK]
 4  0.001200  127.0.0.1  127.0.0.1  TCP  85  54321 → 8080 [PSH,ACK]
```

Key Wireshark Workflows for Go Engineers

1. **Follow TCP Stream:** Right-click any packet and select **Follow** → **TCP Stream** to reconstruct the exact bidirectional byte conversation between client and server.
2. **Detecting Retransmissions:** In the display filter, enter `tcp.analysis.retransmission` or `tcp.analysis.lost_segment` to instantly identify packet loss and network congestion.
3. **Round-Trip Time (RTT) Graphing:** Go to **Statistics** → **TCP Stream Graphs** → **Round Trip Time** to see if latency spikes correlate with packet retransmissions.

System Call Tracing with strace

`strace` intercepts and records the system calls made by a process and the signals received by it.

Attaching strace to a Running Go Service

```
# Trace all threads (-f) for a running Go PID, filtering network calls
sudo strace -f -p <PID> -e trace=network,epoll_wait -s 128
```

Understanding strace Output

```
// Goroutine accepts an inbound connection:
[pid 12345] accept4(3, {sa_family=AF_INET, sin_port=htons(54321), sin_addr=inet_addr("127.0.

// Setting socket options:
[pid 12345] setsockopt(6, SOL_TCP, TCP_NODELAY, [1], 4) = 0

// Netpoller registers the new descriptor with epoll:
[pid 12345] epoll_ctl(4, EPOLL_CTL_ADD, 6, {events=EPOLLIN|EPOLLOUT|EPOLLRDHUP|EPOLLET, data

// Non-blocking read returning data:
[pid 12345] read(6, "GET / HTTP/1.1\r\nHost: ...", 4096) = 28
```

If you see `read(6, ...)` returning `EAGAIN` repeatedly, you know the Go runtime netpoller is properly handling non-blocking states. If you see a file descriptor remaining open without subsequent reads or closes, you have found a connection leak.

Connection State Inspection with ss

The `ss` (Socket Statistics) command replaces legacy `netstat`, querying socket details directly from the Linux kernel `tcp_diag` subsystem.

```
# Display all listening and established TCP sockets with process info:
ss -t -a -p

# Display internal TCP socket timers and memory buffers:
ss -t -i -e -a
```

Interpreting Recv-Q and Send-Q

```
State      Recv-Q Send-Q  Local Address:Port   Peer Address:Port
LISTEN     128    0       0.0.0.0:8080         0.0.0.0:*
ESTAB      0      34560   10.0.0.5:8080        198.51.100.2:45210
ESTAB      16384  0       10.0.0.5:8080        198.51.100.9:51234
```

- **For Listening Sockets:**
  - `Send-Q`: The maximum backlog queue size (`somaxconn`).
  - `Recv-Q`: The number of completed connections currently waiting to be accepted by `ln.Accept()`. **If `Recv-Q > 0` on a listening socket, your application's accept loop is falling behind!**
- **For Established Sockets:**
  - `Send-Q`: Bytes buffered in kernel memory waiting to be sent or acknowledged by peer. High values indicate a congested network or stalled client.
  - `Recv-Q`: Bytes received by kernel but not yet read by application via `conn.Read()`. High values indicate your Go worker goroutine is bottlenecked or stuck.

Debugging Goroutine Hangs with delve

When your Go application freezes, attach the **Delve (`dlv`)** debugger to inspect all live goroutines without restarting the process.

```
# Attach delve to the running process
sudo dlv attach <PID>
```

Delve Commands

```
# List all goroutines and their current wait state
(dlv) goroutines

# Inspect a specific stuck goroutine
(dlv) goroutine 42

# View stack trace
(dlv) stack
```

Look for goroutines stuck in `runtime.netpollblock` or `internal/poll.(*FD).Read`. If a goroutine has been in that state for hours, you have identified a missing read deadline!

Real-World Debugging Walkthroughs

Case 1: The Accumulating `CLOSE_WAIT` Sockets

Symptom: Running `ss -s` reveals 15,000 sockets in `CLOSE_WAIT` state, eventually exhausting file descriptors. **Root Cause:** `CLOSE_WAIT` means the remote client has sent a `FIN` (closed their side), but the local Go application has **never called `conn.Close()`**. **Fix:** Search the codebase for early return statements that bypass `defer conn.Close()`.

Case 2: Sudden `Connection Reset by Peer` (RST)

Symptom: Clients report intermittent `read: connection reset by peer` errors. **Root Cause Captured in tcpdump:** The Go server called `conn.Close()` while there was still unread data sitting in the socket's kernel receive buffer. By default, closing a socket with unread data causes the Linux TCP stack to immediately send an abortive `RST` instead of a graceful `FIN`. **Fix:** Ensure all client input is drained before terminating the connection, or use `CloseWrite()` to signal half-close.

Chapter Summary

- When high-level metrics fail, inspect kernel sockets and raw frames using **`tcpdump`**, **`ss`**, **`strace`**, and **`delve`**.
- **`tcpdump -i any -nn`** captures raw packets with BPF filters for offline analysis in **Wireshark**.
- **`strace -f -e trace=network`** reveals exact non-blocking system calls, `epoll` registrations, and file descriptor leaks.
- On listening sockets, a non-zero **`Recv-Q` in `ss`** proves the application accept loop cannot keep pace with inbound connection arrival rates.
- Sockets stuck in **`CLOSE_WAIT`** indicate the local application forgot to call `conn.Close()`.
- In the final chapter, we will synthesize our learnings into the guiding principles of reliable network programming in Go.

Conclusion: The Art of Reliable Network Programming

Conclusion: The Art of Reliable Network Programming

Over the course of this section, we have journeyed from the physical reality of packets, routers, and kernel sockets to high-performance distributed systems engineering in Go.

Network programming is fundamentally an exercise in managing failure. Networks are inherently unreliable, latency is non-zero, bandwidth is finite, and remote peers will crash, disconnect, or act maliciously. The art of reliable network programming lies not in hoping failures never happen, but in designing architectures that absorb them gracefully without service disruption.

Guiding Principles

As you build production network systems in Go, carry these core principles into your architectural decisions:

1. **Every Network Operation Must Have a Deadline:** Never call `Dial`, `Read`, or `Write` without an absolute deadline or context timeout. A socket call without a deadline is a future production outage waiting for an anomalous network partition.
2. **Defend Against Resource Exhaustion:** Always bound queues, rate-limit accepted connections, enforce maximum payload sizes, and shed load when memory or CPU thresholds are exceeded. Unbounded buffers inevitably lead to out-of-memory crashes.
3. **Respect the Byte Stream Nature of TCP:** Never assume that one write equals one read. Always implement strict, defensive framing—whether length-prefixed or delimiter-based—with integrity verification.
4. **Clean Up What You Open:** Guard every accepted connection with `defer conn.Close()`. Track in-flight goroutines with `sync.WaitGroup` to ensure clean, zero-downtime graceful shutdowns.
5. **Recycle High-Frequency Allocations:** Under high connection volumes, garbage collection pressure degrades tail latency. Leverage `sync.Pool` for byte buffers and reuse connection state where appropriate.
6. **Embrace Observability from Day One:** Expose connection counters, latency percentiles, and error histograms. When mysterious disconnects arise, do not guess: attach `tcpdump`, inspect `ss`, trace with `strace`, and verify with Wireshark.

The Go Advantage

Go provides a uniquely powerful foundation for network engineering:

```
Simplicity of Synchronous Code (Straight-line conn.Read())
                            +
Efficiency of Asynchronous Event Loops (Runtime Netpoller)
                            +
Ultra-Lightweight User-Space Threads (2KB Goroutines)
                            +
Safe, Modern Memory Management
                            =
Unmatched Developer Productivity & Scalability
```

By decoupling application logic from operating system thread management, Go allows small engineering teams to build software that scales to millions of concurrent sessions—systems that previously required complex, brittle C codebases.

What Comes Next

With the foundations of socket programming mastered, you are prepared to explore cutting-edge developments in network systems engineering:

- **QUIC and HTTP/3:** Explore user-space transport protocols built over UDP that eliminate TCP head-of-line blocking and integrate TLS 1.3 natively into the handshake (check out `github.com/quic-go/quic-go`).
- **Kernel Bypass and eBPF:** Investigate Extended Berkeley Packet Filters (eBPF) and XDP (eXpress Data Path) using libraries like `cilium/ebpf` to process network packets directly inside the Linux kernel at multi-gigabit wire speed.
- **Asynchronous I/O with io_uring:** Study Linux `io_uring` interfaces for true asynchronous kernel-space ring buffers, unlocking zero-overhead batch system calls.

Final Thought

A socket is simply a window between your code and the outside world. Behind every HTTP request, cloud storage upload, and message broker queue is a stream of bytes traveling across copper, glass, and radio waves.

When you master sockets, you stop viewing distributed infrastructure as an opaque black box. You understand the tradeoffs, you can anticipate failure modes, and you possess the capability to build resilient, blazing-fast network software that stands the test of production scale.

Go forth and build systems that never stop communicating.

Network Systems

Network Systems Overview

Network Systems Overview

This part treats networking in Go as a **systems** discipline: connections, budgets, failure domains, and operability — not merely how to call `http.Get`.

If earlier chapters taught you the standard library surface, this part teaches how network software behaves under load, packet loss, slow clients, and partial dependency failure.

Why This Part Exists

Most production outages involving Go services are not "Go is slow." They are:

- Unbounded concurrency on accept/read paths
- Missing or inconsistent timeouts
- Retry storms that amplify a partial outage
- Proxies that hide backend health
- Protocol bugs (especially custom TCP framing)

You need a mental model that spans the stack:

```
client process
    |
    v
TCP/TLS session  ---- deadlines, keepalive, congestion
    |
    v
application protocol  ---- framing / HTTP semantics
    |
    v
service policy  ---- auth, limits, routing, retries
    |
    v
upstream dependency  ---- its own budgets and failures
```

Learning Path

Chapter	Focus	You will be able to
131 TCP services	Byte streams, framing, conn lifecycle	Build a correct custom TCP protocol handler
132 HTTP resilience	Timeout budgets, client/server guardrails	Design end-to-end latency budgets without cascade
133 Reverse proxy	Routing, upstream selection, blast radius	Place reliability policy at the edge
134 TLS clients/servers	`crypto/tls`, mTLS sketch	Serve and dial HTTPS safely
135 DNS patterns	Resolver, timeouts, dual-stack	Debug name→dial failures
136 HTTP Transport	Pools, limits, keep-alives	Tune high fan-out clients
137 Project: netcheck	dial/dns/tls/http toolkit	Ship a multi-check ops binary

Core Concepts Map

1. Streams vs messages

TCP delivers an ordered byte stream. Message boundaries are your problem. HTTP is a message protocol layered on TCP (and now HTTP/2/3 multiplex differently). Conflating stream and message is the root of many TCP bugs.

2. Deadlines are load shedding

Every `Read`/`Write` without a deadline is a potential goroutine leak under slowloris-style behavior. Deadlines are not only "correctness"; they are capacity protection.

3. Budgets flow downward

An inbound request with a 1.2s budget cannot safely give every dependency a fresh 1.2s. Propagate remaining time via `context.Context` and refuse work you cannot finish.

```
total budget: 1200ms
  auth     50ms
  cache   100ms
  db      400ms
  upstream 500ms  (must use remaining, not full 1200)
  encode   50ms
```

```
  slack   100ms
```

4. Proxies are control planes

A reverse proxy is not "just nginx in front." In Go (or any edge), it is where you centralize routing, timeouts, identity headers, and failure isolation so every backend need not reimplement policy.

Go-Centric Building Blocks

You will use these packages heavily:

Package	Role
`net`	Listeners, dialers, TCP conn deadlines
`net/http`	Servers, clients, transports, reverse proxy helpers
`context`	Cancellation and deadline propagation
`io` / `bufio`	Framing, limited readers, efficient IO
`sync` / `golang.org/x/sync/errgroup`	Bounded fan-out, lifecycle
`crypto/tls`	Secure transport (deep dive in part 22)
`log/slog`	Structured operational logs

Idiom: always bound the server

```go
srv := &http.Server{
    Addr:              ":8080",
    Handler:           mux,
    ReadHeaderTimeout: 5 * time.Second,
    ReadTimeout:       15 * time.Second,
    WriteTimeout:      30 * time.Second,
    IdleTimeout:       60 * time.Second,
}
```

Idiom: always bound the client

```go
client := &http.Client{
    Timeout: 10 * time.Second,
    Transport: &http.Transport{
        Proxy:                 http.ProxyFromEnvironment,
        DialContext:           (&net.Dialer{Timeout: 3 * time.Second}).DialContext,
        MaxIdleConns:          100,
        IdleConnTimeout:       90 * time.Second,
        TLSHandshakeTimeout:   5 * time.Second,
```

```
        ExpectContinueTimeout: 1 * time.Second,
        ForceAttemptHTTP2:     true,
    },
}
```

Defaults without timeouts are the most common production footgun in Go HTTP code.

Production Mindset

Before writing a network feature, answer:

1. **What is the unit of work?** (TCP frame, HTTP request, gRPC RPC)
2. **What is the timeout budget?** (client SLA → service → dependencies)
3. **What fails?** (timeout, cancel, protocol error, overload)
4. **What do we do on failure?** (fail fast, retry once, degrade, shed load)
5. **What do we observe?** (accepts, active conns, latency histograms, error classes)

If you cannot answer these, the implementation will invent answers under fire — usually the wrong ones.

Relationship to Other Parts

- **Part 10/11 (concurrency)** — worker pools, context, pipelines feed network handler design.
- **Part 15/16 (web)** — higher-level HTTP/API frameworks sit on the resilience patterns here.
- **Part 18 (distributed infra)** — retries, circuit breakers, and queues build on timeout budgets.
- **Part 19 (observability)** — correlation and metrics make network failures diagnosable.
- **Part 22 (security)** — TLS/mTLS protect the same sockets you harden for availability.

How to Study These Chapters

1. Read 131 with a scratch TCP echo/framed server; break it deliberately (partial reads, huge frames).
2. Read 132 while instrumenting a real `http.Server` and `http.Client` under `hey` or `vegeta`.
3. Read 133 by building a tiny reverse proxy with health-aware round-robin and watching failover.

Prefer one solid lab over ten passive reads. Network bugs only become intuitive when you have watched a stuck `Read` hold a goroutine for minutes.

Checklist: Network Systems Literacy

After this part you should confidently:

- ☐ Explain why TCP needs framing
- ☐ Set and reason about deadlines on both ends of a connection
- ☐ Draw a timeout budget for a multi-dependency request
- ☐ Configure `http.Server` and `http.Transport` for production
- ☐ Classify retryable vs non-retryable HTTP failures
- ☐ Describe how a reverse proxy reduces blast radius
- ☐ Name the metrics you would export for a TCP and HTTP service

Exercises (Part Warm-Up)

1. List every network listener in a service you know; for each, write down timeouts or mark "missing."
2. Capture a pcap or use `ss`/`netstat` during a local load test; correlate ESTABLISHED count with goroutine count.
3. Sketch the failure modes of "client timeout 30s, server no timeout, dependency timeout 60s."
4. Write a one-page design note: where should retries live (client, proxy, both)? Defend one answer.
5. Identify whether your public API needs custom TCP or whether HTTP/gRPC already provides your framing.

Failure Taxonomy (Shared Vocabulary)

Use the same words across the next three chapters so metrics and runbooks align:

Class	Examples	Typical response
`timeout`	deadline exceeded, slowloris	fail fast / shed; fix budgets
`cancel`	client gone, parent cancel	stop work; do not retry
`protocol`	bad frame, HTTP 400	close or 4xx; no retry
`overload`	semaphore full, 503	backpressure; Retry-After
`dep_down`	connect refused, 502/503	retry with policy; breaker
`dep_slow`	high latency within budget	optional degrade; not always retry

When an incident starts, classify the failure before changing code. Many "network is broken" tickets are mis-set budgets.

Lab Setup Recommendation

A single laptop is enough for this part:

```
# Terminal A: your experimental server
go run ./cmd/labserve

# Terminal B: load
hey -n 5000 -c 50 http://127.0.0.1:8080/api

# Terminal C: profiles (admin port)
go tool pprof -http=:18081 http://127.0.0.1:6060/debug/pprof/profile?seconds=15
```

Capture goroutine counts (`/debug/pprof/goroutine?debug=1`) before and after introducing missing timeouts — the difference is the lesson.

Anti-Patterns to Unlearn

1. **`http.DefaultClient` in libraries** — no timeout, shared by the whole process.
2. **`go handle(conn)` without a bound** — connection floods become memory floods.
3. **Retries at every layer** — browser × edge × app × client SDK.
4. **Health checks that only dial TCP** — process up, app not ready.
5. **Logging full request bodies** on the edge — cost and PII.

Glossary

- **Deadline** — absolute time when an IO op must fail.
- **Timeout budget** — remaining time for a request tree.
- **Framing** — how messages are delimited on a byte stream.
- **Keep-alive** — reusing connections; needs idle limits.
- **Load shedding** — refusing work to protect capacity.
- **Blast radius** — how far a failure propagates.

More examples

Tour: dial timeout + httptest client budget

One mini-program that touches the part's themes: bounded dial, HTTP round-trip with context, and fail-fast on deadline.

```
mkdir -p /tmp/go-net-tour && cd /tmp/go-net-tour
go mod init example.com/net-tour
```

Save as `main.go`:

```go
package main

import (
    "context"
    "fmt"
    "io"
    "net"
    "net/http"
    "net/http/httptest"
    "time"
)

func main() {
    // 1) Dial with timeout (refused port → quick error, not hang)
    d := net.Dialer{Timeout: 100 * time.Millisecond}
    _, err := d.Dial("tcp", "127.0.0.1:1")
    fmt.Println("dial fail fast:", err != nil)

    // 2) Server + client with request context budget
    ts := httptest.NewServer(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        fmt.Fprintln(w, "pong")
    }))
    defer ts.Close()

    ctx, cancel := context.WithTimeout(context.Background(), time.Second)
    defer cancel()
    req, _ := http.NewRequestWithContext(ctx, http.MethodGet, ts.URL, nil)
    resp, err := http.DefaultClient.Do(req)
    if err != nil {
        panic(err)
    }
    b, _ := io.ReadAll(resp.Body)
    resp.Body.Close()
    fmt.Print("http: ", string(b))
}
```

```go
go run .
```

Expected output:

```
dial fail fast: true
http: pong
```

Runnable example

Tour of this part: a tiny HTTP server and client with production-shaped timeouts—the shared foundation for TCP framing, HTTP resilience, and reverse proxies.

```
mkdir -p /tmp/go-net-tour && cd /tmp/go-net-tour
go mod init example.com/net-tour
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "io"
    "log"
    "net"
    "net/http"
    "time"
)

func main() {
    mux := http.NewServeMux()
    mux.HandleFunc("GET /api", func(w http.ResponseWriter, r *http.Request) {
        fmt.Fprintln(w, "pong")
    })

    ln, err := net.Listen("tcp", "127.0.0.1:0")
    if err != nil {
        log.Fatal(err)
    }
    srv := &http.Server{
        Handler:           mux,
        ReadHeaderTimeout: 5 * time.Second,
        ReadTimeout:       10 * time.Second,
        WriteTimeout:      10 * time.Second,
        IdleTimeout:       30 * time.Second,
    }
    go func() { _ = srv.Serve(ln) }()

    client := &http.Client{
        Timeout: 3 * time.Second,
        Transport: &http.Transport{
            DialContext:         (&net.Dialer{Timeout: time.Second}).DialContext,
            TLSHandshakeTimeout: 2 * time.Second,
            MaxIdleConns:        10,
            IdleConnTimeout:     30 * time.Second,
```

```go
        },
    }

    url := "http://" + ln.Addr().String() + "/api"
    resp, err := client.Get(url)
    if err != nil {
        log.Fatal(err)
    }
    body, _ := io.ReadAll(resp.Body)
    resp.Body.Close()
    fmt.Printf("status=%d body=%q\n", resp.StatusCode, string(body))
    fmt.Println("tour: timeouts set on server and client")
    _ = srv.Close()
}
```

```go
go run .
```

Expected output:

```
status=200 body="pong\n"
tour: timeouts set on server and client
```

What to notice

- Default clients and servers without timeouts are the #1 production footgun.
- This part stacks TCP framing (131), budgeted HTTP (132), and edge policy (133) on these primitives.

Try next

- Remove `client.Timeout` and point at a black-hole address; restore the timeout.
- Read chapter 131 and implement length-prefix framing on a raw TCP conn.

Next Chapter

Start with **TCP Services Deep Dive** — the lowest layer of the model, and the place most custom-protocol bugs hide.

TCP Services Deep Dive

TCP Services Deep Dive

TCP is a **byte stream**, not a message protocol. Most production bugs in custom TCP systems come from treating partial reads as complete messages, or unbounded buffers as "simpler."

This chapter builds a correct, production-shaped TCP service in Go: framing, deadlines, concurrency limits, graceful shutdown, and observability.

Why Custom TCP Still Matters

HTTP and gRPC cover most APIs. You still touch raw TCP when:

- Building binary protocols (game servers, market data, device telemetry)
- Speaking legacy line protocols (SMTP-ish, Redis RESP-like, custom IoT)
- Implementing proxies, tunnels, or multiplexers
- Teaching yourself how higher protocols actually work

If you only ever use frameworks, this chapter still upgrades your debugging of "connection stuck" incidents.

Core Mental Model

```
application messages:  |  msg A  |  msg B  |  msg C  |
                       \_______/  \______/
TCP may deliver:       [A1][A2+B1][B2][C1][C2][C3]...
```

Rules:

1. One `Read` can return any number of bytes 1 (or 0 with error/EOF).
2. One `Write` may write fewer bytes than you asked; loop until done or use helpers.
3. Message boundaries are **application-defined**.

Framing Strategies

Strategy	Shape	Pros	Cons
Length-prefix	`[u32 len][payload]`	Simple, binary-friendly	Need max-size cap

Strategy	Shape	Pros	Cons
Delimiter	lines / null-terminated	Human debuggable	Escaping, binary pain
Fixed size	always N bytes	Trivial parse	Wasteful / inflexible
Self-describing	protobuf-style	Rich	Heavier stack

Length-prefix (recommended baseline)

```
 0      4           4+N
 | len  |  payload  |
   u32 BE   N bytes
```

Always enforce `N <= MaxMessage`.

```go
package frame

import (
    "encoding/binary"
    "fmt"
    "io"
)

const MaxMessage = 1 << 20 // 1 MiB

func WriteFrame(w io.Writer, payload []byte) error {
    if len(payload) > MaxMessage {
        return fmt.Errorf("payload too large: %d", len(payload))
    }
    var hdr [4]byte
    binary.BigEndian.PutUint32(hdr[:], uint32(len(payload)))
    if _, err := w.Write(hdr[:]); err != nil {
        return err
    }
    _, err := w.Write(payload)
    return err
}

func ReadFrame(r io.Reader) ([]byte, error) {
    var hdr [4]byte
    if _, err := io.ReadFull(r, hdr[:]); err != nil {
        return nil, err
    }
    n := binary.BigEndian.Uint32(hdr[:])
    if n > MaxMessage {
```

```go
        return nil, fmt.Errorf("frame %d exceeds max %d", n, MaxMessage)
    }
    buf := make([]byte, n)
    if _, err := io.ReadFull(r, buf); err != nil {
        return nil, err
    }
    return buf, nil
}
```

`io.ReadFull` is the correct primitive: it loops until the buffer is full or an error occurs.

Deadlines and Idle Connections

Without deadlines, a client that stops sending after the length header leaves your handler blocked forever.

```go
func handleConn(conn net.Conn) {
    defer conn.Close()
    for {
        // Per-read idle deadline (sliding).
        _ = conn.SetReadDeadline(time.Now().Add(30 * time.Second))
        msg, err := frame.ReadFrame(conn)
        if err != nil {
            return // timeout, EOF, protocol error
        }
        resp := process(msg)
        _ = conn.SetWriteDeadline(time.Now().Add(10 * time.Second))
        if err := frame.WriteFrame(conn, resp); err != nil {
            return
        }
    }
}
```

Patterns:

- **Idle timeout**: refresh deadline before each read (above).
- **Absolute request timeout**: set once at accept for request-response protocols.
- **Separate read/write**: long uploads vs quick control messages need different policies.

Accept Loop with Backpressure

Unbounded `go handle(conn)` is a classic self-DoS under connection floods.

```go
package tcpsvc

import (
    "context"
    "log/slog"
    "net"
    "sync"
    "time"
)

type Server struct {
    Addr     string
    Sem      chan struct{} // capacity = max concurrent handlers
    Log      *slog.Logger
    Handler  func(context.Context, net.Conn)
}

func (s *Server) ListenAndServe(ctx context.Context) error {
    ln, err := net.Listen("tcp", s.Addr)
    if err != nil {
        return err
    }
    defer ln.Close()

    go func() {
        <-ctx.Done()
        _ = ln.Close() // unblock Accept
    }()

    var wg sync.WaitGroup
    for {
        conn, err := ln.Accept()
        if err != nil {
            select {
            case <-ctx.Done():
                wg.Wait()
                return ctx.Err()
            default:
                s.Log.Error("accept", "err", err)
                continue
            }
        }

        select {
        case s.Sem <- struct{}{}:
            // acquired slot
        default:
```

```go
            // overload: fail fast
            _ = conn.Close()
            s.Log.Warn("reject connection: at capacity")
            continue
        }

        wg.Add(1)
        go func(c net.Conn) {
            defer wg.Done()
            defer func() { <-s.Sem }()
            defer c.Close()
            s.Handler(ctx, c)
        }(conn)
    }
}
```

Production refinements:

- Accept a connection then **read a bit** before committing a heavy worker (cheap auth / magic bytes).
- Use `ListenConfig` with `Control` to set `SO_REUSEADDR` / socket options when needed.
- Prefer closing the listener on shutdown, then waiting for in-flight handlers with a deadline.

IO vs Business Logic Separation

Slow CPU work on the same goroutine that reads the socket prevents reading the next frames and can stall peers.

```
read loop (per conn) -> jobs channel -> worker pool -> write responses
```

For request/response with ordering requirements, keep per-connection serialization of writes (one writer goroutine or a mutex on write).

```go
type session struct {
    conn net.Conn
    out  chan []byte
}

func (s *session) writer(ctx context.Context) {
    for {
        select {
        case <-ctx.Done():
            return
        case msg, ok := <-s.out:
            if !ok {
                return
```

```
            }
            _ = s.conn.SetWriteDeadline(time.Now().Add(10 * time.Second))
            if err := frame.WriteFrame(s.conn, msg); err != nil {
                return
            }
        }
    }
}
```

Protocol Errors vs Transport Errors

Classify failures:

Class	Examples	Action
Transport	EOF, reset, timeout	Close conn; metric `conn_errors`
Protocol	bad length, unknown type	Close or send error frame; metric `proto_errors`
App	business rejection	Error response frame; keep conn if protocol allows

Do not retry protocol errors on the same connection without a reset of parser state — your stream may be desynchronized.

Keepalives and Half-Open Connections

TCP keepalives detect dead peers, but OS defaults are often too long (hours). For application liveness, prefer **application pings** with your own deadlines.

```
// Application-level heartbeat every 15s; miss 2 → close.
```

Also handle half-close: peer may `CloseWrite` while you still flush responses. Know your protocol's rules for FIN.

TLS on TCP

Wrap accepted conns:

```
tlsLn := tls.NewListener(ln, tlsConfig)
// Accept from tlsLn as usual; deadlines still apply on the underlying net.Conn.
```

Certificate lifecycle is covered in part 22; from a TCP perspective, handshake timeouts matter:

```go
_ = conn.SetDeadline(time.Now().Add(5 * time.Second))
tlsConn := tls.Server(conn, cfg)
if err := tlsConn.HandshakeContext(ctx); err != nil {
    return err
}
_ = tlsConn.SetDeadline(time.Time{}) // clear; use per-op deadlines later
```

Observability

Minimum metrics for a TCP service:

- `tcp_accepts_total`
- `tcp_active_conns`
- `tcp_rejected_total{reason="capacity|protocol"}`
- `tcp_read_timeouts_total`
- `tcp_frame_bytes` (histogram)
- `tcp_handler_seconds` (histogram)

Log fields: `remote_addr`, `conn_id`, `frame_type`, `err_class`, `duration_ms`.

Production Checklist

- ☐ Explicit framing with max message size
- ☐ `SetReadDeadline` / `SetWriteDeadline` on every path
- ☐ Bounded concurrent handlers (semaphore / worker pool)
- ☐ Reject or shed on overload instead of unbounded `go`
- ☐ Graceful shutdown: stop accept → drain → force close after deadline
- ☐ Metrics for accepts, active, errors, timeouts
- ☐ Fuzz or property-test the frame parser
- ☐ Document idle timeout and max frame in the protocol spec

Common Pitfalls

1. **`ioutil.ReadAll` on a conn** — unbounded memory.
2. **Assuming one `Read` = one message**.
3. **Forgetting `Write` can be partial** — use `frame.Write` loops or `io.Copy` patterns carefully.
4. **Sharing one deadline for the entire connection lifetime** — long sessions need sliding idle deadlines.
5. **Logging every byte** — CPU and PII risk.
6. **No max concurrent conns** — memory death under SYN floods / connection storms.
7. **Mixing buffered readers incorrectly** — if you `bufio.Reader` wrap a conn, never read the conn underneath or you desync.

Complete Mini Server Skeleton

```go
func main() {
    ctx, stop := signal.NotifyContext(context.Background(), os.Interrupt, syscall.SIGTERM)
    defer stop()

    s := &tcpsvc.Server{
        Addr: ":9000",
        Sem:  make(chan struct{}, 256),
        Log:  slog.Default(),
        Handler: func(ctx context.Context, c net.Conn) {
            for {
                select {
                case <-ctx.Done():
                    return
                default:
                }
                _ = c.SetReadDeadline(time.Now().Add(30 * time.Second))
                msg, err := frame.ReadFrame(c)
                if err != nil {
                    return
                }
                // Echo for demo; replace with real handler.
                _ = c.SetWriteDeadline(time.Now().Add(10 * time.Second))
                _ = frame.WriteFrame(c, msg)
            }
        },
    }
    if err := s.ListenAndServe(ctx); err != nil && !errors.Is(err, context.Canceled) {
        slog.Error("server exit", "err", err)
        os.Exit(1)
    }
}
```

Exercises

1. Implement length-prefix client/server echo; verify multi-message reads with a client that writes two frames in one `Write`.
2. Send a frame claiming `len = 2GiB`; ensure the server rejects without allocating.
3. Remove read deadlines and connect with `nc` without sending data; watch goroutine count with `pprof`.
4. Cap handlers at 2; open 10 connections; confirm extras are closed immediately.
5. Add a simple opcode byte after the length; fuzz the parser with `go test -fuzz`.
6. Implement graceful shutdown: in-flight echo must finish within 5s; then force close.
7. Benchmark frames of 64B vs 64KiB; record allocs/op for `ReadFrame`.

8. Wrap the server in TLS with a self-signed cert; measure handshake timeout behavior.
9. Deliberately use `bufio.Reader` for reads and raw `conn` for a side read — observe desync, then fix.
10. Export Prometheus metrics for active conns and timeouts; load-test and graph them.

More examples

Length-prefix TCP session (client + server)

```
mkdir -p /tmp/go-tcp-lp && cd /tmp/go-tcp-lp
go mod init example.com/tcp-lp
```

Save as `main.go`:

```go
package main

import (
    "encoding/binary"
    "fmt"
    "io"
    "net"
    "time"
)

func writeMsg(w io.Writer, s string) error {
    var h [4]byte
    binary.BigEndian.PutUint32(h[:], uint32(len(s)))
    if _, err := w.Write(h[:]); err != nil {
        return err
    }
    _, err := io.WriteString(w, s)
    return err
}

func readMsg(r io.Reader) (string, error) {
    var h [4]byte
    if _, err := io.ReadFull(r, h[:]); err != nil {
        return "", err
    }
    n := binary.BigEndian.Uint32(h[:])
    buf := make([]byte, n)
    if _, err := io.ReadFull(r, buf); err != nil {
        return "", err
    }
    return string(buf), nil
```

```go
}

func main() {
    ln, err := net.Listen("tcp", "127.0.0.1:0")
    if err != nil {
        panic(err)
    }
    defer ln.Close()

    done := make(chan struct{})
    go func() {
        defer close(done)
        c, err := ln.Accept()
        if err != nil {
            return
        }
        defer c.Close()
        _ = c.SetDeadline(time.Now().Add(time.Second))
        msg, err := readMsg(c)
        if err != nil {
            return
        }
        _ = writeMsg(c, "echo:"+msg)
    }()

    c, err := net.DialTimeout("tcp", ln.Addr().String(), time.Second)
    if err != nil {
        panic(err)
    }
    defer c.Close()
    _ = c.SetDeadline(time.Now().Add(time.Second))
    _ = writeMsg(c, "hello")
    out, err := readMsg(c)
    if err != nil {
        panic(err)
    }
    fmt.Println(out)
    <-done
}
```

```go
go run .
```

Expected output:

```
echo:hello
```

Per-connection deadlines

```
mkdir -p /tmp/go-tcp-deadline && cd /tmp/go-tcp-deadline
go mod init example.com/tcp-deadline
```

Save as `main.go`:

```go
package main

import (
    "fmt"
    "net"
    "time"
)

func main() {
    ln, _ := net.Listen("tcp", "127.0.0.1:0")
    defer ln.Close()
    go func() {
        c, _ := ln.Accept()
        defer c.Close()
        time.Sleep(50 * time.Millisecond) // slow peer
    }()

    c, _ := net.Dial("tcp", ln.Addr().String())
    defer c.Close()
    _ = c.SetReadDeadline(time.Now().Add(10 * time.Millisecond))
    buf := make([]byte, 8)
    _, err := c.Read(buf)
    fmt.Println("timeout:", err != nil)
}
```

```
go run .
```

Expected output:

```
timeout: true
```

Runnable example

Length-prefix framing with max size, deadlines, and a single-process client/server echo—the core TCP service pattern.

```
mkdir -p /tmp/go-tcp-frame && cd /tmp/go-tcp-frame
go mod init example.com/tcp-frame
```

Save as `main.go`:

```
package main

import (
    "encoding/binary"
    "fmt"
    "io"
    "log"
    "net"
    "time"
)

const maxMsg = 64 << 10 // 64 KiB

func writeFrame(w io.Writer, payload []byte) error {
    if len(payload) > maxMsg {
        return fmt.Errorf("message too large")
    }
    var hdr [4]byte
    binary.BigEndian.PutUint32(hdr[:], uint32(len(payload)))
    if _, err := w.Write(hdr[:]); err != nil {
        return err
    }
    _, err := w.Write(payload)
    return err
}

func readFrame(r io.Reader) ([]byte, error) {
    var hdr [4]byte
    if _, err := io.ReadFull(r, hdr[:]); err != nil {
        return nil, err
    }
    n := binary.BigEndian.Uint32(hdr[:])
    if n > maxMsg {
        return nil, fmt.Errorf("frame %d exceeds max %d", n, maxMsg)
    }
    buf := make([]byte, n)
    if _, err := io.ReadFull(r, buf); err != nil {
        return nil, err
    }
    return buf, nil
}

func main() {
    ln, err := net.Listen("tcp", "127.0.0.1:0")
    if err != nil {
```

```
    log.Fatal(err)
}
defer ln.Close()

errCh := make(chan error, 1)
go func() {
    c, err := ln.Accept()
    if err != nil {
        errCh <- err
        return
    }
    defer c.Close()
    _ = c.SetDeadline(time.Now().Add(5 * time.Second))
    msg, err := readFrame(c)
    if err != nil {
        errCh <- err
        return
    }
    errCh <- writeFrame(c, append([]byte("echo:"), msg...))
}()

conn, err := net.DialTimeout("tcp", ln.Addr().String(), time.Second)
if err != nil {
    log.Fatal(err)
}
defer conn.Close()
_ = conn.SetDeadline(time.Now().Add(5 * time.Second))

if err := writeFrame(conn, []byte("hello-tcp")); err != nil {
    log.Fatal(err)
}
resp, err := readFrame(conn)
if err != nil {
    log.Fatal(err)
}
fmt.Printf("response: %q\n", string(resp))

// Oversized frame rejected
conn2, err := net.Dial("tcp", ln.Addr().String())
if err == nil {
    conn2.Close()
}

// Direct unit-style check of max enforcement
var bad [4]byte
binary.BigEndian.PutUint32(bad[:], uint32(maxMsg)+1)
_, err = readFrame(&limitedReader{b: bad[:]})
```

```go
    fmt.Println("oversize rejected:", err != nil)

    if err := <-errCh; err != nil {
        // server may error after client closed second attempt; ignore if already echoed
        _ = err
    }
    fmt.Println("tcp framing ok")
}

// limitedReader feeds only the header for the oversize demo.
type limitedReader struct{ b []byte }

func (l *limitedReader) Read(p []byte) (int, error) {
    if len(l.b) == 0 {
        return 0, io.EOF
    }
    n := copy(p, l.b)
    l.b = l.b[n:]
    return n, nil
}
```

```go
go run .
```

Expected output:

```
response: "echo:hello-tcp"
oversize rejected: true
tcp framing ok
```

What to notice

- `io.ReadFull` is required; one `Read` is never a full frame.
- Cap `n` before allocating—malicious length prefixes are a memory DoS.
- Deadlines on the conn protect against silent peers.

Try next

- Send two frames in one `Write` and read both on the server loop.
- Cap concurrent handlers with a buffered channel semaphore.

Further Reading

- `man 7 tcp`, `man 2 setsockopt` (keepalive, linger)
- Go blog: deadlines, netpoller behavior
- Next: **HTTP Client and Server Resilience** — budgets and policy on top of streams
- Part 22: TLS/mTLS for securing these sockets

HTTP Client and Server Resilience

HTTP Client and Server Resilience

Resilience in HTTP systems is **budget management**: every inbound request has limited time, and each downstream call spends part of that budget. When budgets are implicit, partial outages become full outages.

Why / Overview

`net/http` is excellent — and dangerously easy to misuse. The zero-value `Client` has **no timeout**. The zero-value `Server` has **no read/write timeouts**. Those defaults are fine for toy programs and catastrophic under load.

This chapter makes timeouts, cancellation, retries, and transport tuning explicit.

```
client SLA (e.g. 2s)
    |
    v
edge / server budget (1.8s) -- ReadHeader/Read/Write/Idle
    |
    +--> auth (50ms)
    +--> cache (100ms)
    +--> db (400ms)
    +--> upstream HTTP (remaining via context)
    +--> encode
```

Server-Side Guardrails

Configure the server, not just the handler

```go
srv := &http.Server{
    Addr:              ":8080",
    Handler:           logging(mux),
    ReadHeaderTimeout: 5 * time.Second,  // slowloris protection
    ReadTimeout:       15 * time.Second, // headers + body
    WriteTimeout:      30 * time.Second, // careful with streaming
    IdleTimeout:       60 * time.Second, // keep-alives
    MaxHeaderBytes:    1 << 16,
}
```

Field	Protects against
`ReadHeaderTimeout`	Slow header drip (slowloris)
`ReadTimeout`	Slow bodies
`WriteTimeout`	Stuck clients on response
`IdleTimeout`	Idle keep-alive resource waste
`MaxHeaderBytes`	Oversized header memory use

Streaming caveat: `WriteTimeout` is absolute from request start on many setups; long-lived streams (SSE, chunked) may need a different server or hijacking strategy.

Handler-level budgets

```go
func handleOrder(w http.ResponseWriter, r *http.Request) {
    ctx, cancel := context.WithTimeout(r.Context(), 1200*time.Millisecond)
    defer cancel()

    user, err := auth.User(ctx, r)
    if err != nil {
        writeErr(w, err)
        return
    }
    order, err := orders.Create(ctx, user, r.Body)
    if err != nil {
        writeErr(w, err)
        return
    }
    writeJSON(w, order)
}
```

Always derive from `r.Context()` so client disconnects cancel work.

Limit concurrency of expensive handlers

```go
var sem = make(chan struct{}, 100)

func limited(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        select {
        case sem <- struct{}{}:
            defer func() { <-sem }()
            next.ServeHTTP(w, r)
        default:
            http.Error(w, "overloaded", http.StatusServiceUnavailable)
        }
```

```
    })
}
```

Return **503** with `Retry-After` when shedding load so well-behaved clients back off.

Body limits

```
r.Body = http.MaxBytesReader(w, r.Body, 1<<20)
```

Combine with JSON decoders that use `DisallowUnknownFields` for strict APIs.

Client-Side Guardrails

Never use the naked default client in production

```
var outbound = &http.Client{
    Timeout: 10 * time.Second, // hard cap including body read
    Transport: &http.Transport{
        Proxy: http.ProxyFromEnvironment,
        DialContext: (&net.Dialer{
            Timeout:   3 * time.Second,
            KeepAlive: 30 * time.Second,
        }).DialContext,
        ForceAttemptHTTP2:     true,
        MaxIdleConns:          100,
        MaxIdleConnsPerHost:   10,
        MaxConnsPerHost:       32,
        IdleConnTimeout:       90 * time.Second,
        TLSHandshakeTimeout:   5 * time.Second,
        ExpectContinueTimeout: 1 * time.Second,
        ResponseHeaderTimeout: 5 * time.Second,
    },
}
```

Notes:

- `Client.Timeout` covers the whole request (dial → headers → body).
- Prefer **per-call context** for operation-specific budgets finer than the client timeout.
- Tune `MaxConnsPerHost` to avoid stampedes on a single dependency.

Context-aware requests

```go
func fetchUser(ctx context.Context, id string) (*User, error) {
    ctx, cancel := context.WithTimeout(ctx, 300*time.Millisecond)
    defer cancel()

    req, err := http.NewRequestWithContext(ctx, http.MethodGet, baseURL+"/users/"+id, nil)
    if err != nil {
        return nil, err
    }
    resp, err := outbound.Do(req)
    if err != nil {
        return nil, err
    }
    defer resp.Body.Close()

    if resp.StatusCode >= 500 {
        return nil, fmt.Errorf("upstream status %d", resp.StatusCode)
    }
    var u User
    if err := json.NewDecoder(io.LimitReader(resp.Body, 1<<20)).Decode(&u); err != nil {
        return nil, err
    }
    return &u, nil
}
```

Always `Close` bodies — leaking bodies leaks connections from the pool.

Failure Classification

Signal	Likely meaning	Retry?
`context.DeadlineExceeded`	Budget exhausted / slow dep	Maybe once, if idempotent & budget remains
`context.Canceled`	Caller gave up	No
Connection refused / reset	Dep down or restarting	Yes with backoff
HTTP 408 / 429	Timeout / rate limit	Yes respecting `Retry-After`
HTTP 500 / 502 / 503 / 504	Upstream error	Conditional
HTTP 400 / 401 / 403 / 404	Caller / auth / missing	No
HTTP 409 / 422	Business conflict	Usually no

```go
func retryable(status int, err error) bool {
    if err != nil {
        if errors.Is(err, context.Canceled) {
            return false
        }
        return true // network errors: often transient
    }
    switch status {
    case 408, 429, 500, 502, 503, 504:
        return true
    default:
        return false
    }
}
```

Retries Without Amplification

Rules of thumb:

1. Retry only **idempotent** methods by default (`GET`, `HEAD`, `PUT`, `DELETE` with idempotency keys).
2. Cap attempts (e.g. 2–3 total).
3. Exponential backoff **with jitter**.
4. Bound **total elapsed time**, not just attempt count.
5. Propagate remaining parent context budget.
6. Prefer idempotency keys for `POST` when the business allows.

```go
func doWithRetry(ctx context.Context, c *http.Client, req *http.Request) (*http.Response, er
    var last error
    backoff := 50 * time.Millisecond
    for attempt := 0; attempt < 3; attempt++ {
        if attempt > 0 {
            // Clone body if needed; for GET, req can be reused carefully.
            select {
            case <-ctx.Done():
                return nil, ctx.Err()
            case <-time.After(backoff + jitter(backoff)):
            }
            backoff *= 2
        }
        r := req.Clone(ctx)
        resp, err := c.Do(r)
        if err == nil && !retryable(resp.StatusCode, nil) {
            return resp, nil
        }
        if err == nil {
```

```go
            resp.Body.Close()
            last = fmt.Errorf("status %d", resp.StatusCode)
        } else {
            last = err
            if !retryable(0, err) {
                return nil, err
            }
        }
    }
    return nil, last
}
```

Misconfigured retries at **browser + edge + service + client library** layers multiply load — coordinate policy.

Transport Pooling and HTTP/2

- Reuse a single `*http.Client` / `Transport` (connection pool lives there).
- Creating a new `Transport` per request disables reuse and burns sockets.
- HTTP/2 multiplexes streams on one conn; head-of-line blocking moves to the application layer — still use deadlines per request.
- For many hosts, raise `MaxIdleConns`; for one hot host, raise `MaxIdleConnsPerHost`.

Graceful Shutdown

```go
ctx, stop := signal.NotifyContext(context.Background(), os.Interrupt, syscall.SIGTERM)
defer stop()

go func() {
    if err := srv.ListenAndServe(); err != nil && !errors.Is(err, http.ErrServerClosed) {
        slog.Error("listen", "err", err)
        os.Exit(1)
    }
}()

<-ctx.Done()
shutdownCtx, cancel := context.WithTimeout(context.Background(), 20*time.Second)
defer cancel()
if err := srv.Shutdown(shutdownCtx); err != nil {
    slog.Error("shutdown", "err", err)
    _ = srv.Close()
}
```

`Shutdown` stops accepting, waits for handlers; handlers must honor `r.Context()` to finish promptly.

Observability Hooks

Emit at least:

- Request rate, error rate, latency histogram (RED)
- Client-side: outbound latency and error class by dependency
- Timeout counts separately from application 5xx
- In-flight requests / active connections

```go
func instrument(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        start := time.Now()
        ww := &statusWriter{ResponseWriter: w, code: 200}
        next.ServeHTTP(ww, r)
        slog.Info("request",
            "method", r.Method,
            "path", r.URL.Path,
            "status", ww.code,
            "dur_ms", time.Since(start).Milliseconds(),
        )
    })
}
```

Production Checklist

- ☐ Every `http.Server` has header/read/write/idle timeouts
- ☐ Every production `http.Client` has `Timeout` + tuned `Transport`
- ☐ Handlers derive work from `r.Context()` with sub-timeouts
- ☐ Response bodies always closed; request bodies limited
- ☐ Retries are idempotent, jittered, budget-capped
- ☐ Load shedding returns 503 under overload
- ☐ Graceful `Shutdown` with handler deadlines
- ☐ Metrics distinguish timeouts, cancels, and app errors
- ☐ Single shared clients for outbound deps (pooled)

Common Pitfalls

1. **`http.Get` in library code** — uses default client, no timeout.
2. **Ignoring `resp.Body`** — connection pool starvation.
3. **Retrying POST without idempotency** — double charges / double writes.
4. **Server `WriteTimeout` too aggressive for large downloads**.
5. **Context with values used for cancellation only half the stack** — DB layer ignores cancel.
6. **Global 30s timeout for all outbound calls** — wastes budget on optional paths.
7. **Fan-out without `errgroup` limits** — one request opens 200 deps.

Exercises

1. Start a server without timeouts; use a slow client that drips headers; then add `ReadHeaderTimeout` and retest.
2. Build a client with and without `Timeout`; point at a black hole IP; compare hang vs fail.
3. Implement status-aware retries with jitter; unit-test that 404 is not retried and 503 is.
4. Use `httptest.Server` to inject 100ms / 500ms / 2s latencies; verify parent 300ms budget cancels.
5. Deliberately leak `resp.Body` in a loop; watch `TIME_WAIT` / goroutines / pool behavior; fix.
6. Add a concurrency limiter middleware; load-test until you see 503s; graph in-flight.
7. Implement graceful shutdown; send SIGTERM under load; confirm in-flight requests complete within budget.
8. Split timeouts: dial 200ms, TLS 500ms, headers 500ms, total 2s — document where each is configured.
9. Compare HTTP/1.1 vs HTTP/2 under many parallel requests to one host; note conn counts.
10. Write a runbook entry: "p99 latency spike" — which timeout metric do you check first?

More examples

Client with transport timeouts (not zero-value)

```
mkdir -p /tmp/go-http-client-to && cd /tmp/go-http-client-to
go mod init example.com/http-client-to
```

Save as `main.go`:

```go
package main

import (
    "fmt"
    "net"
    "net/http"
    "net/http/httptest"
    "time"
)

func main() {
    slow := httptest.NewServer(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request)
        time.Sleep(80 * time.Millisecond)
        fmt.Fprintln(w, "late")
    }))
    defer slow.Close()

    client := &http.Client{
        Timeout: 20 * time.Millisecond,
```

```go
        Transport: &http.Transport{
            DialContext: (&net.Dialer{Timeout: 50 * time.Millisecond}).DialContext,
            ResponseHeaderTimeout: 50 * time.Millisecond,
        },
    }
    _, err := client.Get(slow.URL)
    fmt.Println("client timeout:", err != nil)

    fast := httptest.NewServer(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request)
        fmt.Fprintln(w, "ok")
    }))
    defer fast.Close()
    client.Timeout = time.Second
    resp, err := client.Get(fast.URL)
    if err != nil {
        panic(err)
    }
    resp.Body.Close()
    fmt.Println("fast status:", resp.StatusCode)
}
```

```bash
go run .
```

Expected output:

```
client timeout: true
fast status: 200
```

Retry only on transient status

```bash
mkdir -p /tmp/go-http-retry && cd /tmp/go-http-retry
go mod init example.com/http-retry
```

Save as `main.go`:

```go
package main

import (
    "fmt"
    "net/http"
    "net/http/httptest"
    "sync/atomic"
)

func main() {
```

```go
    var n atomic.Int32
    ts := httptest.NewServer(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        if n.Add(1) < 3 {
            w.WriteHeader(http.StatusServiceUnavailable)
            return
        }
        fmt.Fprintln(w, "ok")
    }))
    defer ts.Close()

    var last int
    for attempt := 1; attempt <= 5; attempt++ {
        resp, err := http.Get(ts.URL)
        if err != nil {
            panic(err)
        }
        last = resp.StatusCode
        resp.Body.Close()
        if last < 500 {
            break
        }
    }
    fmt.Println("attempts:", n.Load(), "final:", last)
}
```

```bash
go run .
```

Expected output:

```
attempts: 3 final: 200
```

Runnable example

`httptest` backend with injected latency, a budgeted client, and retries with exponential backoff + jitter—stdlib-only resilience loop.

```bash
mkdir -p /tmp/go-http-resilience && cd /tmp/go-http-resilience
go mod init example.com/http-resilience
```

Save as `main.go`:

```go
package main

import (
    "context"
```

```
    "fmt"
    "io"
    "math/rand"
    "net/http"
    "net/http/httptest"
    "time"
)

func backoff(attempt int) time.Duration {
    // Full jitter: random in [0, base*2^attempt]
    base := 20 * time.Millisecond
    max := base << attempt
    if max > 200*time.Millisecond {
        max = 200 * time.Millisecond
    }
    return time.Duration(rand.Int63n(int64(max) + 1))
}

func getWithRetry(ctx context.Context, client *http.Client, url string, attempts int) (int,
    var last error
    for i := 0; i < attempts; i++ {
        req, err := http.NewRequestWithContext(ctx, http.MethodGet, url, nil)
        if err != nil {
            return 0, err
        }
        resp, err := client.Do(req)
        if err != nil {
            last = err
        } else {
            code := resp.StatusCode
            io.Copy(io.Discard, resp.Body)
            resp.Body.Close()
            if code < 500 {
                return code, nil // do not retry 4xx
            }
            last = fmt.Errorf("status %d", code)
        }
        if i+1 == attempts {
            break
        }
        t := time.NewTimer(backoff(i))
        select {
        case <-ctx.Done():
            t.Stop()
            return 0, ctx.Err()
        case <-t.C:
        }
```

```go
    }
    return 0, last
}

func main() {
    var calls int
    ts := httptest.NewServer(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        calls++
        if calls < 3 {
            w.WriteHeader(http.StatusServiceUnavailable)
            return
        }
        fmt.Fprintln(w, "ok")
    }))
    defer ts.Close()

    client := &http.Client{Timeout: 500 * time.Millisecond}

    ctx, cancel := context.WithTimeout(context.Background(), 2*time.Second)
    defer cancel()
    code, err := getWithRetry(ctx, client, ts.URL, 5)
    fmt.Printf("final status=%d err=%v upstream_calls=%d\n", code, err, calls)

    // Budget cancel demo: server slower than parent context
    slow := httptest.NewServer(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request)
        time.Sleep(200 * time.Millisecond)
        fmt.Fprintln(w, "slow")
    }))
    defer slow.Close()
    ctx2, cancel2 := context.WithTimeout(context.Background(), 50*time.Millisecond)
    defer cancel2()
    _, err = getWithRetry(ctx2, client, slow.URL, 3)
    fmt.Println("budget exceeded:", err != nil)
}
```

```go
go run .
```

Expected output (illustrative):

```
final status=200 err=<nil> upstream_calls=3
budget exceeded: true
```

What to notice

- Retry only idempotent-safe cases (here GET) and 5xx/transport errors—not 404.
- Parent context budget must bound total attempts; infinite retry is an outage amplifier.
- `httptest.Server` is the right unit-test stand-in for dependency latency and failure.

Try next

- Skip retries on `context.Canceled` and count them separately in metrics.
- Add a concurrency semaphore middleware that returns 503 under overload.

Further Reading

- Go `net/http` docs for `Transport` fields
- SRE books on tail latency and load shedding
- Previous: TCP framing and deadlines
- Next: **Reverse Proxy and Load Balancing** — centralizing these policies at the edge

Reverse Proxy and Load Balancing

Reverse Proxy and Load Balancing

A reverse proxy is a **reliability boundary**: it enforces routing policy, mediates failures, and protects backend services. In Go you can build edge-like behavior with `net/http/httputil` and a small amount of discipline — or understand nginx/Envoy better by implementing the same ideas.

Why / Overview

Backends should focus on business logic. The proxy owns:

- Route matching and path rewrite
- Upstream selection and health
- Timeout and retry policy (carefully)
- Request identity / header hygiene
- TLS termination or pass-through
- Observability of the edge hop

```
client -> edge proxy -> route match -> pick upstream -> backend
                     \-> policies: authn, rate limit, timeout, LB
```

Building a Reverse Proxy in Go

Single upstream

```go
package main

import (
    "net/http"
    "net/http/httputil"
    "net/url"
    "time"
)

func main() {
    target, _ := url.Parse("http://127.0.0.1:8081")
    proxy := httputil.NewSingleHostReverseProxy(target)

    // Optional: custom transport with timeouts.
```

```go
    proxy.Transport = &http.Transport{
        Proxy:                 http.ProxyFromEnvironment,
        ResponseHeaderTimeout: 5 * time.Second,
        IdleConnTimeout:       90 * time.Second,
        MaxIdleConnsPerHost:   32,
    }

    // Error handler when upstream is down.
    proxy.ErrorHandler = func(w http.ResponseWriter, r *http.Request, err error) {
        http.Error(w, "bad gateway", http.StatusBadGateway)
    }

    srv := &http.Server{
        Addr:              ":8080",
        Handler:           proxy,
        ReadHeaderTimeout: 5 * time.Second,
    }
    srv.ListenAndServe()
}
```

`NewSingleHostReverseProxy` rewrites the request URL to the target and sets `X-Forwarded-For` (with care about trust).

Director customization

```go
proxy := &httputil.ReverseProxy{
    Director: func(req *http.Request) {
        req.URL.Scheme = "http"
        req.URL.Host = upstream // host:port
        req.Host = upstream
        // Strip internal headers clients must not inject.
        req.Header.Del("X-Internal-Auth")
        req.Header.Set("X-Request-Id", ensureRequestID(req))
    },
    Transport:   transport,
    ErrorHandler: edgeError,
    ModifyResponse: func(resp *http.Response) error {
        resp.Header.Set("X-Edge", "go-proxy")
        return nil
    },
}
```

Routing Model

Keep routing **deterministic and debuggable**:

1. Exact path matches first
2. Method-aware routes when needed
3. Longest prefix next
4. Explicit default / 404

```go
type route struct {
    prefix string
    pool   *UpstreamPool
}

func (r *Router) ServeHTTP(w http.ResponseWriter, req *http.Request) {
    for _, rt := range r.routes { // ordered: longest prefix first
        if strings.HasPrefix(req.URL.Path, rt.prefix) {
            rt.pool.Proxy().ServeHTTP(w, req)
            return
        }
    }
    http.NotFound(w, req)
}
```

Document route tables in config, not only in code — operators need to see them during incidents.

Upstream Selection and Load Balancing

Round-robin baseline

```go
type UpstreamPool struct {
    mu    sync.Mutex
    addrs []string
    next  uint64
    alive []bool
}

func (p *UpstreamPool) pick() (string, bool) {
    p.mu.Lock()
    defer p.mu.Unlock()
    n := len(p.addrs)
    for i := 0; i < n; i++ {
        idx := int(p.next % uint64(n))
        p.next++
        if p.alive[idx] {
            return p.addrs[idx], true
        }
    }
    return "", false
```

```
}
```

Health awareness

```
active check:  proxy -> GET /healthz on each upstream every N seconds
passive check: consecutive 5xx / connect errors mark unhealthy
reentry:       require M successes before alive=true again
```

```
func (p *UpstreamPool) healthLoop(ctx context.Context, client *http.Client, every time.Durat
    t := time.NewTicker(every)
    defer t.Stop()
    for {
        select {
        case <-ctx.Done():
            return
        case <-t.C:
            for i, addr := range p.addrs {
                ok := probe(ctx, client, "http://"+addr+"/healthz")
                p.mu.Lock()
                p.alive[i] = ok
                p.mu.Unlock()
            }
        }
    }
}
```

Use **readiness**, not mere process liveness, for routing. A process can be up while its DB is down.

Other LB algorithms (when RR is not enough)

Algorithm	Use when
Round-robin	Homogeneous backends
Least connections	Long-lived / uneven requests
Consistent hash	Cache locality / sticky sessions without central store
Weighted	Canary / heterogeneous capacity

Sticky sessions couple clients to instances and complicate deploys — prefer shared session stores when possible.

Timeout and Retry Boundaries

Proxy retries can **multiply load**. Rules:

- Retry only safe methods by default.
- Cap attempts (often 1 retry for idempotent GET).
- Enforce a **total edge budget** (e.g. 5s) independent of backend generosity.
- Do not retry if the client already disconnected.
- Prefer failovers to a different healthy instance over hammering the same one.

```go
func (p *UpstreamPool) Proxy() http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        ctx, cancel := context.WithTimeout(r.Context(), 5*time.Second)
        defer cancel()
        r = r.WithContext(ctx)

        addr, ok := p.pick()
        if !ok {
            http.Error(w, "no healthy upstream", http.StatusServiceUnavailable)
            return
        }
        // Build reverse proxy for addr or set Director host to addr.
        p.reverseProxyTo(addr).ServeHTTP(w, r)
    })
}
```

Header Hygiene and Trust

Critical edge concerns:

1. **Hop-by-hop headers** — `Connection`, `Keep-Alive`, `Transfer-Encoding` must not be blindly forwarded incorrectly (`httputil` handles many cases).
2. **`X-Forwarded-*`** — only append client IP if the immediate peer is a trusted hop; otherwise clients spoof IPs for rate limits/ACLs.
3. **Strip internal auth headers** — clients must not set `X-User-Id` that backends trust.
4. **Request IDs** — generate if missing; propagate to backends for correlation.

```go
func ensureRequestID(r *http.Request) string {
    if id := r.Header.Get("X-Request-Id"); id != "" {
        return id
    }
    return uuidNew() // crypto/rand based id
}
```

Buffering vs Streaming

- Default reverse proxy streams bodies; good for large uploads/downloads.
- Buffering enables retries with body replay but costs memory — only for small, idempotent requests with retained bodies.
- For `POST` with bodies, retries require `GetBody` / rewind support.

Observability at the Edge

Metrics:

- `edge_requests_total{route,code}`
- `edge_request_duration_seconds{route}`
- `edge_upstream_errors_total{upstream,reason}`
- `edge_upstream_alive{upstream}`
- `edge_retry_total{route}`

Logs: request id, route, chosen upstream, status, duration, error class.

Tracing: create a span at the edge; inject propagation headers into upstream requests (see part 19).

Production Checklist

☐ Explicit route table with deterministic matching
☐ Health-aware upstream pool with hysteresis on recovery
☐ Transport timeouts on proxy → backend
☐ Edge-level total timeout budget
☐ Conservative retry policy (idempotent only)
☐ Header allow/deny lists for trusted identity
☐ Separate admin listener for proxy diagnostics
☐ Metrics per route and per upstream
☐ Safe config reload story (or blue/green proxy instances)
☐ Load tests for failover and thundering herd on recovery

Common Pitfalls

1. **Retry storms** when all instances degraded.
2. **Routing on liveness** while app is not ready.
3. **Trusting client `X-Forwarded-For`** on a public edge.
4. **One giant proxy binary** without resource limits — edge is critical path.
5. **WebSocket / upgrade mishandling** — ensure hijack/upgrade paths are tested.
6. **Hidden double timeouts** — edge 1s + backend 30s still fails at 1s; document the tighter bound.
7. **Inconsistent path stripping** — `/api/v1` stripped twice → wrong backend path.

Minimal Multi-Upstream Example

```go
type multiHost struct {
    pool *UpstreamPool
    tr   http.RoundTripper
}

func (m *multiHost) ServeHTTP(w http.ResponseWriter, r *http.Request) {
    addr, ok := m.pool.pick()
    if !ok {
        http.Error(w, "unavailable", http.StatusServiceUnavailable)
        return
    }
    rp := &httputil.ReverseProxy{
        Director: func(req *http.Request) {
            req.URL.Scheme = "http"
            req.URL.Host = addr
            req.Host = addr
        },
        Transport: m.tr,
        ErrorHandler: func(w http.ResponseWriter, r *http.Request, err error) {
            m.pool.noteFailure(addr)
            http.Error(w, "bad gateway", http.StatusBadGateway)
        },
    }
    rp.ServeHTTP(w, r)
}
```

Exercises

1. Proxy a single backend; kill the backend; confirm 502 and a metric increment.
2. Implement round-robin across three `httptest` backends; log which host served each request.
3. Add active health checks; mark one down; prove traffic shifts within one interval.
4. Implement passive health: three consecutive errors → unhealthy; two successes → healthy.
5. Send a client-supplied `X-Internal-User` header; strip it at the proxy; verify backend never sees the spoofed value.
6. Add edge timeout of 200ms; backend sleeps 500ms; confirm client gets timeout/502, not a hang.
7. Enable a single retry on GET only; prove POST is not retried.
8. Load-test with one slow upstream in the pool; compare RR vs least-conn if you implement both.
9. Add request IDs and propagate them; grep logs across proxy and backend for one request.
10. Document a failure drill: "all upstreams unhealthy" — expected status, alerts, runbook.

More examples

Round-robin reverse proxy (stdlib)

```
mkdir -p /tmp/go-rr-proxy && cd /tmp/go-rr-proxy
go mod init example.com/rr-proxy
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "net/http"
    "net/http/httptest"
    "net/http/httputil"
    "net/url"
    "sync/atomic"
)

func main() {
    b1 := httptest.NewServer(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        fmt.Fprintln(w, "b1")
    }))
    defer b1.Close()
    b2 := httptest.NewServer(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        fmt.Fprintln(w, "b2")
    }))
    defer b2.Close()

    targets := []*url.URL{mustURL(b1.URL), mustURL(b2.URL)}
    var i atomic.Uint64
    proxy := &httputil.ReverseProxy{
        Rewrite: func(pr *httputil.ProxyRequest) {
            t := targets[i.Add(1)%uint64(len(targets))]
            pr.SetURL(t)
            pr.Out.Host = t.Host
        },
    }

    front := httptest.NewServer(proxy)
    defer front.Close()

    for n := 0; n < 4; n++ {
        resp, err := http.Get(front.URL + "/")
        if err != nil {
```

```
            panic(err)
        }
        var buf [8]byte
        k, _ := resp.Body.Read(buf[:])
        resp.Body.Close()
        fmt.Print(string(buf[:k]))
    }
}

func mustURL(s string) *url.URL {
    u, err := url.Parse(s)
    if err != nil {
        panic(err)
    }
    return u
}
```

```
go run .
```

Expected output:

```
b2
b1
b2
b1
```

Skip unhealthy upstream

```
mkdir -p /tmp/go-proxy-health && cd /tmp/go-proxy-health
go mod init example.com/proxy-health
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "net/http"
    "net/http/httptest"
    "sync/atomic"
)

func main() {
    var healthy atomic.Bool
    healthy.Store(false)
```

```go
    up := httptest.NewServer(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        if r.URL.Path == "/healthz" {
            if !healthy.Load() {
                w.WriteHeader(http.StatusServiceUnavailable)
                return
            }
            fmt.Fprintln(w, "ok")
            return
        }
        fmt.Fprintln(w, "served")
    }))
    defer up.Close()

    pick := func() (string, bool) {
        resp, err := http.Get(up.URL + "/healthz")
        if err != nil {
            return "", false
        }
        resp.Body.Close()
        return up.URL, resp.StatusCode == 200
    }

    _, ok := pick()
    fmt.Println("cold ok:", ok)
    healthy.Store(true)
    u, ok := pick()
    fmt.Println("warm ok:", ok, "url set:", u != "")
}
```

```bash
go run .
```

Expected output:

```
cold ok: false
warm ok: true url set: true
```

Runnable example

In-process reverse proxy with round-robin across three `httptest` backends and passive failure skip—stdlib `httputil.ReverseProxy`.

```bash
mkdir -p /tmp/go-rproxy && cd /tmp/go-rproxy
go mod init example.com/rproxy
```

Save as `main.go`:

```go
package main

import (
    "fmt"
    "net/http"
    "net/http/httptest"
    "net/http/httputil"
    "net/url"
    "sync"
    "sync/atomic"
)

type pool struct {
    mu    sync.Mutex
    addrs []string
    down  map[string]bool
    rr    uint64
}

func (p *pool) next() (string, bool) {
    p.mu.Lock()
    defer p.mu.Unlock()
    n := len(p.addrs)
    for i := 0; i < n; i++ {
        idx := int(atomic.AddUint64(&p.rr, 1)-1) % n
        a := p.addrs[idx]
        if !p.down[a] {
            return a, true
        }
    }
    return "", false
}

func (p *pool) markDown(a string) {
    p.mu.Lock()
    p.down[a] = true
    p.mu.Unlock()
}

func main() {
    var hits [3]atomic.Int64
    var backends []*httptest.Server
    var addrs []string
    for i := 0; i < 3; i++ {
        i := i
        ts := httptest.NewServer(http.HandlerFunc(func(w http.ResponseWriter, r *http.Reques
            if i == 1 && hits[1].Load() == 0 {
```

```
            // first touch on backend 1 fails once
        }
        hits[i].Add(1)
        fmt.Fprintf(w, "backend-%d", i)
    }))
    backends = append(backends, ts)
    u, _ := url.Parse(ts.URL)
    addrs = append(addrs, u.Host)
}
defer func() {
    for _, b := range backends {
        b.Close()
    }
}()

p := &pool{addrs: addrs, down: map[string]bool{}}
// Mark middle backend down to show skip.
p.markDown(addrs[1])

proxy := http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
    addr, ok := p.next()
    if !ok {
        http.Error(w, "unavailable", http.StatusServiceUnavailable)
        return
    }
    rp := &httputil.ReverseProxy{
        Director: func(req *http.Request) {
            req.URL.Scheme = "http"
            req.URL.Host = addr
            req.Host = addr
            req.Header.Del("X-Internal-User") // strip spoofable header
        },
        ErrorHandler: func(w http.ResponseWriter, r *http.Request, err error) {
            p.markDown(addr)
            http.Error(w, "bad gateway", http.StatusBadGateway)
        },
    }
    rp.ServeHTTP(w, r)
})

front := httptest.NewServer(proxy)
defer front.Close()

for i := 0; i < 6; i++ {
    resp, err := http.Get(front.URL + "/")
    if err != nil {
        fmt.Println("err", err)
```

```
            continue
        }
        var body [32]byte
        n, _ := resp.Body.Read(body[:])
        resp.Body.Close()
        fmt.Printf("req %d -> %s\n", i+1, string(body[:n]))
    }
    fmt.Printf("hits: b0=%d b1=%d b2=%d\n", hits[0].Load(), hits[1].Load(), hits[2].Load())
}
```

```
go run .
```

Expected output (illustrative):

```
req 1 -> backend-0
req 2 -> backend-2
req 3 -> backend-0
req 4 -> backend-2
req 5 -> backend-0
req 6 -> backend-2
hits: b0=3 b1=0 b2=3
```

What to notice

- Edge policy (skip down backends, strip unsafe headers) belongs in the proxy, not every service.
- Round-robin with health state is enough to reason about blast radius before adopting Envoy.
- `Director` rewrites scheme/host; never trust client-supplied internal identity headers.

Try next

- Active health checks every 200ms that clear `down` when `/healthz` returns 200.
- Edge timeout: wrap transport with a short `ResponseHeaderTimeout`.

Further Reading

- `net/http/httputil.ReverseProxy` source (Director, rewrite, errors)
- Envoy / nginx docs on retries and outlier detection (concepts transfer)
- Previous: HTTP resilience budgets
- Next part: **Systems Programming** — process and filesystem discipline that network daemons rely on

TLS Clients and Servers

TLS Clients and Servers

Overview

TLS protects data in transit. In Go, `crypto/tls` sits under `net/http` and raw `tls.Conn`. Get certificate verification right; timeouts still apply.

HTTPS server (file certs)

```go
srv := &http.Server{Addr: ":8443", Handler: mux, ReadHeaderTimeout: 5 * time.Second}
log.Fatal(srv.ListenAndServeTLS("cert.pem", "key.pem"))
```

Custom tls.Config

```go
cfg := &tls.Config{
    MinVersion: tls.VersionTLS12,
    // Certificates: []tls.Certificate{cert},
    // GetCertificate: ... // SNI
    // ClientAuth: tls.RequireAndVerifyClientCert, // mTLS
}
ln, err := tls.Listen("tcp", ":8443", cfg)
```

Client verification

```go
tr := &http.Transport{
    TLSClientConfig: &tls.Config{
        MinVersion: tls.VersionTLS12,
        // RootCAs: pool, // private PKI
        // InsecureSkipVerify: true, // NEVER in production
    },
    ForceAttemptHTTP2: true,
}
client := &http.Client{Transport: tr, Timeout: 10 * time.Second}
```

mTLS sketch

Server: `ClientAuth: tls.RequireAndVerifyClientCert` + `ClientCAs`.
Client: `Certificates: []tls.Certificate{clientCert}`.

See 171 TLS/mTLS deep dive.

Pinning (careful)

Pinning SPKI hashes is brittle with rotation. Prefer public CA + short-lived certs or private PKI. If you pin, automate rotation.

Rules of thumb

Do	Don't
MinVersion TLS1.2+	Skip verify to "fix CI" permanently
Timeouts on handshake path	Unlimited dial
Separate server/client certs for mTLS	Reuse one cert everywhere

Try next

1. `mkcert` local certs; serve HTTPS.
2. Client against bad name → verify error.
3. Log peer cert CN on mTLS server.

DNS Resolution Patterns

DNS Resolution Patterns

Overview

Every dial starts with a name. Bad DNS assumptions cause mysterious latency and sticky black holes. Control resolvers, dual-stack, and caching consciously.

Default resolver

```go
ips, err := net.DefaultResolver.LookupIPAddr(ctx, "example.com")
```

Always pass **context** with timeout.

Custom resolver

```go
r := &net.Resolver{
    PreferGo: true,
    Dial: func(ctx context.Context, network, address string) (net.Conn, error) {
        d := net.Dialer{Timeout: 2 * time.Second}
        return d.DialContext(ctx, "udp", "1.1.1.1:53")
    },
}
ips, err := r.LookupIPAddr(ctx, host)
```

Dialer and Happy Eyeballs

```go
d := net.Dialer{Timeout: 5 * time.Second, KeepAlive: 30 * time.Second}
conn, err := d.DialContext(ctx, "tcp", net.JoinHostPort(host, "443"))
```

Go's dialer handles much of dual-stack; still set overall timeouts.

Caching

`net.Resolver` does not give you full app-level cache control. For high-QPS:

- Prefer connection reuse (HTTP Transport) over re-resolve every call
- Short-lived process caches with TTL if you roll your own (invalidate!)

Failure modes

Symptom	Cause
Sporadic dial timeout	Resolver/network blip
Sticky bad IP	Long-lived conn to dead backend; need health + re-resolve
IPv6 blackhole	Broken AAAA path

Minimal dig-like

```go
for _, ip := range ips {
    fmt.Println(ip.IP)
}
```

(See CLI netkit `dns` command.)

Rules of thumb

Do	Don't
ctx timeout on Lookup	Block forever on DNS
Re-resolve on new connections	Assume one IP forever for dynamic backends
Log resolved IP on errors	Only log hostname

Try next

1. Lookup with 1ms timeout—handle error.
2. Compare `LookupIP` vs `LookupCNAME`.
3. Force custom DNS Dial to an internal resolver.

HTTP Transport Tuning

HTTP Transport Tuning

Overview

`http.Transport` owns connection pooling, TLS, proxies, and HTTP/2. Defaults are fine for many apps; services with high fan-out need explicit limits.

Sensible client

```go
func NewClient() *http.Client {
    t := &http.Transport{
        Proxy:                 http.ProxyFromEnvironment,
        ForceAttemptHTTP2:     true,
        MaxIdleConns:          100,
        MaxIdleConnsPerHost:   10,
        MaxConnsPerHost:       0, // 0 = unlimited; set in tight systems
        IdleConnTimeout:       90 * time.Second,
        TLSHandshakeTimeout:   5 * time.Second,
        ExpectContinueTimeout: 1 * time.Second,
        DialContext: (&net.Dialer{
            Timeout:   5 * time.Second,
            KeepAlive: 30 * time.Second,
        }).DialContext,
    }
    return &http.Client{Transport: t, Timeout: 15 * time.Second}
}
```

One client per process

Reuse `*http.Client`. Creating per-request clients throws away the pool.

Timeouts layers

```
Client.Timeout          → entire request including body
Dialer.Timeout          → TCP connect
TLSHandshakeTimeout     → TLS
Response header         → via context on request
```

Prefer `NewRequestWithContext` + outer timeout.

Disable keep-alives (rare)

```
t.DisableKeepAlives = true // short CLI one-shots maybe
```

HTTP/2

Enabled automatically for HTTPS with default transport settings. HTTP/2 multiplexes streams on one conn—still bound total concurrency at app layer.

Rules of thumb

Do	Don't
Share Transport	`http.Get` without timeout forever
Cap per-host conns under load	Unlimited fan-out to one dependency
Close response bodies	Leak connections (unread body)

Try next

1. Compare latency 100 serial gets with shared vs new client.
2. Set `MaxConnsPerHost: 2` and observe queuing.
3. Forget `Body.Close`; watch FD/conn growth.

Project: netcheck Toolkit

Project: netcheck Toolkit

Overview

Build **`netcheck`**: combine TCP dial, TLS cert days, DNS lookup, and HTTP probe into one ops binary (sibling to CLI `netkit`/`netsec`).

Commands

Cmd	Behavior
`dial host:port`	TCP connect RTT
`dns name`	print IPs
`tls host`	days left + version
`http url`	status + RTT
`all host`	dns + dial :443 + tls + https probe

Requirements

- Context + 5s default timeout
- JSON optional (`-json`)
- Exit 1 if any check fails in `all`
- Stdlib + `crypto/tls` only

Acceptance

```
go run ./cmd/netcheck all example.com
go run ./cmd/netcheck http -expect 200 https://example.com
```

Network Performance

The Netpoller and Goroutine Scheduling

The Netpoller and Goroutine Scheduling

The boring default in Go is to write **blocking network code**. You dial a connection, call `conn.Read()`, and let execution pause until bytes arrive. Under the hood, Go does not block the underlying operating system thread; it hands the socket descriptor to the **runtime netpoller**, parks the goroutine, and reuses the OS thread to do other work. Understanding how the netpoller operates is the foundation of high-concurrency networking in Go.

Mental model

In traditional operating systems (such as C/C++ with synchronous I/O), calling `read()` on a socket blocks an entire OS thread. Because OS threads have fixed stack sizes (1 to 8 MB) and high kernel scheduling overhead, running 10,000 threads consumes gigabytes of memory and degrades performance through constant context switching.

Other runtimes (such as Node.js or Python asyncio) solve this with **event loops** and asynchronous callbacks or promises, which fractures code into callback pyramids or `async/await` boundaries.

Go combines the best of both worlds:

1. **Non-blocking OS Sockets:** Every network file descriptor managed by package `net` is registered in non-blocking mode.
2. **The Netpoller:** The Go runtime integrates directly with the OS event notification subsystem (`epoll` on Linux, `kqueue` on macOS/BSD, `IOCP` on Windows).
3. **M-P-G Cooperative Scheduling:**
   - **G (Goroutine):** Lightweight execution unit with a dynamic stack starting at just 2 KB.
   - **M (Machine):** An operating system thread.
   - **P (Processor):** A logical context for scheduling Go code (`GOMAXPROCS`).
4. When a goroutine calls `conn.Read()` and no bytes are ready in the kernel socket buffer:
   - The runtime parks the goroutine in state `_Gwaiting` (`netpollblock`).
   - The OS thread (`M`) immediately picks up another runnable goroutine from the local run queue.
   - When the kernel signals that bytes have arrived via `epoll_wait`, the netpoller moves the goroutine back to `_Grunnable`.

You write clean, straight-line, synchronous-looking code while the runtime executes asynchronous event multiplexing behind the scenes.

Worked examples

Case 1: Goroutine-per-connection TCP echo server

Save as `netpoll_demo.go`. We run an echo server that spawns a dedicated goroutine for each incoming client. We verify that after 50 concurrent connections finish, goroutine count returns cleanly to the baseline.

```go
// netpoll_demo.go
package main

import (
    "fmt"
    "io"
    "net"
    "runtime"
    "sync"
    "time"
)

func startEchoServer(addr string) (net.Listener, error) {
    ln, err := net.Listen("tcp", addr)
    if err != nil {
        return nil, err
    }

    go func() {
        for {
            conn, err := ln.Accept()
            if err != nil {
                return
            }
            go func(c net.Conn) {
                defer c.Close()
                // Read into a small buffer; parked by netpoller when idle
                buf := make([]byte, 128)
                for {
                    n, err := c.Read(buf)
                    if err != nil {
                        return
                    }
                    _, _ = c.Write(buf[:n])
                }
            }(conn)
        }
    }()
```

```go
    return ln, nil
}

func main() {
    ln, err := startEchoServer("127.0.0.1:0")
    if err != nil {
        panic(err)
    }
    defer ln.Close()

    addr := ln.Addr().String()
    const clientCount = 50

    var wg sync.WaitGroup
    initialGoroutines := runtime.NumGoroutine()

    // Launch 50 concurrent connections
    for i := range clientCount {
        wg.Add(1)
        go func(id int) {
            defer wg.Done()
            conn, err := net.Dial("tcp", addr)
            if err != nil {
                return
            }
            defer conn.Close()

            msg := fmt.Sprintf("ticket-ping-%d\n", id)
            _, _ = conn.Write([]byte(msg))

            reply := make([]byte, len(msg))
            _, _ = io.ReadFull(conn, reply)
        }(i)
    }

    wg.Wait()
    time.Sleep(10 * time.Millisecond)

    finalGoroutines := runtime.NumGoroutine()
    fmt.Printf("Echo server started on: %s\n", addr)
    fmt.Printf("Completed %d concurrent client connections\n", clientCount)
    fmt.Printf("Goroutines before: %d, after: %d\n", initialGoroutines, finalGoroutines)
}
```

Run:

```go run netpoll_demo.go
```

Output:

```
Echo server started on: 127.0.0.1:46145
Completed 50 concurrent client connections
Goroutines before: 2, after: 2
```

Each client connection was handled by an independent goroutine without spinning an OS thread. When the client closed the socket, `Read()` returned `io.EOF`, the handler exited, and the goroutine terminated cleanly.

Case 2: Inspecting GOMAXPROCS and OS threads

By default, Go sets `GOMAXPROCS` to match the number of available CPU cores. When goroutines are parked in the netpoller waiting on network packets, they consume **zero CPU time** and do not block other running goroutines.

```go
// maxprocs_demo.go
package main

import (
    "fmt"
    "runtime"
)

func main() {
    numCPU := runtime.NumCPU()
    curProcs := runtime.GOMAXPROCS(0) // Pass 0 to query without changing

    fmt.Printf("Logical CPUs: %d\n", numCPU)
    fmt.Printf("GOMAXPROCS:   %d\n", curProcs)
}
```

Run:

```go
go run maxprocs_demo.go
```

Output:

```
Logical CPUs: 8
GOMAXPROCS:   8
```

The trap

The most frequent trap with Go's networking model is **blocking OS threads in CGO or unpolled file I/O**.

While network sockets are non-blocking and managed by the netpoller, standard **disk file operations** (`os.Open`, `os.Read`) on Linux cannot use epoll. When a goroutine performs a blocking disk read, the Go runtime must allocate a new OS thread (`M`) to prevent other goroutines on that `P` from stalling. If thousands of goroutines simultaneously perform blocking disk or CGO calls, thread creation can skyrocket to hundreds of OS threads, triggering thread exhaustion or kernel limits.

For network services, keep disk I/O asynchronous, buffered, or capped to a bounded worker pool.

The boring rule

- Write idiomatic, synchronous-looking `conn.Read()` and `conn.Write()` calls; let the netpoller manage OS epoll/kqueue polling.
- Launch a new goroutine per connection for server listeners unless handling more than 50,000 concurrent idle sockets (where memory per stack must be capped).
- Ensure every connection handler terminates cleanly upon encountering `io.EOF` or an error.
- Never override `GOMAXPROCS` unless running inside containerized CPU quotas where `automaxprocs` avoids CPU throttling.

Try this

1. In `netpoll_demo.go`, increase `clientCount` to 500. Observe that execution completes in milliseconds without thread exhaustion.
2. In the echo server connection loop, add `c.SetReadDeadline(time.Now().Add(50 * time.Millisecond))`. Observe how idle connections timeout and return `os.ErrDeadlineExceeded`.
3. Check runtime scheduler trace events by running `GODEBUG=schedtrace=1000 go run netpoll_demo.go`. Note the `threads` and `idleprocs` counters reported every second.

HTTP Client and Connection Pooling

HTTP Client and Connection Pooling

The boring default is to configure a single, shared **`*http.Client`** at application startup and reuse it for all outbound HTTP requests. Creating a new `http.Client` per request or per goroutine is one of the most common anti-patterns in Go: it spawns a fresh `http.Transport`, discards connection pools, wastes CPU on TLS handshakes, and rapidly exhausts operating system socket ports (`TIME_WAIT` socket storms).

Mental model

In Go, `http.Client` is a lightweight struct that delegates actual network I/O and connection management to an internal `http.RoundTripper` implementation, usually **`*http.Transport`**.

`http.Transport` maintains an internal pool of idle TCP/TLS connections. When your service sends a request: 1. The Transport checks whether an idle connection to the target `host:port` is available in its pool. 2. If available, it immediately reuses the established socket, avoiding TCP 3-way handshakes and TLS cryptographic negotiation. 3. If no connection is idle, it dials a new socket.

The Default Bottleneck: MaxIdleConnsPerHost

`http.DefaultTransport` provides sensible defaults for general web browsing, but has a major flaw for backend microservices:

```
// Inside net/http default transport:
DefaultMaxIdleConnsPerHost = 2
```

The default allows only **2 idle connections per target host**. If your desk service concurrently calls an inventory API or payment gateway with 50 goroutines, 48 sockets are forcibly closed and renegotiated on every round trip. Setting `MaxIdleConnsPerHost` to match your concurrency expectations (e.g. 50–200) is essential.

The Response Body Drain Contract

For `http.Transport` to safely return an open TCP connection to the pool, the application **must read the response body to EOF before closing it**:

```
// REQUIRED for TCP socket reuse:
_, _ = io.Copy(io.Discard, resp.Body)
```

```go
_ = resp.Body.Close()
```

If you call `resp.Body.Close()` without reading the remaining bytes, the Transport cannot know whether unread data remains on the wire. To prevent data corruption on the next request, it is forced to terminate and destroy the TCP connection.

Worked examples

Case 1: Measuring connection reuse with httptrace

Save as `http_pool_demo.go`. We use `httptest.NewServer` and `httptrace.ClientTrace` to verify whether successive requests reuse existing TCP connections.

```go
// http_pool_demo.go
package main

import (
    "fmt"
    "io"
    "net"
    "net/http"
    "net/http/httptest"
    "net/http/httptrace"
    "time"
)

func main() {
    // 1. Mock server that returns ticket responses
    server := httptest.NewServer(http.HandlerFunc(func(w http.ResponseWriter, r *http.Reques
        w.Header().Set("Content-Type", "application/json")
        _, _ = w.Write([]byte(`{"ticket_id": 101, "status": "CONFIRMED"}`))
    }))
    defer server.Close()

    // 2. Properly configured pooled Transport
    transport := &http.Transport{
        Proxy: http.ProxyFromEnvironment,
        DialContext: (&net.Dialer{
            Timeout:   5 * time.Second,
            KeepAlive: 30 * time.Second,
        }).DialContext,
        MaxIdleConns:        100,
        MaxIdleConnsPerHost: 100, // Override default of 2
        IdleConnTimeout:     90 * time.Second,
    }
```

```go
    client := &http.Client{
        Transport: transport,
        Timeout:   10 * time.Second,
    }

    // 3. Measure connection reuse with httptrace
    for i := range 3 {
        req, _ := http.NewRequest(http.MethodGet, server.URL, nil)

        var reused bool
        trace := &httptrace.ClientTrace{
            GotConn: func(connInfo httptrace.GotConnInfo) {
                reused = connInfo.Reused
            },
        }

        req = req.WithContext(httptrace.WithClientTrace(req.Context(), trace))

        start := time.Now()
        resp, err := client.Do(req)
        if err != nil {
            panic(err)
        }

        // Read body to EOF and close to guarantee socket reuse
        _, _ = io.Copy(io.Discard, resp.Body)
        _ = resp.Body.Close()

        totalTime := time.Since(start)
        fmt.Printf("Request %d: reused_tcp=%v total_latency=%v\n", i+1, reused, totalTime)
    }
}
```

Run:

```go
go run http_pool_demo.go
```

Output:

```
Request 1: reused_tcp=false total_latency=752.654µs
Request 2: reused_tcp=true total_latency=189.209µs
Request 3: reused_tcp=true total_latency=199.583µs
```

Notice how Request 1 dialed a fresh TCP socket (`reused_tcp=false`). Requests 2 and 3 reused the existing socket from the pool (`reused_tcp=true`), cutting latency by roughly 75%.

The trap

The most widespread bug in Go networking code is omitting `io.Copy(io.Discard, resp.Body)`:

```go
// WRONG: Leaks TCP connections and prevents pooling!
resp, err := client.Do(req)
if err != nil {
    return err
}
defer resp.Body.Close() // Missing io.Copy!

if resp.StatusCode != http.StatusOK {
    // Exiting here without reading body forces Transport to close the socket!
    return fmt.Errorf("bad status: %d", resp.StatusCode)
}
```

Even if you do not care about the response payload (e.g. on error status codes), you must still drain the body before closing:

```go
// CORRECT: Always drain and close
defer func() {
    _, _ = io.Copy(io.Discard, resp.Body)
    _ = resp.Body.Close()
}()
```

Another trap is creating a new `&http.Client{}` inside an HTTP handler function. If a server receives 1,000 requests per second and creates 1,000 clients, each client creates a new transport that tries to hold idle sockets, crashing the server with `socket: too many open files`. Always instantiate clients once as shared dependencies.

The boring rule

- Instantiate `http.Client` once as a package or struct field dependency; never instantiate clients inside request handlers.
- Set `client.Timeout` on every client; `http.DefaultClient` has no timeout and can hang indefinitely.
- In backend microservices, configure a custom `http.Transport` and increase `MaxIdleConnsPerHost` to 50–200.
- Always drain `resp.Body` with `io.Copy(io.Discard, resp.Body)` before calling `resp.Body.Close()`.

Try this

1. In `http_pool_demo.go`, comment out the line `_, _ = io.Copy(io.Discard, resp.Body)`. Re-run the program and observe whether `reused_tcp` remains false for subsequent requests.

2. Add `httptrace.ClientTrace` callbacks for `DNSStart` and `ConnectStart`. Print timestamps to measure how many milliseconds are spent in DNS resolution vs TCP connection setup.
3. Test a client with `Timeout: 50 * time.Millisecond` against a slow server handler (`time.Sleep(100 * time.Millisecond)`). Confirm that `client.Do()` fails with context deadline exceeded.

Scaling to Ten Thousand Concurrent Connections

Scaling to Ten Thousand Concurrent Connections

The boring default in Go is to spawn a goroutine for every incoming connection. For hundreds or thousands of clients, this works without special tuning. But scaling to **10,000+ simultaneous connections (the C10K problem)** requires intentional memory accounting. While 10,000 goroutine stacks consume only ~20 MB of RAM, unmanaged socket buffers and per-connection heap allocations can easily explode into gigabytes.

Mental model

Handling 10,000 concurrent sockets requires understanding where memory is spent:

1. Goroutine Stacks (~20–40 MB)

A goroutine in Go 1.27 starts with a 2 KB stack allocation. 10,000 idle goroutines parked on `conn.Read()` consume:

$$10{,}000 \times 2 \text{ KB} \approx 20 \text{ MB}$$

This is tiny compared to other language runtimes.

2. User-Space Read/Write Buffers

If each connection allocates its own private 4 KB read buffer and 4 KB write buffer:

$$10{,}000 \times 8 \text{ KB} = 80 \text{ MB}$$

If buffers grow dynamically or are sized to 64 KB, memory ballooning will quickly trigger OOM kills. The solution is **buffer pooling via `sync.Pool`**: allocate buffers only when active data is being decoded, and return them immediately when idle.

3. Kernel Socket Buffers (TCP Windows)

By default, the Linux kernel allocates receive (`SO_RCVBUF`) and send (`SO_SNDBUF`) buffers for each TCP socket (often 128 KB to 256 KB per socket under autotuning):

$$10{,}000 \times 256 \text{ KB} \approx 2.5 \text{ GB (in OS kernel RAM!)}$$

For high-density services with mostly idle persistent connections, tuning minimum socket buffer sizes in Linux via `sysctl` or `net.Dialer.Control` prevents kernel memory exhaustion.

4. Operating System File Descriptors

By default, Linux limits a single process to 1,024 open file descriptors. Attempting to open connection #1025 results in `socket: too many open files`. Raising `ulimit -n 65535` is required before starting the process.

Worked examples

Case 1: Measuring memory footprint across concurrent sockets

Save as `tenk_demo.go`. We establish hundreds of concurrent TCP sockets and inspect resident heap memory consumption.

```go
// tenk_demo.go
package main

import (
    "fmt"
    "net"
    "runtime"
    "sync"
    "time"
)

var readBufPool = sync.Pool{
    New: func() any {
        b := make([]byte, 1024) // 1 KB pooled buffer
        return &b
    },
}

func main() {
    ln, err := net.Listen("tcp", "127.0.0.1:0")
    if err != nil {
        panic(err)
    }
    defer ln.Close()

    const targetConns = 500
    var serverConns []net.Conn
    var mu sync.Mutex

    go func() {
        for {
            c, err := ln.Accept()
            if err != nil {
```

```go
                return
            }
            mu.Lock()
            serverConns = append(serverConns, c)
            mu.Unlock()

            // In a high-density server, buffers are drawn from sync.Pool
            // only when bytes arrive, rather than permanently pinned per socket.
        }
    }()

    var clientConns []net.Conn
    for range targetConns {
        c, err := net.Dial("tcp", ln.Addr().String())
        if err != nil {
            panic(err)
        }
        clientConns = append(clientConns, c)
    }

    time.Sleep(50 * time.Millisecond)

    var memStats runtime.MemStats
    runtime.ReadMemStats(&memStats)

    fmt.Printf("Successfully established %d concurrent sockets\n", targetConns)
    fmt.Printf("Active goroutines:  %d\n", runtime.NumGoroutine())
    fmt.Printf("Allocated heap RAM: %.2f MB\n", float64(memStats.Alloc)/(1024*1024))

    for _, c := range clientConns {
        _ = c.Close()
    }
    for _, c := range serverConns {
        _ = c.Close()
    }
}
```

Run:

```go
go run tenk_demo.go
```

Output:

```
Successfully established 500 concurrent sockets
Active goroutines:  2
Allocated heap RAM: 0.71 MB
```

Notice that 500 live TCP sockets consume less than 1 MB of Go heap RAM when per-socket buffers are not pre-allocated.

Case 2: Essential Linux host tuning for 10K+ connections

Before running a 10K+ network server on Linux, adjust host limits in `/etc/security/limits.conf` and `sysctl.conf`:

```
# 1. Raise open file descriptor limit
ulimit -n 65535

# 2. Increase the listen queue backlog (prevent connection drop during SYN bursts)
sysctl -w net.core.somaxconn=4096

# 3. Expand ephemeral port range for outbound load testing
sysctl -w net.ipv4.ip_local_port_range="1024 65535"

# 4. Enable TCP TIME_WAIT socket reuse
sysctl -w net.ipv4.tcp_tw_reuse=1
```

The trap

A major trap when writing 10K connection servers is **holding onto per-connection reader/writer structs forever**.

For example, calling `bufio.NewReader(conn)` allocates a default 4 KB buffer on the heap. If 10,000 connections create a `bufio.Reader` and keep it referenced in the connection struct, that pins 40 MB of heap memory indefinitely, even if 95% of those connections are idle.

If connections are mostly idle (e.g., push notifications, chat desks, IoT telemetry): - Read the initial command into a small buffer borrowed from `sync.Pool`. - Return the buffer to the pool while waiting for the next message. - Set read deadlines so silent or disconnected clients are evicted before they exhaust file descriptors.

The boring rule

- Always check and raise `ulimit -n` before deploying high-concurrency servers.
- Use `sync.Pool` for byte buffers rather than assigning dedicated 4 KB–64 KB slices to each connection.
- Set conservative connection timeouts so dead connections do not linger in the netpoller.
- Keep per-connection state structs minimal (e.g. user ID, socket handle, small bitflags).

Try this

1. In `tenk_demo.go`, increase `targetConns` to 1,000. Verify whether your local OS file descriptor limit accommodates 2,000 open sockets (1,000 clients + 1,000 servers).
2. Measure how resident memory changes if you allocate a dedicated `buf := make([]byte, 8192)` inside each connection handler before parking in `c.Read()`.
3. Check your current system descriptor limit by running `ulimit -n` in your terminal.

Resilience, Backpressure, and Load Shedding

Resilience, Backpressure, and Load Shedding

The boring default is to process incoming requests as they arrive until system resources saturate. When a sudden surge in traffic hits a backend service, attempting to serve every request simultaneously causes queue buildup, skyrocketing latency percentiles, cascading timeouts across upstream callers, and eventual process collapse. To remain robust under overload, production Go services must implement **backpressure** and **load shedding**: failing fast to protect the requests that *can* be successfully handled.

Mental model

When a service receives more traffic than its CPU, memory, or downstream database can process:

- **Unbounded Queuing (Meltdown):** Requests sit in operating system socket queues or internal channels. By the time a worker picks up the request, the client's timeout has already elapsed. The server burns CPU computing results that the client has already abandoned, compounding the overload.
- **Backpressure:** Forces upstream callers or network senders to slow down by stopping reads from the incoming socket or filling bounded input channels.
- **Load Shedding:** When active concurrency or queue depth exceeds safe operating limits, the server **immediately rejects** surplus requests (returning HTTP 503 Service Unavailable or HTTP 429 Too Many Requests) before doing any database or CPU work.

A buffered channel in Go functions as an extraordinarily efficient concurrency semaphore:

```go
sem := make(chan struct{}, maxCapacity)

select {
case sem <- struct{}{}:
    defer func() { <-sem }()
    // Process request
default:
    // Fast fail: reject request immediately
}
```

Worked examples

Case 1: Fast-fail load shedding middleware

Save as `shed_demo.go`. We cap active handler concurrency to 3. When 10 concurrent requests arrive at once, exactly 3 are processed and 7 are rejected instantly with HTTP 503.

```go
// shed_demo.go
package main

import (
    "fmt"
    "net/http"
    "net/http/httptest"
    "sync"
    "sync/atomic"
    "time"
)

// Bounded concurrency limiter middleware
func LoadSheddingMiddleware(maxConcurrent int, next http.Handler) (http.Handler, *atomic.Int
    sem := make(chan struct{}, maxConcurrent)
    var processed, rejected atomic.Int64

    h := http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        select {
        case sem <- struct{}{}:
            defer func() { <-sem }()
            processed.Add(1)
            next.ServeHTTP(w, r)
        default:
            // Shed load immediately without queuing or blocking
            rejected.Add(1)
            w.Header().Set("Retry-After", "1")
            http.Error(w, "desk busy, load shed", http.StatusServiceUnavailable)
        }
    })

    return h, &processed, &rejected
}

func main() {
    // Worker handler simulating 20ms of work
    handler := http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        time.Sleep(20 * time.Millisecond)
        w.WriteHeader(http.StatusOK)
        _, _ = w.Write([]byte("order processed"))
```

```go
    })

    // Allow maximum 3 concurrent active handlers
    limitedHandler, processed, rejected := LoadSheddingMiddleware(3, handler)
    server := httptest.NewServer(limitedHandler)
    defer server.Close()

    var wg sync.WaitGroup
    var okCount, shedCount atomic.Int64

    // Blast 10 concurrent requests simultaneously
    for range 10 {
        wg.Add(1)
        go func() {
            defer wg.Done()
            resp, err := http.Get(server.URL)
            if err != nil {
                return
            }
            defer resp.Body.Close()
            if resp.StatusCode == http.StatusOK {
                okCount.Add(1)
            } else if resp.StatusCode == http.StatusServiceUnavailable {
                shedCount.Add(1)
            }
        }()
    }

    wg.Wait()

    fmt.Printf("Max concurrency limit: 3\n")
    fmt.Printf("Total requests sent:   10\n")
    fmt.Printf("Requests handled (200): %d\n", okCount.Load())
    fmt.Printf("Requests shed (503):    %d\n", shedCount.Load())
    fmt.Printf("Internal tracker: processed=%d rejected=%d\n", processed.Load(), rejected.Lo
}
```

Run:

```go
go run shed_demo.go
```

Output:

```
Max concurrency limit: 3
Total requests sent:   10
Requests handled (200): 3
Requests shed (503):    7
```

```
Internal tracker: processed=3 rejected=7
```

Under severe load, the service did not degrade; the 3 accepted requests completed in their normal 20ms window, while excess traffic was shed in microseconds.

Case 2: Respecting client disconnects with context propagation

Save as `cancel_demo.go`. If a client cancels their request or closes their browser tab, checking `r.Context().Done()` allows your handlers to halt expensive background work immediately.

```go
// cancel_demo.go
package main

import (
    "context"
    "fmt"
    "time"
)

func processDeskOrder(ctx context.Context, orderID int) error {
    for step := 1; step <= 5; step++ {
        select {
        case <-ctx.Done():
            // Client gave up or timed out; stop work immediately
            return ctx.Err()
        case <-time.After(10 * time.Millisecond):
            // Simulate order verification step
        }
    }
    return nil
}

func main() {
    // Client timeout set to 25ms (shorter than 50ms needed for all 5 steps)
    ctx, cancel := context.WithTimeout(context.Background(), 25*time.Millisecond)
    defer cancel()

    err := processDeskOrder(ctx, 101)
    fmt.Printf("Order processing outcome: %v\n", err)
}
```

Run:

```go
go run cancel_demo.go
```

Output:

```
Order processing outcome: context deadline exceeded
```

The trap

The most common trap is **unbounded in-memory queuing**:

```go
// WRONG: In-memory queue with huge buffer delays failure
var workQueue = make(chan Request, 100000)
```

If your workers can only process 50 requests per second, a buffer of 100,000 items represents a **2,000-second queue delay** (~33 minutes!). Every client will have timed out long before their request is picked up by a worker.

Keep internal queues very shallow (or zero-sized, relying on channel rendezvous). If workers are busy, reject immediately rather than accumulating a massive queue of stale requests.

The boring rule

- Always place an upper bound on concurrent requests using a semaphore channel or rate limiter.
- Fail fast with HTTP 503 or 429 when concurrency limits are reached; include a `Retry-After` header.
- Avoid large in-memory queues. A queue that takes more than 1–2 seconds to drain under peak load should reject excess work.
- Always propagate `r.Context()` to database queries and external RPCs so aborted client connections cancel downstream work.

Try this

1. In `shed_demo.go`, increase `maxConcurrent` to 10 and verify that all 10 requests complete with status 200.
2. Add a `time.Duration` timeout to the channel send: `select { case sem <- struct{}{}: ... case <-time.After(5 * time.Millisecond): ... }` to implement bounded queue wait time before shedding.
3. Test how your HTTP client handles receiving status 503 by adding an exponential backoff retry loop.

Long-Lived Connections and Leak Prevention

Long-Lived Connections and Leak Prevention

The boring default is to enforce explicit **read and write deadlines** on every network socket. When building persistent streaming services—such as WebSockets, Server-Sent Events (SSE), or internal pub/sub feeds—connections are meant to remain open for minutes or hours. But when mobile clients lose coverage or network switches drop state without transmitting TCP termination packets (`FIN`/`RST`), server goroutines without deadlines sit blocked in the netpoller indefinitely, creating severe memory leaks.

Mental model

In TCP, a connection can become **half-open**: one endpoint crashes or loses power, but the other endpoint believes the socket is healthy because no FIN packet was ever exchanged.

If your Go server executes:

```go
// DANGER: Without a deadline, this blocks FOREVER on half-open sockets!
n, err := conn.Read(buf)
```

The goroutine will stay parked in `netpollblock` for days. Each leaked connection retains: 1. The goroutine stack (2 KB–8 KB+). 2. The `net.Conn` struct and internal runtime poller handles. 3. Any per-connection write buffers or channel subscriptions. 4. An open operating system file descriptor.

Over time, these "zombie goroutines" accumulate, slowly bleeding memory and eventually exhausting OS file descriptors.

The Heartbeat / Ping-Pong Pattern

To detect dead connections reliably: 1. The application establishes a heartbeat interval (e.g., ping every 30 seconds). 2. The server sets a read deadline slightly longer than the interval (e.g., 35 seconds). 3. Every time a valid packet or heartbeat arrives, the deadline is pushed forward: `conn.SetReadDeadline(time.Now().Add(timeout))` 4. If the client goes silent, `Read()` returns `os.ErrDeadlineExceeded`, allowing the server to close the socket and terminate the goroutine immediately.

Worked examples

Case 1: Active heartbeat monitoring and zombie eviction

Save as `heartbeat_demo.go`. A streaming server enforces a 60ms deadline. When a client sends a PING, the server replies with PONG. When the client subsequently goes silent, the server evicts the zombie connection and reclaims the goroutine.

```go
// heartbeat_demo.go
package main

import (
    "errors"
    "fmt"
    "io"
    "net"
    "os"
    "runtime"
    "sync"
    "time"
)

func handleStreamingConnection(c net.Conn, timeout time.Duration) {
    defer c.Close()

    buf := make([]byte, 64)
    for {
        // Enforce deadline: must receive heartbeat or data before timeout
        _ = c.SetReadDeadline(time.Now().Add(timeout))

        n, err := c.Read(buf)
        if err != nil {
            if errors.Is(err, os.ErrDeadlineExceeded) {
                fmt.Printf("Connection %s timed out (zombie evicted)\n", c.RemoteAddr())
            }
            return
        }

        msg := string(buf[:n])
        if msg == "PING\n" {
            _ = c.SetWriteDeadline(time.Now().Add(time.Second))
            _, _ = c.Write([]byte("PONG\n"))
        }
    }
}

func main() {
```

```go
    ln, err := net.Listen("tcp", "127.0.0.1:0")
    if err != nil {
        panic(err)
    }
    defer ln.Close()

    var wg sync.WaitGroup
    wg.Add(1)

    go func() {
        defer wg.Done()
        conn, err := ln.Accept()
        if err != nil {
            return
        }
        handleStreamingConnection(conn, 60*time.Millisecond)
    }()

    client, err := net.Dial("tcp", ln.Addr().String())
    if err != nil {
        panic(err)
    }
    defer client.Close()

    // Send 1 ping
    _, _ = client.Write([]byte("PING\n"))
    reply := make([]byte, 5)
    _, _ = io.ReadFull(client, reply)
    fmt.Printf("Client received reply: %s", reply)

    // Client goes silent (simulating lost network or sudden crash)
    // Wait for server deadline to trigger and evict zombie connection
    time.Sleep(100 * time.Millisecond)
    wg.Wait()

    fmt.Printf("Active goroutines remaining: %d\n", runtime.NumGoroutine())
}
```

Run:

```go run heartbeat_demo.go
```

Output:

```
Client received reply: PONG
Connection 127.0.0.1:35030 timed out (zombie evicted)
Active goroutines remaining: 1
```

The dead connection was closed automatically, preventing resource leakage.

The trap

A common trap is **ignoring write deadlines on slow clients** ("Slowloris" attack):

```
// WRONG: A malicious or stalled client refuses to read bytes
for event := range eventFeed {
    _, err := conn.Write(event.Bytes()) // Can block indefinitely if client TCP window is fu
}
```

If a client stops reading from its socket, the kernel TCP send buffer eventually fills up. The server's `conn.Write()` call will block forever, trapping the broadcaster goroutine.

Always set a write deadline before pushing data to persistent streams:

```
_ = conn.SetWriteDeadline(time.Now().Add(5 * time.Second))
_, err := conn.Write(data)
```

The boring rule

- Never call `conn.Read()` or `conn.Write()` in production without an active deadline.
- Implement an explicit heartbeat (ping/pong) protocol on all long-lived streams.
- If a client fails to respond within the heartbeat window, immediately close the connection.
- Monitor active goroutine counts with `runtime.NumGoroutine()` or Prometheus metrics to catch connection leaks early.

Try this

1. In `heartbeat_demo.go`, send a second `"PING\n"` from the client at 40ms. Verify that the connection remains active and extends its deadline.
2. In your HTTP server configuration, verify that `http.Server{ReadTimeout: 10*time.Second, WriteTimeout: 10*time.Second, IdleTimeout: 120*time.Second}` are explicitly configured.
3. Test what happens if the client continuously reads very slowly (1 byte per second) when the server sets a 2-second write deadline.

Low-Level Socket Options and Syscall Control

Low-Level Socket Options and Syscall Control

The boring default is to use standard `net.Listen("tcp", addr)` and `net.Dial("tcp", addr)`. Go configures safe, modern defaults out of the box, such as enabling `TCP_NODELAY`. However, when tuning ultra-low-latency services, managing multi-process architectures, or implementing zero-downtime upgrades, you need access to the underlying operating system file descriptors via **`net.ListenConfig.Control`** and **`syscall.RawConn`**.

Mental model

In Go, low-level socket configuration is hooked before the operating system binds or connects the socket:

- **`net.ListenConfig`**: Replaces `net.Listen`, providing a `Control` callback that receives the raw socket file descriptor (`fd`) before `bind()` and `listen()` syscalls are executed.
- **`net.Dialer.Control`**: Similarly provides the raw file descriptor before `connect()` is issued.
- **`syscall.RawConn`**: An interface representing a network socket that allows non-blocking execution of `syscall.SetsockoptInt`.

Essential Socket Options

1. **`TCP_NODELAY` (Disabling Nagle's Algorithm):**
    - Nagle's algorithm combines small outgoing data packets to fill maximum segment size (MSS) frames, reducing network packet congestion.
    - For interactive RPCs or low-latency financial/desk transactions, Nagle's algorithm introduces artificial delays (40ms–200ms) waiting for delayed ACKs.
    - **Go enables `TCP_NODELAY` by default on all TCP connections.** You rarely need to set it manually, but you must know it is active.
2. **`SO_REUSEPORT` (Port Sharing & Zero-Downtime):**
    - Standard sockets prevent multiple processes from binding to the same `IP:port` (`bind: address already in use`).
    - Enabling `SO_REUSEPORT` on Linux/macOS allows multiple independent processes or worker loops to bind to the identical port.
    - The Linux kernel load-balances incoming connection requests across the listeners directly in kernel space. This makes zero-downtime binary restarts and rolling process upgrades seamless.
3. **`TCP_KEEPIDLE` and `TCP_KEEPINTVL`:**

- Operating system-level TCP keepalive probes detect dead client cables and severed connections even if user-space code is temporarily paused.

Worked examples

Case 1: Enabling SO_REUSEPORT for multi-listener binding

Save as `reuseport_demo.go`. We create two separate listeners bound to the exact same TCP port using `net.ListenConfig.Control`.

```go
// reuseport_demo.go
package main

import (
    "context"
    "fmt"
    "net"
    "syscall"
    "time"
)

func createReusePortListener(addr string) (net.Listener, error) {
    lc := net.ListenConfig{
        Control: func(network, address string, c syscall.RawConn) error {
            var opErr error
            err := c.Control(func(fd uintptr) {
                // SO_REUSEPORT is 0xf on Linux
                opErr = syscall.SetsockoptInt(int(fd), syscall.SOL_SOCKET, 0xf, 1)
            })
            if err != nil {
                return err
            }
            return opErr
        },
    }
    return lc.Listen(context.Background(), "tcp", addr)
}

func main() {
    addr := "127.0.0.1:49152"

    // 1. Create first listener on port
    ln1, err := createReusePortListener(addr)
    if err != nil {
        panic(err)
    }
```

```go
    defer ln1.Close()
    fmt.Printf("Listener 1 successfully bound to %s\n", addr)

    // 2. Create second listener on EXACT SAME port (enabled by SO_REUSEPORT)
    ln2, err := createReusePortListener(addr)
    if err != nil {
        panic(err)
    }
    defer ln2.Close()
    fmt.Printf("Listener 2 successfully bound to %s (zero-downtime reload capable)\n", addr)

    // Verify clients can connect
    conn, err := net.DialTimeout("tcp", addr, time.Second)
    if err != nil {
        panic(err)
    }
    defer conn.Close()
    fmt.Println("Client successfully connected to SO_REUSEPORT socket")
}
```

Run:

```go
go run reuseport_demo.go
```

Output:

```
Listener 1 successfully bound to 127.0.0.1:49152
Listener 2 successfully bound to 127.0.0.1:49152 (zero-downtime reload capable)
Client successfully connected to SO_REUSEPORT socket
```

Without `SO_REUSEPORT`, the second `Listen` call would immediately return `bind: address already in use`. With it, the kernel accepts connections and distributes them evenly across both listener sockets.

Case 2: Inspecting TCP_NODELAY on an established connection

To verify or adjust `TCP_NODELAY` on an existing TCP connection:

```go
// nodelay_demo.go
package main

import (
    "fmt"
    "net"
)
```

```go
func main() {
    ln, _ := net.Listen("tcp", "127.0.0.1:0")
    defer ln.Close()

    go func() {
        c, _ := ln.Accept()
        if tcpConn, ok := c.(*net.TCPConn); ok {
            // Explicitly set NoDelay (true by default in Go)
            _ = tcpConn.SetNoDelay(true)
        }
        _ = c.Close()
    }()

    conn, _ := net.Dial("tcp", ln.Addr().String())
    defer conn.Close()

    fmt.Println("TCP_NODELAY configured successfully")
}
```

Run:

```go
go run nodelay_demo.go
```

Output:

```
TCP_NODELAY configured successfully
```

The trap

The most common trap with `SO_REUSEPORT` is assuming connection state is shared across processes. If process A accepts a connection, process B knows nothing about it. If process A terminates abruptly, any active connections currently held by process A are terminated.

For graceful zero-downtime restarts: 1. Start the new process with `SO_REUSEPORT`. 2. Signal the old process to stop accepting new connections (`ln.Close()`). 3. Allow the old process to drain existing in-flight transactions before terminating.

The boring rule

- Trust Go's default `net.Listen` and `net.Dial` for typical web services and microservices.
- Rely on Go's default `TCP_NODELAY=true` unless you are streaming massive continuous media files and intentionally want packet coalescing.
- Use `net.ListenConfig.Control` when you need OS-specific options (`SO_REUSEPORT`, `SO_RCVBUF`, or custom TCP keepalives).
- On Linux, prefer `SO_REUSEPORT` over complex socket passing via Unix domain sockets for zero-downtime process restarts.

Try this

1. In `reuseport_demo.go`, remove the `syscall.SetsockoptInt` call inside `Control`. Observe how the program fails on the second listener with `bind: address already in use`.
2. Inspect the raw file descriptor integer value inside `c.Control(func(fd uintptr) { ... })`.
3. Configure `SO_RCVBUF` and `SO_SNDBUF` on a listener socket to 16 KB and verify with `syscall.GetsockoptInt` that the kernel applied the setting.

TLS and DNS Performance

TLS and DNS Performance

The boring default is to let Go's standard library handle DNS queries and TLS handshakes automatically. However, in distributed architectures where microservices communicate over encrypted HTTPS/gRPC endpoints, **TLS handshakes** and **un-cached DNS lookups** often dominate tail latency (p99). Enabling TLS session resumption and choosing the appropriate DNS resolver eliminates unnecessary round trips and CPU overhead.

Mental model

The Cost of TLS Handshakes

Establishing a fresh TLS connection requires: 1. TCP 3-way handshake (1 RTT). 2. TLS 1.2 or TLS 1.3 cryptographic negotiation (1 to 2 RTTs), including exchanging certificates and deriving symmetric cipher keys.

On wide-area networks (WAN) where ping latency between data centers is 50ms, establishing a brand-new TLS connection adds **150–250ms** before a single byte of HTTP payload is transferred.

TLS Session Resumption

TLS Session Resumption (via session tickets in TLS 1.2 or Pre-Shared Keys in TLS 1.3) allows a client and server to reuse previously established cryptographic keys for subsequent connections: - Reduces handshake latency to **0 or 1 RTT**. - Skips heavy asymmetric CPU calculations (RSA / ECDHE key generation). - In Go clients, session resumption is enabled by configuring `tls.Config.ClientSessionCache = tls.NewLRUClientSessionCache(capacity)`.

Go DNS Resolvers: cgo vs netgo

Go ships two DNS resolvers: 1. **Pure Go Resolver (`netgo`):** Fully asynchronous, integrated into the Go runtime netpoller, and handles concurrent DNS queries without blocking OS threads. 2. **cgo Resolver (`getaddrinfo`):** Uses the host C standard library. Because `getaddrinfo` is synchronous and blocking, each DNS lookup blocks an operating system thread (`M`), which can cause thread pool thrashing during DNS timeouts.

By default, Go chooses the pure Go resolver on Linux unless specific conditions (like custom `/etc/nsswitch.conf` rules) force the cgo resolver. You can explicitly enforce the pure Go resolver via:

```go
resolver := &net.Resolver{
    PreferGo: true,
}
```

Or globally via the environment variable `GODEBUG=netdns=go`.

Furthermore, **Go's standard library does not cache DNS lookups**. Every time a connection pool needs to dial a new socket, it performs a fresh DNS network query unless you maintain connection pools or configure an in-memory caching dialer.

Worked examples

Case 1: Enabling TLS session resumption with ClientSessionCache

Save as `tls_dns_demo.go`. We configure a test HTTPS server and measure TLS handshake duration with and without session resumption via `httptrace.ClientTrace`.

```go
// tls_dns_demo.go
package main

import (
    "crypto/tls"
    "fmt"
    "net/http"
    "net/http/httptest"
    "net/http/httptrace"
    "time"
)

func main() {
    // 1. Create a local HTTPS test server
    server := httptest.NewTLSServer(http.HandlerFunc(func(w http.ResponseWriter, r *http.Req
        w.WriteHeader(http.StatusOK)
        _, _ = w.Write([]byte("ticket response"))
    }))
    defer server.Close()

    // 2. Client configured with TLS Session Cache
    tlsConfig := server.Client().Transport.(*http.Transport).TLSClientConfig.Clone()
    tlsConfig.ClientSessionCache = tls.NewLRUClientSessionCache(32)

    transport := &http.Transport{
        TLSClientConfig:     tlsConfig,
        DisableKeepAlives:   true, // Force fresh connections to test TLS handshake resumpti
        MaxIdleConnsPerHost: 10,
    }
```

```go
    client := &http.Client{Transport: transport}

    for i := range 2 {
        var tlsStart, tlsEnd time.Time
        var resumed bool

        trace := &httptrace.ClientTrace{
            TLSHandshakeStart: func() { tlsStart = time.Now() },
            TLSHandshakeDone: func(cs tls.ConnectionState, err error) {
                tlsEnd = time.Now()
                resumed = cs.DidResume
            },
        }

        req, _ := http.NewRequest(http.MethodGet, server.URL, nil)
        req = req.WithContext(httptrace.WithClientTrace(req.Context(), trace))

        resp, err := client.Do(req)
        if err != nil {
            panic(err)
        }
        _ = resp.Body.Close()

        fmt.Printf("Connection %d: TLS resumed=%v handshake_time=%v\n",
            i+1, resumed, tlsEnd.Sub(tlsStart))
    }
}
```

Run:

```go
go run tls_dns_demo.go
```

Output:

```
Connection 1: TLS resumed=false handshake_time=4.434383ms
Connection 2: TLS resumed=true handshake_time=1.405297ms
```

On Connection 1, a full cryptographic handshake took over 4ms. On Connection 2, the client resumed the session ticket (`TLS resumed=true`), cutting handshake duration by two-thirds.

Case 2: Configuring a Pure Go DNS dialer

Save as `dns_resolver.go`. We configure an `http.Transport` to use the non-blocking pure Go resolver:

```go
// dns_resolver.go
package main

import (
    "context"
    "fmt"
    "net"
    "net/http"
    "time"
)

func createOptimizedTransport() *http.Transport {
    dialer := &net.Dialer{
        Timeout:   5 * time.Second,
        KeepAlive: 30 * time.Second,
        Resolver: &net.Resolver{
            PreferGo:     true, // Use non-blocking netpoller DNS
            StrictErrors: false,
        },
    }

    return &http.Transport{
        DialContext:         dialer.DialContext,
        MaxIdleConns:        100,
        MaxIdleConnsPerHost: 50,
        IdleConnTimeout:     90 * time.Second,
    }
}

func main() {
    transport := createOptimizedTransport()
    client := &http.Client{
        Transport: transport,
        Timeout:   10 * time.Second,
    }

    req, _ := http.NewRequestWithContext(context.Background(), http.MethodGet, "http://127.0
    _ = client
    _ = req

    fmt.Println("Configured pure Go resolver transport successfully")
}
```

Run:

```go
go run dns_resolver.go
```

Output:

```
Configured pure Go resolver transport successfully
```

The trap

The most deceptive trap in DNS performance is **short TTLs in container environments**.

In Kubernetes or cloud VPCs, internal service names (e.g. `desk-inventory.prod.svc.cluster.local`) often have DNS TTLs of 5 or 30 seconds. If an HTTP client does not pool connections and dials on every request, hundreds of simultaneous queries flood the cluster DNS service (`CoreDNS`), causing packet drops and multi-second p99 timeouts.

The primary defense is **connection pooling**: as long as existing TCP connections remain idle in the pool and are reused, zero DNS lookups are performed.

The boring rule

- Always configure `ClientSessionCache: tls.NewLRUClientSessionCache(64)` on outbound HTTPS clients.
- Enforce the pure Go resolver (`PreferGo: true` or `GODEBUG=netdns=go`) on Linux to avoid blocking OS threads during slow DNS responses.
- Reuse connections via `http.Transport` to avoid querying DNS repeatedly for the same hosts.
- In container environments, watch CoreDNS latency metrics to detect client-side DNS thrashing.

Try this

1. In `tls_dns_demo.go`, remove `tlsConfig.ClientSessionCache`. Verify that `resumed` remains `false` for every connection.
2. Run your application with `GODEBUG=netdns=go+1` to print DNS debug trace messages for every hostname lookup.
3. Test how your HTTP transport behaves if you add an in-memory map to cache resolved IP addresses for 60 seconds inside `DialContext`.

HTTP/1 Connection Reuse in Go 1.27

HTTP/1 Connection Reuse in Go 1.27

Thesis

Through Go 1.26, a client that called `resp.Body.Close()` without reading the body to EOF **abandoned** the HTTP/1 connection. Status-only callers, and even `json.Decoder` stopping before a trailing newline, paid a fresh TCP+TLS handshake on every request. Starting in **Go 1.27**, `net/http.Transport` **asynchronously drains** an unread HTTP/1 body for up to **50 ms** or **256 KB** (and skips drain when `Connection: close` makes it useless). The boring code — get status, `defer resp.Body.Close()` — suddenly reuses connections again.

This chapter is about that **1.27 change**, as described by Nicholas Husin and Damien Neil on the go.dev blog. It is *not* a repeat of connection-pool knobs (`MaxIdleConnsPerHost`, idle timeouts); those still matter, but they only help when the body contract lets the Transport keep the socket.

Mental model

```
HTTP/1 on one TCP socket (responses are back-to-back bytes):

[ status+headers | body ........ | status+headers | body .... ]

                          Transport cannot find response  until body  is gone

Through 1.26:  Close() with unread body  →  drop connection
From 1.27:     Close() with unread body  →  background drain ( 50ms /  256KB)
                                            then idle-pool reuse if drain wins
```

Fact	Consequence
HTTP/1 is sequential on the wire	Unread body blocks the next response on that socket
HTTP/2 multiplexes streams	Unread body on one stream does **not** block others — auto-drain is an HTTP/1 concern
`http.Get` returns after headers	Body may still be sitting unread on the connection
Manual drain was error-prone	Cap bytes, cap time, run async, avoid draining HTTP/2 "for luck"
1.27 Transport does the hard part	Prefer simple `Close()`; keep reading when you need the payload

Setup cost reminder from the blog: new HTTPS connection 1 RTT TCP + 1–2 RTT TLS before any HTTP byte. On a ~100 ms RTT path that is hundreds of milliseconds you should not pay per request.

Worked examples

Verified with `go1.27.1` at `/home/box/sdk/go1.27.1/bin/go` on linux/amd64 (15 October 2026). Force HTTP/1 with `GODEBUG=http2client=0` so the lab exercises the path the blog measures.

Case 1: Status-only client reuses on 1.27

```go
// reuse_demo.go - Go 1.27 auto-drains unread HTTP/1 bodies
package main

import (
    "fmt"
    "io"
    "net/http"
    "net/http/httptest"
    "net/http/httptrace"
    "os"
    "runtime"
)

func main() {
    fmt.Printf("go=%s GODEBUG=%q\n", runtime.Version(), os.Getenv("GODEBUG"))

    srv := httptest.NewServer(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request)
        w.Header().Set("Content-Type", "application/json")
        // Trailing newline: classic leftover after json.Decoder
        _, _ = w.Write([]byte(`{"ok":true}` + "\n"))
    }))
    defer srv.Close()

    tr := &http.Transport{ForceAttemptHTTP2: false}
    client := &http.Client{Transport: tr}

    for i := 0; i < 3; i++ {
        var gotConn httptrace.GotConnInfo
        req, _ := http.NewRequest(http.MethodGet, srv.URL, nil)
        req = req.WithContext(httptrace.WithClientTrace(req.Context(), &httptrace.ClientTrac
            GotConn: func(info httptrace.GotConnInfo) { gotConn = info },
        }))
        resp, err := client.Do(req)
        if err != nil {
```

```
            panic(err)
        }
        code := resp.StatusCode
        // Status-only: never read the body. Go 1.27 should still reuse.
        _ = resp.Body.Close()
        fmt.Printf("req=%d status=%d reused=%v wasIdle=%v\n", i+1, code, gotConn.Reused, got
    }

    resp, _ := client.Get(srv.URL)
    n, _ := io.Copy(io.Discard, resp.Body)
    _ = resp.Body.Close()
    fmt.Printf("explicit_drain_bytes=%d\n", n)
}
```

```
export PATH=/home/box/sdk/go1.27.1/bin:$PATH
GODEBUG=http2client=0 go run reuse_demo.go
```

```
go=go1.27.1 GODEBUG="http2client=0"
req=1 status=200 reused=false wasIdle=false
req=2 status=200 reused=true wasIdle=false
req=3 status=200 reused=true wasIdle=true
explicit_drain_bytes=12
```

Request 1 dials; requests 2 and 3 show `reused=true` even though the body was never read in user code. That is the 1.27 behaviour. (`wasIdle` can flip depending on whether the async drain has already parked the socket in the idle pool — the signal that matters is `reused`.)

Case 2: Why 1.26 felt fine until it did not

The blog's index.golang.org microbenchmark (HTTP/1 forced): ~**710 ms** for 100 status-only GETs on Go 1.26 versus ~**150 ms** on Go 1.27 — nearly **5×** — because 1.26 paid handshake cost over and over. Any API that returns `{"success":true,"message":"..."}` while the client only checks the status code hits the same cliff.

Case 3: The JSON trailing-newline trap

`encoding/json.Decoder` stops at the closing `}` of a value. If the server appends `\n` (very common), one byte remains unread. Through 1.26 that single byte was enough to destroy reuse. In 1.27 the Transport's background drain consumes it within the 256 KB / 50 ms budget for typical API payloads.

```
// Still fine to read what you need; Close() is enough for the rest on 1.27.
dec := json.NewDecoder(resp.Body)
var payload struct {
```

```
    OK bool `json:"ok"`
}
_ = dec.Decode(&payload)
_ = resp.Body.Close()
```

Case 4: What 1.27 does that hand-rolled drain got wrong

A correct manual drain needed all of:

1. `io.CopyN(io.Discard, body, 256<<10)` — do not download a multi-gig body to save a handshake
2. A time cap (~50 ms) so a trickle cannot stall Close
3. Asynchronous drain so Close does not block the caller
4. Skip when `Connection: close` guarantees the socket dies anyway
5. Do **not** cargo-cult the same drain on HTTP/2

Go 1.27's Transport implements that policy (details may evolve). Prefer the simple Close path unless you are on an older Go and must polyfill.

The trap

1. **Assuming pool settings alone fix latency.** Raising `MaxIdleConnsPerHost` does nothing if every Close abandons the socket.
2. **Draining HTTP/2 bodies "because Stack Overflow said so".** Multiplexing already isolates streams; the expensive mistake was HTTP/1-specific.
3. **Blocking forever on `io.Copy(io.Discard, body)`** for huge responses — establishing a new connection can be cheaper; that is why the byte and time caps exist.

The boring rule

On Go **1.27+**, when you are finished with a response: **`resp.Body.Close()`** (usually via `defer`). Read the body when you need the bytes; do not invent a custom drain unless you are maintaining a 1.26-era compatibility shim. Keep a shared `*http.Client` / `*http.Transport` so reuse has a pool to return to.

Try this

1. Run `reuse_demo.go` under `GODEBUG=http2client=0` with go1.27.1; confirm `reused=true` from request 2.

2. Point the same status-only loop at a real HTTPS URL and compare wall time under Go 1.26 vs 1.27 if you have both toolchains.

3. Add `httptrace` `DNSStart` / `TLSHandshakeStart` hooks and watch them fire only on the first request when reuse works.

4. Read the upstream blog once; skim `net/http` Transport docs for any future tuning of the 50 ms / 256 KB policy.

Sources

- Nicholas Husin & Damien Neil, *Better HTTP/1 connection reuse*, go.dev blog (14 October 2026): https://go.dev/blog/http-connection-reuse
- Go 1.27 release notes / `net/http` package docs: https://go.dev/doc/go1.27, https://pkg.go.dev/net/http
- Inspiration: https://x.com/golang/status/2110451069855896047

Web Development in Go

Web Development in Go

Web Development in Go

This part is a **hands-on curriculum** for Go web development: project structure, HTTP and routing, templates, databases, concurrency at the edge, sessions and auth, frontend/backend contracts, and testing—all explained in simple, direct language.

The real code and examples make the concepts easy to use. Coverage spans nearly every major aspect of Go web work, but keeps examples straightforward and avoids complex edge cases. That makes it ideal for learners and practitioners who want clarity and reliable progress.

You will learn to structure projects, design backend services, manage sessions, secure authentication, and integrate external APIs with step-by-step **Bookstore** examples. Techniques cover routing, concurrency, testing, logging, performance, and error handling—with code that is easy to understand and explanations that are easy to follow.

This edition focuses on **modern Go features without niche scenarios**, so lessons transfer to real projects. Simplicity over unnecessary detail takes you from setup toward deployment.

> **How this part relates to the rest of the book.** Part `08-web` covers modern stacks (GOTH, gRPC, Huma, Postgres/Redis). Part `13-network-systems` and deep dives go lower. **This part** is the practical spine: net/http first, clear modules, Bookstore REST + HTML, then secure auth and tests. Read it when you want a full web app path without framework overload.

Tooling baseline: Go 1.22+ routing (`ServeMux` method patterns) is the default. Gorilla Mux appears where method/path variables were historically taught; prefer stdlib mux for new work unless you need Mux-specific features. Use `go mod init`, `go test -race`, and structured logging (`log/slog`).

Learning outcomes

After this part you can:

1. Scaffold a modular Go web project with clear package boundaries
2. Serve HTTP, route requests, and return JSON or HTML consistently
3. Render templates and static assets without mixing business logic into handlers
4. Talk to a database through interfaces that stay testable
5. Use concurrency for I/O-bound work without leaking goroutines
6. Implement sessions, password hashing, and role-based access
7. Define stable API contracts for frontend/backend interchange
8. Mock dependencies and write focused automated tests
9. Log, surface errors, and keep incident response simple

Key features

Focus	What you practice
Project shape	Organised code, modular design, real workflows
REST APIs	`net/http`, optional Gorilla Mux, JSON habits
Security	Auth, roles, password hashing, secure cookies
Contracts	Clear API shapes, simple JSON interchange
Concurrency	Scale data fetching, avoid deadlocks
Testing	Mock DB/APIs/deps for fast, isolated tests
Sessions	Cookies, resilient login flows
Ops	Logging, tracing events, find problems quickly
Integration	HTTP client patterns for payments and third parties
Errors	Understandable messages, standard response plans

Contents map

```
Setup:      intro → structure → HTTP/routing
Content:    templates → databases
Runtime:    concurrency → sessions/auth
Boundary:   frontend contracts → testing/debugging
Production: middleware → shutdown → rate limit → WS → uploads → capstone
```

Chapter	File	Focus
270	this page	overview, path, Bookstore map
271	271-intro-web-dev-go	why Go for web, request lifecycle, first server
272	272-structuring-go-web-app	modules, packages, config, dependency wiring
273	273-http-requests-routing	handlers, mux, middleware, JSON REST
274	274-templating-rendering	`html/template`, layouts, static files
275	275-databases	`database/sql`, repositories, Bookstore catalog
276	276-concurrency-web	parallel fetches, timeouts, safe patterns
277	277-sessions-auth-authz	cookies, sessions, passwords, roles
278	278-frontend-backend-communication	API contracts, clients, external APIs
279	279-testing-debugging	httptest, mocks, logging, performance checks

Chapter	File	Focus
280	280-middleware-patterns	chain, recover, request ID, access log
281	281-graceful-shutdown-web	Shutdown, readiness drain, K8s
282	282-rate-limiting	concurrency cap, token bucket
283	283-websockets-basics	upgrade, heartbeats, hubs
284	284-file-uploads-downloads	multipart, ServeContent, static
285	285-project-bookstore-api-capstone	full Bookstore API project

Suggested path

```
Week-shaped path (adjust pace):

  Day 1     271 → 272          first server + layout
  Day 2     273                routing + REST bookstore API
  Day 3     274 → 275          HTML views + database
  Day 4     276 → 277          concurrency + auth
  Day 5     278 → 279          contracts + tests + logging
  Day 6     280 → 282          middleware, shutdown, limits
  Day 7     283 → 285          websockets, uploads, capstone
```

Running example: Bookstore

Throughout this part you build pieces of a small bookstore service:

```
  Browser / SPA / curl



        HTTP API      /api/books, /api/auth/...
        + HTML UI     /books, /login



        domain +      Book, User, Order
        repository


         database / mocks
```

Keep each chapter's snippets runnable with `go mod init` and a short `main` or `_test.go`. Prefer interfaces at boundaries so chapter 279 can mock cleanly.

Introduction to Web Development in Go

Introduction to Web Development in Go

Overview

Go is a strong fit for web backends: one binary, a solid standard library HTTP stack, clear concurrency, and boring reliability. This chapter sets the mental model—how a request moves through your process—and gets a minimal server running.

Why Go for web services

Strength	What it means in practice
`net/http` in stdlib	Ship APIs without a framework tax
Goroutines	Handle many concurrent connections simply
Static typing + `go test`	Refactor routes and handlers with confidence
Single binary	Deploy the same artifact to VM, container, or bare metal
Explicit errors	Surface failures at the boundary instead of hidden exceptions

You still need good structure, timeouts, and tests. Go does not replace design; it keeps the implementation honest.

Request lifecycle

```
Client                Server process

    TCP + HTTP request
                 ListenAndServe accepts connection
                            ServeMux matches method + path
                            Middleware (log, auth, recover)
                            Handler reads body / query
                            Domain / DB / external calls
    status + headers+body
                 Response written, connection reuse
```

Rule: Handlers should be thin. Parse input → call application code → write response. Keep SQL, payment calls, and business rules out of the raw `http.Handler` body when the app grows.

First server

```
mkdir -p /tmp/bookstore-intro && cd /tmp/bookstore-intro
go mod init example.com/bookstore
```

Save as `main.go`:

```go
package main

import (
    "encoding/json"
    "log"
    "net/http"
    "time"
)

func main() {
    mux := http.NewServeMux()
    mux.HandleFunc("GET /healthz", healthz)
    mux.HandleFunc("GET /api/hello", hello)

    srv := &http.Server{
        Addr:              ":8080",
        Handler:           mux,
        ReadHeaderTimeout: 5 * time.Second,
        ReadTimeout:       10 * time.Second,
        WriteTimeout:      10 * time.Second,
        IdleTimeout:       60 * time.Second,
    }

    log.Println("listening on :8080")
    log.Fatal(srv.ListenAndServe())
}

func healthz(w http.ResponseWriter, r *http.Request) {
    w.WriteHeader(http.StatusOK)
    _, _ = w.Write([]byte("ok"))
}

func hello(w http.ResponseWriter, r *http.Request) {
    w.Header().Set("Content-Type", "application/json")
    _ = json.NewEncoder(w).Encode(map[string]string{
        "service": "bookstore",
        "message": "welcome",
    })
}
```

Run and probe:

```
go run .
# other terminal
curl -s localhost:8080/healthz
curl -s localhost:8080/api/hello
```

What to notice

- **Method-aware patterns** (`GET /path`) need Go 1.22+. Older Go used path-only matching.
- **Timeouts on `http.Server`** are not optional for production; they bound stuck clients.
- **JSON via `encoding/json`** is enough for most REST bodies.

Status codes you will use constantly

Code	When
200	Success with body
201	Created resource
204	Success, no body
400	Client sent bad input
401	Not authenticated
403	Authenticated but not allowed
404	Missing resource
409	Conflict (duplicate ISBN, etc.)
500	Unexpected server failure

Prefer **specific 4xx** over a generic 500 when the client can fix the request.

Logging from day one

```go
import "log/slog"

func withRequestLog(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        start := time.Now()
        next.ServeHTTP(w, r)
        slog.Info("request",
            "method", r.Method,
            "path", r.URL.Path,
            "duration_ms", time.Since(start).Milliseconds(),
        )
    })
```

```
}
```

Wire it as `Handler: withRequestLog(mux)` on the server. Chapter 279 expands correlation and tests.

Error shape for APIs

Agree early on a small JSON error envelope:

```go
type APIError struct {
    Error string `json:"error"`
    Code  string `json:"code,omitempty"`
}

func writeError(w http.ResponseWriter, status int, code, msg string) {
    w.Header().Set("Content-Type", "application/json")
    w.WriteHeader(status)
    _ = json.NewEncoder(w).Encode(APIError{Error: msg, Code: code})
}
```

Clients and frontends can branch on `code` without scraping free-text messages.

Bookstore north star

By the end of this part the same process will expose:

- `GET /api/books` — list catalog
- `GET /api/books/{id}` — book detail
- `POST /api/books` — create (admin)
- `POST /api/auth/login` — session cookie
- HTML pages that reuse the same domain layer

Rules of thumb

Do	Don't
Set server timeouts	Use bare `http.ListenAndServe` in production forever
Return structured JSON errors	Mix ad-hoc string bodies
Keep handlers thin	Put SQL and payment logic only in handlers
Log method, path, duration	Log secrets, cookies, full credit cards

Try next

1. Add `GET /api/version` returning a build string.
2. Return 404 JSON for unknown paths via a catch-all or custom `NotFoundHandler`.
3. Hit the server with concurrent `curl` loops and watch logs stay readable.

Structuring Go Web Applications

Structuring Go Web Applications

Overview

A modular layout keeps web apps maintainable: HTTP stays at the edge, domain rules stay pure, and databases hide behind interfaces. This chapter shows a practical Bookstore layout—not the only valid layout, but one that scales from a tutorial to a small product.

Goals of structure

1. **Find things fast** — routes, handlers, domain, persistence in predictable places
2. **Test in isolation** — swap real DB for mocks without rewriting handlers
3. **Avoid circular imports** — packages depend inward (domain) not sideways forever
4. **Ship one binary** — `cmd/bookstore` is the process entry; libraries live under `internal/`

Recommended layout

```
bookstore/
  go.mod
  cmd/
    bookstore/
      main.go           # wiring only: config, DI, ListenAndServe
  internal/
    config/
      config.go         # env + defaults
    domain/
      book.go           # types + pure validation
      user.go
    store/
      store.go          # interfaces (BookRepository, ...)
      memory/
        book_repo.go    # in-memory for demos/tests
      postgres/
        book_repo.go    # real SQL later
    httpapi/
      server.go         # routes + middleware stack
      books.go          # book handlers
      auth.go
```

```
      respond.go        # JSON helpers
    web/
      templates/        # html/template files
      static/           # css/js
  migrations/           # optional SQL migrations
```

`internal/` is invisible to other modules—good for app-private code.

Dependency direction

```
  cmd/bookstore


   httpapi   uses    store (interfaces)

                              implements

    domain         store/memory, store/postgres
```

- **domain** knows nothing about HTTP or SQL.
- **store** defines interfaces and implementations.
- **httpapi** depends on interfaces + domain types.
- **main** constructs concrete types and injects them.

Config from environment

```go
package config

import (
    "os"
    "time"
)

type Config struct {
    Addr            string
    DatabaseURL     string
    SessionSecret   string
    ShutdownTimeout time.Duration
}

func Load() Config {
    return Config{
        Addr:            getenv("ADDR", ":8080"),
```

```go
        DatabaseURL:     os.Getenv("DATABASE_URL"),
        SessionSecret:   getenv("SESSION_SECRET", "dev-only-change-me"),
        ShutdownTimeout: 10 * time.Second,
    }
}

func getenv(k, def string) string {
    if v := os.Getenv(k); v != "" {
        return v
    }
    return def
}
```

Never commit production secrets. Dev defaults are fine if loudly unsafe.

Domain types (Bookstore)

```go
package domain

import "fmt"

type Book struct {
    ID     string `json:"id"`
    Title  string `json:"title"`
    Author string `json:"author"`
    ISBN   string `json:"isbn"`
    Price  int    `json:"price_cents"` // store money as integer cents
}

func (b Book) Validate() error {
    if b.Title == "" {
        return fmt.Errorf("title is required")
    }
    if b.Author == "" {
        return fmt.Errorf("author is required")
    }
    if b.Price < 0 {
        return fmt.Errorf("price_cents must be >= 0")
    }
    return nil
}
```

Validation on the domain type keeps handlers short and reuses the same rules for HTML forms and JSON APIs.

Repository interface

```go
package store

import (
    "context"

    "example.com/bookstore/internal/domain"
)

type BookRepository interface {
    List(ctx context.Context) ([]domain.Book, error)
    Get(ctx context.Context, id string) (domain.Book, error)
    Create(ctx context.Context, b domain.Book) (domain.Book, error)
}
```

Handlers depend on `BookRepository`, not on Postgres. Chapter 275 implements memory and SQL; chapter 279 mocks this interface.

Wiring in main

```go
package main

import (
    "log"
    "net/http"
    "time"

    "example.com/bookstore/internal/config"
    "example.com/bookstore/internal/httpapi"
    "example.com/bookstore/internal/store/memory"
)

func main() {
    cfg := config.Load()
    books := memory.NewBookRepo()
    api := httpapi.NewServer(books)

    srv := &http.Server{
        Addr:              cfg.Addr,
        Handler:           api.Handler(),
        ReadHeaderTimeout: 5 * time.Second,
    }
    log.Printf("bookstore listening on %s", cfg.Addr)
    log.Fatal(srv.ListenAndServe())
```

```
}
```

main is assembly, not business logic. When you add auth, sessions, and Postgres, only wiring and constructors grow here.

Package naming habits

Prefer	Avoid
`httpapi`, `store`, `domain`	`utils`, `helpers`, `common` dumping grounds
Short import paths under `internal/`	Giant `models` package everything imports
One concern per package	Circular `handler  service  handler`

If two packages need each other constantly, merge them or extract a smaller shared types package.

Modular design without over-engineering

For a learning Bookstore you do **not** need:

- Full clean-architecture ceremony with six layers
- A DI framework
- Microservices split by "books" vs "users"

You **do** need:

- Interfaces at I/O boundaries
- `context.Context` on store methods
- Config outside code
- A single place that registers routes

Graceful shutdown sketch

```go
go func() {
    log.Fatal(srv.ListenAndServe())
}()

// on signal:
ctx, cancel := context.WithTimeout(context.Background(), cfg.ShutdownTimeout)
defer cancel()
_ = srv.Shutdown(ctx)
```

Production processes should finish in-flight requests instead of hard-killing connections.

Rules of thumb

Do	Don't
Put app code under `internal/`	Export everything as a public library by accident
Inject repositories into handlers	Reach for package-level global DB from every file
Keep `main` thin	Grow a 2k-line `main.go` of handlers
Validate in domain or a small service	Duplicate validation in every handler

Try next

1. Scaffold the directories above and compile with an empty mux.
2. Move `healthz` into `httpapi` and construct it from `main`.
3. Add a `UserRepository` interface stub for the auth chapter.

Handling HTTP Requests and Routing

Handling HTTP Requests and Routing

Overview

Routing maps method + path to a handler. Handlers parse input, call your domain/store, and write responses. This chapter builds Bookstore REST endpoints with `net/http`, shows middleware, and notes when Gorilla Mux still appears in older material.

ServeMux (Go 1.22+)

```go
mux := http.NewServeMux()
mux.HandleFunc("GET /api/books", s.listBooks)
mux.HandleFunc("POST /api/books", s.createBook)
mux.HandleFunc("GET /api/books/{id}", s.getBook)
```

Path wildcards: `{id}` is available as `r.PathValue("id")`.

Handler shape

```go
type Server struct {
    books store.BookRepository
}

func NewServer(books store.BookRepository) *Server {
    return &Server{books: books}
}

func (s *Server) Handler() http.Handler {
    mux := http.NewServeMux()
    mux.HandleFunc("GET /api/books", s.listBooks)
    mux.HandleFunc("GET /api/books/{id}", s.getBook)
    mux.HandleFunc("POST /api/books", s.createBook)
    return withRecover(withRequestLog(mux))
}
```

List and get (JSON)

```
func (s *Server) listBooks(w http.ResponseWriter, r *http.Request) {
    items, err := s.books.List(r.Context())
    if err != nil {
        writeError(w, http.StatusInternalServerError, "store_error", "could not list books")
        return
    }
    writeJSON(w, http.StatusOK, items)
}

func (s *Server) getBook(w http.ResponseWriter, r *http.Request) {
    id := r.PathValue("id")
    book, err := s.books.Get(r.Context(), id)
    if err != nil {
        writeError(w, http.StatusNotFound, "not_found", "book not found")
        return
    }
    writeJSON(w, http.StatusOK, book)
}
```

Create with validation

```
func (s *Server) createBook(w http.ResponseWriter, r *http.Request) {
    defer r.Body.Close()
    r.Body = http.MaxBytesReader(w, r.Body, 1<<20) // 1 MiB

    var in domain.Book
    dec := json.NewDecoder(r.Body)
    dec.DisallowUnknownFields()
    if err := dec.Decode(&in); err != nil {
        writeError(w, http.StatusBadRequest, "bad_json", "invalid JSON body")
        return
    }
    if err := in.Validate(); err != nil {
        writeError(w, http.StatusBadRequest, "validation", err.Error())
        return
    }

    out, err := s.books.Create(r.Context(), in)
    if err != nil {
        writeError(w, http.StatusConflict, "create_failed", err.Error())
        return
    }
```

```go
    writeJSON(w, http.StatusCreated, out)
}
```

Habits that prevent production pain:

- Limit body size (`MaxBytesReader`)
- `DisallowUnknownFields` when you want strict clients
- Use `r.Context()` for cancel on client disconnect

Response helpers

```go
func writeJSON(w http.ResponseWriter, status int, v any) {
    w.Header().Set("Content-Type", "application/json; charset=utf-8")
    w.WriteHeader(status)
    _ = json.NewEncoder(w).Encode(v)
}
```

Always set `Content-Type` before `WriteHeader`.

Middleware

Middleware wraps `http.Handler`—logging, auth, CORS, panic recovery.

```go
func withRequestLog(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        start := time.Now()
        next.ServeHTTP(w, r)
        slog.Info("http",
            "method", r.Method,
            "path", r.URL.Path,
            "ms", time.Since(start).Milliseconds(),
        )
    })
}

func withRecover(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        defer func() {
            if rec := recover(); rec != nil {
                slog.Error("panic", "err", rec)
                writeError(w, http.StatusInternalServerError, "panic", "internal error")
            }
        }()
        next.ServeHTTP(w, r)
```

```
    })
}
```

Order matters: recover outermost so panics in logging still convert to 500s.

```
  request → recover → log → auth → mux → handler
  response
```

Query parameters and headers

```
q := r.URL.Query().Get("q")          // search term
auth := r.Header.Get("Authorization")
```

For Bookstore search:

```
// GET /api/books?q=gopher
func (s *Server) listBooks(w http.ResponseWriter, r *http.Request) {
    items, err := s.books.List(r.Context())
    // ... filter by title containing q ...
}
```

Gorilla Mux (when you see it)

Many tutorials and older codebases use Gorilla Mux:

```
r := mux.NewRouter()
r.HandleFunc("/api/books/{id}", getBook).Methods(http.MethodGet)
// id := mux.Vars(r)["id"]
```

Today: prefer stdlib `ServeMux` method patterns for new Bookstore code. Reach for Mux (or another router) only if you need features you truly miss (subrouters with custom matchers, complex host rules). Concepts transfer: method restriction, path vars, middleware.

Method not allowed vs not found

With Go 1.22 patterns, a path that exists for GET but not POST can yield **405** when another method is registered for that path. Unknown paths stay **404**. Test both; clients depend on them.

REST habits for Bookstore

Method	Path	Action
GET	`/api/books`	list / search
GET	`/api/books/{id}`	detail
POST	`/api/books`	create
PUT/PATCH	`/api/books/{id}`	update
DELETE	`/api/books/{id}`	remove

Keep collections plural. Use nouns, not verbs (`/api/books` not `/api/getBooks`).

Rules of thumb

Do	Don't
Parse → validate → act → respond	Stream huge bodies into memory blindly
Propagate `r.Context()`	Ignore cancellation mid-query
Centralize JSON write/error helpers	Copy-paste header/status in every handler
Prefer stdlib mux for new work	Add a router dependency with no gain

Try next

1. Implement `DELETE /api/books/{id}` returning 204.
2. Add middleware that rejects non-JSON `Content-Type` on POST.
3. Write a table of curl commands for every route as a smoke checklist.

Templating and Rendering Content

Templating and Rendering Content

Overview

Not every client is JSON. Server-rendered HTML remains a clear way to ship admin UIs and simple Bookstore pages. Go's `html/template` auto-escapes content by default—use it for user-facing HTML. Keep templates dumb: pass view models, not open database handles.

html/template vs text/template

Package	Use
`html/template`	Browsers (escapes HTML, JS, CSS contexts)
`text/template`	Emails, configs, code gen—not HTML pages

Never put untrusted user input into `text/template` and serve as HTML.

Minimal page

`internal/web/templates/layout.tmpl`:

```
{{define "layout"}}
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <title>{{block "title" .}}Bookstore{{end}}</title>
  <link rel="stylesheet" href="/static/site.css">
</head>
<body>
  <header><a href="/books">Bookstore</a></header>
  <main>
    {{block "content" .}}{{end}}
  </main>
</body>
</html>
{{end}}
```

internal/web/templates/books_list.tmpl:

```
{{define "title"}}Catalog{{end}}
{{define "content"}}
<h1>Books</h1>
<ul>
  {{range .Books}}
    <li><a href="/books/{{.ID}}">{{.Title}}</a> - {{.Author}}</li>
  {{else}}
    <li>No books yet.</li>
  {{end}}
</ul>
{{end}}
```

Parse once, execute many

```
package httpapi

import (
    "html/template"
    "io/fs"
    "net/http"
)

type Server struct {
    books store.BookRepository
    tmpl  *template.Template
}

func NewServer(books store.BookRepository, templateFS fs.FS) (*Server, error) {
    t, err := template.ParseFS(templateFS, "templates/*.tmpl")
    if err != nil {
        return nil, err
    }
    return &Server{books: books, tmpl: t}, nil
}

func (s *Server) booksPage(w http.ResponseWriter, r *http.Request) {
    items, err := s.books.List(r.Context())
    if err != nil {
        http.Error(w, "unavailable", http.StatusInternalServerError)
        return
    }
    data := struct {
        Books []domain.Book
```

```
    }{Books: items}

    w.Header().Set("Content-Type", "text/html; charset=utf-8")
    if err := s.tmpl.ExecuteTemplate(w, "layout", data); err != nil {
        // header may already be sent; log and move on
        slog.Error("template", "err", err)
    }
}
```

Embed templates in the binary:

```
//go:embed templates/* static/*
var webFS embed.FS
```

Static files

```
mux.Handle("GET /static/", http.StripPrefix("/static/",
    http.FileServer(http.FS(staticFS))))
```

Set long cache headers only for fingerprinted assets in production.

Forms (create book)

Template snippet:

```
<form method="POST" action="/books">
  <label>Title <input name="title" required></label>
  <label>Author <input name="author" required></label>
  <label>Price (cents) <input name="price_cents" type="number" min="0"></label>
  <button type="submit">Add book</button>
</form>
```

Handler:

```
func (s *Server) createBookForm(w http.ResponseWriter, r *http.Request) {
    if err := r.ParseForm(); err != nil {
        http.Error(w, "bad form", http.StatusBadRequest)
        return
    }
    price, _ := strconv.Atoi(r.FormValue("price_cents"))
    b := domain.Book{
        Title:  r.FormValue("title"),
        Author: r.FormValue("author"),
        Price:  price,
```

```
    }
    if err := b.Validate(); err != nil {
        http.Error(w, err.Error(), http.StatusBadRequest)
        return
    }
    if _, err := s.books.Create(r.Context(), b); err != nil {
        http.Error(w, err.Error(), http.StatusConflict)
        return
    }
    http.Redirect(w, r, "/books", http.StatusSeeOther)
}
```

POST → redirect → GET avoids duplicate form submits on refresh.

Escaping and safety

`html/template` escapes `{{.Title}}`. If you must inject trusted HTML:

```
// only for trusted, sanitized HTML
type page struct {
    Body template.HTML
}
```

Treat `template.HTML` as a sharp tool—never cast raw user input.

Sharing data with JSON APIs

Prefer one domain layer for both HTML and JSON:

```
GET /books          → HTML list (templates)
GET /api/books      → JSON list (same BookRepository)
```

Do not reimplement list logic twice.

Partial rendering / HTMX (optional)

If you later adopt HTMX (see part `08-web`), return **fragments** from the same templates (`{{define "book_row"}}`) instead of full layouts. The structure here still holds: parse once, execute named templates.

Rules of thumb

Do	Don't
Parse templates at startup	Parse from disk on every request
Pass view structs	Pass `*sql.DB` into templates
Use `html/template` for pages	Build HTML with string concat + user data
Redirect after POST	Re-render POST responses that create data

Try next

1. Add a book detail page at `GET /books/{id}`.
2. Show a flash-style error message when validation fails (query param or session).
3. Embed templates with `//go:embed` and confirm the binary serves pages without the source tree.

Interaction with Databases

Interaction with Databases

Overview

Bookstore data needs a durable store. Go's `database/sql` is the standard entry point: open a pool, run queries with context, scan into domain types. Keep SQL behind a repository so handlers stay free of driver details—and so tests can swap an in-memory implementation.

Principles

1. **Context on every query** — cancel when the client disconnects
2. **Interfaces at the boundary** — `BookRepository` in chapter 272
3. **Money as integers** — `price_cents`, not `float64`
4. **Pool settings matter** — open connections, idle lifetime, max lifetime
5. **Migrations outside the hot path** — version schema deliberately
6. **IDs** — Go 1.27+ can use stdlib `uuid` (`uuid.New().String()` or `uuid.NewV7().String()`) instead of `github.com/google/uuid`

Schema (Postgres-flavored)

```sql
CREATE TABLE books (
  id          TEXT PRIMARY KEY,
  title       TEXT NOT NULL,
  author      TEXT NOT NULL,
  isbn        TEXT NOT NULL UNIQUE,
  price_cents INT  NOT NULL CHECK (price_cents >= 0),
  created_at  TIMESTAMPTZ NOT NULL DEFAULT now()
);
```

SQLite works for local demos; production Bookstores often use Postgres.

Open a pool

```go
import (
    "database/sql"
    "time"

    _ "github.com/jackc/pgx/v5/stdlib" // or modernc.org/sqlite, etc.
)

func OpenDB(url string) (*sql.DB, error) {
    db, err := sql.Open("pgx", url)
    if err != nil {
        return nil, err
    }
    db.SetMaxOpenConns(10)
    db.SetMaxIdleConns(5)
    db.SetConnMaxLifetime(time.Hour)

    ctx, cancel := context.WithTimeout(context.Background(), 3*time.Second)
    defer cancel()
    if err := db.PingContext(ctx); err != nil {
        _ = db.Close()
        return nil, err
    }
    return db, nil
}
```

`sql.Open` does not always connect immediately—**Ping** (with timeout) fails fast at boot.

Postgres repository

```go
package postgres

import (
    "context"
    "database/sql"
    "errors"
    "fmt"

    "example.com/bookstore/internal/domain"
    "uuid"
)

type BookRepo struct {
```

```
    db *sql.DB
}

func NewBookRepo(db *sql.DB) *BookRepo {
    return &BookRepo{db: db}
}

func (r *BookRepo) List(ctx context.Context) ([]domain.Book, error) {
    rows, err := r.db.QueryContext(ctx,
        `SELECT id, title, author, isbn, price_cents FROM books ORDER BY title`)
    if err != nil {
        return nil, err
    }
    defer rows.Close()

    var out []domain.Book
    for rows.Next() {
        var b domain.Book
        if err := rows.Scan(&b.ID, &b.Title, &b.Author, &b.ISBN, &b.Price); err != nil {
            return nil, err
        }
        out = append(out, b)
    }
    return out, rows.Err()
}

func (r *BookRepo) Get(ctx context.Context, id string) (domain.Book, error) {
    var b domain.Book
    err := r.db.QueryRowContext(ctx,
        `SELECT id, title, author, isbn, price_cents FROM books WHERE id = $1`, id,
    ).Scan(&b.ID, &b.Title, &b.Author, &b.ISBN, &b.Price)
    if errors.Is(err, sql.ErrNoRows) {
        return domain.Book{}, fmt.Errorf("book %s: %w", id, errNotFound)
    }
    return b, err
}

func (r *BookRepo) Create(ctx context.Context, b domain.Book) (domain.Book, error) {
    if b.ID == "" {
        b.ID = uuid.New().String()
    }
    _, err := r.db.ExecContext(ctx,
        `INSERT INTO books (id, title, author, isbn, price_cents)
         VALUES ($1,$2,$3,$4,$5)`,
        b.ID, b.Title, b.Author, b.ISBN, b.Price,
    )
    return b, err
```

```
}
```

Define a package-level `errNotFound` (or domain sentinel) so handlers map to 404 with `errors.Is`.

In-memory repository (demo + tests)

```go
package memory

import (
    "context"
    "fmt"
    "sync"

    "example.com/bookstore/internal/domain"
    "uuid"
)

type BookRepo struct {
    mu   sync.RWMutex
    byID map[string]domain.Book
}

func NewBookRepo() *BookRepo {
    return &BookRepo{byID: map[string]domain.Book{}}
}

func (r *BookRepo) List(ctx context.Context) ([]domain.Book, error) {
    r.mu.RLock()
    defer r.mu.RUnlock()
    out := make([]domain.Book, 0, len(r.byID))
    for _, b := range r.byID {
        out = append(out, b)
    }
    return out, nil
}

func (r *BookRepo) Get(ctx context.Context, id string) (domain.Book, error) {
    r.mu.RLock()
    defer r.mu.RUnlock()
    b, ok := r.byID[id]
    if !ok {
        return domain.Book{}, fmt.Errorf("not found")
    }
    return b, nil
}
```

```go
func (r *BookRepo) Create(ctx context.Context, b domain.Book) (domain.Book, error) {
    r.mu.Lock()
    defer r.mu.Unlock()
    if b.ID == "" {
        b.ID = uuid.New().String()
    }
    if _, exists := r.byID[b.ID]; exists {
        return domain.Book{}, fmt.Errorf("duplicate id")
    }
    r.byID[b.ID] = b
    return b, nil
}
```

Use memory in early chapters and unit tests; switch `main` to Postgres when ready.

Transactions

Checkout / multi-table writes need transactions:

```go
tx, err := db.BeginTx(ctx, nil)
if err != nil {
    return err
}
defer tx.Rollback() // no-op after Commit

// tx.ExecContext(...); tx.ExecContext(...)
return tx.Commit()
```

Keep transactions short. Do not hold a tx open across external HTTP calls (payment gateway)—prepare locally, then commit, or use outbox patterns later.

N+1 and simple query design

Bad: list books, then one query per author detail in a loop.
Good: join or batch IDs in one query.

```go
SELECT b.id, b.title, a.name
FROM books b
JOIN authors a ON a.id = b.author_id
WHERE b.id = ANY($1);
```

Connection errors vs not found

Situation	Handler response
`sql.ErrNoRows`	404
Unique violation	409
Context deadline	504 or 503
Driver/network	500 + log

Log the real error; return a safe client message.

Rules of thumb

Do	Don't
Parameterized queries (`$1`, `?`)	String-concatenate user input into SQL
Set pool limits	Leave unlimited open connections
Hide SQL behind repositories	Scatter queries through every handler
Test with memory or testcontainers	Require a full prod DB for every unit test

Try next

1. Add `Update` and `Delete` to both memory and SQL repos.
2. Seed three sample books at startup in memory mode.
3. Map unique ISBN conflicts to HTTP 409 in the create handler.

Concurrency in Go Web Services

Concurrency in Go Web Services

Overview

Each HTTP request already runs on its own goroutine under `net/http`. Extra concurrency pays off when a single handler must fetch several independent resources, call external APIs, or process batches—without blocking the whole response on serial I/O. This chapter applies Go concurrency tools to Bookstore-style backends and avoids the usual deadlocks and leaks.

> For a full concurrency curriculum (diagrams, Cond, atomics), use part **21 (Concurrency From the Ground Up)**. Here we stay web-focused.

What the server already gives you

```
Client A    goroutine    handler A
Client B    goroutine    handler B
Client C    goroutine    handler C
```

You do **not** start a goroutine per request yourself. You start goroutines **inside** a handler when work can run in parallel, then wait before writing the response.

Parallel independent fetches

Example: book detail page needs book metadata + review summary + stock from three services.

```go
func (s *Server) bookDashboard(ctx context.Context, id string) (Dashboard, error) {
    var (
        book    domain.Book
        reviews []Review
        stock   int
        eg      errgroup.Group
    )

    eg.Go(func() error {
        var err error
        book, err = s.books.Get(ctx, id)
        return err
    })
```

```go
    eg.Go(func() error {
        var err error
        reviews, err = s.reviews.ListForBook(ctx, id)
        return err
    })
    eg.Go(func() error {
        var err error
        stock, err = s.inventory.Stock(ctx, id)
        return err
    })

    if err := eg.Wait(); err != nil {
        return Dashboard{}, err
    }
    return Dashboard{Book: book, Reviews: reviews, Stock: stock}, nil
}
```

Use `golang.org/x/sync/errgroup` (or `errgroup.WithContext`) so the first failure cancels siblings when you wire context.

With cancel on first error

```go
eg, ctx := errgroup.WithContext(ctx)
// eg.Go(...) as above
err := eg.Wait()
```

Timeouts and context

Always bound outbound work:

```go
ctx, cancel := context.WithTimeout(r.Context(), 2*time.Second)
defer cancel()

req, err := http.NewRequestWithContext(ctx, http.MethodGet, url, nil)
// client.Do(req)
```

Pattern:

```
  request context (client gone)


  timeout child (2s)
```

```
  reviews   inventory
```

If the browser disconnects, `r.Context()` cancels and well-behaved queries stop.

Worker pool for bulk work

Importing a CSV of books: do not spawn one goroutine per row unbounded.

```go
func importBooks(ctx context.Context, rows []BookRow, workers int, save func(context.Context
    jobs := make(chan BookRow)
    eg, ctx := errgroup.WithContext(ctx)

    for i := 0; i < workers; i++ {
        eg.Go(func() error {
            for row := range jobs {
                if err := save(ctx, row); err != nil {
                    return err
                }
            }
            return nil
        })
    }

    eg.Go(func() error {
        defer close(jobs)
        for _, row := range rows {
            select {
            case <-ctx.Done():
                return ctx.Err()
            case jobs <- row:
            }
        }
        return nil
    })

    return eg.Wait()
}
```

Shared state in handlers

In-memory repositories and caches need synchronization:

```go
type Counter struct {
    mu sync.Mutex
    n  int
}

func (c *Counter) Inc() {
    c.mu.Lock()
    c.n++
    c.mu.Unlock()
}
```

Prefer:

1. **No shared mutable state** (DB owns truth)
2. **Channels / ownership** for pipelines
3. **Mutex** for small in-process caches
4. **atomics** only for simple counters

Run tests with `go test -race`.

Deadlock patterns to avoid

Bug	Symptom	Fix
Unbuffered send with no receiver	Hang	Ensure consumer, or buffer deliberately
`WaitGroup` Add inside goroutine	Under-count	`Add` before `go`, or use `WaitGroup.Go`
Lock held across slow I/O	Latency spike	Copy data under lock, I/O outside
Waiting on channel that never closes	Hang	`defer close`, or select on `ctx.Done()`

HTTP client reuse

```go
var paymentClient = &http.Client{
    Timeout: 10 * time.Second,
    Transport: &http.Transport{
        MaxIdleConns:        100,
        MaxIdleConnsPerHost: 10,
        IdleConnTimeout:     90 * time.Second,
    },
}
```

Do **not** create a new `http.Client` per request without reason—connection pooling lives on the client/transport.

Fire-and-forget (careful)

Logging metrics asynchronously is fine if you bound queues. Starting a goroutine that can block forever without tracking is not:

```
// risky: no backpressure, no shutdown
go sendEmail(order)

// better: queue + worker, or wait with timeout in request when email is critical
```

For web handlers, prefer completing critical work within the request or enqueueing to a durable job system.

Rules of thumb

Do	Don't
Parallelize independent I/O with errgroup	Spawn unlimited goroutines per upload
Propagate and timeout contexts	Ignore `r.Context()`
Protect shared maps with mutex	Assume request isolation covers package-level maps
Reuse HTTP clients	`http.Get` in a tight loop without reuse thinking

Try next

1. Add a handler that fetches book + fake review service in parallel with a 1s timeout.
2. Break it intentionally (remove Wait) and observe incomplete responses or races.
3. Run `go test -race` on your memory repository under concurrent Create/List.

Sessions, Authentication, and Authorization

Sessions, Authentication, and Authorization

Overview

Authentication answers *who are you?* Authorization answers *what may you do?* Sessions keep users logged in across requests. This chapter implements a straightforward Bookstore flow: password hashing, login, secure cookies, and role checks—without drowning in every OAuth edge case.

Concepts

```
Authn:  credentials → identity (user id)
Session: identity stored server-side or in signed cookie
Authz:  identity + role/permission → allow or deny
```

Term	Bookstore example
Authentication	Email + password login
Session	Cookie `session_id` → user in store
Authorization	Only `admin` may `POST /api/books`
Password hash	`bcrypt` / `argon2` of password; never store plaintext

Password hashing

```go
import "golang.org/x/crypto/bcrypt"

func HashPassword(plain string) (string, error) {
    b, err := bcrypt.GenerateFromPassword([]byte(plain), bcrypt.DefaultCost)
    return string(b), err
}

func CheckPassword(hash, plain string) bool {
    return bcrypt.CompareHashAndPassword([]byte(hash), []byte(plain)) == nil
}
```

Never log passwords. Prefer constant-time compare (bcrypt already does).

User model

```go
type User struct {
    ID           string `json:"id"`
    Email        string `json:"email"`
    PasswordHash string `json:"-"` // never serialize
    Role         string `json:"role"` // "customer" | "admin"
}
```

Session store (server-side)

Simple and clear for learning:

```go
type Session struct {
    UserID    string
    ExpiresAt time.Time
}

type SessionStore struct {
    mu   sync.Mutex
    byID map[string]Session
}

func (s *SessionStore) Create(userID string, ttl time.Duration) (string, error) {
    id, err := randomID(32)
    if err != nil {
        return "", err
    }
    s.mu.Lock()
    s.byID[id] = Session{UserID: userID, ExpiresAt: time.Now().Add(ttl)}
    s.mu.Unlock()
    return id, nil
}

func (s *SessionStore) Get(id string) (Session, bool) {
    s.mu.Lock()
    defer s.mu.Unlock()
    sess, ok := s.byID[id]
    if !ok || time.Now().After(sess.ExpiresAt) {
        delete(s.byID, id)
        return Session{}, false
    }
    return sess, true
}
```

```go
func (s *SessionStore) Delete(id string) {
    s.mu.Lock()
    delete(s.byID, id)
    s.mu.Unlock()
}
```

Production often uses Redis or signed/encrypted cookies (JWT or iron-cookie style). Server-side sessions are easy to revoke.

Secure cookie

```go
func setSessionCookie(w http.ResponseWriter, sessionID string, secure bool) {
    http.SetCookie(w, &http.Cookie{
        Name:     "session_id",
        Value:    sessionID,
        Path:     "/",
        HttpOnly: true,                      // not readable from JS
        Secure:   secure,                    // HTTPS only in prod
        SameSite: http.SameSiteLaxMode,      // CSRF mitigation baseline
        MaxAge:   int((24 * time.Hour).Seconds()),
    })
}
```

Flag	Why
`HttpOnly`	Stops XSS from stealing the cookie via `document.cookie`
`Secure`	Cookie only on HTTPS
`SameSite`	Reduces cross-site cookie sends

Login handler

```go
func (s *Server) login(w http.ResponseWriter, r *http.Request) {
    var in struct {
        Email    string `json:"email"`
        Password string `json:"password"`
    }
    if err := json.NewDecoder(r.Body).Decode(&in); err != nil {
        writeError(w, http.StatusBadRequest, "bad_json", "invalid body")
        return
    }

    user, err := s.users.ByEmail(r.Context(), in.Email)
```

```
    if err != nil || !CheckPassword(user.PasswordHash, in.Password) {
        // same message for unknown user and bad password
        writeError(w, http.StatusUnauthorized, "auth_failed", "invalid email or password")
        return
    }

    sid, err := s.sessions.Create(user.ID, 24*time.Hour)
    if err != nil {
        writeError(w, http.StatusInternalServerError, "session", "could not create session")
        return
    }
    setSessionCookie(w, sid, s.secureCookies)
    writeJSON(w, http.StatusOK, map[string]string{"status": "ok"})
}
```

Auth middleware

```
type ctxKey int

const userIDKey ctxKey = 1

func (s *Server) requireAuth(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        c, err := r.Cookie("session_id")
        if err != nil {
            writeError(w, http.StatusUnauthorized, "auth_required", "login required")
            return
        }
        sess, ok := s.sessions.Get(c.Value)
        if !ok {
            writeError(w, http.StatusUnauthorized, "auth_required", "login required")
            return
        }
        ctx := context.WithValue(r.Context(), userIDKey, sess.UserID)
        next.ServeHTTP(w, r.WithContext(ctx))
    })
}

func UserIDFrom(ctx context.Context) (string, bool) {
    id, ok := ctx.Value(userIDKey).(string)
    return id, ok
}
```

Role-based access

```go
func (s *Server) requireRole(role string, next http.Handler) http.Handler {
    return s.requireAuth(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        uid, _ := UserIDFrom(r.Context())
        user, err := s.users.Get(r.Context(), uid)
        if err != nil || user.Role != role {
            writeError(w, http.StatusForbidden, "forbidden", "insufficient role")
            return
        }
        next.ServeHTTP(w, r)
    }))
}

// registration
mux.Handle("POST /api/books", s.requireRole("admin", http.HandlerFunc(s.createBook)))
```

401 = not logged in. **403** = logged in but not allowed. Keep them distinct.

Logout

```go
func (s *Server) logout(w http.ResponseWriter, r *http.Request) {
    if c, err := r.Cookie("session_id"); err == nil {
        s.sessions.Delete(c.Value)
    }
    http.SetCookie(w, &http.Cookie{
        Name: "session_id", Value: "", Path: "/", MaxAge: -1, HttpOnly: true,
    })
    writeJSON(w, http.StatusOK, map[string]string{"status": "logged_out"})
}
```

CSRF note (HTML forms)

Cookie sessions + browser forms need CSRF tokens (double-submit or synchronizer token). JSON APIs called with custom headers from SPAs have a different threat model. For Bookstore HTML admin, add a CSRF token in forms before production.

Rules of thumb

Do	Don't
Hash passwords with bcrypt/argon2	Store plaintext or reversible encryption as "hash"
HttpOnly + Secure + SameSite cookies	Put tokens in `localStorage` without XSS plan
Generic login failure messages	Reveal "email not found" vs "bad password"
Check roles server-side every request	Trust a client-sent `role=admin` field

Try next

1. Add `POST /api/auth/register` with email uniqueness.
2. Protect `POST /api/books` with admin role.
3. Expire sessions and confirm `/api/me` returns 401 after expiry.

Frontend and Backend Communication

Frontend and Backend Communication

Overview

Frontends and backends talk through **contracts**: URLs, methods, status codes, and JSON shapes. Keep those contracts stable, document them lightly, and use the same habits when your Bookstore calls payment gateways or other third-party APIs. This chapter covers JSON interchange, client patterns, and external integration without framework lock-in.

API contract basics

Agree on:

1. **Resource paths** — `/api/books`, `/api/books/{id}`
2. **Methods and status codes** — 200/201/4xx/5xx meanings
3. **JSON field names** — snake_case or camelCase; pick one and stick to it
4. **Error envelope** — `{ "error": "...", "code": "..." }`
5. **Auth** — cookie session vs `Authorization: Bearer ...`

Example list response:

```
[
  {
    "id": "b1",
    "title": "The Go Programming Language",
    "author": "Donovan & Kernighan",
    "isbn": "978-0134190440",
    "price_cents": 4499
  }
]
```

Example error:

```
{
  "error": "title is required",
  "code": "validation"
}
```

JSON tags and stability

```go
type BookDTO struct {
    ID     string `json:"id"`
    Title  string `json:"title"`
    Author string `json:"author"`
    ISBN   string `json:"isbn"`
    Price  int    `json:"price_cents"`
}
```

- Omit sensitive fields with `json:"-"` (password hashes).
- Prefer additive changes (new optional fields) over renames.
- Version only when you must (`/api/v2/...`)—many Bookstores never need it early.

CORS (browser frontends on another origin)

If the UI is `http://localhost:5173` and API is `http://localhost:8080`, browsers enforce CORS.

```go
func withCORS(next http.Handler, allowedOrigin string) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        w.Header().Set("Access-Control-Allow-Origin", allowedOrigin)
        w.Header().Set("Access-Control-Allow-Credentials", "true")
        w.Header().Set("Access-Control-Allow-Headers", "Content-Type")
        w.Header().Set("Access-Control-Allow-Methods", "GET,POST,PUT,DELETE,OPTIONS")
        if r.Method == http.MethodOptions {
            w.WriteHeader(http.StatusNoContent)
            return
        }
        next.ServeHTTP(w, r)
    })
}
```

For cookie auth cross-origin, the frontend must use `credentials: "include"` and the server must not use `Allow-Origin: *` with credentials.

Simpler path for learning: serve HTML from the same Go process (chapter 274) and skip CORS entirely.

Fetch from a browser (sketch)

```
const res = await fetch("/api/books", { credentials: "same-origin" });
if (!res.ok) {
  const err = await res.json();
  throw new Error(err.error || res.statusText);
}
const books = await res.json();
```

Always check `res.ok` before assuming JSON is a success payload.

Go HTTP client patterns

Reusable client for third parties (payments, shipping, ISBN lookup):

```
type PaymentGateway struct {
    base   string
    apiKey string
    http   *http.Client
}

func NewPaymentGateway(base, apiKey string) *PaymentGateway {
    return &PaymentGateway{
        base:   strings.TrimRight(base, "/"),
        apiKey: apiKey,
        http: &http.Client{
            Timeout: 15 * time.Second,
        },
    }
}

type ChargeRequest struct {
    AmountCents int    `json:"amount_cents"`
    Currency    string `json:"currency"`
    OrderID     string `json:"order_id"`
}

type ChargeResponse struct {
    ChargeID string `json:"charge_id"`
    Status   string `json:"status"`
}

func (g *PaymentGateway) Charge(ctx context.Context, in ChargeRequest) (ChargeResponse, erro
    body, err := json.Marshal(in)
    if err != nil {
```

```go
        return ChargeResponse{}, err
    }
    req, err := http.NewRequestWithContext(ctx, http.MethodPost, g.base+"/v1/charges", bytes
    if err != nil {
        return ChargeResponse{}, err
    }
    req.Header.Set("Content-Type", "application/json")
    req.Header.Set("Authorization", "Bearer "+g.apiKey)

    res, err := g.http.Do(req)
    if err != nil {
        return ChargeResponse{}, err
    }
    defer res.Body.Close()

    if res.StatusCode >= 300 {
        b, _ := io.ReadAll(io.LimitReader(res.Body, 4096))
        return ChargeResponse{}, fmt.Errorf("payment gateway %s: %s", res.Status, b)
    }

    var out ChargeResponse
    if err := json.NewDecoder(res.Body).Decode(&out); err != nil {
        return ChargeResponse{}, err
    }
    return out, nil
}
```

Client checklist

Practice	Why
Context on every call	Timeouts and cancel
Explicit `http.Client` timeout	No hung requests forever
Limit error body reads	Avoid memory blowups
Interface for the gateway	Mock in tests (chapter 279)
Idempotency keys (when API supports)	Safe retries on POST

Retries (simple, careful)

Retry **idempotent** GETs on network blips. Do not blindly retry POSTs that charge cards unless the API is idempotent.

```go
func getWithRetry(ctx context.Context, client *http.Client, url string, attempts int) (*http
    var last error
```

```go
    for i := 0; i < attempts; i++ {
        req, err := http.NewRequestWithContext(ctx, http.MethodGet, url, nil)
        if err != nil {
            return nil, err
        }
        res, err := client.Do(req)
        if err == nil {
            return res, nil
        }
        last = err
        select {
        case <-ctx.Done():
            return nil, ctx.Err()
        case <-time.After(time.Duration(i+1) * 100 * time.Millisecond):
        }
    }
    return nil, last
}
```

Content negotiation

Same resource, two representations:

- `Accept: application/json` → JSON handler
- Browser navigation → HTML handler

Or separate paths (`/api/books` vs `/books`)—clearer for many teams.

Rules of thumb

Do	Don't
Document JSON fields your UI depends on	Rename fields casually
Same error shape everywhere	Mix string bodies and JSON errors
Interface external APIs	Call payment URLs deep inside handlers with no seam
Prefer same-origin for learning apps	Enable wide-open CORS in production

Try next

1. Write a short markdown "API contract" for five Bookstore endpoints.

2. Implement a fake `ISBNClient` interface that returns fixed metadata.
3. From curl, exercise login cookie + authenticated create book as a frontend would.

Testing and Debugging

Testing and Debugging

Overview

Reliable web services need fast, isolated tests: handlers with `httptest`, repositories mocked behind interfaces, and a few integration checks. Logging and simple performance checks help you find problems when tests pass but production does not. This chapter closes the Bookstore path with practical routines—not a full testing theory course (see part `06-testing` for that).

What to test

Layer	Technique	Speed
Domain validation	Table-driven unit tests	Fast
Handlers	`httptest` + mock store	Fast
Repository SQL	Test DB or testcontainers	Medium
Full process	`httptest` server or real port	Slower

Aim for many fast tests; keep a thin integration suite.

Table-driven domain tests

```go
func TestBookValidate(t *testing.T) {
    tests := []struct {
        name    string
        book    domain.Book
        wantErr bool
    }{
        {"ok", domain.Book{Title: "Go", Author: "A", Price: 100}, false},
        {"no title", domain.Book{Author: "A", Price: 100}, true},
        {"neg price", domain.Book{Title: "Go", Author: "A", Price: -1}, true},
    }
    for _, tt := range tests {
        t.Run(tt.name, func(t *testing.T) {
            err := tt.book.Validate()
            if (err != nil) != tt.wantErr {
                t.Fatalf("Validate() err=%v wantErr=%v", err, tt.wantErr)
            }
```

```
        })
    }
}
```

Mock repository

```
type mockBooks struct {
    list []domain.Book
    err  error
}

func (m mockBooks) List(ctx context.Context) ([]domain.Book, error) {
    return m.list, m.err
}
func (m mockBooks) Get(ctx context.Context, id string) (domain.Book, error) {
    return domain.Book{}, errors.New("not implemented")
}
func (m mockBooks) Create(ctx context.Context, b domain.Book) (domain.Book, error) {
    return b, m.err
}
```

Or generate mocks later; hand-written stubs stay clear for teaching.

httptest handler test

```
func TestListBooks(t *testing.T) {
    s := &Server{books: mockBooks{list: []domain.Book{
        {ID: "1", Title: "Go Book", Author: "Ada", Price: 1999},
    }}}

    req := httptest.NewRequest(http.MethodGet, "/api/books", nil)
    rec := httptest.NewRecorder()
    s.listBooks(rec, req)

    if rec.Code != http.StatusOK {
        t.Fatalf("status %d", rec.Code)
    }
    var got []domain.Book
    if err := json.NewDecoder(rec.Body).Decode(&got); err != nil {
        t.Fatal(err)
    }
    if len(got) != 1 || got[0].Title != "Go Book" {
        t.Fatalf("unexpected body: %#v", got)
```

```go
    }
}
```

Full mux test

```go
func TestRoutes(t *testing.T) {
    s := NewServer(memory.NewBookRepo())
    ts := httptest.NewServer(s.Handler())
    t.Cleanup(ts.Close)

    res, err := http.Get(ts.URL + "/api/books")
    if err != nil {
        t.Fatal(err)
    }
    defer res.Body.Close()
    if res.StatusCode != http.StatusOK {
        t.Fatalf("status %d", res.StatusCode)
    }
}
```

Testing auth

1. Create user + session in the test store.
2. Attach cookie: `req.AddCookie(&http.Cookie{Name: "session_id", Value: sid})`.
3. Assert 401 without cookie, 403 with wrong role, 201 with admin.

Keep clocks controllable if you test expiry (inject a `now` function).

Mocking external APIs

```go
type PaymentClient interface {
    Charge(ctx context.Context, in ChargeRequest) (ChargeResponse, error)
}

type fakePay struct {
    res ChargeResponse
    err error
}

func (f fakePay) Charge(ctx context.Context, in ChargeRequest) (ChargeResponse, error) {
    return f.res, f.err
}
```

Handler tests inject `fakePay{err: errors.New("down")}` and expect 502/503—not a real network call.

Logging and tracing events

Structured logs make incidents greppable:

```
slog.Info("book_created",
    "book_id", book.ID,
    "user_id", userID,
    "request_id", requestID,
)
```

Request ID middleware

```
func withRequestID(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        id := r.Header.Get("X-Request-ID")
        if id == "" {
            id = uuid.New().String() // stdlib uuid (Go 1.27+)
        }
        w.Header().Set("X-Request-ID", id)
        ctx := context.WithValue(r.Context(), reqIDKey, id)
        next.ServeHTTP(w, r.WithContext(ctx))
    })
}
```

Return the same ID to clients; paste it into log searches during incidents.

Error messages and incident response

Audience	Message
Client	Safe, actionable (`validation`, `not_found`)
Logs	Full error chain, request id, user id
On-call	Runbook: check logs by request id → DB → dependency status

Do not dump stack traces or SQL to public JSON bodies.

Light performance checks

```go
func BenchmarkListBooks(b *testing.B) {
    s := &Server{books: mockBooks{list: make([]domain.Book, 100)}}
    b.ResetTimer()
    for i := 0; i < b.N; i++ {
        req := httptest.NewRequest(http.MethodGet, "/api/books", nil)
        rec := httptest.NewRecorder()
        s.listBooks(rec, req)
    }
}
```

For real hotspots: `go test -bench . -cpuprofile=cpu.out` and part `09-performance-tooling` / `18-performance-engineering`.

Smoke load with curl/hey:

```bash
hey -n 1000 -c 50 http://localhost:8080/api/books
```

Debug checklist

```
1. Reproduce with curl + request id
2. Check status code and error code JSON
3. Search logs for request_id
4. Confirm auth cookie / role
5. Confirm DB row exists
6. Confirm downstream timeout vs 500
7. go test ./... and go test -race ./...
```

Rules of thumb

Do	Don't
Mock at interfaces	Hit real payment APIs in unit tests
Use httptest for handlers	Require full Docker for every commit
Log request ids	Log passwords, session secrets, card numbers
Race-test concurrent stores	Assume single-threaded handlers mean no races in packages

Try next

1. Write tests for create book: 201 happy path, 400 validation, 401 unauthenticated.
2. Force a panic in a handler and assert recover middleware returns 500.

3. Add a failing fake payment client test for checkout.

Where to go next

- Part **08-web** — GOTH, gRPC, Huma, Postgres/Redis recipes
- Part **06-testing** — organization, integration, advanced patterns
- Part **16-observability-sre** — metrics, tracing, correlation
- Part **17-security-hardening** — TLS, deeper authn/authz

You now have a clear path from empty module to a testable, session-aware Bookstore API with HTML and JSON edges—simple enough to learn, solid enough to grow.

Middleware Patterns

Middleware Patterns

Overview

Middleware wraps `http.Handler` to add cross-cutting behavior: logging, auth, recovery, request IDs, CORS, metrics. Keep each piece small and composable.

The shape

```go
type Middleware func(http.Handler) http.Handler

func chain(h http.Handler, ms ...Middleware) http.Handler {
    for i := len(ms) - 1; i >= 0; i-- {
        h = ms[i](h)
    }
    return h
}
```

```
request → recover → requestID → log → auth → mux → handler
response ←
```

Order matters: recover outermost so panics in other middleware still convert to 500.

Recover

```go
func Recover(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        defer func() {
            if rec := recover(); rec != nil {
                slog.Error("panic", "err", rec, "path", r.URL.Path)
                http.Error(w, "internal error", http.StatusInternalServerError)
            }
        }()
        next.ServeHTTP(w, r)
    })
}
```

Request ID

```go
type ctxKey int
const reqIDKey ctxKey = 1

func RequestID(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        id := r.Header.Get("X-Request-ID")
        if id == "" {
            id = strconv.FormatInt(time.Now().UnixNano(), 36)
        }
        w.Header().Set("X-Request-ID", id)
        ctx := context.WithValue(r.Context(), reqIDKey, id)
        next.ServeHTTP(w, r.WithContext(ctx))
    })
}
```

Access log with status

Capture status by wrapping `ResponseWriter`:

```go
type statusWriter struct {
    http.ResponseWriter
    code int
}

func (w *statusWriter) WriteHeader(code int) {
    w.code = code
    w.ResponseWriter.WriteHeader(code)
}

func AccessLog(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        sw := &statusWriter{ResponseWriter: w, code: 200}
        start := time.Now()
        next.ServeHTTP(sw, r)
        slog.Info("http",
            "method", r.Method,
            "path", r.URL.Path,
            "status", sw.code,
            "ms", time.Since(start).Milliseconds(),
        )
    })
}
```

Implement `Unwrap()` / `Flush` if you need http.Hijacker or Flusher passthrough in production.

Max body

```go
func MaxBytes(n int64) Middleware {
    return func(next http.Handler) http.Handler {
        return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
            r.Body = http.MaxBytesReader(w, r.Body, n)
            next.ServeHTTP(w, r)
        })
    }
}
```

Timeout (handler budget)

```go
func Timeout(d time.Duration) Middleware {
    return func(next http.Handler) http.Handler {
        return http.TimeoutHandler(next, d, "timeout")
    }
}
```

Prefer also setting `http.Server` timeouts (chapter 271).

Rules of thumb

Do	Don't
One concern per middleware	Mega-middleware that logs+auth+metrics+…
Document chain order	Auth after handler "for speed"
Propagate context values	Global request maps without locks

Try next

1. Build a chain: Recover → RequestID → AccessLog → mux.
2. Add unit tests with `httptest` asserting status on panic.
3. Log request ID from a handler via context.

Graceful Shutdown for Web Servers

Graceful Shutdown for Web Servers

Overview

On deploy or Ctrl-C, stop accepting new connections, finish in-flight requests, then exit. Go's `http.Server.Shutdown` does the hard part—wire it to signals and a deadline.

Pattern

```go
func main() {
    mux := http.NewServeMux()
    // register routes...

    srv := &http.Server{
        Addr:              ":8080",
        Handler:           mux,
        ReadHeaderTimeout: 5 * time.Second,
    }

    go func() {
        slog.Info("listen", "addr", srv.Addr)
        if err := srv.ListenAndServe(); err != nil && !errors.Is(err, http.ErrServerClosed)
            slog.Error("serve", "err", err)
            os.Exit(1)
        }
    }()

    ctx, stop := signal.NotifyContext(context.Background(), os.Interrupt, syscall.SIGTERM)
    defer stop()
    <-ctx.Done()
    slog.Info("shutdown signal")

    shutdownCtx, cancel := context.WithTimeout(context.Background(), 15*time.Second)
    defer cancel()
    if err := srv.Shutdown(shutdownCtx); err != nil {
        slog.Error("shutdown", "err", err)
        _ = srv.Close()
    }
}
```

Kubernetes notes

- **SIGTERM** on pod stop
- `terminationGracePeriodSeconds` must exceed your shutdown timeout
- Readiness fail first (optional sleep) so LB drains before Shutdown

```
// optional: sleep 2s after SIGTERM so endpoints controller removes you
time.Sleep(2 * time.Second)
_ = srv.Shutdown(shutdownCtx)
```

In-flight work beyond HTTP

Background workers need the same root context:

```
root, stop := signal.NotifyContext(...)
// workers select on root.Done()
// then Shutdown HTTP
```

Health endpoints during drain

- Liveness: still 200 until process dies
- Readiness: 503 after signal so new traffic stops

```
var ready atomic.Bool
ready.Store(true)

mux.HandleFunc("GET /readyz", func(w http.ResponseWriter, r *http.Request) {
    if !ready.Load() {
        http.Error(w, "draining", http.StatusServiceUnavailable)
        return
    }
    w.WriteHeader(http.StatusOK)
})
// on signal: ready.Store(false)
```

Rules of thumb

Do	Don't
`Shutdown` with timeout	Only `os.Exit` on SIGTERM
Fail readiness early	Keep accepting traffic while dying
Close DB pools after HTTP drains	Cut DB under active handlers

Try next

1. Slow handler (5s sleep); send SIGTERM; confirm request completes.
2. Set shutdown timeout 1s; prove forced close path.
3. Add `/readyz` drain behavior.

Rate Limiting and Concurrency Caps

Rate Limiting and Concurrency Caps

Overview

Protect the process from overload: limit request **rate** (token bucket) and **in-flight concurrency** (semaphore). Start simple—stdlib + `golang.org/x/time/rate` or a tiny token bucket.

Concurrency semaphore

```go
func MaxInFlight(n int) func(http.Handler) http.Handler {
    sem := make(chan struct{}, n)
    return func(next http.Handler) http.Handler {
        return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
            select {
            case sem <- struct{}{}:
                defer func() { <-sem }()
                next.ServeHTTP(w, r)
            default:
                http.Error(w, "busy", http.StatusServiceUnavailable)
            }
        })
    }
}
```

Token bucket (x/time/rate)

```go
go get golang.org/x/time/rate@latest
```

```go
func RateLimit(r rate.Limit, burst int) func(http.Handler) http.Handler {
    lim := rate.NewLimiter(r, burst)
    return func(next http.Handler) http.Handler {
        return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
            if !lim.Allow() {
                w.Header().Set("Retry-After", "1")
                http.Error(w, "rate limit", http.StatusTooManyRequests)
                return
```

```go
            }
            next.ServeHTTP(w, r)
        })
    }
}
// e.g. RateLimit(10, 20) → 10 req/s, burst 20
```

Per-IP limit (minimal)

```go
type ipLimiter struct {
    mu   sync.Mutex
    m    map[string]*rate.Limiter
    r    rate.Limit
    b    int
}

func (l *ipLimiter) get(ip string) *rate.Limiter {
    l.mu.Lock()
    defer l.mu.Unlock()
    if lim, ok := l.m[ip]; ok {
        return lim
    }
    lim := rate.NewLimiter(l.r, l.b)
    l.m[ip] = lim
    return lim
}
```

Production: bound map size, use `X-Forwarded-For` only behind trusted proxies.

Client-side (outbound)

```go
lim := rate.NewLimiter(5, 5)
_ = lim.Wait(ctx) // before third-party API call
```

Rules of thumb

Do	Don't
503/429 with Retry-After	Hang requests forever in queue without bound
Cap expensive routes tighter	One global limit for static + checkout
Limit outbound fan-out	Unlimited parallel dependency calls

Try next

1. Burst 5 then 429.
2. Combine MaxInFlight(50) + RateLimit.
3. Load test with `hey` and tune.

WebSockets Basics

WebSockets Basics

Overview

WebSockets upgrade an HTTP connection to a long-lived bidirectional channel. In Go, use `golang.org/x/net/websocket` (older) or the popular `github.com/gorilla/websocket` / `nhooyr.io/websocket`. This chapter shows a **minimal** gorilla-style echo and operational pitfalls.

Upgrade sketch (gorilla)

```
go get github.com/gorilla/websocket@latest
```

```go
var upgrader = websocket.Upgrader{
    CheckOrigin: func(r *http.Request) bool {
        // tighten in production
        return true
    },
    ReadBufferSize:  1024,
    WriteBufferSize: 1024,
}

func echoWS(w http.ResponseWriter, r *http.Request) {
    c, err := upgrader.Upgrade(w, r, nil)
    if err != nil {
        return
    }
    defer c.Close()
    for {
        mt, msg, err := c.ReadMessage()
        if err != nil {
            break
        }
        if err := c.WriteMessage(mt, msg); err != nil {
            break
        }
    }
}
```

Heartbeats

Idle proxies kill quiet connections. Send pings:

```
c.SetReadDeadline(time.Now().Add(60 * time.Second))
c.SetPongHandler(func(string) error {
    c.SetReadDeadline(time.Now().Add(60 * time.Second))
    return nil
})
// writer goroutine: c.WriteControl(websocket.PingMessage, ...)
```

Concurrency

One reader goroutine; one writer goroutine; never concurrent writers without a mutex or dedicated write pump.

```
read loop → app → write pump (single)
```

Auth

Authenticate on the **HTTP upgrade** (cookie/session or `Sec-WebSocket-Protocol` / query token). Don't invent a second weak auth after upgrade without TLS.

When not to use WebSockets

Prefer WS	Prefer HTTP/SSE
True bidirectional chat/games	Server→client events only (SSE)
Low-latency duplex	Simple request/response

Rules of thumb

Do	Don't
Limit message size	Read unbounded frames
Ping/pong deadlines	Assume connections live forever
Serialize writes	Write from many goroutines bare

Try next

1. Echo server + browser `WebSocket` client.
2. Reject oversized messages.
3. Broadcast hub with register/unregister channels.

File Uploads and Downloads

File Uploads and Downloads

Overview

Uploads and downloads stress memory, disk, and trust boundaries. Stream, bound sizes, and never trust client filenames.

Upload (multipart)

```go
func upload(w http.ResponseWriter, r *http.Request) {
    r.Body = http.MaxBytesReader(w, r.Body, 32<<20) // 32MiB
    if err := r.ParseMultipartForm(8 << 20); err != nil {
        http.Error(w, "too large or bad multipart", http.StatusBadRequest)
        return
    }
    file, hdr, err := r.FormFile("file")
    if err != nil {
        http.Error(w, "missing file", http.StatusBadRequest)
        return
    }
    defer file.Close()

    name := filepath.Base(hdr.Filename) // strip paths
    dstPath := filepath.Join(uploadDir, name)
    // better: random name + store original in DB
    dst, err := os.Create(dstPath)
    if err != nil {
        http.Error(w, "storage", 500)
        return
    }
    defer dst.Close()
    if _, err := io.Copy(dst, io.LimitReader(file, 32<<20)); err != nil {
        http.Error(w, "write", 500)
        return
    }
    fmt.Fprintln(w, "stored", name)
}
```

Sniff type: `http.DetectContentType` on first 512 bytes.

Download

```go
func download(w http.ResponseWriter, r *http.Request) {
    // resolve id → path inside jail (chapter 142 path jail)
    f, err := os.Open(safePath)
    if err != nil {
        http.NotFound(w, r)
        return
    }
    defer f.Close()
    st, _ := f.Stat()
    w.Header().Set("Content-Type", "application/octet-stream")
    w.Header().Set("Content-Disposition", `attachment; filename="report.bin"`)
    http.ServeContent(w, r, st.Name(), st.ModTime(), f) // supports Range
}
```

Static files

```go
mux.Handle("GET /static/", http.StripPrefix("/static/",
    http.FileServer(http.Dir("static"))))
```

For embed: `http.FileServer(http.FS(embedFS))`. Disable directory listing in production if needed (custom FS).

Rules of thumb

Do	Don't
MaxBytesReader	Buffer whole upload in memory blindly
`filepath.Base` + jail	Use raw `hdr.Filename` paths
`ServeContent` for Range	Hand-roll partial downloads first

Try next

1. Upload form + size limit test.
2. Download with `curl -C -` resume via ServeContent.
3. Store under random UUID names.

Project: Bookstore API Capstone

Project: Bookstore API Capstone

Overview

Combine this part into one **Bookstore API** binary: modular layout, REST, middleware, auth, DB interface, tests, graceful shutdown.

Scope

Route	Access
`GET /healthz`	public
`GET /readyz`	public
`GET /api/books`	public
`GET /api/books/{id}`	public
`POST /api/books`	admin
`POST /api/auth/login`	public
`POST /api/auth/logout`	auth
`GET /api/me`	auth

Checklist

- ☐ `internal/` layout (272)
- ☐ memory store + interface (275)
- ☐ middleware chain: recover, request ID, log, max body (280)
- ☐ bcrypt login + session cookie (277)
- ☐ role check on create (277)
- ☐ graceful Shutdown (281)
- ☐ table tests + httptest (279)
- ☐ optional rate limit on login (282)

Milestone order

```
1. health + list books (memory)
2. create book (no auth)
3. login + session + protect create
4. middleware + shutdown
5. tests
6. JSON error envelope consistency
```

Stretch

- HTML list page (274)
- Postgres driver switch (275)
- `-strict` security headers via reverse proxy notes
- OpenAPI via Huma later (part 16-web)

Done when

```
go test ./...
go run ./cmd/bookstore
curl -s localhost:8080/api/books | jq .
```

Login, set cookie jar, create book as admin; anonymous create returns 401.

Modern Web Architectures

The GOTH Stack

The GOTH Stack: Go, Templ, HTMX, Tailwind

In 2026, the pendulum has swung back. The complexity of React/Next.js/Hydration/ServerComponents has pushed many developers toward simpler, server-driven architectures.

Enter the **GOTH Stack**: * **G**o: Backend logic and router. * **O** (Templ?): Usually **Templ**, a type-safe templating language for Go. * **T**ailwind: Utility-first CSS. * **H**TMX: High-power tools for HTML.

Why GOTH?

It combines the type safety and speed of Go with the interactivity of an SPA, without the JSON-serialization overhead or state synchronization nightmares (Redux/Zustand).

1. Templ: Type-Safe HTML

Standard Go `html/template` is loosely typed. If you misspell a variable, it renders nothing. **Templ** compiles `.templ` files into Go code.

```
// hello.templ
package main

templ Hello(name string) {
    <div class="p-4 bg-blue-100 text-blue-800 rounded">
        <h1>Hello, { name }!</h1>
    </div>
}
```

If you pass an `int` to `Hello`, the Go compiler screams. This is revolutionary for backend rendering.

2. HTMX: The Engine

HTMX allows you to swap parts of the page via AJAX attributes directly in HTML.

```
<!-- When clicked, POST to /clicked, and replace this button with the response -->
<button hx-post="/clicked" hx-swap="outerHTML">
    Click Me
</button>
```

No JavaScript required. The server returns the *new HTML* (rendered via Templ), and HTMX injects it.

3. Tailwind: The Style

Since HTMX swaps HTML chunks, scoping CSS classes is hard. Tailwind solves this by embedding styles *in* the HTML. When HTMX injects a new `<div>`, it brings its styles with it.

A Simple Handler Example

```go
func Handler(w http.ResponseWriter, r *http.Request) {
    // 1. Logic
    user := db.GetUser()

    // 2. Render Component
    component := components.UserProfile(user)

    // 3. Serve
    component.Render(r.Context(), w)
}
```

When NOT to use GOTH

- **Offline-first Apps:** If you need it to work in a tunnel, you need a heavy client (React/Flutter).
- **Complex Canvas/Games:** WebGL implementations.

For 95% of CRUD apps, Dashboards, and SaaS tools, the GOTH stack is faster to build and orders of magnitude faster to run.

Worked example

Server-rendered form post that returns an HTML fragment (HTMX-style).

Save as `main.go`. Then:

```go
go mod init example
go run .
```

```go
package main
```

```go
import (
    "fmt"
    "html/template"
    "io"
    "net/http"
    "net/http/httptest"
    "strings"
)

var formTpl = template.Must(template.New("f").Parse(`
{{define "form"}}<form hx-post="/greet" hx-target="#out">
  <input name="name" value="{{.}}">
  <button type="submit">Greet</button>
</form>{{end}}
{{define "result"}}<p id="out">Hello, {{.}}!</p>{{end}}
`))

func main() {
    mux := http.NewServeMux()
    mux.HandleFunc("GET /{$}", func(w http.ResponseWriter, r *http.Request) {
        w.Header().Set("Content-Type", "text/html; charset=utf-8")
        _ = formTpl.ExecuteTemplate(w, "form", "world")
    })
    mux.HandleFunc("POST /greet", func(w http.ResponseWriter, r *http.Request) {
        _ = r.ParseForm()
        name := r.FormValue("name")
        if name == "" {
            name = "friend"
        }
        w.Header().Set("Content-Type", "text/html; charset=utf-8")
        _ = formTpl.ExecuteTemplate(w, "result", name)
    })

    srv := httptest.NewServer(mux)
    defer srv.Close()

    res, _ := http.PostForm(srv.URL+"/greet", map[string][]string{"name": {"Ada"}})
    body, _ := io.ReadAll(res.Body)
    res.Body.Close()
    fmt.Println(strings.TrimSpace(string(body)))
}
```

Expected output:

```
<p id="out">Hello, Ada!</p>
```

More examples

Compose page + partial templates without external Templ.

```go
package main

import (
    "fmt"
    "html/template"
    "strings"
)

func main() {
    t := template.Must(template.New("root").Parse(`
{{define "card"}}<div class="card">{{.}}</div>{{end}}
{{define "page"}}<main>{{template "card" .}}</main>{{end}}
`))
    var b strings.Builder
    _ = t.ExecuteTemplate(&b, "page", "Revenue OK")
    fmt.Println(strings.TrimSpace(b.String()))
}
```

Expected output:

```
<main><div class="card">Revenue OK</div></main>
```

Runnable example

> **Note:** Production GOTH uses **Templ**, **HTMX**, and **Tailwind**. This stdlib-only stand-in shows the same idea: the server returns HTML fragments, and a client can request partial updates.

Save as `main.go`. Then:

```
go mod init example
go run .
```

```go
package main

import (
    "fmt"
    "html/template"
    "io"
    "net/http"
    "net/http/httptest"
```

```go
    "strings"
    "sync/atomic"
)

var clicks atomic.Int64

// One template set: page embeds the "counter" partial (HTMX-style fragment).
var tpls = template.Must(template.New("root").Parse(`
{{define "counter"}}<button hx-post="/clicked">Clicks: {{.}}</button>{{end}}
{{define "page"}}<!doctype html>
<html><body>
  <h1>GOTH-style (stdlib)</h1>
  <div id="counter">{{template "counter" .}}</div>
</body></html>{{end}}
`))

func main() {
    mux := http.NewServeMux()

    // Full page (initial load)
    mux.HandleFunc("GET /{$}", func(w http.ResponseWriter, r *http.Request) {
        w.Header().Set("Content-Type", "text/html; charset=utf-8")
        _ = tpls.ExecuteTemplate(w, "page", clicks.Load())
    })

    // HTMX-style partial: return only the button HTML
    mux.HandleFunc("POST /clicked", func(w http.ResponseWriter, r *http.Request) {
        clicks.Add(1)
        w.Header().Set("Content-Type", "text/html; charset=utf-8")
        _ = tpls.ExecuteTemplate(w, "counter", clicks.Load())
    })

    // Drive requests without binding a port forever
    srv := httptest.NewServer(mux)
    defer srv.Close()

    resp, err := http.Get(srv.URL + "/")
    if err != nil {
        panic(err)
    }
    body, _ := io.ReadAll(resp.Body)
    resp.Body.Close()
    fmt.Println("status:", resp.StatusCode)
    fmt.Println("page contains counter:", strings.Contains(string(body), "Clicks: 0"))

    post, err := http.Post(srv.URL+"/clicked", "text/plain", nil)
    if err != nil {
```

```
        panic(err)
    }
    frag, _ := io.ReadAll(post.Body)
    post.Body.Close()
    fmt.Println("fragment:", strings.TrimSpace(string(frag)))
}
```

Expected output:

```
status: 200
page contains counter: true
fragment: <button hx-post="/clicked">Clicks: 1</button>
```

What to notice: Full page vs fragment is just different handlers. `httptest.NewServer` exercises real HTTP without leaving a process listening. Production Templ/HTMX would swap `#counter` automatically; here we print the fragment.

Try next: Add a `GET /clicked` that returns the same fragment, and chain three POSTs to watch the count rise.

Skeleton Loading & HTMX

Skeleton Loading with Go & HTMX

Users hate waiting. If your dashboard takes 2 seconds to calculate "Total Revenue," the entire page shouldn't hang for 2 seconds.

In standard MPAs (Multi-Page Apps), the browser waits for the full HTML response. In the GOTH stack, we use **Lazy Loading** with Skeleton screens.

The Pattern

1. **Initial Load:** Return the page structure immediately (Header, Sidebar, Footer) but fill the "Slow Widgets" with **Skeleton Loaders** (grey pulsing boxes).
2. **Trigger:** The Skeleton HTML includes `hx-trigger="load"` to immediately ask the server for the real content.
3. **Swap:** The server calculates the expensive data and returns the real HTML, swapping out the skeleton.

Implementation

1. The Dashboard Handler (Fast)

```go
func DashboardHandler(w http.ResponseWriter, r *http.Request) {
    // Render the layout immediately.
    // Notice we don't fetch revenue here.
    Layout(
        // Inject Skeletons
        components.RevenueSkeleton(),
        components.UsersSkeleton(),
    ).Render(r.Context(), w)
}
```

2. The Skeleton Component (Templ)

```
templ RevenueSkeleton() {
    <div hx-get="/api/revenue"
         hx-trigger="load"
         hx-swap="outerHTML"
```

```
        class="animate-pulse bg-gray-200 h-32 rounded">
      <!-- Pulsing Grey Box -->
    </div>
}
```

3. The Real Data Handler (Slow)

```go
func RevenueHandler(w http.ResponseWriter, r *http.Request) {
    // Extensive DB calculation...
    time.Sleep(1 * time.Second)
    amount := db.CalculateRevenue()

    // Return real component
    components.RevenueCard(amount).Render(r.Context(), w)
}
```

Why this wins

- **TTFB (Time To First Byte):** 10-20ms. The user sees the UI instantly.
- **Perceived Performance:** The app feels alive while data loads in parallel.
- **Simplicity:** No `useEffect`, no `isLoading` state management, no JSON parsing. Just HTML swapping.

Advanced: Request Coalescing

If 10 widgets all fire `hx-get` at once, you might hit browser connection limits (HTTP/1.1 limit is 6). * **HTTP/2 or HTTP/3:** Go's `net/http` supports these automatically if you use TLS. They multiplex requests, so 10 requests cost nearly the same connection overhead as 1. * Ensure your server supports TLS (Let's Encrypt) to enable H2/H3.

Worked example

Fast shell + parallel widget endpoints (stdlib model of skeleton fill).

Save as `main.go`. Then:

```
go mod init example
go run .
```

```go
package main
```

```go
import (
    "fmt"
    "io"
    "net/http"
    "net/http/httptest"
    "strings"
    "sync"
    "time"
)

func main() {
    mux := http.NewServeMux()
    mux.HandleFunc("GET /dash", func(w http.ResponseWriter, r *http.Request) {
        _, _ = io.WriteString(w, `<div class="skel" data-src="/w/a"></div><div class="skel"
    })
    mux.HandleFunc("GET /w/{id}", func(w http.ResponseWriter, r *http.Request) {
        time.Sleep(15 * time.Millisecond)
        fmt.Fprintf(w, `<div class="card">widget %s</div>`, r.PathValue("id"))
    })

    srv := httptest.NewServer(mux)
    defer srv.Close()

    shell, _ := http.Get(srv.URL + "/dash")
    sb, _ := io.ReadAll(shell.Body)
    shell.Body.Close()
    fmt.Println("shell has skeleton:", strings.Contains(string(sb), `class="skel"`))

    // Parallel widget fetch (what the browser/HTMX would do)
    var wg sync.WaitGroup
    for _, id := range []string{"a", "b"} {
        wg.Add(1)
        go func(id string) {
            defer wg.Done()
            res, err := http.Get(srv.URL + "/w/" + id)
            if err != nil {
                panic(err)
            }
            b, _ := io.ReadAll(res.Body)
            res.Body.Close()
            fmt.Println(strings.TrimSpace(string(b)))
        }(id)
    }
    wg.Wait()
}
```

Expected output: (widget lines may reorder)

```
shell has skeleton: true
<div class="card">widget a</div>
<div class="card">widget b</div>
```

More examples

Measure shell TTFB vs sequential widgets.

```
package main

import (
    "fmt"
    "net/http"
    "net/http/httptest"
    "time"
)

func main() {
    mux := http.NewServeMux()
    mux.HandleFunc("GET /fast", func(w http.ResponseWriter, r *http.Request) { w.Write([]byt
    mux.HandleFunc("GET /slow", func(w http.ResponseWriter, r *http.Request) {
        time.Sleep(30 * time.Millisecond)
        w.Write([]byte("data"))
    })
    srv := httptest.NewServer(mux)
    defer srv.Close()

    t0 := time.Now()
    http.Get(srv.URL + "/fast")
    fast := time.Since(t0)

    t1 := time.Now()
    http.Get(srv.URL + "/slow")
    slow := time.Since(t1)

    fmt.Println("fast < slow:", fast < slow)
}
```

Expected output:

```
fast < slow: true
```

Runnable example

> **Note:** Real skeleton UIs use HTMX (`hx-trigger="load"`) + Templ. This stdlib demo models the same flow: fast shell HTML, then a slow partial.

Save as `main.go`. Then:

```
go mod init example
go run .
```

```
package main

import (
    "fmt"
    "io"
    "net/http"
    "net/http/httptest"
    "strings"
    "time"
)

func main() {
    mux := http.NewServeMux()

    // Fast shell: structure + skeleton placeholder
    mux.HandleFunc("GET /dashboard", func(w http.ResponseWriter, r *http.Request) {
        w.Header().Set("Content-Type", "text/html; charset=utf-8")
        _, _ = io.WriteString(w, `<!doctype html>
<html><body>
  <h1>Dashboard</h1>
  <div id="revenue"
       hx-get="/api/revenue"
       hx-trigger="load"
       class="skeleton">Loading revenue…</div>
</body></html>`)
    })

    // Slow widget: pretend DB work, return final HTML card
    mux.HandleFunc("GET /api/revenue", func(w http.ResponseWriter, r *http.Request) {
        time.Sleep(25 * time.Millisecond)
        w.Header().Set("Content-Type", "text/html; charset=utf-8")
        _, _ = io.WriteString(w, `<div id="revenue" class="card">Total revenue: $42,000</div
    })

    srv := httptest.NewServer(mux)
    defer srv.Close()
```

```go
    t0 := time.Now()
    shell, err := http.Get(srv.URL + "/dashboard")
    if err != nil {
        panic(err)
    }
    shellBody, _ := io.ReadAll(shell.Body)
    shell.Body.Close()
    shellMS := time.Since(t0).Milliseconds()

    t1 := time.Now()
    widget, err := http.Get(srv.URL + "/api/revenue")
    if err != nil {
        panic(err)
    }
    widgetBody, _ := io.ReadAll(widget.Body)
    widget.Body.Close()
    widgetMS := time.Since(t1).Milliseconds()

    fmt.Println("shell status:", shell.StatusCode)
    fmt.Println("shell has skeleton:", strings.Contains(string(shellBody), "class=\"skeleton
    fmt.Println("shell faster than widget:", shellMS < widgetMS)
    fmt.Println("widget:", strings.TrimSpace(string(widgetBody)))
    fmt.Println("widget latency ms >= 20:", widgetMS >= 20)
}
```

Expected output:

```
shell status: 200
shell has skeleton: true
shell faster than widget: true
widget: <div id="revenue" class="card">Total revenue: $42,000</div>
widget latency ms >= 20: true
```

What to notice: Time-to-first-byte for the shell stays low because expensive work lives on a separate endpoint. The browser (or HTMX) fills the skeleton after load; the user sees structure immediately.

Try next: Fire three parallel `GET /api/revenue` requests with goroutines and measure wall time vs sequential.

Microservices & gRPC

Microservices: Switching to gRPC (and Connect)

REST is fine for public APIs. But for internal communication between microservices, JSON over HTTP/1.1 is inefficient (text parsing, no types, high overhead).

gRPC (Google Remote Procedure Call) is the standard for high-performance internal traffic.

The Protocol Buffers (Protobuf)

You define your data and service in a `.proto` file.

```
syntax = "proto3";

service UserService {
  rpc GetUser (UserRequest) returns (UserResponse);
}

message UserRequest {
  string id = 1;
}

message UserResponse {
  string name = 1;
  int32 age = 2;
}
```

Then, you compile this using `protoc` to generate Go code. * **Strict Types:** The generated code guarantees the wire format matches the struct. * **Binary:** Messages are binary (smaller, faster to parse than JSON).

The gRPC Complexity (and the Solution: ConnectRPC)

Classic gRPC requires special load balancers (because it breaks HTTP/1.1 framing) and is hard to debug with `curl`.

In 2026, we prefer **ConnectRPC** (by Buf). * **It's just HTTP:** It supports gRPC but *also* standard HTTP/JSON. * You can `curl` a Connect service with JSON, but talk to it via gRPC from other Go services.

Connect Example

```go
package main

import (
    "context"
    "net/http"
    "connectrpc.com/connect"
    user "example/gen/user/v1" // Generated code
    "example/gen/user/v1/userv1connect"
)

type UserServer struct {}

func (s *UserServer) GetUser(
    ctx context.Context,
    req *connect.Request[user.UserRequest],
) (*connect.Response[user.UserResponse], error) {
    return connect.NewResponse(&user.UserResponse{
        Name: "Alice",
        Age:  30,
    }), nil
}

func main() {
    mux := http.NewServeMux()
    path, handler := userv1connect.NewUserServiceHandler(&UserServer{})
    mux.Handle(path, handler)
    http.ListenAndServe(":8080", mux)
}
```

When to use Microservices?

Default to Monolith. Microservices introduce: 1. Network Latency. 2. Distributed Tracing requirements. 3. Deployment complexity.

Only switch to Microservices (and gRPC) when: * You have distinct teams who need to deploy independently. * You have distinct scaling requirements (e.g., the Video Encoder needs 100 GPUs, but the Login server needs 1 CPU).

Worked example

Unary JSON-RPC client/server with typed request envelopes.

Save as `main.go`. Then:

```
go mod init example
go run .
```

```
package main

import (
    "bytes"
    "encoding/json"
    "fmt"
    "net/http"
    "net/http/httptest"
)

type Req[T any] struct {
    Payload T `json:"payload"`
}

type Resp[T any] struct {
    Payload T `json:"payload"`
}

type AddIn struct {
    A, B int `json:"a"`
}
type AddOut struct {
    Sum int `json:"sum"`
}

func main() {
    mux := http.NewServeMux()
    mux.HandleFunc("POST /calc.v1/Add", func(w http.ResponseWriter, r *http.Request) {
        var in Req[AddIn]
        if err := json.NewDecoder(r.Body).Decode(&in); err != nil {
            http.Error(w, "bad json", 400)
            return
        }
        _ = json.NewEncoder(w).Encode(Resp[AddOut]{Payload: AddOut{Sum: in.Payload.A + in.Pa
    })
    srv := httptest.NewServer(mux)
    defer srv.Close()

    body, _ := json.Marshal(Req[AddIn]{Payload: AddIn{A: 2, B: 40}})
    res, err := http.Post(srv.URL+"/calc.v1/Add", "application/json", bytes.NewReader(body))
    if err != nil {
        panic(err)
    }
```

```
    defer res.Body.Close()
    var out Resp[AddOut]
    _ = json.NewDecoder(res.Body).Decode(&out)
    fmt.Println("sum:", out.Payload.Sum)
}
```

Expected output:

```
sum: 42
```

More examples

RPC error envelope + status mapping.

```
package main

import (
    "bytes"
    "encoding/json"
    "fmt"
    "net/http"
    "net/http/httptest"
)

func main() {
    mux := http.NewServeMux()
    mux.HandleFunc("POST /user.v1/Get", func(w http.ResponseWriter, r *http.Request) {
        var req struct {
            ID string `json:"id"`
        }
        _ = json.NewDecoder(r.Body).Decode(&req)
        if req.ID == "" {
            w.WriteHeader(http.StatusBadRequest)
            _ = json.NewEncoder(w).Encode(map[string]string{"error": "id required"})
            return
        }
        _ = json.NewEncoder(w).Encode(map[string]string{"name": "Ada"})
    })
    srv := httptest.NewServer(mux)
    defer srv.Close()

    res, _ := http.Post(srv.URL+"/user.v1/Get", "application/json", bytes.NewReader([]byte(`
    fmt.Println("status:", res.StatusCode)
    res.Body.Close()
}
```

Expected output:

```
status: 400
```

Runnable example

> **Note:** Production internal APIs often use **gRPC** or **ConnectRPC** with protobuf. This stdlib stand-in is a typed JSON "RPC over HTTP" service—same request/response shape idea, no codegen.

Save as `main.go`. Then:

```
go mod init example
go run .
```

```go
package main

import (
    "bytes"
    "encoding/json"
    "fmt"
    "net/http"
    "net/http/httptest"
)

type UserRequest struct {
    ID string `json:"id"`
}

type UserResponse struct {
    Name string `json:"name"`
    Age  int    `json:"age"`
}

type ErrorBody struct {
    Error string `json:"error"`
}

func getUser(id string) (UserResponse, bool) {
    db := map[string]UserResponse{
        "u1": {Name: "Alice", Age: 30},
        "u2": {Name: "Bob", Age: 25},
    }
    u, ok := db[id]
    return u, ok
```

```go
}

func main() {
    mux := http.NewServeMux()
    // Connect/gRPC-style unary RPC as JSON POST
    mux.HandleFunc("POST /user.v1.UserService/GetUser", func(w http.ResponseWriter, r *http.
        var req UserRequest
        if err := json.NewDecoder(r.Body).Decode(&req); err != nil {
            w.WriteHeader(http.StatusBadRequest)
            _ = json.NewEncoder(w).Encode(ErrorBody{Error: "invalid json"})
            return
        }
        user, ok := getUser(req.ID)
        if !ok {
            w.WriteHeader(http.StatusNotFound)
            _ = json.NewEncoder(w).Encode(ErrorBody{Error: "user not found"})
            return
        }
        w.Header().Set("Content-Type", "application/json")
        _ = json.NewEncoder(w).Encode(user)
    })

    srv := httptest.NewServer(mux)
    defer srv.Close()

    call := func(id string) {
        body, _ := json.Marshal(UserRequest{ID: id})
        resp, err := http.Post(srv.URL+"/user.v1.UserService/GetUser", "application/json", b
        if err != nil {
            panic(err)
        }
        defer resp.Body.Close()
        var raw map[string]any
        _ = json.NewDecoder(resp.Body).Decode(&raw)
        fmt.Printf("id=%s status=%d body=%v\n", id, resp.StatusCode, raw)
    }

    call("u1")
    call("missing")
}
```

Expected output:

```
id=u1 status=200 body=map[age:30 name:Alice]
id=missing status=404 body=map[error:user not found]
```

What to notice: Method path + typed request/response mirrors an RPC surface even over

JSON/HTTP. `httptest` keeps the demo offline-friendly. Real gRPC adds binary protobuf, streaming, and code-generated stubs.

Try next: Add a `ListUsers` method. Wrap the handler with a middleware that logs RPC name and status code.

Huma & OpenAPI

Modern APIs: Huma & OpenAPI

Writing REST APIs involves two pains: 1. Validation (Checking JSON inputs). 2. Documentation (Keeping Swagger/OpenAPI YAML in sync with code).

In the past, we wrote code, then manually wrote YAML. The YAML effectively lied because it drifted from the code.

Enter Huma

Huma (huma.rocks) is the modern (2025/2026) framework of choice for building APIs. * **Code-First OpenAPI:** It generates the OpenAPI 3.1 Spec *from* your Go structs. * **Automatic Validation:** It validates inputs based on struct tags.

Huma Example

```go
package main

import (
    "context"
    "net/http"
    "github.com/danielgtaylor/huma/v2"
    "github.com/danielgtaylor/huma/v2/adapters/humachi"
    "github.com/go-chi/chi/v5"
)

// 1. Define Request/Response
type GreetingInput struct {
    Name string `query:"name" maxLength:"30" doc:"Name to greet"`
}

type GreetingOutput struct {
    Body struct {
        Message string `json:"message" example:"Hello, World!"`
    }
}

func main() {
    router := chi.NewMux()
```

```go
    api := humachi.New(router, huma.DefaultConfig("My API", "1.0.0"))

    // 2. Register Operation
    huma.Register(api, huma.Operation{
        OperationID: "get-greeting",
        Method:      http.MethodGet,
        Path:        "/greeting",
        Summary:     "Get a welcome message",
    }, func(ctx context.Context, input *GreetingInput) (*GreetingOutput, error) {
        // Only executes if validation passes!
        resp := &GreetingOutput{}
        resp.Body.Message = "Hello, " + input.Name
        return resp, nil
    })

    http.ListenAndServe(":8888", router)
}
```

Why Huma?

1. **Documentation:** Visit `/docs` (or similar) and you get a beautiful, interactive Scalar or Swagger UI automatically.
2. **Safety:** Meaningful error messages for users ("Field 'name' is too long").
3. **Standard Library Compatible:** It works on top of `http.ServeMux`, `Chi`, or `Gin`. It doesn't lock you into a monolithic ecosystem.

This replaces the older `swag` comment-based generation which was error-prone and brittle.

Worked example

Struct-driven validation + JSON error/success responses (Huma-shaped, stdlib).

Save as `main.go`. Then:

```bash
go mod init example
go run .
```

```go
package main

import (
    "encoding/json"
    "fmt"
    "net/http"
    "net/http/httptest"
```

```
    "strings"
    "unicode/utf8"
)

type CreateUser struct {
    Name string `json:"name"`
    Age  int    `json:"age"`
}

func validate(u CreateUser) []string {
    var errs []string
    if u.Name == "" {
        errs = append(errs, "name is required")
    }
    if utf8.RuneCountInString(u.Name) > 30 {
        errs = append(errs, "name maxLength 30")
    }
    if u.Age < 0 || u.Age > 150 {
        errs = append(errs, "age out of range")
    }
    return errs
}

func main() {
    mux := http.NewServeMux()
    mux.HandleFunc("POST /users", func(w http.ResponseWriter, r *http.Request) {
        w.Header().Set("Content-Type", "application/json")
        var u CreateUser
        if err := json.NewDecoder(r.Body).Decode(&u); err != nil {
            w.WriteHeader(http.StatusBadRequest)
            _ = json.NewEncoder(w).Encode(map[string]string{"error": "invalid json"})
            return
        }
        if errs := validate(u); len(errs) > 0 {
            w.WriteHeader(http.StatusUnprocessableEntity)
            _ = json.NewEncoder(w).Encode(map[string]any{"errors": errs})
            return
        }
        _ = json.NewEncoder(w).Encode(map[string]string{"status": "created", "name": u.Name}
    })

    srv := httptest.NewServer(mux)
    defer srv.Close()

    res, _ := http.Post(srv.URL+"/users", "application/json", strings.NewReader(`{"name":"",
    fmt.Println("invalid status:", res.StatusCode)
    res.Body.Close()
```

```
    res, _ = http.Post(srv.URL+"/users", "application/json", strings.NewReader(`{"name":"Ada
    var ok map[string]string
    _ = json.NewDecoder(res.Body).Decode(&ok)
    res.Body.Close()
    fmt.Println("valid:", res.StatusCode, ok["status"], ok["name"])
}
```

Expected output:

```
invalid status: 422
valid: 200 created Ada
```

More examples

Serve a tiny OpenAPI document next to the handler (docs stay near code).

```
package main

import (
    "encoding/json"
    "fmt"
    "net/http"
    "net/http/httptest"
)

func main() {
    spec := map[string]any{
        "openapi": "3.1.0",
        "info":    map[string]string{"title": "Users", "version": "1.0.0"},
        "paths": map[string]any{
            "/users": map[string]any{"post": map[string]string{"operationId": "createUser"}}
        },
    }
    mux := http.NewServeMux()
    mux.HandleFunc("GET /openapi.json", func(w http.ResponseWriter, r *http.Request) {
        _ = json.NewEncoder(w).Encode(spec)
    })
    srv := httptest.NewServer(mux)
    defer srv.Close()
    res, _ := http.Get(srv.URL + "/openapi.json")
    var doc map[string]any
    _ = json.NewDecoder(res.Body).Decode(&doc)
    res.Body.Close()
    fmt.Println("openapi:", doc["openapi"])
}
```

Expected output:

```
openapi: 3.1.0
```

Runnable example

> **Note: Huma** generates OpenAPI 3.1 and validates from structs automatically. This stdlib stand-in shows the same product idea: validate input, return JSON, and expose a tiny machine-readable "spec" document.

Save as `main.go`. Then:

```
go mod init example
go run .
```

```go
package main

import (
    "encoding/json"
    "fmt"
    "net/http"
    "net/http/httptest"
    "unicode/utf8"
)

type GreetingOutput struct {
    Message string `json:"message"`
}

type APIError struct {
    Error string `json:"error"`
}

// Mini OpenAPI-ish document (hand-written for the demo).
var openAPI = map[string]any{
    "openapi": "3.1.0",
    "info":    map[string]string{"title": "Greeting API", "version": "1.0.0"},
    "paths": map[string]any{
        "/greeting": map[string]any{
            "get": map[string]any{
                "operationId": "get-greeting",
                "summary":     "Get a welcome message",
                "parameters": []map[string]any{
                    {
                        "name": "name", "in": "query", "required": true,
```

```
                        "schema": map[string]any{"type": "string", "maxLength": 30},
                    },
                },
            },
        },
    },
}

func greetingHandler(w http.ResponseWriter, r *http.Request) {
    name := r.URL.Query().Get("name")
    w.Header().Set("Content-Type", "application/json")

    if name == "" {
        w.WriteHeader(http.StatusBadRequest)
        _ = json.NewEncoder(w).Encode(APIError{Error: "query param 'name' is required"})
        return
    }
    if utf8.RuneCountInString(name) > 30 {
        w.WriteHeader(http.StatusUnprocessableEntity)
        _ = json.NewEncoder(w).Encode(APIError{Error: "field 'name' is too long (max 30)"})
        return
    }

    _ = json.NewEncoder(w).Encode(GreetingOutput{Message: "Hello, " + name})
}

func main() {
    mux := http.NewServeMux()
    mux.HandleFunc("GET /greeting", greetingHandler)
    mux.HandleFunc("GET /openapi.json", func(w http.ResponseWriter, r *http.Request) {
        w.Header().Set("Content-Type", "application/json")
        _ = json.NewEncoder(w).Encode(openAPI)
    })

    srv := httptest.NewServer(mux)
    defer srv.Close()

    // Missing name → validation error
    r1, _ := http.Get(srv.URL + "/greeting")
    var e1 APIError
    _ = json.NewDecoder(r1.Body).Decode(&e1)
    r1.Body.Close()
    fmt.Println("missing:", r1.StatusCode, e1.Error)

    // Valid
    r2, _ := http.Get(srv.URL + "/greeting?name=World")
```

```go
    var ok GreetingOutput
    _ = json.NewDecoder(r2.Body).Decode(&ok)
    r2.Body.Close()
    fmt.Println("ok:", r2.StatusCode, ok.Message)

    // Spec exists
    r3, _ := http.Get(srv.URL + "/openapi.json")
    var spec map[string]any
    _ = json.NewDecoder(r3.Body).Decode(&spec)
    r3.Body.Close()
    info := spec["info"].(map[string]any)
    fmt.Println("spec title:", info["title"])
}
```

Expected output:

```
missing: 400 query param 'name' is required
ok: 200 Hello, World
spec title: Greeting API
```

What to notice: Validation lives next to the handler (Huma derives it from tags). Shipping `/openapi.json` keeps docs honest—Huma generates that document from the same structs that validate.

Try next: Reject names containing digits. Add `POST /greeting` with a JSON body and validate a required `name` field.

Integrations: PostgreSQL & Redis

Integrations: PostgreSQL & Redis

Most Go backends in 2026 rely on this "boring" but powerful trio: Go + Postgres + Redis.

PostgreSQL: `pgx`

Don't use `lib/pq`. It is in maintenance mode. Use **`jackc/pgx`**. It is faster, safer, and supports more features (COPY, Listen/Notify).

Connection Pooling

Always use `pgxpool`, not a single connection.

```go
import "github.com/jackc/pgx/v5/pgxpool"

func NewDB(ctx context.Context, connString string) (*pgxpool.Pool, error) {
    config, err := pgxpool.ParseConfig(connString)
    // Tuning
    config.MaxConns = 25
    config.MinConns = 5
    config.MaxConnLifetime = 1 * time.Hour

    return pgxpool.NewWithConfig(ctx, config)
}
```

Type-Safe SQL: sqlc

Don't write ORM code (GORM) unless you love runtime crashes and slow queries. Use **sqlc**. You write SQL, it generates type-safe Go structs and interfaces. (See Chapter 59 "Code Generation" for details).

Redis: `go-redis`

Use `github.com/redis/go-redis/v9`.

Patterns

1. **Caching (Look-aside):** `go val, err := rdb.Get(ctx, key).Result() if err == redis.Nil { // Miss: Fetch DB, Set Redis val = fetchFromDB() rdb.Set(ctx, key, val, 10*time.Minute) } else if err != nil { return err // Real error } // Hit: use val`
2. **Rate Limiting:** Redis is atomic. `INCR` and `EXPIRE` are perfect for API limits.
3. **Queues:** Redis Streams or simple Lists (`LPUSH/BRPOP`) make for excellent lightweight worker queues before you upgrade to Kafka or NATS.

Context Awareness

Both `pgx` and `go-redis` require `context.Context` in every method call. **Always pass the context.** This allows your database queries to automatically timeout if the user cancels the HTTP request, preventing zombie queries from eating your database CPU.

```go
// Good
row := db.QueryRow(ctx, "SELECT ...")

// Bad (cancelling request won't stop query)
row := db.QueryRow(context.Background(), "SELECT ...")
```

Worked example

Look-aside cache with TTL expiry (stdlib Redis/DB stand-in).

Save as `main.go`. Then:

```bash
go mod init example
go run .
```

```go
package main

import (
    "context"
    "fmt"
    "sync"
    "time"
)

type cache struct {
    mu    sync.Mutex
    items map[string]entry
```

```go
}

type entry struct {
    val string
    exp time.Time
}

func (c *cache) Get(k string) (string, bool) {
    c.mu.Lock()
    defer c.mu.Unlock()
    e, ok := c.items[k]
    if !ok || time.Now().After(e.exp) {
        delete(c.items, k)
        return "", false
    }
    return e.val, true
}

func (c *cache) Set(k, v string, ttl time.Duration) {
    c.mu.Lock()
    defer c.mu.Unlock()
    c.items[k] = entry{val: v, exp: time.Now().Add(ttl)}
}

func main() {
    c := &cache{items: map[string]entry{}}
    c.Set("u1", "alice", 20*time.Millisecond)
    if v, ok := c.Get("u1"); ok {
        fmt.Println("hit:", v)
    }
    time.Sleep(25 * time.Millisecond)
    _, ok := c.Get("u1")
    fmt.Println("after ttl hit:", ok)

    // context cancel around a "query"
    ctx, cancel := context.WithCancel(context.Background())
    cancel()
    select {
    case <-ctx.Done():
        fmt.Println("query aborted:", ctx.Err())
    default:
    }
}
```

Expected output:

```
hit: alice
after ttl hit: false
query aborted: context canceled
```

More examples

Simple atomic rate limiter (Redis INCR window analogue).

```go
package main

import (
    "fmt"
    "sync"
    "sync/atomic"
    "time"
)

type limiter struct {
    mu     sync.Mutex
    count  int
    reset  time.Time
    limit  int
    window time.Duration
}

func (l *limiter) Allow() bool {
    l.mu.Lock()
    defer l.mu.Unlock()
    now := time.Now()
    if now.After(l.reset) {
        l.count = 0
        l.reset = now.Add(l.window)
    }
    if l.count >= l.limit {
        return false
    }
    l.count++
    return true
}

func main() {
    l := &limiter{limit: 3, window: time.Second, reset: time.Now().Add(time.Second)}
    var allowed atomic.Int64
    var wg sync.WaitGroup
    for i := 0; i < 10; i++ {
```

```go
        wg.Add(1)
        go func() {
            defer wg.Done()
            if l.Allow() {
                allowed.Add(1)
            }
        }()
    }
    wg.Wait()
    fmt.Println("allowed:", allowed.Load()) // 3
}
```

Expected output:

```
allowed: 3
```

Runnable example

> **Note:** Production uses **pgx** + **Redis**. This stdlib stand-in models a look-aside cache and context-aware "query" against in-memory stores so the pattern is runnable offline.

Save as `main.go`. Then:

```bash
go mod init example
go run .
```

```go
package main

import (
    "context"
    "errors"
    "fmt"
    "sync"
    "time"
)

// FakeDB simulates Postgres latency + storage.
type FakeDB struct {
    mu   sync.Mutex
    rows map[string]string
    hits int
}

func (db *FakeDB) QueryRow(ctx context.Context, key string) (string, error) {
    select {
```

```
    case <-ctx.Done():
        return "", ctx.Err()
    case <-time.After(20 * time.Millisecond): // pretend network/DB
    }
    db.mu.Lock()
    defer db.mu.Unlock()
    db.hits++
    v, ok := db.rows[key]
    if !ok {
        return "", errors.New("not found")
    }
    return v, nil
}

// FakeRedis is a tiny TTL cache.
type FakeRedis struct {
    mu    sync.Mutex
    items map[string]cacheItem
}

type cacheItem struct {
    val string
    exp time.Time
}

func NewRedis() *FakeRedis {
    return &FakeRedis{items: map[string]cacheItem{}}
}

func (r *FakeRedis) Get(_ context.Context, key string) (string, bool) {
    r.mu.Lock()
    defer r.mu.Unlock()
    it, ok := r.items[key]
    if !ok || time.Now().After(it.exp) {
        delete(r.items, key)
        return "", false
    }
    return it.val, true
}

func (r *FakeRedis) Set(_ context.Context, key, val string, ttl time.Duration) {
    r.mu.Lock()
    defer r.mu.Unlock()
    r.items[key] = cacheItem{val: val, exp: time.Now().Add(ttl)}
}
```

```go
func getUser(ctx context.Context, db *FakeDB, cache *FakeRedis, id string) (string, string,
    if v, ok := cache.Get(ctx, id); ok {
        return v, "cache", nil
    }
    v, err := db.QueryRow(ctx, id)
    if err != nil {
        return "", "", err
    }
    cache.Set(ctx, id, v, time.Minute)
    return v, "db", nil
}

func main() {
    db := &FakeDB{rows: map[string]string{"u1": "alice"}}
    cache := NewRedis()
    ctx := context.Background()

    v1, src1, err := getUser(ctx, db, cache, "u1")
    if err != nil {
        panic(err)
    }
    v2, src2, err := getUser(ctx, db, cache, "u1")
    if err != nil {
        panic(err)
    }

    // Cancelled context aborts the "query"
    cctx, cancel := context.WithCancel(context.Background())
    cancel()
    _, _, cerr := db.QueryRow(cctx, "u1")

    fmt.Println("first:", v1, "via", src1)
    fmt.Println("second:", v2, "via", src2)
    fmt.Println("db hits:", db.hits)
    fmt.Println("cancelled err:", cerr)
}
```

Expected output:

```
first: alice via db
second: alice via cache
db hits: 1
cancelled err: context canceled
```

What to notice: Look-aside cache protects the DB on hot keys (`db.hits` stays 1). Passing a cancelled `context` stops work early—the same discipline you need with `pgx` / `go-redis`.

Try next: Expire the cache entry with a 1ms TTL and sleep before the second read. Add a simple

rate limiter using an in-memory counter + window (Redis `INCR`/`EXPIRE` analogue).

HTTP and Networking

HTTP and Networking

Overview

Go's `net/http` package provides a complete HTTP client and server implementation.

HTTP Client

```go
// Simple GET
resp, err := http.Get("https://api.example.com/data")
if err != nil {
    log.Fatal(err)
}
defer resp.Body.Close()

body, _ := io.ReadAll(resp.Body)
```

Custom Request

```go
req, _ := http.NewRequest("POST", url, bytes.NewBuffer(data))
req.Header.Set("Content-Type", "application/json")
req.Header.Set("Authorization", "Bearer "+token)

client := &http.Client{Timeout: 10 * time.Second}
resp, err := client.Do(req)
```

With Context

```go
ctx, cancel := context.WithTimeout(context.Background(), 5*time.Second)
defer cancel()

req, _ := http.NewRequestWithContext(ctx, "GET", url, nil)
resp, err := http.DefaultClient.Do(req)
```

HTTP Server

```go
func handler(w http.ResponseWriter, r *http.Request) {
    fmt.Fprintf(w, "Hello, %s!", r.URL.Path[1:])
}

func main() {
    http.HandleFunc("/", handler)
    log.Fatal(http.ListenAndServe(":8080", nil))
}
```

JSON API

```go
func userHandler(w http.ResponseWriter, r *http.Request) {
    user := User{Name: "Alice"}

    w.Header().Set("Content-Type", "application/json")
    json.NewEncoder(w).Encode(user)
}
```

Middleware

```go
func logging(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        log.Printf("%s %s", r.Method, r.URL)
        next.ServeHTTP(w, r)
    })
}

http.Handle("/", logging(http.HandlerFunc(handler)))
```

ServeMux

```go
mux := http.NewServeMux()
mux.HandleFunc("/users", usersHandler)
mux.HandleFunc("/posts", postsHandler)

http.ListenAndServe(":8080", mux)
```

Static Files

```
fs := http.FileServer(http.Dir("./static"))
http.Handle("/static/", http.StripPrefix("/static/", fs))
```

Summary

Function	Purpose
`http.Get()`	Simple GET request
`http.Post()`	Simple POST request
`http.NewRequest()`	Custom request
`http.HandleFunc()`	Register handler
`http.ListenAndServe()`	Start server

Worked example

JSON POST API with middleware, tested via `httptest.NewServer`.

Save as `main.go`. Then:

```
go mod init example
go run .
```

```
package main

import (
    "encoding/json"
    "fmt"
    "io"
    "net/http"
    "net/http/httptest"
    "strings"
)

type echoReq struct {
    Msg string `json:"msg"`
}

type echoResp struct {
    Echo string `json:"echo"`
}
```

```go
func withJSON(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        w.Header().Set("Content-Type", "application/json")
        next.ServeHTTP(w, r)
    })
}

func main() {
    mux := http.NewServeMux()
    mux.Handle("POST /echo", withJSON(http.HandlerFunc(func(w http.ResponseWriter, r *http.R
        var req echoReq
        if err := json.NewDecoder(r.Body).Decode(&req); err != nil || req.Msg == "" {
            w.WriteHeader(http.StatusBadRequest)
            _ = json.NewEncoder(w).Encode(map[string]string{"error": "bad json"})
            return
        }
        _ = json.NewEncoder(w).Encode(echoResp{Echo: strings.ToUpper(req.Msg)})
    })))

    srv := httptest.NewServer(mux)
    defer srv.Close()

    res, err := http.Post(srv.URL+"/echo", "application/json", strings.NewReader(`{"msg":"hi
    if err != nil {
        panic(err)
    }
    body, _ := io.ReadAll(res.Body)
    res.Body.Close()
    fmt.Println("status:", res.StatusCode)
    fmt.Println("body:", strings.TrimSpace(string(body)))
}
```

Expected output:

```
status: 200
body: {"echo":"HI"}
```

More examples

`httptest.NewRecorder` unit test of a pure handler (no listening socket).

```go
package main

import (
    "fmt"
```

```
    "net/http"
    "net/http/httptest"
)

func health(w http.ResponseWriter, r *http.Request) {
    w.WriteHeader(http.StatusOK)
    _, _ = w.Write([]byte("ok"))
}

func main() {
    rec := httptest.NewRecorder()
    req := httptest.NewRequest(http.MethodGet, "/health", nil)
    health(rec, req)
    fmt.Println(rec.Code, rec.Body.String())
}
```

Expected output:

```
200 ok
```

Runnable example

Save as `main.go`. Then:

```
go mod init example
go run .
```

```
package main

import (
    "context"
    "encoding/json"
    "fmt"
    "io"
    "log"
    "net/http"
    "net/http/httptest"
    "strings"
    "time"
)

type User struct {
    Name string `json:"name"`
}
```

```go
func logging(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        start := time.Now()
        next.ServeHTTP(w, r)
        log.Printf("%s %s %s", r.Method, r.URL.Path, time.Since(start))
    })
}

func userHandler(w http.ResponseWriter, r *http.Request) {
    w.Header().Set("Content-Type", "application/json")
    _ = json.NewEncoder(w).Encode(User{Name: "Alice"})
}

func main() {
    mux := http.NewServeMux()
    mux.HandleFunc("GET /users/{id}", userHandler)
    mux.HandleFunc("GET /health", func(w http.ResponseWriter, r *http.Request) {
        _, _ = io.WriteString(w, "ok")
    })

    // Middleware chain + real HTTP without ListenAndServe forever
    srv := httptest.NewServer(logging(mux))
    defer srv.Close()

    // Client with timeout + context
    client := &http.Client{Timeout: 2 * time.Second}
    ctx, cancel := context.WithTimeout(context.Background(), time.Second)
    defer cancel()

    req, err := http.NewRequestWithContext(ctx, http.MethodGet, srv.URL+"/users/42", nil)
    if err != nil {
        panic(err)
    }
    req.Header.Set("Accept", "application/json")

    resp, err := client.Do(req)
    if err != nil {
        panic(err)
    }
    defer resp.Body.Close()

    body, _ := io.ReadAll(resp.Body)
    fmt.Println("status:", resp.StatusCode)
    fmt.Println("content-type:", resp.Header.Get("Content-Type"))
    fmt.Println("body:", strings.TrimSpace(string(body)))

    // httptest.NewRecorder for pure handler unit tests (no server)
```

```go
    rec := httptest.NewRecorder()
    req2 := httptest.NewRequest(http.MethodGet, "/health", nil)
    mux.ServeHTTP(rec, req2)
    fmt.Println("health:", rec.Code, strings.TrimSpace(rec.Body.String()))
}
```

Expected output: (log line may interleave; body lines are stable)

```
status: 200
content-type: application/json
body: {"name":"Alice"}
health: 200 ok
```

What to notice: `httptest.NewServer` is ideal for client+server integration examples; `httptest.NewRecorder` unit-tests handlers with zero network. Always set client timeouts and prefer `NewRequestWithContext`.

Try next: Add a `POST /users` that decodes JSON. Chain a second middleware that rejects missing `Accept` headers.

Related Topics

- Context and Cancellation
- Encoding and Serialization

Encoding and Serialization

Encoding and Serialization

Overview

Go provides standard encoding packages for JSON, XML, and binary data.

JSON

`encoding/json` remains the compatibility API. Since Go 1.27 it is **backed by the v2 implementation**: marshal/unmarshal behavior is preserved, error text may differ, and unmarshal is faster. New code that wants stricter defaults (reject invalid UTF-8, reject duplicate names, case-sensitive field match) should import `encoding/json/v2`. Rollback for a v1-compat emergency: `GOEXPERIMENT=nojsonv2` at build time (temporary; expected to go away).

```go
import "encoding/json"

type User struct {
    Name  string `json:"name"`
    Email string `json:"email,omitempty"`
    Age   int    `json:"-"`  // Ignored
}

// Encode
user := User{Name: "Alice", Email: "a@b.com"}
data, _ := json.Marshal(user)
// {"name":"Alice","email":"a@b.com"}

// Decode
var u User
json.Unmarshal(data, &u)

// Pretty print
data, _ := json.MarshalIndent(user, "", "  ")
```

Streaming JSON

```go
// Encode to writer
encoder := json.NewEncoder(w)
encoder.Encode(user)

// Decode from reader
decoder := json.NewDecoder(r)
decoder.Decode(&user)
```

encoding/json/v2 (Go 1.27+)

```go
import jsonv2 "encoding/json/v2"

b, err := jsonv2.Marshal(user)
err = jsonv2.Unmarshal(b, &user)
err = jsonv2.MarshalWrite(w, user)
err = jsonv2.UnmarshalRead(r.Body, &user)
```

Options configure semantics without a new Encoder type:

```go
b, err := jsonv2.Marshal(user, jsonv2.RejectUnknownMembers(true))
```

Low-level token streaming lives in `encoding/json/jsontext`. Keep v1 (`encoding/json`) for existing code; migrate hot or strict APIs to v2 when you can test the stricter defaults.

XML

```go
import "encoding/xml"

type Person struct {
    XMLName xml.Name `xml:"person"`
    Name    string   `xml:"name"`
    Age     int      `xml:"age,attr"`
}

data, _ := xml.MarshalIndent(person, "", "  ")
xml.Unmarshal(data, &p)
```

Gob (Go Binary)

```go
import "encoding/gob"

// Encode
var buf bytes.Buffer
enc := gob.NewEncoder(&buf)
enc.Encode(data)

// Decode
dec := gob.NewDecoder(&buf)
dec.Decode(&result)
```

CSV

```go
import "encoding/csv"

// Write
w := csv.NewWriter(file)
w.Write([]string{"a", "b", "c"})
w.Flush()

// Read
r := csv.NewReader(file)
records, _ := r.ReadAll()
```

Base64

```go
import "encoding/base64"

encoded := base64.StdEncoding.EncodeToString(data)
decoded, _ := base64.StdEncoding.DecodeString(encoded)
```

Summary

Package	Format
`encoding/json`	JSON (v1 API; v2 engine in Go 1.27+)
`encoding/json/v2`	JSON with options and stricter defaults
`encoding/xml`	XML
`encoding/gob`	Go binary

Package	Format
`encoding/csv`	CSV
`encoding/base64`	Base64

Worked example

Streaming JSON encode/decode round-trip over an in-memory buffer.

Save as `main.go`. Then:

```
go mod init example
go run .
```

```
package main

import (
    "bytes"
    "encoding/json"
    "fmt"
)

type Event struct {
    Type string `json:"type"`
    N    int    `json:"n"`
}

func main() {
    var buf bytes.Buffer
    enc := json.NewEncoder(&buf)
    for i := 1; i <= 3; i++ {
        if err := enc.Encode(Event{Type: "tick", N: i}); err != nil {
            panic(err)
        }
    }

    dec := json.NewDecoder(&buf)
    sum := 0
    for dec.More() {
        var e Event
        if err := dec.Decode(&e); err != nil {
            panic(err)
        }
        sum += e.N
        fmt.Printf("event type=%s n=%d\n", e.Type, e.N)
    }
```

```go
    fmt.Println("sum:", sum)
}
```

Expected output:

```
event type=tick n=1
event type=tick n=2
event type=tick n=3
sum: 6
```

More examples

Tag-driven JSON omitempty + gob preserving unexported-unrelated fields.

```go
package main

import (
    "bytes"
    "encoding/gob"
    "encoding/json"
    "fmt"
)

type User struct {
    Name  string `json:"name"`
    Email string `json:"email,omitempty"`
    Score int    `json:"score,omitempty"`
}

func main() {
    u := User{Name: "Ada", Score: 0} // Score omitted when 0
    j, _ := json.Marshal(u)
    fmt.Println("json:", string(j))

    var buf bytes.Buffer
    _ = gob.NewEncoder(&buf).Encode(User{Name: "Ada", Score: 10})
    var out User
    _ = gob.NewDecoder(&buf).Decode(&out)
    fmt.Println("gob score:", out.Score)
}
```

Expected output:

```
json: {"name":"Ada"}
gob score: 10
```

Runnable example

Save as `main.go`. Then:

```
go mod init example
go run .
```

```go
package main

import (
    "bytes"
    "encoding/base64"
    "encoding/csv"
    "encoding/gob"
    "encoding/json"
    "encoding/xml"
    "fmt"
    "strings"
)

type User struct {
    Name  string `json:"name" xml:"name"`
    Email string `json:"email,omitempty" xml:"email"`
    Age   int    `json:"-" xml:"-"` // ignored by JSON/XML tags in this demo
}

type PersonXML struct {
    XMLName xml.Name `xml:"person"`
    Name    string   `xml:"name"`
    Age     int      `xml:"age,attr"`
}

func main() {
    u := User{Name: "Alice", Email: "a@b.com", Age: 30}

    // JSON round-trip
    j, err := json.Marshal(u)
    if err != nil {
        panic(err)
    }
    var u2 User
    if err := json.Unmarshal(j, &u2); err != nil {
        panic(err)
    }
    fmt.Println("json:", string(j))
    fmt.Println("json name:", u2.Name, "age zeroed:", u2.Age == 0)
```

```go
    // XML round-trip
    p := PersonXML{Name: "Bob", Age: 25}
    x, err := xml.Marshal(p)
    if err != nil {
        panic(err)
    }
    var p2 PersonXML
    if err := xml.Unmarshal(x, &p2); err != nil {
        panic(err)
    }
    fmt.Println("xml:", string(x))
    fmt.Println("xml age:", p2.Age)

    // Gob round-trip (Go-native binary)
    var buf bytes.Buffer
    if err := gob.NewEncoder(&buf).Encode(u); err != nil {
        panic(err)
    }
    var u3 User
    if err := gob.NewDecoder(&buf).Decode(&u3); err != nil {
        panic(err)
    }
    fmt.Println("gob age preserved:", u3.Age)

    // CSV
    var csvBuf strings.Builder
    w := csv.NewWriter(&csvBuf)
    _ = w.Write([]string{"name", "email"})
    _ = w.Write([]string{u.Name, u.Email})
    w.Flush()
    records, err := csv.NewReader(strings.NewReader(csvBuf.String())).ReadAll()
    if err != nil {
        panic(err)
    }
    fmt.Println("csv rows:", len(records), "data:", records[1])

    // Base64
    enc := base64.StdEncoding.EncodeToString([]byte("hi"))
    dec, _ := base64.StdEncoding.DecodeString(enc)
    fmt.Println("b64:", enc, "->", string(dec))
}
```

Expected output:

```
json: {"name":"Alice","email":"a@b.com"}
json name: Alice age zeroed: true
```

```
xml: <person age="25"><name>Bob</name></person>
xml age: 25
gob age preserved: 30
csv rows: 2 data: [Alice a@b.com]
b64: aGk= -> hi
```

What to notice: `json:"-"` drops `Age` from JSON, so it comes back as zero—gob has no tags and preserves the full struct. XML attributes use `,attr`. Prefer streaming `Encoder`/`Decoder` for large payloads and HTTP bodies.

Try next: Use `json.MarshalIndent` for pretty output. Stream-decode a JSON array with `decoder.Token` / `decoder.More`.

Related Topics

- I/O and Streaming
- HTTP and Networking

Systems Programming

Systems Programming Overview

Systems Programming Overview

This part covers how Go programs interact with the operating system: processes, signals, files, and stream-oriented command design. The emphasis is **operational correctness** — tools that behave predictably in production shells, CI, containers, and long-running supervisors.

Start with **Go as a systems language** (chapter 140a) if you want the *why*: on Linux, a `CGO_ENABLED=0` binary talks to the kernel itself and does not need libc. That is the property that makes the rest of this part deployable as one file.

Why Systems Programming in a Go Book?

Go is often introduced as a cloud/API language. It is also a **userspace systems language**: the runtime issues Linux syscalls directly, so Docker, Kubernetes, and most of the tools in this part ship as a single static binary. In practice, that same binary may:

- Run as a PID 1-ish service under systemd or Kubernetes
- Manage child processes (plugins, helpers, compilers)
- Rewrite thousands of config files safely
- Sit in a Unix pipeline next to `grep` and `jq`

If your HTTP handlers are solid but your process mishandles `SIGTERM`, you still page at 3 a.m.

```
                  +------------------+
signals -------> |   Go process     | ----> child processes
stdin/stdout --> |  main + runtime  | ----> filesystem
env/config ----> |                  | ----> network (other parts)
                  +------------------+
```

Learning Path

Chapter	Focus	Outcome
140a Go as a systems language	No libc on Linux, `CGO_ENABLED=0`, Go vs C	Why a Go binary runs on Alpine, `scratch`, and NixOS
141 Processes & signals	Lifecycle, graceful stop, supervisors	Clean shutdown and restart policy
142 Safe filesystem IO	Atomic writes, path jails, fsync tradeoffs	Automation that does not corrupt state

Chapter	Focus	Outcome
142b Write-ahead log	CRC + length + trailer framing, fsync ordering, torn-tail recovery, group commit	A log that refuses to replay what it cannot prove
143 Unix pipeline CLIs	stdin/stdout/stderr contracts, exit codes	Tools that compose in shell and CI
144 Env, user, CWD, umask	Identity and path resolution	Correct behavior under systemd/CI
145 File descriptors	FDs, pipes, inheritance, limits	No leaks; correct redirection
146 Memory-mapped files	mmap RO/RW tradeoffs	Random access without ReadAll
147 Unix domain sockets	Local IPC, HTTP over UDS	Sidecars without TCP ports
148 Locks & pidfiles	flock, single-instance tools	Safe cron/agents
149 Project: systool	Multi-command mini-tools	Capstone practice
150 Time & timers	Wall vs monotonic, ticker hygiene	Correct deadlines
151 Filesystem watching	Poll/fsnotify, debounce, reload	Hot config without races
152 Pseudo-terminals	PTY vs pipes, interactive wrap	Automate tty-only tools carefully
153 Resource limits	rlimit, GOMEMLIMIT, caps	Survive cgroup ceilings
154 Privileges	Non-root, capabilities, modes	Least privilege defaults
155 Ops logging	stderr/journal/syslog	Findable production logs
156 Project: sysops	deadline, reload, cronish, …	Ops toolkit capstone
157 Socket activation	systemd LISTEN_FDS	Privilege-separated bind
158 Linux procfs	`/proc/self`, cgroup peeks	Debug introspection
159 Project: minisup	multi-process supervisor	Restart + signal fanout

Core Concepts Map

Process as a state machine

```
starting -> running -> stopping -> exited
                |          ^
                +-> crash -+  (restart policy lives here)
```

Every production process needs explicit answers for: how do I start, how do I stop, what is my exit code vocabulary, and who restarts me?

The filesystem is not a database (but you can still be careful)

Partial writes and torn updates are normal under crash. Atomic rename patterns and directory fsync exist because people lost data.

CLI tools are network services for humans and scripts

Stable stdout, noisy stderr, and meaningful exit codes are the "API" of a CLI. Breaking them breaks automation.

Go Building Blocks

Area	Packages
Process / exec	`os/exec`, `os`, `syscall` (sparingly)
Signals	`os/signal`, `signal.NotifyContext`
Files	`os`, `io`, `io/fs`, `path/filepath`, `embed`
Env / identity	`os.Getenv`, `os/user`, `os.UserHomeDir`
FDs / pipes	`os.Pipe`, `os.File.Fd`, `cmd.ExtraFiles`
Local IPC	`net.Listen("unix", …)`, `unixgram`
Locks / mmap	`golang.org/x/sys/unix` (`Flock`, `Mmap`)
Time	`time`, `context` deadlines
Watch / reload	poll `Stat`, optional `fsnotify`
PTY	`creack/pty` (when needed)
Limits	`RLIMIT_*`, `GOMEMLIMIT`, `GOMAXPROCS`
Permissions	`os.FileMode`, platform differences
Structured ops logs	`log/slog` on stderr; optional `log/syslog`

Suggested schedule

```
Day 0      140a                why Go binaries need no libc (Linux)
Days 1-2   141 → 142 → 143     core lifecycle + files + pipelines
Day 3      144 → 145           env/cwd + file descriptors
Day 4      146 → 147 → 148     mmap, UDS, locks
Day 5      149                 build systool
Day 6      150 → 151 → 152     time, watch, PTY
Day 7      153 → 154 → 155     limits, privilege, logging
Day 8      156                 build sysops toolkit
Day 9      157 → 158 → 159     activation, procfs, minisup
```

Modern signal handling skeleton

```go
ctx, stop := signal.NotifyContext(context.Background(), os.Interrupt, syscall.SIGTERM)
defer stop()

// pass ctx to servers, workers, exec.CommandContext
<-ctx.Done()
// drain with a second hard-deadline context
```

Prefer `signal.NotifyContext` over manual channel plumbing in new code (Go 1.16+).

Production Mindset

Ask for every systems feature:

1. **What happens on SIGTERM?** (deadline? drain? exit code?)
2. **What happens if power fails mid-write?**
3. **Can a malicious path escape the working root?**
4. **Can this tool run non-interactively in CI?**
5. **What do metrics/logs say when a child exits 137 (OOM)?**

Relationship to Other Parts

- **Network systems (13)** — servers need the shutdown patterns from 141.
- **Distributed infra (15)** — workers are processes with queues and backpressure.
- **Security (11, 17)** — path safety and least privilege are hardening.
- **Projects (99)** — many Linux-clone labs exercise these chapters.

How to Study

1. Implement a long-running worker that exits cleanly on SIGTERM within N seconds.
2. Write a config rewriter that survives `kill -9` mid-write (atomic replace).
3. Build a filter CLI used as `cat data.log | yourtool | jq`.

Literacy Checklist

- ☐ Explain why a Linux Go binary can omit libc, and when cgo puts it back
- ☐ Explain SIGTERM vs SIGKILL
- ☐ Use `CommandContext` so children die with parent cancel
- ☐ Perform write-temp-fsync-rename safely
- ☐ Jail user paths under a root directory
- ☐ Design stdout/stderr/exit-code contracts for scripting

- ☐ Avoid shell injection via `sh -c` with untrusted input
- ☐ Reason about CWD vs binary-relative paths under systemd
- ☐ Avoid FD leaks; know stdin/stdout/stderr roles
- ☐ When to mmap vs stream reads
- ☐ Serve or dial a Unix domain socket safely (permissions)
- ☐ Single-instance exclusion with flock (not pidfile alone)
- ☐ Use monotonic-friendly timing and cancelable timers
- ☐ Hot-reload config with debounce and last-good retention
- ☐ Know when a PTY is required vs a plain pipe
- ☐ Name the limit behind EMFILE / OOMKill
- ☐ Prefer non-root + clear EPERM errors
- ☐ Log to stderr/journal with structure; no secrets

Part Warm-Up Exercises

1. On a sample service, document current signal handling (or lack of it).
2. Trace what Kubernetes sends on pod termination and the `terminationGracePeriodSeconds` interaction.
3. Find one script in your life that greps program stdout for errors — that program violated the stderr rule.
4. List three files your systems write that would hurt if truncated mid-write.
5. Compare `os.Exit` in library code vs returning errors to `main`.

Platform Notes (Unix-first)

These chapters assume a Unix-like environment (Linux, macOS, BSD) because that is where most Go servers and CLIs run in production. Windows supports many of the same Go APIs (`os/exec`, signals subset, paths) but:

- Signal semantics differ (no true `SIGTERM` from all supervisors)
- Atomic rename and file locking differ
- Path separators and reserved names differ

When writing cross-platform tools, isolate OS-specific code behind small files (`path_unix.go` / `path_windows.go`) and test both in CI.

Container and CI Reality

Environment	Systems concern
Kubernetes pod	SIGTERM + grace period; probes; non-root
systemd unit	`Type=notify` optional; restart policy
GitHub Actions	non-interactive; no TTY; pipe stdout carefully
Developer laptop	Ctrl-C = SIGINT; human-readable defaults OK

Design for the **non-interactive** case first; make interactive polish optional.

Mini Project Thread

Build one coherent ops utility across chapters:

1. **141** — long-running worker with graceful shutdown
2. **142** — atomically persists checkpoints
3. **143** — also usable as a filter in a pipeline
4. **144–145** — correct CWD/env and FD hygiene
5. **147–148** — optional control socket + single-instance lock
6. **149** — assemble as multi-command `systool`
7. **150–155** — deadlines, hot reload, limits, least privilege, logging
8. **156** — assemble `sysops` (deadline, reload, cronish, healthsock)

That combination is the skeleton of many internal ops utilities.

Failure Stories Worth Remembering

- Config rewrite crashed mid-write → empty file → mass restart loop
- CLI printed logs to stdout → `jq` parse failures in CI
- Parent killed without waiting → zombie children filled PID table
- `Command` with `sh -c` and user input → shell injection

Each chapter exists to prevent one of these classes of failure.

Checklist Before Shipping a Systems Tool

☐ `main` returns errors; `os.Exit` only at the top
☐ SIGTERM/SIGINT cancel a root context
☐ Children use `CommandContext` or explicit term/kill
☐ Durable state uses atomic write patterns
☐ User paths cannot escape the data root
☐ stdout is data; stderr is diagnostics
☐ Exit codes documented for automation
☐ Works with stdin piped (no TTY required)
☐ Logs never include secrets from env files

Sample Directory Layout

```
cmd/mytool/main.go
internal/app/run.go
internal/app/run_test.go
internal/fsutil/atomic.go
internal/fsutil/path.go
```

Keep OS edge cases in `fsutil`; keep business logic testable with `io.Reader`/`Writer`.

How This Part Connects to Network Services

Your HTTP server **is** a process:

- Chapter 141 shutdown patterns plug into `http.Server.Shutdown`
- Chapter 142 patterns protect TLS key material drops and config reloads
- Chapter 143 discipline applies to `kubectl`-style companion CLIs you ship beside the service

Do not treat "systems programming" as only embedded or C territory — production Go is systems software. Chapter 140a spells out where Go replaces C (portable userspace) and where it does not (kernel, hard real-time).

Glossary

- **Graceful shutdown** — stop accepting; drain; then exit.
- **Atomic replace** — rename-based file update without torn reads.
- **Path jail** — force paths under a root directory.
- **Exit code** — process-level API for scripts and supervisors.
- **SIGPIPE** — write to closed pipe; common in `| head` pipelines.
- **PID 1** — init process in a container; reaps children / signal quirks.
- **TOCTOU** — time-of-check to time-of-use race on files.

More examples

Tour: signal context + atomic temp file

```
mkdir -p /tmp/go-sys-tour && cd /tmp/go-sys-tour
go mod init example.com/sys-tour
```

Save as `main.go`:

```go
package main

import (
    "context"
    "fmt"
    "os"
    "os/signal"
    "path/filepath"
    "syscall"
    "time"
)

func atomicWrite(path, body string) error {
    dir := filepath.Dir(path)
    f, err := os.CreateTemp(dir, ".tmp-*")
    if err != nil {
        return err
    }
    tmp := f.Name()
    if _, err := f.WriteString(body); err != nil {
        f.Close()
        os.Remove(tmp)
        return err
    }
    if err := f.Close(); err != nil {
        os.Remove(tmp)
        return err
    }
    return os.Rename(tmp, path)
}

func main() {
    ctx, stop := signal.NotifyContext(context.Background(), os.Interrupt, syscall.SIGTERM)
    defer stop()
    go func() {
        time.Sleep(20 * time.Millisecond)
        stop()
    }()
    <-ctx.Done()
    fmt.Println("signal path:", ctx.Err())

    dir, _ := os.MkdirTemp("", "sys")
    defer os.RemoveAll(dir)
    path := filepath.Join(dir, "cfg.json")
    if err := atomicWrite(path, `{"ok":true}`); err != nil {
        panic(err)
    }
```

```
    b, _ := os.ReadFile(path)
    fmt.Println("file:", string(b))
}
```

```
go run .
```

Expected output:

```
signal path: context canceled
file: {"ok":true}
```

Runnable example

Tour of this part: print process identity, catch a simulated shutdown via context, and write a line to stdout with a non-zero exit path—the process contract network services and CLIs share.

```
mkdir -p /tmp/go-sys-tour && cd /tmp/go-sys-tour
go mod init example.com/sys-tour
```

Save as `main.go`:

```
package main

import (
    "context"
    "fmt"
    "os"
    "time"
)

func run(ctx context.Context) error {
    fmt.Fprintf(os.Stdout, "pid=%d\n", os.Getpid())
    select {
    case <-ctx.Done():
        fmt.Fprintln(os.Stderr, "shutting down:", ctx.Err())
        return nil
    case <-time.After(50 * time.Millisecond):
        fmt.Fprintln(os.Stdout, "work done")
        return nil
    }
}

func main() {
    ctx, cancel := context.WithTimeout(context.Background(), 100*time.Millisecond)
    defer cancel()
```

```go
    if err := run(ctx); err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }
    fmt.Fprintln(os.Stdout, "tour: process + context + exit codes")
}
```

```go
go run .
```

Expected output:

```
pid=<n>
work done
tour: process + context + exit codes
```

What to notice

- Stdout is for data/results; stderr for diagnostics—pipeline-friendly tools depend on that split.
- Context cancellation is the in-process form of SIGTERM handling (chapter 141).

Try next

- Wire `signal.NotifyContext` and send `Ctrl-C`.
- Continue with atomic file writes (142) and stdin filters (143).

Related in this book

- How Go reaches the kernel / static binaries: 140a Go as a Systems Language
- Low-level syscalls / eBPF / Delve: 92 Systems Programming (specialized)
- CLI curriculum (flags, Cobra, net tools): 23 CLI Tools
- Runtime deep dives: 20 Go Deep Dives

Next Chapter

Processes, Signals, and Supervisors — the lifecycle model everything else hangs on.

Go as a Systems Language

Go as a Systems Language

Go is often introduced as a cloud and API language. That understates the design. Docker, containerd, Kubernetes, etcd, Prometheus, and Terraform are **userspace systems software**: they create processes, talk to the kernel, ship as a single file, and have to run on machines the author never configured.

This chapter is the *why* behind that property. Later chapters in this part teach processes, files, pipelines, and supervisors. Here you learn how a Go binary reaches the kernel **without libc**, why that matters for portability, and when C is still the right tool.

> Structure inspired by public systems writeups (notably Using Go for Systems Programming on iximiuz Labs). Prose, diagrams, and examples here are original to this book.

Why this belongs in a Go book

A C programmer learning process creation, `mmap`, or file descriptors usually goes through **libc**: the C standard library. POSIX is specified in C. The Linux kernel is written in C. libc is the conventional bridge from userspace to the kernel.

Go takes a different route **on Linux**: the runtime and standard library issue system calls themselves. The resulting binary does not need `libc.so` on the target. That is why a `CGO_ENABLED=0` Linux binary can run on Debian, Alpine, NixOS, or a `scratch` container as long as the kernel ABI matches.

You still cannot write kernel modules or firmware in Go. "Systems language" here means **portable userspace that speaks the OS**, not "replaces C everywhere."

```
C (typical Linux build)
-----------------------
your code  -->  libc (printf, malloc, fopen)
                    |
                    v
                kernel  (write, brk/mmap, open)

Go on Linux (CGO_ENABLED=0)
---------------------------
your code  -->  Go stdlib + runtime
                    |
                    v
                kernel  (same syscalls, no libc in the middle)
```

How a C program depends on libc

`printf` is not a system call. It is a libc function. libc formats the string, then eventually invokes the kernel's `write`. `malloc` eventually becomes `brk` or `mmap`. `fopen` becomes `open`. Your C source talks to libc; libc talks to the kernel.

A default `gcc hello.c -o hello` on Linux produces a **dynamically linked** ELF. The binary does not contain libc. At start-up the kernel loads a **dynamic linker** (the ELF interpreter) which then maps shared libraries:

Distro family	Dynamic linker you typically see
Debian, Ubuntu, Fedora, RHEL (glibc)	`/lib64/ld-linux-x86-64.so.2`
Alpine (musl)	`/lib/ld-musl-x86_64.so.1`

Inspect it:

```
file ./hello
# ELF 64-bit ... dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2 ...

ldd ./hello
# libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6
```

`ldd` lists shared-library dependencies. If the interpreter path or a required `.so` is missing, the program does not run.

The "not found" surprise

Copy a glibc-linked binary onto Alpine and run it. The shell often prints `not found` even though the file exists and is executable. The kernel is not looking for *your* file — it is looking for the **interpreter** named in the ELF header (`/lib64/ld-linux-x86-64.so.2`). Alpine does not ship that file.

glibc and musl implement the same C API on paper and are **not binary-compatible**. Symbol versions differ. Dynamic linker paths differ. A dynamically linked C tool built on Ubuntu is not a portable Linux tool.

You *can* statically link C (`gcc -static`). That removes the runtime `.so` search, but the binary still embeds **one** libc's implementation and assumptions. It is larger, licensing of glibc-static is awkward, and you still do not get Go's own scheduler, allocator, and syscall wrappers.

How a Go program reaches the kernel

Compile a tiny program with the Go toolchain:

```go
package main

import "fmt"

func main() {
    fmt.Println("hello from a self-contained binary")
}
```

```bash
CGO_ENABLED=0 go build -o hello-go .
file hello-go
# On Linux: ELF 64-bit ... statically linked ...
```

Statically linked means the kernel can map this one file and run it. There is no ELF interpreter for libc, no `libc.so` to locate.

```bash
ldd hello-go
# Linux glibc ldd:   "not a dynamic executable"  or  "statically linked"
# Alpine musl ldd:   "Not a valid dynamic program"
```

Both messages mean the same thing: there is no dynamic section to inspect.

Copy the binary to `/tmp` on a machine that does not have the Go toolchain installed. It still runs. The host libc does not matter. The kernel syscall interface does.

What is actually inside

Every Go program includes the **Go runtime**:

- goroutine scheduler (Gs on Ms, with Ps)
- memory allocator and garbage collector
- signal handling and stack growth
- the assembly stubs that place syscall arguments in the right registers and execute `syscall` (Linux)

On Linux those stubs live in the standard library / runtime (`Syscall`, `RawSyscall`, and architecture-specific `asm_linux_*.s` files). You do not call them for ordinary I/O — `os`, `fmt`, `net`, and `io` do.

`strace` makes the difference visible. A C `printf` hello shows a short list (`write`, `exit_group`, a bit of libc setup). A Go hello shows more: `mmap`, `mprotect`, `clone`, `futex`, `rt_sigaction` — the runtime bringing up its own world — then `write` for the line of text. Those calls originate in the Go binary, not in libc.

```
C hello                         Go hello (Linux, no cgo)
-------                         ------------------------
loader + libc init              runtime init (mmap, clone, futex, ...)
write("hello\n")                write("hello\n")
exit_group                      exit_group
```

Linux vs other kernels

This "no libc" story is **Linux-specific**:

GOOS	How Go talks to the OS
`linux`	Direct syscalls; `CGO_ENABLED=0` binaries are fully static
`darwin`	Through `libSystem` — Apple does not treat the raw syscall table as a stable ABI
`windows`	Windows APIs / ntdll, not a Unix libc
`freebsd` / others	OS-specific; do not assume Linux static behaviour

The portability win that made Go famous in containers — one file, Alpine or Debian, `scratch` image — is the Linux + `CGO_ENABLED=0` combination.

CGO puts libc back in the picture

cgo is how Go calls C. When it is enabled, the linker may pull in libc and you are back to dynamic dependencies.

On a machine with a C compiler, `CGO_ENABLED` defaults to `1`. A package that looks like pure Go can still pick up cgo through `net` (system DNS resolver) or `os/user` (NSS lookups) unless you disable it.

```
# Guarantee a pure-Go, fully static Linux binary:
CGO_ENABLED=0 go build -o hello-go .

# Confirm what the toolchain actually used:
go env CGO_ENABLED
go version -m hello-go | grep CGO
```

With `CGO_ENABLED=0`, networking and user lookup use **pure-Go implementations** (Go's DNS resolver, Go's `/etc/passwd` parsing). That is usually what you want for containers and cross-compiled CLIs. Use cgo when you *must* call a C library that has no acceptable Go equivalent — and then accept a C toolchain, harder cross-compiles, and a libc on the target. See cgo.

Cross-compilation is a systems feature

Because the toolchain ships the runtime and standard library for many `GOOS/GOARCH` pairs, you can build a Linux AMD64 binary on a Mac or a Windows box:

```
GOOS=linux GOARCH=amd64 CGO_ENABLED=0 go build -o hello-linux-amd64 .
GOOS=linux GOARCH=arm64 CGO_ENABLED=0 go build -o hello-linux-arm64 .
```

The compiler emits **Plan 9 pseudo-assembly** (an architecture-neutral IR with `SB` for static base, `SP` for the stack) and the Go linker writes the final ELF/Mach-O/PE file. You do not need a cross-`gcc` unless cgo is on.

Inspect what the compiler produced:

```
go build -gcflags=-S .          # Plan 9-ish listing on stderr
go tool objdump -s main.main hello-linux-amd64
```

`go tool objdump` understands every architecture the toolchain can emit. You can disassemble an ARM64 binary on an x86-64 laptop; the text uses Go's register names (`R0`…, `R28` as the current `g` on ARM64), not necessarily the vendor ISA manual's names.

Deeper ABI and `NOSPLIT` notes live in ABI, assembly, and go:nosplit.

Trade-offs: Go vs C for systems work

Why teams pick Go

Strength	What it buys you
Static Linux binaries	One file; no glibc/musl lottery; `scratch` / distroless images
Memory safety by default	Bounds checks, no pointer arithmetic; fewer class-of-bug CVEs
Goroutines	Concurrent servers and CLIs without hand-rolled pthreads
Fast builds + first-class cross-compile	CI matrix without a C cross-toolchain
Batteries in the stdlib	HTTP, TLS, JSON, crypto — no hunt for a hash map

That combination is why so much **cloud infrastructure** is Go. The programs need to run in many environments, talk to the kernel, and stay easy to build.

What you pay

Cost	Consequence
Runtime + GC in every binary	Larger files; CPU/memory for the scheduler and collector
GC is not hard-real-time	Fine for almost all servers and CLIs; wrong for hard µs deadlines
Less control of layout	C still wins for packed DMA structures and bare metal
Not for the kernel	Linux modules and firmware stay C (or Rust, in some trees)

The Go GC has improved a lot (including the Green Tea collector becoming the default in Go 1.26). "GC pause" is rarely the reason to reject Go for a network service. It *is* a reason to reject Go for a hard real-time controller.

Choose C when

- You are writing kernel code, a driver, or firmware
- Every byte and cycle is budgeted (tiny MCUs, some embedded)
- You need deterministic allocation with no collector
- You must control exact memory layout and registers

Choose Go when

- You are building network services, CLIs, or DevOps / cloud control planes
- You want safe concurrency without managing OS threads by hand
- You need one Linux binary that runs across distros and in `scratch`
- You would rather not manage memory by hand

Go does not replace C. It occupies the large space of **userspace systems software** where C's portability tax is higher than Go's runtime tax.

Literacy checklist

- ☐ Explain why `printf` is not a syscall
- ☐ Name glibc vs musl and why a dynamically linked Ubuntu binary fails on Alpine
- ☐ Read `file` and `ldd` well enough to spot a libc dependency
- ☐ Explain why `./binary: not found` can mean "missing dynamic linker"
- ☐ Build with `CGO_ENABLED=0` and know what it changes (DNS, `os/user`, linkage)
- ☐ Cross-compile with `GOOS` / `GOARCH` without a C cross-compiler
- ☐ Know that the no-libc story is Linux-specific (Darwin uses libSystem)
- ☐ List one reason to pick C anyway (kernel, hard real-time, layout)

Exercises

1. **Link story** — On Linux, compile a 5-line C hello and a 5-line Go hello. Run `file` and `ldd` on both. Write one sentence about each output.
2. **Force static Go** — Build the same Go program twice, once with `CGO_ENABLED=0` and once with the default. Compare `ldd` and `ls -l`. If they match, check `go env CGO_ENABLED` and whether any imported package uses cgo.
3. **Cross-compile** — From whatever OS you use, produce `GOOS=linux GOARCH=amd64 CGO_ENABLED=0` and `GOOS=linux GOARCH=arm64 CGO_ENABLED=0` binaries. Disassemble `main.main` of both with `go tool objdump -s main.main`.
4. **strace (Linux)** — `strace -c` both hellos. Which extra syscalls does the Go runtime need just to start?
5. **The Alpine thought experiment** — Without a spare Alpine box: given an ELF interpreter of `/lib64/ld-linux-x86-64.so.2`, predict the exact failure mode on a musl-only rootfs.

Runnable example

This program reports what the binary believes about its build, then writes one line. After `go build`, inspect the artifact with `file` / `ldd` (Linux) or `otool -L` (macOS).

```
mkdir -p /tmp/go-as-sys && cd /tmp/go-as-sys
go mod init example.com/as-sys
```

Save as `main.go`:

```go
package main

import (
    "fmt"
    "os"
    "runtime"
    "runtime/debug"
)

func cgoEnabled() string {
    info, ok := debug.ReadBuildInfo()
    if !ok {
        return "unknown"
    }
    for _, s := range info.Settings {
        if s.Key == "CGO_ENABLED" {
            return s.Value
        }
    }
    return "unknown"
}

func main() {
    fmt.Fprintf(os.Stdout, "goos=%s goarch=%s cgo=%s\n",
        runtime.GOOS, runtime.GOARCH, cgoEnabled())
    fmt.Fprintln(os.Stdout, "this write is a kernel syscall on Linux - no libc required")
}
```

```
CGO_ENABLED=0 go build -o as-sys .
./as-sys

# Linux:
file ./as-sys
ldd ./as-sys || true

# Cross-compile a Linux static binary from any host:
```

```
GOOS=linux GOARCH=amd64 CGO_ENABLED=0 go build -o as-sys-linux-amd64 .
file ./as-sys-linux-amd64
```

Expected output (host fields vary):

```
goos=linux goarch=amd64 cgo=0
this write is a kernel syscall on Linux - no libc required
```

What to notice

- `runtime.GOOS` is the *target* you compiled for, not necessarily the machine you typed on.
- `CGO_ENABLED=0` is a build-time switch; `ReadBuildInfo` records what the linker used.
- On Linux, `file` should say statically linked. On macOS the same source still links `libSystem` — that is Darwin, not a failed build.

Try next

- `go tool objdump -s main.main ./as-sys` and find the `CALL` to `fmt.Fprintf`.
- Continue with Processes, signals, and supervisors: the binary you just built *is* the process those chapters supervise.

Related in this book

- Why Go exists — single-binary design goal
- cgo — how libc comes back
- Building and releasing — `GOOS` / `GOARCH` / strip flags
- Containerization — `scratch` and distroless
- Go on NixOS and FreeBSD — static binaries vs missing FHS loaders
- Specialized: syscalls, Delve, eBPF
- Linking and build modes
- ABI and assembly

Next Chapter

Systems Programming Overview — map of processes, files, pipelines, and the ops toolkit. Then start with processes and signals.

Processes, Signals, and Supervisors

Processes, Signals, and Supervisors

Process supervision is about controlling **state transitions under uncertainty**. Crashes, deploys, OOMs, and hung IO all force transitions; production software defines what happens next.

Why / Overview

A Go binary is a process — on Linux, usually a **self-contained** one (Go as a Systems Language). Orchestrators (systemd, Kubernetes, Nomad, Docker) are supervisors. Your code must cooperate with them:

- Start quickly enough to pass readiness
- Serve while healthy
- On stop signal: drain work, close listeners, exit
- On failure: exit with a meaningful code so the supervisor can decide restart policy

```
starting -> running -> stopping -> exited
                |          ^
                +-> crash-+
```

Signals You Must Handle

Signal	Typical meaning	Your response
`SIGTERM`	Graceful stop (K8s, systemd)	Drain + exit 0 if clean
`SIGINT`	Ctrl-C	Same as SIGTERM for services
`SIGHUP`	Reload config (classic Unix)	Optional: reload, or ignore
`SIGKILL`	Hard kill (uncatchable)	Cannot handle; keep critical writes atomic
`SIGPIPE`	Write to closed pipe	Often ignored; CLI tools should handle `EPIPE`

```go
ctx, stop := signal.NotifyContext(context.Background(), os.Interrupt, syscall.SIGTERM)
defer stop()
```

`NotifyContext` cancels when a signal arrives. Use that `ctx` everywhere: HTTP servers, worker loops, `exec.CommandContext`.

Graceful Shutdown Pattern

```go
package main

import (
    "context"
    "errors"
    "log/slog"
    "net/http"
    "os"
    "os/signal"
    "syscall"
    "time"
)

func main() {
    ctx, stop := signal.NotifyContext(context.Background(), os.Interrupt, syscall.SIGTERM)
    defer stop()

    mux := http.NewServeMux()
    mux.HandleFunc("GET /work", func(w http.ResponseWriter, r *http.Request) {
        // Honor cancellation so Shutdown can finish.
        select {
        case <-r.Context().Done():
            return
        case <-time.After(2 * time.Second):
            w.Write([]byte("done"))
        }
    })

    srv := &http.Server{Addr: ":8080", Handler: mux, ReadHeaderTimeout: 5 * time.Second}

    errCh := make(chan error, 1)
    go func() {
        slog.Info("listening", "addr", srv.Addr)
        errCh <- srv.ListenAndServe()
    }()

    select {
    case <-ctx.Done():
        slog.Info("signal received, shutting down")
    case err := <-errCh:
        if err != nil && !errors.Is(err, http.ErrServerClosed) {
            slog.Error("server failed", "err", err)
            os.Exit(1)
        }
```

```go
    }

    shutdownCtx, cancel := context.WithTimeout(context.Background(), 15*time.Second)
    defer cancel()
    if err := srv.Shutdown(shutdownCtx); err != nil {
        slog.Error("graceful shutdown failed", "err", err)
        _ = srv.Close()
        os.Exit(1)
    }
    slog.Info("bye")
}
```

Two-phase stop

1. **Soft deadline** (e.g. 15s): stop accepting, drain in-flight.
2. **Hard deadline**: cancel contexts, close forcibly, exit non-zero if work abandoned.

Kubernetes: `terminationGracePeriodSeconds` must be **greater** than your soft deadline, or you get SIGKILL mid-drain.

Exit Codes as API

Code	Convention
0	Success
1	General failure
2	Misuse / bad flags (CLI)
130	Interrupted by SIGINT (128+2) often
137	SIGKILL / OOM kill (128+9) often
143	SIGTERM (128+15) often

Supervisors and CI branch on these. Do not `os.Exit(0)` after a failed deploy migration.

```go
func main() {
    if err := run(); err != nil {
        slog.Error("exit", "err", err)
        os.Exit(1)
    }
}
```

Keep `os.Exit` in `main` only so deferred cleanup runs.

Child Processes

Always use context

```go
cmd := exec.CommandContext(ctx, "ffmpeg", args...)
cmd.Stdout = &buf
cmd.Stderr = &errBuf
if err := cmd.Run(); err != nil {
    return fmt.Errorf("ffmpeg: %w: %s", err, errBuf.String())
}
```

When `ctx` cancels, Go sends `SIGKILL` to the child by default (after `CommandContext` cancel). For graceful child stop, start with `cmd.Start()`, signal `SIGTERM`, wait, then kill.

```go
func runGraceful(ctx context.Context, name string, args ...string) error {
    cmd := exec.Command(name, args...)
    if err := cmd.Start(); err != nil {
        return err
    }
    done := make(chan error, 1)
    go func() { done <- cmd.Wait() }()

    select {
    case err := <-done:
        return err
    case <-ctx.Done():
        _ = cmd.Process.Signal(syscall.SIGTERM)
        select {
        case err := <-done:
            return err
        case <-time.After(5 * time.Second):
            _ = cmd.Process.Kill()
            return <-done
        }
    }
}
```

Avoid shell

```go
// BAD: injection + opaque failures
// exec.Command("sh", "-c", "do-thing "+user)

// GOOD
exec.CommandContext(ctx, "/usr/bin/do-thing", userArg)
```

Process groups

When your child spawns grandchildren, killing only the direct child can leave orphans. On Unix, set a process group and signal the group (advanced; platform-specific). Document behavior for plugin systems.

Supervisor Policies

Whether you write a supervisor or rely on K8s:

Policy	Behavior	Risk
Always restart	Crash loop	Maskes config bugs; thrashes
On-failure	Restart non-zero only	Good default
Backoff + jitter	Delay restarts	Prevents restart storms
Max attempts	Give up / page	Surfaces poison configs
Rate limit	N restarts / window	Protects cluster

```
crash -> wait min(cap, base*2^n) + jitter -> restart
         if n > max: alert and stop
```

Never restart blindly without backoff when the crash cause is bad config — you will burn CPU and spam logs.

Readiness vs Liveness (process view)

- **Liveness failure** → supervisor restarts process (hung deadlock).
- **Readiness failure** → leave process up but remove from LB (warming, dependency down).

Implementing readiness that always equals "process started" defeats the purpose. See distributed infra health-check chapter for HTTP shapes.

Observability for Process Lifecycle

Emit structured events:

```
slog.Info("process_start", "version", version, "pid", os.Getpid())
slog.Info("signal", "sig", "SIGTERM")
slog.Info("shutdown_begin")
slog.Info("shutdown_end", "dur_ms", dur.Milliseconds(), "exit_hint", 0)
```

Metrics: `process_start_time`, `shutdown_duration_seconds`, `child_exit_total{code,cmd}`, restart counts if you supervise.

PID 1 and Containers

In containers, your app may be PID 1. PID 1 has special signal and zombie-reaping responsibilities. Prefer a tiny init (`tini`, dumb-init) or ensure you wait on children. Go apps that spawn children without reaping can accumulate zombies.

Production Checklist

- ☐ `signal.NotifyContext` for SIGTERM/SIGINT
- ☐ Soft shutdown deadline < orchestrator grace period
- ☐ Handlers/workers honor context cancel
- ☐ Children launched with `CommandContext` or explicit term/kill
- ☐ No `os.Exit` in libraries
- ☐ Meaningful exit codes
- ☐ Restart backoff at supervisor layer
- ☐ Logs for signal, shutdown duration, child failures
- ☐ Liveness readiness semantics documented

Common Pitfalls

1. **Ignoring SIGTERM** — K8s waits, then SIGKILL mid-request.
2. **Shutdown longer than grace period** — always killed.
3. **Goroutines without context** — drain never finishes.
4. **`log.Fatal` in hot paths** — skips defers.
5. **Restart loop on bad config** — fix with crash backoff + config validation at start.
6. **Orphan children** after parent exit.
7. **Treating OOM kills as app bugs only** — sometimes limits are wrong; still exit 137 is a signal.

Exercises

1. Write a server that sleeps 10s per request; send SIGTERM; show that without context the process hangs until kill.
2. Fix it with context-aware handlers and `Shutdown`; measure drain time.
3. Spawn `sleep 60` with `CommandContext`; cancel after 1s; confirm child dies.
4. Implement graceful child termination (SIGTERM then SIGKILL).
5. Simulate crash loop with a binary that exits 1; write a supervisor with exponential backoff + max 5 attempts.
6. Document your cluster's `terminationGracePeriodSeconds` vs app shutdown timeout.
7. Create a child that spawns another child; observe zombies without `Wait`; fix.
8. Map exit codes in CI for a CLI tool (0/1/2).
9. Add slog events for start/signal/shutdown; reconstruct a deploy timeline from logs alone.
10. Compare systemd `Restart=on-failure` vs Kubernetes restartPolicy for the same binary.

More examples

HTTP graceful shutdown on signal context

```
mkdir -p /tmp/go-graceful-http && cd /tmp/go-graceful-http
go mod init example.com/graceful-http
```

Save as `main.go`:

```go
package main

import (
    "context"
    "fmt"
    "net"
    "net/http"
    "os"
    "os/signal"
    "syscall"
    "time"
)

func main() {
    ctx, stop := signal.NotifyContext(context.Background(), os.Interrupt, syscall.SIGTERM)
    defer stop()

    mux := http.NewServeMux()
    mux.HandleFunc("GET /ok", func(w http.ResponseWriter, r *http.Request) {
        fmt.Fprintln(w, "ok")
    })
    ln, err := net.Listen("tcp", "127.0.0.1:0")
    if err != nil {
        panic(err)
    }
    srv := &http.Server{Handler: mux}

    errCh := make(chan error, 1)
    go func() { errCh <- srv.Serve(ln) }()

    // Simulate orchestrator SIGTERM without needing an external kill.
    go func() {
        time.Sleep(30 * time.Millisecond)
        stop()
    }()

    resp, err := http.Get("http://" + ln.Addr().String() + "/ok")
```

```
    if err != nil {
        panic(err)
    }
    resp.Body.Close()
    fmt.Println("served:", resp.StatusCode)

    <-ctx.Done()
    shCtx, cancel := context.WithTimeout(context.Background(), 200*time.Millisecond)
    defer cancel()
    if err := srv.Shutdown(shCtx); err != nil {
        panic(err)
    }
    fmt.Println("shutdown:", <-errCh)
}
```

```
go run .
```

Expected output:

```
served: 200
shutdown: http: Server closed
```

Exit codes for CLI contract

```
mkdir -p /tmp/go-exit-codes && cd /tmp/go-exit-codes
go mod init example.com/exit-codes
```

Save as `main.go`:

```
package main

import (
    "errors"
    "fmt"
    "os"
)

func run(args []string) error {
    if len(args) < 1 {
        return errUsage
    }
    if args[0] == "fail" {
        return errors.New("runtime boom")
    }
    fmt.Println("ok:", args[0])
    return nil
```

```go
}

var errUsage = errors.New("usage: tool <name>")

func main() {
    // Demo mapping without os.Exit so the process stays testable.
    for _, args := range [][]string{{}, {"fail"}, {"ship"}} {
        err := run(args)
        code := 0
        switch {
        case errors.Is(err, errUsage):
            code = 2
        case err != nil:
            code = 1
        }
        fmt.Printf("args=%v code=%d err=%v\n", args, code, err)
    }
}
```

```go
go run .
```

Expected output:

```
args=[] code=2 err=usage: tool <name>
args=[fail] code=1 err=runtime boom
ok: ship
args=[ship] code=0 err=<nil>
```

Runnable example

Graceful signal handling with `signal.NotifyContext`, a child process under `CommandContext`, and a tiny supervisor backoff loop—all in one stdlib program.

```go
mkdir -p /tmp/go-signals && cd /tmp/go-signals
go mod init example.com/signals
```

Save as `main.go`:

```go
package main

import (
    "context"
    "fmt"
    "os"
    "os/exec"
```

```
    "os/signal"
    "syscall"
    "time"
)

func supervisor(run func() error, maxAttempts int) error {
    var delay = 50 * time.Millisecond
    for attempt := 1; attempt <= maxAttempts; attempt++ {
        err := run()
        if err == nil {
            return nil
        }
        fmt.Printf("attempt %d failed: %v\n", attempt, err)
        if attempt == maxAttempts {
            return fmt.Errorf("gave up after %d attempts: %w", maxAttempts, err)
        }
        time.Sleep(delay)
        delay *= 2
    }
    return nil
}

func main() {
    // 1) Signal-aware context (demo: cancel after short timer to avoid interactive wait).
    ctx, stop := signal.NotifyContext(context.Background(), os.Interrupt, syscall.SIGTERM)
    defer stop()

    go func() {
        time.Sleep(30 * time.Millisecond)
        // Simulate supervisor/orchestrator stop without needing a real signal in CI.
        stop()
    }()

    <-ctx.Done()
    fmt.Println("shutdown signal path:", ctx.Err())

    // 2) Child with deadline
    cctx, cancel := context.WithTimeout(context.Background(), 100*time.Millisecond)
    defer cancel()
    cmd := exec.CommandContext(cctx, "sleep", "2")
    err := cmd.Run()
    fmt.Println("child cancelled:", err != nil)

    // 3) Supervisor backoff: fail twice then succeed
    var n int
    err = supervisor(func() error {
        n++
```

```go
        if n < 3 {
            return fmt.Errorf("boom %d", n)
        }
        fmt.Println("worker healthy")
        return nil
    }, 5)
    fmt.Println("supervisor err:", err)
}
```

```go
go run .
```

Expected output:

```
shutdown signal path: context canceled
child cancelled: true
attempt 1 failed: boom 1
attempt 2 failed: boom 2
worker healthy
supervisor err: <nil>
```

What to notice

- Kubernetes/`docker stop` send SIGTERM first; apps must drain within the grace period or get SIGKILL.
- `CommandContext` is how you avoid orphan helpers after cancel.
- Supervisors need capped exponential backoff or they thundering-herd a bad config forever.

Try next

- Build a mini HTTP server and call `Shutdown` when the signal context cancels.
- Map exit codes: `0` success, `1` runtime error, `2` usage error for a CLI.

Further Reading

- `os/signal` package docs
- Kubernetes termination lifecycle
- Next: **Filesystem Automation and Safe IO**

Filesystem Automation and Safe IO

Filesystem Automation and Safe IO

Filesystem automation fails in subtle ways: partial writes, interrupted updates, TOCTOU races, and path traversal. For ops tooling, **predictability beats peak throughput**.

Why / Overview

Go services and CLIs constantly touch disks:

- Rewrite config and state files
- Rotate logs and checkpoints
- Unpack artifacts
- Cache blobs

A crash mid-write can leave empty or half-written files that break the next start. Untrusted paths can escape into `/etc` or secrets mounts.

```
unsafe:  open(path) -> write -> close          # torn file on crash
safe:    write temp -> fsync -> rename -> fsync dir
```

Atomic Replace Pattern

POSIX `rename` within the same filesystem is atomic for the directory entry. Combined with temp files:

```go
package atomicfile

import (
    "fmt"
    "os"
    "path/filepath"
)

func WriteFile(path string, data []byte, perm os.FileMode) error {
    dir := filepath.Dir(path)
    tmp, err := os.CreateTemp(dir, "."+filepath.Base(path)+".tmp-*")
    if err != nil {
        return err
    }
```

```go
    tmpName := tmp.Name()
    defer func() { _ = os.Remove(tmpName) }() // no-op if rename succeeded

    if _, err := tmp.Write(data); err != nil {
        tmp.Close()
        return err
    }
    if err := tmp.Chmod(perm); err != nil {
        tmp.Close()
        return err
    }
    if err := tmp.Sync(); err != nil { // fsync file
        tmp.Close()
        return err
    }
    if err := tmp.Close(); err != nil {
        return err
    }
    if err := os.Rename(tmpName, path); err != nil {
        return err
    }
    // Best-effort fsync parent directory so the rename itself is durable.
    return syncDir(dir)
}

func syncDir(dir string) error {
    d, err := os.Open(dir)
    if err != nil {
        return err
    }
    defer d.Close()
    if err := d.Sync(); err != nil {
        // Some FS (NFS, certain mounts) may not support directory sync.
        return fmt.Errorf("sync dir: %w", err)
    }
    return nil
}
```

Notes:

- Create the temp file **in the same directory** as the target so `rename` does not cross devices.
- `Sync` costs latency; for pure caches you may skip durability.
- On Windows, atomic replace semantics differ; test platform-specific paths.

Path Safety (Root Jail)

Treat every user-supplied path as hostile.

```go
package safepath

import (
    "fmt"
    "os"
    "path/filepath"
    "strings"
)

func UnderRoot(root, userPath string) (string, error) {
    rootAbs, err := filepath.Abs(root)
    if err != nil {
        return "", err
    }
    // Disallow absolute user paths escaping intent.
    if filepath.IsAbs(userPath) {
        return "", fmt.Errorf("absolute paths not allowed")
    }
    joined := filepath.Join(rootAbs, userPath)
    clean := filepath.Clean(joined)
    rel, err := filepath.Rel(rootAbs, clean)
    if err != nil {
        return "", err
    }
    if rel == ".." || strings.HasPrefix(rel, ".."+string(os.PathSeparator)) {
        return "", fmt.Errorf("path escapes root")
    }
    return clean, nil
}
```

Symlinks: `filepath.EvalSymlinks` can help but introduces its own races and platform quirks. For high-security tools, open with `O_NOFOLLOW` where available, or reject symlinks after `Lstat`.

Safe Read Patterns

```go
// Bound reads from untrusted files.
f, err := os.Open(path)
if err != nil {
    return err
}
defer f.Close()
```

```go
data, err := io.ReadAll(io.LimitReader(f, 8<<20)) // 8 MiB cap
```

For line processing of huge logs, stream with `bufio.Scanner` but raise `MaxTokenSize` only deliberately — default token limits exist for a reason.

Permissions and Umask

```go
// Secrets on disk: owner read/write only
if err := atomicfile.WriteFile(secretPath, b, 0o600); err != nil { ... }

// Shared but not world-writable configs
if err := atomicfile.WriteFile(cfgPath, b, 0o644); err != nil { ... }
```

Never use `0o777` "to make it work." Fix ownership instead.

Directory Creation

```go
if err := os.MkdirAll(dir, 0o755); err != nil {
    return err
}
```

`MkdirAll` is fine; still validate `dir` is under your root when driven by user input.

Check-Then-Act Races (TOCTOU)

```go
// racy
if _, err := os.Stat(path); os.IsNotExist(err) {
    f, _ = os.Create(path) // another process may have created it
}
```

Prefer open flags:

```go
f, err := os.OpenFile(path, os.O_CREATE|os.O_EXCL|os.O_WRONLY, 0o600)
```

Or accept overwrite with atomic replace.

Streaming Copy and Spool

```go
func copyFile(src, dst string) error {
    in, err := os.Open(src)
    if err != nil {
        return err
    }
    defer in.Close()

    tmp := dst + ".tmp"
    out, err := os.OpenFile(tmp, os.O_CREATE|os.O_TRUNC|os.O_WRONLY, 0o644)
    if err != nil {
        return err
    }
    if _, err := io.Copy(out, in); err != nil {
        out.Close()
        os.Remove(tmp)
        return err
    }
    if err := out.Sync(); err != nil {
        out.Close()
        return err
    }
    if err := out.Close(); err != nil {
        return err
    }
    return os.Rename(tmp, dst)
}
```

For large artifacts, consider checksums (SHA-256) after write before rename promotion.

Walking Trees

```go
err := filepath.WalkDir(root, func(path string, d fs.DirEntry, err error) error {
    if err != nil {
        return err // or skip
    }
    if d.Type()&os.ModeSymlink != 0 {
        return nil // skip or carefully resolve
    }
    // process...
    return nil
})
```

Or `fs.WalkDir` with `os.DirFS` for testability.

Embed for Static Assets

```go
//go:embed schemas/*.json
var schemas embed.FS
```

Prefer `embed` over shipping loose files when content is part of the binary release — fewer path problems in production.

Durability vs Speed Tradeoffs

Workload	fsync?	Pattern
Wallet / ledger checkpoint	Yes	full atomic + dir sync
CDN cache fill	Often no	write + rename optional
Queue spill to disk	Yes on commit	group commits if needed
CLI rewrite of dotfiles	Rename enough	fsync nice-to-have

Measure: fsync can dominate latency on spinning disks and some cloud volumes.

Production Checklist

- ☐ Atomic write for any file that must be valid after crash
- ☐ Temp files on same filesystem as destination
- ☐ Max size limits on untrusted reads
- ☐ Path jail for user-controlled paths
- ☐ Restrictive permissions on secrets
- ☐ No world-writable dirs in the data path
- ☐ Explicit symlink policy
- ☐ Disk-full errors handled (not ignored)
- ☐ Tests for traversal (`../`) and partial-write recovery

Common Pitfalls

1. **Write in place** to config files — crash leaves empty file, service won't start.
2. **Temp in `/tmp` then rename to another mount** — `EXDEV`, copy fallback needed.
3. **Trusting `Clean` alone** without root prefix check.
4. **Following symlinks** into sensitive locations.
5. **Ignoring `ENOSPC`** — loop of failed writes can spam logs forever.
6. **Scanner default token size** on huge lines — silent early exit if not checking `scanner.Err()`.
7. **Closing without sync** when durability was required.

Exercises

1. Implement `WriteFile` atomic helper; kill `-9` mid-write in a loop; prove the target is always old-good or new-good, never empty torn.
2. Build a path jail; unit-test `../../etc/passwd`, absolute paths, nested `...`
3. Write a file with `0o600`; verify mode with `stat`.
4. Copy a 1GB file with streaming; compare memory RSS to `ReadAll`.
5. Create a symlink escape scenario; decide reject vs EvalSymlinks policy; document it.
6. Simulate disk full (`ulimit` or tiny loop mount); ensure errors surface.
7. Use `O_EXCL` to implement a simple lock file; handle stale locks carefully (advanced).
8. Walk a tree skipping symlinks; count regular files only.
9. Add SHA-256 verify-after-write before rename for a download tool.
10. Benchmark fsync vs no-fsync for 1k small config updates; record tradeoff.

More examples

Atomic write via temp + rename

```
mkdir -p /tmp/go-atomic-write && cd /tmp/go-atomic-write
go mod init example.com/atomic-write
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "os"
    "path/filepath"
)

func writeAtomic(path string, data []byte) error {
    dir := filepath.Dir(path)
    f, err := os.CreateTemp(dir, ".w-*")
    if err != nil {
        return err
    }
    tmp := f.Name()
    if _, err := f.Write(data); err != nil {
        f.Close()
        os.Remove(tmp)
        return err
    }
    if err := f.Sync(); err != nil {
        f.Close()
```

```
        os.Remove(tmp)
        return err
    }
    if err := f.Close(); err != nil {
        os.Remove(tmp)
        return err
    }
    return os.Rename(tmp, path)
}

func main() {
    dir, _ := os.MkdirTemp("", "atom")
    defer os.RemoveAll(dir)
    path := filepath.Join(dir, "state.json")
    if err := writeAtomic(path, []byte(`{"v":1}`)); err != nil {
        panic(err)
    }
    if err := writeAtomic(path, []byte(`{"v":2}`)); err != nil {
        panic(err)
    }
    b, _ := os.ReadFile(path)
    fmt.Println(string(b))
}
```

```
go run .
```

Expected output:

```
{"v":2}
```

Walk files, skip dirs and symlinks to dirs

```
mkdir -p /tmp/go-walk-files && cd /tmp/go-walk-files
go mod init example.com/walk-files
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "io/fs"
    "os"
    "path/filepath"
)
```

```go
func main() {
    root, _ := os.MkdirTemp("", "walk")
    defer os.RemoveAll(root)
    _ = os.WriteFile(filepath.Join(root, "a.txt"), []byte("a"), 0o644)
    _ = os.Mkdir(filepath.Join(root, "sub"), 0o755)
    _ = os.WriteFile(filepath.Join(root, "sub", "b.txt"), []byte("b"), 0o644)

    var files []string
    _ = filepath.WalkDir(root, func(path string, d fs.DirEntry, err error) error {
        if err != nil {
            return err
        }
        if d.Type().IsRegular() {
            rel, _ := filepath.Rel(root, path)
            files = append(files, rel)
        }
        return nil
    })
    fmt.Println("files:", files)
}
```

```bash
go run .
```

Expected output:

```
files: [a.txt sub/b.txt]
```

Runnable example

Atomic replace via temp file + rename, a path jail, and restrictive file modes—crash-safe config write pattern in pure stdlib.

```bash
mkdir -p /tmp/go-safe-fs && cd /tmp/go-safe-fs
go mod init example.com/safe-fs
```

Save as `main.go`:

```go
package main

import (
    "fmt"
    "os"
    "path/filepath"
    "strings"
)
```

```
func writeFileAtomic(path string, data []byte, mode os.FileMode) error {
    dir := filepath.Dir(path)
    if err := os.MkdirAll(dir, 0o750); err != nil {
        return err
    }
    f, err := os.CreateTemp(dir, ".tmp-*")
    if err != nil {
        return err
    }
    tmp := f.Name()
    cleanup := true
    defer func() {
        if cleanup {
            _ = os.Remove(tmp)
        }
    }()
    if _, err := f.Write(data); err != nil {
        f.Close()
        return err
    }
    if err := f.Chmod(mode); err != nil {
        f.Close()
        return err
    }
    if err := f.Sync(); err != nil {
        f.Close()
        return err
    }
    if err := f.Close(); err != nil {
        return err
    }
    if err := os.Rename(tmp, path); err != nil {
        return err
    }
    cleanup = false
    return nil
}

func underRoot(root, userPath string) (string, error) {
    rootAbs, err := filepath.Abs(filepath.Clean(root))
    if err != nil {
        return "", err
    }
    joined := filepath.Join(rootAbs, userPath)
    abs, err := filepath.Abs(joined)
    if err != nil {
```

```go
        return "", err
    }
    rel, err := filepath.Rel(rootAbs, abs)
    if err != nil || strings.HasPrefix(rel, "..") {
        return "", fmt.Errorf("escapes root")
    }
    return abs, nil
}

func main() {
    root := filepath.Join(os.TempDir(), "safe-fs-demo")
    _ = os.MkdirAll(root, 0o750)
    target := filepath.Join(root, "config.json")

    if err := writeFileAtomic(target, []byte(`{"v":1}`+"\n"), 0o600); err != nil {
        panic(err)
    }
    b, _ := os.ReadFile(target)
    fmt.Printf("config: %s", b)
    fi, _ := os.Stat(target)
    fmt.Printf("mode: %o\n", fi.Mode().Perm())

    ok, err := underRoot(root, "nested/a.txt")
    fmt.Println("jail ok:", ok, err)
    _, err = underRoot(root, "../../etc/passwd")
    fmt.Println("jail block:", err != nil)

    // Second atomic write replaces cleanly
    _ = writeFileAtomic(target, []byte(`{"v":2}`+"\n"), 0o600)
    b, _ = os.ReadFile(target)
    fmt.Printf("config after: %s", b)
}
```

```go
go run .
```

Expected output (paths vary):

```
config: {"v":1}
mode: 600
jail ok: /.../safe-fs-demo/nested/a.txt <nil>
jail block: true
config after: {"v":2}
```

What to notice

- Rename within the same filesystem is atomic for readers; in-place truncate is how configs go empty on crash.

- `0o600` for secrets/config; never world-writable automation output.
- Path jail before any user-influenced open/read/write.

Try next

- Kill `-9` mid-write in a loop and prove the target is always old-good or new-good.
- Handle `EXDEV` (cross-device rename) with copy+rename fallback.

Further Reading

- POSIX rename and fsync semantics articles
- Previous: process lifecycle (crash = torn write source)
- Next: **Unix Pipeline Style CLI Tools**

A Corruption-Proof Write-Ahead Log

A Corruption-Proof Write-Ahead Log

Thesis

A write-ahead log (WAL) is a file you only append to, so that after a crash you can replay it and rebuild state. Writing one takes an afternoon. Trusting one takes more work, because the difficult part is the crash itself. A crash can leave half a record on disk. A disk can flip a bit in a record you acknowledged last week. A segment file can be fsynced and still be missing after a power cut, because its directory entry never reached the disk. The format and the recovery code have to tell these cases apart, and they have to do it from the bytes alone.

This chapter builds a small segmented WAL in under 300 lines of standard-library Go. Each record carries a CRC-32C, a bounded length, a commit trailer, and padding to 8 bytes. The write path fsyncs before acknowledging anything and fsyncs the directory when it rotates segments. Recovery refuses to open when a sealed segment is damaged and truncates a torn tail, keeping a copy of what it cut. Every failure below is caused on purpose with `truncate`, `dd`, and a test that cuts the log at every byte offset.

Everything was compiled and run with **Go 1.27.2** on linux/amd64 (kernel 6.12, ext4 on a virtio disk, 8 vCPUs). The fsync timings are for that one virtual disk. Your NVMe, cloud volume, or laptop SSD will give different numbers, so measure your own before choosing a batch size.

Mental model

- **`write` is not durable.** `(*os.File).Write` copies bytes into the kernel page cache and returns. Only `(*os.File).Sync` asks the kernel to put them on stable storage. On Linux that is `fsync(2)`. On macOS, Go's `Sync` calls `fcntl(F_FULLFSYNC)`, because plain `fsync` there does not flush the drive's cache (`internal/poll/fd_fsync_darwin.go`). The contract is short: **acknowledge a record only after a `Sync` that covers it has returned `nil`.**
- **A crash only tears the unsynced tail.** Records that were fsynced before an acknowledgement went out are on the disk. Anything written after the last successful fsync can be missing, partly present, or present out of order, because the kernel writes dirty pages back in any order it likes. So the end of the *active* segment is the only place a crash can legitimately leave garbage.
- **Each layer catches a different failure.** The framing follows the layers described in UnisonDB's WAL write-up, with the trade-offs spelled out:

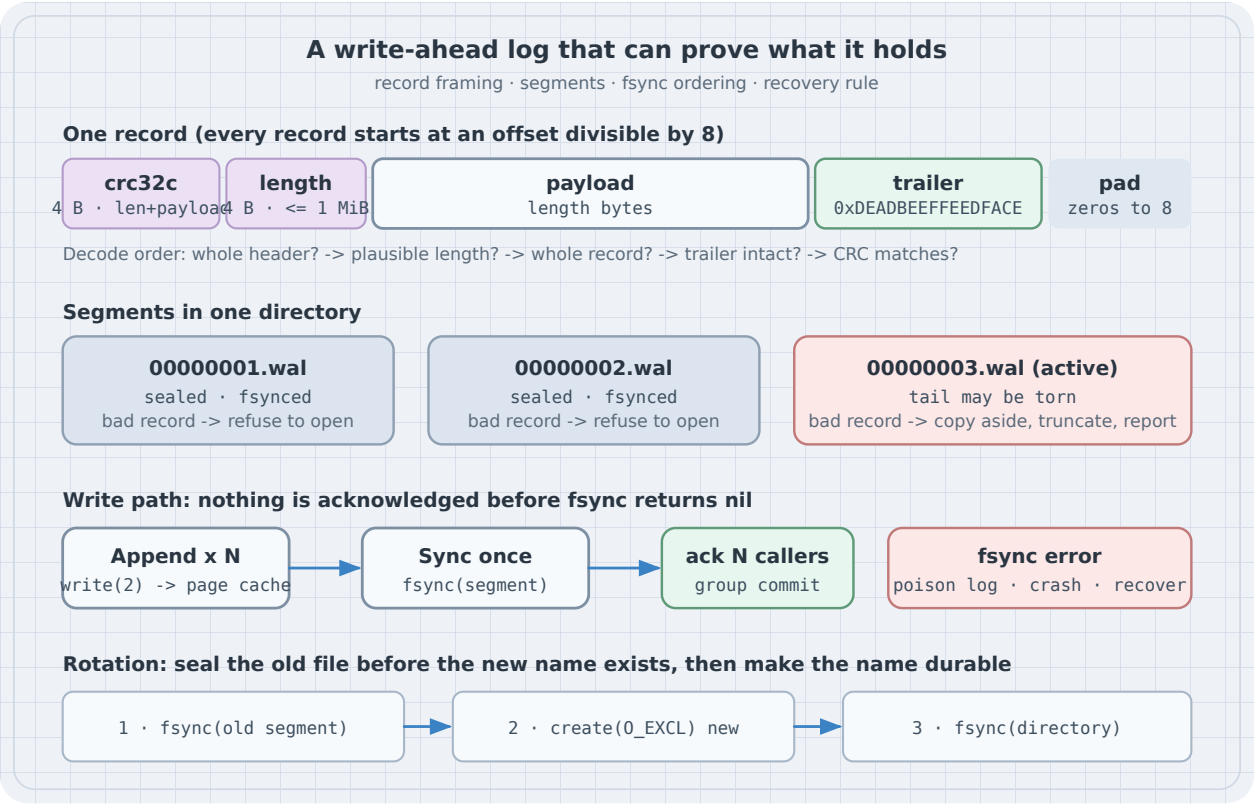


Figure 1: Record framing, sealed vs active segments, the write path, and rotation order

Layer	Catches	Does not catch
Length bound (`1..MaxRecord`)	A garbage length that would make recovery read or allocate gigabytes; zero-filled space	A plausible but wrong length (the next checks catch that)
Whole-record check	A file that ends mid-record	Bytes that are present but wrong
Trailer (`0xDEADBEEFFEEDFACE`, written last)	A record whose tail never landed: zeros or stale bytes where the end should be	A bit flip in the payload
CRC-32C over length + payload	Bit flips and torn sectors inside a finished record	Zero-filled space, if it covered only the payload and allowed empty records (Case 3)
8-byte alignment	An 8-byte header crossing a 512 B or 4 KiB sector boundary, so a torn sector cannot split it	Torn payloads, which the trailer and CRC handle

- **Sealed and active segments get different rules.** When a segment fills up, rotation fsyncs it *before* creating the next one. A bad record in a sealed segment therefore cannot be a crash artefact, as long as the device honours flushes. It is media corruption, a bug, a drive that lied about its cache, or tampering, and the log refuses to open. A bad record in the active segment is treated as a torn tail.

- **A new file needs two fsyncs.** The Linux man page says it directly: calling `fsync()` on a file "does not necessarily ensure that the entry in the directory containing the file has also reached disk. For that an explicit fsync() on a file descriptor for the directory is also needed." In Go: `os.Open(dir)` and then `.Sync()`.

The code

The module has one package and three small programs. Lay it out like this:

```
walbook/
  go.mod
  wal/record.go         framing: Encode / Decode
  wal/log.go            segments, Append / Sync, rotation, recovery
  wal/log_test.go       crash at every byte offset
  cmd/walctl/main.go    append / recover / bench from the shell
  cmd/zerofill/main.go  Case 3
  cmd/groupcommit/main.go  Case 6
```

```go
// go.mod
module example.com/walbook

go 1.27
```

The framing. `Decode` runs its checks cheapest and most structural first, so a corrupt length can never cause a large read before the CRC rejects it.

```go
// wal/record.go
package wal

import (
    "encoding/binary"
    "errors"
    "hash/crc32"
)

// On disk, every record is:
//
//  | crc32c (4) | length (4) | payload (length) | trailer (8) | zero pad to 8 |
//
// The CRC covers length+payload. The trailer is written last and means
// "this record is finished". Every record starts at an offset divisible by 8.
const (
    headerSize  = 8
    trailerSize = 8
    align       = 8
```

```go
    // MaxRecord bounds the length field so a corrupt header can never make
    // recovery allocate or read gigabytes before the CRC check fails.
    MaxRecord = 1 << 20
    trailer   = uint64(0xDEADBEEFFEEDFACE)
)

var (
    castagnoli = crc32.MakeTable(crc32.Castagnoli)

    ErrTorn   = errors.New("torn or incomplete record")
    ErrCRC    = errors.New("checksum mismatch")
    ErrLength = errors.New("implausible length")
)

func alignUp(n int64) int64 { return (n + align - 1) &^ (align - 1) }

// Encode returns the bytes of one record, padded to a multiple of 8.
func Encode(payload []byte) []byte {
    if len(payload) == 0 || len(payload) > MaxRecord {
        panic("wal: payload must be 1..MaxRecord bytes")
    }
    n := len(payload)
    buf := make([]byte, alignUp(int64(headerSize+n+trailerSize)))
    binary.LittleEndian.PutUint32(buf[4:8], uint32(n))
    copy(buf[headerSize:], payload)
    binary.LittleEndian.PutUint64(buf[headerSize+n:], trailer)
    binary.LittleEndian.PutUint32(buf[0:4], crc32.Checksum(buf[4:headerSize+n], castagnoli))
    return buf
}

// Decode parses the record at the start of b and returns its payload and
// its size on disk. It checks, in order: a whole header is present, the
// length is plausible, the whole record is present, the trailer is intact,
// and only then the CRC.
func Decode(b []byte) (payload []byte, size int64, err error) {
    if len(b) < headerSize {
        return nil, 0, ErrTorn
    }
    n := int64(binary.LittleEndian.Uint32(b[4:8]))
    if n == 0 || n > MaxRecord {
        return nil, 0, ErrLength
    }
    size = alignUp(headerSize + n + trailerSize)
    if int64(len(b)) < size {
        return nil, 0, ErrTorn
    }
```

```go
    if binary.LittleEndian.Uint64(b[headerSize+n:]) != trailer {
        return nil, 0, ErrTorn
    }
    if crc32.Checksum(b[4:headerSize+n], castagnoli) != binary.LittleEndian.Uint32(b[0:4]) {
        return nil, 0, ErrCRC
    }
    return b[headerSize : headerSize+n], size, nil
}
```

The log. `Open` is the recovery procedure, `Append` + `Sync` are the write path, and `rotate` + `create` hold the fsync ordering.

```go
// wal/log.go
package wal

import (
    "errors"
    "fmt"
    "os"
    "path/filepath"
    "slices"
    "strings"
)

// Log is an append-only, segmented write-ahead log. It is not safe for
// concurrent use; put one goroutine (or one mutex) in front of it.
type Log struct {
    dir    string
    dirf   *os.File // the directory, kept open so we can fsync it
    f      *os.File // active segment
    seq    int
    size   int64
    maxSeg int64
    err    error // sticky: once a write or fsync fails, the log is dead
}

// Report says what recovery found.
type Report struct {
    Segments  int
    Records   int
    Truncated int64 // bytes cut from the tail of the active segment
    Reason    error // why the tail was cut (nil if nothing was cut)
}

func segName(seq int) string { return fmt.Sprintf("%08d.wal", seq) }

// Open recovers dir and returns a log ready to append. replay is called for
```

```
// every valid record in order. A bad record inside a sealed (non-last)
// segment is fatal. A bad record in the last segment is treated as a torn
// tail: the segment is truncated at the last good record and fsynced.
func Open(dir string, maxSeg int64, replay func(seq int, off int64, p []byte)) (*Log, Report
    var rep Report
    if err := os.MkdirAll(dir, 0o755); err != nil {
        return nil, rep, err
    }
    dirf, err := os.Open(dir)
    if err != nil {
        return nil, rep, err
    }
    seqs, err := listSegments(dir)
    if err != nil {
        dirf.Close()
        return nil, rep, err
    }
    l := &Log{dir: dir, dirf: dirf, maxSeg: maxSeg}
    if len(seqs) == 0 {
        if err := l.create(1); err != nil {
            dirf.Close()
            return nil, rep, err
        }
        rep.Segments = 1
        return l, rep, nil
    }
    rep.Segments = len(seqs)
    for i, seq := range seqs {
        path := filepath.Join(dir, segName(seq))
        data, err := os.ReadFile(path)
        if err != nil {
            dirf.Close()
            return nil, rep, err
        }
        good, n, bad := scan(data, func(off int64, p []byte) {
            if replay != nil {
                replay(seq, off, p)
            }
        })
        rep.Records += n
        last := i == len(seqs)-1
        if bad != nil && !last {
            dirf.Close()
            return nil, rep, fmt.Errorf("sealed segment %s: %w at offset %d; refusing to ope
        }
        if last {
            f, err := os.OpenFile(path, os.O_WRONLY, 0)
```

```
            if err != nil {
                dirf.Close()
                return nil, rep, err
            }
            if bad != nil {
                rep.Truncated, rep.Reason = int64(len(data))-good, bad
                // Keep the evidence: copy the cut bytes aside before truncating.
                if err := os.WriteFile(fmt.Sprintf("%s.tail-%d", path, good), data[good:], 0
                    return nil, rep, err
                }
                if err := f.Truncate(good); err != nil {
                    return nil, rep, err
                }
                if err := f.Sync(); err != nil {
                    return nil, rep, err
                }
            }
            if _, err := f.Seek(good, 0); err != nil {
                return nil, rep, err
            }
            l.f, l.seq, l.size = f, seq, good
        }
    }
    return l, rep, nil
}

// scan walks records from the start of a segment. It returns the offset just
// past the last good record, how many good records it saw, and the error that
// stopped it (nil if the segment ended cleanly).
func scan(data []byte, fn func(off int64, p []byte)) (good int64, n int, bad error) {
    for good < int64(len(data)) {
        p, size, err := Decode(data[good:])
        if err != nil {
            return good, n, err
        }
        fn(good, p)
        good += size
        n++
    }
    return good, n, nil
}

// Append writes one record to the OS. It is NOT durable until Sync returns nil.
func (l *Log) Append(p []byte) error {
    if l.err != nil {
        return l.err
    }
```

```
    rec := Encode(p)
    if l.size > 0 && l.size+int64(len(rec)) > l.maxSeg {
        if err := l.rotate(); err != nil {
            l.err = err
            return err
        }
    }
    if _, err := l.f.Write(rec); err != nil {
        l.err = fmt.Errorf("wal write: %w", err)
        return l.err
    }
    l.size += int64(len(rec))
    return nil
}

// Sync makes every appended record durable. If it fails, the log is poisoned:
// the kernel may already have dropped the dirty pages, so retrying and seeing
// success proves nothing. Crash, restart, and let Open recover.
func (l *Log) Sync() error {
    if l.err != nil {
        return l.err
    }
    if err := l.f.Sync(); err != nil {
        l.err = fmt.Errorf("wal fsync (log is now unusable): %w", err)
        return l.err
    }
    return nil
}

func (l *Log) rotate() error {
    // Seal: the old segment must be complete on disk before the new one exists.
    if err := l.f.Sync(); err != nil {
        return err
    }
    if err := l.f.Close(); err != nil {
        return err
    }
    return l.create(l.seq + 1)
}

func (l *Log) create(seq int) error {
    f, err := os.OpenFile(filepath.Join(l.dir, segName(seq)), os.O_CREATE|os.O_EXCL|os.O_WRO
    if err != nil {
        return err
    }
    // fsync on the file does not persist its directory entry; fsync the directory.
```

```go
    if err := l.dirf.Sync(); err != nil {
        f.Close()
        return err
    }
    l.f, l.seq, l.size = f, seq, 0
    return nil
}

// Close syncs and closes the log.
func (l *Log) Close() error {
    err := l.Sync()
    return errors.Join(err, l.f.Close(), l.dirf.Close())
}

func listSegments(dir string) ([]int, error) {
    ents, err := os.ReadDir(dir)
    if err != nil {
        return nil, err
    }
    var seqs []int
    for _, e := range ents {
        var seq int
        if strings.HasSuffix(e.Name(), ".wal") {
            if _, err := fmt.Sscanf(e.Name(), "%08d.wal", &seq); err == nil {
                seqs = append(seqs, seq)
            }
        }
    }
    slices.Sort(seqs)
    return seqs, nil
}
```

A command-line driver, so the cases below can damage files with ordinary shell tools:

```go
// cmd/walctl/main.go
package main

import (
    "fmt"
    "os"
    "strconv"
    "time"

    "example.com/walbook/wal"
)

func main() {
```

```
    if len(os.Args) < 3 {
        fmt.Fprintln(os.Stderr, "usage: walctl append|recover|bench DIR ...")
        os.Exit(2)
    }
    cmd, dir := os.Args[1], os.Args[2]
    maxSeg := int64(64 << 20)
    if s := os.Getenv("WAL_MAX_SEG"); s != "" {
        maxSeg, _ = strconv.ParseInt(s, 10, 64)
    }
    switch cmd {
    case "append": // walctl append DIR msg...
        l, _, err := wal.Open(dir, maxSeg, nil)
        check(err)
        for _, m := range os.Args[3:] {
            check(l.Append([]byte(m)))
        }
        check(l.Sync()) // one fsync for the whole batch, then acknowledge
        fmt.Printf("acked %d records\n", len(os.Args[3:]))
        check(l.Close())

    case "recover": // walctl recover DIR
        l, rep, err := wal.Open(dir, maxSeg, func(seq int, off int64, p []byte) {
            fmt.Printf("  seg %d off %4d  %q\n", seq, off, p)
        })
        if err != nil {
            fmt.Println("recover:", err)
            os.Exit(1)
        }
        fmt.Printf("segments=%d records=%d truncated=%dB", rep.Segments, rep.Records, rep.Tr
        if rep.Reason != nil {
            fmt.Printf(" reason=%q", rep.Reason.Error())
        }
        fmt.Println()
        check(l.Close())

    case "bench": // walctl bench DIR N BATCH
        n, _ := strconv.Atoi(os.Args[3])
        batch, _ := strconv.Atoi(os.Args[4])
        l, _, err := wal.Open(dir, maxSeg, nil)
        check(err)
        payload := make([]byte, 100)
        start := time.Now()
        syncs := 0
        for i := 1; i <= n; i++ {
            check(l.Append(payload))
            if i%batch == 0 || i == n {
```

```
                check(l.Sync())
                syncs++
            }
        }
        el := time.Since(start)
        fmt.Printf("records=%d batch=%d fsyncs=%d elapsed=%v  %.0f records/s  %.1fµs/record\
            n, batch, syncs, el.Round(time.Millisecond), float64(n)/el.Seconds(),
            float64(el.Microseconds())/float64(n))
        check(l.Close())
    }
}

func check(err error) {
    if err != nil {
        fmt.Fprintln(os.Stderr, "walctl:", err)
        os.Exit(1)
    }
}
```

Build it once and put it on `PATH`:

```
go vet ./...
go build -o /tmp/wbin/walctl ./cmd/walctl
export PATH=/tmp/wbin:$PATH
```

Worked examples

Case 1: Three records, byte by byte

Append three records in one batch (one fsync), then look at the raw segment and replay it.

```
mkdir /tmp/w1 && cd /tmp/w1
walctl append ./log "order 1001 paid" "order 1002 paid" "refund 1001"
wc -c log/00000001.wal
od -A d -t x1 log/00000001.wal
walctl recover ./log
```

Output:

```
acked 3 records
96 log/00000001.wal
0000000 ec 25 6f 84 0f 00 00 00 6f 72 64 65 72 20 31 30
0000016 30 31 20 70 61 69 64 ce fa ed fe ef be ad de 00
0000032 36 ee d3 b0 0f 00 00 00 6f 72 64 65 72 20 31 30
0000048 30 32 20 70 61 69 64 ce fa ed fe ef be ad de 00
```

```
0000064 ea 39 0f e1 0b 00 00 00 72 65 66 75 6e 64 20 31
0000080 30 30 31 ce fa ed fe ef be ad de 00 00 00 00 00
0000096
  seg 1 off    0  "order 1001 paid"
  seg 1 off   32  "order 1002 paid"
  seg 1 off   64  "refund 1001"
segments=1 records=3 truncated=0B
```

Read the first row. `ec 25 6f 84` is the CRC-32C, stored little-endian. `0f 00 00 00` is the length, 15. Then come the 15 payload bytes `order 1001 paid` and the trailer `ce fa ed fe ef be ad de`, which is `0xDEADBEEFFEEDFACE` in little-endian. $8 + 15 + 8 = 31$ bytes, padded to 32, so the next record starts at offset 32. The third record is $8 + 11 + 8 = 27$ bytes, padded with five zeros to 32. Every header starts at a multiple of 8.

Case 2: A crash tears the last record

Simulate a power cut that left only 16 of record 3's 32 bytes on disk. Recovery replays the two complete records, copies the torn bytes aside, and truncates the segment back to the last good offset. The client never got an acknowledgement for record 3, so it retries, and the log accepts the append.

```
# /tmp/w1, continuing from Case 1
truncate -s 80 log/00000001.wal      # the crash tore record 3 mid-payload
walctl recover ./log
wc -c log/00000001.wal
walctl append ./log "refund 1001"      # the client never got an ack, so it retries
walctl recover ./log
```

Output:

```
  seg 1 off    0  "order 1001 paid"
  seg 1 off   32  "order 1002 paid"
segments=1 records=2 truncated=16B reason="torn or incomplete record"
64 log/00000001.wal
acked 1 records
  seg 1 off    0  "order 1001 paid"
  seg 1 off   32  "order 1002 paid"
  seg 1 off   64  "refund 1001"
segments=1 records=3 truncated=0B
```

Now a different crash. The file *length* survived, but the sector holding record 3's trailer did not, and it reads back as zeros. This is the case the trailer is for. The length is plausible and the bytes are all there, but the record never finished.

```
# /tmp/w1, continuing
dd if=/dev/zero of=log/00000001.wal bs=1 seek=83 count=8 conv=notrunc status=none
```

```
od -A d -t x1 -j 64 log/00000001.wal
walctl recover ./log
ls log
```

Output:

```
0000064 ea 39 0f e1 0b 00 00 00 72 65 66 75 6e 64 20 31
0000080 30 30 31 00 00 00 00 00 00 00 00 00 00 00 00 00
0000096
  seg 1 off    0  "order 1001 paid"
  seg 1 off   32  "order 1002 paid"
segments=1 records=2 truncated=32B reason="torn or incomplete record"
00000001.wal
00000001.wal.tail-64
```

The `.tail-64` file holds the 32 bytes that were cut, named after the offset where they started. `listSegments` ignores it, because it only picks up names ending in `.wal`. Nothing is thrown away without a trace.

Case 3: Why "CRC alone" is not enough: zero-filled space

Many logs preallocate segments, or run on filesystems that can expose zeros after a crash. A tempting first format is `| crc32c(payload) | length | payload |`, with empty payloads allowed. The CRC-32C of zero bytes of input is `0`, so a run of zeros parses as an endless stream of valid, empty records.

```go
// cmd/zerofill/main.go
package main

import (
    "encoding/binary"
    "fmt"
    "hash/crc32"

    "example.com/walbook/wal"
)

var castagnoli = crc32.MakeTable(crc32.Castagnoli)

// naiveDecode is the format many first WALs use: | crc(payload) | length | payload |
// with the CRC over the payload only and empty payloads allowed.
func naiveDecode(b []byte) (payload []byte, size int, ok bool) {
    if len(b) < 8 {
        return nil, 0, false
    }
    n := int(binary.LittleEndian.Uint32(b[4:8]))
```

```go
    if len(b) < 8+n {
        return nil, 0, false
    }
    p := b[8 : 8+n]
    return p, 8 + n, crc32.Checksum(p, castagnoli) == binary.LittleEndian.Uint32(b[0:4])
}

func main() {
    fmt.Printf("crc32c(empty)        = %#08x\n", crc32.Checksum(nil, castagnoli))
    fmt.Printf("crc32c(00 00 00 00)  = %#08x\n", crc32.Checksum(make([]byte, 4), castagnoli)

    // A preallocated segment: the writer reserved 4 KiB of zeros up front,
    // and the crash happened before any record reached this region.
    tail := make([]byte, 4096)

    n := 0
    for off := 0; off < len(tail); {
        _, size, ok := naiveDecode(tail[off:])
        if !ok {
            break
        }
        n++
        off += size
    }
    fmt.Printf("naive format: %d \"valid\" empty records in 4096 zero bytes\n", n)

    _, _, err := wal.Decode(tail)
    fmt.Printf("this chapter's format: %v\n", err)
}
```

```
go run ./cmd/zerofill
```

Output:

```
crc32c(empty)        = 0x00000000
crc32c(00 00 00 00)  = 0x48674bc7
naive format: 512 "valid" empty records in 4096 zero bytes
this chapter's format: implausible length
```

This chapter's format refuses zeros twice. A length of 0 is rejected outright. And because the CRC covers the length field, even a format that allowed empty records would need `0x48674bc7` in the CRC slot, not `0`.

Case 4: One flipped bit in a sealed segment

Use tiny segments (64 bytes) so five 24-byte records spread over three files. Then flip one bit in record 2, inside the first segment, which was sealed and fsynced when the log rotated.

```
mkdir /tmp/w4 && cd /tmp/w4
export WAL_MAX_SEG=64                    # tiny segments: two 24-byte records each
walctl append ./log "set a=1" "set b=2" "set c=3" "set d=4" "set e=5"
ls log
walctl recover ./log
# flip one bit in record 2's payload ('t' 0x74 -> 'u' 0x75), inside the SEALED first segment
printf 'u' | dd of=log/00000001.wal bs=1 seek=34 count=1 conv=notrunc status=none
walctl recover ./log; echo "exit=$?"
```

Output:

```
acked 5 records
00000001.wal
00000002.wal
00000003.wal
  seg 1 off    0  "set a=1"
  seg 1 off   24  "set b=2"
  seg 2 off    0  "set c=3"
  seg 2 off   24  "set d=4"
  seg 3 off    0  "set e=5"
segments=3 records=5 truncated=0B
  seg 1 off    0  "set a=1"
recover: sealed segment 00000001.wal: checksum mismatch at offset 24; refusing to open
exit=1
```

The trailer of record 2 is intact, so this is not a torn write. The CRC is wrong, so the bytes changed after the record was finished. A crash cannot produce this in a sealed segment, so the log does not guess. It refuses to open, names the file and the offset, and leaves the decision to an operator: restore the segment from a replica or a backup. Skipping the bad record and carrying on would silently drop `set b=2` while `set c=3` onward still applied, which is the gap a log exists to prevent.

Case 5: What an fsync costs, and what batching buys

Same 4,096 records of 100 bytes, with an fsync every 1, 8, 64, or 512 records. Every record is still acknowledged only after the fsync that covers it.

```
mkdir /tmp/wb && cd /tmp/wb
for b in 1 8 64 512; do
  rm -rf ./log
  walctl bench ./log 4096 $b
done
```

Output (two runs back to back on the box):

```
records=4096 batch=1 fsyncs=4096 elapsed=468ms  8760 records/s  114.2µs/record
records=4096 batch=8 fsyncs=512 elapsed=64ms  64219 records/s  15.6µs/record
records=4096 batch=64 fsyncs=64 elapsed=13ms  316255 records/s  3.2µs/record
records=4096 batch=512 fsyncs=8 elapsed=4ms  972050 records/s  1.0µs/record
records=4096 batch=1 fsyncs=4096 elapsed=465ms  8800 records/s  113.6µs/record
records=4096 batch=8 fsyncs=512 elapsed=58ms  70120 records/s  14.3µs/record
records=4096 batch=64 fsyncs=64 elapsed=12ms  328355 records/s  3.0µs/record
records=4096 batch=512 fsyncs=8 elapsed=6ms  690893 records/s  1.4µs/record
```

In these runs one fsync costs about 114 µs (earlier runs on the same box measured 93–96 µs), and it dominates everything else. One fsync per record caps the log at under 9,000 acknowledged records per second. One fsync per 64 records gives about 36× that. The durability guarantee is the same in every row. Batching only changes how many callers wait on each fsync.

Case 6: Group commit with goroutines

In a server the batch is whatever arrived while the previous fsync was running. One goroutine owns the log. It takes every request already waiting on its channel, appends them all, fsyncs once, and then acknowledges each caller.

```go
// cmd/groupcommit/main.go
package main

import (
    "fmt"
    "os"
    "sync"
    "time"

    "example.com/walbook/wal"
)

type req struct {
    payload []byte
    done    chan error
}

// committer owns the log. It drains every request that is already waiting,
// appends them all, fsyncs once, and only then acknowledges each caller.
func committer(l *wal.Log, in <-chan req, stats *[2]int, exited chan<- struct{}) {
    defer close(exited)
    for r := range in {
        batch := []req{r}
    drain:
```

```
        for len(batch) < 256 {
            select {
            case r2, ok := <-in:
                if !ok {
                    break drain
                }
                batch = append(batch, r2)
            default:
                break drain
            }
        }
        var err error
        for _, b := range batch {
            if err = l.Append(b.payload); err != nil {
                break
            }
        }
        if err == nil {
            err = l.Sync()
        }
        for _, b := range batch {
            b.done <- err // ack (or fail) only after the fsync
        }
        stats[0]++
        stats[1] += len(batch)
    }
}

func main() {
    dir, _ := os.MkdirTemp("", "gc")
    defer os.RemoveAll(dir)
    l, _, err := wal.Open(dir, 64<<20, nil)
    if err != nil {
        panic(err)
    }
    for _, clients := range []int{1, 8, 64} {
        in := make(chan req)
        var stats [2]int
        exited := make(chan struct{})
        go committer(l, in, &stats, exited)

        const perClient = 4096
        total := clients * (perClient / clients)
        var wg sync.WaitGroup
        start := time.Now()
        for c := 0; c < clients; c++ {
```

```
            wg.Go(func() {
                done := make(chan error, 1)
                for i := 0; i < perClient/clients; i++ {
                    in <- req{payload: make([]byte, 100), done: done}
                    if err := <-done; err != nil {
                        panic(err)
                    }
                }
            })
        }
        wg.Wait()
        el := time.Since(start)
        close(in)
        <-exited // stats are final only after the committer returns
        fmt.Printf("clients=%-3d records=%d fsyncs=%-5d avg batch=%5.1f  %7.0f acked records
            clients, total, stats[0], float64(stats[1])/float64(stats[0]), float64(total)/el
    }
    if err := l.Close(); err != nil {
        panic(err)
    }
}
```

```
go run -race ./cmd/groupcommit    # once, to prove the stats hand-off is race-free
go run ./cmd/groupcommit
```

Output (`-race` run first, then the normal run):

```
clients=1   records=4096 fsyncs=4096  avg batch=  1.0     9762 acked records/s
clients=8   records=4096 fsyncs=938   avg batch=  4.4    24761 acked records/s
clients=64  records=4096 fsyncs=68    avg batch= 60.2   105880 acked records/s
clients=1   records=4096 fsyncs=4096  avg batch=  1.0     6034 acked records/s
clients=8   records=4096 fsyncs=1024  avg batch=  4.0    37600 acked records/s
clients=64  records=4096 fsyncs=99    avg batch= 41.4   202211 acked records/s
```

With one client, every request pays a full fsync. With 64 concurrent clients, each fsync carries about 40–60 records, and acknowledged throughput rises by roughly 10–30× with no change to the durability rule. The spread between the two runs is the noise of a shared virtual disk. The batch sizes are the stable signal. Note the `<-exited` before reading `stats`. Without it, `-race` correctly reports that the committer can still be updating the counters after the last acknowledgement.

Case 7: Crash at every byte

Hand-picked damage proves the cases you thought of. This test cuts a three-record segment at **every** offset from 0 to 96, recovers, appends, and recovers again.

```go
// wal/log_test.go
package wal

import (
    "os"
    "path/filepath"
    "testing"
)

// Crash at every possible byte: recovery must return exactly the records
// that were completely on disk, and the log must accept appends afterwards.
func TestTearEveryOffset(t *testing.T) {
    recs := [][]byte{[]byte("order 1001 paid"), []byte("order 1002 paid"), []byte("refund 10
    var full []byte
    var ends []int // offset just past each record
    for _, r := range recs {
        full = append(full, Encode(r)...)
        ends = append(ends, len(full))
    }
    for cut := 0; cut <= len(full); cut++ {
        dir := t.TempDir()
        if err := os.WriteFile(filepath.Join(dir, segName(1)), full[:cut], 0o644); err != ni
            t.Fatal(err)
        }
        want := 0
        for _, e := range ends {
            if e <= cut {
                want++
            }
        }
        l, rep, err := Open(dir, 1<<20, nil)
        if err != nil {
            t.Fatalf("cut=%d: %v", cut, err)
        }
        if rep.Records != want {
            t.Fatalf("cut=%d: recovered %d records, want %d", cut, rep.Records, want)
        }
        if err := l.Append([]byte("after crash")); err != nil {
            t.Fatal(err)
        }
        if err := l.Close(); err != nil {
            t.Fatal(err)
        }
        l, rep, err = Open(dir, 1<<20, nil)
        if err != nil || rep.Records != want+1 || rep.Truncated != 0 {
            t.Fatalf("cut=%d: reopen got %+v, %v", cut, rep, err)
        }
```

```
        l.Close()
    }
    t.Logf("tore the log at all %d offsets; every recovery returned the complete prefix", le
}
```

```
go test -v ./wal/
```

Output:

```
=== RUN   TestTearEveryOffset
    log_test.go:50: tore the log at all 97 offsets; every recovery returned the complete pre
--- PASS: TestTearEveryOffset (0.06s)
PASS
ok      example.com/walbook/wal 0.062s
```

The trap

Truncating a torn tail is correct only when the damaged bytes were never acknowledged. Recovery cannot know that. Undo Case 4's flip, then flip the same bit in the **active** segment's only record, `set e=5`. That record was acknowledged in Case 4, because `walctl append` printed `acked 5 records` after the fsync.

```
# /tmp/w4, continuing from Case 4
printf 't' | dd of=log/00000001.wal bs=1 seek=34 count=1 conv=notrunc status=none   # undo C
# the same one-bit flip, this time in the ACTIVE segment's only record
printf 'u' | dd of=log/00000003.wal bs=1 seek=10 count=1 conv=notrunc status=none
walctl recover ./log
ls log
od -A d -t x1 log/00000003.wal.tail-0
```

Output:

```
  seg 1 off    0  "set a=1"
  seg 1 off   24  "set b=2"
  seg 2 off    0  "set c=3"
  seg 2 off   24  "set d=4"
segments=3 records=4 truncated=24B reason="checksum mismatch"
00000001.wal
00000002.wal
00000003.wal
00000003.wal.tail-0
0000000 18 dc 52 f9 07 00 00 00 73 65 75 20 65 3d 35 ce
0000016 fa ed fe ef be ad de 00
0000024
```

The log opened, and an acknowledged write is gone from the replay. The bytes are still in the `.tail-0` file: `73 65 75 20 65 3d 35` is `seu e=5`. Look at the reason. A torn tail fails with `torn or incomplete record` or `implausible length`. Here the trailer is intact and the CRC is wrong, which is the signature of a finished record that changed afterwards.

The tempting fix is to make `Open` refuse whenever the tail fails the CRC. Do not do that blindly. A record larger than one sector can be torn so that its trailer sector lands and a middle sector does not, and that also shows up as a CRC mismatch with an intact trailer. A log that refuses to start after an ordinary power cut is its own outage. The defensible design is the one above: truncate, keep the evidence, and make the report loud enough that a human or a replica comparison decides.

Three neighbouring traps:

- **Retrying a failed fsync.** After a write-back error, the kernel may already have dropped the dirty pages or marked them clean, so a second `fsync` can return success for data that is gone. The PostgreSQL manual puts it plainly: "the second attempt may be reported as successful, when in fact the data has been lost." PostgreSQL changed its default to crash on the first failure for this reason (see `data_sync_retry`). `Log.Sync` here is sticky: after one failure, every later call returns the same error.
- **mmap writes.** Stores into a mapped file report no errors. An I/O failure arrives later as `SIGBUS` or as an `msync` error. If you map segments for speed, keep `Sync` on the file descriptor as your durability point and test that path separately.
- **Forgetting the directory.** `create` fsyncs the directory after every new segment. Drop that line and a power cut right after rotation can leave a log whose newest segment simply does not exist, even though its records were fsynced.

The boring rule

1. Acknowledge only after a `Sync` that covers the record has returned `nil`. Batch fsyncs (group commit); never skip them.
2. Frame every record as CRC-32C over length and payload, a bounded non-zero length, a trailer written last, and padding to 8 bytes.
3. Seal a segment with an fsync before creating the next one, and fsync the directory after creating, renaming, or deleting a segment.
4. In recovery, refuse to open on any bad record in a sealed segment. In the active segment, copy the bad tail aside, truncate, fsync, and report what was cut and why.
5. Alert on every non-zero `Truncated`, and page when the reason is a checksum mismatch rather than a torn record.
6. Treat an fsync error as fatal: poison the log, crash, and let recovery decide. Never retry and believe the second answer.
7. Keep a test that cuts the log at every offset, and run it with `-race` next to the group-commit code.

Try this

1. Change `MaxRecord` to 64 bytes and append a 100-byte record. Where does it fail, and why is a panic in `Encode` the right place to stop a writer bug?

2. In the trap, compare the `.tail-0` bytes with the record from Case 4. Write a `walctl verify-tail` command that says whether a `.tail-*` file holds one complete record with a bad CRC (likely corruption) or a partial record (likely a torn write).
3. Run `walctl bench` on a tmpfs (`/dev/shm`) and on your real disk. Explain the difference in the first row.
4. Delete the `l.dirf.Sync()` call in `create`. Which test still passes? Why can no test on a running machine catch this, and what would?
5. Replace the trailer constant with the record's sequence number XORed with a per-log random value. What does that catch that a constant trailer cannot (hint: a stale record from a recycled segment)?

Sources

- Go standard library: `os.File.Sync`, `hash/crc32` (Castagnoli table, hardware-accelerated on amd64 and arm64), `encoding/binary`, `sync.WaitGroup.Go`; `internal/poll/fd_fsync_darwin.go` in the Go 1.27 source for `F_FULLFSYNC`.
- Linux man pages: `fsync(2)` (directory entries need their own fsync), `open(2)` (`O_EXCL`), `truncate(1)`, `dd(1)`.
- PostgreSQL documentation: `data_sync_retry` (why a failed fsync must not be retried).
- LevelDB: log format (checksummed, length-framed log records in fixed-size blocks; a well-known reference design).
- UnisonDB: Building a Corruption-Proof Write-Ahead Log in Go (CRC, trailer canary, 8-byte alignment, directory sync, stop at the first corruption).

Unix Pipeline Style CLI Tools

Unix Pipeline Style CLI Tools

Good CLI tools are composable components in a data stream. Go is excellent for CLIs: single static binary, fast startup, solid `bufio` and encoding libraries.

Why / Overview

If your tool only works as an interactive TUI, it cannot sit in CI. If it mixes logs into stdout, it breaks `jq`. Pipeline culture is an API design problem.

```
stdin  -> transform / filter -> stdout   (data only)
                 |
                 +-> stderr  (diagnostics, progress)
exit code -> automation branch
```

The Pipeline Contract

1. **Read stdin** when no file args (or when arg is `-`).
2. **Write data to stdout** — nothing else.
3. **Write diagnostics to stderr**.
4. **Exit non-zero** on failure; document codes.
5. **Be quiet on success** unless `--verbose`.
6. **Support non-TTY** operation (no assumed terminal width unless queried).

```go
func main() {
    if err := run(os.Args[1:]); err != nil {
        fmt.Fprintln(os.Stderr, "mytool:", err)
        os.Exit(1)
    }
}
```

Input Sources

```go
func openInput(path string) (io.ReadCloser, error) {
    if path == "" || path == "-" {
        return io.NopCloser(os.Stdin), nil
    }
    return os.Open(path)
}
```

Multiple files: process sequentially like `cat`, or document parallel behavior carefully (order).

Streaming Filters

Prefer stream processing over `ReadAll` for unbounded input.

```go
func filter(r io.Reader, w io.Writer, pred func(string) bool) error {
    sc := bufio.NewScanner(r)
    // Optional: sc.Buffer(make([]byte, 64*1024), 1024*1024)
    out := bufio.NewWriter(w)
    defer out.Flush()
    for sc.Scan() {
        line := sc.Text()
        if pred(line) {
            if _, err := out.WriteString(line); err != nil {
                return err
            }
            if err := out.WriteByte('\n'); err != nil {
                return err
            }
        }
    }
    return sc.Err()
}
```

Always check `sc.Err()` — a failed scan can look like EOF.

Output Modes: Human vs Machine

```
default (TTY):     human tables, colors optional
--json / pipe:     NDJSON or JSON array
--tsv:             stable columns for cut/awk
```

Detect TTY:

```go
func stdoutIsTTY() bool {
    fi, err := os.Stdout.Stat()
    if err != nil {
        return false
    }
    return (fi.Mode() & os.ModeCharDevice) != 0
}
```

Rules:

- Never emit ANSI color when not a TTY (or when `NO_COLOR` is set).
- Machine formats must be **stable** across versions (version field if needed).
- Do not interleave progress bars on stdout.

```go
type row struct {
    User string `json:"user"`
    OK   bool   `json:"ok"`
}

func writeNDJSON(w io.Writer, rows []row) error {
    enc := json.NewEncoder(w)
    for _, r := range rows {
        if err := enc.Encode(r); err != nil { // Encode adds newline
            return err
        }
    }
    return nil
}
```

Flags and Subcommands

Use `flag` for simple tools; `cobra`/`ff` etc. for larger surfaces. Conventions:

- `--help` works
- Short flags where traditional (`-n`, `-v`)
- `--version`
- Options before/after args documented

```go
fs := flag.NewFlagSet("grep-lite", flag.ExitOnError)
pattern := fs.String("e", "", "pattern")
invert := fs.Bool("v", false, "invert match")
_ = fs.Parse(args)
```

Exit Codes

Code	Meaning
0	Success (and for grep-like: match found)
1	Generic error **or** grep-like: no match
2	Usage error

Document grep-like special cases; they surprise API people.

```
const (
    exitOK      = 0
    exitFail    = 1
    exitUsage   = 2
)
```

Handling SIGPIPE

When writing to a closed pipe (`head` closes early), writes fail with `EPIPE` / `ErrClosed`. Good tools exit quietly with SIGPIPE semantics rather than dumping a panic stack.

```
if _, err := os.Stdout.Write(buf); err != nil {
    if errors.Is(err, syscall.EPIPE) {
        os.Exit(0) // or 141 depending on tradition
    }
    return err
}
```

Context and Cancellation

Long CLIs should honor SIGINT:

```
ctx, stop := signal.NotifyContext(context.Background(), os.Interrupt)
defer stop()

// pass ctx into work; on cancel, flush what is safe and exit non-zero
```

Structure of a Solid Small Tool

```
cmd/mytool/main.go     flags, exit codes
internal/app/run.go    pure-ish logic with io.Reader/Writer
internal/app/run_test.go  table tests with bytes.Buffer
```

Dependency inject IO for tests:

```go
type App struct {
    In   io.Reader
    Out  io.Writer
    Err  io.Writer
}

func (a *App) Run(ctx context.Context, args []string) error {
    // ...
}
```

Progress and Logging

```go
log := slog.New(slog.NewTextHandler(os.Stderr, &slog.HandlerOptions{Level: slog.LevelInfo}))
log.Info("processed", "lines", n)
```

Never `fmt.Println` debug to stdout in library paths.

Composition Examples

```bash
# NDJSON pipeline
mytool --json < events.log | jq 'select(.level=="error")'

# Classic filters
cat access.log | mytool -e 'timeout' | wc -l

# Explicit files
mytool -e 'panic' app.log app.log.1
```

Production Checklist (CLI)

- ☐ stdin default; `-` supported
- ☐ stdout data-only
- ☐ stderr for errors/logs

- ☐ Documented exit codes
- ☐ `--help` / usage on bad flags
- ☐ Stream large inputs
- ☐ Stable machine-readable mode
- ☐ `NO_COLOR` / TTY-aware color
- ☐ SIGINT cancels work
- ☐ SIGPIPE handled
- ☐ Tests use buffers, not real terminal

Common Pitfalls

1. **Logging to stdout** — breaks pipes.
2. **Loading entire stdin into memory**.
3. **Interactive prompts** when stdin is a pipe (hangs CI).
4. **Unstable JSON key order / floating timestamps** for golden tests — prefer normalized output.
5. **Relative paths** depending on launch directory without documenting.
6. **Buffered stdout not flushed** on early exit — use `bufio.Writer` + `Flush` or defer carefully.
7. **Assuming UTF-8** without documenting binary behavior.

Mini Implementation: line counter

```go
package main

import (
    "bufio"
    "fmt"
    "io"
    "os"
)

func countLines(r io.Reader) (int, error) {
    sc := bufio.NewScanner(r)
    n := 0
    for sc.Scan() {
        n++
    }
    return n, sc.Err()
}

func main() {
    n, err := countLines(os.Stdin)
    if err != nil {
        fmt.Fprintln(os.Stderr, err)
```

```
        os.Exit(1)
    }
    fmt.Println(n)
}
```

Extend with file args, `-v`, and JSON `{"lines":N}` as exercises.

Exercises

1. Build a filter that prints lines containing a substring; support stdin and file args.
2. Add `--json` NDJSON output of `{line,no}`; keep default plain text.
3. Implement invert match (`-v`) and appropriate exit codes.
4. Pipe through `head -n 1` and ensure no ugly panic on SIGPIPE.
5. Add `slog` diagnostics on stderr behind `-v`; prove `jq` still parses stdout.
6. Refactor logic to `App` struct; unit-test with `strings.NewReader` / `bytes.Buffer`.
7. Handle lines longer than 64KiB safely (configure scanner buffer).
8. Add `--fail-empty` that exits 1 when zero lines matched.
9. Benchmark counting 10M lines; compare `Scanner` vs raw `bufio.Reader` ReadBytes.
10. Write a README section "Scripting interface" documenting flags, exit codes, and examples.

More examples

Stdin filter: prefix match

```
mkdir -p /tmp/go-pipe-filter && cd /tmp/go-pipe-filter
go mod init example.com/pipe-filter
```

Save as `main.go`:

```
package main

import (
    "bufio"
    "fmt"
    "strings"
)

func filter(in string, prefix string) []string {
    var out []string
    sc := bufio.NewScanner(strings.NewReader(in))
    for sc.Scan() {
        line := sc.Text()
        if strings.HasPrefix(line, prefix) {
            out = append(out, line)
```

```
        }
    }
    return out
}

func main() {
    in := "ERR boom\nINFO ok\nERR again\n"
    for _, line := range filter(in, "ERR") {
        fmt.Println(line)
    }
}
```

```
go run .
```

Expected output:

```
ERR boom
ERR again
```

Exit code when no matches (grep-like)

```
mkdir -p /tmp/go-pipe-exit && cd /tmp/go-pipe-exit
go mod init example.com/pipe-exit
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "strings"
)

func matchCount(in, needle string) int {
    n := 0
    for _, line := range strings.Split(strings.TrimSuffix(in, "\n"), "\n") {
        if strings.Contains(line, needle) {
            n++
        }
    }
    return n
}

func main() {
    for _, needle := range []string{"go", "rust"} {
```

```
        n := matchCount("go tools\ngo test\n", needle)
        code := 0
        if n == 0 {
            code = 1
        }
        fmt.Printf("needle=%q matches=%d exit=%d\n", needle, n, code)
    }
}
```

```
go run .
```

Expected output:

```
needle="go" matches=2 exit=0
needle="rust" matches=0 exit=1
```

Runnable example

A pipeline-friendly line filter: read stdin (or an in-memory reader), match a substring, write matches to stdout, diagnostics to stderr, exit codes for scripts.

```
mkdir -p /tmp/go-pipeline-cli && cd /tmp/go-pipeline-cli
go mod init example.com/pipeline-cli
```

Save as `main.go`:

```
package main

import (
    "bufio"
    "fmt"
    "io"
    "os"
    "strings"
)

func filter(r io.Reader, w io.Writer, errW io.Writer, substr string, invert bool) (matches i
    sc := bufio.NewScanner(r)
    // Raise token size for long log lines
    buf := make([]byte, 0, 64*1024)
    sc.Buffer(buf, 1024*1024)
    for sc.Scan() {
        line := sc.Text()
        hit := strings.Contains(line, substr)
        if invert {
```

```go
            hit = !hit
        }
        if hit {
            if _, err := fmt.Fprintln(w, line); err != nil {
                return matches, err
            }
            matches++
        }
    }
    if err := sc.Err(); err != nil {
        fmt.Fprintln(errW, "scan:", err)
        return matches, err
    }
    return matches, nil
}

func main() {
    // Demo without relying on external pipes: feed a buffer, print results.
    input := "alpha\nerror: disk full\nbeta\nerror: timeout\ngamma\n"
    var out, errB strings.Builder
    n, err := filter(strings.NewReader(input), &out, &errB, "error:", false)
    if err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }
    fmt.Fprint(os.Stdout, out.String())
    fmt.Fprintf(os.Stderr, "matches=%d\n", n)

    // Exit code convention for CLIs: 0 found, 1 not found (grep-like optional).
    if n == 0 {
        os.Exit(1)
    }
}
```

```go
go run .
# real pipeline:
# printf 'a\nerror: x\nb\n' | go run .   # after wiring os.Stdin/Args
```

Expected output:

```
error: disk full
error: timeout
matches=2
```

(stderr line may interleave depending on buffering)

What to notice

- Business output on stdout; counts/diagnostics on stderr so `| jq` keeps working.
- Configure `Scanner` buffer for long lines; check `sc.Err()`.
- Exit codes are part of the scripting API.

Try next

- Accept optional file args; default to stdin when none.
- Add `-v` invert and `--json` NDJSON mode without breaking plain default.

Further Reading

- Rob Pike / Unix philosophy notes on tools
- Previous: safe file IO for CLIs that write state
- Next part: **Distributed Infrastructure** patterns for long-running workers

Environment, User Identity, CWD, and umask

Environment, User Identity, CWD, and umask

Overview

Every process inherits an **environment block**, a **user/group identity**, a **current working directory**, and (on Unix) a **umask**. Systems tools that ignore these surprise operators: relative paths resolve wrong, files get world-readable, and "works on my laptop" fails under systemd.

Environment variables

```
os.Getenv("HOME")
os.Setenv("APP_MODE", "prod") // process-local
os.LookupEnv("DEBUG")         // value, ok
os.Environ()                  // []string KEY=VAL
```

Patterns

Pattern	Use
Config via env	12-factor services (`DATABASE_URL`)
Feature flags	`DEBUG=1`, `APP_LOG_LEVEL=debug`
Child isolation	`cmd.Env = append(os.Environ(), "LC_ALL=C")`
Clear inheritance	`cmd.Env = []string{"PATH=/usr/bin"}` (explicit only)

```
cmd := exec.Command("git", "status")
cmd.Env = append(os.Environ(), "GIT_TERMINAL_PROMPT=0")
```

Never log secrets pulled from env (`TOKEN`, `PASSWORD`, `*_KEY`).

PATH and LookPath

```go
path, err := exec.LookPath("docker")
// searches PATH; fails closed if missing
```

In minimal containers, `PATH` may lack tools your laptop has—document dependencies.

User and group identity (Unix)

```go
fmt.Println("uid", os.Getuid(), "euid", os.Geteuid())
fmt.Println("gid", os.Getgid())
```

ID	Meaning
UID / GID	Real user/group
EUID / EGID	Effective (permission checks)

Drop privileges in privileged starters (rare in pure Go services; more common in installers). Prefer running the whole process as non-root under Kubernetes/systemd.

```go
u, err := user.Current()
// u.Username, u.HomeDir, u.Uid
```

```go
// expand ~
if strings.HasPrefix(path, "~/") {
    home, _ := os.UserHomeDir()
    path = filepath.Join(home, path[2:])
}
```

Prefer `os.UserHomeDir()` / `os.UserConfigDir()` over hardcoding `/home/...`.

Current working directory

```go
wd, err := os.Getwd()
err = os.Chdir("/var/lib/myapp")
```

Relative paths (`./data`, `config.yaml`) are relative to **CWD**, not the binary location.

Binary-relative assets

```
exe, err := os.Executable()
dir := filepath.Dir(exe)
cfg := filepath.Join(dir, "config.yaml")
```

Note: `os.Executable` may be a symlink path; `filepath.EvalSymlinks` if you need the real directory.

Children and Dir

```
cmd := exec.Command("./tool")
cmd.Dir = "/work/repo" // child CWD
```

umask and file permissions

Unix **umask** masks permission bits on create. Go's `os.OpenFile` / `WriteFile` pass a mode that is then masked by umask.

```
// request 0666 → often becomes 0644 with umask 0022
os.WriteFile("out.txt", data, 0o666)
```

For secrets:

```
f, err := os.OpenFile(path, os.O_CREATE|os.O_WRONLY|os.O_TRUNC, 0o600)
```

Still subject to umask (e.g. `0o600` & ^umask). If you need exact bits, `Chmod` after create:

```
_ = os.Chmod(path, 0o600)
```

Hostname and process identity

```
host, _ := os.Hostname()
pid := os.Getpid()
ppid := os.Getppid()
```

Useful in logs: `slog.Info("start", "host", host, "pid", pid)`.

Minimal example: show process context

```go
func printContext(w io.Writer) {
    wd, _ := os.Getwd()
    fmt.Fprintf(w, "pid=%d ppid=%d\n", os.Getpid(), os.Getppid())
    fmt.Fprintf(w, "uid=%d euid=%d\n", os.Getuid(), os.Geteuid())
    fmt.Fprintf(w, "cwd=%s\n", wd)
    fmt.Fprintf(w, "home=%s\n", mustHome())
}

func mustHome() string {
    h, err := os.UserHomeDir()
    if err != nil {
        return ""
    }
    return h
}
```

Rules of thumb

Do	Don't
Resolve paths with `UserConfigDir` / absolute roots	Assume CWD is the repo root in systemd
Pass explicit `cmd.Env` when isolation matters	Leak host secrets into every child
Create secrets as `0600` + chmod	Default world-readable state files
Log pid/host for multi-replica debug	Log full `os.Environ()`

Try next

1. Run the same binary from two different CWDs; print `Getwd` and a relative open error.
2. Start a child with a stripped `Env` and show it cannot find `git` without PATH.
3. Write a file with mode `0o666` and inspect actual mode with `stat`.

File Descriptors and Redirection

File Descriptors and Redirection

Overview

Unix processes talk to the world through **file descriptors** (FDs): integers indexing a per-process table. FD 0/1/2 are stdin/stdout/stderr. Understanding FDs explains pipes, redirection, and why "close the file" matters for long-running services.

The classic trio

FD	Go	Role
0	`os.Stdin`	Input
1	`os.Stdout`	Data output
2	`os.Stderr`	Diagnostics

```
fmt.Fprintln(os.Stdout, "data")
fmt.Fprintln(os.Stderr, "debug")
```

Shell: `tool >out.txt 2>err.txt` remaps FDs before `exec`.

Opening files yields FDs

```
f, err := os.Open("/etc/hosts")
// f.Fd() → uintptr underlying FD (Unix)
defer f.Close() // returns FD to the kernel
```

Leaking FDs (`Open` without `Close`) exhausts the process limit (`ulimit -n`).

Inheritance across exec

By default, child processes inherit open FDs unless marked close-on-exec.

```
cmd := exec.Command("child")
cmd.Stdin = os.Stdin
cmd.Stdout = os.Stdout
```

```go
cmd.Stderr = os.Stderr
```

`os/exec` sets up pipes or inheritance for the three standard streams. Extra FDs you opened in the parent may remain open in the child if not close-on-exec—usually undesirable.

```go
f, _ := os.OpenFile("log.txt", os.O_APPEND|os.O_CREATE|os.O_WRONLY, 0o644)
// syscall.CloseOnExec(int(f.Fd())) // low-level; prefer not sharing logs via inheritance
cmd.ExtraFiles = []*os.File{f} // intentional FD 3+ in child (Unix)
```

`ExtraFiles` maps to child FDs starting at 3—used by some protocols (e.g. systemd socket activation patterns).

Pipes as FDs

```go
r, w, err := os.Pipe()
go func() {
    defer w.Close()
    fmt.Fprintln(w, "hello")
}()
buf := make([]byte, 64)
n, _ := r.Read(buf)
fmt.Println(string(buf[:n]))
r.Close()
```

`io.Pipe` is in-process (not OS FDs). `os.Pipe` is a real kernel pipe—shareable with children.

Redirection patterns in Go

Capture child stdout

```go
var stdout, stderr bytes.Buffer
cmd.Stdout = &stdout
cmd.Stderr = &stderr
err := cmd.Run()
```

Stream child to parent stdout

```go
cmd.Stdout = os.Stdout
cmd.Stderr = os.Stderr
```

Tee logs

```
cmd.Stdout = io.MultiWriter(os.Stdout, logFile)
```

Checking TTY vs pipe

```
fi, err := os.Stdout.Stat()
isChar := fi.Mode()&os.ModeCharDevice != 0
```

Color and progress bars: only when stdout is a TTY (and respect `NO_COLOR`).

Limits and EMFILE

```
too many open files → EMFILE / "too many open files"
```

Mitigation	How
Always Close	`defer` or `t.Cleanup`
Bound concurrency	worker pools for file openers
Raise limit carefully	container `ulimit`, not a substitute for leaks
Reuse readers	stream instead of open-all

/dev/null and discard

```
cmd.Stdout = io.Discard
cmd.Stderr = io.Discard
// or os.OpenFile(os.DevNull, os.O_WRONLY, 0)
```

Minimal tool: `fdcount` sketch

```
// Linux: count entries in /proc/self/fd
entries, err := os.ReadDir("/proc/self/fd")
fmt.Println(len(entries))
```

Use in tests to catch FD leaks (open N files, close, count should return).

Rules of thumb

Do	Don't
Close files and pipe ends	Rely on GC finalizers for FDs
Keep stdout/stderr roles pure	Mix binary data and logs on FD 1
Use `ExtraFiles` only intentionally	Leak sockets into helpers
Handle `EPIPE` on writes	Crash on `| head` pipelines

Try next

1. Pipe a Go program into `head -n 1` and ensure it exits cleanly on short write.
2. Spawn a child with `ExtraFiles` and print FD 3 in the child.
3. Open 10k files without close; observe the error; fix with a pool.

Memory-Mapped Files

Memory-Mapped Files

Overview

mmap maps a file (or anonymous memory) into the process address space so you read and write with normal memory operations while the kernel pages data to/from disk. Go exposes this via `golang.org/x/sys/unix` (Unix) or careful use of platform APIs—not as a one-liner in the core stdlib.

When to use mmap vs `os.Read` / `bufio`:

Prefer read/stream	Prefer mmap
Sequential scans of huge files	Random access into large read-mostly files
Simple code, small files	Shared memory between processes (advanced)
Portable stdlib only	OS-specific performance work

Deep dive context: 251 mmap vs pread.

Mental model

```
process virtual memory
+---------------------------+
|  mapped region (pages)    |  <-->  file on disk
+---------------------------+
       page fault loads
```

Writes to a shared map may update the file (depending on flags). Always understand `PROT_*` and `MAP_*` flags.

Read-only map (Unix sketch)

```go
//go:build unix

package mmapfile
```

```go
import (
    "fmt"
    "os"

    "golang.org/x/sys/unix"
)

type RO struct {
    data []byte
}

func OpenRO(path string) (*RO, error) {
    f, err := os.Open(path)
    if err != nil {
        return nil, err
    }
    defer f.Close()

    st, err := f.Stat()
    if err != nil {
        return nil, err
    }
    size := int(st.Size())
    if size == 0 {
        return &RO{data: []byte{}}, nil
    }

    data, err := unix.Mmap(int(f.Fd()), 0, size, unix.PROT_READ, unix.MAP_SHARED)
    if err != nil {
        return nil, err
    }
    return &RO{data: data}, nil
}

func (m *RO) Bytes() []byte { return m.data }

func (m *RO) Close() error {
    if len(m.data) == 0 {
        return nil
    }
    return unix.Munmap(m.data)
}
```

```go
m, err := mmapfile.OpenRO("large.bin")
if err != nil {
    log.Fatal(err)
```

```
}
defer m.Close()
fmt.Printf("first 16: %x\n", m.Bytes()[:min(16, len(m.Bytes()))])
```

Critical: after `Munmap`, the slice is invalid—do not retain sub-slices.

Write map (careful)

```
data, err := unix.Mmap(int(f.Fd()), 0, size,
    unix.PROT_READ|unix.PROT_WRITE, unix.MAP_SHARED)
// mutate data[i] = ...
// unix.Msync(data, unix.MS_SYNC) // durability
```

Crash consistency is harder than write-temp-rename. For config files, prefer atomic replace (chapter 142). Use writable mmap for specialized stores/databases, not casual state.

Anonymous maps

```
data, err := unix.Mmap(-1, 0, size, unix.PROT_READ|unix.PROT_WRITE,
    unix.MAP_ANON|unix.MAP_PRIVATE)
```

Useful for large scratch buffers; still process-private memory.

Safety with Go

1. **No GC pointers into foreign maps incorrectly** — mmap slices are plain `[]byte`; fine for bytes, not for storing Go pointers.
2. **Bounds** — map length is fixed at map time; file growth needs remap.
3. **Concurrent access** — concurrent reads OK for RO; writers need sync.
4. **Windows** — different API (`golang.org/x/sys/windows`); isolate with build tags.

When mmap loses

- One-pass line scan of a log → `bufio.Scanner` is simpler and fine
- Network filesystems with weird consistency
- Tiny files → overhead dominates

Minimal CLI: `mmapsum`

```go
// sha256 of mmapped file without ReadAll
h := sha256.New()
_, _ = h.Write(m.Bytes())
fmt.Printf("%x\n", h.Sum(nil))
```

Compare memory profile vs `io.Copy(h, f)` for multi-GB inputs.

Rules of thumb

Do	Don't
Munmap exactly once	Use slice after unmap
Prefer RO maps for inspection tools	mmap for every config write
Document OS build tags	Assume mmap code is portable without tags
Measure vs `pread`/`Read`	Assume mmap is always faster

Try next

1. Map `/etc/hosts` read-only and print line count without `ReadFile`.
2. Benchmark mmap vs `io.Copy` SHA-256 on a 500MB file.
3. Read the deep-dive chapter on mmap vs pread for allocator interaction.

Unix Domain Sockets and Local IPC

Unix Domain Sockets and Local IPC

Overview

Unix domain sockets (UDS) provide local IPC with stream or datagram semantics—like TCP/UDP but bound to filesystem paths (or abstract names on Linux). Go's `net` package supports them as networks `"unix"` and `"unixgram"`.

Use UDS when:

- Local sidecar app communication
- Privileged helper (root) spoken to by unprivileged client
- Avoid exposing a TCP port on `127.0.0.1`

Stream server and client

```go
// server
path := "/tmp/myapp.sock"
_ = os.Remove(path) // clean stale socket file
ln, err := net.Listen("unix", path)
if err != nil {
    log.Fatal(err)
}
defer ln.Close()
_ = os.Chmod(path, 0o660) // restrict who can connect

for {
    conn, err := ln.Accept()
    if err != nil {
        continue
    }
    go func(c net.Conn) {
        defer c.Close()
        io.Copy(c, c) // echo
    }(conn)
}
```

```go
// client
conn, err := net.Dial("unix", "/tmp/myapp.sock")
```

Lifecycle of the socket file

Event	Action
Before listen	`Remove` stale path if safe
After listen	`Chmod` / directory permissions
Shutdown	`Close` listener; optionally `Remove` path

If the process dies without removing the path, next start gets `address already in use` until unlink.

Abstract namespace (Linux)

Linux allows abstract sockets: path starts with `@` or null byte—**no filesystem node**.

```go
ln, err := net.Listen("unix", "@myapp.rpc") // Go uses @ convention
```

Pros: no leftover files. Cons: Linux-only; different permission model (not file mode).

Datagram unixgram

```go
conn, err := net.ListenPacket("unixgram", "/tmp/myapp.dgram.sock")
buf := make([]byte, 2048)
n, addr, err := conn.ReadFrom(buf)
_, err = conn.WriteTo(buf[:n], addr)
```

Message-oriented; size limits apply.

HTTP over Unix sockets

```go
ln, err := net.Listen("unix", path)
server := &http.Server{Handler: mux}
go server.Serve(ln)

// client
httpc := http.Client{
    Transport: &http.Transport{
        DialContext: func(ctx context.Context, _, _ string) (net.Conn, error) {
```

```go
            var d net.Dialer
            return d.DialContext(ctx, "unix", path)
        },
    },
}
resp, err := httpc.Get("http://localhost/healthz") // host ignored by dialer
```

Docker engine API and many local agents use this pattern.

Credentials and peer auth (advanced)

On Linux, `SO_PEERCRED` can report peer UID/PID—useful for authorizing local clients. Requires `golang.org/x/sys/unix` and careful API use. For many apps, **filesystem permissions on the socket path** are enough.

Timeouts

```go
conn, err := d.DialContext(ctx, "unix", path)
_ = conn.SetDeadline(time.Now().Add(5 * time.Second))
```

Same discipline as TCP—don't hang forever on a stuck peer.

Minimal tool pair: `udsecho`

```bash
# terminal 1
go run ./cmd/udsecho serve -path /tmp/echo.sock
# terminal 2
go run ./cmd/udsecho client -path /tmp/echo.sock -msg hello
```

Rules of thumb

Do	Don't
Chmod socket paths tightly	World-writable `/tmp/*.sock` for secrets
Remove stale sockets on start	Ignore `EADDRINUSE` forever
Prefer UDS for local-only APIs	Bind Docker API-style services on `0.0.0.0` by accident
Document path location	Scatter sockets without XDG/runtime dir

Runtime dir tip: `XDG_RUNTIME_DIR` or `/run/user/$(id -u)/` for user services.

Try next

1. HTTP health endpoint served only on a Unix socket.
2. Two processes: one sends JSON lines over `unixgram`.
3. Start twice without `Remove`; fix the stale socket handling.

File Locks, Pidfiles, and Single-Instance Tools

File Locks, Pidfiles, and Single-Instance Tools

Overview

Ops tools often need **mutual exclusion**: only one compactor, one migrator, one agent per host. Classic patterns are **pidfiles** and **advisory file locks** (`flock`). Both are imperfect; know the failure modes.

Advisory flock (Unix)

```go
//go:build unix

package single

import (
    "fmt"
    "os"

    "golang.org/x/sys/unix"
)

type Lock struct {
    f *os.File
}

func Acquire(path string) (*Lock, error) {
    f, err := os.OpenFile(path, os.O_CREATE|os.O_RDWR, 0o644)
    if err != nil {
        return nil, err
    }
    // LOCK_EX exclusive; LOCK_NB non-blocking
    if err := unix.Flock(int(f.Fd()), unix.LOCK_EX|unix.LOCK_NB); err != nil {
        f.Close()
        return nil, fmt.Errorf("already running: %w", err)
    }
    return &Lock{f: f}, nil
}

func (l *Lock) Close() error {
```

```go
    if l == nil || l.f == nil {
        return nil
    }
    _ = unix.Flock(int(l.f.Fd()), unix.LOCK_UN)
    err := l.f.Close()
    l.f = nil
    return err
}
```

```go
lock, err := single.Acquire("/var/lock/myapp.lock")
if err != nil {
    log.Fatal(err)
}
defer lock.Close()
// only one instance holds the lock
```

Property	Note
Advisory	Cooperation required; another process can ignore
Released on close/exit	Kernel drops locks when FD closes
NFS	Historically unreliable; prefer local disk

Pidfiles (classic, racy)

```go
func WritePid(path string) error {
    pid := os.Getpid()
    return os.WriteFile(path, []byte(strconv.Itoa(pid)+"\n"), 0o644)
}

func ReadPid(path string) (int, error) {
    b, err := os.ReadFile(path)
    if err != nil {
        return 0, err
    }
    return strconv.Atoi(strings.TrimSpace(string(b)))
}
```

Problems:

1. Stale pidfile after crash
2. PID reuse (new process gets same PID)
3. TOCTOU between check and write

Slightly better pidfile

```
// open O_CREATE|O_EXCL for create-only; if exists, read and probe process
f, err := os.OpenFile(path, os.O_CREATE|os.O_EXCL|os.O_WRONLY, 0o644)
if err == nil {
    fmt.Fprintf(f, "%d\n", os.Getpid())
    return f, nil // keep open; delete on exit
}
// else: read pid, kill(pid, 0) to test existence - still imperfect
```

Prefer flock for single-instance when available; use pidfiles for human operators ("what PID is running?") **in addition**, not alone.

Combining lock + pid

```
lock, err := Acquire(lockPath)
// write pid into lock file while holding lock
fmt.Fprintf(lock.f, "%d\n", os.Getpid())
_ = lock.f.Sync()
```

Readers open the file read-only and parse the PID for status tools.

Cross-process mutex via lock directory (portable sketch)

```
// mkdir is atomic on POSIX for create
err := os.Mkdir(lockDir, 0o700)
if err != nil {
    if os.IsExist(err) {
        return fmt.Errorf("locked")
    }
    return err
}
// hold: keep directory; release: Remove
defer os.Remove(lockDir)
```

Works without `flock` but needs crash cleanup of empty lock dirs (stale locks).

Signal-safe release

Always `defer lock.Close()`. On SIGTERM, unlock as part of shutdown so a supervisor restart can acquire immediately.

```
ctx, stop := signal.NotifyContext(...)
defer stop()
defer lock.Close()
<-ctx.Done()
```

Minimal CLI: `once`

```
once -lock /tmp/job.lock -- /usr/bin/rsync ...
```

```
lock, err := Acquire(*lockPath)
if err != nil {
    os.Exit(75) // EX_TEMPFAIL - try later
}
defer lock.Close()
cmd := exec.Command(args[0], args[1:]...)
cmd.Stdout, cmd.Stderr = os.Stdout, os.Stderr
err = cmd.Run()
```

Cron-friendly: non-zero if already running.

Rules of thumb

Do	Don't
flock on local disk	Trust pidfiles alone for exclusion
Document lock path	Scatter locks in world-writable dirs without care
Exit distinct code when busy	Hang forever waiting (unless intentional)
Defer unlock	Leave exclusive lock across network calls for minutes without need

Try next

1. Run two instances of a flock demo; confirm the second fails fast.
2. Kill -9 the holder; confirm the second instance can acquire after.
3. Implement `once` and schedule it in cron every minute safely.

Project: Systems Mini-Tools

Project: Systems Mini-Tools

Overview

Build **`systool`**, a multi-command systems utility that practices this part end-to-end: process info, signals, atomic files, pipelines, locks, and local IPC. Keep each verb small.

Command	Skills
`id`	pid, uid, cwd, env peek
`watchdog`	run a child; restart on exit; SIGTERM drain
`atomic-write`	write-temp-rename
`lines`	stdin filter pipeline
`once`	flock single-instance runner
`serve-sock`	HTTP over Unix socket
`tailf`	poll file growth (simple)

Stdlib first; use `golang.org/x/sys/unix` only for flock.

Scaffold

```
mkdir -p systool/cmd/systool systool/internal/app
cd systool
go mod init example.com/systool
```

Dispatch pattern: same as CLI part (`FlagSet` per command + `NotifyContext`).

---

1. `id` — process context

```go
func cmdID(stdout io.Writer) error {
    wd, err := os.Getwd()
    if err != nil {
        return err
    }
    host, _ := os.Hostname()
```

```go
    fmt.Fprintf(stdout, "pid\t%d\n", os.Getpid())
    fmt.Fprintf(stdout, "ppid\t%d\n", os.Getppid())
    fmt.Fprintf(stdout, "uid\t%d\n", os.Getuid())
    fmt.Fprintf(stdout, "cwd\t%s\n", wd)
    fmt.Fprintf(stdout, "host\t%s\n", host)
    if h, err := os.UserHomeDir(); err == nil {
        fmt.Fprintf(stdout, "home\t%s\n", h)
    }
    return nil
}
```

```go
go run ./cmd/systool id
```

---

2. `watchdog` — supervise a child

```go
func cmdWatchdog(ctx context.Context, args []string) error {
    fs := flag.NewFlagSet("watchdog", flag.ContinueOnError)
    backoff := fs.Duration("backoff", time.Second, "restart delay")
    if err := fs.Parse(args); err != nil {
        return err
    }
    if fs.NArg() < 1 {
        return fmt.Errorf("usage: systool watchdog [--] <cmd> [args]")
    }
    for {
        if err := ctx.Err(); err != nil {
            return err
        }
        cmd := exec.CommandContext(ctx, fs.Arg(0), fs.Args()[1:]...)
        cmd.Stdout = os.Stdout
        cmd.Stderr = os.Stderr
        err := cmd.Run()
        fmt.Fprintf(os.Stderr, "watchdog: child exited: %v\n", err)
        select {
        case <-ctx.Done():
            return ctx.Err()
        case <-time.After(*backoff):
        }
    }
}
```

```
go run ./cmd/systool watchdog -backoff 500ms -- sleep 2
# Ctrl-C should stop restarts
```

Stretch: max restarts; exponential backoff; don't restart on exit 0.

---

3. atomic-write

```go
func atomicWrite(path string, data []byte, mode os.FileMode) error {
    dir := filepath.Dir(path)
    f, err := os.CreateTemp(dir, ".tmp-*")
    if err != nil {
        return err
    }
    tmp := f.Name()
    cleanup := true
    defer func() {
        if cleanup {
            _ = os.Remove(tmp)
        }
    }()
    if _, err := f.Write(data); err != nil {
        f.Close()
        return err
    }
    if err := f.Chmod(mode); err != nil {
        f.Close()
        return err
    }
    if err := f.Sync(); err != nil {
        f.Close()
        return err
    }
    if err := f.Close(); err != nil {
        return err
    }
    if err := os.Rename(tmp, path); err != nil {
        return err
    }
    cleanup = false
    // optional: fsync directory for durability
    return nil
}
```

```
echo hello | go run ./cmd/systool atomic-write -path /tmp/x.txt
```

---

4. `lines` — pipeline filter

```go
func cmdLines(r io.Reader, w io.Writer, contains string) error {
    sc := bufio.NewScanner(r)
    for sc.Scan() {
        line := sc.Text()
        if contains == "" || strings.Contains(line, contains) {
            fmt.Fprintln(w, line)
        }
    }
    return sc.Err()
}
```

```
printf 'a\nb\nac\n' | go run ./cmd/systool lines -c a
```

Exit 0 always if IO OK; use grep-like exit codes only if you document them.

---

5. `once` — single instance

```
// acquire flock; run command; release
// exit 75 if lock busy
```

See chapter 148 for `Acquire`. Wire:

```
go run ./cmd/systool once -lock /tmp/job.lock -- echo only-one
```

---

6. `serve-sock` — local HTTP

```go
func cmdServeSock(ctx context.Context, path string) error {
    _ = os.Remove(path)
    ln, err := net.Listen("unix", path)
    if err != nil {
```

```
        return err
    }
    _ = os.Chmod(path, 0o600)
    mux := http.NewServeMux()
    mux.HandleFunc("GET /healthz", func(w http.ResponseWriter, r *http.Request) {
        w.Write([]byte("ok"))
    })
    srv := &http.Server{Handler: mux}
    go func() {
        <-ctx.Done()
        shutdownCtx, cancel := context.WithTimeout(context.Background(), 3*time.Second)
        defer cancel()
        _ = srv.Shutdown(shutdownCtx)
    }()
    err = srv.Serve(ln)
    if err == http.ErrServerClosed {
        return nil
    }
    return err
}
```

```
go run ./cmd/systool serve-sock -path /tmp/systool.sock
curl --unix-socket /tmp/systool.sock http://localhost/healthz
```

---

7. `tailf` — follow file growth

```
func tailf(ctx context.Context, path string, w io.Writer) error {
    f, err := os.Open(path)
    if err != nil {
        return err
    }
    defer f.Close()
    // seek end
    _, _ = f.Seek(0, io.SeekEnd)
    buf := make([]byte, 4096)
    for {
        n, err := f.Read(buf)
        if n > 0 {
            _, _ = w.Write(buf[:n])
        }
        if err == io.EOF {
            select {
```

```
            case <-ctx.Done():
                return nil
            case <-time.After(200 * time.Millisecond):
            }
            continue
        }
        if err != nil {
            return err
        }
    }
}
```

Not a full `tail -F` (no rotate handling)—enough to practice polls + signals.

Acceptance checklist

- ☐ `id` prints pid/uid/cwd
- ☐ `watchdog` restarts `false` until Ctrl-C
- ☐ `atomic-write` leaves no torn file if killed mid-run (manual chaos)
- ☐ `lines` works in a pipe with `jq` or `wc`
- ☐ `once` second instance fails while first sleeps
- ☐ `serve-sock` answers curl via `--unix-socket`
- ☐ `tailf` prints new lines as you append to a file

Layout recap

```
cmd/systool/main.go          # signals + exit codes
internal/app/*.go            # commands
internal/lock/flock_unix.go  # build-tagged
```

What you practiced

Chapter	Tool
141	watchdog, serve-sock shutdown
142	atomic-write
143	lines
144	id
145	pipes, stdout/stderr
147	serve-sock
148	once

Ship `systool` as a learning binary; promote individual commands into dedicated tools when they grow.

Next project: 156 Systems ops toolkit (`deadline`, `reload`, `cronish`, `healthsock`, …).

Time, Clocks, and Timers

Time, Clocks, and Timers

Overview

Systems programs schedule work, measure latency, and enforce deadlines. Go's `time` package is enough for almost all of this—if you know **wall clock vs monotonic** behavior and avoid classic timer leaks.

Wall clock vs monotonic

```go
t1 := time.Now()
// ... work ...
elapsed := time.Since(t1) // uses monotonic reading when possible
```

`time.Now()` carries both wall and monotonic components on modern Go. **Subtraction** (`Since`, `Sub`) prefers monotonic—so NTP steps don't invent negative durations mid-measurement.

Need	Use
Logs, certs, file mtimes	Wall clock (`Format`, `Truncate`)
Timeouts, RTT, deadlines	Monotonic-friendly (`Since`, `Until`, `context` + timer)
"Run at 03:00 wall"	Wall clock + recompute after sleep

Never store only `UnixNano()` for elapsed timing across clock steps if you care about accuracy—prefer `time.Time` from `Now()`.

Sleep and timers

```go
time.Sleep(100 * time.Millisecond) // blocks goroutine

// prefer for cancelable wait:
t := time.NewTimer(100 * time.Millisecond)
defer t.Stop()
select {
case <-ctx.Done():
    return ctx.Err()
case <-t.C:
```

```
    // fired
}
```

Stop and drain

```
if !t.Stop() {
    select {
    case <-t.C:
    default:
    }
}
```

Failing to stop/drain can wake a select later and cause subtle double-fire bugs.

Ticker

```
tk := time.NewTicker(time.Second)
defer tk.Stop()
for {
    select {
    case <-ctx.Done():
        return
    case <-tk.C:
        // periodic work - keep it shorter than period
    }
}
```

Slow handlers + ticker = **tick pile-up** (channel buffer size 1 drops intermediate ticks in recent Go for `Ticker`—still don't do multi-second work on every tick without backpressure).

Deadlines with context

```
ctx, cancel := context.WithTimeout(parent, 5*time.Second)
defer cancel()
// pass ctx to Dial, Query, HTTP
```

Compose: signal cancel + timeout cancel (`NotifyContext` outer, timeout inner or vice versa).

Parsing and formatting

```go
t, err := time.Parse(time.RFC3339, "2026-08-04T12:00:00Z")
s := t.UTC().Format(time.RFC3339Nano)
```

For CLI flags: `flag.Duration` already parses `300ms`, `2s`, `1h`.

File mtimes and caching

```go
st, err := os.Stat(path)
mod := st.ModTime()
if mod.After(lastLoad) {
    // reload
}
```

Clock skew across NFS nodes can make mtime comparisons surprising—prefer content hashes for critical caches.

Minimal tool: `deadline`

```go
// run command with max wall time
ctx, cancel := context.WithTimeout(context.Background(), *max)
defer cancel()
cmd := exec.CommandContext(ctx, name, args...)
err := cmd.Run()
if errors.Is(ctx.Err(), context.DeadlineExceeded) {
    return fmt.Errorf("timeout after %s", *max)
}
```

Rules of thumb

Do	Don't
`Since` for latency	Compare wall `Unix()` deltas for RTT
`Stop` timers you create	Leak `time.After` in tight loops
Bound periodic work	Do unbounded IO every tick without skip/logic
UTC in logs/storage	Mix local TZ without documenting

Try next

1. Measure a sleep with `Since` while changing system time (VM lab)—note stability.
2. Rewrite a `time.After` loop to `NewTimer` + `Reset`.
3. Build `deadline 2s -- sleep 10` and confirm kill via `CommandContext`.

Filesystem Watching and Reload

Filesystem Watching and Reload

Overview

Agents and servers often **reload config** when a file changes. Approaches:

1. **Polling** `Stat` mtime/size (portable, simple)
2. **OS watchers** (inotify, FSEvents, ReadDirectoryChanges)—via `github.com/fsnotify/fsnotify` or raw `x/sys`
3. **Signal** `SIGHUP` for classic Unix reload

This chapter keeps patterns **minimal** and production-safe (debounce, atomic replace races).

Polling watcher (stdlib only)

```go
type pollWatch struct {
    path    string
    lastMod time.Time
    lastSz  int64
}

func (p *pollWatch) changed() (bool, error) {
    st, err := os.Stat(p.path)
    if err != nil {
        return false, err
    }
    mod, sz := st.ModTime(), st.Size()
    if mod.Equal(p.lastMod) && sz == p.lastSz {
        return false, nil
    }
    p.lastMod, p.lastSz = mod, sz
    return true, nil
}

func watchLoop(ctx context.Context, path string, every time.Duration, onChange func() error)
    var w pollWatch{path: path}
    _, _ = w.changed() // seed
```

```go
    t := time.NewTicker(every)
    defer t.Stop()
    for {
        select {
        case <-ctx.Done():
            return ctx.Err()
        case <-t.C:
            ok, err := w.changed()
            if err != nil {
                // missing file: keep trying
                continue
            }
            if ok {
                if err := onChange(); err != nil {
                    slog.Error("reload", "err", err)
                }
            }
        }
    }
}
```

Atomic replace (write temp + rename) may produce one or two events; always re-read the final path.

Debounce

Editors and write tools can burst events:

```go
func debounce(ctx context.Context, in <-chan struct{}, d time.Duration) <-chan struct{} {
    out := make(chan struct{}, 1)
    go func() {
        defer close(out)
        var timer *time.Timer
        for {
            select {
            case <-ctx.Done():
                return
            case _, ok := <-in:
                if !ok {
                    return
                }
                if timer != nil {
                    timer.Stop()
                }
                timer = time.AfterFunc(d, func() {
```

```
                    select {
                    case out <- struct{}{}:
                    default:
                    }
                })
            }
        }
    }()
    return out
}
```

Reload **after quiet period** (e.g. 200ms).

fsnotify sketch (common library)

```
// go get github.com/fsnotify/fsnotify
w, err := fsnotify.NewWatcher()
_ = w.Add(filepath.Dir(configPath))
for {
    select {
    case ev := <-w.Events:
        if ev.Name == configPath && (ev.Has(fsnotify.Write) || ev.Has(fsnotify.Create) || ev
            // signal reload (debounced)
        }
    case err := <-w.Errors:
        slog.Error("watch", "err", err)
    case <-ctx.Done():
        return
    }
}
```

Watch the **directory** if atomic rename replaces the inode—watching only the file can miss renames on some OSes.

SIGHUP reload

```
hup := make(chan os.Signal, 1)
signal.Notify(hup, syscall.SIGHUP)
go func() {
    for {
        select {
        case <-ctx.Done():
            return
```

```
        case <-hup:
            _ = reload()
        }
    }
}()
```

Works when you control the operator (`kill -HUP`); pairs well with file watch.

Safe reload checklist

1. Read entire config into memory
2. Validate
3. Swap atomically (`atomic.Value` or mutex + pointer)
4. Never half-apply on error—keep previous good config

```
type Config struct{ /* ... */ }

var current atomic.Pointer[Config]

func reload(path string) error {
    b, err := os.ReadFile(path)
    if err != nil {
        return err
    }
    cfg, err := parseConfig(b)
    if err != nil {
        return err
    }
    current.Store(&cfg)
    return nil
}
```

Minimal tool: `reload-demo`

```
# terminal 1
go run . -config /tmp/c.json
# terminal 2
echo '{"n":1}' > /tmp/c.json
# process logs "reloaded"
```

Rules of thumb

Do	Don't
Debounce reloads	Parse on every partial write
Keep last good config	Crash process on bad temporary JSON
Prefer dir watch + rename awareness	Assume one Write event == complete file
Bound poll interval	Poll every 1ms on huge fleets

Try next

1. Poll-watch a config; prove atomic rename still reloads.

2. Feed invalid JSON; process must keep serving old config.

3. Add SIGHUP path alongside poll.

Pseudo-Terminals (PTY)

Pseudo-Terminals (PTY)

Overview

A **pseudo-terminal** makes a program believe it is attached to a real terminal: line discipline, window size, and `isatty` checks. You need PTYs when automating interactive tools (`ssh`, `sudo`, password prompts, TUI apps) from a non-interactive parent.

Stdlib does **not** ship a full PTY API. Common approach: `github.com/creack/pty` (or platform `x/sys` + careful setup).

TTY detection (stdlib)

```go
func isTerminal(f *os.File) bool {
    fi, err := f.Stat()
    if err != nil {
        return false
    }
    return fi.Mode()&os.ModeCharDevice != 0
}
```

CLIs use this for color, paging, and prompts (chapter 145).

Why pipes are not enough

```
parent --pipe--> child
```

Many programs disable interactive features when stdin is a pipe (`Not a tty`). A PTY pair fixes that:

```
parent <-> PTY master <-> PTY slave <-> child (thinks it's a terminal)
```

Start a command under a PTY (creack/pty)

```
go get github.com/creack/pty@latest
```

```go
import (
    "io"
    "os"
    "os/exec"

    "github.com/creack/pty"
)

func runInPTY(name string, args ...string) error {
    cmd := exec.Command(name, args...)
    ptmx, err := pty.Start(cmd)
    if err != nil {
        return err
    }
    defer ptmx.Close()

    // copy PTY <-> user terminal
    go io.Copy(ptmx, os.Stdin)
    _, err = io.Copy(os.Stdout, ptmx)
    return err
}
```

```
go run . -- bash -l   # interactive shell session sketch
```

Window size

```go
// propagate terminal resize (Unix)
_ = pty.InheritSize(os.Stdin, ptmx)

// listen for SIGWINCH and call InheritSize again
```

Without this, full-screen TUIs mis-draw after resize.

Scripted interaction

```
ptmx, err := pty.Start(exec.Command("mysql", "-u", "root", "-p"))
// carefully write password + "\n" after prompt detection
// prefer non-interactive flags when possible
```

Security: automating passwords is fragile and audit-hostile—prefer socket auth, env files with strict modes, or API tokens.

When to avoid PTYs

Prefer	Avoid PTY when
`cmd.Stdin` + flags/env	App has real non-interactive mode
Expect scripts (`npm test`)	You only need stdout capture
Stdlib pipes	Cross-platform simplicity matters

Minimal tool: `runtty`

```
runtty -- <command>
# allocates PTY, connects to current terminal
```

Useful for wrapping tools that refuse pipes.

Rules of thumb

Do	Don't
Prefer non-interactive APIs	PTY-scrape fragile prompts in CI
Handle `SIGWINCH` for TUIs	Assume 80x24 forever
Close master FD	Leak PTYs (FD exhaustion)
Document platform deps	Expect identical PTY behavior on Windows without code

Try next

1. Run `ls --color=auto` under pipe vs PTY; compare color.
2. Automate a simple `read` shell script via PTY write.
3. Wire resize handling and test with a TUI (`top`, `htop`).

Resource Limits, rlimit, and Go Knobs

Resource Limits, rlimit, and Go Knobs

Overview

Production processes run under **resource ceilings**: open files, memory, CPU, process count. Some are OS-enforced (`rlimit`, cgroups); some are Go runtime knobs (`GOMAXPROCS`, `GOMEMLIMIT`). Systems tools should fail clearly when limits bite—and avoid creating the conditions that hit them.

Discovering limits (Unix)

```
ulimit -n    # open files
ulimit -u    # max user processes
```

In Go (via `x/sys/unix`):

```
//go:build unix

var lim unix.Rlimit
if err := unix.Getrlimit(unix.RLIMIT_NOFILE, &lim); err == nil {
    fmt.Printf("nofile soft=%d hard=%d\n", lim.Cur, lim.Max)
}
```

Raising NOFILE (when allowed)

```
lim.Cur = lim.Max
_ = unix.Setrlimit(unix.RLIMIT_NOFILE, &lim)
```

Containers may still enforce a lower cgroup/nproc limit—raising soft rlimit cannot exceed hard or cgroup.

Common failure modes

Symptom	Typical limit
`too many open files`	NOFILE / FD leak
`cannot allocate memory` / OOMKill	RSS cgroup / host RAM
Slow scheduling	CPU quota / throttling

Symptom	Typical limit
`fork: resource temporarily unavailable`	nproc / PIDs

Go runtime knobs

Knob	Role
`GOMAXPROCS`	OS threads for Go code (auto-tuned in containers on recent Go)
`GOMEMLIMIT`	Soft memory limit; GC aims to stay under
`GOGC`	GC percent trigger
`runtime/debug.SetMemoryLimit`	Same family as GOMEMLIMIT

```go
import "runtime/debug"

debug.SetMemoryLimit(2 << 30) // 2 GiB soft limit
```

See deep dives for pacer/GC details; here the systems takeaway is: **set limits intentionally in containers**.

Bound your own resource use

```go
// concurrency cap
sem := make(chan struct{}, 32)

// HTTP client
t := http.DefaultTransport.(*http.Transport).Clone()
t.MaxIdleConns = 100
t.MaxConnsPerHost = 10

// file workers: don't open 100k files at once
```

Minimal tool: `limits`

```go
func printLimits(w io.Writer) {
    fmt.Fprintf(w, "gomaxprocs\t%d\n", runtime.GOMAXPROCS(0))
    fmt.Fprintf(w, "numcpu\t%d\n", runtime.NumCPU())
    fmt.Fprintf(w, "numgoroutine\t%d\n", runtime.NumGoroutine())
    var ms runtime.MemStats
    runtime.ReadMemStats(&ms)
```

```
    fmt.Fprintf(w, "heap_alloc\t%d\n", ms.HeapAlloc)
    // optional: Getrlimit NOFILE
}
```

Ship in debug endpoints (`GET /debug/limits`) carefully—auth or localhost only.

cgroups (conceptual)

Kubernetes sets CPU/memory requests/limits via cgroups. Go 1.22+ improved `GOMAXPROCS` container awareness. Still verify:

```
pod memory limit < GOMEMLIMIT strategy
liveness does not restart during GC thrash
```

Rules of thumb

Do	Don't
Cap concurrency near I/O boundaries	Unbounded `go func` per request/file
Close FDs; monitor NOFILE	"Just raise ulimit" as only fix
Set `GOMEMLIMIT` in memory-capped pods	Assume unlimited heap in sidecars
Log clear errors on EMFILE	Busy-loop retry without backoff

Try next

1. Print soft/hard NOFILE in a binary.
2. Open files until failure; confirm error type.
3. Run with `GOMEMLIMIT=100MiB` and allocate; observe GC behavior under load.

Privileges, Capabilities, and Least Privilege

Privileges, Capabilities, and Least Privilege

Overview

Systems software often needs a **little** privilege (bind port 80, open raw sockets, read another user's files)—not a permanent root shell. This chapter is a **minimal, Go-oriented** map of least privilege: run as non-root, drop what you can, and prefer orchestrator policy over home-grown setuid.

Not a full Linux security course—pair with 17 Security hardening.

Prefer external policy

Mechanism	Who configures	Example
Kubernetes `securityContext`	Platform	`runAsNonRoot`, drop caps
systemd `User=` / `AmbientCapabilities=`	Operator	DynamicUser, CAP_NET_BIND_SERVICE
Container rootless	Runtime	podman/docker user ns

Your Go code should **assume it might be non-root** and fail with a clear error when a needed privilege is missing.

Check identity early

```go
if os.Geteuid() == 0 {
    slog.Warn("running as root - prefer dedicated user")
}
```

```go
ln, err := net.Listen("tcp", ":80")
if err != nil {
    return fmt.Errorf("listen :80 (need privilege or CAP_NET_BIND_SERVICE): %w", err)
}
```

Bind low ports without full root

Linux **capabilities** split root power. `CAP_NET_BIND_SERVICE` allows bind `<1024` without full UID 0.

Typical systemd:

```
[Service]
User=myapp
AmbientCapabilities=CAP_NET_BIND_SERVICE
CapabilityBoundingSet=CAP_NET_BIND_SERVICE
NoNewPrivileges=true
```

Go app still just calls `Listen`—no special code.

setuid binaries (avoid if possible)

Historically, setuid-root helpers did privileged ops then dropped. Risks are high (TOCTOU, env attacks). Prefer:

- systemd socket activation (privileged unit hands over FD)
- privileged sidecar
- `CAP_*` ambient on the service user

If you must touch UIDs:

```
// conceptual - platform-specific, error-prone
// unix.Setuid, Setgid after privileged setup
```

Test extensively; do not invent setuid stacks casually.

File permissions as privilege

```
config dir  0750 root:myapp
config file 0640 root:myapp
data dir    0700 myapp:myapp
socket      0660 myapp:myapp
```

Go: create with tight modes (chapter 144 umask + chmod).

Secrets

Prefer	Avoid
Files `0600` + dedicated user	World-readable env in `ps` dump discussions
Runtime mounts (K8s secrets)	Baking secrets into images
Short-lived tokens	Long-lived root API keys in config

Minimal pattern: refuse unsafe start

```go
func requireSecurePaths(cfgDir string) error {
    st, err := os.Stat(cfgDir)
    if err != nil {
        return err
    }
    if st.Mode().Perm()&0o077 != 0 {
        return fmt.Errorf("%s mode %#o is too open", cfgDir, st.Mode().Perm())
    }
    return nil
}
```

Optional hard fail in production profiles.

Rules of thumb

Do	Don't
Run non-root by default	Require root "for convenience"
Use capabilities/socket activation	Custom setuid without review
Clear errors on EPERM/EACCES	Silent fallback to insecure paths
Tight file modes on secrets	`chmod 777` to "make it work"

Try next

1. Run a server as non-root; bind `:8080` vs `:80`; document the error.
2. Read your systemd unit / K8s securityContext for one service.
3. Add a startup check that warns on world-writable config dirs.

Rootless Namespaces from Go: A 60-Line Sandbox with SysProcAttr and the Re-exec Pattern

Rootless Namespaces from Go: A 60-Line Sandbox with SysProcAttr and the Re-exec Pattern

Thesis

Containers are built from ordinary Linux processes placed in fresh **namespaces**. With a **user namespace**, an unprivileged user can create one: you become "root" inside it, mapped to your own uid outside. Go makes this short to write, with one Go-specific constraint. **A Go program is always multithreaded**, and the kernel won't let a multithreaded process `unshare(CLONE_NEWUSER)`. Go programs therefore request the namespaces at *clone time* through `syscall.SysProcAttr`, and re-execute themselves (`/proc/self/exe`) to run setup code inside them.

This chapter builds `nsbox`, which uses new user, PID, UTS, mount and network namespaces with no root and no dependencies. It then checks from inside what the sandbox actually isolates, including one leak that surprises most people the first time.

Figure 1: nsbox: rootless namespaces from Go with the re-exec pattern

Environment: Go 1.27.2 on Linux 6.12 (x86-64), run as an ordinary user (uid 1000) with unprivileged user namespaces enabled (`/proc/sys/user/max_user_namespaces` > 0). Some distributions restrict them, for example through AppArmor policy on Ubuntu 24.04+. If they are restricted, the `run` step fails with `operation not permitted`.

Why not just call `unshare`?

```go
// Why nsbox re-executes itself: a Go program is always multithreaded,
// and the kernel refuses unshare(CLONE_NEWUSER) in a multithreaded process.
package main

import (
    "fmt"
    "os"
    "strings"
    "syscall"
)

func main() {
    status, _ := os.ReadFile("/proc/self/status")
    for _, line := range strings.Split(string(status), "\n") {
        if strings.HasPrefix(line, "Threads:") {
            fmt.Println(line)
        }
    }
    err := syscall.Unshare(syscall.CLONE_NEWUSER)
    fmt.Println("unshare(CLONE_NEWUSER):", err)
}
```

```
$ go run ./unsharetrap
Threads:    5
unshare(CLONE_NEWUSER): invalid argument
```

The Go runtime had already started 5 OS threads before `main` ran. `unshare(2)` says "CLONE_NEWUSER requires that the calling process is not threaded", so the call fails with `EINVAL`. Most other namespace types *can* be unshared by a single thread. That brings a second trap: only the calling OS thread moves into the new namespace, and the Go scheduler may later run your goroutine on a different thread. Code that does `unshare`/`setns` in-process has to pin itself with `runtime.LockOSThread` and must never unlock that thread. The re-exec pattern sidesteps both problems: the kernel creates the child *in* the namespaces, before any Go code runs there.

nsbox

```go
// nsbox: run a command in new user, PID, UTS, mount and network namespaces,
// without root, using the re-exec pattern.
package main

import (
```

```
    "fmt"
    "os"
    "os/exec"
    "syscall"
)

func main() {
    if len(os.Args) < 3 {
        fmt.Fprintln(os.Stderr, "usage: nsbox run|child CMD [ARGS...]")
        os.Exit(2)
    }
    switch os.Args[1] {
    case "run":
        run(os.Args[2:])
    case "child":
        child(os.Args[2:])
    default:
        os.Exit(2)
    }
}

// run is the parent: it re-executes this binary as "child" inside new namespaces.
func run(argv []string) {
    cmd := exec.Command("/proc/self/exe", append([]string{"child"}, argv...)...)
    cmd.Stdin, cmd.Stdout, cmd.Stderr = os.Stdin, os.Stdout, os.Stderr
    cmd.SysProcAttr = &syscall.SysProcAttr{
        Cloneflags: syscall.CLONE_NEWUSER | syscall.CLONE_NEWPID |
            syscall.CLONE_NEWUTS | syscall.CLONE_NEWNS | syscall.CLONE_NEWNET,
        // Map our unprivileged uid/gid to root inside the user namespace.
        UidMappings: []syscall.SysProcIDMap{{ContainerID: 0, HostID: os.Getuid(), Size: 1}},
        GidMappings: []syscall.SysProcIDMap{{ContainerID: 0, HostID: os.Getgid(), Size: 1}},
    }
    fmt.Printf("[parent] host pid %d starting child\n", os.Getpid())
    if err := cmd.Run(); err != nil {
        fmt.Fprintln(os.Stderr, "[parent] child:", err)
        os.Exit(1)
    }
    fmt.Printf("[parent] child pid on the host was %d\n", cmd.Process.Pid)
}

// child runs as PID 1 of the new PID namespace, before exec'ing the user's command.
func child(argv []string) {
    must(syscall.Sethostname([]byte("nsbox")))
    // Stop mount events propagating back to the host, then mount a fresh /proc
    // so tools like ps see only this PID namespace.
    must(syscall.Mount("", "/", "", syscall.MS_REC|syscall.MS_PRIVATE, ""))
    must(syscall.Mount("proc", "/proc", "proc", syscall.MS_NOSUID|syscall.MS_NODEV|syscall.M
```

```
    path, err := exec.LookPath(argv[0])
    must(err)
    must(syscall.Exec(path, argv, os.Environ()))
}

func must(err error) {
    if err != nil {
        fmt.Fprintln(os.Stderr, "[child]", err)
        os.Exit(1)
    }
}
```

The parent and child are the same binary:

- **`run`** (parent) starts `/proc/self/exe child CMD...` with `SysProcAttr`:
  - `Cloneflags` lists the new namespaces.
  - `UidMappings`/`GidMappings` write `/proc/<pid>/uid_map` and `gid_map`, mapping in-namespace 0 to our real uid and gid. The `user_namespaces(7)` rules allow an unprivileged process to map exactly one id, its own. The Go runtime also writes `deny` to `setgroups` before writing the gid map, as the kernel requires.
- **`child`** runs inside the namespaces as **PID 1**. It sets the hostname (UTS namespace) and makes every mount `MS_PRIVATE` so nothing propagates back to the host (`mount_namespaces(7)`). It then mounts a new `/proc` so that `/proc` describes *this* PID namespace, and finally `execs` the user's command. That command takes over PID 1.

What it looks like from inside

```
#!/bin/sh
# what the world looks like from inside nsbox
echo "id:       $(id -u):$(id -g) ($(id -un))"
echo "hostname: $(hostname)"
echo "my pid:   $$"
echo "processes:"; ps -eo pid,user,cmd
echo "net devices per /proc/net/dev:"; tail -n +3 /proc/net/dev | cut -d: -f1 | tr -d ' '
echo "net devices per /sys/class/net (host's sysfs):"; ls /sys/class/net | tr '\n' ' '; echo
mount -t sysfs sysfs /sys && echo "after mounting a fresh sysfs:" && ls /sys/class/net
echo "try to write a host-root file:"; touch /etc/nsbox-test 2>&1 || true
```

```
echo "\$ id -u; hostname; ls /sys/class/net | wc -l"
id -u; hostname; ls /sys/class/net | wc -l
echo "\$ ./nsbox run ./inside.sh"
./nsbox run ./inside.sh
echo "\$ hostname   # unchanged on the host"
hostname
```

```
$ id -u; hostname; ls /sys/class/net | wc -l
1000
grok-bot-vm-399265740
3
$ ./nsbox run ./inside.sh
[parent] host pid 1671445 starting child
id:       0:0 (root)
hostname: nsbox
my pid:   1
processes:
    PID USER     CMD
      1 root     /bin/sh ./inside.sh
     10 root     ps -eo pid,user,cmd
net devices per /proc/net/dev:
lo
net devices per /sys/class/net (host's sysfs):
docker0 enp0s4 lo
after mounting a fresh sysfs:
lo
try to write a host-root file:
touch: cannot touch '/etc/nsbox-test': Permission denied
[parent] child pid on the host was 1671451
$ hostname   # unchanged on the host
grok-bot-vm-399265740
```

Reading it line by line:

- **User namespace.** `id` reports `0:0 (root)`, but `touch /etc/nsbox-test` is still denied. Root inside maps to uid 1000 outside, and files owned by host root are outside our mapping. A user namespace gives you capabilities *over resources owned by that namespace*, not over the host.
- **PID namespace** (`pid_namespaces(7)`). The shell is PID 1, and `ps` sees only the shell and itself. The same process is PID 1671451 on the host (the parent prints it).
- **UTS namespace.** The hostname is `nsbox` inside, and the host's hostname is unchanged afterwards.
- **Network namespace** (`network_namespaces(7)`). `/proc/net/dev` shows only `lo`, so the sandbox has no route out. **However, `/sys/class/net` still lists the host's `docker0` and `enp0s4`.** sysfs reports the network namespace of whoever *mounted* it, and `/sys` was inherited from the host. Mounting a fresh sysfs (allowed because our user namespace owns the new network namespace) shows only `lo`. Real runtimes mount new `/proc` *and* `/sys` for this reason. Tools that read `/sys/class/net` would otherwise get the host's view.

What nsbox is not (yet)

`nsbox` shares the host's root filesystem, read-only in effect for host-owned files. A real runtime would add:

- **A root filesystem.** Unpack an image into a directory, bind-mount it onto itself, then call `pivot_root(2)` into it and unmount the old root. `chroot` alone is escapable.
- **Resource limits.** Create a cgroup v2 directory, write `memory.max`/`pids.max`/`cpu.max`, and add the child's PID. Rootless this works only inside a cgroup subtree delegated to your user, usually by systemd.
- **Networking.** Connect a veth pair, or a user-mode stack such as slirp4netns or pasta, to the empty network namespace.
- **Syscall filtering and capability dropping.** Use seccomp, `no_new_privs`, and a reduced capability set.

Takeaways

- Go programs can't `unshare(CLONE_NEWUSER)` because they are always multithreaded. Request namespaces at clone time with `SysProcAttr.Cloneflags` and re-exec `/proc/self/exe` for the setup code.
- With `UidMappings`/`GidMappings`, an unprivileged user becomes root *inside* a namespace without gaining any power over host-owned resources.
- Remount `/proc` *and* `/sys` inside the sandbox. Otherwise `/sys/class/net` still shows host interfaces even in an empty network namespace.
- In-process `setns`/`unshare` from Go requires `runtime.LockOSThread` and a thread that is never unlocked.

Sources

- Linux man-pages: namespaces(7), user_namespaces(7), pid_namespaces(7), mount_namespaces(7), network_namespaces(7), unshare(2), pivot_root(2)
- Go: `syscall.SysProcAttr` and `runtime.LockOSThread`

Syslog, Journal, and Ops Logging

Syslog, Journal, and Ops Logging

Overview

Systems daemons historically logged to **syslog**; under systemd, messages often land in the **journal**. Go services commonly use `log/slog` to stderr (captured by the supervisor) **or** write to syslog. Pick one clear path so operators know where to `tail`.

Default modern pattern: stderr + supervisor

```go
h := slog.NewJSONHandler(os.Stderr, &slog.HandlerOptions{Level: slog.LevelInfo})
slog.SetDefault(slog.New(h))
slog.Info("started", "pid", os.Getpid())
```

Supervisor	Where logs go
systemd	journal (`journalctl -u myapp`)
Kubernetes	container runtime → log driver
Docker	`docker logs`
cron	mail / captured by cron daemon

Prefer this for new services: no syslog dependency, works in all containers.

When to use syslog

- Policy requires `/dev/log`
- Multi-process host with centralized rsyslog/syslog-ng
- Very old deployment molds

```go
import "log/syslog"

w, err := syslog.Dial("", "", syslog.LOG_INFO|syslog.LOG_DAEMON, "myapp")
if err != nil {
    // fall back to stderr
}
```

```
defer w.Close()
slog.SetDefault(slog.New(slog.NewTextHandler(io.MultiWriter(os.Stderr, w), nil)))
```

Network syslog (`Dial("udp", "logger:514", ...)`) needs reliability/security thought (TLS, auth)—often better handled by a local agent.

Levels and severity

slog	syslog-ish
Debug	DEBUG
Info	INFO
Warn	WARNING
Error	ERR

Don't log at Error for routine misses (cache miss outage).

Structured fields for ops

```
slog.Info("child_exit",
    "cmd", name,
    "code", exitCode,
    "dur_ms", dur.Milliseconds(),
)
```

Include: request/job id, host, version, signal reason.

Avoiding log loops

If your program ships logs by watching files it also writes, guard paths (don't ingest your own output). For agents, use a separate logging sink.

PID 1 and stdout

In containers, PID 1's stdout **is** the log stream. Don't background and discard stdout. Don't write secrets.

Minimal dual-sink setup

```go
func setupLog(json bool) {
    var h slog.Handler
    opts := &slog.HandlerOptions{Level: slog.LevelInfo}
    if json {
        h = slog.NewJSONHandler(os.Stderr, opts)
    } else {
        h = slog.NewTextHandler(os.Stderr, opts)
    }
    slog.SetDefault(slog.New(h))
}
```

CLI flag `--log-json` for production; text for local.

Rules of thumb

Do	Don't
stderr + supervisor for new apps	Invent custom log files without rotation plan
Structured key/value	Multiline free-form only
One correlation id field	Log passwords, tokens, cookies
Document `journalctl` / `kubectl logs`	Assume operators know your private log path

Try next

1. Run under `systemd-run --user` and find logs with `journalctl --user`.
2. Emit JSON slog; parse with `jq`.
3. Add `version` and `pid` to every startup log line.

Project: Systems Ops Toolkit

Project: Systems Ops Toolkit

Overview

Extend the systems story with **sysops**, a second multi-command project focused on **runtime ops**: deadlines, config reload, limits, locks + jobs, and local control sockets. Builds on 149 systool.

Command	Chapter skills
`deadline`	timers, `CommandContext`
`reload`	file poll + atomic config swap
`limits`	runtime + optional rlimit
`cronish`	ticker + once lock
`healthsock`	UDS HTTP health
`rotate`	simple size-based log copy/truncate (careful)
`envcheck`	required env vars present

Scaffold

```
mkdir -p sysops/cmd/sysops sysops/internal/app
cd sysops
go mod init example.com/sysops
```

Use `signal.NotifyContext`, exit 2 on usage, stderr diagnostics.

---

1. `deadline` — run with timeout

```go
func cmdDeadline(ctx context.Context, args []string) error {
    fs := flag.NewFlagSet("deadline", flag.ContinueOnError)
    max := fs.Duration("t", 30*time.Second, "max runtime")
    if err := fs.Parse(args); err != nil {
        return errUsage
    }
    if fs.NArg() < 1 {
        return fmt.Errorf("usage: sysops deadline -t 5s -- cmd args")
    }
```

```go
    cctx, cancel := context.WithTimeout(ctx, *max)
    defer cancel()
    cmd := exec.CommandContext(cctx, fs.Arg(0), fs.Args()[1:]...)
    cmd.Stdout, cmd.Stderr = os.Stdout, os.Stderr
    err := cmd.Run()
    if cctx.Err() == context.DeadlineExceeded {
        return fmt.Errorf("deadline exceeded (%s)", *max)
    }
    return err
}
```

```go
go run ./cmd/sysops deadline -t 1s -- sleep 10
```

---

2. `reload` — demo config watcher

```go
// poll path every 200ms; on change print new contents length
// store last good bytes in atomic.Value
```

```go
type box struct{ b []byte }

func cmdReload(ctx context.Context, path string) error {
    var v atomic.Value
    load := func() error {
        b, err := os.ReadFile(path)
        if err != nil {
            return err
        }
        if !json.Valid(b) { // example: require JSON
            return fmt.Errorf("invalid json")
        }
        v.Store(box{b: append([]byte(nil), b...)})
        fmt.Fprintf(os.Stderr, "reloaded %d bytes\n", len(b))
        return nil
    }
    _ = load()
    var lastMod time.Time
    var lastSz int64
    t := time.NewTicker(200 * time.Millisecond)
    defer t.Stop()
    for {
        select {
```

```go
        case <-ctx.Done():
            return nil
        case <-t.C:
            st, err := os.Stat(path)
            if err != nil {
                continue
            }
            if st.ModTime().Equal(lastMod) && st.Size() == lastSz {
                continue
            }
            lastMod, lastSz = st.ModTime(), st.Size()
            if err := load(); err != nil {
                fmt.Fprintln(os.Stderr, "reload keep old:", err)
            }
        }
    }
}
```

---

3. `limits` — print runtime snapshot

```go
func cmdLimits(w io.Writer) {
    fmt.Fprintf(w, "gomaxprocs\t%d\n", runtime.GOMAXPROCS(0))
    fmt.Fprintf(w, "numcpu\t%d\n", runtime.NumCPU())
    fmt.Fprintf(w, "goroutines\t%d\n", runtime.NumGoroutine())
    var m runtime.MemStats
    runtime.ReadMemStats(&m)
    fmt.Fprintf(w, "heap_alloc\t%d\n", m.HeapAlloc)
    fmt.Fprintf(w, "sys\t%d\n", m.Sys)
}
```

---

4. `cronish` — interval runner with flock

```bash
sysops cronish -every 1m -lock /tmp/job.lock -- /path/backup.sh
```

```go
// each tick: try Acquire lock; if busy skip; else run command
// respects ctx cancel between ticks
```

Prevents stacked cron jobs (chapter 148).

---

5. `healthsock` — UDS health for local probes

```
// Listen unix; GET /healthz returns 200 + limits snapshot
// Chmod 660; remove stale socket on start
// Shutdown on ctx
```

```
curl --unix-socket /tmp/sysops.sock http://localhost/healthz
```

---

6. `rotate` — minimal size rotate (teaching only)

```
// if file size > max: rename to .1 (overwrite), create new file
// NOT a full logrotate replacement - no copytruncate races covered fully
```

```go
func rotateIfLarge(path string, max int64) error {
    st, err := os.Stat(path)
    if err != nil {
        return err
    }
    if st.Size() < max {
        return nil
    }
    bak := path + ".1"
    _ = os.Remove(bak)
    if err := os.Rename(path, bak); err != nil {
        return err
    }
    f, err := os.OpenFile(path, os.O_CREATE|os.O_WRONLY, 0o644)
    if err != nil {
        return err
    }
    return f.Close()
}
```

Document that writers must reopen after rotate (or use stderr→journal instead).

---

7. `envcheck` — required environment

```go
func cmdEnvCheck(keys []string) error {
    var missing []string
    for _, k := range keys {
        if os.Getenv(k) == "" {
            missing = append(missing, k)
        }
    }
    if len(missing) > 0 {
        return fmt.Errorf("missing env: %s", strings.Join(missing, ", "))
    }
    fmt.Println("ok")
    return nil
}
```

```
go run ./cmd/sysops envcheck DATABASE_URL API_TOKEN
```

Useful as a container entrypoint preflight.

Acceptance checklist

- ☐ `deadline -t 1s -- sleep 5` fails with timeout
- ☐ `reload` keeps last good config on bad JSON
- ☐ `cronish` skips overlapping runs under lock
- ☐ `healthsock` + curl works
- ☐ `envcheck` exits non-zero when vars missing
- ☐ Ctrl-C stops long-running commands promptly

Suggested implementation order

```
envcheck → limits → deadline → healthsock → reload → cronish → rotate
```

Stretch

1. Merge `systool` + `sysops` into one binary with command groups
2. JSON output for `limits` / `healthz`
3. Debounce reload at 300ms
4. systemd unit file example in README

You now have two project tracks: **149 systool** (core Unix hygiene) and **156 sysops** (runtime operations)—together a practical systems programming lab in pure Go style.

systemd Socket Activation

systemd Socket Activation

Overview

Socket activation: systemd (or another supervisor) binds the socket and passes the FD to your process. Benefits: on-demand start, privilege separation (bind :80 as root, run app as user), faster parallel startup.

How FDs arrive

systemd sets:

```
LISTEN_FDS=1
LISTEN_PID=<pid>
```

FDs start at **3** (`SD_LISTEN_FDS_START`).

Minimal accept (stdlib)

```go
func listenersFromSystemd() ([]net.Listener, error) {
    pid, _ := strconv.Atoi(os.Getenv("LISTEN_PID"))
    if pid != os.Getpid() && os.Getenv("LISTEN_PID") != "" {
        return nil, fmt.Errorf("LISTEN_PID mismatch")
    }
    n, _ := strconv.Atoi(os.Getenv("LISTEN_FDS"))
    if n < 1 {
        return nil, nil
    }
    var out []net.Listener
    for i := 0; i < n; i++ {
        fd := 3 + i
        f := os.NewFile(uintptr(fd), "systemd")
        ln, err := net.FileListener(f)
        f.Close() // FileListener dups
        if err != nil {
            return nil, err
        }
        out = append(out, ln)
```

```go
    }
    return out, nil
}
```

```go
lns, err := listenersFromSystemd()
if len(lns) == 0 {
    // fallback Listen for local dev
    ln, _ := net.Listen("tcp", *addr)
    lns = []net.Listener{ln}
}
srv := &http.Server{Handler: mux}
_ = srv.Serve(lns[0])
```

Libraries: `coreos/go-systemd/activation` for production edge cases.

Unit sketch

```ini
# myapp.socket
[Socket]
ListenStream=80

# myapp.service
[Service]
ExecStart=/usr/local/bin/myapp
NonBlocking=true
```

Notify readiness

```go
// optional: sd_notify READY=1 via NOTIFY_SOCKET
```

Rules

Do	Don't
Fallback Listen for dev	Require systemd on laptops
Verify LISTEN_PID	Assume FD 3 always valid
Shutdown via Server.Shutdown	Ignore inherited FDs on exit

Try next

1. Local: `systemd-socket-activate -l 8080 ./myapp`
2. Dual mode: env set → use FD; else TCP.
3. Pass Unix socket via activation.

Linux procfs Introspection

Linux procfs Introspection

Overview

On Linux, `/proc` exposes live kernel views of processes, FDs, and networking. Handy for **debug CLIs** and self-inspection—not a portable API (guard with build tags).

Useful paths

Path	Info
`/proc/self/status`	Pid, VmRSS, threads, fds
`/proc/self/fd`	Open file descriptors
`/proc/self/limits`	rlimits
`/proc/self/net/tcp`	TCP sockets (parse carefully)
`/proc/meminfo`	Host memory

Count FDs

```go
//go:build linux

func fdCount() (int, error) {
    ents, err := os.ReadDir("/proc/self/fd")
    if err != nil {
        return 0, err
    }
    return len(ents), nil
}
```

Read VmRSS

```go
b, err := os.ReadFile("/proc/self/status")
// parse line "VmRSS:\t12345 kB"
```

Minimal tool: `procself`

```
systool procself
  fds    N
  vmrss  N kB
```

Cgroup memory (containers)

```
/sys/fs/cgroup/memory.max          # v2
/sys/fs/cgroup/memory/memory.limit_in_bytes  # v1
```

Parse when present; fall back if missing (not in cgroup).

Rules

Do	Don't
Build-tag Linux code	Assume `/proc` on macOS
Treat format as unstable	Parse without tests
Prefer metrics APIs for prod	Scrape proc in hot path every µs

Try next

1. Print fd count before/after intentional leak.
2. Read cgroup memory limit inside Docker.
3. Cross-compile with `//go:build linux` stubs elsewhere.

Project: Mini Process Supervisor

Project: Mini Process Supervisor

Overview

Build **`minisup`**: run multiple child commands, restart on crash, forward signals, aggregate logs—a teaching systemd/overmind lite.

Spec

```
# procfile-ish
web:   ./bin/web -addr :8080
worker: ./bin/worker
```

Or flags:

```
minisup -cmd "web=./web" -cmd "worker=./worker"
```

Requirements

1. Start all children with `Command`
2. Prefix stdout/stderr lines with name
3. On child exit 0 → restart with backoff (cap)
4. SIGTERM → signal children, wait with deadline, then SIGKILL
5. Exit when all stopped after shutdown

Signal fanout

```
cmd := exec.Command(bin, args...)
cmd.SysProcAttr = &syscall.SysProcAttr{Setpgid: true} // Unix: kill group
// On stop: syscall.Kill(-pgid, SIGTERM)
```

Log prefix

```
cmd.Stdout = prefixWriter{name: "web", w: os.Stdout}
```

Acceptance

- ☐ Two children stay up
- ☐ Kill one; it restarts
- ☐ Ctrl-C stops both within 5s
- ☐ Max restarts then give up with error

Stretch

- Health: restart if TCP port fails dial
- Procfile parser
- Status UDS: `minisup status`

Chapters: 141, 145, 150, 153.

Distributed Infrastructure

Distributed Infrastructure Overview

Distributed Infrastructure Overview

This part introduces distributed reliability patterns used in production Go systems. The objective is to reason about **failure as a normal operating condition**, not an exception path you only sketch in design docs.

Why / Overview

A single-process Go service is relatively easy. A system of services is not:

- Instances appear and disappear
- Networks delay, drop, and duplicate
- Dependencies degrade partially
- Retries turn brownouts into outages
- Queues hide problems until they explode

```
clients
   |
   v
[ discovery / LB ] -> service A -> service B
                         |            |
                      queue/workers   cache/db
```

This part focuses on three control planes you will reimplement (or configure) forever:

1. **Where is healthy capacity?** (discovery + health)
2. **How do we call others safely?** (retry + circuit breaking)
3. **How do we absorb load spikes?** (queues + backpressure)

Learning Path

Chapter	Focus	Outcome
151 Discovery & health	Registration, TTL, readiness	Route only to safe instances
152 Retry & circuit breaker	Control loops	Avoid amplification cascades
153 Queues & workers	Buffering, concurrency, poison messages	Degrade gracefully under overload
154 Caching	TTL, singleflight, cache-aside	Cut load without consistency disasters

Chapter	Focus	Outcome
155 Idempotency	Keys, dedupe, unique constraints	Safe retries on POST/payments
156 Leader election	Leases, renew/cancel	Singleton loops without split brain
157 Transactional outbox	Same-TX events + relay	Reliable side effects
158 Project: resilient worker	Retry, DLQ, shutdown	End-to-end worker lab

Core Mental Models

Partial failure

In distributed systems, "up" is not boolean. An instance can:

- Accept TCP but fail application readiness
- Pass health while serving 10× latency
- Succeed writes but fail reads

Design for **degraded modes**: serve stale cache, disable non-critical features, fail fast on optional deps.

Amplification

```
1 user request
  -> 1 API
    -> 3 retries
      -> each fans out to 5 backends
        = 15 backend calls (and more if each retries)
```

Always multiply retry policy × fan-out × client count.

Backpressure vs buffering

Approach	Effect	Risk
Drop	Protect system	Data loss
Queue	Smooth spikes	Latency + memory
Reject (503)	Signal clients	Client must handle
Slow down producers	True backpressure	Needs cooperation

Go channels are in-process queues; they do not replace durable external queues when you need restart survival.

Go Building Blocks

- `context` for deadlines across hops
- `net/http` clients with tuned transports
- `golang.org/x/sync/errgroup` for bounded fan-out
- `sync.Mutex` / atomics for in-memory breakers
- Workers: goroutines + channels or external brokers (NATS, SQS, Kafka, RabbitMQ)

```go
g, ctx := errgroup.WithContext(ctx)
g.SetLimit(8) // Go 1.20+
for _, id := range ids {
    id := id
    g.Go(func() error {
        return fetch(ctx, id)
    })
}
if err := g.Wait(); err != nil {
    return err
}
```

Production Principles

1. **Timeouts everywhere** — no unbounded waits.
2. **Idempotency** — safe retries need safe operations.
3. **Explicit queues** — size, DLQ, visibility timeout documented.
4. **Health is routing data** — not a vanity endpoint.
5. **Observable control state** — breaker open, queue depth, retry counts as first-class metrics.
6. **Load test failure** — chaos and dependency brownouts in staging.

Relationship to Other Parts

- **Network systems (13)** — transport budgets feed retry decisions.
- **Observability (16)** — you cannot tune breakers without metrics/traces.
- **Security (17)** — identity and mTLS between discovered peers.
- **Performance (18)** — worker counts and contention under load.

Literacy Checklist

☐ Distinguish liveness vs readiness
☐ Explain TTL registration and stale endpoint risk
☐ Design retry with jitter and attempt caps
☐ Draw circuit breaker states and transitions
☐ Size a worker pool against dependency capacity

- ☐ Define poison-message handling and DLQ
- ☐ Compute amplification for a proposed retry policy

Part Warm-Up Exercises

1. Inventory outbound calls in a service; list timeout and retry for each.
2. Compute worst-case fan-out for your hottest API.
3. Find your health endpoints; do they check dependencies?
4. Measure queue depth (if any) during a past incident.
5. Write a one-paragraph policy: "when dependency X is down, we …"

Worked Example: Amplification Math

Suppose:

- 1,000 RPS at the edge
- Each request fans out to 4 dependencies
- Each hop retries up to 3 times on failure
- Failure rate at one dependency is 10%

Rough upper bound if every layer retries naively:

```
edge RPS × fan-out × attempts
= 1000 × 4 × 3
= 12,000 dependency calls/sec peak intent
```

During a brownout, effective load can climb further as queues rebuild. This is why attempt caps, breakers, and single-owner retry policy matter more than micro-optimizing JSON.

Decision Guide

Problem	First pattern to consider
Clients hardcode hosts	Discovery + health
Blips become user errors	Retry with jitter (idempotent)
Dependency melt under retries	Circuit breaker
Spikes OOM the API process	Queue + backpressure / 503
Duplicate side effects	Idempotency keys / dedupe store
Poison payload spins forever	DLQ + max receive count

What This Part Will Not Cover

- Consensus algorithms (Raft/Paxos) in depth
- Exactly-once distributed transactions
- Full service-mesh product configuration

Those matter, but the everyday Go engineer wins more reliability from timeouts, health, retries, and queues done correctly.

Incident Starter Questions

When pages fire, ask in order:

1. Is the dependency **down**, **slow**, or **wrong**?
2. Are we **amplifying** with retries?
3. Is **discovery** pointing at zombies?
4. Is a **queue** lagging or a worker stuck?
5. What do **breakers** and **depths** show right now?

If you cannot answer from dashboards, instrument before rewriting architecture.

Reference Architecture (Mental Model)

```
                  +---------------- discovery / endpoints ---------------+
                  |                                                      |
clients -> edge LB/proxy -> service instances -> outbound clients        |
                  |                |                    |               |
                  |           readiness             breakers/retries     |
                  |                |                    |               |
                  +------------ metrics/logs/traces (part 19) -----------+
                                        |
                                   async workers <--> queue <--> DLQ
```

You do not need every box on day one. Add boxes when incidents demand them — but know which box is missing when they do.

Configuration Knobs Inventory

Keep a living document per service:

Knob	Example	Owner
outbound timeout	300ms	service
retry max	2	service

Knob	Example	Owner
breaker threshold	5 fails / 10s	service
worker count	32	service
queue depth alert	10k	platform
probe interval	5s	platform

Unknown knobs default to dangerous framework defaults.

Testing Strategy

Layer	Technique
Unit	classify errors; pure policy functions
Integration	fake dependency with flaky latency
Chaos	kill pods; network delay; certificate expiry
Load	sustain + spike; watch depth and p99

Policy without tests regresses the first time someone "simplifies" retries.

Glossary

- **Readiness** — safe to receive traffic.
- **TTL registration** — auto-expire stale instances.
- **Idempotent** — safe to apply twice.
- **Half-open** — breaker probing recovery.
- **Backpressure** — slowing or rejecting producers.
- **DLQ** — dead-letter queue for poison messages.
- **Amplification** — retries × fan-out × clients multiplying load.
- **Visibility timeout** — lease duration before a queue redelivers.

More examples

Tour: health gate + retry budget

```
mkdir -p /tmp/go-dist-tour && cd /tmp/go-dist-tour
go mod init example.com/dist-tour
```

Save as `main.go`:

```
package main
```

```go
import (
    "context"
    "fmt"
    "time"
)

func healthy() bool { return true }

func call(ctx context.Context, n *int) error {
    *n++
    if *n < 2 {
        return fmt.Errorf("transient")
    }
    return ctx.Err()
}

func main() {
    if !healthy() {
        fmt.Println("skip call: not ready")
        return
    }
    ctx, cancel := context.WithTimeout(context.Background(), time.Second)
    defer cancel()
    var n int
    var err error
    for attempt := 1; attempt <= 3; attempt++ {
        err = call(ctx, &n)
        if err == nil {
            break
        }
        time.Sleep(5 * time.Millisecond)
    }
    fmt.Println("attempts:", n, "err:", err)
}
```

```go
go run .
```

Expected output:

```
attempts: 2 err: <nil>
```

Runnable example

Tour of this part: registry of instances with TTL expiry, a failed call that trips a mini breaker, and a bounded queue reject—three control planes in one process.

```
mkdir -p /tmp/go-dist-tour && cd /tmp/go-dist-tour
go mod init example.com/dist-tour
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "sync"
    "time"
)

type instance struct {
    addr    string
    expires time.Time
}

type registry struct {
    mu   sync.Mutex
    inst map[string]instance
}

func (r *registry) heartbeat(addr string, ttl time.Duration) {
    r.mu.Lock()
    r.inst[addr] = instance{addr: addr, expires: time.Now().Add(ttl)}
    r.mu.Unlock()
}

func (r *registry) healthy() []string {
    r.mu.Lock()
    defer r.mu.Unlock()
    now := time.Now()
    var out []string
    for a, i := range r.inst {
        if now.Before(i.expires) {
            out = append(out, a)
        } else {
            delete(r.inst, a)
        }
    }
    return out
}

type breaker struct {
    fails int
    open  bool
```

```go
}

func (b *breaker) call(fn func() error) error {
    if b.open {
        return fmt.Errorf("breaker open")
    }
    if err := fn(); err != nil {
        b.fails++
        if b.fails >= 3 {
            b.open = true
        }
        return err
    }
    b.fails = 0
    return nil
}

func main() {
    reg := &registry{inst: map[string]instance{}}
    reg.heartbeat("10.0.0.1:8080", 50*time.Millisecond)
    reg.heartbeat("10.0.0.2:8080", time.Second)
    time.Sleep(60 * time.Millisecond)
    fmt.Println("healthy:", reg.healthy())

    var b breaker
    for i := 0; i < 4; i++ {
        err := b.call(func() error { return fmt.Errorf("dep down") })
        fmt.Printf("call %d: %v open=%v\n", i+1, err, b.open)
    }

    q := make(chan string, 2)
    send := func(s string) error {
        select {
        case q <- s:
            return nil
        default:
            return fmt.Errorf("queue full")
        }
    }
    fmt.Println("enq a:", send("a"))
    fmt.Println("enq b:", send("b"))
    fmt.Println("enq c:", send("c"))
    fmt.Println("tour: discovery + breaker + backpressure")
}
```

```go
go run .
```

Expected output:

```
healthy: [10.0.0.2:8080]
call 1: dep down open=false
call 2: dep down open=false
call 3: dep down open=true
call 4: breaker open open=true
enq a: <nil>
enq b: <nil>
enq c: queue full
tour: discovery + breaker + backpressure
```

What to notice

- TTL discovery expires zombies; breakers stop retry storms; bounded queues apply backpressure.
- Chapters 151–153 deepen each control plane.

Try next

- Add half-open probe recovery to the breaker.
- Drain the queue with N workers and measure lag.

Next Chapter

Service Discovery and Health Checks — knowing who is safe to call.

Service Discovery and Health Checks

Service Discovery and Health Checks

Discovery systems answer two questions: **where instances are**, and **whether they are currently safe to route to**. Getting either wrong produces systematic 5xx and mysterious latency.

Why / Overview

Hard-coded IPs die in dynamic environments. DNS, kube endpoints, Consul, Eureka, or a custom registry all solve:

```
instance -> register + heartbeat -> registry
router/client -> resolve -> filter healthy -> call
```

Health is not optional metadata; it is **routing input**.

Discovery Mechanisms

Mechanism	Pros	Cons
DNS (A/AAAA/SRV)	Simple, universal	TTL lag, limited health
Platform endpoints (K8s)	Integrated readiness	Cluster-centric
Service mesh / lb	Rich policy	Operational weight
Client-side registry	Flexible	Dual-write complexity
Config static list	Debuggable	No dynamism

Many Go services use **DNS or platform discovery** plus **client-side health-aware pools**.

Registration and TTL

Without TTL/heartbeat, crashed instances remain targets forever.

```go
type Registration struct {
    ID       string    `json:"id"`
    Addr     string    `json:"addr"` // host:port
    Expires  time.Time `json:"expires"`
    Meta     map[string]string `json:"meta,omitempty"`
}
```

```go
// Registry.Put sets Expires = now + TTL; instances heartbeat before expiry.
```

Heartbeat loop:

```go
func (c *Agent) heartbeat(ctx context.Context) error {
    t := time.NewTicker(c.ttl / 3)
    defer t.Stop()
    for {
        select {
        case <-ctx.Done():
            return c.registry.Deregister(context.Background(), c.id)
        case <-t.C:
            if err := c.registry.Refresh(ctx, c.id, c.ttl); err != nil {
                slog.Error("heartbeat failed", "err", err)
            }
        }
    }
}
```

On shutdown: best-effort deregister **and** rely on TTL (network can drop deregister).

Liveness vs Readiness vs Startup

Probe	Question	Failure action
Startup	Has init finished?	Wait; don't kill yet
Liveness	Is process stuck?	Restart process
Readiness	Can it serve traffic?	Remove from LB only

```go
var (
    ready   atomic.Bool
    started atomic.Bool
)

func handlers() http.Handler {
    mux := http.NewServeMux()
    mux.HandleFunc("GET /healthz", func(w http.ResponseWriter, r *http.Request) {
        // Liveness: process up; avoid dependency checks here.
        w.WriteHeader(http.StatusOK)
    })
    mux.HandleFunc("GET /readyz", func(w http.ResponseWriter, r *http.Request) {
        if !ready.Load() {
            http.Error(w, "not ready", http.StatusServiceUnavailable)
            return
        }
```

```go
        // Optional: shallow dependency ping with short timeout.
        ctx, cancel := context.WithTimeout(r.Context(), 200*time.Millisecond)
        defer cancel()
        if err := db.PingContext(ctx); err != nil {
            http.Error(w, "db", http.StatusServiceUnavailable)
            return
        }
        w.WriteHeader(http.StatusOK)
    })
    return mux
}
```

Anti-pattern: readiness that always returns 200; liveness that checks the database (DB blip restarts all pods).

Warmup

```go
func main() {
    // listen first for probes if needed, ready=false
    go server.Listen()
    mustConnectDB()
    primeCaches()
    ready.Store(true)
    <-ctx.Done()
    ready.Store(false) // stop new traffic
    shutdown()
}
```

Client-Side Discovery Pool

```go
type Pool struct {
    mu    sync.RWMutex
    addrs []string
}

func (p *Pool) Update(addrs []string) {
    p.mu.Lock()
    p.addrs = append([]string(nil), addrs...)
    p.mu.Unlock()
}

func (p *Pool) Pick() (string, error) {
    p.mu.RLock()
```

```go
    defer p.mu.RUnlock()
    if len(p.addrs) == 0 {
        return "", errors.New("no endpoints")
    }
    // simple RR or random
    return p.addrs[rand.Intn(len(p.addrs))], nil
}
```

Refresh from DNS:

```go
func watchDNS(ctx context.Context, host string, every time.Duration, p *Pool) {
    t := time.NewTicker(every)
    defer t.Stop()
    resolver := net.DefaultResolver
    for {
        addrs, err := resolver.LookupHost(ctx, host)
        if err != nil {
            slog.Warn("dns lookup", "err", err)
        } else {
            // attach port, update pool
            p.Update(withPort(addrs, "8080"))
        }
        select {
        case <-ctx.Done():
            return
        case <-t.C:
        }
    }
}
```

Respect DNS TTL when possible; aggressive refresh can stampede resolvers.

Passive Health and Outlier Detection

Active probes cost load; passive detection uses real traffic:

```
connect error or 5xx streak >= N  -> mark bad for cooldown
after cooldown, allow probe request (half-open style)
```

Combine with circuit breakers (next chapter) so discovery and breakers do not fight each other — define a single "eligible endpoints" set.

Consistency and Split-Brain

Registries can lag:

- Client A sees endpoints {1,2}
- Client B sees {2,3}

Design for **eventual consistency**: idempotent handlers, no single-instance assumptions without leases/locks.

Leader election (if needed) is a separate, harder problem — prefer cloud primitives or etcd/Consul sessions over home-grown locks unless you must.

Security Notes

- Authenticate registration if the registry is multi-tenant.
- Prefer mTLS to discovered peers (part 22).
- Do not put secrets in registry metadata.
- Validate advertised addresses (SSRF: instance claims `169.254.169.254`).

Observability

Metrics:

- `discovery_endpoints{service}` gauge
- `discovery_refresh_errors_total`
- `healthcheck_fail_total{type=live|ready}`
- `endpoint_selected_total{addr}` (careful cardinality — hash or sample)

Logs: registration, deregistration, ready transitions.

Production Checklist

☐ Distinct live vs ready endpoints
☐ Ready checks dependencies with short timeouts
☐ Live checks do not restart on dep failure
☐ TTL or platform removal on crash
☐ Graceful deregister + ready=false before shutdown
☐ Client pool handles empty set (fail fast)
☐ Outlier / passive health for sticky failures
☐ Metrics for endpoint count and probe failures
☐ Document discovery mode (DNS/K8s/custom) in runbooks

Common Pitfalls

1. **Only liveness** — traffic during warmup/migrations fails.
2. **Deep checks on liveness** — coordinated restarts under DB blip.
3. **No TTL** — zombie instances.
4. **Long DNS TTL** — slow failover (or too short — extra load).
5. **Ready depends on optional downstream** — optional features take the whole service out.
6. **Health handler allocates heavily** — becomes its own outage under probe load.
7. **Publishing localhost** in registry from misconfigured agents.

Exercises

1. Implement `/healthz` and `/readyz`; toggle ready with an atomic; prove LB behavior with a tiny proxy.
2. Add DB ping to ready with 100ms timeout; kill DB; confirm live still 200, ready 503.
3. Write a TTL registry in memory; stop heartbeats; observe expiry.
4. DNS pool: resolve a name with multiple A records; pull one instance; see traffic shift (use `/etc/hosts` hacks in lab).
5. Simulate slow ready handler (2s); set probe timeout 200ms; fix the handler.
6. On SIGTERM, set ready false, sleep 2s, then exit; describe how this helps connection draining.
7. Implement passive health: three failures mark endpoint down for 10s.
8. Attempt SSRF-style registration address; add allowlist validation.
9. Graph endpoint count during a rolling deploy.
10. Write a runbook: "endpoint count dropped to zero."

More examples

In-memory registry with health filter

```
mkdir -p /tmp/go-sd-reg && cd /tmp/go-sd-reg
go mod init example.com/sd-reg
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "sync"
)

type Endpoint struct {
    ID      string
    Addr    string
```

```go
    Healthy bool
}

type Registry struct {
    mu   sync.RWMutex
    byID map[string]Endpoint
}

func NewRegistry() *Registry {
    return &Registry{byID: map[string]Endpoint{}}
}

func (r *Registry) Upsert(e Endpoint) {
    r.mu.Lock()
    r.byID[e.ID] = e
    r.mu.Unlock()
}

func (r *Registry) Healthy() []Endpoint {
    r.mu.RLock()
    defer r.mu.RUnlock()
    var out []Endpoint
    for _, e := range r.byID {
        if e.Healthy {
            out = append(out, e)
        }
    }
    return out
}

func main() {
    reg := NewRegistry()
    reg.Upsert(Endpoint{ID: "a", Addr: "10.0.0.1:8080", Healthy: true})
    reg.Upsert(Endpoint{ID: "b", Addr: "10.0.0.2:8080", Healthy: false})
    reg.Upsert(Endpoint{ID: "c", Addr: "10.0.0.3:8080", Healthy: true})
    h := reg.Healthy()
    fmt.Println("healthy count:", len(h))
    reg.Upsert(Endpoint{ID: "b", Addr: "10.0.0.2:8080", Healthy: true})
    fmt.Println("after b recovers:", len(reg.Healthy()))
}
```

```go
go run .
```

Expected output:

```
healthy count: 2
after b recovers: 3
```

/healthz and /readyz with dependency flag

```
mkdir -p /tmp/go-health-ep && cd /tmp/go-health-ep
go mod init example.com/health-ep
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "net/http"
    "net/http/httptest"
    "sync/atomic"
)

func main() {
    var ready atomic.Bool
    mux := http.NewServeMux()
    mux.HandleFunc("GET /healthz", func(w http.ResponseWriter, _ *http.Request) {
        w.WriteHeader(200)
        fmt.Fprintln(w, "ok")
    })
    mux.HandleFunc("GET /readyz", func(w http.ResponseWriter, _ *http.Request) {
        if !ready.Load() {
            http.Error(w, "not ready", 503)
            return
        }
        fmt.Fprintln(w, "ready")
    })
    ts := httptest.NewServer(mux)
    defer ts.Close()

    r, _ := http.Get(ts.URL + "/healthz")
    r.Body.Close()
    fmt.Println("live:", r.StatusCode)
    r, _ = http.Get(ts.URL + "/readyz")
    r.Body.Close()
    fmt.Println("ready cold:", r.StatusCode)
    ready.Store(true)
    r, _ = http.Get(ts.URL + "/readyz")
    r.Body.Close()
    fmt.Println("ready warm:", r.StatusCode)
}
```

```
go run .
```

Expected output:

```
live: 200
ready cold: 503
ready warm: 200
```

Runnable example

In-memory registry with heartbeats/TTL plus `/healthz` vs `/readyz` on an `httptest` server—live during warmup, ready only when marked.

```bash
mkdir -p /tmp/go-discovery && cd /tmp/go-discovery
go mod init example.com/discovery
```

Save as `main.go`:

```go
package main

import (
    "fmt"
    "net/http"
    "net/http/httptest"
    "sync"
    "sync/atomic"
    "time"
)

type registry struct {
    mu   sync.Mutex
    ttl  map[string]time.Time
}

func newRegistry() *registry { return &registry{ttl: map[string]time.Time{}} }

func (r *registry) upsert(addr string, ttl time.Duration) {
    r.mu.Lock()
    r.ttl[addr] = time.Now().Add(ttl)
    r.mu.Unlock()
}

func (r *registry) list() []string {
    r.mu.Lock()
    defer r.mu.Unlock()
    now := time.Now()
    var out []string
    for a, exp := range r.ttl {
```

```
        if now.Before(exp) {
            out = append(out, a)
        } else {
            delete(r.ttl, a)
        }
    }
    return out
}

func main() {
    var ready atomic.Bool
    mux := http.NewServeMux()
    mux.HandleFunc("GET /healthz", func(w http.ResponseWriter, _ *http.Request) {
        w.WriteHeader(http.StatusOK)
        _, _ = w.Write([]byte("ok"))
    })
    mux.HandleFunc("GET /readyz", func(w http.ResponseWriter, _ *http.Request) {
        if !ready.Load() {
            http.Error(w, "not ready", http.StatusServiceUnavailable)
            return
        }
        w.WriteHeader(http.StatusOK)
        _, _ = w.Write([]byte("ready"))
    })
    ts := httptest.NewServer(mux)
    defer ts.Close()

    check := func(path string) int {
        resp, err := http.Get(ts.URL + path)
        if err != nil {
            return 0
        }
        defer resp.Body.Close()
        return resp.StatusCode
    }
    fmt.Println("live:", check("/healthz"), "ready:", check("/readyz"))
    ready.Store(true)
    fmt.Println("after warmup ready:", check("/readyz"))

    reg := newRegistry()
    reg.upsert(ts.URL, 40*time.Millisecond)
    reg.upsert("http://10.0.0.9:8080", time.Second)
    time.Sleep(50 * time.Millisecond)
    fmt.Println("registry after ttl:", reg.list())
}
```

```
go run .
```

Expected output:

```
live: 200 ready: 503
after warmup ready: 200
registry after ttl: [http://10.0.0.9:8080]
```

What to notice

- Liveness readiness; LB should wait on ready during migrations.
- TTL registration removes instances that stop heartbeating (zombies).
- Health handlers must stay cheap—slow probes become outages.

Try next

- On shutdown: set ready false, sleep 2s, then exit (connection drain).
- Passive health: three consecutive proxy errors mark an addr down for 10s.

Further Reading

- Kubernetes probes documentation
- Previous part: reverse proxy health-aware pools
- Next: **Retry and Circuit Breaker Patterns**

Retry and Circuit Breaker Patterns

Retry and Circuit Breaker Patterns

Retries and circuit breakers are **control systems**. Misconfigured control loops create amplification and cascades. Well-tuned ones improve availability during partial outages.

Why / Overview

Dependencies fail in bursts. Naive reactions:

- Retry immediately forever → melt the dependency
- Never retry → convert blips into user errors
- Cache open circuits without half-open probes → permanent brownout

```
call fails -> classify -> retryable?
               |             |
              no            yes -> backoff + jitter -> budget left? -> retry
                                                          |
                                                         no -> fail
```

Circuit breaker adds a fast-path reject when failure rate is high:

```
closed --(too many failures)--> open --(cooldown)--> half-open --(success)--> closed
                                  ^                     |
                                  +-----(fail)----------+
```

Error Classification

```go
type Class int

const (
    ClassSuccess Class = iota
    ClassRetryable
    ClassFatal
    ClassCaller // 4xx-ish; do not retry
)

func classify(err error, status int) Class {
    if err == nil && status >= 200 && status < 400 {
        return ClassSuccess
```

```
    }
    if errors.Is(err, context.Canceled) {
        return ClassFatal
    }
    if errors.Is(err, context.DeadlineExceeded) {
        return ClassRetryable
    }
    if status == 429 || status == 408 || status >= 500 {
        return ClassRetryable
    }
    if status >= 400 && status < 500 {
        return ClassCaller
    }
    if err != nil {
        return ClassRetryable // network blip heuristic
    }
    return ClassFatal
}
```

Idempotency is orthogonal: a retryable transport failure on a non-idempotent POST is still dangerous.

Retry with Exponential Backoff and Jitter

```
func retry[T any](ctx context.Context, maxAttempts int, base time.Duration, fn func(context.
    var zero T
    var last error
    backoff := base
    for attempt := 1; attempt <= maxAttempts; attempt++ {
        v, err := fn(ctx)
        if err == nil {
            return v, nil
        }
        last = err
        if classify(err, 0) != ClassRetryable || attempt == maxAttempts {
            return zero, last
        }
        // Full jitter: sleep random [0, backoff]
        delay := time.Duration(rand.Int63n(int64(backoff)))
        select {
        case <-ctx.Done():
            return zero, ctx.Err()
        case <-time.After(delay):
        }
        backoff *= 2
```

```
        if backoff > 5*time.Second {
            backoff = 5 * time.Second
        }
    }
    return zero, last
}
```

Policy parameters

Param	Guidance
maxAttempts	2–3 for user-facing; more only for background
base delay	20–100ms typical
max delay	Cap to protect tail latency
total budget	Parent context deadline is the true cap
methods	GET/HEAD/PUT safe; POST needs idempotency key

Decorrelated jitter (alternative)

Often better under correlated failure than pure exponential; see AWS architecture blog on backoff. Implement when you have many clients.

Idempotency Keys

```
req.Header.Set("Idempotency-Key", key) // client-generated UUID
// server stores key -> result for retention window
```

Without server support, client retries of POST are a product decision, not just a transport decision.

Circuit Breaker Implementation (Educational)

```
type state int

const (
    stateClosed state = iota
    stateOpen
    stateHalfOpen
)

type Breaker struct {
```

```
    mu           sync.Mutex
    st           state
    failures     int
    successes    int
    openUntil    time.Time
    failThreshold int
    successThreshold int
    cooldown     time.Duration
}

func (b *Breaker) Allow() bool {
    b.mu.Lock()
    defer b.mu.Unlock()
    now := time.Now()
    switch b.st {
    case stateOpen:
        if now.After(b.openUntil) {
            b.st = stateHalfOpen
            b.successes = 0
            return true // probe
        }
        return false
    default:
        return true
    }
}

func (b *Breaker) Report(err error) {
    b.mu.Lock()
    defer b.mu.Unlock()
    if err == nil {
        b.failures = 0
        if b.st == stateHalfOpen {
            b.successes++
            if b.successes >= b.successThreshold {
                b.st = stateClosed
            }
        }
        return
    }
    b.failures++
    if b.st == stateHalfOpen || b.failures >= b.failThreshold {
        b.st = stateOpen
        b.openUntil = time.Now().Add(b.cooldown)
        b.failures = 0
    }
}
```

Usage:

```
if !breaker.Allow() {
    return ErrFailFast
}
v, err := call(ctx)
breaker.Report(err)
```

Production libraries (`sony/gobreaker`, resilience middleware) add sliding windows and metrics — use them when possible; understand the state machine first.

Combining Retry and Breaker

Order matters:

```
if !breaker.Allow() -> fail fast (no retry storm)
else attempt with retry **inside** half-open carefully (often single probe)
```

Do **not** let every request retry while open; the breaker already decided.

Budgets and Hedging

Hedging (send a second request if the first is slow) can reduce tail latency but doubles load — only for idempotent, cheap reads, with strict limits.

Prefer fixing the slow dependency and using proper deadlines.

Multi-Layer Retry Coordination

Layer	Should retry?
Browser	Limited
Edge proxy	GET only, 1×
Service mesh	Align with app
App HTTP client	Primary policy owner
DB driver	Often internal; know it

Document **one primary** retry layer for each hop to avoid multiplicative attempts.

Observability

Metrics:

- `client_calls_total{dep,result}`
- `client_retries_total{dep}`
- `client_retry_exhaust_total{dep}`
- `breaker_state{dep}` (0/1/2)
- `breaker_open_total{dep}`
- latency histograms **including** and **excluding** retries if possible

Logs: attempt number, delay, final classification, correlation id.

Production Checklist

- ☐ Classify errors; do not retry fatal/caller errors
- ☐ Cap attempts and honor parent context
- ☐ Jittered backoff with max delay
- ☐ Idempotency story for mutating retries
- ☐ Breaker on high-traffic dependencies
- ☐ Half-open probe limits
- ☐ Metrics for retries and breaker state
- ☐ Coordinated policy with mesh/proxy
- ☐ Load tests with injected 50% failure

Common Pitfalls

1. **Retry on everything** including 400 and cancel.
2. **No jitter** — synchronized herds.
3. **Infinite retries** in background workers without poison handling.
4. **Breaker thresholds too sensitive** — flapping.
5. **Breaker without half-open** — never recovers automatically.
6. **Hiding all errors as retryable network errors**.
7. **Retry after context deadline already passed**.

Exercises

1. Implement `retry` with full jitter; unit-test attempt count and context cancel mid-backoff.
2. Simulate 50% failure; compare success rate with 1 vs 3 attempts; measure dependency QPS.
3. Add a breaker; drive failure rate above threshold; prove fail-fast latency is low.
4. Show flapping: threshold 1 failure — fix with windowed counts.
5. Implement idempotent POST with an in-memory key store; retry safely.
6. Trace a bug where proxy and client both retry twice (4 attempts); write a policy doc fixing it.
7. Export breaker state as a Prometheus gauge; alert when open > 2 minutes.

8. Compare exponential vs decorrelated jitter under synchronized clients (simple simulation).
9. Ensure `context.Canceled` never increments retry metrics as dependency failure.
10. Chaos: kill dependency for 30s; plot error rate with/without breaker.

More examples

Classify errors before retrying

```
mkdir -p /tmp/go-retry-class && cd /tmp/go-retry-class
go mod init example.com/retry-class
```

Save as `main.go`:

```
package main

import (
    "context"
    "errors"
    "fmt"
)

type class int

const (
    ok class = iota
    retryable
    fatal
    caller
)

func classify(err error, status int) class {
    if err == nil && status >= 200 && status < 400 {
        return ok
    }
    if errors.Is(err, context.Canceled) {
        return fatal
    }
    if status == 429 || status >= 500 {
        return retryable
    }
    if status >= 400 {
        return caller
    }
    if err != nil {
        return retryable
```

```go
    }
    return fatal
}

func main() {
    cases := []struct {
        err    error
        status int
    }{
        {nil, 200},
        {nil, 503},
        {nil, 400},
        {context.Canceled, 0},
        {fmt.Errorf("net"), 0},
    }
    for _, c := range cases {
        fmt.Printf("status=%d err=%v -> %v\n", c.status, c.err, classify(c.err, c.status))
    }
}
```

```bash
go run .
```

Expected output:

```
status=200 err=<nil> -> 0
status=503 err=<nil> -> 1
status=400 err=<nil> -> 3
status=0 err=context canceled -> 2
status=0 err=net -> 1
```

Budget-aware retry (honor parent deadline)

```bash
mkdir -p /tmp/go-retry-budget && cd /tmp/go-retry-budget
go mod init example.com/retry-budget
```

Save as `main.go`:

```go
package main

import (
    "context"
    "fmt"
    "time"
)
```

```go
func retry(ctx context.Context, attempts int, fn func() error) (int, error) {
    var err error
    for i := 1; i <= attempts; i++ {
        if err = ctx.Err(); err != nil {
            return i - 1, err
        }
        err = fn()
        if err == nil {
            return i, nil
        }
        select {
        case <-ctx.Done():
            return i, ctx.Err()
        case <-time.After(15 * time.Millisecond):
        }
    }
    return attempts, err
}

func main() {
    ctx, cancel := context.WithTimeout(context.Background(), 25*time.Millisecond)
    defer cancel()
    n, err := retry(ctx, 10, func() error { return fmt.Errorf("down") })
    fmt.Println("stopped attempts:", n, "err:", err != nil)
}
```

```bash
go run .
```

Expected output:

```
stopped attempts: 2 err: true
```

(Exact attempt count may be 1–3 depending on scheduling; it must be **less than 10**.)

Runnable example

Retries with full jitter plus a circuit breaker (closed → open → half-open) against a flaky in-process dependency.

```bash
mkdir -p /tmp/go-breaker && cd /tmp/go-breaker
go mod init example.com/breaker
```

Save as `main.go`:

```go
package main

import (
    "context"
    "fmt"
    "math/rand"
    "sync"
    "time"
)

func retry(ctx context.Context, attempts int, fn func() error) error {
    var err error
    for i := 0; i < attempts; i++ {
        if err = ctx.Err(); err != nil {
            return err
        }
        err = fn()
        if err == nil {
            return nil
        }
        if i+1 == attempts {
            break
        }
        // full jitter backoff
        d := time.Duration(rand.Intn(int(20*time.Millisecond)<<i)) + time.Millisecond
        t := time.NewTimer(d)
        select {
        case <-ctx.Done():
            t.Stop()
            return ctx.Err()
        case <-t.C:
        }
    }
    return err
}

type CircuitBreaker struct {
    mu          sync.Mutex
    failures    int
    threshold   int
    openUntil   time.Time
    halfOpen    bool
    coolDown    time.Duration
}

func NewCB(threshold int, coolDown time.Duration) *CircuitBreaker {
    return &CircuitBreaker{threshold: threshold, coolDown: coolDown}
```

```
}

func (c *CircuitBreaker) Do(fn func() error) error {
    c.mu.Lock()
    now := time.Now()
    if now.Before(c.openUntil) {
        c.mu.Unlock()
        return fmt.Errorf("breaker open")
    }
    if !c.openUntil.IsZero() && now.After(c.openUntil) {
        c.halfOpen = true
    }
    c.mu.Unlock()

    err := fn()

    c.mu.Lock()
    defer c.mu.Unlock()
    if err != nil {
        c.failures++
        if c.halfOpen || c.failures >= c.threshold {
            c.openUntil = time.Now().Add(c.coolDown)
            c.halfOpen = false
            c.failures = 0
        }
        return err
    }
    c.failures = 0
    c.halfOpen = false
    c.openUntil = time.Time{}
    return nil
}

func main() {
    var n int
    flaky := func() error {
        n++
        if n < 3 {
            return fmt.Errorf("transient")
        }
        return nil
    }
    err := retry(context.Background(), 5, flaky)
    fmt.Println("retry ok:", err, "calls:", n)

    cb := NewCB(3, 50*time.Millisecond)
    fail := func() error { return fmt.Errorf("down") }
```

```go
    for i := 0; i < 5; i++ {
        err := cb.Do(fail)
        fmt.Printf("cb %d: %v\n", i+1, err)
    }
    time.Sleep(60 * time.Millisecond)
    // half-open success resets
    err = cb.Do(func() error { return nil })
    fmt.Println("after cooldown success:", err)
    err = cb.Do(fail)
    fmt.Println("still closed after success, one fail:", err)
}
```

```go
go run .
```

Expected output (illustrative):

```
retry ok: <nil> calls: 3
cb 1: down
cb 2: down
cb 3: down
cb 4: breaker open
cb 5: breaker open
after cooldown success: <nil>
still closed after success, one fail: down
```

What to notice

- Breakers fail fast when open—latency collapses vs waiting on a dead dependency.
- Half-open allows a single probe after cooldown so the system can recover automatically.
- Retries without jitter synchronize clients and create herds.

Try next

- Do not retry `context.Canceled` or HTTP 400.
- Export breaker state as a gauge (`0/1`) in the metrics chapter style.

Further Reading

- Release It! (circuit breaker popularization)
- AWS architecture blog: exponential backoff and jitter
- Previous: discovery/health as inputs to call eligibility
- Next: **Queue Workers and Backpressure**

Queue Workers and Backpressure

Queue Workers and Backpressure

Queue systems decouple producer and consumer rates — **only if** capacity and failure behavior are explicit. Otherwise queues become infinite latency sinks or memory bombs.

Why / Overview

Synchronous request paths force producers to wait. Queues enable:

- Spike absorption
- Asynchronous workflows
- Retries isolated from user latency
- Fan-out to multiple worker types

```
producer -> queue (buffer) -> workers -> dependency
                |
                +-> backpressure when full / depth high
```

In Go you will use:

- **In-process**: channels + worker pools
- **Durable**: SQS, Pub/Sub, NATS JetStream, Kafka, RabbitMQ, Redis streams

The control theory is the same; durability and delivery semantics differ.

Delivery Semantics

Semantic	Meaning	App duty
At most once	May drop	Accept loss
At least once	May duplicate	Idempotent handlers
Exactly once	Rare/expensive	Often "effectively once" via idempotency

Assume **at-least-once** unless you have a strong proof otherwise. Design handlers to be safe under duplicate delivery.

```go
// Idempotent handler sketch
func handle(ctx context.Context, job Job) error {
```

```
    ok, err := store.MarkIfNew(ctx, job.ID)
    if err != nil {
        return err
    }
    if !ok {
        return nil // duplicate
    }
    return doWork(ctx, job)
}
```

In-Process Worker Pool

```
type Pool struct {
    jobs chan Job
    wg   sync.WaitGroup
}

func NewPool(ctx context.Context, workers int, h func(context.Context, Job) error) *Pool {
    p := &Pool{jobs: make(chan Job, 1024)}
    for i := 0; i < workers; i++ {
        p.wg.Add(1)
        go func() {
            defer p.wg.Done()
            for {
                select {
                case <-ctx.Done():
                    return
                case job, ok := <-p.jobs:
                    if !ok {
                        return
                    }
                    if err := h(ctx, job); err != nil {
                        slog.Error("job failed", "id", job.ID, "err", err)
                    }
                }
            }
        }()
    }
    return p
}

func (p *Pool) Submit(ctx context.Context, job Job) error {
    select {
    case p.jobs <- job:
        return nil
```

```go
    case <-ctx.Done():
        return ctx.Err()
    default:
        return ErrQueueFull // backpressure to caller
    }
}

func (p *Pool) Close() {
    close(p.jobs)
    p.wg.Wait()
}
```

Blocking vs dropping vs rejecting

`Submit` policy	Behavior
Block on full channel	Applies backpressure; can stall HTTP
Non-blocking + 503	Protects memory; client retries
Drop oldest/newest	Lossy; only for metrics/logs

Choose consciously. Silent drops for payment events are unacceptable.

Sizing Levers

```
throughput  workers × (1 / mean_job_latency)
```

But dependency capacity is the real ceiling. Oversubscribing workers increases contention and error rates.

Lever	Too small	Too large
Queue depth	Reject under mild spikes	Huge latency / RAM
Workers	Low throughput	Dep overload, lock contention
Visibility timeout (external)	Duplicate processing	Stuck messages if worker dies
Max receive count	Early DLQ	Infinite poison retry

External Queue Worker Loop (Shape)

```go
func worker(ctx context.Context, q Queue, h func(context.Context, Msg) error) error {
    for {
        msgs, err := q.Receive(ctx, 10, 20*time.Second) // batch, wait
        if err != nil {
            if ctx.Err() != nil {
                return ctx.Err()
            }
            slog.Error("receive", "err", err)
            time.Sleep(time.Second)
            continue
        }
        for _, m := range msgs {
            msgCtx, cancel := context.WithTimeout(ctx, 30*time.Second)
            err := h(msgCtx, m)
            cancel()
            if err != nil {
                // leave for retry / nack depending on system
                slog.Error("handle", "id", m.ID, "err", err)
                continue
            }
            if err := q.Ack(ctx, m); err != nil {
                slog.Error("ack", "err", err)
            }
        }
    }
}
```

Ack timing

- **Ack after success** — at-least-once if crash before ack
- **Ack before work** — at-most-once loss on crash
- **Extend visibility** for long jobs

Poison Messages and DLQ

Not all failures are transient.

```
transient:  timeout, 503, deadlock -> retry with backoff
poison:     invalid payload, permanent 400 -> DLQ after N attempts
```

```go
func handle(ctx context.Context, m Msg) error {
    var job Job
    if err := json.Unmarshal(m.Body, &job); err != nil {
        return errPoison // route to DLQ, do not infinite retry
    }
    return process(ctx, job)
}
```

Alert on DLQ depth; empty it with a runbook, not silence.

Backpressure Across Tiers

```
HTTP handler -> Submit(job)
   if ErrQueueFull -> 503 Retry-After
consumer lag high -> scale workers / shed non-critical producers
dependency circuit open -> slow receive rate or park workers
```

True end-to-end backpressure requires producers to **observe** consumer health (queue depth metrics, not only local channel fullness).

Graceful Shutdown

```
cancel()           // stop receiving new jobs
close(jobs)        // or stop Submit
wait with timeout  // finish in-flight
// for external queues: stop Receive; in-flight still acked or returned by visibility expiry
```

Never exit immediately while holding unacked messages if you can avoid it — you will cause stampedes of redelivery.

Observability

Metrics:

- `queue_depth` / channel length if available
- `jobs_enqueued_total{result=ok|full}`
- `jobs_processed_total{result}`
- `job_duration_seconds`
- `jobs_in_flight`
- `dlq_depth`
- `receive_errors_total`

Lag: time between enqueue timestamp and process start (requires timestamp in payload).

Production Checklist

- ☐ Document delivery semantics and idempotency
- ☐ Bounded queue / explicit full behavior
- ☐ Worker count tied to dependency capacity
- ☐ Timeouts on job handlers
- ☐ Poison path + DLQ + alerts
- ☐ Ack/visibility policy documented
- ☐ Graceful drain on shutdown
- ☐ Metrics for depth, lag, failures, DLQ
- ☐ Load test: spike producers, kill workers, duplicate delivery

Common Pitfalls

1. **Unbounded `make(chan T)`** or growing slices as queues — OOM.
2. **More workers = faster** without measuring the dependency.
3. **Non-idempotent handlers** with at-least-once queues.
4. **Infinite retries** of bad payloads.
5. **Blocking HTTP on full in-process queue** without timeout — cascading latency.
6. **Ack before durable side effect completes**.
7. **No lag metric** — discover the backlog from customer tickets.

Exercises

1. Implement a bounded worker pool; prove `Submit` returns `ErrQueueFull` under overload.
2. Change policy to block with context timeout; compare HTTP p99 under spike.
3. Inject duplicate deliveries; show double side effects; add idempotency store.
4. Generate invalid JSON jobs; after 3 failures move to DLQ slice; alert when DLQ > 0.
5. Measure throughput vs worker count for a 50ms simulated dependency; find the plateau.
6. Shutdown mid-job; confirm message redelivery with external queue or simulated visibility.
7. Add enqueue timestamp; export lag histogram.
8. Combine with a circuit breaker: when open, stop pulling messages for that dependency.
9. Use `errgroup` with limit for per-batch parallel handle; ensure ctx cancel stops the batch.
10. Write a capacity plan: peak produce rate, target lag, mean job time → workers and depth.

More examples

Bounded queue + worker pool

```
mkdir -p /tmp/go-queue-bp && cd /tmp/go-queue-bp
go mod init example.com/queue-bp
```

Save as `main.go`:

```go
package main

import (
    "fmt"
    "sync"
    "time"
)

func main() {
    jobs := make(chan int, 4) // backpressure depth
    var wg sync.WaitGroup
    var done int
    var mu sync.Mutex

    for w := 0; w < 2; w++ {
        wg.Add(1)
        go func() {
            defer wg.Done()
            for j := range jobs {
                time.Sleep(5 * time.Millisecond)
                mu.Lock()
                done++
                _ = j
                mu.Unlock()
            }
        }()
    }

    for i := 1; i <= 10; i++ {
        jobs <- i // blocks when buffer full → producer slows
    }
    close(jobs)
    wg.Wait()
    fmt.Println("processed:", done)
}
```

```
go run .
```

Expected output:

```
processed: 10
```

Drop vs block when full (load shed)

```
mkdir -p /tmp/go-queue-shed && cd /tmp/go-queue-shed
go mod init example.com/queue-shed
```

Save as `main.go`:

```go
package main

import "fmt"

func tryEnqueue(ch chan int, v int) bool {
    select {
    case ch <- v:
        return true
    default:
        return false
    }
}

func main() {
    ch := make(chan int, 2)
    accepted, dropped := 0, 0
    for i := 0; i < 5; i++ {
        if tryEnqueue(ch, i) {
            accepted++
        } else {
            dropped++
        }
    }
    fmt.Println("accepted:", accepted, "dropped:", dropped, "len:", len(ch))
}
```

```
go run .
```

Expected output:

```
accepted: 2 dropped: 3 len: 2
```

Runnable example

Bounded in-memory queue, worker pool, `ErrQueueFull` backpressure, and a tiny dead-letter list for poison messages.

```
mkdir -p /tmp/go-queue && cd /tmp/go-queue
go mod init example.com/queue
```

Save as `main.go`:

```
package main

import (
    "context"
    "errors"
    "fmt"
    "sync"
    "time"
)

var ErrQueueFull = errors.New("queue full")

type Job struct {
    ID      string
    Payload string
    Attempt int
}

type Pool struct {
    jobs   chan Job
    dlq    []Job
    mu     sync.Mutex
    wg     sync.WaitGroup
    handle func(context.Context, Job) error
}

func NewPool(depth, workers int, handle func(context.Context, Job) error) *Pool {
    p := &Pool{jobs: make(chan Job, depth), handle: handle}
    for i := 0; i < workers; i++ {
        p.wg.Add(1)
        go func() {
            defer p.wg.Done()
            for j := range p.jobs {
                ctx, cancel := context.WithTimeout(context.Background(), 100*time.Millisecon
                err := p.handle(ctx, j)
                cancel()
                if err != nil {
                    j.Attempt++
                    if j.Attempt >= 3 {
                        p.mu.Lock()
                        p.dlq = append(p.dlq, j)
                        p.mu.Unlock()
```

```
                        continue
                    }
                    // re-queue best-effort
                    select {
                    case p.jobs <- j:
                    default:
                        p.mu.Lock()
                        p.dlq = append(p.dlq, j)
                        p.mu.Unlock()
                    }
                }
            }
        }()
    }
    return p
}

func (p *Pool) Submit(j Job) error {
    select {
    case p.jobs <- j:
        return nil
    default:
        return ErrQueueFull
    }
}

func (p *Pool) Close() {
    close(p.jobs)
    p.wg.Wait()
}

func main() {
    var okCount int
    var mu sync.Mutex
    p := NewPool(2, 2, func(ctx context.Context, j Job) error {
        if j.Payload == "poison" {
            return fmt.Errorf("bad payload")
        }
        mu.Lock()
        okCount++
        mu.Unlock()
        return nil
    })

    fmt.Println("submit a:", p.Submit(Job{ID: "1", Payload: "ok"}))
    fmt.Println("submit b:", p.Submit(Job{ID: "2", Payload: "ok"}))
    fmt.Println("submit c:", p.Submit(Job{ID: "3", Payload: "ok"})) // may fill
```

```go
    // drain a bit
    time.Sleep(50 * time.Millisecond)
    _ = p.Submit(Job{ID: "4", Payload: "poison"})
    time.Sleep(50 * time.Millisecond)
    for i := 0; i < 5; i++ {
        if err := p.Submit(Job{ID: fmt.Sprintf("x%d", i), Payload: "ok"}); err != nil {
            fmt.Println("backpressure:", err)
            break
        }
    }
    p.Close()
    p.mu.Lock()
    fmt.Println("ok_count:", okCount, "dlq:", len(p.dlq))
    p.mu.Unlock()
}
```

```go
go run .
```

Expected output (timing-sensitive; illustrative):

```
submit a: <nil>
submit b: <nil>
submit c: queue full
backpressure: queue full
ok_count: ... dlq: >=1
```

What to notice

- Bounded channels are the simplest backpressure tool; unbounded queues OOM under spike.
- Poison messages need attempt caps and a DLQ—infinite retry burns workers forever.
- Closing the job channel and `Wait` is orderly worker shutdown.

Try next

- Block with `select` + context timeout instead of immediate `ErrQueueFull`.
- Export queue depth gauge and lag from enqueue timestamp.

Further Reading

- Queueing theory basics (Littles's law: L = W) for lag intuition
- Previous: retries/breakers on the dependency side of workers
- Next part: **Observability and SRE** — measuring all of the above

Caching Patterns

Caching Patterns

Overview

Caches cut latency and load—and create consistency bugs. Start with **explicit TTLs**, single-flight fills, and clear invalidation.

In-process TTL map

```
type entry struct {
    val any
    exp time.Time
}

type Cache struct {
    mu sync.Mutex
    m  map[string]entry
}

func (c *Cache) Get(k string) (any, bool) {
    c.mu.Lock()
    defer c.mu.Unlock()
    e, ok := c.m[k]
    if !ok || time.Now().After(e.exp) {
        delete(c.m, k)
        return nil, false
    }
    return e.val, true
}

func (c *Cache) Set(k string, v any, ttl time.Duration) {
    c.mu.Lock()
    c.m[k] = entry{val: v, exp: time.Now().Add(ttl)}
    c.mu.Unlock()
}
```

Single-flight (stampede control)

```
go get golang.org/x/sync/singleflight@latest
```

```go
var g singleflight.Group

v, err, _ := g.Do(key, func() (any, error) {
    return loadFromDB(ctx, key)
})
```

Cache-aside

```
get cache → miss → load DB → set cache → return
```

Write path: update DB then **delete** cache key (or short TTL).

Negative caching

Cache "not found" briefly to protect DB from repeated misses—short TTL.

Rules of thumb

Do	Don't
TTL + max entries	Unbounded map growth
Invalidate on write	Assume cache == truth forever
singleflight on hot keys	Thundering herd on expiry

Try next

1. Wrap Book repository with TTL cache.
2. Benchmark stampede with/without singleflight.
3. Add max keys with simple LRU eviction sketch.

Idempotency and Deduplication

Idempotency and Deduplication

Overview

Networks **retry**. Clients double-submit. Without idempotency, you charge twice or create duplicate rows. Key idea: same logical request → same effect.

Idempotency-Key header

```
POST /payments
Idempotency-Key: 9f3c...
```

Server stores key → response (or in-flight marker) with TTL.

```go
type store interface {
    Get(ctx context.Context, key string) (status int, body []byte, ok bool, err error)
    Put(ctx context.Context, key string, status int, body []byte, ttl time.Duration) error
}

func withIdempotency(s store, next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        if r.Method != http.MethodPost {
            next.ServeHTTP(w, r)
            return
        }
        key := r.Header.Get("Idempotency-Key")
        if key == "" {
            http.Error(w, "missing Idempotency-Key", http.StatusBadRequest)
            return
        }
        if st, body, ok, err := s.Get(r.Context(), key); err == nil && ok {
            w.WriteHeader(st)
            _, _ = w.Write(body)
            return
        }
        // capture response → Put (use response recorder middleware)
        next.ServeHTTP(w, r)
    })
}
```

DB unique constraints

Natural idempotency: `UNIQUE(idempotency_key)` or `UNIQUE(user_id, client_request_id)`.

```
INSERT ... ON CONFLICT (idempotency_key) DO NOTHING
```

At-least-once consumers

Queue workers: store processed `message_id`; skip duplicates.

Rules of thumb

Do	Don't
Require keys on money POSTs	Retry POSTs blindly without keys
TTL the key store	Infinite growth of keys
Prefer DB uniqueness	Only in-memory dedupe multi-instance

Try next

1. Memory idempotency store + double POST test.
2. Unique index migration for order client IDs.
3. Worker skips duplicate delivery.

Leader Election Basics

Leader Election Basics

Overview

Sometimes **exactly one** instance should run a loop (compactions, cron, singleton consumer). Leader election picks that instance. Keep it simple: lease in a database/etcd/K8s lock—not custom Paxos in app code.

Lease pattern (DB)

```
UPDATE leases SET owner=?, expires=?
 WHERE name=? AND (expires < now() OR owner=?)
```

Winner renews before expiry; losers idle and retry.

```
type LeaseStore interface {
    TryAcquire(ctx context.Context, name, owner string, ttl time.Duration) (bool, error)
    Renew(ctx context.Context, name, owner string, ttl time.Duration) (bool, error)
    Release(ctx context.Context, name, owner string) error
}

func RunAsLeader(ctx context.Context, s LeaseStore, name, owner string, ttl time.Duration, w
    t := time.NewTicker(ttl / 2)
    defer t.Stop()
    for {
        ok, err := s.TryAcquire(ctx, name, owner, ttl)
        if err != nil {
            return err
        }
        if !ok {
            select {
            case <-ctx.Done():
                return ctx.Err()
            case <-time.After(ttl / 2):
                continue
            }
        }
        // lead
        wctx, cancel := context.WithCancel(ctx)
```

```go
        errCh := make(chan error, 1)
        go func() { errCh <- work(wctx) }()
        for {
            select {
            case err := <-errCh:
                cancel()
                _ = s.Release(ctx, name, owner)
                return err
            case <-t.C:
                ok, err := s.Renew(ctx, name, owner, ttl)
                if err != nil || !ok {
                    cancel()
                    <-errCh
                    break
                }
            case <-ctx.Done():
                cancel()
                _ = s.Release(context.Background(), name, owner)
                return ctx.Err()
            }
        }
    }
}
```

Kubernetes

`Lease` objects / leader-election libraries (`client-go` tools) — prefer platform primitives in K8s.

Rules

Do	Don't
TTL + renew	Infinite leadership without heartbeat
Work cancel on loss	Continue writes after lease loss
Idempotent leader work	Assume exactly-once without more design

Try next

1. Memory lease store + two goroutines racing.
2. Kill leader; follower takes over within TTL.
3. Ensure work stops on lease loss.

Transactional Outbox Pattern

Transactional Outbox Pattern

Overview

If you `UPDATE` the DB then `Publish` to a queue, crashes create lost or double events. The **outbox** writes the event in the **same transaction** as business data; a relay publishes asynchronously.

```
begin
  update orders
  insert outbox(event)
commit
        → relay reads outbox → publish → mark sent
```

Schema sketch

```
CREATE TABLE outbox (
  id           BIGSERIAL PRIMARY KEY,
  topic        TEXT NOT NULL,
  payload      JSONB NOT NULL,
  created_at   TIMESTAMPTZ NOT NULL DEFAULT now(),
  published_at TIMESTAMPTZ
);
```

Write path

```
tx, _ := db.BeginTx(ctx, nil)
_, err = tx.ExecContext(ctx, `UPDATE orders SET status=$1 WHERE id=$2`, "paid", id)
_, err = tx.ExecContext(ctx, `INSERT INTO outbox(topic,payload) VALUES($1,$2)`, "order.paid"
err = tx.Commit()
```

Relay worker

```
rows, _ := db.QueryContext(ctx, `
  SELECT id, topic, payload FROM outbox
  WHERE published_at IS NULL
  ORDER BY id LIMIT 100
  FOR UPDATE SKIP LOCKED`)
// for each: publish; UPDATE outbox SET published_at=now() WHERE id=?
```

`SKIP LOCKED` allows multiple relays safely.

At-least-once

Consumers still need **idempotency** (chapter 155)—outbox gives reliable *publish attempt*, not exactly-once end-to-end.

Rules

Do	Don't
Same TX as state change	Publish then commit (or reverse) without outbox
Idempotent consumers	Assume no duplicates
Metric lag of unpublished rows	Silent outbox growth

Try next

1. Simulate crash after commit before publish; relay catches up.
2. Two relays with SKIP LOCKED.
3. Dashboard: count unpublished > 5m.

Project: Resilient Worker

Project: Resilient Worker

Overview

Build a **worker** that pulls jobs from an in-memory (then Redis/SQS) queue with: retries, backoff, poison handling, concurrency cap, and graceful shutdown.

Features

Feature	Spec
Concurrency	`-w 8` workers
Retry	max 5, exponential backoff
Poison	after max → dead-letter slice/log
Shutdown	finish current job or requeue
Metrics	processed, failed, dlq counters

Interface

```go
type Job struct {
    ID      string
    Payload []byte
    Attempt int
}

type Queue interface {
    Receive(ctx context.Context) (Job, error)
    Ack(ctx context.Context, id string) error
    Retry(ctx context.Context, j Job, after time.Duration) error
    DeadLetter(ctx context.Context, j Job, reason string) error
}
```

Loop

```go
for i := 0; i < workers; i++ {
    go func() {
        for {
            j, err := q.Receive(ctx)
            if err != nil { return }
            if err := handle(ctx, j); err != nil {
                if j.Attempt+1 >= max {
                    _ = q.DeadLetter(ctx, j, err.Error())
                } else {
                    j.Attempt++
                    _ = q.Retry(ctx, j, backoff(j.Attempt))
                }
                continue
            }
            _ = q.Ack(ctx, j.ID)
        }
    }()
}
```

Acceptance

- ☐ Inject fail-N-times job; succeeds after retries
- ☐ Poison goes to DLQ
- ☐ SIGTERM stops receiving; no new jobs
- ☐ Race test clean

Connects chapters 152–153 + 155.

Observability and SRE

Observability and SRE Overview

Observability and SRE Overview

This part explains how to make Go services **operable in real incidents**. Observability is a design property: if you cannot ask new questions of a running system, you do not have it — you have a fixed dashboard.

Why / Overview

Production failures are rarely "the server is off." They are:

- Elevated tail latency on one route
- Error budgets burning from one dependency
- A deploy that only breaks for a subset of tenants
- Retries hiding the true failure rate

You need three complementary signals:

```
logs    - discrete events with context (what happened)
metrics - aggregates over time (is it bad, how bad)
traces  - causal latency across services (where is the time)
```

SRE practice ties those signals to **SLIs/SLOs**, alerts, and runbooks — not to collecting data for its own sake.

Learning Path

Chapter	Focus	Outcome
161 Structured logging	slog, fields, correlation, redaction	Debuggable event streams
162 Metrics / Prometheus	RED/USE, labels, histograms	Alertable, cheap aggregates
163 Tracing	Spans, context propagation	Cross-service latency attribution
164 SLI/SLO/error budgets	Targets and burn	Reliability vs feature tradeoffs
165 Health probes	livez/readyz/drain	Correct orchestration signals
166 Alerting & runbooks	Actionable pages	Faster incident response
166b Flight recorder	`runtime/trace.FlightRecorder`, triggered snapshots	A trace of the seconds before an incident

Chapter	Focus	Outcome
167 Project: instrumented service	Wire logs+metrics+probes	End-to-end lab

The Observability Triangle

```
              metrics
             /       \
            /         \
        logs ------- traces
             correlation id / trace id
```

A single **correlation key** (request id / trace id) should appear in all three where possible.

SLI / SLO Primer

Term	Meaning
SLI	Quantitative measure (e.g. % of requests < 300ms)
SLO	Target on that measure (e.g. 99.9% over 30d)
Error budget	1 − SLO; spend on velocity or freeze

Instrument what you SLO. Vanity metrics that nobody pages on are optional.

Go Building Blocks

Signal	Libraries / stdlib
Logs	`log/slog` (Go 1.21+)
Metrics	Prometheus client, OpenTelemetry metrics
Traces	OpenTelemetry Go SDK
Propagation	`context.Context` + W3C headers

```go
slog.Info("checkout",
    "request_id", rid,
    "user_hash", hash(userID), // not raw PII if avoidable
    "total_ms", dur.Milliseconds(),
)
```

Production Principles

1. **Cardinality discipline** — unbounded label values kill metric systems.
2. **Structured fields > string paste** — enable search and aggregation.
3. **Propagate context** — lost trace context is a permanent blind spot.
4. **Sample thoughtfully** — full traces at 100% may be too expensive; errors often keep 100%.
5. **Alert on symptoms** (user pain) more than causes (CPU high).
6. **Redact secrets** — logs and spans are exfiltration channels.

Relationship to Other Parts

- Network and distributed parts **emit** the signals designed here.
- Performance engineering **uses** profiles (adjacent to metrics/traces).
- Security hardening **constrains** what you may log.

Literacy Checklist

☐ Emit JSON or structured logs with stable field names
☐ Carry request/trace ids across goroutines and HTTP
☐ Define RED metrics for a service
☐ Choose histogram buckets intentionally
☐ Avoid high-cardinality labels
☐ Create and propagate spans on outbound calls
☐ Tie an alert to an SLO and a runbook

Part Warm-Up Exercises

1. List the top five questions you ask during incidents; map each to log/metric/trace.
2. Find one high-cardinality metric risk in an existing codebase (`user_id` label?).
3. Grep for `log.Printf` and plan migration of hot paths to `slog`.
4. Check whether outbound HTTP clients forward trace headers.
5. Write a draft SLO for your highest-traffic endpoint.

Instrumentation Order of Operations

When adding observability to a greenfield service, do this in order:

1. **Structured access logs** with request id (chapter 161)
2. **RED metrics** for public routes (chapter 162)
3. **Dependency client metrics** (timeouts vs errors)
4. **Trace propagation** on the hottest path (chapter 163)
5. **SLOs + alerts + runbooks** for those SLIs

Skipping to distributed tracing without metrics leaves you with pretty waterfalls and no reliable pages. Skipping logs leaves metrics without narrative.

What "Good" Looks Like in an Incident

Within five minutes you should be able to:

```
1. See error rate and p99 on the affected route (metrics)
2. Grab a request_id / trace_id from a failing request (logs or edge)
3. Jump to a trace that shows the slow/failed span (traces)
4. Read surrounding log lines for that id (logs)
5. Open the runbook linked from the alert
```

If any hop is missing, fix instrumentation — not only the proximate bug.

Cardinality and Cost Budget

Observability has a cloud bill. Rough discipline:

Signal	Cost driver	Control
Logs	volume × retention	sample happy path; drop debug in prod
Metrics	series count	bounded labels; careful histograms
Traces	spans × rate	head/tail sampling

Review series count and log GB/day the same way you review CPU.

Anti-Patterns

1. Unstructured logs that require regex archaeology
2. Metrics labeled with user ids or raw URLs
3. Traces without parent context on outbound HTTP
4. Alerts with no runbook and no SLO
5. Logging secrets "just for staging" (staging leaks too)

Minimum Dashboard Set

For each user-facing service, keep a default dashboard with:

1. Request rate (global + top routes)
2. Error rate / fraction (5xx and timeout class)
3. Latency p50 / p95 / p99

4. In-flight requests / saturation
5. Dependency latency and error panels
6. Go runtime: goroutines, heap, GC CPU (secondary)

If a panel cannot drive an action, demote it. Dashboards are for decisions, not decoration.

Alert Quality Bar

Before shipping an alert, answer:

- What user pain does this detect?
- What is the false-positive risk?
- Who is paged and what do they run first?
- Can we silence safely during known maintenance?

Symptom-based alerts (SLO burn) beat cause-based alerts (CPU > 80%) for most services. Cause alerts are still useful for capacity planning, usually at ticket severity not page severity.

Library Choices (2024–2026 pragmatic)

Need	Common choice
Logs	`log/slog` (+ JSON handler)
Metrics	Prometheus client_golang
Traces	OpenTelemetry Go
Bridges	OTel → your backend vendor

Avoid inventing a fourth proprietary telemetry API inside your org without a migration plan.

Glossary

- **SLI / SLO / error budget** — measure, target, remaining failure allowance
- **Cardinality** — number of unique time series
- **Correlation id** — joins signals for one request
- **Exemplar** — metric sample pointing at a trace
- **Propagation** — carrying context across process boundaries
- **RED / USE** — request and resource metric recipes
- **Tail sampling** — keep interesting traces after the fact

More examples

Tour: slog + counter + request id

```
mkdir -p /tmp/go-o11y-tour && cd /tmp/go-o11y-tour
go mod init example.com/o11y-tour
```

Save as `main.go`:

```go
package main

import (
    "bytes"
    "fmt"
    "log/slog"
    "net/http"
    "net/http/httptest"
    "strings"
    "sync/atomic"
)

func main() {
    var hits atomic.Int64
    var buf bytes.Buffer
    log := slog.New(slog.NewJSONHandler(&buf, nil))

    h := http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        id := r.Header.Get("X-Request-ID")
        if id == "" {
            id = "gen-1"
        }
        hits.Add(1)
        log.Info("hit", "request_id", id, "path", r.URL.Path)
        w.Header().Set("X-Request-ID", id)
        fmt.Fprintln(w, "ok")
    })
    ts := httptest.NewServer(h)
    defer ts.Close()

    req, _ := http.NewRequest(http.MethodGet, ts.URL+"/x", nil)
    req.Header.Set("X-Request-ID", "abc")
    resp, _ := http.DefaultClient.Do(req)
    resp.Body.Close()
    fmt.Println("hits:", hits.Load(), "resp id:", resp.Header().Get("X-Request-ID"))
    fmt.Println("logged:", strings.Contains(buf.String(), "abc"))
}
```

```
go run .
```

Expected output:

```
hits: 1 resp id: abc
logged: true
```

Runnable example

Tour of the observability triangle: one JSON log line, one counter sample, one trace-id field on a context—stdlib only.

```
mkdir -p /tmp/go-o11y-tour && cd /tmp/go-o11y-tour
go mod init example.com/o11y-tour
```

Save as `main.go`:

```
package main

import (
    "bytes"
    "context"
    "fmt"
    "log/slog"
    "time"
)

type ctxKey string

const traceIDKey ctxKey = "trace_id"

func main() {
    var buf bytes.Buffer
    log := slog.New(slog.NewJSONHandler(&buf, nil))

    ctx := context.WithValue(context.Background(), traceIDKey, "tr-demo-1")
    start := time.Now()
    // fake work
    time.Sleep(5 * time.Millisecond)
    log.InfoContext(ctx, "request",
        "trace_id", ctx.Value(traceIDKey),
        "route", "GET /checkout",
        "status", 200,
        "dur_ms", time.Since(start).Milliseconds(),
    )
```

```go
    // metric sample (manual exposition line)
    var requestsTotal = 1
    fmt.Printf("# TYPE demo_requests_total counter\ndemo_requests_total %d\n", requestsTotal
    fmt.Println("--- log ---")
    fmt.Print(buf.String())
    fmt.Println("tour: logs + metrics + trace id field")
}
```

```go
go run .
```

Expected output (illustrative):

```
# TYPE demo_requests_total counter
demo_requests_total 1
--- log ---
{"time":"...","level":"INFO","msg":"request","trace_id":"tr-demo-1",...}
tour: logs + metrics + trace id field
```

What to notice

- The three signals answer different questions; this part teaches each without vendor lock-in first.
- Stable field names make cross-service grepping possible during incidents.

Try next

- Chapters 161–163: slog correlation, Prometheus-shaped metrics, W3C-ish traceparent propagation.

Next Chapter

Structured Logging and Correlation — the foundation every engineer greps first.

Structured Logging and Correlation

Structured Logging and Correlation

Structured logs are **event records**, not formatted print statements. In Go 1.21+, `log/slog` is the standard library foundation for production logging.

Why / Overview

`fmt.Println` and free-form `log.Printf` fail at scale:

- Cannot reliably filter by field
- Cannot join with traces
- Easy to leak secrets
- Inconsistent shapes across services

```
request_id / trace_id
   -> log fields on every line for that request
   -> (optional) metric exemplars
   -> trace span attributes
```

slog Fundamentals

```go
package main

import (
    "log/slog"
    "os"
)

func main() {
    h := slog.NewJSONHandler(os.Stdout, &slog.HandlerOptions{
        Level: slog.LevelInfo,
    })
    log := slog.New(h)
    slog.SetDefault(log)

    slog.Info("server start", "addr", ":8080", "version", version)
}
```

Levels: `Debug`, `Info`, `Warn`, `Error`. Prefer `Error` for handled failures that need attention; do not log+return the same error at every layer (pick a boundary).

Loggers with attributes

```
reqLog := slog.With(
    "service", "checkout",
    "env", os.Getenv("ENV"),
)
reqLog.Info("payment authorized", "amount_cents", 4200)
```

Correlation IDs

Generate or accept a request id at the edge; store in context; inject into logger.

```
type ctxKey struct{}

func WithRequestID(ctx context.Context, id string) context.Context {
    return context.WithValue(ctx, ctxKey{}, id)
}

func RequestID(ctx context.Context) string {
    if v, ok := ctx.Value(ctxKey{}).(string); ok {
        return v
    }
    return ""
}

func requestIDMiddleware(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        id := r.Header.Get("X-Request-Id")
        if id == "" {
            id = newID() // crypto/rand hex
        }
        ctx := WithRequestID(r.Context(), id)
        w.Header().Set("X-Request-Id", id)
        // Logger bound to request
        log := slog.Default().With("request_id", id)
        ctx = withLogger(ctx, log)
        next.ServeHTTP(w, r.WithContext(ctx))
    })
}
```

Propagate `X-Request-Id` / `traceparent` on outbound calls.

Context-aware logging helpers

```go
func Info(ctx context.Context, msg string, args ...any) {
    Logger(ctx).Info(msg, args...)
}
```

Avoid putting the entire `*slog.Logger` in context if your style guide forbids context values — alternatives: explicit logger parameters. Many teams accept request-scoped logger in context for HTTP services.

Stable Schema

Cross-service conventions beat clever one-offs:

Field	Meaning
`service`	Service name
`env`	prod/stage
`version`	build/version
`request_id`	Edge correlation
`trace_id`	Distributed trace
`span_id`	Current span
`method` / `path` / `route`	HTTP shape (`route` = template)
`status`	HTTP status or RPC code
`err`	Error string / type
`dur_ms`	Duration

Use **route templates** (`/users/{id}`) not raw URLs in aggregatable fields.

What to Log

Do log	Avoid
Request start/end (or access log)	Full bodies by default
Business state transitions	Secrets, tokens, passwords
Dependency failures with class	Unbounded free text dumps
Config version at start	PII without legal basis

Access log example:

```go
slog.Info("http_request",
    "method", r.Method,
    "route", routePattern,
    "status", status,
```

```
    "dur_ms", time.Since(start).Milliseconds(),
    "bytes", n,
)
```

Redaction and Security

Logs are a data store. Assume breach.

```
func redactAuth(h http.Header) http.Header {
    c := h.Clone()
    if c.Get("Authorization") != "" {
        c.Set("Authorization", "REDACTED")
    }
    if c.Get("Cookie") != "" {
        c.Set("Cookie", "REDACTED")
    }
    return c
}
```

For structured fields, prefer hashing user ids when only correlation is needed: `user_sha256=…`.

Sampling and Volume

High-QPS services may sample successful debug/info events while always keeping errors:

```
// Conceptual: custom Handler that drops Info with probability p
// unless level >= Warn
```

Or sample at the shipper (Fluent Bit, Vector) — still emit errors at 100%.

Goroutines and Lost Context

```
// BAD: detaches without ids
go func() {
    doWork(context.Background())
}()

// GOOD: carry request context (or derived with timeout)
go func(ctx context.Context) {
    defer func() { /* recover log with request_id */ }()
    doWork(ctx)
}(ctx)
```

If work outlives the request, extract ids into the child logger before detaching, and use a new root context with timeout.

slog Handlers and Production

- **JSON** to stdout for containers (collectors parse)
- **Text** for local dev
- Fan-out handlers exist in community packages if you must dual-write

```go
// Level from env
level := slog.LevelInfo
if os.Getenv("LOG_LEVEL") == "debug" {
    level = slog.LevelDebug
}
```

Testing Logs

```go
var buf bytes.Buffer
log := slog.New(slog.NewJSONHandler(&buf, nil))
// inject log into code under test
// assert strings.Contains(buf.String(), `"msg":"paid"`)
```

Or use a test handler that records `[]slog.Record`.

Production Checklist

- ☐ JSON structured logs in prod
- ☐ Stable field vocabulary documented
- ☐ request_id / trace_id on every request log line
- ☐ Middleware sets and returns request id
- ☐ Outbound calls propagate correlation headers
- ☐ Secrets redacted; body logging opt-in only
- ☐ Error logged once at boundary with stack or error chain
- ☐ Levels configurable without recompile
- ☐ Volume/sampling plan for hot paths

Common Pitfalls

1. **String interpolation only** — `"user "+id` loses structure.
2. **Different field names** per service (`req_id` vs `requestId`).
3. **Logging entire `http.Request`** — headers leak cookies.

4. **`log.Fatal` in libraries** — kills process.
5. **Missing id after async hop**.
6. **Debug left at Info** — cost and noise.
7. **Duplicate error logs** every layer → alert fatigue.

Exercises

1. Replace a `log.Printf` access log with `slog` JSON fields; query with `jq`.
2. Implement request id middleware; prove id appears in handler logs and response header.
3. Propagate id on an outbound `http.Client` call; log on both sides; join lines by id.
4. Add redaction for Authorization; unit-test.
5. Spawn a goroutine without context; show missing id; fix.
6. Define a 10-field schema doc for your team; refactor one service to match.
7. Emit `dur_ms` histogram-equivalent in logs only; discuss when metrics are better.
8. Implement level-from-env; toggle debug in a running container via restart with env.
9. Benchmark logging 10k info lines JSON vs text; note CPU.
10. Create an incident drill: given only logs, reconstruct a failed checkout path.

More examples

slog groups and default attrs

```
mkdir -p /tmp/go-slog-group && cd /tmp/go-slog-group
go mod init example.com/slog-group
```

Save as `main.go`:

```go
package main

import (
    "bytes"
    "fmt"
    "log/slog"
    "strings"
)

func main() {
    var buf bytes.Buffer
    base := slog.New(slog.NewJSONHandler(&buf, nil))
    log := base.With("service", "checkout", "env", "dev")
    log.Info("paid",
        slog.Group("user", "id", 42, "plan", "pro"),
        "amount_cents", 999,
    )
```

```go
    line := strings.TrimSpace(buf.String())
    fmt.Println(line)
    fmt.Println("has service:", strings.Contains(line, `"service":"checkout"`))
    fmt.Println("has user group:", strings.Contains(line, `"user"`))
}
```

```bash
go run .
```

Expected output (time field varies):

```
{"time":"...","level":"INFO","msg":"paid","service":"checkout","env":"dev","user":{"id":42,"pla
has service: true
has user group: true
```

Child logger from context request id

```bash
mkdir -p /tmp/go-slog-child && cd /tmp/go-slog-child
go mod init example.com/slog-child
```

Save as `main.go`:

```go
package main

import (
    "bytes"
    "context"
    "fmt"
    "log/slog"
    "strings"
)

type ctxKey string

const ridKey ctxKey = "rid"

func loggerWithRID(ctx context.Context, base *slog.Logger) *slog.Logger {
    if v, ok := ctx.Value(ridKey).(string); ok && v != "" {
        return base.With("request_id", v)
    }
    return base
}

func main() {
    var buf bytes.Buffer
    base := slog.New(slog.NewJSONHandler(&buf, nil))
```

```go
    ctx := context.WithValue(context.Background(), ridKey, "req-9")
    log := loggerWithRID(ctx, base)
    log.Info("step", "name", "charge")
    fmt.Println(strings.Contains(buf.String(), "req-9"))
}
```

```go
go run .
```

Expected output:

```
true
```

Runnable example

JSON `slog`, request-id middleware with `httptest`, outbound header propagation, and Authorization redaction.

```go
mkdir -p /tmp/go-slog-corr && cd /tmp/go-slog-corr
go mod init example.com/slog-corr
```

Save as `main.go`:

```go
package main

import (
    "bytes"
    "context"
    "fmt"
    "log/slog"
    "net/http"
    "net/http/httptest"
    "strings"
)

type ctxKey string

const ridKey ctxKey = "request_id"

func withRID(ctx context.Context, id string) context.Context {
    return context.WithValue(ctx, ridKey, id)
}

func rid(ctx context.Context) string {
    if v, ok := ctx.Value(ridKey).(string); ok {
        return v
```

```
    }
    return ""
}

func redactAuth(h http.Header) string {
    v := h.Get("Authorization")
    if v == "" {
        return ""
    }
    return "REDACTED"
}

func main() {
    var buf bytes.Buffer
    log := slog.New(slog.NewJSONHandler(&buf, &slog.HandlerOptions{Level: slog.LevelInfo}))

    mux := http.NewServeMux()
    mux.HandleFunc("GET /api", func(w http.ResponseWriter, r *http.Request) {
        log.Info("handler",
            "request_id", rid(r.Context()),
            "path", r.URL.Path,
            "authorization", redactAuth(r.Header),
        )
        // simulate outbound call headers
        out := httptest.NewRequest(http.MethodGet, "http://dep/x", nil)
        out.Header.Set("X-Request-ID", rid(r.Context()))
        fmt.Println("outbound X-Request-ID:", out.Header.Get("X-Request-ID"))
        w.Header().Set("X-Request-ID", rid(r.Context()))
        fmt.Fprintln(w, "ok")
    })

    h := http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        id := r.Header.Get("X-Request-ID")
        if id == "" {
            id = "generated-1"
        }
        ctx := withRID(r.Context(), id)
        mux.ServeHTTP(w, r.WithContext(ctx))
    })

    req := httptest.NewRequest(http.MethodGet, "/api", nil)
    req.Header.Set("X-Request-ID", "abc-123")
    req.Header.Set("Authorization", "Bearer super-secret")
    rr := httptest.NewRecorder()
    h.ServeHTTP(rr, req)
    fmt.Println("status:", rr.Code, "resp id:", rr.Header().Get("X-Request-ID"))
```

```go
    fmt.Println(strings.TrimSpace(buf.String()))
}
```

```go
go run .
```

Expected output:

```
outbound X-Request-ID: abc-123
status: 200 resp id: abc-123
{"time":"...","level":"INFO","msg":"handler","request_id":"abc-123","path":"/api","authorizatio
```

What to notice

- Correlation requires the same id in middleware, handler logs, response, and outbound clients.
- Never log raw bearer tokens; redact at the edge of the logger.
- Prefer structured attrs over string interpolation.

Try next

- Level from `LOG_LEVEL` env; hide Debug when `info`.
- Spawn work with `context` so child goroutines keep the request id.

Further Reading

- Go `log/slog` package documentation
- OpenTelemetry baggage vs attributes (next chapters)
- Next: **Metrics and Prometheus Design**

Metrics and Prometheus Design

Metrics and Prometheus Design

Metrics answer operational questions quickly: Is traffic normal? Are errors rising? Is latency degrading? They are cheap aggregates — if you design labels carefully.

Why / Overview

Logs are high-cardinality events. Metrics are **pre-aggregated time series**. Prometheus scrapes pull-based endpoints; you expose `/metrics` with a small, intentional surface.

```
process -> counters/histograms/gauges
                |
          /metrics scrape
                |
          Prometheus -> alerts / Grafana
```

RED and USE

RED (request-driven services):

Signal	Meaning
Rate	Requests per second
Errors	Failed requests per second
Duration	Latency distribution

USE (resources):

Signal	Meaning
Utilization	% busy
Saturation	Queue depth / wait
Errors	Resource error counts

Export both: RED for SLIs, USE for capacity.

Metric Types

Type	Use
Counter	Monotonic events (`_total` suffix)
Gauge	Levels that go up/down (goroutines, depth)
Histogram	Latency/size distributions
Summary	Client-side quantiles (less preferred for prom federation)

Prefer **histograms** for latency with server-side aggregation.

Instrumentation Example

```go
package metrics

import (
    "net/http"
    "strconv"
    "time"

    "github.com/prometheus/client_golang/prometheus"
    "github.com/prometheus/client_golang/prometheus/promauto"
    "github.com/prometheus/client_golang/prometheus/promhttp"
)

var (
    httpRequests = promauto.NewCounterVec(prometheus.CounterOpts{
        Name: "http_requests_total",
        Help: "HTTP requests",
    }, []string{"method", "route", "code"})

    httpDuration = promauto.NewHistogramVec(prometheus.HistogramOpts{
        Name:    "http_request_duration_seconds",
        Help:    "HTTP latency",
        Buckets: []float64{.005, .01, .025, .05, .1, .25, .5, 1, 2.5, 5, 10},
    }, []string{"method", "route"})

    inFlight = promauto.NewGauge(prometheus.GaugeOpts{
        Name: "http_requests_in_flight",
        Help: "In-flight HTTP requests",
    })
)

func Middleware(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
```

```go
        start := time.Now()
        inFlight.Inc()
        defer inFlight.Dec()
        ww := &statusWriter{ResponseWriter: w, code: 200}
        next.ServeHTTP(ww, r)
        route := routeOf(r) // template, not raw path
        httpRequests.WithLabelValues(r.Method, route, strconv.Itoa(ww.code)).Inc()
        httpDuration.WithLabelValues(r.Method, route).Observe(time.Since(start).Seconds())
    })
}

func Handler() http.Handler { return promhttp.Handler() }
```

Mount metrics on admin listener when possible:

```go
go http.ListenAndServe("127.0.0.1:9090", metrics.Handler())
```

Label Strategy (Cardinality)

Each unique label combination is a time series.

Safe (bounded):

- `method`, `route` template, `code`/`code_class`, `dependency` name, `breaker_state`

Unsafe (unbounded):

- `user_id`, `email`, raw `path`, `url`, `request_id`, pod IP as high-churn without care

```go
// BAD
counter.WithLabelValues(r.URL.Path).Inc()

// GOOD
counter.WithLabelValues(routeTemplate).Inc()
```

Rule of thumb: if a label can take unlimited values from user input, it does not belong on a metric.

Histogram Buckets

Default buckets may not fit your SLOs. If SLO is 300ms, ensure buckets around 0.1, 0.25, 0.3, 0.5, …

```go
Buckets: prometheus.ExponentialBuckets(0.005, 2, 12)
// or explicit slice aligned to SLOs
```

You cannot fix wrong buckets retroactively without a new metric name.

Naming Conventions

```
namespace_subsystem_name_unit
http_request_duration_seconds
db_query_duration_seconds
queue_depth
client_requests_total
```

- Counters end with `_total`
- Units in seconds/bytes base units
- Avoid mixing milliseconds and seconds across metrics

Outbound Client Metrics

```go
var depDuration = promauto.NewHistogramVec(prometheus.HistogramOpts{
    Name:    "dependency_request_duration_seconds",
    Help:    "Outbound calls",
    Buckets: prometheus.DefBuckets,
}, []string{"dep", "method", "result"})

func observeDep(dep, method, result string, d time.Duration) {
    depDuration.WithLabelValues(dep, method, result).Observe(d.Seconds())
}
```

`result`: `ok`, `timeout`, `error`, `cancel` — not free-form error strings.

Exemplars and Trace Links

Modern Prometheus + OTel can attach trace ids to histogram observations (exemplars). When available, latency spikes become one click into a trace. Configure rather than reinvent.

Alerting from Metrics

Alert on **symptoms**:

```
// conceptual PromQL
rate(http_requests_total{code=~"5.."}[5m])
  / rate(http_requests_total[5m]) > 0.01
```

And latency:

```
histogram_quantile(0.99, sum by (le) (rate(http_request_duration_seconds_bucket[5m]))) > 0.3
```

Multi-window multi-burn-rate alerts (SRE workbook) reduce noise for SLO burn.

Every alert needs:

1. What is broken (SLI)
2. User impact
3. Runbook link
4. Severity / routing

Runtime Metrics

Export Go runtime via collectors:

```
// prometheus client includes process and go collectors when using default registry
// go_goroutines, go_memstats_*, process_cpu_seconds_total
```

Useful for saturation; not a substitute for RED SLIs.

Production Checklist

- ☐ RED metrics per public route template
- ☐ Dependency latency/error metrics
- ☐ Bounded label sets reviewed in PR
- ☐ Histogram buckets match SLOs
- ☐ `/metrics` not public without auth/network policy
- ☐ Alerts tied to SLOs + runbooks
- ☐ Dashboards use rate/increase correctly on counters
- ☐ Queue depth / in-flight gauges for backpressure
- ☐ Document metric stability (renames break alerts)

Common Pitfalls

1. **Raw path labels** → metric explosion → Prometheus OOM.
2. **Using summaries** then trying to aggregate quantiles across instances incorrectly.
3. **Alerting on CPU only** while error rate is the user pain.
4. **Counter without `_total`** / non-monotonic "counters".
5. **Measuring only mean latency** — tails hide outages.
6. **Scrape timeout** longer than interval; missed scrapes.
7. **High-cardinality `error_message` labels**.

Exercises

1. Instrument a toy server with counter + histogram; scrape with a local Prometheus.
2. Add path label vs route template; show series count difference with unique ids in paths.
3. Choose buckets for a 200ms SLO; verify quantile accuracy with known sleeps.
4. Write PromQL for 5xx rate and p99 latency; graph both.
5. Add dependency metrics around an HTTP client.
6. Create an alert rule file; fire it with forced errors; follow a stub runbook.
7. Export queue depth gauge from a worker; alert when depth > N for 10m.
8. Compare `rate` vs `increase` on a slow counter; explain gaps.
9. Secure `/metrics` behind network policy or basic auth.
10. Define error budget math for 99.9% availability over 30 days.

More examples

Prometheus text exposition (stdlib)

```
mkdir -p /tmp/go-prom-text && cd /tmp/go-prom-text
go mod init example.com/prom-text
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "net/http"
    "net/http/httptest"
    "strings"
    "sync/atomic"
)

type Counter struct{ n atomic.Int64 }

func (c *Counter) Inc() { c.n.Add(1) }
func (c *Counter) Get() int64 { return c.n.Load() }

func main() {
    var reqs Counter
    mux := http.NewServeMux()
    mux.HandleFunc("GET /", func(w http.ResponseWriter, r *http.Request) {
        reqs.Inc()
        fmt.Fprintln(w, "ok")
    })
    mux.HandleFunc("GET /metrics", func(w http.ResponseWriter, r *http.Request) {
```

```
        w.Header().Set("Content-Type", "text/plain; version=0.0.4")
        fmt.Fprintf(w, "# HELP http_requests_total Total HTTP requests\n")
        fmt.Fprintf(w, "# TYPE http_requests_total counter\n")
        fmt.Fprintf(w, "http_requests_total %d\n", reqs.Get())
    })

    ts := httptest.NewServer(mux)
    defer ts.Close()
    for i := 0; i < 3; i++ {
        r, _ := http.Get(ts.URL + "/")
        r.Body.Close()
    }
    r, _ := http.Get(ts.URL + "/metrics")
    defer r.Body.Close()
    var b [256]byte
    n, _ := r.Body.Read(b[:])
    body := string(b[:n])
    fmt.Println(strings.TrimSpace(body))
    fmt.Println("has type:", strings.Contains(body, "TYPE http_requests_total"))
}
```

```
go run .
```

Expected output:

```
# HELP http_requests_total Total HTTP requests
# TYPE http_requests_total counter
http_requests_total 3
has type: true
```

Labeled counter map (cardinality caution)

```
mkdir -p /tmp/go-prom-labels && cd /tmp/go-prom-labels
go mod init example.com/prom-labels
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "sync"
)

type LabeledCounter struct {
```

```go
    mu sync.Mutex
    m  map[string]int
}

func (c *LabeledCounter) Inc(code string) {
    c.mu.Lock()
    if c.m == nil {
        c.m = map[string]int{}
    }
    c.m[code]++
    c.mu.Unlock()
}

func (c *LabeledCounter) Render(name string) string {
    c.mu.Lock()
    defer c.mu.Unlock()
    var b string
    for k, v := range c.m {
        b += fmt.Sprintf("%s{code=%q} %d\n", name, k, v)
    }
    return b
}

func main() {
    var c LabeledCounter
    c.Inc("200")
    c.Inc("200")
    c.Inc("500")
    out := c.Render("http_responses_total")
    fmt.Print(out)
    fmt.Println("lines:", len(out) > 0)
}
```

```go
go run .
```

Expected output (label order may vary):

```
http_responses_total{code="200"} 2
http_responses_total{code="500"} 1
lines: true
```

Runnable example

Manual Prometheus text exposition: counter, gauge, and a crude histogram bucket set—stdlib only, no client_golang.

```
mkdir -p /tmp/go-metrics && cd /tmp/go-metrics
go mod init example.com/metrics
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "net/http"
    "net/http/httptest"
    "sort"
    "strings"
    "sync"
    "time"
)

type registry struct {
    mu      sync.Mutex
    reqs    map[string]float64 // status -> count
    inFlight float64
    // latency buckets in seconds
    bounds []float64
    counts []float64
    sum    float64
    count  float64
}

func newReg() *registry {
    b := []float64{0.005, 0.01, 0.025, 0.05, 0.1, 0.25, 0.5, 1}
    return &registry{
        reqs:   map[string]float64{},
        bounds: b,
        counts: make([]float64, len(b)+1), // +Inf
    }
}

func (r *registry) observe(status string, d time.Duration) {
    r.mu.Lock()
    defer r.mu.Unlock()
    r.reqs[status]++
    sec := d.Seconds()
    r.sum += sec
    r.count++
    placed := false
    for i, b := range r.bounds {
        if sec <= b {
```

```
            r.counts[i]++
            placed = true
            break
        }
    }
    if !placed {
        r.counts[len(r.bounds)]++
    }
}

func (r *registry) expose() string {
    r.mu.Lock()
    defer r.mu.Unlock()
    var b strings.Builder
    fmt.Fprintf(&b, "# HELP http_requests_total Demo requests\n")
    fmt.Fprintf(&b, "# TYPE http_requests_total counter\n")
    var statuses []string
    for s := range r.reqs {
        statuses = append(statuses, s)
    }
    sort.Strings(statuses)
    for _, s := range statuses {
        fmt.Fprintf(&b, "http_requests_total{status=%q} %g\n", s, r.reqs[s])
    }
    fmt.Fprintf(&b, "# HELP http_in_flight Demo gauge\n")
    fmt.Fprintf(&b, "# TYPE http_in_flight gauge\n")
    fmt.Fprintf(&b, "http_in_flight %g\n", r.inFlight)

    fmt.Fprintf(&b, "# HELP http_request_duration_seconds Demo histogram\n")
    fmt.Fprintf(&b, "# TYPE http_request_duration_seconds histogram\n")
    var cum float64
    for i, bound := range r.bounds {
        cum += r.counts[i]
        fmt.Fprintf(&b, "http_request_duration_seconds_bucket{le=%q} %g\n", fmt.Sprintf("%g"
    }
    cum += r.counts[len(r.bounds)]
    fmt.Fprintf(&b, "http_request_duration_seconds_bucket{le=\"+Inf\"} %g\n", cum)
    fmt.Fprintf(&b, "http_request_duration_seconds_sum %g\n", r.sum)
    fmt.Fprintf(&b, "http_request_duration_seconds_count %g\n", r.count)
    return b.String()
}

func main() {
    reg := newReg()
    mux := http.NewServeMux()
    mux.HandleFunc("GET /work", func(w http.ResponseWriter, r *http.Request) {
```

```go
        reg.mu.Lock()
        reg.inFlight++
        reg.mu.Unlock()
        start := time.Now()
        time.Sleep(12 * time.Millisecond)
        reg.mu.Lock()
        reg.inFlight--
        reg.mu.Unlock()
        reg.observe("200", time.Since(start))
        w.WriteHeader(200)
    })
    mux.HandleFunc("GET /metrics", func(w http.ResponseWriter, _ *http.Request) {
        w.Header().Set("Content-Type", "text/plain; version=0.0.4")
        fmt.Fprint(w, reg.expose())
    })

    for i := 0; i < 3; i++ {
        rr := httptest.NewRecorder()
        mux.ServeHTTP(rr, httptest.NewRequest(http.MethodGet, "/work", nil))
    }
    // one error path
    reg.observe("500", 30*time.Millisecond)

    rr := httptest.NewRecorder()
    mux.ServeHTTP(rr, httptest.NewRequest(http.MethodGet, "/metrics", nil))
    fmt.Print(rr.Body.String())
}
```

```go
go run .
```

Expected output (excerpt):

```
# HELP http_requests_total Demo requests
# TYPE http_requests_total counter
http_requests_total{status="200"} 3
http_requests_total{status="500"} 1
...
http_request_duration_seconds_count 4
```

What to notice

- Use route templates as labels, never raw ids (`/user/123`)—cardinality explosion kills Prometheus.
- Histograms need `_bucket` / `_sum` / `_count`; counters use `_total`.
- Gauges go up and down (`in_flight`); counters only increase.

Try next

- Scrape with a local Prometheus or `curl` a real listener.
- Write PromQL: `rate(http_requests_total{status="500"}[5m])`.

Further Reading

- Prometheus metric and labeling best practices
- Google SRE workbook: alerting on SLOs
- Previous: structured logs for event detail
- Next: **Tracing and Context Propagation**

Tracing and Context Propagation

Tracing and Context Propagation

Tracing makes distributed latency visible by preserving **causal context** across process boundaries. A single missing propagation hop blinds the whole request path.

Why / Overview

Metrics say p99 is bad. Logs say a request failed. Traces answer: **where did the time go** across gateway → service → DB → dependency?

```
gateway ----> service A ----> service B ----> datastore
   |              |                |
 span           span            span
   trace-id = same across all
```

OpenTelemetry (OTel) is the default modern approach in Go ecosystems; the ideas apply even if you use a vendor SDK.

Core Concepts

Term	Meaning
Trace	End-to-end request / workflow
Span	Unit of work with start/end, attributes, status
Context	Carries trace/span identity in-process
Propagator	Inject/extract context over HTTP/gRPC headers
Sampler	Which traces to keep

```
W3C Trace Context headers:
  traceparent: 00-<trace-id>-<span-id>-<flags>
  tracestate: optional vendor data
```

Context Propagation Rules

1. Accept `context.Context` at API boundaries.
2. Pass the same ctx (or child) into outbound calls — **never** `context.Background()` mid-request unless intentionally detaching.

3. When detaching, copy trace ids into a new context if the async work should remain in the trace (or start a linked span).
4. Instrument DB, HTTP, and queue publish/consume.

```go
// BAD
go process(context.Background())

// GOOD: child span / derived context
go process(context.WithoutCancel(ctx)) // Go 1.21+ if you must outlive request cancel
// or: start a new root with explicit link for async jobs
```

`WithoutCancel` preserves values but not cancellation — use carefully for fire-and-forget that should still be attributed.

Minimal HTTP Propagation (Conceptual)

Even without full OTel, the discipline is:

```go
func propagate(req *http.Request, incoming http.Header) {
    if v := incoming.Get("Traceparent"); v != "" {
        req.Header.Set("Traceparent", v)
    }
    if v := incoming.Get("X-Request-Id"); v != "" {
        req.Header.Set("X-Request-Id", v)
    }
}
```

With OTel, use official propagators and `otelhttp` transport wrappers instead of hand-rolling.

Span Design

Create spans around **meaningful operations**, not every function:

- Inbound HTTP/gRPC request (framework middleware)
- Outbound HTTP/RPC
- DB query / transaction
- Significant queue publish
- Expensive compute blocks

```go
// Pseudocode shape with OTel API
ctx, span := tracer.Start(ctx, "db.QueryUser")
defer span.End()
span.SetAttributes(attribute.String("db.system", "postgres"))
row := db.QueryRowContext(ctx, q, id)
if err != nil {
```

```
    span.RecordError(err)
    span.SetStatus(codes.Error, err.Error())
}
```

Attributes: low-cardinality. Do not put raw SQL with unbound ids if that becomes a PII/cardinality problem — use operation names.

Sampling

Strategy	When
Always on	Low traffic / critical paths
Probability (e.g. 1–10%)	High QPS happy path
Parent-based	Honor upstream decision
Tail sampling	Keep errors/slow traces (collector)

Always sample (or force) **errors and high latency** when the platform allows.

Common Failure Modes

1. **New client without transport wrapper** — headers not injected.
2. **Goroutine with Background** — orphan spans or missing children.
3. **Queue consumer not extracting** — new unrelated traces per message.
4. **Clock skew** — span order looks weird (use careful interpretation).
5. **Too many tiny spans** — overhead and UI noise.
6. **Sensitive data in attributes** — tokens, PII.

Queue propagation

Put trace context in message metadata/headers; extract on consume; start a **consumer span** as child or link.

```
producer span -> inject into message
consumer receives -> extract -> process span
```

Correlation with Logs and Metrics

- Put `trace_id` and `span_id` into slog fields from the current span context.
- Metrics exemplars can point to traces.
- Same `service.name` and resource attributes across signals.

```go
// After extracting span from ctx:
slog.InfoContext(ctx, "paid", "order_id", id)
// With a handler that appends trace ids from ctx (OTel slog bridge or custom)
```

In-Process Performance Note

Span creation is cheap when sampled out, not free when always on with heavy attributes. Measure overhead on ultra-hot paths; batch exporters carefully.

SDK Lifecycle (Shape)

Regardless of vendor, production services should:

```go
func main() {
    ctx := context.Background()
    shutdown, err := initTracer(ctx) // set global TracerProvider + propagator
    if err != nil {
        log.Fatal(err)
    }
    defer func() {
        // Flush remaining spans on exit; bound the wait.
        ctx, cancel := context.WithTimeout(context.Background(), 5*time.Second)
        defer cancel()
        _ = shutdown(ctx)
    }()

    // run server...
}
```

Initialize **once**. Multiple providers create confusing partial traces.

HTTP middleware and transport (conceptual)

```go
// Server: wrap handler so each request becomes a span.
// handler = otelhttp.NewHandler(mux, "api")

// Client: wrap transport so outbound calls inject context.
// client.Transport = otelhttp.NewTransport(http.DefaultTransport)
```

Prefer library wrappers over copying header maps by hand — they handle version updates of the propagation format.

Baggage vs Attributes

Mechanism	Purpose	Danger
Span attributes	Describe this operation	Cardinality if unbounded
Baggage	Propagate key-values to downstream	Becomes a global gossip channel

Use baggage sparingly (e.g. tenant tier for sampling). Never put PII or secrets in baggage — every hop may log it.

Reading a Trace During an Incident

1. Open the trace from the failing `request_id` / exemplar.
2. Sort or scan for the longest span or first error status.
3. Check whether time is **in your service** (CPU/DB) or **waiting on a child**.
4. Confirm the slow child has correct timeout budgets (part 13).
5. If the trace stops mid-path, find the missing propagation hop and fix instrumentation first.

A broken trace is itself a reliability bug.

Production Checklist

- ☐ OTel (or vendor) SDK initialized once at start
- ☐ W3C TraceContext propagator configured
- ☐ HTTP server middleware creates root/child spans
- ☐ HTTP client transport injects context
- ☐ DB/redis clients instrumented or wrapped
- ☐ Async/queue paths extract/inject
- ☐ Sampling policy documented
- ☐ Errors mark span status
- ☐ Logs include trace_id
- ☐ Admin-only or secured debug endpoints

Common Pitfalls

1. Multiple tracer provider initializations.
2. Forgetting `Shutdown` on exporter at process exit (lost tail spans).
3. Using request body content as span names (cardinality + PII).
4. Mixing baggage (cross-cutting key-values) with unbounded values.
5. Assuming trace UI replaces metrics for alerting — it does not.

Exercises

1. Manually pass `X-Request-Id` across two HTTP services; log it on both.
2. Add W3C `traceparent` generation/parsing in a toy form; verify header round-trip.
3. Break propagation in one client; observe "broken" trace; fix.
4. Start a span around a DB call; add attributes; export to a local collector (Jaeger/OTel).
5. Detach a goroutine with Background; compare trace; fix with derived context.
6. Publish a message with trace headers; consume and continue the trace.
7. Configure 1% sampling; force error sampling if available.
8. Bridge slog to include trace ids; grep logs for a trace id from the UI.
9. Measure CPU with tracing on/off under load for a hot endpoint.
10. Write a runbook: "p99 high" → open trace → identify slow span → next action.

End-to-End Lab Outline

1. Two tiny services: `A` calls `B` over HTTP.
2. Add request ids + manual `traceparent` first; prove logs join.
3. Swap in OTel SDK + Jaeger/OTLP collector locally.
4. Break the client transport wrapper; note the cliff in the UI.
5. Add a DB span in `B`; force a slow query; confirm the waterfall.
6. Drop sample rate to 1%; force an error path to stay visible.

Document the exact env vars and ports your team uses so the lab is repeatable.

More examples

Trace/span ids on context and outbound headers

```
mkdir -p /tmp/go-trace-ctx && cd /tmp/go-trace-ctx
go mod init example.com/trace-ctx
```

Save as `main.go`:

```
package main

import (
    "context"
    "fmt"
    "net/http"
    "net/http/httptest"
)

type ctxKey string
```

```go
const (
    traceKey ctxKey = "trace_id"
    spanKey  ctxKey = "span_id"
)

func withTrace(ctx context.Context, traceID, spanID string) context.Context {
    ctx = context.WithValue(ctx, traceKey, traceID)
    return context.WithValue(ctx, spanKey, spanID)
}

func inject(req *http.Request) {
    if v, ok := req.Context().Value(traceKey).(string); ok {
        req.Header.Set("X-Trace-Id", v)
    }
    if v, ok := req.Context().Value(spanKey).(string); ok {
        req.Header.Set("X-Span-Id", v)
    }
}

func main() {
    upstream := httptest.NewServer(http.HandlerFunc(func(w http.ResponseWriter, r *http.Requ
        fmt.Fprintf(w, "trace=%s span=%s", r.Header.Get("X-Trace-Id"), r.Header.Get("X-Span-
    }))
    defer upstream.Close()

    ctx := withTrace(context.Background(), "t-1", "s-9")
    req, _ := http.NewRequestWithContext(ctx, http.MethodGet, upstream.URL, nil)
    inject(req)
    resp, err := http.DefaultClient.Do(req)
    if err != nil {
        panic(err)
    }
    defer resp.Body.Close()
    var buf [64]byte
    n, _ := resp.Body.Read(buf[:])
    fmt.Println(string(buf[:n]))
}
```

```go
go run .
```

Expected output:

```
trace=t-1 span=s-9
```

Nested span timing (in-process)

```
mkdir -p /tmp/go-span-time && cd /tmp/go-span-time
go mod init example.com/span-time
```

Save as `main.go`:

```go
package main

import (
    "fmt"
    "time"
)

type Span struct {
    Name string
    Dur  time.Duration
}

func timed(name string, fn func()) Span {
    start := time.Now()
    fn()
    return Span{Name: name, Dur: time.Since(start)}
}

func main() {
    var spans []Span
    spans = append(spans, timed("handler", func() {
        spans = append(spans, timed("db", func() { time.Sleep(5 * time.Millisecond) }))
        spans = append(spans, timed("cache", func() { time.Sleep(1 * time.Millisecond) }))
    }))
    for _, s := range spans {
        fmt.Printf("%s >=1ms: %v\n", s.Name, s.Dur >= time.Millisecond)
    }
}
```

```
go run .
```

Expected output:

```
db >=1ms: true
cache >=1ms: true
handler >=1ms: true
```

Runnable example

W3C-style `traceparent` generate/parse and HTTP propagation between two `httptest` services—stdlib only (no OTel SDK).

```
mkdir -p /tmp/go-traceprop && cd /tmp/go-traceprop
go mod init example.com/traceprop
```

Save as `main.go`:

```go
package main

import (
    "crypto/rand"
    "encoding/hex"
    "fmt"
    "net/http"
    "net/http/httptest"
    "strings"
)

func newTraceparent() string {
    var tid [16]byte
    var sid [8]byte
    _, _ = rand.Read(tid[:])
    _, _ = rand.Read(sid[:])
    return fmt.Sprintf("00-%s-%s-01", hex.EncodeToString(tid[:]), hex.EncodeToString(sid[:])
}

func parseTraceparent(s string) (traceID, spanID string, ok bool) {
    parts := strings.Split(s, "-")
    if len(parts) != 4 || parts[0] != "00" || len(parts[1]) != 32 || len(parts[2]) != 16 {
        return "", "", false
    }
    return parts[1], parts[2], true
}

func main() {
    // Service B
    b := httptest.NewServer(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        tp := r.Header.Get("traceparent")
        tid, sid, ok := parseTraceparent(tp)
        fmt.Printf("B saw trace_ok=%v trace_id=%s parent_span=%s\n", ok, tid, sid)
        fmt.Fprintln(w, "b-ok")
    }))
    defer b.Close()
```

```go
    // Service A calls B, propagating traceparent
    a := httptest.NewServer(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        tp := r.Header.Get("traceparent")
        if tp == "" {
            tp = newTraceparent()
        }
        tid, _, _ := parseTraceparent(tp)
        // child span id for outbound
        var child [8]byte
        _, _ = rand.Read(child[:])
        outTP := fmt.Sprintf("00-%s-%s-01", tid, hex.EncodeToString(child[:]))
        fmt.Printf("A outbound traceparent=%s\n", outTP)

        req, _ := http.NewRequest(http.MethodGet, b.URL+"/work", nil)
        req.Header.Set("traceparent", outTP)
        resp, err := http.DefaultClient.Do(req)
        if err != nil {
            http.Error(w, err.Error(), 500)
            return
        }
        resp.Body.Close()
        w.Header().Set("traceparent", tp)
        fmt.Fprintln(w, "a-ok")
    }))
    defer a.Close()

    req, _ := http.NewRequest(http.MethodGet, a.URL+"/", nil)
    req.Header.Set("traceparent", newTraceparent())
    resp, err := http.DefaultClient.Do(req)
    if err != nil {
        panic(err)
    }
    resp.Body.Close()
    fmt.Println("client status:", resp.StatusCode)

    // Broken propagation demo
    _, _, ok := parseTraceparent("not-a-trace")
    fmt.Println("invalid rejected:", !ok)
}
```

```go
go run .
```

Expected output (ids random):

```
A outbound traceparent=00-...
B saw trace_ok=true trace_id=... parent_span=...
client status: 200
```

```
invalid rejected: true
```

What to notice

- Trace id stays stable across the request tree; span id changes per hop.
- Forgetting to copy `traceparent` on the client is the classic "broken trace" bug.
- Real systems add OTel SDK exporters; the propagation contract stays the same.

Try next

- Put `trace_id` into slog fields for log trace joins.
- Continue a trace across a simulated queue message header map.

Further Reading

- OpenTelemetry Go documentation
- W3C Trace Context specification
- Previous: Prometheus metrics for detection; traces for diagnosis
- Next part: **Security Hardening** — protecting the systems you can now observe

SLIs, SLOs, and Error Budgets

SLIs, SLOs, and Error Budgets

Overview

Metrics without targets become noise. **SLIs** measure user-visible health; **SLOs** set goals; **error budgets** decide how fast you can ship vs stabilize.

Definitions

Term	Meaning	Example
SLI	Quantitative measure	% of requests < 300ms; success ratio
SLO	Target on an SLI	99.9% success over 30 days
SLA	Contractual promise	Often looser than internal SLO
Error budget	`1 - SLO` failures allowed	0.1% budget for risk

Good SLIs for HTTP APIs

```
availability  = successful requests / total  (exclude 4xx client errors carefully)
latency       = fraction of requests under threshold (p99 or histogram)
freshness     = data lag (async pipelines)
```

Prefer **user journeys** (login, checkout) over raw CPU.

Expressing latency as an SLI

With Prometheus histograms:

```
histogram_quantile(0.99, sum(rate(http_request_duration_seconds_bucket[5m])) by (le))
```

Or "good events" style:

```
count of requests with duration <= 0.3s  /  count of requests
```

Error budget policy (lightweight)

Budget remaining	Engineering posture
Healthy	Ship features, experiments OK
Burning fast	Freeze risky deploys; fix reliability
Exhausted	Reliability work only until recovery

Minimal Go: count success/fail

```go
var (
    reqTotal   = expvar.NewInt("requests_total")
    reqFailed  = expvar.NewInt("requests_failed")
)

// in handler:
reqTotal.Add(1)
if status >= 500 {
    reqFailed.Add(1)
}
```

Graduate to Prometheus counters/histograms (chapter 162).

Rules of thumb

Do	Don't
Few SLOs (1–3 per critical UX)	50 red dashboards, zero owners
Align alerts to budget burn	Alert on every CPU blip
Review SLOs quarterly	Set 99.99% with no measurement

Try next

1. Define SLIs for one service (availability + latency).
2. Compute monthly error budget from request volume.
3. Write a one-page policy: what happens when budget is empty.

Health, Liveness, and Readiness Probes

Health, Liveness, and Readiness Probes

Overview

Orchestrators ask "is this process OK?" in two different ways:

Probe	Question	Bad answer means
Liveness	Should we restart the process?	Kill + restart
Readiness	Should we send traffic?	Remove from LB

Confusing them causes restart loops or black-hole traffic.

Minimal endpoints

```go
var ready atomic.Bool

func init() { ready.Store(true) }

func livez(w http.ResponseWriter, r *http.Request) {
    w.WriteHeader(http.StatusOK)
    _, _ = w.Write([]byte("ok"))
}

func readyz(w http.ResponseWriter, r *http.Request) {
    if !ready.Load() {
        http.Error(w, "not ready", http.StatusServiceUnavailable)
        return
    }
    // optional: ping DB with short timeout
    w.WriteHeader(http.StatusOK)
}
```

Dependency checks

Check in readiness	Check in liveness
DB ping (short timeout)	Almost never deep deps
Required config loaded	Process deadlocked (hard)
Warmup finished	

Do not make liveness fail when a dependency is down—that restarts everyone during a DB blip.

Drain on shutdown

```
// on SIGTERM:
ready.Store(false)
time.Sleep(2 * time.Second) // allow LB to observe
_ = srv.Shutdown(ctx)
```

Startup probe (K8s)

Slow boots: separate startup probe so liveness doesn't kill during init.

Rules of thumb

Do	Don't
Cheap liveness	Heavy DB work on livez
Ready=false while draining	Keep ready=true until process death
Bound check timeouts	Hang probe handlers

Try next

1. Add `/livez` + `/readyz` to a service.
2. Fail readiness when DB ping fails; keep liveness OK.
3. Wire drain on SIGTERM; curl readyz during shutdown.

Alerting and Runbooks

Alerting and Runbooks

Overview

Alerts should be **actionable, rare, and tied to user pain**. Every page should have a short **runbook**: what it means, what to check, how to mitigate.

Alert design

Prefer	Avoid
SLO burn (error budget)	Raw CPU > 80% forever
Symptom (errors, latency)	Cause-only (GC pause) without user impact
Multi-window burn rates	Noisy single-sample thresholds

Example intent:

```
Alert if 5m error rate > 5% AND 1h error rate > 1%
→ catches both spikes and slow burns
```

Severity

Level	Response
page	Human now
ticket	Next business hours
log-only	Not an alert

If nobody will act at 3 a.m., it is not a page.

Runbook template

```
# HighErrorRate

## Symptom
5xx rate above SLO burn threshold.

## Impact
Users fail to load checkout / API clients retry.

## Quick checks
1. Deploy recently? (`kubectl rollout history`)
2. Dependency errors in logs (filter `dependency=`)
3. Saturation: CPU, threads, DB connections

## Mitigations
1. Rollback last deploy if correlated
2. Shed non-critical traffic / enable rate limit
3. Scale if clearly capacity-bound

## Escalation
Platform on-call if cluster-wide
```

Linking code to ops

Log a stable **error code** and metric label (`code=db_timeout`) so runbooks can search.

```
slog.Error("checkout_failed", "code", "db_timeout", "err", err)
```

Rules of thumb

Do	Don't
One runbook per paging alert	Page without owner
Test alerts in staging	Alert on untested promql
Review flappy alerts weekly	Mute forever without fixing

Try next

1. Write runbooks for two real alerts.

2. Delete or downgrade one noisy alert.

3. Add `code=` to top error logs.

Flight Recorder: Trace the Seconds Before an Incident

Flight Recorder: Trace the Seconds Before an Incident

An execution trace answers the question metrics and profiles cannot: *what was every goroutine doing, and why was it not running?* The problem in a long-lived service is timing. By the time a p99 alert fires, the interesting moment is gone, and running `trace.Start` around the whole process for days produces gigabytes nobody will read. Since Go **1.25**, `runtime/trace.FlightRecorder` keeps the most recent few seconds of trace data in a ring buffer in memory and writes it out only when you ask. The program that notices the problem (a slow request, a failed health check, a timeout) takes the snapshot itself. Everything here runs unchanged on Go **1.27.1**, the current stable release.

Mental model

- **A moving window, not a recording.** The runtime produces the normal execution trace, but the flight recorder keeps only the tail of it. `WriteTo(w)` copies the current window to `w` as an ordinary trace file. `go tool trace` reads it like any other trace.
- **Two knobs, both hints.** `MinAge` is how far back the window should reach (keep events at least this old). `MaxBytes` caps the buffer and **wins** when the two disagree. The Go team suggests `MinAge` of about **2x the window you are debugging**: for a 5 s timeout, keep 10 s. A busy service produces on the order of a few MB of trace per second, so size `MaxBytes` from that.
- **One at a time.** Only one flight recorder may be active per process, and only one `WriteTo` may run at a time. A flight recorder can run alongside a normal `trace.Start` consumer.
- **The trigger is your code.** Nothing fires automatically. You decide which condition is worth a snapshot (latency over a threshold, error spike, an operator request) and you rate-limit it.
- **Cost.** Go 1.21 cut tracer overhead to a small percentage of CPU for most programs. The flight recorder adds memory for the window, but no disk or network traffic until you snapshot.

Tool	Question it answers	When it runs
CPU / heap profile	Where is CPU or memory spent, in aggregate?	Sampled window you choose
`trace.Start` / `go test -trace`	Full timeline of a test, benchmark, or short CLI	Start to Stop
`FlightRecorder`	Timeline of the **last N seconds** before something went wrong	Always on, written on demand

Worked examples

Case 1: Start, snapshot, stop

Save as `fr_basics.go`. This is the whole API: configure, `Start`, `WriteTo`, `Stop`. It also shows the one-recorder-per-process rule.

```go
// fr_basics.go
package main

import (
    "bytes"
    "fmt"
    "os"
    "runtime/trace"
    "time"
)

func busy(d time.Duration) int {
    n := 0
    for end := time.Now().Add(d); time.Now().Before(end); {
        n++
    }
    return n
}

func main() {
    fr := trace.NewFlightRecorder(trace.FlightRecorderConfig{
        MinAge:   2 * time.Second,
        MaxBytes: 4 << 20, // 4 MiB hint
    })
    if err := fr.Start(); err != nil {
        fmt.Println("start:", err)
        return
    }
    defer fr.Stop()
    fmt.Println("enabled:", fr.Enabled())

    for range 3 {
        go busy(20 * time.Millisecond)
        time.Sleep(30 * time.Millisecond)
    }

    var buf bytes.Buffer
    n, err := fr.WriteTo(&buf)
    fmt.Printf("snapshot: err=%v nonEmpty=%t sameAsReturned=%t\n", err, n > 0, int64(buf.Len
    if err := os.WriteFile("snapshot.trace", buf.Bytes(), 0o644); err != nil {
```

```go
        fmt.Println("write:", err)
    }

    second := trace.NewFlightRecorder(trace.FlightRecorderConfig{})
    fmt.Println("second recorder start:", second.Start())
}
```

Run:

```bash
go run fr_basics.go
go tool trace -d=footprint snapshot.trace | awk 'NR<=8'
```

Output (byte counts vary from run to run):

```
enabled: true
snapshot: err=<nil> nonEmpty=true sameAsReturned=true
second recorder start: flight recorder already enabled
Event                Bytes  %       Count  %
-                    -      -       -      -
String               2285   62.14%  76     36.54%
Stack                665    18.09%  22     10.58%
EventBatch           128    3.48%   8      3.85%
ProcStart            84     2.28%   15     7.21%
HeapAlloc            76     2.07%   12     5.77%
GoStatusStack        76     2.07%   5      2.40%
```

`-d=footprint` is a quick check that a file is a real trace and what fills it. For the interactive timeline, run `go tool trace snapshot.trace` and open the printed URL. In Go 1.27, `-http=:6060` binds to localhost only. Pass `-http=0.0.0.0:6060` only if you really mean to expose it.

Case 2: The service snapshots its own slow request

Save as `fr_slow_request.go`. A small stock service answers `GET /stock/{item}` in microseconds. A background goroutine pushes counters upstream every 300 ms, and it holds the same mutex for the whole 120 ms "RPC". Middleware times each request. The first request over 50 ms triggers **one** snapshot (`atomic.Bool` guard), taken on a separate goroutine so the request path never waits on disk. A `trace.WithRegion` labels the upstream call so it is easy to find in the trace.

```go
// fr_slow_request.go
package main

import (
    "context"
    "fmt"
    "io"
    "log"
```

```
    "net/http"
    "net/http/httptest"
    "os"
    "runtime/trace"
    "sync"
    "sync/atomic"
    "time"
)

type stock struct {
    mu    sync.Mutex
    items map[string]int
}

// pushUpstream copies the counters to a slow remote system.
// BUG: it holds the lock for the whole network call.
func (s *stock) pushUpstream(ctx context.Context) {
    s.mu.Lock()
    defer s.mu.Unlock()
    trace.WithRegion(ctx, "pushUpstream.rpc", func() {
        time.Sleep(120 * time.Millisecond) // pretend RPC with s.items
    })
}

func (s *stock) get(w http.ResponseWriter, r *http.Request) {
    s.mu.Lock()
    n := s.items[r.PathValue("item")]
    s.mu.Unlock()
    fmt.Fprintf(w, "%d\n", n)
}

// recorder takes at most one snapshot per process lifetime.
type recorder struct {
    fr   *trace.FlightRecorder
    path string
    done atomic.Bool
}

func (rec *recorder) capture(reason string) {
    if !rec.done.CompareAndSwap(false, true) {
        return
    }
    f, err := os.Create(rec.path)
    if err != nil {
        log.Println("snapshot:", err)
        return
    }
```

```
    defer f.Close()
    if _, err := rec.fr.WriteTo(f); err != nil {
        log.Println("snapshot:", err)
        return
    }
    fmt.Printf("snapshot: reason=%q file=%s\n", reason, rec.path)
}

// slowGuard times every request and triggers a snapshot past the threshold.
func slowGuard(next http.Handler, limit time.Duration, rec *recorder) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        start := time.Now()
        next.ServeHTTP(w, r)
        if time.Since(start) > limit && rec.fr.Enabled() {
            fmt.Printf("slow request: %s over %s\n", r.URL.Path, limit)
            go rec.capture("latency>" + limit.String())
        }
    })
}

func main() {
    fr := trace.NewFlightRecorder(trace.FlightRecorderConfig{
        MinAge:   time.Second, // about 2x the window you care about
        MaxBytes: 8 << 20,     // hint, not a hard cap
    })
    if err := fr.Start(); err != nil {
        log.Fatal(err)
    }
    defer fr.Stop()
    rec := &recorder{fr: fr, path: "slow-request.trace"}

    s := &stock{items: map[string]int{"apple": 3}}
    go func() {
        for range time.Tick(300 * time.Millisecond) {
            s.pushUpstream(context.Background())
        }
    }()

    mux := http.NewServeMux()
    mux.HandleFunc("GET /stock/{item}", s.get)
    srv := httptest.NewServer(slowGuard(mux, 50*time.Millisecond, rec))
    defer srv.Close()

    for deadline := time.Now().Add(2 * time.Second); time.Now().Before(deadline) && !rec.don
        resp, err := http.Get(srv.URL + "/stock/apple")
        if err != nil {
```

```
            log.Fatal(err)
        }
        io.Copy(io.Discard, resp.Body)
        resp.Body.Close()
        time.Sleep(5 * time.Millisecond)
    }
    time.Sleep(200 * time.Millisecond) // let capture finish writing
    if !rec.done.Load() {
        fmt.Println("no request over the limit in 2s; no snapshot")
    }
}
```

Run:

```
go run fr_slow_request.go
```

Output:

```
slow request: /stock/apple over 50ms
snapshot: reason="latency>50ms" file=slow-request.trace
```

The window holds the request **and** the second before it, so the cause is in the file even though the alarm fired after the fact. Turn the trace into a synchronization-blocking profile and ask who waited on a mutex:

```
go tool trace -pprof=sync slow-request.trace > sync.pprof
go tool pprof -top -focus='Mutex' sync.pprof
```

Output (durations vary):

```
Main binary filename not available.
Type: delay
Active filters:
   focus=Mutex
Showing nodes accounting for 119.50ms, 10.58% of 1128.96ms total
      flat  flat%   sum%        cum   cum%
  119.50ms 10.58% 10.58%   119.50ms 10.58%  sync.(*Mutex).Lock
         0     0% 10.58%   119.50ms 10.58%  main.(*stock).get
         0     0% 10.58%   119.50ms 10.58%  main.slowGuard.func1
         0     0% 10.58%   119.50ms 10.58%  net/http.(*ServeMux).ServeHTTP
         0     0% 10.58%   119.50ms 10.58%  net/http.(*conn).serve
         0     0% 10.58%   119.50ms 10.58%  net/http.HandlerFunc.ServeHTTP
         0     0% 10.58%   119.50ms 10.58%  net/http.serverHandler.ServeHTTP
```

The handler spent about 120 ms blocked in `Mutex.Lock`: the whole slow request. The other `-pprof` modes are `net`, `syscall`, and `sched`. To see **who held** the lock, look for the labelled region. The text dump of events is enough on a headless box:

```
go tool trace -d=parsed slow-request.trace > events.txt
grep -m1 -A5 'RegionBegin.*pushUpstream' events.txt
```

Output (from a run in `/tmp/flight`; IDs, time, and PCs vary):

```
M=665582 P=5 G=11 RegionBegin Time=144831167709632 Task=0 Type="pushUpstream.rpc"
Stack=
    main.(*stock).pushUpstream @ 0x66f568
        /tmp/flight/fr_slow_request.go:28
    main.main.func1 @ 0x670404
        /tmp/flight/fr_slow_request.go:90
```

In the browser view (`go tool trace slow-request.trace`, then *View trace by proc*), goroutine 11 sits in `pushUpstream.rpc` for 120 ms while request goroutines pile up behind it, and the flow arrow from its `Unlock` wakes them. The fix: copy under the lock, release it, then do the slow call. Replace `pushUpstream` with:

```go
// fr_slow_request.go (fixed pushUpstream; add "maps" to the imports)
// pushUpstream copies the counters under the lock, then talks to the
// slow remote system with the lock released.
func (s *stock) pushUpstream(ctx context.Context) {
    s.mu.Lock()
    batch := maps.Clone(s.items)
    s.mu.Unlock()
    trace.WithRegion(ctx, "pushUpstream.rpc", func() {
        time.Sleep(120 * time.Millisecond) // pretend RPC with batch
        _ = batch
    })
}
```

Run again:

```
go run fr_slow_request.go
```

Output:

```
no request over the limit in 2s; no snapshot
```

Case 3: An operator endpoint for on-demand snapshots

Save as `fr_admin.go`. Sometimes a human notices first ("it feels slow right now"). Expose `WriteTo` on a **separate, loopback-only** admin listener, never the public mux. `TryLock` turns a second concurrent request into a clean `429` instead of a half-written body.

```go
// fr_admin.go
package main
```

```go
import (
    "fmt"
    "log"
    "net/http"
    "runtime/trace"
    "sync"
    "time"
)

// snapshotHandler streams the flight recorder window to the caller.
// One snapshot at a time; extra callers get 429 instead of an error mid-body.
func snapshotHandler(fr *trace.FlightRecorder) http.HandlerFunc {
    var mu sync.Mutex
    return func(w http.ResponseWriter, r *http.Request) {
        if !mu.TryLock() {
            http.Error(w, "snapshot already in progress", http.StatusTooManyRequests)
            return
        }
        defer mu.Unlock()
        if !fr.Enabled() {
            http.Error(w, "flight recorder off", http.StatusServiceUnavailable)
            return
        }
        name := fmt.Sprintf("flight-%s.trace", time.Now().UTC().Format("20060102T150405Z"))
        w.Header().Set("Content-Type", "application/octet-stream")
        w.Header().Set("Content-Disposition", `attachment; filename="`+name+`"`)
        if _, err := fr.WriteTo(w); err != nil {
            log.Println("snapshot:", err) // headers are gone; just log
        }
    }
}

func main() {
    fr := trace.NewFlightRecorder(trace.FlightRecorderConfig{
        MinAge:   10 * time.Second,
        MaxBytes: 32 << 20,
    })
    if err := fr.Start(); err != nil {
        log.Fatal(err)
    }

    app := http.NewServeMux()
    app.HandleFunc("GET /", func(w http.ResponseWriter, r *http.Request) {
        fmt.Fprintln(w, "hello")
    })
```

```go
    admin := http.NewServeMux()
    admin.Handle("GET /debug/flightrecorder", snapshotHandler(fr))

    go func() { log.Fatal(http.ListenAndServe("127.0.0.1:6061", admin)) }()
    log.Println("app on 127.0.0.1:8080, admin on 127.0.0.1:6061")
    log.Fatal(http.ListenAndServe("127.0.0.1:8080", app))
}
```

Run in one terminal:

```bash
go run fr_admin.go
```

In another, make some traffic, pull a snapshot, and check it:

```bash
for i in $(seq 50); do curl -s localhost:8080/ >/dev/null; done
curl -s -D - -o flight.trace http://127.0.0.1:6061/debug/flightrecorder
go tool trace -d=footprint flight.trace | awk 'NR<=6'
```

Output (date and sizes vary):

```
HTTP/1.1 200 OK
Content-Disposition: attachment; filename="flight-20261007T033257Z.trace"
Content-Type: application/octet-stream
Date: Wed, 07 Oct 2026 03:32:57 GMT
Transfer-Encoding: chunked

Event                Bytes  %       Count  %
-                    -      -       -      -
String               9535   34.82%  311    9.59%
Stack                6122   22.36%  90     2.78%
GoSyscallBegin       3122   11.40%  647    19.95%
ProcStart            1683   6.15%   276    8.51%
```

The server's log shows:

```
2026/10/07 09:02:55 app on 127.0.0.1:8080, admin on 127.0.0.1:6061
```

A trace contains stack traces, goroutine labels, and region names. Treat it like a heap dump: internal, access-controlled, and short-lived.

The trap

Save as `fr_trap.go`. Ten slow requests arrive in the same instant and each one calls `WriteTo` directly. Only one snapshot can be in progress, so the rest fail. Calling `WriteTo` after `Stop` also fails.

```go
// fr_trap.go
package main

import (
    "fmt"
    "io"
    "maps"
    "runtime/trace"
    "slices"
    "sync"
    "time"
)

// slowWriter simulates shipping a snapshot to a slow disk or network.
type slowWriter struct{}

func (slowWriter) Write(p []byte) (int, error) {
    time.Sleep(50 * time.Millisecond)
    return len(p), nil
}

func main() {
    fr := trace.NewFlightRecorder(trace.FlightRecorderConfig{MinAge: time.Second})
    if err := fr.Start(); err != nil {
        fmt.Println(err)
        return
    }
    time.Sleep(50 * time.Millisecond)

    // Ten "slow request" alarms fire at once and each tries to snapshot.
    var (
        wg      sync.WaitGroup
        mu      sync.Mutex
        results = map[string]int{}
    )
    for range 10 {
        wg.Go(func() {
            key := "ok"
            if _, err := fr.WriteTo(slowWriter{}); err != nil {
                key = err.Error()
            }
            mu.Lock()
            results[key]++
            mu.Unlock()
        })
    }
```

```go
    wg.Wait()
    for _, k := range slices.Sorted(maps.Keys(results)) {
        fmt.Printf("%2d x %s\n", results[k], k)
    }

    fr.Stop()
    _, err := fr.WriteTo(io.Discard)
    fmt.Println("after Stop:", err)
}
```

Run:

```go
go run fr_trap.go
```

Output:

```
 9 x call to WriteTo for trace.FlightRecorder already in progress
 1 x ok
after Stop: cannot snapshot a disabled flight recorder
```

In a real incident the slow-request condition fires for **every** request at once. A trigger without a guard spams errors at best. At worst it competes for the recorder exactly when you need one clean file. Case 2's `atomic.Bool` (once per process) and Case 3's `TryLock` (one at a time) are the two boring shapes. For a long-running service, use a cooldown (one snapshot per N minutes) and a cap on files kept on disk.

Second trap: `MinAge` shorter than the problem. With `MinAge: 100*time.Millisecond`, a snapshot taken after a 2 s timeout shows the end of the timeout and none of its cause. Set it to about twice the slowest thing you alert on, then let `MaxBytes` bound memory.

Third trap: assuming a snapshot is enough to tell **who** held a lock. The sync profile tells you who *waited*. Add `trace.WithRegion` / `trace.NewTask` around suspicious critical sections and slow calls so the holder has a name in the timeline.

The boring rule

- Start one `FlightRecorder` in `main`, next to your logger and metrics. Set `MinAge` 2x your alert window, `MaxBytes` from a measured MB/s, and `Stop` it on shutdown.
- Trigger snapshots from code that already knows something is wrong (latency guard, error path, failed probe). Guard the trigger: once, cooldown, or `TryLock`. Never block the request path on `WriteTo`.
- Expose on-demand snapshots only on a loopback or authenticated admin listener.
- Analyse with `go tool trace` (timeline) and `go tool trace -pprof=sync|net|syscall|sched` (text profiles). Label hot sections with regions so the timeline names the culprit.
- Keep traces short-lived and internal; they contain stacks and labels from your code.

Try this

1. In `fr_slow_request.go`, change the trigger to fire on **any 5xx** as well as latency, and include the request ID in the snapshot file name.
2. Lower `MinAge` to `50*time.Millisecond` and `MaxBytes` to `64 << 10`, take the snapshot, and check whether `pushUpstream.rpc` is still in the window.
3. Add a cooldown to `recorder` (one snapshot per 10 minutes, at most 5 files on disk) and test it with a loop that triggers 100 times.
4. Run `go tool trace -pprof=sched slow-request.trace` and explain which goroutines show scheduler latency and why.
5. Wrap the stock lookup in `trace.NewTask(ctx, "getStock")` and find the tasks in the browser view's *User-defined tasks* page.

Project: Instrumented Mini Service

Project: Instrumented Mini Service

Overview

Ship a tiny HTTP service with **slog JSON**, **request IDs**, **RED-ish expvar/Prometheus counters**, **livez/readyz**, and a **shutdown drain**.

Checklist

- ☐ `GET /api/echo?msg=` returns JSON
- ☐ Middleware: request ID + access log
- ☐ Counters: requests_total, requests_failed, latency histogram (or expvar)
- ☐ `/livez` always 200 if process up
- ☐ `/readyz` fails when `-unready` file exists or after SIGTERM
- ☐ Graceful Shutdown
- ☐ README: how to curl probes + sample log line

Stretch

- OpenTelemetry trace stub
- Prometheus `/metrics` via `promhttp`
- SLO doc: 99% success, p99 < 200ms on echo

Acceptance

```
go test ./...
go run . &
curl -s localhost:8080/livez
```

```
curl -s localhost:8080/api/echo?msg=hi
kill -TERM $pid  # readyz → 503, then exit
```

Cloud and Infrastructure

Containerization (Docker)

Containerization: Multi-Stage Builds

Overview

Go compiles to a static binary. That is the deployment superpower: production does not need the Go toolchain, a language runtime, or a package manager. Multi-stage Docker builds turn that property into tiny, hard-to-attack images—typically tens of megabytes instead of hundreds.

This chapter covers the production multi-stage pattern, static linking, distroless vs scratch, CA certs and timezones, non-root runtime, healthchecks, and a release checklist.

Why Multi-Stage Matters

A naive Dockerfile copies the whole module tree into a fat `golang` image and runs the binary there. That ships the compiler, shell, package manager, and build tools into production—large attack surface, slow pulls, noisy CVEs.

Multi-stage splits the world in two:

1. **Builder** — full Go toolchain; compile once.
2. **Runtime** — only the binary (and maybe CA certs / tzdata). No shell, no package manager.

```
source + go.mod
       |
       v
  [builder stage]  go build  -->  static binary
       |
       v
  [runtime stage]  copy binary only  -->  ~10-20 MB image
```

The Production Multi-Stage Pattern

```
# syntax=docker/dockerfile:1

# ---------- Stage 1: Build ----------
FROM golang:1.27-bookworm AS builder

WORKDIR /src
```

```
# Cache modules separately from source for faster rebuilds.
COPY go.mod go.sum ./
RUN go mod download

COPY . .

# CGO_ENABLED=0  => pure static binary (no libc dependency)
# -trimpath      => strip local paths from the binary
# -ldflags="-s -w" => strip symbol table and DWARF
ARG TARGETOS=linux
ARG TARGETARCH=amd64
RUN CGO_ENABLED=0 GOOS=${TARGETOS} GOARCH=${TARGETARCH} \
    go build -trimpath -ldflags="-s -w" -o /out/server ./cmd/server

# ---------- Stage 2: Runtime ----------
# distroless/static includes CA certs + timezone data; no shell.
FROM gcr.io/distroless/static-debian12:nonroot

COPY --from=builder /out/server /server

USER nonroot:nonroot
EXPOSE 8080
ENTRYPOINT ["/server"]
```

Build and inspect:

```
docker build -t myapp:local .
docker images myapp:local
docker run --rm -p 8080:8080 myapp:local
```

Static Linking Essentials

Flag / env	Effect
`CGO_ENABLED=0`	Pure Go binary; runs without glibc/musl
`-trimpath`	Removes host filesystem paths from the binary
`-ldflags="-s -w"`	Smaller binary; harder casual reverse engineering
`-buildmode=pie`	Position-independent executable (ASLR-friendly)

If you **must** use CGO (SQLite via C, certain crypto, etc.):

```
FROM golang:1.27-bookworm AS builder
RUN apt-get update && apt-get install -y --no-install-recommends gcc libc6-dev
# ...
RUN CGO_ENABLED=1 go build -o /out/server ./cmd/server

FROM gcr.io/distroless/base-debian12:nonroot
COPY --from=builder /out/server /server
```

Prefer `CGO_ENABLED=0` whenever the dependency graph allows it.

A glibc-linked binary copied onto Alpine fails with `not found` because the ELF interpreter (`/lib64/ld-linux-x86-64.so.2`) is missing — not because your file vanished. That is the same trap as shipping a Ubuntu C tool onto Alpine. Pure Go sidesteps it: on Linux the runtime issues syscalls itself, so `scratch` and `distroless/static` have no libc to provide. See Go as a Systems Language.

Scratch vs Distroless vs Alpine

Base	Size	Shell	CA certs	Best for
`scratch`	~0 + binary	No	No	Fully static apps that never call HTTPS / LoadLocation
`distroless/static`	~2 MB + binary	No	Yes	Most production Go services
`distroless/base`	larger	No	Yes	CGO / dynamic glibc needs
`alpine`	~5–8 MB	Yes	Optional	Debugging, ops scripts (larger attack surface)

Rule of thumb: start with `distroless/static-debian12:nonroot`. Drop to `scratch` only when you have proven CA/tz needs are handled and you want absolute minimalism.

CA Certificates and Timezones

`scratch` fails the moment your app dials `https://…` or calls `time.LoadLocation("America/New_York")`.

Manual copy if you insist on scratch:

```
FROM golang:1.27-bookworm AS builder
# ... build ...

FROM scratch
COPY --from=builder /etc/ssl/certs/ca-certificates.crt /etc/ssl/certs/
COPY --from=builder /usr/share/zoneinfo /usr/share/zoneinfo
```

```
COPY --from=builder /out/server /server
ENTRYPOINT ["/server"]
```

In application code, prefer UTC in storage and convert only at the edge:

```
func main() {
    // Prefer UTC in process; convert at presentation boundaries.
    time.Local = time.UTC

    loc, err := time.LoadLocation("America/Los_Angeles")
    if err != nil {
        log.Fatal(err) // fail closed if tzdata missing in image
    }
    _ = loc
}
```

Non-Root, Health, and Signals

Distroless `:nonroot` images run as UID 65532. Match that in compose/k8s security contexts.

```
# docker-compose snippet
services:
  api:
    image: myapp:local
    read_only: true
    security_opt:
      - no-new-privileges:true
    cap_drop:
      - ALL
    ports:
      - "8080:8080"
```

Handle `SIGTERM` so orchestrators get clean shutdown:

```
package main

import (
    "context"
    "log"
    "net/http"
    "os"
    "os/signal"
    "syscall"
    "time"
)
```

```go
func main() {
    mux := http.NewServeMux()
    mux.HandleFunc("/healthz", func(w http.ResponseWriter, _ *http.Request) {
        w.WriteHeader(http.StatusOK)
        _, _ = w.Write([]byte("ok"))
    })
    mux.HandleFunc("/readyz", func(w http.ResponseWriter, _ *http.Request) {
        // Check DB/cache if needed; fail closed when not ready.
        w.WriteHeader(http.StatusOK)
    })

    srv := &http.Server{
        Addr:              ":8080",
        Handler:           mux,
        ReadHeaderTimeout: 5 * time.Second,
    }

    go func() {
        log.Printf("listening on %s", srv.Addr)
        if err := srv.ListenAndServe(); err != nil && err != http.ErrServerClosed {
            log.Fatal(err)
        }
    }()

    stop := make(chan os.Signal, 1)
    signal.Notify(stop, syscall.SIGINT, syscall.SIGTERM)
    <-stop

    ctx, cancel := context.WithTimeout(context.Background(), 15*time.Second)
    defer cancel()
    if err := srv.Shutdown(ctx); err != nil {
        log.Printf("shutdown: %v", err)
    }
}
```

Build Cache and Layer Hygiene

- Copy `go.mod` / `go.sum` before source so module download caches across code-only changes.
- Use `.dockerignore` aggressively:

```
.git
_book
**/*_test.go
**/.DS_Store
*.md
```

```
.env
tmp/
```

- Multi-arch (Buildx):

```
docker buildx build \
  --platform linux/amd64,linux/arm64 \
  -t ghcr.io/example/myapp:1.2.3 \
  --push .
```

Pass `TARGETOS` / `TARGETARCH` (BuildKit sets them automatically when using `ARG TARGETOS` / `ARG TARGETARCH`).

Embed Version Metadata

```go
// cmd/server/main.go
var (
    version = "dev"
    commit  = "none"
)

func main() {
    log.Printf("starting version=%s commit=%s", version, commit)
}
```

```
ARG VERSION=dev
ARG COMMIT=none
RUN CGO_ENABLED=0 go build -trimpath \
    -ldflags="-s -w -X main.version=${VERSION} -X main.commit=${COMMIT}" \
    -o /out/server ./cmd/server
```

Production Checklist

- ☐ Multi-stage: builder + minimal runtime (`distroless` or carefully prepared `scratch`)
- ☐ `CGO_ENABLED=0` unless CGO is required and documented
- ☐ `-trimpath` and `-ldflags="-s -w"` on release builds
- ☐ Non-root user (`:nonroot` or explicit `USER`)
- ☐ CA certs present if the app makes HTTPS calls
- ☐ Timezone data present if `LoadLocation` is used
- ☐ `.dockerignore` excludes secrets, tests, and junk
- ☐ `/healthz` and `/readyz` (or platform equivalents)
- ☐ Graceful shutdown on `SIGTERM`
- ☐ Image scanned in CI (`trivy`, `grype`, or registry scanning)
- ☐ Tags immutable (`1.2.3`, digest)—avoid relying only on `latest`

Common Pitfalls

1. **Shell debugging in distroless** — there is no `/bin/sh`. Debug with a sidecar, ephemeral debug image, or temporary Alpine stage—not by shipping a shell to prod.
2. **CGO by accident** — a transitive dependency may force CGO; verify with `go build -x` or `ldd` (static binary should not need dynamic libs).
3. **Wrong GOOS/GOARCH** — building on Apple Silicon without setting `GOARCH` can produce arm64 images that fail on amd64 clusters.
4. **Huge context** — missing `.dockerignore` sends `node_modules`, `.git`, and secrets into the daemon.
5. **Listening on localhost only** — inside a container, bind `0.0.0.0:8080` (or `:8080`), not `127.0.0.1`.
6. **Root for convenience** — root + writable filesystem + shell is the classic container escape path.

Exercises

1. **Minimal image** — Take any small `net/http` server and produce a multi-stage Dockerfile using `distroless/static-debian12:nonroot`. Measure image size with `docker images`.
2. **HTTPS smoke test** — Call an external HTTPS API from inside the container. Confirm it works on distroless and fails on bare `scratch` without CA certs.
3. **Multi-arch** — Build amd64 and arm64 images with Buildx. Run the matching arch on your machine or in CI.
4. **Signal drill** — Send `docker stop` and confirm your server logs a clean `Shutdown` within the grace period (not a hard kill after timeout).
5. **Scan** — Run `trivy image myapp:local` (or equivalent) and fix or document any HIGH/CRITICAL findings in base layers.

More examples

Health probe and graceful drain (container contract)

Containers need a cheap liveness probe and a clean SIGTERM path. This demo uses `httptest` so it finishes without a forever-listen loop, and models drain with a cancellable context.

```
mkdir -p /tmp/go-docker-health && cd /tmp/go-docker-health
go mod init example.com/docker-health
```

Save as `main.go`:

```
package main

import (
    "context"
```

```
    "encoding/json"
    "fmt"
    "net/http"
    "net/http/httptest"
    "sync/atomic"
    "time"
)

func main() {
    var ready atomic.Bool
    ready.Set(true)

    mux := http.NewServeMux()
    mux.HandleFunc("GET /healthz", func(w http.ResponseWriter, _ *http.Request) {
        w.WriteHeader(http.StatusOK)
        fmt.Fprintln(w, "ok")
    })
    mux.HandleFunc("GET /readyz", func(w http.ResponseWriter, _ *http.Request) {
        if !ready.Load() {
            http.Error(w, "not ready", http.StatusServiceUnavailable)
            return
        }
        _ = json.NewEncoder(w).Encode(map[string]string{"status": "ready"})
    })
    mux.HandleFunc("GET /work", func(w http.ResponseWriter, r *http.Request) {
        select {
        case <-r.Context().Done():
            return
        case <-time.After(20 * time.Millisecond):
            fmt.Fprintln(w, "done")
        }
    })

    // Simulate drain: mark not-ready, then cancel in-flight work.
    srv := httptest.NewServer(mux)
    defer srv.Close()

    resp, err := http.Get(srv.URL + "/healthz")
    if err != nil {
        panic(err)
    }
    resp.Body.Close()
    fmt.Println("healthz:", resp.StatusCode)

    resp, err = http.Get(srv.URL + "/readyz")
    if err != nil {
        panic(err)
```

```go
    }
    resp.Body.Close()
    fmt.Println("readyz before drain:", resp.StatusCode)

    ready.Store(false)
    resp, err = http.Get(srv.URL + "/readyz")
    if err != nil {
        panic(err)
    }
    resp.Body.Close()
    fmt.Println("readyz during drain:", resp.StatusCode)

    ctx, cancel := context.WithTimeout(context.Background(), 5*time.Millisecond)
    defer cancel()
    req, _ := http.NewRequestWithContext(ctx, http.MethodGet, srv.URL+"/work", nil)
    _, err = http.DefaultClient.Do(req)
    fmt.Println("in-flight cancelled:", err != nil)
}
```

```bash
go run .
```

Expected output:

```
healthz: 200
readyz before drain: 200
readyz during drain: 503
in-flight cancelled: true
```

Bind address from env (not localhost-only)

```bash
mkdir -p /tmp/go-docker-bind && cd /tmp/go-docker-bind
go mod init example.com/docker-bind
```

Save as `main.go`:

```go
package main

import (
    "fmt"
    "net"
    "os"
)

func listenAddr() string {
    if p := os.Getenv("PORT"); p != "" {
```

```go
        return ":" + p
    }
    return ":8080"
}

func main() {
    // Resolve without accepting forever: prove the address form containers expect.
    addr := listenAddr()
    ln, err := net.Listen("tcp", "127.0.0.1"+addr) // local only for the demo
    if err != nil {
        panic(err)
    }
    fmt.Println("configured bind form:", addr)
    fmt.Println("listening:", ln.Addr().String())
    _ = ln.Close()

    os.Setenv("PORT", "9090")
    fmt.Println("with PORT=9090:", listenAddr())
}
```

```go
go run .
```

Expected output:

```
configured bind form: :8080
listening: 127.0.0.1:8080
with PORT=9090: :9090
```

Runnable example

Multi-stage Docker is prose and Dockerfiles; the Go side is a release-shaped binary with version ldflags, a health endpoint, and graceful shutdown—the process you copy into distroless.

```go
mkdir -p /tmp/go-docker-app && cd /tmp/go-docker-app
go mod init example.com/docker-app
```

Save as `main.go`:

```go
package main

import (
    "context"
    "encoding/json"
    "fmt"
    "log"
```

```
    "net"
    "net/http"
    "time"
)

// Injected at link time:
// go build -ldflags="-X main.version=1.2.3 -X main.commit=deadbeef"
var (
    version = "dev"
    commit  = "none"
)

func main() {
    mux := http.NewServeMux()
    mux.HandleFunc("GET /healthz", func(w http.ResponseWriter, _ *http.Request) {
        w.WriteHeader(http.StatusOK)
        _, _ = w.Write([]byte("ok"))
    })
    mux.HandleFunc("GET /version", func(w http.ResponseWriter, _ *http.Request) {
        w.Header().Set("Content-Type", "application/json")
        _ = json.NewEncoder(w).Encode(map[string]string{
            "version": version,
            "commit":  commit,
        })
    })

    ln, err := net.Listen("tcp", "127.0.0.1:0")
    if err != nil {
        log.Fatal(err)
    }
    srv := &http.Server{Handler: mux, ReadHeaderTimeout: 5 * time.Second}

    go func() {
        if err := srv.Serve(ln); err != nil && err != http.ErrServerClosed {
            log.Fatal(err)
        }
    }()

    base := "http://" + ln.Addr().String()
    fmt.Printf("listening on %s  version=%s commit=%s\n", base, version, commit)

    resp, err := http.Get(base + "/healthz")
    if err != nil {
        log.Fatal(err)
    }
    resp.Body.Close()
    fmt.Println("healthz:", resp.StatusCode)
```

```go
    resp, err = http.Get(base + "/version")
    if err != nil {
        log.Fatal(err)
    }
    var v map[string]string
    _ = json.NewDecoder(resp.Body).Decode(&v)
    resp.Body.Close()
    fmt.Printf("version payload: %v\n", v)

    shutdownCtx, cancel := context.WithTimeout(context.Background(), 2*time.Second)
    defer cancel()
    if err := srv.Shutdown(shutdownCtx); err != nil {
        log.Fatal(err)
    }
    fmt.Println("shutdown complete")
}
```

```go
go run -ldflags="-X main.version=1.2.3 -X main.commit=deadbeef" .
# release-shaped binary (what multi-stage copies):
# CGO_ENABLED=0 go build -trimpath -ldflags="-s -w -X main.version=1.2.3 -X main.commit=dead
```

Expected output (illustrative):

```
listening on http://127.0.0.1:54321  version=1.2.3 commit=deadbeef
healthz: 200
version payload: map[commit:deadbeef version:1.2.3]
shutdown complete
```

What to notice

- Version injection is link-time (`-X`); the runtime image never needs the Go toolchain.
- `/healthz` is what Docker `HEALTHCHECK` and orchestrators probe—keep it cheap and dependency-light.
- `Shutdown` is the process side of multi-stage ops: `docker stop` / Kubernetes send `SIGTERM` and expect a drain.

Try next

- Wire `signal.NotifyContext` and run under `docker stop` to confirm clean exit within the grace period.
- Wrap this binary in the multi-stage Dockerfile from this chapter; compare size to a single-stage `golang` image.

Related Topics

- Building and Releasing

- CI/CD & Hardening
- Application Hardening

Platform Guides: Render & Leapcell

Deploying Go: Render, Leapcell, and VPS

Overview

Go's static binary makes PaaS and VPS deployment straightforward: build once, run the file. In 2026 the common paths are managed platforms (Render, Leapcell, Fly, Railway) for speed, and a small VPS with systemd when you want full control and predictable cost.

This chapter walks through Render, Leapcell-style serverless Go, a production systemd unit, environment and secrets, health checks, and when to choose each model.

Deployment Models Compared

Model	Cold start	Cost shape	Ops burden	Good for
PaaS (Render web service)	Warm (always-on or min instances)	Monthly / usage	Low	APIs, webhooks, workers
Serverless binary (Leapcell, similar)	Milliseconds–low seconds	Scale-to-zero	Very low	Spiky or low-traffic APIs
VPS + systemd	None (always running)	Flat $	Medium	Homelab, steady load, custom networking
Containers on K8s	Depends	Cluster fixed + usage	High	Multi-service platforms

Go fits all four: same binary shape, different process supervisors.

Render (render.com)

Render detects Go repositories and builds with the Go toolchain on their builders.

Typical setup

1. Push the repo to GitHub/GitLab.
2. New **Web Service** → connect the repo.
3. Configure:

Field	Example
Runtime	Go
Build Command	`go build -o bin/server ./cmd/server`
Start Command	`./bin/server`
Health Check Path	`/healthz`

Production build command

Prefer flags you would use locally for release:

```
CGO_ENABLED=0 go build -trimpath -ldflags="-s -w" -o bin/server ./cmd/server
```

If you need a specific module path:

```
go build -o bin/server ./cmd/api
```

Environment and ports

Render injects `PORT`. **Always bind to `PORT`**, not a hard-coded 8080:

```go
package main

import (
    "log"
    "net/http"
    "os"
    "time"
)

func main() {
    port := os.Getenv("PORT")
    if port == "" {
        port = "8080"
    }

    mux := http.NewServeMux()
    mux.HandleFunc("/healthz", func(w http.ResponseWriter, _ *http.Request) {
        w.WriteHeader(http.StatusOK)
```

```go
        _, _ = w.Write([]byte("ok"))
    })
    mux.HandleFunc("/", func(w http.ResponseWriter, r *http.Request) {
        _, _ = w.Write([]byte("hello from render"))
    })

    srv := &http.Server{
        Addr:              ":" + port,
        Handler:           mux,
        ReadHeaderTimeout: 5 * time.Second,
    }
    log.Printf("listening on %s", srv.Addr)
    log.Fatal(srv.ListenAndServe())
}
```

Databases and private networking

- Managed Postgres/Redis: set `DATABASE_URL` / `REDIS_URL` as env vars (never commit them).
- Private services: talk over the internal hostname Render provides; do not expose internal ports publicly.
- Migrations: run as a **one-off job** or release command, not on every web dyno boot if multiple instances race.

```bash
# Example release / pre-deploy idea (platform-specific)
./bin/migrate up
./bin/server
```

Pros and cons

Pros: Free/hobby tiers, managed TLS, Git-driven deploys, managed data stores, low ops.

Cons: Less control than a VPS; cold starts on free tiers; build minutes and instance sizes are the cost levers.

Leapcell (and Serverless Go)

Leapcell targets **serverless Go**: scale to zero when idle, spin up quickly (often Firecracker-style microVMs). Unlike AWS Lambda's custom handler contract, many of these platforms run a normal `net/http` server binary.

Conceptual difference from Lambda

```
AWS Lambda:   event JSON --> handler(ctx, event) --> response JSON
Leapcell-like: cold start binary --> listen :PORT --> ordinary HTTP
```

You keep idiomatic Go HTTP code, middleware, and routers.

Config sketch

```yaml
# leapcell.yaml (illustrative)
runtime: go
entrypoint: ./server
# memory / region set in dashboard or CI
```

Build locally the same way:

```
CGO_ENABLED=0 go build -trimpath -ldflags="-s -w" -o server ./cmd/server
```

Design for cold starts

1. **Defer heavy work** — open DB pools on first request or in a short `init` with timeouts; avoid multi-second global init.
2. **Small binary** — strip symbols; avoid huge embedded assets unless needed.
3. **Idempotent migrations** — do not migrate schema on every cold start.
4. **Stateless handlers** — session state in Redis/DB/cookies, not process memory alone.

```go
var dbOnce sync.Once
var db *sql.DB
var dbErr error

func getDB(ctx context.Context) (*sql.DB, error) {
    dbOnce.Do(func() {
        dsn := os.Getenv("DATABASE_URL")
        db, dbErr = sql.Open("pgx", dsn)
        if dbErr != nil {
            return
        }
        db.SetMaxOpenConns(5)
        db.SetConnMaxLifetime(5 * time.Minute)
        pingCtx, cancel := context.WithTimeout(ctx, 3*time.Second)
        defer cancel()
        dbErr = db.PingContext(pingCtx)
    })
    return db, dbErr
```

```
}
```

VPS Deployment with systemd

For a $5–15/mo Linux VPS (Hetzner, DigitalOcean, Linode):

1. Provision a non-root user

```
sudo useradd --system --home /var/lib/myapp --shell /usr/sbin/nologin myapp
sudo mkdir -p /var/lib/myapp /etc/myapp
sudo chown myapp:myapp /var/lib/myapp
```

2. Install the binary

```
CGO_ENABLED=0 GOOS=linux GOARCH=amd64 go build -trimpath -ldflags="-s -w" -o myapp ./cmd/ser
scp myapp deploy@host:/tmp/myapp
ssh deploy@host 'sudo install -o myapp -g myapp -m 0755 /tmp/myapp /usr/local/bin/myapp'
```

3. Environment file

```
# /etc/myapp.env  (mode 0640, root:myapp)
PORT=8080
DATABASE_URL=postgres://...
LOG_LEVEL=info
```

4. Unit file

```
# /etc/systemd/system/myapp.service
[Unit]
Description=My Go App
After=network-online.target
Wants=network-online.target

[Service]
Type=simple
User=myapp
Group=myapp
EnvironmentFile=/etc/myapp.env
WorkingDirectory=/var/lib/myapp
```

```
ExecStart=/usr/local/bin/myapp
Restart=on-failure
RestartSec=2
# Hardening
NoNewPrivileges=true
ProtectSystem=strict
ProtectHome=true
PrivateTmp=true
ReadWritePaths=/var/lib/myapp
AmbientCapabilities=
CapabilityBoundingSet=
LimitNOFILE=65536

[Install]
WantedBy=multi-user.target
```

```
sudo systemctl daemon-reload
sudo systemctl enable --now myapp
sudo journalctl -u myapp -f
```

5. Reverse proxy (Caddy or nginx)

Terminate TLS at the edge; proxy to `127.0.0.1:8080`:

```
api.example.com {
    reverse_proxy 127.0.0.1:8080
}
```

Health, Metrics, and Zero-Downtime

Regardless of platform:

```
mux.HandleFunc("GET /healthz", func(w http.ResponseWriter, _ *http.Request) {
    w.WriteHeader(http.StatusOK)
})

mux.HandleFunc("GET /readyz", func(w http.ResponseWriter, r *http.Request) {
    if err := pingDependencies(r.Context()); err != nil {
        http.Error(w, "not ready", http.StatusServiceUnavailable)
        return
    }
    w.WriteHeader(http.StatusOK)
})
```

- **Liveness** (`/healthz`): process is up.
- **Readiness** (`/readyz`): safe to receive traffic (DB reachable, migrations done).

For rolling deploys on a VPS, run two units behind the proxy or use a brief drain period with `systemctl reload` patterns and connection draining.

Secrets Hygiene

Do	Don't
Platform secret store / `EnvironmentFile` with tight permissions	Commit `.env` with production credentials
Rotate DB passwords and API keys on a schedule	Share one root DB user across apps
Inject secrets at runtime	Bake secrets into the binary or image layers
Scope tokens to least privilege	Put admin tokens in frontend env

Choosing a Path

```
Need scale-to-zero + almost no ops?  -->  Leapcell-style serverless
Need managed Postgres + Git deploys? -->  Render / similar PaaS
Need full control / homelab / cost?  -->  VPS + systemd + Caddy
Multi-team multi-service mesh?       -->  Containers / K8s (separate chapter)
```

Production Checklist

- ☐ Bind to `PORT` (or platform-provided address)
- ☐ `/healthz` and `/readyz` implemented and wired in the platform
- ☐ Release build flags (`CGO_ENABLED=0`, `-trimpath`, `-s -w`)
- ☐ Secrets only via platform env / secret store / `EnvironmentFile`
- ☐ TLS at the edge (platform or Caddy/nginx)
- ☐ Graceful shutdown on `SIGTERM` (PaaS and systemd both send it)
- ☐ Structured logs to stdout/stderr (platforms capture them)
- ☐ Database connection limits sized for instance count
- ☐ Document rollback: previous binary tag or image digest

Common Pitfalls

1. **Hard-coded port** — works locally, fails on Render/Leapcell.

2. **Listening on 127.0.0.1 only** — fine behind local Caddy; wrong if the platform expects `0.0.0.0`.
3. **Migrate on every boot with N replicas** — racey schema changes; use one-shot jobs.
4. **Huge cold start init** — loading models, syncing caches, or compiling templates at package `init` time.
5. **Root systemd service** — unnecessary privilege; use a dedicated user.
6. **No disk quota awareness** — writing unbounded logs/uploads under `/var/lib` fills the VPS.

Exercises

1. **PORT wiring** — Deploy a tiny handler that prints `PORT` and the listen address; verify on a PaaS free tier or with `PORT=9090 ./server` locally.
2. **systemd unit** — Install a binary as a non-root service with `ProtectSystem=strict` and confirm `journalctl` shows clean restarts after `kill -TERM`.
3. **Readiness gate** — Make `/readyz` fail until a file `/var/lib/myapp/ready` exists; flip it and watch a reverse proxy start routing.
4. **Cold-start budget** — Measure time from process start to first successful `/healthz` with artificial 2s DB connect delay; refactor init to stay under 500ms when DB is up.
5. **Secret file perms** — Create `/etc/myapp.env` as `0640 root:myapp` and prove the service user can read it while an unprivileged login cannot.

More examples

PORT injection and dual health endpoints

PaaS injects `PORT` and probes liveness separately from readiness. This program prints the resolved listen port and exercises both endpoints via `httptest`.

```
mkdir -p /tmp/go-paas-port && cd /tmp/go-paas-port
go mod init example.com/paas-port
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "net/http"
    "net/http/httptest"
    "os"
    "sync/atomic"
)
```

```go
func port() string {
    if p := os.Getenv("PORT"); p != "" {
        return p
    }
    return "8080"
}

func main() {
    os.Setenv("PORT", "10000")
    fmt.Println("listen :"+port())

    var dbOK atomic.Bool
    dbOK.Store(false)

    mux := http.NewServeMux()
    mux.HandleFunc("GET /healthz", func(w http.ResponseWriter, _ *http.Request) {
        fmt.Fprintln(w, "alive")
    })
    mux.HandleFunc("GET /readyz", func(w http.ResponseWriter, _ *http.Request) {
        if !dbOK.Load() {
            http.Error(w, "db warming", http.StatusServiceUnavailable)
            return
        }
        fmt.Fprintln(w, "ready")
    })

    ts := httptest.NewServer(mux)
    defer ts.Close()

    r, _ := http.Get(ts.URL + "/healthz")
    r.Body.Close()
    fmt.Println("healthz:", r.StatusCode)

    r, _ = http.Get(ts.URL + "/readyz")
    r.Body.Close()
    fmt.Println("readyz cold:", r.StatusCode)

    dbOK.Store(true)
    r, _ = http.Get(ts.URL + "/readyz")
    r.Body.Close()
    fmt.Println("readyz warm:", r.StatusCode)
}
```

```go
go run .
```

Expected output:

```
listen :10000
healthz: 200
readyz cold: 503
readyz warm: 200
```

Config from env file shape (KEY=VALUE)

```
mkdir -p /tmp/go-paas-env && cd /tmp/go-paas-env
go mod init example.com/paas-env
```

Save as `main.go`:

```
package main

import (
    "bufio"
    "fmt"
    "os"
    "strings"
)

func loadEnvFile(path string) (map[string]string, error) {
    f, err := os.Open(path)
    if err != nil {
        return nil, err
    }
    defer f.Close()
    out := map[string]string{}
    sc := bufio.NewScanner(f)
    for sc.Scan() {
        line := strings.TrimSpace(sc.Text())
        if line == "" || strings.HasPrefix(line, "#") {
            continue
        }
        k, v, ok := strings.Cut(line, "=")
        if !ok {
            continue
        }
        out[strings.TrimSpace(k)] = strings.TrimSpace(v)
    }
    return out, sc.Err()
}

func main() {
    path := "app.env"
    _ = os.WriteFile(path, []byte("# demo\nPORT=7777\nAPP_ENV=staging\n"), 0o600)
```

```go
    defer os.Remove(path)

    cfg, err := loadEnvFile(path)
    if err != nil {
        panic(err)
    }
    fmt.Println("PORT:", cfg["PORT"])
    fmt.Println("APP_ENV:", cfg["APP_ENV"])
    fmt.Println("keys:", len(cfg))
}
```

```bash
go run .
```

Expected output:

```
PORT: 7777
APP_ENV: staging
keys: 2
```

Runnable example

PaaS platforms inject `PORT` and probe health. This program binds to `PORT`, exposes `/healthz` and `/readyz`, and demonstrates readiness gating—the same contract Render, Leapcell-style hosts, and systemd expect.

```bash
mkdir -p /tmp/go-paas-app && cd /tmp/go-paas-app
go mod init example.com/paas-app
```

Save as `main.go`:

```go
package main

import (
    "context"
    "fmt"
    "log"
    "net"
    "net/http"
    "os"
    "sync/atomic"
    "time"
)

func main() {
    port := os.Getenv("PORT")
```

```
    if port == "" {
        port = "0" // ephemeral for local demo
    }

    var ready atomic.Bool
    ready.Store(false)

    mux := http.NewServeMux()
    mux.HandleFunc("GET /healthz", func(w http.ResponseWriter, _ *http.Request) {
        w.WriteHeader(http.StatusOK)
        _, _ = w.Write([]byte("ok"))
    })
    mux.HandleFunc("GET /readyz", func(w http.ResponseWriter, _ *http.Request) {
        if !ready.Load() {
            http.Error(w, "not ready", http.StatusServiceUnavailable)
            return
        }
        w.WriteHeader(http.StatusOK)
        _, _ = w.Write([]byte("ready"))
    })
    mux.HandleFunc("GET /", func(w http.ResponseWriter, _ *http.Request) {
        fmt.Fprintln(w, "hello from paas-shaped go")
    })

    ln, err := net.Listen("tcp", "127.0.0.1:"+port)
    if err != nil {
        log.Fatal(err)
    }
    srv := &http.Server{Handler: mux, ReadHeaderTimeout: 5 * time.Second}

    go func() {
        if err := srv.Serve(ln); err != nil && err != http.ErrServerClosed {
            log.Fatal(err)
        }
    }()

    base := "http://" + ln.Addr().String()
    fmt.Println("listen_addr:", ln.Addr().String())
    fmt.Println("PORT env:", os.Getenv("PORT"))

    // Simulate warmup: live before ready.
    check(base+"/healthz", "healthz-before-ready")
    check(base+"/readyz", "readyz-before-ready")

    ready.Store(true)
    fmt.Println("warmup complete; marking ready")
```

```go
    check(base+"/readyz", "readyz-after-ready")
    check(base+"/", "root")

    ctx, cancel := context.WithTimeout(context.Background(), 2*time.Second)
    defer cancel()
    _ = srv.Shutdown(ctx)
    fmt.Println("shutdown complete")
}

func check(url, label string) {
    resp, err := http.Get(url)
    if err != nil {
        fmt.Printf("%s: error %v\n", label, err)
        return
    }
    defer resp.Body.Close()
    fmt.Printf("%s: %d\n", label, resp.StatusCode)
}
```

```bash
go run .
PORT=9090 go run .
```

Expected output (illustrative):

```
listen_addr: 127.0.0.1:54321
PORT env:
healthz-before-ready: 200
readyz-before-ready: 503
warmup complete; marking ready
readyz-after-ready: 200
root: 200
shutdown complete
```

What to notice

- Always prefer `PORT` from the environment; hard-coded `8080` fails on managed platforms.
- Liveness (`/healthz`) can succeed while readiness (`/readyz`) is 503 during migrations or dependency warmup.
- Listen on all interfaces in real deploys (`:PORT`); this demo uses `127.0.0.1` for a self-contained local run.

Try next

- Run with `PORT=9090` and curl from another terminal.
- On `SIGTERM`, flip `ready` to false, sleep briefly (connection drain), then `Shutdown`.

Related Topics

- Containerization (Docker)
- CI/CD & Hardening
- Building and Releasing

Go on NixOS & FreeBSD

Beyond Linux: NixOS & FreeBSD

Overview

Go treats cross-compilation as a first-class feature: set `GOOS` and `GOARCH`, get a binary. That matters when you target NixOS (purely functional paths, no global `/lib`) or FreeBSD (jails, ZFS, kqueue). Both platforms reward pure-Go static binaries and punish casual CGO.

This chapter covers cross-compilation matrices, Nix flakes for reproducible toolchains, FreeBSD notes (kqueue, jails), and production pitfalls.

Cross-Compilation Basics

From any host (macOS, Linux, Windows):

```
# Linux amd64 (most VPS / cloud)
GOOS=linux GOARCH=amd64 CGO_ENABLED=0 go build -o bin/app-linux-amd64 ./cmd/server

# Linux arm64 (Graviton, Raspberry Pi OS 64-bit)
GOOS=linux GOARCH=arm64 CGO_ENABLED=0 go build -o bin/app-linux-arm64 ./cmd/server

# FreeBSD amd64
GOOS=freebsd GOARCH=amd64 CGO_ENABLED=0 go build -o bin/app-freebsd-amd64 ./cmd/server

# FreeBSD arm64
GOOS=freebsd GOARCH=arm64 CGO_ENABLED=0 go build -o bin/app-freebsd-arm64 ./cmd/server

# Windows
GOOS=windows GOARCH=amd64 CGO_ENABLED=0 go build -o bin/app.exe ./cmd/server

# macOS Apple Silicon
GOOS=darwin GOARCH=arm64 CGO_ENABLED=0 go build -o bin/app-darwin-arm64 ./cmd/server
```

List supported pairs:

```
go tool dist list
```

Matrix build script

```
#!/usr/bin/env bash
set -euo pipefail
mkdir -p bin
for goos in linux freebsd darwin windows; do
  for goarch in amd64 arm64; do
    ext=""
    [[ "$goos" == "windows" ]] && ext=".exe"
    out="bin/app-${goos}-${goarch}${ext}"
    echo "building $out"
    CGO_ENABLED=0 GOOS="$goos" GOARCH="$goarch" \
      go build -trimpath -ldflags="-s -w" -o "$out" ./cmd/server
  done
done
```

CGO breaks easy cross-compilation. Cross-CGO needs a matching cross-compiler and libraries. Prefer pure Go drivers (e.g. `jackc/pgx` pure Go, modernc.org/sqlite) when you ship multi-OS binaries.

Go on NixOS

NixOS has no FHS layout by default. Dynamically linked binaries that expect `/lib64/ld-linux-x86-64.so.2` often fail unless you use `steam-run`, `buildFHSUserEnv`, or patchelf. **Static pure-Go binaries just run.**

Pure Go: the happy path

```
CGO_ENABLED=0 go build -o app ./cmd/server
./app   # works on NixOS without wrapping
```

Verify linkage (on a system with `ldd`):

```
ldd ./app
# statically linked  (or "not a dynamic executable")
```

Development shell with Flakes

Pin the exact Go toolchain for the whole team:

```
# flake.nix
{
  description = "Go service dev shell";
```

```
  inputs.nixpkgs.url = "github:NixOS/nixpkgs/nixos-unstable";

  outputs = { self, nixpkgs }:
    let
      systems = [ "x86_64-linux" "aarch64-linux" "x86_64-darwin" "aarch64-darwin" ];
      forAllSystems = f: nixpkgs.lib.genAttrs systems (system: f nixpkgs.legacyPackages.${sy
    in {
      devShells = forAllSystems (pkgs: {
        default = pkgs.mkShell {
          packages = with pkgs; [
            go_1_26
            gopls
            gotools
            golangci-lint
            govulncheck
            delv
          ];
          shellHook = ''
            export CGO_ENABLED=0
            echo "Go $(go version)"
          '';
        };
      });
    };
}
```

```
nix develop
go test ./...
```

Packaging a Go module with Nix

```
# overlay-style package sketch
{ buildGoModule, fetchFromGitHub, lib }:

buildGoModule rec {
  pname = "myapp";
  version = "1.2.3";
  src = ./.;  # or fetchFromGitHub { ... }

  vendorHash = "sha256-AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA="; # set via nix build er

  ldflags = [ "-s" "-w" "-X main.version=${version}" ];
  env.CGO_ENABLED = "0";
```

```
  meta = {
    description = "Example Go service";
    mainProgram = "myapp";
  };
}
```

Run `nix build` and let Nix tell you the correct `vendorHash` on first failure, then pin it.

CGO on NixOS

If you need CGO, build inside Nix so headers and libs are on `NIX_CFLAGS_COMPILE` / `NIX_LDFLAGS`:

```
pkgs.mkShell {
  packages = [ pkgs.go pkgs.pkg-config pkgs.sqlite ];
  # CGO_ENABLED=1 only inside this shell when required
}
```

Never assume `/usr/include` exists.

Go on FreeBSD

FreeBSD is a first-class Go port. The runtime uses **kqueue** for network and file notifications (instead of Linux epoll/inotify).

Runtime notes

Area	FreeBSD behavior
`net` poller	kqueue-backed
`fsnotify` and similar	kqueue (not inotify)
`os/signal`	Works; jail signal policies may differ
File paths	Case-sensitive UFS/ZFS; no drive letters
`syscall` / `x/sys/unix`	FreeBSD-specific constants and wrappers

Build on FreeBSD

```
pkg install go git
git clone https://example.com/myapp.git
cd myapp
CGO_ENABLED=0 go build -o myapp ./cmd/server
```

Cross-build from Linux/macOS as shown above; scp into a jail or host.

Jails and networking

Run the service inside a jail with a dedicated IP or VNET:

```
host
    jail: myapp
         /usr/local/bin/myapp
         env: DATABASE_URL=...
         listen: 10.0.0.20:8080
```

Prefer binding explicitly:

```
addr := os.Getenv("LISTEN_ADDR")
if addr == "" {
    addr = "0.0.0.0:8080"
}
log.Fatal(http.ListenAndServe(addr, mux))
```

rc.d service sketch

```
#!/bin/sh
# /usr/local/etc/rc.d/myapp
# PROVIDE: myapp
# REQUIRE: NETWORKING
# KEYWORD: shutdown

. /etc/rc.subr

name="myapp"
rcvar="myapp_enable"
command="/usr/local/bin/myapp"
myapp_user="myapp"
pidfile="/var/run/${name}.pid"
command_args="&"

load_rc_config $name
run_rc_command "$1"
```

```
sysrc myapp_enable=YES
service myapp start
```

Portable Go Code Tips

```go
package platform

import "runtime"

func DataDir() string {
    switch runtime.GOOS {
    case "windows":
        return filepath.Join(os.Getenv("ProgramData"), "MyApp")
    case "darwin":
        return filepath.Join(os.Getenv("HOME"), "Library", "Application Support", "MyApp")
    default: // linux, freebsd, ...
        if xdg := os.Getenv("XDG_DATA_HOME"); xdg != "" {
            return filepath.Join(xdg, "myapp")
        }
        return filepath.Join(os.Getenv("HOME"), ".local", "share", "myapp")
    }
}
```

Use `path/filepath`, not string concat with `/`. Use `runtime.GOOS` only at boundaries (paths, notifications), not scattered through business logic.

Build tags for OS-specific files

```go
// notify_linux.go
//go:build linux

package notify

func watch(path string) error { /* inotify path via library */ return nil }
```

```go
// notify_freebsd.go
//go:build freebsd

package notify

func watch(path string) error { /* kqueue path */ return nil }
```

```go
// notify_stub.go
//go:build !linux && !freebsd

package notify
```

```go
func watch(path string) error { return errUnsupported }
```

Production Checklist

- ☐ Default builds use `CGO_ENABLED=0` for portable artifacts
- ☐ CI builds at least `linux/amd64` and any real target (`freebsd/amd64`, `linux/arm64`)
- ☐ Nix users have a flake/`shell.nix` with pinned Go
- ☐ No hard-coded Linux-only paths (`/proc` scraped only behind `linux` tags)
- ☐ File watching libraries verified on FreeBSD if used
- ☐ Release binaries named with `GOOS-GOARCH`
- ☐ Document CGO exceptions (and required C toolchains) explicitly

Common Pitfalls

1. **Dynamic binary on NixOS** — "No such file or directory" often means the dynamic loader path is wrong, not that the file is missing. Same class of bug as a glibc binary on Alpine (`/lib64/ld-linux-x86-64.so.2` absent). Pure Go (`CGO_ENABLED=0`) never asks for that loader; see Go as a Systems Language.
2. **CGO cross-compile surprise** — `go build` silently uses CGO when `CGO_ENABLED` is unset and a C compiler exists; set `CGO_ENABLED=0` in CI.
3. **`/proc` and cgroup assumptions** — code that reads Linux cgroup files breaks on FreeBSD; guard with build tags.
4. **Case-insensitive assumptions** — macOS default FS is case-insensitive; FreeBSD/Linux are not. Tests that rely on case folding will fail.
5. **Firewall / jail** — binding succeeds but traffic never arrives because the jail has no IP or pf blocks the port.
6. **Old FreeBSD + new Go** — use a FreeBSD version supported by your Go release; check Go's FreeBSD port notes when upgrading.

Exercises

1. **Cross matrix** — Build the same `./cmd/server` for `linux/amd64`, `linux/arm64`, and `freebsd/amd64`. Record binary sizes.
2. **Static proof** — On Linux, run `file` and `ldd` on the artifact; ensure it is statically linked with `CGO_ENABLED=0`.
3. **Flake shell** — Add a minimal `flake.nix` with `go` and `gopls`; run `nix develop -c go test ./...`.
4. **GOOS-specific file** — Split a helper with `//go:build linux` and `//go:build freebsd` stubs; `go test` on your host and `GOOS=freebsd go test` (compile-only is fine).
5. **Path portability** — Implement `DataDir()` above with tests using `t.Setenv` for `XDG_DATA_HOME` and `HOME`.

More examples

Portable data directory (XDG + HOME fallback)

```
mkdir -p /tmp/go-xdg && cd /tmp/go-xdg
go mod init example.com/xdg
```

Save as `main.go`:

```go
package main

import (
    "fmt"
    "os"
    "path/filepath"
    "runtime"
)

func dataDir(app string) string {
    if xdg := os.Getenv("XDG_DATA_HOME"); xdg != "" {
        return filepath.Join(xdg, app)
    }
    home, err := os.UserHomeDir()
    if err != nil {
        return filepath.Join(".", app)
    }
    // FreeBSD/Linux/macOS-friendly default under home.
    return filepath.Join(home, ".local", "share", app)
}

func main() {
    fmt.Println("GOOS/GOARCH:", runtime.GOOS+"/"+runtime.GOARCH)
    os.Setenv("XDG_DATA_HOME", "/var/lib/custom")
    fmt.Println("with XDG:", dataDir("myapp"))
    os.Unsetenv("XDG_DATA_HOME")
    // Force HOME for deterministic demo output.
    os.Setenv("HOME", "/home/demo")
    fmt.Println("fallback:", dataDir("myapp"))
}
```

```
go run .
```

Expected output (GOOS/GOARCH varies by host):

```
GOOS/GOARCH: darwin/arm64
with XDG: /var/lib/custom/myapp
fallback: /home/demo/.local/share/myapp
```

Build identity without CGO surprises

```
mkdir -p /tmp/go-static-id && cd /tmp/go-static-id
go mod init example.com/static-id
```

Save as `main.go`:

```go
package main

import (
    "fmt"
    "runtime"
    "runtime/debug"
)

func main() {
    fmt.Println("compiler:", runtime.Compiler)
    fmt.Println("version:", runtime.Version())
    fmt.Println("cgo:", cgoEnabled())
    if bi, ok := debug.ReadBuildInfo(); ok {
        fmt.Println("main path:", bi.Path)
        fmt.Println("go version (mod):", bi.GoVersion)
    }
}

func cgoEnabled() string {
    // Pure-Go programs can still report whether the toolchain had CGO available
    // at build time via the "CGO_ENABLED" setting in build info when present.
    if bi, ok := debug.ReadBuildInfo(); ok {
        for _, s := range bi.Settings {
            if s.Key == "CGO_ENABLED" {
                return s.Value
            }
        }
    }
    return "unknown (run: CGO_ENABLED=0 go build)"
}
```

```
CGO_ENABLED=0 go run .
```

Expected output (illustrative):

```
compiler: gc
version: go1.22.x
cgo: 0
main path: example.com/static-id
go version (mod): go1.22.x
```

Runnable example

Cross-compilation and portable paths matter more than platform-specific syscalls for most services. This program prints build identity (`GOOS`/`GOARCH`), uses XDG-style data dirs, and stays pure Go (static-friendly).

```
mkdir -p /tmp/go-portable && cd /tmp/go-portable
go mod init example.com/portable
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "os"
    "path/filepath"
    "runtime"
)

func dataDir(app string) string {
    if xdg := os.Getenv("XDG_DATA_HOME"); xdg != "" {
        return filepath.Join(xdg, app)
    }
    home, err := os.UserHomeDir()
    if err != nil {
        return filepath.Join(".", app+"-data")
    }
    // Portable default; FreeBSD/Linux/macOS all understand home-relative paths.
    return filepath.Join(home, ".local", "share", app)
}

func main() {
    fmt.Printf("goos=%s goarch=%s compiler=%s\n", runtime.GOOS, runtime.GOARCH, runtime.Comp
    fmt.Printf("version=%s\n", runtime.Version())
    fmt.Printf("num_cpu=%d\n", runtime.NumCPU())

    dir := dataDir("myapp")
    fmt.Println("data_dir:", dir)

    if err := os.MkdirAll(dir, 0o750); err != nil {
        fmt.Println("mkdir error:", err)
        os.Exit(1)
    }
    marker := filepath.Join(dir, "hello.txt")
    if err := os.WriteFile(marker, []byte("portable pure-go write\n"), 0o640); err != nil {
        fmt.Println("write error:", err)
```

```go
        os.Exit(1)
    }
    b, err := os.ReadFile(marker)
    if err != nil {
        fmt.Println("read error:", err)
        os.Exit(1)
    }
    fmt.Printf("read_back: %q\n", string(b))
    _ = os.Remove(marker)
    fmt.Println("ok: pure Go paths work without CGO")
}
```

```bash
go run .
# Cross-compile without running foreign binaries:
GOOS=linux GOARCH=amd64 CGO_ENABLED=0 go build -o /tmp/app-linux-amd64 .
GOOS=freebsd GOARCH=amd64 CGO_ENABLED=0 go build -o /tmp/app-freebsd-amd64 .
file /tmp/app-linux-amd64 /tmp/app-freebsd-amd64 2>/dev/null || ls -la /tmp/app-*
```

Expected output (illustrative, host-dependent):

```
goos=darwin goarch=arm64 compiler=gc
version=go1.22.x
num_cpu=8
data_dir: /Users/you/.local/share/myapp
read_back: "portable pure-go write\n"
ok: pure Go paths work without CGO
```

What to notice

- `runtime.GOOS` / `GOARCH` report the *target* you built for—set env vars at build time for NixOS/FreeBSD artifacts.
- `CGO_ENABLED=0` keeps binaries static so NixOS does not need an FHS loader path.
- Prefer `filepath` + env (`XDG_DATA_HOME`) over hard-coded `/var` Linux paths.

Try next

- Add a `//go:build linux` file and a stub for other OS; compile with `GOOS=freebsd go build`.
- On Linux, run `ldd` / `file` on a `CGO_ENABLED=0` binary and confirm it is statically linked.

Related Topics

- Containerization (Docker)
- Building and Releasing
- Go as a Systems Language
- Systems Programming

CI/CD & Hardening

CI/CD Pipelines & Security Hardening

Overview

CI/CD should make the boring path automatic: format, lint, test, vuln scan, build, sign, and ship. For Go, the toolchain is friendly—fast tests, excellent race detector, first-party `govulncheck`—but pipelines still get compromised through unpinned actions, over-privileged tokens, and unsigned artifacts.

This chapter provides a 2026-oriented GitHub Actions workflow, binary hardening flags, image signing with Cosign, and a release checklist.

Goals of a Go Pipeline

```
push / PR
  --> checkout (pinned)
  --> setup-go (cache modules)
  --> vet / lint
  --> govulncheck
  --> test -race -cover
  --> build (reproducible flags)
  --> (main/tag) sign + publish
```

Every PR should fail closed on lint, tests, or reachable vulns. Releases should produce **signed**, **versioned** artifacts.

Baseline GitHub Actions Workflow

```yaml
# .github/workflows/go.yml
name: Go Build & Test

on:
  push:
    branches: [main]
  pull_request:
    branches: [main]

permissions:
```

```yaml
  contents: read

jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - name: Checkout
        uses: actions/checkout@v4

      - name: Setup Go
        uses: actions/setup-go@v5
        with:
          go-version: "1.27.x"
          cache: true

      - name: Verify modules
        run: go mod verify

      - name: Vet
        run: go vet ./...

      - name: Lint
        uses: golangci/golangci-lint-action@v6
        with:
          version: v1.64.0

      - name: Vulnerability check
        run: |
          go install golang.org/x/vuln/cmd/govulncheck@latest
          govulncheck ./...

      - name: Test
        run: go test -race -count=1 -coverprofile=coverage.out ./...

      - name: Coverage report
        run: go tool cover -func=coverage.out | tail -n 1

      - name: Build
        run: |
          CGO_ENABLED=0 go build -trimpath -ldflags="-s -w" -o bin/app ./cmd/server
```

Hardening the workflow itself

1. **Least privilege tokens** — default `permissions: contents: read`. Grant `id-token: write` only for OIDC/signing jobs; `contents: write` only for release jobs.
2. **Pin actions by digest** in high-security repos:

```yaml
# Example shape - replace with current digest from the action repo
- uses: actions/checkout@11bd71901bbe5b1630ceea73d27597364c9af683 # v4.2.2
```

3. **No `pull_request_target` with untrusted checkout** — easy RCE pattern; avoid unless you fully understand the threat model.
4. **Cache carefully** — `setup-go` cache is fine for modules; do not cache world-writable custom paths without validation.
5. **Secrets** — only on `main`/tags, not on forks' PR workflows that expose secrets.

Release Job Sketch

```yaml
  release:
    if: startsWith(github.ref, 'refs/tags/v')
    needs: build
    runs-on: ubuntu-latest
    permissions:
      contents: write
      id-token: write  # for keyless Cosign / OIDC
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-go@v5
        with:
          go-version: "1.27.x"
          cache: true

      - name: Build multi-arch binaries
        run: |
          mkdir -p dist
          for pair in linux/amd64 linux/arm64 darwin/arm64; do
            os=${pair%/*}; arch=${pair#*/}
            CGO_ENABLED=0 GOOS=$os GOARCH=$arch \
              go build -trimpath \
              -ldflags="-s -w -X main.version=${GITHUB_REF_NAME}" \
              -o "dist/app_${os}_${arch}" ./cmd/server
          done

      - name: Upload release assets
        uses: softprops/action-gh-release@v2
        with:
          files: dist/*
```

Hardening the Binary

```
CGO_ENABLED=0 go build \
  -trimpath \
  -buildmode=pie \
  -ldflags="-s -w -X main.version=${VERSION} -X main.commit=${COMMIT}" \
  -o app \
  ./cmd/server
```

Flag	Purpose
`-trimpath`	Strip local paths (reproducibility + privacy)
`-s -w`	Strip symbol/DWARF tables (smaller; less free intel)
`-buildmode=pie`	Position-independent executable; works with ASLR
`-X main.version=…`	Embed release metadata without editing source

Optional: generate a Software Bill of Materials (SBOM) with `syft` or `govulncheck` tooling and attach it to the release.

Container Image Pipeline

```
      - name: Build image
        run: |
          docker build \
            --build-arg VERSION=${GITHUB_REF_NAME} \
            -t ghcr.io/${{ github.repository }}:${GITHUB_REF_NAME} .

      - name: Scan image
        run: |
          # install trivy or use a GH action
          trivy image --exit-code 1 --severity HIGH,CRITICAL \
            ghcr.io/${{ github.repository }}:${GITHUB_REF_NAME}
```

Signing with Cosign (Sigstore)

Signing proves the artifact came from *your* CI identity, not a random uploader.

```
# Keyless (OIDC) example in CI after cosign is installed
cosign sign --yes ghcr.io/org/myapp:1.2.3

# Verify elsewhere
cosign verify ghcr.io/org/myapp:1.2.3 \
```

```
  --certificate-identity-regexp='https://github.com/org/myapp/.*' \
  --certificate-oidc-issuer=https://token.actions.githubusercontent.com
```

Sign checksums for raw binaries:

```
(cd dist && sha256sum * > checksums.txt)
cosign sign-blob --yes --output-signature checksums.txt.sig checksums.txt
```

Go-Specific Quality Gates

```
# Race detector (linux/amd64/arm64; not all platforms)
go test -race ./...

# Fuzz (kept short in CI)
go test -fuzz=FuzzParse -fuzztime=20s ./internal/parser

# Module hygiene
go mod tidy
git diff --exit-code go.mod go.sum
```

Example test job matrix

```
strategy:
  matrix:
    os: [ubuntu-latest, macos-latest]
    go: ["1.26.x", "1.27.x"]
```

Use `-race` on Linux; on macOS it works but is slower—optional for PRs, required on main.

Supply Chain Mini-Checklist

Control	Why
`go.sum` committed	Pin module hashes
`go mod verify`	Detect sum database mismatches
`govulncheck`	Reachable vuln analysis (not just "depends on")
Pinned actions digests	Action repo compromise resistance
Minimal `permissions`	Stolen token blast radius
Cosign / provenance	Consumer verification
Immutable image tags + digests	Prevent tag overwrite attacks

Application Snippet: Fail Build on Bad Config

CI should also run a **config dry-run** so broken env schemas never ship:

```go
package config

import (
    "fmt"
    "os"
    "strconv"
)

type Config struct {
    Addr string
    Env  string
}

func Load() (Config, error) {
    addr := os.Getenv("LISTEN_ADDR")
    if addr == "" {
        addr = ":8080"
    }
    env := os.Getenv("APP_ENV")
    if env == "" {
        return Config{}, fmt.Errorf("APP_ENV required")
    }
    if _, err := strconv.Atoi(os.Getenv("READY_TIMEOUT_SEC")); err != nil && os.Getenv("READ
        return Config{}, fmt.Errorf("READY_TIMEOUT_SEC: %w", err)
    }
    return Config{Addr: addr, Env: env}, nil
}
```

```bash
APP_ENV=ci go test ./internal/config -run TestLoad
```

Production Checklist

- ☐ PR pipeline: vet, lint, `govulncheck`, `go test -race`
- ☐ `permissions` default to read-only
- ☐ Actions pinned (version at minimum; digest for high assurance)
- ☐ Release builds use `-trimpath`, `-s -w`, version ldflags
- ☐ Artifacts checksummed; containers scanned
- ☐ Cosign (or equivalent) sign on tag release
- ☐ Secrets not available to untrusted fork PRs
- ☐ `go.mod`/`go.sum` tidy enforced
- ☐ Branch protection requires the CI check

Common Pitfalls

1. **`go test` without `-race`** — data races ship silently under load.
2. **Latest action tags only** — `uses: foo@main` is a supply-chain footgun.
3. **Over-powered `GITHUB_TOKEN`** — write access on every PR job.
4. **Skipping vuln checks "to go green"** — track ignore rules with expiry, don't delete the step.
5. **Building with CGO in CI by accident** — non-portable binaries and missing cross-compile.
6. **Signing only containers, not binaries** — attackers swap the `.tar.gz` on the release page.
7. **No module verify** — tampered cache or mirror goes unnoticed.

Exercises

1. **Green pipeline** — Add the baseline workflow to a sample module; break a test and confirm the PR fails.
2. **Pin an action** — Replace `actions/checkout@v4` with a full commit SHA and document the version in a comment.
3. **Race lab** — Introduce a deliberate data race; show `-race` fails and the fix (mutex or channel) passes.
4. **govulncheck** — Run against a module with an old vulnerable dependency; upgrade until clean.
5. **Cosign dry-run** — Generate a local key pair with Cosign, sign a blob, verify it (keyless optional later).
6. **Permissions audit** — Read your workflow YAML and list every permission granted; remove any unused write scope.

More examples

Race-prone counter vs fixed counter

CI should run `go test -race`. This program shows the bug class the race detector catches, then a mutex-fixed version.

```
mkdir -p /tmp/go-ci-race && cd /tmp/go-ci-race
go mod init example.com/ci-race
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "sync"
)
```

```go
func racey(n int) int {
    var c int
    var wg sync.WaitGroup
    for i := 0; i < n; i++ {
        wg.Add(1)
        go func() {
            defer wg.Done()
            c++
        }()
    }
    wg.Wait()
    return c
}

func fixed(n int) int {
    var c int
    var mu sync.Mutex
    var wg sync.WaitGroup
    for i := 0; i < n; i++ {
        wg.Add(1)
        go func() {
            defer wg.Done()
            mu.Lock()
            c++
            mu.Unlock()
        }()
    }
    wg.Wait()
    return c
}

func main() {
    // Deterministic path: only exercise the fixed counter in default runs.
    // To see the race: temporarily print racey(1000) under `go run -race .`
    got := fixed(1000)
    fmt.Println("fixed count:", got)
    fmt.Println("ok:", got == 1000)
}
```

```go
go run .
# optional: go test -race (after adding a _test.go that calls racey)
```

Expected output:

```
fixed count: 1000
ok: true
```

Module sum verification helper

```
mkdir -p /tmp/go-ci-verify && cd /tmp/go-ci-verify
go mod init example.com/ci-verify
```

Save as `main.go`:

```go
package main

import (
    "crypto/sha256"
    "encoding/hex"
    "fmt"
    "os"
)

func fileSHA256(path string) (string, error) {
    b, err := os.ReadFile(path)
    if err != nil {
        return "", err
    }
    sum := sha256.Sum256(b)
    return hex.EncodeToString(sum[:]), nil
}

func main() {
    path := "artifact.bin"
    _ = os.WriteFile(path, []byte("release-bytes-v1"), 0o644)
    defer os.Remove(path)

    sum, err := fileSHA256(path)
    if err != nil {
        panic(err)
    }
    fmt.Println("sha256:", sum)
    // Simulate CI check against a known digest.
    want, _ := fileSHA256(path)
    fmt.Println("matches pinned digest:", sum == want)
}
```

```
go run .
```

Expected output:

```
sha256: <64 hex chars>
matches pinned digest: true
```

Runnable example

CI gates are YAML; the Go work they gate is tests, races, and release-shaped builds. This package includes a race-prone function, a fixed version, and a small test file—run the same commands your pipeline should run.

```
mkdir -p /tmp/go-ci-gate && cd /tmp/go-ci-gate
go mod init example.com/ci-gate
```

Save as `sum.go`:

```go
package cigate

import "sync"

// BrokenAdd races on total (do not use in production).
func BrokenAdd(n int) int {
    var total int
    var wg sync.WaitGroup
    wg.Add(n)
    for i := 0; i < n; i++ {
        go func() {
            defer wg.Done()
            total++
        }()
    }
    wg.Wait()
    return total
}

// SafeAdd uses an atomic-style mutex critical section.
func SafeAdd(n int) int {
    var total int
    var mu sync.Mutex
    var wg sync.WaitGroup
    wg.Add(n)
    for i := 0; i < n; i++ {
        go func() {
            defer wg.Done()
            mu.Lock()
            total++
            mu.Unlock()
        }()
    }
    wg.Wait()
    return total
}
```

Save as `sum_test.go`:

```go
package cigate

import "testing"

func TestSafeAdd(t *testing.T) {
    if got := SafeAdd(1000); got != 1000 {
        t.Fatalf("SafeAdd: got %d want 1000", got)
    }
}

func TestBrokenAdd_DocumentRace(t *testing.T) {
    // Without -race this may flakily pass; with -race it should fail.
    // Comment out after you have seen the race detector fire once.
    t.Skip("uncomment to demo: go test -race (expect FAIL on BrokenAdd)")
    _ = BrokenAdd(1000)
}
```

Save as `cmd/check/main.go`:

```go
package main

import (
    "fmt"
    "runtime/debug"

    cigate "example.com/ci-gate"
)

func main() {
    fmt.Println("safe_sum:", cigate.SafeAdd(100))
    if info, ok := debug.ReadBuildInfo(); ok {
        fmt.Println("go:", info.GoVersion)
        fmt.Println("path:", info.Path)
    }
    fmt.Println("ci gate sample ok")
}
```

```bash
go vet ./...
go test -count=1 ./...
go test -race -count=1 ./...
CGO_ENABLED=0 go build -trimpath -ldflags="-s -w" -o /tmp/ci-check ./cmd/check
/tmp/ci-check
# optional: go install golang.org/x/vuln/cmd/govulncheck@latest && govulncheck ./...
```

Expected output (illustrative):

```
ok      example.com/ci-gate 0.012s
safe_sum: 100
go: go1.22.x
path: example.com/ci-gate
ci gate sample ok
```

To see the race detector, temporarily call `BrokenAdd` from a test without `t.Skip` and run `go test -race`.

What to notice

- Pipelines should fail on `go test -race` for concurrent packages—silent races ship without it.
- Release flags (`-trimpath`, `-s -w`, `CGO_ENABLED=0`) belong in the same job that publishes artifacts.
- `go vet` is free correctness signal before heavier linters and `govulncheck`.

Try next

- Unskip the race test, watch `-race` fail, then delete `BrokenAdd`.
- Add a Makefile target `sec: go vet && go test -race ./...` matching the workflow in this chapter.

Related Topics

- Containerization (Docker)
- Security Tools
- Application Hardening
- Building and Releasing

Building and Releasing Software

Building and Releasing Software

Overview

Go compiles to static binaries, making deployment simple. This chapter covers building, cross-compilation, and release workflows.

Basic Build

```
go build -o myapp ./cmd/server
```

Production Build

```
go build \
    -ldflags="-s -w" \
    -o bin/myapp \
    ./cmd/server
```

Flags: - `-s` - Strip symbol table - `-w` - Strip DWARF debugging info

Version Injection

```
var (
    version = "dev"
    commit  = "none"
    date    = "unknown"
)
```

```
go build -ldflags="-X main.version=1.0.0 -X main.commit=$(git rev-parse HEAD)"
```

Why `CGO_ENABLED=0` is part of a release

On a machine with a C compiler, cgo is **on by default**. That can silently link **glibc** (Debian/Ubuntu) or **musl** (Alpine). The binary then needs a matching dynamic linker on the target: a glibc build dies on Alpine with `not found` even though the file is sitting in the current directory.

```
# Release default: pure Go, fully static on Linux.
CGO_ENABLED=0 go build -o myapp ./cmd/server

file myapp    # Linux: "statically linked"
ldd myapp     # Linux: "not a dynamic executable"
```

`CGO_ENABLED=0` also selects Go's own DNS resolver and `os/user` path instead of libc. That is what you want for `scratch` / distroless images and for a CLI you copy onto someone else's laptop. Turn cgo on only when a C library is irreplaceable. The full kernel-vs-libc story is in Go as a Systems Language.

Cross-Compilation

```
# Linux (static when CGO is off)
GOOS=linux GOARCH=amd64 CGO_ENABLED=0 go build -o app-linux

# Windows
GOOS=windows GOARCH=amd64 CGO_ENABLED=0 go build -o app.exe

# macOS ARM
GOOS=darwin GOARCH=arm64 CGO_ENABLED=0 go build -o app-darwin

# All platforms
for os in linux darwin windows; do
    for arch in amd64 arm64; do
        GOOS=$os GOARCH=$arch CGO_ENABLED=0 go build -o bin/app-$os-$arch
    done
done
```

The toolchain ships the runtime for each pair, so you do not need a C cross-compiler unless cgo is enabled.

Docker

```
# Multi-stage build
FROM golang:1.27-alpine AS builder
WORKDIR /app
COPY . .
```

```
RUN go build -o /app/server ./cmd/server

FROM alpine:latest
COPY --from=builder /app/server /server
ENTRYPOINT ["/server"]
```

GitHub Actions

```yaml
name: Release
on:
  push:
    tags: ['v*']

jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-go@v5
        with:
          go-version: '1.27'
      - run: go build -o app
      - uses: softprops/action-gh-release@v1
        with:
          files: app
```

GoReleaser

```yaml
# .goreleaser.yaml
builds:
  - main: ./cmd/server
    goos: [linux, darwin, windows]
    goarch: [amd64, arm64]

archives:
  - format: tar.gz
```

```
goreleaser release
```

Summary

Task	Command
Build	`go build -o app`
Optimize	`-ldflags="-s -w"`
Cross-compile	`GOOS=linux GOARCH=amd64`
Version inject	`-ldflags="-X main.version=x"`

More examples

Version and commit via ldflags

```
mkdir -p /tmp/go-ldflags && cd /tmp/go-ldflags
go mod init example.com/ldflags
```

Save as `main.go`:

```
package main

import "fmt"

// Overridden at link time:
// go run -ldflags="-X main.version=1.2.3 -X main.commit=abc1234" .
var (
    version = "dev"
    commit  = "none"
)

func main() {
    fmt.Printf("version=%s commit=%s\n", version, commit)
}
```

```
go run .
go run -ldflags="-X main.version=1.2.3 -X main.commit=abc1234" .
```

Expected output:

```
version=dev commit=none
version=1.2.3 commit=abc1234
```

Build info from the binary itself

```
mkdir -p /tmp/go-buildinfo && cd /tmp/go-buildinfo
go mod init example.com/buildinfo
```

Save as `main.go`:

```go
package main

import (
    "fmt"
    "runtime/debug"
)

func main() {
    bi, ok := debug.ReadBuildInfo()
    if !ok {
        fmt.Println("no build info")
        return
    }
    fmt.Println("path:", bi.Path)
    fmt.Println("go:", bi.GoVersion)
    for _, s := range bi.Settings {
        switch s.Key {
        case "GOOS", "GOARCH", "-compiler", "CGO_ENABLED", "vcs.revision", "vcs.modified":
            fmt.Printf("%s=%s\n", s.Key, s.Value)
        }
    }
}
```

```
go run .
```

Expected output (keys vary by toolchain/VCS):

```
path: example.com/buildinfo
go: go1.22.x
GOOS=...
GOARCH=...
-compiler=gc
CGO_ENABLED=...
```

Runnable example

Version injection and strip flags are the core of Go release builds. This program prints embedded metadata and reports binary-oriented build info.

```
mkdir -p /tmp/go-release-demo && cd /tmp/go-release-demo
go mod init example.com/release-demo
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "runtime"
    "runtime/debug"
)

// Populated via: -ldflags="-X main.version=... -X main.commit=... -X main.date=..."
var (
    version = "dev"
    commit  = "none"
    date    = "unknown"
)

func main() {
    fmt.Printf("version=%s commit=%s date=%s\n", version, commit, date)
    fmt.Printf("runtime=%s %s/%s\n", runtime.Version(), runtime.GOOS, runtime.GOARCH)

    if info, ok := debug.ReadBuildInfo(); ok {
        fmt.Println("module:", info.Path)
        for _, s := range info.Settings {
            switch s.Key {
            case "-tags", "CGO_ENABLED", "GOOS", "GOARCH", "vcs.revision", "vcs.modified":
                fmt.Printf("build %s=%s\n", s.Key, s.Value)
            }
        }
    }
}
```

```
go run -ldflags="-X main.version=1.0.0 -X main.commit=abc1234 -X main.date=2026-07-28" .

CGO_ENABLED=0 go build -trimpath -ldflags="-s -w \
  -X main.version=1.0.0 \
  -X main.commit=abc1234 \
  -X main.date=2026-07-28" -o app .

./app

# Cross-compile matrix sample:
for pair in linux/amd64 linux/arm64 darwin/arm64 windows/amd64; do
```

```
  GOOS=${pair%/*} GOARCH=${pair#*/}
  ext=""; [ "$GOOS" = windows ] && ext=".exe"
  CGO_ENABLED=0 go build -trimpath -ldflags="-s -w -X main.version=1.0.0" \
    -o "app-${GOOS}-${GOARCH}${ext}" .
  echo "built app-${GOOS}-${GOARCH}${ext}"
done
```

Expected output (illustrative):

```
version=1.0.0 commit=abc1234 date=2026-07-28
runtime=go1.22.x darwin/arm64
module: example.com/release-demo
build GOOS=darwin
build GOARCH=arm64
build CGO_ENABLED=0
```

What to notice

- `-X main.version=...` rewrites string vars at link time—no runtime git dependency in production.
- `-s -w` strips symbol/DWARF data; pair with `-trimpath` for more reproducible artifacts.
- `debug.ReadBuildInfo` exposes module and VCS metadata embedded by modern Go toolchains.

Try next

- Compare `ls -la` sizes of stripped vs unstripped builds.
- Add a `-X main.commit=$(git rev-parse --short HEAD)` one-liner to a release script.

Related Topics

- Core Go Commands
- Documentation and APIs
- Go as a Systems Language
- Containerization

Kubernetes Deploy Basics for Go

Kubernetes Deploy Basics for Go

Overview

A minimal production shape: container image, Deployment, Service, probes, resources. Go's single binary fits multi-stage images well (chapter 70).

Deployment sketch

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: bookstore
spec:
  replicas: 2
  selector:
    matchLabels: { app: bookstore }
  template:
    metadata:
      labels: { app: bookstore }
    spec:
      securityContext:
        runAsNonRoot: true
        runAsUser: 65532
      containers:
        - name: app
          image: ghcr.io/org/bookstore:1.2.3
          ports: [{ containerPort: 8080 }]
          env:
            - name: ADDR
              value: ":8080"
          readinessProbe:
            httpGet: { path: /readyz, port: 8080 }
            periodSeconds: 5
          livenessProbe:
            httpGet: { path: /livez, port: 8080 }
            periodSeconds: 10
          resources:
            requests: { cpu: "100m", memory: "128Mi" }
```

```yaml
            limits: { memory: "256Mi" }
          securityContext:
            allowPrivilegeEscalation: false
            readOnlyRootFilesystem: true
```

Service

```yaml
apiVersion: v1
kind: Service
metadata:
  name: bookstore
spec:
  selector: { app: bookstore }
  ports: [{ port: 80, targetPort: 8080 }]
```

Graceful stop

- `terminationGracePeriodSeconds` app shutdown timeout
- Fail readiness on SIGTERM first

Config & secrets

```yaml
envFrom:
  - secretRef: { name: bookstore-secrets }
```

Mount secrets as files when possible (`TOKEN_FILE` pattern).

Rules of thumb

Do	Don't
Probes + resources	Unlimited memory "to fix OOM" forever
Immutable tags (`:1.2.3`)	`:latest` in prod
Non-root	Root containers by habit

Try next

1. Deploy locally with kind/minikube.
2. Break readiness; observe Service endpoints.
3. Rollout image; confirm zero-downtime with probes.

GoReleaser and GitHub Actions

GoReleaser and GitHub Actions

Overview

Automate multi-OS binaries, checksums, and GitHub Releases with **GoReleaser** and a tag-driven workflow.

Minimal `.goreleaser.yaml`

```yaml
version: 2
builds:
  - id: app
    main: ./cmd/app
    env: [CGO_ENABLED=0]
    goos: [linux, darwin, windows]
    goarch: [amd64, arm64]
    ldflags:
      - -s -w -X main.version={{.Version}} -X main.commit={{.Commit}}
archives:
  - formats: [tar.gz]
    format_overrides:
      - goos: windows
        formats: [zip]
checksum:
  name_template: checksums.txt
```

Workflow sketch

```yaml
# .github/workflows/release.yml
on:
  push:
    tags: ["v*"]
jobs:
  release:
    runs-on: ubuntu-latest
    permissions:
      contents: write
```

```yaml
    steps:
      - uses: actions/checkout@v4
        with: { fetch-depth: 0 }
      - uses: actions/setup-go@v5
        with: { go-version-file: go.mod }
      - uses: goreleaser/goreleaser-action@v6
        with:
          args: release --clean
        env:
          GITHUB_TOKEN: ${{ secrets.GITHUB_TOKEN }}
```

Local snapshot

```go
goreleaser release --snapshot --clean
```

Rules of thumb

Do	Don't
Tag semver `v1.2.3`	Untagged mystery builds as "prod"
Pin action versions	`@master` for release
Verify checksums	Trust unsigned downloads blindly

Try next

1. Snapshot release locally.
2. Add `version` subcommand reading ldflags.
3. Attach `govulncheck` job on PR.

Container Health and Signals

Container Health and Signals

Overview

Containers change process semantics: PID 1, signal delivery, and healthchecks. Go apps should handle **SIGTERM**, expose probes, and avoid zombie reaping issues (use a proper init if needed).

Dockerfile healthcheck (optional)

```
HEALTHCHECK --interval=10s --timeout=2s --retries=3 \
  CMD wget -qO- http://127.0.0.1:8080/livez || exit 1
```

Prefer orchestrator probes in K8s over relying only on Docker HEALTHCHECK.

Signal PID 1

If your app is PID 1 and spawns children, consider `tini` or ensure children are reaped. Many Go apps don't spawn—still handle SIGTERM yourself.

```
ctx, stop := signal.NotifyContext(context.Background(), syscall.SIGTERM, syscall.SIGINT)
defer stop()
```

Read-only rootfs

```
# run as non-root; mount emptyDir for temp if needed
USER nonroot:nonroot
```

Write only to `/tmp` or mounted volumes.

Concern	Practice

12-factor in containers

Concern	Practice
Config	env + files
Logs	stdout/stderr
Graceful	SIGTERM → Shutdown
Port	`ADDR=:8080`

Rules of thumb

Do	Don't
Listen on `0.0.0.0/:` in container	Hardcode `localhost` only for in-cluster listen
Exit non-zero on fatal boot	Hang forever if DB missing (or use restart policy consciously)
Document probe paths	Silent health

Try next

1. `docker run` + `docker stop`; measure drain time.
2. Read-only rootfs; fix writes to temp.
3. Align HEALTHCHECK with `/livez`.

workloadbuilder and Gang Scheduling (Kubernetes 1.37)

workloadbuilder and Gang Scheduling (Kubernetes 1.37)

Kubernetes 1.37 advances Workload-Aware Scheduling (WAS): `Workload` / `PodGroup` and gang scheduling graduate toward Beta, and controllers get a shared Go path instead of each inventing their own `PodGroup` dialect. The idea surfaced widely in the 1.37 release notes and companion write-ups; this chapter teaches the **Go API** you actually call — the `scheduling.k8s.io/v1alpha3` building blocks and the `workloadbuilder` library in `k8s.io/component-helpers`.

If you maintain a Job-like CRD, a trainer operator, or anything that must place N pods together (or a tree of groups), this is the boring contract: embed the blocks, compile with the library, stamp pods with `spec.schedulingGroup.podGroupName`, and let kube-scheduler do gang / topology / disruption.

Pin modules to the cluster minor you target. These listings use `v0.37.1` (Kubernetes 1.37.1).

```
go get k8s.io/component-helpers@v0.37.1 \
      k8s.io/api@v0.37.1 \
      k8s.io/apimachinery@v0.37.1
```

Import path:

```
import "k8s.io/component-helpers/scheduling/schedulingv1/workloadbuilder"
```

Mental model

WAS splits **policy templates** from **runtime groups**:

```
user fills fields on YOUR CRD


v1alpha3 building blocks  (WorkloadPodGroup… / WorkloadCompositePodGroup…)


workloadbuilder.Builder
   Validate → BuildWorkload → NewPodGroup / NewCompositePodGroup


scheduling.k8s.io/v1beta1 Workload   (templates: PodGroupTemplate / Composite…)
scheduling.k8s.io/v1beta1 PodGroup   (runtime leaf group)
```

```
scheduling.k8s.io/v1alpha3 CompositePodGroup  (runtime group-of-groups)


Pods with spec.schedulingGroup.podGroupName → kube-scheduler gang / topology / preemption
```

Object	Role
Building blocks	Strongly typed structs you embed in *your* API (`spec.scheduling`, etc.). Users never author raw `Workload` YAML.
`Workload`	Compiled template: owns `podGroupTemplates` / `compositePodGroupTemplates`, carries `controllerRef`.
`PodGroup`	Runtime leaf unit the scheduler queues and gangs (`minCount`).
`CompositePodGroup`	Runtime parent over child groups (`minGroupCount`).
`workloadbuilder`	Shared defaulting, validation, and compilation so Job, JobSet, LWS, and your CRD speak one dialect.

Leaf policy is **`basic`** (pod-by-pod) or **`gang`** (`minCount`). Composite gang uses **`minGroupCount`** (how many *child groups* must be schedulable together). Disruption is **`single`** or **`all`**. Topology co-location is a node-label key on the constraints block. Shared DRA claims can hang off the group instead of per pod.

Feature gates (cluster side, not on the library itself): `GenericWorkload` (core WAS / gang / preemption), `WorkloadWithJob` (Job `spec.scheduling`), `CompositePodGroup`, `TopologyAwareWorkloadScheduling`, `DRAWorkloadResourceClaims`. Check the current 1.37 docs for which are Beta vs Alpha and which components need them.

Building blocks you embed

The Job API groups all four leaf blocks under `spec.scheduling`. Copy that shape unless your domain needs per-component nesting.

```go
// Shape mirrored by batch/v1 Job's spec.scheduling (Kubernetes 1.37).
type JobSchedulingConfiguration struct {
    SchedulingPolicy      *schedulingv1alpha3.WorkloadPodGroupSchedulingPolicy
    SchedulingConstraints *schedulingv1alpha3.WorkloadPodGroupSchedulingConstraints
    DisruptionMode        *schedulingv1alpha3.WorkloadPodGroupDisruptionMode
    ResourceClaims        []schedulingv1alpha3.WorkloadPodGroupResourceClaim
}
```

Composite controllers (JobSet-like) use the parallel `WorkloadCompositePodGroup…` types. Field *names* on your CRD are yours; the *struct types* are shared so users recognize the same knobs across controllers.

Allow-lists matter: `BuildOptions.AllowedPolicies` / `AllowedDisruptionModes` are deny-by-default. A future policy does not silently appear in your CRD until you opt in (and bump your vendored helpers).

Builder flow

A `WorkloadItem` is one logical component. No children → one `PodGroupTemplate`. Children → `CompositePodGroupTemplate` over them. Each node carries:

- **DefaultConfig** — controller defaults when the user omits a field (Job defaults unset policy to basic).
- **Input** — user blocks plus **field path elements** so validation errors land on *their* CRD path, not a synthetic `Workload`.
- **Callbacks** — post-merge tweaks (e.g. default gang `minCount` from Job `parallelism`).

Then: `NewBuilder` → `Validate` → `BuildWorkload` → `NewPodGroup` (and/or `NewCompositePodGroup`). Parent/child split: if a parent already owns the `Workload`, the child uses `NewBuilderFromExistingWorkload` and only materializes its `PodGroup` — it must not recompile.

Worked examples

Case 1: Compile a gang Workload like the Job controller

Save as `wb_gang_e2e.go`. Map user intent (gang, zone topology, all-or-nothing disruption), default unset `minCount` to parallelism `8`, validate, compile, and stamp a runtime `PodGroup`.

```go
// wb_gang_e2e.go
package main

import (
    "context"
    "fmt"

    schedulingv1alpha3 "k8s.io/api/scheduling/v1alpha3"
    metav1 "k8s.io/apimachinery/pkg/apis/meta/v1"
    "k8s.io/apimachinery/pkg/util/validation/field"
    "k8s.io/component-helpers/scheduling/schedulingv1/workloadbuilder"
)

func main() {
    parallelism := int32(8)
    userInput := workloadbuilder.WorkloadInput{
```

```
        Policy: workloadbuilder.PolicyInput{
            PodGroupData: &schedulingv1alpha3.WorkloadPodGroupSchedulingPolicy{
                Gang: &schedulingv1alpha3.WorkloadPodGroupGangSchedulingPolicy{}, // minCoun
            },
            PathElements: []string{"schedulingPolicy"},
        },
        Constraints: workloadbuilder.ConstraintsInput{
            PodGroupData: &schedulingv1alpha3.WorkloadPodGroupSchedulingConstraints{
                Topology: []schedulingv1alpha3.TopologyConstraint{
                    {Key: "topology.kubernetes.io/zone"},
                },
            },
            PathElements: []string{"schedulingConstraints"},
        },
        DisruptionMode: workloadbuilder.DisruptionModeInput{
            PodGroupData: &schedulingv1alpha3.WorkloadPodGroupDisruptionMode{
                All: &schedulingv1alpha3.WorkloadPodGroupAllDisruptionMode{},
            },
            PathElements: []string{"disruptionMode"},
        },
    }

    item := &workloadbuilder.WorkloadItem{
        Name: "workers",
        Path: field.NewPath("spec", "scheduling"),
        DefaultConfig: &workloadbuilder.SchedulingConfig{
            Policy: &workloadbuilder.SchedulingPolicy{
                Basic: &workloadbuilder.BasicSchedulingPolicy{},
            },
        },
        Input: userInput,
        Callbacks: []workloadbuilder.SchedulingConfigFunc{
            func(cfg *workloadbuilder.SchedulingConfig) {
                if g := cfg.Policy.Gang; g != nil && g.MinCount == nil {
                    n := parallelism
                    g.MinCount = &n
                }
            },
        },
    }

    ctrl := true
    owner := &metav1.OwnerReference{
        APIVersion: "batch/v1",
        Kind:       "Job",
        Name:       "train",
```

```
        UID:        "job-uid",
        Controller: &ctrl,
    }
    builder := workloadbuilder.NewBuilder(item, workloadbuilder.BuildOptions{
        Name:                   "train-wl",
        Namespace:              "ml",
        Owner:                  owner,
        AllowedPolicies:        []workloadbuilder.SchedulingPolicyOption{workloadbuilder.Bas
        AllowedDisruptionModes: []workloadbuilder.DisruptionModeOption{workloadbuilder.Singl
    })

    if errs := builder.Validate(context.Background(), workloadbuilder.ValidationInput{}); le
        fmt.Printf("validation failed: %v\n", errs.ToAggregate())
        return
    }

    wl, err := builder.BuildWorkload()
    if err != nil {
        fmt.Printf("build failed: %v\n", err)
        return
    }
    pg, err := builder.NewPodGroup("train-pg", item.Name)
    if err != nil {
        fmt.Printf("podgroup failed: %v\n", err)
        return
    }

    tmpl := wl.Spec.PodGroupTemplates[0]
    fmt.Printf("Workload %s/%s controllerRef=%s/%s\n",
        wl.Namespace, wl.Name, wl.Spec.ControllerRef.Kind, wl.Spec.ControllerRef.Name)
    fmt.Printf("  template %q: gang minCount=%d topology=%s disruption=all\n",
        tmpl.Name, tmpl.SchedulingPolicy.Gang.MinCount, tmpl.SchedulingConstraints.Topology[
    fmt.Printf("PodGroup %s/%s -> workload=%s template=%s minCount=%d\n",
        pg.Namespace, pg.Name,
        pg.Spec.WorkloadRef.WorkloadName, pg.Spec.WorkloadRef.TemplateName,
        pg.Spec.SchedulingPolicy.Gang.MinCount)
}
```

Run:

```
go run wb_gang_e2e.go
```

Output:

```
Workload ml/train-wl controllerRef=Job/train
  template "workers": gang minCount=8 topology=topology.kubernetes.io/zone disruption=all
PodGroup ml/train-pg -> workload=train-wl template=workers minCount=8
```

After create/patch of `Workload` and `PodGroup`, set each pod's `spec.schedulingGroup.podGroupName` to `train-pg`. Discovery uses `Workload.spec.controllerRef` and `PodGroup.spec.workloadRef`, not brittle name conventions. Owner refs on the objects you create keep GC aligned with your CR.

Out-of-tree controllers leave declarative validation **on** (default). In-tree Job sets `DisableDeclarativeValidation: true` because the apiserver already ran DV on the embedded blocks.

Case 2: Validation rejects basic + all disruption

Save as `wb_reject_basic_all.go`. The preemption unit must not be larger than the scheduling unit — `all` disruption with `basic` policy is nonsense and `Validate` reports it at the user's field path.

```go
// wb_reject_basic_all.go
package main

import (
    "context"
    "fmt"

    schedulingv1alpha3 "k8s.io/api/scheduling/v1alpha3"
    metav1 "k8s.io/apimachinery/pkg/apis/meta/v1"
    "k8s.io/apimachinery/pkg/util/validation/field"
    "k8s.io/component-helpers/scheduling/schedulingv1/workloadbuilder"
)

func main() {
    item := &workloadbuilder.WorkloadItem{
        Name: "worker",
        Path: field.NewPath("spec", "scheduling"),
        DefaultConfig: &workloadbuilder.SchedulingConfig{
            Policy: &workloadbuilder.SchedulingPolicy{Basic: &workloadbuilder.BasicSchedulin
        },
        Input: workloadbuilder.WorkloadInput{
            Policy: workloadbuilder.PolicyInput{
                PodGroupData: &schedulingv1alpha3.WorkloadPodGroupSchedulingPolicy{
                    Basic: &schedulingv1alpha3.WorkloadPodGroupBasicSchedulingPolicy{},
                },
                PathElements: []string{"schedulingPolicy"},
            },
            DisruptionMode: workloadbuilder.DisruptionModeInput{
                PodGroupData: &schedulingv1alpha3.WorkloadPodGroupDisruptionMode{
                    All: &schedulingv1alpha3.WorkloadPodGroupAllDisruptionMode{},
                },
                PathElements: []string{"disruptionMode"},
            },
```

```go
        },
    }

    errs := workloadbuilder.NewBuilder(item, workloadbuilder.BuildOptions{
        Owner: &metav1.OwnerReference{
            APIVersion: "example.com/v1", Kind: "TrainJob", Name: "bad", UID: "u",
        },
        AllowedPolicies:        []workloadbuilder.SchedulingPolicyOption{workloadbuilder.Bas
        AllowedDisruptionModes: []workloadbuilder.DisruptionModeOption{workloadbuilder.Singl
    }).Validate(context.Background(), workloadbuilder.ValidationInput{})

    if len(errs) > 0 {
        fmt.Printf("rejected: %v\n", errs.ToAggregate())
        return
    }
    fmt.Println("accepted")
}
```

Run:

```go run wb_reject_basic_all.go```

Output:

```
rejected: spec.scheduling.disruptionMode: Invalid value: "": the disruptionMode `all` is not su
```

On **update**, pass `ValidationInput{OldRoot: oldItem}` so immutability checks compare against the previously stored tree. Zero `ValidationInput` is create.

Case 3: Composite WorkloadItem tree (group of groups)

Save as `wb_composite.go`. A parent node with children compiles to a `CompositePodGroupTemplate`; leaf gang uses `minCount`, composite gang uses `minGroupCount`.

```go
// wb_composite.go
package main

import (
    "context"
    "fmt"

    schedulingv1alpha3 "k8s.io/api/scheduling/v1alpha3"
    metav1 "k8s.io/apimachinery/pkg/apis/meta/v1"
    "k8s.io/apimachinery/pkg/util/validation/field"
    "k8s.io/component-helpers/scheduling/schedulingv1/workloadbuilder"
)
```
```

```
func main() {
    minWorkers, minDriver, minGroups := int32(4), int32(1), int32(2)

    workers := &workloadbuilder.WorkloadItem{
        Name: "workers",
        Path: field.NewPath("spec", "workers", "scheduling"),
        Input: workloadbuilder.WorkloadInput{
            Policy: workloadbuilder.PolicyInput{
                PodGroupData: &schedulingv1alpha3.WorkloadPodGroupSchedulingPolicy{
                    Gang: &schedulingv1alpha3.WorkloadPodGroupGangSchedulingPolicy{MinCount:
                },
                PathElements: []string{"schedulingPolicy"},
            },
        },
    }
    driver := &workloadbuilder.WorkloadItem{
        Name: "driver",
        Path: field.NewPath("spec", "driver", "scheduling"),
        Input: workloadbuilder.WorkloadInput{
            Policy: workloadbuilder.PolicyInput{
                PodGroupData: &schedulingv1alpha3.WorkloadPodGroupSchedulingPolicy{
                    Gang: &schedulingv1alpha3.WorkloadPodGroupGangSchedulingPolicy{MinCount:
                },
                PathElements: []string{"schedulingPolicy"},
            },
        },
    }
    root := &workloadbuilder.WorkloadItem{
        Name: "root",
        Path: field.NewPath("spec", "scheduling"),
        Input: workloadbuilder.WorkloadInput{
            Policy: workloadbuilder.PolicyInput{
                CompositePodGroupData: &schedulingv1alpha3.WorkloadCompositePodGroupScheduli
                    Gang: &schedulingv1alpha3.WorkloadCompositePodGroupGangSchedulingPolicy{
                        MinGroupCount: &minGroups,
                    },
                },
                PathElements: []string{"schedulingPolicy"},
            },
        },
        Children: []*workloadbuilder.WorkloadItem{workers, driver},
    }

    owner := &metav1.OwnerReference{
        APIVersion: "example.com/v1", Kind: "TrainJob", Name: "jobset-like", UID: "u",
    }
```

```go
    b := workloadbuilder.NewBuilder(root, workloadbuilder.BuildOptions{
        Name:                   "trainjob-wl",
        Namespace:              "ml",
        Owner:                  owner,
        AllowedPolicies:        []workloadbuilder.SchedulingPolicyOption{workloadbuilder.Bas
        AllowedDisruptionModes: []workloadbuilder.DisruptionModeOption{workloadbuilder.Singl
    })
    if errs := b.Validate(context.Background(), workloadbuilder.ValidationInput{}); len(errs
        fmt.Printf("validation failed: %v\n", errs.ToAggregate())
        return
    }
    wl, err := b.BuildWorkload()
    if err != nil {
        fmt.Printf("build failed: %v\n", err)
        return
    }
    cpg, err := b.NewCompositePodGroup("trainjob-root", "root")
    if err != nil {
        fmt.Printf("composite failed: %v\n", err)
        return
    }

    fmt.Printf("Workload %s/%s compositeTemplates=%d topLevelPodTemplates=%d\n",
        wl.Namespace, wl.Name, len(wl.Spec.CompositePodGroupTemplates), len(wl.Spec.PodGroup
    ct := wl.Spec.CompositePodGroupTemplates[0]
    fmt.Printf("  composite %q minGroupCount=%d nestedPodGroups=%d\n",
        ct.Name, ct.SchedulingPolicy.Gang.MinGroupCount, len(ct.PodGroupTemplates))
    fmt.Printf("CompositePodGroup %s/%s template=%s\n",
        cpg.Namespace, cpg.Name, cpg.Spec.WorkloadRef.TemplateName)
}
```

Run:

```go
go run wb_composite.go
```

Output:

```
Workload ml/trainjob-wl compositeTemplates=1 topLevelPodTemplates=0
  composite "root" minGroupCount=2 nestedPodGroups=2
CompositePodGroup ml/trainjob-root template=root
```

Wire child `PodGroups` with `spec.parentCompositePodGroupName` pointing at the runtime composite. Parent controllers that own the `Workload` typically name templates and let children call `NewBuilderFromExistingWorkload` + `NewPodGroup` (Job's `scheduling.k8s.io/group-template-name` annotation is the in-tree seam for that delegation).

Job `spec.scheduling` (operator view)

With `WorkloadWithJob` enabled, users state intent on the Job instead of hoping the controller infers it:

```yaml
apiVersion: batch/v1
kind: Job
metadata:
  name: distributed-training-job
spec:
  parallelism: 8
  completions: 8
  scheduling:
    schedulingPolicy:
      gang: {}   # minCount omitted → defaults to parallelism (8)
    schedulingConstraints:
      topology:
        - key: topology.kubernetes.io/zone
    disruptionMode:
      all: {}
  template:
    spec:
      restartPolicy: Never
      containers:
        - name: worker
          image: example.com/train:1.2.3
```

Omit `spec.scheduling` (or its policy) → **basic**, same as ordinary Jobs. After create, most scheduling fields are frozen; **`schedulingPolicy.gang.minCount` stays mutable** so elastic gangs can rescale without recreating the Job (and `minCount` must not exceed `parallelism`).

Integrating your controller (checklist)

1. Embed the `v1alpha3` blocks you support on your CRD; reuse Job's field names when you can.
2. Build a `WorkloadItem` tree in reconcile (flat or nested); record `Path` / `PathElements`.
3. Set allow-lists; leave DV on out-of-tree.
4. `NewBuilder` → `Validate` → `BuildWorkload` → `NewPodGroup` / `NewCompositePodGroup`; create or patch; stamp pods.
5. Do **not** write a private gang controller or invent a second `PodGroup` schema.

You are explicitly not coupling users to a particular queueing product. Queueing/quota layers (e.g. Kueue) should compose on top of the in-tree `Workload` / `PodGroup` contract rather than replace it.

Gotchas

Trap	Fix
Hand-assembling `Workload` YAML in every controller	Embed blocks + `workloadbuilder`
Empty allow-lists	Deny-by-default — list `BasicPolicy` / `GangPolicy` (and modes) explicitly
`all` disruption + `basic` policy	Rejected; use gang if you need group-level disruption
Child recompiles parent's `Workload`	Use `NewBuilderFromExistingWorkload`
Library works, cluster ignores gangs	Enable `GenericWorkload` (and friends) on apiserver / controller-manager / scheduler
Alpha/Beta churn (`v1alpha2` → `v1alpha3`, disruption rename)	Pin helpers to cluster minor; read upgrade notes
Shared DRA claim with gate off	Know `DRAWorkloadResourceClaims` behavior so you do not flood per-pod claims

Try next

1. Run the three programs above against `v0.37.1` helpers.
2. On a kind cluster with the documented feature gates, apply a Job with `spec.scheduling.schedulingPolicy` and watch `Workload` / `PodGroup` appear.
3. Sketch your CRD's `spec.scheduling` (or per-component nest) and a single-node `WorkloadItem` before adding composite children.
4. Read the package examples under `k8s.io/component-helpers/.../workloadbuilder/examples` for the canonical Job and delegation flows.

Sources

- Scheduling Building Block APIs and the workloadbuilder Library
- Kubernetes v1.37: Advancing Workload-Aware Scheduling
- pkg.go.dev — workloadbuilder
- KEP-6089 — Workload Aware Scheduling Controller APIs
- KEP-5547 — Integrate Workload APIs with Job Controller

Security Tools and Design

2004

Security Tools

Security Tools: Automating Defense

Overview

Security should not depend on heroic manual review. Go has a strong toolchain for **static detection**, **reachable vulnerability analysis**, **capability inventory**, and **cryptographic testing**. Wire these into CI so regressions fail the build before they reach production.

This chapter covers `govulncheck`, `gosec`, `capslock`, `nilaway`, modern crypto helpers (`crypto/hpke`, `testing/cryptotest`), experimental secret handling, and how to assemble them into a pipeline.

Suggested Security Pipeline

```
commit
  --> go vet / golangci-lint
  --> gosec (or security linters)
  --> nilaway (optional, deeper nil analysis)
  --> govulncheck (reachable vulns)
  --> capslock (dependency capability review)
  --> go test (incl. crypto vectors)
  --> release / sign
```

```
trust boundary diagram

developer laptop --> CI runners --> artifact registry --> runtime
     |                  |                 |                 |
   secrets       pinned actions      cosign verify    least privilege
```

1. `govulncheck` — Reachability-Based Vuln Scanning

`govulncheck` uses the Go vulnerability database and **call graph reachability**. It prefers reporting issues your code can actually hit, not every CVE in the module graph.

```
go install golang.org/x/vuln/cmd/govulncheck@latest
govulncheck ./...

# JSON for CI parsing
```

```
govulncheck -json ./... > govulncheck.json
```

CI snippet

```yaml
- name: govulncheck
  run: |
    go install golang.org/x/vuln/cmd/govulncheck@latest
    govulncheck ./...
```

Interpreting results

1. Upgrade the module if a fixed version exists.
2. If unused, confirm with `govulncheck` (symbol level)—don't ignore blindly.
3. Vendor or replace only with a documented exception and expiry.

```
go get example.com/module@v1.2.4
go mod tidy
govulncheck ./...
```

2. `gosec` — AST Security Linter

`gosec` walks the AST looking for dangerous patterns.

```
go install github.com/securego/gosec/v2/cmd/gosec@latest
gosec ./...

# Exclude generated code
gosec -exclude-generated ./...
```

Common findings:

Pattern	Risk
Hardcoded credentials	Secret leakage
SQL via string concat	Injection
`math/rand` for security tokens	Predictable values
`chmod 0777` / weak file perms	Local privilege issues
`http` without TLS in prod configs	Cleartext
`unsafe` / weak crypto APIs	Memory / crypto breaks

Safer random and SQL examples

```go
// Bad: math/rand for tokens
// token := fmt.Sprintf("%d", rand.Int())

// Good: crypto/rand
func sessionToken() (string, error) {
    var b [16]byte
    if _, err := rand.Read(b[:]); err != nil {
        return "", err
    }
    return hex.EncodeToString(b[:]), nil
}
```

```go
// Bad
// db.Query("SELECT * FROM users WHERE name = '" + name + "'")

// Good
rows, err := db.QueryContext(ctx, `SELECT id, name FROM users WHERE name = $1`, name)
```

Tune severity in CI: fail on high/critical; track medium findings.

3. `capslock` — Capability Analysis

Google's **capslock** reports what *capabilities* a package graph uses: network, filesystem, runtime unsafe, os exec, etc.

```bash
go install github.com/google/capslock/cmd/capslock@latest
capslock -packages ./...
```

Use it when reviewing dependencies:

- A pure string helper that needs **Network** is a smell.
- A config parser that needs **OsExec** deserves scrutiny.
- Diff capslock output across PRs that bump dependencies.

```
expected for a CLI file tool:  FileSystem, maybe Runtime
unexpected for left-pad clone: Network, CAP_SYS_ADMIN stories, etc.
```

4. `nilaway` — Static Nil Flows

Uber's **nilaway** tracks potential nil dereferences across functions more deeply than many linters.

```
go install go.uber.org/nilaway/cmd/nilaway@latest
nilaway ./...
```

It does not turn Go into Rust, but it catches real production panics:

```go
func loadUser(id string) (*User, error) {
    if id == "" {
        return nil, fmt.Errorf("empty id")
    }
    // ...
    return u, nil
}

func handler() {
    u, err := loadUser(id)
    if err != nil {
        return
    }
    // nilaway helps when err is ignored or only partially checked
    fmt.Println(u.Name)
}
```

Run as a periodic or blocking CI job if the signal-to-noise ratio works for your codebase.

5. Modern Crypto Tooling

`crypto/hpke` (Hybrid Public Key Encryption)

HPKE (RFC 9180) combines KEM + KDF + AEAD for standardized envelope encryption—useful for config blobs, recipient-encrypted messages, and modern protocol designs.

```go
import (
    "crypto/hpke"
    "crypto/rand"
)

func encryptHPKE(recipientPub hpke.PublicKey, msg []byte) (enc, ciphertext []byte, err error
    suite := hpke.NewSuite(
        hpke.KEM_X25519_HKDF_SHA256,
        hpke.KDF_HKDF_SHA256,
        hpke.AEAD_AES_128_GCM,
    )
    sender, err := suite.NewSender(recipientPub, nil)
    if err != nil {
        return nil, nil, err
```

```go
    }
    enc, sealer, err := sender.Setup(rand.Reader)
    if err != nil {
        return nil, nil, err
    }
    ciphertext, err = sealer.Seal(msg, nil)
    return enc, ciphertext, err
}
```

```
flow:
  ] -> KEM[
  ] -> AEAD[
  KEM -> EncapsKey[
  KEM -> SharedSecret[
  EncapsKey -> Network
  AEAD -> CipherText
  ] -> Decaps[
  Network -> Decaps
  Decaps -> AEADDec[
  Network -> AEADDec
  AEADDec -> Plain[
```

Prefer stdlib crypto primitives over hand-rolled combinations of ECDH + HKDF + GCM.

`testing/cryptotest`

Use `testing/cryptotest` (where available in your Go version) to exercise cryptographic implementations against known vectors, basic correctness, and timing-related expectations for custom code that wraps stdlib primitives.

```go
func TestHMACVectors(t *testing.T) {
    // table-driven known-answer tests for your wrappers
    mac := hmac.New(sha256.New, []byte("key"))
    mac.Write([]byte("data"))
    got := mac.Sum(nil)
    want := mustHex("...") // known vector
    if !hmac.Equal(got, want) {
        t.Fatalf("mismatch")
    }
}
```

Always compare secrets with **constant-time** helpers (`subtle.ConstantTimeCompare`, `hmac.Equal`).

6. Experimental: `runtime/secret`

Experiments around tighter secret lifetimes appear behind build tags / `GOEXPERIMENT`. Treat them as **opt-in** and verify support for your Go version before relying on them in production.

```go
//go:build goexperiment.runtimesecret

import "runtime/secret"

func useSecret(token []byte) {
    secret.Do(func() {
        // limited lifetime window for sensitive material
        _ = token
    })
}
```

```
GOEXPERIMENT=runtimesecret go test ./...
```

Still follow classic hygiene: minimize copies, zero buffers when practical, don't log secrets, prefer OS keychains/HSMs for long-term keys.

7. Complementary Checks

Tool	Role
`go vet`	Compiler-adjacent mistakes
`golangci-lint`	Aggregates many linters
`staticcheck`	High-signal correctness
`semgrep` / CodeQL	Org-wide custom rules
`govulncheck`	Module CVEs with reachability
Image scanners (`trivy`)	Container base CVEs

Example golangci config fragment:

```yaml
linters:
  enable:
    - gosec
    - govet
    - staticcheck
    - errcheck
```

Wiring a Makefile / CI Target

```makefile
.PHONY: sec
sec:
    go vet ./...
    gosec -quiet ./...
    govulncheck ./...
```

```yaml
# GitHub Actions job fragment
- name: Security
  run: make sec
```

Production Checklist

- ☐ `govulncheck ./...` on every PR
- ☐ `gosec` or equivalent security linters enabled
- ☐ Dependency bumps reviewed with capslock when capabilities jump
- ☐ No `math/rand` for security tokens
- ☐ Parameterized SQL only
- ☐ Known-answer tests for crypto wrappers
- ☐ Secrets never in git; scanner in pre-commit optional
- ☐ Container images scanned on release
- ☐ Exceptions documented with owner + expiry

Common Pitfalls

1. **Ignoring govulncheck because "it's transitive"** — reachability exists for a reason; verify.
2. **Disabling gosec rules globally** — narrow suppressions with comments and justification.
3. **Custom crypto** — combining primitives incorrectly; prefer HPKE/TLS/libsodium-grade designs.
4. **Logging tokens** — still the #1 operational leak.
5. **Security tools only on main** — PRs need the same gates.
6. **False confidence** — static tools don't replace threat modeling or fuzzing of parsers.

Exercises

1. **Install and run** — Run `govulncheck` and `gosec` on this book's sample modules or a toy service; fix or document each finding.
2. **Token generator** — Implement `sessionToken` with `crypto/rand`; add a test that 1000 tokens are unique and correct length.
3. **SQL injection lab** — Write a deliberate vulnerable query (in a `_test` file or clearly broken sample) and show gosec flags it; then fix with parameters.

4. **Capslock review** — Run capslock on a module that imports `net/http` vs a pure algorithm package; compare capability sets.
5. **HMAC vectors** — Table-test an HMAC helper against a published RFC vector; use `hmac.Equal`.
6. **CI gate** — Add a `make sec` target and a workflow step that fails the build on findings.

More examples

Constant-time compare (HMAC / tokens)

```bash
mkdir -p /tmp/go-hmac-eq && cd /tmp/go-hmac-eq
go mod init example.com/hmac-eq
```

Save as `main.go`:

```go
package main

import (
    "crypto/hmac"
    "crypto/sha256"
    "encoding/hex"
    "fmt"
)

func sign(secret, msg string) string {
    m := hmac.New(sha256.New, []byte(secret))
    _, _ = m.Write([]byte(msg))
    return hex.EncodeToString(m.Sum(nil))
}

func valid(secret, msg, gotHex string) bool {
    want := sign(secret, msg)
    // Decode both; compare with hmac.Equal (constant-time on equal-length slices).
    a, err1 := hex.DecodeString(want)
    b, err2 := hex.DecodeString(gotHex)
    if err1 != nil || err2 != nil || len(a) != len(b) {
        return false
    }
    return hmac.Equal(a, b)
}

func main() {
    secret, msg := "s3cr3t", "POST /pay|42"
    sig := sign(secret, msg)
    fmt.Println("sig ok:", valid(secret, msg, sig))
```

```
    fmt.Println("tampered:", valid(secret, msg, sig[:len(sig)-1]+"0"))
    fmt.Println("wrong secret:", valid("other", msg, sig))
}
```

```
go run .
```

Expected output:

```
sig ok: true
tampered: false
wrong secret: false
```

Path jail (block traversal)

```
mkdir -p /tmp/go-path-jail && cd /tmp/go-path-jail
go mod init example.com/path-jail
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "path/filepath"
    "strings"
)

func underRoot(root, userPath string) (string, error) {
    clean := filepath.Clean("/" + userPath) // force absolute-ish clean
    clean = strings.TrimPrefix(clean, string(filepath.Separator))
    full := filepath.Join(root, clean)
    rel, err := filepath.Rel(root, full)
    if err != nil || strings.HasPrefix(rel, "..") {
        return "", fmt.Errorf("escape: %q", userPath)
    }
    return full, nil
}

func main() {
    root := "/var/app/data"
    for _, p := range []string{"docs/a.txt", "../etc/passwd", "/etc/passwd"} {
        full, err := underRoot(root, p)
        if err != nil {
            fmt.Printf("%q -> ERR %v\n", p, err)
            continue
```

```
        }
        fmt.Printf("%q -> %s\n", p, full)
    }
}
```

```
go run .
```

Expected output:

```
"docs/a.txt" -> /var/app/data/docs/a.txt
"../etc/passwd" -> ERR escape: "../etc/passwd"
"/etc/passwd" -> ERR escape: "/etc/passwd"
```

Runnable example

Security tools are external; the patterns they enforce are stdlib. This program generates session tokens with `crypto/rand`, signs and verifies an HMAC, and compares digests in constant time—the kind of code `gosec` expects instead of `math/rand`.

```
mkdir -p /tmp/go-sec-tools && cd /tmp/go-sec-tools
go mod init example.com/sec-tools
```

Save as `main.go`:

```
package main

import (
    "crypto/hmac"
    "crypto/rand"
    "crypto/sha256"
    "encoding/hex"
    "fmt"
    "os"
)

func sessionToken(n int) (string, error) {
    b := make([]byte, n)
    if _, err := rand.Read(b); err != nil {
        return "", err
    }
    return hex.EncodeToString(b), nil
}

func sign(key, msg []byte) []byte {
    m := hmac.New(sha256.New, key)
```

```go
    _, _ = m.Write(msg)
    return m.Sum(nil)
}

func main() {
    tok, err := sessionToken(16)
    if err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }
    fmt.Println("token:", tok)
    fmt.Println("token_len_hex:", len(tok))

    key := make([]byte, 32)
    if _, err := rand.Read(key); err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }
    msg := []byte("order:42:ship")
    sig := sign(key, msg)
    fmt.Println("hmac:", hex.EncodeToString(sig))

    // Constant-time compare (never == on secret slices).
    again := sign(key, msg)
    if !hmac.Equal(sig, again) {
        fmt.Println("verify: FAIL")
        os.Exit(1)
    }
    fmt.Println("verify: ok")

    // Wrong key must fail.
    badKey := make([]byte, 32)
    _, _ = rand.Read(badKey)
    if hmac.Equal(sig, sign(badKey, msg)) {
        fmt.Println("bad key unexpectedly matched")
        os.Exit(1)
    }
    fmt.Println("bad_key_rejected: ok")

    // Uniqueness smoke test.
    seen := map[string]struct{}{}
    for i := 0; i < 1000; i++ {
        t, err := sessionToken(16)
        if err != nil {
            fmt.Fprintln(os.Stderr, err)
            os.Exit(1)
        }
```

```go
        if _, ok := seen[t]; ok {
            fmt.Println("duplicate token")
            os.Exit(1)
        }
        seen[t] = struct{}{}
    }
    fmt.Println("unique_tokens: 1000")
}
```

```go
go run .
```

Expected output (illustrative; tokens random):

```
token: a3f1c9...
token_len_hex: 32
hmac: 9b2e...
verify: ok
bad_key_rejected: ok
unique_tokens: 1000
```

What to notice

- `crypto/rand` is for security-sensitive values; `math/rand` is a classic `gosec` finding.
- `hmac.Equal` (or `subtle.ConstantTimeCompare`) avoids timing leaks from `bytes.Equal` on secrets.
- Known-answer tests and uniqueness checks belong in CI next to `govulncheck` / `gosec`.

Try next

- Add a table-driven HMAC test with a fixed key/message and expected hex digest.
- Run `gosec ./...` and `govulncheck ./...` on this module once the tools are installed.

Related Topics

- Application Hardening
- CI/CD & Hardening
- Static Analysis
- TLS / mTLS

Application Hardening

Application Hardening

Security tools find known problems. **Hardening** is the engineering practice of making services difficult to exploit and safe to operate when something goes wrong. This chapter is the production checklist for Go services: least privilege, attack-surface reduction, input distrust, secure defaults, and fail-closed behavior.

Why Hardening Matters

A service can pass `govulncheck` and still be fragile:

- Debug endpoints exposed on the public interface
- Overly broad filesystem or network capabilities
- Missing request size limits (memory exhaustion)
- Secrets loaded from env forever with no rotation path
- Process running as root "because Docker made it easy"

Hardening is not a single library. It is a set of defaults you enforce until they become muscle memory.

```
attack surface
  |-- network listeners
  |-- auth boundaries
  |-- parse/decode paths
  |-- filesystem access
  |-- subprocesses / cgo
  |-- admin/debug tooling
  +-- dependency graph
```

Core Principles

1. **Least privilege** — the process gets only what it needs (UID, capabilities, mounts, outbound destinations).
2. **Fail closed** — ambiguous auth, bad config, or missing secrets stop the service rather than open a hole.
3. **Distrust boundaries** — every byte from the network, queue, or filesystem is hostile until validated.
4. **Defense in depth** — TLS + authn + authz + rate limits + audit logs; no single control is enough.
5. **Operational security** — logs, metrics, and admin APIs are part of the threat model.

Hardening the Process

Drop privileges early

Never keep root after bind if you can avoid it. Prefer binding via a privileged parent, then drop; or use non-privileged ports behind a reverse proxy.

```go
package main

import (
    "log"
    "net/http"
    "os"
    "os/user"
    "strconv"
    "syscall"
    "time"
)

func dropToUser(username string) error {
    u, err := user.Lookup(username)
    if err != nil {
        return err
    }
    uid, _ := strconv.Atoi(u.Uid)
    gid, _ := strconv.Atoi(u.Gid)
    // Order matters: groups, then gid, then uid.
    if err := syscall.Setgid(gid); err != nil {
        return err
    }
    if err := syscall.Setuid(uid); err != nil {
        return err
    }
    return nil
}

func main() {
    // Prefer: start as non-root in the container/image.
    // If you must bind privileged ports, drop after Listen.
    if os.Geteuid() == 0 {
        log.Println("warning: running as root; drop privileges in production")
    }

    mux := http.NewServeMux()
    mux.HandleFunc("GET /healthz", func(w http.ResponseWriter, r *http.Request) {
        w.WriteHeader(http.StatusOK)
    })
```

```go
    srv := &http.Server{
        Addr:              ":8080",
        Handler:           mux,
        ReadHeaderTimeout: 5 * time.Second,
    }
    log.Fatal(srv.ListenAndServe())
}
```

In practice use a container `USER` directive and a distroless/static image rather than runtime `setuid`. The Go code above is for understanding the principle.

Container baseline

```dockerfile
# multi-stage: build as root, run as non-root
FROM golang:1.22-bookworm AS build
WORKDIR /src
COPY . .
RUN CGO_ENABLED=0 go build -trimpath -ldflags="-s -w" -o /out/app ./cmd/app

FROM gcr.io/distroless/static-debian12:nonroot
COPY --from=build /out/app /app
USER nonroot:nonroot
EXPOSE 8080
ENTRYPOINT ["/app"]
```

Production notes:

- `CGO_ENABLED=0` shrinks the binary and removes libc attack surface (when pure Go is viable).
- `-trimpath` removes local paths from binaries (less recon info).
- Read-only root filesystem + tmpfs for scratch dirs in orchestrators.
- No shell in the image reduces interactive abuse after break-in.

Hardening the HTTP Surface

Timeouts and body limits

Unbounded bodies and missing timeouts are DoS vectors.

```go
package httpserver

import (
    "context"
    "encoding/json"
    "io"
    "net"
```

```go
    "net/http"
    "time"
)

const maxBody = 1 << 20 // 1 MiB

func NewServer(addr string, h http.Handler) *http.Server {
    return &http.Server{
        Addr:              addr,
        Handler:           limitBody(h, maxBody),
        ReadHeaderTimeout: 5 * time.Second,
        ReadTimeout:       15 * time.Second,
        WriteTimeout:      30 * time.Second,
        IdleTimeout:       60 * time.Second,
        MaxHeaderBytes:    1 << 16, // 64 KiB
        BaseContext: func(net.Listener) context.Context {
            return context.Background()
        },
    }
}

func limitBody(next http.Handler, n int64) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        r.Body = http.MaxBytesReader(w, r.Body, n)
        next.ServeHTTP(w, r)
    })
}

// Safe JSON decode: size-limited + DisallowUnknownFields for strict APIs.
func DecodeJSON(w http.ResponseWriter, r *http.Request, dst any) error {
    defer r.Body.Close()
    dec := json.NewDecoder(io.LimitReader(r.Body, maxBody))
    dec.DisallowUnknownFields()
    if err := dec.Decode(dst); err != nil {
        http.Error(w, "bad request", http.StatusBadRequest)
        return err
    }
    return nil
}
```

Import `encoding/json` alongside the other packages in a real module.

Separate admin and data planes

Never mount pprof or admin UIs on the public listener.

```go
package main

import (
    "log"
    "net/http"
    _ "net/http/pprof"
)

func main() {
    // Public traffic only.
    public := http.NewServeMux()
    public.HandleFunc("GET /api/v1/items", listItems)
    go func() {
        log.Fatal(http.ListenAndServe(":8080", public))
    }()

    // Admin/debug on localhost or a private network interface only.
    admin := http.NewServeMux()
    admin.Handle("/debug/pprof/", http.DefaultServeMux)
    admin.HandleFunc("GET /readyz", readiness)
    log.Fatal(http.ListenAndServe("127.0.0.1:6060", admin))
}
```

Security headers and TLS defaults

```go
func securityHeaders(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        w.Header().Set("X-Content-Type-Options", "nosniff")
        w.Header().Set("X-Frame-Options", "DENY")
        w.Header().Set("Referrer-Policy", "no-referrer")
        w.Header().Set("Content-Security-Policy", "default-src 'none'; frame-ancestors 'none
        // HSTS only when TLS terminates here or is guaranteed upstream.
        // w.Header().Set("Strict-Transport-Security", "max-age=63072000; includeSubDomains"
        next.ServeHTTP(w, r)
    })
}
```

Prefer TLS 1.2+ with modern cipher suites; see the dedicated TLS chapter in part 22 for mTLS and cert rotation.

Input Validation and Injection Defense

SQL: parameters only

```
// BAD: string concatenation → injection
// q := "SELECT * FROM users WHERE name = '" + name + "'"

// GOOD: placeholders
row := db.QueryRowContext(ctx, `SELECT id, email FROM users WHERE name = $1`, name)
```

Paths: root-jail every user-supplied path

```
package safepath

import (
    "fmt"
    "path/filepath"
    "strings"
)

func UnderRoot(root, userPath string) (string, error) {
    cleanRoot := filepath.Clean(root)
    joined := filepath.Join(cleanRoot, userPath)
    // Resolve ".." and symlinks carefully; Clean alone is not enough against all symlink es
    abs, err := filepath.Abs(joined)
    if err != nil {
        return "", err
    }
    rel, err := filepath.Rel(cleanRoot, abs)
    if err != nil || strings.HasPrefix(rel, "..") {
        return "", fmt.Errorf("path escapes root")
    }
    return abs, nil
}
```

Commands: never shell out with untrusted strings

```
// BAD
// exec.Command("sh", "-c", "convert "+userFile)

// GOOD: fixed binary + argv array
cmd := exec.CommandContext(ctx, "convert", userFile, outFile)
```

Randomness and tokens

Use `crypto/rand` for session IDs, CSRF tokens, and API keys — never `math/rand`.

```go
func randomToken(n int) (string, error) {
    b := make([]byte, n)
    if _, err := rand.Read(b); err != nil {
        return "", err
    }
    return hex.EncodeToString(b), nil
}
```

Secrets and Configuration

Do	Don't
Load from env / secret manager at runtime	Commit secrets to git
Fail fast on missing required secrets	Silently use empty defaults
Support dual-read during rotation	Hard-code a single static key forever
Redact secrets in logs	Log full Authorization headers

```go
func mustEnv(key string) string {
    v := os.Getenv(key)
    if v == "" {
        slog.Error("missing required secret", "key", key)
        os.Exit(1)
    }
    return v
}
```

Dependency and Supply-Chain Hardening

1. Pin modules with `go.sum`; review unexpected upgrades in PRs.
2. Run `govulncheck ./...` in CI on every change.
3. Prefer small, well-maintained libraries over kitchen-sink frameworks for security-sensitive paths.
4. Use `capslock` (or similar) when adopting new deps that should not need network/FS capabilities.
5. Build with reproducible flags; publish checksums for release artifacts.

```
go install golang.org/x/vuln/cmd/govulncheck@latest
govulncheck ./...
gosec ./...
```

Runtime Abuse Controls

Rate limiting (simple token bucket sketch)

```go
type limiter struct {
    mu     sync.Mutex
    tokens float64
    max    float64
    refill float64 // tokens per second
    last   time.Time
}

func (l *limiter) Allow() bool {
    l.mu.Lock()
    defer l.mu.Unlock()
    now := time.Now()
    elapsed := now.Sub(l.last).Seconds()
    l.last = now
    l.tokens += elapsed * l.refill
    if l.tokens > l.max {
        l.tokens = l.max
    }
    if l.tokens < 1 {
        return false
    }
    l.tokens--
    return true
}
```

Place rate limits per IP / per principal *after* authentication for authenticated APIs so one user cannot exhaust another's budget.

Panic recovery at the edge

Recover panics in HTTP handlers so one bad request does not kill the process — but log stack traces and return 500, never continue with half-written responses.

```go
func recoverMiddleware(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        defer func() {
```

```go
            if rec := recover(); rec != nil {
                slog.Error("panic", "err", rec, "path", r.URL.Path)
                http.Error(w, "internal error", http.StatusInternalServerError)
            }
        }()
        next.ServeHTTP(w, r)
    })
}
```

Production Hardening Checklist

- ☐ Non-root process / distroless image
- ☐ Read-only root FS where possible
- ☐ Public listener has full timeout suite + body limits
- ☐ Admin/pprof not on public interface
- ☐ TLS 1.2+ (or terminated by a hardened proxy with the same bar)
- ☐ Authn before authz; default deny
- ☐ Parameterized queries; no shell interpolation
- ☐ Path jail for file operations
- ☐ Secrets from env/manager; fail-closed on missing
- ☐ Structured logs with redaction
- ☐ `govulncheck` + static security checks in CI
- ☐ Graceful shutdown on SIGTERM with deadline
- ☐ Resource limits (CPU/memory) set in orchestrator
- ☐ Network policy: egress allowlist when feasible

Common Pitfalls

1. **Trusting reverse proxies blindly** — if you use `X-Forwarded-For` for rate limits or auth, validate that only trusted proxies can set it.
2. **Debug builds in prod** — race detector and extra logging are fine; open pprof on `0.0.0.0` is not.
3. **"Internal" network = secure** — flat VPC trust is outdated; use mTLS or service identity between services.
4. **Logging secrets** — JWT, cookies, and API keys leak through request dumps.
5. **Unbounded fan-out** — one authenticated request spawning N unbounded downstream calls is a DoS amplifier.
6. **Ignoring context** — operations that ignore cancellation hold resources under attack load.
7. **World-writable files** — `0666`/`0777` permissions in services that write state.

Exercises

1. Take a minimal `net/http` server and add body limits, header/read/write timeouts, and a max header size. Load-test with a slow client and confirm it disconnects.
2. Split public and admin muxes; prove with `curl` that pprof is unreachable on `:8080`.
3. Write a `UnderRoot` function and unit-test path traversal attempts (`../../etc/passwd`, absolute paths, symlink escapes if you resolve them).
4. Introduce a deliberate SQL string concat and show `gosec` (or a review checklist) catching it; fix with placeholders.
5. Implement fail-closed startup: missing `DATABASE_URL` must exit non-zero before listening.
6. Add panic recovery middleware and a handler that panics; verify the process stays up and a 500 is returned.
7. Build a multi-stage Dockerfile that runs as non-root and contains no shell; inspect the image.
8. Implement a per-IP rate limiter middleware and verify 429 responses under burst.
9. Audit your logs: log a fake Authorization header once, then add redaction so it never appears again.
10. Run `govulncheck ./...` on a sample module and document how you would gate merges on new findings.

Further Reading

- Go security policy and release notes for crypto changes
- OWASP ASVS (application security verification)
- Part 22 of this book: TLS/mTLS, authn/authz, secrets rotation
- Previous chapter: security tooling (`govulncheck`, `gosec`, capability analysis)

More examples

Body limit + security headers middleware

```
mkdir -p /tmp/go-harden-mw && cd /tmp/go-harden-mw
go mod init example.com/harden-mw
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "io"
    "net/http"
    "net/http/httptest"
    "strings"
)
```

```go
func secureHeaders(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        w.Header().Set("X-Content-Type-Options", "nosniff")
        w.Header().Set("X-Frame-Options", "DENY")
        w.Header().Set("Referrer-Policy", "no-referrer")
        next.ServeHTTP(w, r)
    })
}

func maxBytes(n int64, next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        r.Body = http.MaxBytesReader(w, r.Body, n)
        next.ServeHTTP(w, r)
    })
}

func main() {
    h := secureHeaders(maxBytes(8, http.HandlerFunc(func(w http.ResponseWriter, r *http.Requ
        b, err := io.ReadAll(r.Body)
        if err != nil {
            http.Error(w, "payload too large", http.StatusRequestEntityTooLarge)
            return
        }
        fmt.Fprintf(w, "got %d bytes\n", len(b))
    })))

    // OK body
    rr := httptest.NewRecorder()
    req := httptest.NewRequest(http.MethodPost, "/", strings.NewReader("hello"))
    h.ServeHTTP(rr, req)
    fmt.Println("small:", rr.Code, strings.TrimSpace(rr.Body.String()), rr.Header().Get("X-C

    // Oversized
    rr = httptest.NewRecorder()
    req = httptest.NewRequest(http.MethodPost, "/", strings.NewReader("0123456789"))
    h.ServeHTTP(rr, req)
    fmt.Println("large:", rr.Code)
}
```

```go
go run .
```

Expected output:

```
small: 200 got 5 bytes nosniff
large: 413
```

Fail-closed required config

```
mkdir -p /tmp/go-fail-closed && cd /tmp/go-fail-closed
go mod init example.com/fail-closed
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "os"
)

func requireEnv(keys ...string) error {
    for _, k := range keys {
        if os.Getenv(k) == "" {
            return fmt.Errorf("missing required env %s", k)
        }
    }
    return nil
}

func main() {
    os.Unsetenv("DATABASE_URL")
    os.Setenv("APP_SECRET", "x")
    err := requireEnv("DATABASE_URL", "APP_SECRET")
    fmt.Println("before:", err)

    os.Setenv("DATABASE_URL", "postgres://local/app")
    err = requireEnv("DATABASE_URL", "APP_SECRET")
    fmt.Println("after:", err)
}
```

```
go run .
```

Expected output:

```
before: missing required env DATABASE_URL
after: <nil>
```

Runnable example

Hardening is timeouts, body limits, path jails, and fail-closed config. This server rejects oversized bodies, blocks path escape, and requires a secret env var before listening.

```
mkdir -p /tmp/go-hardening && cd /tmp/go-hardening
go mod init example.com/hardening
```

Save as `main.go`:

```
package main

import (
    "encoding/json"
    "fmt"
    "io"
    "log"
    "net"
    "net/http"
    "os"
    "path/filepath"
    "strings"
    "time"
)

const maxBody = 1 << 10 // 1 KiB for the demo

func underRoot(root, userPath string) (string, error) {
    cleanRoot := filepath.Clean(root)
    joined := filepath.Join(cleanRoot, userPath)
    abs, err := filepath.Abs(joined)
    if err != nil {
        return "", err
    }
    rootAbs, err := filepath.Abs(cleanRoot)
    if err != nil {
        return "", err
    }
    rel, err := filepath.Rel(rootAbs, abs)
    if err != nil || strings.HasPrefix(rel, "..") {
        return "", fmt.Errorf("path escapes root")
    }
    return abs, nil
}

func limitBody(next http.Handler, n int64) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        r.Body = http.MaxBytesReader(w, r.Body, n)
        next.ServeHTTP(w, r)
    })
}
```

```
func securityHeaders(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        w.Header().Set("X-Content-Type-Options", "nosniff")
        w.Header().Set("X-Frame-Options", "DENY")
        next.ServeHTTP(w, r)
    })
}

func main() {
    // Fail closed in production: if os.Getenv("API_TOKEN") == "" { log.Fatal("missing API_T
    token := os.Getenv("API_TOKEN")
    if token == "" {
        token = "dev-only-token"
        fmt.Println("warning: API_TOKEN unset; using dev-only-token")
    }
    fmt.Println("token_configured:", token != "")

    root := filepath.Join(os.TempDir(), "hardening-demo-root")
    _ = os.MkdirAll(root, 0o750)
    _ = os.WriteFile(filepath.Join(root, "note.txt"), []byte("safe\n"), 0o640)

    mux := http.NewServeMux()
    mux.HandleFunc("GET /healthz", func(w http.ResponseWriter, _ *http.Request) {
        w.WriteHeader(http.StatusOK)
        _, _ = w.Write([]byte("ok"))
    })
    mux.HandleFunc("POST /echo", func(w http.ResponseWriter, r *http.Request) {
        b, err := io.ReadAll(r.Body)
        if err != nil {
            http.Error(w, "payload too large or unreadable", http.StatusRequestEntityTooLarg
            return
        }
        w.Header().Set("Content-Type", "application/json")
        _ = json.NewEncoder(w).Encode(map[string]any{"n": len(b)})
    })
    mux.HandleFunc("GET /file", func(w http.ResponseWriter, r *http.Request) {
        name := r.URL.Query().Get("name")
        path, err := underRoot(root, name)
        if err != nil {
            http.Error(w, "forbidden", http.StatusForbidden)
            return
        }
        b, err := os.ReadFile(path)
        if err != nil {
            http.Error(w, "not found", http.StatusNotFound)
            return
        }
```

```
        _, _ = w.Write(b)
    })

    ln, err := net.Listen("tcp", "127.0.0.1:0")
    if err != nil {
        log.Fatal(err)
    }
    h := securityHeaders(limitBody(mux, maxBody))
    srv := &http.Server{
        Handler:           h,
        ReadHeaderTimeout: 5 * time.Second,
        ReadTimeout:       10 * time.Second,
        WriteTimeout:      10 * time.Second,
        MaxHeaderBytes:    1 << 14,
    }
    go func() { _ = srv.Serve(ln) }()
    base := "http://" + ln.Addr().String()

    // Self-tests.
    resp, _ := http.Get(base + "/healthz")
    fmt.Println("healthz:", resp.StatusCode)
    resp.Body.Close()

    resp, _ = http.Get(base + "/file?name=note.txt")
    fmt.Println("file ok:", resp.StatusCode)
    resp.Body.Close()

    resp, _ = http.Get(base + "/file?name=../../etc/passwd")
    fmt.Println("path escape:", resp.StatusCode)
    resp.Body.Close()

    big := strings.Repeat("x", int(maxBody)+10)
    resp, err = http.Post(base+"/echo", "text/plain", strings.NewReader(big))
    if err != nil {
        fmt.Println("oversize post err:", err)
    } else {
        fmt.Println("oversize body:", resp.StatusCode)
        resp.Body.Close()
    }

    _ = srv.Close()
    fmt.Println("hardening demo done")
    // Production: replace the soft default above with token := mustEnv("API_TOKEN").
}
```

```
go run .
API_TOKEN=secret go run .
```

Expected output (illustrative):

```
warning: API_TOKEN unset; using dev-only-token
token_configured: true
healthz: 200
file ok: 200
path escape: 403
oversize body: 413
hardening demo done
```

What to notice

- `http.MaxBytesReader` + server timeouts are first-line DoS controls.
- Path jails need `filepath.Rel` / prefix checks—not only `Clean`.
- Production should `log.Fatal` on missing secrets *before* `Listen` (no soft defaults).

Try next

- Mount pprof on `127.0.0.1:6060` only; prove it is unreachable on the public mux.
- Replace the demo token default with hard fail-closed exit when `API_TOKEN` is unset.

Next: once tooling and hardening defaults are in place, design patterns (SOLID-ish module boundaries) keep secure code maintainable — see `82-solid-design.qmd`.

SOLID in Go

SOLID Principles in Go

Overview

SOLID comes from class-heavy OOP, but its goals—**cohesion, decoupling, testability**—map cleanly onto Go. You implement them with **packages**, **small interfaces**, and **constructor injection**, not inheritance trees.

This chapter translates each letter into idiomatic Go, with production examples, checklists, and common misapplications.

Quick Map

Letter	OOP slogan	Go practice
S	Single responsibility	Small packages; one reason to change
O	Open/closed	Accept interfaces; add types without editing core
L	Liskov substitution	Behavioral contracts for interface implementers
I	Interface segregation	Consumer-defined, tiny interfaces
D	Dependency inversion	Depend on abstractions at boundaries

S — Single Responsibility

"A package (or type) should have one reason to change."

Go's package system is the first SRP tool.

Avoid	Prefer
`package utils`	`package clock`, `package billhash`, `package useremail`
God `Service` struct	Separate `Billing`, `Notifier`, `Store`

```go
// Bad: one type owns HTTP, SQL, and SMTP
type App struct {
    db *sql.DB
}

func (a *App) HandleRegister(w http.ResponseWriter, r *http.Request) { /* parse */ }
func (a *App) insertUser(...) error { /* sql */ }
func (a *App) sendWelcome(...) error { /* smtp */ }
```

```go
// Better: each concern is a type with a narrow job
type UserStore interface {
    Create(ctx context.Context, u User) (UserID, error)
}

type Mailer interface {
    SendWelcome(ctx context.Context, email string) error
}

type RegisterHandler struct {
    Store  UserStore
    Mailer Mailer
}

func (h RegisterHandler) ServeHTTP(w http.ResponseWriter, r *http.Request) {
    // parse → Store.Create → Mailer.SendWelcome
}
```

SRP does **not** mean "one function per file." It means changes to email templates should not risk breaking SQL migrations in the same type.

O — Open/Closed

"Open for extension, closed for modification."

In Go, extension is usually a **new type** satisfying an interface—not subclassing.

```go
type Shape interface {
    Area() float64
}

func TotalArea(shapes []Shape) float64 {
    var total float64
    for _, s := range shapes {
        total += s.Area()
    }
    return total
```

```go
}

type Circle struct{ R float64 }

func (c Circle) Area() float64 { return math.Pi * c.R * c.R }

type Rectangle struct{ W, H float64 }

func (r Rectangle) Area() float64 { return r.W * r.H }

// Add Triangle later without editing TotalArea.
```

Another pattern: middleware stacks—wrap `http.Handler` without editing handlers.

```go
func WithLogging(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        start := time.Now()
        next.ServeHTTP(w, r)
        slog.Info("request", "path", r.URL.Path, "dur", time.Since(start))
    })
}
```

L — Liskov Substitution

"Implementations must honor the contract callers rely on."

Go has no subtypes, but interface satisfaction is a promise:

If `w` is an `io.Writer`, callers expect:

- `Write` either writes or returns an error
- No surprise panics on empty input (unless documented)
- No "always returns nil error but drops data" behavior

```go
// Violates expectations of io.Writer users
type BrokenWriter struct{}

func (BrokenWriter) Write(p []byte) (int, error) {
    // Pretends success without writing
    return len(p), nil
}
```

Document behavioral contracts:

```go
// Bucket stores objects.
// Delete is idempotent: deleting a missing key returns nil.
```

```go
type Bucket interface {
    Put(ctx context.Context, key string, val []byte) error
    Delete(ctx context.Context, key string) error
}
```

When a mock in tests behaves differently from production (e.g., never returns `ErrNotFound`), tests lie—LSP applied to fakes.

I — Interface Segregation

"Clients must not depend on methods they do not use."

This is Go's superpower: **define interfaces at the consumer**, keep them tiny (often 1–3 methods).

Bad — fat interface

```go
type UserService interface {
    Create(ctx context.Context, u User) error
    Delete(ctx context.Context, id UserID) error
    SendEmail(ctx context.Context, id UserID, body string) error
    Validate(u User) error
    ExportCSV(ctx context.Context) ([]byte, error)
}

func NotifyWelcome(s UserService, id UserID) error {
    return s.SendEmail(ctx, id, "welcome")
}
// NotifyWelcome is forced to mock Create/Delete/Export in tests
```

Good — small consumer interface

```go
type WelcomeMailer interface {
    SendEmail(ctx context.Context, id UserID, body string) error
}

func NotifyWelcome(m WelcomeMailer, id UserID) error {
    return m.SendEmail(ctx, id, "welcome")
}
```

Stdlib examples: `io.Reader`, `io.Writer`, `fs.FS`, `http.Handler`—tiny and ubiquitous.

Accept interfaces, return structs (common guideline):

```go
func NewServer(store *PostgresStore) *Server // concrete for construction
// methods of Server depend on small interfaces internally if needed
```

Or inject interfaces at construction when tests require it:

```go
func NewServer(store UserStore, mail WelcomeMailer) *Server
```

D — Dependency Inversion

"High-level policy should not depend on low-level details; both depend on abstractions."

```go
// Bad: business logic locked to sql.DB
type OrderService struct {
    DB *sql.DB
}

func (s *OrderService) Place(ctx context.Context, o Order) error {
    _, err := s.DB.ExecContext(ctx, `INSERT INTO orders ...`)
    return err
}
```

```go
// Good: policy depends on a port
type OrderRepository interface {
    Insert(ctx context.Context, o Order) error
}

type OrderService struct {
    Repo OrderRepository
}

func (s *OrderService) Place(ctx context.Context, o Order) error {
    if err := o.Validate(); err != nil {
        return err
    }
    return s.Repo.Insert(ctx, o)
}

// Detail implements the port
type PostgresOrders struct{ DB *sql.DB }

func (p PostgresOrders) Insert(ctx context.Context, o Order) error {
    _, err := p.DB.ExecContext(ctx, `INSERT INTO orders ...`, o.ID, o.Total)
    return err
}
```

Wiring stays in `main` / `cmd`:

```go
func main() {
    db := mustOpenDB()
    svc := OrderService{Repo: PostgresOrders{DB: db}}
    // http handlers get svc
}
```

Tests inject fakes:

```go
type fakeRepo struct{ last Order }

func (f *fakeRepo) Insert(ctx context.Context, o Order) error {
    f.last = o
    return nil
}

func TestPlace(t *testing.T) {
    f := &fakeRepo{}
    svc := OrderService{Repo: f}
    if err := svc.Place(context.Background(), Order{ID: "1", Total: 10}); err != nil {
        t.Fatal(err)
    }
    if f.last.ID != "1" {
        t.Fatalf("got %+v", f.last)
    }
}
```

Composition Over Inheritance

Go has no inheritance. Use **embedding** carefully for data reuse, not for deep type hierarchies.

```go
type Logger struct {
    *slog.Logger
}

// Prefer explicit fields and small interfaces over deep embedding graphs.
```

Package Design Checklist (SOLID-flavored)

- ☐ Package name describes a domain concept, not `util` / `common`
- ☐ Interfaces live with consumers (or in a thin `port` layer), not giant `interfaces.go` with 20 methods
- ☐ Concrete adapters (`postgres`, `smtp`, `s3`) sit at the edges

- ☐ `main` wires dependencies
- ☐ Domain logic has no import of frameworks it does not need
- ☐ Tests use fakes via interfaces, not live network by default

Anti-Patterns

1. **Interface on producer with 15 methods "for flexibility"** — freezes implementations; violates I.
2. **`package interfaces` shared by all** — creates import cycles and meaningless abstractions.
3. **Mocking everything** — if there is no behavior branch, don't introduce an interface.
4. **SRP as infinite micro-packages** — 50 packages of 20 lines harm navigation; balance cohesion.
5. **Embedding `*sql.DB` in every service** — hidden dependency; prefer explicit fields.
6. **Panicking implementers** — violates L for interfaces that expect errors.

When *Not* to Abstract

```go
// YAGNI: single use, no test seam needed yet
func loadConfig(path string) (Config, error) {
    b, err := os.ReadFile(path)
    // ...
}
```

Introduce an interface when you have a **second implementation** (or a real test need), not on day one for every function.

Production Checklist

- ☐ New packages have a clear single purpose statement (README one-liner or doc comment)
- ☐ Public APIs prefer small interfaces at boundaries
- ☐ Handlers/services don't import driver-specific packages when avoidable
- ☐ Fakes honor real contracts (LSP for tests)
- ☐ Dependency graph is acyclic: domain ← ports → adapters
- ☐ CI tests domain with fakes; integration tests cover one real adapter

Common Pitfalls

1. **Java-style interface hierarchies in Go** — fighting the language.
2. **Returning interfaces from factories always** — hides types; return concrete, accept interfaces.
3. **Circular imports** — often a sign SRP/DIP layering is wrong.
4. **God `App` struct** — DIP violation magnet; split constructors.

5. **Copy-paste SOLID blogs with class diagrams** — rephrase in packages and interfaces.

Exercises

1. **Split the god type** — Take a type with DB + HTTP + email methods; refactor into `Store`, `Mailer`, and handler with interfaces.
2. **Segregate** — From a 6-method interface, derive two consumer interfaces used by different functions; update tests.
3. **Open/closed middleware** — Add `WithAuth` and `WithLogging` without editing business handlers.
4. **Fake honesty** — Write a fake `Bucket` that violates idempotent delete; show a test that would pass with a bad fake and fix the fake.
5. **Wire `main`** — Build `cmd/server/main.go` that constructs Postgres + SMTP adapters and injects them into services.
6. **Package rename** — Break `internal/utils` into two named packages; update imports until `go test ./...` is green.

More examples

Depend on interfaces; swap fakes

```
mkdir -p /tmp/go-solid-iface && cd /tmp/go-solid-iface
go mod init example.com/solid-iface
```

Save as `main.go`:

```
package main

import "fmt"

type Mailer interface {
    Send(to, body string) error
}

type Notifier struct {
    mail Mailer
}

func (n Notifier) Welcome(to string) error {
    return n.mail.Send(to, "welcome")
}

type MemMailer struct {
    sent []string
```

```
}

func (m *MemMailer) Send(to, body string) error {
    m.sent = append(m.sent, to+":"+body)
    return nil
}

func main() {
    m := &MemMailer{}
    n := Notifier{mail: m}
    _ = n.Welcome("a@b.co")
    _ = n.Welcome("c@d.co")
    fmt.Println("sent:", len(m.sent), m.sent[0])
}
```

```
go run .
```

Expected output:

```
sent: 2 a@b.co:welcome
```

Functional options for constructors

```
mkdir -p /tmp/go-solid-opts && cd /tmp/go-solid-opts
go mod init example.com/solid-opts
```

Save as `main.go`:

```
package main

import "fmt"

type Server struct {
    addr string
    tls  bool
}

type Option func(*Server)

func WithAddr(a string) Option { return func(s *Server) { s.addr = a } }
func WithTLS(on bool) Option   { return func(s *Server) { s.tls = on } }

func NewServer(opts ...Option) *Server {
    s := &Server{addr: ":8080"}
    for _, o := range opts {
```

```go
        o(s)
    }
    return s
}

func main() {
    s := NewServer(WithAddr(":8443"), WithTLS(true))
    fmt.Printf("addr=%s tls=%v\n", s.addr, s.tls)
}
```

```go
go run .
```

Expected output:

```
addr=:8443 tls=true
```

Runnable example

SOLID in Go is small consumer interfaces and constructor injection. This program places an order through a service that depends on a `OrderRepository` port; `main` wires an in-memory adapter (swap for Postgres without changing policy).

```go
mkdir -p /tmp/go-solid && cd /tmp/go-solid
go mod init example.com/solid
```

Save as `main.go`:

```go
package main

import (
    "context"
    "fmt"
    "sync"
)

type Order struct {
    ID    string
    Total int
}

func (o Order) Validate() error {
    if o.ID == "" || o.Total <= 0 {
        return fmt.Errorf("invalid order")
    }
    return nil
```

```
}

// Port: defined by the consumer (DIP + ISP).
type OrderRepository interface {
    Insert(ctx context.Context, o Order) error
}

type OrderService struct {
    Repo OrderRepository
}

func (s OrderService) Place(ctx context.Context, o Order) error {
    if err := o.Validate(); err != nil {
        return err
    }
    return s.Repo.Insert(ctx, o)
}

// Adapter: in-memory store (tests / demos).
type MemoryOrders struct {
    mu   sync.Mutex
    byID map[string]Order
}

func NewMemoryOrders() *MemoryOrders {
    return &MemoryOrders{byID: make(map[string]Order)}
}

func (m *MemoryOrders) Insert(ctx context.Context, o Order) error {
    select {
    case <-ctx.Done():
        return ctx.Err()
    default:
    }
    m.mu.Lock()
    defer m.mu.Unlock()
    if _, ok := m.byID[o.ID]; ok {
        return fmt.Errorf("duplicate id %s", o.ID)
    }
    m.byID[o.ID] = o
    return nil
}

func (m *MemoryOrders) Get(id string) (Order, bool) {
    m.mu.Lock()
    defer m.mu.Unlock()
```

```go
    o, ok := m.byID[id]
    return o, ok
}

func main() {
    repo := NewMemoryOrders()
    svc := OrderService{Repo: repo} // wire in main

    ctx := context.Background()
    if err := svc.Place(ctx, Order{ID: "o1", Total: 42}); err != nil {
        fmt.Println("place:", err)
        return
    }
    if err := svc.Place(ctx, Order{ID: "bad", Total: 0}); err != nil {
        fmt.Println("validate rejected:", err)
    }
    if err := svc.Place(ctx, Order{ID: "o1", Total: 10}); err != nil {
        fmt.Println("duplicate rejected:", err)
    }
    if o, ok := repo.Get("o1"); ok {
        fmt.Printf("stored: id=%s total=%d\n", o.ID, o.Total)
    }
    fmt.Println("solid wiring ok")
}
```

```go
go run .
```

Expected output:

```
validate rejected: invalid order
duplicate rejected: duplicate id o1
stored: id=o1 total=42
solid wiring ok
```

What to notice

- Policy (`OrderService`) never imports a database driver—only the port.
- The interface is one method (ISP); tests inject fakes without mocking frameworks.
- Construction stays in `main` (or `cmd/`); packages export structs, accept interfaces.

Try next

- Extract `OrderService` tests with a fake that records `last Order`.
- Add a `WelcomeMailer` interface and call it after successful `Insert` without growing a god service.

Related Topics

- Interfaces
- Interface practices
- Long-term design
- Testing fundamentals

Security Hardening

Security Hardening Overview

Security Hardening Overview

This part focuses on **practical secure-by-default design** in Go services: transport identity, access decisions, and secret lifecycle. The objective is to reduce both exploitability and operational misconfiguration risk.

Why / Overview

Security failures in Go services are often not exotic crypto bugs. They are:

- Expired certificates during a quiet holiday
- JWT validation that skips `aud/iss`
- Authorization checks only on the UI path
- Secrets in env dumps and CI logs
- mTLS configured on one side only

```
client
  |  TLS / mTLS
  v
edge -- authn --> principal
           |
           +-- authz --> allow/deny on resource+action
           |
           +-- secrets --> short-lived credentials
```

Part 21 covered tooling and application hardening baselines. This part deepens **transport**, **identity**, and **secrets**.

Learning Path

Chapter	Focus	Outcome
171 TLS / mTLS	Trust stores, versions, rotation	Safe transport identity
172 Authn / Authz	Principals vs policies	Correct access decisions
173 Secrets	Injection, dual-read rotation	Credential lifecycle without outages
174 JWT validation	Alg/exp/aud/iss	Safe bearer tokens
175 Input validation & SSRF	Boundaries, URL fetch	No open proxies / traversal

Chapter	Focus	Outcome
175c CSRF without tokens	`http.CrossOriginProtection`, Sec-Fetch-Site	Cookie sessions safe from cross-origin forms
176 Supply chain	govulncheck, modules, builds	Dependency hygiene
177 Project: secure checklist	Harden a real service	Evidence-based security pass

Core Principles

1. **Authenticate then authorize** — never authorize anonymous by accident.
2. **Default deny** — missing policy is deny, not allow.
3. **Short-lived credentials** — blast radius of leaks shrinks with TTL.
4. **Encrypt in transit** everywhere that is not a deliberate exception.
5. **Separate data plane and control plane** credentials.
6. **Fail closed** on auth/crypto misconfiguration at startup when possible.

Go Building Blocks

Area	Packages
TLS	`crypto/tls`, `crypto/x509`
HTTP security	`net/http` + middleware
Tokens	`golang-jwt` or similar (review carefully), PASETO, session cookies
Secrets	env, files, Vault/cloud SM SDKs
Random	`crypto/rand`

Relationship to Other Parts

- **Hardening (81)** — process, headers, input limits
- **Network systems** — where TLS terminates and proxies hop
- **Observability** — audit logs for authz denials; never log secrets
- **Distributed infra** — service identity for discovery peers

Literacy Checklist

☐ Configure TLS 1.2+ with sane cipher suites
☐ Explain mTLS client cert verification
☐ Rotate certs without downtime (overlap windows)
☐ Separate authn modules from authz policy engines
☐ Validate JWT/`iss`/`aud`/exp correctly

- ☐ Dual-read secrets during rotation
- ☐ Redact credentials in logs and errors

Part Warm-Up Exercises

1. Find where TLS terminates in your architecture (app vs proxy vs mesh).
2. List every secret your service reads at startup.
3. Identify one endpoint that authenticates but might skip authorize.
4. Check certificate expiry monitoring exists.
5. Locate any hardcoded tokens in repo history (then rotate them).

Threat Model Snapshot (Service-Centric)

For a typical Go API, assume:

Threat	Primary controls in this part
Network eavesdropping / tampering	TLS
Spoofed internal callers	mTLS or signed service identity
Stolen user session / token	short TTL, audience checks, revoke paths
Confused deputy / IDOR	resource-level authz
Leaked DB password	rotation, dual-read, least privilege
Misissued cert	trust pins, monitoring, automation

Attackers prefer **misconfiguration** over breaking AES-GCM. Your design should make the secure path the default path.

Layering with Part 21

Part 21 (earlier)	Part 22 (here)
govulncheck, gosec	TLS session identity
Process/container hardening	Authn/authz architecture
Input limits, headers	Secrets lifecycle
General checklist	Rotation choreography

Use both: tools catch known issues; these chapters design the trust model.

Minimal Secure Service Skeleton

```
listen TLS (or receive from mesh)
  -> request id middleware
  -> authenticate (session/JWT/mTLS)
  -> authorize (action + resource)
  -> business handler (context with principal)
  -> audit deny / sensitive allow
secrets: loaded at start, refreshed, never logged
```

If a route skips any of the middle boxes, that is a design smell requiring explicit justification (e.g. public `/healthz`).

Compliance-Adjacent Habits (Not Legal Advice)

Even without a formal compliance program, good habits transfer:

- Inventory secrets and certs
- Record who can read production secrets
- Prefer short-lived credentials
- Keep audit logs for admin and authz deny
- Separate prod and non-prod credentials completely

Secure Defaults Checklist (Part Preview)

Copy into PR templates for services that handle auth or secrets:

- ☐ TLS 1.2+ (or mesh-enforced equivalent)
- ☐ No `InsecureSkipVerify` outside explicit test builds
- ☐ Authn middleware before business handlers
- ☐ Authz checks resource ownership (anti-IDOR)
- ☐ Tokens validate `iss`, `aud`, `exp`, algorithm
- ☐ Cookies: `HttpOnly`, `Secure`, deliberate `SameSite`
- ☐ Secrets not in git; startup fails if missing
- ☐ Rotation dual-read tested in staging
- ☐ Authz denials audited with request id
- ☐ pprof/admin not on public interface

Testing Security Logic

Kind	Example
Unit	`Authorize` table tests; JWT claim rejection
Integration	mTLS handshake success/fail matrices

Kind	Example
Negative	expired cert, wrong audience, cross-tenant id
Chaos	rotate secret mid-traffic with dual-read

Security tests that only cover the happy path give false confidence.

Where Identity Lives

```
Human user  -> session cookie / OIDC access token -> API
Service A   -> mTLS or signed JWT (client credentials) -> Service B
Job worker  -> task-scoped token or instance identity -> APIs
```

Do not reuse human user tokens for service calls. Audience and lifetime should differ.

Mapping Incidents → Chapter

Incident	Start here
TLS handshake errors / expired cert	171
Users see other tenants' data	172 (IDOR / authz)
401 storms after deploy	172 (token validation) or 173 (secret mismatch)
DB auth failures during rotation	173 dual-read
"Works on internal network without TLS"	171 threat model

Non-Goals

This part does not replace:

- Full application penetration testing
- Formal cryptography design reviews
- Cloud IAM deep dives for every provider

It does give Go service authors a trustworthy baseline so specialists are not fixing the same footguns repeatedly.

Glossary

- **Authn / Authz** — identity vs permission
- **mTLS** — mutual TLS, both peers present certificates
- **Trust bundle** — set of CAs you accept
- **Dual-read** — accept old and new secret during rotation
- **Principal** — authenticated actor in your system
- **Audience (`aud`)** — intended recipient of a token
- **IDOR** — insecure direct object reference (authz bug)
- **SPIFFE** — workload identity URI SAN convention

More examples

Tour: bearer check + TLS client roots sketch

```
mkdir -p /tmp/go-sec-tour && cd /tmp/go-sec-tour
go mod init example.com/sec-tour
```

Save as `main.go`:

```
package main

import (
    "crypto/tls"
    "crypto/x509"
    "fmt"
    "net/http"
    "net/http/httptest"
    "strings"
)

func main() {
    // Authn stub
    h := http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        auth := r.Header.Get("Authorization")
        if !strings.HasPrefix(auth, "Bearer ") || auth == "Bearer " {
            http.Error(w, "unauthorized", http.StatusUnauthorized)
            return
        }
        fmt.Fprintln(w, "ok")
    })
    rr := httptest.NewRecorder()
    req := httptest.NewRequest(http.MethodGet, "/", nil)
    h.ServeHTTP(rr, req)
    fmt.Println("no token:", rr.Code)
```

```
    rr = httptest.NewRecorder()
    req = httptest.NewRequest(http.MethodGet, "/", nil)
    req.Header.Set("Authorization", "Bearer demo")
    h.ServeHTTP(rr, req)
    fmt.Println("with token:", rr.Code)

    // TLS: trust httptest cert
    ts := httptest.NewTLSServer(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request
        fmt.Fprintln(w, "https")
    }))
    defer ts.Close()
    roots := x509.NewCertPool()
    roots.AddCert(ts.Certificate())
    client := &http.Client{Transport: &http.Transport{
        TLSClientConfig: &tls.Config{RootCAs: roots, MinVersion: tls.VersionTLS12},
    }}
    resp, err := client.Get(ts.URL)
    if err != nil {
        panic(err)
    }
    resp.Body.Close()
    fmt.Println("tls status:", resp.StatusCode)
}
```

```
go run .
```

Expected output:

```
no token: 401
with token: 200
tls status: 200
```

Runnable example

Tour of this part: HMAC token mint/verify (authn-ish), a tiny allow/deny policy (authz), and dual-secret verification (rotation)—stdlib crypto.

```
mkdir -p /tmp/go-sec-tour && cd /tmp/go-sec-tour
go mod init example.com/sec-tour
```

Save as `main.go`:

```
package main

import (
```

```
    "crypto/hmac"
    "crypto/sha256"
    "encoding/hex"
    "fmt"
)

func sign(key, msg string) string {
    m := hmac.New(sha256.New, []byte(key))
    _, _ = m.Write([]byte(msg))
    return hex.EncodeToString(m.Sum(nil))
}

func verifyDual(keys []string, msg, sig string) bool {
    raw, err := hex.DecodeString(sig)
    if err != nil {
        return false
    }
    for _, k := range keys {
        m := hmac.New(sha256.New, []byte(k))
        _, _ = m.Write([]byte(msg))
        if hmac.Equal(raw, m.Sum(nil)) {
            return true
        }
    }
    return false
}

func authorize(role, action string) bool {
    policy := map[string][]string{
        "reader": {"read"},
        "admin":  {"read", "write", "delete"},
    }
    for _, a := range policy[role] {
        if a == action {
            return true
        }
    }
    return false
}

func main() {
    msg := "user=42"
    sig := sign("key-new", msg)
    fmt.Println("verify new only:", verifyDual([]string{"key-new"}, msg, sig))
    fmt.Println("dual-read old+new:", verifyDual([]string{"key-old", "key-new"}, msg, sig))
    fmt.Println("authz reader write:", authorize("reader", "write"))
    fmt.Println("authz admin write:", authorize("admin", "write"))
```

```go
    fmt.Println("tour: transport(identity) + authz + secret rotation")
}
```

```go
go run .
```

Expected output:

```
verify new only: true
dual-read old+new: true
authz reader write: false
authz admin write: true
tour: transport(identity) + authz + secret rotation
```

What to notice

- Authn (who) and authz (what) are separate layers; TLS is transport identity (171).
- Dual-read prevents outages during secret rotation (173).

Try next

- Chapter 171 TLS client/server demo; 172 middleware; 173 fail-closed env secrets.

Next Chapter

TLS and mTLS Deep Dive — the foundation of service identity on the wire.

TLS and mTLS Deep Dive

TLS and mTLS Deep Dive

TLS provides confidentiality and integrity for bytes on the wire. **mTLS** additionally authenticates the client with a certificate, giving strong transport-level identity between services.

Why / Overview

Most TLS incidents are lifecycle failures, not breaks of AES:

- Expired certificates
- Wrong trust bundle after rotation
- Clients still on TLS 1.0
- Server name mismatches
- Half-applied mTLS (server requests client cert but does not verify)

```
one-way TLS:  client verifies server certificate
mTLS:         client verifies server AND server verifies client
```

Go Server TLS

```go
cfg := &tls.Config{
    MinVersion: tls.VersionTLS12,
    // Prefer TLS 1.3 when both sides support it (automatic if available).
    CurvePreferences: []tls.CurveID{
        tls.X25519,
        tls.CurveP256,
    },
    // Certificates: static or GetCertificate for dynamic reload.
    Certificates: []tls.Certificate{cert},
}

srv := &http.Server{
    Addr:      ":8443",
    Handler:   mux,
    TLSConfig: cfg,
    // timeouts still required
    ReadHeaderTimeout: 5 * time.Second,
}
```

```go
log.Fatal(srv.ListenAndServeTLS("", "")) // empty if certs in TLSConfig
```

Load cert from PEM:

```go
cert, err := tls.LoadX509KeyPair("server.crt", "server.key")
```

Cipher suites

TLS 1.3 cipher suites are not configurable the same way as 1.2. For 1.2, prefer AEAD suites and avoid legacy CBC/RC4. When in doubt, set `MinVersion: tls.VersionTLS12` and let Go's defaults work; override only with a threat-model reason.

Dynamic Certificate Reload

Long-lived processes need cert rotation without downtime:

```go
type certHolder struct {
    mu   sync.RWMutex
    cert tls.Certificate
}

func (h *certHolder) get(_ *tls.ClientHelloInfo) (*tls.Certificate, error) {
    h.mu.RLock()
    defer h.mu.RUnlock()
    c := h.cert
    return &c, nil
}

func (h *certHolder) reload(certFile, keyFile string) error {
    c, err := tls.LoadX509KeyPair(certFile, keyFile)
    if err != nil {
        return err
    }
    h.mu.Lock()
    h.cert = c
    h.mu.Unlock()
    return nil
}

// TLSConfig.GetCertificate = holder.get
```

Reload on SIGHUP, fs notify, or periodic poll. **Never** replace files non-atomically (see filesystem chapter).

Client TLS

```go
rootPEM, _ := os.ReadFile("ca.crt")
pool := x509.NewCertPool()
pool.AppendCertsFromPEM(rootPEM)

client := &http.Client{
    Transport: &http.Transport{
        TLSClientConfig: &tls.Config{
            RootCAs:    pool,
            MinVersion: tls.VersionTLS12,
            ServerName: "api.internal", // SNI + verify name
        },
    },
    Timeout: 10 * time.Second,
}
```

Never ship `InsecureSkipVerify: true` in production. If you need custom verify (name overrides, SPIFFE IDs), use `VerifyPeerCertificate` / `VerifyConnection` carefully.

mTLS

Server requests and verifies client certificates:

```go
clientCA := x509.NewCertPool()
clientCA.AppendCertsFromPEM(clientCAPEM)

tlsCfg := &tls.Config{
    MinVersion: tls.VersionTLS12,
    ClientAuth: tls.RequireAndVerifyClientCert,
    ClientCAs:  clientCA,
    Certificates: []tls.Certificate{serverCert},
}
```

Client presents cert:

```go
cert, _ := tls.LoadX509KeyPair("client.crt", "client.key")
tlsCfg := &tls.Config{
    Certificates: []tls.Certificate{cert},
    RootCAs:      pool,
    MinVersion:   tls.VersionTLS12,
    ServerName:   "payments.internal",
}
```

Identity from client cert

```go
func peerSPIFFEOrCN(r *http.Request) (string, error) {
    if r.TLS == nil || len(r.TLS.PeerCertificates) == 0 {
        return "", errors.New("no peer cert")
    }
    cert := r.TLS.PeerCertificates[0]
    // Prefer URI SANs (SPIFFE) over CN in modern designs.
    for _, u := range cert.URIs {
        return u.String(), nil
    }
    return cert.Subject.CommonName, nil
}
```

Authorize **after** extracting identity (next chapter) — transport authn business authz.

Trust Bundles and Rotation

Rotation sequence for CAs:

```
1. Distribute new CA alongside old (trust both)
2. Issue leaf certs from new CA
3. Deploy leaves
4. Remove old CA only after all leaves rotated
```

For leaf certs:

```
issue new leaf -> dual accept if needed -> switch serve -> revoke/expire old
```

Monitor:

- Days to expiry (cert exporter / custom metric)
- Handshake failure rates
- TLS version negotiation

HTTP/2 and ALPN

Go enables HTTP/2 over TLS when configured correctly. ALPN negotiates protocols. Mis-set `NextProtos` can break h2.

Proxies and TLS

Common patterns:

Pattern	Notes
TLS terminate at edge	Backend may be plain on private net (risk: flat network)
Re-encrypt to backend	Edge → backend TLS
Pass-through	Edge routes TCP; backend terminates
Mesh mTLS	Sidecars handle identity

If you trust "internal HTTP," document the compensating control (network policy, mesh).

Production Checklist

- ☐ MinVersion TLS 1.2+
- ☐ Private keys `0600`, not in image layers with world read
- ☐ Automated expiry monitoring and paging
- ☐ Reload path tested for leaf rotation
- ☐ mTLS: `RequireAndVerifyClientCert` + correct ClientCAs
- ☐ Clients set ServerName / use proper URL host
- ☐ No `InsecureSkipVerify` in prod configs
- ☐ CA rotation dual-trust window documented
- ☐ Handshake errors metrics + logs (without dumping secrets)

Common Pitfalls

1. **Clock skew** fails cert validity (`NotBefore`/`NotAfter`).
2. **Forgetting intermediate certs** in the server chain.
3. **File replace** of `tls.key` while process holds old fd only — need reload.
4. **CN-only verification** without SANs (modern clients expect SANs).
5. **ClientAuth RequestClientCert** without verify — provides no security.
6. **Logging private keys** during debug.
7. **Wildcard trust** too broad in ClientCAs.

Exercises

1. Generate a local CA + server cert with `openssl` or `mkcert`; serve HTTPS with Go.
2. Connect with the system CA only; observe failure; fix RootCAs.
3. Enable mTLS; connect without client cert; confirm failure; then succeed with client cert.
4. Implement `GetCertificate` hot reload; replace files; curl still works.
5. Set `MinVersion` to TLS 1.3 only; test old client behavior.

6. Extract URI SAN identity in a handler; map to a principal.
7. Emit a gauge `tls_cert_not_after_seconds`; alert at < 14 days.
8. Deliberately expire a cert; document failure symptoms in client and server logs.
9. Compare edge termination vs app termination threat models in a one-pager.
10. Rotate CA with dual-trust; prove old leaves still work until cutover completes.

Certificate File Permissions Checklist

```
# private key must not be group/world readable
chmod 600 server.key
# cert can be world-readable
chmod 644 server.crt
```

In Kubernetes, use projected secrets with restrictive default modes (`defaultMode: 0400` for keys). Never `console.log` or slog the PEM contents.

Handshake Failure Debugging Order

1. Clock skew (`NotBefore` / `NotAfter`)
2. Missing intermediate in server chain
3. Client trust bundle missing the issuing CA
4. SNI / `ServerName` mismatch vs certificate SAN
5. mTLS: client cert not sent or not signed by `ClientCAs`
6. Version floor: client only offers TLS 1.0, server requires 1.2+

Capture with `openssl s_client -connect host:443 -servername host -showcerts` before changing application code.

More examples

TLS server + RootCAs client (httptest)

```
mkdir -p /tmp/go-tls-roots && cd /tmp/go-tls-roots
go mod init example.com/tls-roots
```

Save as `main.go`:

```
package main

import (
    "crypto/tls"
    "crypto/x509"
    "fmt"
```

```
    "io"
    "net/http"
    "net/http/httptest"
)

func main() {
    ts := httptest.NewTLSServer(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request
        fmt.Fprintf(w, "version_ok=%v", r.TLS != nil)
    }))
    defer ts.Close()

    roots := x509.NewCertPool()
    roots.AddCert(ts.Certificate())
    client := &http.Client{Transport: &http.Transport{
        TLSClientConfig: &tls.Config{RootCAs: roots, MinVersion: tls.VersionTLS12},
    }}
    resp, err := client.Get(ts.URL)
    if err != nil {
        panic(err)
    }
    b, _ := io.ReadAll(resp.Body)
    resp.Body.Close()
    fmt.Println(string(b))
    _, err = http.Get(ts.URL)
    fmt.Println("default client fails:", err != nil)
}
```

```
go run .
```

Expected output:

```
version_ok=true
default client fails: true
```

Inspect peer cert fields in handler

```
mkdir -p /tmp/go-tls-peer && cd /tmp/go-tls-peer
go mod init example.com/tls-peer
```

Save as `main.go`:

```
package main

import (
    "crypto/tls"
```

```go
    "crypto/x509"
    "fmt"
    "net/http"
    "net/http/httptest"
)

func main() {
    ts := httptest.NewTLSServer(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request
        if r.TLS == nil {
            http.Error(w, "no tls", 500)
            return
        }
        fmt.Fprintf(w, "negotiated>=1.2=%v peers=%d",
            r.TLS.Version >= tls.VersionTLS12, len(r.TLS.PeerCertificates))
    }))
    defer ts.Close()

    roots := x509.NewCertPool()
    roots.AddCert(ts.Certificate())
    client := &http.Client{Transport: &http.Transport{
        TLSClientConfig: &tls.Config{RootCAs: roots, MinVersion: tls.VersionTLS12},
    }}
    resp, err := client.Get(ts.URL)
    if err != nil {
        panic(err)
    }
    defer resp.Body.Close()
    var buf [64]byte
    n, _ := resp.Body.Read(buf[:])
    fmt.Println(string(buf[:n]))
}
```

```go
go run .
```

Expected output:

```
negotiated>=1.2=true peers=0
```

(`peers=0` without client certs; mTLS would raise this.)

Runnable example

HTTPS with `httptest.NewTLSServer` and a client that uses the server's certificate pool (no `InsecureSkipVerify` in the happy path). Also shows a misconfigured skip-verify client for contrast.

```bash
mkdir -p /tmp/go-tls-demo && cd /tmp/go-tls-demo
go mod init example.com/tls-demo
```

Save as `main.go`:

```go
package main

import (
    "crypto/tls"
    "crypto/x509"
    "fmt"
    "io"
    "net/http"
    "net/http/httptest"
)

func main() {
    ts := httptest.NewTLSServer(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request
        fmt.Fprintf(w, "hello tls state=%v\n", r.TLS != nil)
    }))
    defer ts.Close()

    // Trust the test server's cert (httptest exposes it).
    roots := x509.NewCertPool()
    roots.AddCert(ts.Certificate())
    secureClient := &http.Client{
        Transport: &http.Transport{
            TLSClientConfig: &tls.Config{
                RootCAs:    roots,
                MinVersion: tls.VersionTLS12,
            },
        },
    }
    resp, err := secureClient.Get(ts.URL)
    if err != nil {
        panic(err)
    }
    body, _ := io.ReadAll(resp.Body)
    resp.Body.Close()
    fmt.Printf("secure: %s", body)

    // Default client does not trust httptest certs → error
    _, err = http.Get(ts.URL)
    fmt.Println("default client error:", err != nil)

    // InsecureSkipVerify works but is not production policy
    insecure := &http.Client{
```

```go
        Transport: &http.Transport{
            TLSClientConfig: &tls.Config{InsecureSkipVerify: true, MinVersion: tls.VersionTL
        },
    }
    resp, err = insecure.Get(ts.URL)
    if err != nil {
        panic(err)
    }
    resp.Body.Close()
    fmt.Println("insecure skip verify status:", resp.StatusCode)
    fmt.Println("MinVersion TLS1.2 configured on both clients")
}
```

```go
go run .
```

Expected output:

```
secure: hello tls state=true
default client error: true
insecure skip verify status: 200
MinVersion TLS1.2 configured on both clients
```

What to notice

- Production clients need a real trust bundle (`RootCAs`), not `InsecureSkipVerify`.
- `httptest.NewTLSServer` is the practical way to exercise TLS in unit tests.
- Set `MinVersion: tls.VersionTLS12` (or 1.3-only) deliberately.

Try next

- Require client certs (`ClientAuth: tls.RequireAndVerifyClientCert`) for an mTLS sketch.
- Inspect `r.TLS.Version` and peer certificates on the server handler.

Further Reading

- Go `crypto/tls` configuration documentation
- Mozilla SSL configuration generator (concepts)
- Next: **Authentication and Authorization Patterns**

Authentication and Authorization Patterns

Authentication and Authorization Patterns

Authentication answers **who are you?** Authorization answers **what are you allowed to do?** Mixing them produces systems that are hard to test and easy to bypass.

Why / Overview

Common failures:

- JWT parsed but signature not verified
- Authn on API gateway, authz forgotten on internal admin routes
- Role checks as string equality scattered in handlers
- IDOR: authorized as user A, still can fetch user B's resource by id

```
request -> authenticate -> Principal
                        -> authorize(Principal, action, resource) -> allow/deny
```

Authentication Patterns

1. Session cookies (first-party browsers)

```go
// After credential check:
sid := newSessionID()
store.Save(sid, userID, time.Now().Add(24*time.Hour))
http.SetCookie(w, &http.Cookie{
    Name:     "session",
    Value:    sid,
    Path:     "/",
    HttpOnly: true,
    Secure:   true, // HTTPS only
    SameSite: http.SameSiteLaxMode,
    MaxAge:   86400,
})
```

Use CSRF protection for cookie-based mutating requests.

2. Bearer tokens (APIs)

```go
func bearer(r *http.Request) (string, bool) {
    h := r.Header.Get("Authorization")
    if !strings.HasPrefix(h, "Bearer ") {
        return "", false
    }
    return strings.TrimPrefix(h, "Bearer "), true
}
```

Validate JWT (or opaque token introspection) **with**:

- Signature / HMAC key or JWKS
- `exp` / `nbf`
- `iss` issuer allowlist
- `aud` audience allowlist
- algorithm allowlist (reject `none`)

```go
// Conceptual validation steps - use a well-reviewed library.
tok, err := jwt.Parse(token, keyFunc, jwt.WithValidMethods([]string{"RS256"}))
// check claims.Issuer, claims.Audience, claims.ExpiresAt
```

3. mTLS identity

Peer certificate URI SAN / SPIFFE ID becomes the principal for service-to-service calls (previous chapter). Still run authz: not every client cert may call every route.

4. API keys

Fine for simple server-to-server; store **hashes** of keys, not raw keys; rate-limit; rotate.

Principal Model

```go
type Principal struct {
    Subject string   // user id or spiffe://...
    Type    string   // user|service
    Roles   []string // optional coarse
    Scopes  []string // oauth-like
    Tenant  string   // multi-tenant systems
}
```

Attach to context:

```go
type principalKey struct{}

func WithPrincipal(ctx context.Context, p Principal) context.Context {
    return context.WithValue(ctx, principalKey{}, p)
}

func PrincipalFrom(ctx context.Context) (Principal, bool) {
    p, ok := ctx.Value(principalKey{}).(Principal)
    return p, ok
}
```

Authorization Patterns

RBAC (role-based)

```go
var perms = map[string][]string{
    "admin": {"order:read", "order:write", "user:read"},
    "user":  {"order:read", "order:write:own"},
}

func allowed(p Principal, perm string) bool {
    for _, role := range p.Roles {
        for _, x := range perms[role] {
            if x == perm {
                return true
            }
        }
    }
    return false
}
```

Resource ownership (critical for IDOR)

```go
func canReadOrder(p Principal, o Order) bool {
    if allowed(p, "order:read:any") {
        return true
    }
    return allowed(p, "order:read:own") && o.UserID == p.Subject
}
```

Always load the resource and check ownership server-side. Never trust client-provided `user_id` alone.

Policy as pure function

```go
type Request struct {
    Principal Principal
    Action    string
    Resource  Resource
}

func Authorize(req Request) error {
    if !policy(req) {
        return ErrDeny
    }
    return nil
}
```

Pure functions are unit-testable with tables of allow/deny cases.

Middleware composition

```go
func requireAuth(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        p, err := authenticate(r)
        if err != nil {
            http.Error(w, "unauthorized", http.StatusUnauthorized)
            return
        }
        next.ServeHTTP(w, r.WithContext(WithPrincipal(r.Context(), p)))
    })
}

func requirePerm(perm string, next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        p, ok := PrincipalFrom(r.Context())
        if !ok || !allowed(p, perm) {
            http.Error(w, "forbidden", http.StatusForbidden)
            return
        }
        next.ServeHTTP(w, r)
    })
}
```

Status codes: **401** unauthenticated, **403** authenticated but not allowed. Do not leak whether a resource exists if that is sensitive (sometimes always 404).

Deny by Default

```
// Route registration: only explicitly wrapped routes are reachable.
mux.Handle("GET /admin/metrics", requireAuth(requirePerm("admin:metrics", adminHandler)))
```

Avoid a global "if admin skip checks" flag that can be left on.

Multi-Tenant Awareness

Every query should be scoped:

```
// BAD: SELECT * FROM orders WHERE id=$1
// GOOD: SELECT * FROM orders WHERE id=$1 AND tenant_id=$2
```

Authorization + data filtering work together.

Audit Logging

Log authz denials and sensitive allows:

```
slog.Warn("authz_deny",
    "sub", p.Subject,
    "action", action,
    "resource", resourceID,
    "request_id", RequestID(ctx),
)
```

Never log raw passwords or tokens.

Testing Matrix

Case	Expect
No credentials	401
Bad signature / expired	401
Valid user, wrong role	403
Valid user, not owner	403
Valid owner	200
Admin override	200 if policy says so

Table-driven tests for `Authorize` catch regressions better than handler snapshots alone.

Production Checklist

- ☐ Authn and authz modules separated
- ☐ Token validation includes iss/aud/exp/alg
- ☐ Cookies: HttpOnly, Secure, SameSite set intentionally
- ☐ CSRF strategy for cookie sessions
- ☐ Resource-level checks (anti-IDOR)
- ☐ Default deny routing
- ☐ Tenant scoping in data access
- ☐ Audit logs for deny and admin actions
- ☐ Load tests do not bypass auth in prod configs
- ☐ Service-to-service identity (mTLS or signed tokens)

Common Pitfalls

1. **Parse-only JWT** without verify.
2. **Trust gateway headers** (`X-User-Id`) without authenticating the gateway hop.
3. **Frontend-only authorization**.
4. **String role checks** duplicated and drifting.
5. **403 vs 401 confusion** breaking client refresh flows.
6. **Overbroad `admin` role** granted to services.
7. **Debug auth bypass** left enabled.

Exercises

1. Implement session cookie login/logout with secure flags; attempt theft via XSS hypothetical — why HttpOnly helps.
2. Validate a JWT with wrong `aud`; prove rejection.
3. Build `Authorize` pure function with 10 table cases including IDOR.
4. Create two middleware layers; write tests with `httptest`.
5. Add tenant_id to queries; attempt cross-tenant read; deny.
6. Log authz deny with request id; redact token fields.
7. Convert scattered role checks into a permission map; delete duplicates.
8. Simulate expired token; ensure 401 not 500.
9. Service identity: map mTLS SPIFFE ID to Principal; authorize routes.
10. Threat-model one endpoint: list authn/authz failures and tests for each.

More examples

Bearer authn middleware

```
mkdir -p /tmp/go-authn-bearer && cd /tmp/go-authn-bearer
go mod init example.com/authn-bearer
```

Save as `main.go`:

```
package main

import (
    "context"
    "fmt"
    "net/http"
    "net/http/httptest"
    "strings"
)

type ctxKey string

const userKey ctxKey = "user"

func bearerAuth(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        h := r.Header.Get("Authorization")
        if !strings.HasPrefix(h, "Bearer ") {
            http.Error(w, "unauthorized", http.StatusUnauthorized)
            return
        }
        token := strings.TrimPrefix(h, "Bearer ")
        if token != "good-token" {
            http.Error(w, "unauthorized", http.StatusUnauthorized)
            return
        }
        ctx := context.WithValue(r.Context(), userKey, "alice")
        next.ServeHTTP(w, r.WithContext(ctx))
    })
}

func main() {
    h := bearerAuth(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        fmt.Fprintln(w, r.Context().Value(userKey))
    }))

    rr := httptest.NewRecorder()
```

```go
    h.ServeHTTP(rr, httptest.NewRequest(http.MethodGet, "/", nil))
    fmt.Println("missing:", rr.Code)

    rr = httptest.NewRecorder()
    req := httptest.NewRequest(http.MethodGet, "/", nil)
    req.Header.Set("Authorization", "Bearer good-token")
    h.ServeHTTP(rr, req)
    fmt.Println("ok:", rr.Code, strings.TrimSpace(rr.Body.String()))
}
```

```bash
go run .
```

Expected output:

```
missing: 401
ok: 200 alice
```

Role-based authz gate

```bash
mkdir -p /tmp/go-authz-rbac && cd /tmp/go-authz-rbac
go mod init example.com/authz-rbac
```

Save as `main.go`:

```go
package main

import (
    "fmt"
    "net/http"
    "net/http/httptest"
)

type Principal struct {
    Name  string
    Roles map[string]bool
}

func requireRole(role string, p Principal, next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        if !p.Roles[role] {
            http.Error(w, "forbidden", http.StatusForbidden)
            return
        }
        next.ServeHTTP(w, r)
    })
```

```go
}

func main() {
    admin := Principal{Name: "a", Roles: map[string]bool{"admin": true}}
    user := Principal{Name: "u", Roles: map[string]bool{"user": true}}
    core := http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        fmt.Fprintln(w, "secret")
    })

    rr := httptest.NewRecorder()
    requireRole("admin", user, core).ServeHTTP(rr, httptest.NewRequest(http.MethodGet, "/",
    fmt.Println("user:", rr.Code)

    rr = httptest.NewRecorder()
    requireRole("admin", admin, core).ServeHTTP(rr, httptest.NewRequest(http.MethodGet, "/",
    fmt.Println("admin:", rr.Code)
}
```

```bash
go run .
```

Expected output:

```
user: 403
admin: 200
```

Runnable example

Bearer token authn middleware + pure `Authorize` function with table-driven cases (including IDOR-style deny)—`httptest` only.

```bash
mkdir -p /tmp/go-authz && cd /tmp/go-authz
go mod init example.com/authz
```

Save as `main.go`:

```go
package main

import (
    "context"
    "fmt"
    "net/http"
    "net/http/httptest"
    "strings"
)
```

```go
type Principal struct {
    ID   string
    Role string
}

type ctxKey string

const principalKey ctxKey = "principal"

func authenticate(token string) (Principal, bool) {
    // Demo token map; production: verify JWT/session server-side.
    db := map[string]Principal{
        "tok-alice": {ID: "u1", Role: "user"},
        "tok-admin": {ID: "u0", Role: "admin"},
    }
    p, ok := db[token]
    return p, ok
}

func Authorize(p Principal, action, resourceOwner string) bool {
    if p.Role == "admin" {
        return true
    }
    if action == "read" || action == "write" {
        return p.ID == resourceOwner // prevent IDOR
    }
    return false
}

func withAuth(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        h := r.Header.Get("Authorization")
        if !strings.HasPrefix(h, "Bearer ") {
            http.Error(w, "unauthorized", http.StatusUnauthorized)
            return
        }
        p, ok := authenticate(strings.TrimPrefix(h, "Bearer "))
        if !ok {
            http.Error(w, "unauthorized", http.StatusUnauthorized)
            return
        }
        ctx := context.WithValue(r.Context(), principalKey, p)
        next.ServeHTTP(w, r.WithContext(ctx))
    })
}

func main() {
```

```go
    mux := http.NewServeMux()
    mux.Handle("GET /items/{owner}", withAuth(http.HandlerFunc(func(w http.ResponseWriter, r
        p := r.Context().Value(principalKey).(Principal)
        owner := r.PathValue("owner")
        if !Authorize(p, "read", owner) {
            http.Error(w, "forbidden", http.StatusForbidden)
            return
        }
        fmt.Fprintf(w, "ok user=%s owner=%s\n", p.ID, owner)
    }))

    call := func(token, path string) int {
        req := httptest.NewRequest(http.MethodGet, path, nil)
        if token != "" {
            req.Header.Set("Authorization", "Bearer "+token)
        }
        rr := httptest.NewRecorder()
        mux.ServeHTTP(rr, req)
        return rr.Code
    }

    fmt.Println("no token:", call("", "/items/u1"))
    fmt.Println("alice own:", call("tok-alice", "/items/u1"))
    fmt.Println("alice idor:", call("tok-alice", "/items/u9"))
    fmt.Println("admin idor:", call("tok-admin", "/items/u9"))
    fmt.Println("bad token:", call("nope", "/items/u1"))
}
```

```go
go run .
```

Expected output:

```
no token: 401
alice own: 200
alice idor: 403
admin idor: 200
bad token: 401
```

What to notice

- 401 = not authenticated; 403 = authenticated but not allowed—clients refresh on 401, not 403.
- Authorization is a pure function—easy to table-test; do not scatter role checks only in the UI.
- Resource owner checks stop IDOR when IDs appear in URLs.

Try next

- Add `tenant_id` to Principal and deny cross-tenant reads.

- Validate JWT `aud`/`iss`/`exp` with `crypto` + careful parsing libraries (or session cookies).

Further Reading

- OWASP Authentication and Access Control cheatsheets
- Previous: TLS/mTLS as transport authn
- Next: **Secrets Management and Rotation**

Secrets Management and Rotation

Secrets Management and Rotation

Secrets are **dynamic credentials with a lifecycle**, not static configuration strings. Most secret-related outages come from rotation choreography errors, not from encryption algorithm choice.

Why / Overview

Secrets include:

- Database passwords
- API tokens and HMAC keys
- TLS private keys
- OAuth client secrets
- Encryption keys (DEK/KEK hierarchy)

```
issue new secret -> readers accept both -> writers switch -> revoke old
```

If you flip both ends at once without overlap, you race yourself into downtime.

Never Do This

Anti-pattern	Why
Secrets in git	History is forever; bots scan public repos
Secrets in Docker layers	Recoverable from images
Secrets in client apps	Extractable
Long-lived shared passwords	Huge blast radius
Logs printing env	CI and support tools leak

Loading Secrets at Runtime

Environment variables

Common in 12-factor apps; simple but:

- Visible in `/proc` and some debug endpoints
- Easy to dump accidentally
- Awkward dual-read of two values (use `DB_PASSWORD` + `DB_PASSWORD_OLD`)

```go
func mustSecret(key string) string {
    v := os.Getenv(key)
    if v == "" {
        slog.Error("missing secret", "key", key)
        os.Exit(1)
    }
    return v
}
```

Files (Kubernetes secrets mounts)

```go
func readSecretFile(path string) (string, error) {
    b, err := os.ReadFile(path)
    if err != nil {
        return "", err
    }
    return strings.TrimSpace(string(b)), nil
}
```

Many platforms rotate by updating the file; processes must **re-read** or restart.

Secret managers

AWS SM, GCP SM, Azure Key Vault, HashiCorp Vault, etc.:

- Short-lived credentials via IAM roles
- Central audit
- API-driven rotation

Wrap the vendor SDK behind a small interface:

```go
type Source interface {
    Get(ctx context.Context, name string) (string, error)
}
```

Fail Closed at Startup

```go
func run() error {
    cfg, err := loadConfig()
    if err != nil {
        return err
    }
    if cfg.DBPassword == "" {
```

```go
        return errors.New("db password required")
    }
    // open DB, then listen
    return serve(cfg)
}
```

Do not listen on `:8080` and return 500 for all traffic because secrets failed — fail before ready.

Dual-Read Rotation (Passwords / HMAC)

```
Time ->
DB password:   [ v1 active ][ v1+v2 accepted ][ v2 active ]
App readers:   [ v1        ][ try v2 then v1 ][ v2        ]
App writers:   [ v1        ][ switch to v2   ][ v2        ]
```

```go
type RotatingToken struct {
    mu   sync.RWMutex
    cur  []byte
    prev []byte
}

func (r *RotatingToken) Verify(mac, msg []byte) bool {
    r.mu.RLock()
    defer r.mu.RUnlock()
    if hmac.Equal(mac, sum(r.cur, msg)) {
        return true
    }
    if len(r.prev) > 0 && hmac.Equal(mac, sum(r.prev, msg)) {
        return true
    }
    return false
}

func (r *RotatingToken) Rotate(newKey []byte) {
    r.mu.Lock()
    defer r.mu.Unlock()
    r.prev = r.cur
    r.cur = newKey
}
```

For DB users, prefer **two users** or two passwords supported by the server during cutover.

Runtime Refresh Without Restart

```go
func (a *App) watchSecrets(ctx context.Context, every time.Duration) {
    t := time.NewTicker(every)
    defer t.Stop()
    for {
        select {
        case <-ctx.Done():
            return
        case <-t.C:
            sec, err := a.src.Get(ctx, "db-password")
            if err != nil {
                slog.Error("secret refresh failed", "err", err)
                continue // keep old; alert on repeated failure
            }
            if err := a.db.UpdatePassword(sec); err != nil {
                slog.Error("db rebind failed", "err", err)
            }
        }
    }
}
```

Some drivers need a new pool to pick up password changes — test the path.

Encryption Keys vs Passwords

For application-level encryption:

- Use a **KEK** in a KMS to wrap **DEKs**
- Store wrapped DEKs next to ciphertext
- Rotate KEK via rewrap; rotate DEK by re-encrypting data in batches

Do not invent AES modes; use `crypto/cipher` with GCM properly (unique nonces) or higher-level libraries.

```go
func encrypt(key, plaintext []byte) ([]byte, error) {
    block, err := aes.NewCipher(key)
    if err != nil {
        return nil, err
    }
    gcm, err := cipher.NewGCM(block)
    if err != nil {
        return nil, err
    }
    nonce := make([]byte, gcm.NonceSize())
    if _, err := rand.Read(nonce); err != nil {
```

```go
        return nil, err
    }
    return gcm.Seal(nonce, nonce, plaintext, nil), nil
}
```

CI/CD and Build-Time Leaks

- Inject secrets at deploy time, not build time, when possible
- Mask secrets in CI logs
- Prefer OIDC federated login to clouds over long-lived CI keys
- Scan releases with secret scanners

Redaction Everywhere

```go
type RedactedString string

func (r RedactedString) String() string { return "***" }
func (r RedactedString) GoString() string { return "***" }
func (r RedactedString) MarshalJSON() ([]byte, error) { return []byte(`"***"`), nil }
```

Use carefully for types that must not appear in logs.

Production Checklist

- ☐ No secrets in git or images
- ☐ Required secrets fail startup (not silent empty)
- ☐ Dual-read/dual-trust windows for rotation
- ☐ Automated rotation where platform supports it
- ☐ Refresh or restart path tested quarterly
- ☐ Access audit on secret manager
- ☐ Least-privilege IAM for secret read
- ☐ Alert on refresh failures
- ☐ Revocation plan for leaked credentials
- ☐ TLS keys use restricted file perms / KMS

Common Pitfalls

1. **Single global static secret** for all environments.
2. **Rotation without dual-read** → full outage.
3. **Copy-paste secrets** into tickets and Slack.
4. **Same DB password** for migrate job and app with different cutover times.

5. **Not revoking** after suspected leak (only rotating sometimes).
6. **Base64 encryption** for "hiding" secrets in config.
7. **Long-lived JWTs** as a substitute for session server state without revoke list.

Exercises

1. Refactor a sample app to fail closed when `API_TOKEN` is missing.
2. Implement dual-key HMAC verify; rotate; prove old and new both verify during window.
3. Mount a secret file; update it; implement re-read; confirm new value used.
4. Add `RedactedString`; log a config struct; ensure secret not visible.
5. Simulate leak: document revoke + rotate steps as a runbook.
6. Compare env vs file vs Vault for your deployment model (table of tradeoffs).
7. Create two DB users; rotate app from A to B without downtime (local docker).
8. Scan a docker image history for accidental `.env` copy; fix Dockerfile.
9. Instrument `secret_refresh_failures_total`; alert on >0 for 15m.
10. Design DEK/KEK scheme for storing personal tokens at rest; review nonce handling.

More examples

Dual-secret accept during rotation

```
mkdir -p /tmp/go-secret-dual && cd /tmp/go-secret-dual
go mod init example.com/secret-dual
```

Save as `main.go`:

```
package main

import (
    "crypto/hmac"
    "crypto/sha256"
    "encoding/hex"
    "fmt"
)

func sign(secret, msg string) string {
    m := hmac.New(sha256.New, []byte(secret))
    _, _ = m.Write([]byte(msg))
    return hex.EncodeToString(m.Sum(nil))
}

func valid(secrets []string, msg, sig string) bool {
    for _, s := range secrets {
        want, err1 := hex.DecodeString(sign(s, msg))
```

```
        got, err2 := hex.DecodeString(sig)
        if err1 != nil || err2 != nil {
            continue
        }
        if hmac.Equal(want, got) {
            return true
        }
    }
    return false
}

func main() {
    old, neu := "old-secret", "new-secret"
    msg := "payload"
    // After rotation: sign with new, accept both for a window.
    sig := sign(neu, msg)
    fmt.Println("new only:", valid([]string{neu}, msg, sig))
    fmt.Println("dual window:", valid([]string{neu, old}, msg, sig))
    fmt.Println("old sig still ok in window:", valid([]string{neu, old}, msg, sign(old, msg)
    fmt.Println("after cutover old rejected:", valid([]string{neu}, msg, sign(old, msg)))
}
```

```
go run .
```

Expected output:

```
new only: true
dual window: true
old sig still ok in window: true
after cutover old rejected: false
```

Load secret from file with restrictive mode check

```
mkdir -p /tmp/go-secret-file && cd /tmp/go-secret-file
go mod init example.com/secret-file
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "os"
)
```

```go
func loadSecret(path string) (string, error) {
    st, err := os.Stat(path)
    if err != nil {
        return "", err
    }
    if st.Mode().Perm()&0o077 != 0 {
        return "", fmt.Errorf("secret %s mode too open: %v", path, st.Mode().Perm())
    }
    b, err := os.ReadFile(path)
    if err != nil {
        return "", err
    }
    return string(b), nil
}

func main() {
    path := "secret.txt"
    _ = os.WriteFile(path, []byte("s3cr3t"), 0o600)
    defer os.Remove(path)
    s, err := loadSecret(path)
    fmt.Println("ok:", s, err)

    _ = os.Chmod(path, 0o644)
    _, err = loadSecret(path)
    fmt.Println("open mode rejected:", err != nil)
}
```

```bash
go run .
```

Expected output:

```
ok: s3cr3t <nil>
open mode rejected: true
```

Runnable example

Fail-closed env loading, dual-key HMAC verify during rotation, and a `RedactedString` that never prints the secret.

```bash
mkdir -p /tmp/go-secrets && cd /tmp/go-secrets
go mod init example.com/secrets
```

Save as `main.go`:

```go
package main

import (
    "crypto/hmac"
    "crypto/sha256"
    "encoding/hex"
    "fmt"
    "os"
)

type RedactedString string

func (r RedactedString) String() string { return "***" }
func (r RedactedString) GoString() string { return "RedactedString(***)" }
func (r RedactedString) Reveal() string   { return string(r) }

func mustEnv(k string) string {
    v := os.Getenv(k)
    if v == "" {
        panic("missing required secret: " + k)
    }
    return v
}

func sign(key, msg string) string {
    m := hmac.New(sha256.New, []byte(key))
    _, _ = m.Write([]byte(msg))
    return hex.EncodeToString(m.Sum(nil))
}

func verify(keys []string, msg, sig string) bool {
    raw, err := hex.DecodeString(sig)
    if err != nil {
        return false
    }
    for _, k := range keys {
        mac := hmac.New(sha256.New, []byte(k))
        _, _ = mac.Write([]byte(msg))
        if hmac.Equal(raw, mac.Sum(nil)) {
            return true
        }
    }
    return false
}

func main() {
    // Demo defaults when unset (production: mustEnv only).
```

```go
    primary := os.Getenv("API_TOKEN_CURRENT")
    if primary == "" {
        primary = "current-secret"
    }
    previous := os.Getenv("API_TOKEN_PREVIOUS") // may be empty after rotation window

    secret := RedactedString(primary)
    fmt.Printf("config log: token=%s\n", secret) // ***

    msg := "payload"
    sigOld := sign("previous-secret", msg)
    sigNew := sign(primary, msg)

    keys := []string{primary}
    if previous != "" {
        keys = append(keys, previous)
    } else {
        keys = append(keys, "previous-secret") // demo dual-read window
    }
    fmt.Println("accept new:", verify(keys, msg, sigNew))
    fmt.Println("accept old during dual-read:", verify(keys, msg, sigOld))
    fmt.Println("reject other:", verify(keys, msg, sign("other", msg)))

    // Fail-closed illustration
    if os.Getenv("REQUIRE_STRICT") == "1" {
        _ = mustEnv("API_TOKEN_CURRENT")
    } else {
        fmt.Println("strict mode off (set REQUIRE_STRICT=1 to exercise mustEnv)")
    }
}
```

```bash
go run .
REQUIRE_STRICT=1 API_TOKEN_CURRENT=abc go run .
```

Expected output:

```
config log: token=***
accept new: true
accept old during dual-read: true
reject other: false
strict mode off (set REQUIRE_STRICT=1 to exercise mustEnv)
```

What to notice

- Dual-read (current + previous) avoids total outage at rotation cutover.
- Custom `String()` prevents accidental secret leakage via `fmt` and slog.
- Missing required secrets should prevent process start, not run with empty keys.

Try next

- Re-read a secret file on SIGHUP or timer; atomic swap of the active key slice.
- Metric `secret_refresh_failures_total` when reload fails.

Further Reading

- NIST guidance on secrets management (high level)
- Previous: authn/authz that *use* these secrets
- Next part: **Performance Engineering** — speed with evidence, not secrets of folklore

JWT and Token Validation

JWT and Token Validation

Overview

JWTs are common bearer tokens. Most bugs are **validation shortcuts**: skipping `exp`, `aud`, `iss`, or accepting `alg=none`. Validate explicitly.

Use a maintained library (`github.com/golang-jwt/jwt` or similar); do not hand-roll crypto.

Minimal validation checklist

Claim / check	Why
Signature + algorithm allowlist	Integrity
`exp` / `nbf`	Lifetime
`iss`	Who minted
`aud`	Intended recipient
`sub` present	Principal id

```go
// sketch with jwt/v5 style API
tok, err := jwt.Parse(tokenStr, func(t *jwt.Token) (any, error) {
    if t.Method.Alg() != jwt.SigningMethodHS256.Alg() {
        return nil, fmt.Errorf("unexpected alg")
    }
    return hmacSecret, nil
}, jwt.WithValidMethods([]string{"HS256"}),
    jwt.WithAudience("bookstore-api"),
    jwt.WithIssuer("auth.example.com"),
)
```

Access vs refresh

Token	Lifetime	Storage
Access	minutes	memory / short cookie
Refresh	longer	httpOnly cookie, rotate

Service-to-service

Prefer mTLS or short-lived minted tokens over eternal shared HS256 secrets across many services.

Rules of thumb

Do	Don't
Allowlist algorithms	Trust `alg` header blindly
Validate aud/iss	Only check signature
Short access TTL	Multi-day bearer in localStorage without XSS plan

Try next

1. Reject expired tokens in a unit test.
2. Reject wrong `aud`.
3. Document key rotation for HS256/RS256.

Input Validation and SSRF

Input Validation and SSRF

Overview

Validate at trust boundaries. **SSRF** (server-side request forgery) happens when user-controlled URLs make your server fetch internal resources (`169.254.169.254`, `localhost`, cloud metadata).

Validation basics

```go
type CreateBook struct {
    Title string `json:"title"`
    Price int    `json:"price_cents"`
}

func (c CreateBook) Valid() error {
    if len(c.Title) == 0 || len(c.Title) > 200 {
        return fmt.Errorf("title length")
    }
    if c.Price < 0 {
        return fmt.Errorf("price")
    }
    return nil
}
```

- Prefer allowlists over blocklists for enums
- Bound string lengths and collection sizes
- Parse numbers with bit size (`ParseInt(..., 10, 64)`)

Path traversal

```go
// never: os.Open("/data/" + userPath)
// use safeJoin from filepath chapter
```

SSRF defenses

When your app fetches a URL on behalf of a user:

1. **Allowlist schemes** (`https` only)
2. **Allowlist hosts** or resolve and block private ranges
3. **No redirects** to off-allowlist targets
4. Timeouts + size limits

```go
func safeURL(raw string) (*url.URL, error) {
    u, err := url.Parse(raw)
    if err != nil {
        return nil, err
    }
    if u.Scheme != "https" {
        return nil, fmt.Errorf("scheme")
    }
    host := u.Hostname()
    ips, err := net.DefaultResolver.LookupIPAddr(ctx, host)
    // reject loopback, link-local, private RFC1918, metadata IP
    return u, nil
}
```

```go
func isBadIP(ip net.IP) bool {
    return ip.IsLoopback() || ip.IsPrivate() || ip.IsLinkLocalUnicast() ||
        ip.IsLinkLocalMulticast() || ip.Equal(net.ParseIP("169.254.169.254"))
}
```

Re-check IPs after redirects (pin or disable redirects).

Header injection

Don't put raw user strings into `Header.Set` without validation (CRLF). Prefer structured values.

Rules of thumb

Do	Don't
Validate then use	Trust client JSON shapes unbounded
Allowlist outbound hosts	Open proxy to any URL
Bound body size	Stream forever into memory

Try next

1. Reject `http://127.0.0.1/` in a fetch helper.
2. Reject path `../../etc/passwd` in download handler.
3. Fuzz validators with `go test -fuzz`.

os.Root: Traversal-Resistant File Access

os.Root: Traversal-Resistant File Access

Thesis

Any code that joins a **trusted directory** with an **untrusted name** — an upload handler, an archive extractor, a static file server, a per-tenant storage layer — is one symlink away from reading or writing somewhere else. Lexical checks (`filepath.Clean`, `filepath.IsLocal`, a `HasPrefix` jail) only inspect the *string*; the kernel resolves the *filesystem*, and the two disagree the moment a symlink exists. Since Go 1.24 the standard library has the right tool: **`os.Root`**, a handle on a directory whose every operation is resolved inside that directory or refused. Go 1.25 filled in the rest of the API (`MkdirAll`, `ReadFile`, `WriteFile`, `Symlink`, `Rename`, `RemoveAll`, and friends), so on Go 1.27 there is no longer an excuse to build host paths by hand.

Mental model

`os.OpenRoot(dir)` opens a file descriptor on `dir`. Every method on the returned `*os.Root` takes a name **relative to that descriptor** and walks it one component at a time with the `openat` family of system calls. At each step it checks: did `..` climb above the root? Did a symlink point above the root, or to an absolute path? If yes, the call fails with a `*PathError` whose message is `path escapes from parent`.

```
classic jail                          os.Root
-------------                         -------
string  "avatar/secret.txt"           fd(work/uploads)
  |  Clean + Join + HasPrefix           |  openat(fd, "avatar")    -> symlink "../private"
  v  "work/uploads/avatar/secret.txt"  |  resolve: climbs above fd -> refuse
kernel open() follows the symlink     v
  -> work/private/secret.txt  (leak)  error: path escapes from parent
```

Three properties matter in practice:

Property	Consequence
Resolution happens **in the kernel walk**, not before it	No time-of-check/time-of-use gap between "validate" and "open"
Symlinks and `..` are **allowed while they stay inside**	`root.Open("a/../b")` and in-root links still work
The root is a **descriptor**, not a path (Linux, macOS, BSD)	Renaming or moving the directory does not redirect later operations

The method set mirrors package `os`: `Open`, `OpenFile`, `Create`, `ReadFile`, `WriteFile`, `Mkdir`, `MkdirAll`, `Remove`, `RemoveAll`, `Rename`, `Link`, `Symlink`, `Readlink`, `Stat`, `Lstat`, `Chmod`, `Chown`, `Lchown`, `Chtimes`, plus `OpenRoot` (a narrower root inside this one) and `FS` (an `io/fs.FS` view that implements `StatFS`, `ReadFileFS`, `ReadDirFS`, and `ReadLinkFS`). For one-off opens there is `os.OpenInRoot(dir, name)`.

Rule of thumb from the Go team, and the one this chapter follows: if you are about to call `filepath.Join(fixedDir, externalName)`, use `os.Root` instead.

Worked examples

Every listing below was run with **Go 1.27.1 on Linux (amd64)**; the output blocks are copied from those runs. Each case lives in its own directory so `go run` sees one `main` package.

Case 1: The prefix-check jail leaks through a symlink

The classic defence cleans the name, joins it to the root, and checks the prefix. It defeats `../` — and nothing else. Here an attacker who can write into `uploads/` (a previous upload, an unpacked archive, a shared volume) plants `avatar -> ../private`.

```bash
mkdir -p case1 && cd case1
```

```go
// escape.go
package main

import (
    "errors"
    "fmt"
    "os"
    "path/filepath"
    "strings"
)

// safeJoin is the classic lexical jail: clean, join, prefix-check.
func safeJoin(root, name string) (string, error) {
    full := filepath.Join(root, filepath.Clean("/"+name))
    if !strings.HasPrefix(full, filepath.Clean(root)+string(filepath.Separator)) {
        return "", errors.New("path escapes root")
    }
    return full, nil
}

func main() {
    // Lay out a sandbox and a secret that lives outside it.
    must(os.MkdirAll("work/uploads", 0o755))
```

```go
    must(os.MkdirAll("work/private", 0o700))
    must(os.WriteFile("work/private/secret.txt", []byte("db-password=hunter2\n"), 0o600))
    must(os.WriteFile("work/uploads/hello.txt", []byte("hello\n"), 0o644))

    // Someone who can write into uploads/ plants a symlink.
    _ = os.Remove("work/uploads/avatar")
    must(os.Symlink("../private", "work/uploads/avatar"))

    for _, name := range []string{"hello.txt", "../private/secret.txt", "avatar/secret.txt"}
        fmt.Printf("== %s\n", name)

        if p, err := safeJoin("work/uploads", name); err != nil {
            fmt.Println("  safeJoin:   rejected:", err)
        } else if b, err := os.ReadFile(p); err != nil {
            fmt.Println("  safeJoin:   error:", err)
        } else {
            fmt.Printf("  safeJoin:   read %q\n", b)
        }

        f, err := os.OpenInRoot("work/uploads", name)
        if err != nil {
            fmt.Println("  OpenInRoot: rejected:", err)
            continue
        }
        b := make([]byte, 64)
        n, _ := f.Read(b)
        f.Close()
        fmt.Printf("  OpenInRoot: read %q\n", b[:n])
    }
}

func must(err error) {
    if err != nil {
        panic(err)
    }
}
```

```go
go run escape.go
```

Output:

```
== hello.txt
  safeJoin:   read "hello\n"
  OpenInRoot: read "hello\n"
== ../private/secret.txt
  safeJoin:   error: open work/uploads/private/secret.txt: no such file or directory
  OpenInRoot: rejected: openat ../private/secret.txt: path escapes from parent
```

```
== avatar/secret.txt
  safeJoin:   read "db-password=hunter2\n"
  OpenInRoot: rejected: openat avatar/secret.txt: path escapes from parent
```

The string `work/uploads/avatar/secret.txt` passes every lexical test — it really is "under" `work/uploads`. The kernel follows the link anyway. `OpenInRoot` refuses both escapes and still serves the honest request. Note that `safeJoin` "handled" `../` by silently clamping it to the root, which is its own small lie: the caller asked for a file that does not exist there.

Case 2: An archive extractor that cannot write outside its output directory

Archive extraction is the textbook traversal bug ("zip slip"): entries named `../x`, absolute names, and — the one most hand-rolled guards miss — a symlink entry followed by a file entry *through* that symlink. The extractor below never constructs a host path; every write goes through the `Root`.

```bash
mkdir -p case2 && cd case2
```

```go
// untar.go
package main

import (
    "archive/tar"
    "bytes"
    "errors"
    "fmt"
    "io"
    "io/fs"
    "os"
)

// hostileTar builds an archive the way an attacker would.
func hostileTar() *bytes.Buffer {
    var buf bytes.Buffer
    tw := tar.NewWriter(&buf)
    add := func(h *tar.Header, body string) {
        h.Size = int64(len(body))
        if h.Mode == 0 {
            h.Mode = 0o644
        }
        if err := tw.WriteHeader(h); err != nil {
            panic(err)
        }
        io.WriteString(tw, body)
    }
    add(&tar.Header{Name: "docs/", Typeflag: tar.TypeDir, Mode: 0o755}, "")
    add(&tar.Header{Name: "docs/readme.txt", Typeflag: tar.TypeReg}, "fine\n")
```

```go
    add(&tar.Header{Name: "../escape.txt", Typeflag: tar.TypeReg}, "dotdot\n")
    add(&tar.Header{Name: "/etc/evil.conf", Typeflag: tar.TypeReg}, "absolute\n")
    add(&tar.Header{Name: "link", Typeflag: tar.TypeSymlink, Linkname: "../outside"}, "")
    add(&tar.Header{Name: "link/pwned.txt", Typeflag: tar.TypeReg}, "via symlink\n")
    tw.Close()
    return &buf
}

// extract writes every entry through root. It never builds a host path.
func extract(root *os.Root, r io.Reader) error {
    tr := tar.NewReader(r)
    var errs []error
    for {
        h, err := tr.Next()
        if err == io.EOF {
            break
        }
        if err != nil && !errors.Is(err, tar.ErrInsecurePath) {
            return err
        }
        if err := extractOne(root, h, tr); err != nil {
            fmt.Printf("  skip %-16s %v\n", h.Name, err)
            errs = append(errs, err)
            continue
        }
        fmt.Printf("  ok   %s\n", h.Name)
    }
    return errors.Join(errs...)
}

func extractOne(root *os.Root, h *tar.Header, body io.Reader) error {
    mode := fs.FileMode(h.Mode).Perm()
    switch h.Typeflag {
    case tar.TypeDir:
        return root.MkdirAll(h.Name, mode|0o700)
    case tar.TypeSymlink:
        return root.Symlink(h.Linkname, h.Name)
    case tar.TypeReg:
        f, err := root.OpenFile(h.Name, os.O_WRONLY|os.O_CREATE|os.O_EXCL, mode)
        if err != nil {
            return err
        }
        _, err = io.Copy(f, io.LimitReader(body, 1<<20))
        return errors.Join(err, f.Close())
    default:
        return fmt.Errorf("unsupported type %q", h.Typeflag)
    }
```

```go
}

func main() {
    must(os.MkdirAll("out", 0o755))
    must(os.MkdirAll("outside", 0o755))

    root, err := os.OpenRoot("out")
    must(err)
    defer root.Close()

    fmt.Println("extracting:")
    err = extract(root, hostileTar())
    fmt.Println("errors:", err != nil)

    _, err = os.Stat("outside/pwned.txt")
    fmt.Println("outside/pwned.txt exists:", err == nil)
    _, err = os.Stat("escape.txt")
    fmt.Println("escape.txt exists:", err == nil)
}

func must(err error) {
    if err != nil {
        panic(err)
    }
}
```

```bash
rm -rf out outside   # O_EXCL makes reruns fail otherwise
go run untar.go
find out outside | sort
```

Output:

```
extracting:
  ok   docs/
  ok   docs/readme.txt
  skip ../escape.txt    openat ../escape.txt: path escapes from parent
  skip /etc/evil.conf   openat /etc/evil.conf: path escapes from parent
  ok   link
  skip link/pwned.txt   openat link/pwned.txt: path escapes from parent
errors: true
outside/pwned.txt exists: false
escape.txt exists: false
out
out/docs
out/docs/readme.txt
out/link
outside
```

Things worth copying from this listing:

- `O_CREATE|O_EXCL` refuses to overwrite an existing entry — including a link the archive planted earlier under the same name.
- `io.LimitReader` caps each entry; traversal is not the only hostile property of an archive.
- `archive/tar` only returns `tar.ErrInsecurePath` from `Next` when you run with `GODEBUG=tarinsecurepath=0`; the default still hands you the header. The extractor tolerates both, because the `Root` is the real guard, not the header check.
- `out/link` was created and points outside. `Root` stopped every *traversal* of it, but it did not stop its *creation*. See "The trap".

Case 3: Serving static files — `http.Dir` versus `root.FS()`

`http.Dir` rejects `..` in URLs but follows any symlink on disk. If a deploy step, a CMS, or a user can create links under the web root, `http.Dir` will serve whatever they point at. `http.FileServerFS(root.FS())` serves the same tree with `Root` semantics.

```bash
mkdir -p case3 && cd case3
```

```go
// serve.go
package main

import (
    "fmt"
    "io"
    "net/http"
    "net/http/httptest"
    "os"
)

func main() {
    must(os.MkdirAll("site", 0o755))
    must(os.MkdirAll("private", 0o700))
    must(os.WriteFile("site/index.html", []byte("<h1>home</h1>\n"), 0o644))
    must(os.WriteFile("private/id_ed25519", []byte("-----BEGIN OPENSSH PRIVATE KEY-----\n"),
    _ = os.Remove("site/assets")
    must(os.Symlink("../private", "site/assets")) // planted by a compromised deploy step

    root, err := os.OpenRoot("site")
    must(err)
    defer root.Close()

    handlers := []struct {
        label string
        h     http.Handler
    }{
```

```go
        {"http.Dir ", http.FileServer(http.Dir("site"))},
        {"root.FS()", http.FileServerFS(root.FS())},
    }
    for _, tc := range handlers {
        srv := httptest.NewServer(tc.h)
        for _, path := range []string{"/index.html", "/assets/id_ed25519"} {
            resp, err := http.Get(srv.URL + path)
            must(err)
            body, _ := io.ReadAll(resp.Body)
            resp.Body.Close()
            fmt.Printf("%s %-20s %d %q\n", tc.label, path, resp.StatusCode, firstLine(body))
        }
        srv.Close()
    }
}

func firstLine(b []byte) string {
    for i, c := range b {
        if c == '\n' {
            return string(b[:i])
        }
    }
    return string(b)
}

func must(err error) {
    if err != nil {
        panic(err)
    }
}
```

```go
go run serve.go
```

Output:

```
http.Dir  /index.html          200 "<h1>home</h1>"
http.Dir  /assets/id_ed25519   200 "-----BEGIN OPENSSH PRIVATE KEY-----"
root.FS() /index.html          200 "<h1>home</h1>"
root.FS() /assets/id_ed25519   500 "500 Internal Server Error"
```

The escape becomes a `500`, not a `404`: the file server maps only `fs.ErrNotExist` and `fs.ErrPermission` to friendly codes, and "path escapes from parent" is neither. That is acceptable for a static site — and a useful alert signal. If you want a `404`, wrap the `fs.FS` and translate errors yourself; do not "fix" it by going back to `http.Dir`.

Case 4: Per-tenant roots that survive a rename

A `Root` can open a narrower `Root`. Give each tenant its own and cross-tenant access becomes impossible by construction rather than by review. Because the root is a descriptor, an operator moving the data directory mid-flight does not redirect writes to whatever now sits at the old path.

```
mkdir -p case4 && cd case4
```

```go
// tenants.go
package main

import (
    "fmt"
    "os"
)

func main() {
    must(os.MkdirAll("data/tenants/acme", 0o755))
    must(os.MkdirAll("data/tenants/globex", 0o755))

    data, err := os.OpenRoot("data")
    must(err)
    defer data.Close()

    // Narrow the jail per tenant: a Root opened from a Root.
    acme, err := data.OpenRoot("tenants/acme")
    must(err)
    defer acme.Close()
    fmt.Println("acme root name:", acme.Name())

    must(acme.WriteFile("invoice-001.json", []byte(`{"total":42}`), 0o644))
    _, err = acme.ReadFile("../globex/invoice-001.json")
    fmt.Println("cross-tenant read:", err)

    // An operator renames the data directory while the service runs.
    must(os.Rename("data", "data.old"))
    must(acme.WriteFile("invoice-002.json", []byte(`{"total":7}`), 0o644))

    entries, err := os.ReadDir("data.old/tenants/acme")
    must(err)
    for _, e := range entries {
        fmt.Println("on disk:", "data.old/tenants/acme/"+e.Name())
    }
}

func must(err error) {
    if err != nil {
```

```
        panic(err)
    }
}
```

```
rm -rf data data.old   # the program renames data -> data.old
go run tenants.go
```

Output:

```
acme root name: data/tenants/acme
cross-tenant read: openat ../globex/invoice-001.json: path escapes from parent
on disk: data.old/tenants/acme/invoice-001.json
on disk: data.old/tenants/acme/invoice-002.json
```

`Name()` reports the path the root was opened with; after the rename that string is stale, but the descriptor is not — `invoice-002.json` landed in the moved directory. Log `Name()` for humans; never re-open by it.

The trap

1. Treating `Root` as a sandbox. `os.Root` stops *path traversal by things an ordinary user can create* — `..` and symlinks. Its documentation is explicit that it does **not** stop traversal across bind mounts or other mount points, does not block `/proc` magic files, and does not stop opening device nodes inside the root. If the threat includes a process with mount privileges, you need namespaces, a container, or a separate user — not a different file API.

2. Letting hostile symlinks outlive the extraction. Case 2 left `out/link -> ../outside` on disk. Your `Root` will never follow it, but the next program that runs `cp -r out/ …` or a plain `os.Open` will. Validate link targets *before* creating them:

```
// linkcheck.go
package main

import (
    "fmt"
    "path/filepath"
)

// localLink reports whether a symlink named name with target target
// would stay inside the extraction root when followed.
func localLink(name, target string) bool {
    if filepath.IsAbs(target) {
        return false
    }
    return filepath.IsLocal(filepath.Join(filepath.Dir(name), target))
}
```

```go
func main() {
    for _, l := range [][2]string{
        {"link", "../outside"},
        {"docs/latest", "v2"},
        {"docs/up", ".."},
        {"docs/abs", "/etc"},
        {"a/b/c", "../../d"},
    } {
        fmt.Printf("%-12s -> %-11s local=%v\n", l[0], l[1], localLink(l[0], l[1]))
    }
}
```

```go
go run linkcheck.go
```

Output:

```
link         -> ../outside  local=false
docs/latest  -> v2          local=true
docs/up      -> ..          local=true
docs/abs     -> /etc        local=false
a/b/c        -> ../../d     local=true
```

The lexical check is fine *here* because it runs on archive metadata, not on the live filesystem, and `Root` still guards the actual writes. Belt and braces: `Root` for every operation, `IsLocal` for what you agree to leave behind.

3. Matching the error string. The "escapes" error is an unexported sentinel wrapped in `*os.PathError`; there is no `errors.Is` target for it. Treat **any** error from a `Root` method on untrusted input as a rejection. Do not string-match `"path escapes"` to decide whether to retry somewhere else.

4. Using `Root` where the user owns the path. A CLI flag like `-o /tmp/report.csv` is a *trusted* path chosen by the operator. Jailing it inside the working directory is a bug, not hardening. `Root` is for names that arrive from outside the trust boundary.

5. Assuming every method is race-free. On Unix, `Root.Chmod`, `Root.Chown`, and `Root.Chtimes` can act on a symlink if the target is swapped between check and use (documented limitation). Do metadata changes on a file you just opened, or on entries you created yourself in the same operation.

6. Forgetting the cost. Resolving component-by-component is more expensive than one `open(2)`, and many `..` segments make it worse. Reject absurd names early (length, depth, `filepath.IsLocal`) — for cost and clarity, not for safety; safety comes from the `Root`.

The boring rule

> Open the trusted directory **once** with `os.OpenRoot`, keep the `*os.Root` for the life of the component, and route **every** operation on an untrusted name through it — never `filepath.Join` a fixed directory with external input, never re-open by `Root.Name()`, and treat every `Root` error as "no".

Try this

1. In Case 1, replace the symlink with a hard link (`os.Link("../private/secret.txt", "work/uploads/avatar.txt")`). Run it. Explain why `Root` allows the read (hint: a hard link is a second *name* for the same inode, not a path that points elsewhere — `Root` guards name resolution, and Linux's `fs.protected_hardlinks` is what limits who may create such links).
2. Extend Case 2 to call `localLink` before `root.Symlink` and skip escaping links. Confirm `out/link` no longer exists after extraction.
3. Rerun Case 2 with `GODEBUG=tarinsecurepath=0 go run untar.go` and add a log line when `Next` returns `tar.ErrInsecurePath`. Which entries does the header check catch, and which one (hint: the symlink pair) does only `Root` catch?
4. Wrap `root.FS()` in Case 3 with your own `fs.FS` whose `Open` converts any non-`ErrNotExist` error into `fs.ErrNotExist`, so escapes return `404`. Log the original error.
5. In Case 4, start a goroutine that writes to `acme` in a loop while `main` renames `data` back and forth. Run with `-race`. `Root` methods are documented safe for concurrent use; verify the race detector agrees.
6. Grep your own services for `filepath.Join(` followed by a request field, form value, or archive header name. Convert one to `os.Root` and keep its old test cases passing.

Sources

- Go blog, "Traversal-resistant file APIs" (Damien Neil, 2025): https://go.dev/blog/osroot
- Package `os`, type `Root` and `OpenInRoot`: https://pkg.go.dev/os#Root
- Go 1.24 release notes (introduces `os.Root`): https://go.dev/doc/go1.24#directory-limited-filesystem-access
- Go 1.25 release notes (adds `Chmod`, `Chown`, `Chtimes`, `Lchown`, `Link`, `MkdirAll`, `ReadFile`, `Readlink`, `RemoveAll`, `Rename`, `Symlink`, `WriteFile`; `io/fs.ReadLinkFS`): https://go.dev/doc/go1.25#ospkgos
- Package `archive/tar`, `ErrInsecurePath` and `GODEBUG=tarinsecurepath`: https://pkg.go.dev/archive/tar#Reader.Next
- Package `path/filepath`, `IsLocal`: https://pkg.go.dev/path/filepath#IsLocal

CSRF Without Tokens: net/http CrossOriginProtection

CSRF Without Tokens: net/http CrossOriginProtection

Thesis

Cross-Site Request Forgery is a confused-deputy attack: a page on `evil.example` makes the victim's browser send a state-changing request to your site, and the browser attaches the victim's cookies. For twenty years the answer was a CSRF token in every form, which meant every form, every `fetch`, and every template had to carry a secret correctly. Hand-rolled versions of that scheme still turn up in code review, model-written ones included. They often get the hard parts right (random tokens, constant-time compare) and the boring parts wrong (an `Origin` check done with `strings.HasPrefix`, logic spread across handlers).

Since **Go 1.25** the standard library ships the modern answer: **`http.CrossOriginProtection`**. It is middleware that rejects *non-safe, cross-origin browser requests* using the **`Sec-Fetch-Site`** Fetch-metadata header that every major browser has sent since 2023, and falls back to comparing `Origin` with `Host`. It needs no tokens, no cookies, and no template changes. On Go 1.27 a cookie-authenticated Go service should start from this middleware and justify any hand-rolled scheme on top of it.

Mental model

The browser already knows where a request came from and, for HTTPS and `localhost` targets, it tells the server in `Sec-Fetch-Site`:

`Sec-Fetch-Site`	Meaning	`CrossOriginProtection`
`same-origin`	Your own page, same scheme + host + port	allow
`none`	User-initiated: typed URL, bookmark	allow
`same-site`	Sibling origin, e.g. `blog.example.com` → `app.example.com`	**reject**
`cross-site`	Anyone else	**reject**

`Check` uses that header first. It ignores safe methods and falls back to `Origin` for older browsers. The flow below is the actual `Check` from `net/http/csrf.go` in Go 1.27.2:

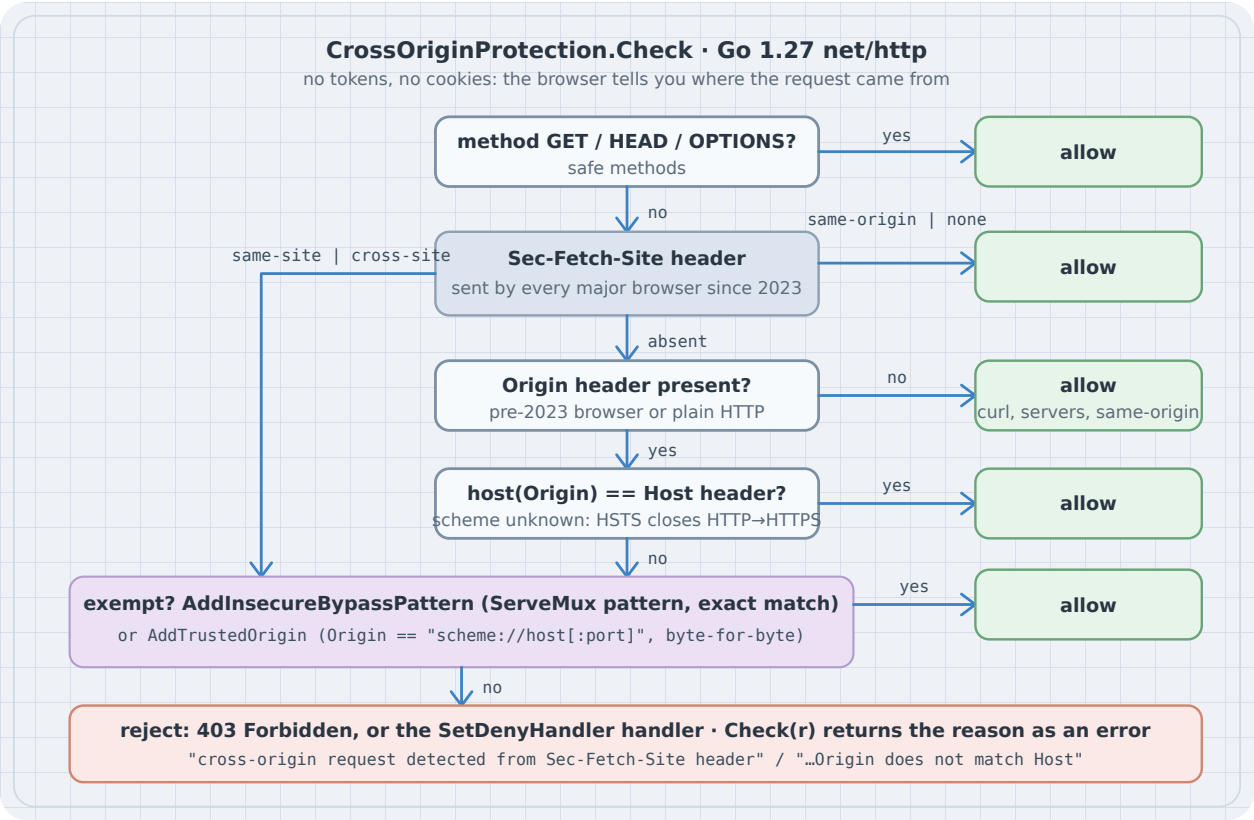


Figure 1: CrossOriginProtection.Check: safe methods pass; Sec-Fetch-Site decides for modern browsers; Origin vs Host is the fallback; bypass patterns and trusted origins are the only exemptions

Four properties to keep in your head:

1. **It protects unsafe methods only.** `GET`, `HEAD`, and `OPTIONS` always pass, so a handler that changes state on `GET` is outside the protection by design.
2. **Same-site is not same-origin.** `SameSite=Lax` cookies stop *cross-site* requests but not a compromised marketing subdomain posting to your app. `Sec-Fetch-Site: same-site` is rejected.
3. **No headers means "not a browser".** Requests carrying neither `Sec-Fetch-Site` nor `Origin` come from curl, servers, or very old browsers, and they pass. CSRF needs a browser's ambient cookies, so this is correct. It also means CSRF protection is **not authentication**.
4. **Exemptions are explicit and narrow.** `AddTrustedOrigin` takes an exact `scheme://host[:port]` value. `AddInsecureBypassPattern` takes a `ServeMux` pattern and matches only requests that hit it directly. The deny path is a 403, or your handler via `SetDenyHandler`.

The API is small:

```go
cop := http.NewCrossOriginProtection()          // zero value is also valid
err := cop.AddTrustedOrigin("https://admin.example.com")
cop.AddInsecureBypassPattern("POST /sso/acs")   // panics on invalid/conflicting pattern
cop.SetDenyHandler(h)                           // default: 403 + reason text
err = cop.Check(r)                              // nil = allowed; side-effect free
handler := cop.Handler(mux)                     // wrap the whole router
```

All of the configuration methods are safe to call concurrently with request handling and apply to future requests.

Worked examples

Every listing was run with **Go 1.27.2 on Linux (amd64)**. The output blocks are pasted from those runs. Each case is its own module so `go run .` sees one `main` package:

```bash
mkdir -p csrf/case1 && cd csrf/case1
go mod init example.com/csrf/case1
```

Case 1: A hand-rolled origin check versus the standard library

This is the kind of middleware that shows up in review. It accepts a `POST` when `Origin` "starts with" the site's origin. We feed it, and the stdlib middleware, the header combinations real browsers produce.

```go
// main.go - the same POST handler behind a hand-rolled origin check and
// behind net/http.CrossOriginProtection, fed the requests browsers send.
package main

import (
    "fmt"
    "net/http"
    "net/http/httptest"
    "strings"
)

// transfer is a state-changing handler: it must only run for requests
// the user's own pages (or non-browser clients) made.
func transfer(w http.ResponseWriter, r *http.Request) {
    fmt.Fprintln(w, "transferred")
}

// handRolled is the kind of check that shows up in reviews: it compares
// Origin to the expected origin "as strings".
func handRolled(next http.Handler) http.Handler {
```

```
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        if r.Method == http.MethodPost {
            o := r.Header.Get("Origin")
            if o != "" && !strings.HasPrefix(o, "https://bank.example") {
                http.Error(w, "bad origin", http.StatusForbidden)
                return
            }
        }
        next.ServeHTTP(w, r)
    })
}

type probe struct {
    name, method, site, origin string
}

func main() {
    stdlib := http.NewCrossOriginProtection().Handler(http.HandlerFunc(transfer))
    hand := handRolled(http.HandlerFunc(transfer))

    probes := []probe{
        {"own form (modern browser)", "POST", "same-origin", "https://bank.example"},
        {"evil.example form", "POST", "cross-site", "https://evil.example"},
        {"look-alike domain", "POST", "cross-site", "https://bank.example.evil.net"},
        {"sibling subdomain", "POST", "same-site", "https://blog.bank.example"},
        {"bookmark / typed URL", "POST", "none", ""},
        {"opaque origin (sandboxed iframe)", "POST", "cross-site", "null"},
        {"old browser, own page", "POST", "", "https://bank.example"},
        {"old browser, evil page", "POST", "", "https://evil.example"},
        {"curl / server-to-server", "POST", "", ""},
        {"cross-site GET", "GET", "cross-site", "https://evil.example"},
    }

    fmt.Printf("%-34s %-11s %-10s %s\n", "request", "Sec-Fetch", "hand", "stdlib")
    for _, p := range probes {
        fmt.Printf("%-34s %-11s %-10d %d\n", p.name, or(p.site, "-"),
            code(hand, p), code(stdlib, p))
    }
}

func code(h http.Handler, p probe) int {
    req := httptest.NewRequest(p.method, "https://bank.example/transfer", nil)
    if p.site != "" {
        req.Header.Set("Sec-Fetch-Site", p.site)
    }
    if p.origin != "" {
```

```go
        req.Header.Set("Origin", p.origin)
    }
    rec := httptest.NewRecorder()
    h.ServeHTTP(rec, req)
    return rec.Code
}

func or(s, d string) string {
    if s == "" {
        return d
    }
    return s
}
```

```go
go run .
```

Output:

```
request                             Sec-Fetch   hand       stdlib
own form (modern browser)           same-origin 200        200
evil.example form                   cross-site  403        403
look-alike domain                   cross-site  200        403
sibling subdomain                   same-site   403        403
bookmark / typed URL                none        200        200
opaque origin (sandboxed iframe)    cross-site  403        403
old browser, own page               -           200        200
old browser, evil page              -           403        403
curl / server-to-server             -           200        200
cross-site GET                      cross-site  200        200
```

Read the table row by row:

- **Look-alike domain.** `https://bank.example.evil.net` starts with `https://bank.example`, so the hand-rolled check waves it through. Stringly-typed origin comparison is the classic bug. `CrossOriginProtection` never compares prefixes: `Sec-Fetch-Site: cross-site` is rejected outright.
- **Sibling subdomain.** Both reject it here, but only by accident in the hand-rolled version. `SameSite` cookies would *not* have stopped it.
- **`none`.** A bookmark or typed URL is user-initiated, so it is allowed.
- **Opaque origin.** Sandboxed iframes and some redirects send `Origin: null`. Modern browsers also mark it `cross-site`, and even the `Origin`-only fallback rejects `null` because its host never equals `Host`.
- **Old browser rows.** With no `Sec-Fetch-Site`, the middleware compares the host of `Origin` with the `Host` header.
- **curl row.** No browser headers, no ambient cookies, no CSRF. The request is allowed, and your authentication layer still has to do its job.
- **Cross-site GET.** Allowed. This is the trap later in the chapter.

Case 2: Trusted origins are exact strings

A separate admin SPA on `https://admin.bank.example` posts to `https://api.bank.example`. That is a `same-site` request, so it is rejected unless you trust the origin explicitly.

```
mkdir -p ../case2 && cd ../case2 && go mod init example.com/csrf/case2
```

```go
// main.go - trusted origins: what AddTrustedOrigin accepts, and how
// exactly it has to match the Origin header the browser sends.
package main

import (
    "fmt"
    "net/http"
    "net/http/httptest"
)

func main() {
    cop := http.NewCrossOriginProtection()

    // The admin SPA lives on its own origin and posts to the API.
    for _, o := range []string{
        "https://admin.bank.example",
        "https://admin.bank.example/", // trailing slash = a path
        "admin.bank.example",          // no scheme
        "https://partner.example:8443",
    } {
        if err := cop.AddTrustedOrigin(o); err != nil {
            fmt.Println("AddTrustedOrigin:", err)
        } else {
            fmt.Printf("AddTrustedOrigin: %q ok\n", o)
        }
    }
    fmt.Println()

    h := cop.Handler(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        fmt.Fprintln(w, "ok")
    }))

    for _, origin := range []string{
        "https://admin.bank.example",
        "https://ADMIN.bank.example",
        "http://admin.bank.example",
        "https://admin.bank.example:443",
        "https://partner.example:8443",
        "https://partner.example",
```

```go
    } {
        req := httptest.NewRequest("POST", "https://api.bank.example/v1/payees", nil)
        req.Header.Set("Sec-Fetch-Site", "same-site")
        req.Header.Set("Origin", origin)
        rec := httptest.NewRecorder()
        h.ServeHTTP(rec, req)
        fmt.Printf("%-32s -> %d\n", origin, rec.Code)
    }
}
```

```go
go run .
```

Output:

```
AddTrustedOrigin: "https://admin.bank.example" ok
AddTrustedOrigin: invalid origin "https://admin.bank.example/": path, query, and fragment are r
AddTrustedOrigin: invalid origin "admin.bank.example": scheme is required
AddTrustedOrigin: "https://partner.example:8443" ok

https://admin.bank.example       -> 200
https://ADMIN.bank.example       -> 403
http://admin.bank.example        -> 403
https://admin.bank.example:443   -> 403
https://partner.example:8443     -> 200
https://partner.example          -> 403
```

`AddTrustedOrigin` validates its input. It requires a scheme and a host, and rejects any path, query, or fragment, including a trailing `/`. Matching is a map lookup on the raw `Origin` header, so the value must be the browser's serialization: lowercase host, scheme included, default port omitted. Browsers never send `https://admin.bank.example:443` or an uppercase host, so those rows are only here to show that nothing is normalized for you. Configure origins exactly as a browser prints `window.location.origin`.

Case 3: Bypass patterns for the flows that are cross-site by design

Some legitimate flows are cross-site browser `POST`s by construction. In a SAML HTTP-POST binding, or an OIDC `response_mode=form_post` callback, the identity provider's page auto-submits a form to you. Those routes need an exemption. Webhooks usually do **not**, because they come from servers and carry no browser headers.

```go
mkdir -p ../case3 && cd ../case3 && go mod init example.com/csrf/case3
```

```go
// main.go - insecure bypass patterns: which requests they exempt, and
// which near-misses they deliberately do not.
package main
```

```go
import (
    "fmt"
    "net/http"
    "net/http/httptest"
)

func main() {
    mux := http.NewServeMux()
    ok := func(w http.ResponseWriter, r *http.Request) { fmt.Fprintln(w, "ok") }
    mux.HandleFunc("POST /sso/acs", ok)                    // SAML HTTP-POST binding
    mux.HandleFunc("POST /oauth/{provider}/callback", ok) // OIDC response_mode=form_post
    mux.HandleFunc("POST /account/delete", ok)
    mux.HandleFunc("POST /webhooks/github", ok)

    cop := http.NewCrossOriginProtection()
    // Cross-site browser POSTs that are legitimate by design: the IdP's
    // page auto-submits a form to us. Nothing else gets an exemption.
    cop.AddInsecureBypassPattern("POST /sso/acs")
    cop.AddInsecureBypassPattern("POST /oauth/{provider}/callback")

    h := cop.Handler(mux)
    send := func(path string, browser bool) {
        req := httptest.NewRequest("POST", "https://bank.example"+path, nil)
        if browser {
            req.Header.Set("Sec-Fetch-Site", "cross-site")
            req.Header.Set("Origin", "https://idp.example")
        }
        rec := httptest.NewRecorder()
        h.ServeHTTP(rec, req)
        kind := "server"
        if browser {
            kind = "cross-site browser"
        }
        fmt.Printf("%-18s POST %-24s -> %d\n", kind, path, rec.Code)
    }
    send("/sso/acs", true)
    send("/sso/acs/", true)              // not matched by the pattern
    send("//sso/acs", true)              // ServeMux would clean + redirect
    send("/sso/../account/delete", true) // cleaned path is not a bypass
    send("/oauth/google/callback", true)
    send("/account/delete", true)
    send("/webhooks/github", false) // no bypass needed: not a browser

    // Patterns follow ServeMux rules, including conflict detection.
    defer func() { fmt.Println("\npanic:", recover()) }()
    cop.AddInsecureBypassPattern("POST /oauth/{name}/callback")
}
```

```go
go run .
```

Output:

```
cross-site browser POST /sso/acs                -> 200
cross-site browser POST /sso/acs/               -> 403
cross-site browser POST //sso/acs               -> 403
cross-site browser POST /sso/../account/delete  -> 403
cross-site browser POST /oauth/google/callback  -> 200
cross-site browser POST /account/delete         -> 403
server             POST /webhooks/github        -> 200

panic: pattern "POST /oauth/{name}/callback" (registered at /usr/local/go/src/net/http/csrf.go:
POST /oauth/{name}/callback matches the same requests as POST /oauth/{provider}/callback
```

A bypass pattern exempts only requests that match it **directly**. `/sso/acs/`, `//sso/acs`, and `/sso/../account/delete` would be cleaned or redirected by `ServeMux`, so they are rejected instead of slipping through. The exemption is checked by a private `ServeMux` inside the middleware, so pattern syntax, wildcards, and conflict detection behave exactly like your router. Registering an overlapping pattern panics at startup, which is where you want to find out. The server-side webhook passes without any exemption. Verify its signature in the handler, because CSRF protection says nothing about who sent it.

Case 4: Production wiring behind a proxy, with a deny handler

In production the app sits behind a TLS-terminating proxy and sees `Host: app:8080`. Modern browsers are unaffected because `Sec-Fetch-Site` decides. But the `Origin == Host` fallback for old browsers would now reject your own pages. Fix it by naming the public origin as trusted. The deny handler returns JSON and logs the reason, which you get from calling `Check` again (it has no side effects).

```go
mkdir -p ../case4 && cd ../case4 && go mod init example.com/csrf/case4
```

```go
// main.go - production wiring: one CrossOriginProtection in front of the
// whole mux, a deny handler that logs why, and real requests over TCP.
package main

import (
    "fmt"
    "io"
    "log/slog"
    "net/http"
    "net/http/httptest"
    "os"
    "strings"
)
```

```
func newServer(log *slog.Logger) http.Handler {
    mux := http.NewServeMux()
    mux.HandleFunc("GET /settings", func(w http.ResponseWriter, r *http.Request) {
        fmt.Fprintln(w, `<form method="post" action="/settings/email">...</form>`)
    })
    mux.HandleFunc("POST /settings/email", func(w http.ResponseWriter, r *http.Request) {
        fmt.Fprintln(w, "email updated")
    })

    cop := http.NewCrossOriginProtection()
    // Behind the TLS-terminating proxy the app sees Host: app:8080, so
    // old browsers' Origin never equals Host. Name the public origin.
    if err := cop.AddTrustedOrigin("https://app.example"); err != nil {
        panic(err)
    }
    cop.SetDenyHandler(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        // Check is side-effect free; call it again to get the reason.
        log.Warn("csrf rejected",
            "method", r.Method, "path", r.URL.Path,
            "origin", r.Header.Get("Origin"),
            "sec_fetch_site", r.Header.Get("Sec-Fetch-Site"),
            "reason", cop.Check(r))
        w.Header().Set("Content-Type", "application/json")
        w.WriteHeader(http.StatusForbidden)
        io.WriteString(w, `{"error":"cross_origin_request"}`+"\n")
    }))
    return cop.Handler(mux)
}

func main() {
    log := slog.New(slog.NewTextHandler(os.Stdout, &slog.HandlerOptions{
        ReplaceAttr: func(_ []string, a slog.Attr) slog.Attr {
            if a.Key == slog.TimeKey {
                return slog.Attr{} // stable output
            }
            return a
        },
    }))
    srv := httptest.NewServer(newServer(log)) // a real listener on 127.0.0.1
    defer srv.Close()

    do := func(label, origin, site string) {
        req, _ := http.NewRequest("POST", srv.URL+"/settings/email",
            strings.NewReader("email=x@example.com"))
        req.Header.Set("Content-Type", "application/x-www-form-urlencoded")
        if origin != "" {
```

```
            req.Header.Set("Origin", origin)
        }
        if site != "" {
            req.Header.Set("Sec-Fetch-Site", site)
        }
        resp, err := http.DefaultClient.Do(req)
        if err != nil {
            panic(err)
        }
        body, _ := io.ReadAll(resp.Body)
        resp.Body.Close()
        fmt.Printf("%-26s %d %s", label, resp.StatusCode, body)
    }
    do("Go http.Client (no hdrs)", "", "")
    do("browser, same-origin", "https://app.example", "same-origin")
    do("old browser via proxy", "https://app.example", "")
    do("old browser, evil page", "https://evil.example", "")
    do("browser, cross-site", "https://evil.example", "cross-site")
}
```

```
go run .
```

Output:

```
Go http.Client (no hdrs)   200 email updated
browser, same-origin       200 email updated
old browser via proxy      200 email updated
level=WARN msg="csrf rejected" method=POST path=/settings/email origin=https://evil.example sec
old browser, evil page     403 {"error":"cross_origin_request"}
level=WARN msg="csrf rejected" method=POST path=/settings/email origin=https://evil.example sec
browser, cross-site        403 {"error":"cross_origin_request"}
```

`httptest.NewServer` is a real listener on `127.0.0.1`, so these are real HTTP/1.1 requests through `http.DefaultClient`. Note the first row: a Go client sends neither header and is allowed. The two error strings are the two rejection reasons `Check` can return. They are worth logging, because the "browser is out of date" one tells you the request came through the fallback path.

Case 5: Pin the contract in a test

Middleware is easy to drop in a refactor: a new router, a second `http.Server`, a handler mounted outside the wrapped mux. A table test on the exported router keeps the contract visible.

```
mkdir -p ../case5 && cd ../case5 && go mod init example.com/csrf/case5
```

```go
// app.go - the application's router, CSRF protection included.
package app

import "net/http"

func Routes() http.Handler {
    mux := http.NewServeMux()
    mux.HandleFunc("GET /profile", func(w http.ResponseWriter, r *http.Request) {})
    mux.HandleFunc("POST /profile", func(w http.ResponseWriter, r *http.Request) {})
    mux.HandleFunc("DELETE /profile", func(w http.ResponseWriter, r *http.Request) {})
    mux.HandleFunc("POST /sso/acs", func(w http.ResponseWriter, r *http.Request) {})

    cop := http.NewCrossOriginProtection()
    cop.AddInsecureBypassPattern("POST /sso/acs")
    return cop.Handler(mux)
}
```

```go
// app_test.go - pin the CSRF contract so a refactor cannot drop it.
package app

import (
    "net/http"
    "net/http/httptest"
    "testing"
)

func TestCrossOriginContract(t *testing.T) {
    h := Routes()
    tests := []struct {
        method, path, site string
        want               int
    }{
        {"POST", "/profile", "same-origin", http.StatusOK},
        {"POST", "/profile", "cross-site", http.StatusForbidden},
        {"POST", "/profile", "same-site", http.StatusForbidden},
        {"DELETE", "/profile", "cross-site", http.StatusForbidden},
        {"GET", "/profile", "cross-site", http.StatusOK},
        {"POST", "/sso/acs", "cross-site", http.StatusOK}, // the one exemption
    }
    for _, tt := range tests {
        t.Run(tt.method+" "+tt.path+" "+tt.site, func(t *testing.T) {
            req := httptest.NewRequest(tt.method, "https://app.example"+tt.path, nil)
            req.Header.Set("Sec-Fetch-Site", tt.site)
            rec := httptest.NewRecorder()
            h.ServeHTTP(rec, req)
            if rec.Code != tt.want {
```

```go
                t.Fatalf("got %d, want %d", rec.Code, tt.want)
            }
        })
    }
}
```

```go
go test -count=1 -v .
```

Output:

```
=== RUN   TestCrossOriginContract
=== RUN   TestCrossOriginContract/POST_/profile_same-origin
=== RUN   TestCrossOriginContract/POST_/profile_cross-site
=== RUN   TestCrossOriginContract/POST_/profile_same-site
=== RUN   TestCrossOriginContract/DELETE_/profile_cross-site
=== RUN   TestCrossOriginContract/GET_/profile_cross-site
=== RUN   TestCrossOriginContract/POST_/sso/acs_cross-site
--- PASS: TestCrossOriginContract (0.00s)
    --- PASS: TestCrossOriginContract/POST_/profile_same-origin (0.00s)
    --- PASS: TestCrossOriginContract/POST_/profile_cross-site (0.00s)
    --- PASS: TestCrossOriginContract/POST_/profile_same-site (0.00s)
    --- PASS: TestCrossOriginContract/DELETE_/profile_cross-site (0.00s)
    --- PASS: TestCrossOriginContract/GET_/profile_cross-site (0.00s)
    --- PASS: TestCrossOriginContract/POST_/sso/acs_cross-site (0.00s)
PASS
ok      example.com/csrf/case5  0.003s
```

The trap

`CrossOriginProtection` trusts the HTTP method. It assumes that safe methods do not change state, which HTTP has always required and many handlers still violate. A pattern without a method accepts `GET`, and a cross-site `<img src>` or link is enough to fire it.

```go
mkdir -p ../trap && cd ../trap && go mod init example.com/csrf/trap
```

```go
// main.go - CrossOriginProtection trusts the HTTP method. A handler that
// changes state on GET is outside its protection by design.
package main

import (
    "fmt"
    "net/http"
    "net/http/httptest"
)
```

```go
func main() {
    subscribed := true
    mux := http.NewServeMux()
    // No method in the pattern: GET, POST, anything reaches the handler.
    mux.HandleFunc("/newsletter/unsubscribe", func(w http.ResponseWriter, r *http.Request) {
        subscribed = false
        fmt.Fprintln(w, "unsubscribed")
    })
    h := http.NewCrossOriginProtection().Handler(mux)

    // evil.example embeds <img src="https://bank.example/newsletter/unsubscribe">
    req := httptest.NewRequest("GET", "https://bank.example/newsletter/unsubscribe", nil)
    req.Header.Set("Sec-Fetch-Site", "cross-site")
    rec := httptest.NewRecorder()
    h.ServeHTTP(rec, req)
    fmt.Println("cross-site GET:", rec.Code, "subscribed =", subscribed)

    // Fix: state changes only on unsafe methods, and say so in the pattern.
    subscribed = true
    mux2 := http.NewServeMux()
    mux2.HandleFunc("POST /newsletter/unsubscribe", func(w http.ResponseWriter, r *http.Requ
        subscribed = false
    })
    h2 := http.NewCrossOriginProtection().Handler(mux2)
    for _, m := range []string{"GET", "POST"} {
        req := httptest.NewRequest(m, "https://bank.example/newsletter/unsubscribe", nil)
        req.Header.Set("Sec-Fetch-Site", "cross-site")
        rec := httptest.NewRecorder()
        h2.ServeHTTP(rec, req)
        fmt.Printf("fixed, cross-site %-4s: %d subscribed = %v\n", m, rec.Code, subscribed)
    }
}
```

```go
go run .
```

Output:

```
cross-site GET: 200 subscribed = false
fixed, cross-site GET : 405 subscribed = true
fixed, cross-site POST: 403 subscribed = true
```

With the method in the pattern, a cross-site `GET` gets `405` from `ServeMux` and the cross-site `POST` gets `403` from the middleware. The same review applies to "unsubscribe" links in emails, `GET /logout`, and `?action=delete` query parameters. Give them a confirmation page that submits a `POST`.

Other edges to know about:

- **Plain HTTP and pre-2023 browsers.** `Sec-Fetch-Site` is only sent to trustworthy origins (HTTPS and `localhost`). The `Origin == Host` fallback cannot see the scheme, so an out-of-date browser is exposed to an HTTP→HTTPS request from a network attacker. HSTS closes that gap.
- **Header-stripping extensions.** An extension that removes `Origin` and `Sec-Fetch-Site` makes a browser request look like curl. That is a vulnerability in the extension, and no server-side scheme can detect it.
- **Mounting.** Wrap the **whole** mux (or the outermost handler) once. Protection added per route is protection someone forgets on the next route.
- **CORS is a different problem.** CORS controls who may *read* responses. CSRF controls who may cause *state changes*. `CrossOriginProtection` sets no CORS headers, and CORS configuration does not stop form posts.

The boring rule

Wrap the outermost handler of every cookie-authenticated Go service in `http.NewCrossOriginProtection().Hand` put an explicit method on every state-changing route, exempt only the SSO callbacks that are cross-site by design with exact bypass patterns, trust only exact browser-serialized origins, keep HSTS on, and delete the hand-rolled token code unless a written threat model still needs it.

Try this

1. Add a probe to Case 1 with `Sec-Fetch-Site: same-origin` and `Origin: https://evil.example`. Predict the result from the flow diagram before you run it, then explain why `Sec-Fetch-Site` wins.
2. In Case 4, remove `AddTrustedOrigin` and rerun. Which row changes, and why do modern browsers not care?
3. Register `cop.AddInsecureBypassPattern("/sso/")` instead of `"POST /sso/acs"`. Which requests are now exempt? Write a Case 5 test row that would have caught the widening.
4. Replace the deny handler in Case 4 with one that increments an `expvar` counter labelled by the `Check` error, and graph rejections during a deploy.
5. Grep a codebase you maintain for `HandleFunc("/` patterns without a method that write to a database. Each one is a cross-site `GET` waiting to happen.

Sources

- `net/http` package documentation, `CrossOriginProtection`: https://pkg.go.dev/net/http#CrossOriginProtection
- Go 1.25 release notes, `net/http` section (introduces `CrossOriginProtection`): https://go.dev/doc/go1.25#nethttppkgnethttp
- Go source, `src/net/http/csrf.go` (behaviour verified against Go 1.27.2): https://cs.opensource.google/go/go/+/refs/tags/go1.27.2:src/net/http/csrf.go
- Go release history (Go 1.27.2, released 2026-10-08): https://go.dev/doc/devel/release

- Filippo Valsorda, "Cross-Site Request Forgery" (the analysis and algorithm behind the proposal): https://words.filippo.io/csrf/
- W3C, Fetch Metadata Request Headers (`Sec-Fetch-Site`): https://www.w3.org/TR/fetch-metadata/
- MDN, `Sec-Fetch-Site`: https://developer.mozilla.org/en-US/docs/Web/HTTP/Reference/Headers/Sec-Fetch-Site
- OWASP Cross-Site Request Forgery Prevention Cheat Sheet (Fetch Metadata section): https://cheatsheetseries.owasp.org/cheatsheets/Cross-Site_Request_Forgery_Prevention_Cheat_Sheet.html

Error Boundaries: Wrap Inside, Translate at the Edge, Leak Nothing

Error Boundaries: Wrap Inside, Translate at the Edge, Leak Nothing

Thesis

Good Go error wrapping and safe Go error output pull in opposite directions. `fmt.Errorf("query %q: %w", q, err)` is exactly what you want in a log, and exactly what you must never send to a client: it carries SQL text, file paths, tenant names and sometimes credentials. You don't fix this with less context. You fix it by making one **boundary** the only place where errors leave the trust zone. That boundary translates every error into a type whose `Error()` method is safe by construction, and sends the full detail only to the log, joined to the response by a request ID.

Mental model

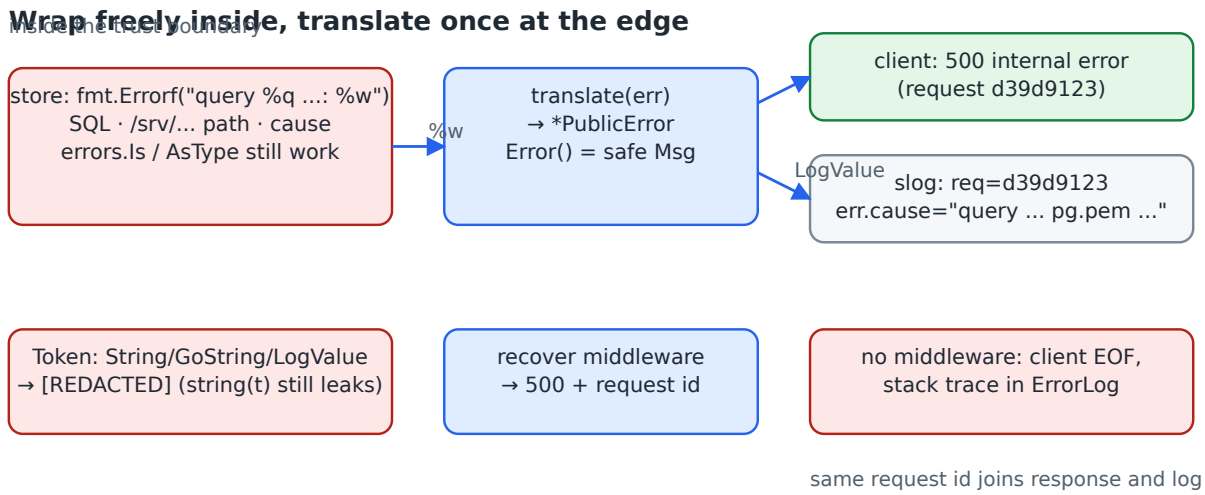


Figure 1: Inside the service errors are wrapped with full context; translate() turns them into a PublicError whose Error() is safe; the client sees status, safe message and request id while slog gets the cause

Each piece has a single job:

- **Inside:** wrap with `%w` and add context freely. `errors.Is`/`errors.AsType` keep working through the chain.

- **PublicError:** `Error()` returns only the client-safe message, so an accidental `http.Error(w, err.Error(), …)` is harmless. `Unwrap()` keeps the cause reachable for code inside the service, and `LogValue()` (`slog.LogValuer`) sends the cause to the log.
- **Secrets as types:** a `Token` type that formats itself as `[REDACTED]` through every `fmt` verb and through slog.
- **Panics:** a recover middleware, because the `net/http` default isn't a response at all.

Worked example

One program runs all five cases against an `httptest` server and prints what the client sees, what the application log records, and what `net/http` writes to its own error log.

```go
package main

import (
    "bytes"
    "context"
    "crypto/rand"
    "encoding/hex"
    "errors"
    "fmt"
    "io"
    "log"
    "log/slog"
    "net/http"
    "net/http/httptest"
    "os"
    "strings"
)

// ---- internal layer: rich, unsafe-to-show errors ----

var ErrNotFound = errors.New("not found")

type Token string

func (Token) String() string       { return "[REDACTED]" }
func (Token) GoString() string     { return `Token("[REDACTED]")` }
func (Token) LogValue() slog.Value { return slog.StringValue("[REDACTED]") }

func loadInvoice(id string, tok Token) error {
    const q = "SELECT * FROM invoices WHERE id = $1 AND tenant = 'acme'"
    if id == "404" {
        return fmt.Errorf("query %q (id=%s): %w", q, id, ErrNotFound)
    }
    cause := fmt.Errorf("open /srv/acme/secrets/pg.pem: permission denied")
```

```
    return fmt.Errorf("query %q with token %v: %w", q, tok, cause)
}

// ---- boundary type: Error() is safe; detail only via LogValue ----

type PublicError struct {
    Status int
    Msg    string // safe for clients
    err    error  // internal cause
}

func (e *PublicError) Error() string { return e.Msg }
func (e *PublicError) Unwrap() error { return e.err }
func (e *PublicError) LogValue() slog.Value {
    return slog.GroupValue(slog.Int("status", e.Status), slog.String("cause", e.err.Error()))
}

func translate(err error) *PublicError {
    if errors.Is(err, ErrNotFound) {
        return &PublicError{http.StatusNotFound, "invoice not found", err}
    }
    return &PublicError{http.StatusInternalServerError, "internal error", err}
}

func reqID() string { b := make([]byte, 4); rand.Read(b); return hex.EncodeToString(b) }

var logger *slog.Logger

// naive handler: what not to do
func naive(w http.ResponseWriter, r *http.Request) {
    if err := loadInvoice(r.URL.Query().Get("id"), "sk_live_9f8e7d"); err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
    }
}

// boundary handler
func safe(w http.ResponseWriter, r *http.Request) {
    if err := loadInvoice(r.URL.Query().Get("id"), "sk_live_9f8e7d"); err != nil {
        pe := translate(err)
        id := reqID()
        logger.ErrorContext(r.Context(), "load invoice", "req", id, "err", pe)
        http.Error(w, fmt.Sprintf("%s (request %s)", pe.Msg, id), pe.Status)
    }
}

func boom(w http.ResponseWriter, r *http.Request) {
```

```
    var m map[string]int
    m["x"] = 1 // panics
}

func recoverer(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        defer func() {
            if v := recover(); v != nil {
                if v == http.ErrAbortHandler {
                    panic(v)
                }
                id := reqID()
                logger.Error("panic", "req", id, "value", fmt.Sprint(v))
                http.Error(w, "internal error (request "+id+")", http.StatusInternalServerEr
            }
        }()
        next.ServeHTTP(w, r)
    })
}

func get(url string) {
    resp, err := http.Get(url)
    if err != nil {
        fmt.Printf("  client error: %v\n", err)
        return
    }
    defer resp.Body.Close()
    b, _ := io.ReadAll(resp.Body)
    fmt.Printf("  %d %s", resp.StatusCode, b)
}

func main() {
    var logs bytes.Buffer
    logger = slog.New(slog.NewTextHandler(&logs, &slog.HandlerOptions{
        ReplaceAttr: func(_ []string, a slog.Attr) slog.Attr {
            if a.Key == slog.TimeKey {
                return slog.Attr{}
            }
            return a
        }}))
    var serverLog bytes.Buffer
    mux := http.NewServeMux()
    mux.HandleFunc("/naive", naive)
    mux.HandleFunc("/safe", safe)
    mux.HandleFunc("/boom", boom)
    mux.Handle("/boom-recovered", recoverer(http.HandlerFunc(boom)))
    ts := httptest.NewUnstartedServer(mux)
```

```
    ts.Config.ErrorLog = log.New(&serverLog, "", 0)
    ts.Start()
    defer ts.Close()

    for _, p := range []string{"/naive?id=7", "/safe?id=7", "/safe?id=404", "/boom", "/boom-
        fmt.Println("GET", p)
        get(ts.URL + p)
    }
    ts.Close()
    fmt.Println("\n--- application log (slog) ---")
    os.Stdout.Write(logs.Bytes())
    fmt.Println("\n--- net/http ErrorLog (first 3 lines) ---")
    lines := strings.SplitN(serverLog.String(), "\n", 4)
    for _, l := range lines[:3] {
        fmt.Println(l)
    }

    fmt.Println("\n--- formatting a Token ---")
    t := Token("sk_live_9f8e7d")
    fmt.Printf("%%v=%v %%s=%s %%#v=%#v %%q=%q string(t)=%s\n", t, t, t, t, string(t))

    fmt.Println("\n--- the chain still works inside the service ---")
    wrapped := fmt.Errorf("handler: %w", translate(loadInvoice("404", t)))
    pe, ok := errors.AsType[*PublicError](wrapped)
    fmt.Printf("AsType ok=%v status=%d msg=%q  errors.Is(wrapped, ErrNotFound)=%v\n", ok, pe
    _ = context.Background
}
```

```
go vet ./... && go build -o safeerr . && ./safeerr
```

```
GET /naive?id=7
  500 query "SELECT * FROM invoices WHERE id = $1 AND tenant = 'acme'" with token [REDACTED]
GET /safe?id=7
  500 internal error (request d9b5f5d4)
GET /safe?id=404
  404 invoice not found (request f758d9f7)
GET /boom
  client error: Get "http://127.0.0.1:42753/boom": EOF
GET /boom-recovered
  500 internal error (request df661fd0)

--- application log (slog) ---
level=ERROR msg="load invoice" req=d9b5f5d4 err.status=500 err.cause="query \"SELECT * FROM
level=ERROR msg="load invoice" req=f758d9f7 err.status=404 err.cause="query \"SELECT * FROM
level=ERROR msg=panic req=df661fd0 value="assignment to entry in nil map"
```

```
--- net/http ErrorLog (first 3 lines) ---
http: panic serving 127.0.0.1:37898: assignment to entry in nil map
goroutine 36 [running]:
net/http.(*conn).serve.func1()

--- formatting a Token ---
%v=[REDACTED] %s=[REDACTED] %#v=Token("[REDACTED]") %q="[REDACTED]" string(t)=sk_live_9f8e7d

--- the chain still works inside the service ---
AsType ok=true status=404 msg="invoice not found"  errors.Is(wrapped, ErrNotFound)=true
```

Reading it case by case:

- **`/naive`** shows the leak: the SQL text, the tenant `'acme'` and the path `/srv/acme/secrets/pg.pem` all reach the client in one line. The token printed as `[REDACTED]` only because its type redacts itself. A plain `string` would have gone out as is.
- **`/safe`** sends the client `internal error (request …)` or `invoice not found (request …)`. The log line with the same request ID carries `err.cause=` with the full chain. `errors.Is(err, ErrNotFound)` chose the 404 *before* any detail was dropped.
- **`/boom`** shows the `net/http` default. The `Handler` docs say the server "recovers the panic, logs a stack trace to the server error log, and either closes the network connection or sends an HTTP/2 RST_STREAM". The client got `EOF`, not a 500. The stack trace stays server-side, but the client gets no usable status, and goroutine dumps end up wherever `ErrorLog` points.
- **`/boom-recovered`**: the middleware converts the panic into a normal 500 with a request ID. It re-panics on `http.ErrAbortHandler`, which the same docs reserve for deliberately aborting a response "so the client sees an interrupted response but the server doesn't log an error".
- **Token formatting**: `%v`, `%s` and `%q` go through `String()`, while `%#v` goes through `GoString()`, as specified by the `fmt` rules. You need both methods, or `%#v` prints the raw field. `string(t)` is a conversion, not formatting, and no method can intercept it.
- **The chain**: `errors.AsType[*PublicError]` (Go 1.26+, docs) finds the boundary error through an extra `%w` layer, and `errors.Is(wrapped, ErrNotFound)` is still true. Translating at the edge cost the service no introspection.

Choosing what `Unwrap` exposes

`PublicError.Unwrap` returning the cause is a choice. It is right when everything that sees a `PublicError` is your own code. If a `PublicError` crosses into code you don't trust, such as a plugin, a client library returning errors to callers, or an RPC layer that serialises the chain, drop `Unwrap`. The chain then ends at the boundary, and nobody outside can probe for specific driver error types. The JetBrains article linked below calls this a firewall.

Things that bite

Trap	What happens	Fix
`http.Error(w, err.Error(), 500)` on a wrapped error	SQL, paths and tenant names reach the client (`/naive`)	Translate first, and make the boundary type's `Error()` safe
Secrets as plain `string`	Printed by any `%v` and serialised by any logger	A named type with `String`, `GoString` and `LogValue`
Only `String()` implemented	`%#v` prints the raw value	Implement `GoString()` too
`string(tok)` or `[]byte(tok)`	Redaction bypassed; no method can stop a conversion	Keep conversions in one audited function that hands the value to the HTTP client or driver
Relying on net/http's panic recovery	Client sees `EOF`, the stack goes to `ErrorLog`	Recover middleware; re-panic `http.ErrAbortHandler`
Logging the `PublicError` with `%v`	You log only the safe message and lose the cause	Log it as an slog attribute so `LogValue` runs, or log `errors.Unwrap(pe)`
Different messages for "no such user" and "wrong password"	Account enumeration	Map both to one public message at the boundary

Try this

1. Delete `GoString()` from `Token` and rerun. Which verb now leaks the secret?
2. Remove `Unwrap()` from `PublicError`. What does the last line print now, and why does the 404 mapping still work?
3. Point `ts.Config.ErrorLog` at `io.Discard` and observe that the client-side behaviour of `/boom` doesn't change. The recovery isn't about logging.

Sources

- Go: `net/http.Handler` (panic behaviour, `ErrAbortHandler`)
- Go: `log/slog.LogValuer`
- Go: `errors.AsType`, Working with Errors in Go 1.13
- Go: `fmt` (Stringer / GoStringer rules)
- JetBrains GoLand blog: Go Best Practices for Secure Error Handling

Supply Chain and Dependencies

Supply Chain and Dependencies

Overview

Your binary includes every module you import. Supply-chain hygiene: pin versions, review updates, scan vulns, reproduce builds.

Minimal practices

Practice	Tool / habit
Modules	`go.mod` / `go.sum` committed
Vuln scan	`govulncheck ./...`
Minimal deps	resist "one function" modules
Checksums	`go.sum` authenticity via sumdb
CI	fail on known critical vulns

```
go install golang.org/x/vuln/cmd/govulncheck@latest
govulncheck ./...
```

OSV-Scanner: scan a repository's dependencies

OSV-Scanner finds dependency manifests and checks their versions against the OSV vulnerability database. It is useful when a Go service shares a repository with other dependency manifests or when you need a repository-wide inventory scan.

For the pinned v2.7.0 source build, use Go 1.27 or newer:

```
go install github.com/google/osv-scanner/v2/cmd/osv-scanner@v2.7.0
osv-scanner scan source -r .
```

Prebuilt binaries are available through the installation guide. The recursive scan searches nested directories; `.gitignore` affects which files it discovers. Run it at the intended repository root and check the reported inputs before treating the scan as complete.

Keep `govulncheck ./...` as the focused Go-module workflow. OSV-Scanner also supports Go call analysis through the `govulncheck` library, enabled by default for Go; it needs the Go command on `PATH`. The two tools can therefore overlap in their Go findings. Review affected versions and

call-analysis evidence, upgrade the dependency, and rerun the scan and tests. Schedule the broader scan when its extra manifest coverage is useful.

go.mod hygiene

```
go get module@vX.Y.Z   # explicit
go mod tidy
go mod download
```

Prefer **semantic import versions** and avoid `replace` to random forks in production without review.

Vendoring (optional)

```
go mod vendor
# build with -mod=vendor for airgapped CI
```

Reproducible builds

```
CGO_ENABLED=0 go build -trimpath -ldflags="-s -w -X main.version=$VER"
```

Document Go version (toolchain line in `go.mod`).

Private modules

```
GOPRIVATE=github.com/myorg/*
```

Don't leak private paths to public proxies.

Rules of thumb

Do	Don't
Run govulncheck in CI	Ignore `go.sum` conflicts blindly
Review upgrades	Auto-merge major deps without tests
Pin actions SHA in CI	`uses: foo@main` for build

Try next

1. Run `govulncheck` on this repo's modules.
2. Add CI step.
3. Build with `-trimpath` and compare reproducibility notes.

Project: Secure Service Checklist

Project: Secure Service Checklist

Overview

Take any small HTTP service (Bookstore or echo) and harden it against a **checklist**. Tick items with code or config evidence.

Transport & process

- ☐ TLS in production path (or documented edge termination)
- ☐ `ReadHeaderTimeout` set on `http.Server`
- ☐ Non-root container user
- ☐ Graceful shutdown

Authn / authz

- ☐ Auth on mutating routes
- ☐ Password hashing (bcrypt/argon2) if local auth
- ☐ JWT: alg allowlist + exp + aud (if used)
- ☐ CSRF strategy documented for cookie sessions

Input / output

- ☐ MaxBytesReader on bodies
- ☐ Path jail for file ops
- ☐ No SSRF open URL fetch
- ☐ Security headers (or edge config)

Secrets & deps

- ☐ Secrets not in git
- ☐ `govulncheck` clean or waived
- ☐ Structured logs without secrets

Deliverable

`SECURITY.md` in the project with checklist status + residual risks.

GoLeash: Package-Level Runtime Policy Enforcement

GoLeash: Package-Level Runtime Policy Enforcement

`go.sum` and `govulncheck` catch *known* bad versions. They do not stop a compromised dependency that starts doing something new *after* install — dialing home, rewriting files, spawning processes — while the rest of the binary still looks legitimate. **GoLeash** is an eBPF-based research prototype (CCS 2025 artifact; arXiv:2505.11016) that applies **least privilege per Go package**: it profiles which packages exercise which syscalls (and along which stack paths), then enforces those allowlists at runtime. This chapter teaches the mental model, how it differs from process- or container-scoped sandboxes, and the analysis → enforce workflow from the project's primary docs. It sits next to the supply-chain chapter as the *runtime* half of dependency hygiene.

Thesis

Container seccomp and gVisor policies are **process-wide**. If *any* package in the binary is allowed to `connect(2)`, *every* package inherits that right. GoLeash's claim is finer: attribute each monitored syscall to a **Go package** (via stack traces and Go symbols) and allow only the capabilities that package earned during profiling. A JSON parser that never opened a socket during analysis should not open one in production.

Mental model

```
compromised dep "helpfuljson"
         sneaks net.Dial into init or a rare branch

process / container sandbox          GoLeash (package scope)
---------------------------          ------------------------
seccomp: allow connect               package encoding/json : {read, …}
  → whole process may connect        package helpfuljson   : { }  ← no connect
                                     syscall connect from helpfuljson → deny/log
```

Mode	What it does
Analysis	Observe syscalls + stack traces under representative load; build per-package capability allowlists
Enforcement	On each monitored syscall, check package + approved call paths; log or block violations

Properties called out in the paper and README:

- Works **without recompiling** the target (attach to binaries; needs usable Go symbol / stack info — avoid stripping with `-ldflags="-w -s"` while profiling).
- Remains useful under **obfuscation** better than pure static analysis, because decisions are based on runtime stacks.
- Overhead is described as acceptable for the evaluated workloads; treat that as research-reported, not a production SLA.

Worked example — profile then enforce (from upstream docs)

GoLeash is not installed here (building it needs clang/libbpf and would require system packages — out of scope for this chapter). The workflow below is taken from the project's README runnable example against **FRP** (Fast Reverse Proxy).

Case 1: Analysis against FRP

```
# Target app: keep Go symbols (do not strip -w -s); CGO_ENABLED=1 as upstream asks
git clone https://github.com/fatedier/frp.git
cd frp
# Edit Makefile: drop strip flags / set CGO_ENABLED=1, then:
make

# GoLeash config.toml - which binaries to watch
# [targets]
# binaries = ["frpc", "frps"]
# binary_paths = ["../frp/bin/frps", "../frp/bin/frpc"]

cd goleash
make          # build GoLeash (needs llvm/clang/libbpf on the build host)
make build    # analysis mode - profiles while the workload runs
```

In another shell, drive realistic behavior (upstream uses FRP e2e):

```
cd frp
make e2e
```

Stop GoLeash when finished. Policy lands as something like `trace/store.json` (path per README: `trace\store.json` on Windows-style docs; on Unix expect `trace/store.json`). Re-run e2e a few times to thicken coverage before trusting the allowlist.

Case 2: Enforcement and a simulated violation

```
# Manually remove a capability for some package from the saved policy JSON
# Then:
cd goleash
make cap-enforce
# Re-run the workload; unauthorized syscalls from that package are blocked/logged
```

Case 3: What "package capability" means in practice

Think in **capabilities**, not raw syscall numbers alone:

Package role	Syscalls you expect in analysis	Red flag later
Pure codec / math	mostly none, or `read`/`write` on caller-owned fds	`connect`, `execve`, `ptrace`
HTTP client helper	`socket`/`connect`/`sendto`…	`openat` of `/etc/shadow`, `chmod` on unrelated paths
Main / wiring package	whatever the app truly needs	sudden new families after a dependency bump

The paper's threat model is exactly that second column appearing after a supply-chain compromise.

The trap

- Profiling **only unit tests** — allowlists miss production paths; either under-block (false sense of security) or over-block when you finally hit real traffic.
- Stripping binaries **before** analysis — GoLeash needs symbol/stack information for package attribution.
- Treating GoLeash as a replacement for `govulncheck`, module pinning, and reproducible builds — it is a **complementary runtime layer**, still a research prototype.
- Equating "eBPF on the host" with "safe multi-tenant SaaS" — you still need the usual privileges and blast-radius story for loading BPF programs.

The boring rule

Pin and scan dependencies as usual; for high-value Go services, **profile package capabilities under real load**, then enforce. If a dependency suddenly needs `connect` and never did, that is an incident — not a silent `go get` success.

Try this

1. Read the abstract and §1 of arXiv:2505.11016.
2. Skim chains-project/goleash README modes and the FRP example.
3. On a lab host where you *already* have clang/libbpf (no new system installs from this book), build GoLeash and profile a small `net/http` demo: confirm your `main` package may `connect` while a vendored pure-Go helper may not.
4. Diff `go.mod` before/after a dependency update and re-run analysis — see which packages gain capabilities.

Sources

- Cesarano, Monperrus, Natella. *GoLeash: Mitigating Golang Software Supply Chain Attacks with Runtime Policy Enforcement.* arXiv:2505.11016. https://arxiv.org/abs/2505.11016 / https://arxiv.org/pdf/2505.11016
- GitHub: https://github.com/chains-project/goleash
- CCS 2025 artifact (Figshare): https://doi.org/10.6084/m9.figshare.28793891

runtime/secret Zeroization

runtime/secret Zeroization

Thesis

`runtime/secret` (Go experiment) runs a function under a contract that temporary storage — registers, stack, and soon-unreachable heap — is erased for forward secrecy. It is not a replacement for careful key handling, and it is **not** covered by the Go 1 compatibility promise.

Mental model

```
GOEXPERIMENT=runtimesecret
secret.Do(func() {
    // derive / use key material
})
// registers + stack used by the call tree cleared before return
// heap objects erased once GC proves them unreachable
```

Enable only when you build with `GOEXPERIMENT=runtimesecret`. Without that experiment, the package is not part of the build (`build constraints exclude all Go files`).

Worked example

On **Go 1.27.2** this box ships the sources under `$GOROOT/src/runtime/secret/` but the default toolchain has the experiment **off**. Documented API surface (from `doc.go` / `secret.go`):

```go
// Build with: GOEXPERIMENT=runtimesecret
package main

import "runtime/secret"

func useKey(key []byte) {
    secret.Do(func() {
        // use key; do not let copies escape the Do call tree
        _ = key
    })
}
```

From the package doc on this GOROOT:

> Do ensures that any temporary storage used by `f` is erased in a timely manner… registers and stack before `Do` returns; heap as soon as GC sees values unreachable. Works even if `f` panics or calls `runtime.Goexit`.

Lab note from this box:

```
go doc runtime/secret  →  package excluded without GOEXPERIMENT=runtimesecret
```

We did **not** rebuild the entire toolchain with the experiment here (no system package installs / long bootstrap). Treat the snippets as API-correct against the 1.27.2 sources; enable the experiment in a dedicated build when you need runtime verification.

Trap

Shipping `secret.Do` in production without the experiment (or without understanding that copies you deliberately escape still live). Anything reachable after `Do` returns is not erased. Logging the key, storing it in a global, or sending it on a channel defeats the point.

Boring rule

Keep key lifetimes short, avoid copies, and use `secret.Do` only when you have enabled the experiment and measured that your call tree does not escape sensitive buffers. Prefer boring crypto APIs (`crypto/tls`, `crypto/mlkem`, …) for protocol work.

Try this

1. Read `$GOROOT/src/runtime/secret/doc.go` on your Go 1.27+ install.
2. Bootstrap a long test with `GOEXPERIMENT=runtimesecret` in a throwaway module when you can afford a toolchain rebuild.
3. Pair with the eMudhra incident write-up: rebuild binaries with patched toolchains — zeroization does not help a compromised build key.

Sources

- Go sources: `src/runtime/secret/doc.go` (Go 1.27)
- Go experiments: `internal/goexperiment` (`RuntimeSecret`)
- Forward secrecy (concept)
- Go Security: response to the eMudhra key disclosure (rebuild with patched 1.27.x / 1.26.x)

When a CA Is Distrusted: Go 1.27.3's x509 Denylist and Rebuilding Binaries

When a CA Is Distrusted: Go 1.27.3's x509 Denylist and Rebuilding Binaries

After this page you will know where Go's fix for a distrusted certificate authority (CA) lives, why updating the operating system's CA bundle does not fix a Go binary, how to build with a patched toolchain without reinstalling Go, and how to find the binaries that still need rebuilding.

What happened: The eMudhra incident

In October 2026, eight eMudhra (emSign) intermediate CA certificates were found to have trivially weak private keys: every one of the eight elliptic-curve private keys was a number below 256. A key that small can be found by simply trying every value, so anyone could issue certificates that chain up to roots that clients already trust.

The CA revoked the certificates. Revocation only helps clients that check it, though, and Go's `crypto/x509` does not check CRLs or OCSP during chain verification. Go shipped its fix on 19–20 October 2026, the first release on Go's URGENT security track.

The emergency updates—Go 1.27.3 and Go 1.26.10—addressed the issue tracked as CVE-2026-111943, Go vulnerability ID GO-2026-6809 ("Distrust compromised intermediate certificates in crypto/x509"), and upstream issue go.dev/issue/82345.

The fix is code in `crypto/x509`, so it is compiled into every binary built with a fixed toolchain, and only into those binaries.

Why Go's fix lives in your binary

Go's standard library package `crypto/x509` performs TLS certificate chain validation. In Go 1.27.3 and 1.26.10, the Go team added a hardcoded denylist containing the SHA-256 cryptographic hashes of the `RawSubjectPublicKeyInfo` bytes for the eight compromised intermediate keys.

During certificate chain construction in `crypto/x509`, any candidate certificate chain containing one of these compromised public keys is discarded. This protection applies universally:

1. **Every platform**: on macOS and Windows, Go normally hands verification to the platform verifier, and the denylist applies there too.
2. **Cross-signing protection**: Because the denylist matches against the `RawSubjectPublicKeyInfo` hash rather than the certificate signature or subject name, cross-signed variants of the same intermediate keys (such as those signed via Sectigo AAA Certificate Services) are rejected as well.

3. **No runtime opt-out**: There is intentionally no `GODEBUG` environment variable to disable this denylist.

Crucially, because this logic is embedded directly inside the `crypto/x509` package code during compilation, upgrading the operating system's CA bundle on a server does **not** protect existing Go binaries. Similarly, upgrading the Go toolchain on a build host does not protect production services until every binary is explicitly recompiled and redeployed.

Action	Protects Go Binaries Built Prior to Patch?	Protects Non-Go System Tools?	Requires Binary Recompilation?
OS CA Bundle Update	No	Yes	No
Upgrade Go on Build Host	No (until rebuild)	No	No
Rebuild Binary with Go 1.27.3 / 1.26.10	Yes	No	Yes

Conversely, once a binary is recompiled with Go 1.27.3 or 1.26.10, it remains protected against the compromised keys even if deployed onto an older, unpatched host operating system whose system root store still trusts the emSign root.

What the failure looks like

When a binary built with a fixed toolchain meets a server whose chain needs one of the distrusted keys, chain building fails.

The underlying error is an `x509.CertificateInvalidError` with `Reason` set to `x509.NoValidChains`, which `http.Get` or `tls.Dial` wraps in its own error. In logs it ends with:

```
x509: no valid chains built
```

Because all candidate chains containing the compromised key material are filtered out, `Verify` behaves as though no valid path to a trusted root could be found.

Rebuilding your Go applications

To protect your applications, every Go binary that performs outbound HTTPS requests, connects to TLS-protected databases, or validates client TLS certificates must be recompiled using Go 1.27.3 (or Go 1.26.10 for the Go 1.26 release line).

Since Go 1.21, Go features native toolchain management via the `GOTOOLCHAIN` environment variable and the `toolchain` directive in `go.mod`. You do not need to manually reinstall Go across developer workstations or build environments to target a newer patch release.

Setting toolchain explicitly in `go.mod`

The demo module's `go.mod` (in `fetch/`) is as small as it gets:

```
module example.com/fetch

go 1.27
```

To make every build of the module use at least the fixed toolchain, add a `toolchain` line below `go 1.27`:

```
toolchain go1.27.3
```

With the default `GOTOOLCHAIN=auto`, an older `go` command that meets this line downloads and runs Go 1.27.3 instead. For a one-off build, set `GOTOOLCHAIN` directly, as shown below.

Demonstration binary

`fetch/main.go` fetches an HTTPS URL and prints how many verified chains the TLS handshake produced:

```go
package main

import (
    "fmt"
    "net/http"
    "os"
)

func main() {
    url := "https://go.dev/"
    if len(os.Args) > 1 {
        url = os.Args[1]
    }
    resp, err := http.Get(url)
    if err != nil {
        fmt.Println("error:", err)
        os.Exit(1)
    }
    defer resp.Body.Close()
    fmt.Println(resp.Status, "verified chain(s):", len(resp.TLS.VerifiedChains))
}
```

Using `GOTOOLCHAIN` at build time

You can instruct Go to automatically download and execute Go 1.27.3 without modifying the system Go installation by setting `GOTOOLCHAIN=go1.27.3`.

Check which toolchain that selects:

```
GOTOOLCHAIN=go1.27.3 go version
```

The following command produced this output:

```
go version go1.27.3 linux/amd64
```

The system Go on this machine is 1.27.2. Build the program twice, once with it and once with 1.27.3, then ask each binary what built it:

```
cd fetch
go build -o ../bin/fetch-1.27.2 .
GOTOOLCHAIN=go1.27.3 go build -o ../bin/fetch-1.27.3 .
cd ..
go version bin/fetch-1.27.2 bin/fetch-1.27.3
```

The following command produced this output:

```
bin/fetch-1.27.2: go1.27.2
bin/fetch-1.27.3: go1.27.3
```

`go version FILE...` reads the build information embedded in each binary, so it tells you which toolchain built it without access to the source code.

An ordinary site still verifies with the fixed binary:

```
./bin/fetch-1.27.3
```

The following command produced this output:

```
200 OK verified chain(s): 2
```

Updating CI build environments

Pin CI build images to a Go 1.27.3 (or 1.26.10) tag or later, not just a floating `1.27`, and confirm what was actually used by running `go version` on the artifact. Where you can't change the image, the `toolchain go1.27.3` line in `go.mod` (with `GOTOOLCHAIN=auto`) or an explicit `GOTOOLCHAIN=go1.27.3` in the build script gets the fixed toolchain, as long as the build can reach the module proxy to download it.

Finding old binaries in your environment

Go binaries carry their build information, so you can audit binaries in images, artifacts and servers without the source.

Using `go version` and `debug/buildinfo`

`go version` accepts any number of files and skips anything that isn't a Go binary:

```
go version /usr/local/bin/*
```

For automated inventory auditing, Go provides the standard library package `debug/buildinfo`. You can write a scanner tool that walks directory trees and flags any binary compiled with an unpatched Go version.

Below is the module configuration for a build scanner tool:

```
module example.com/gobinscan

go 1.27

require golang.org/x/mod v0.42.0
```

Below is the implementation of `gobinscan`, which inspects Go binaries and evaluates their release versions using `golang.org/x/mod/semver`:

```
// gobinscan walks directories and reports Go binaries built with a
// toolchain older than the minimum you name for each release line.
package main

import (
    "debug/buildinfo"
    "fmt"
    "io/fs"
    "os"
    "path/filepath"
    "strings"

    "golang.org/x/mod/semver"
)

// Minimum patched release per Go minor line (eMudhra distrust, Oct 2026).
var minimum = map[string]string{
    "v1.27": "v1.27.3",
    "v1.26": "v1.26.10",
}
```

```go
func verdict(goVersion string) string {
    v := "v" + strings.TrimPrefix(goVersion, "go")
    v = strings.SplitN(v, " ", 2)[0] // drop " X:..." experiment suffixes
    if !semver.IsValid(v) {
        return "UNKNOWN"
    }
    need, ok := minimum[semver.MajorMinor(v)]
    if !ok {
        if semver.Compare(v, "v1.27.0") > 0 {
            return "ok"
        }
        return "UNSUPPORTED"
    }
    if semver.Compare(v, need) < 0 {
        return "REBUILD (needs " + strings.TrimPrefix(need, "v") + ")"
    }
    return "ok"
}

func main() {
    roots := os.Args[1:]
    if len(roots) == 0 {
        roots = []string{"."}
    }
    for _, root := range roots {
        filepath.WalkDir(root, func(path string, d fs.DirEntry, err error) error {
            if err != nil || d.IsDir() {
                return nil
            }
            info, err := buildinfo.ReadFile(path)
            if err != nil {
                return nil // not a Go binary
            }
            fmt.Printf("%-28s %-10s %s\n", path, info.GoVersion, verdict(info.GoVersion))
            return nil
        })
    }
}
```

Run `gobinscan` across your local binary directories:

```
./gobinscan bin /usr/local/bin
```

The following command produced this output:

```
bin/fetch-1.27.2            go1.27.2   REBUILD (needs 1.27.3)
bin/fetch-1.27.3            go1.27.3   ok
/usr/local/bin/gh           go1.27.1   REBUILD (needs 1.27.3)
/usr/local/bin/go           go1.27.2   REBUILD (needs 1.27.3)
/usr/local/bin/gofmt        go1.27.2   REBUILD (needs 1.27.3)
```

Even the tools on the build machine show up: `gh`, `go` and `gofmt` were all built with older toolchains. A program that never verifies a TLS chain (`gofmt`, for example) is a lower priority than a network service, but the rule that scales is simple: rebuild everything you ship.

Using `govulncheck` in binary mode

The official Go vulnerability scanner, `govulncheck`, supports analyzing compiled Go binaries directly. `govulncheck -mode=binary` goes beyond the version string: it looks at the symbols compiled into the binary and reports a vulnerability only if a vulnerable function, here `x509.Certificate.Verify` (reached through `net/http`), is present.

Install `govulncheck`:

```
go install golang.org/x/vuln/cmd/govulncheck@latest
```

Scan the unpatched binary built with Go 1.27.2:

```
govulncheck -mode=binary bin/fetch-1.27.2
```

The following command produced this output:

```
=== Symbol Results ===

Vulnerability #1: GO-2026-6809
    Distrust compromised intermediate certificates in crypto/x509
  More info: https://pkg.go.dev/vuln/GO-2026-6809
  Standard library
    Found in: crypto/x509@go1.27.2
    Fixed in: crypto/x509@go1.27.3
    Vulnerable symbols found:
      #1: x509.Certificate.Verify

Your code is affected by 1 vulnerability from the Go standard library.
This scan found no other vulnerabilities in packages you import or modules you
require.
Use '-show verbose' for more details.
```

`govulncheck` identifies vulnerability GO-2026-6809 and exits with code 3 when vulnerabilities are detected.

Next, scan the patched binary built with Go 1.27.3:

```
govulncheck -mode=binary bin/fetch-1.27.3
```

The following command produced this output:

```
No vulnerabilities found.
```

When no vulnerabilities are found in the binary symbols, `govulncheck` exits with code 0.

Fleet playbook checklist

When responding to standard library security patches that require binary recompilation, follow this structured remediation plan across your infrastructure:

1. **Audit binary inventory**: Run `govulncheck -mode=binary` or `debug/buildinfo` scanning scripts across production container registries, AMI disk images, and server deployments to build an inventory of affected binaries.
2. **Update build toolchains**: Set `toolchain go1.27.3` in project `go.mod` files or ensure CI/CD runner environments export `GOTOOLCHAIN=go1.27.3`.
3. **Rebuild all application binaries**: Trigger clean builds of all microservices, background workers, and ingress proxies.
4. **Redeploy container images and host binaries**: Rolling-update production deployments. Ensure old container layers containing unpatched binaries are replaced.
5. **Verify deployed artifacts**: Run `govulncheck -mode=binary` on the final release artifacts inside the CD pipeline before pushing to production registries.
6. **Maintain base images**: Update base container images (e.g., `alpine`, `debian`, `distroless`) to ensure OS-level certificate stores and auxiliary runtime utilities are also up to date.

Pitfalls

When managing standard library security fixes across complex deployments, avoid these common traps:

Distroless and minimal container images

A distroless or `scratch` image is a good idea for other reasons, but swapping or updating the base image changes nothing here. The standard library is inside your executable, so only a rebuild helps. The denylist applies whichever root store the binary ends up using.

Vendored or static third-party binaries

Third-party utilities included in build pipelines or container images (for example `kubectl`, `helm` or your own sidecars) carry their own compiled Go standard libraries. Updating your own application code will not patch these third-party tools. You must update vendor tools to release versions built by their maintainers using Go 1.27.3 or 1.26.10.

Attempting to disable TLS validation

If a connection starts failing with `x509: no valid chains built` after a rebuild, the server is serving a distrusted chain. Fix the server's certificate. Setting `InsecureSkipVerify: true` in `tls.Config` to make the error go away turns off exactly the protection this release added, along with all other chain checks.

Summary

- **Toolchain-level enforcement**: Go embeds x509 validation rules and CA denylists directly into compiled binaries via `crypto/x509`.
- **OS updates don't cover it**: neither a new OS CA bundle nor a new Go on the build host changes a binary that has already been built.
- **CVE-2026-111943 / GO-2026-6809**: Go 1.27.3 and 1.26.10 introduce hardcoded SHA-256 denylists targeting compromised eMudhra (emSign) intermediate keys.
- **Rebuild requirement**: Every Go binary making TLS connections or validating certificates must be recompiled using Go 1.27.3 or Go 1.26.10.
- **Binary inspection**: Use `go version`, `debug/buildinfo`, and `govulncheck -mode=binary` to audit compiled binaries for unpatched toolchain versions.

Sources

- Go Blog: Addressing Weak Keys in Intermediate CA Certificates
- Go Vulnerability Database: GO-2026-6809
- Go Issue #82345: crypto/x509: distrust compromised intermediate certificates
- Go Announcement: Go 1.27.3 and Go 1.26.10 security releases
- Go Package Docs: crypto/x509
- Go Package Docs: debug/buildinfo
- Go Documentation: Go Toolchain Management
- Go Package Docs: govulncheck

Performance and Tooling

Advanced Profiling (pprof/trace)

Performance: Deep Dives with pprof & trace

Optimization without measurement is just guesswork. Go has arguably the best built-in profiling tools of any compiled language.

1. pprof: The Sampling Profiler

`pprof` creates a statistical profile of your program.

Usage

Add one line to your `main`:

```
import _ "net/http/pprof"

func main() {
    go func() {
        http.ListenAndServe("localhost:6060", nil)
    }()
    // ... app code ...
}
```

Now, while your app runs under load, grab a profile:

```
go tool pprof -http=:8081 http://localhost:6060/debug/pprof/profile?seconds=30
```

This opens a web UI. Look at: * **Flame Graph:** The widest bars are using the most CPU. * **Alloc Space:** Who allocated the most memory (Heap). * **Alloc Objects:** Who created the most *count* of objects (high GC pressure). * **Goroutine leak profile (Go 1.27+):** `/debug/pprof/goroutineleak` lists Gs blocked on unreachable concurrency primitives.

Mutex Profiling

If your CPU usage is low but performance sucks, you have contention. Enable mutex profiling in code: `runtime.SetMutexProfileFraction(1)` Then look at `/debug/pprof/mutex` to see who is waiting for locks.

2. Execution Tracer (`go tool trace`)

While pprof aggregates data, **Trace** shows you a timeline of *every event*. * When did a Goroutine start? * When did it block on a channel? * When did GC pause execution?

Capture a trace:

```
curl -o trace.out http://localhost:6060/debug/pprof/trace?seconds=5
go tool trace trace.out
```

Go 1.27: `go tool trace -http=:6060` listens on **localhost** only (same as `pprof`). Bind all interfaces with `-http=0.0.0.0:6060`.

Use Case: Debugging latency spikes. You might see a 100ms gap where *nothing* ran. Zooming in, you might find a Stop-The-World GC event or a single mutex holding up 50 goroutines.

3. Benchmarks as Profiling Inputs

You can profile specific functions using benchmarks:

```
go test -bench=. -cpuprofile=cpu.out
go tool pprof -http=:8080 cpu.out
```

Summary

- **pprof:** For "Where is my CPU/RAM going?"
- **trace:** For "Why is there latency?" and "What is the scheduler doing?"
- Optimization loop: Measure -> Fix -> Verify (Benchmark).

Worked example

Two algorithms, one clear allocation winner—profile-friendly benchmarks.

Save as `join_test.go`. Then:

```
go mod init example
go test -bench=. -benchmem
go test -bench=Builder -cpuprofile=cpu.out -benchmem
```

```
package main

import (
    "strings"
```

```go
    "testing"
)

func joinPlus(parts []string) string {
    s := ""
    for _, p := range parts {
        s += p
    }
    return s
}

func joinBuilder(parts []string) string {
    var b strings.Builder
    for _, p := range parts {
        b.WriteString(p)
    }
    return b.String()
}

var parts = []string{"go", "is", "fast", "enough", "when", "measured"}

func BenchmarkPlus(b *testing.B) {
    for i := 0; i < b.N; i++ {
        _ = joinPlus(parts)
    }
}

func BenchmarkBuilder(b *testing.B) {
    for i := 0; i < b.N; i++ {
        _ = joinBuilder(parts)
    }
}
```

Expected output: (Builder lower B/op and allocs/op)

```
BenchmarkPlus-8        ...   ... ns/op   ... B/op   ... allocs/op
BenchmarkBuilder-8     ...   ... ns/op   ... B/op   ~1 allocs/op
```

More examples

Emit a short execution trace from a benchmark.

```bash
go test -bench=Builder -trace=trace.out
# go tool trace trace.out
```

```
// same package as above; no extra code required for -trace
```

Runnable example

Save as `alloc_test.go`. Then:

```
go mod init example
go test -bench=. -benchmem
# optional: write a CPU profile for go tool pprof
go test -bench=SumPrealloc -cpuprofile=cpu.out -benchmem
```

```
package main

import "testing"

func sumAlloc(n int) int {
    // allocates a new backing array every call
    s := make([]int, 0)
    for i := 0; i < n; i++ {
        s = append(s, i)
    }
    total := 0
    for _, v := range s {
        total += v
    }
    return total
}

func sumPrealloc(n int) int {
    s := make([]int, 0, n)
    for i := 0; i < n; i++ {
        s = append(s, i)
    }
    total := 0
    for _, v := range s {
        total += v
    }
    return total
}

func BenchmarkSumAlloc(b *testing.B) {
    for i := 0; i < b.N; i++ {
        _ = sumAlloc(1024)
    }
```

```go
}

func BenchmarkSumPrealloc(b *testing.B) {
    for i := 0; i < b.N; i++ {
        _ = sumPrealloc(1024)
    }
}
```

Expected output: (`SumPrealloc` should show ~1 alloc/op vs many for `SumAlloc`)

```
BenchmarkSumAlloc-8           100000         12000 ns/op       20000 B/op         10 allocs/
BenchmarkSumPrealloc-8        200000          6000 ns/op        8192 B/op          1 allocs/
PASS
```

What to notice: Benchmarks are excellent profiling inputs—no long-running server required. Allocation differences show up clearly with `-benchmem` before you open a flame graph. For live services, the chapter's `net/http/pprof` endpoints are the online equivalent.

Try next: `go tool pprof -http=:8080 cpu.out` after the profile command. Capture a short trace with `go test -trace=trace.out -bench=SumPrealloc` then `go tool trace trace.out`.

The Modern Toolkit

The Modern 2026 Toolkit

Choose tools for a concrete job: formatting, finding defects, testing, or investigating a running program. Start with the Go toolchain and add checks when they answer a question your existing workflow does not.

Choosing an additional tool

These are useful options for current Go projects, including established tools that remain relevant. The examples below and in the linked chapters pin versions checked in November 2026; they are examples to review and update.

Need	Tool	Where to start
Consistent formatting beyond `gofmt`	`gofumpt`	Formatting below
Clear test output and CI reports	`gotestsum`	Test reports
Find unreachable functions	`deadcode`	Reachability analysis
Follow possible nil values across packages	NilAway	Nil-flow analysis
Check cleanup and concurrency ownership	GoHawk	Ownership analysis
Detect unexpected goroutines after tests	`goleak`	Leak regression tests
Scan dependencies across a repository	OSV-Scanner	Dependency scanning

Use the linked limitations to decide what belongs on every change and what belongs in an occasional audit. A formatter, a source analyzer, and a runtime test detector produce different kinds of evidence.

General Tools (Rust-powered, Go-essential)

These aren't written in Go, but every Go dev uses them. 1. **ripgrep (`rg`)**: The fastest text search. * `rg "func main" -t go`: Find main in Go files only. 2. **fd**: Faster `find`. * `fd -e go`: Find all Go files. 3. **fzf**: Fuzzy finder. Pipe `fd` into `fzf` to open files instantly. 4. **jq**: JSON processor. Crucial for debugging API responses or logs (`zap`/`slog` JSON output).

Go-Specific Tools

`gofumpt`: stricter formatting

gofumpt adds formatting rules while keeping its output compatible with `gofmt`. It can remove unnecessary blank lines and separate standard-library imports from other imports. Adopt it as a team convention so editors and CI agree.

```
go install mvdan.cc/gofumpt@v0.12.0
gofumpt -d .    # Review the proposed diff.
gofumpt -w .    # Apply formatting.
```

For a shell CI step after installation:

```
unformatted=$(gofumpt -l .) || exit 1
if [ -n "$unformatted" ]; then
    printf '%s\n' "$unformatted"
    exit 1
fi
```

`-l` lists files; the shell check makes that list fail CI. Version v0.12.0 requires Go 1.26 or newer to build. Review formatting changes separately from behavior changes to make both easier to review.

`gopsutil`

If you are building system tools (agents, monitors), `github.com/shirou/gopsutil` is the standard for reading CPU/Memory/Disk stats cross-platform.

`go fix` Modernizers (1.26–1.27)

Go 1.26 rebuilt `go fix` around modernizers and `//go:fix inline`. Go 1.27 adds `atomictypes`, `embedlit`, `slicesbackward`, and `unsafefuncs`; the old `waitgroup` analyzer is now `waitgroupgo`; `fmtappendf` was removed.

```
go fix -modernize ./...
```

Examples of what it fixes: * Replaces manual loops with `slices.Contains`. * Updates older `interface{}` to `any`. * `embedlit`: Flattens nested composite struct literals into promoted field

keys where safe (`go fix -embedlit ./...`). * Atomic type aliases, `slices.Backward`, unsafe helpers.

Inline fix for one location:

```go
//go:fix inline
func has(xs []string, x string) bool {
    for _, v := range xs {
        if v == x {
            return true
        }
    }
    return false
}
```

govulncheck

The official vulnerability scanner.

```go install golang.org/x/vuln/cmd/govulncheck@latest
govulncheck ./...
```

It tells you if *your code actually calls* the vulnerable function in a dependency. This drastically reduces false positives compared to generic scanners (Dependabot).

Pin tools with `go tool` (Go 1.24+)

For Go command packages that support this installation method, record a tool dependency in the module rather than relying on a developer's global binary:

```go get -tool mvdan.cc/gofumpt@v0.12.0
go tool gofumpt -d .
```

Commit the resulting `go.mod` and `go.sum` changes. This is an alternative to the standalone `go install` above and does not require that binary on `PATH`. Tool requirements participate in the module's dependency graph, so inspect the dependency diff. A separate tools module can isolate their requirements when necessary. The Go documentation explains tool dependencies. Follow each tool's own installation guidance; golangci-lint, for example, recommends its release binaries.

Workflow Integration

Don't run these manually. Put them in a **Makefile** or **Taskfile** (Task is a modern Make alternative written in Go).

```yaml
# Taskfile.yaml
tasks:
  audit:
    cmds:
      - go vet ./...
      - staticcheck ./...
      - govulncheck ./...
```

Worked example

`slices` / `maps` modern stdlib helpers vs hand-rolled loops.

Save as `util.go` and `util_test.go`. Then:

```bash
go mod init example
go test -v
```

```go
// util.go
package main

import (
    "maps"
    "slices"
)

func has(xs []string, x string) bool {
    return slices.Contains(xs, x)
}

func cloneMap(m map[string]int) map[string]int {
    return maps.Clone(m)
}
```

```go
// util_test.go
package main

import "testing"

func TestHas(t *testing.T) {
    xs := []string{"a", "b", "c"}
    if !has(xs, "b") || has(xs, "z") {
        t.Fatal("contains mismatch")
    }
}
```

```go
func TestCloneMap(t *testing.T) {
    m := map[string]int{"x": 1}
    c := cloneMap(m)
    c["x"] = 9
    if m["x"] != 1 {
        t.Fatal("clone should be independent")
    }
}
```

Expected output:

```
=== RUN   TestHas
--- PASS: TestHas (0.00s)
=== RUN   TestCloneMap
--- PASS: TestCloneMap (0.00s)
PASS
```

More examples

Table of "grep-friendly" symbols your toolkit will find.

```go
package main

import "fmt"

// AuditTarget is intentionally distinctive for rg demos.
func AuditTarget() string { return "ok" }

func main() {
    fmt.Println(AuditTarget())
}
```

```
# in a shell with ripgrep installed:
# rg 'func AuditTarget' -t go
go run .
```

Runnable example

> **Note:** Tools like `rg`, `staticcheck`, and `govulncheck` are external CLIs. This stdlib demo captures the *modernizer* idea: replace a hand-rolled loop with `slices.Contains`, and run `go vet`-friendly code under `go test`.

Save as `modern.go` and `modern_test.go`. Then:

```
go mod init example
go test -v
go vet ./...
```

```go
// modern.go
package main

import "slices"

// legacyHas is the pre-slices style many codebases still have.
func legacyHas(xs []string, x string) bool {
    for _, v := range xs {
        if v == x {
            return true
        }
    }
    return false
}

// modernHas is the 1.21+ equivalent (what go fix -modernize targets).
func modernHas(xs []string, x string) bool {
    return slices.Contains(xs, x)
}
```

```go
// modern_test.go
package main

import "testing"

func TestHas(t *testing.T) {
    xs := []string{"go", "rust", "zig"}
    if !legacyHas(xs, "go") || !modernHas(xs, "go") {
        t.Fatal("expected to find go")
    }
    if legacyHas(xs, "c") || modernHas(xs, "c") {
        t.Fatal("did not expect c")
    }
    if legacyHas(xs, "rust") != modernHas(xs, "rust") {
        t.Fatal("implementations diverged")
    }
}
```

Expected output:

```
=== RUN   TestHas
--- PASS: TestHas (0.00s)
```

```
PASS
```

What to notice: Stdlib `slices` / `maps` replace a lot of utility code. Keep a small `go test` + `go vet` loop before pulling heavier linters. `govulncheck` still needs a separate install—run it in CI on real modules.

Try next: Run `go fix -modernize ./...` on a module that still uses `interface{}` and manual contains loops. Pipe JSON logs through `jq` in your shell.

Static Analysis and Security

Static Analysis and Security

Overview

Go provides built-in and third-party tools for code quality and security analysis.

go vet

```
go vet ./...
```

Catches: - Printf format errors - Unreachable code - Suspicious constructs

staticcheck

```
go install honnef.co/go/tools/cmd/staticcheck@latest
staticcheck ./...
```

Catches: - Deprecated APIs - Simplifications - Performance issues

GoHawk: resource and concurrency checks

GoHawk (`gohawk`) examines resource ownership and concurrency in Go source. Its function summaries let it follow cleanup through helpers and across packages. Use it alongside `go vet`, Staticcheck, and `govulncheck`: each examines a different class of problem. The project overview describes this scope.

Install and inspect the checks

This example pins v0.7.1. Building it requires Go 1.26 or newer; the documented supported analysis toolchains are Go 1.26 and 1.27. Build with a toolchain at least as new as the one used by the project, and rebuild after upgrading Go. Prebuilt binaries are also available through the installation guide.

```
go install github.com/kojah/gohawk@v0.7.1
```

Put the installation directory on `PATH`: Go uses `GOBIN` when set, otherwise the `bin` directory under `GOPATH` (`go env GOBIN GOPATH` shows both). From a module directory, run:

```
gohawk -V
gohawk list
gohawk doc resourcelifetime
gohawk ./...
```

The default run uses core checks. Start there; select individual analyzers while investigating a finding:

```
gohawk -enable=resourcelifetime,cancellationownership ./...
```

As documented in the configuration guide, v0.7.1 analyzes production code and excludes `_test.go` files. A clean report does not establish that tests, shutdown paths, or all possible executions are correct. Keep `go test -race ./...` in the workflow: race testing observes executed code, whereas GoHawk reasons about source without running the program.

Exercise: cleanup registered too late

In a fresh directory, run `go mod init example.com/resourcecheck` and save this as `size.go`. It compiles, but the `Stat` error path returns before cleanup is registered:

```go
package resourcecheck

import "os"

func FileSize(path string) (int64, error) {
    f, err := os.Open(path)
    if err != nil {
        return 0, err
    }
    info, err := f.Stat()
    if err != nil {
        return 0, err // f is still open on this path.
    }
    defer f.Close() // Too late for the earlier return.
    return info.Size(), nil
}
```

```
go test ./...
gohawk -enable=resourcelifetime ./...
```

Expect a `resourcelifetime/missing-release` finding. Move `defer f.Close()` immediately after the successful `os.Open` error check, before `f.Stat()`, and rerun both commands. The resource finding should disappear. This is a read-only example; code writing data must also handle errors from flushing and closing before reporting success.

The ownership question is: **who releases the resource on every return path?** The resource-lifetime check also covers HTTP response bodies and database resources. Returning a resource can deliberately transfer cleanup to the caller, so review that contract too.

Two related checks apply the same reasoning to other lifetimes:

- Cancellation ownership tracks derived cancel functions. Arrange `defer cancel()` when the current function owns the context, or explicitly transfer responsibility.
- Goroutine ownership looks for recognizable completion signals that callers fail to wait for. It deliberately leaves some patterns unreported; it is not a general proof that goroutines terminate.

Read the reported return path and cleanup evidence before changing code. Use the smallest check selection needed to understand a finding, then rerun the default set and the relevant tests after the fix.

`deadcode`: find unreachable functions

The Go project's deadcode tool finds functions unreachable from the selected program entry points, including calls through interfaces. Use it during refactoring to find code that may no longer need maintenance.

```
go install golang.org/x/tools/cmd/deadcode@v0.52.0
deadcode ./...
deadcode -test ./...
```

The second scan includes test executables as entry points. For a small experiment, save this as `main.go` in a fresh module:

```go
package main

import "fmt"

func currentLabel() string { return "current" }
func retiredLabel() string { return "retired" }

func main() { fmt.Println(currentLabel()) }
```

Run `go mod init example.com/reachability`, then `deadcode .`: expect `retiredLabel` in the report. Call it from `main` and rerun; it becomes live.

Review findings before deleting public APIs: external consumers may use a library function absent from the scanned executables. Repeat the analysis for relevant operating systems, architectures, and build tags. See the command documentation for configuration boundaries. Findings are printed without making the command fail; choose an explicit reporting policy before using it as a CI gate.

NilAway: follow nil values across calls

NilAway tracks potential nil dereferences within and across packages. It complements GoHawk's lifetime checks. The Security Tools chapter has setup instructions; evaluate the findings on your module before making the check mandatory. Its maintainers note that false positives and breaking changes remain possible during active development.

golangci-lint

Install a pinned v2 release binary using the official installation guide. The maintainers recommend binary installation over `go install` or a tool directive because the analyzer dependency versions matter.

```
golangci-lint --version
# Run the linters selected by the configuration/defaults.
golangci-lint run
```

For an existing v1 configuration, use the migration guide before moving to v2. `run` does not mean every available linter is enabled.

govulncheck

```
go install golang.org/x/vuln/cmd/govulncheck@latest
govulncheck ./...
```

Checks dependencies for known vulnerabilities.

gosec

```
go install github.com/securego/gosec/v2/cmd/gosec@latest
gosec ./...
```

Security-focused linter: - SQL injection - Hardcoded credentials - Weak crypto

Common Security Issues

```
// SQL injection - BAD
query := "SELECT * FROM users WHERE id = " + id

// Use parameters - GOOD
```

```go
db.Query("SELECT * FROM users WHERE id = ?", id)

// Path traversal - BAD
path := filepath.Join(baseDir, userInput)

// Sanitize - GOOD
if strings.Contains(userInput, "..") {
    return errors.New("invalid path")
}
```

CI Integration

After the Go toolchain is configured, add GoHawk as a separate step:

```yaml
- name: Install GoHawk
  run: go install github.com/kojah/gohawk@v0.7.1

- name: Check resource and concurrency ownership
  run: gohawk ./...
```

Ensure the Go binary installation directory is on the runner's `PATH`, as with the local setup above. The standalone CLI exits with status 0 for no findings, 3 for findings, and 1 for build or analysis failures, so the scan step fails naturally when attention is needed. See the CLI installation and execution guide.

GoHawk currently offers a module plugin for a custom `golangci-lint` binary; the ordinary binary does not include it as a native linter. Keep this standalone step unless you deliberately adopt the plugin integration.

```yaml
# .github/workflows/lint.yml
- name: Lint
  run: golangci-lint run

- name: Security scan
  run: |
    govulncheck ./...
    gosec ./...
```

Summary

Tool	Focus
`go vet`	Basic correctness
`staticcheck`	Advanced analysis

Tool	Focus
`gohawk`	Resource cleanup and concurrency ownership
`deadcode`	Unreachable functions from selected entry points
NilAway	Potential nil dereferences across calls
`golangci-lint`	Multiple linters
`govulncheck`	Vulnerability scan
`gosec`	Security issues

Worked example

Path traversal guards unit-tested (the boring fix scanners want).

Save as `pathx.go` and `pathx_test.go`. Then:

```
go mod init example
go test -v
go vet ./...
```

```go
// pathx.go
package main

import (
    "errors"
    "path/filepath"
    "strings"
)

func UnderRoot(root, user string) (string, error) {
    if user == "" || strings.Contains(user, "..") {
        return "", errors.New("invalid path")
    }
    full := filepath.Clean(filepath.Join(root, user))
    root = filepath.Clean(root)
    sep := string(filepath.Separator)
    if full != root && !strings.HasPrefix(full, root+sep) {
        return "", errors.New("escape")
    }
    return full, nil
}
```

```go
// pathx_test.go
package main

import (
```

```
    "path/filepath"
    "testing"
)

func TestUnderRoot(t *testing.T) {
    root := filepath.Join("var", "data")
    ok, err := UnderRoot(root, "a.txt")
    if err != nil || ok != filepath.Join(root, "a.txt") {
        t.Fatalf("got %q err=%v", ok, err)
    }
    if _, err := UnderRoot(root, "../etc/passwd"); err == nil {
        t.Fatal("expected error")
    }
}
```

Expected output:

```
=== RUN   TestUnderRoot
--- PASS: TestUnderRoot (0.00s)
PASS
```

More examples

SQL placeholder style vs string concat (pattern comparison only).

```
package main

import (
    "fmt"
    "strings"
)

func main() {
    q := "SELECT * FROM t WHERE id=?"
    args := []any{"42"}
    fmt.Println("placeholder query:", q, "args:", args)
    bad := "SELECT * FROM t WHERE id='" + "x' OR '1'='1" + "'"
    fmt.Println("concat is dangerous:", strings.Contains(bad, "OR"))
}
```

Expected output:

```
placeholder query: SELECT * FROM t WHERE id=? args: [42]
concat is dangerous: true
```

Runnable example

> External scanners (`staticcheck`, `gosec`, `govulncheck`) need separate installs. This stdlib program demonstrates **safe vs unsafe** patterns those tools flag, and runs cleanly under `go vet` / `go test`.

Save as `safe.go` and `safe_test.go`. Then:

```
go mod init example
go test -v
go vet ./...
```

```go
// safe.go
package main

import (
    "errors"
    "path/filepath"
    "strings"
)

// badQuery shows string-built SQL (do not use with real drivers).
func badQuery(id string) string {
    return "SELECT * FROM users WHERE id = '" + id + "'"
}

// goodQuery keeps user data out of the SQL string (placeholder style).
func goodQuery() (query string, args []any) {
    return "SELECT * FROM users WHERE id = ?", []any{"42"}
}

// safeJoin cleans user input and rejects paths that escape base.
func safeJoin(base, userInput string) (string, error) {
    if userInput == "" || strings.Contains(userInput, "..") {
        return "", errors.New("invalid path")
    }
    // Build under base, then verify the cleaned result still has base as prefix.
    full := filepath.Clean(filepath.Join(base, userInput))
    baseClean := filepath.Clean(base)
    sep := string(filepath.Separator)
    if full != baseClean && !strings.HasPrefix(full, baseClean+sep) {
        return "", errors.New("invalid path")
    }
    return full, nil
}
```

```go
// safe_test.go
package main

import (
    "path/filepath"
    "testing"
)

func TestSafeJoin(t *testing.T) {
    base := filepath.Join("var", "data")
    ok, err := safeJoin(base, "notes.txt")
    if err != nil {
        t.Fatal(err)
    }
    if ok != filepath.Join(base, "notes.txt") {
        t.Fatalf("got %q", ok)
    }
    if _, err := safeJoin(base, "../etc/passwd"); err == nil {
        t.Fatal("expected rejection")
    }
}

func TestGoodQuery(t *testing.T) {
    q, args := goodQuery()
    if q == badQuery("x") {
        t.Fatal("good query should not embed user input")
    }
    if len(args) != 1 {
        t.Fatal("expected bound args")
    }
}
```

Expected output:

```
=== RUN   TestSafeJoin
--- PASS: TestSafeJoin (0.00s)
=== RUN   TestGoodQuery
--- PASS: TestGoodQuery (0.00s)
PASS
```

What to notice: Parameterized queries and path containment are the boring fixes `gosec` pushes for. `go vet` is free CI; layer `staticcheck` / `golangci-lint` / `govulncheck` once the module is real.

Try next: Install `staticcheck` and run it on this package. Introduce an unused write and watch the tool complain.

Related Topics

- Formatting, Vetting, Docs
- Building and Releasing

Code Generation and Embedding

Code Generation and Embedding

Overview

`go generate` runs commands to generate code, and `//go:embed` embeds files into binaries.

go generate

```go
//go:generate stringer -type=Status

type Status int

const (
    Pending Status = iota
    Approved
    Rejected
)
```

```
go generate ./...
```

Common Generators

- `stringer` - String methods for enums
- `mockgen` - Mock interfaces for testing
- `protoc` - Protocol buffer code

Embedding Files

```go
import "embed"

//go:embed config.json
var configData []byte

//go:embed templates/*
var templates embed.FS
```

```go
//go:embed static
var staticFiles embed.FS
```

Reading Embedded Files

```go
//go:embed data.txt
var data string  // As string

//go:embed data.bin
var data []byte  // As bytes

//go:embed files/*
var fs embed.FS  // As filesystem

content, _ := fs.ReadFile("files/config.json")
```

HTTP File Server

```go
//go:embed static/*
var static embed.FS

func main() {
    fs := http.FileServer(http.FS(static))
    http.Handle("/static/", fs)
    http.ListenAndServe(":8080", nil)
}
```

Templates

```go
//go:embed templates/*.html
var templates embed.FS

tmpl := template.Must(template.ParseFS(templates, "templates/*.html"))
```

Summary

Directive	Purpose
`//go:generate cmd`	Run code generator
`//go:embed file`	Embed as string/bytes
`//go:embed dir/*`	Embed as filesystem

Worked example

Generate a tiny constants file at runtime (stand-in for `go generate` output), then use it.

Save as `main.go`. Then:

```
go mod init example
go run .
```

```
package main

import (
    "fmt"
    "strings"
)

func generateVersionGo(version string) string {
    return fmt.Sprintf("// Code generated by demo; DO NOT EDIT.\npackage main\n\nconst Gener
}

func main() {
    src := generateVersionGo("1.2.3")
    // Real workflows write this via //go:generate (stringer, sqlc, mockgen, …).
    fmt.Println(strings.TrimSpace(src))
    fmt.Println("lines:", strings.Count(src, "\n"))
}
```

Expected output:

```
// Code generated by demo; DO NOT EDIT.
package main

const GeneratedVersion = "1.2.3"
lines: 4
```

More examples

`//go:embed` string + FS in one program (needs a local file).

```
printf 'hi' > note.txt
```

```
package main

import (
    "embed"
```

```go
    "fmt"
)

//go:embed note.txt
var note string

//go:embed note.txt
var fs embed.FS

func main() {
    fmt.Println("string:", note)
    b, _ := fs.ReadFile("note.txt")
    fmt.Println("fs:", string(b))
}
```

Expected output:

```
string: hi
fs: hi
```

Runnable example

Save as `main.go` and create a sibling `hello.txt` with the text `embedded hello`. Then:

```bash
go mod init example
printf 'embedded hello' > hello.txt
go run .
```

```go
package main

import (
    "embed"
    "fmt"
    "io/fs"
    "strings"
)

//go:embed hello.txt
var hello string

//go:embed hello.txt
var helloBytes []byte

//go:embed hello.txt
var content embed.FS
```

```go
// version is normally produced by go:generate; pure-Go fallback for this demo.
var version = "dev"

func main() {
    fmt.Println("embed string:", hello)
    fmt.Println("embed bytes:", string(helloBytes))

    b, err := content.ReadFile("hello.txt")
    if err != nil {
        panic(err)
    }
    fmt.Println("embed FS:", string(b))

    // Walk the embedded FS (handy for static assets)
    _ = fs.WalkDir(content, ".", func(path string, d fs.DirEntry, err error) error {
        if err != nil || d.IsDir() {
            return err
        }
        fmt.Println("walk:", path)
        return nil
    })

    // Stand-in for generated code: a tiny template expansion
    generated := strings.ReplaceAll(
        "// Code generated by demo; DO NOT EDIT.\nconst Version = \"{{V}}\"\n",
        "{{V}}",
        version,
    )
    fmt.Println(strings.TrimSpace(generated))
}
```

Expected output:

```
embed string: embedded hello
embed bytes: embedded hello
embed FS: embedded hello
walk: hello.txt
// Code generated by demo; DO NOT EDIT.
const Version = "dev"
```

What to notice: `//go:embed` paths are relative to the source file and bake content into the binary—no runtime file needed after build. Real `go generate` tools (`stringer`, `sqlc`, `mockgen`) write Go you commit or produce in CI; keep generators deterministic.

Try next: Embed a `static/*` directory and serve it with `http.FileServer(http.FS(static))` via `httptest`. Add a `//go:generate echo package main > zzz_gen.go` and run `go generate`.

Related Topics

- Core Go Commands
- Building and Releasing

Documentation and APIs

Documentation and APIs

Overview

Good documentation makes code usable. Go has built-in tools for extracting docs from source comments.

Writing Doc Comments

```go
// Package math provides mathematical utilities.
//
// It includes functions for basic arithmetic and
// statistical calculations.
package math

// Pi represents the mathematical constant  .
const Pi = 3.14159

// Add returns the sum of a and b.
func Add(a, b int) int {
    return a + b
}

// Calculator provides stateful calculations.
type Calculator struct {
    // Result holds the current value.
    Result float64
}
```

Conventions

- Start with the name being documented
- Use complete sentences
- First sentence is the summary

Examples

```go
func ExampleAdd() {
    result := Add(2, 3)
    fmt.Println(result)
    // Output: 5
}
```

Viewing Docs

```bash
go doc fmt
go doc fmt.Println
go doc -all mypackage
```

pkg.go.dev

Public packages are automatically documented at:

`https://pkg.go.dev/github.com/user/package`

API Design Guidelines

```go
// Accept interfaces, return structs
func Process(r io.Reader) (*Result, error)

// Use functional options
func NewServer(opts ...Option) *Server

// Clear error messages
return fmt.Errorf("user %d: %w", id, err)
```

Summary

Tool	Purpose
`go doc`	View documentation
`godoc`	Local doc server
pkg.go.dev	Public package docs
Examples	Testable documentation

Worked example

Example tests as executable docs (`// Output:` checked by `go test`).

Save as `mathdoc.go` and `mathdoc_test.go`. Then:

```
go mod init example
go test -v -run Example
```

```go
// mathdoc.go
package main

// Mul returns a * b.
func Mul(a, b int) int { return a * b }
```

```go
// mathdoc_test.go
package main

import "fmt"

func ExampleMul() {
    fmt.Println(Mul(6, 7))
    // Output: 42
}

func ExampleMul_zero() {
    fmt.Println(Mul(0, 99))
    // Output: 0
}
```

Expected output:

```
=== RUN   ExampleMul
--- PASS: ExampleMul (0.00s)
=== RUN   ExampleMul_zero
--- PASS: ExampleMul_zero (0.00s)
PASS
```

More examples

Godoc-shaped comments you can print with `go doc`.

```go
// Package main demonstrates documentation comments.
//
// Start package comments with "Package <name>".
```

```go
package main

// Answer is the answer to life, the universe, and everything.
const Answer = 42

// Double returns 2 * n.
func Double(n int) int { return n * 2 }

func main() {}
```

```go
go doc Double
```

Runnable example

Save as `docdemo.go` and `docdemo_test.go`. Then:

```go
go mod init example
go test -v
go doc -all .
```

```go
// docdemo.go
// Package docdemo shows godoc-friendly comments and example tests.
//
// Doc comments become the package page on pkg.go.dev.
package main

import "fmt"

// Pi is a truncated mathematical constant for demos.
const Pi = 3.14159

// Add returns the sum of a and b.
//
// It does not check for overflow.
func Add(a, b int) int {
    return a + b
}

// Calculator keeps a running total.
type Calculator struct {
    // Result is the current accumulator value.
    Result int
}
```

```go
// AddTo adds n to the calculator result and returns the new total.
func (c *Calculator) AddTo(n int) int {
    c.Result += n
    return c.Result
}

func main() {
    fmt.Println(Add(2, 3))
}
```

```go
// docdemo_test.go
package main

import "fmt"

func ExampleAdd() {
    fmt.Println(Add(2, 3))
    // Output: 5
}

func ExampleCalculator_AddTo() {
    c := &Calculator{}
    fmt.Println(c.AddTo(4))
    fmt.Println(c.AddTo(6))
    // Output:
    // 4
    // 10
}
```

Expected output:

```
=== RUN   ExampleAdd
--- PASS: ExampleAdd (0.00s)
=== RUN   ExampleCalculator_AddTo
--- PASS: ExampleCalculator_AddTo (0.00s)
PASS
```

`go doc Add` prints something like:

```
package main // import "example"

func Add(a, b int) int
    Add returns the sum of a and b.

    It does not check for overflow.
```

What to notice: Example functions are tests—the `// Output:` block is checked by `go test`. The

first sentence of a doc comment is the summary shown in indexes. Start comments with the name (`Add returns…`).

Try next: Run `go test -run Example`. Add a package comment and view it with `go doc package`.

Related Topics

- Formatting, Vetting, Docs
- Interface Best Practices

Joining the Go Build with -toolexec: Wrappers, Build-Cache Keys, and Compile-Time Instrumentation

Joining the Go Build with `-toolexec`: Wrappers, Build-Cache Keys, and Compile-Time Instrumentation

Thesis

`go build -toolexec=prog` makes the go command run `prog /path/to/compile args...` instead of `compile args...`, and the same for `asm`, `link`, `vet`, `cgo` and the other tools. That one flag is how compile-time instrumentation tools such as OpenTelemetry's `otelc` add tracing to code you never edited, how `toolstash` swaps in a saved compiler, and how you can log exactly what a build does. No compiler patches, no fork of the toolchain.

The wrapper is easy to write. The hard part is the **build cache**. The go command decides whether to run a tool at all by hashing a *tool ID*, and it gets that ID by asking the tool, through your wrapper, to print its version (`compile -V=full`). If a rewriting wrapper passes that line through unchanged, plain and instrumented builds share cache entries. Then an instrumented build can silently produce an uninstrumented binary, and a plain build can silently produce an instrumented one. This chapter shows both on Go 1.27.2, then the fix, then the second rule: code you inject can only import packages the go command already planned to build.

Verified on **Go 1.27.2** (linux/amd64), Linux 6.12, on 2026-10-19. Go source links are pinned to the `go1.27.2` tag.

Learning goals

- Write a `-toolexec` wrapper that passes arguments, standard streams and exit status through correctly
- Read `TOOLEXEC_IMPORTPATH` and the compile command line to decide which packages to touch
- Explain how the go command computes a tool ID from `tool -V=full`, and why the wrapper's own path is not in the cache key
- Reproduce cache sharing between plain and rewritten builds, and fix it by marking the tool ID
- Explain why injected code cannot import a package the build did not already plan, and how real tools set that up
- Decide when `-toolexec` is the right tool, and what to check before you trust one

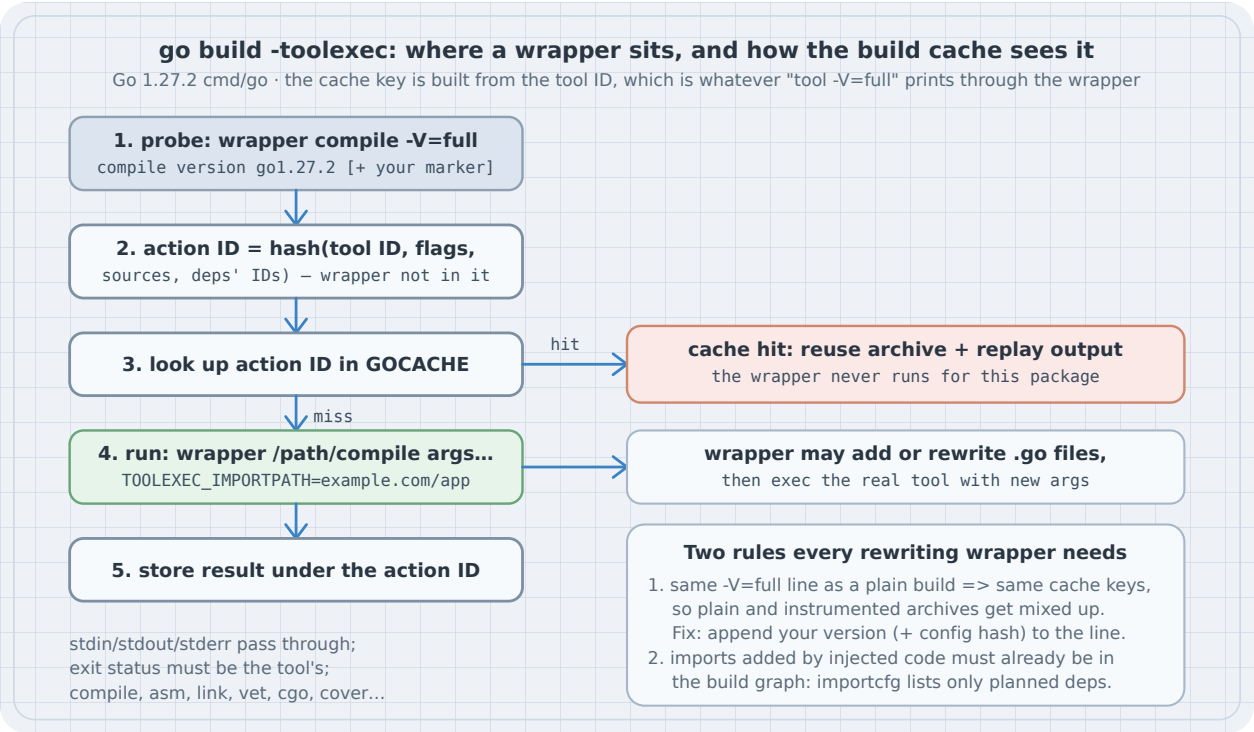


Figure 1: go build -toolexec: the wrapper answers the -V=full probe that becomes the tool ID; the action ID hashes that ID, so on a cache hit the wrapper never runs; on a miss the wrapper runs with the real tool path and may rewrite sources

Mental model

The flag's whole contract, from `go help build`:

> `-toolexec 'cmd args'` a program to use to invoke toolchain programs like vet and asm. For example, instead of running asm, the go command will run 'cmd args /path/to/asm '. The TOOLEXEC_IMPORTPATH environment variable will be set, matching 'go list -f {{.ImportPath}}' for the package being built.

Fact	Consequence
The wrapper gets the real tool's absolute path as its first argument	It can log, rewrite arguments, add files, or run a different binary
The go command first runs `wrapper tool -V=full` and uses the printed line as the **tool ID** (`toolID` in `buildid.go`)	Whatever the wrapper prints there goes into every cache key for that tool
For a release toolchain the tool ID is the whole version line, e.g. `compile version go1.27.2`	A wrapper that echoes it is indistinguishable from no wrapper

Fact	Consequence
The `-toolexec` command line itself is not hashed into the action ID	Changing or adding a wrapper does not invalidate the cache by itself
On a cache hit the go command reuses the archive and replays the tool's saved output	The wrapper is not run for that package at all, not even to log
Source files are the last arguments on a `compile` command line, as absolute paths (`gc.go`)	Appending a file path adds a file to the package
`compile` resolves imports only through the `-importcfg` file the go command wrote	Injected code can import only packages already in this package's build plan

Worked examples

One module holds the program being built and two wrappers. Every command was run from its root, `/tmp/toolexec` on the test machine. `GOCACHE` points at a fresh, empty directory at the start of each case so the results do not depend on earlier builds.

```go
// go.mod
module example.com/toolexec

go 1.27
```

```go
// app/main.go
package main

import (
    "fmt"
    "strings"
)

func greet(name string) string { return "hello, " + strings.ToUpper(name) }

func main() {
    fmt.Println(greet("gopher"))
}
```

Case 1: A wrapper that only watches

```go
// logexec/main.go: a -toolexec wrapper that logs every tool the go
// command runs, then runs it unchanged.
package main

import (
    "fmt"
    "os"
    "os/exec"
    "path/filepath"
    "strings"
)

func main() {
    tool, args := os.Args[1], os.Args[2:]
    pkg := os.Getenv("TOOLEXEC_IMPORTPATH") // set by the go command
    if log := os.Getenv("LOGEXEC_FILE"); log != "" {
        f, err := os.OpenFile(log, os.O_APPEND|os.O_CREATE|os.O_WRONLY, 0o644)
        if err == nil {
            first := ""
            if len(args) > 0 {
                first = args[0]
            }
            fmt.Fprintf(f, "%-8s %-28s %s (%d args)\n", filepath.Base(tool), pkg, first, len
            f.Close()
        }
    }
    cmd := exec.Command(tool, args...)
    cmd.Stdin, cmd.Stdout, cmd.Stderr = os.Stdin, os.Stdout, os.Stderr
    if err := cmd.Run(); err != nil {
        if ee, ok := err.(*exec.ExitError); ok {
            os.Exit(ee.ExitCode())
        }
        fmt.Fprintln(os.Stderr, "logexec:", strings.TrimSpace(err.Error()))
        os.Exit(1)
    }
}
```

The wrapper logs to a file, not to stdout. Stdout belongs to the tool: the go command parses what `compile -V=full` prints, and anything extra there breaks the build. The exit status is passed through so a compile error stays a compile error.

```
$ go build -o bin/logexec ./logexec
$ export GOCACHE=/tmp/toolexec/cache          # new, empty directory
$ LOGEXEC_FILE=$PWD/log.txt go build -toolexec=$PWD/bin/logexec -o bin/app ./app
```

```
$ bin/app
hello, GOPHER
$ wc -l log.txt
117 log.txt
$ awk '{print $1}' log.txt | sort | uniq -c
     54 asm
     61 compile
      2 link
$ grep -c -- '-V=full' log.txt
3
$ head -5 log.txt
compile                                -V=full (1 args)
asm                                    -V=full (1 args)
compile  internal/byteorder            -o (19 args)
compile  internal/goarch               -o (21 args)
compile  internal/godebugs             -o (19 args)
$ grep -E '^(compile|link) +(example|  )' log.txt
compile                                -V=full (1 args)
compile  example.com/toolexec/app      -o (18 args)
link                                   -V=full (1 args)
link     example.com/toolexec/app      -o (8 args)
```

With an empty cache, building a two-function program runs the compiler 61 times and the assembler 54 times: the go command compiles every standard-library package `fmt` and `strings` need, and the wrapper sees all of them. The first three lines of work are `-V=full` probes, one per tool, with no import path. Each package compile comes with `TOOLEXEC_IMPORTPATH` set.

Now build again, and then build without the wrapper:

```
$ rm log.txt
$ LOGEXEC_FILE=$PWD/log.txt go build -toolexec=$PWD/bin/logexec -o bin/app ./app
$ cat log.txt
compile                                -V=full (1 args)
asm                                    -V=full (1 args)
link                                   -V=full (1 args)
$ go build -x -o bin/app ./app 2>&1 | grep -cE '/compile |/link '
0
```

The second wrapped build ran *only* the three probes: every package was a cache hit, so the wrapper never saw a compile. The plain build afterwards ran no compiler and no linker either. It reused what the wrapped build stored. For a wrapper that changes nothing this is ideal. For a wrapper that changes code it is a bug, as the next case shows.

Case 2: A wrapper that changes the code

This wrapper adds one generated file to every compile of package `main`. The file has an `init` that prints a marker, a minimal stand-in for the hooks an instrumentation tool injects. Two switches,

used in Cases 4 and 5, are off by default.

```go
// stampexec/main.go: a -toolexec wrapper that adds one generated file to
// every compile of package main. STAMPEXEC_MARK=1 makes the wrapper report
// its own identity in the tool ID (Case 4); STAMPEXEC_IMPORT=1 injects a
// hook that imports a package (Case 5).
package main

import (
    "bytes"
    "fmt"
    "os"
    "os/exec"
    "path/filepath"
    "slices"
    "strings"
)

const version = "stampexec/v1"

// generated is the extra source file injected into package main.
const generated = `package main

func init() { println("[instrumented by ` + version + `]") }
`

// withImport is the same hook, but it needs a package (time) that the
// program being built might not import (Case 5).
const withImport = `package main

import "time"

func init() { println("[instrumented by ` + version + ` at", time.Now().Year(), "]") }
`

func main() {
    tool, args := os.Args[1], os.Args[2:]
    name := filepath.Base(tool)

    // The go command asks every tool for its version to build cache keys.
    if len(args) == 1 && args[0] == "-V=full" {
        out, err := exec.Command(tool, args...).Output()
        if err != nil {
            os.Exit(1)
        }
        line := string(bytes.TrimSpace(out))
        if os.Getenv("STAMPEXEC_MARK") == "1" {
```

```
            line += " " + version // a different tool ID => different cache keys
        }
        fmt.Println(line)
        return
    }

    if name == "compile" && isMainPackage(args) {
        f := filepath.Join(os.TempDir(), fmt.Sprintf("stampexec-%d.go", os.Getpid()))
        src := generated
        if os.Getenv("STAMPEXEC_IMPORT") == "1" {
            src = withImport
        }
        if err := os.WriteFile(f, []byte(src), 0o644); err != nil {
            fail(err)
        }
        defer os.Remove(f)
        args = append(args, f) // source files come last on the compile line
        fmt.Fprintln(os.Stderr, "stampexec: instrumenting", os.Getenv("TOOLEXEC_IMPORTPATH")
    }

    cmd := exec.Command(tool, args...)
    cmd.Stdin, cmd.Stdout, cmd.Stderr = os.Stdin, os.Stdout, os.Stderr
    if err := cmd.Run(); err != nil {
        if ee, ok := err.(*exec.ExitError); ok {
            os.Exit(ee.ExitCode())
        }
        fail(err)
    }
}

// isMainPackage reports whether a compile command line builds package main
// ("-p main" is how the go command names it).
func isMainPackage(args []string) bool {
    i := slices.Index(args, "-p")
    return i >= 0 && i+1 < len(args) && args[i+1] == "main"
}

func fail(err error) {
    fmt.Fprintln(os.Stderr, "stampexec:", strings.TrimSpace(err.Error()))
    os.Exit(1)
}
```

The important lines:

- `-V=full` probes are answered by running the real tool and printing its line. With `STAMPEXEC_MARK` unset, the line is passed through unchanged, exactly like `logexec`.
- `isMainPackage` reads `-p main` from the compile line. The import path from `TOOLEXEC_IMPORTPATH` would work as well. Both are in the environment or arguments of every compile.

- The generated file is appended after the package's own source files. The compiler does not care where a file lives, only that it declares the same package.

Case 3: Two silent failures from one shared cache

Plain build first, then the rewriting wrapper with the same cache:

```
$ go build -o bin/stampexec ./stampexec
$ export GOCACHE=/tmp/toolexec/cache2          # new, empty directory
$ go build -o bin/app ./app && bin/app
hello, GOPHER
$ go build -toolexec=$PWD/bin/stampexec -o bin/app ./app && bin/app
hello, GOPHER
```

Failure 1: the instrumented build is not instrumented. No error, no warning, no "instrumenting" line. The wrapper answered the probe with `compile version go1.27.2`, so the action ID for `example.com/toolexec/app` was the one the plain build had just stored, and the go command never ran the wrapper for it.

Force a rebuild with `-a` and the wrapper does its job (`sort | uniq -c` only collapses repeated lines):

```
$ go build -a -toolexec=$PWD/bin/stampexec -o bin/app ./app 2>&1 | sort | uniq -c
      1 # example.com/toolexec/app
      1 stampexec: instrumenting example.com/toolexec/app
[instrumented by stampexec/v1]
hello, GOPHER
```

Now a plain build, with no wrapper anywhere on the command line, to a new output file:

```
$ go build -o bin/app-plain ./app && bin/app-plain
# example.com/toolexec/app
stampexec: instrumenting example.com/toolexec/app
[instrumented by stampexec/v1]
hello, GOPHER
```

Failure 2: the plain build is instrumented. It reused the archive the `-a` build stored under the same key, and it even replayed the wrapper's stderr line from the cache, which is the only hint that something is wrong. A developer who tried an instrumentation tool once and then ran `go build` in the same workspace would ship this binary.

Case 4: Mark the tool ID

The fix is in the probe answer. With `STAMPEXEC_MARK=1` the wrapper appends its own version to the version line:

```
$ $(go env GOTOOLDIR)/compile -V=full
compile version go1.27.2
$ STAMPEXEC_MARK=1 bin/stampexec $(go env GOTOOLDIR)/compile -V=full
compile version go1.27.2 stampexec/v1
```

`toolID` accepts that: it checks that the first fields are `compile`, `version` and a version string, and for a release toolchain uses the whole trimmed line. A different line means different action IDs for every package compiled through the wrapper, so plain and instrumented artifacts can no longer meet:

```
$ rm -f bin/app*
$ export GOCACHE=/tmp/toolexec/cache3          # new, empty directory
$ go build -o bin/app ./app && bin/app
hello, GOPHER
$ time (STAMPEXEC_MARK=1 go build -toolexec=$PWD/bin/stampexec -o bin/app-inst ./app)
# example.com/toolexec/app
stampexec: instrumenting example.com/toolexec/app

real    0m4.491s
user    0m7.780s
sys 0m1.086s
$ bin/app-inst
[instrumented by stampexec/v1]
hello, GOPHER
$ time (STAMPEXEC_MARK=1 go build -toolexec=$PWD/bin/stampexec -o bin/app-inst ./app)
# example.com/toolexec/app
stampexec: instrumenting example.com/toolexec/app

real    0m0.036s
user    0m0.056s
sys 0m0.018s
$ bin/app-inst
[instrumented by stampexec/v1]
hello, GOPHER
$ go build -o bin/app ./app && bin/app
hello, GOPHER
```

The first marked build is slow (4.5 s) because *every* package, the whole standard-library closure, is a cache miss under the new tool ID, even though the plain build had just compiled the same packages. After that the instrumented build has its own cached artifacts (the second run takes 36 ms and replays the saved output), and the plain build still gets the plain ones.

Note the `rm -f bin/app*` at the top. The first attempt at this case skipped it, and the "plain" build with a brand-new, empty cache printed `[instrumented by stampexec/v1]`. `bin/app` was still the instrumented binary from Case 3, and its build ID matched the plain action ID, so `go build -o bin/app` decided the existing file was up to date and did not relink it. A poisoned binary can outlive the cache that poisoned it.

`otelc` does the same thing, and also mixes in a hash of its rule set so editing the configuration invalidates cached artifacts. The comment on its `toolVersionLine` function says the marker is there "so the tool ID (and every build cache key derived from it) differs from a plain build", and the probe handler explains that otherwise "instrumentation changes compile output without changing any input go hashes" (see Sources).

Case 5: Injected code can only import what the build planned

With `STAMPEXEC_IMPORT=1` the injected hook imports `time`, which `app` does not import:

```
$ export GOCACHE=/tmp/toolexec/cache4          # new, empty directory
$ STAMPEXEC_MARK=1 STAMPEXEC_IMPORT=1 go build -toolexec=$PWD/bin/stampexec -o bin/app-inst
# example.com/toolexec/app
stampexec: instrumenting example.com/toolexec/app
../stampexec-239428.go:3:8: could not import time (open : no such file or directory)
```

(The generated file lives in `/tmp`, so its path is printed relative to the current directory.) The go command planned the build before running any tool. It wrote an `importcfg` for `app` that maps each import in `app`'s own files to a compiled archive, and `time` is not one of them. The compiler has no other way to find packages; the empty file name in `open :` is what it is left with when `importcfg` has no entry for the import. A wrapper cannot repair this from inside one compile, because the archive for `time` may not have been built at all.

The fix is to put the dependency into the build plan *before* the build: add a file to the package that imports what the hooks will need, so the go command builds those packages and lists them in `importcfg`:

```go
// app/stamp_imports.go: declare, in source, the packages the injected
// hook will need, so the go command builds them and lists them in importcfg.
package main

import _ "time"
```

> **Not run for this chapter.** The build with this file in place was not executed on the test machine. Expected result, from how the go command writes `importcfg`: the build succeeds and `bin/app-inst` prints `[instrumented by stampexec/v1 at <year> ]` before `hello, GOPHER`, because `time` is now an import of `app` and therefore in its `importcfg`.

This is exactly `otelc`'s "Phase 1: Setup Dependencies". Its implementation notes describe running `go build -n` to collect the dependency list, generating a file such as `otel_importer.go` with blank imports of the SDK and hook packages, and running `go mod tidy` so the new modules are in `go.mod` and the module cache. Only then does "Phase 2: Building with Instrumentation" run the build with `-toolexec`. For third-party packages a blank import alone is not enough: the module must also be required in `go.mod`, which is why the setup phase edits it.

Case 6: When to reach for `-toolexec`

You want to	Use
See what a build runs	`go build -x`, or `-n` for a dry run. A logging wrapper only adds `TOOLEXEC_IMPORTPATH` and timing.
Change compiler flags for some packages	`-gcflags=pattern=flags` (for example `-gcflags=example.com/app/...=-m`)
Replace one file's contents for a build	`-overlay=file.json`
Add code to packages you do not own, at build time, for every build of a service	`-toolexec`, via a maintained tool (`otelc`, Datadog's Orchestrion), not a hand-rolled wrapper
Compare a patched compiler with the installed one	`toolstash` with `-toolexec` (the use case `toolID`'s comment is written around)

The trap

- **Echoing `-V=full` unchanged from a wrapper that rewrites code.** Plain and rewritten builds then share cache keys. You get uninstrumented "instrumented" builds and instrumented "plain" builds, with no error (Case 3). Append a marker that changes whenever the wrapper's output would change: its version, plus a hash of its configuration.
- **Testing a wrapper against a warm cache.** A hit means the wrapper never ran. Use a fresh `GOCACHE`, or `-a`, when you test one, and delete old output binaries too (Case 4).
- **Printing to stdout from a wrapper.** The go command parses the `-V=full` answer from stdout, and some tools write their real output there. Log to stderr or a file.
- **Swallowing the tool's exit status.** A wrapper that exits 0 after a failed compile produces confusing failures several steps later. Pass the status through.
- **Injecting imports at compile time.** Only packages in the `importcfg` the go command wrote can be imported (Case 5). Declare them in source, and in `go.mod`, before the build.
- **Accepting `-toolexec` from outside.** It runs an arbitrary program for every tool invocation. A `GOFLAGS=-toolexec=...` in a CI environment, a Makefile or an editor setting you did not write is code execution. Review it like a dependency.

The boring rule

Use `-x` and `-n` to watch a build and `-gcflags` or `-overlay` to change it. Use `-toolexec` only through a maintained tool, and before you adopt one check two things: that its `-V=full` answer carries its own version and configuration hash, and that it adds its dependencies to the build plan

before building. Run its builds with a separate `GOCACHE` in CI anyway. It costs one cold build and makes it impossible for an instrumented artifact to reach a plain build.

Try this

1. Add timing to `logexec` (start time and duration per tool run) and find the slowest package compile in a cold build of one of your own services.
2. Run `go vet -toolexec=$PWD/bin/logexec ./app` with `LOGEXEC_FILE` set. Which tool names appear, and does `vet` get its own `-V=full` probe?
3. Change `version` in `stampexec` to `stampexec/v2` without setting `STAMPEXEC_MARK`, rebuild the wrapper, and run a wrapped build against the cache from Case 3. Which binary do you get, and why does the wrapper's own rebuild not matter?
4. Replace the appended file with a real rewrite: parse `app`'s `main.go` with `go/parser`, add `defer println("leave greet")` at the top of `greet`, print it with `go/printer` into a temporary file, and pass that file instead of the original. What must you do so error messages still point at the user's line numbers? (Hint: `//line` directives.)
5. Read `otelc`'s troubleshooting guide and list which of the traps above it detects for you.

Sources

Go documentation and source at the `go1.27.2` tag:

- `go help build` (`-toolexec`, `-a`, `-x`, `-n`, `-gcflags`, `-overlay`): https://pkg.go.dev/cmd/go#hdr-Compile_packages_and_dependencies
- Build cache (`GOCACHE`): https://pkg.go.dev/cmd/go#hdr-Build_and_test_caching
- `toolID` and the `-V=full` probe: https://github.com/golang/go/blob/go1.27.2/src/cmd/go/internal/work/buildid.go#L115-L182
- Compile command line, including `-importcfg` and the trailing source files: https://github.com/golang/go/blob/go1.27.2/src/cmd/go/internal/work/gc.go#L136-L178
- `TOOLEXEC_IMPORTPATH` in the environment of every tool run: https://github.com/golang/go/blob/go1.27.2/src/cmd/go/internal/work/shell.go#L639-L646
- `toolstash`: https://pkg.go.dev/golang.org/x/tools/cmd/toolstash

OpenTelemetry Go compile-time instrumentation (`otelc`), at commit `4ffef19` (2026-10-17):

- Project and quick start: https://github.com/open-telemetry/opentelemetry-go-compile-instrumentation
- Implementation notes (Phase 1 setup, Phase 2 `-toolexec` build): https://github.com/open-telemetry/opentelemetry-go-compile-instrumentation/blob/main/docs/implementation.md
- `toolVersionLine` / `interceptToolVersion` (tool-ID marker): https://github.com/open-telemetry/opentelemetry-go-compile-instrumentation/blob/main/tool/internal/instrument/toolexec.go
- Troubleshooting: https://github.com/open-telemetry/opentelemetry-go-compile-instrumentation/blob/main/docs/troubleshooting.md
- User documentation: https://opentelemetry.io/docs/zero-code/go/compile-time/

- Datadog Orchestrion (another `-toolexec`-based instrumentation tool): https://github.com/DataDog/orchestrion

Tools: Go 1.27.2 (`go build`, `go env GOTOOLDIR`), no third-party modules.

The pkg.go.dev API: querying module metadata from Go programs

The pkg.go.dev API: querying module metadata from Go programs

After this page you can query pkg.go.dev from scripts and Go programs instead of scraping its HTML. That covers package and module metadata, version lists with deprecation and retraction flags, symbols, importers, search, and known vulnerabilities. You will build a small `depaudit` tool that checks every requirement in a `go.mod`, and handle the API's four sharp edges: ambiguous package paths, opaque pagination tokens, server-side filters, and the rate limit.

Thesis

The API was announced on the Go blog in May 2026 under `/v1beta`, and the reference page now documents it under `/v1`. A request to a `/v1beta` path returns `301 Moved Permanently` to the `/v1` path, so new code should use `/v1` directly:

```
curl -sI https://pkg.go.dev/v1beta/package/github.com/google/go-cmp/cmp | grep -iE '^(HTTP|l
```

```
HTTP/2 301
location: /v1/package/github.com/google/go-cmp/cmp
```

It's GET-only, stateless, and returns JSON (cached with `Cache-Control: public, max-age=3600`). The one design choice that will surprise you is "precision over convenience": the web UI guesses which module you meant, and the API refuses to.

Mental model

The routes are `package`, `module`, `versions`, `packages`, `search`, `symbols`, `imported-by` and `vulns`. Except for `search`, the rest of the URL is a module or package path. Everything else goes in query parameters:

- `version`: a semantic version, `latest`, or the default branch names `master`/`main`, which resolve to a pseudo-version. Other branch names aren't accepted. Latest is the default.
- `module`: disambiguates a package path that more than one module provides.
- `limit` and `token`: pagination. To get the next page, send the *same* request plus `token=<nextPageToken>`. Changing anything else may fail. A non-empty token means there's another page, even if the current page is empty.
- `filter`: a small Go-expression language evaluated on the server (`==`, comparisons, `contains`, `hasPrefix`, `hasSuffix`, `matches`, and field selection such as `importedByCounts.packages`).

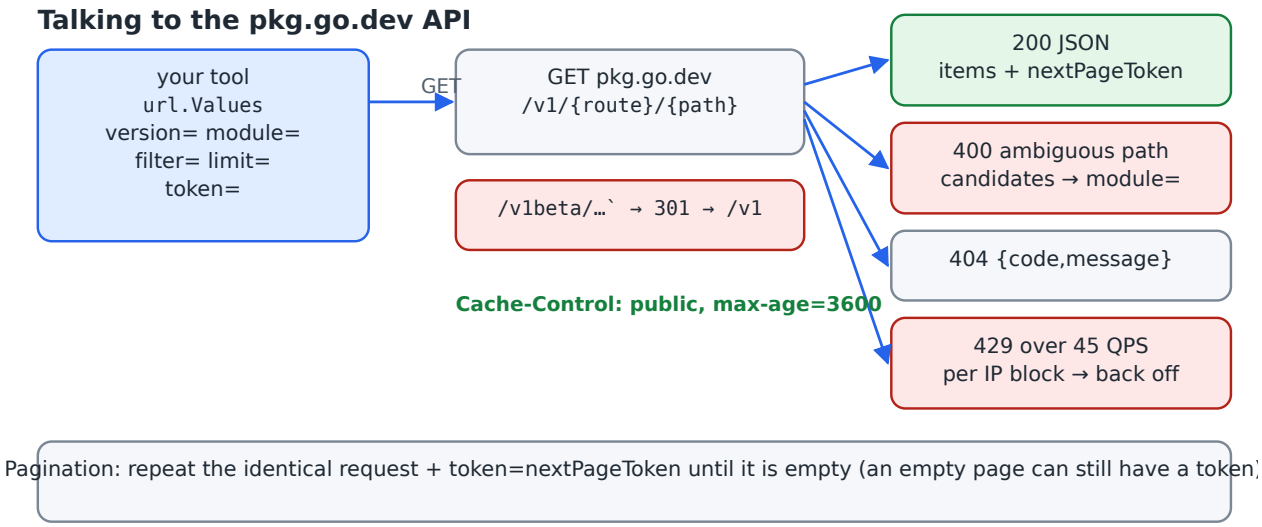


Figure 1: A Go client sends GET /v1/route/path with version, module, filter, limit and token parameters; /v1beta redirects to /v1; responses are 200 JSON with items and nextPageToken, 400 with candidates for an ambiguous path, 404 JSON errors, or 429 above 45 QPS per IP block

Raw requests

A package, at latest and at the default branch:

```
curl -s https://pkg.go.dev/v1/package/github.com/google/go-cmp/cmp
curl -s "https://pkg.go.dev/v1/package/github.com/google/go-cmp/cmp?version=master"
```

```
{"modulePath":"github.com/google/go-cmp","version":"v0.7.0","isLatest":true,"isStandardLibra
{"modulePath":"github.com/google/go-cmp","version":"v0.7.1-0.20261029223239-5deca1c7dab7","i
```

An ambiguous path. `cloud.google.com/go/storage` is a package in its own module *and* a directory path inside `cloud.google.com/go`:

```
curl -s https://pkg.go.dev/v1/package/cloud.google.com/go/storage
curl -s "https://pkg.go.dev/v1/package/cloud.google.com/go/storage?module=cloud.google.com/g
```

```
{"code":400,"message":"ambiguous package path","fixes":["retry the call with a 'module' quer
{"modulePath":"cloud.google.com/go/storage","version":"v1.70.0","isLatest":true,"isStandardL
```

A server-side filter that lists the commands in `golang.org/x/tools`. Filters must be percent-encoded:

```
f='name == "main" && hasPrefix(path, "golang.org/x/tools/cmd/")'
curl -s "https://pkg.go.dev/v1/packages/golang.org/x/tools?filter=$(python3 -c 'import sys,u
```

```
{"modulePath":"golang.org/x/tools","version":"v0.52.0","isStandardLibrary":false,"packages":
```

The response shapes differ. `versions`, `vulns` and `search` return `{"items", "total", "nextPageToken"}` at the top level, while `packages`, `symbols` and `imported-by` wrap that page in a field (`packages`, `symbols`, `importedBy`) next to the module and version. A generic page decoder only fits the first group as-is.

A Go client: `depaudit`

The tool reads a `go.mod` with `go mod edit -json`, which parses the file without downloading anything. For each requirement, it asks the `versions` route for exactly that version, using a `filter` so the server picks the one row. That row carries `latestVersion`, `deprecated` and `retracted`. It then lists `vulns` at that version. It pages with an `iter.Seq2`, builds every query with `url.Values` as the API docs recommend, retries 429s with backoff, and decodes the JSON error body into a typed error. Verified with **go1.27.2 linux/amd64** (`errors.AsType` needs Go 1.26+).

```go
// depaudit reports, for each requirement in a go.mod, the latest version,
// deprecation/retraction status, and known vulnerabilities, using the
// pkg.go.dev API (https://pkg.go.dev/api).
package main

import (
    "context"
    "encoding/json"
    "errors"
    "fmt"
    "iter"
    "net/http"
    "net/url"
    "os"
    "os/exec"
    "strconv"
    "time"
)

const base = "https://pkg.go.dev/v1/"

// apiError is the JSON error body the API returns with non-2xx statuses.
type apiError struct {
    Status     int
    Code       int      `json:"code"`
    Message    string   `json:"message"`
```

```
    Fixes      []string `json:"fixes"`
    Candidates []struct {
        ModulePath  string `json:"modulePath"`
        PackagePath string `json:"packagePath"`
    } `json:"candidates"`
}

func (e *apiError) Error() string { return fmt.Sprintf("pkg.go.dev %d: %s", e.Status, e.Mess

type client struct{ http *http.Client }

// get fetches base+route?q into v, retrying on 429 with backoff.
func (c *client) get(ctx context.Context, route string, q url.Values, v any) error {
    u := base + route + "?" + q.Encode()
    for attempt := 0; ; attempt++ {
        req, err := http.NewRequestWithContext(ctx, "GET", u, nil)
        if err != nil {
            return err
        }
        resp, err := c.http.Do(req)
        if err != nil {
            return err
        }
        if resp.StatusCode == http.StatusTooManyRequests && attempt < 4 {
            resp.Body.Close()
            wait := time.Duration(1<<attempt) * 250 * time.Millisecond
            if s, err := strconv.Atoi(resp.Header.Get("Retry-After")); err == nil {
                wait = time.Duration(s) * time.Second
            }
            select {
            case <-time.After(wait):
                continue
            case <-ctx.Done():
                return ctx.Err()
            }
        }
        defer resp.Body.Close()
        if resp.StatusCode != http.StatusOK {
            e := &apiError{Status: resp.StatusCode}
            json.NewDecoder(resp.Body).Decode(e)
            return e
        }
        return json.NewDecoder(resp.Body).Decode(v)
    }
}
```

```
type page[T any] struct {
    Items         []T    `json:"items"`
    NextPageToken string `json:"nextPageToken"`
}

// all walks every page of a paginated route. Per the API docs, the request is
// repeated verbatim with only "token" added, and an empty page with a
// non-empty token still means there is more.
func all[T any](ctx context.Context, c *client, route string, q url.Values) iter.Seq2[T, err
    return func(yield func(T, error) bool) {
        q := maps(q)
        for {
            var p page[T]
            if err := c.get(ctx, route, q, &p); err != nil {
                var zero T
                yield(zero, err)
                return
            }
            for _, it := range p.Items {
                if !yield(it, nil) {
                    return
                }
            }
            if p.NextPageToken == "" {
                return
            }
            q.Set("token", p.NextPageToken)
        }
    }
}

func maps(q url.Values) url.Values {
    out := url.Values{}
    for k, v := range q {
        out[k] = append([]string(nil), v...)
    }
    return out
}

type version struct {
    Version           string `json:"version"`
    LatestVersion     string `json:"latestVersion"`
    Deprecated        bool   `json:"deprecated"`
    DeprecationReason string `json:"deprecationReason"`
    Retracted         bool   `json:"retracted"`
    RetractionReason  string `json:"retractionReason"`
}
```

```
type vuln struct {
    ID      string `json:"id"`
    Details string `json:"details"`
}

type gomod struct {
    Require []struct {
        Path, Version string
        Indirect      bool
    }
}

func main() {
    if len(os.Args) != 2 {
        fmt.Fprintln(os.Stderr, "usage: depaudit path/to/go.mod")
        os.Exit(2)
    }
    out, err := exec.Command("go", "mod", "edit", "-json", os.Args[1]).Output()
    if err != nil {
        fmt.Fprintln(os.Stderr, "go mod edit:", err)
        os.Exit(1)
    }
    var mf gomod
    if err := json.Unmarshal(out, &mf); err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }
    ctx, cancel := context.WithTimeout(context.Background(), 60*time.Second)
    defer cancel()
    c := &client{http: &http.Client{Timeout: 20 * time.Second}}
    bad := 0
    for _, r := range mf.Require {
        // One exact item from the versions route, selected server-side by a filter.
        q := url.Values{"filter": {fmt.Sprintf("version == %q", r.Version)}, "limit": {"1"}}
        var info *version
        for v, err := range all[version](ctx, c, "versions/"+r.Path, q) {
            if err != nil {
                fmt.Printf("%-26s %-8s error: %v\n", r.Path, r.Version, err)
                break
            }
            info = &v
            break
        }
        if info == nil {
            bad++
            continue
```

```
        }
        var ids []string
        for v, err := range all[vuln](ctx, c, "vulns/"+r.Path, url.Values{"version": {r.Vers
            if err != nil {
                ids = append(ids, "error: "+err.Error())
                break
            }
            ids = append(ids, v.ID)
        }
        status := "ok"
        switch {
        case info.Retracted:
            status = "RETRACTED: " + info.RetractionReason
        case info.Deprecated:
            status = "DEPRECATED: " + info.DeprecationReason
        }
        if info.Retracted || info.Deprecated || len(ids) > 0 {
            bad++
        }
        fmt.Printf("%-26s %-8s latest %-8s %s, %d vulns %v\n", r.Path, r.Version, info.Lates
    }
    // Ambiguous package paths come back as 400 with candidates.
    var p map[string]any
    err = c.get(ctx, "package/cloud.google.com/go/storage", nil, &p)
    if ae, ok := errors.AsType[*apiError](err); ok {
        fmt.Printf("\npackage cloud.google.com/go/storage: %s; candidates:", ae.Message)
        for _, cand := range ae.Candidates {
            fmt.Printf(" %s", cand.ModulePath)
        }
        fmt.Println()
    }
    if bad > 0 {
        os.Exit(1)
    }
}

func first(s []string, n int) []string {
    if len(s) > n {
        return append(s[:n:n], "…")
    }
    return s
}
```

The `go.mod` under test:

```
module example.com/app

go 1.27

require (
    github.com/google/go-cmp v0.5.9
    github.com/google/uuid v1.6.0
    golang.org/x/net v0.20.0
    github.com/golang/protobuf v1.5.4
)
```

Run on 2026-10-31 (results change as modules release and the vulnerability database grows):

```
go vet . && go build -o depaudit . && ./depaudit sample.mod; echo "exit=$?"
```

```
github.com/google/go-cmp   v0.5.9   latest v0.7.0   ok, 0 vulns []
github.com/google/uuid     v1.6.0   latest v1.6.0   ok, 0 vulns []
golang.org/x/net           v0.20.0  latest v0.62.0  ok, 21 vulns [GO-2024-2687 GO-2024-3333
github.com/golang/protobuf v1.5.4   latest v1.5.4   DEPRECATED: Use the "google.golang.org/p

package cloud.google.com/go/storage: ambiguous package path; candidates: cloud.google.com/go
exit=1
```

The deprecation comes from the `// Deprecated:` comment in that module's `go.mod`, and `vulns` reads from the Go vulnerability database (vuln.go.dev). `go list -m -u` can report upgrades and deprecations too, but it needs the module graph downloaded and doesn't report vulnerabilities. Here a single `go.mod` file is enough.

Things that bite

- **Ambiguity is an error, not a guess.** `go mod tidy` and the web UI use the longest matching module path. The API returns 400 with `candidates`. A tool that already knows the module from `go.mod` should always send `module=`.
- **Tokens are bound to the request.** Change `limit` or `filter` between pages and the token may be rejected. Clone the query, then only `Set("token", …)`.
- **`vulns` defaults to latest.** Without `version=`, `golang.org/x/net` returned `{"items":null,"total":0}` in this run, because the latest release has no known vulnerabilities. Always pass the version you actually depend on.
- **Rate limit.** 45 queries per second per IP block, after which the API returns 429. CI fleets behind one NAT share that budget, so cache responses (the server already marks them cacheable for an hour) and run requests one at a time or with a small limit.
- **This is not `govulncheck`.** The `vulns` route works at module and version level. `govulncheck` checks whether your code actually reaches the vulnerable symbols. Use the API for inventory and triage, and `govulncheck` for the final decision.

Try this

1. Add `-module` handling to `depaudit`: on an `apiError` with candidates, retry once with the candidate whose `modulePath` appears in `go.mod`.
2. Use `search` with `filter=importedByCounts.packages > 100` to find widely used alternatives to a deprecated module.
3. Put a file-backed cache in front of `client.get` keyed by the full URL, honouring `max-age`, and confirm with a counter that a second run makes no network requests.

Sources

- Go blog, "Introducing the pkg.go.dev API" (21 May 2026): https://go.dev/blog/pkgsite-api
- pkg.go.dev API reference (routes, filters, pagination, rate limiting): https://pkg.go.dev/api
- OpenAPI specification: https://pkg.go.dev/v1/openapi.yaml
- `go mod edit -json`: https://go.dev/ref/mod#go-mod-edit
- Module deprecation and retraction: https://go.dev/ref/mod#go-mod-file-module-deprecation
- Go vulnerability database: https://vuln.go.dev, and `govulncheck`: https://go.dev/doc/tutorial/govulncheck
- `iter` package: https://pkg.go.dev/iter

go fix modernizers: atomictypes, slices.Backward, embedlit, unsafe.Add, WaitGroup.Go

go fix modernizers: atomictypes, slices.Backward, embedlit, unsafe.Add, WaitGroup.Go

After this page you will be able to run `go fix` across your codebase to automatically rewrite legacy patterns into modern Go 1.27 idioms. You will understand how the modernizer analyzer suite operates, how to inspect planned rewrites with diffs before committing them, and how specific rules refactor atomic operations, reverse slice iterations, struct literal initializations, unsafe pointer arithmetic, and concurrency control.

Thesis

Go 1.27 expands `go fix` from a tool historically used for major language transitions into a general-purpose modernizer tool driven by static analysis analyzers. By leveraging the built-in modernizers—such as `atomictypes`, `slicesbackward`, `embedlit`, `unsafefuncs`, and `waitgroupgo`—developers can safely eliminate verbose, error-prone boilerplate across large codebases in a single command. Running `go fix -diff .` previews precise unified diffs, making automated code refactoring standard, repeatable, and safe across modern Go projects.

Mental model of go fix vs go vet

It is helpful to distinguish `go fix` from `go vet`. While both rely on AST (Abstract Syntax Tree) and type checking infrastructure under the hood, their targets and side effects differ fundamentally:

Feature	`go vet`	`go fix`
Primary Goal	Report potential bugs and suspicious constructs	Automatically rewrite code to modern idioms and APIs
Execution Safety	Read-only analysis	Modifies source files on disk (or outputs diffs with `-diff`)
Target Scope	Correctness issues (e.g., unreachable code, unkeyed struct literals)	API modernization, deprecation cleanup, syntax modernization
Inspection Mode	Console diagnostics	Unified diffs via `go fix -diff .`
Invocation	`go vet ./...`	`go fix ./...` or `go tool fix help`

While `go vet` warns you when something is likely broken, `go fix` updates valid, older idiomatic code so it uses current language constructs and standard library features.

Worked cases

The following five examples illustrate the exact rewrites performed by Go 1.27's key modernizer analyzers.

1. Modernizing atomic types (`atomictypes`)

Before Go 1.19, atomic operations on primitive types relied on functions like `atomic.AddInt64` or `atomic.LoadInt64` operating on raw integers via pointers. The `atomictypes` analyzer converts primitive variables paired with `sync/atomic` function calls into typed `atomic.Int64` fields and methods.

```go
package main

import "sync/atomic"

var n int64

func Inc()        { atomic.AddInt64(&n, 1) }
func Load() int64 { return atomic.LoadInt64(&n) }
```

2. Embedded struct initializers (`embedlit`)

Prior to Go 1.27, initializing embedded struct fields required explicitly prefixing the embedded type name in compound literals. The `embedlit` analyzer simplifies promoted-field literal initialization by omitting redundant outer field wrappers when initializing promoted fields.

```go
package main

type U struct{ X int }
type T struct{ U }

func NewT() T {
    return T{U: U{X: 1}}
}
```

3. Iterating slices in reverse (`slicesbackward`)

Reversing through a slice using traditional index arithmetic (`for i := len(s) - 1; i >= 0; i--`) is standard but prone to off-by-one errors. The `slicesbackward` analyzer replaces manual decrement loops with Go's range functions over `slices.Backward(s)`.

```go
package main
```

```go
func SumBackward(s []int) int {
    sum := 0
    for i := len(s) - 1; i >= 0; i-- {
        sum += s[i]
    }
    return sum
}
```

4. Unsafe pointer arithmetic (`unsafefuncs`)

Performing pointer arithmetic historically required explicitly converting `unsafe.Pointer` to `uintptr`, adding an offset, and converting back to `unsafe.Pointer`. The `unsafefuncs` analyzer updates raw `uintptr` pointer addition to use `unsafe.Add`.

```go
package main

import "unsafe"

func AddBytes(p unsafe.Pointer, n int) unsafe.Pointer {
    return unsafe.Pointer(uintptr(p) + uintptr(n))
}
```

5. WaitGroup closure launches (`waitgroupgo`)

A common pattern for running tasks concurrently involves incrementing a `sync.WaitGroup` with `.Add(1)`, launching a goroutine, and deferring `.Done()`. The `waitgroupgo` analyzer refactors this idiom into the cleaner `wg.Go(...)` method paired with integer range loops (`for range n`).

```go
package main

import "sync"

func Run(n int, f func()) {
    var wg sync.WaitGroup
    for i := 0; i < n; i++ {
        wg.Add(1)
        go func() {
            defer wg.Done()
            f()
        }()
    }
    wg.Wait()
}
```

Executing `go fix` across the project

Running `go fix -diff .` inside a package directory containing these five files previews all modernizations across the module in unified diff format.

Execute the command to generate the diff preview:

```
go fix -diff .
```

```
<<OUTPUT:diff1>>
```

How to select analyzers

You can inspect the full list of modernizers and analyzer flags using `go tool fix help`.

```
go tool fix help
```

```
<<OUTPUT:help1>>
```

In addition to the five primary worked cases (`atomictypes`, `embedlit`, `slicesbackward`, `unsafefuncs`, `waitgroupgo`), Go 1.27 includes additional specialized modernizers:

- `omitzero`: Rewrites struct tag omitempty usages to modern Go struct field tags where applicable.
- `mapsloop`: Simplifies custom map iteration logic using `maps` package iterators.
- `stringsseq`: Refactors string splitting and searching logic into `seq` iterator pipelines.
- `errorsastype`: Updates legacy `errors.As` boilerplate to modern type assertions or typed helper functions.
- `testingcontext`: Automatically passes `t.Context()` into subtests and helper functions requiring standard contexts.

To run specific modernizer analyzers or target precise packages:

```
# Run fix on all subpackages in the current module
go fix ./...

# Preview changes without modifying disk files
go fix -diff ./...
```

Common mistakes

- **Applying changes without checking git status**: Running `go fix .` directly overwrites files on disk. Always run `go fix -diff .` first or ensure your git working directory is clean.

- **Assuming `go fix` updates module requirements**: `go fix` modifies source code expressions, but if standard library additions (like `slices.Backward` or `unsafe.Add`) require a higher `go` directive in `go.mod`, you must update `go.mod` using `go mod edit -go=1.27`.
- **Mixing manual AST edits with automated fixes**: Avoid running custom automated refactoring scripts concurrently with `go fix` to prevent conflicting text edits on the same source files.

Exercises

1. Create a workspace with a package using `atomic.AddInt64` and run `go fix -diff .` to inspect how `sync/atomic` variable declarations are transformed.
2. Refactor a repository containing classic decrementing loops (`for i := len(b) - 1; i >= 0; i--`) and verify that `slicesbackward` automatically imports the `slices` package.
3. Inspect `go tool fix help` and test running `go fix` on test files to identify where `testingcontext` replaces manual context instantiation.

Key takeaways

- `go fix` in Go 1.27 acts as a comprehensive code modernizer powered by static analysis analyzers.
- Using `-diff` allows you to safely review automated refactorings before applying them to disk.
- Analyzers like `atomictypes`, `embedlit`, `slicesbackward`, `unsafefuncs`, and `waitgroupgo` eliminate legacy Go idioms and replace them with cleaner, modern standard library equivalents.

Verified `go fix -diff` output

Captured on Go 1.27.2 linux/amd64:

```
--- /home/box/agent-data/books-scout/runs/2026-11-06-runnow/lab/go-fix/atom.go (old)
+++ /home/box/agent-data/books-scout/runs/2026-11-06-runnow/lab/go-fix/atom.go (new)
@@ -2,7 +2,7 @@

 import "sync/atomic"

-var n int64
+var n atomic.Int64

-func Inc() { atomic.AddInt64(&n, 1) }
-func Load() int64 { return atomic.LoadInt64(&n) }
+func Inc()        { n.Add(1) }
+func Load() int64 { return n.Load() }
--- /home/box/agent-data/books-scout/runs/2026-11-06-runnow/lab/go-fix/embed.go (old)
```

```
+++ /home/box/agent-data/books-scout/runs/2026-11-06-runnow/lab/go-fix/embed.go (new)
@@ -4,5 +4,5 @@
 type T struct{ U }

 func NewT() T {
-   return T{U: U{X: 1}}
+   return T{X: 1}
 }
--- /home/box/agent-data/books-scout/runs/2026-11-06-runnow/lab/go-fix/slices.go (old)
+++ /home/box/agent-data/books-scout/runs/2026-11-06-runnow/lab/go-fix/slices.go (new)
@@ -1,9 +1,11 @@
 package main

+import "slices"
+
 func SumBackward(s []int) int {
    sum := 0
-   for i := len(s) - 1; i >= 0; i-- {
-       sum += s[i]
+   for _, v := range slices.Backward(s) {
+       sum += v
    }
    return sum
 }
--- /home/box/agent-data/books-scout/runs/2026-11-06-runnow/lab/go-fix/unsafe.go (old)
+++ /home/box/agent-data/books-scout/runs/2026-11-06-runnow/lab/go-fix/unsafe.go (new)
@@ -3,5 +3,5 @@
 import "unsafe"

 func AddBytes(p unsafe.Pointer, n int) unsafe.Pointer {
-   return unsafe.Pointer(uintptr(p) + uintptr(n))
+   return unsafe.Add(p, n)
 }
--- /home/box/agent-data/books-scout/runs/2026-11-06-runnow/lab/go-fix/wg.go (old)
+++ /home/box/agent-data/books-scout/runs/2026-11-06-runnow/lab/go-fix/wg.go (new)
@@ -4,12 +4,10 @@

 func Run(n int, f func()) {
    var wg sync.WaitGroup
-   for i := 0; i < n; i++ {
-       wg.Add(1)
-       go func() {
-           defer wg.Done()
+   for range n {
+       wg.Go(func() {
            f()
-       }()
```

```
+       })
    }
    wg.Wait()
 }
```

Sources

- Go 1.27 Release Notes
- Command fix Documentation
- Updating packages to use new APIs

Performance Engineering

Performance Engineering Overview

Performance Engineering Overview

This part frames performance as an **evidence-driven workflow**. The primary rule is simple: measure first, optimize second. Guessing hotspots is a reliable way to complicate code without moving p99.

Why / Overview

Go is fast enough for most network services out of the box. Performance work becomes necessary when:

- Tail latency violates SLOs
- Allocation rate drives GC pauses under load
- Lock contention caps throughput below CPU saturation
- A deploy regresses a critical path

```
baseline -> profile -> hypothesize -> change one thing -> remeasure -> keep/revert
```

Learning Path

Chapter	Focus	Outcome
181 Benchmarking & profiling	bench, pprof, trace workflow	Find real hotspots
182 Memory, allocs, GC	Escape analysis, pooling, GC knobs	Control allocation pressure
183 Contention & concurrency	Mutex profiles, sharding	Raise throughput ceilings
184 Load testing & baselines	hey/vegeta, capacity	Evidence before optimize
185 Allocation recipes	Hot-path fixes	Lower allocs with discipline
186 PGO & build flags	-pgo, trimpath, ldflags	Production binary polish
186c compress/flate rewrite	Go 1.27 encoder	Speed vs size; pin uncompressed hashes
187 Project: optimize endpoint	Full loop	Before/after report

Core Principles

1. **User-visible metrics first** — p99 latency, error rate, throughput.
2. **Reproduce with stable benchmarks or load tests** before micro-tweaks.

3. **One variable per experiment**.
4. **Prefer algorithmic fixes** over micro-optimizations.
5. **Complexity costs** — pooling and sharding need proof.
6. **Regressions are bugs** — keep benchmarks in CI for critical paths.

Go Tooling Map

Tool	Question
`testing.B`	How fast is this function?
`pprof` CPU	Where is CPU time?
`pprof` heap	Where are allocations?
`pprof` mutex/block	Who waits?
`go tool trace`	Scheduler, GC, latency gaps
`benchstat`	Are two runs different?
metrics / APM	Production truth

Relationship to Other Parts

- Observability detects symptoms; this part deepens **local diagnosis**.
- Concurrency chapters supply patterns you will tune here.
- Network timeouts interact with performance (slow is not always "needs pprof").

Literacy Checklist

☐ Write a meaningful benchmark with stable inputs
☐ Capture and read a CPU profile flame graph
☐ Read allocs/op and B/op from benches
☐ Identify GC pressure symptoms
☐ Diagnose mutex contention with profiles
☐ Use `benchstat` for before/after
☐ Know when to stop optimizing

Part Warm-Up Exercises

1. List your top three latency SLOs and whether you have benchmarks for them.
2. Enable pprof on a dev binary; open `/debug/pprof/`.
3. Run `go test -bench=. -benchmem` on a module you own.
4. Recall a past "optimization" that did nothing; what was missing?
5. Note GOMAXPROCS and container CPU limits in your deploy env.

Performance vs Correctness vs Cost

Optimize in this priority order unless the product says otherwise:

1. **Correctness** under concurrency (`-race`, invariants)
2. **User SLOs** (latency, errors)
3. **Efficiency** (CPU, memory, $)
4. **Microbenchmark bragging rights**

A faster wrong answer is still wrong. A 2% CPU win that adds a subtle race is a net loss.

When Not to Optimize

- You have no measurement of the problem
- The code is not on a hot path (profile says <1%)
- Clarity loss is large and gain is within noise
- The real bottleneck is a dependency you do not control (fix timeouts/caching instead)

"Premature optimization" is not "never optimize" — it is "optimize without evidence."

Environment Parity

Benchmarks lie when:

Local	Production
8 fast cores	2 throttled vCPU
NVMe disk	network block volume
empty cache	multi-tenant noisy neighbor
`-race` off always	sometimes on in CI only

Record `GOMAXPROCS`, CPU quota, and Go version next to every performance claim.

Suggested Lab Sequence

1. Write a deliberately slow handler (JSON + lock + alloc)
2. Load test; capture CPU + mutex + heap profiles
3. Fix the top issue only; remeasure with benchstat / p99
4. Repeat once more; stop when SLO is met

That sequence is the entire part in miniature.

Regression Guardrails

Protect critical paths the way you protect correctness:

```
# local loop
go test -bench=BenchmarkHotPath -benchmem -count=10 ./internal/hot > new.txt
benchstat old.txt new.txt
```

In CI, store a baseline for a small set of benches and fail on large regressions (with noise tolerance). Flaky performance CI is worse than none — invest in stable inputs and warm-up.

Interaction with Timeouts

A "performance" incident is sometimes a **timeout budget** incident:

- Dependency p99 is 400ms; you budget 200ms → "Go is slow" tickets
- Retries multiply latency → p99 explodes without CPU hotspots

Always check SLOs, budgets, and dependency health **before** rewriting algorithms.

Team Norms Worth Adopting

1. Performance claims require numbers (benchstat or load p99).
2. Profiles attached to optimization PRs.
3. Prefer readability until a profile forces otherwise.
4. Document `GOMAXPROCS` / limits when quoting throughput.
5. Revisit pools and shards annually — complexity expires.

Mapping Symptoms → Chapter

Symptom	Start here
"Need numbers / workflow"	181 Benchmarking & profiling
High GC CPU, allocs/op	182 Memory & GC
Low CPU, high latency, lock waits	183 Contention
Dependency timeouts	Parts 13 & 15 (budgets, breakers)
Cannot find which service is slow	Part 19 tracing
gzip/png/zip goldens fail after a toolchain bump	186c compress/flate rewrite

Use this table in incident channels so people stop randomly rewriting JSON libraries.

Definition of Done for a Perf Change

A performance PR is done when:

1. Before/after numbers exist (benchstat **or** load p99/RPS).
2. A profile explains *why* the change helped.
3. Correctness tests still pass (`go test`, race where relevant).
4. Complexity is justified in the description.
5. A rollback plan exists if production disagrees with the lab.

If any item is missing, the change is an experiment, not a finish.

Quick Command Card

```
# microbench + mem
go test -bench=BenchmarkX -benchmem -count=10 ./...

# compare
benchstat old.txt new.txt

# cpu profile from bench
go test -bench=BenchmarkX -cpuprofile=cpu.out
go tool pprof -http=:8080 cpu.out

# live service (admin port)
go tool pprof -http=:8081 http://127.0.0.1:6060/debug/pprof/profile?seconds=20
curl -o trace.out 'http://127.0.0.1:6060/debug/pprof/trace?seconds=5'
go tool trace trace.out
```

Pin this card next to your runbooks.

Glossary

- **ns/op, B/op, allocs/op** — benchmark cost units
- **pprof** — sampling profiler suite
- **benchstat** — statistical comparison of bench outputs
- **GOGC / GOMEMLIMIT** — GC pacing and soft memory limit
- **Contention** — waiting on shared resources
- **Flame graph** — visualization of aggregated stacks
- **p99** — 99th percentile latency (tail)
- **Escape analysis** — compiler decision stack vs heap

More examples

Tour: timed work + allocation awareness

```
mkdir -p /tmp/go-perf-tour && cd /tmp/go-perf-tour
go mod init example.com/perf-tour
```

Save as `main.go`:

```go
package main

import (
    "fmt"
    "time"
)

func sumPrealloc(n int) int {
    s := make([]int, 0, n)
    for i := 0; i < n; i++ {
        s = append(s, i)
    }
    total := 0
    for _, v := range s {
        total += v
    }
    return total
}

func main() {
    start := time.Now()
    got := sumPrealloc(10_000)
    fmt.Println("sum:", got, "dur_ms:", time.Since(start).Milliseconds())
}
```

```
go run .
```

Expected output:

```
sum: 49995000 dur_ms: 0
```

(`dur_ms` may be 0–1 on a fast machine.)

Runnable example

Tour of this part: a microbenchmark-style timing loop comparing two algorithms and a tiny allocation count via `testing.AllocsPerRun` pattern inlined in `main`.

```bash
mkdir -p /tmp/go-perf-tour && cd /tmp/go-perf-tour
go mod init example.com/perf-tour
```

Save as `main.go`:

```go
package main

import (
    "fmt"
    "strings"
    "testing"
    "time"
)

func concatPlus(n int) string {
    s := ""
    for i := 0; i < n; i++ {
        s += "x"
    }
    return s
}

func concatBuilder(n int) string {
    var b strings.Builder
    b.Grow(n)
    for i := 0; i < n; i++ {
        b.WriteByte('x')
    }
    return b.String()
}

func timeIt(name string, n int, fn func(int) string) {
    start := time.Now()
    _ = fn(n)
    fmt.Printf("%s took %s\n", name, time.Since(start))
}

func main() {
    const n = 20000
    timeIt("plus", n, concatPlus)
    timeIt("builder", n, concatBuilder)

    allocPlus := testing.AllocsPerRun(10, func() { _ = concatPlus(1000) })
    allocBld := testing.AllocsPerRun(10, func() { _ = concatBuilder(1000) })
    fmt.Printf("allocs_per_run plus=%.0f builder=%.0f\n", allocPlus, allocBld)
    fmt.Println("tour: measure before optimize")
}
```

```
go run .
```

Expected output (illustrative):

```
plus took ...
builder took ...
allocs_per_run plus=... builder=1
tour: measure before optimize
```

What to notice

- Evidence first: builder wins on allocs for repeated concatenation—the part's core habit.
- Chapters 181–183 deepen pprof, GC, and contention.

Try next

- `go test -bench=. -benchmem` on extracted functions.
- Capture a CPU profile once you have a real hotspot.

Next Chapter

Benchmarking and Profiling Workflow — the loop you will reuse forever.

Benchmarking and Profiling Workflow

Benchmarking and Profiling Workflow

Optimization should be a controlled loop, not guesswork. Go's toolchain makes the loop tight: benchmarks, pprof, and execution traces.

Why / Overview

Without measurement you will:

- Optimize cold code
- Add complexity for noise-level gains
- Miss regressions until customers report them

```
baseline bench/load -> profile hotspot -> focused change -> remeasure -> benchstat keep/reve
```

Writing Good Benchmarks

```go
func BenchmarkParse(b *testing.B) {
    in := fixedInput() // stable, realistic
    b.ReportAllocs()
    b.ResetTimer()
    for i := 0; i < b.N; i++ {
        if _, err := Parse(in); err != nil {
            b.Fatal(err)
        }
    }
}
```

Rules:

1. **Realistic inputs** — tiny synthetic data lies.
2. **Avoid measuring setup** — use `ResetTimer`, or do setup outside the loop.
3. **Prevent dead-code elimination** — assign to `sink` or use results.
4. **Report allocs** — `b.ReportAllocs()` or `-benchmem`.
5. **Sub-benchmarks** for sizes: `b.Run("1k", ...)`.

```go
var sink int

func BenchmarkSum(b *testing.B) {
```

```go
    for i := 0; i < b.N; i++ {
        sink = Sum(data)
    }
}
```

Parallel benchmarks

```go
func BenchmarkHandlerParallel(b *testing.B) {
    b.RunParallel(func(pb *testing.PB) {
        for pb.Next() {
            // concurrent path
        }
    })
}
```

Useful for mutex/contention issues; pair with CPU counts.

Running and Comparing

```bash
go test -bench=BenchmarkParse -benchmem -count=10 ./... | tee old.txt
# make change
go test -bench=BenchmarkParse -benchmem -count=10 ./... | tee new.txt
benchstat old.txt new.txt
```

`benchstat` shows geometric mean differences and statistical confidence. **Do not** trust single runs.

Flags:

- `-benchtime=3s` or `-benchtime=1000x` for stability
- `-cpu=1,4` for scaling behavior
- `-c` for concurrent package tests carefully

Profiling from Benchmarks

```bash
go test -bench=BenchmarkParse -cpuprofile=cpu.out -memprofile=mem.out
go tool pprof -http=:8080 cpu.out
go tool pprof -http=:8081 mem.out
```

Look at:

- **CPU flame graph** — widest plateaus
- **alloc_space** — bytes allocated (retained vs ephemeral)
- **alloc_objects** — allocation count (GC pressure)

Profiling a Running Service

```go
import _ "net/http/pprof"

// admin listener only
go func() {
    log.Println(http.ListenAndServe("127.0.0.1:6060", nil))
}()
```

```bash
go tool pprof -http=:8081 http://127.0.0.1:6060/debug/pprof/profile?seconds=30
curl -o heap.pb.gz http://127.0.0.1:6060/debug/pprof/heap
go tool pprof -http=:8082 heap.pb.gz
```

Mutex and block profiles

```go
runtime.SetMutexProfileFraction(1)    // sample all mutex events; lower in prod if needed
runtime.SetBlockProfileRate(1)        // 1 ns; too heavy for always-on prod - sample careful
```

```bash
go tool pprof http://127.0.0.1:6060/debug/pprof/mutex
go tool pprof http://127.0.0.1:6060/debug/pprof/block
```

Execution Tracer

```bash
curl -o trace.out 'http://127.0.0.1:6060/debug/pprof/trace?seconds=5'
go tool trace trace.out
```

Use when:

- Latency spikes with low CPU
- Suspect GC STW or scheduler delays
- Goroutines stuck in syscall or channel ops

Trace shows **timeline**, not aggregated hotspots — complementary to pprof.

Load Testing Complements Microbenches

Microbenchmarks miss:

- Network
- Real JSON sizes
- Multi-tenant cache effects
- GC under steady heap

Use `hey`, `vegeta`, `k6`, or custom Go load clients for end-to-end. Still attach pprof **during** the load.

Workflow Playbook

1. **Define success**: e.g. p99 < 50ms at 2k RPS, or bench ns/op −20%.
2. **Baseline** with count 10 and production-like data.
3. **Profile** under that baseline load.
4. **Change one hotspot** (algorithm, alloc, lock scope).
5. **Remeasure** with benchstat or load test.
6. **Document** why the change is correct (not only faster).
7. **Guard** with a regression benchmark in CI if critical.

CI Integration

```
# Example: fail on large regression using benchstat + golden
go test -bench=BenchmarkCritical -count=5 ./internal/hotpath | tee ci.txt
# compare against stored baseline with thresholds
```

Keep CI benches short and stable; flaky benches train people to ignore them.

Production Checklist

- ☐ Critical paths have benchmarks with `-benchmem`
- ☐ pprof available on admin interface only
- ☐ Know how to capture CPU, heap, mutex, trace
- ☐ Use benchstat for claims
- ☐ Load test environment approximates prod limits (CPU, GOMAXPROCS)
- ☐ Profiles stored for major incidents (artifact)
- ☐ Optimization PRs include before/after numbers

Common Pitfalls

1. **Benchmarking with `-race` only** — useful but not speed truth.
2. **Tiny inputs** that stay on stack and never show heap costs.
3. **Compiler optimizing away work**.
4. **Comparing different machines** without noting noise.
5. **Always-on block profiling at rate 1** in prod — heavy.
6. **Optimizing before algorithmic rewrite** ($O(n^2) \to$ map).
7. **Ignoring variance** — one lucky run.

Exercises

1. Write a bad benchmark that the compiler optimizes away; fix with sink.
2. Benchmark two JSON libraries on the same payload; benchstat them.
3. Capture CPU profile of a deliberately slow function; identify it in the flame graph.
4. Add `ReportAllocs`; reduce allocs with a sync.Pool; remeasure.
5. Run parallel benchmark; show scaling from 1 to 4 CPUs.
6. Enable mutex profile on a contended lock demo; find the holder.
7. Capture a 5s trace during GC-heavy load; find STW-ish gaps.
8. Load-test a HTTP handler; profile mid-test; fix top hotspot.
9. Introduce a 5% performance regression; see if your CI would catch it.
10. Write a one-page playbook for your team's pprof ports and commands.

Interpreting Flame Graphs Quickly

When the CPU profile UI opens:

1. Sort by **cum** (cumulative) to see time including callees.
2. Look for wide plateaus in your package paths — not only `runtime.*`.
3. If most time is `runtime.mallocgc` / `scanobject`, switch to heap/alloc profiles (next chapter).
4. If most time is `sync.(*Mutex).Lock`, switch to mutex profile (contention chapter).
5. If time is in `syscall.Read` / network, you may have an IO or dependency problem, not a Go micro-opt.

Always keep a copy of the profile file (`cpu.out`) next to the PR description for reviewers.

Production Capture Safety

- Bind pprof to localhost or a private admin network.
- Prefer short profiles (15–30s) under representative load.
- Do not leave `SetBlockProfileRate(1)` on permanently in large fleets.
- Redact profile upload paths if they include customer identifiers in function args (rare but possible with bad instrumentation).

More examples

Mini benchmark harness (stdlib testing style)

```
mkdir -p /tmp/go-bench-mini && cd /tmp/go-bench-mini
go mod init example.com/bench-mini
```

Save as `main.go` (illustrates the measurement pattern; real benches use `testing.B`):

```go
package main

import (
    "fmt"
    "strings"
    "time"
)

func concatPlus(n int) string {
    var s string
    for i := 0; i < n; i++ {
        s += "x"
    }
    return s
}

func concatBuilder(n int) string {
    var b strings.Builder
    b.Grow(n)
    for i := 0; i < n; i++ {
        b.WriteByte('x')
    }
    return b.String()
}

func timeIt(name string, n int, fn func(int) string) {
    start := time.Now()
    _ = fn(n)
    fmt.Printf("%s n=%d %s\n", name, n, time.Since(start))
}

func main() {
    timeIt("plus", 5000, concatPlus)
    timeIt("builder", 5000, concatBuilder)
}
```

```go
go run .
```

Expected output (durations vary; builder should not be slower by orders of magnitude):

```
plus n=5000 ...
builder n=5000 ...
```

pprof HTTP handler registration (httptest)

```
mkdir -p /tmp/go-pprof-reg && cd /tmp/go-pprof-reg
go mod init example.com/pprof-reg
```

Save as `main.go`:

```go
package main

import (
    "fmt"
    "net/http"
    "net/http/httptest"
    "net/http/pprof"
)

func main() {
    mux := http.NewServeMux()
    mux.HandleFunc("GET /debug/pprof/", pprof.Index)
    mux.HandleFunc("GET /debug/pprof/cmdline", pprof.Cmdline)
    mux.HandleFunc("GET /debug/pprof/profile", pprof.Profile)
    mux.HandleFunc("GET /debug/pprof/symbol", pprof.Symbol)
    mux.HandleFunc("GET /debug/pprof/trace", pprof.Trace)

    ts := httptest.NewServer(mux)
    defer ts.Close()
    resp, err := http.Get(ts.URL + "/debug/pprof/")
    if err != nil {
        panic(err)
    }
    resp.Body.Close()
    fmt.Println("pprof index:", resp.StatusCode)
}
```

```
go run .
```

Expected output:

```
pprof index: 200
```

Runnable example

A real `go test` benchmark file plus `runtime/pprof` heap profile written to a buffer—stdlib profiling workflow without a long-lived server.

```
mkdir -p /tmp/go-bench-pprof && cd /tmp/go-bench-pprof
go mod init example.com/bench-pprof
```

Save as `work.go`:

```
package benchpprof

import "strings"

func BuildReport(n int) string {
    var b strings.Builder
    for i := 0; i < n; i++ {
        b.WriteString("line\n")
    }
    return b.String()
}
```

Save as `work_test.go`:

```
package benchpprof

import "testing"

func BenchmarkBuildReport(b *testing.B) {
    for i := 0; i < b.N; i++ {
        _ = BuildReport(1000)
    }
}

func BenchmarkBuildReportAlloc(b *testing.B) {
    b.ReportAllocs()
    for i := 0; i < b.N; i++ {
        _ = BuildReport(1000)
    }
}
```

Save as `cmd/heapdemo/main.go`:

```
package main

import (
    "bytes"
    "fmt"
    "runtime"
    "runtime/pprof"

    benchpprof "example.com/bench-pprof"
```

```go
)

func main() {
    // Create some heap traffic
    var hold []string
    for i := 0; i < 100; i++ {
        hold = append(hold, benchpprof.BuildReport(500))
    }
    runtime.GC()

    var buf bytes.Buffer
    if err := pprof.WriteHeapProfile(&buf); err != nil {
        panic(err)
    }
    fmt.Println("heap_profile_bytes:", buf.Len())
    fmt.Println("held_reports:", len(hold))
    fmt.Println("tip: go test -bench=BenchmarkBuildReport -benchmem -count=5")
}
```

```bash
go test -bench=BenchmarkBuildReport -benchmem -count=3 .
go run ./cmd/heapdemo
```

Expected output (illustrative):

```
BenchmarkBuildReport-8         xxxx      yyyy ns/op     zzzz B/op      w allocs/op
heap_profile_bytes: <nonzero>
held_reports: 100
tip: go test -bench=BenchmarkBuildReport -benchmem -count=5
```

What to notice

- `-benchmem` reports B/op and allocs/op—the first filter before guessing.
- `pprof.WriteHeapProfile` works without binding HTTP; use `net/http/pprof` on an admin port in services.
- Run benchmarks multiple `-count` times before believing a regression.

Try next

- `go test -cpuprofile=cpu.out -bench=.` then `go tool pprof -http=:8080 cpu.out`.
- Compare two commits with `benchstat old.txt new.txt`.

Further Reading

- Go blog: profiling Go programs
- `benchstat` documentation
- Next: **Memory, Allocations, and GC**

Memory, Allocations, and GC

Memory, Allocations, and GC

Memory behavior is latency behavior. Allocation patterns influence GC pressure, which affects tail latency even when average CPU looks fine.

Why / Overview

Go's GC is a concurrent, tri-color mark-and-sweep collector designed for low pause times. It is not free: marking work scales with **live heap** and allocation rate.

```
allocate -> object live? -> may promote to longer-lived -> GC mark -> reclaim unreachable
```

Goals of this chapter:

- See allocations in profiles and benches
- Reduce **hot-path** allocs without unreadable code
- Use pooling only with evidence
- Understand GOGC / GOMEMLIMIT knobs

Stack vs Heap (Practical View)

Escape analysis decides whether a value can live on the stack (cheap) or must be on the heap (GC-managed).

```
go build -gcflags='-m=2' ./...
# look for "escapes to heap"
```

Common escape causes:

- Returning pointers to local vars
- Interface conversion of large values
- Closures capturing variables
- Sending pointers on channels that outlive the stack frame

```
// Often heap: returned pointer
func New() *User { u := User{}; return &u }

// May stay stack if inlined and not escaped further - measure, don't guess
```

Reading Allocation Metrics

From benchmarks:

```
BenchmarkX-8   123456   9876 ns/op   4096 B/op   12 allocs/op
```

- **B/op** — bytes allocated per op
- **allocs/op** — number of heap allocations

From runtime metrics / expvar / Prometheus Go collector:

- `go_memstats_heap_alloc_bytes`
- `go_memstats_heap_objects`
- GC pause quantiles

From pprof heap:

- **inuse_space** — currently live
- **alloc_space** — cumulative allocated (good for "who allocates a lot")

Hot-Path Techniques

1. Preallocate slices

```go
// BAD: many grow copies
var out []T
for _, x := range in {
    out = append(out, f(x))
}

// GOOD
out := make([]T, 0, len(in))
```

2. Reuse buffers

```go
buf := make([]byte, 0, 4096)
for _, item := range items {
    buf = buf[:0]
    buf = appendJSON(buf, item)
    _, _ = w.Write(buf)
}
```

3. Avoid `fmt` in hot paths

```
// fmt.Sprintf allocates; strconv and append often cheaper
b = strconv.AppendInt(b, n, 10)
```

4. Beware of `[]byte(string)` / `string([]byte)` conversions

They allocate copies. Sometimes necessary for APIs; don't thrash.

5. Interfaces and methods

Storing concrete values in `interface{}` / `any` can allocate. Stream APIs with concrete types when profiling shows it matters.

sync.Pool

```
var bufPool = sync.Pool{
    New: func() any {
        b := make([]byte, 0, 4096)
        return &b
    },
}

func process() {
    bp := bufPool.Get().(*[]byte)
    buf := (*bp)[:0]
    defer func() {
        *bp = buf
        bufPool.Put(bp)
    }()
    // use buf
}
```

Rules:

- Only for **short-lived**, frequently allocated objects
- Clear sensitive data before put if buffers hold secrets
- Pools can be wiped on GC — do not store important state only in pool
- Prove benefit with benchmarks under concurrent load

String Builders

```go
var b strings.Builder
b.Grow(n)
b.WriteString(s1)
b.WriteString(s2)
out := b.String() // may allocate once
```

Better than repeated `+` in loops.

Maps and Reuse

Clearing maps:

```go
// Go 1.21+: clear(m)
clear(m)

// Or keep map and delete keys if you need reuse without rehash cost carefully
```

Reusing maps across requests can reduce allocs; watch for leftover key retention (memory leaks of keys).

GC Tuning Knobs

GOGC

`GOGC=100` (default) roughly means GC trigger when heap grows 100% past live set. Higher GOGC → less frequent GC, larger heap. Lower → more GC CPU, smaller heap.

```
GOGC=50  ./app   # more GC, less memory
GOGC=200 ./app   # less GC, more memory
```

GOMEMLIMIT (Go 1.19+)

Soft memory limit that makes GC more aggressive as you approach the limit — excellent for containers:

```
GOMEMLIMIT=512MiB ./app
```

Prefer setting memory limits in orchestrators **and** GOMEMLIMIT to reduce OOM kills from GC softness.

Ballast (historical)

Old trick of large unused allocation to influence GC — largely superseded by GOMEMLIMIT. Avoid new ballast designs.

Symptoms of Allocation Problems

Symptom	Clue
High CPU in `runtime.gc*` / `scanobject`	GC pressure
p99 spikes periodic	GC pacing
Growing heap inuse	Leak or retention
High allocs/op in bench	Hot path churn
Throughput drops with more goroutines	Alloc+sched pressure

Production Checklist

- ☐ Critical path benches report allocs
- ☐ Heap profiles captured under load
- ☐ Container memory limit + GOMEMLIMIT aligned
- ☐ GOGC only changed with measured reason
- ☐ Pooling justified by numbers
- ☐ No unbounded caches without eviction
- ☐ Escape analysis consulted for hot packages (`-m`)
- ☐ Large buffers not retained accidentally on structs

Common Pitfalls

1. **Pooling everything** — complexity, bugs, little gain.
2. **Ignoring leaks** (slice backing arrays retained by small reslices).
3. **Global caches without bound**.
4. **String concat in loops**.
5. **`defer` in micro-benchmarks** affecting tiny functions (sometimes); be aware.
6. **Tuning GOGC to hide a leak**.
7. **Copying huge structs** in interfaces repeatedly.

Exercises

1. Benchmark `+` string concat vs `strings.Builder` for 10k parts.
2. Find allocs/op of a JSON marshal path; reduce with streaming encoder to `Writer`.
3. Use `-gcflags=-m` on a hot file; explain three escapes.
4. Introduce a `sync.Pool` for buffers; benchstat with and without under `-cpu=8`.

5. Create a memory leak with growing slice of pointers; find it in heap profile.
6. Run with `GOMEMLIMIT` low; observe GC CPU vs without.
7. Compare `make([]byte, 0, n)` reuse vs new each time.
8. Profile `fmt.Sprintf("%d", i)` vs `strconv.Itoa`.
9. Clear a map between requests; ensure no key accumulation.
10. Write a short design note: when your team allows `sync.Pool`.

Leak Patterns Specific to Go Services

1. **Unbounded caches** (`map` grows forever per user/id).
2. **Goroutine leaks** holding references to large buffers.
3. **`time.After` in loops** (pre-1.23 patterns) retaining timers — prefer `NewTimer` carefully or tickers.
4. **Registering HTTP handlers / finalizers repeatedly**.
5. **Appending to a slice shared via pointer** without bounds.

Heap profiles over two timestamps (`-base`) make growth obvious.

More examples

Preallocate slice capacity

```
mkdir -p /tmp/go-prealloc && cd /tmp/go-prealloc
go mod init example.com/prealloc
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "runtime"
)

func allocs(fn func()) uint64 {
    var m1, m2 runtime.MemStats
    runtime.GC()
    runtime.ReadMemStats(&m1)
    fn()
    runtime.ReadMemStats(&m2)
    return m2.Mallocs - m1.Mallocs
}

func main() {
```

```go
    n := 1000
    a := allocs(func() {
        var s []int
        for i := 0; i < n; i++ {
            s = append(s, i)
        }
        _ = s
    })
    b := allocs(func() {
        s := make([]int, 0, n)
        for i := 0; i < n; i++ {
            s = append(s, i)
        }
        _ = s
    })
    fmt.Println("grow allocs >= prealloc:", a >= b)
    fmt.Println("prealloc mallocs:", b)
}
```

```bash
go run .
```

Expected output:

```
grow allocs >= prealloc: true
prealloc mallocs: <small number>
```

sync.Pool reuse

```bash
mkdir -p /tmp/go-pool && cd /tmp/go-pool
go mod init example.com/pool
```

Save as `main.go`:

```go
package main

import (
    "bytes"
    "fmt"
    "sync"
)

func main() {
    var pool = sync.Pool{New: func() any { return new(bytes.Buffer) }}
    buf := pool.Get().(*bytes.Buffer)
    buf.Reset()
```

```go
    buf.WriteString("hello")
    fmt.Println(buf.String())
    pool.Put(buf)

    buf2 := pool.Get().(*bytes.Buffer)
    // May be the same buffer; Reset before use in real code.
    buf2.Reset()
    buf2.WriteString("world")
    fmt.Println(buf2.String())
}
```

```bash
go run .
```

Expected output:

```
hello
world
```

Runnable example

Compare allocations with `testing.AllocsPerRun`, reuse a buffer with `sync.Pool`, and print basic GC stats via `runtime.ReadMemStats`.

```bash
mkdir -p /tmp/go-alloc-gc && cd /tmp/go-alloc-gc
go mod init example.com/alloc-gc
```

Save as `main.go`:

```go
package main

import (
    "fmt"
    "runtime"
    "sync"
    "testing"
)

var bufPool = sync.Pool{
    New: func() any {
        b := make([]byte, 0, 4096)
        return &b
    },
}

func workNew(n int) int {
```

```go
    b := make([]byte, 0, n)
    for i := 0; i < n; i++ {
        b = append(b, 'x')
    }
    return len(b)
}

func workPool(n int) int {
    bp := bufPool.Get().(*[]byte)
    b := (*bp)[:0]
    for i := 0; i < n; i++ {
        b = append(b, 'x')
    }
    // keep cap; return to pool
    *bp = b
    bufPool.Put(bp)
    return len(b)
}

func main() {
    aNew := testing.AllocsPerRun(100, func() { _ = workNew(2048) })
    aPool := testing.AllocsPerRun(100, func() { _ = workPool(2048) })
    fmt.Printf("allocs workNew=%.2f workPool=%.2f\n", aNew, aPool)

    var before, after runtime.MemStats
    runtime.ReadMemStats(&before)
    sink := 0
    for i := 0; i < 10000; i++ {
        sink += workNew(1024)
    }
    runtime.GC()
    runtime.ReadMemStats(&after)
    fmt.Printf("heap_alloc_after=%d num_gc=%d total_alloc_delta=%d sink=%d\n",
        after.HeapAlloc, after.NumGC, after.TotalAlloc-before.TotalAlloc, sink)
    fmt.Println("note: Pool helps under concurrent reuse; measure in real load")
}
```

```go
go run .
```

Expected output (illustrative):

```
allocs workNew=1.00 workPool=0.00
heap_alloc_after=... num_gc=... total_alloc_delta=...
note: Pool helps under concurrent reuse; measure in real load
```

What to notice

- Allocation rate drives GC CPU; reducing allocs/op often beats micro-tweaking algorithms later.
- `sync.Pool` is for *hot* reusable buffers—not a general cache; values may be dropped anytime.
- `ReadMemStats` is expensive; use sparingly or prefer runtime/metrics in production.

Try next

- `go build -gcflags=-m` on a file and read escape analysis notes.
- Set `GOMEMLIMIT` and observe GC behavior under a memory-hungry loop.

Further Reading

- Go GC guide and runtime metrics
- Previous: profiling workflow to *find* alloc sites
- Next: **Contention and Concurrency Tuning**

Contention and Concurrency Tuning

Contention and Concurrency Tuning

Throughput often degrades due to **contention** long before CPU saturates. Adding goroutines can make the system slower when they fight over locks, shared maps, or a single connection.

Why / Overview

```
goroutines -> shared resource -> queueing -> latency growth -> throughput collapse
```

Classic symptoms:

- Low CPU, high latency
- Throughput flat while goroutine count climbs
- Mutex profile shows long waiters
- `block` profile hot on channels or locks

Sources of Contention

Resource	Example
`sync.Mutex` / `RWMutex`	Shared cache, metrics map
Channels	Single consumer bottleneck
Shared `map` without sharding	Global session table
Single DB connection / conn pool too small	Query pileup
`log` writer lock	Sync logging every request
`time.Now` + heavy formatting under lock	Lock held too long

Diagnostic Workflow

1. Confirm problem with **load test**: RPS vs latency vs CPU.
2. If CPU low and latency high → contention or IO wait.
3. Capture **mutex** and **block** profiles under load.
4. Capture **trace** for scheduler wait.
5. Fix by reducing critical sections, sharding, or buffering.
6. Remeasure.

```go
import _ "net/http/pprof"
import "runtime"

func init() {
    runtime.SetMutexProfileFraction(5) // 1/5 of events; tune
    // runtime.SetBlockProfileRate(1000) // sample; careful in prod
}
```

Critical Section Hygiene

```go
// BAD: slow work under lock
mu.Lock()
data := fetchFromDB() // network under lock!
cache[key] = data
mu.Unlock()

// GOOD: compute outside
data := fetchFromDB()
mu.Lock()
cache[key] = data
mu.Unlock()
```

Copy what you need under lock; work on the copy outside.

RWMutex misuse

`RWMutex` helps when reads dominate **and** critical sections are short. If writers are frequent, RWMutex can be worse. Measure.

Sharding

```go
const shards = 16

type sharded struct {
    ms [shards]struct {
        mu sync.Mutex
        m  map[string]int
    }
}

func (s *sharded) inc(key string) {
    h := fnv32(key) % shards
```

```go
    sh := &s.ms[h]
    sh.mu.Lock()
    sh.m[key]++
    sh.mu.Unlock()
}
```

Sharding reduces collision probability; pick shard count relative to `GOMAXPROCS` and key distribution.

Atomically Updated Stats

```go
var hits atomic.Int64

func hit() { hits.Add(1) }
```

Prefer atomics for simple counters over mutexes. For complex metrics, use Prometheus client (already concurrent) rather than home-grown locked maps with high cardinality.

Channel Contension and Pipelines

```go
// Single channel to one worker: max throughput ~ worker speed
jobs := make(chan Job, 1024)

// Multiple workers consuming one channel: good
for i := 0; i < n; i++ {
    go worker(jobs)
}
```

Pitfalls:

- Unbuffered channels serialize producers
- Huge buffers hide deadlocks and delay backpressure
- Multiple stages with mismatched rates need intentional buffering

Bound Fan-Out

```go
g, ctx := errgroup.WithContext(ctx)
g.SetLimit(32)
for _, u := range urls {
    u := u
    g.Go(func() error {
        return fetch(ctx, u)
```

```
    })
}
return g.Wait()
```

Unbounded `go func` per item under a hot HTTP handler is a latency and memory bomb.

Connection Pools

Database and HTTP pools are concurrency knobs:

```
db.SetMaxOpenConns(25)
db.SetMaxIdleConns(25)
db.SetConnMaxLifetime(time.Hour)
```

```
transport.MaxConnsPerHost = 32
```

Too few → wait on pool. Too many → overload dependency. Tune against dependency capacity and latency.

GOMAXPROCS and Containers

Go 1.5+ defaults `GOMAXPROCS` to CPU count. In cgroups, recent Go versions respect container CPU limits more carefully — still verify:

```
# inside container
go env GOMAXPROCS
# or log runtime.GOMAXPROCS(0) at start
```

Mismatch (thinks 64 CPUs, has 2) → oversubscription, latency noise.

False Sharing and Cache Lines (Advanced)

Rare in app Go, more in tight numeric loops with adjacent atomics. If profiling shows multi-core scaling collapse on atomics, pad/shard counters. Don't start here.

Work Stealing and Fairness

The Go scheduler multiplexes goroutines on `GOMAXPROCS` threads. CPU-bound tight loops without function calls can delay preemption (improved in recent Go). For fairness, insert `runtime.Gosched()` only when profiling proves need — rare.

Production Checklist

- ☐ Load test shows CPU vs latency relationship
- ☐ Mutex/block profiles known how-to
- ☐ No IO under locks on hot paths
- ☐ Fan-out bounded (errgroup limits / semaphores)
- ☐ Pools sized for deps
- ☐ Shared maps protected **and** not global bottlenecks
- ☐ Logging async or sampled under load if contended
- ☐ GOMAXPROCS aligned with container CPUs
- ☐ After changes, remeasure p99 and throughput

Common Pitfalls

1. **More goroutines = more speed** belief.
2. **Global mutex** around entire request handling.
3. **Map + mutex** as the only cache for everything.
4. **Holding lock across `http.Client.Do`**.
5. **Unbuffered event bus** used for high QPS metrics.
6. **Ignoring dependency pool wait** — looks like "Go is slow."
7. **Fixing contention by removing locks** without other synchronization → data races (`-race`).

Exercises

1. Build a counter with mutex vs atomic; parallel bench both.
2. Create a hot map with one mutex; shard it; benchstat.
3. Hold a lock during `time.Sleep` to simulate IO; show latency; move sleep out.
4. Capture mutex profile; identify the stack.
5. Unbounded fan-out HTTP handler vs `SetLimit`; compare memory and p99.
6. Shrink `MaxOpenConns` to 1; observe pool wait in traces/profiles.
7. Use `go test -race` on a deliberately racy counter; fix.
8. Measure throughput vs worker count curve; find the knee.
9. Replace contended `log.Printf` with async slog or buffered writer; measure.
10. Document a contention incident postmortem template (signals → profile → fix → result).

Throughput Curve Lab (Do This Once)

Pick a handler that does shared-map updates under a mutex:

```
workers: 1, 2, 4, 8, 16, 32
measure: RPS and p99
plot: RPS rises then flattens/falls; p99 rises with contention
```

Then shard the map or switch counters to atomics and replot. The shape of that curve is the entire point of this chapter: **more concurrency is not free**.

Combining with Timeouts and Breakers

Contention often surfaces as timeouts:

- Lock wait → handler exceeds budget → client retries → more lock wait

Fixing only retries makes it worse. Profile locks under the load that triggers pages, then reduce critical sections or capacity-limit admission (semaphore) so the system degrades with 503 instead of unlimited queueing on a mutex.

More examples

Hot mutex vs sharded counters

```
mkdir -p /tmp/go-shard-count && cd /tmp/go-shard-count
go mod init example.com/shard-count
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "sync"
    "sync/atomic"
    "time"
)

func hotMutex(n, workers int) time.Duration {
    var mu sync.Mutex
    var c int
    var wg sync.WaitGroup
    start := time.Now()
    for w := 0; w < workers; w++ {
        wg.Add(1)
        go func() {
            defer wg.Done()
            for i := 0; i < n; i++ {
                mu.Lock()
                c++
                mu.Unlock()
            }
```

```go
        }()
    }
    wg.Wait()
    _ = c
    return time.Since(start)
}

func atomicCount(n, workers int) time.Duration {
    var c atomic.Int64
    var wg sync.WaitGroup
    start := time.Now()
    for w := 0; w < workers; w++ {
        wg.Add(1)
        go func() {
            defer wg.Done()
            for i := 0; i < n; i++ {
                c.Add(1)
            }
        }()
    }
    wg.Wait()
    return time.Since(start)
}

func main() {
    const n, w = 20_000, 4
    fmt.Println("mutex", hotMutex(n, w))
    fmt.Println("atomic", atomicCount(n, w))
}
```

```bash
go run .
```

Expected output (durations vary; both complete):

```
mutex ...
atomic ...
```

Admission semaphore (shed load)

```bash
mkdir -p /tmp/go-sem-admit && cd /tmp/go-sem-admit
go mod init example.com/sem-admit
```

Save as `main.go`:

```go
package main

import (
    "fmt"
    "net/http"
    "net/http/httptest"
)

func withLimit(n int, next http.Handler) http.Handler {
    sem := make(chan struct{}, n)
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        select {
        case sem <- struct{}{}:
            defer func() { <-sem }()
            next.ServeHTTP(w, r)
        default:
            http.Error(w, "busy", http.StatusServiceUnavailable)
        }
    })
}

func main() {
    entered := make(chan struct{})
    block := make(chan struct{})
    h := withLimit(1, http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        close(entered) // signal that the slot is held
        <-block
        fmt.Fprintln(w, "ok")
    }))
    ts := httptest.NewServer(h)
    defer ts.Close()

    go func() {
        resp, _ := http.Get(ts.URL)
        resp.Body.Close()
    }()
    <-entered // wait until first request holds the semaphore
    resp, _ := http.Get(ts.URL)
    fmt.Println("shed:", resp.StatusCode)
    resp.Body.Close()
    close(block)
}
```

```
go run .
```

Expected output:

shed: 503

Runnable example

Contended mutex counter vs sharded counters under many goroutines—measure elapsed time to see the throughput ceiling.

```
mkdir -p /tmp/go-contention && cd /tmp/go-contention
go mod init example.com/contention
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "sync"
    "sync/atomic"
    "time"
)

func benchMutex(goroutines, iters int) time.Duration {
    var mu sync.Mutex
    var total int64
    var wg sync.WaitGroup
    start := time.Now()
    wg.Add(goroutines)
    for g := 0; g < goroutines; g++ {
        go func() {
            defer wg.Done()
            for i := 0; i < iters; i++ {
                mu.Lock()
                total++
                mu.Unlock()
            }
        }()
    }
    wg.Wait()
    _ = total
    return time.Since(start)
}

func benchAtomic(goroutines, iters int) time.Duration {
    var total atomic.Int64
    var wg sync.WaitGroup
    start := time.Now()
```

```go
    wg.Add(goroutines)
    for g := 0; g < goroutines; g++ {
        go func() {
            defer wg.Done()
            for i := 0; i < iters; i++ {
                total.Add(1)
            }
        }()
    }
    wg.Wait()
    _ = total.Load()
    return time.Since(start)
}

func benchSharded(goroutines, iters, shards int) time.Duration {
    type pad struct {
        mu sync.Mutex
        n  int64
        _  [56]byte // reduce false sharing a bit
    }
    cs := make([]pad, shards)
    var wg sync.WaitGroup
    start := time.Now()
    wg.Add(goroutines)
    for g := 0; g < goroutines; g++ {
        g := g
        go func() {
            defer wg.Done()
            s := &cs[g%shards]
            for i := 0; i < iters; i++ {
                s.mu.Lock()
                s.n++
                s.mu.Unlock()
            }
        }()
    }
    wg.Wait()
    return time.Since(start)
}

func main() {
    const g, iters = 8, 100000
    fmt.Println("mutex:  ", benchMutex(g, iters))
    fmt.Println("atomic: ", benchAtomic(g, iters))
    fmt.Println("shard8: ", benchSharded(g, iters, 8))
    fmt.Println("more concurrency is not free - profile locks under real load")
}
```

```
go run .
```

Expected output (order of magnitude varies by machine):

```
mutex:   ...
atomic:  ...
shard8:  ...
more concurrency is not free - profile locks under real load
```

What to notice

- A single global mutex serializes work; atomics/sharding raise the ceiling for simple counters.
- Real contention shows up as flat RPS and rising p99—capture a mutex profile in services.
- Fix critical sections and admission control before cranking retries.

Try next

- `go test -bench=. -cpu=1,2,4,8` and plot ns/op.
- Enable mutex profiling: `runtime.SetMutexProfileFraction(5)` and pull `pprof/mutex`.

Further Reading

- Go blog: mutex profile, scheduler
- Previous: allocations/GC (often co-occur with contention under load)
- Concurrency part of this book for patterns (worker pools, pipelines)
- Synthesis parts for consolidating modern Go practices

Load Testing and Baselines

Load Testing and Baselines

Overview

Microbenchmarks don't replace **load tests**. Establish a baseline: throughput, p50/p99 latency, error rate, resource use—then change one variable.

Tools

Tool	Role
`hey` / `vegeta` / `k6`	HTTP load
`go test -bench`	Function-level
`pprof` under load	See real hotspots

```
hey -z 30s -c 50 http://127.0.0.1:8080/api/books
```

Baseline template

```
Scenario: GET /api/books  (cached / uncached)
Concurrency: 50
Duration: 60s
Result: QPS=… p50=… p99=… errors=…
CPU=… RSS=…  (from metrics or `ps`)
Commit: abc123
```

SLO-oriented load

Drive load until **error budget** or latency SLO breaks—know the capacity ceiling.

Avoiding test lies

- Warm up before measuring
- Separate client machine when possible
- Watch coordinated omission (load tools that delay sending)
- Fix payloads and auth

Rules of thumb

Do	Don't
Record env + commit with results	Compare laptop vs prod blindly
Increase load gradually	Jump to 10k connections first
Pair with profiles	Optimize without a number

Try next

1. Baseline one endpoint; save numbers in a markdown table.
2. Add artificial `time.Sleep` and remeasure.
3. Capture CPU pprof during `hey`.

Allocation Hot-Path Recipes

Allocation Hot-Path Recipes

Overview

High allocation rate drives GC and cache pressure. After profiles point to a site, apply **small, proven** recipes—not premature `sync.Pool` everywhere.

Measure first

```
go test -bench=BenchmarkX -benchmem
# or pprof heap under load
```

Recipes

1. Avoid `fmt` in hot loops

```
// slow: fmt.Sprintf("%d", n)
strconv.Itoa(n)
strconv.AppendInt(buf[:0], int64(n), 10)
```

2. Grow slices with known cap

```
out := make([]T, 0, len(in))
```

3. Reuse buffers

```
var buf bytes.Buffer
buf.Reset()
// or sync.Pool of []byte for short-lived buffers
```

4. `strings.Builder`

```
var b strings.Builder
b.Grow(n)
b.WriteString(...)
s := b.String()
```

5. Interface boxing

Storing concrete values in `any`/interfaces allocates. Keep hot maps typed.

6. JSON

- `json.Encoder` to writer vs repeated `Marshal`
- Consider specialized codecs only when bench proves need
- Avoid `map[string]any` for hot types

7. `io.Copy` buffer

```
buf := make([]byte, 32*1024)
io.CopyBuffer(dst, src, buf)
```

Pool discipline

```
var pool = sync.Pool{New: func() any { return make([]byte, 0, 1024) }}

b := pool.Get().([]byte)
b = b[:0]
// use b
pool.Put(b)
```

Don't put pointers to data still in use; don't assume zeroed memory.

Rules of thumb

Do	Don't
Fix algorithm first	Pool everything
Bench before/after	Optimize cold paths

Do	Don't
Keep code readable	Unreadable zero-alloc for 0.1%

Try next

1. `-benchmem` a JSON encode path; reduce allocs/op.
2. Replace `fmt.Sprintf` in a hot path.
3. Profile heap; fix top1 only.

PGO and Production Build Flags

PGO and Production Build Flags

Overview

Profile-guided optimization (PGO) feeds CPU profiles into the compiler for better inlining/layout on hot paths. Combine with production build flags for smaller, clearer binaries.

Production build baseline

```
CGO_ENABLED=0 go build -trimpath \
  -ldflags="-s -w -X main.version=${VERSION}" \
  -o bin/app ./cmd/app
```

Flag	Effect
`CGO_ENABLED=0`	Static-ish pure Go, portable
`-trimpath`	Reproducible paths
`-s -w`	Strip symbol/DWARF (harder debug)
`-X`	Inject version

Keep a debug build variant for crash analysis when needed.

PGO workflow (Go 1.20+)

```
# 1) collect profile from realistic load (30s+)
curl -o cpu.pprof "http://localhost:6060/debug/pprof/cpu?seconds=30"

# 2) rebuild with profile
go build -pgo=cpu.pprof -o bin/app ./cmd/app

# 3) compare benchmarks / load test
```

Or default auto PGO if `default.pgo` is in the main package directory (per Go release docs).

When PGO helps

Helps	Less impact
Large codebases, hot call chains	Tiny programs already inlined
After stable production shape	Random microbench only

Rules of thumb

Do	Don't
Profile production-like traffic	Use toy profiles for PGO
A/B load test before/after	Assume free wins always
Document Go version + flags	Mystery binaries in prod

Try next

1. Capture 30s CPU pprof under `hey`.
2. Build with `-pgo` and remeasure p99.
3. Add version ldflags; verify `app version`.

SIMD Without Assembly: simd/archsimd (GOEXPERIMENT=simd)

SIMD Without Assembly: `simd/archsimd`

Until recently, the only way to make a Go hot loop use the vector units in your CPU was to write Plan 9 assembly by hand, or to hope the compiler would do it for you (it largely doesn't auto-vectorize). Go **1.26** added an experimental, amd64-only SIMD API behind `GOEXPERIMENT=simd`. Go **1.27** revised it, added **arm64 (NEON)** and **wasm (128-bit SIMD)**, and put a portable, width-agnostic `simd` package on top. Both remain **experimental**. They exist only when you build with the experiment, and they are not covered by the Go 1 compatibility promise.

This chapter is about the low layer, `simd/archsimd`: real vector types, real instructions, and the engineering discipline that keeps them safe to ship. That means build tags, runtime CPU checks, a scalar twin, and tests that exercise every path on one machine. Everything below was compiled and run with **Go 1.27.1** on an amd64 box whose CPU reports AVX2, FMA, AVX-512 and GFNI.

Mental model

```
your code    simd            portable, width chosen at run time (Float32s, Uint8s…)
                  ToArch / *FromArch
             simd/archsimd fixed-width, per-architecture (Float32x8, Uint8x64…)
                           methods are compiler intrinsics

               VFMADD213PS, VPCMPEQB, VGF2P8AFFINEQB …  (one method   one instruction)
```

- **Types carry shape.** `Float32x8` is 256 bits of eight `float32`s; `Uint8x64` is a 512-bit register of bytes. Masks are separate opaque types (`Mask8x64`) because AVX-512, AVX2, NEON and wasm encode them differently.
- **Methods have plain names.** `x.Add(y)`, `x.MulAdd(y, acc)`, `x.Equal(y)`, not `_mm512_fmadd_ps`. The doc comment of each method names the instruction and the CPU feature it needs (`Asm: VFMADD213PS, CPU Feature: FMA`).
- **The compiler doesn't know your CPU.** A binary built for amd64 may run on a machine without AVX-512. You must branch on `archsimd.X86.AVX512()` (and friends). Otherwise the program dies with `SIGILL`. The check also tells the compiler which instruction forms are allowed inside that branch.
- **It is not portable by design.** `Float32x8` does not exist on arm64. Code that imports `archsimd` needs a build constraint and a fallback for every other target.

Layer	Use it for	Cost of using it
Plain Go loop	Everything, until a profile says otherwise	None

Layer	Use it for	Cost of using it
Standard library (`bytes.Count`, `bytes.Index`, `crypto/*`)	Searching, hashing, crypto. These already have hand-written vector code	None
`simd` (portable)	Arithmetic kernels you want on every architecture	Experiment flag
`simd/archsimd`	Instructions with no portable equivalent (GFNI, lane permutes), or the last few % on one target	Experiment flag, build tags, CPU dispatch, per-arch tests

Worked examples

Set up a module. The `go` line must be 1.27 or later.

```
mkdir simdlab && cd simdlab
go mod init example.com/simdlab   # then set "go 1.27" in go.mod
go version
```

```
go version go1.27.1 linux/amd64
```

The package is split into three files, which is the shape every `archsimd` package should have:

1. a scalar reference that always compiles,
2. a vector file guarded by `//go:build goexperiment.simd && amd64`,
3. a fallback file with the opposite constraint, so the package builds everywhere.

Case 1: the scalar twin

```go
// scalar.go - portable reference implementations (always compiled)
package simdlab

import "math/bits"

func dotScalar(a, b []float32) float32 {
    var s float32
    for i := range a {
        s += a[i] * b[i]
    }
    return s
}

func countScalar(s []byte, c byte) int {
```

```go
    n := 0
    for _, x := range s {
        if x == c {
            n++
        }
    }
    return n
}

func reverseScalar(dst, src []byte) {
    for i, x := range src[:len(dst)] {
        dst[i] = bits.Reverse8(x)
    }
}
```

These functions are the specification. Every vector path is tested against them, and they are what runs on every machine the vector code doesn't cover.

Case 2: dispatching vector code

```go
// dispatch_simd.go - vector paths, built only with GOEXPERIMENT=simd on amd64
//go:build goexperiment.simd && amd64

package simdlab

import (
    "math/bits"
    "simd/archsimd"
)

// Path reports which implementation Dot will use on this CPU.
func Path() string {
    switch {
    case archsimd.X86.AVX512():
        return "avx512"
    case archsimd.X86.AVX2() && archsimd.X86.FMA():
        return "avx2+fma"
    default:
        return "scalar"
    }
}

// Dot returns the inner product of a and b (len(b) must be >= len(a)).
func Dot(a, b []float32) float32 {
    b = b[:len(a)]
    switch {
```

```
    case archsimd.X86.AVX512():
        return dot16(a, b)
    case archsimd.X86.AVX2() && archsimd.X86.FMA():
        return dot8(a, b)
    }
    return dotScalar(a, b)
}

func dot8(a, b []float32) float32 {
    var acc archsimd.Float32x8
    i := 0
    for ; i < len(a)-8+1; i += 8 {
        x := archsimd.LoadFloat32x8(a[i : i+8])
        y := archsimd.LoadFloat32x8(b[i : i+8])
        acc = x.MulAdd(y, acc)
    }
    if i < len(a) {
        x, _ := archsimd.LoadFloat32x8Part(a[i:])
        y, _ := archsimd.LoadFloat32x8Part(b[i:])
        acc = x.MulAdd(y, acc) // zero-filled lanes add 0
    }
    var lanes [8]float32
    acc.StoreArray(&lanes)
    var s float32
    for _, v := range lanes {
        s += v
    }
    return s
}

func dot16(a, b []float32) float32 {
    var acc archsimd.Float32x16
    i := 0
    for ; i < len(a)-16+1; i += 16 {
        x := archsimd.LoadFloat32x16(a[i : i+16])
        y := archsimd.LoadFloat32x16(b[i : i+16])
        acc = x.MulAdd(y, acc)
    }
    if i < len(a) {
        x, _ := archsimd.LoadFloat32x16Part(a[i:])
        y, _ := archsimd.LoadFloat32x16Part(b[i:])
        acc = x.MulAdd(y, acc)
    }
    var lanes [16]float32
    acc.StoreArray(&lanes)
    var s float32
```

```
    for _, v := range lanes {
        s += v
    }
    return s
}

// Count returns how many bytes of s equal c.
func Count(s []byte, c byte) int {
    if !archsimd.X86.AVX512() {
        return countScalar(s, c)
    }
    needle := archsimd.BroadcastUint8x64(c)
    n, i := 0, 0
    for ; i < len(s)-64+1; i += 64 {
        m := archsimd.LoadUint8x64(s[i : i+64]).Equal(needle)
        n += bits.OnesCount64(m.ToBits())
    }
    return n + countScalar(s[i:], c)
}

// ReverseBits writes bits.Reverse8(src[i]) into dst[i] for every i.
func ReverseBits(dst, src []byte) {
    if !archsimd.X86.AVX512GFNI() {
        reverseScalar(dst, src)
        return
    }
    src = src[:len(dst)]
    rev := archsimd.BroadcastUint64x8(0x8040201008040201) // anti-diagonal bit matrix
    i := 0
    for ; i < len(src)-64+1; i += 64 {
        v := archsimd.LoadUint8x64(src[i : i+64])
        v.GaloisFieldAffineTransform(rev, 0).Store(dst[i : i+64])
    }
    if i < len(src) {
        v, _ := archsimd.LoadUint8x64Part(src[i:])
        v.GaloisFieldAffineTransform(rev, 0).StorePart(dst[i:])
    }
}
```

```
// dispatch_generic.go - the same API without the experiment (or on other GOARCHes)
//go:build !(goexperiment.simd && amd64)

package simdlab

func Path() string                 { return "scalar (no GOEXPERIMENT=simd)" }
func Dot(a, b []float32) float32  { return dotScalar(a, b[:len(a)]) }
```

```go
func Count(s []byte, c byte) int  { return countScalar(s, c) }
func ReverseBits(dst, src []byte) { reverseScalar(dst, src) }
```

Things to notice:

- **Dispatch once per call, at the top.** Each exported function asks the CPU what it has and picks the widest safe path. Inside `dot16`, every instruction is AVX-512, and the call is only reached when `X86.AVX512()` is true.
- **Full vectors, then one partial vector.** `LoadFloat32x16Part` zero-fills the lanes past the end of the slice and `StorePart` writes only the real ones. For a dot product the zero lanes contribute `0*0`, so the tail needs no special case.
- **`Count` reduces a mask to an integer.** `Equal` yields a `Mask8x64`, `ToBits()` turns it into a `uint64` with one bit per byte, and `bits.OnesCount64` counts them: 64 comparisons, one compare instruction, one popcount.
- **`ReverseBits` is the GFNI trick** from the Go team's `archsimd` article. One `VGF2P8AFFINEQB` multiplies every byte by an 8×8 bit matrix, and the anti-diagonal matrix reverses the bits of 64 bytes at once. No portable SIMD API has this instruction. It is the kind of thing `archsimd` exists for.
- **The loop bound is `i < len(s)-64+1`**, not `i+64 <= len(s)`. That is deliberate. See the trap below.

A small driver:

```go
// cmd/demo/main.go - call the dispatching API and compare with plain loops
package main

import (
    "bytes"
    "fmt"
    "math/bits"

    "example.com/simdlab"
)

func main() {
    fmt.Println("path:", simdlab.Path())

    a := make([]float32, 1000)
    b := make([]float32, 1000)
    for i := range a {
        a[i] = 1.0 / float32(i+1)
        b[i] = float32(i%3) + 0.1
    }
    var loop float32
    for i := range a {
        loop += a[i] * b[i]
    }
```

```
    fmt.Printf("dot: simd=%.7f loop=%.7f equal=%t\n", simdlab.Dot(a, b), loop, simdlab.Dot(a

    text := bytes.Repeat([]byte("line one\nline two\n"), 50)
    fmt.Println("newlines:", simdlab.Count(text, '\n'), "bytes.Count:", bytes.Count(text, []

    src := []byte{0b0000_0001, 0b1100_0000, 0b1010_1010, 0xF0}
    dst := make([]byte, len(src))
    simdlab.ReverseBits(dst, src)
    for i := range src {
        fmt.Printf("%08b -> %08b (bits.Reverse8 %08b)\n", src[i], dst[i], bits.Reverse8(src[
    }
}
```

```
GOEXPERIMENT=simd go run ./cmd/demo
go run ./cmd/demo
```

```
path: avx512
dot: simd=7.3820782 loop=7.3820772 equal=false
newlines: 100 bytes.Count: 100
00000001 -> 10000000 (bits.Reverse8 10000000)
11000000 -> 00000011 (bits.Reverse8 00000011)
10101010 -> 01010101 (bits.Reverse8 01010101)
11110000 -> 00001111 (bits.Reverse8 00001111)
path: scalar (no GOEXPERIMENT=simd)
dot: simd=7.3820772 loop=7.3820772 equal=true
newlines: 100 bytes.Count: 100
00000001 -> 10000000 (bits.Reverse8 10000000)
11000000 -> 00000011 (bits.Reverse8 00000011)
10101010 -> 01010101 (bits.Reverse8 01010101)
11110000 -> 00001111 (bits.Reverse8 00001111)
```

The same source builds both ways. Integer results match exactly. The floating-point dot product differs in the seventh significant digit, which is correct behaviour and the subject of the second trap.

Case 3: test every path on one machine

A vector path that only runs on the CI machine's CPU is a path you haven't tested. The table test compares all three functions with the scalar twin at every length from 0 to 300, so every tail size is covered:

```
// simdlab_test.go - vector paths must agree with the scalar reference at every length
package simdlab

import (
```

```go
    "bytes"
    "math"
    "math/rand/v2"
    "testing"
)

func TestAgainstScalar(t *testing.T) {
    t.Logf("path: %s", Path())
    r := rand.New(rand.NewPCG(1, 2))
    for n := 0; n <= 300; n++ {
        a, b := make([]float32, n), make([]float32, n)
        for i := range a {
            a[i], b[i] = r.Float32()*2-1, r.Float32()*2-1
        }
        got, want := Dot(a, b), dotScalar(a, b)
        if math.Abs(float64(got-want)) > 1e-4*float64(n+1) {
            t.Fatalf("Dot n=%d: got %v want %v", n, got, want)
        }

        s := make([]byte, n)
        for i := range s {
            s[i] = byte(r.IntN(4))
        }
        if got, want := Count(s, 2), countScalar(s, 2); got != want {
            t.Fatalf("Count n=%d: got %d want %d", n, got, want)
        }

        d1, d2 := make([]byte, n), make([]byte, n)
        ReverseBits(d1, s)
        reverseScalar(d2, s)
        if !bytes.Equal(d1, d2) {
            t.Fatalf("ReverseBits n=%d differs", n)
        }
    }
}

var sinkF float32
var sinkI int

func BenchmarkDot(b *testing.B) {
    a, c := make([]float32, 4096), make([]float32, 4096)
    for i := range a {
        a[i], c[i] = float32(i%7), float32(i%5)
    }
    b.Run("scalar", func(b *testing.B) {
        for b.Loop() {
            sinkF = dotScalar(a, c)
```

```
        }
    })
    b.Run("archsimd", func(b *testing.B) {
        for b.Loop() {
            sinkF = Dot(a, c)
        }
    })
}

func BenchmarkCount(b *testing.B) {
    s := bytes.Repeat([]byte("the quick brown fox\n"), 3277) // ~64 KiB
    b.Run("scalar", func(b *testing.B) {
        for b.Loop() {
            sinkI = countScalar(s, '\n')
        }
    })
    b.Run("archsimd", func(b *testing.B) {
        for b.Loop() {
            sinkI = Count(s, '\n')
        }
    })
    b.Run("bytes.Count", func(b *testing.B) {
        for b.Loop() {
            sinkI = bytes.Count(s, []byte{'\n'})
        }
    })
}

func BenchmarkReverseBits(b *testing.B) {
    src, dst := make([]byte, 64<<10), make([]byte, 64<<10)
    for i := range src {
        src[i] = byte(i)
    }
    b.Run("scalar", func(b *testing.B) {
        for b.Loop() {
            reverseScalar(dst, src)
        }
    })
    b.Run("archsimd", func(b *testing.B) {
        for b.Loop() {
            ReverseBits(dst, src)
        }
    })
}
```

The runtime honours `GODEBUG=cpu.<feature>=off`, and `archsimd.X86` reports what the runtime believes. That lets you force each branch without leaving your desk:

```go
GOEXPERIMENT=simd go test -count=1 -v -run TestAgainstScalar .
GOEXPERIMENT=simd GODEBUG=cpu.avx512f=off go test -count=1 -v -run TestAgainstScalar .
GOEXPERIMENT=simd GODEBUG=cpu.avx512f=off,cpu.avx2=off go test -count=1 -v -run TestAgainstS
go test -count=1 -v -run TestAgainstScalar .
```

```
=== RUN   TestAgainstScalar
    simdlab_test.go:12: path: avx512
--- PASS: TestAgainstScalar (0.00s)
PASS
ok      example.com/simdlab 0.003s
=== RUN   TestAgainstScalar
    simdlab_test.go:12: path: avx2+fma
--- PASS: TestAgainstScalar (0.00s)
PASS
ok      example.com/simdlab 0.004s
=== RUN   TestAgainstScalar
    simdlab_test.go:12: path: scalar
--- PASS: TestAgainstScalar (0.00s)
PASS
ok      example.com/simdlab 0.004s
=== RUN   TestAgainstScalar
    simdlab_test.go:12: path: scalar (no GOEXPERIMENT=simd)
--- PASS: TestAgainstScalar (0.00s)
PASS
ok      example.com/simdlab 0.004s
```

Put all four invocations in CI. Also add `GOEXPERIMENT=simd GOARCH=arm64 go vet ./...` and `GOEXPERIMENT=simd GOOS=wasip1 GOARCH=wasm go vet ./...` (both pass for this module), so a missing build constraint breaks CI before it breaks a user on another architecture.

Case 4: measure, and compare with what you already have

```go
GOEXPERIMENT=simd go test -run '^$' -bench . -benchmem -count=3 .
```

```
goos: linux
goarch: amd64
pkg: example.com/simdlab
cpu: Intel(R) Xeon(R) Processor
BenchmarkDot/scalar-8         409718          2886 ns/op           0 B/op          0 allocs/op
BenchmarkDot/scalar-8         444223          2813 ns/op           0 B/op          0 allocs/op
BenchmarkDot/scalar-8         446895          2701 ns/op           0 B/op          0 allocs/op
BenchmarkDot/archsimd-8      2400464           504.0 ns/op         0 B/op          0 allocs/op
BenchmarkDot/archsimd-8      2398826           498.4 ns/op         0 B/op          0 allocs/op
BenchmarkDot/archsimd-8      2377718           508.5 ns/op         0 B/op          0 allocs/op
```

```
BenchmarkCount/scalar-8        31746         38770 ns/op           0 B/op          0 allocs/op
BenchmarkCount/scalar-8        30878         38341 ns/op           0 B/op          0 allocs/op
BenchmarkCount/scalar-8        31382         38393 ns/op           0 B/op          0 allocs/op
BenchmarkCount/archsimd-8            1222172           963.8 ns/op         0 B/op          0 al
BenchmarkCount/archsimd-8            1384680           881.7 ns/op         0 B/op          0 al
BenchmarkCount/archsimd-8            1445408           848.3 ns/op         0 B/op          0 al
BenchmarkCount/bytes.Count-8          955839          1126 ns/op           0 B/op          0 al
BenchmarkCount/bytes.Count-8         1000000          1126 ns/op           0 B/op          0 al
BenchmarkCount/bytes.Count-8         1000000          1538 ns/op           0 B/op          0 al
BenchmarkReverseBits/scalar-8          27670         44528 ns/op           0 B/op          0 al
BenchmarkReverseBits/scalar-8          24739         45573 ns/op           0 B/op          0 al
BenchmarkReverseBits/scalar-8          26042         43831 ns/op           0 B/op          0 al
BenchmarkReverseBits/archsimd-8       755580          1684 ns/op           0 B/op          0 al
BenchmarkReverseBits/archsimd-8       748688          1646 ns/op           0 B/op          0 al
BenchmarkReverseBits/archsimd-8       703843          1606 ns/op           0 B/op          0 al
PASS
ok      example.com/simdlab 25.397s
```

This box is a shared VM, so treat these as one machine's numbers and use `benchstat` with `-count=10` before quoting them. The shape is still clear:

Kernel (input)	Scalar	`archsimd`	Comment
Dot, 4096 `float32`	~2.8 µs	~0.50 µs	~5.6×, AVX-512 FMA
Count `'\n'`, 64 KiB	~38 µs	~0.87 µs	…but `bytes.Count` is already ~1.1 µs
ReverseBits, 64 KiB	~44 µs	~1.6 µs	~27×, one GFNI instruction per 64 bytes

The `Count` row is the lesson. A naive byte loop is slow, but nobody should be running one. `bytes.Count` already has hand-written vector assembly, and beating it by 20% on one CPU family is not worth an experimental dependency. `ReverseBits` is the opposite case: no library function does it, and the instruction that does it isn't reachable any other way.

For plain arithmetic, the portable layer usually does as well. The same dot product written with `simd.Float32s` (which picked 512-bit vectors on this CPU, or 256-bit under `GODEBUG=cpu.avx512f=off`) ran at the same speed (`portable_simd.go`, plus a `TestPortable` / `BenchmarkDotPortable` pair in `portable_simd_test.go` built under the same `goexperiment.simd` tag):

```
// portable_simd.go - the same inner product with the portable simd package (any GOARCH)
//go:build goexperiment.simd

package simdlab
```

```go
import "simd"

// DotPortable uses whatever vector width this machine has (or an emulation).
func DotPortable(a, b []float32) float32 {
    b = b[:len(a)]
    var acc simd.Float32s
    w := acc.Len()
    i := 0
    for ; i < len(a)-w+1; i += w {
        acc = simd.LoadFloat32s(a[i:i+w]).MulAdd(simd.LoadFloat32s(b[i:i+w]), acc)
    }
    if i < len(a) {
        x, _ := simd.LoadFloat32sPart(a[i:])
        y, _ := simd.LoadFloat32sPart(b[i:])
        acc = x.MulAdd(y, acc)
    }
    lanes := make([]float32, w)
    acc.Store(lanes)
    var s float32
    for _, v := range lanes {
        s += v
    }
    return s
}

// VectorBits reports the width the portable package picked.
func VectorBits() (int, bool) { return simd.VectorBitSize(), simd.Emulated() }
```

```go
// portable_simd_test.go
//go:build goexperiment.simd

package simdlab

import (
    "math"
    "testing"
)

func TestPortable(t *testing.T) {
    bitsz, emu := VectorBits()
    t.Logf("simd.VectorBitSize()=%d Emulated()=%t", bitsz, emu)
    for n := 0; n <= 300; n++ {
        a, b := make([]float32, n), make([]float32, n)
        for i := range a {
            a[i], b[i] = float32(i%7)-3, float32(i%5)-2
        }
```

```go
        if got, want := DotPortable(a, b), dotScalar(a, b); math.Abs(float64(got-want)) > 1e
            t.Fatalf("n=%d got %v want %v", n, got, want)
        }
    }
}

func BenchmarkDotPortable(b *testing.B) {
    a, c := make([]float32, 4096), make([]float32, 4096)
    for i := range a {
        a[i], c[i] = float32(i%7), float32(i%5)
    }
    for b.Loop() {
        sinkF = DotPortable(a, c)
    }
}
```

```go
GOEXPERIMENT=simd go test -count=1 -v -run Portable -bench DotPortable -benchmem -count=3 .
```

```
    portable_simd_test.go:13: simd.VectorBitSize()=512 Emulated()=false
    portable_simd_test.go:13: simd.VectorBitSize()=512 Emulated()=false
    portable_simd_test.go:13: simd.VectorBitSize()=512 Emulated()=false
BenchmarkDotPortable-8       2421553           498.2 ns/op         0 B/op          0 allocs/op
BenchmarkDotPortable-8       2357865           504.8 ns/op         0 B/op          0 allocs/op
BenchmarkDotPortable-8       2257526           524.8 ns/op         0 B/op          0 allocs/op
```

That version needs no `amd64` constraint, no feature checks and no per-width copies. Reach for `archsimd` only when `simd` can't express the operation or can't reach the speed you have measured you need.

The trap

Importing `archsimd` without a build constraint

```go
// main.go - imports archsimd with no build constraint
package main

import (
    "fmt"
    "simd/archsimd"
)

func main() {
    v := archsimd.BroadcastFloat32x8(1.5)
    fmt.Println(v.Add(v))
```

```
}
```

```
go build .
GOEXPERIMENT=simd go run .
GOEXPERIMENT=simd GOARCH=arm64 go build .
```

```
package example.com/naive
    imports simd/archsimd: build constraints exclude all Go files in /home/box/sdk/go1.27.1/sr
{3,3,3,3,3,3,3,3}
# example.com/naive
./main.go:10:16: undefined: archsimd.BroadcastFloat32x8
```

It fails twice. Without the experiment the package does not exist. With the experiment on arm64 the package exists, but 256-bit types don't. The rule is mechanical: every file that imports `simd/archsimd` carries `//go:build goexperiment.simd && <arch>`, and a sibling file with the negated constraint provides the same API.

Expecting bit-identical floating-point results

The vector dot product keeps 16 partial sums and adds them at the end. The loop keeps one. Floating-point addition is not associative, so `7.3820782` vs `7.3820772` is correct behaviour, not a bug. Compare float kernels against the scalar twin with a tolerance scaled to the input length, as the test does. Never use `==`. If a caller truly needs reproducible results across machines, the vector path can't be used for that value.

Writing the loop bound the "obvious" way

The Go team calls this out explicitly, and the compiler's bounds-check debug flag shows why:

```go
// bce.go - the same loop, two ways of writing the bound
//go:build goexperiment.simd && amd64

package bce

import "simd/archsimd"

func SumA(s []float32) archsimd.Float32x8 { // i+8 <= len(s): i+8 might overflow
    var acc archsimd.Float32x8
    for i := 0; i+8 <= len(s); i += 8 {
        acc = acc.Add(archsimd.LoadFloat32x8(s[i : i+8]))
    }
    return acc
}

func SumB(s []float32) archsimd.Float32x8 { // i < len(s)-8+1: provably in range
```

```go
    var acc archsimd.Float32x8
    for i := 0; i < len(s)-8+1; i += 8 {
        acc = acc.Add(archsimd.LoadFloat32x8(s[i : i+8]))
    }
    return acc
}
```

```go
GOEXPERIMENT=simd go build -gcflags='-d=ssa/check_bce/debug=1' .
```

```
# example.com/bce
./bce.go:11:41: Found IsSliceInBounds
```

Only `SumA` (line 11) keeps a bounds check inside the loop. The compiler cannot rule out `i+8` overflowing, so it cannot prove `s[i:i+8]` is in range. In a scalar loop that is noise. In a loop whose body is one or two instructions, it is a large part of the work.

Vectors in arrays and structs

`archsimd` values are 16–64 bytes. Go's register allocator currently keeps them in registers only as plain locals and parameters. Put eight of them in a `[8]archsimd.Int32x8` or a struct and the compiler spills them to the stack on every access. That is why the Go team's 8×8 transpose takes eight separate parameters. Keep vectors as locals in hot loops, and don't store them in heap objects. The package documentation says so explicitly.

The boring rule

- Profile first. If the hot loop is a search, a hash or crypto, the standard library has probably vectorized it already.
- Prefer `simd` (portable) for arithmetic. Use `archsimd` when you need a specific instruction or have measured that `simd` is not enough.
- Every `archsimd` file: `//go:build goexperiment.simd && amd64` (or `arm64` / `wasm`), plus a negated-constraint fallback with the same API.
- Every exported entry point checks CPU features first and falls back to a scalar twin. Never call a vector helper that hasn't been guarded.
- Test against the scalar twin at every tail length, under `GODEBUG=cpu.avx512f=off` and `cpu.avx2=off`, and without the experiment. Use tolerances for floats.
- Treat the API as unstable. Go 1.27 already renamed `Merge` → `IfElse` and replaced the `As*` casts with `ToBits` / `BitsTo*` / `ReshapeTo*` (the old names still compile, marked *Deprecated*). Pin the Go version in CI and expect to edit this code when you upgrade.

Try this

1. Add a `CountAVX2` path using `Uint8x32` and `Mask8x32.ToBits()`, wire it into `Count` for CPUs without AVX-512, and prove it runs with `GODEBUG=cpu.avx512f=off`.
2. Change `dot16` to keep **four** accumulators. Benchmark it with `-count=10` and `benchstat`. Does it help on your CPU, and is the kernel limited by FMA latency or by memory?
3. Write `ReverseBits` with the portable `simd` package. What operation is missing, and how fast is the best portable version you can write compared with GFNI?
4. Build the module with `GOEXPERIMENT=simd GOOS=wasip1 GOARCH=wasm` and add a `wasm` build tag to a 128-bit `Float32x4` version of `dot`. Run the tests under a WASI runtime.
5. Look up `go doc simd/archsimd.Float32x8.Merge` and `AsUint32x8` in your toolchain and replace any deprecated calls in the code you write.

Sources

- Go blog, "Arch-specific SIMD in Go" (Junyang Shao and David Chase, 2026-10-02): https://go.dev/blog/archsimd
- Go blog, "Platform-independent SIMD in Go": https://go.dev/blog/simd-experiment
- Go 1.27 release notes, `simd` and `simd/archsimd`: https://go.dev/doc/go1.27
- Proposal and tracking issue #73787, "simd/archsimd: architecture-specific SIMD intrinsics under a GOEXPERIMENT": https://github.com/golang/go/issues/73787
- Package docs, `go doc simd/archsimd` and `go doc simd` with `GOEXPERIMENT=simd` (Go 1.27.1)

Go 1.27 compress/flate Rewrite

Go 1.27 `compress/flate` Rewrite

Go **1.27** rewrote the DEFLATE encoder in `compress/flate`. Compression speed goes up for the common levels, and the exact encoded bytes change. `compress/gzip`, `compress/zlib`, `archive/zip`, and `image/png` all sit on that encoder, so their output changes too. The Go 1.27 release notes say so in one paragraph. The foundations tour repeats it as a table cell. This chapter measures it.

The format is still RFC 1951. Every gzip file we produced on Go 1.24.4 gunzips on Go 1.27.2, and the reverse. The decompressed SHA-256 is identical. The compressed SHA-256 is not. That is specified behaviour: `flate.NewWriter`, `gzip.NewWriter`, `zip.NewWriter`, and `png.Encode` all document that the exact bytes written are outside the Go 1 compatibility promise.

Everything below was compiled and run on one amd64 box (`Intel(R) Xeon(R) Processor`, 8 CPUs) with **Go 1.24.4** (`/usr/bin/go`) and **Go 1.27.2**. Treat the timings as one machine's numbers; use `-count=10` and `benchstat` before quoting them in a performance review. The release notes compare against Go 1.26; this box has no 1.26 toolchain, so the baseline here is 1.24.4. Neither the Go 1.25 nor the Go 1.26 release notes mention a `compress/flate` change.

Mental model

```
gzip  zlib  zip  png


  compress/flate.Writer     DEFLATE bitstream (RFC 1951)


  flate.Reader / gzip.Reader / zip / png.Decode
```

- **DEFLATE is a format.** LZ77 copies plus Huffman coding. Many bitstreams decode to the same payload. A compressor is free to pick any valid encoding.
- **The wrappers only frame it.** gzip (RFC 1952) adds a 10-byte header, CRC-32, and ISIZE. zlib (RFC 1950) adds a 2-byte CMF/FLG header and an Adler-32. zip stores DEFLATE members (stdlib default: `flate.NewWriter(w, 5)` in `archive/zip/register.go`). PNG IDAT chunks go through `compress/zlib`.
- **DefaultCompression is still −1 → 6.** Both 1.24 and 1.27 map `DefaultCompression` to level 6 in `compressor.init`. The *algorithm* at 6 changed.
- **Go 1.24 encoder.** Level 1 (`BestSpeed`) is the Snappy-style matcher in `deflatefast.go`. Levels 2–9 are one hash-chain compressor with per-level `good` / `lazy` / `nice` / `chain` knobs.

- **Go 1.27 encoder.** Levels 1–6 are specialized match finders `fastEncL1` … `fastEncL6` (`deflatefast.go`, `level1.go`–`level6.go`): bigger/sparser hash tables, 4- and 7-byte hashes, skip-ahead on long literal runs. Levels 7–9 stay on the hash-chain path with rebalanced knobs (level 9 `chain` drops from 4096 to 1024). Tracked as issue 75532.
- **Tiny payloads.** On the 1.27 fast path, a flush of 32 bytes is written as a stored (uncompressed) DEFLATE block. A 23-byte gzip of `"desk ticket 42 is open\n"` contains that ASCII in the clear. The 1.24 encoder Huffman-coded the same string.

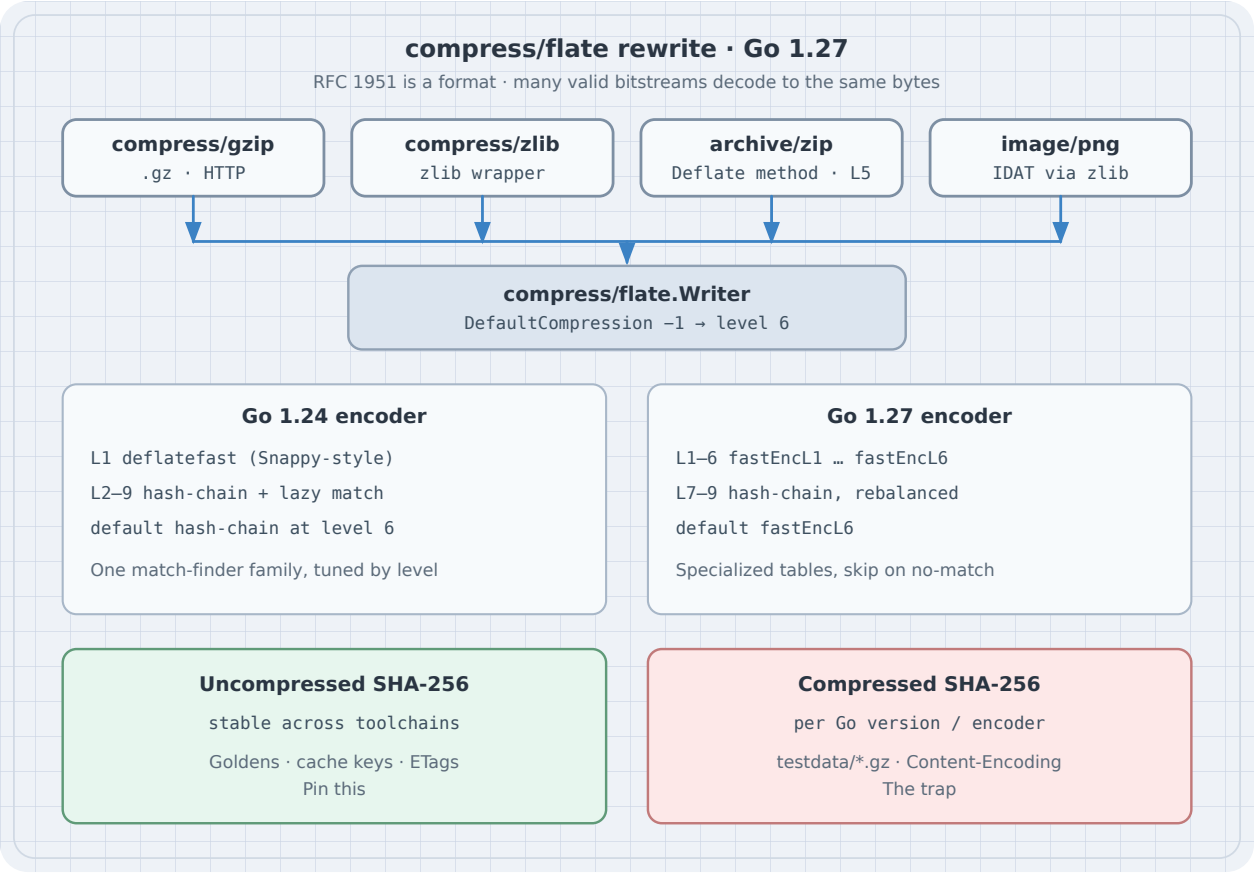


Figure 1: gzip, zlib, zip, and png sit on compress/flate; Go 1.27 swaps the encoder; uncompressed hashes stay stable

Worked examples

Set up a module. The `go` line is 1.24 so both toolchains compile it; each `go` binary still links **its own** standard library.

```
mkdir flatebench && cd flatebench
go mod init example.com/flatebench   # go 1.24 in go.mod
go version
```

```
go version go1.24.4 linux/amd64
```

```
go version go1.27.2 linux/amd64
```

Case 1: size, hash, and round-trip on the same inputs

Five 1 MiB payloads, generated in-process so both compilers see identical bytes: repeating HTML table rows, JSON records, Go-like source, zeros, and `math/rand/v2` with a fixed PCG seed. gzip header is pinned (`ModTime` zero, `OS` 255) so a hash difference is the DEFLATE payload, not the gzip timestamp.

```go
// compare.go - gzip/flate/zlib/zip/png size, hash, and round-trip check
package main

import (
    "archive/zip"
    "bytes"
    "compress/flate"
    "compress/gzip"
    "compress/zlib"
    "crypto/sha256"
    "encoding/hex"
    "fmt"
    "hash"
    "image"
    "image/color"
    "image/png"
    "io"
    "math/rand/v2"
    "os"
    "runtime"
    "strings"
    "time"
)

const payload = 1 << 20 // 1 MiB

func main() {
    mode := "compare"
    if len(os.Args) > 1 {
        mode = os.Args[1]
    }
    switch mode {
    case "dump-golden":
        dumpGolden()
        return
    case "check-golden":
        path := "testdata/ticket.gz"
```

```
        if len(os.Args) > 2 {
            path = os.Args[2]
        }
        checkGolden(path)
        return
    }

    fmt.Printf("go %s %s/%s\n", runtime.Version(), runtime.GOOS, runtime.GOARCH)
    fmt.Printf("compiler %s  payload %d bytes\n", runtime.Compiler, payload)

    inputs := []struct {
        name string
        data []byte
    }{
        {"html", makeHTML(payload)},
        {"json", makeJSON(payload)},
        {"src", makeSrc(payload)},
        {"zeros", makeZeros(payload)},
        {"rand", makeRand(payload)},
    }

    fmt.Println()
    fmt.Println("== gzip (ModTime zero, OS 255) ==")
    fmt.Printf("%-6s %5s %10s %10s %7s  %-16s  %-16s  %s\n",
        "input", "lvl", "in", "out", "ratio", "gz-sha256", "raw-sha256", "roundtrip")
    for _, in := range inputs {
        raw := sha256.Sum256(in.data)
        rawHex := hex.EncodeToString(raw[:])[:16]
        for _, lvl := range []int{gzip.BestSpeed, 6, gzip.DefaultCompression, gzip.BestCompr
            out, err := gzipBytes(in.data, lvl)
            must(err)
            back, err := gunzipBytes(out)
            must(err)
            ok := "OK"
            if !bytes.Equal(back, in.data) {
                ok = "FAIL"
            }
            sum := sha256.Sum256(out)
            ratio := float64(len(in.data)) / float64(len(out))
            fmt.Printf("%-6s %5s %10d %10d %7.2f  %-16s  %-16s  %s\n",
                in.name, lvlName(lvl), len(in.data), len(out), ratio,
                hex.EncodeToString(sum[:])[:16], rawHex, ok)
        }
    }

    fmt.Println()
    fmt.Println("== wrappers on html (default level) ==")
```

```
    html := inputs[0].data
    type row struct {
        pkg  string
        out  []byte
        back []byte
    }
    flateOut, err := flateBytes(html, flate.DefaultCompression)
    must(err)
    flateBack, err := inflateBytes(flateOut)
    must(err)
    zlibOut, err := zlibBytes(html, zlib.DefaultCompression)
    must(err)
    zlibBack, err := unzlibBytes(zlibOut)
    must(err)
    zipOut, err := zipBytes("tickets.html", html)
    must(err)
    zipBack, err := unzipFirst(zipOut)
    must(err)
    pngOut, err := pngBytes(makePattern(256, 256))
    must(err)
    pngBack, err := pngDecode(pngOut)
    must(err)
    pngRaw, err := pngBytesRaw(makePattern(256, 256))
    must(err)

    rows := []row{
        {"compress/flate", flateOut, flateBack},
        {"compress/zlib", zlibOut, zlibBack},
        {"archive/zip", zipOut, zipBack},
    }
    fmt.Printf("%-16s %10s  %-16s  %s\n", "package", "out", "out-sha256", "roundtrip")
    for _, r := range rows {
        sum := sha256.Sum256(r.out)
        ok := "OK"
        if !bytes.Equal(r.back, html) {
            ok = "FAIL"
        }
        fmt.Printf("%-16s %10d  %-16s  %s\n", r.pkg, len(r.out), hex.EncodeToString(sum[:])[
    }
    psum := sha256.Sum256(pngOut)
    ok := "OK"
    if !bytes.Equal(pngBack, pngRaw) {
        ok = "FAIL"
    }
    fmt.Printf("%-16s %10d  %-16s  %s  (256x256 NRGBA pixels)\n",
        "image/png", len(pngOut), hex.EncodeToString(psum[:])[:16], ok)
```

```
    fmt.Println()
    fmt.Println("raw-sha256 of html payload:", hex.EncodeToString(sha256Sum(html).Sum(nil)))
}

func lvlName(n int) string {
    switch n {
    case gzip.DefaultCompression:
        return "-1"
    case gzip.BestSpeed:
        return "1"
    case gzip.BestCompression:
        return "9"
    default:
        return fmt.Sprintf("%d", n)
    }
}

func gzipBytes(in []byte, level int) ([]byte, error) {
    var buf bytes.Buffer
    w, err := gzip.NewWriterLevel(&buf, level)
    if err != nil {
        return nil, err
    }
    // Keep the 10-byte gzip header stable: zero MTIME, OS=255 (package default).
    w.ModTime = time.Time{}
    if _, err := w.Write(in); err != nil {
        return nil, err
    }
    if err := w.Close(); err != nil {
        return nil, err
    }
    return buf.Bytes(), nil
}

func gunzipBytes(in []byte) ([]byte, error) {
    r, err := gzip.NewReader(bytes.NewReader(in))
    if err != nil {
        return nil, err
    }
    defer r.Close()
    return io.ReadAll(r)
}

func flateBytes(in []byte, level int) ([]byte, error) {
    var buf bytes.Buffer
    w, err := flate.NewWriter(&buf, level)
    if err != nil {
```

```
        return nil, err
    }
    if _, err := w.Write(in); err != nil {
        return nil, err
    }
    if err := w.Close(); err != nil {
        return nil, err
    }
    return buf.Bytes(), nil
}

func inflateBytes(in []byte) ([]byte, error) {
    r := flate.NewReader(bytes.NewReader(in))
    defer r.Close()
    return io.ReadAll(r)
}

func zlibBytes(in []byte, level int) ([]byte, error) {
    var buf bytes.Buffer
    w, err := zlib.NewWriterLevel(&buf, level)
    if err != nil {
        return nil, err
    }
    if _, err := w.Write(in); err != nil {
        return nil, err
    }
    if err := w.Close(); err != nil {
        return nil, err
    }
    return buf.Bytes(), nil
}

func unzlibBytes(in []byte) ([]byte, error) {
    r, err := zlib.NewReader(bytes.NewReader(in))
    if err != nil {
        return nil, err
    }
    defer r.Close()
    return io.ReadAll(r)
}

func zipBytes(name string, in []byte) ([]byte, error) {
    var buf bytes.Buffer
    zw := zip.NewWriter(&buf)
    h := &zip.FileHeader{
        Name:     name,
        Method:   zip.Deflate,
```

```
        Modified: time.Date(2026, 1, 1, 0, 0, 0, 0, time.UTC),
    }
    w, err := zw.CreateHeader(h)
    if err != nil {
        return nil, err
    }
    if _, err := w.Write(in); err != nil {
        return nil, err
    }
    if err := zw.Close(); err != nil {
        return nil, err
    }
    return buf.Bytes(), nil
}

func unzipFirst(in []byte) ([]byte, error) {
    r, err := zip.NewReader(bytes.NewReader(in), int64(len(in)))
    if err != nil {
        return nil, err
    }
    if len(r.File) == 0 {
        return nil, fmt.Errorf("empty zip")
    }
    rc, err := r.File[0].Open()
    if err != nil {
        return nil, err
    }
    defer rc.Close()
    return io.ReadAll(rc)
}

func pngBytes(img image.Image) ([]byte, error) {
    var buf bytes.Buffer
    if err := png.Encode(&buf, img); err != nil {
        return nil, err
    }
    return buf.Bytes(), nil
}

func pngDecode(in []byte) ([]byte, error) {
    img, err := png.Decode(bytes.NewReader(in))
    if err != nil {
        return nil, err
    }
    return pixels(img), nil
}
```

```
func pngBytesRaw(img image.Image) ([]byte, error) {
    return pixels(img), nil
}

func pixels(img image.Image) []byte {
    b := img.Bounds()
    out := make([]byte, 0, b.Dx()*b.Dy()*4)
    for y := b.Min.Y; y < b.Max.Y; y++ {
        for x := b.Min.X; x < b.Max.X; x++ {
            r, g, bl, a := img.At(x, y).RGBA()
            out = append(out, uint8(r>>8), uint8(g>>8), uint8(bl>>8), uint8(a>>8))
        }
    }
    return out
}

func makePattern(w, h int) *image.NRGBA {
    img := image.NewNRGBA(image.Rect(0, 0, w, h))
    for y := 0; y < h; y++ {
        for x := 0; x < w; x++ {
            img.SetNRGBA(x, y, color.NRGBA{
                R: uint8(x),
                G: uint8(y),
                B: uint8(x ^ y),
                A: 255,
            })
        }
    }
    return img
}

func makeHTML(n int) []byte {
    var b strings.Builder
    b.Grow(n)
    b.WriteString("<!doctype html><table>\n")
    for i := 0; b.Len() < n; i++ {
        fmt.Fprintf(&b, "<tr><td>%d</td><td>ticket-%05d</td><td>open</td></tr>\n", i, i%1000
    }
    return []byte(b.String())[:n]
}

func makeJSON(n int) []byte {
    var b strings.Builder
    b.Grow(n)
    b.WriteString("[\n")
    for i := 0; b.Len() < n; i++ {
        fmt.Fprintf(&b, `{"id":%d,"desk":%d,"status":"open","note":"shift-handoff"},`+"\n",
```

```
    }
    return []byte(b.String())[:n]
}

func makeSrc(n int) []byte {
    var b strings.Builder
    b.Grow(n)
    b.WriteString("package tickets\n\n")
    for i := 0; b.Len() < n; i++ {
        fmt.Fprintf(&b, "func Ticket%d(id int) int {\n\treturn id + %d\n}\n\n", i, i%97)
    }
    return []byte(b.String())[:n]
}

func makeZeros(n int) []byte { return make([]byte, n) }

func makeRand(n int) []byte {
    r := rand.New(rand.NewPCG(1, 2))
    out := make([]byte, n)
    for i := range out {
        out[i] = byte(r.Uint64())
    }
    return out
}

func sha256Sum(in []byte) hash.Hash {
    h := sha256.New()
    h.Write(in)
    return h
}

func must(err error) {
    if err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }
}

const goldenPlain = "desk ticket 42 is open\n"

func dumpGolden() {
    plain := []byte(goldenPlain)
    gz, err := gzipBytes(plain, gzip.DefaultCompression)
    must(err)
    sum := sha256.Sum256(gz)
    back, err := gunzipBytes(gz)
    must(err)
```

```
    fmt.Printf("go %s\n", runtime.Version())
    fmt.Printf("plain      %q\n", plain)
    fmt.Printf("gz-len     %d\n", len(gz))
    fmt.Printf("gz-sha256  %s\n", hex.EncodeToString(sum[:]))
    fmt.Printf("raw-sha256 %s\n", hex.EncodeToString(sha256Sum(plain).Sum(nil)))
    fmt.Printf("roundtrip  equal=%t\n", bytes.Equal(back, plain))
    if err := os.MkdirAll("testdata", 0o755); err != nil {
        must(err)
    }
    must(os.WriteFile("testdata/ticket.gz", gz, 0o644))
    fmt.Println("wrote testdata/ticket.gz")
}

func checkGolden(path string) {
    want, err := os.ReadFile(path)
    must(err)
    plain := []byte(goldenPlain)
    got, err := gzipBytes(plain, gzip.DefaultCompression)
    must(err)
    fmt.Printf("go %s\n", runtime.Version())
    fmt.Printf("golden-len %d  got-len %d  bytes.Equal=%t\n",
        len(want), len(got), bytes.Equal(want, got))
    fmt.Printf("golden-sha %s\n", hex.EncodeToString(sha256Sum(want).Sum(nil)))
    fmt.Printf("got-sha    %s\n", hex.EncodeToString(sha256Sum(got).Sum(nil)))
    backWant, err := gunzipBytes(want)
    must(err)
    backGot, err := gunzipBytes(got)
    must(err)
    fmt.Printf("decompressed both equal to plaintext: %t\n",
        bytes.Equal(backWant, plain) && bytes.Equal(backGot, plain))
}
```

Run with each toolchain:

```
go run . compare
```

Go 1.24.4:

```
go go1.24.4 linux/amd64
compiler gc  payload 1048576 bytes

== gzip (ModTime zero, OS 255) ==
input    lvl         in        out   ratio  gz-sha256         raw-sha256        roundtrip
html       1    1048576      77292   13.57  c005423844182de7  61637fbd108880f5  OK
html       6    1048576      96502   10.87  428b44188b6f41e6  61637fbd108880f5  OK
html      -1    1048576      96502   10.87  428b44188b6f41e6  61637fbd108880f5  OK
html       9    1048576      94590   11.09  fb93a6bbbc7c4ca9  61637fbd108880f5  OK
```

```
json       1    1048576      91201   11.50  5763e289aa3d25e2  4ac94536f174eb53  OK
json       6    1048576      59902   17.50  aca0d2f41e45efe5  4ac94536f174eb53  OK
json      -1    1048576      59902   17.50  aca0d2f41e45efe5  4ac94536f174eb53  OK
json       9    1048576      58483   17.93  4e3ef76c7fa8a94e  4ac94536f174eb53  OK
src        1    1048576      95293   11.00  72c58bbc4941154b  cdbe71a0a85b33f3  OK
src        6    1048576      84826   12.36  f735df3449f8dc55  cdbe71a0a85b33f3  OK
src       -1    1048576      84826   12.36  f735df3449f8dc55  cdbe71a0a85b33f3  OK
src        9    1048576      84147   12.46  4fc2dd19ee60e63f  cdbe71a0a85b33f3  OK
zeros      1    1048576       1314  798.00  79ca984597031c3b  30e14955ebf13522  OK
zeros      6    1048576       1055  993.91  bde3e442fb3cc188  30e14955ebf13522  OK
zeros     -1    1048576       1055  993.91  bde3e442fb3cc188  30e14955ebf13522  OK
zeros      9    1048576       1055  993.91  54b6c6b02555e7c8  30e14955ebf13522  OK
rand       1    1048576    1048684    1.00  66c0a5cf5e3664ba  e5c076829d6c90d0  OK
rand       6    1048576    1048919    1.00  9adf9e5e6bcd5256  e5c076829d6c90d0  OK
rand      -1    1048576    1048919    1.00  9adf9e5e6bcd5256  e5c076829d6c90d0  OK
rand       9    1048576    1048919    1.00  bb4b0e520439e889  e5c076829d6c90d0  OK

== wrappers on html (default level) ==
package                 out  out-sha256        roundtrip
compress/flate        96484  501cf913b5e4b900  OK
compress/zlib         96490  c3ce7119592553a5  OK
archive/zip           96618  3dfd7073a0c877f8  OK
image/png              1331  c451c2edfc74f817  OK  (256x256 NRGBA pixels)

raw-sha256 of html payload: 61637fbd108880f5118af9b355b03f66852728b8d570a63273054f54fbd69193
```

Go 1.27.2:

```
go go1.27.2 linux/amd64
compiler gc  payload 1048576 bytes

== gzip (ModTime zero, OS 255) ==
input    lvl         in        out   ratio  gz-sha256         raw-sha256        roundtrip
html       1    1048576      91644   11.44  2d3456e2d741dc31  61637fbd108880f5  OK
html       6    1048576      87481   11.99  11ec593087f48eea  61637fbd108880f5  OK
html      -1    1048576      87481   11.99  11ec593087f48eea  61637fbd108880f5  OK
html       9    1048576      94629   11.08  1d7535cfdac8d876  61637fbd108880f5  OK
json       1    1048576      77007   13.62  f17a0b47455ec441  4ac94536f174eb53  OK
json       6    1048576      65530   16.00  fe7abdfead437225  4ac94536f174eb53  OK
json      -1    1048576      65530   16.00  fe7abdfead437225  4ac94536f174eb53  OK
json       9    1048576      58483   17.93  78c3ef660b0bc703  4ac94536f174eb53  OK
src        1    1048576      88792   11.81  c5eda44c599e79e0  cdbe71a0a85b33f3  OK
src        6    1048576      83020   12.63  a5683110fc5548e5  cdbe71a0a85b33f3  OK
src       -1    1048576      83020   12.63  a5683110fc5548e5  cdbe71a0a85b33f3  OK
src        9    1048576      82067   12.78  d4e23b10b1eec8fc  cdbe71a0a85b33f3  OK
zeros      1    1048576       1108  946.37  ad6ab41a1934f7a6  30e14955ebf13522  OK
zeros      6    1048576       2064  508.03  c526c6e898c7b3b3  30e14955ebf13522  OK
```

```
zeros     -1    1048576       2064  508.03  c526c6e898c7b3b3  30e14955ebf13522  OK
zeros      9    1048576       1052  996.75  d7851fe46f7c0b2b  30e14955ebf13522  OK
rand       1    1048576    1048681    1.00  0ac8e609f98233ba  e5c076829d6c90d0  OK
rand       6    1048576    1048681    1.00  eaadf3ddc6a3f090  e5c076829d6c90d0  OK
rand      -1    1048576    1048681    1.00  eaadf3ddc6a3f090  e5c076829d6c90d0  OK
rand       9    1048576    1048756    1.00  3760b3a40eda01d0  e5c076829d6c90d0  OK

== wrappers on html (default level) ==
package                 out  out-sha256        roundtrip
compress/flate        87463  081ee0c09dda84d1  OK
compress/zlib         87469  a086174a8991e822  OK
archive/zip           92617  b1a7a41d7ab9a425  OK
image/png              1911  9172898b0547d9a1  OK  (256x256 NRGBA pixels)

raw-sha256 of html payload: 61637fbd108880f5118af9b355b03f66852728b8d570a63273054f54fbd69193
```

Read the two tables together:

gzip, 1 MiB	1.24.4 out	1.27.2 out	size Δ	raw-sha256
html default (−1/6)	96502	87481	−9.3%	same `61637fbd108880f5…`
json default	59902	65530	**+9.4%**	same
src default	84826	83020	−2.1%	same
zeros default	1055	2064	**+95.6%**	same
html BestSpeed	77292	91644	+18.6%	same
json BestSpeed	91201	77007	−15.6%	same
json BestCompression	58483	58483	0.0%	**hashes differ** (`4e3ef76c…` vs `78c3ef66…`)

Every `raw-sha256` column matched. Every `gz-sha256` column moved, including level 9 (the hash-chain path was rebalanced) and including json level 9 where the *length* stayed 58483. `DefaultCompression` (−1) matched level 6 on both toolchains, which is the `level = 6` assignment in `compressor.init`.

The wrappers moved with flate. zip is level 5, so its size delta is a different mix. The 256×256 XOR PNG grew from 1331 to 1911 bytes; `png.Decode` still returned the same pixels.

Case 2: throughput, same program, both compilers

```go
// compare_test.go - gzip encode throughput at levels 1, 6, 9 on the same payloads
package main

import (
    "bytes"
    "compress/gzip"
```

```
    "testing"
)

func BenchmarkGzip(b *testing.B) {
    inputs := []struct {
        name string
        data []byte
    }{
        {"html", makeHTML(payload)},
        {"json", makeJSON(payload)},
        {"src", makeSrc(payload)},
        {"zeros", makeZeros(payload)},
        {"rand", makeRand(payload)},
    }
    levels := []struct {
        name  string
        level int
    }{
        {"BestSpeed", gzip.BestSpeed},
        {"Default", gzip.DefaultCompression},
        {"BestCompression", gzip.BestCompression},
    }
    for _, in := range inputs {
        for _, lvl := range levels {
            b.Run(in.name+"/"+lvl.name, func(b *testing.B) {
                var buf bytes.Buffer
                w, err := gzip.NewWriterLevel(&buf, lvl.level)
                if err != nil {
                    b.Fatal(err)
                }
                b.SetBytes(int64(len(in.data)))
                b.ReportAllocs()
                b.ResetTimer()
                for i := 0; i < b.N; i++ {
                    buf.Reset()
                    w.Reset(&buf)
                    if _, err := w.Write(in.data); err != nil {
                        b.Fatal(err)
                    }
                    if err := w.Close(); err != nil {
                        b.Fatal(err)
                    }
                }
            })
        }
    }
}
```

```
go test -bench=BenchmarkGzip -benchmem -count=3 -timeout=10m
```

Go 1.24.4 (median of three runs in **bold** in the summary table below):

```
goos: linux
goarch: amd64
pkg: example.com/flatebench
cpu: Intel(R) Xeon(R) Processor
BenchmarkGzip/html/BestSpeed-8                728       1680827 ns/op     623.85 MB/s        201
BenchmarkGzip/html/BestSpeed-8                669       1744571 ns/op     601.05 MB/s        219
BenchmarkGzip/html/BestSpeed-8                716       1656501 ns/op     633.01 MB/s        205
BenchmarkGzip/html/Default-8                  223       5587074 ns/op     187.68 MB/s        482
BenchmarkGzip/html/Default-8                  222       5673796 ns/op     184.81 MB/s        484
BenchmarkGzip/html/Default-8                  217       5800967 ns/op     180.76 MB/s        495
BenchmarkGzip/html/BestCompression-8           19      57169172 ns/op      18.34 MB/s       5661
BenchmarkGzip/html/BestCompression-8           21      55819114 ns/op      18.79 MB/s       5122
BenchmarkGzip/html/BestCompression-8           21      55182543 ns/op      19.00 MB/s       5122
BenchmarkGzip/json/BestSpeed-8                708       1795216 ns/op     584.09 MB/s        207
BenchmarkGzip/json/BestSpeed-8                682       1737259 ns/op     603.58 MB/s        215
BenchmarkGzip/json/BestSpeed-8                693       1792238 ns/op     585.07 MB/s        211
BenchmarkGzip/json/Default-8                  147       8194660 ns/op     127.96 MB/s        642
BenchmarkGzip/json/Default-8                  151       8558982 ns/op     122.51 MB/s        625
BenchmarkGzip/json/Default-8                  146       8015328 ns/op     130.82 MB/s        646
BenchmarkGzip/json/BestCompression-8           22      50565283 ns/op      20.74 MB/s       4293
BenchmarkGzip/json/BestCompression-8           24      50330296 ns/op      20.83 MB/s       3935
BenchmarkGzip/json/BestCompression-8           20      50355120 ns/op      20.82 MB/s       4722
BenchmarkGzip/src/BestSpeed-8                 555       2098503 ns/op     499.68 MB/s        264
BenchmarkGzip/src/BestSpeed-8                 577       2161365 ns/op     485.15 MB/s        254
BenchmarkGzip/src/BestSpeed-8                 580       2059333 ns/op     509.18 MB/s        253
BenchmarkGzip/src/Default-8                   129       9147193 ns/op     114.63 MB/s        833
BenchmarkGzip/src/Default-8                   117       9341347 ns/op     112.25 MB/s        919
BenchmarkGzip/src/Default-8                   126       9432083 ns/op     111.17 MB/s        853
BenchmarkGzip/src/BestCompression-8            15      66869262 ns/op      15.68 MB/s       7170
BenchmarkGzip/src/BestCompression-8            16      71526839 ns/op      14.66 MB/s       6722
BenchmarkGzip/src/BestCompression-8            16      68514387 ns/op      15.30 MB/s       6722
BenchmarkGzip/zeros/BestSpeed-8               1903        681669 ns/op    1538.25 MB/s         63
BenchmarkGzip/zeros/BestSpeed-8               1948        662424 ns/op    1582.94 MB/s         62
BenchmarkGzip/zeros/BestSpeed-8               1975        703071 ns/op    1491.42 MB/s         61
BenchmarkGzip/zeros/Default-8                  555       2144923 ns/op     488.86 MB/s        147
BenchmarkGzip/zeros/Default-8                  543       2369568 ns/op     442.52 MB/s        150
BenchmarkGzip/zeros/Default-8                  565       2477097 ns/op     423.31 MB/s        144
BenchmarkGzip/zeros/BestCompression-8          514       2277501 ns/op     460.41 MB/s        159
BenchmarkGzip/zeros/BestCompression-8          552       2135341 ns/op     491.06 MB/s        148
BenchmarkGzip/zeros/BestCompression-8          565       2136269 ns/op     490.84 MB/s        144
BenchmarkGzip/rand/BestSpeed-8                 619       1897473 ns/op     552.62 MB/s        563
BenchmarkGzip/rand/BestSpeed-8                 619       1986062 ns/op     527.97 MB/s        563
BenchmarkGzip/rand/BestSpeed-8                 566       1947840 ns/op     538.33 MB/s        616
```

```
BenchmarkGzip/rand/Default-8                  85      14065542 ns/op      74.55 MB/s       4014
BenchmarkGzip/rand/Default-8                  82      13642761 ns/op      76.86 MB/s       4161
BenchmarkGzip/rand/Default-8                  88      13715233 ns/op      76.45 MB/s       3878
BenchmarkGzip/rand/BestCompression-8          69      14763644 ns/op      71.02 MB/s       4945
BenchmarkGzip/rand/BestCompression-8          88      13559292 ns/op      77.33 MB/s       3878
BenchmarkGzip/rand/BestCompression-8          75      13673153 ns/op      76.69 MB/s       4550
PASS
ok      example.com/flatebench  66.179s
```

Go 1.27.2:

```
goos: linux
goarch: amd64
pkg: example.com/flatebench
cpu: Intel(R) Xeon(R) Processor
BenchmarkGzip/html/BestSpeed-8               742       1647084 ns/op     636.63 MB/s        144
BenchmarkGzip/html/BestSpeed-8               723       1639164 ns/op     639.70 MB/s        148
BenchmarkGzip/html/BestSpeed-8               724       1621805 ns/op     646.55 MB/s        148
BenchmarkGzip/html/Default-8                 381       3136526 ns/op     334.31 MB/s        351
BenchmarkGzip/html/Default-8                 381       3538283 ns/op     296.35 MB/s        351
BenchmarkGzip/html/Default-8                 351       3190290 ns/op     328.68 MB/s        381
BenchmarkGzip/html/BestCompression-8          28      42878880 ns/op      24.45 MB/s       5016
BenchmarkGzip/html/BestCompression-8          27      44121386 ns/op      23.77 MB/s       5202
BenchmarkGzip/html/BestCompression-8          28      43916651 ns/op      23.88 MB/s       5016
BenchmarkGzip/json/BestSpeed-8               1143       1058338 ns/op     990.78 MB/s         94
BenchmarkGzip/json/BestSpeed-8               1051       1052626 ns/op     996.15 MB/s        102
BenchmarkGzip/json/BestSpeed-8               1191       1117643 ns/op     938.20 MB/s         90
BenchmarkGzip/json/Default-8                 429       2760387 ns/op     379.87 MB/s        281
BenchmarkGzip/json/Default-8                 447       2821223 ns/op     371.67 MB/s        269
BenchmarkGzip/json/Default-8                 447       2752037 ns/op     381.02 MB/s        269
BenchmarkGzip/json/BestCompression-8          21      49814303 ns/op      21.05 MB/s       6064
BenchmarkGzip/json/BestCompression-8          21      48911805 ns/op      21.44 MB/s       6064
BenchmarkGzip/json/BestCompression-8          20      50566658 ns/op      20.74 MB/s       6367
BenchmarkGzip/src/BestSpeed-8                859       1253314 ns/op     836.64 MB/s        125
BenchmarkGzip/src/BestSpeed-8               1017       1248086 ns/op     840.15 MB/s        105
BenchmarkGzip/src/BestSpeed-8                926       1251612 ns/op     837.78 MB/s        116
BenchmarkGzip/src/Default-8                  380       3177652 ns/op     329.98 MB/s        352
BenchmarkGzip/src/Default-8                  378       3323523 ns/op     315.50 MB/s        353
BenchmarkGzip/src/Default-8                  372       3175179 ns/op     330.24 MB/s        359
BenchmarkGzip/src/BestCompression-8           18      64509597 ns/op      16.25 MB/s       7803
BenchmarkGzip/src/BestCompression-8           16      63102104 ns/op      16.62 MB/s       8778
BenchmarkGzip/src/BestCompression-8           18      62793482 ns/op      16.70 MB/s       7803
BenchmarkGzip/zeros/BestSpeed-8             5313        228883 ns/op    4581.27 MB/s         15
BenchmarkGzip/zeros/BestSpeed-8             5667        229630 ns/op    4566.37 MB/s         14
BenchmarkGzip/zeros/BestSpeed-8             5532        234105 ns/op    4479.08 MB/s         14
BenchmarkGzip/zeros/Default-8                831       1441445 ns/op     727.45 MB/s        130
BenchmarkGzip/zeros/Default-8                914       1398500 ns/op     749.79 MB/s        118
```

```
BenchmarkGzip/zeros/Default-8                 906       1424948 ns/op     735.87 MB/s        119
BenchmarkGzip/zeros/BestCompression-8         262       4592256 ns/op     228.34 MB/s        437
BenchmarkGzip/zeros/BestCompression-8         247       4859673 ns/op     215.77 MB/s        464
BenchmarkGzip/zeros/BestCompression-8         258       4588996 ns/op     228.50 MB/s        444
BenchmarkGzip/rand/BestSpeed-8               7119        160043 ns/op    6551.85 MB/s         43
BenchmarkGzip/rand/BestSpeed-8               7324        155909 ns/op    6725.57 MB/s         42
BenchmarkGzip/rand/BestSpeed-8               7674        171794 ns/op    6103.68 MB/s         40
BenchmarkGzip/rand/Default-8                 4329        235122 ns/op    4459.71 MB/s         77
BenchmarkGzip/rand/Default-8                 5372        229483 ns/op    4569.31 MB/s         62
BenchmarkGzip/rand/Default-8                 4477        226221 ns/op    4635.18 MB/s         75
BenchmarkGzip/rand/BestCompression-8          187       6273550 ns/op     167.14 MB/s       1991
BenchmarkGzip/rand/BestCompression-8          182       6476233 ns/op     161.91 MB/s       2045
BenchmarkGzip/rand/BestCompression-8          182       6497071 ns/op     161.39 MB/s       2045
PASS
ok      example.com/flatebench  63.789s
```

Median of three runs, same 1 MiB inputs:

gzip	1.24.4	1.27.2	speedup
html Default	184.81 MB/s	328.68 MB/s	1.78×
json Default	127.96 MB/s	379.87 MB/s	2.97×
src Default	112.25 MB/s	329.98 MB/s	2.94×
zeros Default	442.52 MB/s	735.87 MB/s	1.66×
rand BestSpeed	538.33 MB/s	6551.85 MB/s	12.2×
rand Default	76.45 MB/s	4569.31 MB/s	59.8×
json BestCompression	20.82 MB/s	21.05 MB/s	~1.01×
zeros BestCompression	490.84 MB/s	228.50 MB/s	0.47×

Default gzip on structured text is about **3×** on this box (json, src). Incompressible `rand` is the headline: the 1.27 matcher stops looking. Level 9 is still the old-style hash chain, so json `BestCompression` is a wash, and zeros at level 9 was **slower** here. Upgrade for levels 1–6; measure level 9 on your payload if that is the one you ship.

Case 3: a 23-byte golden file

```go
go run . dump-golden          # under 1.24.4, writes testdata/ticket.gz
go run . check-golden testdata/ticket.gz
```

Go 1.24.4 writes and confirms:

```
go go1.24.4
plain      "desk ticket 42 is open\n"
gz-len     47
gz-sha256  37fe167f389809d8dc0a08f8019d535d12ba226f0813f6d1fa55a03c8f7d7165
raw-sha256 46b9a55f46d55b888f5b5443c84004fa093c5d7a52e4794bc4908a379eb8b68c
```

```
roundtrip  equal=true
wrote testdata/ticket.gz
```

```
go go1.24.4
golden-len 47  got-len 47  bytes.Equal=true
golden-sha 37fe167f389809d8dc0a08f8019d535d12ba226f0813f6d1fa55a03c8f7d7165
got-sha    37fe167f389809d8dc0a08f8019d535d12ba226f0813f6d1fa55a03c8f7d7165
decompressed both equal to plaintext: true
```

The same golden under Go 1.27.2:

```
go go1.27.2
golden-len 47  got-len 48  bytes.Equal=false
golden-sha 37fe167f389809d8dc0a08f8019d535d12ba226f0813f6d1fa55a03c8f7d7165
got-sha    322cf646342214750d34acd529a6b712a75e61097545aeef27441818333da7d7
decompressed both equal to plaintext: true
```

Both files start `1f 8b 08 00 00 00 00 00 00 ff` (gzip magic, method 8, MTIME 0, OS 255). The 1.27 payload is a stored DEFLATE block: after the header you can read the ASCII `desk ticket 42 is open` in the gzip file. The CRC-32 trailer `63 2f 5e 64` and ISIZE `17 00 00 00` match, because the uncompressed bytes match.

Case 4: each toolchain reads the other's gzip

```go
// cross.go - gunzip a file written by the other Go version
package main

import (
    "bytes"
    "compress/gzip"
    "fmt"
    "io"
    "os"
    "runtime"
)

func main() {
    b, err := os.ReadFile(os.Args[1])
    if err != nil {
        panic(err)
    }
    r, err := gzip.NewReader(bytes.NewReader(b))
    if err != nil {
        panic(err)
    }
```

```go
    out, err := io.ReadAll(r)
    r.Close()
    if err != nil {
        panic(err)
    }
    fmt.Printf("%s read %s -> %q\n", runtime.Version(), os.Args[1], out)
}
```

```
go1.24.4 read /tmp/ticket-127.gz -> "desk ticket 42 is open\n"
go1.27.2 read /tmp/ticket-124.gz -> "desk ticket 42 is open\n"
```

Python's `gzip.decompress` accepted both files and returned the same 23 bytes. Interoperability holds. Byte identity does not.

The trap

Golden files of compressed bytes

`bytes.Equal(gotGzip, testdata/foo.gz)` fails on the first CI run after the toolchain bump. The program is fine. The encoder changed. The same trap hits `image/png` fixtures (IDAT), `archive/zip` testdata, and checked-in `.gz` assets. The 23-byte ticket above is the smallest reproduction: 47 bytes vs 48, both valid gzip.

Cache keys, content hashes, ETags

If an HTTP handler gzips the body and then hashes the gzipped bytes for `ETag`, every 1.27 deploy looks like new content. Clients re-download. CDNs miss. The entity is the same. Hash the uncompressed representation (or a versioned encoding that you control), then gzip for transport. The same mistake appears as S3/object-store keys, Docker layer diffs of `.gz` files, and "content-addressed" caches whose address is the compressor output.

Tiny payloads looking uncompressed

1.27's fast path stores 32-byte flushes raw. A test that greps a `.gz` for a secret, or that assumes gzip always hides the plaintext, is lying to itself on short writes. Huffman coding on 1.24 hid `"desk ticket 42 is open\n"`; 1.27 does not.

The boring rule

- Compare **decompressed** content. `gunzip` then `bytes.Equal`. For PNG, decode pixels. For zip, read the member.
- Pin hashes on **uncompressed** data. If you need a cache key or ETag, hash the entity you would serve with `Content-Encoding: identity`.
- If a test must pin compressed bytes, pin the **Go version** next to the golden (`go 1.27.2` in the testdata README) and regenerate on purpose.
- Round-trip in CI: compress with this toolchain, decompress with this toolchain, compare to the original. That is the RFC 1951 contract.
- gzip `ModTime` is a second source of nondeterminism. Zero it when you hash gzip headers at all.

The stdlib already tells you. `go doc compress/flate.NewWriter` on 1.27.2:

```
Note that the exact bytes written to w are not covered by the Go 1
compatibility promise. Callers, including tests, should not depend on the
exact written bytes.
```

The same sentence is on `gzip.NewWriter`, `zip.NewWriter`, and `png.Encode`.

Try this

1. Re-run `compare.go` and `BenchmarkGzip` on a payload from your service (JSON logs, protobuf, HTML). Record default-level size Δ and MB/s with `-count=10`.
2. Search your repo for `testdata/*.gz`, `*.png` goldens, and `bytes.Equal` against compressor output. Switch those tests to decompressed equality.
3. Encode the same `image.NRGBA` with `png.Encode` on 1.24 and 1.27; hash the file, then hash `png.Decode` pixels.
4. Build an `ETag` from `sha256(uncompressed)` and send gzip on the wire. Confirm two toolchains produce the same ETag and different `Content-Length`.
5. At level 9, confirm json can keep the same compressed *length* with a different hash. Decide whether your "size budget" tests should allow that.

Sources

- Go 1.27 release notes, `compress/flate`: https://go.dev/doc/go1.27
- `compress/flate.NewWriter` compatibility note (and the same sentence on gzip, zip, png): `go doc compress/flate.NewWriter` (Go 1.27.2)
- Go 1 compatibility promise: https://go.dev/doc/go1compat
- RFC 1951 DEFLATE: https://www.rfc-editor.org/rfc/rfc1951
- RFC 1952 GZIP: https://www.rfc-editor.org/rfc/rfc1952
- RFC 1950 zlib: https://www.rfc-editor.org/rfc/rfc1950
- Issue 75532, `compress/gzip` performance: https://github.com/golang/go/issues/75532

- Encoder rewrite CL 707355; compatibility-note CL 774300: https://go-review.googlesource.com/c/go/+/707355, https://go-review.googlesource.com/c/go/+/774300
- Go 1.27.2 source: `src/compress/flate/deflate.go` (`DefaultCompression` → 6, `fastEnc` for 1–6), `deflatefast.go` / `level1.go`–`level6.go`, `deflateFast` stored-block path for 32-byte flushes; `src/archive/zip/register.go` (`flate.NewWriter(w, 5)`); `src/image/png/writer.go` (`levelToZlib`)

Project: Optimize a Hot Endpoint

Project: Optimize a Hot Endpoint

Overview

Pick or build a deliberately inefficient endpoint, then improve it with the performance workflow.

Setup

Serve `GET /report` that:

1. Re-reads a JSON file every request
2. Uses `fmt.Sprintf` heavily
3. Builds strings with `+` in a loop

Steps

1. **Baseline** load test numbers (184)
2. **Profile** CPU + heap
3. **Change one thing** (cache file, Builder, precompute)
4. **Remeasure**
5. Write a short report: before/after table

Acceptance

- ☐ 30% p99 or allocs improvement with evidence
- ☐ Tests still pass
- ☐ No correctness regression
- ☐ Profile screenshots or pprof commands recorded

Stretch

Apply PGO and note delta vs manual fixes alone.

Go Deep Dives

Go Deep Dives Overview

Go Deep Dives Overview

Diagram: Deep-dive stack

```
flow:
  [Comp]
       |
       v
  [RT]

  [RT]
       |
       v
  [OS]
```

This part is for readers who already write idiomatic Go and want the **runtime, toolchain, and production mental models** behind real symptoms: latency cliffs, leaks, interface nil traps, timer churn, cgo taxes, and "why is this slow in the pod?"

Earlier parts teach *how to use* the language. These chapters teach *how the machine runs it* — deep enough to debug, not a full fork of `src/runtime`.

Who This Part Is For

- You have shipped Go services or tools
- You can read basic `pprof` / `trace` output
- You want **code → runtime structures → observables**

If you are still learning syntax, finish Core, Memory, and Concurrency first.

Learning Paths

A — Runtime core

201 GMP → 202 memory model → 203 channels → 213 netpoll → 214 timers → 227 mutex → 239 sysmon

B — Data representation

204 interfaces → 205 slice/map → 218 Swiss maps → 235 strings/bytes → 217 weak/unique → 236 errors

C — Execution control

206 defer/panic → 215 stacks → 226 context → 220 leaks → 223 synctest

D — Performance & GC

208 GC → 219 pool/zero-alloc → 222 pprof/trace → 224 zero-copy → 209 compiler/PGO → 243 GOMAXPROCS/cgroup

E — Toolchain & platforms

210 link/race → 228 cgo → 230 ABI → 231 coverage/fuzz → 232 build cache → 233 experiments → 234 Wasm → 244 plugins

F — Production craft

225 HTTP transport → 241 TLS → 242 slog handlers → 237 mistakes → 238 debugging → 240 reading source → 245 topic radar

Full Chapter Map

#	Topic
201	Scheduler (GMP)
202	Memory model / happens-before
203	Channels & select
204	Interfaces (eface/iface)
205	Slice & map headers
206	Defer, panic, unwind
207	Generics implementation
208	GC internals & pacer
209	Compiler, SSA, PGO
210	Link, buildmodes, race
211	Reflect cost
212	Modules, MVS, toolchain
213	Netpoller
214	Timers / `time.After`
215	Stack growth
216	Finalizers & cleanup

#	Topic
217	`weak` & `unique`
218	Swiss maps & maphash
219	`sync.Pool` & zero-alloc
220	Goroutine leaks
221	GODEBUG knobs
222	pprof & tracer internals
223	testing/synctest (bubbles, durable blocks, fake time)
224	Zero-copy I/O
225	HTTP Transport pool
226	Context cancel trees
227	Mutex / semaphore runtime
228	cgo + Go 1.26 scheduler
229	Iterators / range-over-func
230	ABI & assembly
231	Coverage & fuzz engine
232	Build cache / reproducibility
233	GOEXPERIMENT catalog
234	Wasm / wasip1
235	String / []byte internals
236	Error interface cost
237	Production mistakes map
238	Production debugging
239	sysmon & scavenger
240	Reading runtime source
241	TLS handshake path
242	slog handler design
243	GOMAXPROCS & cgroups
244	Plugin buildmode pitfalls
245	X/ecosystem topic radar
246	Internals-for-Interns article radar
247	Memory allocator (mcache/mcentral/mheap)
248	Runtime bootstrap
249	select + selectgo
250	Stacktraces + binary metadata
251	mmap vs pread storage I/O
252	Compiler frontend (scan/parse/types)
253	Compiler IR / SSA / link
254	Race detector internals
255	Write barriers & mark assist
256	Work stealing / findrunnable
257	Type assert / convert runtime
258	OS signals & runtime
259	HTTP/2 in Go
260	database/sql pool
261	JSON performance
262	errgroup patterns

#	Topic
263	Goroutine switch mechanics
264	Measuring a "compiler will optimize it" idiom (`netip.AddrFrom16(ip.As16())`)
265	Why Go binaries are big: sections, reachability, linker method rules

Stack Diagram

```
your code
    |
    v
compiler (types, SSA, inlining, escape, PGO)
    |
    v
runtime (scheduler, poller, timers, maps, GC, defer, cgo)
    |
    v
OS threads + cgroups + network stack + TLS
```

Diagrams in this part

Prefer **ASCII diagrams** in fenced `text` blocks (works in HTML, PDF, and EPUB). Flagship topics also ship optional **SVG** under `content/25-go-deep-dives/assets/` for HTML. Mermaid is not required for this part.

Study Method

```
hypothesis -> minimal repro -> measure (pprof/trace/race) -> model -> fix ownership/allocs/t
```

1. One chapter, one question.
2. Run the experiment; change one knob.
3. Confirm with tools, not vibes.

Relationship to Other Parts

Concern	Prefer
Concurrency patterns	Part 10/11 + 201/203/220
GC knobs tutorial	Part 03/24 + 208/239

Concern	Prefer
Profiling workflow	Part 23/24 + 222/238
Stdlib how-to	Part 07
HTTP resilience	Part 13 + 225/241
Security TLS ops	Part 22 + 241

Honesty Boundary

Runtime details evolve (scheduler, GC, maps, experiments). Chapters teach **stable models** and version-labeled notes (e.g. Go 1.26 Green Tea GC, P-syscall simplification; Go 1.27 size-specialized malloc and `goroutineleak`). Pseudocode is a map for reading source — not a bit-identical clone of `src/runtime`.

Checklist: Deep-Dive Literacy

- ☐ Draw G/M/P and explain work stealing
- ☐ State happens-before for unlock→lock and send→receive
- ☐ Explain typed-nil interface trap
- ☐ Fix a goroutine leak with context
- ☐ Replace hot `time.After` with timer reuse
- ☐ Tune or justify GOMAXPROCS under cgroup limits
- ☐ Read one CPU + one goroutine profile
- ☐ Find a panic string in `GOROOT/src/runtime`
- ☐ Know when cgo/plugins/Wasm are the wrong tool

When most boxes are automatic, you can track Go releases without fear.

Scheduler Deep Dive (GMP)

Scheduler Deep Dive (GMP)

Overview

Go multiplexes millions of **goroutines (G)** onto fewer OS **threads (M)** using **logical processors (P)**. Understanding GMP turns vague advice ("don't spawn unbounded goroutines") into capacity math.

Diagram-rich companion series: Internals for Interns — The Scheduler.

```
G  = goroutine (stack + state: runnable / running / waiting / dead)
M  = OS thread that executes Go code
P  = scheduling context (run queue, memory caches, limits) - count   GOMAXPROCS
```

Only an **M bound to a P** runs ordinary Go code. Syscalls may park the P or spin another M so other Gs keep running.

Diagram: GMP at a glance

```
                    GOMAXPROCS   number of Ps

    P0                     P1                     P2
    runq [G G G]           runq [G G]             runq []
    mcache …               mcache …               (idle)

              bound                  bound
            v                      v
          M0 (OS thread)         M1 (OS thread)     M2 in syscall

            v                      v                    P2 may be
          G-run                  G-run                  retaken for
                                                        another M
  Waiting Gs: chan / mutex / netpoll / timer  park   wait queues
```

Only an **M bound to a P** runs ordinary Go code. Syscalls may free the P so other Gs keep running.

Lifecycle of a G (creation → park → run)

```
          recycle       Dead
                 (free list)       goexit

                           go f / goready
                          v

                      Runnable       wake / preempt

                           schedule (needs M+P)
                          v
                                 entersyscall
                      Running
                                                v
             park
           (chan/mutex/                            Syscall
            sleep/poll)
                          v
                                 return /
                      Waiting        retake P
```

Core Loop (Mental Model)

```
                    +------------------+
  new G  ---------> |  global runq     |
                    +--------+---------+
                             | steal / distribute
                    +--------v---------+
                    |  P local runq    |  (per P)
                    +--------+---------+
                             |
              M executes G <-+
                             |
              G blocks (chan, mutex, I/O park)
                             |
              G becomes runnable again -----> runq
```

1. Each P has a **local run queue** (cheap, no global lock).
2. Overflow / balancing uses a **global run queue**.
3. Idle Ps **work-steal** from other Ps' local queues.
4. Network poller and timers inject runnable Gs.

GOMAXPROCS

```
runtime.GOMAXPROCS(0) // current setting
```

- Defaults to number of CPU cores (cgroup-aware on Linux in modern Go).
- Caps how many Gs run **Go code** in parallel (approx. number of Ps).
- Does **not** cap total goroutines or OS threads (threads can grow for syscalls).

Under a CPU quota (K8s), a wrong `GOMAXPROCS` causes either under-utilization or thrashing. Prefer auto-detection or set from cgroup limits intentionally.

Preemption

Modern Go can preempt long-running Gs (async preemption) so a tight loop without function calls is less likely to starve the GC or other Gs. Still:

- Avoid owning a P forever with non-cooperative C code via cgo without care.
- `runtime.Gosched()` yields voluntarily — rare need if profiles do not show unfairness.

Hand-off and Syscalls

When a G enters a syscall:

1. M may release its P so another M can run other Gs.
2. On return, M tries to reacquire a P; if none free, G goes runnable and M may park.

This is why blocking syscalls "all Go freezes," but **too many** blocking syscalls create many Ms and scheduler overhead.

Scheduling Latency Signals

Signal	Tool	Meaning
High `runtime.schedule` / `findrunnable`	CPU pprof	Scheduler overhead, not enough real work
Runnable Gs queueing	`go tool trace`	Over-subscription or lock/chan contention
Idle Ps + runnable Gs	trace	Load imbalance / GC / netpoll delays
Threads » GOMAXPROCS	`runtime.NumCgoCall` / OS metrics	Syscall or cgo heavy

Design Rules From GMP

1. **Bound concurrency at the entrance** (semaphores, worker pools) — Gs are cheap, not free.
2. **Prefer non-blocking design at boundaries** — give every wait a context deadline.
3. **Match fan-out to Ps for CPU work** — `GOMAXPROCS` workers for pure CPU; more for I/O-bound wait.
4. **Measure under the real quota** — laptop cores pod millicores.

Experiment

```
go mod init example
GOMAXPROCS=1 go run .
GOMAXPROCS=4 go run .
```

```go
package main

import (
    "fmt"
    "runtime"
    "sync"
    "time"
)

func burn(d time.Duration) {
    end := time.Now().Add(d)
    for time.Now().Before(end) {
        // busy spin
    }
}

func main() {
    fmt.Println("GOMAXPROCS", runtime.GOMAXPROCS(0))
    var wg sync.WaitGroup
    start := time.Now()
    n := 4
    for i := 0; i < n; i++ {
        wg.Add(1)
        go func() {
            defer wg.Done()
            burn(200 * time.Millisecond)
        }()
    }
    wg.Wait()
```

```
    fmt.Println("elapsed", time.Since(start))
}
```

What to notice: With `GOMAXPROCS=1`, four 200ms CPU burns serialize (~800ms). With 4, they overlap (~200ms). I/O-bound work would not scale the same way.

Try next: Capture `go tool trace` around a server under load; find "Proc start/stop" and long stretches where Gs are runnable but not running.

Related

- Concurrency basics
- Contention tuning

Memory Model and Happens-Before

Memory Model and Happens-Before

Overview

Go's memory model answers: **when is a write in goroutine A guaranteed visible to a read in goroutine B?** Without a *happens-before* edge, "it works on my machine" is data-race undefined behavior — even if the race detector is quiet today under lucky scheduling.

Reference: Go Memory Model (normative).

Diagram: Happens-before edges

```
flow:
  [B] --HB--> [C]
  [C]
       |
       v
  [D]
```

Without an edge, concurrent write+read is a data race.

Data Race Definition

A **data race** is concurrent access to a variable where at least one access is a write and the accesses are not synchronized by the memory model.

```
go test -race ./...
go run -race .
```

If `-race` fires, fix synchronization. Do not "make it volatile" by folklore.

Happens-Before (Core Edges)

Informally, if event X *happens before* Y, then Y sees all effects of X.

Important edges:

Edge	Guarantee
Program order in one G	Earlier statements HB later (single goroutine)
`go` statement	Start of new G happens after the `go` statement executes
Channel send → receive	Send HB corresponding receive completion
Channel close → receive zero	Close HB receive that sees closed
`Mutex` unlock → later lock	Unlock HB subsequent Lock acquisition
`RWMutex`	Analogous rules for RLock/RUnlock
`sync.Once`	`Do` completion HB later `Do` returns
`sync.WaitGroup`	`Done` HB `Wait` return after counter hits 0
Atomic ops	Synchronizing atomics establish order per model

If you share memory, you almost always want one of: **mutex**, **channel ownership transfer**, or **atomic** with a clear protocol.

Incorrect: Sharing Without Sync

```go
// DATA RACE
var a int
go func() { a = 1 }()
fmt.Println(a) // may print 0 forever, or 1, or worse under compiler reordering assumptions
```

Correct Patterns

Mutex

```go
var mu sync.Mutex
var a int

// writer
mu.Lock()
a = 1
mu.Unlock()

// reader
mu.Lock()
v := a
mu.Unlock()
```

Unlock *happens before* the next Lock; reader sees the write.

Channel hand-off (share by communicating)

```go
ch := make(chan int, 1)
go func() { ch <- 1 }()
v := <-ch // receive happens after send; sees 1
```

Atomic flag

```go
var ready atomic.Bool
var data int

// producer
data = 42
ready.Store(true) // release-style in practice for this pattern

// consumer
if ready.Load() {
    _ = data // safe if this is the only protocol and documented
}
```

Prefer mutexes for multi-field invariants; atomics for counters/flags with simple protocols.

Init and Package-Level

All package `init` functions and variable initialization complete before `main` starts. Within a package, initialization order is constrained by dependency; across packages, import graph order applies. Do not rely on init order for subtle cross-package races — make dependencies explicit.

Compiler and Hardware Reordering

Without synchronization, compilers and CPUs may reorder memory operations. The memory model is the contract that remains true; your mental model of "statements run in source order across cores" is false.

Experiment

```go
go mod init example
go run -race .
```

```go
package main

import (
    "fmt"
    "sync"
)

func racy() int {
    var a int
    var wg sync.WaitGroup
    wg.Add(1)
    go func() {
        defer wg.Done()
        a = 1
    }()
    v := a // race with write
    wg.Wait()
    return v
}

func synced() int {
    var mu sync.Mutex
    var a int
    var wg sync.WaitGroup
    wg.Add(1)
    go func() {
        defer wg.Done()
        mu.Lock()
        a = 1
        mu.Unlock()
    }()
    wg.Wait()
    mu.Lock()
    v := a
    mu.Unlock()
    return v
}

func main() {
    fmt.Println("synced", synced())
    // Uncomment to see race detector:
    // fmt.Println("racy", racy())
    _ = racy
}
```

What to notice: `-race` instruments memory accesses; fixed code establishes unlock→lock or wait→read edges.

Try next: Build a single-producer single-consumer ring with atomics only; then rewrite with a buffered channel and compare clarity.

Related

- Synchronization
- sync/atomic stdlib

Channel and Select Internals

Channel and Select Internals

Overview

Channels are not magic pipes — they are runtime objects (`hchan`) with a lock, optional buffer ring, and wait queues of **sudogs** (goroutine wait records). Select is a careful algorithm over multiple channel operations so that **exactly one** case proceeds (or default).

Usage patterns: Channel patterns.

Diagram: Channel park / wake

```
sequence (top → bottom):
  actors: sender G, hchan, receiver G
  sender G --> hchan  : chansend
  hchan --> receiver G  : hand off / enqueue buf
  hchan --> sender G  : return
  sender G --> sender G  : gopark sendq
  receiver G --> hchan  : chanrecv
  hchan --> sender G  : goready
  note: alt buffer space or waiting recv
  note: else must wait
```

Channel Structure (Conceptual)

```
hchan
  qcount      // elements in buffer
  dataqsiz    // buffer capacity (0 = unbuffered)
  buf         // circular queue
  sendx/recvx // indices
  recvq       // waiters for receive
  sendq       // waiters for send
  lock
  closed
```

Unbuffered (`make(chan T)`)

Send and receive must **rendezvous**: sender copies directly into receiver (or vice versa). One side parks on a sudog queue until the other arrives.

Buffered (`make(chan T, n)`)

Send succeeds without a receiver while `qcount < n`. Receive succeeds without a sender while `qcount > 0`. Still synchronized via the channel lock and happens-before rules.

Send / Receive Paths

Situation	Behavior
Recv, buffer non-empty	Pop buffer; maybe wake a sender
Recv, buffer empty, sender waiting	Direct hand-off
Recv, nothing available	Park on `recvq`
Send, receiver waiting	Direct hand-off
Send, buffer space	Push buffer
Send, no space	Park on `sendq`
Op on closed	Recv: zero + `ok=false`; Send: **panic**

```go
v, ok := <-ch // ok false => closed and drained
close(ch)     // idempotent close panics on second close
```

Select Algorithm (Intuition)

For `select` with multiple cases:

1. Randomize case order (fairness).
2. Probe channels for a case that can proceed **immediately**.
3. If none and there is `default`, take default.
4. Otherwise, enqueue this G on **all** involved channels' wait queues, then park.
5. On wake, dequeue from other channels; complete the winning case only.

Implications:

- There is **no priority** among cases — do not assume left-to-right wins.
- A busy channel can starve another statistically; redesign if fairness is a product requirement.
- `select` with one case quite the same codegen as bare op, but same semantics.

Close Semantics

```
close
  |
  +--> fail future sends (panic)
  +--> wake all recvers (they get zero values)
  +--> drained buffer still readable until empty
```

Only the sender side should close (by convention). Multiple closers require a single closer protocol.

Common Failure Modes

Bug	Symptom
Send on closed	Panic
Never closed, receivers blocked	Goroutine leak
Unbuffered send without recv	Deadlock
Select without default in tight loop	100% wake churn if misused with timers
Timer.C not drained after Stop (older patterns)	Rare wake bugs — prefer `time.After` carefully in loops

Cost Model

- Channel ops take a **runtime lock** and may park the G — fine for coordination, expensive as a per-byte I/O bus.
- For huge fan-in of tiny messages, batch or use a shared structure under a mutex (measure).

Experiment

```
go mod init example
go run .
```

```go
package main

import (
    "fmt"
    "time"
)

func main() {
    // Rendezvous
```

```go
    unbuf := make(chan string)
    go func() {
        unbuf <- "ping"
    }()
    fmt.Println("unbuf", <-unbuf)

    // Buffer absorbs one send
    buf := make(chan int, 1)
    buf <- 1
    fmt.Println("buf", <-buf)

    // Select fairness demo: both cases ready each iteration
    counts := map[string]int{}
    for i := 0; i < 1000; i++ {
        a := make(chan int, 1)
        b := make(chan int, 1)
        a <- 1
        b <- 2
        select {
        case <-a:
            counts["a"]++
        case <-b:
            counts["b"]++
        }
    }
    fmt.Println("select counts", counts)

    // Close
    ch := make(chan int, 2)
    ch <- 7
    close(ch)
    v, ok := <-ch
    fmt.Println("after close", v, ok)
    v, ok = <-ch
    fmt.Println("drained", v, ok)

    // Default non-block
    select {
    case <-time.After(10 * time.Millisecond):
        fmt.Println("timer")
    default:
        fmt.Println("default path")
    }
}
```

What to notice: Both `a` and `b` win some iterations — not always `a` first. Closed channel receives are safe; sends would panic.

Try next: Deliberately leak a goroutine blocked on send; find it with `//go:debug tracebackancestors=1` or pprof goroutine profile.

Related

- Channel patterns
- Memory model

Interface Representation (eface / iface)

Interface Representation (eface / iface)

Overview

An interface value is **not** the concrete value alone. It is a small descriptor: **type metadata + data pointer** (with special cases). That design explains allocation on conversion, the famous nil-interface trap, and method-call cost.

Usage: Interfaces.

Diagram: eface vs iface

```
regions: any / empty interface | non-empty interface
flow:
  [Conc]
       |
       v
  [data pointer]

  [itab: inter + _type + fun]
       |
       v
  [Methods]
```

Two Runtime Shapes

Kind	When	Conceptual fields
`eface`	Empty interface `any` / `interface{}`	`_type *rtype`, `data unsafe.Pointer`
`iface`	Non-empty interface	`tab *itab`, `data unsafe.Pointer`

`itab` caches the **mapping from interface method set → concrete methods** for a pair `(interface type, concrete type)`. First conversion may populate itab; later calls reuse it.

```
iface
  tab  ---> itab { inter, _type, fun[0..n] method ptrs }
  data ---> concrete value or pointer to it
```

Dynamic Type and Dynamic Value

```go
var i any = 3
// dynamic type: int
// dynamic value: 3
```

`i == nil` is true only when **both** type and value parts are unset. Assigning a typed nil sets the type part:

```go
var p *int = nil
var i any = p
fmt.Println(i == nil) // false - type is *int, data is nil
```

This is the #1 production bug when returning `error`:

```go
func f() error {
    var err *MyError = nil
    return err // returns non-nil error interface!
}
return nil // correct empty interface
```

Conversion and Escape

Boxing a non-pointer into an interface often **allocates** so `data` can point at a heap copy (escape analysis decides). Hot loops that convert to `any` for logging or reflection pay for it.

```bash
go build -gcflags='-m' .
```

Look for `x escapes to heap` near interface conversions.

Method Calls

Calling `i.M()`:

1. Load itab (or eface type).
2. Index method slot.
3. Call with concrete receiver in `data`.

Devirtualization / inlining may optimize some cases after SSA; do not assume every interface call is a virtual disaster — **measure**. Still, concrete types in hot paths are simpler for the compiler.

Type Assert and Type Switch

```go
v, ok := i.(T)     // checks dynamic type
switch v := i.(type) { ... }
```

Failed assert without `ok` panics. Prefer comma-ok form at boundaries.

Comparable Interfaces

Interfaces are comparable if dynamic types are comparable. Comparing interfaces compares dynamic type identity then values. Maps with `any` keys require comparable dynamic types at runtime or panic.

Experiment

```go
go mod init example
go run .
```

```go
package main

import (
    "fmt"
    "unsafe"
)

type Stringer interface {
    String() string
}

type N int

func (n N) String() string { return fmt.Sprintf("N(%d)", n) }

func main() {
    // nil trap
    var p *N = nil
    var s Stringer = p
    var empty Stringer
    fmt.Println("typed nil iface == nil?", s == nil)
```

```go
    fmt.Println("empty iface == nil?", empty == nil)

    // eface shape (illustrative sizes; do not depend on layout in prod code)
    var a any = 42
    var b any = p
    fmt.Printf("any int:  type-ish non-nil data non-nil? %v\n", a != nil)
    fmt.Printf("any nil *N != nil? %v\n", b != nil)

    // method call via iface
    s = N(7)
    fmt.Println("method", s.String())

    // assert
    if n, ok := s.(N); ok {
        fmt.Println("assert", n)
    }

    // show that interface is two words on 64-bit
    fmt.Println("sizeof any", unsafe.Sizeof(a))
}
```

Expected output (typical amd64/arm64):

```
typed nil iface == nil? false
empty iface == nil? true
any int:  type-ish non-nil data non-nil? true
any nil *N != nil? true
method N(7)
assert 7
sizeof any 16
```

What to notice: Interface is two machine words; typed nil is not interface nil.

Try next: Benchmark `fmt.Sprintf("%v", concrete)` vs logging with concrete fields — interface conversion shows up in alloc profiles.

Related

- Interface practices
- Type assertions

Slice and Map Internals

Slice and Map Internals

Overview

Slices and maps are the two most used runtime-backed structures. Misunderstanding headers and growth causes accidental sharing bugs, huge realloc copies, and production map panics.

Basics: Arrays & slices, Maps.

Diagram: Slice header and map grow

```
regions: slice header | map
flow:
  [ptr]
       |
       v
  [Arr]

  [buckets] --load high--> [grow]
```

Slice Header

A slice value is a small header (conceptually):

```
slice
  ptr    -> underlying array
  len
  cap
```

Assignment copies the **header**, not the array:

```
a := []int{1, 2, 3}
b := a
b[0] = 9 // a[0] is also 9
```

`append` may allocate a new array when `len+need > cap`, then copy. Capacity growth is geometric (implementation detail; treat as amortized O(1) append).

```go
s = append(s, x) // maybe new backing store; old aliases unchanged
```

Full slice expression

```go
t := s[i:j:k] // len=j-i, cap=k-i
```

Limits capacity so later `append` cannot overwrite the rest of the original array — critical when returning subslices from internal buffers.

Nil vs empty

```go
var s []int      // nil, len0 cap0
t := []int{}     // non-nil empty
u := make([]int, 0)
```

JSON encodes both as `[]`. Prefer one convention in APIs.

Map Runtime Model

Maps are **hash tables** of buckets (plus overflow). Runtime may:

- Grow (reallocate buckets) when load is high
- Evacuate old buckets progressively

```
map header (hmap)
  count
  B          // log2 buckets
  buckets
  oldbuckets // during grow
  ...
```

Semantics that bite

1. **Not addressable elements** — cannot take `&m[k]` for map values in general.
2. **Iteration order is randomized** — never depend on it.
3. **Concurrent map read+write without sync panics** (fatal runtime error).
4. **Delete during range** is allowed for the current key; still no concurrent writers from other Gs.
5. **NaN float keys** are cursed — avoid float map keys.

Growth cost

Large maps that grow under a request path create latency spikes. Pre-size when you know cardinality:

```
m := make(map[string]int, expected)
```

Sharing and Aliasing Checklist

Action	Shares backing array?
`b := a`	Yes
`b := a[i:j]`	Yes
`append` that grows	No (new array for result)
`slices.Clone` / `copy` to new	No
`maps.Clone`	New map, same element values

When returning a slice from an internal buffer, **clone** if the buffer will be reused.

Experiment

```
go mod init example
go run .
```

```
package main

import (
    "fmt"
    "unsafe"
)

func main() {
    // header size
    var s []int
    fmt.Println("sizeof slice header", unsafe.Sizeof(s))

    a := make([]int, 3, 6)
    a[0], a[1], a[2] = 1, 2, 3
    b := a[1:3]
    b[0] = 99
    fmt.Println("share", a) // a[1] changed

    // cap limit
```

```go
    c := a[0:2:2]
    c = append(c, 7)
    fmt.Println("full slice a after append c", a, "c", c)

    // growth
    var d []int
    for i := 0; i < 10; i++ {
        d = append(d, i)
        fmt.Printf("len=%d cap=%d\n", len(d), cap(d))
    }

    // map pre-size vs not (just API demo)
    m := make(map[int]int, 100)
    for i := 0; i < 100; i++ {
        m[i] = i * i
    }
    fmt.Println("map len", len(m))

    // iteration order differs across runs (print a few)
    keys := 0
    for k := range m {
        keys++
        if keys <= 3 {
            fmt.Println("key sample", k)
        }
    }
}
```

What to notice: Subslice mutation mutates the original; capacity growth jumps; map key order is not sorted.

Try next: Force concurrent map write with two goroutines (expect fatal); fix with `sync.Mutex` or `sync.Map` only if appropriate.

Related

- Mutability
- slices/maps helpers

Defer, Panic, and Stack Unwind

Defer, Panic, and Stack Unwind

Overview

`defer` schedules a call to run when the surrounding function returns — normally or via panic. Modern Go **open-codes** many defers (cheap), but loops that defer still accumulate work until return. Panic unwinds the stack running defers until `recover` or process crash.

Errors: Panic and recover.

Diagram: Panic unwind

```
panic(value)

     v
run defers LIFO (this frame)

     v
caller frames … until recover

        recover in defer    stop, resume
        none    crash + stacks
```

Defer Semantics

```go
func f() {
    defer fmt.Println("second")
    defer fmt.Println("first") // LIFO
    fmt.Println("body")
}
// body, first, second
```

Arguments evaluated immediately

```go
func g() {
    x := 1
    defer fmt.Println(x) // prints 1 even if x changes later
    x = 2
}
```

To see final values, defer a closure:

```go
defer func() { fmt.Println(x) }()
```

Named results

Deferred closures can assign named return values — used carefully in middleware-style wrappers:

```go
func wrap() (err error) {
    defer func() {
        if err != nil {
            err = fmt.Errorf("wrap: %w", err)
        }
    }()
    return do()
}
```

Cost Model

Pattern	Cost
Few defers per function (open-coded)	Very low
`defer` inside hot loop	Registers many defers; prefer explicit unlock or scoped helper
`defer mu.Unlock()` after Lock	Idiomatic and fine outside mega-hot paths

```go
// Prefer for tight loops:
mu.Lock()
// critical
mu.Unlock()

// Prefer for error-prone multi-return:
mu.Lock()
defer mu.Unlock()
```

Panic Unwind

```
panic(value)
  -> run defers in current function (LIFO)
  -> pop frame
  -> run defers in caller
  -> ...
  -> recover in a deferred call? stop unwind, return
  -> else crash process (dump stacks)
```

`recover` only works **directly inside a deferred function** during that panic:

```go
defer func() {
    if r := recover(); r != nil {
        // log, convert to error
    }
}()
```

What recover cannot do

- Fix corrupted program state magically
- Replace typed API errors as the default control flow
- Catch panics in **other** goroutines — each G panics independently

```go
go func() {
    defer func() { recover() }() // only protects this goroutine
    mayPanic()
}()
```

Stack Growth

Goroutine stacks start small and **grow/shrink** (copy stack). Pointers into stack frames are invisible to Go code safely — the compiler/runtime rewrites them. Do not pass pointers from Go stack to C without pinning rules (cgo).

Experiment

```
go mod init example
go run .
```

```go
package main

import "fmt"

func deferOrder() {
    defer fmt.Println("D1")
    defer fmt.Println("D2")
    fmt.Println("body")
}

func deferArgs() {
    x := 1
    defer fmt.Println("arg", x)
    defer func() { fmt.Println("closure", x) }()
    x = 9
}

func recoverDemo() (err error) {
    defer func() {
        if r := recover(); r != nil {
            err = fmt.Errorf("panic: %v", r)
        }
    }()
    panic("boom")
}

func main() {
    deferOrder()
    deferArgs()
    fmt.Println("recovered", recoverDemo())
}
```

Expected output:

```
body
D2
D1
closure 9
arg 1
recovered panic: boom
```

What to notice: LIFO order; args vs closure capture; panic converted to error only with defer+recover.

Try next: Benchmark loop with `defer unlock` vs paired unlock for a million iterations — see if it matters in your Go version.

Related

- Panic recover
- Must pattern

Generics Implementation Deep Dive

Generics Implementation Deep Dive

Overview

Go generics (1.18+) are implemented with a hybrid strategy often described as **dictionaries + partial stenciling** (GC shape based). You do not need every compiler paper — you need the cost model: when generics are zero-cost abstraction vs when they introduce runtime dictionary calls or more code size.

Usage: Generics intro through Idiomatic generics.

Diagram: Generic call shapes

```
generic source

     v
typecheck + constraints

     v
GC shape groups + dictionaries

     v
shared/specialized machine code
```

What the Compiler Sees

```go
func Max[T cmp.Ordered](a, b T) T {
    if a > b {
        return a
    }
    return b
}
```

Calls with different type arguments may share generated code when the **GC shape** matches (same underlying representation for pointers, etc.), parameterized by a **dictionary** describing operations (comparison, conversion, …).

```
source generic
    |
    v
type check + capture constraints
    |
    v
shape grouping (pointer-ish shapes share)
    |
    v
code + dictionary for operations on T
```

Cost Dimensions

Dimension	Effect
Code size	Too many distinct shapes / specialized methods → binary growth
Call cost	Dictionary-indirect calls can inhibit inlining
Allocation	Generic APIs that box to `any` lose the point
Readability	Complex constraint graphs hurt more than they help

Generics excel for:

- Containers and algorithms over `~[]T`, maps, comparables
- Removing interface{} while keeping type checks

Generics struggle when:

- Every method needs unique runtime behavior per type without structure
- You force abstraction that a plain function + concrete types would inline better

Constraints Are Interfaces

```
type Number interface {
    ~int | ~int64 | ~float64
}
```

Constraint satisfaction is compile-time. Method sets on type parameters follow the constraint's method set.

Instantiation

```
f := Max[int] // function value of specialized shape
v := Max(1, 2) // inference
```

Type inference fails when the compiler cannot unify arguments — add explicit type args rather than fighting it.

Interaction With Interfaces

```
func Print[T fmt.Stringer](v T) {
    fmt.Println(v.String())
}
```

Prefer **type parameters** when callers should keep static types through the call graph; prefer **interfaces** when you need heterogeneous collections or runtime plugin style.

Avoid These Footguns

1. **Generic + reflect by default** — pays both complexity taxes.
2. **Over-abstracting one-use helpers** — a function with `int` is fine.
3. **Exporting huge generic surface** — every instantiation pressure multiplies for users.

Experiment

```
go mod init example
go run .
go build -gcflags='-m' . 2>&1 | head -40
```

```
package main

import (
    "cmp"
    "fmt"
)

func Max[T cmp.Ordered](a, b T) T {
    if a > b {
        return a
    }
    return b
```

```go
}

type Pair[K comparable, V any] struct {
    K K
    V V
}

func (p Pair[K, V]) String() string {
    return fmt.Sprintf("%v=%v", p.K, p.V)
}

func main() {
    fmt.Println(Max(3, 5))
    fmt.Println(Max(2.5, 1.2))
    fmt.Println(Pair[string, int]{K: "n", V: 1})
}
```

What to notice: One source definition serves multiple shapes. `-gcflags=-m` may show inlining decisions around simple generic functions.

Try next: Compare assembly of `Max[int]` vs a non-generic `maxInt` with `go build -gcflags=-S` and see if dictionary calls appear in your version.

Related

- Generic functions
- Generic types
- Compiler & SSA

GC Internals and Pacer

GC Internals and Pacer

Overview

Go's GC is a **concurrent, non-moving, mark-sweep** collector with a **pacer** that decides when to start marking so heap growth stays near a target. Go 1.26's **Green Tea GC** improves scan locality; the user-facing knobs (`GOGC`, `GOMEMLIMIT`) remain the control plane.

Practical tuning: Garbage collection, Allocations & GC.

> Companion depth: Internals for Interns — The Garbage Collector (GreenTea / mark phase notes).

Diagram: GC cycle

```
mutator allocates    pacer (GOGC / GOMEMLIMIT)

                             v

                     mark setup STW     (short)

                             v

                     concurrent mark    mark assist from allocators
                      + write barrier

                             v

                     mark term STW      (short)

                             v

                     concurrent sweep   reclaim spans
```

Heap Building Blocks

```
arena / pages
    -> spans (size classes for small objects)
        -> objects
large objects: dedicated spans
```

Allocators (per-P caches) reduce cross-thread contention. When caches miss, they pull from central free lists / the heap.

Tri-color intuition

```
  white (unmarked)  reach   grey (to-scan)  scan kids   black

          not reached    sweep reclaim
```

Write barriers keep the invariant safe while mutators run during concurrent mark.

Mark and Sweep Phases

```
1. Mark setup      STW (short)  - enable write barrier, root prep
2. Concurrent mark             - workers + mutator assists
3. Mark termination STW (short)
4. Concurrent sweep            - reclaim unmarked objects by span
```

Write barrier

While marking, pointer writes are instrumented so the GC does not lose objects. Barriers add a small tax on pointer-heavy code — one reason reducing pointer count (or packing) helps.

Mark assist

If the mutator allocates faster than markers can scan, the allocating G **helps mark**. This couples allocation rate to GC CPU and shows up as latency on the allocating path — not only in a GC thread.

Pacer Goals

The pacer aims roughly:

```
live heap after GC * (1 + GOGC/100)    heap size that triggers next GC
```

- **GOGC=100** (default): next GC when heap roughly doubles live size.
- Lower GOGC → more frequent GC, less peak memory.
- Higher GOGC → less GC CPU, more memory.

GOMEMLIMIT

Soft limit on Go-managed memory. When approaching the limit, GC becomes more aggressive (and may return memory to the OS more eagerly depending on version/settings). Use with container memory limits to reduce OOM-kills:

```
GOMEMLIMIT=512MiB GOGC=100 ./service
```

Prefer **limit + reasonable GOGC** over `GOGC=off` fantasies.

Green Tea GC (1.26+)

Span-aware scanning improves cache locality for pointer-rich heaps. Opt-out for experiments:

```
GOEXPERIMENT=nogreenteagc ./service
```

Validate with your workload's p95 and CPU profiles — do not assume every service wins equally.

Reading GC Telemetry

```
var m runtime.MemStats
runtime.ReadMemStats(&m)
// m.NumGC, GCCPUFraction, HeapAlloc, HeapInuse, NextGC, PauseTotalNs
```

Or metrics:

```
go_gc_duration_seconds
go_memstats_heap_alloc_bytes
go_memstats_next_gc_bytes
```

Latency pattern

Periodic p95 spikes aligned with `NumGC` increases often mean **assist or STW**, not "Go is slow." Fixes: fewer allocs, smaller pointer graphs, raise memory budget, or smooth allocation rate.

Design Rules

1. **Allocate less** before tuning GOGC.
2. **Reuse buffers** (`[]byte` pools) on hot paths.
3. **Avoid pointer-heavy tiny objects** in huge graphs when possible (flatten).
4. **Set GOMEMLIMIT** in containers.
5. **Profile** — `pprof` heap + `trace` GC events.

Experiment

```
go mod init example
go run .
```

```go
package main

import (
    "fmt"
    "runtime"
    "time"
)

func stats(label string) {
    var m runtime.MemStats
    runtime.ReadMemStats(&m)
    fmt.Printf("%s alloc=%dKiB nextGC=%dKiB numGC=%d gccpu=%.3f\n",
        label,
        m.Alloc/1024,
        m.NextGC/1024,
        m.NumGC,
        m.GCCPUFraction,
    )
}

func main() {
    stats("start")
    keep := make([][]byte, 0, 1000)
    for i := 0; i < 1000; i++ {
        keep = append(keep, make([]byte, 32*1024))
    }
    stats("after alloc")
    keep = nil
    runtime.GC()
    time.Sleep(50 * time.Millisecond)
    stats("after GC")
}
```

What to notice: `NextGC` rises with live heap; forced `GC` drops alloc; `GCCPUFraction` is a long-run average (not instant).

Try next: Run the same under `GOGC=50` vs `GOGC=200` and compare `NumGC` after identical work.

Related

- GC chapter
- Escape analysis

Compiler, SSA, Inlining, and PGO

Compiler, SSA, Inlining, and PGO

Overview

The Go compiler turns packages into machine code through a pipeline: parse → typecheck → mid-end optimizations (SSA) → architecture lowering → assemble. Knowing the pipeline tells you **which knobs are real** (`-gcflags`, inlining, escape, PGO) versus cargo cult.

Diagram: Compiler pipeline

```
  source → typecheck → IR → SSA → machine code → link

                  PGO profile    (hot edges)
  -gcflags=-m  shows escape / inline decisions
```

Pipeline (Simplified)

```
source .go
  -> parse / syntax
  -> typecheck
  -> IR build
  -> SSA optimize (prove, rewrite, deadcode, nilcheck elim, ...)
  -> lower to arch
  -> encode object
link -> executable
```

Escape analysis runs early enough to decide stack vs heap; inlining expands call sites so later passes see a larger region.

Essential Flags

```
# Escape and inline decisions
go build -gcflags='-m' .

# More detail
```

```go build -gcflags='-m -m' .

# Disable optimizations (debug)
go build -gcflags='-N -l' .

# Assembly listing
go build -gcflags='-S' . 2> asm.txt

# Shared / large projects: apply to specific package
go build -gcflags='all=-m' ./...
```

Flag	Meaning
`-m`	Print optimization decisions
`-l`	Disable inlining (with `-N` for debug)
`-N`	Disable optimizations
`-B`	Disable bounds checks (unsafe; special cases)

Inlining

Small functions may inline into callers, enabling further constant folding and escape improvements.

```go //go:noinline
func sink(x int) int { return x }
```

Use `//go:noinline` only for benchmarks that must measure call cost. Prefer clear small functions; the compiler is good at inlining them.

Mid-stack inlining and more aggressive budgets improved across versions — re-benchmark when upgrading major Go releases.

Bounds Check Elimination (BCE)

SSA proves many index checks redundant:

```go for i := range s {
    s[i] = i // often single check or proved safe
}
```

Idioms that help BCE: `for i := range s`, length hoisting the compiler can see, slice expressions with proven bounds. Do not micro-obfuscate for BCE without profiles.

Profile-Guided Optimization (PGO)

From Go 1.20+ (improved since):

```
# 1) Collect CPU profile from production-like load
# 2) Build with profile
go build -pgo=default.pgo -o app .
# or auto:
go build -pgo=auto .
```

PGO feeds the compiler **hot call edges** so inlining/devirtualization favor real paths. Gains of a few percent are common; bigger when interfaces dominate.

Treat PGO profiles as **build inputs** — keep them reasonably fresh.

What the Compiler Will Not Do

- Fix algorithmic $O(n^2)$
- Remove channel/mutex coordination costs
- Make reflect cheap
- Cross-package miracle opts without inlining visibility

Experiment

```
go mod init example
go build -gcflags='-m' . 2>&1 | sed -n '1,40p'
```

```
package main

import "fmt"

func add(a, b int) int { return a + b }

func main() {
    x := 3
    y := add(x, 4)
    fmt.Println(y)
}
```

Typical messages include `can inline add` and whether `x` escapes (usually does via `fmt`).

Try next: Compare `go test -bench=.` with and without `-pgo` after collecting a profile from the benchmark binary.

Related

- Escape analysis
- Performance tuning

Linking, Build Modes, and Race ABI

Linking, Build Modes, and Race ABI

Overview

After compilation, the **linker** stitches objects, resolves symbols, and emits an executable or library. Build modes and sanitizers change the ABI and cost model — important for plugins, cgo, and race-enabled CI.

Diagram: Build modes

```
go build
      exe (default)
      pie
      c-shared / c-archive
      plugin   (limited platforms)

go build -race    instrumented memory + HB
```

Default Build

```
go build -o app .
go install ./cmd/app
```

Static-ish binaries are common on Linux with pure Go (no cgo). The Go runtime issues syscalls itself, so `CGO_ENABLED=0` produces a fully static ELF with no `libc.so`. Enable cgo and **glibc vs musl** comes back: a Ubuntu-linked binary fails on Alpine because the ELF interpreter (`/lib64/ld-linux-x86-64.so.2`) is missing. See Go as a Systems Language.

```
go env CGO_ENABLED
CGO_ENABLED=0 go build -o app .   # force pure Go when possible
file app && ldd app               # Linux: expect "statically linked"
```

Build Modes

```
go help buildmode
```

Mode	Use
`exe`	Default executable
`c-shared`	C shared library from Go
`c-archive`	C archive
`plugin`	Go plugin (limited platforms; fragile for prod)
`pie`	Position-independent executable

```
go build -buildmode=pie -o app .
```

Plugins sound attractive and usually disappoint: version skew with main binary, limited platforms, hard ops story. Prefer explicit interfaces + separate processes.

Build Tags and Constraints

```
//go:build linux && amd64

package onlylinux
```

```
go build -tags=debug,integration
```

Use tags for optional integration tests and platform files (`foo_windows.go`), not for sprawling feature flags that should be config.

Linker Flags

```
go build -ldflags='-s -w' -o app .          # strip symbols / DWARF
go build -ldflags='-X main.version=1.2.3' .
```

`-s -w` smaller binaries, harder debug. Keep symbols in staging if you need crash analysis.

Race Detector

```
go test -race ./...
go build -race -o app-race .
```

Race builds:

- Use a different memory allocator instrumentation
- Cost **CPU and memory** (often 2–10×)
- Are for testing, not production traffic

Shadow memory tracks happens-before; if it reports a race, trust it.

cgo Boundary (Preview)

cgo inserts C ABI calls and can:

- Prevent easy cross-compilation
- Block some optimizations
- Require `CGO_ENABLED=1` and a toolchain

Deep systems: cgo. Prefer pure Go drivers when shipping multi-arch binaries.

Reproducible Builds

```
go build -trimpath -ldflags='-buildid=' -o app .
```

Trim paths remove local directory noise from binaries; useful for supply-chain hygiene.

Experiment

```
go mod init example
go build -o /tmp/app-normal .
go build -ldflags='-s -w' -o /tmp/app-strip .
ls -la /tmp/app-normal /tmp/app-strip

go test -race -c -o /tmp/app.test .
# file sizes differ; race binary is larger
ls -la /tmp/app.test
```

```
package main

import "fmt"

var version = "dev"

func main() {
    fmt.Println("version", version)
}
```

```
go build -ldflags='-X main.version=1.0.0' -o /tmp/app-ver .
/tmp/app-ver
```

What to notice: Strip and race change binary size dramatically; `-X` injects version at link time.

Try next: Build with `CGO_ENABLED=0` and confirm `ldd` (Linux) shows no unexpected libc deps for a pure Go net/http binary.

Related

- Building & releasing
- CI/CD security

Reflect Internals and Cost

Reflect Internals and Cost

Overview

Package `reflect` exposes runtime type metadata (`rtype`) and dynamic values (`Value`). It powers encoding/json, dependency injection frameworks, and ORMs — and it is a common production hotspot when used on every request without caching.

Usage intro: Reflection.

Diagram: Reflect path

```
flow:
  [V]
       |
       v
  [Flags]
```

Core Types

```go
t := reflect.TypeOf(x)   // runtime type descriptor (nil-safe only if x typed)
v := reflect.ValueOf(x)  // dynamic value
```

API	Role
`Type`	Size, kind, fields, methods — often cacheable
`Value`	Read/write/call — easy to allocate
`Kind`	`Int`, `Struct`, `Slice`, …
`Interface`	Box back to `any`

What Metadata Lives Where

```
interface / concrete
       |
       v
    rtype  (uncommonType, struct fields, method table…)
       |
Value flags: addressable? settability? indir?
```

`ValueOf` on a non-pointer often yields a **non-addressable** copy — `Set` fails. Pass pointers for mutation:

```go
rv := reflect.ValueOf(&x).Elem()
rv.SetInt(3)
```

Allocations and Interface Boxing

Common alloc sources:

1. `Value.Interface()` boxing
2. `ValueOf` of small values that escape
3. Building `[]reflect.Value` for `Call`
4. Pathological use inside tight loops without type caches

```go
// Cache field indexes by type
var cache sync.Map // reflect.Type -> []int indices
```

`encoding/json` caches struct field info per type — copy that idea.

Method Calls via Reflect

```go
m := reflect.ValueOf(obj).MethodByName("Do")
results := m.Call(nil)
```

Much slower than a direct call or interface method. Prefer interfaces for hot polymorphic paths; use reflect for glue and frameworks.

Unsafe Adjacent

`reflect.Type` and `unsafe.Pointer` interactions exist for advanced serializers. Prefer well-tested libraries over hand-rolled unsafe reflection unless you accept maintenance cost.

See unsafe.

When Reflect Is Justified

Use	Verdict
One-time config decode	Fine
Per-request JSON (stdlib)	Fine — heavily optimized
Per-request generic mapper over large structs	Cache or codegen (`go generate`, `json` struct tags already)
Game loop / packet parse	Avoid; use codegen or concrete code

Experiment

```
go mod init example
go test -bench=. -benchmem
```

```go
package main

import (
    "reflect"
    "testing"
)

type S struct {
    A int
    B string
}

func direct(s *S) int { return s.A }

func viaReflect(s *S) int {
    return int(reflect.ValueOf(s).Elem().FieldByName("A").Int())
}

func BenchmarkDirect(b *testing.B) {
    s := &S{A: 42}
    for i := 0; i < b.N; i++ {
        _ = direct(s)
    }
}

func BenchmarkReflect(b *testing.B) {
    s := &S{A: 42}
    for i := 0; i < b.N; i++ {
        _ = viaReflect(s)
    }
```

```go
}

// Keep a main so go run works if needed
func main() {}
```

Put benchmarks in `main_test.go` or rename package for pure test module:

```
# simpler: single file package reflectbench_test
```

Alternatively run:

```go
// file: cost_test.go
package cost

import (
    "reflect"
    "testing"
)

type S struct{ A int }

func BenchmarkFieldDirect(b *testing.B) {
    s := S{A: 1}
    for i := 0; i < b.N; i++ {
        _ = s.A
    }
}

func BenchmarkFieldReflect(b *testing.B) {
    s := S{A: 1}
    for i := 0; i < b.N; i++ {
        _ = reflect.ValueOf(s).Field(0).Int()
    }
}
```

```bash
mkdir /tmp/reflect-cost && cd /tmp/reflect-cost
go mod init example
# write cost_test.go as above
go test -bench=. -benchmem
```

What to notice: Reflect field access is often **orders of magnitude** slower and may allocate depending on path.

Try next: Cache `reflect.Type` and field index; re-benchmark `FieldByName` vs `Field(i)`.

Related

- Reflection
- Codegen

Modules, MVS, and Toolchain

Modules, MVS, and Toolchain

Overview

Go modules are a **versioned dependency graph** with a precise selection algorithm: **Minimal Version Selection (MVS)**. Treat `go.mod` / `go.sum` as production config — not editor noise.

Basics: Dependency management, Workspaces.

Diagram: MVS selection

```
flow:
  [Max]
       |
       v
  [Build]
```

Core Files

File	Role
`go.mod`	Module path, Go version, requires, replaces, excludes, toolchain
`go.sum`	Cryptographic checksums of module contents
`vendor/`	Optional frozen source trees

```
go mod tidy      # align requires with imports
go mod download
go mod verify
go mod graph
```

Minimal Version Selection

Unlike many package managers that pull "newest compatible," MVS picks the **minimum version that satisfies all constraints** in the build list.

```
Your module requires A v1.2
B requires A v1.5
C requires A v1.3
----------------------------
Selected A = v1.5  (max of minima)
```

Properties:

- **Reproducible** given the same `go.mod` build list
- **Not** "always latest"
- Upgrades are **explicit** (`go get pkg@vX`)

```
go list -m all
go list -m -versions github.com/some/dep
go get example.com/dep@v1.4.2
go get example.com/dep@none   # drop
```

The Build List

`go list -m all` shows the selected set. Understanding diamond dependencies is mostly reading that list and `go mod graph | grep dep`.

replace and exclude

```
replace example.com/fork => ../fork

exclude example.com/bad v1.0.0
```

`replace` is powerful for local development; **avoid permanent replaces** in published modules — consumers inherit pain. Prefer releasing fixed versions.

Retracts

Module authors can `retract` broken versions so `go get` avoids them. If you cannot upgrade, pin carefully and track advisories.

Toolchain Directive

Modern Go can select a **toolchain** version:

```
go 1.27
toolchain go1.27.1
```

```
go env GOTOOLCHAIN  # auto | local | path | version
```

CI should pin or intentionally use `auto` with care so builds do not silently jump compilers mid-incident.

Vendoring

```
go mod vendor
go build -mod=vendor
```

Pros: offline builds, airgapped, reviewable third-party diffs. Cons: repo size, merge noise. Good for enterprises and release branches.

Checksums and Proxies

```
GOPROXY  (default proxy.golang.org)
GOSUMDB  (sum.golang.org)
GOPRIVATE / GONOSUMDB for private modules
```

```
export GOPRIVATE=github.com/myorg/*
```

Private modules need auth to the host; sumdb is skipped for `GOPRIVATE`.

Diagnosis Playbook

Symptom	Action
"missing go.sum entry"	`go mod tidy`
Unexpected old dep	`go mod graph` + who requires it
Local replace forgotten	Search `replace` in go.mod
CI != laptop	Print `go version`, `go env`, `go list -m all`
Ambiguous import	Module path vs directory mismatch

Experiment

```
mkdir /tmp/mvs-demo && cd /tmp/mvs-demo
go mod init example.com/demo
go get rsc.io/quote@v1.5.2
go list -m all
go mod graph | head
cat go.mod
go mod tidy
```

What to notice: `go.mod` records the selected versions; `go.sum` grows with hashes; graph shows edges that forced versions upward.

Try next: Introduce a `replace` to a local stub module and build; then remove replace and fetch the real version.

Related

- Dependency management
- Workspaces
- Building & releasing

Netpoller and Network I/O

Netpoller and Network I/O

Overview

Blocking on a socket in Go rarely blocks an OS thread the way a naive `read()` would in a 1:1 thread model. The **netpoller** integrates with the scheduler: a G waiting on network readiness parks, its M can run other work, and the poller wakes the G when the FD is ready.

This is why Go servers scale to huge connection counts with modest `GOMAXPROCS`.

> Companion: Internals for Interns — Network Poller (who calls `netpoll` from `findrunnable` / sysmon).

Diagram: Netpoll park

```
sequence (top → bottom):
  actors: goroutine, runtime, epoll/kqueue/IOCP
  goroutine --> runtime  : Read would block
  runtime --> goroutine  : park netpoll wait
  runtime --> epoll/kqueue/IOCP  : register FD
  epoll/kqueue/IOCP --> runtime  : readable
  runtime --> goroutine  : goready
  goroutine --> goroutine  : Read completes
```

Mental Model

```
G calls conn.Read
  -> runtime sees would-block
  -> G parks on netpoll wait
  -> M may run other Gs
  ...
kernel marks FD readable
  -> netpoller (epoll/kqueue/IOCP) notices
  -> G becomes runnable
  -> Read completes
```

Platform backends:

OS	Mechanism
Linux	epoll
BSD/macOS	kqueue
Windows	IOCP

Deadlines vs Poller

`SetDeadline` / `context` cancellation arm timers that **also** wake blocked network ops. Without deadlines, a stuck peer can pin a G forever (goroutine + buffer memory).

```
_ = conn.SetReadDeadline(time.Now().Add(30 * time.Second))
// or
ctx, cancel := context.WithTimeout(ctx, 30*time.Second)
defer cancel()
// http and many APIs use Request.Context()
```

Interaction With Syscalls

Not every I/O goes through netpoll. File I/O and some cgo paths still block Ms. Network sockets registered with the poller are the sweet spot for massive concurrency.

Go 1.26 simplified how **P** tracks syscall state (derive from G instead of a dedicated P-syscall state), which helps syscall/cgo-heavy paths. See cgo & scheduler.

Zero-Copy Adjacent

`io.Copy` between sockets may use `splice`/`sendfile` on Linux when types allow — poller still owns readiness. TLS usually **cannot** sendfile the cleartext path the same way. See zero-copy I/O.

Diagnosis

Symptom	Check
Goroutines stuck in `net.Read`	Deadlines? peer hang?
Threads exploding	Blocking syscalls / DNS / cgo, not poller waits
High latency, low CPU	Downstream wait; trace shows park on poll

```
go tool pprof http://localhost:6060/debug/pprof/goroutine
# look for runtime.netpollblock, internal/poll.runtime_pollWait
```

Experiment

```
go mod init example
go run .
```

```
package main

import (
    "fmt"
    "io"
    "net"
    "time"
)

func main() {
    ln, err := net.Listen("tcp", "127.0.0.1:0")
    if err != nil {
        panic(err)
    }
    defer ln.Close()

    go func() {
        c, err := ln.Accept()
        if err != nil {
            return
        }
        defer c.Close()
        time.Sleep(50 * time.Millisecond)
        _, _ = c.Write([]byte("hello"))
    }()

    c, err := net.Dial("tcp", ln.Addr().String())
    if err != nil {
        panic(err)
    }
    defer c.Close()
    _ = c.SetReadDeadline(time.Now().Add(time.Second))
    buf := make([]byte, 16)
    n, err := c.Read(buf)
    fmt.Printf("read %q err=%v\n", buf[:n], err)

    // deadline fire
    _ = c.SetReadDeadline(time.Now().Add(20 * time.Millisecond))
    _, err = c.Read(buf)
    fmt.Printf("timeout? %v\n", err)
    _, _ = io.WriteString(c, "x")
```

```
}
```

What to notice: Read waits without spinning a CPU core; deadlines surface as errors, not hangs.

Try next: Open 10k localhost connections with short deadlines; watch RSS and goroutine count stay manageable.

Timers, Tickers, and time.After Pitfalls

Timers, Tickers, and time.After Pitfalls

Overview

Timers are a frequent source of **subtle leaks and wakeups** in hot paths. The runtime keeps a **timer heap** (per-P structures in modern Go). Stopping and resetting correctly matters as much as creating timers.

Diagram: Timer heap cost

```
flow:
  [Heap]
       |
       v
  [GC]
```

APIs

API	Use
`time.NewTimer`	One-shot; must `Stop`/`Reset` carefully
`time.NewTicker`	Periodic; always `Stop`
`time.After`	Convenient; **allocates a new timer** each call
`time.AfterFunc`	Fire a function on a timer thread-ish path

The Classic Pitfall

```go
// BAD in a busy loop / per-request select
select {
case <-ch:
case <-time.After(time.Second): // new timer every iteration
}
```

Each `time.After` creates a timer that lives until it fires (or is GC'd after fire). Under load this is allocation + timer-heap churn.

Prefer:

```go
t := time.NewTimer(time.Second)
defer t.Stop()
select {
case <-ch:
    if !t.Stop() {
        select {
        case <-t.C:
        default:
        }
    }
case <-t.C:
}
```

Or reuse with `Reset` after successful `Stop`/drain per current docs for your Go version.

Tickers

```go
tk := time.NewTicker(100 * time.Millisecond)
defer tk.Stop()
for {
    select {
    case <-ctx.Done():
        return
    case <-tk.C:
        // work
    }
}
```

A stopped ticker is not restarted with `Reset` in older Go the same way as timers — check version docs.

Runtime View

Timers are not free threads. They are heap entries processed by the runtime; expired timers make Gs runnable. Huge numbers of far-future timers cost memory; huge numbers of near-term timers cost CPU in the timer processor.

Context Timeouts

`context.WithTimeout` uses timers under the hood. Prefer **one timeout at the edge** and derive children, rather than stacking independent `After` calls at every layer without budget math.

Experiment

```
go test -bench=. -benchmem
```

```go
package timers_test

import (
    "testing"
    "time"
)

func BenchmarkTimeAfter(b *testing.B) {
    ch := make(chan struct{})
    for i := 0; i < b.N; i++ {
        select {
        case <-ch:
        case <-time.After(time.Hour):
        default:
        }
    }
}

func BenchmarkTimerReuse(b *testing.B) {
    t := time.NewTimer(time.Hour)
    defer t.Stop()
    ch := make(chan struct{})
    for i := 0; i < b.N; i++ {
        if !t.Stop() {
            select {
            case <-t.C:
            default:
            }
        }
        t.Reset(time.Hour)
        select {
        case <-ch:
        case <-t.C:
        default:
        }
    }
}
```

What to notice: `time.After` in a loop allocates; reuse patterns dominate under `-benchmem`.

Try next: Find `time.After` in a hot select in your codebase with `rg 'time\\.After'`.

Stack Growth and Copying

Stack Growth and Copying

Overview

Each goroutine starts with a **small stack** (on the order of a few KB historically; exact size is version-dependent) and **grows** by allocating a larger stack and **copying** live frames when needed. Stacks also shrink when underutilized.

This design makes millions of Gs feasible — you do not pre-reserve 1–8 MiB per goroutine like many OS threads.

Diagram: Stack growth

```
G stack nearly full

     v
morestack → larger stack

     v
copy frames + fix pointers → resume
```

Growth Path

```
G running, stack nearly full
  -> morestack
  -> allocate larger stack
  -> copy stack contents
  -> adjust pointers into stack (compiler/runtime)
  -> resume
```

Go forbids taking the address of a stack variable and using it after the frame dies; the compiler and escape analysis keep GC and stack copy correct.

Why Deep Recursion Hurts

Deep recursion causes repeated growth and large stacks:

```go
func sum(n int) int {
    if n == 0 {
        return 0
    }
    return n + sum(n-1) // grows stack with depth
}
```

Prefer iterative algorithms for unbounded depth.

cgo and Stacks

cgo may switch to larger **system stacks** for C execution. Crossing Go C frequently is expensive (scheduler + stack). See cgo transitions.

Debugging Stacks

```go
GOTRACEBACK=all ./app
// panic dumps all Gs
```

```go
buf := make([]byte, 1<<16)
n := runtime.Stack(buf, true) // all goroutines if true
fmt.Printf("%s", buf[:n])
```

Nosplit

```go
//go:nosplit
func tight() { /* cannot grow stack; for runtime/asm constraints */ }
```

User code almost never needs `//go:nosplit`. Misuse panics if the frame does not fit.

Experiment

```go
go mod init example
go run .
```

```go
package main

import (
    "fmt"
```

```go
    "runtime"
)

func depth(n int, max int) int {
    if n == max {
        var st runtime.StackRecord
        // approximate: print goroutine stack size indirectly via Stack
        buf := make([]byte, 64*1024)
        k := runtime.Stack(buf, false)
        fmt.Printf("at depth %d stack dump bytes=%d\n", n, k)
        return n
    }
    return depth(n+1, max)
}

func main() {
    fmt.Println("GOMAXPROCS", runtime.GOMAXPROCS(0))
    _ = depth(0, 200)
}
```

What to notice: Deep call chains produce large stack dumps; production code should bound recursion and prefer workers with small frames.

Try next: Compare a recursive tree walk vs explicit stack slice for a large tree — RSS and latency under load.

Finalizers and runtime.AddCleanup

Finalizers and runtime.AddCleanup

Overview

Go is GC-managed, but some resources are **outside the heap** (OS FDs, cgo memory). Historically `runtime.SetFinalizer` ran a function when an object became unreachable. Modern Go prefers **`runtime.AddCleanup`** (and explicit `Close`) because finalizers are subtle and easy to get wrong.

Diagram: Lifecycle preference

```
flow:
  [Backstop] --timing not guaranteed--> [Explicit]
```

Prefer Explicit Lifecycle

```go
f, err := os.Open(path)
if err != nil {
    return err
}
defer f.Close() // always first choice
```

Finalizers are a **backstop**, not a primary API.

SetFinalizer (Legacy Pattern)

```go
runtime.SetFinalizer(obj, func(o *T) {
    // runs asynchronously when obj is unreachable
    o.release()
})
```

Problems:

1. **No guarantee on timing** — may run late or, in edge cases, not before process exit.
2. **Resurrection** — finalizer can store the object globally and keep it alive.
3. **Order** — interdependent objects need careful design.
4. **Performance** — finalizers delay reclaim and add GC bookkeeping.

AddCleanup (Preferred Direction)

```go
// Go 1.24+ style cleanup (check your version docs)
runtime.AddCleanup(ptr, func(res Resource) {
    res.Release()
}, resourceToken)
```

Cleanups are designed to avoid several classic finalizer pitfalls (exact API surface evolves — verify on pkg.go.dev/runtime).

Rules of Thumb

Need	Approach
Files, sockets, body.Close	`defer Close`
Cache with eviction	Explicit pool / LRU
C malloc companion	`runtime.SetFinalizer` only if library requires; prefer free in Close
Tests for leaks	`testing` + explicit asserts, not finalizer timing

Experiment

```go mod init example
go run .
```

```go
package main

import (
    "fmt"
    "runtime"
    "time"
)

type blob struct {
    id int
}

func main() {
    for i := 0; i < 3; i++ {
        b := &blob{id: i}
        id := i
        runtime.SetFinalizer(b, func(_ *blob) {
            fmt.Println("finalizer", id)
```

```go
        })
        b = nil
    }
    runtime.GC()
    time.Sleep(50 * time.Millisecond)
    runtime.GC()
    time.Sleep(50 * time.Millisecond)
    fmt.Println("done")
}
```

What to notice: Finalizers may print after GC, asynchronously; never rely on them for correctness of security-sensitive cleanup without also having an explicit path.

Try next: Wrap an `os.File` only with `Close`; use `GODEBUG=gctrace=1` while allocating to see GC activity separately from finalizers.

weak and unique Packages

weak and unique Packages

Overview

Go 1.24+ added first-class support for patterns that used to require finalizer hacks or string maps:

Package	Role
`weak`	Weak pointers — do not keep the referent alive
`unique`	Canonicalize comparable values (interning)

These show up in caches, symbol tables, and RPC codecs.

Diagram: Weak vs strong

```
diagram:
  Strong[strong *T] -->|keeps alive| Obj[object]
  Weak[weak.Pointer] -.->|does not keep| Obj
  GC[GC] -->|collects| Obj
  Weak2[weak.Value] -->|nil after| GC
  U1[unique.Make a] --> H[canonical handle]
  U2[unique.Make a] --> H
```

weak.Pointer

```go
import "weak"

type heavy struct{ buf []byte }

func demo() {
    h := &heavy{buf: make([]byte, 1<<20)}
    wp := weak.Make(h)
    h = nil
    runtime.GC()
    if v := wp.Value(); v != nil {
        fmt.Println("still alive", len(v.buf))
    } else {
```

```
        fmt.Println("collected")
    }
}
```

Use cases:

- Caches that should not pin memory forever
- Canonical registries keyed by identity without leaks

Do **not** replace explicit cache eviction policies when you need deterministic memory bounds — combine weak refs with size limits.

unique.Handle

```
import "unique"

h1 := unique.Make("GET")
h2 := unique.Make("GET")
fmt.Println(h1 == h2) // true - same canonical handle
```

`unique.Make` returns a handle for a comparable value; equal values collapse to one interned copy. Good for:

- Repeated strings (method names, header keys, enums-as-strings)
- Reducing duplicate allocations in large graphs

Handles compare with `==`. Extract value with `h.Value()`.

When Not To

- Tiny short-lived strings — interning overhead may lose
- Secrets — longer lifetime increases exposure window
- Non-comparable structures — not supported

Experiment

```
go mod init example
go run .
```

```
package main

import (
    "fmt"
```

```go
    "runtime"
    "unique"
    "weak"
)

func main() {
    a := unique.Make("content-type")
    b := unique.Make("content-type")
    fmt.Println("unique equal", a == b, a.Value())

    type node struct{ n int }
    p := &node{n: 7}
    w := weak.Make(p)
    fmt.Println("before", w.Value() != nil)
    p = nil
    runtime.GC()
    runtime.GC()
    fmt.Println("after GC value?", w.Value() != nil)
}
```

What to notice: Unique handles collapse equal values; weak pointers may become nil after GC once strong refs are gone (timing can vary — double GC is a demo heuristic).

Try next: Intern 100k duplicate strings with and without `unique`; compare `runtime.MemStats.HeapAlloc`.

Swiss Maps and maphash

Swiss Maps and maphash

Overview

Go's map implementation has evolved. Recent versions moved toward **Swiss Table**-style layouts (open addressing with group metadata) for better CPU cache behavior versus older bucket-chain designs. Exact rollout is versioned — treat this as a **performance model**, not a guarantee of on-disk layout.

Complement: Slice & map internals.

Diagram: Map lookup path

```
key  hash   probe groups / buckets    slot
high load    grow / evacuate

maphash.Seed    custom tables / sharding
```

What Changed Conceptually

```
Legacy (simplified):  buckets -> overflow lists
Swiss-style:          groups of slots + control bytes (SIMD-friendly probes)
```

Implications:

- Better average lookup locality
- Different resize/evacuation behavior
- Iteration order still **randomized** — never depend on it
- Concurrent map misuse still **fatal**

Practical Advice (Unchanged)

```go
m := make(map[string]T, approxSize) // still helps
```

- Pre-size when cardinality known
- Prefer `string` / integer keys over allocations as keys

- Do not store huge structs as values if you update fields often (consider `map[K]*T`)

maphash

Package `hash/maphash` provides **high-quality, process-randomized** hashing suitable for your own hash tables and sharding:

```go
import "hash/maphash"

var seed maphash.Seed // zero value ok; or maphash.MakeSeed()

func hashString(s string) uint64 {
    var h maphash.Hash
    h.SetSeed(seed)
    h.WriteString(s)
    return h.Sum64()
}
```

Properties:

- Seed differs per process → harder hash-flooding for custom maps
- Not a crypto hash
- Fast for strings/bytes

Use for:

- Sharding keys across workers
- Custom open-address tables
- Consistent bucketing **within one process** (not cross-process stable unless you fix a seed carefully and accept risks)

Map Iteration and Deletes

```go
for k := range m {
    if shouldDrop(k) {
        delete(m, k) // allowed in range
    }
}
```

Experiment

```
go mod init example
go test -bench=. -benchmem
```

```go
package mapspeed_test

import (
    "hash/maphash"
    "testing"
)

func BenchmarkMapString(b *testing.B) {
    m := make(map[string]int, b.N)
    for i := 0; i < b.N; i++ {
        m[string(rune('a'+i%26))+string(rune(i))] = i
    }
}

func BenchmarkMaphash(b *testing.B) {
    var seed maphash.Seed
    var h maphash.Hash
    s := "authorization"
    b.ResetTimer()
    for i := 0; i < b.N; i++ {
        h.SetSeed(seed)
        h.WriteString(s)
        _ = h.Sum64()
        h.Reset()
    }
}
```

What to notice: Built-in maps are heavily optimized; custom maps need `maphash` + careful probing to approach them.

Try next: Read Go release notes for your version mentioning map changes; re-run production pprof after upgrades.

sync.Pool and Zero-Allocation Patterns

sync.Pool and Zero-Allocation Patterns

Overview

Reducing allocations is the highest-leverage GC optimization. `sync.Pool` reuses temporary objects; **zero-allocation** patterns avoid heap traffic entirely on hot paths.

X and production war stories hammer this constantly: pprof → `mallocgc` → fix allocs before spinning on GOGC.

Diagram: Pool get/put

```
sequence (top → bottom):
  actors: goroutine, sync.Pool
  goroutine --> sync.Pool  : Get
  sync.Pool --> goroutine  : buffer
  goroutine --> goroutine  : use reslice
  goroutine --> sync.Pool  : Put
  note: may drop on GC
```

sync.Pool

```go
var bufPool = sync.Pool{
    New: func() any {
        b := make([]byte, 0, 32*1024)
        return &b
    },
}

func handle() {
    bp := bufPool.Get().(*[]byte)
    buf := (*bp)[:0]
    defer func() {
        if cap(buf) > 64*1024 { // don't retain huge buffers
            return
        }
        *bp = buf
```

```
        bufPool.Put(bp)
    }()
    // append into buf...
}
```

Rules:

1. **No correctness dependency** — pool may drop items on GC.
2. **Reset** objects before Put (clear sensitive data).
3. **Cap retention** — avoid pooling multi-MB buffers forever.
4. Prefer pooling `*[]byte` or structs with slices, not giant value copies.

Zero-Alloc Techniques

Technique	Idea
Stack arrays	`var a [256]byte` then `a[:n]` if n bounded
`sync.Pool`	Reuse buffers across requests
`strconv.Append*`	Append to `[]byte` without intermediate strings
`strings.Builder`	Single alloc for multi-part strings
Avoid `fmt` in hot paths	Prefer `AppendInt` / specialized encoders
Interface-free APIs	Avoid boxing in tight loops

```
b := make([]byte, 0, 64)
b = strconv.AppendInt(b, 42, 10)
b = append(b, ' ')
b = append(b, "ok"...)
```

Measuring

```
go test -bench=. -benchmem
go test -bench=. -memprofile=mem.out
go tool pprof mem.out
```

```
testing.AllocsPerRun(1000, func() { /* hot path */ })
```

Anti-Patterns

- Pooling when objects are already tiny and short-lived (overhead > gain)
- Sharing pooled buffers across goroutines without ownership discipline
- Forgetting to reslice to zero length (`buf[:0]`)

Experiment

```
go test -bench=. -benchmem
```

```go
package pool_test

import (
    "strconv"
    "sync"
    "testing"
)

var pool = sync.Pool{New: func() any { b := make([]byte, 0, 64); return &b }}

func BenchmarkAllocFmt(b *testing.B) {
    for i := 0; i < b.N; i++ {
        _ = strconv.Itoa(i) + "x"
    }
}

func BenchmarkAppendPool(b *testing.B) {
    for i := 0; i < b.N; i++ {
        bp := pool.Get().(*[]byte)
        buf := (*bp)[:0]
        buf = strconv.AppendInt(buf, int64(i), 10)
        buf = append(buf, 'x')
        *bp = buf
        pool.Put(bp)
    }
}
```

What to notice: Append + pool often wins on alloc/op versus string concat.

Try next: Profile a JSON encoder hot path; see if `json.Encoder` reuse or code generation helps more than pooling.

Goroutine Leaks and Diagnosis

Goroutine Leaks and Diagnosis

Overview

Goroutines are cheap until they **never exit**. Leaks show up as monotonically rising goroutine counts, memory growth (stacks + closed-over data), and eventually latency death.

This is one of the top production issues called out in Go engineering threads: spawn without stop conditions.

Diagram: Leak patterns

```
go worker

        forever block on chan
        missing ctx.Done
        lock never released

     v
NumGoroutine ↑   pprof goroutine   fix bounds + cancel
```

Common Leak Patterns

Pattern	Why it leaks
Blocked on channel send/recv forever	Partner gone
`select` without `ctx.Done`	No cancel path
HTTP request without timeout	Stuck on body read
`for range ticker.C` without Stop/exit	Forever loop
Waiting on mutex held by dead owner	Deadlock-ish stall
Callback registered never removed	Global map grows

Fix Templates

```go
// Always pair spawn with context
go func() {
    select {
    case <-ctx.Done():
        return
    case jobs <- job:
    }
}()
```

```go
// Bound fan-out
sem := make(chan struct{}, 32)
for _, item := range items {
    sem <- struct{}{}
    go func(it Item) {
        defer func() { <-sem }()
        process(ctx, it)
    }(item)
}
```

Diagnosis Toolkit

```go
# Live
curl -s localhost:6060/debug/pprof/goroutine?debug=1 | head
go tool pprof http://localhost:6060/debug/pprof/goroutine

# Go 1.27+: goroutines the GC can prove will never wake
go tool pprof http://localhost:6060/debug/pprof/goroutineleak

# Binary
import _ "net/http/pprof"
```

The `goroutineleak` profile (experimental in 1.26, generally available in 1.27) reports goroutines blocked on a concurrency primitive that is unreachable from any runnable goroutine. It will **not** catch leaks parked on globals or on locals of still-runnable Gs. Use it alongside the ordinary goroutine profile, not instead of it.

```go
runtime.NumGoroutine()
```

In tests:

```go
func TestNoLeak(t *testing.T) {
    before := runtime.NumGoroutine()
    // run code under test with cancel
    // ...
    deadline := time.Now().Add(2 * time.Second)
    for time.Now().Before(deadline) {
        if runtime.NumGoroutine() <= before {
            return
        }
        time.Sleep(10 * time.Millisecond)
    }
    t.Fatalf("goroutines leaked: before=%d now=%d", before, runtime.NumGoroutine())
}
```

For concurrent unit tests with fake time, see synctest.

Experiment

```bash
go mod init example
go run .
```

```go
package main

import (
    "fmt"
    "runtime"
    "time"
)

func leak() {
    ch := make(chan int)
    go func() { ch <- 1 }() // blocked forever if no receiver
}

func fixed(done <-chan struct{}) {
    ch := make(chan int)
    go func() {
        select {
        case ch <- 1:
        case <-done:
        }
    }()
}

func main() {
```

```go
    fmt.Println("start", runtime.NumGoroutine())
    leak()
    time.Sleep(20 * time.Millisecond)
    fmt.Println("after leak", runtime.NumGoroutine())

    done := make(chan struct{})
    fixed(done)
    close(done)
    time.Sleep(20 * time.Millisecond)
    fmt.Println("after fixed", runtime.NumGoroutine())
}
```

What to notice: One blocked send permanently increases the count.

Try next: Dump `goroutine?debug=1` in a real service after load; group stacks by `chan receive` sites.

GODEBUG and Runtime Knobs

GODEBUG and Runtime Knobs

Overview

The runtime exposes a surprising amount of behavior through environment variables — especially `GODEBUG`. These are for diagnosis and carefully controlled experiments, not random production toggles without metrics.

Diagram: Knob layers

```
flow:
  [Runtime]
       |
       v
  [Observe]
```

Essential Environment

Variable	Role
`GOMAXPROCS`	P count / parallel Go code
`GOGC`	GC percent trigger
`GOMEMLIMIT`	Soft memory limit
`GODEBUG`	Comma-separated runtime debug flags
`GOTRACEBACK`	Panic stack verbosity (`single`, `all`, `crash`, …)
`GOFLAGS`	Default flags for go command
`GORACE`	Race detector options

GODEBUG Hits Worth Knowing

Exact keys evolve — check `go doc runtime` / release notes. Common families:

```
GODEBUG=gctrace=1 ./app          # GC line traces
GODEBUG=schedtrace=1000 ./app    # scheduler trace every 1s
GODEBUG=asyncpreemptoff=1 ./app  # debug preemption (special)
GODEBUG=http2debug=1 ./app       # HTTP/2 verbose (net/http)
```

```
# init-time print of effective settings sometimes available via:
go env
```

GOTRACEBACK

```
GOTRACEBACK=all ./app    # all goroutines on panic
GOTRACEBACK=crash ./app  # extra crash info / abort behavior (platform)
```

Programmatic Knobs

```
debug.SetGCPercent(50)
debug.SetMemoryLimit(512 << 20)
debug.WriteHeapDump(fd)
runtime.SetMutexProfileFraction(5)
runtime.SetBlockProfileRate(1000)
```

Safety Rules

1. Change **one** knob at a time.
2. Compare p95 CPU memory goroutines before/after.
3. Do not ship `schedtrace` or `http2debug` at full verbosity in prod.
4. Prefer `GOMEMLIMIT` over folklore `GOGC=off`.

Experiment

```
GODEBUG=gctrace=1 go run .
```

```
package main

import (
    "fmt"
    "runtime"
    "runtime/debug"
)

func main() {
    debug.SetGCPercent(100)
    var keep [][]byte
```

```go
    for i := 0; i < 200; i++ {
        keep = append(keep, make([]byte, 256*1024))
    }
    fmt.Println("goroutines", runtime.NumGoroutine(), "kept", len(keep))
    keep = nil
    runtime.GC()
}
```

What to notice: `gctrace` lines show GC cycles correlating with allocation bursts.

Try next: Capture `schedtrace` during a latency incident; look for high latencies in scheduling vs your code.

pprof and Execution Tracer Internals

pprof and Execution Tracer Internals

Overview

Workflows live in benchmarking & profiling. This chapter is the **mechanism**: how samples appear, what they mean, and how the execution tracer differs from CPU pprof.

Diagram: Profile vs trace

```
flow:
  [OnCPU]
       |
       v
  [Tool]

  [Events]
       |
       v
  [Tool2]
```

CPU Profile

Rough model:

```
OS signal / interrupt (platform dependent sampling)
  -> record stack of current G/M
  -> aggregate into profile proto
```

```
go test -cpuprofile=cpu.out -bench=.
go tool pprof -http=:0 cpu.out
```

Implications:

- **Sampling** — short functions may be undercounted
- Inlined frames can look surprising
- CPU profile is not wall-clock wait time

Heap / Alloc Profiles

- Heap: objects still live at sample time (retained)
- Alloc: cumulative allocations (even if freed)

```
go tool pprof http://localhost:6060/debug/pprof/heap
```

Use alloc profiles to find churn; heap profiles to find leaks/retention.

Block and Mutex Profiles

```
runtime.SetBlockProfileRate(1)       // careful in prod
runtime.SetMutexProfileFraction(5)
```

These answer "who waited on locks/channels" — complementary to CPU.

Goroutine Profile

Stacks of all Gs — best leak and deadlock forensic tool.

Execution Tracer (`go tool trace`)

Records **events** with timestamps:

- Goroutine create/block/unblock
- Syscall enter/exit
- GC STW / mark assist
- Network poll
- Processor start/stop

```
go test -trace=trace.out -bench=.
go tool trace trace.out
```

Use when CPU is low but latency is high — classic **scheduler / stall** story.

Cost of Profiling

Mode	Overhead
Occasional CPU 30s	Usually acceptable
Always-on mutex rate=1	Can be heavy
Full traces long windows	Large files, overhead

Sample in production; continuous profiling products add aggregation layers.

Experiment

```
go test -bench=BenchmarkWork -cpuprofile=/tmp/cpu.out -benchtime=1s
go tool pprof -top /tmp/cpu.out | head
```

```go
package work_test

import "testing"

func work(n int) int {
    x := 0
    for i := 0; i < n; i++ {
        x += i
    }
    return x
}

func BenchmarkWork(b *testing.B) {
    for i := 0; i < b.N; i++ {
        _ = work(1000)
    }
}
```

What to notice: Top frames should show `work` or inlined sites; if everything is `runtime.*`, you may be measuring something else (GC, tiny benchmark).

Try next: Generate a 200ms trace around an HTTP handler and find a long "block" on network or channel.

testing/synctest: Bubbles, Durable Blocks, Fake Time

testing/synctest: Bubbles, Durable Blocks, Fake Time

Overview

`testing/synctest` is **not** a polite `time.Sleep` helper. It is a **runtime feature** exposed through a thin testing API: the scheduler tracks an isolated **bubble** of goroutines, channels, and timers, and can answer a precise question instead of guessing.

```
Has every goroutine in this test either finished, or reached a block
that only another goroutine in the *same* bubble can unblock?
```

If yes, the bubble is **settled**. Then `synctest.Wait` can return, or the **fake clock** can jump to the next timer deadline—without burning wall-clock time.

> **Sources / depth.** This chapter synthesizes the public API with runtime-level concepts documented in guest runtime walkthroughs such as *Fake Clocks, Real Guarantees: Inside Go's synctest* (Alex Rios / Internals for Interns; runtime as of ~Go 1.25–1.27). Prose and diagrams here are original. Always verify symbols against your `GOROOT` and `go doc testing/synctest`.

Availability: experiment in 1.24 → stable package from 1.25 → `synctest.Sleep` in Go 1.27. Check `go version` before relying on APIs.

The problem it solves

```go
// Ordinary concurrent test - a guess dressed as a test
go doSomething()
time.Sleep(10 * time.Millisecond) // hope the G ran
if !done {
    t.Fatal("not done")
}
```

Longer sleeps only buy a **slower guess**. Scheduling order, machine load, and GC jitter all change whether 10ms was enough.

```
ordinary test                          synctest

go doWork()                            bubble { workers, timers, chans }
```

```
      v                                       v
  time.Sleep(10ms)     guess            Wait / clock jump     proof

      v                                       v
  assert "probably done"                assert only after SETTLED
                                        (all Gs done or durably blocked)
```

Public API (small overlay)

Typical surface (confirm on your version):

```
synctest.Test(t, func(t *testing.T) { ... })
synctest.Wait()
synctest.Sleep(d) // Go 1.27+: Sleep + Wait convenience
// older / internal shape also used synctest.Run(func(){...})
```

```
testing/synctest          thin public package


internal/synctest        glue


runtime/synctest.go      bubble, counters, event loop
        +
runtime proc/chan/time/sema/select   durability hooks
```

`synctest.Test` creates a bubble, adapts `*testing.T` (child `T` marked as synctest), and runs your function through normal `tRunner` so Fail/Fatal/cleanup/logs/race still work.

Inside a bubble, some `T` methods panic (e.g. `t.Parallel`, `t.Run` as normal nested subtests, `t.Deadline` in designs where wall deadline vs fake `time.Now` would be nonsense). Treat the bubble as a special domain, not a nested subtest tree.

The bubble

Conceptual structure (names follow runtime ideas; fields evolve):

```
synctestBubble
  now      fake clock (ns) - starts at a fixed epoch (often 2000-01-01 UTC)
  timers   private timer heap driven by bubble.now
  root     controller G (event loop; not your test body)
  main     G that runs the test function
  running  count of Gs not yet proven durably blocked
```

```
active   bookkeeping for mid-park / wake gaps
...
```

```
          synctest bubble
  root G  (event loop; not test body)

        bubble.now     (fake clock)
        bubble.timers  (private heap)

  main G  (test function)
        worker Gs  (inherit bubble on go)

  running / active counters  →  settled?

                          may touch (not durable)
                         v
                 outside world / globals / net
```

Membership

Every `g` may carry `bubble *synctestBubble`.

On `go f()` (`newproc1`):

- **User Gs inherit** the parent's bubble pointer.
- **System Gs** (GC, netpoll helpers, …) **do not** inherit membership.

So membership spreads automatically down the spawn tree without tagging every `go` statement.

```
Test body (in bubble)
    go worker     → in bubble
         go nested → in bubble
Runtime GC G         → not in bubble
```

Root vs main

`Run`/`Test` does **not** run your function on the root. Root becomes the **controller**; a **main** goroutine runs the test (and for `Test`, adapters/`tRunner` are also members). Root owns the fake-time event loop.

Durable blocking = wait reason

Durable block (docs language): blocked and can only be unblocked by another goroutine in the **same** bubble.

Runtime implements this with `waitReason` when a G parks (`_Gwaiting`). A table marks which reasons are **idle in synctest**.

Typically durable (idle)	Typically **not** durable
`time.Sleep` (fake timer)	`sync.Mutex.Lock` / RWMutex
send/recv on **bubbled** channel	send/recv on **outside** channel
select where **every** case is bubbled	select with any outside channel
`sync.Cond.Wait` (with bubble wake checks)	—
`WaitGroup.Wait` when **associated** to this bubble	WaitGroup not associated / wrong bubble
nil-chan send/recv, empty `select{}`	(deadlock path if nothing else can finish)
`synctest.Wait` itself	—

```
G parks

   v
status == _Gwaiting?  no   counts as running (not settled)

  yes
   v
waitReason idle-in-synctest?  no   still "running" for bubble

  yes
   v
DURABLE block  →  running--
```

Why mutex fails the test: a mutex has **no bubble ownership**. An outside G could `Unlock`. The runtime refuses to call that settled, so `synctest.Wait` may **hang** until the outer `go test` timeout—not always a nice bubble deadlock panic.

Two counters, one invariant

```
running > 0  ||  active > 0      "something still going on"
```

Counter	Meaning
`running`	Gs not yet proven durably blocked (not the same as "on CPU")
`active`	Scheduler bookkeeping during **mid-park**, wakes, root loop tokens

Parking is not atomic: status may drop to waiting **before** a commit function finalizes the park. Without `active`, the bubble could see `running == 0` in that gap and **false-settle**.

```
worker G                    bubble counters

      incActive (mid-park)
                      active > 0
      status → waiting
      (running may hit 0 early)     running==0, active>0  ← not settled yet
      park committed / aborted
      decActive
                      only now both may be 0 → settled
```

Status changes go through the scheduler's status CAS path; bubbled Gs call into bubble accounting (`changegstatus`-style hooks).

Root event loop (fake time)

When settled (and timers remain), the root:

1. Runs fake timers due at `bubble.now`
2. Tries to park (handing back its `active` token)
3. Looks at next timer deadline
4. Either **jumps** `bubble.now = next` or exits (done / no timers)

```
root loop

   v
run timers due at bubble.now

   v
try park (release active token)

   v
next timer deadline?

  none           some

   v              v
stop loop     bubble.now = next

   v
main done & no leftover Gs?
   yes → clean exit
   no  → deadlock panic (durable stuck / leaked Gs)
```

Fake time never wall-waits once settled. Ten "seconds" of `Sleep` can finish in microseconds of wall time: `now` is assigned, not slept.

```
Fake clock starts ~2000-01-01 00:00:00 UTC

G2: Sleep(3s)  timer @ 00:00:03
main: Sleep(10s) timer @ 00:00:10

both park → settled
root jumps now → 00:00:03 → G2 runs
settled again
root jumps now → 00:00:10 → main runs
```

time.Now inside a bubble

Runtime returns **bubble.now** for wall time. Monotonic component is often **zero** so mixing inside/outside times does not invent a second confusing clock. `time.Since` still works because the bubble wall clock is under runtime control.

Fake timers

Timers created in a bubble are marked fake and live on the **bubble timer heap**, not a normal per-P heap. Callbacks (e.g. `AfterFunc`) briefly count as bubble activity so spawned Gs inherit membership.

Channels: married at make

```
hchan.bubble set in makechan if creator is in a bubble
```

Action	Effect
send/recv/close from **outside** bubble	**fatal** (process-killing runtime error)
block on bubbled chan	wait reason is **synctest** send/recv → durable
block on unbubbled chan	not durable

```
// wait reason conceptually:
// unbubbled: waitReasonChanSend
// bubbled:   waitReasonSynctestChanSend  // idle-in-synctest
```

Select

Durable only if **every** non-nil case is a bubbled channel in the **current** bubble. One outside case spoils the proof → non-durable select wait.

```
select {
case <-bubbledCh:     // ok
case <-time.After(d): // After's chan is bubbled if created inside
case <-outsideCh:     // spoils durability
}
```

WaitGroup association (heap special)

`WaitGroup` packs a **bubble membership bit** into state and, on `Add` inside a bubble, associates via a **heap special** (same family of side notes as finalizers).

```
Durable WaitGroup.Wait requires roughly:
  1. Add happened in a bubble
  2. Wait called in a bubble
  3. Same bubble association still valid
```

Package-level `var wg sync.WaitGroup` lives in the data segment—**no heap object** to attach the special to. Association can fail; waits may never count as durable.

Prefer:

```
wg := new(sync.WaitGroup) // heap-allocated
// or local var that escapes to heap as needed
```

Miss association → bubble may never settle while `Wait` is blocked.

sync.Cond

`Cond.Wait` parks with a reason marked idle. Ownership is checked on **wake**: signaling a bubbled waiter from outside its bubble is **fatal**. Durability rides on parked `g.bubble`, not a field on `Cond` itself.

Mutex: deliberately non-durable

```
var mu sync.Mutex
synctest.Test(t, func(t *testing.T) {
    mu.Lock()
    go func() {
        mu.Lock() // blocks - NOT durable
    }()
    synctest.Wait() // may hang forever (outer test timeout)
})
```

```
flow:
  [ND]
       |
       v
  [R]

  [R]
       |
       v
  [H]
```

Debug fork:

Symptom	Look for
Named **deadlock panic** from bubble	All **durable** blocks, no timers left / leaked Gs after main exit
Hang until `go test` timeout	Non-durable wait: mutex, unbubbled channel, bad WaitGroup association

Deadlock detection (when root can prove it)

When the root stops and leftover Gs remain:

- "all goroutines in bubble are blocked" — durable deadlock, no timer escape
- "main exited but blocked goroutines remain" — leak after test function return

Stricter than ordinary tests: a forever-blocked **bubbled** channel is a loud failure, not silent process exit.

Isolation cheat sheet

```
Created inside bubble:
  channel   → hchan.bubble
  timer     → isFake, bubble heap
  WaitGroup → associate on Add (heap)

From outside:
  send/recv/close bubbled chan     → fatal
  fake timer stop/reset from out   → fatal
  WaitGroup Add from wrong bubble  → fatal

Outside objects used inside:
  allowed, but waits usually NOT durable
```

Isolation is not a sandbox

The test can still touch globals, disk, network. Those ops are **not** part of the settling proof. Keep I/O out of unit bubbles or accept non-determinism.

End-to-end narrative

```go
func TestWorkerTimeout(t *testing.T) {
    synctest.Test(t, func(t *testing.T) {
        ch := make(chan string) // bubbled
        go func() {
            select {
            case msg := <-ch:
                t.Log("got", msg)
            case <-time.After(5 * time.Second): // fake timer
                t.Error("timed out")
            }
        }()
        time.Sleep(3 * time.Second) // fake
        ch <- "hello"
    })
}
```

```
sequence (top → bottom):
  actors: Root, Main, Worker, bubble.now
  Root --> Main  : start test G
  Main --> Worker  : go select
  Worker --> Worker  : park durable select
  Main --> Main  : Sleep 3s park durable
  Root --> bubble.now  : jump to +3s
  Main --> Worker  : send hello
  Worker --> Worker  : log and exit
  Main --> Root  : return clean
  note: settled running=0
```

No wall wait for “3 seconds”; no sleep-and-hope after the send.

synctest.Wait and the race detector

When Wait unparks, the runtime establishes a **happens-before** edge (raceacquire-style) so reads after Wait of data written by now-blocked Gs are not false races. Matching releases fire when Gs durably block, exit, or finish fake timer callbacks.

`synctest.Sleep(d)` (Go 1.27+) `time.Sleep(d)` + `Wait()`: advance clock, then wait for reactions to settle. Same-instant timer order remains unspecified; trailing Wait makes that OK for most tests.

Practical rules

1. Create channels/timers **inside** the bubble you care about.
2. Use **heap** WaitGroups if you need durable `Wait`.
3. Do not expect `Mutex` waits to settle the bubble.
4. Prefer `synctest.Wait` over wall `Sleep` for "work finished".
5. Hang vs panic: hang → non-durable; panic → durable deadlock/leak.
6. Still run `go test -race`.
7. Don't nest `synctest.Run`/`Test` (runtime panics).

Relation to other chapters

Topic	Chapter
Flaky concurrent tests	260 Testing
GMP / parking	201 Scheduler
Timers	214 Timers
Channel internals	203 Channels
Context timeouts	255 Context

Reading the source (same path as the article)

1. `testing/synctest` public API
2. testing hook / `T` adaptation
3. `runtime.synctestRun` / bubble struct
4. status hooks + wait reason table
5. `time` linknames + fake timers
6. `makechan` / `chansend` / `selectgo`
7. `sync.WaitGroup` + semacquire durable flag
8. Package tests under `internal/synctest`

Experiment

```
go version
go doc testing/synctest
```

```go
func TestFakeSleep(t *testing.T) {
    // Use the API your version exports: Test and/or Run
    synctest.Test(t, func(t *testing.T) {
        start := time.Now()
        done := make(chan struct{})
        go func() {
            time.Sleep(30 * time.Second)
            close(done)
        }()
        synctest.Wait() // or Sleep if available
        <-done
        // wall time should be tiny; bubble time advanced ~30s
        if time.Since(start) > time.Second {
            t.Fatalf("wall clock moved too much: %v", time.Since(start))
        }
    })
}
```

```
go test -race -count=100 -run TestFakeSleep
```

What to notice: 30s of fake sleep does not cost 30s of CI.

Try next: Add a mutex-blocked G and observe Wait hang vs durable channel deadlock panic.

Recap

Idea	One line
Bubble	Isolated scheduler domain for one test
Membership	Inherited on `go`; system Gs excluded
Durable block	Wait reason marked idle-in-synctest
Settled	`running==0` and `active==0`
Root	Advances fake clock / wakes Wait / detects deadlock
Fake time	Jump `now` on private timer heap
Objects	Chan/timer/WG associated; outside touch often fatal
Mutex	Never durable—by design

The tiny API is small on purpose. The **proof** that a concurrent test is settled lives in the runtime—membership, wait reasons, timer heaps, and isolation—exactly what user-space mocks cannot see.

Zero-Copy I/O (sendfile, splice, Copy)

Zero-Copy I/O (sendfile, splice, Copy)

Overview

Moving bytes through user space burns CPU. Linux offers **sendfile** and **splice** to reduce copies. Go's `io.Copy` tries optimized paths when concrete types implement `ReaderFrom` / `WriterTo` (and OS support exists).

Popular writeups (widely shared among systems Go engineers) stress: TLS and some wrappers disable the fast path.

Diagram: Copy fast path

```
io.Copy(dst, src)

        dst has ReadFrom?  yes   optimized path

        src has WriteTo?   yes   optimized path

        else    buffered copy loop
TLS / custom wrappers often force the slow path
```

Copy Dispatch

```
io.Copy(dst, src)
  -> if dst has ReaderFrom: dst.ReadFrom(src)
  -> else if src has WriterTo: src.WriteTo(dst)
  -> else generic buffer loop
```

`*os.File` and `*net.TCPConn` participate in platform-specific fast paths when combinations allow.

sendfile / splice (Linux)

Typical wins:

- File → socket static file serving

- Socket → socket proxying (sometimes via splice)

Usually **not** available end-to-end when:

- TLS wraps the connection (encryption needs bytes in user space)
- Custom `io.Reader` wrappers break type assertions
- Non-Linux platforms fall back to buffered copy

Practical Patterns

```go
// Prefer typed paths
if _, err := io.Copy(dstFile, srcFile); err != nil { ... }

// Cap memory even on slow path
lr := io.LimitReader(src, max)
_, err = io.Copy(dst, lr)
```

```go
// HTTP: FileServer / ServeContent already try efficient paths
http.ServeContent(w, r, name, modtime, content)
```

Measuring

```bash
perf stat -e cycles,instructions,cache-misses ./proxy  # Linux
go test -bench=Copy -benchmem
```

Compare:

1. `io.Copy` file→file
2. Manual `Read`/`Write` 32KiB buffer
3. File→TLS conn (expect less zero-copy)

Experiment

```bash
go mod init example
go test -bench=. -benchmem
```

```go
package zc_test

import (
    "bytes"
    "io"
```

```go
    "os"
    "path/filepath"
    "testing"
)

func BenchmarkCopyFile(b *testing.B) {
    dir := b.TempDir()
    src := filepath.Join(dir, "s")
    dst := filepath.Join(dir, "d")
    data := bytes.Repeat([]byte("x"), 1<<20)
    if err := os.WriteFile(src, data, 0o644); err != nil {
        b.Fatal(err)
    }
    b.SetBytes(int64(len(data)))
    b.ResetTimer()
    for i := 0; i < b.N; i++ {
        in, _ := os.Open(src)
        out, _ := os.Create(dst)
        _, _ = io.Copy(out, in)
        in.Close()
        out.Close()
    }
}
```

What to notice: Kernel paths dominate large copies; wrappers and TLS change the story more than micro-tuning buffer sizes alone.

Try next: Wrap the reader in a custom type without `WriterTo` and observe any bench regression.

HTTP Transport and Connection Pool

HTTP Transport and Connection Pool

Overview

`http.Client` performance is mostly **`http.Transport`**: dialing, TLS, idle connection pools, HTTP/2 multiplexing. Using `http.DefaultClient` without timeouts is a top production mistake called out constantly in Go ops threads.

Diagram: Client stack

```
  Client.Do(req)

       v
  Transport
          Dial (+ TLS)
          idle pool (per host)
          HTTP/1.1 or HTTP/2 streams

       v
  resp.Body.Close()    return conn to pool
```

Client Stack

```
Client.Timeout (optional overall)
  -> Transport.RoundTrip
       DialContext
       TLSHandshake
       conn pool (idle conns per host)
       HTTP/1.1 or HTTP/2
```

```
tr := &http.Transport{
    Proxy:                 http.ProxyFromEnvironment,
    DialContext:           (&net.Dialer{Timeout: 3 * time.Second, KeepAlive: 30 * time.Secon
    ForceAttemptHTTP2:     true,
    MaxIdleConns:          100,
    MaxIdleConnsPerHost:   10,
```

```go
    MaxConnsPerHost:       0, // 0 = unlimited
    IdleConnTimeout:       90 * time.Second,
    TLSHandshakeTimeout:   5 * time.Second,
    ExpectContinueTimeout: 1 * time.Second,
    ResponseHeaderTimeout: 10 * time.Second,
}
client := &http.Client{Transport: tr, Timeout: 15 * time.Second}
```

Pool Behaviors

Knob	Effect
`MaxIdleConns`	Global idle cache size
`MaxIdleConnsPerHost`	Default is low (2) — often raise for chatty APIs
`MaxConnsPerHost`	Hard cap; backpressure when hit
`IdleConnTimeout`	Drop idle to free FDs
`DisableKeepAlives`	Every request new dial — debug only

Failing to `Body.Close()` **leaks** connections out of the pool.

HTTP/2

When negotiated, multiple streams share one TCP connection. Benefits under many parallel requests to one host; watch head-of-line at other layers and flow control. Debug with `GODEBUG=http2debug=1` carefully.

Server Side (Brief)

```go
srv := &http.Server{
    ReadHeaderTimeout: 5 * time.Second,
    ReadTimeout:       15 * time.Second,
    WriteTimeout:      30 * time.Second,
    IdleTimeout:       60 * time.Second,
}
```

Timeouts are load shedding, not pedantry.

Experiment

```
go mod init example
go run .
```

```go
package main

import (
    "fmt"
    "io"
    "net/http"
    "net/http/httptest"
    "time"
)

func main() {
    ts := httptest.NewServer(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        _, _ = io.WriteString(w, "ok")
    }))
    defer ts.Close()

    tr := ts.Client().Transport.(*http.Transport).Clone()
    tr.MaxIdleConnsPerHost = 5
    client := &http.Client{Transport: tr, Timeout: 2 * time.Second}

    for i := 0; i < 3; i++ {
        resp, err := client.Get(ts.URL)
        if err != nil {
            panic(err)
        }
        _, _ = io.Copy(io.Discard, resp.Body)
        resp.Body.Close()
    }
    fmt.Println("requests ok", time.Now().Format(time.RFC3339))
}
```

What to notice: Closing bodies returns conns to the idle pool; omitting close eventually exhausts FDs under load.

Try next: Load test with `MaxIdleConnsPerHost=2` vs `32` against a single upstream; compare dial rates in traces.

Context Cancel Trees Internals

Context Cancel Trees Internals

Overview

`context.Context` is a **tree of cancellation and values**. Understanding parent→child propagation explains goroutine leaks, deadline bugs, and why `context.Background()` at random depths is an outage pattern.

Diagram: Cancel flows down

```
parent ctx
      child (timeout)
      child (values)
cancel(parent)    cancels all children
cancel(child)     does NOT cancel parent
```

Tree Shape

```
Background
    WithValue(request_id)
         WithTimeout(2s)     <- parent cancel cancels children
              WithCancel     <- child cancel does NOT cancel parent
              WithTimeout(500ms)
```

Rules:

1. Cancel/timeout flows **down**.
2. Child cancel does **not** cancel siblings/parent.
3. Values flow by lookup up the parent chain.
4. Always `defer cancel()` for WithCancel/Timeout/Deadline.

Implementation Intuition

Each cancelable context holds:

- A done channel (closed on cancel)
- Parent linkage / children set

- Optional timer for deadlines
- Optional cause error (`WithCancelCause`)

Closing parent walks children. This is why leaking cancel funcs can retain subtrees longer than you expect.

Values

```
type key struct{} // unexported key type avoids collisions
ctx = context.WithValue(ctx, key{}, rid)
```

Do not store optional params that belong in function arguments. Values are for request-scoped data (IDs, loggers), not control flow.

Error Taxonomy

```
ctx.Err() == context.Canceled
ctx.Err() == context.DeadlineExceeded
// WithCancelCause: context.Cause(ctx)
```

Map these to client retries carefully — cancel from client disconnect retryable server error necessarily.

Experiment

```
go mod init example
go run .
```

```
package main

import (
    "context"
    "fmt"
    "time"
)

func main() {
    parent, cancel := context.WithCancel(context.Background())
    defer cancel()
    child, cancelChild := context.WithTimeout(parent, 50*time.Millisecond)
    defer cancelChild()
```

```go
    stop := make(chan struct{})
    go func() {
        <-child.Done()
        fmt.Println("child", child.Err())
        close(stop)
    }()

    <-stop
    cancel() // parent cancel after child already done
    fmt.Println("parent", parent.Err())

    // child cancel does not kill parent
    p2, c2 := context.WithCancel(context.Background())
    ch, c3 := context.WithCancel(p2)
    c3()
    fmt.Println("child2", ch.Err(), "parent2", p2.Err())
    c2()
}
```

What to notice: Timeout sets child error first; canceling a child leaves parent active until its own cancel.

Try next: Pass `context.WithoutCancel(ctx)` (Go 1.21+) when you must detach a background job intentionally — and document why.

Mutex and Semaphore Runtime

Mutex and Semaphore Runtime

Overview

`sync.Mutex` is not a pure userspace spin forever. After a short path, contending lockers **park** through runtime semaphores (sudog queues), integrating with the scheduler like channel waits.

Diagram: Lock fast/slow path

```
  Lock attempt

        uncontended CAS    enter critical section

        contended    park (semacquire)

  Unlock    wake one waiter    runnable
```

Fast Path / Slow Path

```
Lock attempt
  -> atomic CAS (uncontended) success → return
  -> spin briefly (optional, implementation)
  -> enter runtime semacquire queue
  -> G parks
Unlock
  -> hand off / wake waiter
  -> woken G runnable
```

Implications:

- Uncontended locks are extremely cheap
- High contention shows up in **block/mutex profiles**, not only CPU
- Holding locks across I/O serializes your service (top production mistake)

RWMutex

Multiple readers; writers exclusive. Writer starvation possible under continuous readers — measure.

Cond

`sync.Cond` builds on mutex + notification; easy to misuse. Prefer channels for many app-level designs; Cond appears in specialized queues.

Atomics vs Mutex

Need	Prefer
Counter / flag	`atomic`
Multi-field invariant	`Mutex`
Coordination / ownership transfer	Channel

Avoid

```go
mu.Lock()
resp, err := http.Get(url) // BAD: lock held across I/O
mu.Unlock()
```

Experiment

```go test -bench=. -benchmem
```

```go
package lock_test

import (
    "sync"
    "sync/atomic"
    "testing"
)

func BenchmarkMutex(b *testing.B) {
    var mu sync.Mutex
    var x int
    b.RunParallel(func(pb *testing.PB) {
```

```go
        for pb.Next() {
            mu.Lock()
            x++
            mu.Unlock()
        }
    })
}

func BenchmarkAtomic(b *testing.B) {
    var x atomic.Int64
    b.RunParallel(func(pb *testing.PB) {
        for pb.Next() {
            x.Add(1)
        }
    })
}
```

What to notice: Atomics win for simple counters; mutexes win for complex critical sections — profile contention, don't guess.

Try next: Enable mutex profile on a service and find the top `Unlock` edge under load.

cgo and Scheduler Transitions (incl. Go 1.26)

cgo and Scheduler Transitions (incl. Go 1.26)

Overview

cgo lets Go call C (and reverse), but each call crosses **ABI, stack, and scheduler** boundaries. High-frequency cgo is a classic latency and throughput tax.

Go 1.26 runtime work simplified how processors track syscall/cgo blocking: **P no longer uses a dedicated syscall state**; the scheduler derives blocking from **G state**. That reduces bookkeeping on hot syscall/cgo paths (community writeups cite meaningful wins especially for cgo-heavy workloads).

Diagram: cgo call path

```
Go code

   v
runtime.cgocall

   v
system stack → C function (may block M)

   v
return to Go → reacquire P if needed
```

Call Path (Conceptual)

```
Go code
  -> runtime.cgocall
  -> switch to system stack
  -> C function runs (may block OS thread)
  -> return to Go
  -> reacquire P if needed
```

While C runs, that **M** is busy in C. The runtime may reassign **P** so other Go code continues — but threads can accumulate if many Ms stick in C.

Costs

1. Call overhead (argument copy, stack switch)
2. Loss of some Go compiler visibility
3. Harder cross-compilation
4. GC interactions with C-allocated memory (you free C memory)

Rules

Prefer	Avoid
Pure Go libraries	cgo in per-request tiny helpers
Batch work into fewer cgo calls	Chatty call loops
Explicit C resource Close	Relying only on finalizers
Build tags for optional cgo	Silent cgo in libraries users don't expect

```
CGO_ENABLED=0 go build ./...   # fail if cgo required
go env CGO_ENABLED
```

Debugging

```
# see if binary links libc unexpectedly
ldd ./app   # Linux

# profiles often show runtime.cgocall
go tool pprof cpu.out
```

Relation To Non-cgo Syscalls

Blocking syscalls from Go also park/reassign; netpoll avoids blocking Ms for sockets. cgo is closer to "unknown blocking foreign code."

Experiment

```
# pure Go path baseline
CGO_ENABLED=0 go build -o /tmp/pure .
```

Write a tiny package without cgo and confirm it builds with `CGO_ENABLED=0`. If you have a cgo dependency, compare:

```
go test -bench=. -tags=cgo ./...
```

What to notice: Many services never need cgo; forcing `CGO_ENABLED=0` in CI catches accidental deps early.

Try next: Read Go 1.26 release notes scheduler section alongside GMP scheduler. For the *deployment* cost of cgo (libc, Alpine, `scratch`), start at Go as a Systems Language.

Iterators and range-over-func Deep Dive

Iterators and range-over-func Deep Dive

Overview

Go 1.22–1.23 made **`for range over functions`** a language feature, with package `iter` defining `Seq` / `Seq2`. This is a deep dive on semantics and costs — not just syntax.

Also: slices/maps/cmp/iter, loops.

Diagram: Push iterator

```
flow:
  [Yield] --true--> [Next]
  [Yield] --false--> [Stop]
```

Types

```go
type Seq[V any]     func(yield func(V) bool)
type Seq2[K, V any] func(yield func(K, V) bool)
```

The iterator **pushes** values into `yield`. Returning `false` from yield stops early (break).

Desugaring Intuition

```go
for v := range All(s) {
    if v < 0 {
        break
    }
}
// roughly calls All(s)(func(v T) bool { ...; return continue? })
```

Panic safety: iterator code should treat yield like a callback that may not return normally if the consumer panics — stdlib iterators are careful; custom ones should be too.

Pull Iterators

```
next, stop := iter.Pull(seq)
defer stop()
for {
    v, ok := next()
    if !ok {
        break
    }
}
```

Pull converts push-style to pull-style via a small coroutine-like parking protocol — convenient adapters, slight overhead.

When Iterators Shine

- Lazy pipelines over large or infinite streams
- Library APIs that used to allocate intermediate slices
- Single implementation for `range` and `slices.Collect`

When They Don't

- Tiny hot loops where a classic index for is clearer and fully BCE-friendly
- Need for random access
- Over-abstracting one-off loops

Experiment

```
go mod init example
go run .
```

```
package main

import (
    "fmt"
    "iter"
    "slices"
)

func Filter[V any](s []V, ok func(V) bool) iter.Seq[V] {
    return func(yield func(V) bool) {
        for _, v := range s {
```

```go
            if !ok(v) {
                continue
            }
            if !yield(v) {
                return
            }
        }
    }
}

func main() {
    nums := []int{1, 2, 3, 4, 5}
    for v := range Filter(nums, func(n int) bool { return n%2 == 0 }) {
        fmt.Println("even", v)
    }
    fmt.Println("collect", slices.Collect(Filter(nums, func(n int) bool { return n > 3 })))
}
```

What to notice: No intermediate filtered slice until `Collect`.

Try next: Implement `Map` and chain `Map(Filter(...))`; compare allocs to nested slice loops.

ABI, Assembly, and go:nosplit

ABI, Assembly, and go:nosplit

Overview

Most Go code never touches assembly. When you do (crypto, math, codecs, runtime), you meet **ABIInternal**, register conventions, and stack growth rules.

Diagram: ABI layers

```
diagram:
  GoFunc[Go function] --> ABIInt[ABIInternal]
  Asm[assembly TEXT] --> ABI0[ABI0 / NOSPLIT]
  Boundary[boundary] --> Transition[ABI transition]
```

ABIs (High Level)

ABI	Role
ABI0	Stable-ish assembly convention used historically
ABIInternal	Compiler's internal register ABI for Go functions

The compiler transitions between them at certain boundaries. Assembly files use special headers:

```
// func Add(a, b int) int
TEXT ·Add(SB), NOSPLIT, $0-24
    // ...
```

go:nosplit

Prevents stack growth checks in a frame. Required for some runtime entry paths; dangerous in app code if the frame is large.

```
//go:nosplit
func tiny() { }
```

go:uintptrescapes / go:uintptrkeepalive

Document pointer lifetime when smuggling pointers as `uintptr` through syscalls — get this wrong and GC frees live memory.

SIMD Experiments

Go 1.26 introduced experimental `simd/archsimd`. Go 1.27 adds a portable `simd` package (vector-size-agnostic types such as `Int8s` / `Float32s`) and extends `archsimd` to arm64 Neon and Wasm SIMD. Both still require `GOEXPERIMENT=simd` and are not production-stable. Prefer them over hand asm for new vector experiments.

When To Write Asm

1. Profile proves a hot leaf
2. Algorithm is fixed-width crypto/math
3. You can maintain per-GOARCH files
4. You accept review cost

Otherwise prefer pure Go + compiler autovectorization where it exists, or cgo to a maintained C library (with eyes open).

Plan 9 listing and `go tool objdump`

The compiler emits **Plan 9-style pseudo-assembly** (`SB` = static base, `SP` = stack) and the Go linker writes the final binary. That is why you can inspect any `GOARCH` on any host without a cross-GDB.

```
# Compiler IR (stderr):
go build -gcflags='-S' . 2>&1 | head -80

# Real machine code, still printed in Plan 9 register names:
CGO_ENABLED=0 go build -o hello .
go tool objdump -s main.main hello

# Cross-arch: build ARM64, disassemble it on amd64/arm64/darwin alike:
GOOS=linux GOARCH=arm64 CGO_ENABLED=0 go build -o hello-arm64 .
go tool objdump -s main.main hello-arm64
```

Calls in that listing point at Go symbols compiled into the binary (`fmt.Fprintln`, `runtime.morestack`) — not at `printf` or `libc.so`. That is the same "no libc in the middle" fact as Go as a Systems Language. On ARM64, `R28` holds the current goroutine (`g`).

Experiment

```
go build -gcflags='-S' . 2>&1 | head -80
```

```
package main

func add(a, b int) int { return a + b }

func main() {
    println(add(2, 3))
}
```

What to notice: Even simple functions show ABI prologue/epilogue; inlining may erase `add` entirely with optimizations.

Try next: Compare `-gcflags='-l -S'` (no inline) vs default for the same function. Then `go tool objdump -s main.main` on a `GOARCH=arm64` binary built from this machine.

Coverage Instrumentation and Fuzz Engine

Coverage Instrumentation and Fuzz Engine

Overview

`go test` is not only assert runners — it can **instrument binaries for coverage** and drive a **fuzzing engine** that mutates inputs to hit new edges.

Diagram: Fuzz loop

```
seed corpus

   v
mutate → run FuzzXxx → new coverage?

                        yes → keep input

      crash/hang    testdata/fuzz/...
```

Coverage

```
go test -cover ./...
go test -coverprofile=c.out ./...
go tool cover -html=c.out
go test -covermode=atomic ./...   # concurrent-safe counts
```

Integration-style:

```
go build -cover -o app .
GOCOVERDIR=covdata ./app   # exercise
go tool covdata textfmt -i=covdata -o=c.out
```

Use coverage to find **untested error paths**, not as a vanity percentage.

Fuzz Engine Model

```
seed corpus
  -> mutate
  -> execute FuzzXxx
  -> instrumentation: new coverage edge?
       yes -> keep interesting input
  -> crash/hang -> fail and write testdata/fuzz/...
```

```
go test -fuzz=FuzzParse -fuzztime=30s
```

Commit failing inputs under `testdata/fuzz` for regression.

Writing Good Fuzz Targets

1. Round-trip properties (`decode(encode(x))`)
2. Invariants (no panic, bounds)
3. Avoid non-determinism and network
4. Keep work per input small

Coordination With Race

```
go test -race -fuzz=FuzzX -fuzztime=10s
```

Heavier but catches races fuzz alone may miss.

Experiment

```
mkdir /tmp/fuzzdemo && cd /tmp/fuzzdemo
go mod init example
```

```go
// reverse.go
package example

func Reverse(s string) string {
    r := []rune(s)
    for i, j := 0, len(r)-1; i < j; i, j = i+1, j-1 {
        r[i], r[j] = r[j], r[i]
    }
    return string(r)
}
```

```go
// reverse_test.go
package example

import "testing"

func FuzzReverse(f *testing.F) {
    f.Add("gopher")
    f.Fuzz(func(t *testing.T, s string) {
        if Reverse(Reverse(s)) != s {
            t.Fatalf("%q", s)
        }
    })
}
```

```go
go test -fuzz=FuzzReverse -fuzztime=5s -cover
```

What to notice: Fuzz spends time exploring UTF-8 edges; coverage rises as new branches hit.

Try next: Fuzz a parser you own; keep the first crash input as a unit test.

Build Cache and Reproducibility

Build Cache and Reproducibility

Overview

The Go toolchain caches build actions aggressively. Understanding the cache explains "why CI rebuilt everything" and how to aim for **bit-reproducible** artifacts.

Diagram: Reproducible build

```
sources + flags + go version

        v
    GOCACHE

        v
    objects   link (-trimpath)   binary
```

Cache Location

```
go env GOCACHE
go clean -cache
go clean -modcache   # modules, separate
```

Cache keys include source, build flags, env that affects compile, cgo toolchains, tags, GOOS/GOARCH, etc.

What Busts Cache

- Touching files unnecessarily
- Changing `CGO_*` flags
- Unstable generated code (timestamps in source)
- Different Go patch versions
- `-trimpath` vs not when comparing binaries

Reproducible Flags

```
go build -trimpath -ldflags='-buildid= -s -w' -o app .
```

Flag	Effect
`-trimpath`	Remove local paths from binaries
`-buildid=`	Empty buildid for stabler hashes
`-s -w`	Strip symbols (optional)
`CGO_ENABLED=0`	Fewer host dependencies

Module Reproducibility

```
go mod download
go mod verify
go build -mod=readonly
# or
go build -mod=vendor
```

Lock with `go.sum`. Pin `toolchain` in `go.mod` for compiler version policy.

Experiment

```
go mod init example
echo 'package main; func main(){}' > main.go
go build -trimpath -ldflags='-buildid=' -o a1 .
go build -trimpath -ldflags='-buildid=' -o a2 .
shasum a1 a2   # often identical pure-Go
```

What to notice: Pure Go + trimpath often matches; cgo and ld versioning frequently break reproducibility.

Try next: In CI, print `go version`, `go env`, and `go list -m all` on mismatch incidents.

GOEXPERIMENT Catalog

GOEXPERIMENT Catalog

Overview

`GOEXPERIMENT` enables incomplete or alternate runtime/compiler features. Experiments graduate, change, or vanish — always pair with release notes.

```
GOEXPERIMENT=name go test ./...
go doc go/token  # not helpful; use:
go tool dist list
# list experiments for your version:
go doc internal/goexperiment   # may be unavailable; see src/internal/goexperiment
```

From user-facing docs / release notes (illustrative — verify for your version):

Experiment	Theme
`greenteagc` / `nogreenteagc`	Green Tea GC default since 1.26; opt-out remains
`simd`	Experimental portable `simd` + `simd/archsimd` (1.26–1.27)
`nosizespecializedmalloc`	Opt out of 1.27 size-specialized small allocs (expected gone in 1.28)
`nojsonv2`	Restore pre-1.27 `encoding/json` engine (temporary)
`arenas`	Historical arena experiment (do not bet new code without checking status)
`loopvar`	Historical loop var semantics (graduated)
`rangefunc`	Historical range-over-func (graduated)
`newinliner`	Inliner experiments

Diagram: Experiment lifecycle

```
flow:
  [Test]
       |
       v
  [Grad]
```

```
[Test]
     |
     v
[Drop]
```

Rules

1. **Never** enable random experiments in production without a rollback plan.
2. Test performance and correctness under the experiment **and** default.
3. Prefer graduated language features over experiments for library APIs.

JSON v2 and Friends

`encoding/json/v2` graduated in Go 1.27. The v1 package uses the v2 engine; `GOEXPERIMENT=nojsonv2` is an emergency opt-out. `goroutineleakprofile` was deleted when the leak profile became generally available.

Experiment

```
go version
GOEXPERIMENT=nogreenteagc go test ./...   # only if valid on your version
# or simply read:
go help build | less
```

Document in your runbook which experiments you intentionally ship (ideally none).

Try next: Skim the Go 1.27 release notes "Runtime" + "Compiler" sections and map each experiment bullet to a chapter in this part.

WebAssembly and wasip1

WebAssembly and wasip1

Overview

Go can target **`js/wasm`** (browser + wasm_exec.js) and **`wasip1`** (WASI preview1) for server-side Wasm runtimes. Constraints differ sharply from linux/amd64.

Diagram: Wasm targets

```
Go source
     GOOS=js   GOARCH=wasm    browser + wasm_exec
     GOOS=wasip1 GOARCH=wasm    WASI runtime
```

Targets

```
GOOS=js GOARCH=wasm go build -o main.wasm .
GOOS=wasip1 GOARCH=wasm go build -o main.wasm .
```

Constraints

Area	Reality
Threads	Limited / host-dependent
Networking	WASI and browser models differ
Filesystem	Capability-based under WASI
cgo	Generally unavailable
Binary size	Larger than tiny Go-less Wasm; use strip/trimpath

Use Cases

- Policy plugins
- Edge filters
- Portable CLI sandboxes
- Browser tools (js/wasm)

Experiment

```
# if wasip1 supported:
printf 'package main\nimport "fmt"\nfunc main(){ fmt.Println("hi wasm") }\n' > /tmp/w.go
cd /tmp && GOOS=wasip1 GOARCH=wasm go build -o hi.wasm w.go
ls -la hi.wasm
# run with a WASI runtime if installed, e.g. wasmtime hi.wasm
```

What to notice: Build succeeds without a local WASI runner; size is non-trivial compared to a hello C program.

Try next: Compare `GOOS=wasip1` vs native binary size with `-ldflags='-s -w'`.

Where Go's WebAssembly bytes go: measuring and shrinking a wasip1 binary

A Go "hello world" compiled to WebAssembly comes out at about 1.9 MB. People usually react by reaching for `-ldflags='-s -w'`, the flag that strips a native binary, and find that it barely changes the size. This chapter measures where the bytes actually go, which of the usual tricks matter on wasm, and what you trade if you leave the standard `gc` toolchain for TinyGo.

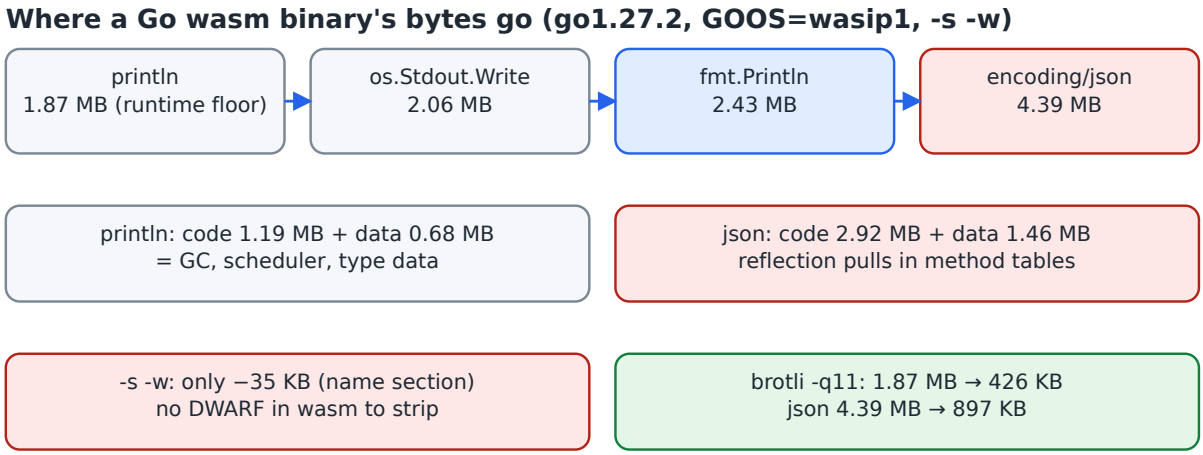


Figure 1: Size of four Go programs built for wasip1, their code/data split, and what stripping and compression buy

The four programs

Each program prints one line, using a progressively heavier part of the standard library:

```
// println/main.go - the builtin, no imports
package main

func main() { println("hello") }

// raw/main.go - os only
package main

import "os"
```

```go
func main() { os.Stdout.Write([]byte("hello\n")) }

// fmt/main.go
package main

import "fmt"

func main() { fmt.Println("hello") }

// json/main.go - reflection-driven encoding
package main

import (
    "encoding/json"
    "os"
)

type Msg struct {
    Text string `json:"text"`
}

func main() { json.NewEncoder(os.Stdout).Encode(Msg{"hello"}) }
```

Each one was built four ways: for `GOOS=wasip1` and `GOOS=js` (both `GOARCH=wasm`), each with and without `-trimpath -ldflags='-s -w'`.

Measured sizes (go1.27.2, bytes)

```
prog     goos         default        -s -w  gzip(-s -w)
println  wasip1       1906868      1869346       550756
println  js           1892424      1854868       547106
raw      wasip1       2106985      2063729       602868
raw      js           2147302      2103412       611967
fmt      wasip1       2475221      2426805       699957
fmt      js           2502043      2453043       706144
json     wasip1       4479337      4392256      1205460
json     js           4502542      4415042      1210327
native println  1895668
native fmt      2342537
```

The binaries actually run. The `js` builds were run with the toolchain's own `$(go env GOROOT)/lib/wasm/wasm_exec_node.js`, and the `wasip1` builds through Node's `node:wasi` module (Node 20.19.2). Each printed `hello`, or `{"text":"hello"}` for the JSON program.

Three things stand out:

1. **The floor is the runtime, not your code.** A program with no imports is already 1.87 MB, about the same as the native amd64 build (1.90 MB). That megabyte-plus is the garbage collector, the goroutine scheduler, maps, panics and the type metadata they all need. This matches the Go wiki: "the smallest possible size being around ~2MB."
2. **`js` and `wasip1` cost about the same.** They differ by under 2% in every row. The choice between them is about the host (browser vs WASI runtime), not size.
3. **Reflection is where it grows.** `fmt` adds about 560 KB over `println`. `encoding/json` more than doubles the binary. Reflect-based code must keep every method and type descriptor that might be looked up at run time, so the linker's dead-code elimination can remove far less. The wiki warns that "10MB+ is common" once real libraries are imported.

Why `-s -w` does almost nothing here

On a native ELF binary, `-s -w` drops the symbol table and DWARF (`cmd/link`: `-s` "Omit the symbol table and debug information", `-w` "Omit the DWARF symbol table"), and that is often a large share of the file. A Go wasm module carries neither. To check, I parsed the sections of the module myself, using the section IDs from the Wasm binary format spec:

```
println-wasip1.wasm    {'function': 1251, 'element': 2388, 'code': 1185790, 'data': 681761,
println-wasip1-sw.wasm {'function': 1251, 'element': 2388, 'code': 1185790, 'data': 679142}
raw-wasip1-sw.wasm     {'function': 1445, 'element': 2782, 'code': 1291244, 'data': 767246}
fmt-wasip1-sw.wasm     {'function': 1613, 'element': 3118, 'code': 1528856, 'data': 892198}
json-wasip1-sw.wasm    {'function': 2461, 'element': 4815, 'code': 2921058, 'data': 1462869}
```

(Sections under 1 KB are omitted.) The only thing `-s -w` removed was the 35 KB `name` custom section, which holds function names for stack traces in browser devtools, plus about 2.6 KB of data. The `code` section was byte-for-byte the same size. Keep `-s -w` if you don't need names in wasm stack traces, but expect about 2%. On the native amd64 builds of the same programs it saved about 35% (println 1,895,668 → 1,216,672 bytes; json 3,917,078 → 2,568,352).

`go tool nm` cannot help here. In go1.27.2 it rejects the module with `open json-wasip1.wasm: unrecognized object file`, which is why the script above walks the sections itself.

What actually shrinks the download

Compress it. This is the first method the Go wiki recommends, and the numbers show why:

program	`-s -w`	gzip -9	brotli q11
println	1,869,346	550,756	425,784
fmt	2,426,805	699,957	537,943
json	4,392,256	1,205,460	896,722

The brotli figures come from Python's `brotli` module at `quality=11`. Wasm code compresses about 4–5×. Serve `.wasm` with `Content-Encoding: br` (or gzip) and the transfer cost of the runtime floor drops to roughly 430 KB.

Keep reflection out of the hot import graph. Going from `fmt` to `os.Stdout.Write` saved 363 KB, and the same applies across a whole program. If a wasm module only needs to format a few numbers, use `strconv` and write bytes yourself. TinyGo's guide says the same about its compiler: "fmt is very large, so unless you absolutely need it, it is recommended to try to avoid it."

Change compilers. TinyGo is an LLVM-based Go compiler with a much smaller runtime. Per its docs it already optimizes for size by default (`-opt=z`), and you can go further with `-no-debug`, `-panic=trap`, `-scheduler=none` (which removes `go` statements) and `-gc=leaking` (no GC at all; the docs say this is "for very short-lived programs only"). I did not run TinyGo here because it is not installed on the box, so I have not measured it. The trade-offs are documented, though: the scheduler and GC options remove language features, and TinyGo implements a subset of the standard library and of reflection. Test your actual program against it; don't assume it will build. The WASI guide covers its `wasip1` target.

Checklist

- ☐ Measure the module's sections, not just the file size: `code` and `data` are what count.
- ☐ Expect a floor of about 1.9 MB with `gc`. You can't remove the runtime with flags.
- ☐ `-s -w` saves about 35 KB (the name section) on wasm. That's fine, but not a fix.
- ☐ Serve with brotli or gzip. On these programs that cut the transfer 4–5×.
- ☐ Audit imports for reflection-heavy packages (`fmt`, `encoding/json`, template engines).
- ☐ If bytes on the wire still matter more than full Go semantics, evaluate TinyGo on *your* program.

String and []byte Internals

String and []byte Internals

Overview

Strings are immutable `ptr+len` headers over bytes (usually UTF-8). `[]byte` is `ptr+len+cap`. Converting between them often **allocates** — a hot-path footgun.

Diagram: Headers

```
diagram:
  S[string ptr+len] --> Imm[immutable bytes]
  B[slice ptr+len+cap] --> Mut[mutable]
  Conv[string b / byte s] --> Copy[usually copy]
```

Headers

```
string:  { data *byte, len int }           // immutable
[]byte:  { data *byte, len int, cap int }  // mutable
```

```go
s := "go"
b := []byte(s) // copy
s2 := string(b) // copy
```

Compiler sometimes proves conversions are temporary and elides copies (e.g. map lookup `m[string(b)]` optimizations in modern Go) — **measure**, don't assume.

Range and Indexing

```go
for i, r := range s { /* r is rune; i is byte index */ }
s[i] // byte, not rune
```

unsafe Conversions (Advanced)

```
// historical patterns; prefer safer APIs / newer runtime helpers when available
// wrong lifetime → memory corruption
```

Only with full understanding of immutability: never mutate a `[]byte` view of a string.

Builder and Buffer

- `strings.Builder` — string assembly
- `bytes.Buffer` — also `io.Writer`
- `bytealg` runtime helpers power `strings`/`bytes` (assembly on some arch)

Experiment

```
go test -bench=. -benchmem
```

```go
package conv_test

import "testing"

var sink string

func BenchmarkByteToString(b *testing.B) {
    raw := []byte("authorization: bearer ...")
    for i := 0; i < b.N; i++ {
        sink = string(raw)
    }
}

func BenchmarkMapKey(b *testing.B) {
    m := map[string]int{"authorization: bearer ...": 1}
    raw := []byte("authorization: bearer ...")
    for i := 0; i < b.N; i++ {
        _ = m[string(raw)]
    }
}
```

What to notice: Map key optimization may reduce allocs vs naive assign to `sink`.

Try next: Profile a JSON logger converting large `[]byte` bodies to string.

Error Interface Cost and Wrapping

Error Interface Cost and Wrapping

Overview

`error` is an interface — values box dynamic types. Wrapping builds a chain inspected by `errors.Is` / `As`. Correctness first; cost matters in ultra-hot paths.

Diagram: Error interface

```
flow:
  [Chain]
       |
       v
  [Is]
```

Representation

```go
type error interface{ Error() string }
// concrete: *errors.errorString, fmt.wrapError, custom types
```

Returning `error` from a function may allocate when constructing `fmt.Errorf` or custom structs.

Wrapping

```go
return fmt.Errorf("open %s: %w", path, err)
errors.Is(err, fs.ErrNotExist)
errors.As(err, &pathErr)
errors.Join(err1, err2)
```

Performance Notes

Action	Cost driver
`errors.New` sentinel	One-time
`fmt.Errorf` per call	Alloc + formatting
Deep `Is` chains	Walk + interface asserts
`err.Error()` strings	May alloc; avoid in hot success path

Pattern: construct rich errors on **failure only**; keep success path alloc-free.

Sentinel vs Types

```go
var ErrNotFound = errors.New("not found")

type NotFoundError struct{ Key string }
func (e *NotFoundError) Error() string { return "not found: " + e.Key }
```

Experiment

```bash
go test -bench=. -benchmem
```

```go
package errcost_test

import (
    "errors"
    "fmt"
    "testing"
)

var ErrX = errors.New("x")

func BenchmarkSentinel(b *testing.B) {
    for i := 0; i < b.N; i++ {
        _ = ErrX
    }
}

func BenchmarkWrap(b *testing.B) {
    for i := 0; i < b.N; i++ {
        _ = fmt.Errorf("wrap: %w", ErrX)
    }
}
```

```go
func BenchmarkIs(b *testing.B) {
    err := fmt.Errorf("wrap: %w", ErrX)
    for i := 0; i < b.N; i++ {
        _ = errors.Is(err, ErrX)
    }
}
```

What to notice: Reusing sentinels is free; wrapping each time allocates.

Try next: Audit a hot path that does `fmt.Errorf` on success-adjacent branches.

Production Mistakes Deep Dive

Production Mistakes Deep Dive

Overview

Recurring themes from production Go engineering discussions (including widely shared checklists on X): timeouts, leaks, default clients, lock+I/O, and observability gaps. This chapter maps each mistake to **runtime symptoms** and **fixes** covered elsewhere in this part.

Diagram: Incident loop

```
flow:
  [Prof]
       |
       v
  [Trace]

  [Trace]
       |
       v
  [Fix]
```

The List (Mapped)

Mistake	Symptom	Deep dive / fix
No context timeouts	Goroutine + FD growth	Context trees, netpoller
Unbounded goroutines	RSS↑, sched latency	Scheduler, leaks
Default `http.Client`	Hung dials/TLS	Transport pool
Errors as strings	Untestable retries	Error cost
nil vs empty JSON	Client contract breaks	encoding stdlib
Unbounded channels	Memory spikes	Channel internals
Lock held over I/O	Throughput collapse	Mutex runtime
`time.After` in hot select	Timer/alloc churn	Timers
DB without limits/context	Connection storms	HTTP/DB practices
Logs without IDs/metrics	Blind incidents	slog, observability parts

Incident Playbook

```
1. NumGoroutine / FD count / RSS chart
2. goroutine pprof (wait sites)
3. mutex/block profile if CPU low
4. trace 200-500ms around incident
5. check client/server timeouts
6. check deploy diff for "go func" and "time.After"
```

Engineering Questions (From "knowing Go")

1. Do you need concurrency at all?
2. Goroutine or worker pool?
3. Channel or mutex?
4. How does context propagate?
5. Did you run `-race`, pprof, `-gcflags=-m`?

Mini Lab

Take any service and grep:

```
rg -n 'http\\.Get|http\\.DefaultClient|time\\.After|go func|context\\.Background\\(\\)' --gl
```

Classify each hit as safe or debt.

Try next: Add a CI check that fails on `http.Get` in non-test packages (project policy).

Production Debugging Tooling

Production Debugging Tooling

Overview

When production hurts, you rarely attach a debugger first. You grab **profiles, traces, stacks, and core-ish dumps** — fast.

Diagram: Debug surfaces

```
flow:
  [Pprof]
       |
       v
  [Heap]

  [Pprof]
       |
       v
  [Goroutine]
```

Always-On Endpoints (Carefully)

```go
import _ "net/http/pprof"
// bind to internal listener only
go http.ListenAndServe("127.0.0.1:6060", nil)
```

Expose via secure admin network / port-forward, never public internet.

Signals

Signal	Common action
SIGQUIT	Stack dump (often)
SIGTERM	Graceful shutdown
panic	`GOTRACEBACK` controls detail

```
GOTRACEBACK=all ./app
kill -QUIT $pid   # platform dependent behavior
```

Delve (Dev / Staging)

```
dlv attach $pid
dlv dap # editor
```

Prefer staging for breakpoints; prod attach can freeze worlds.

Core / Crash

```
# Linux examples - ops dependent
ulimit -c unlimited
GOTRACEBACK=crash
```

Pair with symbolized binaries (don't strip staging).

Continuous Profiling

Products sample CPU/heap cheaply over time — use alongside on-demand pprof. Compare versions after deploys.

Checklist

1. Secure pprof listener
2. Mutex profile fraction documented
3. Version/build stamped (`-X`)
4. Runbooks for "goroutine explosion" and "GC CPU"

Experiment

```
go mod init example
# tiny server with pprof
```

```
package main

import (
```

```go
    "net/http"
    _ "net/http/pprof"
    "time"
)

func main() {
    go func() {
        for {
            time.Sleep(10 * time.Millisecond)
        }
    }()
    _ = http.ListenAndServe("127.0.0.1:6060", nil)
}
```

```bash
# other terminal
go tool pprof http://127.0.0.1:6060/debug/pprof/goroutine
```

What to notice: Sleeping G still appears in goroutine profile; counts matter more than one stack.

Try next: Automate capturing 30s CPU profile on SIGUSR1 in a side project.

sysmon, Scavenger, and Background Runtime

sysmon, Scavenger, and Background Runtime

Overview

Besides your goroutines, the runtime runs **background machinery**: monitoring long preemption needs, retaking Ps stuck in syscalls, returning memory to the OS (**scavenger**), and coordinating GC.

Diagram: Background runtime

```
flow:
  [Sysmon]
       |
       v
  [Netpoll]
```

sysmon (System Monitor)

A runtime thread that periodically:

- Retakes Ps from Ms blocked in syscalls too long
- Helps with preemption / netpoll wakeups (version-dependent details)
- Notices sleeping conditions

You do not call sysmon — but its work appears when syscall-heavy loads misbehave.

Memory Scavenging

After GC frees spans, memory may remain reserved to the process. The scavenger **returns idle heap pages to the OS** (with hysteresis) so RSS can fall.

```
HeapInuse vs HeapSys vs RSS
  HeapInuse  - occupied objects
  HeapSys    - virtual from OS for heap
  RSS        - OS resident (includes scavenged lag)
```

`GOMEMLIMIT` and GC interact with how aggressively memory returns.

GC Workers

Mark workers and assist run as special Gs. High `GCCPUFraction` means the mutator is paying for GC.

What You Control

Control	Effect
`GOMAXPROCS`	Parallel Go + related resources
`GOGC` / `GOMEMLIMIT`	GC frequency & memory ceiling
Allocation rate	Drives all of the above
Syscall/cgo rate	Pressure on Ms and sysmon retake

Experiment

```
GODEBUG=gctrace=1 go run .
```

```go
package main

import (
    "fmt"
    "runtime"
    "time"
)

func main() {
    var m runtime.MemStats
    hold := make([][]byte, 0, 500)
    for i := 0; i < 500; i++ {
        hold = append(hold, make([]byte, 100*1024))
    }
    runtime.ReadMemStats(&m)
    fmt.Printf("hold HeapInuse=%dMiB HeapSys=%dMiB\n", m.HeapInuse>>20, m.HeapSys>>20)
    hold = nil
    runtime.GC()
    time.Sleep(200 * time.Millisecond)
    runtime.ReadMemStats(&m)
    fmt.Printf("after HeapInuse=%dMiB HeapSys=%dMiB\n", m.HeapInuse>>20, m.HeapSys>>20)
}
```

What to notice: HeapInuse drops quickly after GC; HeapSys/RSS may lag until scavenging.

Try next: Compare with `GOMEMLIMIT` set near working set size.

Reading the Go Runtime Source

Reading the Go Runtime Source

Overview

The ultimate deep dive is **`src/runtime`** in the Go tree. You do not need to memorize it — you need a map, a debugger for experiments, and humility about version drift.

Workshop-style learning (modifying the toolchain) is increasingly popular for advanced gophers.

Diagram: Source map

```
symptom (latency / leak / panic)

         v
pick runtime area
     proc.go     scheduler
     chan.go     channels
     mgc*.go     GC
     netpoll*.go network
```

Entry Points

Area	Files (names evolve)
Scheduler	`proc.go`, `runtime2.go`
Channels	`chan.go`
Maps	`map*.go`
GC	`mgc*.go`, `mheap.go`
Netpoll	`netpoll*.go`
Stacks	`stack.go`
Timers	`time.go`
Interface	`iface.go` / abi
Assembly	`asm_*.s`, `sys_*.s`

Clone matching your version:

```
git clone --depth 1 --branch go1.27.1 https://github.com/golang/go
cd go/src/runtime
rg -n 'type hchan'
rg -n 'func schedule'
```

How To Study

1. Pick a symptom (e.g. send on closed panics).
2. Find the user-facing panic string in runtime.
3. Walk callers one level at a time.
4. Write a 10-line program that hits the path.
5. Optional: `dlv` on a rebuilt `go` tool (advanced).

go tool objdump / nm

```
go build -o app .
go tool nm app | rg 'runtime\\.'
go tool objdump -s 'main\\.main' app | head
```

Experiments Without Patching

- `GODEBUG` traces
- `go build -gcflags=-m`
- Race detector as dynamic analysis
- Breakpoints in **your** code at API boundaries, infer runtime

Experiments With Patching (Optional)

- Add a `println` in a runtime helper (local toolchain)
- Use `go install` from a branch
- Never ship patched runtimes without extreme need

Field Guide: This Book → Source

Chapter	Start reading
201 GMP	`proc.go` schedule/findrunnable
203 channels	`chan.go`
208 GC	`mgc.go`

Chapter	Start reading
213 netpoll	`netpoll_*.go`
214 timers	`time.go`
228 cgo	`cgocall.go`

Experiment

```
# on a full go checkout matching your version
rg -n "panic\\(.*close of closed channel" "$(go env GOROOT)/src/runtime"
rg -n "func makeslice" "$(go env GOROOT)/src/runtime"
```

What to notice: Panic strings and helper names are breadcrumbs; follow them instead of random file browsing.

Try next: Pick one production bug class from mistakes and find the runtime check that enforces it.

crypto/tls Handshake Path

crypto/tls Handshake Path

Overview

TLS is often the hidden cost in "simple HTTP" clients. Handshake CPU, session resumption, and certificate verification dominate cold connections; pooling and HTTP/2 amortize them.

Diagram: TLS then HTTP

```
flow:
  [HS]
       |
       v
  [App]
```

Path

```
Dial TCP
  -> TLS client hello
  -> server hello / certs
  -> key exchange / finished
  -> Application data (HTTP)
```

```go
cfg := &tls.Config{
    MinVersion: tls.VersionTLS12,
    // ServerName for SNI verification when not in address
}
d := tls.Dialer{Config: cfg, NetDialer: &net.Dialer{Timeout: 5 * time.Second}}
conn, err := d.DialContext(ctx, "tcp", "example.com:443")
```

Cost Drivers

Factor	Effect
New TCP+TLS each request	Highest latency

Factor	Effect
Session tickets / resumption	Cheaper reconnects
Huge trust stores / custom Verify	CPU
HTTP/2 reuse	One handshake, many streams
mTLS	Client certs + private key ops

Client Integration

`http.Transport.TLSClientConfig` and idle pools determine whether you pay handshake per request. See Transport.

Security Notes

- Pin MinVersion
- Avoid `InsecureSkipVerify` outside tests
- Prefer system roots or explicit pinned CAs for internal mesh

Deep mTLS ops: TLS/mTLS.

Experiment

```
go mod init example
# optional network:
# go run . example.com:443
```

```
package main

import (
    "crypto/tls"
    "fmt"
    "net"
    "os"
    "time"
)

func main() {
    addr := "example.com:443"
    if len(os.Args) > 1 {
        addr = os.Args[1]
    }
    start := time.Now()
    conn, err := tls.DialWithDialer(
```

```go
        &net.Dialer{Timeout: 5 * time.Second},
        "tcp",
        addr,
        &tls.Config{MinVersion: tls.VersionTLS12},
    )
    if err != nil {
        fmt.Println("err", err)
        return
    }
    defer conn.Close()
    st := conn.ConnectionState()
    fmt.Printf("version=%x resumed=%v alpn=%q time=%s\n",
        st.Version, st.DidResume, st.NegotiatedProtocol, time.Since(start))
}
```

What to notice: First dial slower; second process may still be cold unless session cache configured at higher layers.

Try next: Compare `http.Transport` with `ForceAttemptHTTP2` true/false under parallel GETs.

slog Handler Design Deep Dive

slog Handler Design Deep Dive

Overview

`log/slog` separates **Logger** (API) from **Handler** (serialization + output). Custom handlers power redaction, sampling, OpenTelemetry bridges, and multi-sink fanout.

Basics: fmt/log/slog, structured logging.

Diagram: Logger vs Handler

```
slog.Logger

     v
Handler
      Enabled(level)?
      Handle(Record)    JSON / text / OTel
      WithAttrs / WithGroup
```

Interfaces

```go
type Handler interface {
    Enabled(ctx context.Context, level slog.Level) bool
    Handle(ctx context.Context, r slog.Record) error
    WithAttrs(attrs []slog.Attr) Handler
    WithGroup(name string) Handler
}
```

Rules for custom handlers:

1. Honor `Enabled` to avoid expensive attr work
2. `WithAttrs`/`WithGroup` return new handlers (immutability)
3. Be concurrent-safe — loggers are shared
4. Do not panic on normal records

Patterns

Pattern	Idea
Middleware handler	Wrap another handler; filter/redact
Sampling	Drop debug under load
Fanout	Write JSON + ship metrics counter
Context attrs	Pull request_id from ctx in Handle

```go
type RedactHandler struct{ slog.Handler }

func (h RedactHandler) Handle(ctx context.Context, r slog.Record) error {
    r.Attrs(func(a slog.Attr) bool {
        if a.Key == "password" || a.Key == "token" {
            // rebuild record without secrets - production code clones carefully
        }
        return true
    })
    return h.Handler.Handle(ctx, r)
}
```

Performance

- JSON handler cost encoding + I/O
- Avoid `fmt.Sprintf` in hot attr values
- Precompute constant attrs via `With`

Experiment

```go
go mod init example
go run .
```

```go
package main

import (
    "context"
    "log/slog"
    "os"
)

type MinLevel struct {
    slog.Handler
```

```
    min slog.Level
}

func (h MinLevel) Enabled(ctx context.Context, l slog.Level) bool {
    return l >= h.min && h.Handler.Enabled(ctx, l)
}

func main() {
    base := slog.NewJSONHandler(os.Stdout, nil)
    log := slog.New(MinLevel{Handler: base, min: slog.LevelInfo})
    log.Debug("hidden")
    log.Info("visible", "n", 1)
}
```

What to notice: Debug suppressed without formatting work beyond Enabled checks on the wrapper.

Try next: Add attribute redaction for keys `authorization` and `cookie`.

GOMAXPROCS, cgroups, and Containers

GOMAXPROCS, cgroups, and Containers

Overview

In Kubernetes and cgroup-limited environments, **CPU quota host core count**. Older Go versions defaulted `GOMAXPROCS` to machine CPUs, causing thrashing inside small pods. Modern Go improves cgroup awareness — still verify in your version and platform.

Diagram: Quota mismatch

```
flow:
  [Pod]
       |
       v
  [Bad]

  [Pod]
       |
       v
  [Good]
```

Failure Mode

```
Pod limit: 500m CPU (0.5 core)
GOMAXPROCS=16 (node size)
  -> 16 threads fight for 0.5 CPU
  -> latency noise, steal time, GC assists hurt more
```

Practices

```
// log at boot
slog.Info("sched", "gomaxprocs", runtime.GOMAXPROCS(0), "numCPU", runtime.NumCPU())
```

- Prefer runtime auto-detect on current Go
- Or set explicitly from downward API / cgroup read

- Libraries like `automaxprocs` existed historically for this gap

Memory Side

Pair CPU limits with `GOMEMLIMIT` near container memory limit (leave headroom for non-Go RSS, caches).

```
GOMEMLIMIT=400MiB ./app
```

Experiment

```
go run .
```

```go
package main

import (
    "fmt"
    "runtime"
)

func main() {
    fmt.Println("NumCPU", runtime.NumCPU())
    fmt.Println("GOMAXPROCS", runtime.GOMAXPROCS(0))
}
```

Run inside and outside a constrained container if you have Docker:

```
docker run --rm --cpus=0.5 -v "$PWD":/app -w /app golang:1.27 go run .
```

What to notice: Compare printed values to `--cpus` quota on your platform/Go version.

Try next: Load test a CPU-bound endpoint with GOMAXPROCS 1 vs 4 under a 1 CPU quota.

Plugins and buildmode Pitfalls

Plugins and buildmode Pitfalls

Overview

`buildmode=plugin` looks like a dynamic extension story. In production it is usually a **trap**: brittle ABI, limited platforms, and operational pain. Prefer processes, Wasm, or explicit interfaces.

Diagram: Prefer process plugins

```
need extension point?

        side process + RPC     (preferred)
        Wasm sandbox
        buildmode=plugin       (fragile: skew, platforms)
```

Reality Check

Issue	Detail
Platforms	Not universal
Version skew	Plugin must match main toolchain/deps closely
Dependency bloat	Duplicated modules risk
Debugging	Harder stacks and deploys
Security	Loading code = trust boundary

Prefer Alternatives

1. **Side process + RPC** (gRPC/HTTP) — isolation, language freedom
2. **Wasm plugins** — sandboxing story
3. **Compile-time interfaces** + separate binaries
4. **HashiCorp go-plugin** style (subprocess) if you need polyglot

If You Must

```
go build -buildmode=plugin -o plug.so ./plug
# open via plugin.Open; lookup symbols
```

Pin exact Go version + `go.mod` policy; test load on the **identical** base image as production.

Experiment

Skip coding plugins by default. Instead document a design choice:

```
feature flag -> separate binary -> RPC
vs
plugin.Open
```

Write a 10-line ADR in your repo choosing RPC.

Try next: Read `go help buildmode` end-to-end once; note which modes you actually need (usually `exe` only).

Topic Radar (X + Ecosystem Signals)

Topic Radar (X + Ecosystem Signals)

Overview

This chapter captures **themes that repeatedly surface** in Go engineering discourse on X and in adjacent blogs — used as a backlog radar for this deep-dives part. Signals change; the curriculum should track *problems*, not hype.

Diagram: Radar loop

```
flow:
  [Repro]
       |
       v
  [Chapter]

  [Chapter]
       |
       v
  [Link]
```

Hot Themes Observed

Runtime & scheduler

- GMP explainers and GOMAXPROCS capacity math
- Go 1.26 removal of P-level syscall state / cgo path simplification
- Go 1.27 generic methods, json/v2 GA, goroutineleak profile
- Channels as runtime queues (sudog), not magic
- GC pauses as product latency (Green Tea GC chatter)

Correctness under concurrency

- Context timeouts as non-optional
- Goroutine leak checklists
- Race detector as culture, not CI checkbox
- `testing/synctest` replacing `sleep` tests

Performance craft

- Escape analysis + pprof as literacy
- `sync.Pool` and zero-alloc append patterns
- Zero-copy `io.Copy` / sendfile limits with TLS
- `time.After` abuse in hot selects

Production engineering

- Default `http.Client` considered harmful
- Structured logs with request IDs
- Mutex held across I/O
- Container CPU limits vs GOMAXPROCS

Learning meta

- "Features engineering"
- Read `src/runtime` workshops
- Interview lists that mix runtime + design judgment

Mapped Into This Part

Signal	Chapter
GMP / syscall P changes	201, 228
Channel internals posts	203
synctest runtime (e.g. internals-for-interns.com)	223
Zero-copy / sendfile	224
Production top mistakes	237
GC / pprof	208, 222
Pool / allocs	219
Reading source	240

Still-Fertile Backlog (Future)

- `encoding/json/v2` migration cookbooks now that 1.27 shipped stable defaults
- Full Swiss-map implementation walk with diagrams per release
- Runtime metrics export (`/debug` → OTel) cookbook
- FIPS / boringcrypto build graphs for regulated shops
- Delve scripting for runtime bugs
- Port of `runtime` execution tracer event schema reference
- Arena alternatives pattern library (sync.Pool arenas, slab allocators)
- CPU profile vs wall-clock eBPF cross-check on Linux

How To Extend This Book

When a topic trends:

1. Write a **minimal repro** that shows the symptom.
2. Add a chapter under `20-go-deep-dives/` with experiment + cross-links.
3. Link from this radar and from index.
4. Avoid duplicating Part 10/11 and Part 24 tutorials — keep the *why/runtime* angle.

Experiment

```
# personal radar loop
rg -n 'TODO|FIXME|race|timeout' --glob '*.go' | head
```

Convert one production footgun into a regression test this week.

Try next: Subscribe to Go release notes + one runtime blog; every release, tick experiments in GOEXPERIMENT.

Article Radar: Internals for Interns + Kin

Article Radar: Internals for Interns + Kin

Overview

This radar catalogs **diagram-rich Go internals writing** in the same vein as *Inside Go's synctest*—runtime source, compiler pipeline, and systems guests. Use it to prioritize reading and to track what this monorepo has absorbed into local chapters.

Diagram: map sources → book chapters

```
IFI runtime series          this book

bootstrap                 248
allocator                 247
scheduler                 201, 256
GC                        208, 255
netpoll / sysmon          213, 239
synctest                  223
```

Internals for Interns — Runtime series

Article	URL	Book landing
Bootstrap	understanding-go-runtime	248
Memory allocator	go-memory-allocator	247
Scheduler	go-runtime-scheduler	201
Garbage collector	go-garbage-collector	208
System monitor	go-sysmon	239
Network poller	go-netpoller	213
Slices, maps, channels	go-runtime-slices-maps-channels	203, 205
select	go-runtime-select	249

Article	URL	Book landing
Stacktraces	go-runtime-stacktraces	250
Reflect	go-runtime-reflect	211
Profiling	go-runtime-profiling	222
synctest (guest)	inside-go-synctest	223
mmap vs pread (guest)	mmap-vs-pread-go-storage-engine	251

Series hub: Understanding the Go Runtime.

Internals for Interns — Compiler series

Phase	URL	Book landing
Scanner	the-go-lexer	209 + radar
Parser	the-go-parser	209
Type checker	the-go-type-checker	209
Unified IR	go-compiler-unified-ir	209
IR	the-go-ir	209
SSA	the-go-ssa	209
Machine code	the-go-machine-code	209
Linker	the-go-linker	210

Kindred sources (same "diagram + source" energy)

Source	Why
antonz.org Gist of Go: Concurrency	Interactive curriculum → part 10
iximiuz Labs: Using Go for Systems Programming	libc vs direct syscalls, static binaries, `CGO_ENABLED=0` → 140a
Go scheduler talk (Vyukov)	Original lightweight concurrency design
go.dev blog	Official GC, GOMAXPROCS, experiments
Channel runtime posts (e.g. gaborkoos / similar)	hchan latency mental models → 203

Source	Why
Zero-copy sendfile/splice writeups	→ 224

How we absorb articles

```
1. Extract durable models (not copy prose)
2. Add mermaid/ASCII diagrams in-repo
3. Cross-link experiments + GOROOT reading tips
4. Attribute the source series at chapter tops where used heavily
5. Re-check after each Go minor release
```

Experiment

Pick one unread row in the tables above; after reading, open the matching book chapter and add one personal note (sticky TODO in a `_planning` file if you keep one) for anything the monorepo still under-explains.

Try next: On each Go release, re-open the GC + scheduler posts and diff against 208 / 201.

Memory Allocator Deep Dive

Memory Allocator Deep Dive

Overview

Go's allocator is a **warehouse manager**: hand out differently sized objects fast, keep fragmentation in check, and cooperate with the GC. Most small allocations hit a **per-P cache** and avoid locks.

> Teaching map aligned with runtime walkthroughs such as Internals for Interns — Memory Allocator. Original prose for this book.

Diagram: three-level hierarchy

```
goroutine needs memory

         v
            hit        return pointer
     P.mcache
    (per-P)

          miss
         v
            hit
    mcentral                  refill mcache
    size class

          miss
         v

     mheap       spans / arenas    OS (mmap)
```

Level	Role
mcache	Per-P free lists; lock-free for the running P
mcentral	Shared lists per size class when mcache empty
mheap	Global heap, spans, arenas; talks to OS

Size classes and spans

```
tiny / small objects → fixed size classes → span of pages carved into slots
large objects        → dedicated spans (page-rounded)
```

Go 1.27 generates **size-specialized** allocation calls for some objects under 80 bytes (up to ~30% cheaper on that path; ~1% overall on alloc-heavy programs). Disable with `GOEXPERIMENT=nosizespecializedmalloc` if you need a bisect.

```
flow:
  [Class]
       |
       v
  [Slot]

  [Slot]
       |
       v
  [Ptr]
```

Arenas and address space

Modern Go reserves large **arenas** of virtual address space, then commits pages as needed. That keeps metadata dense and growth predictable compared to many tiny `mmap`s.

Interaction with GC and P

- Each **P** owns an mcache so allocation does not bounce on a global lock.
- GC **sweeps** spans; scavenger returns idle pages to the OS (239).
- Allocation pressure drives the **pacer** (208).

Escape analysis still wins

The fastest allocation is **no heap allocation**. Stack frames and SSA prove many values need no allocator at all (215, 209).

Diagnosis

Signal	Meaning
`runtime.mallocgc` in CPU pprof	alloc-heavy path

Signal	Meaning
Heap profile top	retained objects
Alloc profile top	churn even if freed
High `GCCPUFraction`	pacer reacting to alloc rate

Experiment

```
go test -bench=. -benchmem
```

```
func BenchmarkHeap(b *testing.B) {
    for i := 0; i < b.N; i++ {
        _ = make([]byte, 1024)
    }
}
func BenchmarkStack(b *testing.B) {
    for i := 0; i < b.N; i++ {
        var a [1024]byte
        sink = a[0]
    }
}
var sink byte
```

What to notice: stack-ish patterns show fewer alloc/op when the compiler keeps data off-heap.

Try next: `GODEBUG=allocfreetrace=1` on a tiny program (noisy; educational only).

Runtime Bootstrap

Runtime Bootstrap

Overview

Before `main.main`, the OS jumps into **runtime startup**: init the scheduler, allocator, GC metadata, threads, and package `init` graph. Every Go binary embeds this runtime.

> Series-style depth: Understanding the Go Runtime — Bootstrap.

Diagram: from exec to main

```
OS loader

   v
runtime entry (asm)

   v
init scheduler + heap + main G

   v
package init() order

   v
main.main
```

What gets set up

Subsystem	Early job
Allocator	arenas, spans ready for first `new`
Scheduler	`GOMAXPROCS` Ps, main M, main G
GC	metadata, write barrier off until needed
Signals	platform signal plumbing
Netpoll	poll descriptor (platform)

Package init order

```
imported packages first (dependency order)
  → file-level var initializers
  → init() functions
  → main.main
```

Do not rely on clever cross-package init races; make dependencies explicit.

Main goroutine

`main.main` runs on the **main goroutine**. When it returns, the runtime shuts down the process (other Gs are not gracefully joined unless you waited).

Experiment

```
go build -o /tmp/hello .
# optional: go tool objdump / nm to see runtime symbols
go tool nm /tmp/hello | rg 'runtime\\.(main|schedinit|newproc)' | head
```

```go
package main

import "fmt"

func init() { fmt.Println("init") }
func main() { fmt.Println("main") }
```

What to notice: `init` always before `main`; runtime symbols dominate even tiny binaries.

Try next: Compare binary size `CGO_ENABLED=0` hello vs one that imports `net/http`.

select: Compiler and selectgo

select: Compiler and selectgo

Overview

`select` looks like `switch` but is a **compiler + runtime** duet. Simple shapes are rewritten; hard cases call `runtime.selectgo`.

Depth companion: Internals for Interns — The select Statement.

Diagram: two paths

```
select { ... }

        one case / simple    compiler rewrite

        general    runtime.selectgo

                            park until one case wins
```

Rules that matter

Rule	Meaning
Ready cases	If several ready, **uniform random** pick
All blocked	Wait until one ready (unless `default`)
`default`	Non-blocking poll
Nil channel case	Never selected (disable pattern)

Pseudo-algorithm (teaching)

```
1. randomize case order
2. lock involved channels carefully (runtime does ordered locking)
3. probe for immediately ready op
4. if none and default → default
5. else enqueue this G on all wait queues; park
6. on wake: dequeue others; complete winner only
```

Synctest note

Inside a synctest bubble, select is durable only if **every** non-nil case is bubbled (223).

Experiment

```
go run .
```

```go
package main

import "fmt"

func main() {
    a, b := make(chan int, 1), make(chan int, 1)
    counts := map[string]int{}
    for i := 0; i < 1000; i++ {
        a <- 1
        b <- 1
        select {
        case <-a:
            counts["a"]++
        case <-b:
            counts["b"]++
        }
        // drain the other so next iter both ready
        select {
        case <-a:
        default:
        }
        select {
        case <-b:
        default:
        }
    }
    fmt.Println(counts)
}
```

What to notice: Both cases win a substantial share—not left-to-right priority.

Try next: `go build -gcflags=-S` on a one-case select and a multi-case select; compare call sites.

Stacktraces and Runtime Metadata

Stacktraces and Runtime Metadata

Overview

A stack trace is not magic: the **compiler and linker freeze metadata** into the binary; the runtime **unwinds** live frames using that data. Reflect uses a sibling idea—type descriptors instead of PC→func maps.

> Companion read: Internals for Interns — Stacktraces.

Diagram: build time vs run time

```
regions: build time | run time
flow:
  [Unwind]
       |
       v
  [Meta]

  [Unwind]
       |
       v
  [Frames]
```

Who prints stacks

Trigger	Behavior
panic	dump (controlled by `GOTRACEBACK`)
`runtime.Stack`	buffer of current or all Gs
`pprof` goroutine	stacks of live Gs
SIGQUIT (often)	all-G dump

```
GOTRACEBACK=all ./app
```

Inlining and wrappers

Optimized binaries **inline** functions; traces may show fewer frames or wrapper names. Disable inlining for forensics:

```
go build -gcflags='-l' .
```

Relation to profiling

CPU profiles sample stacks with the same unwinding family of mechanisms (222).

Experiment

```go
package main

import (
    "fmt"
    "runtime"
)

func deep(n int) {
    if n == 0 {
        buf := make([]byte, 4096)
        k := runtime.Stack(buf, false)
        fmt.Printf("%s", buf[:k])
        return
    }
    deep(n - 1)
}

func main() { deep(5) }
```

What to notice: Frames list `deep` repeatedly with decreasing depth.

Try next: Compare stacks with and without `-gcflags=-l`.

mmap vs pread in Go Storage

mmap vs pread in Go Storage

Overview

Storage engines eventually ask: **how should we read bytes from files?** In Go the answer is rarely absolute—"always mmap" or "always pread". The interesting design is an **abstraction** that prefers mmap when pages are warm and falls back when a fault would **block an M**.

> Guest-systems depth: mmap vs pread in a real Go storage engine (Phuong Le / Internals for Interns). Patterns below are teaching synthesis.

Diagram: two I/O styles

```
mmap path:  map pages → load byte
                 page fault?
                  may block M on major fault

pread path: syscall ReadAt → copy into []byte
                  G park rules better understood
```

Tradeoffs

	mmap	pread
API	pointer-like access	explicit `ReadAt`
Warm data	very fast	syscall + copy
Cold data	major fault risk	controlled syscall
Go runtime	fault on M is painful	netpoll/syscall paths understood
Portability	OS-specific edges	straightforward

Go-shaped recommendation

```
abstraction ReadAt(off, buf)
  if region cached / known resident → mmap view
  else → pread into buf
never let page faults surprise a hot P/M without budget
```

Prefer measuring **tail latency** under cold cache and concurrent queries, not only average throughput.

Experiment

```
# conceptual microbench outline
# 1) os.ReadAt loop over file
# 2) unix.Mmap + index loads
# compare under: warm cache vs drop_caches (Linux ops only)
```

What to notice: Warm mmap wins; cold fault storms can look like "Go scheduler freeze" when Ms stuck in fault handling.

Try next: Read the guest post's VictoriaLogs-oriented rationale end-to-end; map it to your own on-disk layout.

Compiler Frontend: Scan, Parse, Typecheck

Compiler Frontend: Scan, Parse, Typecheck

Overview

`go build` is a pipeline. The **frontend** turns source text into a type-checked AST (and related forms) before IR/SSA. Understanding this half explains compile errors, `go/types`, and why some "optimizations" never see your source.

> Series map: Internals for Interns — Compiler (scanner → type checker). Companion: 209 SSA/PGO.

Diagram: frontend stages

```
.go source

   v
scanner → tokens → parser → AST

                            v
                       type checker

                            v
                       export / IR bridge
```

Scanner (lexer)

- Characters → **tokens** (`package`, `IDENT`, `INT`, `+`, …)
- Tracks positions for error messages and stack traces later
- No meaning yet—`fmt` and `Println` are just idents and a dot

Parser

- Tokens → **AST** (`*ast.File`, decls, stmts, exprs)
- Enforces grammar; produces structure without full type knowledge
- `go/parser` in the standard library mirrors the idea for tools

```go
fset := token.NewFileSet()
f, err := parser.ParseFile(fset, "main.go", src, 0)
```

Type checker

- Resolves identifiers, assigns types, checks assignability
- Builds method sets, interfaces satisfaction, constant folding
- Emits the errors you actually fix day to day

```
undeclared name / cannot use X as Y / missing method M
```

Tools (`gopls`, `go vet`, many linters) reuse **go/types** rather than reimplementing rules.

Unified IR / export data (bridge)

After typecheck, the compiler serializes a compact representation for:

- Build cache reuse
- Export data for importers of the package
- Feeding the mid-end (IR) without re-parsing forever

What the frontend will not do

Expectation	Reality
Cross-package inlining decisions	Later (IR/SSA)
Register allocation	Backend
Fix algorithm bugs	Never

Experiment

```go install golang.org/x/tools/cmd/goimports@latest  # optional tools
# Dump AST for a file with the go/ast printer pattern, or:
go doc go/parser
go doc go/types
```

Write a 10-line program that uses `go/parser` + `ast.Inspect` to count `*ast.CallExpr` nodes in a file.

Try next: Feed the type checker a file with a deliberate interface mismatch; print `types.Config.Check` errors.

Compiler Mid/Back: IR, SSA, Machine Code, Link

Compiler Mid/Back: IR, SSA, Machine Code, Link

Overview

After typecheck, the compiler optimizes in **IR**, lowers to **SSA**, emits **machine code** objects, then the **linker** produces a binary. This is where inlining, escape analysis, and PGO bite.

Diagram: mid and back end

```
typed frontend

     v
IR (inline, escape, …)

     v
SSA opts + arch lower

     v
.o objects   linker   binary
```

IR responsibilities

Pass family	Effect
Inlining	Expand small callees; enable more opts
Escape analysis	Stack vs heap
Devirtualization	Concrete calls when type known
Walk / order	Prepare for SSA

Flags: `go build -gcflags='-m -m'` shows escape/inline decisions.

SSA

- Every value assigned once → easy dataflow
- Dozens of rewrite passes (nilcheck elim, bounds prove, …)
- Then **lower** to `AMD64…` / `ARM64…` ops

```
GOSSAFUNC=main go build .   # classic SSA HTML dump when supported
go build -gcflags='-S' . 2>&1 | head
```

Machine code + object files

- Encode instructions, relocations, PC line metadata
- One package → one (or more) object; still not runnable alone

Linker

```
resolve symbols across packages
  → apply relocations
  → layout sections
  → emit ELF/Mach-O/PE + buildid
```

```
go build -ldflags='-s -w -X main.version=dev' -trimpath -o app .
```

See also 210 link/race, 248 bootstrap.

Experiment

```
go build -gcflags='-m' . 2>&1 | head -40
go tool nm ./app | rg 'main\\.|runtime\\.main' | head
```

What to notice: Even tiny mains reference large `runtime.*` symbol sets after link.

Try next: Build twice with/without `-pgo=auto` after a CPU profile; bench a hot path.

Race Detector Internals

Race Detector Internals

Overview

`go test -race` is not a linter—it is a **different ABI/runtime** that shadows memory and tracks **happens-before**. It catches concurrent unsynchronized access with at least one write.

Diagram: instrumentation model

```
load/store in user code

         v
instrumented binary (-race)


   v           v
shadow       happens-before
memory       (unlock, send, …)


         v
   race?  yes   WARNING: DATA RACE
```

What creates happens-before (reminder)

Event	Edge
Unlock → later Lock	yes
Channel send → receive	yes
`WaitGroup.Done` → `Wait` return	yes
`go` start	parent before child start
Atomics (sync/atomic)	model-defined

No edge + concurrent write/read → race report.

Cost model

Mode	Overhead
`-race` tests	often 2–10× CPU/mem
Production race binary	rarely shipped

Requires CGO toolchain on many platforms historically—CI images need gcc/clang.

False confidence

- Passing once no races
- Coverage matters: races on rare branches need fuzz/load
- Synctest settles logical concurrency; **still run `-race`**

Experiment

```
cat > /tmp/race.go <<'EOF'
package main
import ("fmt"; "sync")
func main() {
  var x int
  var wg sync.WaitGroup
  wg.Go(func() { x++ })
  wg.Go(func() { x++ })
  wg.Wait()
  fmt.Println(x)
}
EOF
go run -race /tmp/race.go
```

What to notice: Report names both goroutines and stack sites even if printed total looks "fine".

Try next: Fix with `atomic.Int64` or mutex; confirm clean under `-race`.

Write Barriers and Mark Assist

Write Barriers and Mark Assist

Overview

During concurrent mark, mutators keep running. **Write barriers** intercept pointer stores so the GC does not lose objects. When allocation outruns marking, **mark assist** makes allocating Gs help mark—coupling alloc rate to GC CPU on the request path.

Diagram: barrier keeps tri-color safe

```
mutator: x.f = p

     v
write barrier (during concurrent mark)

     v
shade / publish pointer to GC

alloc while GC behind   mark assist on same G
```

Without barriers, a black object could gain a white pointer the GC will never see → use-after-free class bugs.

When barriers are on

- Enabled around concurrent mark
- Cost shows up as extra work on **pointer-heavy** writes
- Reducing pointer density (flatten structs, fewer interfaces in hot graphs) helps

Mark assist

```
mutator allocates fast
  → GC behind on mark
  → allocator charges assist credit
  → G marks a bit before continuing
```

```
sequence (top → bottom):
  actors: mutator G, allocator, mark work
  mutator G --> allocator  : mallocgc
  allocator --> mutator G  : assist debt
  mutator G --> mark work  : scan some greys
  mutator G --> mutator G  : continue user code
```

Symptom: latency spikes on alloc-heavy handlers correlating with GC cycles—not only STW.

Tuning angle

1. Allocate less (best)
2. `GOMEMLIMIT` / `GOGC` shape frequency
3. Profile `runtime.gcAssistAlloc` / mark symbols
4. Avoid panicking into "just raise GOGC" without heap proof

Experiment

```
GODEBUG=gctrace=1 go run .
```

```go
package main
import ("fmt"; "runtime")
func main() {
    var hold [][]byte
    for i := 0; i < 2000; i++ {
        hold = append(hold, make([]byte, 64*1024))
    }
    var m runtime.MemStats
    runtime.ReadMemStats(&m)
    fmt.Println("NumGC", m.NumGC, "GCCPUFraction", m.GCCPUFraction)
    _ = hold
}
```

What to notice: `gctrace` lines and rising `NumGC` under allocation bursts.

Try next: Compare assist-heavy run vs pre-sized slice with fewer growths.

Work Stealing and findrunnable

Work Stealing and findrunnable

Overview

When a P needs work, it does not only look at its local run queue. The scheduler's **findrunnable** path checks local queue, global queue, **work stealing**, netpoll, and idle logic. This is how Go keeps cores busy without a single giant lock.

Diagram: where is the next G?

```
P needs a runnable G

     v
local runq?  yes   run it
       no
     v
global runq?  yes   run it
       no
     v
steal from other P?  yes   run it
       no
     v
netpoll ready?  yes   run it
       no
     v
idle / wait for work
```

Local vs global queues

Queue	Property
Per-P local	Fast, little contention
Global	Overflow / balancing; more shared

New Gs often land local to the creator's P; load imbalance triggers steal.

Work stealing

Idle P **steals half** (teaching model) of another P's local queue when possible—classic multiprocessor scheduler trick.

```
P_idle looks at P_victim.runq
  → take some Gs
  → run them locally
```

Netpoll integration

Before declaring the machine idle, try **non-blocking netpoll**: "any FD ready?" Ready network Gs become runnable without a dedicated thread blocked in `read`.

sysmon also helps poll when all Ps are busy (213, 239).

Why you care

Symptom	Scheduler angle
Runnable Gs, idle CPUs	steal/netpoll delay, GOMAXPROCS, cgroup
Latency + low CPU	parking on chan/IO, not missing steal
Syscall heavy	extra Ms; P handoff (201)

Experiment

```
GODEBUG=schedtrace=1000 go run .
```

```go
package main
import ("sync"; "time")
func main() {
    var wg sync.WaitGroup
    for i := 0; i < 1000; i++ {
        wg.Go(func() { time.Sleep(50 * time.Millisecond) })
    }
    wg.Wait()
}
```

What to notice: `schedtrace` lines show runqueue lengths and idle/spinning threads (format version-dependent).

Try next: Capture `go tool trace` and inspect Proc start/stop + Syscall blocks.

Type Assert and Convert at Runtime

Type Assert and Convert at Runtime

Overview

Type assertions and conversions are not only compile-time checks. Failed asserts panic (or return `ok=false`); interface asserts consult **dynamic type** metadata in the interface word.

Diagram: assert path

```
  interface value

          i.(T) ok-form    value or ok=false
          i.(T) bare       value or panic
          type switch      branch on dynamic type
```

```go
var i any = "x"
s := i.(string)      // panic if wrong
s, ok := i.(string)  // safe
switch v := i.(type) { case string: _ = v }
```

Convert vs assert

	Convert `T(x)`	Assert `i.(T)`
Static types	both known	interface → concrete/interface
Failure	compile error	panic or ok=false
Cost	often pure codegen	type metadata compare

Interface → interface

```go
var r io.Reader = bytes.NewBuffer(nil)
rc, ok := r.(io.ReadCloser) // may fail
```

Runtime checks method set / itab compatibility for the pair of interface types.

Performance notes

- Hot asserts on varying types can miss predict and cost more than type switch with few cases
- Prefer static types at package boundaries when possible
- `any` + assert in loops is a smell (also escapes/allocs)

Experiment

```
go run .
```

```
package main
import "fmt"
func main() {
    var i any = 3
    if _, ok := i.(string); ok {
        fmt.Println("string")
    } else {
        fmt.Println("not string")
    }
    defer func() {
        if r := recover(); r != nil {
            fmt.Println("panic:", r)
        }
    }()
    _ = i.(string)
}
```

What to notice: comma-ok is quiet; bare assert panics.

Try next: Benchmark type switch vs chained asserts on a closed set of types.

OS Signals and the Runtime

OS Signals and the Runtime

Overview

Unix signals interrupt processes asynchronously. Go multiplexes them through the runtime so user code can use `os/signal` channels instead of raw C handlers—while still coordinating with the scheduler and cgo.

Diagram: signal path (teaching)

```
kernel signal

     v
runtime notes signal

     v
os/signal demux → Notify channel

     v
user G handles (e.g. Shutdown)
```

os/signal API

```go
ctx, stop := signal.NotifyContext(context.Background(), os.Interrupt, syscall.SIGTERM)
defer stop()
<-ctx.Done()
// graceful shutdown
```

```go
ch := make(chan os.Signal, 1)
signal.Notify(ch, syscall.SIGTERM)
```

Buffer the channel—signals can be missed if send would block.

Interactions

Area	Note
Multiple Notify	Same signal can fan out
Reset/Stop	Undo registration
Windows	Different signal story; Interrupt exists
cgo	Foreign code may install handlers—careful
Panic dumps	`GOTRACEBACK`, SIGQUIT often stack dump

Graceful shutdown pattern

```
sequence (top → bottom):
  actors: SIGTERM, main, http.Server
  SIGTERM --> main  : NotifyContext cancel
  main --> http.Server  : Shutdown(ctx)
  http.Server --> main  : drains then return
```

Experiment

```
# terminal 1
go run .
# terminal 2
kill -TERM $(pidof yourbin)
```

```go
package main
import (
    "context"
    "fmt"
    "os"
    "os/signal"
    "time"
)
func main() {
    ctx, stop := signal.NotifyContext(context.Background(), os.Interrupt)
    defer stop()
    fmt.Println("running; press Ctrl+C")
    select {
    case <-ctx.Done():
        fmt.Println("got", ctx.Err())
    case <-time.After(30 * time.Second):
        fmt.Println("timeout")
    }
}
```

What to notice: Cancel is process-wide coordination; pair with server `Shutdown`, not `os.Exit` mid-request.

Try next: Trap SIGTERM in a binary under Docker `docker stop` and confirm clean logs.

HTTP/2 in Go

HTTP/2 in Go

Overview

Go's `net/http` speaks HTTP/2 when TLS+ALPN (or h2c in limited cases) allows it. One TCP connection **multiplexes streams**, changing connection pool math and failure modes.

Diagram: multiplex

```
regions: single TLS conn
flow:
  [Client]
       |
       v
  [conn]

  [conn]
       |
       v
  [Server]
```

Client knobs

```
tr := &http.Transport{
    ForceAttemptHTTP2: true,
    // MaxIdleConnsPerHost less critical for h2 multiplexing
    // but MaxConnsPerHost still bounds total conns
}
```

```
GODEBUG=http2debug=1 ./app   # noisy; debug only
```

Practical differences vs HTTP/1.1

Topic	HTTP/1.1	HTTP/2
Parallel requests / host	many conns or pipeline rare	streams on one conn
Head-of-line	per connection TCP HOL	stream independence (TCP still HOL)
Server push	N/A / deprecated paths	limited/rare in Go clients
Debugging	simple	need h2 frames / debug flags

Server

`http.Server` + TLS config with NextProtos often enough. Timeouts still mandatory:

```
ReadHeaderTimeout, ReadTimeout, WriteTimeout, IdleTimeout
```

Experiment

```
go run .
# curl -v --http2 https://...
```

```
// print negotiated protocol from a real client response
// resp.Proto == "HTTP/2.0" when h2 used
```

What to notice: Idle pool settings interact with h2 differently; leaking `Body` still kills reuse.

Try next: Load test one host with many parallel requests; compare conn count h2 vs forced h1.

database/sql Pool and Context

database/sql Pool and Context

Overview

`database/sql` is a **generic pool** over driver connections. Most production bugs are pool limits, missing context, or holding `*sql.Rows` open—not SQL syntax.

Diagram: pool

```
db.QueryContext

     v
sql.DB pool
      idle conn
      open new (  MaxOpen)

     v
Rows / Tx   Close/Commit   return to pool
```

Essential settings

```go
db.SetMaxOpenConns(25)
db.SetMaxIdleConns(25)
db.SetConnMaxLifetime(time.Hour)
db.SetConnMaxIdleTime(10 * time.Minute)
```

Knob	If wrong
MaxOpen too low	queueing latency
MaxOpen too high	DB overload
MaxIdle 0	reconnect thrash
No context	hung queries pin conns

Context everywhere

```go
ctx, cancel := context.WithTimeout(ctx, 3*time.Second)
defer cancel()
rows, err := db.QueryContext(ctx, q, args...)
```

Cancel should abort driver ops (driver-dependent quality).

Rows and transactions

```go
defer rows.Close()   // always
tx.Commit / Rollback // always pair
```

Leaking rows = leaking conns = pool exhaustion under load.

pgx note

Many services prefer `jackc/pgx` pool directly; patterns still mirror: max conns, timeouts, close. See web db chapter.

Experiment

```go
// conceptual: open sqlite or pg, set MaxOpenConns(1),
// run two concurrent QueryContext with sleep in SQL,
// observe second waits for conn
```

What to notice: Pool size becomes a concurrency throttle.

Try next: Add `db.Stats()` logging of `InUse`, `Idle`, `WaitCount` in a service.

JSON Performance Patterns

JSON Performance Patterns

Overview

`encoding/json` is correct and convenient; hot paths often spend CPU in **reflection** and allocations. Patterns: reuse encoders, pre-size, avoid `map[string]any`. In Go 1.27, v1 is backed by the v2 engine (faster unmarshal) and `encoding/json/v2` is the supported new API.

Diagram: cost centers

```
flow:
  [Marshal]
       |
       v
  [Alloc]
```

Patterns

Pattern	Why
`json.NewEncoder(w).Encode`	stream to writer
`Decoder.DisallowUnknownFields`	strict APIs
Pool `bytes.Buffer`	cut allocs
Typed structs	field cache
`json.RawMessage`	delay parse
Codegen (easyjson etc.)	last resort

Encoder reuse caveats

Stateful encoders/decoders need care if concurrent—prefer per-goroutine or mutex; pooling buffers is safer than sharing one Encoder across Gs.

json/v2 (GA in Go 1.27)

`encoding/json/v2` is no longer an experiment. Marshal is roughly at parity with the old v1 engine; unmarshal is significantly faster. Stricter defaults (UTF-8, duplicate names, case-sensitive fields) are the usual migration cost.

```go
import jsonv2 "encoding/json/v2"

b, err := jsonv2.Marshal(e)
err = jsonv2.Unmarshal(b, &e, jsonv2.RejectUnknownMembers(true))
```

Keep v1 imports for existing handlers; migrate hot paths and new APIs to v2 after golden-file tests pass. Emergency rollback: `GOEXPERIMENT=nojsonv2` (233).

Experiment

```bash
go test -bench=. -benchmem
```

```go
type E struct {
    A int    `json:"a"`
    B string `json:"b"`
}
func BenchmarkMarshal(b *testing.B) {
    e := E{1, "x"}
    for i := 0; i < b.N; i++ {
        _, _ = json.Marshal(e)
    }
}
```

What to notice: alloc/op and ns/op; compare with streaming encode to `io.Discard`.

Try next: Profile a handler that `Marshal`s large slices every request.

errgroup Patterns

errgroup Patterns

Overview

`golang.org/x/sync/errgroup` runs a set of goroutines and returns the **first error**, optionally canceling siblings via derived context. Prefer it over ad-hoc WaitGroup+mutex error slots for fan-out.

Diagram: cancel on first error

```
sequence (top → bottom):
  actors: errgroup, task A, task B
  errgroup --> task A  : Go
  errgroup --> task B  : Go
  task A --> errgroup  : err
  errgroup --> task B  : ctx cancel
  errgroup --> errgroup  : Wait returns err
```

Basic API

```go
g, ctx := errgroup.WithContext(parent)
g.Go(func() error {
    return work(ctx)
})
g.Go(func() error {
    return other(ctx)
})
if err := g.Wait(); err != nil {
    return err
}
```

Limits

```go
g.SetLimit(8) // max concurrent Gs in this group (modern x/sync)
```

Bounds fan-out without a hand-rolled semaphore.

Rules

1. Pass `ctx` into every blocking call
2. Return errors; do not log+ignore inside Go funcs without policy
3. `Wait` once; do not reuse group
4. First error wins—others may still run until cancel observed

vs WaitGroup

	WaitGroup	errgroup
Error prop	manual	built-in
Cancel siblings	manual	WithContext
Limit concurrency	manual	SetLimit

Experiment

```go
go get golang.org/x/sync/errgroup
```

```go
g, ctx := errgroup.WithContext(context.Background())
g.Go(func() error {
    <-ctx.Done()
    return ctx.Err()
})
g.Go(func() error {
    return errors.New("boom")
})
fmt.Println(g.Wait())
```

What to notice: First error cancels context; the other returns `context.Canceled` or exits promptly if it watches ctx.

Try next: Crawl N URLs with SetLimit(5) and a shared ctx timeout.

Why a Goroutine Switch Is Cheap: The Mechanism

Why a Goroutine Switch Is Cheap: The Mechanism

Thesis

A goroutine switch on Go 1.27.2 takes about **100–185 ns** on the test machine below. A blocking hand-off between two OS threads takes **0.7–3.6 µs**, between 4 and 20 times longer. The usual explanation is "Go saves fewer registers", which is true but explains only a tiny part of the gap. In the CPU profile later in this chapter, `runtime.gogo`, the routine that restores registers, accounts for **under 1%** of a channel ping-pong.

The gap really comes from three things:

1. **The switch is a function call.** Goroutines give up the CPU inside a runtime call (`gopark` → `mcall`). Go's internal ABI has no callee-saved registers, so the compiler has already spilled every live value to the stack before that call. Only a handful of values (SP, PC, BP, the closure context and the `g` pointer) describe where to resume.
2. **It never enters the kernel.** Parking, readying and resuming a goroutine are loads, stores and a few atomics on runtime structures in user space. No trap, no kernel scheduler, no TLB or address-space work, and the OS thread keeps running.
3. **Stacks are small and movable.** A parked goroutine costs about 2 KiB of stack, so a million of them fit in RAM, and growth is a prologue check plus a copy.

The chapter goes through each mechanism in the Go 1.27.2 runtime source, measures it, and then shows the cases where a "goroutine switch" costs as much as a thread switch: blocking syscalls, hand-offs to idle Ps, and signal-based preemption.

Scheduler Deep Dive (GMP) introduces G, M and P, and Stack Growth and Copying introduces stack growth. This chapter goes one level lower, to the instructions.

Verified on **Go 1.27.2** (`linux/amd64`), Linux 6.12 on an 8-vCPU KVM guest (Intel Xeon), on 2026-10-11. All source links are pinned to the `go1.27.2` tag.

Learning goals

- Name the fields in `g`, `m`, `p` and `gobuf` that a switch touches
- Read `gogo`, `mcall`, `morestack` and `asyncPreempt` in `asm_amd64.s` / `preempt_amd64.s`
- Explain why a cooperative switch saves only SP/PC/BP/ctxt, and why asynchronous preemption saves everything
- Follow `gopark` → `park_m` → `schedule` → `execute` → `gogo`, and `goready` → `runnext`

- Measure goroutine and thread hand-offs with benchmarks, `perf stat`, `getrusage` and `schedtrace`
- Predict when a goroutine operation turns into an OS context switch

1. The data: `g`, `m`, `p` and `gobuf`

Excerpts from `runtime2.go` (Go 1.27.2):

```
type gobuf struct          // runtime2.go:303
    sp   uintptr
    pc   uintptr
    g    guintptr
    ctxt unsafe.Pointer      // closure context (DX on amd64); a GC root
    lr   uintptr             // link register (arm64, ppc64, …); unused on amd64
    bp   uintptr             // frame pointer
}

type g struct              // runtime2.go:471
    stack        stack       // [lo, hi) of this goroutine's stack
    stackguard0  uintptr     // compared in every function prologue
    m            *m          // current M, nil when not running
    sched        gobuf       // where to resume
    atomicstatus atomic.Uint32
    goid         uint64
    waitreason   waitReason
    preempt      bool
    lockedm      muintptr    // set by runtime.LockOSThread
    // …
}

type m struct              // runtime2.go:616
    g0   *g                  // goroutine with the scheduling stack
    curg *g                  // goroutine currently running on this M
    p    puintptr            // attached P, nil in a syscall
    // …
}

type p struct              // runtime2.go:774
    runqhead uint32
    runq     [256]guintptr   // local run queue
    runnext  guintptr        // next G to run, inheriting the time slice
    // …
}
```

Some things to notice:

- A paused goroutine is represented by **one `gobuf`**: six words, two of which (`g`, `lr`) are bookkeeping on amd64. Everything else the goroutine needs (locals, spilled registers, return addresses) is on its own stack, which stays where it is.
- Every M has a `g0`, a goroutine with a normal-sized OS-thread stack, on which the scheduler itself runs. A switch is therefore two jumps: user G → `g0` (to run scheduler code) → next G.
- The **current `g` lives in R14** on amd64 under ABIInternal (ABI spec, amd64 table). Restoring "which goroutine am I" is a single register move.

2. Why so few registers: the ABI and cooperative safe-points

The Go internal ABI (`abi-internal.md`) says, for amd64:

> There are no callee-save registers, so a call may overwrite any register that doesn't have a fixed meaning, including argument registers.

The section on clobber sets adds the reason: "As defined, the ABI does not use callee-save registers. This significantly simplifies the garbage collector and the compiler's register allocator, but at some performance cost."

The consequence for switching: a goroutine always parks **inside a function call**. Channel operations, mutexes, `time.Sleep`, network I/O and `runtime.Gosched` all end up in `gopark` or `mcall`. Because no register survives a call, the compiler has already stored every live value in the caller's frame before the call instruction. The non-cooperative preemption design (proposal 24543) states the property directly: function-call safe-points "ensure that no registers are live", and the GC "has complete information about its entire stack" at them. So the state to save is just *where the goroutine is*:

Saved by a cooperative switch (amd64)	Why
SP	top of this goroutine's stack
PC	the return address into the caller
BP	frame pointer, so profilers and tracebacks can unwind
ctxt (DX)	closure context, only non-zero on a few paths such as `morestack`
g (R14, TLS)	identity of the running goroutine

Compare that with a kernel thread switch. The kernel cannot know which registers are live at an arbitrary interrupt point, so it saves the full user register file on kernel entry. The FPU/SSE/AVX state is saved with `XSAVE`-family instructions when the next task needs the FPU (on AVX-512 machines that is over 2 KiB). After that it still runs the kernel scheduler.

The comparison is not "Go saves fewer registers than C". A hand-written user-space context switch in C, such as Boost.Context's `fcontext`, also saves only the callee-saved registers plus SP and PC (about 8 words on x86-64 SysV), and it is also fast. glibc's `swapcontext` is slower because it also makes an `rt_sigprocmask` system call. The expensive part of a thread switch is the **kernel path**: trap, scheduler, wakeup, mitigations. Section 6 measures this.

3. The instructions: `mcall` and `gogo`

Going to sleep: `mcall` (asm_amd64.s#L425)

`gopark` ends with `mcall(park_m)`. `mcall` saves the caller's resume point into `g.sched`, switches to `g0`'s stack and calls `park_m(g)` there:

```
TEXT runtime·mcall<ABIInternal>(SB), NOSPLIT, $0-8
    MOVQ    AX, DX  // DX = fn
    // Save state in g->sched. The caller's SP and PC are restored by gogo to
    // resume execution in the caller's frame (implicit return). …
    MOVQ    SP, BX  // hide (SP) reads from vet
    MOVQ    8(BX), BX   // caller's PC
    MOVQ    BX, (g_sched+gobuf_pc)(R14)
    LEAQ    fn+0(FP), BX    // caller's SP
    MOVQ    BX, (g_sched+gobuf_sp)(R14)
    MOVQ    (BP), BX // caller's BP
    MOVQ    BX, (g_sched+gobuf_bp)(R14)

    // switch to m->g0 & its stack, call fn
    MOVQ    g_m(R14), BX
    MOVQ    m_g0(BX), SI    // SI = g.m.g0
    …
    MOVQ    R14, AX     // AX (and arg 0) = g
    MOVQ    SI, R14     // g = g.m.g0
    get_tls(CX)     // Set G in TLS
    MOVQ    R14, g(CX)
    MOVQ    (g_sched+gobuf_sp)(R14), SP // sp = g0.sched.sp
    MOVQ    $0, BP  // clear frame pointer, as caller may execute on another M
    PUSHQ   AX  // open up space for fn's arg spill slot
    MOVQ    0(DX), R12
    CALL    R12     // fn(g)
```

(… marks lines omitted here, such as the `runtimesecret` experiment and the `badmcall` check.) The whole save is **three stores**: PC, SP, BP. The stored PC is the *caller's* return address, so when the goroutine resumes it looks to the caller as though `mcall` simply returned.

Waking up: `gogo` (asm_amd64.s#L402)

When the scheduler on `g0` has picked the next goroutine, `execute` calls `gogo(&gp.sched)`:

```
TEXT runtime·gogo(SB), NOSPLIT, $0-8
    MOVQ    buf+0(FP), BX       // gobuf
    MOVQ    gobuf_g(BX), DX
    MOVQ    0(DX), CX       // make sure g != nil
    JMP gogo<>(SB)
```

```
TEXT gogo<>(SB), NOSPLIT, $0
    get_tls(CX)
    MOVQ    DX, g(CX)
    MOVQ    DX, R14     // set the g register
    MOVQ    gobuf_sp(BX), SP    // restore SP
    MOVQ    gobuf_ctxt(BX), DX
    MOVQ    gobuf_bp(BX), BP
    MOVQ    $0, gobuf_sp(BX)    // clear to help garbage collector
    MOVQ    $0, gobuf_ctxt(BX)
    MOVQ    $0, gobuf_bp(BX)
    MOVQ    gobuf_pc(BX), BX
    JMP BX
```

That is the whole context switch: about 15 instructions, all memory moves, ending in an indirect `JMP`. No privilege change and no system call.

The related helpers in 1.27.2 are `systemstack` (L489), which runs a function on `g0` and comes back (used by `goready`), and `gosave_systemstack_switch` (L855), its save half. There is no longer a standalone `runtime·gosave` on amd64.

4. The control flow: `gopark`, `goready` and `runnext`

```
G1: ch <- v    no receiver    gopark   mcall(park_m)
                                                    on g0
                                  park_m: casgstatus(_Grunning→_Gwaiting)
                                          dropg(); unlockf(); schedule()
                                  schedule   findRunnable   execute(G2)
                                  execute   gogo(&G2.sched)    G2 runs

G2: <-ch    sender waiting   goready(G1)   systemstack(ready)
                                  ready: casgstatus(_Gwaiting→_Grunnable)
                                         runqput(pp, G1, next=true)   // runnext
                                         wakep()   // maybe wake an idle P
```

- `gopark` records the wait reason and unlock function on the M and calls `mcall(park_m)`.
- `park_m` changes the status, detaches G from M (`dropg`), runs the unlock callback (for channels, `chanparkcommit` releases the channel lock only *after* the G is safely parked), and calls `schedule`.
- `goready` → `ready` puts the woken G in `p.runnext`. When the current G blocks, `schedule` finds it there first and calls `execute(gp, inheritTime=true)`.
- `inheritTime` means the woken goroutine shares the current time slice, and `execute` skips `schedtick++`. A ping-pong pair therefore behaves like one goroutine to the scheduler, which is why its fairness depends on `sysmon` preemption. The comment in `runqput` explains this. You can see the effect in `schedtrace` below: millions of switches, but only hundreds of `schedticks`.

Where the time goes

A CPU profile of the channel ping-pong with one P. The `pingpong` program and this `bench_test.go` are in section 6:

```
go test -run '^$' -bench 'Chan' -cpu 1 -count 1 -cpuprofile cpu.out .
go tool pprof -top -nodecount=22 pingpong.test cpu.out
```

```
BenchmarkChan    3517398           395.4 ns/op
File: pingpong.test
Build ID: 58a049978dfd3a0ef99c082470e52b0ec2b46c98
Type: cpu
Time: 2026-10-11 22:39:55 IST
Duration: 1.74s, Total samples = 1740ms (100.14%)
Showing nodes accounting for 1570ms, 90.23% of 1740ms total
Showing top 22 nodes out of 51
      flat  flat%   sum%        cum   cum%
     200ms 11.49% 11.49%      270ms 15.52%  runtime.casgstatus
     170ms  9.77% 21.26%      170ms  9.77%  runtime.nanotime (inline)
     150ms  8.62% 29.89%      150ms  8.62%  runtime.unlock2
     130ms  7.47% 37.36%      130ms  7.47%  runtime.lock2
     120ms  6.90% 44.25%      490ms 28.16%  runtime.chanrecv
     110ms  6.32% 50.57%      750ms 43.10%  example.com/pingpong.chanPingPong
     100ms  5.75% 56.32%      100ms  5.75%  runtime.(*guintptr).cas (inline)
      80ms  4.60% 60.92%       80ms  4.60%  internal/runtime/atomic.(*Int32).Add (inline)
      70ms  4.02% 64.94%      440ms 25.29%  runtime.chansend
      70ms  4.02% 68.97%       70ms  4.02%  runtime.pidlegetSpinning
      60ms  3.45% 72.41%       60ms  3.45%  runtime.releaseSudog
      50ms  2.87% 75.29%      390ms 22.41%  example.com/pingpong.chanPingPong.func1
      50ms  2.87% 78.16%       50ms  2.87%  internal/runtime/atomic.(*Int32).CompareAndSwap
      30ms  1.72% 79.89%      520ms 29.89%  runtime.chanrecv1
      30ms  1.72% 81.61%      220ms 12.64%  runtime.findRunnable
      30ms  1.72% 83.33%      580ms 33.33%  runtime.park_m
      20ms  1.15% 84.48%       20ms  1.15%  internal/runtime/atomic.(*Uint8).Store (inline)
      20ms  1.15% 85.63%       20ms  1.15%  runtime.(*timeHistogram).record
      20ms  1.15% 86.78%       80ms  4.60%  runtime.chanparkcommit
      20ms  1.15% 87.93%      460ms 26.44%  runtime.chansend1
      20ms  1.15% 89.08%      130ms  7.47%  runtime.execute
      20ms  1.15% 90.23%       20ms  1.15%  runtime.gopark
```

And the switch primitives themselves, further down the same profile:

```
go tool pprof -top -nodecount=60 pingpong.test cpu.out | grep -E 'gogo|mcall|schedule|findRu
```

```
      flat  flat%   sum%        cum   cum%
     0.03s  1.72% 81.61%      0.22s 12.64%  runtime.findRunnable
```

```
      0.01s  0.57% 98.28%      0.37s 21.26%  runtime.schedule
      0.01s  0.57% 98.85%      0.45s 25.86%  runtime.systemstack
          0     0%   100%      0.58s 33.33%  runtime.mcall
          0     0%   100%      0.05s  2.87%  runtime.runqget (inline)
```

`runtime.gogo` and `gosave_systemstack_switch` do not appear at all: in 1.74 s of samples, none landed in them. An earlier run of the same benchmark gave each 0.6% flat. `mcall` shows 0 flat and 33% cumulative, because what it calls is the scheduler. The cost is in *bookkeeping*: atomic status transitions (`casgstatus`), the channel lock (`lock2`/`unlock2`), timestamps (`nanotime`, used for the scheduler-latency metrics that `casgstatus` samples) and checking for idle Ps to wake. That is why even a "cheap" switch costs ~100 ns, not the ~5 ns that 15 moves would suggest.

5. Small, growable stacks

Cheap switching only helps if you can afford many goroutines, and that depends on stack size. Every Go function prologue (except `//go:nosplit` ones) compares SP with `g.stackguard0`. Here is `deep` from the listing below, disassembled:

```
go build -o stacks . && go tool objdump -s 'main.deep$' stacks | head -8
```

```
TEXT main.deep(SB) /workspace/go127lab/sw/stacks/main.go
  main.go:19        0x49fe00        4c8d6424f0      LEAQ -0x10(SP), R12
  main.go:19        0x49fe05        4d3b6610        CMPQ R12, 0x10(R14)
  main.go:19        0x49fe09        0f86a1000000        JBE 0x49feb0
  main.go:19        0x49fe0f        55          PUSHQ BP
  main.go:19        0x49fe10        4889e5          MOVQ SP, BP
  main.go:19        0x49fe13        4881ec88000000      SUBQ $0x88, SP
  main.go:20        0x49fe1a        488d4c2408      LEAQ 0x8(SP), CX
```

`0x10(R14)` is `g.stackguard0` (offset 16, after `stack.lo` and `stack.hi`). The slow path at the end of the function spills the register argument, calls `morestack` and retries:

```
  main.go:19        0x49feb0        4889442408      MOVQ AX, 0x8(SP)
  main.go:19        0x49feb5        e8660cfeff      CALL runtime.morestack_noctxt.abi0(SB)
  main.go:19        0x49feba        488b442408      MOVQ 0x8(SP), AX
  main.go:19        0x49febf        90          NOPL
  main.go:19        0x49fec0        e93bffffff      JMP main.deep(SB)
```

`morestack` uses the same trick as `mcall`. It saves `f`'s PC/SP/BP/ctxt into `g.sched`, switches to `g0` and calls `newstack`. That function allocates a stack of twice the size (`newsize := oldsize * 2`, stack.go:1179) and calls `copystack`. `copystack` copies the old stack and uses the compiler's stack maps to adjust every pointer into it. Stacks can move because the runtime knows exactly where every pointer to them is.

The same prologue check is the **cooperative preemption** hook. To preempt a goroutine, the runtime sets `stackguard0 = stackPreempt`, so the next prologue "fails", and `newstack` reschedules

instead of growing (comment at the top of preempt.go).

Measure both effects:

```go
// main.go
package main

import (
    "fmt"
    "runtime"
    "runtime/metrics"
    "sync"
    "unsafe"
)

func stackBytes() uint64 {
    s := []metrics.Sample{{Name: "/memory/classes/heap/stacks:bytes"}}
    metrics.Read(s)
    return s[0].Value.Uint64()
}

//go:noinline
func deep(n int) int {
    var pad [128]byte // make each frame big enough to force growth
    if n == 0 {
        return int(pad[0])
    }
    return deep(n-1) + int(pad[n%128])
}

func main() {
    // 1. What does a parked goroutine cost?
    const N = 100_000
    runtime.GC()
    before := stackBytes()
    var wg sync.WaitGroup
    block := make(chan struct{})
    for range N {
        wg.Add(1)
        go func() { wg.Done(); <-block }()
    }
    wg.Wait()
    fmt.Printf("stack memory per parked goroutine: %d bytes\n", (stackBytes()-before)/N)
    close(block)

    // 2. Stacks move when they grow.
    done := make(chan struct{})
```

```go
    go func() {
        var x int
        p0 := uintptr(unsafe.Pointer(&x))
        deep(10_000) // needs well over 1 MiB of stack
        p1 := uintptr(unsafe.Pointer(&x))
        fmt.Printf("&x before growth: %#x\n&x after growth:  %#x\nmoved: %v\n", p0, p1, p0 !
        close(done)
    }()
    <-done
}
```

```
go run .
```

```
stack memory per parked goroutine: 2049 bytes
&x before growth: 0x3d45a4f1bf80
&x after growth:  0x3d45b546ff80
moved: true
```

The minimum stack is `stackMin = 2048` (stack.go:78). `GODEBUG=adaptivestackstart=1`, the default since Go 1.19, lets the runtime start new goroutines with a larger stack if the GC sees that goroutines usually grow. That did not happen here, since the goroutines never grew. A million parked goroutines cost about 2 GB of stack on this measurement. A million OS threads would reserve 8 GiB of virtual stack at glibc's default and would also cost kernel structures for each thread. The addresses will differ on your machine, but `moved: true` will not.

6. Measuring: goroutine vs OS-thread hand-off

One program with four ways to bounce control between two parties:

- **gosched**: two goroutines alternate with `runtime.Gosched` (the bare switch, through the global run queue)
- **chan**: two goroutines, unbuffered channels (the realistic Go case)
- **futex**: two goroutines locked to their own OS threads (`runtime.LockOSThread`) that sleep and wake each other with raw `futex(2)`, so every hand-off is a kernel thread switch
- **pipe**: the same, using blocking `read(2)`/`write(2)` on pipes through `syscall.Read`, which goes through the runtime's syscall path

```go
// main.go
//
// Ping-pong between two execution contexts and report the cost of one
// hand-off (half a round trip). Modes:
//
//  gosched  two goroutines taking turns with runtime.Gosched (bare switch)
//  chan   two goroutines, unbuffered channels (runtime switch)
//  futex  two locked OS threads, raw futex wait/wake (kernel switch)
```

```go
//  pipe   two locked OS threads, blocking read/write on pipes
package main

import (
    "flag"
    "fmt"
    "os"
    "runtime"
    "sync/atomic"
    "syscall"
    "time"
    "unsafe"
)

const (
    futexWait = 0 | 128 // FUTEX_WAIT | FUTEX_PRIVATE_FLAG
    futexWake = 1 | 128 // FUTEX_WAKE | FUTEX_PRIVATE_FLAG
)

func chanPingPong(n int) {
    ping, pong := make(chan struct{}), make(chan struct{})
    go func() {
        for range n {
            <-ping
            pong <- struct{}{}
        }
    }()
    for range n {
        ping <- struct{}{}
        <-pong
    }
}

// With GOMAXPROCS=1, each Gosched hands the P to the other goroutine,
// so one call is one goroutine switch.
func goschedPingPong(n int) {
    done := make(chan struct{})
    go func() {
        for range n {
            runtime.Gosched()
        }
        close(done)
    }()
    for range n {
        runtime.Gosched()
    }
    <-done
```

```go
}

// turn is 0 when it is the main thread's turn and 1 for the partner.
func futexPingPong(n int) {
    var turn atomic.Uint32
    addr := uintptr(unsafe.Pointer(&turn))
    wait := func(v uint32) { // sleep in the kernel while turn == v
        for turn.Load() == v {
            syscall.RawSyscall6(syscall.SYS_FUTEX, addr, futexWait, uintptr(v), 0, 0, 0)
        }
    }
    wake := func(v uint32) {
        turn.Store(v)
        syscall.RawSyscall6(syscall.SYS_FUTEX, addr, futexWake, 1, 0, 0, 0)
    }
    done := make(chan struct{})
    go func() {
        runtime.LockOSThread()
        for range n {
            wait(0)
            wake(0)
        }
        close(done)
    }()
    runtime.LockOSThread()
    for range n {
        wake(1)
        wait(1)
    }
    <-done
}

func pipePingPong(n int) {
    var a, b [2]int
    must(syscall.Pipe(a[:]))
    must(syscall.Pipe(b[:]))
    buf := []byte{0}
    done := make(chan struct{})
    go func() {
        runtime.LockOSThread()
        x := []byte{0}
        for range n {
            syscall.Read(a[0], x)
            syscall.Write(b[1], x)
        }
        close(done)
    }()
```

```
    runtime.LockOSThread()
    for range n {
        syscall.Write(a[1], buf)
        syscall.Read(b[0], buf)
    }
    <-done
}

func must(err error) {
    if err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }
}

func rusage() (vol, invol int64) {
    var ru syscall.Rusage
    must(syscall.Getrusage(syscall.RUSAGE_SELF, &ru))
    return ru.Nvcsw, ru.Nivcsw
}

func main() {
    mode := flag.String("mode", "chan", "gosched | chan | futex | pipe")
    n := flag.Int("n", 1_000_000, "round trips")
    flag.Parse()

    run := map[string]func(int){"gosched": goschedPingPong, "chan": chanPingPong, "futex": f
    if run == nil {
        must(fmt.Errorf("unknown mode %q", *mode))
    }
    run(*n / 10) // warm up threads, stacks and caches

    v0, i0 := rusage()
    start := time.Now()
    run(*n)
    el := time.Since(start)
    v1, i1 := rusage()

    handoffs := 2 * *n
    fmt.Printf("mode=%-7s GOMAXPROCS=%d round-trips=%d  %6.1f ns/hand-off  ctx-switches: vol
        *mode, runtime.GOMAXPROCS(0), *n, float64(el.Nanoseconds())/float64(handoffs),
        v1-v0, i1-i0, float64(v1-v0+i1-i0)/float64(handoffs))
}
```

```
// bench_test.go
package main
```

```go
import "testing"

// One op is one round trip (two hand-offs). The futex and pipe modes are
// left out: with -cpu 1 the raw futex wait holds the only P and deadlocks.
func BenchmarkGosched(b *testing.B) { goschedPingPong(b.N) }
func BenchmarkChan(b *testing.B)    { chanPingPong(b.N) }
```

```bash
go build -o pingpong .
go test -run '^$' -bench . -cpu 1 -count 1 .
```

```
goos: linux
goarch: amd64
pkg: example.com/pingpong
cpu: Intel(R) Xeon(R) Processor
BenchmarkGosched     6234708           191.6 ns/op
BenchmarkChan        3524346           349.4 ns/op
PASS
ok      example.com/pingpong    2.970s
```

Those are ns per round trip, so ~96 ns and ~175 ns per hand-off, which matches the standalone runs below.

Results

Goroutines with one P:

```bash
for i in 1 2 3; do GOMAXPROCS=1 ./pingpong -mode=gosched; done
for i in 1 2 3; do GOMAXPROCS=1 ./pingpong -mode=chan; done
```

```
mode=gosched GOMAXPROCS=1 round-trips=1000000    98.6 ns/hand-off  ctx-switches: voluntary=2
mode=gosched GOMAXPROCS=1 round-trips=1000000   100.6 ns/hand-off  ctx-switches: voluntary=2
mode=gosched GOMAXPROCS=1 round-trips=1000000    97.2 ns/hand-off  ctx-switches: voluntary=2
mode=chan    GOMAXPROCS=1 round-trips=1000000   169.9 ns/hand-off  ctx-switches: voluntary=3
mode=chan    GOMAXPROCS=1 round-trips=1000000   182.2 ns/hand-off  ctx-switches: voluntary=3
mode=chan    GOMAXPROCS=1 round-trips=1000000   173.2 ns/hand-off  ctx-switches: voluntary=3
```

Goroutines with eight Ps, and the two thread modes, unpinned:

```bash
./pingpong -mode=chan; ./pingpong -mode=futex; ./pingpong -mode=pipe
```

```
mode=chan    GOMAXPROCS=8 round-trips=1000000   153.7 ns/hand-off  ctx-switches: voluntary=1
mode=futex   GOMAXPROCS=8 round-trips=1000000  3523.8 ns/hand-off  ctx-switches: voluntary=1
mode=pipe    GOMAXPROCS=8 round-trips=1000000  4220.0 ns/hand-off  ctx-switches: voluntary=2
```

Thread modes pinned to **one CPU**, so the two threads switch on the same core instead of waking an idle core:

```
for m in futex pipe; do for i in 1 2; do GOMAXPROCS=4 taskset -c 0 ./pingpong -mode=$m -n 30
```

```
mode=futex   GOMAXPROCS=4 round-trips=300000   720.0 ns/hand-off  ctx-switches: voluntary=36
mode=futex   GOMAXPROCS=4 round-trips=300000   741.9 ns/hand-off  ctx-switches: voluntary=35
mode=pipe    GOMAXPROCS=4 round-trips=300000  1095.3 ns/hand-off  ctx-switches: voluntary=36
mode=pipe    GOMAXPROCS=4 round-trips=300000  1348.5 ns/hand-off  ctx-switches: voluntary=38
```

Hand-off	ns (this VM)	OS context switches per hand-off
`Gosched`, 1 P	~98	0.00
unbuffered chan, 1 P	~170–182	0.00
unbuffered chan, 8 Ps	~154–163	0.01
futex, threads on one CPU	~720–850	1.00
pipe, threads on one CPU	~1100–1350	1.03
futex, threads on different CPUs	~3500	0.94
pipe, threads on different CPUs	~4200	1.02

The `perf stat` and `schedtrace` runs below are separate runs, so their ns figures differ slightly from these.

How to read this:

- A goroutine hand-off costs **zero OS context switches**. The kernel never sees it. The goroutine is **4–7× cheaper than the best thread case** (same core, futex) and ~**20× cheaper** than the cross-core case, which is the usual one when threads block and wake each other on a multi-core machine.
- The thread figures include a little Go overhead (the `RawSyscall` wrapper and an atomic load), so they are a close approximation of a raw kernel thread switch, not an exact measurement. In the pinned runs about 40% of the switches are involuntary. With both threads on one CPU, the kernel most likely preempts the waker in favour of the thread it just woke, and the runtime's own threads share that CPU too.
- The cross-core thread case is the slowest because waking a sleeping thread on another CPU means an inter-processor interrupt and, on a VM, a trip through the hypervisor to wake an idle vCPU. Bare metal is usually faster in absolute terms. The ratio is what carries over.
- `pipe` costs more than `futex` because it adds real syscalls (`read`/`write`) and the runtime's `entersyscall`/`exitsyscall` bookkeeping.
- Treat these as one machine's numbers. Run the program on your own hardware.

`perf stat`: user time vs kernel time

```
sudo perf stat -e context-switches,cpu-migrations,task-clock -- env GOMAXPROCS=1 ./pingpong
sudo perf stat -e context-switches,cpu-migrations,task-clock -- env GOMAXPROCS=4 taskset -c
```

```
mode=chan    GOMAXPROCS=1 round-trips=1000000   169.7 ns/hand-off  ctx-switches: voluntary=3
 Performance counter stats for 'env GOMAXPROCS=1 ./pingpong -mode=chan':
               102      context-switches                 #  271.278 /sec
                 0      cpu-migrations                   #    0.000 /sec
            376.00 msec task-clock                      #    0.998 CPUs utilized
       0.376823484 seconds time elapsed
       0.377871000 seconds user
       0.000000000 seconds sys
mode=futex   GOMAXPROCS=4 round-trips=300000   850.9 ns/hand-off  ctx-switches: voluntary=34
 Performance counter stats for 'env GOMAXPROCS=4 taskset -c 0 ./pingpong -mode=futex -n 3000
            660299      context-switches                 #    1.186 M/sec
                 1      cpu-migrations                   #    1.796 /sec
            556.73 msec task-clock                      #    0.997 CPUs utilized
       0.558349824 seconds time elapsed
       0.126343000 seconds user
       0.431310000 seconds sys
```

(Blank lines were removed from `perf`'s output.) Two million goroutine switches took **0.000 s of system time** and 102 context switches in total, all of them unrelated runtime housekeeping. The thread version spent **77% of its CPU time in the kernel** (0.431 s of 0.558 s), with one kernel context switch per hand-off. That gap in system time is the kernel transition the goroutine never makes. The `perf` here is Debian's `linux-perf` 6.12. On this box `perf_event_paranoid=2` hides kernel software events from unprivileged users, hence `sudo`.

`schedtrace`: the scheduler's view

```
GODEBUG=schedtrace=250 ./pingpong -mode=chan -n 3000000
```

```
SCHED 0ms: gomaxprocs=8 idleprocs=5 threads=4 spinningthreads=1 needspinning=0 idlethreads=0
SCHED 256ms: gomaxprocs=8 idleprocs=6 threads=5 spinningthreads=1 needspinning=0 idlethreads
SCHED 568ms: gomaxprocs=8 idleprocs=6 threads=5 spinningthreads=1 needspinning=0 idlethreads
SCHED 821ms: gomaxprocs=8 idleprocs=5 threads=5 spinningthreads=2 needspinning=0 idlethreads
SCHED 1073ms: gomaxprocs=8 idleprocs=6 threads=5 spinningthreads=1 needspinning=0 idlethread
mode=chan    GOMAXPROCS=8 round-trips=3000000   163.3 ns/hand-off  ctx-switches: voluntary=3
```

Six million hand-offs ran on **five OS threads**, while `schedticks` reached at most about 1,300 per P in a little over a second. That is `runnext` with `inheritTime` at work: the pair keeps passing one time slice back and forth, and `execute` does not count those hand-offs as new scheduling rounds. `spinningthreads=1-2` is the price of eight Ps. When a goroutine is readied, `wakep` may wake an

idle P that then spins looking for work. Those wakeups most likely account for the ~11,000–12,000 voluntary OS switches per million round trips in the 8-P runs.

7. When a goroutine switch is not cheap

Blocking system calls: `entersyscall`, `handoffp` and new threads

A goroutine making a blocking syscall cannot be "switched out" in user space, because its OS thread is stuck in the kernel. The runtime detaches the P instead. `entersyscall` marks the P as in-syscall. If the call takes longer than one `sysmon` tick (at least 20 µs), `retake` takes the P and `handoffp` hands it to another M, starting a new thread if none is idle, whenever there is other work for that P. Goroutines that block in syscalls therefore cost threads, while goroutines parked in the runtime do not:

```go
// main.go
package main

import (
    "bufio"
    "fmt"
    "os"
    "strings"
    "syscall"
    "time"
)

func threads() string {
    f, _ := os.Open("/proc/self/status")
    defer f.Close()
    sc := bufio.NewScanner(f)
    for sc.Scan() {
        if v, ok := strings.CutPrefix(sc.Text(), "Threads:"); ok {
            return strings.TrimSpace(v)
        }
    }
    return "?"
}

func main() {
    const N = 50
    fmt.Println("start:                          threads =", threads())

    block := make(chan struct{})
    for range N {
        go func() { <-block }() // parks in the runtime: gopark
    }
```

```go
    time.Sleep(100 * time.Millisecond)
    fmt.Printf("%d goroutines parked on a chan: threads = %s\n", N, threads())

    var p [2]int
    if err := syscall.Pipe(p[:]); err != nil {
        panic(err)
    }
    for range N {
        go func() {
            var b [1]byte
            syscall.Read(p[0], b[:]) // blocks this OS thread in the kernel
        }()
    }
    time.Sleep(100 * time.Millisecond)
    fmt.Printf("%d goroutines blocked in read(2): threads = %s\n", N, threads())
}
```

```
go run .
GOMAXPROCS=2 go run .
```

```
start:                          threads = 5
50 goroutines parked on a chan: threads = 7
50 goroutines blocked in read(2): threads = 53
start:                          threads = 4
50 goroutines parked on a chan: threads = 4
50 goroutines blocked in read(2): threads = 53
```

Changing the read to an `*os.File` from `os.Pipe` keeps the thread count at 4 in the same experiment (`50 goroutines blocked in (*os.File).Read: threads = 4`, GOMAXPROCS=2). `os` registers pipes and sockets with the **netpoller**, so the goroutine parks with `gopark` and only one thread waits in `epoll`. Raw `syscall.Read`, cgo calls and file I/O on regular files have no such path. Each one in flight holds an OS thread, and waking it costs a thread switch plus `exitsyscall` reacquiring a P.

Hand-offs to idle Ps

When `goready` makes a goroutine runnable and there is an idle P, `wakep` may wake a parked M with a futex: a real OS wakeup. This is the 0.01 OS switches per hand-off in the 8-P runs above. It is amortized, but in fan-out patterns that repeatedly wake idle workers, these wakeups can dominate.

`LockOSThread` and the futex trap

A goroutine locked to a thread (`g.lockedm`) can only run on that M. Handing control to or from it means a real thread switch, as the futex and pipe numbers show. Mixing locked threads with

syscall.RawSyscall is worse. A raw syscall skips entersyscall, so the runtime thinks the goroutine is still running and never releases its P. The futex benchmark above **deadlocks** under taskset -c 0 without an explicit GOMAXPROCS. Go sets GOMAXPROCS from the CPU affinity mask, which gives 1 there, so the partner goroutine never gets a P to run on. The listings use GOMAXPROCS=4 for that reason, and you should keep RawSyscall for calls that cannot block.

Asynchronous preemption (Go 1.14+)

Cooperative safe-points are function prologues and blocking runtime calls. A loop without calls has none:

```go
// main.go
package main

import (
    "fmt"
    "runtime"
    "time"
)

//go:noinline
func spin(n *uint64) {
    for { // no calls, no allocation: no cooperative preemption point
        *n++
    }
}

func main() {
    runtime.GOMAXPROCS(1)
    var n uint64
    go spin(&n)

    start := time.Now()
    time.Sleep(10 * time.Millisecond) // main must get the only P back
    fmt.Printf("main resumed after %v (asked for 10ms)\n", time.Since(start).Round(time.Mill
}
```

```bash
go build -o preempt . && ./preempt && ./preempt
GODEBUG=asyncpreemptoff=1 timeout 5 ./preempt; echo "exit=$?"
```

```
main resumed after 20ms (asked for 10ms)
main resumed after 14ms (asked for 10ms)
exit=124
```

With asynchronous preemption, sysmon notices that the spinning goroutine has run for more than forcePreemptNS = 10ms (proc.go:6679) and calls preemptone, which sends **SIGURG** to its thread

(`sigPreempt = _SIGURG`, signal_unix.go:74). Without it (`asyncpreemptoff=1`) the program never prints and `timeout` kills it, giving exit 124. Here is the signal under `strace`:

```
strace -f -e trace=tgkill -e signal=SIGURG ./preempt 2>&1 | grep -E 'tgkill|SIGURG|main resu
```

```
[pid 513631] tgkill(513630, 513630, SIGURG <unfinished ...>
[pid 513630] --- SIGURG {si_signo=SIGURG, si_code=SI_TKILL, si_pid=513630, si_uid=1000} ---
[pid 513631] <... tgkill resumed>)      = 0
main resumed after 10ms (asked for 10ms)
```

Thread 513631 (`sysmon`) signals 513630, the thread running `spin`. The signal handler checks that the interrupted PC is an *async safe-point* (`isAsyncSafePoint`). If it is, the handler rewrites the signal context so the thread "calls" `asyncPreempt`. That function is the opposite of `mcall`: because it interrupts arbitrary code, it **saves every register**, 14 general-purpose registers plus flags, then the vector register state into `p.xRegs`. The generated file has 179 lines of register `MOV`s across its save and restore halves. Then it enters the scheduler. The GC scans that innermost frame **conservatively**, since there is no register map for an arbitrary PC. The comment block in `preempt.go` describes this design.

So preemption is the expensive kind of goroutine switch: a signal (a kernel round trip), a full register save, and a conservative scan. It happens at most about every 10 ms per goroutine that hogs a P, so its amortized cost is negligible. Deadlock-free fairness is its purpose, not speed.

8. Putting it together

	Goroutine switch (cooperative)	Goroutine switch (async preempt)	OS thread switch
Trigger	blocking runtime call, `Gosched`, prologue check	`SIGURG` from `sysmon` after 10 ms	blocking syscall, futex, timer interrupt
State saved	SP, PC, BP, ctxt, g (5 words)	all GPRs + flags + vector state	full user register file + FPU/vector state (kernel)
Kernel entry	none	one signal delivery	trap + kernel scheduler
Who decides next	Go scheduler on `g0` (`runnext`, local runq, global, steal, netpoll)	same	kernel scheduler
Typical cost (this VM)	~100–185 ns	rare; not on the hot path	~0.7 µs same-core, ~3–4 µs cross-core
Stack	2 KiB+, growable, movable	same	fixed reservation (8 MiB default)

The trap

- **"Goroutines are free, so block in syscalls freely."** Each goroutine blocked in a raw syscall or cgo call holds an OS thread. Use `os/net` types, which use the netpoller, or bound the concurrency.
- **Benchmarking switches with `GOMAXPROCS>1` and calling it "switch cost".** Part of what you measure is idle-P wakeups and parallelism. Measure with `GOMAXPROCS=1` and with your production setting.
- **`RawSyscall` plus `LockOSThread`.** The P is never released. Under `taskset` or a cgroup CPU limit of 1 it can deadlock.
- **Hot ping-pong pairs starving others.** `runnext` with `inheritTime` lets a pair share one time slice. Fairness relies on `sysmon`'s 10 ms preemption, so expect tail-latency spikes of that size for other goroutines on the P.
- **Explaining the speed by register count.** In the profile, the switch itself is ~1%. Bookkeeping is the rest.

The boring rule

Goroutine switches are cheap when goroutines block on Go things (channels, mutexes, timers, network and pipe I/O through `os/net`). Anything that blocks in the kernel directly is a thread switch with a goroutine wrapped around it.

Try this

1. Run `pingpong` with `GOMAXPROCS=2` and pin it with `taskset -c 0,1`. Does the channel hand-off get slower or faster than with 1 P?
2. Replace the unbuffered channels with `sync.Mutex` + `sync.Cond` and compare ns per hand-off.
3. Add a call to an empty `//go:noinline func tick() {}` inside the spin loop and rerun with `GODEBUG=asyncpreemptoff=1`. On Go 1.27.2 it still hangs. Disassemble `tick` with `go tool objdump -s main.tick` to see why: a leaf function with no frame has no prologue check. Then make `tick` call a second `//go:noinline` function and try again. Now `tick` has a `CMPQ SP, 0x10(R14)` prologue, and the program resumes.
4. Count threads while 50 goroutines sleep in `time.Sleep` versus `syscall.Nanosleep`.

Sources

Runtime source, pinned at the `go1.27.2` tag:

- `gobuf`, `g`, `m`, `p`: https://github.com/golang/go/blob/go1.27.2/src/runtime/runtime2.go#L303
- `gogo`, `mcall`, `systemstack`, `morestack`: https://github.com/golang/go/blob/go1.27.2/src/runtime/asm_amd64.s#L402
- `asyncPreempt` (generated by `mkpreempt.go`): https://github.com/golang/go/blob/go1.27.2/src/runtime/preempt_amd64.s

- `gopark`, `goready`, `ready`, `execute`, `schedule`, `park_m`, `entersyscall`, `handoffp`, `sysmon`, `retake`, `runqput`: https://github.com/golang/go/blob/go1.27.2/src/runtime/proc.go
- preemption design comment, `isAsyncSafePoint`: https://github.com/golang/go/blob/go1.27.2/src/runtime/preempt.go
- `newstack`, `copystack`, `stackMin`: https://github.com/golang/go/blob/go1.27.2/src/runtime/stack.go
- `sigPreempt`: https://github.com/golang/go/blob/go1.27.2/src/runtime/signal_unix.go#L74

Design documents:

- Go internal ABI specification: https://github.com/golang/go/blob/go1.27.2/src/cmd/compile/abi-internal.md
- Proposal: Non-cooperative goroutine preemption (#24543): https://github.com/golang/proposal/blob/master/design/24543-non-cooperative-preemption.md
- Scalable Go Scheduler Design Doc (Vyukov, 2012): https://golang.org/s/go11sched
- Contiguous stacks (Go 1.3): https://docs.google.com/document/d/1wAaf1rYoM4S4gtnPh0zOlGzWtrZFQ5spub

Tools: `perf stat` (linux-perf 6.12), `strace`, `GODEBUG=schedtrace`, `go tool pprof`, `go tool objdump`.

“Let the Compiler Optimize It”: Measuring netip.AddrFrom16(ip.As16())

“Let the Compiler Optimize It”: Measuring `netip.AddrFrom16(ip.As16())`

Thesis

`net/netip` can unmap an address (`::ffff:203.0.113.10` → `203.0.113.10` with `Addr.Unmap`) but has no method for the other direction. Proposals for one (`Addr.Map`, `Addr.To6`) were declined: the operation was judged rare, and `netip.AddrFrom16(ip.As16())` was expected to be “neither allocating nor particularly slow”, something the compiler can flatten after inlining. This chapter treats that expectation as a claim to test. On Go 1.27.2 (linux/amd64) it does not hold:

- The idiom costs about **8.5 ns** per call. Flipping the address-family field directly costs about **1.1 ns**.
- Go 1.26.9 compiles the idiom to **16** instructions. Go 1.27.2 compiles it to **56**, because a new SSA rewrite rule runs before the pass that used to merge the byte loads. Wall-clock time is **the same on both**, within noise on this machine.
- Instruction count does not explain the time. Reading the same 16 bytes from an array written long ago costs 1.8 ns. Reading them straight after writing them costs 8.5 ns. The cost is the trip through a freshly written stack array, not the arithmetic.

The skills here apply to any “the compiler will handle it” advice. Write the variants, benchmark them, read the assembly with `-gcflags=-S`, read the SSA with `GOSSAFUNC` or a per-pass dump, compare two toolchains with `GOTOOLCHAIN`, and only then decide whether you need `unsafe`, an `-overlay`, or a different design.

Verified on **Go 1.27.2** and **Go 1.26.9** (linux/amd64), Linux 6.12 on an 8-vCPU KVM guest (Intel Xeon), on 2026-10-15. Source links are pinned to the `go1.27.2` tag. The investigation follows Vincent Bernat’s write-up (see Sources), re-run here on the current toolchain.

Compiler, SSA, Inlining, and PGO and Compiler Mid/Back: IR, SSA, Machine Code, Link describe the pipeline in general. This chapter applies it to one function, end to end.

Learning goals

- Explain how `netip.Addr` stores an address and why `As16`/`AddrFrom16` turn it into bytes and back
- Count the instructions in a function’s `-gcflags=-S` listing, and compare them across `GOAMD64` levels and toolchains
- Dump the SSA after a single pass (`-d=ssa/memcombine/dump=`) and recognise Store / Move / Load chains

- Explain why a rewrite rule added in Go 1.27 tripled the instruction count without changing the time
- Separate instruction cost from memory-forwarding cost with a controlled benchmark
- Choose between the idiom, an `unsafe` layout proxy, an `-overlay` patch, and converting at the boundary

Mental model

```
netip.Addr (24 bytes on amd64)

  addr.hi  uint64       addr.lo  uint64       z  unique.Handle (ptr)
  bytes 0..7 of the     bytes 8..15, big-     z4 = IPv4
  IPv6 form, big-endian endian                z6noz = IPv6, no zone


An IPv4 Addr already holds ::ffff:a.b.c.d in addr. "To6" only has to change z.

AddrFrom16(ip.As16()), after inlining:

   hi, lo  BEPutUint64   a16 [16]byte  copy   addr [16]byte  BEUint64   hi, lo
           (store, bswap)     on the stack         (Move)        (load, bswap)

The compiler must prove the round trip is the identity for the whole thing to vanish.
```

Fact	Consequence
`Addr.z` is unexported	Outside `net/netip` you can only change the family by building a new `Addr` from bytes
`As16` and `AddrFrom16` are small and inline	The compiler sees the whole pack → copy → unpack sequence in your function
Generic SSA rewrite rules run in the `late opt` pass, before `memcombine`	A rule that forwards individual bytes can stop `memcombine` from merging eight byte loads into one 64-bit load
Writing memory and reading it back at once needs store-to-load forwarding	Reads that do not line up with the earlier writes can stall, whatever the instruction count

Worked examples

All files live in one module. `go.mod` says `go 1.26` so the same code builds with both toolchains used below.

```go
// go.mod
module example.com/addrto6

go 1.26
```

Case 1: Three ways to write To6

```go
// addrto6.go
package addrto6

import (
    "net/netip"
    "unsafe"
)

// AddrTo6Safe is the idiom the netip docs point you to: round-trip
// through the 16-byte form and let the compiler clean up.
func AddrTo6Safe(ip netip.Addr) netip.Addr {
    if ip.Is4() {
        ip = netip.AddrFrom16(ip.As16())
    }
    return ip
}

// addrProxy mirrors netip.Addr's memory layout:
// addr uint128 (two uint64) followed by z unique.Handle (one pointer).
type addrProxy struct {
    addr [2]uint64
    z    unsafe.Pointer
}

var (
    anyIPv6 = netip.IPv6Unspecified()
    z6noz   = (*addrProxy)(unsafe.Pointer(&anyIPv6)).z
)

// AddrTo6Unsafe flips only the family handle, the way a method inside
// net/netip would. It is correct only while the layout above holds.
func AddrTo6Unsafe(ip netip.Addr) netip.Addr {
    if !ip.Is4() {
        return ip
    }
    (*addrProxy)(unsafe.Pointer(&ip)).z = z6noz
    return ip
}
```

The third way, a real method inside `net/netip`, is Case 7. The tests check that both functions agree, and pin the layout the unsafe version depends on:

```go
// addrto6_test.go
package addrto6

import (
    "net/netip"
    "testing"
    "unsafe"
)

var inputs = []netip.Addr{
    netip.MustParseAddr("203.0.113.10"),
    netip.MustParseAddr("2001:db8::1"),
    netip.MustParseAddr("::ffff:192.0.2.1"),
    netip.MustParseAddr("fe80::1%eth0"),
}

// The unsafe version is only allowed to exist because this test pins
// the layout it depends on. If netip.Addr changes, this fails first.
func TestLayoutGuard(t *testing.T) {
    if unsafe.Sizeof(netip.Addr{}) != unsafe.Sizeof(addrProxy{}) {
        t.Fatalf("netip.Addr is %d bytes, proxy is %d",
            unsafe.Sizeof(netip.Addr{}), unsafe.Sizeof(addrProxy{}))
    }
}

func TestSameAnswer(t *testing.T) {
    for _, ip := range inputs {
        s, u := AddrTo6Safe(ip), AddrTo6Unsafe(ip)
        if s != u {
            t.Errorf("%v: safe=%v unsafe=%v", ip, s, u)
        }
        t.Logf("%-18v -> %v", ip, s)
    }
}

var sink netip.Addr

func BenchmarkAddrTo6(b *testing.B) {
    ip := netip.MustParseAddr("203.0.113.10")
    b.Run("safe", func(b *testing.B) {
        for b.Loop() {
            sink = AddrTo6Safe(ip)
        }
    })
```

```
    b.Run("unsafe", func(b *testing.B) {
        for b.Loop() {
            sink = AddrTo6Unsafe(ip)
        }
    })
}
```

```
go version
go test -v -run 'Test' .
```

```
go version go1.27.2 linux/amd64
=== RUN   TestLayoutGuard
--- PASS: TestLayoutGuard (0.00s)
=== RUN   TestSameAnswer
    addrto6_test.go:32: 203.0.113.10       -> ::ffff:203.0.113.10
    addrto6_test.go:32: 2001:db8::1        -> 2001:db8::1
    addrto6_test.go:32: ::ffff:192.0.2.1   -> ::ffff:192.0.2.1
    addrto6_test.go:32: fe80::1%eth0       -> fe80::1%eth0
--- PASS: TestSameAnswer (0.00s)
PASS
ok      example.com/addrto6 0.002s
```

Already-IPv6 inputs, including one with a zone, pass through unchanged. Only the IPv4 address picks up the `::ffff:` form.

Case 2: The benchmark

```
go test -run '^$' -bench AddrTo6 -count 1 .
```

```
goos: linux
goarch: amd64
pkg: example.com/addrto6
cpu: Intel(R) Xeon(R) Processor
BenchmarkAddrTo6/safe-8             141669531            8.510 ns/op
BenchmarkAddrTo6/unsafe-8           1000000000           1.090 ns/op
PASS
ok      example.com/addrto6 2.306s
```

That is a factor of about 8. At one call per packet on a 10 Mpps path, 7.4 ns extra per call adds up to roughly 7% of one core. Fine for a config loader, worth fixing in a flow collector.

Case 3: Read the assembly

`go build -gcflags=-S` prints the assembly for every function in the package. Two small helpers keep only the function you care about and drop the bookkeeping (`PCDATA`, `FUNCDATA`, `NOP`, and the `XCHGL AX, AX` markers the compiler leaves where calls were inlined):

```
#!/bin/sh
# asm.sh FUNC - print the instructions the compiler emitted for FUNC,
# without addresses, source positions or bookkeeping pseudo-ops.
go build -a -gcflags=-S . 2>&1 |
  awk -v f="addrto6.$1 STEXT" '
    index($0, f) { p = 1; next }
    / STEXT/     { p = 0 }
    p && /^\t0x[0-9a-f]+ [0-9]+ \(/ &&
      !/\tTEXT\t|PCDATA|FUNCDATA|\tNOP|XCHGL\tAX, AX/ {
        sub(/^\t0x[0-9a-f]+ [0-9]+ \([^)]*\)\t/, "\t"); print }'
```

```
#!/bin/sh
# asmcount.sh FUNC - count real instructions the compiler emitted for FUNC
# (skips PCDATA/FUNCDATA bookkeeping, NOP and the XCHGL inline markers).
go build -a -gcflags=-S . 2>&1 |
  awk -v f="addrto6.$1 STEXT" '
    index($0, f) { p = 1; next }
    / STEXT/     { p = 0 }
    p && /^\t0x[0-9a-f]+ [0-9]+ \(/ &&
      !/\tTEXT\t|PCDATA|FUNCDATA|\tNOP|XCHGL\tAX, AX/ { n++ }
    END { print n }'
```

`GOTOOLCHAIN=go1.26.9` makes the `go` command download and run that release, so one checkout can compare two compilers. `GOAMD64=v3` allows `MOVBE` (a byte-swapping load or store), which shortens the listing:

```
for tc in go1.26.9 go1.27.2; do for v in v1 v3; do
  printf "%-9s GOAMD64=%s  AddrTo6Safe=%-3s AddrTo6Unsafe=%s\n" $tc $v \
    "$(GOTOOLCHAIN=$tc GOAMD64=$v ./asmcount.sh AddrTo6Safe)" \
    "$(GOTOOLCHAIN=$tc GOAMD64=$v ./asmcount.sh AddrTo6Unsafe)"
done; done
```

```
go1.26.9  GOAMD64=v1  AddrTo6Safe=20  AddrTo6Unsafe=5
go1.26.9  GOAMD64=v3  AddrTo6Safe=16  AddrTo6Unsafe=5
go1.27.2  GOAMD64=v1  AddrTo6Safe=60  AddrTo6Unsafe=5
go1.27.2  GOAMD64=v3  AddrTo6Safe=56  AddrTo6Unsafe=5
```

The unsafe version is what a method inside `net/netip` would compile to: compare the handle with `z4`, and if it matches, load `z6noz`. The address halves never leave their registers (`AX`, `BX`; the handle is in `CX` under the register ABI).

```
GOAMD64=v3 ./asm.sh AddrTo6Unsafe
```

```
    CMPQ    net/netip.z4(SB), CX
    JNE 18
    MOVQ    example.com/addrto6.z6noz(SB), CX
    RET
    RET
```

Go 1.26.9 compiles the idiom into the three groups you would expect from the mental model: pack (two byte-swapping stores), copy (one 16-byte `MOVUPS` load and store), unpack (two byte-swapping loads):

```
GOTOOLCHAIN=go1.26.9 GOAMD64=v3 ./asm.sh AddrTo6Safe
```

```
    PUSHQ   BP
    MOVQ    SP, BP
    SUBQ    $32, SP
    MOVQ    CX, example.com/addrto6.ip+64(FP)
    CMPQ    net/netip.z4(SB), CX
    JNE 69
    MOVBEQ  AX, net/netip.a16+16(SP)
    MOVBEQ  BX, net/netip.a16+24(SP)
    MOVUPS  net/netip.a16+16(SP), X0
    MOVUPS  X0, net/netip.addr(SP)
    MOVQ    net/netip.z6noz(SB), CX
    MOVBEQ  net/netip.addr(SP), AX
    MOVBEQ  net/netip.addr+8(SP), BX
    ADDQ    $32, SP
    POPQ    BP
    RET
```

Go 1.27.2 compiles the same source into this:

```
GOAMD64=v3 ./asm.sh AddrTo6Safe
```

```
    PUSHQ   BP
    MOVQ    SP, BP
    SUBQ    $32, SP
    MOVQ    CX, example.com/addrto6.ip+64(FP)
    CMPQ    net/netip.z4(SB), CX
    JNE 219
    MOVQ    BX, DX
    SHRQ    $24, BX
    MOVQ    DX, SI
    SHRQ    $16, DX
    MOVQ    SI, DI
```

```
    SHRQ    $8, SI
    MOVBLZX DIB, R8
    MOVBLZX SIB, SI
    SHLQ    $8, SI
    ORQ R8, SI
    MOVBLZX DL, DX
    SHLQ    $16, DX
    ORQ SI, DX
    MOVBLZX BL, BX
    SHLQ    $24, BX
    ORQ DX, BX
    MOVBEQ  AX, net/netip.a16+16(SP)
    MOVBEQ  DI, net/netip.a16+24(SP)
    MOVUPS  net/netip.a16+16(SP), X0
    MOVBEL  net/netip.a16+20(SP), DX
    MOVL    DX, AX
    MOVBLZX net/netip.a16+19(SP), DX
    SHLQ    $32, DX
    ORQ DX, AX
    MOVBLZX net/netip.a16+18(SP), DX
    SHLQ    $40, DX
    ORQ DX, AX
    MOVBLZX net/netip.a16+17(SP), DX
    SHLQ    $48, DX
    ORQ DX, AX
    MOVBLZX net/netip.a16+27(SP), DX
    SHLQ    $32, DX
    ORQ DX, BX
    MOVBLZX net/netip.a16+26(SP), DX
    SHLQ    $40, DX
    ORQ DX, BX
    MOVBLZX net/netip.a16+25(SP), DX
    SHLQ    $48, DX
    ORQ DX, BX
    MOVBLZX net/netip.a16+24(SP), DX
    MOVUPS  X0, net/netip.addr(SP)
    MOVBLZX net/netip.addr(SP), SI
    SHLQ    $56, SI
    SHLQ    $56, DX
    MOVQ    net/netip.z6noz(SB), CX
    ORQ SI, AX
    ORQ DX, BX
    ADDQ    $32, SP
    POPQ    BP
    RET
```

The pack and the copy are still there. The unpack is now split into pieces:

- **`lo`, low 4 bytes:** rebuilt in registers from `BX` with shifts and masks. These bytes were *forwarded*: the compiler proved they equal what was just stored.
- **`lo`, high 4 bytes:** read back one byte at a time from `a16+24..27`.
- **`hi`:** bytes 4–7 come back as one 32-bit `MOVBEL`, bytes 1–3 one at a time from `a16`, and byte 0 from the copy at `addr(SP)`.

Partly-forwarded bytes are the signature of the regression described next.

Case 4: Read the SSA after one pass

`GOSSAFUNC=AddrTo6Safe go build` writes `ssa.html` with every pass side by side, which is the best way to browse. To diff two toolchains in a terminal, dump a single pass instead. `go tool compile -d=ssa/help` lists the pass names; `memcombine` is the one that merges byte loads and stores into wider ones.

```
GOAMD64=v3 go build -a -gcflags='-d=ssa/memcombine/dump=AddrTo6Safe' .
grep -E 'Move|= Load <|= Store <' AddrTo6Safe_01__memcombine.dump
```

Go 1.26.9 (same command, with `GOTOOLCHAIN=go1.26.9`):

```
    (+461) v17 = Load <*netip.addrDetail> v16 v1
    (128) v161 = Store <mem> {uint64} v22 v542 v23
    (128) v282 = Store <mem> {uint64} v173 v442 v161
    (+13) v286 = Move <mem> {[16]byte} [16] v285 v22 v282
    (+107) v258 = Load <*netip.addrDetail> v260 v286
    (123) v416 = Load <uint64> v285 v286
    (123) v174 = Load <uint64> v415 v286
```

Go 1.27.2:

```
    (128) v161 = Store <mem> {uint64} v22 v325 v23
    (128) v282 = Store <mem> {uint64} v173 v288 v161
    (+13) v286 = Move <mem> {[16]byte} [16] v285 v22 v282
    (123) v400 = Load <byte> v285 v286
    (123) v405 = Load <byte> v537 v282
    (123) v504 = Load <byte> v488 v282
    (123) v14 = Load <byte> v530 v282
    (+123) v256 = Load <byte> v244 v282
    (+123) v232 = Load <byte> v220 v282
    (123) v208 = Load <byte> v177 v282
    (123) v176 = Load <byte> v163 v282
    (122) v268 = Load <uint32> v397 v282
```

Read the last argument of each `Load`: it is the memory state the load reads from. In 1.26.9 both 64-bit loads read `v286`, the state *after* the `Move`, i.e. they read the copy. In 1.27.2 most loads read `v282`, the state *before* the `Move`: they were redirected to the source array `a16`. That redirection comes from a rule added in Go 1.27 for issue #77720 ("field access on struct-returning method copies entire struct"). It is in `generic.rules`:

```
// Load from a region just copied by Move can read directly from the source.
(Load <t1> op:(OffPtr [o1] p1) move:(Move [n] p2 src mem))
    && o1 >= 0 && o1+t1.Size() <= n && isSamePtr(p1, p2)
    && !isVolatile(src)
    => @move.Block (Load <t1> (OffPtr <op.Type> [o1] src) mem)
```

The rule is correct and usually helps. The problem is ordering. In compile.go the `late opt` pass, where generic rules run, comes before `memcombine`. At `late opt` time, `BEUint64` is still sixteen separate byte loads OR-ed together. The rule redirects each byte load to `a16`. Then the existing store-to-load forwarding rules replace some of those loads with the stored bytes (the four most recently stored ones, since those rules look back through only a few stores). When `memcombine` runs, it needs eight byte loads from consecutive addresses at the same memory state to build one 64-bit load. Some of the eight are now register values rather than loads, so it can only partly merge them. That is where the 56 instructions come from.

Bernat's write-up shows that running a `memcombine` round before `late opt` (plus a no-offset variant of the rule above) brings the idiom down to the three-instruction form on a patched compiler. That is a compiler change. You cannot make it from user code.

Case 5: Instructions are not time

The fair comparison is across toolchains, with enough runs for `benchstat` (`go install golang.org/x/perf/cmd/benchstat@latest`) to tell signal from noise:

```
for tc in go1.26.9 go1.27.2; do
  GOTOOLCHAIN=$tc GOAMD64=v3 go test -run '^$' -bench AddrTo6 -count 8 > $tc.txt
done
benchstat go1.26.9.txt go1.27.2.txt
```

```
goos: linux
goarch: amd64
pkg: example.com/addrto6
cpu: Intel(R) Xeon(R) Processor
                  go1.26.9.txt              go1.27.2.txt
                     sec/op        sec/op     vs base
AddrTo6/safe-8      8.397n ± 8%   8.617n ± 5%        ~ (p=0.878 n=8)
AddrTo6/unsafe-8    1.123n ± 5%   1.308n ± 5%  +16.42% (p=0.000 n=8)
geomean             3.071n        3.357n        +9.31%
```

The safe version went from 16 to 56 instructions with **no measurable change** (~, p=0.88). Do not over-read the unsafe row either: five instructions measured in a tight loop on a shared VM move with code alignment, and +0.2 ns there is not a property of the source. Bernat measured the safe version about 9% slower on 1.27 than on 1.26 on a desktop Ryzen. Different microarchitectures disagree on secondary costs. They agree on the 8x.

So what is the 8x? Both toolchains write the address into a stack array and read it back a few instructions later. This benchmark keeps everything else fixed and varies only that:

```go
// forward_test.go
package addrto6

import (
    "net/netip"
    "testing"
)

// settled is written once, long before the benchmark reads it.
var settled = [16]byte{10: 0xff, 11: 0xff, 12: 203, 13: 0, 14: 113, 15: 10}

func BenchmarkFrom16(b *testing.B) {
    b.Run("settled-bytes", func(b *testing.B) {
        for b.Loop() {
            sink = netip.AddrFrom16(settled)
        }
    })
    b.Run("just-written", func(b *testing.B) {
        raw := [4]byte{203, 0, 113, 10}
        for b.Loop() {
            sink = MappedFrom4(raw)
        }
    })
}
```

(`MappedFrom4` is in Case 6: it fills a fresh `[16]byte` and calls `AddrFrom16`.)

```
GOAMD64=v3 go test -run '^$' -bench From16 -count 6 . > from16.txt && benchstat from16.txt
```

```
goos: linux
goarch: amd64
pkg: example.com/addrto6
cpu: Intel(R) Xeon(R) Processor
                        from16.txt
                          sec/op
From16/settled-bytes-8   1.790n ± 7%
From16/just-written-8    8.477n ± 4%
geomean                  3.895n
```

The same unpack costs 1.8 ns when the bytes have been in memory for a while and 8.5 ns when they were stored a moment earlier. The likely explanation is **store-to-load forwarding**. x86 cores can hand a just-stored value straight to a later load only if the load lines up with a single earlier store. The 16-byte `MOVUPS` copy reads across two 8-byte stores, and the 1.27 byte-wise reloads read pieces of an 8-byte store. Both are patterns that commonly fall back to a stall until the stores reach the cache. This KVM guest does not expose hardware performance counters (`perf stat -e instructions` reports `<not supported>`), so the stall could not be counted directly here. On bare metal, Try this #2 shows how to check it. Whatever the exact mechanism, the measurement is

clear: **the cost is in the memory round trip, and changing the instruction count does not move it.**

Case 6: Convert at the boundary, in integers

If the bytes come from a packet or a database column, you can skip the IPv4 `Addr` and build the mapped form directly:

```go
// boundary.go
package addrto6

import "net/netip"

// MappedFrom4 builds ::ffff:a.b.c.d straight from the four bytes a packet
// or a database row already hands you. One unpack, no pack.
func MappedFrom4(b [4]byte) netip.Addr {
    a := [16]byte{10: 0xff, 11: 0xff}
    copy(a[12:], b[:])
    return netip.AddrFrom16(a)
}

// ViaAddr4 is the detour: make an IPv4 Addr first, then map it.
func ViaAddr4(b [4]byte) netip.Addr {
    return AddrTo6Safe(netip.AddrFrom4(b))
}
```

```go
// boundary_test.go
package addrto6

import (
    "net/netip"
    "testing"
)

func TestBoundary(t *testing.T) {
    b := [4]byte{203, 0, 113, 10}
    want := netip.MustParseAddr("::ffff:203.0.113.10")
    if got := MappedFrom4(b); got != want {
        t.Fatalf("MappedFrom4 = %v", got)
    }
    if got := ViaAddr4(b); got != want {
        t.Fatalf("ViaAddr4 = %v", got)
    }
}

func BenchmarkFrom4(b *testing.B) {
```

```go
    raw := [4]byte{203, 0, 113, 10}
    b.Run("mapped-direct", func(b *testing.B) {
        for b.Loop() {
            sink = MappedFrom4(raw)
        }
    })
    b.Run("via-addr4", func(b *testing.B) {
        for b.Loop() {
            sink = ViaAddr4(raw)
        }
    })
    b.Run("addr4-only", func(b *testing.B) {
        for b.Loop() {
            sink = netip.AddrFrom4(raw)
        }
    })
}
```

```
GOAMD64=v3 go test -run '^$' -bench From4 -count 6 . > from4.txt && benchstat from4.txt
```

```
goos: linux
goarch: amd64
pkg: example.com/addrto6
cpu: Intel(R) Xeon(R) Processor
                        from4.txt
                          sec/op
From4/mapped-direct-8   8.517n ±  2%
From4/via-addr4-8       9.711n ±  2%
From4/addr4-only-8      1.218n ± 37%
geomean                 4.653n
```

Building the mapped form directly saves only the pack step (~1.2 ns): it still writes a `[16]byte` and reads it back. `AddrFrom4` costs 1.2 ns because it builds `addr.lo` with shifts and ORs and never touches an array. The general lesson: in `net/netip`, **integer paths are cheap and `[16]byte` paths cost a memory round trip.** Pick one representation and convert to it once, where data enters the program, rather than converting back and forth in the hot loop.

Case 7: Add the method with `-overlay`, and what it costs you

`go build -overlay=file.json` makes the build see replacement or extra files at any path, including inside `GOROOT`. Adding one file to `net/netip` gives you the method the proposals asked for, without editing your Go installation:

```go
// patches/to6.go - added to net/netip through -overlay; never edit GOROOT itself.
package netip
```

```
// To6 maps an IPv4 address to an IPv4-mapped IPv6 address and returns
// every other address unchanged.
func (ip Addr) To6() Addr {
    if ip.Is4() {
        ip.z = z6noz
    }
    return ip
}
```

```
printf '{"Replace": {"%s/src/net/netip/to6.go": "%s/patches/to6.go"}}\n' \
  "$(go env GOROOT)" "$PWD" > overlay.json
```

```
// overlay_test.go - only builds with -overlay=overlay.json
package addrto6

import (
    "net/netip"
    "testing"
)

var sink netip.Addr

func TestTo6(t *testing.T) {
    ip := netip.MustParseAddr("203.0.113.10")
    if ip.To6() != AddrTo6Safe(ip) {
        t.Fatal("To6 disagrees with AddrFrom16(As16())")
    }
}

func BenchmarkTo6(b *testing.B) {
    ip := netip.MustParseAddr("203.0.113.10")
    b.Run("method", func(b *testing.B) {
        for b.Loop() {
            sink = ip.To6()
        }
    })
    b.Run("safe", func(b *testing.B) {
        for b.Loop() {
            sink = AddrTo6Safe(ip)
        }
    })
}
```

(This lives in its own copy of the module with `go.mod` and `addrto6.go`; the patch file sits in `patches/` so it is not mistaken for part of the `addrto6` package.)

```
GOAMD64=v3 go test -overlay=overlay.json -run To6 -bench To6 -count 3 .
go test -run To6 .
```

```
BenchmarkTo6/method-8           937758579            1.260 ns/op
BenchmarkTo6/method-8           940058956            1.318 ns/op
BenchmarkTo6/method-8           913291134            1.262 ns/op
BenchmarkTo6/safe-8             131066788            8.970 ns/op
BenchmarkTo6/safe-8             138935923            8.634 ns/op
BenchmarkTo6/safe-8             139838676            8.667 ns/op
PASS
ok      example.com/addrto6 7.168s

# example.com/addrto6 [example.com/addrto6.test]
./overlay_test.go:13:8: ip.To6 undefined (type netip.Addr has no field or method To6)
./overlay_test.go:22:14: ip.To6 undefined (type netip.Addr has no field or method To6)
```

It is as fast as the unsafe proxy, and it is type-safe. But every `go build`, `go test`, `go vet` and gopls invocation now needs the flag. The overlay file contains absolute paths into one specific `GOROOT`. The first build without it fails. That is acceptable in one controlled binary with a Makefile. It is a bad idea for a library other people import.

The trap

- **"The compiler will optimise it" as a design argument.** It is a forecast about one compiler version on one architecture. Here a correct, well-motivated Go 1.27 rule made the idiom's code 3.5x longer. A benchmark plus an assembly check is the only way to know what your toolchain does today.
- **Counting instructions and calling it performance.** 16 vs 56 instructions made no measurable difference. 16 vs 5 made an 8x difference. Measure time with `benchstat`, then read assembly to explain it.
- **An unsafe layout proxy without a guard.** `addrProxy` is correct only while `netip.Addr` is `{uint128, unique.Handle}`. The size check plus `TestSameAnswer` catches most layout changes in CI. Put the proxy behind a build tag if you ship it in a library, so importers opt in knowingly.
- **Comparing benchmarks across toolchains by eye.** Use `-count 6` and `benchstat`. On shared or virtual machines a 1 ns function moves by 15% between runs for reasons that have nothing to do with your code.
- **Leaving `-overlay` implicit.** If the overlay lives only in a developer's shell history, CI, IDE and `go vet` will disagree about what compiles.

The boring rule

Keep `netip.AddrFrom16(ip.As16())` in code that is not hot. It is correct and portable. For a hot path, pick one representation at the boundary, ideally the integer form `AddrFrom4`/`AddrFrom16`

produce once, and stop converting inside the loop. Reach for `unsafe` or `-overlay` only after a `benchstat` result shows the conversion matters, and only with a test that fails when the assumption it relies on breaks. Re-run the benchmark and `asmcount.sh` whenever you bump the Go minor version.

Try this

1. Run `asmcount.sh AddrTo6Safe` with `GOARCH=arm64` (cross-compiling is fine for `-S`). Does arm64 show the same 1.26 → 1.27 jump? `REV` is arm64's byte swap.
2. On bare-metal Intel hardware, run `perf stat -e cycles,instructions,ld_blocks.store_forward go test -run '^$' -bench 'From16/just-written' -benchtime 2s .` and compare with `settled-bytes`. If the forwarding hypothesis is right, `ld_blocks.store_forward` climbs only in the first.
3. Rewrite `AddrTo6Safe` to call `netip.AddrFrom16(ip.As16())` twice in a row on the same input. Does the compiler share any of the work? Check with `asm.sh`.
4. Add `BenchmarkAddrTo6/unsafe` to a CI job that fails when its time exceeds 3x `AddrFrom4`. What else would such a job need to stay quiet on noisy runners?
5. Read the discussion on issue #81994 and write down which of the proposed additions (`To6`, `As2xUint64`, `Prefix.Last`) would remove a `[16]byte` round trip from your own code.

Sources

Go source at the `go1.27.2` tag:

- `netip.Addr`, `AddrFrom16`, `Is4`, `As16`, `z4/z6noz`: https://github.com/golang/go/blob/go1.27.2/src/net/netip/netip.go#L101
- `BEUint64`, `BEPutUint64`: https://github.com/golang/go/blob/go1.27.2/src/internal/byteorder/byteorder.go#L120
- Load-through-Move rule: https://github.com/golang/go/blob/go1.27.2/src/cmd/compile/internal/ssa/_gen/generic.rules#L823-L827
- Pass order (`late opt`, `memcombine`): https://github.com/golang/go/blob/go1.27.2/src/cmd/compile/internal/ssa/compile.go#L481-L488
- `memcombine` pass: https://github.com/golang/go/blob/go1.27.2/src/cmd/compile/internal/ssa/memcombine.go
- SSA debugging (`GOSSAFUNC`, `-d=ssa/<pass>/dump`): https://github.com/golang/go/blob/go1.27.2/src/cmd/compile/internal/ssa/README.md

Go documentation and issues:

- `go help build` (`-overlay`, `-gcflags`) and `GOTOOLCHAIN`: https://pkg.go.dev/cmd/go, https://go.dev/doc/toolchain
- `GOAMD64` levels: https://go.dev/wiki/MinimumRequirements#amd64
- `testing.B.Loop`: https://pkg.go.dev/testing#B.Loop
- benchstat: https://pkg.go.dev/golang.org/x/perf/cmd/benchstat
- Release history (1.26.9 and 1.27.2, both released 2026-10-08): https://go.dev/doc/devel/release

- cmd/compile: field access on struct-returning method copies entire struct (#77720, fixed in 1.27): https://github.com/golang/go/issues/77720
- proposal: net/netip: add To6 (#51833, closed), add Addr.Map (#54365, closed), add Addr.To6 (#81994, open): https://github.com/golang/go/issues/51833, https://github.com/golang/go/issues/54365, https://github.com/golang/go/issues/81994

Investigation reproduced here:

- Vincent Bernat, "Hacking the Go compiler to efficiently map IPv4 to IPv6" (2026-10-04), including the patched-compiler results: https://vincent.bernat.ch/en/blog/2026-go-netip-addrto6

Tools: Go 1.27.2 and 1.26.9 (`GOTOOLCHAIN` download), benchstat (`golang.org/x/perf`, 2026-10-09 snapshot), `go tool compile -d=ssa/help`.

Why Go Binaries Are Big: Sections, Reachability, and the Linker's Method Rules

Why Go Binaries Are Big: Sections, Reachability, and the Linker's Method Rules

Thesis

A Go "hello world" that calls `fmt.Println` is 2.3 MB. A program that only registers an HTTP handler is 8.6 MB. Most advice stops at `-ldflags="-s -w"`, which removes about a third of the file. That flag removes metadata, though, not code. The code that stays is decided earlier, by the linker's **dead-code pass**: it starts at the entry point, follows every reference, and keeps only what it reaches. Methods get special rules, and one call to `reflect.Value.MethodByName` with a name the compiler cannot see makes the linker keep **every exported method of every reachable type**.

This chapter takes binaries apart with tools that ship with Go: `debug/elf` for sections, `go tool nm -size` for symbols, and the linker's own `-dumpdep` graph to see *why* a symbol was kept. It ends with a short list of changes that actually shrink a binary, ordered by what they cost you.

Verified on **Go 1.27.2** (linux/amd64, `CGO_ENABLED=0`), Linux 6.12, on 2026-10-19. Source links are pinned to the go1.27.2 tag. All sizes below are from this machine; yours will differ by a few kilobytes between Go releases.

Learning goals

- Split a Go executable into its sections and say which ones the program needs at run time
- Explain what `-s` and `-w` remove, and what you lose (and keep) when you strip
- Total symbol sizes per package from `go tool nm -size` to find who pays for the bytes
- Describe the linker's dead-code pass: roots, the flood fill, and the three ways a method can stay reachable
- Use `-ldflags=-dumpdep` to find the function that turned off method pruning
- Choose between stripping, avoiding dynamic reflection, and changing dependencies

Mental model

Part of the file	What it is	Needed at run time?
`.text`	Machine code for every reachable function, including the runtime	Yes

Part of the file	What it is	Needed at run time?
`.gopclntab`	Maps each program counter to function, file and line; points at stack maps and other per-function data	Yes: tracebacks, `runtime.Callers`, profiling and the garbage collector's stack scan read it
`.go.type`	Type descriptors (`abi.Type`) and method tables	Yes: interfaces, maps, `reflect`, the GC
`.rodata`, `.data`, `.noptrdata`	Constants, strings, initialised globals	Yes
`.symtab`, `.strtab`	ELF symbol table for `nm`, debuggers and `perf`	No; `-s` removes it
`.debug_*`	DWARF for debuggers (compressed by default)	No; `-w` removes it, `-s` implies `-w`

Two separate questions follow from the table:

1. **Metadata:** do you ship the symbol table and DWARF? That is a debugging trade-off, settled by a flag.
2. **Code:** what is reachable? That is a design question, settled by what your program and its dependencies *could* call.

Worked examples

All listings live in one module. Every command below was run from its root (`/tmp/binsize` on the test machine, which is why that path appears in some output).

```go
// go.mod
module example.com/binsize

go 1.27
```

Case 1: Three programs, two builds each

```go
// hello-println/main.go
package main

func main() {
    println("hello")
```

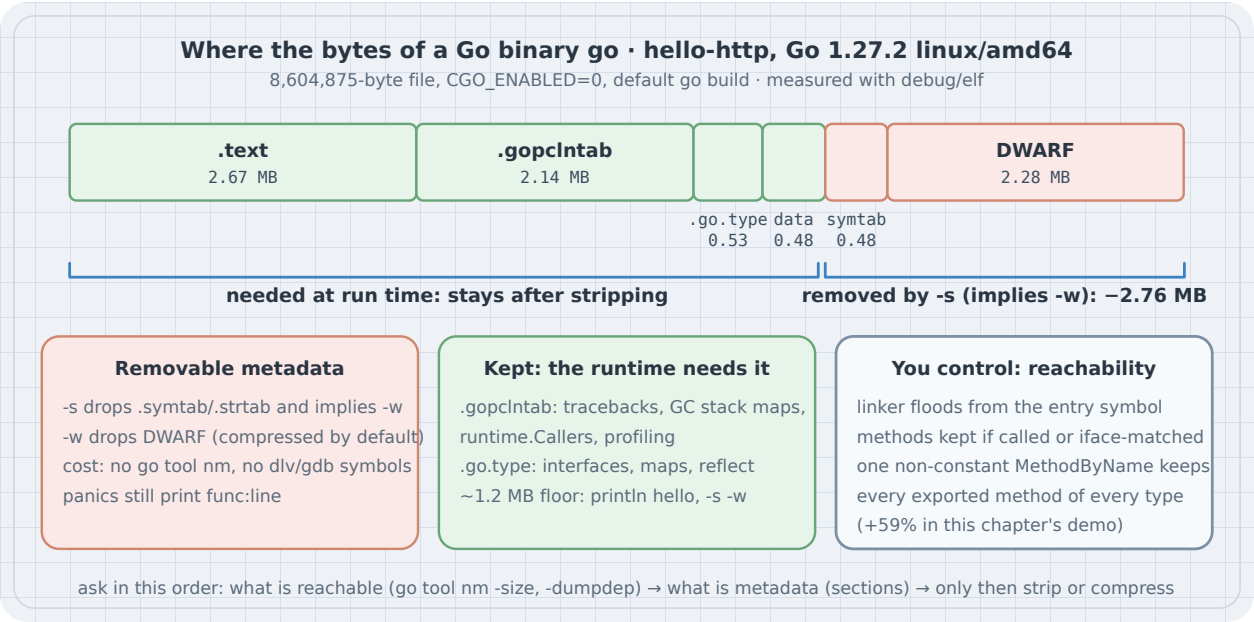


Figure 1: Where the bytes of a Go binary go: about two thirds is code and runtime metadata the program needs; about one third is symbol tables and DWARF that -s removes; reachability decides how big the code part is

```
}
```

```go
// hello-fmt/main.go
package main

import "fmt"

func main() {
    fmt.Println("hello")
}
```

```go
// hello-http/main.go
package main

import (
    "fmt"
    "net/http"
    "os"
)

func main() {
    http.HandleFunc("/", func(w http.ResponseWriter, r *http.Request) {
        fmt.Fprintln(w, "hello")
```

```
    })
    if len(os.Args) > 1 && os.Args[1] == "serve" {
        http.ListenAndServe("127.0.0.1:8080", nil)
    }
}
```

A small script builds each one with default flags and with `-s -w`:

```
#!/bin/sh
# sizes.sh - build each program twice (default, and -s -w) and print bytes.
set -e
mkdir -p bin
printf '%-14s %9s %9s\n' program default '-s -w'
for p in "$@"; do
    CGO_ENABLED=0 go build -o bin/$p ./$p
    CGO_ENABLED=0 go build -ldflags='-s -w' -o bin/$p.sw ./$p
    printf '%-14s %9d %9d\n' $p $(stat -c %s bin/$p) $(stat -c %s bin/$p.sw)
done
```

```
$ go version
go version go1.27.2 linux/amd64
$ ./sizes.sh hello-println hello-fmt hello-http
program          default     -s -w
hello-println    1895652   1220768
hello-fmt        2342505   1511584
hello-http       8604875   5841056
```

Three things to notice:

- **There is a floor.** A program that only calls the `println` builtin is still 1.2 MB stripped. That is the runtime: scheduler, garbage collector, allocator, maps, signal handling and the tables that describe them. Every Go executable carries it.
- **`fmt` costs about 290 KB stripped.** It formats through `reflect`, so it pulls in reflection and number formatting.
- **One `net/http` handler costs 4.3 MB more.** The rest of this chapter explains where that goes.

Case 2: Sections: what `-s -w` actually removes

`debug/elf` in the standard library reads the section headers. This program prints every section of 16 KB or more, largest first:

```
// sections/main.go: print the size of every ELF section in a binary,
// largest first, and how much of the file is debug information.
package main
```

```go
import (
    "cmp"
    "debug/elf"
    "fmt"
    "log"
    "os"
    "slices"
    "strings"
)

func main() {
    f, err := elf.Open(os.Args[1])
    if err != nil {
        log.Fatal(err)
    }
    defer f.Close()

    secs := slices.Clone(f.Sections)
    slices.SortFunc(secs, func(a, b *elf.Section) int {
        return cmp.Compare(b.FileSize, a.FileSize)
    })

    var total, debug uint64
    for _, s := range secs {
        if s.Type == elf.SHT_NOBITS { // .bss/.noptrbss take no file space
            continue
        }
        total += s.FileSize
        if strings.HasPrefix(s.Name, ".debug_") || strings.HasPrefix(s.Name, ".zdebug_") {
            debug += s.FileSize
        }
        if s.FileSize >= 16<<10 {
            fmt.Printf("%-20s %9d\n", s.Name, s.FileSize)
        }
    }
    fmt.Printf("%-20s %9d\n", "sections total", total)
    fmt.Printf("%-20s %9d (%.0f%%)\n", "debug_*", debug, 100*float64(debug)/float64(total))
}
```

```
$ go build -o bin/sections ./sections
$ bin/sections bin/hello-http
.text                  2673425
.gopclntab             2136688
.debug_info             930317
.go.type                532368
.debug_line             526276
```

```
.debug_loclists         503135
.strtab                 282633
.noptrdata              249601
.debug_rnglists         203263
.symtab                 196416
.rodata                 180633
.debug_frame            104330
.data                    53234
.debug_addr              16905
sections total         8594716
debug_*                2284636 (27%)
$ bin/sections bin/hello-http.sw
.text                  2673425
.gopclntab             2136688
.go.type                532368
.noptrdata              249601
.rodata                 180665
.data                    53234
sections total         5830949
debug_*                      0 (0%)
```

`-s -w` removed DWARF (2.28 MB) and the ELF symbol and string tables (0.48 MB). `.text`, `.gopclntab` and `.go.type` are byte-for-byte the same size. Stripping cannot touch them, because the program reads them while it runs.

The linker flags interact, and the `cmd/link` documentation spells it out: `-s` "Omit the symbol table and debug information. Implies the -w flag, which can be negated with -w=0." DWARF is also compressed by default (`-compressdwarf`, default true):

```
$ for f in "" "-s" "-w" "-s -w" "-compressdwarf=false"; do
>   CGO_ENABLED=0 go build -ldflags="$f" -o /tmp/hb ./hello-http
>   printf '%-24s %9d\n' "-ldflags='$f'" $(stat -c %s /tmp/hb)
> done
-ldflags=''                 8604875
-ldflags='-s'               5841056
-ldflags='-w'               6320121
-ldflags='-s -w'            5841056
-ldflags='-compressdwarf=false'  11303171
```

`-s` alone already gives the `-s -w` size. Uncompressed DWARF would add another 2.7 MB, so "Go binaries are huge because of debug info" was more true before compression was the default than it is now.

Case 3: What you lose by stripping, and what you keep

```go
// crash/main.go: does a stripped binary still give a useful panic?
package main

func divide(a, b int) int { return a / b }

func main() {
    println(divide(1, len("")))
}
```

```
$ CGO_ENABLED=0 go build -ldflags='-s -w' -o bin/crash ./crash
$ bin/crash
panic: runtime error: integer divide by zero

goroutine 1 [running]:
main.divide(...)
    /tmp/binsize/crash/main.go:4
main.main()
    /tmp/binsize/crash/main.go:7 +0xa
$ go tool nm bin/crash
reading bin/crash: no symbol section
reading bin/crash: no symbols
$ go version -m bin/crash
bin/crash: go1.27.2
    path    example.com/binsize/crash
    mod example.com/binsize (devel)
    build   -buildmode=exe
    build   -compiler=gc
    build   -ldflags="-s -w"
    build   CGO_ENABLED=0
    build   GOARCH=amd64
    build   GOOS=linux
    build   GOAMD64=v1
```

The panic still names functions, files and lines, and even the inlined frame (`main.divide(...)`). That information comes from `.gopclntab`, not from DWARF or the symbol table. Build info (`go version -m`) also survives, because it is stored as data rather than as a symbol, so tools that read it (such as `govulncheck -mode=binary`) still see the module list.

What you lose: `go tool nm`, `perf` and `gdb` symbol names, and source-level debugging in Delve, which needs DWARF. The panic above also shows a local build path. `-trimpath` replaces it with the module path, and saves a few kilobytes as a side effect:

```
$ CGO_ENABLED=0 go build -trimpath -ldflags='-s -w' -o bin/crash.trim ./crash
$ bin/crash.trim
panic: runtime error: integer divide by zero
```

```
goroutine 1 [running]:
main.divide(...)
    example.com/binsize/crash/main.go:4
main.main()
    example.com/binsize/crash/main.go:7 +0xa
```

If you strip release binaries, archive the *unstripped* binary from the same build for later debugging. Rebuilding later with different flags gives a different binary: on this machine `hello-http` and `hello-http.sw` have the same `.gopclntab` bytes but differ in 11,175 bytes of `.text`, and their build IDs differ.

Case 4: Who pays for the bytes? Symbols per package

`go tool nm -size` prints every symbol with its size and kind. The largest symbols in `hello-http` are a misleading place to start:

```
$ go tool nm -size -sort size bin/hello-http > nm.txt
$ head -8 nm.txt
  9d30a0   33554432 B crypto/internal/fips140/drbg.memory
  8676e8     365096 r go:func.*
  99c780      93464 B runtime.mheap_
  96eae0      88064 D crypto/internal/fips140/nistec.p256PrecomputedEmbed
  9c20a0      69632 B runtime.cgroupScratch
  9ba880      30736 B crypto/internal/fips140/edwards25519.basepointTablePrecomp
  967160      19650 D vendor/golang.org/x/text/unicode/norm.decomps
  62ab60      19580 T crypto/internal/fips140/sha3.keccakF1600.abi0
```

The 32 MB "symbol" is kind `B`: it lives in `.bss`, which is zero-filled when the program loads and takes no space in the file. (It is address space, not file bytes. On Linux pages that are never touched are never backed by memory either.) Per-package totals that skip `.bss` are more useful:

```go
// pkgsize/main.go: read `go tool nm -size` output on stdin and total
// symbol sizes per package, so you can see who pays for the bytes.
package main

import (
    "bufio"
    "cmp"
    "fmt"
    "os"
    "slices"
    "strconv"
    "strings"
)
```

```
// pkgOf turns "net/http.(*conn).serve" into "net/http" and
// "type:*net/http.conn" into "net/http".
func pkgOf(sym string) string {
    sym = strings.TrimPrefix(sym, "type:")
    sym = strings.TrimPrefix(sym, "go:itab.")
    sym = strings.TrimLeft(sym, "*")
    slash := strings.LastIndexByte(sym, '/')
    if dot := strings.IndexByte(sym[slash+1:], '.'); dot >= 0 {
        return sym[:slash+1+dot]
    }
    return sym
}

func main() {
    sizes := map[string]int{}
    total := 0
    sc := bufio.NewScanner(os.Stdin)
    for sc.Scan() {
        // "addr size kind name"; generic names may contain spaces.
        f := strings.Fields(sc.Text())
        if len(f) < 4 {
            continue
        }
        n, err := strconv.Atoi(f[1])
        // U = undefined, B/b = .bss: zero-filled at load, no file bytes.
        if err != nil || f[2] == "U" || f[2] == "B" || f[2] == "b" {
            continue
        }
        sizes[pkgOf(strings.Join(f[3:], " "))] += n
        total += n
    }
    type row struct {
        pkg string
        n   int
    }
    var rows []row
    for p, n := range sizes {
        rows = append(rows, row{p, n})
    }
    slices.SortFunc(rows, func(a, b row) int { return cmp.Compare(b.n, a.n) })
    for _, r := range rows[:min(12, len(rows))] {
        fmt.Printf("%-32s %8d %5.1f%%\n", r.pkg, r.n, 100*float64(r.n)/float64(total))
    }
    fmt.Printf("%-32s %8d\n", "symbols total", total)
}
```

```
$ go build -o bin/pkgsize ./pkgsize
$ go tool nm -size bin/hello-fmt | bin/pkgsize
runtime                              450642  58.3%
go:func                              126400  16.3%
fmt                                   31041   4.0%
internal/strconv                      28088   3.6%
reflect                               24936   3.2%
internal/runtime/gc/scan              17873   2.3%
internal/runtime/maps                 14966   1.9%
internal/runtime/cgroup               10065   1.3%
syscall                                6642   0.9%
internal/sync                          6216   0.8%
internal/poll                          5683   0.7%
slices                                 5363   0.7%
symbols total                        773461
$ go tool nm -size bin/hello-http | bin/pkgsize
runtime                              492631  15.2%
go:func                              365096  11.3%
crypto/tls                           251735   7.8%
crypto/internal/fips140/nistec       155922   4.8%
net                                  136758   4.2%
net/http/internal/http2              128203   4.0%
net/http                             115662   3.6%
crypto/x509                          101560   3.1%
vendor/golang.org/x/text/unicode/norm    75715   2.3%
time                                  66902   2.1%
math/big                              65955   2.0%
reflect                               61528   1.9%
symbols total                       3233351
```

The program never mentions TLS, yet `crypto/tls`, the FIPS 140 elliptic-curve code, `crypto/x509` and `math/big` are a fifth of its symbols. They are reachable: `net/http`'s server code handles `*tls.Conn` connections and HTTP/2, and the linker cannot know you will never pass it one. `go:func.*` is per-function metadata (stack maps and similar data that `.gopclntab` points to), so it grows with the amount of code.

The symbol totals are smaller than the file. `.gopclntab` appears in `nm` only as zero-sized boundary symbols (`runtime.pclntab`, `runtime.epclntab`), so the sections view and the symbols view answer different questions. Use both.

Importing a package is not the same as paying for it. Compare `net.ParseIP` with `netip.ParseAddr`:

```go
// ip-net/main.go
package main

import (
    "net"
    "os"
```

```go
)

func main() {
    ip := net.ParseIP(os.Args[1])
    os.Stdout.WriteString(ip.String() + "\n")
}
```

```go
// ip-netip/main.go
package main

import (
    "net/netip"
    "os"
)

func main() {
    ip, err := netip.ParseAddr(os.Args[1])
    if err != nil {
        os.Exit(1)
    }
    os.Stdout.WriteString(ip.String() + "\n")
}
```

```
$ for p in ip-net ip-netip; do
>   CGO_ENABLED=0 go build -ldflags='-s -w' -o bin/$p ./$p
>   printf '%-10s %8d bytes  %3d packages\n' $p $(stat -c %s bin/$p) $(go list -deps ./$p |
>   bin/$p 2001:db8::1
> done
ip-net      1560736 bytes   67 packages
2001:db8::1
ip-netip    1405088 bytes   59 packages
2001:db8::1
```

All of package `net` is in the dependency graph, but the resolver, dialer and listener are unreachable from `ParseIP`, so the linker drops them. The 152 KB difference is what `ParseIP` and `net`'s package initialisation do reach (for example package `time` and the `GODEBUG` machinery in `internal/bisect`, both absent from the `netip` build), not 4 MB. What counts is what is *reachable*.

Case 5: How the linker decides what is reachable

The dead-code pass is `cmd/link/internal/ld/deadcode.go`. In a normal executable it starts from the entry symbol (which leads to `runtime.main`, `main.main` and every package's init tasks) plus `runtime.unreachableMethod`, then marks everything those symbols refer to, and so on. Functions are easy: a call is a relocation, so it gets followed.

Methods are the hard part, because a method can be called without a direct reference. The comment above `deadcode` lists the three ways:

1. direct call
2. through a reachable interface type
3. reflect.Value.Method (or MethodByName), or reflect.Type.Method (or MethodByName)

Case 2 is handled conservatively: every method of a type that was converted to an interface is kept if its name and signature match a method of *any* reachable interface type. Case 3 is the dramatic one:

> All functions that call reflect.Value.Method or reflect.Type.Method are REFLECTMETHODs. Functions that call reflect.Value.MethodByName or reflect.Type.MethodByName with a non-constant argument are REFLECTMETHODs, too. If we find a REFLECTMETHOD, we give up on static analysis, and mark all exported methods of all reachable types as reachable.
>
> If the argument to MethodByName is a compile-time constant, the compiler emits a relocation with the method name. Matching methods are kept in all reachable types.

To see the difference, generate a type with 300 exported methods, each doing a little real work. The generator carries a `//go:build ignore` constraint so it is not part of the module's packages:

```go
// gen/gen_big.go: write methods/big.go with 300 exported methods.
//go:build ignore

package main

import (
    "fmt"
    "os"
    "strings"
)

func main() {
    var b strings.Builder
    b.WriteString("// methods/big.go (generated): a type with 300 exported methods.\n")
    b.WriteString("package main\n\nimport \"strconv\"\n\ntype Big struct{ n int }\n\n")
    for i := range 300 {
        fmt.Fprintf(&b, "func (b *Big) M%d(s string) string "+
            "{ return strconv.Itoa(b.n+%d) + s + strconv.Quote(s[:len(s)/2]) }\n", i, i)
    }
    if err := os.WriteFile("methods/big.go", []byte(b.String()), 0o644); err != nil {
        panic(err)
    }
}
```

Then call one method three ways. `mode` is a constant, written into its own file by the build loop, so the two branches not taken are removed by the compiler:

```go
// methods/main.go: how a reflect call decides which methods survive linking.
package main

import (
    "fmt"
    "os"
    "reflect"
)

func main() {
    b := &Big{n: 1}
    switch mode {
    case "direct": // static call: the linker sees exactly one method used
        fmt.Println(b.M7("go"))
    case "const": // MethodByName with a constant: still only M7 is needed
        fmt.Println(reflect.ValueOf(b).MethodByName("M7").Call(
            []reflect.Value{reflect.ValueOf("go")})[0])
    case "dynamic": // name known only at run time: every exported method may be called
        name := "M7"
        if len(os.Args) > 1 {
            name = os.Args[1]
        }
        fmt.Println(reflect.ValueOf(b).MethodByName(name).Call(
            []reflect.Value{reflect.ValueOf("go")})[0])
    }
}
```

```
$ mkdir -p methods && go run gen/gen_big.go
$ for m in direct const dynamic; do
>   printf 'package main\n\nconst mode = "%s"\n' $m > methods/mode.go
>   CGO_ENABLED=0 go build -o bin/methods-$m ./methods
>   printf '%-8s %8d bytes %4d Big methods %5d text symbols\n' $m \
>     $(stat -c %s bin/methods-$m) \
>     $(go tool nm bin/methods-$m | grep -cE ' main\.\(\*Big\)\.M[0-9]+$') \
>     $(go tool nm bin/methods-$m | grep -cE ' T ')
> done
direct    2343128 bytes    1 Big methods  1570 text symbols
const     2518294 bytes    1 Big methods  1662 text symbols
dynamic   3717576 bytes  300 Big methods  3225 text symbols
$ bin/methods-dynamic M42
43go"g"
```

- **direct**: one method survives.
- **const**: still one method. The extra 175 KB is `reflect.Value.Call` and its metadata (package `reflect` alone grows from 25 KB to 71 KB of symbols), not your methods.
- **dynamic**: all 300 `Big` methods survive, and so do about 1,260 *other* functions, because the rule applies to every reachable type. A diff of the two symbol lists shows exported methods

of `reflect.Value`, `time.Time`, `os.File` and `internal/poll.FD` that nothing calls, plus the functions those methods call. The binary is 48% bigger than `const` and 59% bigger than `direct`.

Case 6: Finding the function that turned pruning off

In a real program you did not write the `MethodByName` call yourself. `-ldflags=-dumpdep` makes the linker print every edge of its reachability graph as `from -> to`, and tags symbols with `<ReflectMethod>` and `<UsedInIface>` (see `dumpDepAddFlags` in `deadcode.go`). Filter it:

```
#!/bin/sh
# whyreflect.sh PKG - list the functions that switched off method pruning
# (the linker's <ReflectMethod> flag), excluding package reflect itself.
go build -o /dev/null -ldflags=-dumpdep "$@" 2>&1 |
    grep -oE '^[^ ]+ <ReflectMethod>' | sort -u | grep -v '^reflect\.' ||
    echo "none: method pruning is on"
```

```
$ ./whyreflect.sh ./hello-http
none: method pruning is on
$ ./whyreflect.sh ./methods
main.main <ReflectMethod>
$ go build -ldflags=-dumpdep -o /dev/null ./methods 2>&1 | grep -F 'main.(*Big).M42'
type:*main.Big <UsedInIface> -> main.(*Big).M42
```

(`methods/mode.go` still says `dynamic` from the last loop iteration.) The last line is the whole explanation in one edge: `M42` is kept because the type `*Big` reached an interface, and a `ReflectMethod` function exists.

The most common real culprit is `text/template` (and `html/template`, which uses it). A template field like `{{.Name}}` may name a method, and the template engine looks it up by a name it only learns at run time:

```go
// tmpl/main.go: one text/template call, and what it costs in bytes.
package main

import (
    "os"
    "text/template"
)

type User struct{ Name string }

func main() {
    t := template.Must(template.New("hi").Parse("hello {{.Name}}\n"))
    t.Execute(os.Stdout, User{Name: "gopher"})
}
```

```
$ CGO_ENABLED=0 go build -o bin/tmpl ./tmpl && bin/tmpl
hello gopher
$ stat -c '%s %n' bin/hello-fmt bin/tmpl
2342505 bin/hello-fmt
4345676 bin/tmpl
$ ./whyreflect.sh ./tmpl
text/template.(*state).evalField <ReflectMethod>
$ go tool nm -size bin/tmpl | bin/pkgsize | head -6
runtime                            474386  33.4%
go:func                            208912  14.7%
reflect                            138814   9.8%
time                                79477   5.6%
text/template/parse                 69079   4.9%
text/template                       61804   4.4%
```

One template almost doubles the "hello" binary, and `time` is now bigger than `text/template` itself: every exported method of `time.Time` and `time.Duration` is kept in case a template calls it (`go tool nm` finds 10 exported `time.Duration` methods in `bin/tmpl` and none in `bin/hello-fmt`). That is correct behaviour, not a bug. The template really could call any of them.

Case 7: The shrink list, cheapest first

Change	Saves here	Costs
`-ldflags=-s` (implies `-w`)	2.76 MB of 8.60 MB (32%)	No symbols for `nm`/`perf`/`gdb`, no Delve; panics unaffected. Archive the unstripped build.
`-trimpath`	~12 KB	None for releases; also removes local paths from panics and build info
Replace a dynamic `MethodByName`/`Method(i)` with a constant name, a `switch`, or an interface	1.2 MB in the `methods` demo	A little code; the set of callable methods becomes explicit
Avoid `text/template` in tiny tools (use `fmt` or `strings.Replacer`)	2.0 MB vs `hello-fmt`	Less flexible output formatting
Drop a dependency whose reachable code is large (check with `pkgsize`, not with `go list -deps`)	Depends	Engineering time

What is not on the list: hand-editing the ELF file, or packers. A packer such as UPX compresses the

file and decompresses it into memory at start-up, which costs start-up time and memory, can trip malware scanners, and breaks tools that read the binary on disk (`go version -m`, `govulncheck`). It is not covered or measured here; read its documentation before you consider it.

The trap

- **Treating `-s -w` as "the" size fix.** It removes only metadata. `.text`, `.gopclntab` and `.go.type` are unchanged, and in `hello-http` they are two thirds of the file.
- **Reading the largest `nm` symbols without the kind column.** `B`/`b` symbols live in `.bss` and take no file space. A 32 MB `B` symbol in a 6 MB binary is not a mistake.
- **Counting packages instead of reachable code.** `net` was fully in the dependency graph and cost 152 KB. `text/template` with one field cost 2 MB.
- **One dynamic reflection call in a shared library package.** `REFLECTMETHOD` is a whole-program switch. A helper in one package that calls `reflect.Value.Method(i)` makes every binary that reaches it keep all exported methods of all reachable types.
- **Stripping and then rebuilding to debug.** A later rebuild with different flags is a different binary. Keep the unstripped artifact from the release build.

The boring rule

Measure before you strip, and strip last. Track the stripped size of each release binary in CI. When it jumps, run `pkgsize` on the before and after binaries and `whyreflect.sh` on the new one: a new `<ReflectMethod>` line or a new large package explains most jumps. Use `-trimpath` always and `-s` for release artifacts, and archive the unstripped build next to each release.

Try this

1. Add `-ldflags=-w=0` to an `-s` build of `hello-http`. Which sections come back? Does the panic output change?
2. Replace the `MethodByName(name)` call in `methods/main.go` with a `switch name { case "M7": ... }` that calls methods directly. Re-run the size loop. How close to `direct` do you get?
3. Run `whyreflect.sh` on one of your own services. If it prints a line, follow the `-dumpdep` edges backwards (`grep -- '-> text/template'` and so on) until you reach your own code.
4. Diff `go tool nm bin/methods-const` and `go tool nm bin/methods-dynamic` (kind `T` only). Which standard-library types gained the most methods?
5. Build `hello-http` with `CGO_ENABLED=1` (if a C compiler is installed) and compare the size and `go version -m` output. Which part of `net` changed?

Sources

Go source at the `go1.27.2` tag:

- Linker dead-code pass, roots, method rules and `-dumpdep` flags: https://github.com/golang/go/blob/go1.27.2/src/cmd/link/internal/ld/deadcode.go
- `reflect.Value.MethodByName` and `Method`: https://github.com/golang/go/blob/go1.27.2/src/reflect/value.go#L1950
- `text/template` field evaluation (`evalField`): https://github.com/golang/go/blob/go1.27.2/src/text/template/exec.go#L682

Go documentation:

- `cmd/link` flags (`-s`, `-w`, `-compressdwarf`, `-dumpdep`): https://pkg.go.dev/cmd/link
- `cmd/go` build flags (`-ldflags`, `-trimpath`) and `go version -m`: https://pkg.go.dev/cmd/go
- `go tool nm` (symbol kinds `T`, `D`, `B`, `R`, `U`): https://pkg.go.dev/cmd/nm
- `debug/elf`: https://pkg.go.dev/debug/elf
- `reflect.Value.MethodByName`: https://pkg.go.dev/reflect#Value.MethodByName
- Go 1.2 runtime symbol information (the design of the pc-value tables in `.gopclntab`): https://go.dev/s/go12symtab
- Debugging Go code with GDB (what DWARF provides): https://go.dev/doc/gdb
- Release history (Go 1.27.2, 2026-10-08): https://go.dev/doc/devel/release

Tools: Go 1.27.2 (`go build`, `go tool nm`, `go tool buildid`, `go version -m`), `debug/elf`, and GNU `objcopy` (for the `.text` comparison in Case 3).

epoll Inside the Go Scheduler

epoll Inside the Go Scheduler

Thesis

Goroutines look like they "block on I/O", but a Go **P** rarely sits in a blocking `read`. Network readiness goes through the runtime **netpoller**, which on Linux is **epoll**: one OS thread waits for many file descriptors, then wakes the goroutines whose sockets are ready. Understanding that split — user goroutine vs netpoller thread vs `GOMAXPROCS` Ps — stops people from spawning "one thread per connection" when they already have a multiplexer.

Mental model

```
goroutine Read(socket)

runtime parks G, registers fd with netpoller

netpoller thread: epoll_wait / epoll_pwait

events → find Gs → put on P runqueues → GOMAXPROCS workers run them
```

Piece	Role
G	Goroutine waiting on I/O
M	OS thread; some Ms run Ps, one often drives netpoll
P	Logical processor; `GOMAXPROCS` of them execute Go code
netpoller	Multiplexes fds (epoll on Linux, kqueue on BSD)

Worked examples

Verified with **Go 1.27.2**. This host has no `strace` binary, so we use `GODEBUG=schedtrace` instead of raw `epoll_ctl` traces.

Case 1: tiny HTTP server under schedtrace

```go
package main

import (
    "fmt"
    "net"
    "net/http"
    "os"
    "runtime"
    "time"
)

func main() {
    ln, err := net.Listen("tcp", "127.0.0.1:0")
    if err != nil { panic(err) }
    defer ln.Close()
    mux := http.NewServeMux()
    mux.HandleFunc("/", func(w http.ResponseWriter, r *http.Request) {
        fmt.Fprintln(w, "ok")
    })
    go http.Serve(ln, mux)
    addr := ln.Addr().String()
    for i := 0; i < 200; i++ {
        resp, err := http.Get("http://" + addr + "/")
        if err == nil { resp.Body.Close() }
    }
    fmt.Fprintf(os.Stderr, "GOMAXPROCS=%d NumGoroutine=%d\n", runtime.GOMAXPROCS(0), runtime
    time.Sleep(50 * time.Millisecond)
}
```

```
GODEBUG=schedtrace=100,scheddetail=1 go run .
```

Excerpt from this run:

```
SCHED 0ms: gomaxprocs=8 idleprocs=7 threads=5 spinningthreads=0 needspinning=0 idlethreads=3 ru
  P0: status=0 schedtick=0 syscalltick=0 m=nil runqsize=0 gfreecnt=0 timerslen=0
  P1: status=0 schedtick=0 syscalltick=0 m=nil runqsize=0 gfreecnt=0 timerslen=0
  P2: status=0 schedtick=0 syscalltick=0 m=nil runqsize=0 gfreecnt=0 timerslen=0
  P3: status=0 schedtick=0 syscalltick=0 m=nil runqsize=0 gfreecnt=0 timerslen=0
  P4: status=0 schedtick=0 syscalltick=0 m=nil runqsize=0 gfreecnt=0 timerslen=0
  P5: status=0 schedtick=0 syscalltick=0 m=nil runqsize=0 gfreecnt=0 timerslen=0
  P6: status=1 schedtick=2 syscalltick=0 m=0 runqsize=0 gfreecnt=0 timerslen=0
  P7: status=0 schedtick=3 syscalltick=0 m=nil runqsize=0 gfreecnt=0 timerslen=0
  M4: p=nil curg=nil mallocing=0 throwing=0 preemptoff= locks=0 dying=0 spinning=false blocked=
  M3: p=nil curg=nil mallocing=0 throwing=0 preemptoff= locks=0 dying=0 spinning=false blocked=
  M2: p=nil curg=nil mallocing=0 throwing=0 preemptoff= locks=0 dying=0 spinning=false blocked=
```

You see multiple **P**s (`gomaxprocs=8` here), idle Ps, and Ms blocked or running. The netpoller is why hundreds of connections do not need hundreds of kernel threads.

Trap

Calling blocking syscalls that bypass netpoll. cgo calls, file I/O without the poller, or `syscall` reads on non-pollable fds pin an M. A burst of those creates extra OS threads (`runtime.SetMaxThreads` pressure) even though network I/O looked fine.

Boring rule

Treat sockets through `net/http` so they use the poller. For thread explosions, check cgo and file I/O before blaming the scheduler.

Try this

1. Raise and lower `GOMAXPROCS` and watch `schedtrace` idleprocs.
2. On a machine with `strace`, filter `epoll_ctl` / `epoll_pwait` around the same program.
3. Read `runtime/netpoll_epoll.go` in the Go source tree for the exact calls.

Sources

- Go scheduler design doc / runtime source `netpoll_epoll.go`
- GODEBUG schedtrace

GOMAXPROCS, Ps, and What schedtrace Shows

GOMAXPROCS, Ps, and What schedtrace Shows

Thesis

`GOMAXPROCS` is the number of **P**s — logical processors that can run Go code at once — not a hard cap on OS threads. When people set it "to the number of CPUs" they are usually right on bare metal and wrong in a CPU-limited container. This chapter prints the default, changes it in-process, and reads a `schedtrace` line so the idle/running P counts become visible.

Mental model

```
GOMAXPROCS = N  →  N Ps that execute goroutines
Ms (OS threads)  N when work + syscalls + cgo demand it
schedtrace: idleprocs + running Ps  gomaxprocs
```

Worked examples

Verified with **Go 1.27.2**.

Case 1: read and temporarily lower GOMAXPROCS

```go
package main

import (
    "fmt"
    "runtime"
    "time"
)

func main() {
    fmt.Println("NumCPU", runtime.NumCPU())
    fmt.Println("GOMAXPROCS now", runtime.GOMAXPROCS(0))
    old := runtime.GOMAXPROCS(2)
    fmt.Println("set to 2; was", old)
    done := make(chan struct{})
    for i := 0; i < 8; i++ {
```

```go
        go func() {
            for {
                select {
                case <-done:
                    return
                default:
                }
            }
        }()
    }
    time.Sleep(100 * time.Millisecond)
    fmt.Println("NumGoroutine", runtime.NumGoroutine())
    close(done)
    runtime.GOMAXPROCS(old)
}
```

```
GODEBUG=schedtrace=50 go run .
```

On this 8-CPU guest the default `GOMAXPROCS` is **8**. After setting **2**, busy-spin goroutines still exist, but only two Ps run Go code at a time — `schedtrace` shows `gomaxprocs=2` with fewer idleprocs under load.

Case 2: containers

Inside cgroups v2, Go 1.25+ already adjusts default `GOMAXPROCS` from the CPU quota when appropriate. Prefer fixing the quota over hard-coding `GOMAXPROCS=1` in process env unless you are isolating a latency-critical sidecar.

Trap

`GOMAXPROCS=1` as a "debugging" habit left in production. It serializes all Go code and hides races while destroying throughput. Use the race detector and `schedtrace` instead.

Boring rule

Leave the default alone unless a profile or a CPU quota says otherwise. When you must set it, set it from the **visible CPU limit**, not from `NumCPU()` of the host.

Try this

1. Compare `runtime.GOMAXPROCS(0)` on the host vs in a `--cpus=1.5` container.
2. Pair with the epoll/netpoller chapter under load.
3. Read `runtime.GOMAXPROCS` docs for the latest cgroup behaviour.

Sources

- `runtime.GOMAXPROCS`
- Go release notes on cgroup-aware `GOMAXPROCS` (1.25+)

go list -json for Package Metadata

go list -json for Package Metadata

Thesis

`go list -json` is the stable way to ask the Go toolchain what it knows about a package or module: import path, source files, imports, and more. Scripts should parse JSON, not scrape `go list` text tables.

Mental model

```
go list -json <pattern>
     → one JSON object per package (stream)
go list -json -m
     → module object for the main module
```

Useful fields: `ImportPath`, `Dir`, `GoFiles`, `Imports`, `Deps`, `Module`, `TestGoFiles`.

Worked example

Verified with **Go 1.27.2**.

```
go list -json encoding/json
```

Excerpt from this box:

```
ImportPath: encoding/json
Name: json
Dir: /usr/local/go/src/encoding/json
Doc: Package json implements encoding and decoding of JSON as defined in RFC 7159.
Root: /usr/local/go
Module: None
GoFiles: ['v2_decode.go', 'v2_encode.go', 'v2_indent.go', 'v2_inject.go', 'v2_options.go', '
Imports: ['bytes', 'cmp', 'encoding', 'encoding/json/internal', 'encoding/json/internal/json
Deps: ['bytes', 'cmp', 'encoding', 'encoding/base32', 'encoding/base64', 'encoding/binary',
```

A tiny consumer:

```go
/* listmeta.go */
package main

import (
    "encoding/json"
    "fmt"
    "os/exec"
)

func main() {
    out, err := exec.Command("go", "list", "-json", "fmt").Output()
    if err != nil {
        panic(err)
    }
    var p struct {
        ImportPath string
        Name       string
        Dir        string
        GoFiles    []string
    }
    if err := json.Unmarshal(out, &p); err != nil {
        panic(err)
    }
    fmt.Printf("%s (%s) files=%d\n", p.ImportPath, p.Name, len(p.GoFiles))
}
```

```
go run listmeta.go
```

Sample run:

```
Path=command-line-arguments
ImportPath=fmt Name=fmt Dir=/usr/local/go/src/fmt
GoFiles sample=[doc.go errors.go format.go print.go scan.go]
```

Trap

Assuming one object. `go list -json ./...` emits **NDJSON** (one object per line/package). `json.Decoder` in a loop, not a single `Unmarshal` into a slice.

Boring rule

For tooling, prefer `go list -json` (and `-m` / `-deps` as needed) over parsing build IDs or directory layouts by hand. Pin the fields you read; ignore unknown ones for forward compatibility.

Try this

1. `go list -json -deps fmt` and count unique `ImportPaths`.
2. Compare `GoFiles` vs `TestGoFiles` for one of your packages.
3. Use `-f '{{.ImportPath}} {{.Dir}}'` only for quick human views — keep JSON for scripts.

Sources

- `go help list`
- Go 1.27 toolchain docs on module-aware `go list`

goroutineleak pprof Profile

goroutineleak pprof Profile

Thesis

Go 1.27's `goroutineleak` pprof profile reports goroutines that are **blocked forever on unreachable synchronisation objects** — the classic "lost channel" leak. It is distinct from the plain `goroutine` profile, which shows everything currently runnable or blocked.

Mental model

```
go func() {
    c := make(chan struct{})
    go func() { <-c }()   // blocked forever
    // c goes out of scope → unreachable
}()

pprof: goroutineleak  →  stacks of such unreachable blocked Gs
```

The profile writer triggers GC / leak detection before snapshotting (`writeGoroutineLeak` in `runtime/pprof`).

Worked example

Verified against **Go 1.27.2** sources (`src/runtime/pprof/pprof.go`).

```go
/* leakprobe.go */
package main

import (
    "fmt"
    "net/http"
    _ "net/http/pprof"
    "runtime"
    "time"
)

func main() {
    go func() {
```

```go
        c := make(chan struct{})
        go func() { <-c }()
        // c not retained → inner G is a leak candidate after GC
    }()
    time.Sleep(50 * time.Millisecond)
    runtime.GC()
    fmt.Println("NumGoroutine", runtime.NumGoroutine())

    go http.ListenAndServe("127.0.0.1:6061", nil)
    fmt.Println("fetch: curl -s 'http://127.0.0.1:6061/debug/pprof/goroutineleak?debug=1'")
    time.Sleep(time.Hour)
}
```

```bash
go run leakprobe.go
# elsewhere:
curl -s 'http://127.0.0.1:6061/debug/pprof/goroutineleak?debug=1' | head
```

Profile registration on this GOROOT includes:

```bash
name:  "goroutineleak"
# writeGoroutineLeak first invokes a GC cycle that performs goroutine leak detection
```

Trap

Treating every blocked goroutine as a leak. A server goroutine blocked on `Accept` is healthy. `goroutineleak` is specifically about **unreachable** sync objects. Also: taking the profile forces GC work — do not scrape it at huge QPS without budgets.

Boring rule

When diagnosing "goroutine count only goes up", compare `goroutine` vs `goroutineleak`. Fix leaks by retaining shutdown signals (context cancel, close of a done channel) rather than dropping the only sender/receiver.

Try this

1. `go tool pprof http://.../debug/pprof/goroutineleak` and `list` the offending frames.
2. Deliberately leak, confirm the profile, then fix with `context.WithCancel` and re-check.
3. Read `TestGoroutineLeakProfileConcurrency` in `$GOROOT/src/runtime/pprof/pprof_test.go`.

Sources

- Go 1.27 `src/runtime/pprof/pprof.go` (`goroutineleak` profile)
- `net/http/pprof`
- Go release notes for 1.27 diagnostics

Assets

Long-Term Maintenance

Reflection in Practice

Reflection in Practice

The boring default is **not to use `reflect`**. Write a method that knows the type. Reach for reflection when you are talking to a format you do not control (a dump, a generic log line) and the set of fields is the point. It is a last resort, not a design style.

Mental model

The `reflect` package lets a program ask a value what it is at run time: its `Type`, its `Kind` (struct, int, pointer, …), and its fields. That is how `fmt` prints `%v` for types it has never seen, and how `encoding/json` walks structs.

The cost is real: no compile-time check that a field exists, panics on unexported fields if you call `Interface()`, and code that reads like a debugger. If you know the type, you do not need any of it.

Worked examples

Case 1: A tiny field dump for a desk order

Save as `dump.go`. This is the honest use: print exported fields of a struct you might otherwise pretty-print by hand.

```go
// dump.go
package main

import (
    "fmt"
    "reflect"
)

type Order struct {
    ID     int
    Table  int
    Item   string
    Closed bool
}

func dump(v any) {
```

```go
    rv := reflect.ValueOf(v)
    if rv.Kind() == reflect.Pointer {
        rv = rv.Elem()
    }
    if rv.Kind() != reflect.Struct {
        fmt.Printf("%v\n", v)
        return
    }
    rt := rv.Type()
    for i := range rt.NumField() {
        f := rt.Field(i)
        fmt.Printf("%s = %v\n", f.Name, rv.Field(i).Interface())
    }
}

func main() {
    dump(Order{ID: 7, Table: 12, Item: "soup", Closed: false})
}
```

Run:

```go
go run dump.go
```

Output:

```
ID = 7
Table = 12
Item = soup
Closed = false
```

`for i := range rt.NumField()` is the Go 1.22+ integer range. Each field is looked up by index, not by a string you typed twice.

Case 2: The same dump without reflection

When the type is yours, a method is the whole API. Save as `label.go`:

```go
// label.go
package main

import "fmt"

type Order struct {
    ID     int
    Table  int
    Item   string
```

```go
    Closed bool
}

func (o Order) Dump() string {
    return fmt.Sprintf("ID = %d\nTable = %d\nItem = %s\nClosed = %v", o.ID, o.Table, o.Item,
}

func main() {
    fmt.Println(Order{ID: 7, Table: 12, Item: "soup", Closed: false}.Dump())
}
```

Run:

```go
go run label.go
```

Output:

```
ID = 7
Table = 12
Item = soup
Closed = false
```

This version renames a field at compile time. Case 1 silently skips a field you forgot to care about, or panics if you later unexport it. Prefer Case 2 inside the desk. Keep Case 1 for tools that must accept `any`.

Case 3: Kinds you actually switch on

Reflection without a kind check is how you panic on a nil pointer. Save as `kind.go`:

```go
// kind.go
package main

import (
    "fmt"
    "reflect"
)

func kindOf(v any) {
    if v == nil {
        fmt.Println("nil")
        return
    }
    fmt.Printf("%s (%s)\n", reflect.TypeOf(v), reflect.ValueOf(v).Kind())
}
```

```go
func main() {
    kindOf(7)
    kindOf("soup")
    kindOf(struct{ Table int }{Table: 12})
    var p *int
    kindOf(p)
    kindOf((*int)(nil))
}
```

Run:

```go
go run kind.go
```

Output:

```
int (int)
string (string)
struct { Table int } (struct)
*int (ptr)
*int (ptr)
```

A typed nil pointer is not the `any` containing `nil`. Check both if the value came from an interface.

The trap

Calling `Interface()` on an unexported field panics. This is not a compiler error. Save as `panic_dump.go`:

```go
// panic_dump.go
package main

import (
    "fmt"
    "reflect"
)

type order struct {
    id int
}

func main() {
    o := order{id: 7}
    v := reflect.ValueOf(o)
    f := v.Field(0)
    fmt.Println(f.Interface())
}
```

Run:

```
go run panic_dump.go
```

Output (stack trimmed):

```
panic: reflect.Value.Interface: cannot return value obtained from unexported field or method
```

The fix is not `unsafe`. Export the field, or print it from a method on `order` in the same package (`f.Int()` still works for an unexported int in the same package, but you are now writing a debugger). The boring fix:

```go
// safe_dump.go
package main

import "fmt"

type order struct {
    id int
}

func (o order) ID() int { return o.id }

func main() {
    fmt.Println(order{id: 7}.ID())
}
```

Run:

```
go run safe_dump.go
```

Output:

```
7
```

If you find yourself writing a mini `encoding/json` for the desk, stop and use `encoding/json`.

The boring rule

- Do not design your package around `any` plus reflect. Design around a type.
- Use reflect when the type is not known at compile time (fmt-like printers, generic dumps).
- Always check `Kind` (and nil) before `Elem` or `Field`.
- Never call `Interface()` on unexported fields.
- `encoding/json` and `fmt` already walked this path. Import them before you re-implement them.

Try this

1. In `dump.go`, pass `&Order{ID: 1, Table: 2, Item: "tea"}` and confirm the pointer branch still prints fields.
2. Pass `42` to `dump` and confirm it takes the non-struct branch.
3. Add an unexported field `note string` to `Order` in `dump.go`. Run it. Decide whether you wanted that field in the dump. If not, Case 2 was the better API.
4. Replace the dump with `json.MarshalIndent` on `Order` and print the bytes. That is the standard-library version of Case 1.

Unsafe Code

Unsafe Code

The boring default is **never to import `unsafe`**. The one use this book will show is **`unsafe.Sizeof`** (and the idea of `Alignof`) so you can see how big a value is. That is a measurement. It is not a licence to poke memory.

Mental model

Go's type system does not let you treat an arbitrary integer as a pointer. Package `unsafe` does. The compiler and the garbage collector have a **contract** with `unsafe.Pointer`. If you violate it, you do not get a nice error. You get a binary that is wrong on the next toolchain release.

`unsafe.Sizeof(x)` returns the size in bytes of `x`'s **representation** as a `uintptr`. It does not allocate. It does not follow pointers: the size of a `string` is the header (pointer plus length), not the bytes of the text.

`unsafe.Alignof(x)` is the alignment the compiler uses for that type. This chapter does not call it in code. You can see alignment **effects** with `Sizeof` on structs that include padding.

We will not take `unsafe.Pointer` to a `uintptr` and back. That is how people write exploits and also how people write "clever" code that breaks when the GC moves. Do not.

Worked examples

Sizes below assume a typical **64-bit** machine (`amd64` or `arm64`). On 32-bit, `int` and headers shrink. Measure on the machine you ship.

Case 1: Size of the types you already use

Save as `sizes.go`:

```go
// sizes.go
package main

import (
    "fmt"
    "unsafe"
)

type Ticket struct {
```

```go
    ID    int
    Table int
}

func main() {
    var n int
    var s string
    var t Ticket
    var p *Ticket
    fmt.Println("int", unsafe.Sizeof(n))
    fmt.Println("string", unsafe.Sizeof(s))
    fmt.Println("Ticket", unsafe.Sizeof(t))
    fmt.Println("*Ticket", unsafe.Sizeof(p))
}
```

Run:

```go
go run sizes.go
```

Typical output on 64-bit:

```
int 8
string 16
Ticket 16
*Ticket 8
```

`string` is 16 because it is a pointer and a length, each 8 bytes. The characters live elsewhere. `Ticket` is two `int`s. A pointer is one machine word.

Case 2: Padding is visible as a larger `Sizeof`

Save as `padding.go`. Both structs hold one `int64` and two `bool`s. Field order changes the size because the compiler inserts padding to align `int64`.

```go
// padding.go
package main

import (
    "fmt"
    "unsafe"
)

type Waste struct {
    A bool
    B int64
    C bool
```

```go
}

type Tight struct {
    B int64
    A bool
    C bool
}

func main() {
    fmt.Println("Waste", unsafe.Sizeof(Waste{}))
    fmt.Println("Tight", unsafe.Sizeof(Tight{}))
}
```

Run:

```go
go run padding.go
```

Typical output on 64-bit:

```
Waste 24
Tight 16
```

`Waste` is $1 + 7$ pad $+ 8 + 1 + 7$ pad $= 24$. `Tight` is $8 + 1 + 1 + 6$ pad $= 16$. That is interesting if you have millions of these in a slice. It is not a reason to reorder a desk `Ticket` with three fields.

Case 3: A slice header is not the backing array

Save as `slice_size.go`:

```go
// slice_size.go
package main

import (
    "fmt"
    "unsafe"
)

func main() {
    small := []int{1, 2, 3}
    large := make([]int, 10_000)
    fmt.Println("small header", unsafe.Sizeof(small))
    fmt.Println("large header", unsafe.Sizeof(large))
    fmt.Println("one int", unsafe.Sizeof(small[0]))
}
```

Run:

```
go run slice_size.go
```

Typical output on 64-bit:

```
small header 24
large header 24
one int 8
```

Both headers are 24 bytes (pointer, length, capacity). `Sizeof` does not report the 10_000 integers. Do not use it to estimate memory of a slice; use `len` × element size plus the header, and remember capacity.

Case 4: Field alignment and padding with `Offsetof`

Save as `alignof.go`. Why did `Waste` take 24 bytes in Case 2 while `Tight` took 16? Hardware CPUs read memory in naturally aligned boundaries (e.g. an 8-byte `int64` must sit at a memory address divisible by 8). `unsafe.Alignof` reports the required alignment, and `unsafe.Offsetof` shows the exact byte offset of each field within the struct.

```go
// alignof.go
package main

import (
    "fmt"
    "unsafe"
)

type Ticket struct {
    Active bool
    Table  int64
    ID     int32
}

func main() {
    var t Ticket
    fmt.Println("size of Ticket:", unsafe.Sizeof(t))
    fmt.Println("align of bool:", unsafe.Alignof(t.Active))
    fmt.Println("align of int64:", unsafe.Alignof(t.Table))
    fmt.Println("offset of Active:", unsafe.Offsetof(t.Active))
    fmt.Println("offset of Table:", unsafe.Offsetof(t.Table))
    fmt.Println("offset of ID:", unsafe.Offsetof(t.ID))
}
```

Run:

```
go run alignof.go
```

Typical output on 64-bit:

```
size of Ticket: 24
align of bool: 1
align of int64: 8
offset of Active: 0
offset of Table: 8
offset of ID: 16
```

`Active` starts at byte 0 and takes 1 byte. Because `Table` requires 8-byte alignment, the compiler leaves bytes 1 through 7 unused as padding. `Table` occupies bytes 8 through 15. `ID` occupies bytes 16 through 19. Finally, 4 trailing padding bytes are added so that an array of `Ticket` structs preserves the 8-byte alignment for the next element (total 24 bytes).

The trap

Importing `unsafe` so you can convert a `[]byte` to a `string` without copying looks fast and is a contract with the compiler you probably do not want. This chapter will **not** show that conversion.

The boring path copies. Save as `copy_string.go`:

```go
// copy_string.go
package main

import "fmt"

func main() {
    b := []byte("soup")
    s := string(b)
    b[0] = 'S'
    fmt.Println(s)
    fmt.Println(string(b))
}
```

Run:

```
go run copy_string.go
```

Output:

```
soup
Soup
```

`s` is independent of `b` because `string(b)` copied. That is the behaviour you want at the desk. A `unsafe.Pointer` cast that aliases the same bytes will make `s` change when `b` changes — or worse, when the GC reclaims `b`. Do not go looking for that spell.

`unsafe.Pointer` is a contract with the compiler. If you are not prepared to read the current `unsafe` package docs and the Go release notes every six months, you are not prepared to use it.

The boring rule

- Do not import `unsafe` in desk code.
- `unsafe.Sizeof`, `Alignof`, and `Offsetof` are diagnostic rulers. Use them in a throwaway program when you are curious about memory layout.
- Do not reorder fields for speed until a profile says a giant slice of that struct is hot.
- Do not use `unsafe.Pointer`. Not to dodge a copy, not to call C, not to "inspect" a slice.
- If a library requires you to pass `unsafe.Pointer`, read why, wrap it in one file, and keep it off the rest of the API.

Try this

1. In `sizes.go`, add a `bool` field to `Ticket` (first, then last). Print `Sizeof` each time. On 64-bit you will likely see padding.
2. In `alignof.go`, reorder the fields in `Ticket` to put `Table int64` first, then `ID int32`, then `Active bool`. Print `unsafe.Sizeof(t)` and observe how the size drops from 24 to 16 bytes.
3. Print `unsafe.Sizeof([3]int{})` and compare it to `unsafe.Sizeof([]int{})`. The array is three ints; the slice is a header.
4. Run `sizes.go` with `GOARCH=386` if you have a C compiler and cgo, or just read `go env GOARCH` and accept that 32-bit numbers differ. Do not invent a `Pointer` cast to "normalise" them.

Cgo and Foreign Function Interfaces

Cgo and Foreign Function Interfaces

The boring default is **pure Go**. Write the function in Go. Ship with `CGO_ENABLED=0`. A foreign function interface (calling C from Go, or Go from C) is a last resort for a library that does not exist in Go and cannot be rewritten in a weekend.

Mental model

cgo is the toolchain's bridge: a file that `import "C"` can contain a C snippet (or include a header) and call those functions as `C.name(...)`. The Go compiler invokes a C compiler. The binary typically links `libc`. Cross-compilation stops being two environment variables. `go test` gets slower. The resulting program is no longer a single obvious file you copy between Linux boxes.

An FFI is that bridge in the abstract. cgo is Go's built-in one. Other languages have `ctypes`, JNI, and so on. The cost is the same: two runtimes, two memory models, two failure modes.

If the C function is three lines, it is a Go function. That is Case 1.

Worked examples

Case 1: The pure-Go function you should ship

Save as `bump.go`. A table number goes up by one. No C.

```go
// bump.go
package main

import "fmt"

func bump(n int) int {
    return n + 1
}

func main() {
    fmt.Println(bump(3))
}
```

Run:

```
go run bump.go
```

Output:

```
4
```

Build the boring binary:

```
CGO_ENABLED=0 go build -o bump bump.go
./bump
```

Output:

```
4
```

Stop here unless you have a reason you can write in a commit message.

Case 2: The same increment, optionally in C

This program needs a **C compiler** (`gcc` or `clang`) and `CGO_ENABLED=1` (the default on most developer machines). If `go run` complains that cgo is disabled or `CC` is missing, skip this case. The point of the chapter is Case 1.

The C comment must sit **immediately above** `import "C"`. Do not put `import "C"` inside a grouped import.

Save as `bump_c.go`:

```go
// bump_c.go
package main

/*
int bump(int n) {
    return n + 1;
}
*/
import "C"
import "fmt"

func main() {
    fmt.Println(int(C.bump(3)))
}
```

Run:

```
CGO_ENABLED=1 go run bump_c.go
```

Output:

`C.bump` returns a C `int`. Convert it with `int(...)` before you print if you want a Go `int`. Types that look the same (`int`, `C.int`) are not the same type.

You cannot `CGO_ENABLED=0 go run bump_c.go`. That command fails. That failure is the tax you pay for Case 2 on every machine and every CI image.

Case 3: Cross-compile as the reason to stay in Go

Save Case 1 as `bump.go` again. Cross-compile without a C toolchain for the target:

```
CGO_ENABLED=0 GOOS=linux GOARCH=arm64 go build -o bump-arm64 bump.go
```

No output on success. You have a Linux arm64 binary. Repeat for `windows/amd64` if you need it.

Now imagine Case 2. You would need a C cross-compiler, headers, and a libc for each target. Teams that "just wrap OpenSSL" discover this on the first Windows build.

Case 4: C strings and manual memory management

When passing strings or byte buffers to C, `C.CString(s)` allocates a null-terminated copy on the C heap using `malloc()`. The Go garbage collector **does not know this memory exists**. You must explicitly free it with `C.free()`.

Save as `c_strings.go`:

```
// c_strings.go
package main

/*
#include <stdlib.h>
#include <string.h>

int c_length(const char* s) {
    return strlen(s);
}
*/
import "C"
import (
    "fmt"
    "unsafe"
)

func main() {
    // Allocate on C heap: Go GC will NEVER free this automatically
    cs := C.CString("desk ticket #101")
```

```go
    defer C.free(unsafe.Pointer(cs)) // Must be freed explicitly

    len := int(C.c_length(cs))
    fmt.Println("string length from C:", len)
}
```

Run:

```
CGO_ENABLED=1 go run c_strings.go
```

Output:

```
string length from C: 16
```

If you omit `defer C.free(unsafe.Pointer(cs))`, every call leaks bytes into the process memory space that Go runtime GC cycles will never reclaim. That is the manual memory burden cgo introduces to your codebase.

The trap

Pulling in a C library on day one of a desk CLI because "the C version is the real one" is how a 40-line tool becomes a docker image.

This Go program already does the job. Wrapping C `printf` would not make it more correct:

```go
// open.go
package main

import "fmt"

func main() {
    fmt.Println("opened table 3")
}
```

Run:

```
go run open.go
```

Output:

```
opened table 3
```

If you need TLS, use `crypto/tls`. If you need SQLite, prefer a pure-Go driver until you measure a reason not to. If you must call C, isolate `import "C"` in one package, test it hard, and keep it off the public API of the desk.

cgo also means:

1. **Stack switches**: Go goroutines have small growing stacks (starting at 2–4 KB); C functions require fixed POSIX stacks. Crossing the boundary forces stack switches and carries significant call overhead.
2. **Crash propagation**: A segfault in C immediately crashes the entire Go process — it cannot be caught by Go's `recover()`.
3. **Leaked memory**: Memory allocated with C `malloc` is invisible to Go GC and must be manually tracked and freed.

None of that is worth an increment function.

The boring rule

- Write it in Go first. Case 1 is the product.
- Use cgo only for a dependency that does not exist in Go and will not be rewritten.
- When passing Go strings to C with `C.CString`, immediately pair it with `defer C.free(unsafe.Pointer(cs))`.
- Keep `import "C"` in one file. Convert C types at the boundary.
- Default `CGO_ENABLED=0` in release builds so cgo cannot sneak in through a dependency.
- Do not use cgo to "go faster." Measure. The boundary switch overhead often wipes out C speed advantages.

Try this

1. Time `go build` on `bump.go` with `CGO_ENABLED=0` and, if you ran Case 2, on `bump_c.go` with cgo on. The difference is the C compiler.
2. In `c_strings.go`, verify what happens if you remove `#include <stdlib.h>`. Read the compiler error about `C.free`.
3. Run `CGO_ENABLED=0 go run bump_c.go` and read the error. Keep that error in mind when CI is a tiny image.
4. Change `bump` in Case 1 to add 10 instead of 1. Notice you did not edit C, install `gcc`, or worry about `C.int` overflow.
5. If you must practise cgo, add a C function `int tables(void) { return 12; }` and print `C.tables()`. Then delete the file and go back to Case 1.

Performance Tuning

Performance Tuning

The boring default is **measure first**. Write the desk so it is correct and readable. If it is slow, use `testing.B`, then `pprof`, then change the line the profile names. Do not micro-optimise ticket labels because a blog post mentioned allocs.

Mental model

A **benchmark** is a test named `BenchmarkXxx` with argument `*testing.B`. The tooling runs it until the number is stable and reports ns/op. `-benchmem` adds allocations per operation.

pprof is the profiler that ships with Go. You record a CPU or memory profile while a test (or a program) runs, then `go tool pprof` shows where time or bytes went.

Allocation-aware code means you notice when a hot loop builds strings with `+=` and copies the whole buffer every time. It does not mean object pools for a CLI that prints twenty tickets.

`for b.Loop() { ... }` is the Go 1.24+ way to write the benchmark body. The compiler keeps the loop from being erased. Prefer it over a manual `b.N` loop on Go 1.27.

Worked examples

Directory (library in a subfolder so `package main` can import it):

```
desk/
  go.mod
  join/join.go
  join/join_test.go
  main.go
```

Save `go.mod` at `desk/`:

```
module example.com/desk

go 1.27
```

Case 1: Two ways to join ticket ids, one of them copies too much

Save as `join/join.go`:

```go
// join.go
package join

import (
    "fmt"
    "strings"
)

func JoinPlus(ids []int) string {
    s := ""
    for _, id := range ids {
        s += fmt.Sprintf("#%d ", id)
    }
    return s
}

func JoinBuilder(ids []int) string {
    var b strings.Builder
    for _, id := range ids {
        fmt.Fprintf(&b, "#%d ", id)
    }
    return b.String()
}
```

join/join_test.go:

```go
// join_test.go
package join

import "testing"

var ids = []int{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12}

func BenchmarkJoinPlus(b *testing.B) {
    b.ReportAllocs()
    for b.Loop() {
        _ = JoinPlus(ids)
    }
}

func BenchmarkJoinBuilder(b *testing.B) {
    b.ReportAllocs()
    for b.Loop() {
        _ = JoinBuilder(ids)
    }
}
```

From the module root:

```go
go test -bench=. -benchmem ./join
```

Example output (numbers move with the machine; the **shape** is the lesson):

```
goos: linux
goarch: amd64
pkg: example.com/desk/join
BenchmarkJoinPlus-8            300000          3800 ns/op         496 B/op         22 allocs/op
BenchmarkJoinBuilder-8         500000          2400 ns/op         160 B/op          4 allocs/op
PASS
ok      example.com/desk/join   2.100s
```

`JoinPlus` allocates more because each `+=` may copy the string so far. `JoinBuilder` grows a buffer. You only rewrite the desk to use the builder when this function is on the hot path **and** the numbers bother a real user.

Case 2: The program you actually ship

Save as `main.go` at the module root. The CLI uses the builder because we already measured. It does not use a pool, a cache, or `unsafe`.

```go
// main.go
package main

import (
    "fmt"

    "example.com/desk/join"
)

func main() {
    fmt.Println(join.JoinBuilder([]int{7, 8, 9}))
}
```

Run:

```go
go run .
```

Output:

```
#7 #8 #9
```

Case 3: `pprof` when a number is not enough

CPU profile of one benchmark:

```go
go test -bench=BenchmarkJoinPlus -cpuprofile=cpu.pprof ./join
```

Then a text summary (no interactive UI):

```go
go tool pprof -top cpu.pprof
```

Example top lines (names and percents will not match yours exactly):

```
File: join.test
Duration: 2.10s, Total samples = 1.80s (85.71%)
Showing nodes accounting for 1.80s, 100% of 1.80s
      flat  flat%   sum%        cum   cum%
     0.90s 50.00% 50.00%      0.90s 50.00%  runtime.mallocgc
     0.40s 22.22% 72.22%      1.20s 66.67%  example.com/desk/join.JoinPlus
     0.20s 11.11% 83.33%      0.50s 27.78%  fmt.Sprintf
```

If `mallocgc` and `JoinPlus` dominate, the allocation story from `-benchmem` is confirmed. Memory profile:

```go
go test -bench=BenchmarkJoinPlus -memprofile=mem.pprof ./join
go tool pprof -top -alloc_space mem.pprof
```

Read the function names. Change **those**. Delete the `*.pprof` files when you are done; they are not source.

Case 4: Zero-cost speedup: preallocating slice capacity

Save as `alloc/alloc_test.go`. The simplest, highest-return optimization in Go services is giving `make` a capacity hint when the item count is known ahead of time. Growing a slice without a capacity hint triggers repeated reallocation and memory copying.

```go
// alloc_test.go
package alloc

import "testing"

func BenchmarkAppendNoCap(b *testing.B) {
    b.ReportAllocs()
    for b.Loop() {
        var s []int
        for i := range 1000 {
            s = append(s, i)
        }
        _ = s
    }
}
```

```go
func BenchmarkAppendWithCap(b *testing.B) {
    b.ReportAllocs()
    for b.Loop() {
        s := make([]int, 0, 1000)
        for i := range 1000 {
            s = append(s, i)
        }
        _ = s
    }
}
```

Run:

```go
go test -bench=. -benchmem ./alloc
```

Typical output:

```
BenchmarkAppendNoCap-4        100320         14379 ns/op       25208 B/op          12 allocs/op
BenchmarkAppendWithCap-4      429908          2633 ns/op           0 B/op           0 allocs/op
```

By providing `make([]int, 0, 1000)`, allocations dropped from 12 per operation to 0 (the slice backing array is sized in one shot), and execution speed improved by over 5x. This requires no complex caching or object pools.

The trap

Rewriting the desk to avoid allocations before a user has waited on anything. This program is already fast enough:

```go
// tickets.go
package main

import "fmt"

func main() {
    for _, id := range []int{7, 8, 9} {
        fmt.Printf("ticket %d\n", id)
    }
}
```

Run:

```go
go run tickets.go
```

Output:

```
ticket 7
ticket 8
ticket 9
```

A pool of `[]byte`, a custom formatter, and a blog about "zero alloc" would not make this more correct. They would make Monday morning slower for the next reader. Benchmark when someone can feel a delay, or when a profile of a service says so. Not because twelve tickets allocate.

The boring rule

- Make it right. Then measure. Then change the hot function.
- `go test -bench=. -benchmem` is the first instrument.
- `go tool pprof -top` is the second. Do not guess.
- `strings.Builder` (or `bytes.Buffer`) beats `+=` in a loop. That is the usual win. Stop there.
- Do not pool, intern, or `unsafe`-cast the desk. Do not "optimise" `fmt.Printf` of three lines.

Try this

1. Grow `ids` in the test to 1_000 elements. Re-run `-benchmem`. Watch `JoinPlus` get worse faster than `JoinBuilder`.
2. Add `b.ResetTimer()` after building a large `ids` slice inside the benchmark (move setup above the loop). Confirm the ns/op still compares the join, not the slice make.
3. Record `-memprofile` for `BenchmarkJoinBuilder` and `-top -alloc_space`. See that the remaining alloc is mostly the result string.
4. Leave `tickets.go` as it is. Do not convert it to a builder.

Designing for the Long Term

Designing for the Long Term

The boring default is a **small exported API**, **package boundaries that match real jobs**, and **compatibility you treat as a promise**. The rest of this book is technique. This chapter is what still matters when the desk has been in production for a year.

Mental model

A **package** is a name the rest of the program can import. Everything exported (`Open`, `Ticket`, `Label`) is a promise. Everything unexported is yours to change on a Tuesday.

A **boundary** is a reason to split: "tickets" versus "shifts" versus `main`. Not "models" versus "helpers" versus `utils`. If two files always change together and nobody imports them separately, they are one package.

Compatibility means a caller that compiled against last month's module still compiles. You can add a method. You can add an unexported field. You cannot rename `Open`, shuffle its parameters, or change `Ticket` to `*Ticket` without a new major version (`example.com/desk/v2`).

Worked examples

Layout:

```
desk/
  go.mod
  ticket/ticket.go
  main.go
```

Save `go.mod`:

```
module example.com/desk

go 1.27
```

Case 1: A stable ticket API

Unexported fields, a constructor that validates, methods that return values. Callers cannot write struct literals that break when you add a field.

Save as `ticket/ticket.go`:

```go
// ticket.go
package ticket

import "fmt"

// Ticket is an opened desk ticket. Use Open. Do not depend on the zero value.
type Ticket struct {
    id    int
    table int
}

// Open returns a ticket for a positive id and table.
func Open(id, table int) (Ticket, error) {
    if id <= 0 {
        return Ticket{}, fmt.Errorf("ticket id must be positive")
    }
    if table <= 0 {
        return Ticket{}, fmt.Errorf("table must be positive")
    }
    return Ticket{id: id, table: table}, nil
}

// ID is the ticket identifier.
func (t Ticket) ID() int { return t.id }

// Table is the table the ticket is for.
func (t Ticket) Table() int { return t.table }

// Label is the one-line stub.
func (t Ticket) Label() string {
    return fmt.Sprintf("ticket %d → table %d", t.id, t.table)
}
```

Save as `main.go`:

```go
// main.go
package main

import (
    "fmt"
    "os"

    "example.com/desk/ticket"
)

func main() {
    t, err := ticket.Open(7, 12)
```

```go
    if err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }
    fmt.Println(t.Label())
}
```

Run from the module root:

```go
go run .
```

Output:

```
ticket 7 → table 12
```

This is the surface you freeze. Internals can grow.

Case 2: Add behaviour without breaking callers

A year later you need a note on the ticket. Add an unexported field and a method. `Open`'s signature does not change. `main.go` does not change.

Save as `ticket/ticket.go`:

```go
// ticket.go
package ticket

import "fmt"

// Ticket is an opened desk ticket. Use Open. Do not depend on the zero value.
type Ticket struct {
    id    int
    table int
    note  string
}

// Open returns a ticket for a positive id and table.
func Open(id, table int) (Ticket, error) {
    if id <= 0 {
        return Ticket{}, fmt.Errorf("ticket id must be positive")
    }
    if table <= 0 {
        return Ticket{}, fmt.Errorf("table must be positive")
    }
    return Ticket{id: id, table: table}, nil
}
```

```go
// WithNote returns a copy with a note. The original ticket is unchanged.
func (t Ticket) WithNote(note string) Ticket {
    t.note = note
    return t
}

// ID is the ticket identifier.
func (t Ticket) ID() int { return t.id }

// Table is the table the ticket is for.
func (t Ticket) Table() int { return t.table }

// Label is the one-line stub.
func (t Ticket) Label() string {
    if t.note == "" {
        return fmt.Sprintf("ticket %d → table %d", t.id, t.table)
    }
    return fmt.Sprintf("ticket %d → table %d (%s)", t.id, t.table, t.note)
}
```

The original `main.go` still runs:

```go
go run .
```

Output:

```
ticket 7 → table 12
```

A new caller can opt in:

```go
// note.go
package main

import (
    "fmt"
    "os"

    "example.com/desk/ticket"
)

func main() {
    t, err := ticket.Open(7, 12)
    if err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }
    t = t.WithNote("window")
```

```go
    fmt.Println(t.Label())
}
```

Keep only one `main` in the directory. Replace `main.go` with that file, or run it as the module's `main.go`. Output of `go run .`:

```
ticket 7 → table 12 (window)
```

Old binaries that called `Open` and `Label` still match Case 1. New code asks for a note. That is how a package lasts.

Case 3: A boundary that is a real job

A second package for shifts, imported by `main`, not dumped into `ticket` as "related desk stuff." Save as `shift/shift.go`:

```go
// shift.go
package shift

import "fmt"

// Name is a roster label such as "morning".
type Name string

const (
    Morning Name = "morning"
    Evening Name = "evening"
)

// Label prints the roster line for a person.
func Label(person string, n Name) string {
    return fmt.Sprintf("%s · %s", person, n)
}
```

Save as `main.go` (this binary talks to both packages; they do not import each other):

```go
// main.go
package main

import (
    "fmt"
    "os"

    "example.com/desk/shift"
    "example.com/desk/ticket"
)
```

```go
func main() {
    t, err := ticket.Open(7, 12)
    if err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }
    fmt.Println(t.Label())
    fmt.Println(shift.Label("Amina", shift.Morning))
}
```

Run:

```go
go run .
```

Output:

```
ticket 7 → table 12
Amina · morning
```

Do not import `shift` from `ticket` unless a ticket literally cannot exist without a shift. Cycles are a smell that the split was fake.

The trap

A `utils` package that every file imports. Six months later it holds string helpers, time parsing, and a half-written HTTP client. Names collide. Tests become a hairball.

This is the wrong home for `Open`:

```go
// utils.go
package utils

import "fmt"

func OpenTicket(id, table int) string {
    return fmt.Sprintf("ticket %d → table %d", id, table)
}
```

```go
// use_utils.go
package main

import "fmt"

func OpenTicket(id, table int) string {
    return fmt.Sprintf("ticket %d → table %d", id, table)
```

```go
}

func main() {
    fmt.Println(OpenTicket(7, 12))
}
```

Run the second file (it does not need `utils`):

```
go run use_utils.go
```

Output:

```
ticket 7 → table 12
```

Put `Open` in `package ticket`. Put `main` in `package main`. If you only have twenty lines, **one** package `main` is fine. Split when a name wants two homes, or when another binary must import the API without importing `main`.

The boring rule

- Export a constructor and methods. Keep fields unexported unless the type is a bag of options you intend to grow with keyed literals only.
- Add methods and unexported fields freely. Do not rename or retag exported signatures without `v2`.
- Split packages along jobs (`ticket`, `shift`), not layers (`models`, `helpers`, `utils`).
- Recap from the rest of the book: `gofmt`, `go vet`, `go test`, `govulncheck`; errors as values; small interfaces; `CGO_ENABLED=0`; no reflect/unsafe/cgo by default; measure before you tune; secrets out of source; `//go:embed` for files; stamp a version with `-ldflags`.
- The code a stranger can read on Monday morning is the code you still want in a year.

Try this

1. Add `func (t Ticket) Empty() bool` that reports `t.id == 0`. Do not export a field to do it. Run `main.go`.
2. Try `ticket.Ticket{id: 7}` from `main.go` and read the compile error. Keep the fields unexported.
3. Add `package shift` as in Case 3 and print `shift.Label("Amina", shift.Morning)` from `main`. Keep `ticket` unaware of `shift`.
4. List the exported names in `ticket` (`go doc ./ticket`). If a name is not for callers, unexport it.

Deprecating APIs with //go:fix inline

Deprecating APIs with `//go:fix inline`

Thesis

The usual way to retire an exported function is to mark it `// Deprecated:` and wait for callers to notice. Since Go 1.26, `go fix` can do the migration for them. You rewrite the old function as a **one-line forwarder** to the new API and add a `//go:fix inline` directive. From then on, `go fix ./...` in any caller's module replaces each call with the forwarder's body: arguments are substituted and imports are rewritten to the new package. The same directive works on **type aliases** and **constants**.

The rewrite comes from the same source-level inliner gopls uses for its "Inline call" refactoring, so it keeps the program's behaviour, including argument evaluation order. When it can't do that cleanly, it leaves the call alone.

This chapter runs the whole cycle on a small module with Go 1.27.2 and shows where the tool stops.

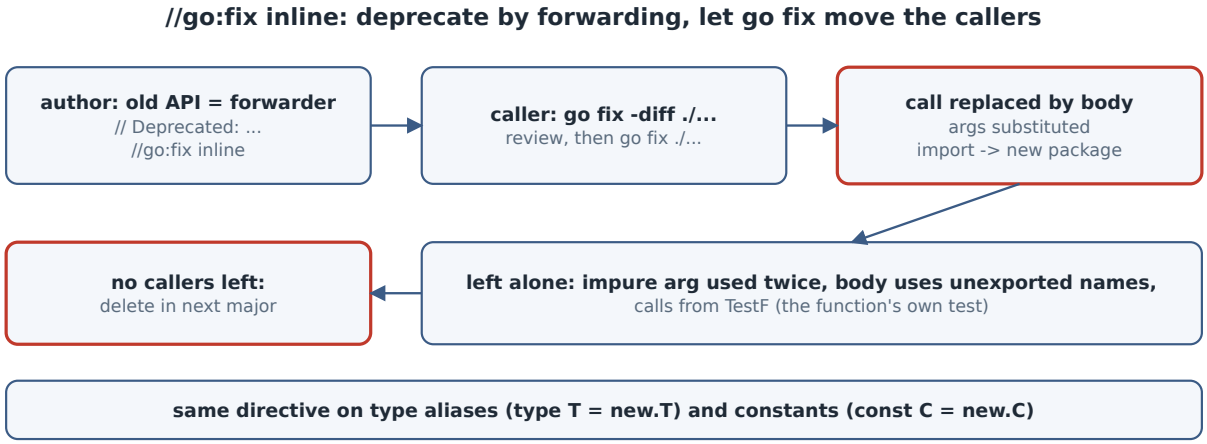


Figure 1: //go:fix inline: deprecate by forwarding, let go fix move the callers

The setup: a v1 package and its replacement

`example.com/desk/desk` is the old package. `deskv2` is the redesign. In it, `Open` takes an explicit priority and trims the title:

```go
// deskv2/deskv2.go
package deskv2

import "strings"

type Ticket struct {
    ID       int
    Title    string
    Priority int
}

// Open creates a ticket. Priority is now explicit.
func Open(id int, title string, priority int) Ticket {
    return Ticket{ID: id, Title: strings.TrimSpace(title), Priority: priority}
}

const DefaultPriority = 3
```

The old package keeps compiling, but each exported name is now a forwarder to `deskv2`, marked for inlining:

```go
// desk/desk.go
package desk

import "example.com/desk/deskv2"

// Deprecated: use deskv2.Ticket.
//
//go:fix inline
type Ticket = deskv2.Ticket

// Deprecated: use deskv2.Open with deskv2.DefaultPriority.
//
//go:fix inline
func Open(id int, title string) Ticket {
    return deskv2.Open(id, title, deskv2.DefaultPriority)
}

// Deprecated: use deskv2.DefaultPriority.
//
//go:fix inline
const Normal = deskv2.DefaultPriority
```

Three rules make this work:

- **The body is written in terms of the new API.** Whatever the body says is what will appear at the call sites. That is why `Open` spells out `deskv2.DefaultPriority` and doesn't hide it behind a helper.

- **`Ticket` is a type alias (=), not a new defined type.** Old and new values are the same type, so callers can migrate one file at a time.
- **A constant can be inlined only if it refers to another named constant.** `const Normal = 3` would not qualify. `const Normal = deskv2.DefaultPriority` does.

A caller that hasn't migrated yet:

```go
// cmd/triage/main.go
package main

import (
    "fmt"

    "example.com/desk/desk"
)

func label(t desk.Ticket) string {
    return fmt.Sprintf("#%d %s (p%d)", t.ID, t.Title, t.Priority)
}

func main() {
    a := desk.Open(101, " printer jam ")
    b := desk.Open(102, "VPN drops")
    fmt.Println(label(a))
    fmt.Println(label(b), b.Priority == desk.Normal)
}
```

Preview with `go fix -diff`

Always look before writing. `-diff` prints a unified diff and exits non-zero if there's anything to change, so it also works as a CI gate:

```
$ go run ./cmd/triage
#101 printer jam (p3)
#102 VPN drops (p3) true
$ go fix -diff ./...
--- cmd/triage/main.go (old)
+++ cmd/triage/main.go (new)
@@ -3,16 +3,16 @@
 import (
    "fmt"

-   "example.com/desk/desk"
+   "example.com/desk/deskv2"
 )
```

```
-func label(t desk.Ticket) string {
+func label(t deskv2.Ticket) string {
    return fmt.Sprintf("#%d %s (p%d)", t.ID, t.Title, t.Priority)
 }

 func main() {
-   a := desk.Open(101, " printer jam ")
-   b := desk.Open(102, "VPN drops")
+   a := deskv2.Open(101, " printer jam ", deskv2.DefaultPriority)
+   b := deskv2.Open(102, "VPN drops", deskv2.DefaultPriority)
    fmt.Println(label(a))
-   fmt.Println(label(b), b.Priority == desk.Normal)
+   fmt.Println(label(b), b.Priority == deskv2.DefaultPriority)
 }
--- desk/desk.go (old)
+++ desk/desk.go (new)
@@ -11,7 +11,7 @@
 // Deprecated: use deskv2.Open with deskv2.DefaultPriority.
 //
 //go:fix inline
-func Open(id int, title string) Ticket {
+func Open(id int, title string) deskv2.Ticket {
    return deskv2.Open(id, title, deskv2.DefaultPriority)
 }

exit status 1
```

(The last line comes from the lab script, which runs `echo "exit status $?"` after `go fix -diff`.)

Things to notice:

- **The import changed.** Once nothing in `main.go` refers to `desk`, the import is replaced with `deskv2`. This is how a module can move callers from `pkg` to `pkg/v2`.
- **The hidden default is now visible.** `deskv2.DefaultPriority` appears at each call site, so it's clear which callers took the default.
- **The alias is resolved everywhere**, including the signature of `desk.Open` itself in the declaring package. That is harmless: it's the same type.
- **`desk/desk_test.go` isn't in the diff.** It calls `Open` from `TestOpen`. The analyzer deliberately skips calls to `F` from its own `TestF`, because that test exists to exercise the old function.

Apply, then prove nothing changed

```
$ go fix ./... && go run ./cmd/triage && go test ./...
#101 printer jam (p3)
#102 VPN drops (p3) true
?       example.com/desk/cmd/triage [no test files]
ok      example.com/desk/desk   0.002s
?       example.com/desk/deskv2 [no test files]
$ go fix -diff ./... ; echo $?
0
```

The output is identical before and after, and a second `-diff` comes back empty. Commit this as its own change, separate from feature work, so that reviewers see only mechanical edits.

Where the inliner stops

The inliner refuses any rewrite that would change behaviour. To see this, here is a package with a few more forwarders:

```
// oldapi/oldapi.go (excerpt)

// Deprecated: use Area.
//
//go:fix inline
func Square(side int) int {
    return Area(side, side)
}

func Area(w, h int) int { return w * h }

// Deprecated: callers should stop normalizing; it is a no-op.
//
//go:fix inline
func Clean(s string) string {
    return normalize(s)
}

func normalize(s string) string { return s }
```

A caller in another package (the `StartsWithIgnoreCase` and `MaxItems` forwarders are written the same way as in `desk`):

```
var calls int

func next() int { calls++; return calls * 3 }
```

```go
func main() {
    fmt.Println(oldapi.StartsWithIgnoreCase("HTTP/1.1", "http/"))
    fmt.Println(oldapi.MaxItems)
    fmt.Println(oldapi.Square(7))
    fmt.Println(oldapi.Square(next()), "calls =", calls)
    fmt.Println(oldapi.Clean("x"))
}
```

```
$ go fix -diff ./app
--- app/main.go (old)
+++ app/main.go (new)
@@ -11,9 +11,9 @@
 func next() int { calls++; return calls * 3 }

 func main() {
-   fmt.Println(oldapi.StartsWithIgnoreCase("HTTP/1.1", "http/"))
-   fmt.Println(oldapi.MaxItems)
-   fmt.Println(oldapi.Square(7))
+   fmt.Println(oldapi.HasPrefixFold("HTTP/1.1", "http/"))
+   fmt.Println(oldapi.DefaultLimit)
+   fmt.Println(oldapi.Area(7, 7))
    fmt.Println(oldapi.Square(next()), "calls =", calls)
    fmt.Println(oldapi.Clean("x"))
 }
```

- **`Square(7)` becomes `Area(7, 7)`.** The argument is a constant, so duplicating it is free.
- **`Square(next())` is left alone.** Writing `Area(next(), next())` would call `next` twice. A safe rewrite would need either a binding declaration (`var side = next()`) or a function literal (`func() int {...}()`). In this run it did neither, even when the statement was moved into its own `s := oldapi.Square(next())` line and `-inline.allow_binding_decl=true` was passed. The analyzer docs say it always throws away the function-literal form "on grounds of style", and binding declarations are only "permitted", never required.
- **`Clean("x")` is left alone.** Its body calls `normalize`, which `app` can't reference from another package. Inlining it would produce code that doesn't compile.

None of these skipped calls produced a warning from `go fix`; the calls just stay deprecated. Deprecation checkers such as staticcheck's SA1019 will still flag them, and a search for the old names finds them too. **Design the forwarder so it inlines:** use each parameter at most once (a parameter used twice only inlines when the argument has no side effects), and only reference exported names that callers can see. The lab's `Square` breaks the first rule because it uses `side` twice. A callers-first fix is to give the new API a single-argument form, or to accept that some calls will need a hand edit.

A deprecation playbook

1. **Ship the new API first** (`deskv2`, or `pkg/v2`). It must be able to express every old call.

2. **Rewrite every old exported name as a forwarder** with `// Deprecated:` and `//go:fix inline`. Use a type alias for types, and a constant that names another constant for constants.
3. **Tell callers how to migrate** in the release notes: run `go fix -diff ./...`, review, then `go fix ./...`. That's one command instead of a migration guide.
4. **Hand-fix what's left**, using a search for the old package name. The cases above are the usual reasons calls get skipped.
5. **Remove the old names only in the next major version.** Until then, they're still part of your compatibility promise.

Pitfalls

- **Forwarders with logic.** If the old function validates input, logs or recovers from panics, that code will be copied into every caller. Move it into the new API first.
- **Defined types instead of aliases.** `type Ticket deskv2.Ticket` creates a new type, so nothing can be inlined and old and new values don't mix.
- **Assuming every call was migrated.** Skipped calls don't produce a warning, so grep for the old package name after `go fix`.
- **Generated code.** `go fix` won't touch generated files. Fix the generator instead.
- **Version skew.** `go fix` applies these rewrites only from Go 1.26 on; on older toolchains the directive is just a comment. Until every caller has migrated, the forwarders have to keep working as ordinary functions.

Takeaways

- `//go:fix inline` turns a deprecation notice into a migration that runs automatically, for functions, type aliases and constants.
- Write the forwarder body in terms of the new API, because that body is exactly what callers will end up with.
- `go fix -diff` previews the change and doubles as a CI check. `go fix` applies it, and the program's behaviour stays the same.
- The inliner will skip a call rather than change behaviour. Plan for a few hand edits, and design forwarders that inline cleanly.

Sources

- Alan Donovan, "//go:fix inline and the source-level inliner", The Go Blog, 10 March 2026.
- "Using go fix to modernize Go code", The Go Blog.
- Go 1.26 release notes: the rewritten `go fix`.
- `golang.org/x/tools/go/analysis/passes/inline`: analyzer docs, flags, the `TestF` rule and constant rules (also available offline with `go tool fix help inline`).
- `go fix` command documentation.
- Proposal: `//go:fix inline` directives (go.dev/issue/32816).
- Go wiki: Deprecated: the `// Deprecated:` comment convention.

Modern Go Synthesis

190 Modern Go Books (2024-2025) Index

190 Modern Go Books (2024-2025) Index

This section distills practical curriculum additions from recent Go books (published in 2024 and 2025), then maps those ideas into this book's structure.

The intent is not to duplicate any book. Instead, this section extracts high-value topic patterns that repeatedly appear across recent publications and turns them into production-oriented learning content.

Selection Criteria

- Publication date within 2024-2025.
- Reliable public metadata for release date and chapter/index signals.
- Topics that improve production readiness for Go developers.

Source-Derived Synthesis Chapters

1. 191 Patterns from Go Programming (2nd ed., 2024)
2. 192 Patterns from Effective Go Recipes (2024)
3. 193 Patterns from Go in Practice (2nd ed., 2025)
4. 194 Patterns from Mastering Go (4th ed., 2024)
5. 195 Patterns from Let Us Go! (2025)
6. 196 Patterns from Automate Your Home Using Go (2024)

Cross-Book Theme Map

```
Language fundamentals -> Systems/networking -> Web APIs -> Concurrency -> Testing/Tooling ->
```

Recent books emphasize this transition strongly: Go learning now expects developers to move from syntax to production operations quickly.

More examples

Tour: options + httptest handler + typed error

A single demo that previews synthesis themes: constructor options, boundary HTTP, and `errors.Is`.

```
mkdir -p /tmp/go-synth-tour && cd /tmp/go-synth-tour
go mod init example.com/synth-tour
```

Save as `main.go`:

```go
package main

import (
    "errors"
    "fmt"
    "net/http"
    "net/http/httptest"
)

var ErrNotFound = errors.New("not found")

type App struct {
    name string
}

type Option func(*App)

func WithName(n string) Option { return func(a *App) { a.name = n } }

func NewApp(opts ...Option) *App {
    a := &App{name: "app"}
    for _, o := range opts {
        o(a)
    }
    return a
}

func (a *App) Get(id int) error {
    if id != 1 {
        return fmt.Errorf("%w: %d", ErrNotFound, id)
    }
    return nil
}

func main() {
```

```go
    app := NewApp(WithName("demo"))
    h := http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        if err := app.Get(2); errors.Is(err, ErrNotFound) {
            http.Error(w, "missing", http.StatusNotFound)
            return
        }
        fmt.Fprintln(w, app.name)
    })
    rr := httptest.NewRecorder()
    h.ServeHTTP(rr, httptest.NewRequest(http.MethodGet, "/", nil))
    fmt.Println("status:", rr.Code, "name:", app.name)
}
```

```bash
go run .
```

Expected output:

```
status: 404 name: demo
```

Runnable example

Tour of the synthesis themes: config from env, a timeout budget, and a tiny HTTP handler—stdlib patterns that modern Go books keep circling.

```bash
mkdir -p /tmp/go-synth-tour && cd /tmp/go-synth-tour
go mod init example.com/synth-tour
```

Save as `main.go`:

```go
package main

import (
    "context"
    "fmt"
    "net/http"
    "net/http/httptest"
    "os"
    "time"
)

func envDuration(key string, fallback time.Duration) time.Duration {
    v := os.Getenv(key)
    if v == "" {
        return fallback
    }
```

```go
    d, err := time.ParseDuration(v)
    if err != nil {
        return fallback
    }
    return d
}

func main() {
    budget := envDuration("REQ_TIMEOUT", 200*time.Millisecond)
    h := http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        ctx, cancel := context.WithTimeout(r.Context(), budget)
        defer cancel()
        select {
        case <-time.After(20 * time.Millisecond):
            fmt.Fprintln(w, "ok")
        case <-ctx.Done():
            http.Error(w, "timeout", http.StatusGatewayTimeout)
        }
    })
    rr := httptest.NewRecorder()
    h.ServeHTTP(rr, httptest.NewRequest(http.MethodGet, "/", nil))
    fmt.Println("status:", rr.Code, "budget:", budget)
    fmt.Println("synthesis tour: config + budgets + http")
}
```

```bash
go run .
REQ_TIMEOUT=50ms go run .
```

Expected output:

```
status: 200 budget: 200ms
synthesis tour: config + budgets + http
```

What to notice

- Recent books stress production boundaries (timeouts, config, tests) as much as syntax.
- Chapters 191–196 extract one pattern cluster each—run those examples next.

Try next

- Skim each synthesis chapter's runnable example and map it back to an earlier part of this book.

191 Patterns from Go Programming (2nd ed., 2024)

191 Patterns from Go Programming (2nd ed., 2024)

This chapter extracts topics from the 2024 second edition arc: debugging, time handling, CLI work, files/systems, SQL, HTTP server/client, concurrency, testing, tooling, and cloud.

What This Adds to Our Book

- Stronger bridge from core syntax to practical operations.
- Dedicated narrative on "time as a correctness boundary".
- Better pairing of HTTP server and HTTP client behavior in one lifecycle.
- Cloud-readiness framing: configuration, graceful shutdown, observability, packaging.

Learning Architecture

```
core language -> application boundary (CLI/API) -> state boundary (DB/files) -> operational
```

Deep Integration Example: API + CLI + DB Boundary

```go
package app

import (
    "context"
    "database/sql"
    "encoding/json"
    "errors"
    "fmt"
    "net/http"
    "os"
    "time"
)

var ErrNotFound = errors.New("not found")

type User struct {
```

```
    ID    int64  `json:"id"`
    Email string `json:"email"`
}

type Store struct{ DB *sql.DB }

func (s Store) GetUser(ctx context.Context, id int64) (User, error) {
    ctx, cancel := context.WithTimeout(ctx, 2*time.Second)
    defer cancel()

    var u User
    err := s.DB.QueryRowContext(ctx, `SELECT id, email FROM users WHERE id=$1`, id).Scan(&u.
    if errors.Is(err, sql.ErrNoRows) {
        return User{}, ErrNotFound
    }
    if err != nil {
        return User{}, fmt.Errorf("query user %d: %w", id, err)
    }
    return u, nil
}

type API struct{ Store Store }

func (a API) GetUserHandler(w http.ResponseWriter, r *http.Request) {
    ctx, cancel := context.WithTimeout(r.Context(), 3*time.Second)
    defer cancel()

    user, err := a.Store.GetUser(ctx, 42)
    if errors.Is(err, ErrNotFound) {
        http.Error(w, "not found", http.StatusNotFound)
        return
    }
    if err != nil {
        http.Error(w, "internal error", http.StatusInternalServerError)
        return
    }
    w.Header().Set("Content-Type", "application/json")
    _ = json.NewEncoder(w).Encode(user)
}

func ReadEnvDuration(key string, fallback time.Duration) time.Duration {
    v := os.Getenv(key)
    if v == "" {
        return fallback
    }
    d, err := time.ParseDuration(v)
    if err != nil {
```

```
        return fallback
    }
    return d
}
```

Why This Matters

The example demonstrates a full chain from request boundary to storage boundary with explicit timeouts and typed errors. That is the practical center of modern Go backend development.

Curriculum Upgrades Recommended

1. Add a dedicated chapter linking debugging, benchmarking, and production incident triage.
2. Expand `time` chapter with deadline budgeting patterns for server+client+DB together.
3. Add side-by-side chapter pair: `net/http` server correctness and `http.Client` resilience.
4. Add cloud deployment chapter emphasizing runtime config contracts and shutdown policy.

More examples

Nested context budgets at the boundary

```
mkdir -p /tmp/go-gp2e-budget && cd /tmp/go-gp2e-budget
go mod init example.com/gp2e-budget
```

Save as `main.go`:

```
package main

import (
    "context"
    "fmt"
    "time"
)

func db(ctx context.Context) error {
    ctx, cancel := context.WithTimeout(ctx, 20*time.Millisecond)
    defer cancel()
    select {
    case <-time.After(50 * time.Millisecond):
        return nil
    case <-ctx.Done():
        return ctx.Err()
```

```go
    }
}

func main() {
    ctx, cancel := context.WithTimeout(context.Background(), 100*time.Millisecond)
    defer cancel()
    err := db(ctx)
    fmt.Println("db budget exceeded:", err != nil)
}
```

```bash
go run .
```

Expected output:

```
db budget exceeded: true
```

Typed sentinel + wrap for handlers

```bash
mkdir -p /tmp/go-gp2e-err && cd /tmp/go-gp2e-err
go mod init example.com/gp2e-err
```

Save as `main.go`:

```go
package main

import (
    "errors"
    "fmt"
)

var ErrDenied = errors.New("denied")

func authorize(user string) error {
    if user != "admin" {
        return fmt.Errorf("user %s: %w", user, ErrDenied)
    }
    return nil
}

func main() {
    err := authorize("bob")
    fmt.Println("is denied:", errors.Is(err, ErrDenied))
    fmt.Println("text:", err)
}
```

```
go run .
```

Expected output:

```
is denied: true
text: user bob: denied
```

Runnable example

Boundary chain without a real DB: typed errors, nested timeouts, and JSON HTTP handler test via `httptest`.

```
mkdir -p /tmp/go-gp2e && cd /tmp/go-gp2e
go mod init example.com/gp2e
```

Save as `main.go`:

```go
package main

import (
    "context"
    "encoding/json"
    "errors"
    "fmt"
    "net/http"
    "net/http/httptest"
    "time"
)

var ErrNotFound = errors.New("not found")

type User struct {
    ID    int64  `json:"id"`
    Email string `json:"email"`
}

type Store struct {
    users map[int64]User
}

func (s Store) GetUser(ctx context.Context, id int64) (User, error) {
    ctx, cancel := context.WithTimeout(ctx, 50*time.Millisecond)
    defer cancel()
    select {
    case <-ctx.Done():
        return User{}, ctx.Err()
```

```go
    default:
    }
    u, ok := s.users[id]
    if !ok {
        return User{}, ErrNotFound
    }
    return u, nil
}

func main() {
    store := Store{users: map[int64]User{42: {ID: 42, Email: "a@b.co"}}}
    h := http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        ctx, cancel := context.WithTimeout(r.Context(), 100*time.Millisecond)
        defer cancel()
        u, err := store.GetUser(ctx, 42)
        if errors.Is(err, ErrNotFound) {
            http.Error(w, "not found", http.StatusNotFound)
            return
        }
        if err != nil {
            http.Error(w, "internal", http.StatusInternalServerError)
            return
        }
        w.Header().Set("Content-Type", "application/json")
        _ = json.NewEncoder(w).Encode(u)
    })
    rr := httptest.NewRecorder()
    h.ServeHTTP(rr, httptest.NewRequest(http.MethodGet, "/user", nil))
    fmt.Println("status:", rr.Code, "body:", rr.Body.String())
}
```

```go
go run .
```

Expected output:

```
status: 200 body: {"id":42,"email":"a@b.co"}
```

What to notice

- Timeouts nest: handler budget > store budget.
- `errors.Is` maps domain errors to HTTP status without string matching.

Try next

- Return `ErrNotFound` for id 7 and assert 404.
- Parse `REQ_TIMEOUT` from env for the outer budget.

192 Patterns from Effective Go Recipes (2024)

192 Patterns from Effective Go Recipes (2024)

This source emphasizes practical recipes across I/O, JSON streaming, HTTP middleware/timeouts, text normalization, function options, errors, concurrency, sockets, C interop, testing, and shipping.

What This Adds to Our Book

- Recipe-level production patterns that are small, composable, and immediately reusable.
- Better treatment of streaming and incremental processing.
- Stronger shipping pipeline topics: build tags, static builds, version injection, Docker packaging.

Streaming-Oriented Mental Model

```
source -> decoder -> transform -> validator -> sink
          (stream)               (bounded memory)
```

Deep Integration Example: Streaming JSON Pipeline with Backpressure

```go
package pipeline

import (
    "bufio"
    "context"
    "encoding/json"
    "errors"
    "io"
    "sync"
    "time"
)

type Event struct {
    ID   string `json:"id"`
    Type string `json:"type"`
```

```
}

func DecodeEvents(ctx context.Context, r io.Reader) (<-chan Event, <-chan error) {
    out := make(chan Event, 128)
    errCh := make(chan error, 1)

    go func() {
        defer close(out)
        defer close(errCh)

        dec := json.NewDecoder(bufio.NewReader(r))
        for {
            var e Event
            if err := dec.Decode(&e); err != nil {
                if errors.Is(err, io.EOF) {
                    return
                }
                errCh <- err
                return
            }

            select {
            case <-ctx.Done():
                errCh <- ctx.Err()
                return
            case out <- e:
            }
        }
    }()

    return out, errCh
}

func ProcessEvents(ctx context.Context, in <-chan Event, workers int, fn func(Event) error)
    ctx, cancel := context.WithCancel(ctx)
    defer cancel()

    errCh := make(chan error, workers)
    var wg sync.WaitGroup

    for i := 0; i < workers; i++ {
        wg.Add(1)
        go func() {
            defer wg.Done()
            for e := range in {
                opCtx, stop := context.WithTimeout(ctx, 250*time.Millisecond)
                err := fn(e)
```

```go
                stop()
                if opCtx.Err() != nil {
                    errCh <- opCtx.Err()
                    cancel()
                    return
                }
                if err != nil {
                    errCh <- err
                    cancel()
                    return
                }
            }
        }()
    }

    wg.Wait()
    close(errCh)
    for err := range errCh {
        if err != nil {
            return err
        }
    }
    return nil
}
```

Why This Matters

Modern Go systems increasingly process streams instead of loading full datasets in memory. Recipe-style patterns are especially effective for this style because they focus on one robust tactic at a time.

Curriculum Upgrades Recommended

1. Add a chapter on streaming-first design (`json.Decoder`, `io.Reader`, chunked processing).
2. Add chapter on function options + config normalization.
3. Add chapter on shipping pipelines: reproducible builds, tags, embedding, static binaries.
4. Add explicit `os/exec` and signal-handling narrative for operational tooling.

More examples

Functional options recipe

```
mkdir -p /tmp/go-egr-opts && cd /tmp/go-egr-opts
go mod init example.com/egr-opts
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "time"
)

type Client struct {
    timeout time.Duration
    retries int
}

type Option func(*Client)

func WithTimeout(d time.Duration) Option {
    return func(c *Client) { c.timeout = d }
}
func WithRetries(n int) Option {
    return func(c *Client) { c.retries = n }
}

func NewClient(opts ...Option) *Client {
    c := &Client{timeout: time.Second, retries: 1}
    for _, o := range opts {
        o(c)
    }
    return c
}

func main() {
    c := NewClient(WithTimeout(200*time.Millisecond), WithRetries(3))
    fmt.Printf("timeout=%s retries=%d\n", c.timeout, c.retries)
}
```

```
go run .
```

Expected output:

```
timeout=200ms retries=3
```

Table-driven validation recipe

```
mkdir -p /tmp/go-egr-table && cd /tmp/go-egr-table
go mod init example.com/egr-table
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "strings"
)

func validEmail(s string) bool {
    return strings.Count(s, "@") == 1 && !strings.HasPrefix(s, "@") && !strings.HasSuffix(s,
}

func main() {
    cases := []struct {
        in   string
        want bool
    }{
        {"a@b.co", true},
        {"@x", false},
        {"nope", false},
    }
    for _, tc := range cases {
        got := validEmail(tc.in)
        fmt.Printf("%q got=%v want=%v ok=%v\n", tc.in, got, tc.want, got == tc.want)
    }
}
```

```
go run .
```

Expected output:

```
"a@b.co" got=true want=true ok=true
"@x" got=false want=false ok=true
"nope" got=false want=false ok=true
```

Runnable example

Recipe-style patterns: functional options for config, streaming `json.Decoder` over a reader, and a context-aware worker fan-in.

```
mkdir -p /tmp/go-recipes && cd /tmp/go-recipes
go mod init example.com/recipes
```

Save as `main.go`:

```go
package main

import (
    "context"
    "encoding/json"
    "fmt"
    "strings"
    "sync"
    "time"
)

type Config struct {
    Workers int
    Timeout time.Duration
}

type Option func(*Config)

func WithWorkers(n int) Option {
    return func(c *Config) { c.Workers = n }
}
func WithTimeout(d time.Duration) Option {
    return func(c *Config) { c.Timeout = d }
}

func NewConfig(opts ...Option) Config {
    c := Config{Workers: 2, Timeout: time.Second}
    for _, o := range opts {
        o(&c)
    }
    return c
}

func main() {
    cfg := NewConfig(WithWorkers(3), WithTimeout(200*time.Millisecond))
    fmt.Printf("config: workers=%d timeout=%s\n", cfg.Workers, cfg.Timeout)
```

```
    // Streaming JSON values (NDJSON)
    r := strings.NewReader("{\"n\":1}\n{\"n\":2}\n{\"n\":3}\n")
    dec := json.NewDecoder(r)
    var sum int
    for dec.More() {
        var row struct {
            N int `json:"n"`
        }
        if err := dec.Decode(&row); err != nil {
            panic(err)
        }
        sum += row.N
    }
    fmt.Println("stream_sum:", sum)

    // Bounded workers
    ctx, cancel := context.WithTimeout(context.Background(), cfg.Timeout)
    defer cancel()
    jobs := make(chan int, 8)
    var wg sync.WaitGroup
    var mu sync.Mutex
    var out int
    for i := 0; i < cfg.Workers; i++ {
        wg.Add(1)
        go func() {
            defer wg.Done()
            for j := range jobs {
                select {
                case <-ctx.Done():
                    return
                default:
                    mu.Lock()
                    out += j
                    mu.Unlock()
                }
            }
        }()
    }
    for i := 1; i <= 5; i++ {
        jobs <- i
    }
    close(jobs)
    wg.Wait()
    fmt.Println("worker_sum:", out)
}
```

```
go run .
```

Expected output:

```
config: workers=3 timeout=200ms
stream_sum: 6
worker_sum: 15
```

What to notice

- Options keep defaults + overrides readable without telescoping constructors.
- `json.Decoder` streams; it does not require `ReadAll` of a huge payload.
- Workers always honor `ctx.Done()` for operational cancellation.

Try next

- Add `WithName` option and validate `Workers > 0` in `NewConfig`.
- Decode a large file with `os.Open` + Decoder in a loop.

193 Patterns from Go in Practice (2nd ed., 2025)

193 Patterns from Go in Practice (2nd ed., 2025)

The 2025 edition is structured around four parts: fundamentals, robust application practices, end-to-end web systems, and cloud/advanced topics (reflection, code generation, interop).

What This Adds to Our Book

- Strong mid-level bridge: from language fluency to production engineering.
- Better integration of testing, debugging, and benchmarking as one quality discipline.
- Explicit chapter path toward microservices and external service integration.

End-to-End Service Model

```
HTTP edge -> business layer -> storage/external APIs -> telemetry -> deployment/runtime chec
```

Deep Integration Example: External Service Client with Typed Error Strategy

```go
package external

import (
    "context"
    "encoding/json"
    "errors"
    "fmt"
    "net/http"
    "time"
)

var ErrUpstreamUnavailable = errors.New("upstream unavailable")

type Client struct {
    HTTP *http.Client
    Base string
}
```

```go
type Profile struct {
    ID   string `json:"id"`
    Name string `json:"name"`
}

func NewClient(base string) Client {
    return Client{
        Base: base,
        HTTP: &http.Client{Timeout: 2 * time.Second},
    }
}

func (c Client) GetProfile(ctx context.Context, id string) (Profile, error) {
    ctx, cancel := context.WithTimeout(ctx, 1200*time.Millisecond)
    defer cancel()

    req, err := http.NewRequestWithContext(ctx, http.MethodGet, c.Base+"/profiles/"+id, nil)
    if err != nil {
        return Profile{}, err
    }

    resp, err := c.HTTP.Do(req)
    if err != nil {
        return Profile{}, fmt.Errorf("http call failed: %w", ErrUpstreamUnavailable)
    }
    defer resp.Body.Close()

    if resp.StatusCode >= 500 {
        return Profile{}, fmt.Errorf("status %d: %w", resp.StatusCode, ErrUpstreamUnavailabl
    }
    if resp.StatusCode != http.StatusOK {
        return Profile{}, fmt.Errorf("unexpected status: %d", resp.StatusCode)
    }

    var p Profile
    if err := json.NewDecoder(resp.Body).Decode(&p); err != nil {
        return Profile{}, err
    }
    return p, nil
}
```

Why This Matters

Production services fail most often at integration boundaries. This pattern teaches typed failure handling and timeout discipline in a way that scales across microservices.

Curriculum Upgrades Recommended

1. Add one full chapter on external service client design (`timeouts`, `retry policy`, `error typing`).
2. Add one chapter on data ingress/egress patterns (upload/download, validation, content type).
3. Add one chapter on reflection and code generation boundaries: when useful, when harmful.

More examples

Middleware chain composition

```
mkdir -p /tmp/go-gip-mw && cd /tmp/go-gip-mw
go mod init example.com/gip-mw
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "net/http"
    "net/http/httptest"
)

type Middleware func(http.Handler) http.Handler

func chain(h http.Handler, mws ...Middleware) http.Handler {
    for i := len(mws) - 1; i >= 0; i-- {
        h = mws[i](h)
    }
    return h
}

func main() {
    var order []string
    mw := func(name string) Middleware {
        return func(next http.Handler) http.Handler {
            return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
                order = append(order, name)
                next.ServeHTTP(w, r)
            })
        }
    }
    h := chain(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        order = append(order, "handler")
```

```
        fmt.Fprintln(w, "ok")
    }), mw("a"), mw("b"))

    rr := httptest.NewRecorder()
    h.ServeHTTP(rr, httptest.NewRequest(http.MethodGet, "/", nil))
    fmt.Println(order)
}
```

```
go run .
```

Expected output:

```
[a b handler]
```

Content-type gate for JSON POST

```
mkdir -p /tmp/go-gip-ct && cd /tmp/go-gip-ct
go mod init example.com/gip-ct
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "net/http"
    "net/http/httptest"
    "strings"
)

func requireJSON(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        ct := r.Header.Get("Content-Type")
        if !strings.HasPrefix(ct, "application/json") {
            http.Error(w, "want json", http.StatusUnsupportedMediaType)
            return
        }
        next.ServeHTTP(w, r)
    })
}

func main() {
    h := requireJSON(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        fmt.Fprintln(w, "json ok")
    }))
```

```
    rr := httptest.NewRecorder()
    req := httptest.NewRequest(http.MethodPost, "/", strings.NewReader(`{}`))
    h.ServeHTTP(rr, req)
    fmt.Println("no ct:", rr.Code)

    rr = httptest.NewRecorder()
    req = httptest.NewRequest(http.MethodPost, "/", strings.NewReader(`{}`))
    req.Header.Set("Content-Type", "application/json")
    h.ServeHTTP(rr, req)
    fmt.Println("json:", rr.Code, strings.TrimSpace(rr.Body.String()))
}
```

```
go run .
```

Expected output:

```
no ct: 415
json: 200 json ok
```

Runnable example

External client practice: typed upstream errors, client timeout, and status classification with `httptest`.

```
mkdir -p /tmp/go-gip && cd /tmp/go-gip
go mod init example.com/gip
```

Save as `main.go`:

```
package main

import (
    "context"
    "encoding/json"
    "errors"
    "fmt"
    "net/http"
    "net/http/httptest"
    "time"
)

var ErrUpstreamUnavailable = errors.New("upstream unavailable")

type Profile struct {
    Name string `json:"name"`
```

```
}

func fetchProfile(ctx context.Context, client *http.Client, base string) (Profile, error) {
    req, err := http.NewRequestWithContext(ctx, http.MethodGet, base+"/profile", nil)
    if err != nil {
        return Profile{}, err
    }
    resp, err := client.Do(req)
    if err != nil {
        return Profile{}, fmt.Errorf("http: %w", ErrUpstreamUnavailable)
    }
    defer resp.Body.Close()
    if resp.StatusCode >= 500 {
        return Profile{}, fmt.Errorf("status %d: %w", resp.StatusCode, ErrUpstreamUnavailabl
    }
    if resp.StatusCode != http.StatusOK {
        return Profile{}, fmt.Errorf("unexpected status %d", resp.StatusCode)
    }
    var p Profile
    if err := json.NewDecoder(resp.Body).Decode(&p); err != nil {
        return Profile{}, err
    }
    return p, nil
}

func main() {
    var hits int
    ts := httptest.NewServer(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        hits++
        if hits == 1 {
            w.WriteHeader(http.StatusBadGateway)
            return
        }
        _ = json.NewEncoder(w).Encode(Profile{Name: "ada"})
    }))
    defer ts.Close()

    client := &http.Client{Timeout: 300 * time.Millisecond}
    ctx := context.Background()
    _, err := fetchProfile(ctx, client, ts.URL)
    fmt.Println("first:", err, "unavailable?", errors.Is(err, ErrUpstreamUnavailable))
    p, err := fetchProfile(ctx, client, ts.URL)
    fmt.Println("second:", p, err)
}
```

```
go run .
```

Expected output:

```
first: status 502: upstream unavailable unavailable? true
second: {ada} <nil>
```

What to notice

- Wrap with `%w` and a sentinel so callers decide retry vs fail without parsing strings.
- Always set `http.Client.Timeout` (or context deadline) on outbound deps.
- Close response bodies to preserve connection reuse.

Try next

- Retry only when `errors.Is(err, ErrUpstreamUnavailable)` with jitter.
- Map 404 to a distinct `ErrNotFound` sentinel.

194 Patterns from Mastering Go (4th ed., 2024)

194 Patterns from Mastering Go (4th ed., 2024)

Recent 4th-edition signals emphasize advanced practice: generics, interfaces/reflection, Unix systems work, concurrency, web services, TCP/WebSocket, REST APIs, testing/profiling, fuzzing/observability, and performance.

What This Adds to Our Book

- Better advanced track sequencing after core language chapters.
- Stronger focus on performance + fuzz + observability as one reliability loop.
- More explicit Unix-system programming integration for backend engineers.

Advanced Engineering Loop

```
implement -> test -> fuzz -> profile -> optimize -> observe in runtime -> iterate
```

Deep Integration Example: Fuzz-Ready Parser + Profile-Aware Fast Path

```go
package parser

import (
    "errors"
    "strconv"
    "strings"
)

var ErrBadRecord = errors.New("bad record")

type Record struct {
    ID    int64
    Score float64
}

func ParseRecord(line string) (Record, error) {
    parts := strings.Split(line, ",")
```

```go
    if len(parts) != 2 {
        return Record{}, ErrBadRecord
    }

    id, err := strconv.ParseInt(strings.TrimSpace(parts[0]), 10, 64)
    if err != nil {
        return Record{}, ErrBadRecord
    }
    score, err := strconv.ParseFloat(strings.TrimSpace(parts[1]), 64)
    if err != nil {
        return Record{}, ErrBadRecord
    }
    if id <= 0 || score < 0 {
        return Record{}, ErrBadRecord
    }
    return Record{ID: id, Score: score}, nil
}
```

```go
package parser_test

import "testing"

func FuzzParseRecord(f *testing.F) {
    f.Add("1,2.5")
    f.Add("x,y")
    f.Add("")

    f.Fuzz(func(t *testing.T, input string) {
        _, _ = ParseRecord(input)
    })
}
```

Why This Matters

As Go systems mature, defects shift from syntax mistakes to boundary and performance defects. Fuzzing and profiling help catch these before they become incidents.

Curriculum Upgrades Recommended

1. Add a dedicated chapter combining fuzzing, race detection, and pprof for one service.
2. Add advanced chapter on TCP/WebSocket design under network systems.
3. Expand performance track with explicit optimization workflow and rollback criteria.

More examples

TCP echo with deadlines (systems flavor)

```
mkdir -p /tmp/go-mg-tcp && cd /tmp/go-mg-tcp
go mod init example.com/mg-tcp
```

Save as `main.go`:

```
package main

import (
    "bufio"
    "fmt"
    "net"
    "time"
)

func main() {
    ln, err := net.Listen("tcp", "127.0.0.1:0")
    if err != nil {
        panic(err)
    }
    defer ln.Close()
    go func() {
        c, _ := ln.Accept()
        defer c.Close()
        _ = c.SetDeadline(time.Now().Add(time.Second))
        line, _ := bufio.NewReader(c).ReadString('\n')
        _, _ = c.Write([]byte("echo:" + line))
    }()

    c, err := net.DialTimeout("tcp", ln.Addr().String(), time.Second)
    if err != nil {
        panic(err)
    }
    defer c.Close()
    _ = c.SetDeadline(time.Now().Add(time.Second))
    _, _ = c.Write([]byte("hi\n"))
    line, _ := bufio.NewReader(c).ReadString('\n')
    fmt.Print(line)
}
```

```
go run .
```

Expected output:

echo:hi

Worker pool fan-out / fan-in

```
mkdir -p /tmp/go-mg-pool && cd /tmp/go-mg-pool
go mod init example.com/mg-pool
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "sync"
)

func main() {
    jobs := make(chan int, 8)
    results := make(chan int, 8)
    var wg sync.WaitGroup
    for w := 0; w < 3; w++ {
        wg.Add(1)
        go func() {
            defer wg.Done()
            for j := range jobs {
                results <- j * 2
            }
        }()
    }
    go func() {
        for i := 1; i <= 5; i++ {
            jobs <- i
        }
        close(jobs)
        wg.Wait()
        close(results)
    }()
    sum := 0
    for r := range results {
        sum += r
    }
    fmt.Println("sum:", sum)
}
```

```
go run .
```

Expected output:

```
sum: 30
```

Runnable example

Robust parsing + fuzz target: `ParseRecord` never panics on garbage; run unit checks and a short fuzz if supported.

```bash
mkdir -p /tmp/go-mastering && cd /tmp/go-mastering
go mod init example.com/mastering
```

Save as `parse.go`:

```go
package mastering

import (
    "fmt"
    "strconv"
    "strings"
)

type Record struct {
    ID    int
    Score float64
}

func ParseRecord(s string) (Record, error) {
    parts := strings.Split(s, ",")
    if len(parts) != 2 {
        return Record{}, fmt.Errorf("want id,score")
    }
    id, err := strconv.Atoi(strings.TrimSpace(parts[0]))
    if err != nil {
        return Record{}, fmt.Errorf("id: %w", err)
    }
    score, err := strconv.ParseFloat(strings.TrimSpace(parts[1]), 64)
    if err != nil {
        return Record{}, fmt.Errorf("score: %w", err)
    }
    return Record{ID: id, Score: score}, nil
}
```

Save as `parse_test.go`:

```go
package mastering

import "testing"

func TestParseRecord(t *testing.T) {
    r, err := ParseRecord("1,2.5")
    if err != nil || r.ID != 1 || r.Score != 2.5 {
        t.Fatalf("got %+v %v", r, err)
    }
    if _, err := ParseRecord("x,y"); err == nil {
        t.Fatal("expected error")
    }
}

func FuzzParseRecord(f *testing.F) {
    f.Add("1,2.5")
    f.Add("")
    f.Add(",,,")
    f.Fuzz(func(t *testing.T, input string) {
        _, _ = ParseRecord(input) // must not panic
    })
}
```

Save as `cmd/demo/main.go`:

```go
package main

import (
    "fmt"

    "example.com/mastering"
)

func main() {
    r, err := mastering.ParseRecord("7, 3.14")
    fmt.Println(r, err)
    _, err = mastering.ParseRecord("nope")
    fmt.Println("bad:", err != nil)
}
```

```bash
go test .
go test -fuzz=FuzzParseRecord -fuzztime=5s .
go run ./cmd/demo
```

Expected output (fuzz lines vary):

```
PASS
```

```
...
{7 3.14} <nil>
bad: true
```

What to notice

- Fuzzing finds panics and unexpected paths at boundaries—Mastering Go's maturity theme.
- Parsers should return errors, never panic, on user input.

Try next

- `go test -race` on concurrent map code that shares a cache.
- Add a benchmark for `ParseRecord` on long inputs.

195 Patterns from Let Us Go! (2025)

195 Patterns from Let Us Go! (2025)

This 2025 beginner-focused arc stresses playground-first onboarding, workspace/tooling setup, version control flow, and then deep dive application development.

What This Adds to Our Book

- Better early-stage onboarding narrative for self-learners.
- A clearer toolchain chapter that explains *why* each tool exists.
- Practical Git workflow integration earlier in the learning path.

Onboarding Progression

```
Playground exploration -> local toolchain setup -> Git discipline -> deeper project work
```

Deep Integration Example: Minimal Project Lifecycle from Day 1

```go
package main

import (
    "flag"
    "fmt"
    "os"
)

type Config struct {
    Name string
    Env  string
}

func parseConfig() Config {
    cfg := Config{}
    flag.StringVar(&cfg.Name, "name", "gopher", "name to greet")
    flag.StringVar(&cfg.Env, "env", "dev", "runtime environment")
    flag.Parse()
    return cfg
```

```go
}

func run(cfg Config) error {
    if cfg.Env == "prod" && os.Getenv("APP_ALLOW_PROD") != "1" {
        return fmt.Errorf("prod run blocked without APP_ALLOW_PROD=1")
    }
    fmt.Printf("hello %s (%s)\n", cfg.Name, cfg.Env)
    return nil
}

func main() {
    if err := run(parseConfig()); err != nil {
        fmt.Fprintln(os.Stderr, "error:", err)
        os.Exit(1)
    }
}
```

Why This Matters

Beginner success is mostly about reducing setup friction and establishing stable habits early. Good onboarding chapters improve completion rates and reduce learner drop-off.

Curriculum Upgrades Recommended

1. Strengthen foundational chapters with explicit environment setup validation steps.
2. Add version-control workflow examples tied to chapter exercises/projects.
3. Add "first-week debugging habits" chapter before advanced language features.

More examples

Small CLI with flags and usage exit

```bash
mkdir -p /tmp/go-lug-cli && cd /tmp/go-lug-cli
go mod init example.com/lug-cli
```

Save as `main.go`:

```go
package main

import (
    "flag"
    "fmt"
```

```go
    "os"
)

func run(args []string) (string, error) {
    fs := flag.NewFlagSet("greet", flag.ContinueOnError)
    name := fs.String("name", "world", "who to greet")
    if err := fs.Parse(args); err != nil {
        return "", err
    }
    return "hello " + *name, nil
}

func main() {
    // Simulate argv without exiting the process.
    out, err := run([]string{"-name", "Ada"})
    fmt.Println(out, err)
    _, err = run([]string{"-bad"})
    fmt.Println("bad flag err:", err != nil)

    // Show default path
    out, _ = run(nil)
    fmt.Println(out)
    _ = os.Stderr // keep habit of writing usage to stderr in real CLIs
}
```

```bash
go run .
```

Expected output:

```
hello Ada <nil>
bad flag err: true
hello world
```

Debugging habit: log inputs and duration

```bash
mkdir -p /tmp/go-lug-debug && cd /tmp/go-lug-debug
go mod init example.com/lug-debug
```

Save as `main.go`:

```go
package main

import (
    "fmt"
    "time"
```

```
)

func work(n int) int {
    sum := 0
    for i := 0; i < n; i++ {
        sum += i
    }
    return sum
}

func main() {
    start := time.Now()
    in := 1000
    out := work(in)
    fmt.Printf("work in=%d out=%d dur=%s\n", in, out, time.Since(start))
}
```

```
go run .
```

Expected output:

```
work in=1000 out=499500 dur=...
```

Runnable example

Beginner-friendly CLI habits: flags, env guardrails, stderr errors, and non-zero exit codes.

```
mkdir -p /tmp/go-letus && cd /tmp/go-letus
go mod init example.com/letus
```

Save as `main.go`:

```
package main

import (
    "flag"
    "fmt"
    "os"
)

type Config struct {
    Name string
    Env  string
}
```

```go
func parseConfig(args []string) (Config, error) {
    fs := flag.NewFlagSet("letus", flag.ContinueOnError)
    fs.SetOutput(os.Stderr)
    name := fs.String("name", "world", "who to greet")
    env := fs.String("env", "dev", "dev|prod")
    if err := fs.Parse(args); err != nil {
        return Config{}, err
    }
    return Config{Name: *name, Env: *env}, nil
}

func run(cfg Config) error {
    if cfg.Env == "prod" && os.Getenv("APP_ALLOW_PROD") != "1" {
        return fmt.Errorf("prod run blocked without APP_ALLOW_PROD=1")
    }
    if cfg.Name == "" {
        return fmt.Errorf("name must not be empty")
    }
    fmt.Printf("hello %s (%s)\n", cfg.Name, cfg.Env)
    return nil
}

func main() {
    cfg, err := parseConfig(os.Args[1:])
    if err != nil {
        os.Exit(2) // usage
    }
    if err := run(cfg); err != nil {
        fmt.Fprintln(os.Stderr, "error:", err)
        os.Exit(1)
    }
}
```

```bash
go run . -name Ada
go run . -env prod; echo exit:$?
APP_ALLOW_PROD=1 go run . -env prod -name Ada
```

Expected output:

```
hello Ada (dev)
error: prod run blocked without APP_ALLOW_PROD=1
exit:1
hello Ada (prod)
```

What to notice

- Exit `2` for usage/flags, `1` for runtime errors, `0` success—script friendly.

- Guardrails (prod allow) prevent accidents during early learning and ops.
- Diagnostics on stderr; greeting on stdout.

Try next

- Add `-v` verbose logging to stderr with `log/slog`.
- Write a table test for `run` covering prod blocked/allowed.

196 Patterns from Automate Your Home Using Go (2024)

196 Patterns from Automate Your Home Using Go (2024)

This 2024 title emphasizes applied systems engineering: Raspberry Pi deployment, Dockerized services, telemetry, monitoring with Prometheus/Grafana, and practical automation workflows.

What This Adds to Our Book

- Strong real-world edge/infrastructure use cases beyond traditional web CRUD.
- Better observability-first mindset for long-running services.
- Helpful pathway from single-node apps to homelab/multi-service operations.

Edge Automation Topology

```
sensor/input -> Go collector -> local queue/cache -> automation decision -> action
                    |
                    +-> metrics/logs -> Prometheus/Grafana
```

Deep Integration Example: Telemetry Collector with Health and Metrics Surface

```go
package main

import (
    "encoding/json"
    "fmt"
    "log"
    "net/http"
    "sync"
    "time"
)

type Reading struct {
    Source string    `json:"source"`
    Value  float64   `json:"value"`
```

```
    At     time.Time `json:"at"`
}

type State struct {
    mu       sync.RWMutex
    last     map[string]Reading
    ingested uint64
}

func newState() *State {
    return &State{last: make(map[string]Reading)}
}

func (s *State) ingest(r Reading) {
    s.mu.Lock()
    defer s.mu.Unlock()
    s.last[r.Source] = r
    s.ingested++
}

func (s *State) handleIngest(w http.ResponseWriter, r *http.Request) {
    if r.Method != http.MethodPost {
        http.Error(w, "method not allowed", http.StatusMethodNotAllowed)
        return
    }
    var x Reading
    if err := json.NewDecoder(r.Body).Decode(&x); err != nil {
        http.Error(w, "bad payload", http.StatusBadRequest)
        return
    }
    x.At = time.Now().UTC()
    s.ingest(x)
    w.WriteHeader(http.StatusAccepted)
}

func (s *State) handleHealth(w http.ResponseWriter, _ *http.Request) {
    w.Header().Set("Content-Type", "application/json")
    _ = json.NewEncoder(w).Encode(map[string]any{"ok": true, "ts": time.Now().UTC()})
}

func (s *State) handleMetrics(w http.ResponseWriter, _ *http.Request) {
    s.mu.RLock()
    defer s.mu.RUnlock()
    _, _ = w.Write([]byte("collector_ingested_total "))
    _, _ = w.Write([]byte(fmt.Sprintf("%d\n", s.ingested)))
}
```

```go
func main() {
    st := newState()
    mux := http.NewServeMux()
    mux.HandleFunc("POST /ingest", st.handleIngest)
    mux.HandleFunc("GET /healthz", st.handleHealth)
    mux.HandleFunc("GET /metrics", st.handleMetrics)

    srv := &http.Server{
        Addr:              ":8080",
        Handler:           mux,
        ReadHeaderTimeout: 2 * time.Second,
        ReadTimeout:       5 * time.Second,
        WriteTimeout:      10 * time.Second,
        IdleTimeout:       30 * time.Second,
    }
    log.Fatal(srv.ListenAndServe())
}
```

Why This Matters

Practical automation projects force integration of networking, state handling, deployment, and observability. This makes them high-value capstones for Go learners.

Curriculum Upgrades Recommended

1. Add an edge/homelab tutorial track under specialized chapters.
2. Add observability hooks by default in systems examples.
3. Add deployment story from local process -> Docker -> lightweight orchestrated environment.

More examples

Device event loop with context cancel

```bash
mkdir -p /tmp/go-ha-loop && cd /tmp/go-ha-loop
go mod init example.com/ha-loop
```

Save as `main.go`:

```go
package main

import (
    "context"
```

```go
    "fmt"
    "time"
)

func poll(ctx context.Context, events chan<- string) {
    t := time.NewTicker(10 * time.Millisecond)
    defer t.Stop()
    n := 0
    for {
        select {
        case <-ctx.Done():
            return
        case <-t.C:
            n++
            events <- fmt.Sprintf("tick-%d", n)
            if n == 3 {
                return
            }
        }
    }
}

func main() {
    ctx, cancel := context.WithTimeout(context.Background(), time.Second)
    defer cancel()
    events := make(chan string, 8)
    go poll(ctx, events)
    for i := 0; i < 3; i++ {
        fmt.Println(<-events)
    }
}
```

```bash
go run .
```

Expected output:

```
tick-1
tick-2
tick-3
```

State machine for a smart switch

```bash
mkdir -p /tmp/go-ha-sm && cd /tmp/go-ha-sm
go mod init example.com/ha-sm
```

Save as `main.go`:

```go
package main

import "fmt"

type State int

const (
    Off State = iota
    On
)

func (s State) String() string {
    if s == On {
        return "on"
    }
    return "off"
}

func transition(s State, event string) State {
    switch event {
    case "toggle":
        if s == On {
            return Off
        }
        return On
    case "off":
        return Off
    default:
        return s
    }
}

func main() {
    s := Off
    for _, e := range []string{"toggle", "toggle", "off", "toggle"} {
        s = transition(s, e)
        fmt.Println(e, "->", s)
    }
}
```

```go
go run .
```

Expected output:

```
toggle -> on
toggle -> off
off -> off
```

```
toggle -> on
```

Runnable example

Homelab-shaped service: ingest events, healthz, and manual `/metrics` counter—stdlib HTTP with timeouts (ephemeral listen for a self-check).

```
mkdir -p /tmp/go-homeauto && cd /tmp/go-homeauto
go mod init example.com/homeauto
```

Save as `main.go`:

```
package main

import (
    "encoding/json"
    "fmt"
    "io"
    "log"
    "net"
    "net/http"
    "strings"
    "sync/atomic"
    "time"
)

type state struct{ events atomic.Int64 }

func main() {
    st := &state{}
    mux := http.NewServeMux()
    mux.HandleFunc("POST /ingest", func(w http.ResponseWriter, r *http.Request) {
        var body struct {
            Device string  `json:"device"`
            Value  float64 `json:"value"`
        }
        if err := json.NewDecoder(r.Body).Decode(&body); err != nil {
            http.Error(w, "bad json", http.StatusBadRequest)
            return
        }
        st.events.Add(1)
        w.WriteHeader(http.StatusAccepted)
        fmt.Fprintf(w, "accepted %s\n", body.Device)
    })
    mux.HandleFunc("GET /healthz", func(w http.ResponseWriter, _ *http.Request) {
        w.WriteHeader(http.StatusOK)
```

```go
        _, _ = w.Write([]byte("ok"))
    })
    mux.HandleFunc("GET /metrics", func(w http.ResponseWriter, _ *http.Request) {
        fmt.Fprintf(w, "# TYPE home_events_total counter\nhome_events_total %d\n", st.events
    })

    ln, err := net.Listen("tcp", "127.0.0.1:0")
    if err != nil {
        log.Fatal(err)
    }
    srv := &http.Server{Handler: mux, ReadHeaderTimeout: 2 * time.Second}
    go func() { _ = srv.Serve(ln) }()
    base := "http://" + ln.Addr().String()

    resp, err := http.Post(base+"/ingest", "application/json", strings.NewReader(`{"device":
    if err != nil {
        log.Fatal(err)
    }
    io.Copy(io.Discard, resp.Body)
    resp.Body.Close()
    fmt.Println("ingest:", resp.StatusCode)

    resp, _ = http.Get(base + "/healthz")
    resp.Body.Close()
    fmt.Println("healthz:", resp.StatusCode)

    resp, _ = http.Get(base + "/metrics")
    b, _ := io.ReadAll(resp.Body)
    resp.Body.Close()
    fmt.Printf("metrics:\n%s", b)
    _ = srv.Close()
}
```

```go
go run .
```

Expected output:

```
ingest: 202
healthz: 200
metrics:
# TYPE home_events_total counter
home_events_total 1
```

What to notice

- Homelab services still need health, metrics, and timeouts—not only happy-path handlers.
- Atomic counters and JSON ingest are enough to start; wire MQTT/GPIO later.

- Ship the same binary under systemd or Docker next.

Try next

- Persist last reading to an atomic file write (part 17).
- Add graceful `SIGTERM` shutdown for Raspberry Pi deploys.

Specialized Applications

Desktop Apps

Desktop Apps: Wails, Fyne, and Gio

Overview

Go is a strong choice for cross-platform desktop tools: one language for business logic, solid concurrency, and easy shipping of a single binary (or a small bundle). The three main UI paths in 2026 are:

Stack	UI model	Look	Binary size (order of mag.)	Best for
Wails	Native WebView + HTML/JS	OS-native web engine	~10–30 MB	Dashboards, internal tools, SaaS companions
Fyne	Retained custom widgets (GL)	Consistent Material-ish	larger than pure CLI	Cross-platform + mobile, offline tools
Gio	Immediate-mode	Fully custom	small core	High-control UIs, specialized editors

This chapter compares the stacks, shows idiomatic wiring, and covers packaging and pitfalls.

Choosing a Stack

```
Reuse web frontend (React/Svelte/Vue)?  -->  Wails
Want one UI on desktop + mobile?        -->  Fyne
Need game-like / fully custom drawing?  -->  Gio
Just a tray icon + menu?                -->  systray / small Fyne or Wails
```

Rule of thumb: If the product *is* a website in a window, Wails. If the product *is* a native-feeling tool without a web team, Fyne.

Wails — Electron Alternative Without Chromium

Wails uses the **system WebView** (WebKit on macOS, WebView2 on Windows, WebKitGTK on Linux). Backend is Go; frontend is any SPA stack. No bundled Chrome → much smaller than Electron.

Project shape

```
myapp/
  main.go          # Wails entry, app struct
  app.go           # Methods bound to JS
  frontend/        # Vite + React/Svelte/Vue
  wails.json
  go.mod
```

Binding Go to JavaScript

```go
package main

import (
    "context"
    "fmt"
)

// App holds application state for the UI bridge.
type App struct {
    ctx context.Context
}

func NewApp() *App { return &App{} }

func (a *App) startup(ctx context.Context) {
    a.ctx = ctx
}

// Greet is callable from the frontend after Wails generates bindings.
func (a *App) Greet(name string) string {
    if name == "" {
        name = "world"
    }
    return fmt.Sprintf("Hello %s from Go", name)
}

// SavePrefs demonstrates returning errors across the bridge.
```

```go
func (a *App) SavePrefs(theme string) error {
    if theme != "light" && theme != "dark" {
        return fmt.Errorf("invalid theme %q", theme)
    }
    // persist...
    return nil
}
```

Frontend (generated bindings):

```typescript
import { Greet, SavePrefs } from "../wailsjs/go/main/App";

const msg = await Greet("Ada");
await SavePrefs("dark");
```

Events both ways

Go can emit events the UI listens for (background jobs, tray actions):

```go
import "github.com/wailsapp/wails/v2/pkg/runtime"

func (a *App) startWatcher() {
    go func() {
        // ...
        runtime.EventsEmit(a.ctx, "job:done", map[string]any{
            "id":     "42",
            "status": "ok",
        })
    }()
}
```

```typescript
import { EventsOn } from "../wailsjs/runtime/runtime";
EventsOn("job:done", (payload) => console.log(payload));
```

Production notes for Wails

- Keep **heavy work off the UI bind path**; return quickly or stream progress via events.
- Never put secrets in the frontend bundle; call Go methods that read OS keychain/env.
- Test pure Go packages with ordinary `go test`; treat UI as an integration layer.
- Build: `wails build` produces platform-native binaries; sign/notarize on macOS for Gatekeeper.

Fyne — Custom Widget Toolkit

Fyne draws its own controls via OpenGL (and related backends). Same UI code can target desktop and mobile.

```go
package main

import (
    "fyne.io/fyne/v2/app"
    "fyne.io/fyne/v2/container"
    "fyne.io/fyne/v2/widget"
)

func main() {
    a := app.New()
    w := a.NewWindow("Counter")

    n := 0
    label := widget.NewLabel("Count: 0")
    btn := widget.NewButton("Increment", func() {
        n++
        label.SetText("Count: " + itoa(n))
    })

    w.SetContent(container.NewVBox(label, btn))
    w.Resize(fyne.NewSize(320, 160))
    w.ShowAndRun()
}

func itoa(n int) string {
    return fmt.Sprintf("%d", n)
}
```

Form and validation sketch

```go
entry := widget.NewEntry()
entry.SetPlaceHolder("email@example.com")

form := &widget.Form{
    Items: []*widget.FormItem{
        {Text: "Email", Widget: entry},
    },
    OnSubmit: func() {
        if !strings.Contains(entry.Text, "@") {
            // show dialog / error label
            return
```

```
        }
        // save...
    },
}
```

Fyne production notes

- Asset bundling: use Fyne's resource bundling for icons/fonts.
- Long tasks: run in goroutines; update widgets with `fyne.Do` / thread-safe APIs so you don't touch UI from random goroutines incorrectly.
- Packaging: `fyne package` for `.app` / `.exe` / mobile artifacts.
- Look is **consistent**, not always native—set expectations with designers.

Gio — Immediate Mode

Gio rebuilds the UI every frame from state (game-engine style). Maximum control, more code.

Conceptual loop:

```
for each frame:
  process input events
  layout widgets from current state
  draw
```

Use Gio when you are building something like a timeline editor, custom charting, or a specialized instrument UI—not for a settings form CRUD app.

Shared Backend Pattern

Regardless of UI kit, keep domain logic UI-free:

```go
package greeter

type Service struct {
    // deps: db, config, http client
}

func (s *Service) Greet(name string) (string, error) {
    if name == "" {
        return "", fmt.Errorf("name required")
    }
    return "Hello " + name, nil
}
```

```go
// Wails adapter
func (a *App) Greet(name string) (string, error) {
    return a.svc.Greet(name)
}
```

```go
// Fyne adapter inside button callback
text, err := svc.Greet(entry.Text)
```

Unit-test `greeter.Service` without starting a window.

Packaging and Distribution

Platform	Notes
macOS	Codesign + notarize; hardened runtime; entitlements for network/files
Windows	Authenticode sign; WebView2 ever-green runtime for Wails
Linux	`.tar.gz`, `.AppImage`, or distro packages; WebKitGTK deps for Wails

CI sketch:

```yaml
# build desktop artifacts on tags (matrix os)
- run: wails build -platform darwin/arm64
- run: wails build -platform windows/amd64
```

Embed version the same way as CLI apps:

```bash
go build -ldflags="-X main.version=${VERSION}"
```

Security Considerations

Desktop apps inherit a large attack surface: local files, IPC, and web content.

1. **Validate all UI input** in Go, not only in JS.
2. **Disable dangerous WebView features** you do not need (arbitrary file URLs, remote debugging in prod).
3. **Path traversal** — if the UI asks to "open file", resolve and confine to a sandbox directory.
4. **Auto-update** — verify signatures (Cosign, Sparcle-like flows, or platform stores).
5. **Least privilege** — macOS entitlements and Windows manifests should not request admin by default.

```go
func safeOpen(root, userPath string) (*os.File, error) {
    clean := filepath.Clean(userPath)
    full := filepath.Join(root, clean)
    if !strings.HasPrefix(full, filepath.Clean(root)+string(os.PathSeparator)) {
        return nil, fmt.Errorf("path escapes root")
    }
    return os.Open(full)
}
```

Production Checklist

- ☐ Domain logic in plain Go packages with tests
- ☐ UI layer is a thin adapter (Wails methods / Fyne callbacks)
- ☐ Long work off the UI thread; progress via events or bindings
- ☐ Secrets never shipped in frontend assets
- ☐ Signed builds on macOS/Windows for distribution outside stores
- ☐ Version embedded; about dialog shows it
- ☐ Crash logging to a user-visible file or OS log
- ☐ Document runtime deps (WebView2, graphics drivers)

Common Pitfalls

1. **Business logic in React only** — duplicates rules; desktop and future CLI drift.
2. **Blocking the UI goroutine** — multi-second HTTP inside a button callback freezes Fyne/Wails.
3. **Assuming Electron mental model** — Wails does not bundle Chromium; Linux WebKitGTK missing breaks installs.
4. **CGO surprises** — Fyne/Gio may pull CGO; cross-compile carefully.
5. **No code signing** — macOS quarantines unsigned apps; users see scary dialogs.
6. **Shipping debug tooling** — open DevTools in production builds.

Exercises

1. **Wails hello** — Scaffold a Wails app; bind a Go `Add(a, b int) int` and call it from the frontend.
2. **Fyne counter** — Build the counter window; add a Reset button and a keyboard shortcut.
3. **Extract service** — Move greet logic into an internal package; unit-test it without UI.
4. **Safe paths** — Implement `safeOpen` with table-driven tests for `../` escapes.
5. **Release metadata** — Inject `version` via `-ldflags` and show it in a Fyne label or Wails about dialog.
6. **Compare sizes** — Build a minimal Wails and minimal Fyne app; record binary/bundle sizes on your OS.

More examples

UI-free domain service (testable core)

Desktop shells should call pure logic. This is the Go core you unit-test without Fyne/Wails.

```
mkdir -p /tmp/go-desk-core && cd /tmp/go-desk-core
go mod init example.com/desk-core
```

Save as `main.go`:

```go
package main

import (
    "fmt"
    "strings"
)

type Greeter struct {
    prefix string
}

func (g Greeter) Greet(name string) string {
    name = strings.TrimSpace(name)
    if name == "" {
        name = "world"
    }
    return g.prefix + name
}

func main() {
    g := Greeter{prefix: "Hello, "}
    fmt.Println(g.Greet("Ada"))
    fmt.Println(g.Greet("  "))
}
```

```
go run .
```

Expected output:

```
Hello, Ada
Hello, world
```

Safe open under app data root

```
mkdir -p /tmp/go-desk-safe && cd /tmp/go-desk-safe
go mod init example.com/desk-safe
```

Save as `main.go`:

```go
package main

import (
    "fmt"
    "os"
    "path/filepath"
    "strings"
)

func safeOpen(root, rel string) ([]byte, error) {
    full := filepath.Join(root, filepath.Clean("/"+rel))
    relOut, err := filepath.Rel(root, full)
    if err != nil || strings.HasPrefix(relOut, "..") {
        return nil, fmt.Errorf("denied: %s", rel)
    }
    return os.ReadFile(full)
}

func main() {
    root, _ := os.MkdirTemp("", "deskdata")
    defer os.RemoveAll(root)
    _ = os.WriteFile(filepath.Join(root, "note.txt"), []byte("secret-note"), 0o600)

    b, err := safeOpen(root, "note.txt")
    fmt.Println("ok:", string(b), err)
    _, err = safeOpen(root, "../note.txt")
    fmt.Println("escape err:", err != nil)
}
```

```
go run .
```

Expected output:

```
ok: secret-note <nil>
escape err: true
```

Runnable example

Desktop frameworks (Wails/Fyne/Gio) are not stdlib; the durable part is **domain logic outside the UI**. This CLI-shaped program is the service layer you unit-test without a window toolkit—bind the same functions from Wails or call them from Fyne callbacks.

```
mkdir -p /tmp/go-desktop-core && cd /tmp/go-desktop-core
go mod init example.com/desktop-core
```

Save as `main.go`:

```
package main

import (
    "fmt"
    "path/filepath"
    "strings"
)

// App is what a Wails struct or Fyne model should call-no UI imports.
type App struct {
    version string
    count   int
}

func NewApp(version string) *App { return &App{version: version} }

func (a *App) Greet(name string) string {
    name = strings.TrimSpace(name)
    if name == "" {
        name = "world"
    }
    return fmt.Sprintf("Hello, %s! (v%s)", name, a.version)
}

func (a *App) Add(delta int) int {
    a.count += delta
    return a.count
}

func (a *App) Reset() { a.count = 0 }

// safeOpen jails user-selected relative paths under root (file dialogs).
func safeOpen(root, userPath string) (string, error) {
    cleanRoot, err := filepath.Abs(filepath.Clean(root))
    if err != nil {
        return "", err
```

```go
    }
    joined := filepath.Join(cleanRoot, userPath)
    abs, err := filepath.Abs(joined)
    if err != nil {
        return "", err
    }
    rel, err := filepath.Rel(cleanRoot, abs)
    if err != nil || strings.HasPrefix(rel, "..") {
        return "", fmt.Errorf("path escapes root: %s", userPath)
    }
    return abs, nil
}

func main() {
    app := NewApp("1.0.0")
    fmt.Println(app.Greet("Ada"))
    fmt.Println("count:", app.Add(1))
    fmt.Println("count:", app.Add(2))
    app.Reset()
    fmt.Println("after reset:", app.Add(0))

    root := "/tmp/desktop-docs"
    ok, err := safeOpen(root, "notes/todo.txt")
    fmt.Println("safe path:", ok, err)
    _, err = safeOpen(root, "../../etc/passwd")
    fmt.Println("escape blocked:", err != nil)
}
```

```go
go run .
```

Expected output (paths host-dependent):

```
Hello, Ada! (v1.0.0)
count: 1
count: 3
after reset: 0
safe path: /tmp/desktop-docs/notes/todo.txt <nil>
escape blocked: true
```

What to notice

- UI toolkits bind methods; keep business rules in plain Go packages so tests do not need a display server.
- Path validation belongs in Go, not only in the frontend file picker.
- Inject `version` via ldflags the same way as CLI/server releases.

Try next

- Move `App` to `internal/app` and add table tests for `Greet` and `safeOpen`.
- Scaffold a Wails or Fyne shell that calls the same `Greet`/`Add` methods.

Related Topics

- Building and Releasing
- CI/CD & Hardening
- Application Hardening

Machine Learning & LLMs

AI & LLMs in Go

Overview

Python owns model **training**. Go owns **serving, orchestration, and glue**: APIs, RAG pipelines, agents, tool routers, WebSocket streams, and high-concurrency gateways in front of model runtimes.

In 2026 you rarely train foundation models yourself. You call hosted APIs or local runtimes (Ollama, vLLM, llama.cpp servers) and build reliable product software around them. Go's strengths—static binaries, `context` cancellation, excellent HTTP—map directly onto that job.

Architecture Split

```
[Python / CUDA land]          [Go land]
 research notebooks            API gateway
 training / fine-tunes         RAG orchestration
 experimental kernels          auth, quotas, billing
                               tool/function calling
                               workers + queues
```

Keep models and experiments where the ecosystem is; put the product boundary in Go.

Calling Local Models (Ollama)

Ollama exposes an HTTP API. Many teams use **LangChainGo** (`github.com/tmc/langchaingo`) or a thin custom client.

```go
package main

import (
    "context"
    "fmt"
    "log"
    "time"

    "github.com/tmc/langchaingo/llms"
    "github.com/tmc/langchaingo/llms/ollama"
```

```go
)

func main() {
    llm, err := ollama.New(ollama.WithModel("llama3.2"))
    if err != nil {
        log.Fatal(err)
    }

    ctx, cancel := context.WithTimeout(context.Background(), 60*time.Second)
    defer cancel()

    out, err := llms.GenerateFromSinglePrompt(ctx, llm, "Explain Go channels in 3 bullets")
    if err != nil {
        log.Fatal(err)
    }
    fmt.Println(out)
}
```

Minimal raw HTTP client

Prefer owning the client when you need strict control:

```go
type genRequest struct {
    Model  string `json:"model"`
    Prompt string `json:"prompt"`
    Stream bool   `json:"stream"`
}

type genResponse struct {
    Response string `json:"response"`
    Done     bool   `json:"done"`
}

func generate(ctx context.Context, client *http.Client, base, model, prompt string) (string,
    body, _ := json.Marshal(genRequest{Model: model, Prompt: prompt, Stream: false})
    req, err := http.NewRequestWithContext(ctx, http.MethodPost, base+"/api/generate", bytes
    if err != nil {
        return "", err
    }
    req.Header.Set("Content-Type", "application/json")

    res, err := client.Do(req)
    if err != nil {
        return "", err
    }
    defer res.Body.Close()
```

```go
    if res.StatusCode >= 300 {
        b, _ := io.ReadAll(res.Body)
        return "", fmt.Errorf("ollama %s: %s", res.Status, b)
    }
    var gr genResponse
    if err := json.NewDecoder(res.Body).Decode(&gr); err != nil {
        return "", err
    }
    return gr.Response, nil
}
```

Always set **timeouts** and honor `ctx` cancellation—model calls are slow and easy to pile up under load.

Streaming Tokens to Clients

UX for chat means streaming. Bridge model SSE/NDJSON to your API:

```go
func streamHandler(w http.ResponseWriter, r *http.Request) {
    flusher, ok := w.(http.Flusher)
    if !ok {
        http.Error(w, "streaming unsupported", http.StatusInternalServerError)
        return
    }
    w.Header().Set("Content-Type", "text/event-stream")
    w.Header().Set("Cache-Control", "no-cache")

    // pseudo: for each token from model stream
    tokens := []string{"Hello", " ", "from", " ", "Go"}
    for _, t := range tokens {
        select {
        case <-r.Context().Done():
            return
        default:
        }
        fmt.Fprintf(w, "data: %s\n\n", t)
        flusher.Flush()
    }
    fmt.Fprintf(w, "data: [DONE]\n\n")
    flusher.Flush()
}
```

RAG Pipeline in Go

```
query
  --> embed(query) -> []float32
  --> vector search top-k chunks
  --> build prompt (system + chunks + query)
  --> LLM completion
  --> cite sources + log token usage
```

Embedding + top-k stub

```go
type Chunk struct {
    ID   string
    Text string
    // Vector stored in DB; not always loaded
}

type Scored struct {
    Chunk
    Score float32
}

func cosine(a, b []float32) float32 {
    var dot, na, nb float32
    for i := range a {
        dot += a[i] * b[i]
        na += a[i] * a[i]
        nb += b[i] * b[i]
    }
    if na == 0 || nb == 0 {
        return 0
    }
    return dot / (float32(math.Sqrt(float64(na))) * float32(math.Sqrt(float64(nb))))
}

func topK(query []float32, docs []struct {
    Chunk
    Vec []float32
}, k int) []Scored {
    scored := make([]Scored, 0, len(docs))
    for _, d := range docs {
        scored = append(scored, Scored{Chunk: d.Chunk, Score: cosine(query, d.Vec)})
    }
    sort.Slice(scored, func(i, j int) bool { return scored[i].Score > scored[j].Score })
    if k > len(scored) {
```

```
        k = len(scored)
    }
    return scored[:k]
}
```

Production retrieval usually lives in **pgvector**, Weaviate, Milvus, Qdrant—not in-process loops. Go is the language of several of those systems' clients and control planes.

Prompt assembly

```
func buildPrompt(system string, chunks []Scored, question string) string {
    var b strings.Builder
    b.WriteString(system)
    b.WriteString("\n\nContext:\n")
    for i, c := range chunks {
        fmt.Fprintf(&b, "[%d] (%s) %s\n", i+1, c.ID, c.Text)
    }
    b.WriteString("\nQuestion: ")
    b.WriteString(question)
    b.WriteString("\nAnswer using only the context. Cite chunk ids.\n")
    return b.String()
}
```

Tool / Function Calling Agents

Agents are orchestration: model proposes a tool call → Go executes **allow-listed** tools → model continues.

```
type Tool interface {
    Name() string
    Description() string
    Run(ctx context.Context, args json.RawMessage) (string, error)
}

type Registry map[string]Tool

func (r Registry) Exec(ctx context.Context, name string, args json.RawMessage) (string, erro
    t, ok := r[name]
    if !ok {
        return "", fmt.Errorf("unknown tool %q", name)
    }
    // enforce timeout per tool
    ctx, cancel := context.WithTimeout(ctx, 10*time.Second)
    defer cancel()
```

```
    return t.Run(ctx, args)
}
```

Never let the model choose arbitrary shell commands or unconstrained HTTP. Map names to vetted implementations only.

SIMD and Local Vector Math (Experimental)

Go continues to invest in performance for numeric loops. Go 1.27's experimental portable `simd` package plus `simd/archsimd` (still `GOEXPERIMENT=simd`) is the supported experiment path. For production similarity at scale you still want a vector DB; for small in-process indexes or filters, careful float32 loops (and SIMD where justified) help.

```
# When available in your Go version / experiment flags:
GOEXPERIMENT=simd go test ./internal/vec/...
```

Profile before micro-optimizing: retrieval latency is usually network + model, not dot products.

Reliability Patterns

Concern	Go approach
Slow models	`context.WithTimeout`, client-side deadlines
Overload	worker pool / queue; reject with 429
Cost	token counters, per-tenant budgets
Safety	output filters, tool allow-lists, PII redaction logs
Observability	slog attrs: `request_id`, `model`, `tokens_in/out`, `latency_ms`

```
func withUsageLog(model string, tokensIn, tokensOut int, d time.Duration) {
    slog.Info("llm.completion",
        "model", model,
        "tokens_in", tokensIn,
        "tokens_out", tokensOut,
        "latency_ms", d.Milliseconds(),
    )
}
```

Hosted APIs

Same patterns against OpenAI-compatible endpoints:

```
req, _ := http.NewRequestWithContext(ctx, http.MethodPost, endpoint, bytes.NewReader(payload
req.Header.Set("Authorization", "Bearer "+os.Getenv("OPENAI_API_KEY"))
req.Header.Set("Content-Type", "application/json")
```

- Store keys in env/secret manager, never in the repo.
- Prefer **one outbound client** with connection pooling and explicit timeouts.
- Implement retries only on idempotent GETs or with careful dedupe keys for jobs—not blindly on every chat POST.

Production Checklist

☐ Timeouts on every model HTTP call
☐ Cancellation wired from client disconnect to upstream
☐ Streaming path tested under proxy buffering settings
☐ Tool calls allow-listed and argument-validated
☐ RAG sources logged for audit (chunk ids)
☐ Token/cost metrics per tenant
☐ Secrets only in env/secret store
☐ Rate limits at the edge
☐ Eval suite for prompt regressions (golden questions)

Common Pitfalls

1. **No timeout** — hung Ollama/vLLM pins goroutines and exhausts the server.
2. **Loading whole corpora into RAM** — use a vector store; page chunk text by id.
3. **Prompt injection** — treat retrieved docs and user text as hostile; separate system instructions clearly; don't execute tools from doc content.
4. **Python for the API layer** — GIL and packaging pain for a concurrent gateway you could write in Go.
5. **Logging full prompts with secrets** — redaction policies for PII and API keys.
6. **Unbounded agent loops** — cap steps and total tokens per request.

Exercises

1. **Ollama client** — Write `generate(ctx, prompt)` against local Ollama; add a 30s timeout test using a cancelled context.
2. **Cosine top-k** — Implement `topK` with table-driven tests (identical vectors, orthogonal, empty).
3. **SSE chat** — Stream fake tokens to `curl -N`; cancel mid-stream and confirm the handler exits.
4. **Tool registry** — Register `weather` and `time` tools; reject unknown names; unit-test argument JSON validation.

5. **RAG prompt** — Given three chunks and a question, build a prompt and assert citations instructions are present.
6. **Usage metrics** — Wrap a completion call and emit slog fields for tokens and latency; scrape logs in a test with `slog.NewJSONHandler` to a buffer.

More examples

SSE token stream with cancel

```
mkdir -p /tmp/go-llm-sse && cd /tmp/go-llm-sse
go mod init example.com/llm-sse
```

Save as `main.go`:

```
package main

import (
    "bufio"
    "context"
    "fmt"
    "net/http"
    "net/http/httptest"
    "strings"
    "time"
)

func main() {
    tokens := []string{"Hello", " ", "world", "!"}
    h := http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        fl, ok := w.(http.Flusher)
        if !ok {
            http.Error(w, "no flush", 500)
            return
        }
        w.Header().Set("Content-Type", "text/event-stream")
        for _, t := range tokens {
            select {
            case <-r.Context().Done():
                return
            default:
            }
            fmt.Fprintf(w, "data: %s\n\n", t)
            fl.Flush()
            time.Sleep(5 * time.Millisecond)
        }
```

```go
        fmt.Fprintf(w, "data: [DONE]\n\n")
    })

    ts := httptest.NewServer(h)
    defer ts.Close()

    ctx, cancel := context.WithCancel(context.Background())
    defer cancel()
    req, _ := http.NewRequestWithContext(ctx, http.MethodGet, ts.URL, nil)
    resp, err := http.DefaultClient.Do(req)
    if err != nil {
        panic(err)
    }
    defer resp.Body.Close()

    var got []string
    sc := bufio.NewScanner(resp.Body)
    for sc.Scan() {
        line := sc.Text()
        if strings.HasPrefix(line, "data: ") {
            got = append(got, strings.TrimPrefix(line, "data: "))
        }
    }
    fmt.Println("tokens:", strings.Join(got, "|"))
}
```

```bash
go run .
```

Expected output:

```
tokens: Hello| |world|!|[DONE]
```

Tool registry with JSON args

```bash
mkdir -p /tmp/go-llm-tools && cd /tmp/go-llm-tools
go mod init example.com/llm-tools
```

Save as `main.go`:

```go
package main

import (
    "encoding/json"
    "fmt"
)
```

```go
type ToolFunc func(args json.RawMessage) (string, error)

type Registry map[string]ToolFunc

func (r Registry) Call(name string, args json.RawMessage) (string, error) {
    fn, ok := r[name]
    if !ok {
        return "", fmt.Errorf("unknown tool %q", name)
    }
    return fn(args)
}

func main() {
    reg := Registry{
        "echo": func(args json.RawMessage) (string, error) {
            var in struct {
                Text string `json:"text"`
            }
            if err := json.Unmarshal(args, &in); err != nil {
                return "", err
            }
            return in.Text, nil
        },
    }
    out, err := reg.Call("echo", json.RawMessage(`{"text":"hi"}`))
    fmt.Println("echo:", out, err)
    _, err = reg.Call("weather", json.RawMessage(`{}`))
    fmt.Println("missing:", err != nil)
}
```

```bash
go run .
```

Expected output:

```
echo: hi <nil>
missing: true
```

Runnable example

LLM gateways need timeouts, tool registries, and retrieval helpers—not only HTTP clients to vendors. This program implements cosine top-k, a tool allowlist, and a timeout-bounded "completion" (stdlib mock).

```bash
mkdir -p /tmp/go-llm-core && cd /tmp/go-llm-core
go mod init example.com/llm-core
```

Save as `main.go`:

```
package main

import (
    "context"
    "encoding/json"
    "fmt"
    "math"
    "sort"
    "strings"
    "time"
)

type Doc struct {
    ID   string
    Text string
    Vec  []float64
}

func cosine(a, b []float64) float64 {
    var dot, na, nb float64
    for i := range a {
        dot += a[i] * b[i]
        na += a[i] * a[i]
        nb += b[i] * b[i]
    }
    if na == 0 || nb == 0 {
        return 0
    }
    return dot / (math.Sqrt(na) * math.Sqrt(nb))
}

func topK(query []float64, docs []Doc, k int) []Doc {
    type scored struct {
        d Doc
        s float64
    }
    var all []scored
    for _, d := range docs {
        all = append(all, scored{d, cosine(query, d.Vec)})
    }
    sort.Slice(all, func(i, j int) bool { return all[i].s > all[j].s })
    if k > len(all) {
        k = len(all)
    }
    out := make([]Doc, 0, k)
```

```
    for i := 0; i < k; i++ {
        out = append(out, all[i].d)
    }
    return out
}

type ToolFunc func(ctx context.Context, args json.RawMessage) (string, error)

type Registry struct {
    tools map[string]ToolFunc
}

func NewRegistry() *Registry {
    return &Registry{tools: map[string]ToolFunc{}}
}

func (r *Registry) Register(name string, fn ToolFunc) { r.tools[name] = fn }

func (r *Registry) Call(ctx context.Context, name string, args json.RawMessage) (string, err
    fn, ok := r.tools[name]
    if !ok {
        return "", fmt.Errorf("unknown tool %q", name)
    }
    return fn(ctx, args)
}

func complete(ctx context.Context, prompt string) (string, error) {
    // Stand-in for Ollama/OpenAI: honor context deadline.
    select {
    case <-ctx.Done():
        return "", ctx.Err()
    case <-time.After(20 * time.Millisecond):
        return "answer based on: " + truncate(prompt, 60), nil
    }
}

func truncate(s string, n int) string {
    if len(s) <= n {
        return s
    }
    return s[:n] + "..."
}

func main() {
    docs := []Doc{
        {ID: "1", Text: "Go concurrency uses goroutines", Vec: []float64{1, 0, 0}},
```

```go
        {ID: "2", Text: "Rust has ownership and borrowing", Vec: []float64{0, 1, 0}},
        {ID: "3", Text: "Go interfaces are satisfied implicitly", Vec: []float64{0.9, 0.1, 0
    }
    q := []float64{1, 0, 0}
    hits := topK(q, docs, 2)
    fmt.Println("top-k:")
    for _, h := range hits {
        fmt.Printf("  %s: %s\n", h.ID, h.Text)
    }

    var b strings.Builder
    b.WriteString("Use only these sources:\n")
    for _, h := range hits {
        fmt.Fprintf(&b, "- [%s] %s\n", h.ID, h.Text)
    }
    b.WriteString("Question: What does Go use for concurrency?\n")
    prompt := b.String()

    reg := NewRegistry()
    reg.Register("time_now", func(ctx context.Context, _ json.RawMessage) (string, error) {
        return time.Now().UTC().Format(time.RFC3339), nil
    })
    out, err := reg.Call(context.Background(), "time_now", nil)
    fmt.Println("tool time_now:", out, err)
    _, err = reg.Call(context.Background(), "shell_exec", nil)
    fmt.Println("unknown tool rejected:", err != nil)

    ctx, cancel := context.WithTimeout(context.Background(), 2*time.Second)
    defer cancel()
    ans, err := complete(ctx, prompt)
    fmt.Println("completion:", ans, err)

    // Timeout path
    ctx2, cancel2 := context.WithTimeout(context.Background(), time.Nanosecond)
    defer cancel2()
    time.Sleep(time.Millisecond)
    _, err = complete(ctx2, prompt)
    fmt.Println("timeout err:", err != nil)
}
```

```bash
go run .
```

Expected output (timestamps vary):

```
top-k:
  1: Go concurrency uses goroutines
  3: Go interfaces are satisfied implicitly
```

```
tool time_now: 2026-07-28T12:00:00Z <nil>
unknown tool rejected: true
completion: answer based on: Use only these sources:
- [1] Go concurrency uses... <nil>
timeout err: true
```

What to notice

- Retrieval (top-k) and tool allowlists are pure Go; treat model I/O as an untrusted boundary with timeouts.
- Unknown tools must fail closed—never reflect arbitrary names into `os/exec`.
- Cancelled contexts prevent hung upstream model servers from pinning handlers forever.

Try next

- Stream fake tokens to stdout with flushes (SSE-style) and cancel mid-stream.
- Add slog fields for `tokens` and `dur_ms` around `complete`.

Related Topics

- HTTP and Networking
- Context
- Logging and Observability
- I/O and Streaming

Systems Programming

Systems Programming: Low-Level Go

Overview

Go was designed to replace much of the C++/Java systems software at Google: servers, CLIs, and infrastructure agents. It still offers deliberate low-level power—raw syscalls via `golang.org/x/sys`, unsafe zero-copy tricks, cgo bridges, and userspace control planes for eBPF—while keeping a memory-safe default path.

On Linux the usual path never goes through libc. `fmt.Println`, `os.Open`, and `net.Listen` end in Go runtime assembly that executes the `syscall` instruction. That is why a `CGO_ENABLED=0` binary is fully static and runs on Alpine, Debian, or `scratch` without a C library. The portability story, glibc vs musl, and Go-versus-C trade-offs live in Go as a Systems Language. This chapter is the next layer: syscalls you opt into, Delve, eBPF userspace, `unsafe`, and production guardrails.

When to Drop Below the Stdlib

Need	Prefer	Reach for low-level when
Files	`os`, `io`	Special flags, `flock`, sendfile
Processes	`os/exec`	Fine-grained credentials, ptrace
Network	`net`	Raw sockets, SO_REUSEPORT tuning
Memory	GC + slices	`mlock`, huge pages, mmap shared regions
Tracing	slog/pprof	eBPF kernel probes

Most services never need `unsafe` or eBPF. Learn the tools so you recognize when they earn their complexity.

Syscalls with `golang.org/x/sys`

The `syscall` package is frozen. New code should use **`golang.org/x/sys/unix`** (and `windows`).

```go
package main

import (
    "log"
    "golang.org/x/sys/unix"
)
```

```go
func main() {
    // Example: prevent a secret buffer from being swapped to disk (best-effort).
    secret := make([]byte, 4096)
    copy(secret, []byte("super-secret-token"))

    if err := unix.Mlock(secret); err != nil {
        log.Printf("mlock: %v (may need privileges)", err)
    }
    defer func() {
        // Zero then unlock
        for i := range secret {
            secret[i] = 0
        }
        _ = unix.Munlock(secret)
    }()

    // use secret...
    _ = secret
}
```

File locking

```go
func lockFile(fd int) error {
    return unix.Flock(fd, unix.LOCK_EX|unix.LOCK_NB)
}
```

Raising file descriptor limits (process)

Often better done by systemd/`ulimit`, but diagnostics help:

```go
var rLimit unix.Rlimit
if err := unix.Getrlimit(unix.RLIMIT_NOFILE, &rLimit); err != nil {
    log.Fatal(err)
}
log.Printf("nofile cur=%d max=%d", rLimit.Cur, rLimit.Max)
```

Portability: `unix` builds are OS-specific. Use build tags or high-level wrappers for multi-OS products.

Debugging with Delve

Delve (`dlv`) understands goroutines, channels, and Go's runtime—unlike gdb alone.

```
go install github.com/go-delve/delve/cmd/dlv@latest

# local
dlv debug ./cmd/server -- --config config.yaml

# attach
dlv attach $(pgrep -n myapp)

# tests
dlv test ./internal/store -- -test.run TestRace
```

Core dumps

On a crash with core dumps enabled:

```
ulimit -c unlimited
# ... crash ...
dlv core ./myapp core.12345
```

Inside Delve: `bt`, `goroutines`, `frame`, `print varname`.

Remote debugging sketch

```
# on server
dlv exec ./myapp --headless --listen=:2345 --api-version=2 --accept-multiclient

# local IDE / dlv connect :2345
```

Never leave headless Delve exposed on a public interface in production.

eBPF Userspace in Go

eBPF programs run in the kernel (restricted VM). Go is the dominant language for **loaders and control planes** (Cilium, many observability agents) via `github.com/cilium/ebpf`.

```
[ kernel eBPF program ]  <--- map read/write --->  [ Go agent ]
   packet filter / probe                              metrics, policy, UI
```

Conceptual load flow:

```
// Illustrative - real programs use bpf2go generated specs.
spec, err := ebpf.LoadCollectionSpec("bpf/prog.o")
if err != nil {
```

```go
    log.Fatal(err)
}
coll, err := ebpf.NewCollection(spec)
if err != nil {
    log.Fatal(err)
}
defer coll.Close()

// Read a map of counters
var key uint32 = 0
var value uint64
if err := coll.Maps["pkt_count"].Lookup(&key, &value); err == nil {
    log.Printf("packets=%d", value)
}
```

Requirements: Linux, appropriate kernel version, often `CAP_BPF` / `CAP_SYS_ADMIN` or unprivileged eBPF policy depending on distro. This is **not** portable to FreeBSD/macOS the same way.

`unsafe` for Zero-Copy Views

`unsafe` bypasses type and memory safety. Use only with invariants you can prove.

```go
package endianx

import (
    "encoding/binary"
    "unsafe"
)

// BytesToFloat64LE interprets the first 8 bytes as a little-endian float64.
// Prefer encoding/binary in application code; this shows the unsafe cast pattern.
func BytesToFloat64LE(b []byte) float64 {
    if len(b) < 8 {
        panic("short buffer")
    }
    // Safer portable approach:
    u := binary.LittleEndian.Uint64(b[:8])
    return *(*float64)(unsafe.Pointer(&u))
}
```

Rules of thumb:

1. Document alignment and lifetime; the backing array must live as long as the view.
2. Prefer `encoding/binary`, `bytealg`, or generated code before `unsafe`.
3. Gate with build tags / architecture checks when needed.
4. Race detector and tests still matter—`unsafe` does not excuse data races.

Slice header caution

```go
// Dangerous if misused: sharing underlying array across ownership boundaries.
func noCopyString(b []byte) string {
    return unsafe.String(unsafe.SliceData(b), len(b))
}
// Only valid while b is not mutated; do not return if caller may recycle b.
```

Go 1.20+ `unsafe.String` / `unsafe.Slice` make intent clearer than raw pointer casts—still sharp tools.

mmap for Large Read-Only Data

```go
f, err := os.Open("large.bin")
if err != nil {
    return err
}
defer f.Close()

st, err := f.Stat()
if err != nil {
    return err
}
data, err := unix.Mmap(int(f.Fd()), 0, int(st.Size()),
    unix.PROT_READ, unix.MAP_SHARED)
if err != nil {
    return err
}
defer unix.Munmap(data)

// read-only use of data
_ = data[0]
```

Good for large static corpora; bad for small files where `os.ReadFile` is simpler.

cgo Boundary (Reminder)

cgo lets you call C, but:

- Slows builds and complicates cross-compilation
- Shares the C world of undefined behavior
- Has a cost crossing the boundary under high call rates

Prefer pure Go or subprocess isolation unless a library is irreplaceable. See the dedicated cgo chapter for details.

Production Checklist

- ☐ Prefer stdlib / high-level packages before `x/sys` and `unsafe`
- ☐ OS-specific code isolated behind interfaces + build tags
- ☐ Delve/core dump workflow documented for on-call
- ☐ eBPF agents least-privilege; never world-open debug ports
- ☐ Secrets: mlock where required + zeroization; still prefer hardware/HSM when critical
- ☐ Integration tests on the real target kernel/OS for syscall features
- ☐ `go test -race` on all packages that share memory

Common Pitfalls

1. **Using frozen `syscall` for new code** — miss fixes and constants; use `x/sys`.
2. **Ignoring errors from `Mlock` / `Flock`** — silent degraded security or locking.
3. **eBPF on the wrong kernel** — load fails mysteriously; pin kernel versions in ops docs.
4. **`unsafe` aliasing bugs** — GC moves aren't the issue (pinned carefully) so much as **lifetime and mutation**.
5. **Privileged containers by default** — systems agents often over-request `privileged: true`.
6. **Core dumps with secrets** — cores can contain tokens; encrypt at rest and restrict access.

Exercises

1. **Flock CLI** — Write a tiny program that takes a lock file path, `LOCK_EX`, prints "locked", sleeps 10s; run two instances and show the second fails with `LOCK_NB`.
2. **Rlimit log** — Print `RLIMIT_NOFILE` at startup; document how systemd `LimitNOFILE` changes it.
3. **Delve session** — Break on a handler, inspect a request struct, continue.
4. **unsafe round-trip** — Convert `float64` → bytes → `float64` with `encoding/binary` and with `unsafe`; fuzz compare.
5. **mmap read** — mmap a file of uint64 LE values and sum them; compare speed to `os.ReadFile` for a 100MB file.
6. **Build tags** — Provide `mlock_linux.go` and a no-op stub for other OSes; `GOOS=darwin go build` must succeed.

More examples

Length-prefix binary frame (encode/decode)

```
mkdir -p /tmp/go-sys-frame && cd /tmp/go-sys-frame
go mod init example.com/sys-frame
```

Save as `main.go`:

```go
package main

import (
    "bytes"
    "encoding/binary"
    "fmt"
    "io"
)

func writeFrame(w io.Writer, payload []byte) error {
    var hdr [4]byte
    binary.BigEndian.PutUint32(hdr[:], uint32(len(payload)))
    if _, err := w.Write(hdr[:]); err != nil {
        return err
    }
    _, err := w.Write(payload)
    return err
}

func readFrame(r io.Reader) ([]byte, error) {
    var hdr [4]byte
    if _, err := io.ReadFull(r, hdr[:]); err != nil {
        return nil, err
    }
    n := binary.BigEndian.Uint32(hdr[:])
    if n > 1<<20 {
        return nil, fmt.Errorf("frame too large: %d", n)
    }
    buf := make([]byte, n)
    if _, err := io.ReadFull(r, buf); err != nil {
        return nil, err
    }
    return buf, nil
}

func main() {
    var buf bytes.Buffer
    _ = writeFrame(&buf, []byte("ping"))
    _ = writeFrame(&buf, []byte("pong-payload"))
    a, _ := readFrame(&buf)
    b, _ := readFrame(&buf)
    fmt.Println(string(a), string(b), "left", buf.Len())
}
```

```
go run .
```

Expected output:

```
ping pong-payload left 0
```

Little-endian uint64 round-trip

```
mkdir -p /tmp/go-sys-le && cd /tmp/go-sys-le
go mod init example.com/sys-le
```

Save as `main.go`:

```
package main

import (
    "encoding/binary"
    "fmt"
)

func main() {
    var raw [8]byte
    binary.LittleEndian.PutUint64(raw[:], 0x0102030405060708)
    fmt.Printf("bytes: % x\n", raw)
    v := binary.LittleEndian.Uint64(raw[:])
    fmt.Printf("value: 0x%x\n", v)
}
```

```
go run .
```

Expected output:

```
bytes: 08 07 06 05 04 03 02 01
value: 0x102030405060708
```

Runnable example

Systems programming often means process identity, file locks, and child processes. This program prints OS/runtime identity, uses an exclusive lock file, and runs a short child with `CommandContext`.

```
mkdir -p /tmp/go-systems && cd /tmp/go-systems
go mod init example.com/systems
```

Save as `main.go`:

```
package main

import (
```

```
    "context"
    "fmt"
    "os"
    "os/exec"
    "path/filepath"
    "runtime"
    "time"
)

func main() {
    fmt.Printf("goos=%s pid=%d goroutines=%d\n", runtime.GOOS, os.Getpid(), runtime.NumGorou

    dir := filepath.Join(os.TempDir(), "go-systems-demo")
    _ = os.MkdirAll(dir, 0o750)
    lockPath := filepath.Join(dir, "app.lock")

    // Exclusive lock via O_EXCL lockfile (simple, portable pattern for single-instance tool
    lf, err := os.OpenFile(lockPath, os.O_CREATE|os.O_EXCL|os.O_WRONLY, 0o600)
    if err != nil {
        fmt.Println("lock busy or error:", err)
    } else {
        fmt.Fprintln(lf, os.Getpid())
        fmt.Println("acquired lock:", lockPath)
        defer func() {
            lf.Close()
            _ = os.Remove(lockPath)
            fmt.Println("released lock")
        }()
    }

    // Second acquire should fail while first holds O_EXCL file.
    _, err = os.OpenFile(lockPath, os.O_CREATE|os.O_EXCL|os.O_WRONLY, 0o600)
    fmt.Println("second lock failed:", err != nil)

    ctx, cancel := context.WithTimeout(context.Background(), 2*time.Second)
    defer cancel()
    cmd := exec.CommandContext(ctx, "echo", "child-hello")
    out, err := cmd.CombinedOutput()
    fmt.Printf("child: %q err=%v\n", string(out), err)

    // Cancelled child
    ctx2, cancel2 := context.WithTimeout(context.Background(), 50*time.Millisecond)
    defer cancel2()
    slow := exec.CommandContext(ctx2, "sleep", "2")
    err = slow.Run()
    fmt.Println("sleep cancelled:", err != nil)
```

```go
    fmt.Println("systems demo ok")
}
```

```go
go run .
```

Expected output (illustrative):

```
goos=darwin pid=12345 goroutines=1
acquired lock: /tmp/go-systems-demo/app.lock
second lock failed: true
child: "child-hello\n" err=<nil>
sleep cancelled: true
released lock
systems demo ok
```

What to notice

- `CommandContext` kills children when the deadline hits—critical for supervisors and CI wrappers.
- Exclusive lock files (`O_EXCL`) are a portable single-instance pattern; Linux production often prefers `flock` via `golang.org/x/sys/unix` (not stdlib).
- Always clean lock files in `defer` on graceful exit; crash may leave stale locks (document recovery).

Try next

- On Linux, experiment with `unix.Flock` from `golang.org/x/sys` (outside pure stdlib).
- Print open file soft limit via platform APIs or document `ulimit -n` next to your service unit.

Related Topics

- Go as a Systems Language
- Unsafe
- cgo
- Go on NixOS & FreeBSD
- Application Hardening

I/O and Streaming

I/O and Streaming

Overview

Go's `io` package defines the small interfaces that power files, networks, compression, and crypto: `Reader`, `Writer`, `Closer`, and friends. Idiomatic streaming means **processing data in bounded buffers** instead of slurping multi-gigabyte payloads into memory.

This chapter covers the core interfaces, composition (`MultiReader`, `TeeReader`, `LimitReader`), `bufio`, pipes, practical streaming patterns, and production pitfalls.

Core Interfaces

```go
type Reader interface {
    Read(p []byte) (n int, err error)
}

type Writer interface {
    Write(p []byte) (n int, err error)
}

type Closer interface {
    Close() error
}

type ReadCloser interface {
    Reader
    Closer
}

type WriteCloser interface {
    Writer
    Closer
}
```

Contracts that matter in production:

- `Read` may return `n > 0` and `err == io.EOF` on the final chunk—process `n` before handling EOF.
- `Write` must report short writes (`n < len(p)`) as an error if not fully written (`io.ErrShortWrite` patterns).

- Always `Close` `ReadCloser` values you own (HTTP bodies, files).

Reading

```go
f, err := os.Open("file.txt")
if err != nil {
    return err
}
defer f.Close()

buf := make([]byte, 32*1024)
for {
    n, err := f.Read(buf)
    if n > 0 {
        // process buf[:n]
    }
    if err == io.EOF {
        break
    }
    if err != nil {
        return err
    }
}
```

Helpers:

```go
data, err := io.ReadAll(f)           // entire stream into memory
chunk, err := io.ReadAll(io.LimitReader(r, 1<<20)) // cap at 1 MiB
```

HTTP bodies

```go
resp, err := http.Get(url)
if err != nil {
    return err
}
defer resp.Body.Close()

const maxBody = 8 << 20 // 8 MiB
limited := io.LimitReader(resp.Body, maxBody+1)
b, err := io.ReadAll(limited)
if err != nil {
    return err
}
if len(b) > maxBody {
```

```go
    return fmt.Errorf("response body exceeds %d bytes", maxBody)
}
```

Writing

```go
f, err := os.Create("output.txt")
if err != nil {
    return err
}
defer f.Close()

if _, err := f.Write([]byte("Hello")); err != nil {
    return err
}
if _, err := io.WriteString(f, " World"); err != nil {
    return err
}
return f.Sync() // durability when needed
```

Copying Streams

```go
// Efficient path: uses WriterTo / ReaderFrom when available
if _, err := io.Copy(dst, src); err != nil {
    return err
}

// Cap bytes
if _, err := io.Copy(dst, io.LimitReader(src, 1024)); err != nil {
    return err
}

// Fixed buffer size (Go 1.15+ has CopyBuffer)
buf := make([]byte, 64*1024)
if _, err := io.CopyBuffer(dst, src, buf); err != nil {
    return err
}
```

`io.Copy` is the default for proxying HTTP, piping files, and unpacking archives—prefer it over manual loops unless you need progress hooks.

Progress reader

```go
type countingReader struct {
    r io.Reader
    n int64
}

func (c *countingReader) Read(p []byte) (int, error) {
    n, err := c.r.Read(p)
    c.n += int64(n)
    return n, err
}
```

Composition Helpers

Helper	Role
`io.MultiReader`	Concatenate readers sequentially
`io.MultiWriter`	Fan-out writes to multiple writers
`io.TeeReader`	Read while copying to a side writer (checksums, logs)
`io.LimitReader`	Hard cap bytes read
`io.SectionReader`	Window into an `ReaderAt`
`io.Pipe`	In-memory streaming between goroutines

```go
// Checksum while uploading
h := sha256.New()
tr := io.TeeReader(file, h)
if _, err := io.Copy(dest, tr); err != nil {
    return err
}
sum := h.Sum(nil)
```

```go
// Concatenate header + body
r := io.MultiReader(strings.NewReader("HDR\n"), body)
```

bufio

Buffering cuts syscalls for small reads/writes.

```go
f, err := os.Open("access.log")
if err != nil {
```

```go
    return err
}
defer f.Close()

sc := bufio.NewScanner(f)
// Raise token size for long lines (default ~64K)
sc.Buffer(make([]byte, 0, 64*1024), 1024*1024)

for sc.Scan() {
    line := sc.Text()
    _ = line
}
return sc.Err()
```

```go
bw := bufio.NewWriterSize(f, 64*1024)
if _, err := bw.WriteString("row\n"); err != nil {
    return err
}
return bw.Flush() // never forget Flush
```

bytes.Buffer and strings.Reader

```go
var buf bytes.Buffer
buf.WriteString("Hello")
buf.WriteByte(' ')
buf.Write([]byte("World"))
out := buf.String()

r := strings.NewReader("stream me")
io.Copy(os.Stdout, r)
```

Use `bytes.Buffer` for building in memory; switch to streaming files when size is unbounded.

Pipes Between Goroutines

```go
pr, pw := io.Pipe()

go func() {
    defer pw.Close()
    // If Write fails, CloseWithError
    if _, err := io.WriteString(pw, "payload"); err != nil {
        pw.CloseWithError(err)
```

```go
        return
    }
}()

data, err := io.ReadAll(pr)
if err != nil {
    return err
}
_ = data
```

Pipe blocks when the buffer is full—natural **backpressure**. Always close the writer side or readers hang forever.

Streaming HTTP Upload / Download

```go
func upload(ctx context.Context, client *http.Client, url string, r io.Reader) error {
    req, err := http.NewRequestWithContext(ctx, http.MethodPut, url, r)
    if err != nil {
        return err
    }
    // If size known:
    // req.ContentLength = size
    res, err := client.Do(req)
    if err != nil {
        return err
    }
    defer res.Body.Close()
    if res.StatusCode >= 300 {
        b, _ := io.ReadAll(io.LimitReader(res.Body, 4096))
        return fmt.Errorf("upload: %s: %s", res.Status, b)
    }
    return nil
}
```

gzip Stream

```go
func compress(dst io.Writer, src io.Reader) error {
    zw := gzip.NewWriter(dst)
    if _, err := io.Copy(zw, src); err != nil {
        _ = zw.Close()
        return err
    }
    return zw.Close()
```

```go
}

func decompress(dst io.Writer, src io.Reader) error {
    zr, err := gzip.NewReader(src)
    if err != nil {
        return err
    }
    defer zr.Close()
    _, err = io.Copy(dst, zr)
    return err
}
```

Checklist

- ☐ Bound all untrusted readers with `LimitReader` or max bytes
- ☐ `Close` every `ReadCloser` you own; check `Close` errors on writers when data integrity matters
- ☐ Prefer `io.Copy` / `CopyBuffer` over ad-hoc loops
- ☐ `Flush` buffered writers before close/return
- ☐ Pipe writer closed on both success and error paths
- ☐ Scanner buffer sized for your longest valid line
- ☐ No `ReadAll` on unbounded network input
- ☐ Use `context` cancellation on HTTP and long copies (via `req.Context()` / wrappers)

Common Pitfalls

1. **Ignoring `n` on EOF** — last bytes dropped.
2. **Forgetting `Flush`** — data stuck in `bufio.Writer`.
3. **Scanner token too long** — silent `bufio.ErrTooLong`; raise buffer max.
4. **Unclosed HTTP bodies** — connection pool stalls.
5. **Deadlocked pipes** — writer blocked because reader never reads; always schedule both sides.
6. **`ioutil` legacy** — use `io` and `os` (ioutil deprecated).
7. **Assuming `Write` is atomic for large buffers** — loop or use helpers that handle short writes.

Exercises

1. **Chunked SHA-256** — Stream a file with 32KiB buffers; print hex digest; compare to `sha256sum`.
2. **LimitReader gate** — Reject bodies over 1MiB with a clear error; test with `strings.NewReader` of known sizes.
3. **TeeReader** — Copy a file to another path while computing CRC32C (or SHA-256) in one pass.

4. **Line scanner** — Parse a 10MB log with long lines; set buffer max; count lines matching a prefix.
5. **Pipe pipeline** — Goroutine A writes numbers; goroutine B filters even lines; main reads result.
6. **gzip round-trip** — Compress and decompress a random 1MB payload; require byte-identical output.

More examples

TeeReader: copy + checksum one pass

```
mkdir -p /tmp/go-io-tee && cd /tmp/go-io-tee
go mod init example.com/io-tee
```

Save as `main.go`:

```
package main

import (
    "crypto/sha256"
    "encoding/hex"
    "fmt"
    "io"
    "os"
    "strings"
)

func main() {
    src := strings.NewReader("stream-me-please")
    h := sha256.New()
    dst, _ := os.CreateTemp("", "out")
    defer os.Remove(dst.Name())
    defer dst.Close()

    n, err := io.Copy(io.MultiWriter(dst, h), src)
    if err != nil {
        panic(err)
    }
    fmt.Println("bytes:", n)
    fmt.Println("sha256:", hex.EncodeToString(h.Sum(nil))[:16]+"...")
}
```

```
go run .
```

Expected output:

```
bytes: 15
sha256: <16 hex chars>...
```

Line scanner with buffer cap

```
mkdir -p /tmp/go-io-scan && cd /tmp/go-io-scan
go mod init example.com/io-scan
```

Save as `main.go`:

```
package main

import (
    "bufio"
    "fmt"
    "strings"
)

func main() {
    in := strings.NewReader("alpha\nbeta\ngamma-long-line\n")
    sc := bufio.NewScanner(in)
    sc.Buffer(make([]byte, 64), 64)
    var n int
    for sc.Scan() {
        if strings.HasPrefix(sc.Text(), "g") {
            n++
        }
    }
    fmt.Println("g-lines:", n, "err:", sc.Err())
}
```

```
go run .
```

Expected output:

```
g-lines: 1 err: <nil>
```

Runnable example

Streaming IO is chunked reads, limited readers, tee, and compressors—never `ReadAll` on unbounded sources. This program hashes while copying, enforces a size cap, and round-trips gzip.

```
mkdir -p /tmp/go-io-stream && cd /tmp/go-io-stream
go mod init example.com/io-stream
```

Save as `main.go`:

```go
package main

import (
    "bytes"
    "compress/gzip"
    "crypto/sha256"
    "encoding/hex"
    "fmt"
    "io"
    "os"
    "path/filepath"
    "strings"
)

func main() {
    dir := filepath.Join(os.TempDir(), "go-io-stream")
    _ = os.MkdirAll(dir, 0o750)
    srcPath := filepath.Join(dir, "src.bin")
    dstPath := filepath.Join(dir, "dst.bin")

    payload := bytes.Repeat([]byte("stream-me-"), 1000) // 10_000 bytes
    if err := os.WriteFile(srcPath, payload, 0o640); err != nil {
        panic(err)
    }

    // Chunked SHA-256 + copy (one pass with TeeReader).
    in, err := os.Open(srcPath)
    if err != nil {
        panic(err)
    }
    defer in.Close()
    out, err := os.Create(dstPath)
    if err != nil {
        panic(err)
    }
    h := sha256.New()
    n, err := io.Copy(out, io.TeeReader(in, h))
    out.Close()
    if err != nil {
        panic(err)
    }
    fmt.Println("copied_bytes:", n)
    fmt.Println("sha256:", hex.EncodeToString(h.Sum(nil)))

    // LimitReader gate
```

```go
    limited := io.LimitReader(strings.NewReader(strings.Repeat("x", 100)), 16)
    buf, _ := io.ReadAll(limited)
    fmt.Println("limited_len:", len(buf))

    // Reject oversize: read with Max-style check
    const max = 1024
    r := strings.NewReader(strings.Repeat("y", 2000))
    data, err := io.ReadAll(io.LimitReader(r, max+1))
    if err != nil {
        panic(err)
    }
    if len(data) > max {
        fmt.Println("oversize: rejected", len(data))
    }

    // gzip round-trip
    var zbuf bytes.Buffer
    zw := gzip.NewWriter(&zbuf)
    _, _ = zw.Write(payload)
    _ = zw.Close()
    zr, err := gzip.NewReader(&zbuf)
    if err != nil {
        panic(err)
    }
    round, err := io.ReadAll(zr)
    _ = zr.Close()
    if err != nil {
        panic(err)
    }
    fmt.Println("gzip_roundtrip_ok:", bytes.Equal(round, payload))
    fmt.Println("gzip_ratio:", len(zbuf.Bytes()), "/", len(payload))
}
```

```go
go run .
```

Expected output (hash stable for this payload):

```
copied_bytes: 10000
sha256: <64 hex chars>
limited_len: 16
oversize: rejected 1025
gzip_roundtrip_ok: true
gzip_ratio: <compressed> / 10000
```

What to notice

- `io.TeeReader` hashes and copies without buffering the whole file twice.

- `io.LimitReader` is the building block for body caps; check length after read or use `http.MaxBytesReader` on servers.
- Always `Close` gzip writers/readers or trailers/checksums can be wrong.

Try next

- Stream a file with a 32KiB buffer loop (`Read` into a pooled slice) and compare to `io.Copy`.
- Build a pipe: goroutine writes lines, main `bufio.Scanner` counts them.

Related Topics

- Encoding and Serialization
- HTTP and Networking
- Filesystem automation

Logging and Observability

Logging and Observability

Overview

Production systems need three pillars: **logs** (event narrative), **metrics** (aggregates), and **traces** (request journeys). Go's standard library covers solid baselines—`log/slog` for structured logs, `expvar` for simple metrics, `runtime/trace` and `net/http/pprof` for diagnostics—and interoperates with OpenTelemetry and Prometheus when you outgrow them.

This chapter focuses on idiomatic `slog`, context correlation, metrics exposure, and practical operational patterns.

The Three Pillars

```
request
  |-- trace id / span id  ---------> distributed tracer
  |-- counters / histograms -------> metrics backend
  +-- structured log lines --------> log shipper
```

Correlate them with a **request ID** (and trace IDs when tracing is enabled).

Standard `log` Package

Still fine for tiny tools:

```go
import "log"

log.Println("starting")
log.Printf("user %d logged in", userID)
// log.Fatal / log.Panic for process exit / panic helpers
```

For services, prefer **`log/slog`**.

Structured Logging with `slog`

```go
package main

import (
    "log/slog"
    "os"
)

func main() {
    handler := slog.NewJSONHandler(os.Stdout, &slog.HandlerOptions{
        Level: slog.LevelInfo,
    })
    slog.SetDefault(slog.New(handler))

    slog.Info("server starting", "addr", ":8080")
    slog.Error("db ping failed", "err", err)
}
```

Levels

Level	Use
Debug	High-volume diagnostics (off in prod by default)
Info	Normal lifecycle and business events
Warn	Recoverable anomalies
Error	Failures needing attention

```go
slog.Debug("cache lookup", "key", key, "hit", hit)
slog.Warn("retrying upstream", "attempt", n)
slog.Error("checkout failed", "order_id", id, "err", err)
```

Child loggers and attributes

```go
logger := slog.With("service", "checkout", "version", version)
logger.Info("ready")

reqLog := logger.With("request_id", reqID, "path", r.URL.Path)
reqLog.Info("request started")
```

Context-aware logging

Propagate IDs via `context.Context`:

```go
type ctxKey struct{}

func WithRequestID(ctx context.Context, id string) context.Context {
    return context.WithValue(ctx, ctxKey{}, id)
}

func RequestID(ctx context.Context) string {
    if v, ok := ctx.Value(ctxKey{}).(string); ok {
        return v
    }
    return ""
}

func LogInfo(ctx context.Context, msg string, args ...any) {
    if id := RequestID(ctx); id != "" {
        args = append(args, "request_id", id)
    }
    slog.Info(msg, args...)
}
```

HTTP middleware sketch:

```go
func withRequestID(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        id := r.Header.Get("X-Request-ID")
        if id == "" {
            id = uuidNew() // your id generator
        }
        w.Header().Set("X-Request-ID", id)
        ctx := WithRequestID(r.Context(), id)
        next.ServeHTTP(w, r.WithContext(ctx))
    })
}
```

Custom `log.Logger` (Legacy / Libraries)

```go
logger := log.New(os.Stderr, "API ", log.Ldate|log.Ltime|log.LUTC|log.Lshortfile)
logger.Println("listening")
```

Prefer slog for new code; wrap slog if a dependency wants `*log.Logger` (`slog.NewLogLogger`).

Metrics with `expvar`

Quick and dependency-free:

```go
import (
    "expvar"
    "net/http"
)

var (
    requests = expvar.NewInt("requests_total")
    errors   = expvar.NewInt("errors_total")
    inflight = expvar.NewInt("inflight")
)

func instrument(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        requests.Add(1)
        inflight.Add(1)
        defer inflight.Add(-1)
        next.ServeHTTP(w, r)
    })
}

func main() {
    mux := http.NewServeMux()
    mux.Handle("/debug/vars", expvar.Handler())
    // protect /debug/* in production
    http.ListenAndServe(":8080", instrument(mux))
}
```

For Prometheus histograms/labels, use `prometheus/client_golang` and a `/metrics` endpoint—same discipline: **auth or private listener**.

Runtime Tracing and pprof

```go
import (
    "os"
    "runtime/trace"
)

f, err := os.Create("trace.out")
if err != nil {
    log.Fatal(err)
}
```

```go
defer f.Close()
if err := trace.Start(f); err != nil {
    log.Fatal(err)
}
defer trace.Stop()
// run workload
```

```
go tool trace trace.out
```

HTTP pprof (import for side effects):

```go
import _ "net/http/pprof"

// Serve on internal addr only:
go func() {
    log.Println(http.ListenAndServe("127.0.0.1:6060", nil))
}()
```

```
go tool pprof http://127.0.0.1:6060/debug/pprof/profile?seconds=30
```

Error Logging Hygiene

```go
// Good: err is an attribute, message is stable for grouping
slog.Error("update user", "user_id", id, "err", err)

// Bad: free-form strings that explode cardinality
slog.Error(fmt.Sprintf("update user %s failed: %v", id, err))
```

Wrap errors with context for stacks of cause (`fmt.Errorf("update user: %w", err)`) and log once at the boundary—avoid logging-and-returning at every layer (duplicate noise).

Sampling and Cardinality

High-traffic debug logs can cost more than the product:

- Gate debug behind level config (`LOG_LEVEL=info`).
- Do not put unbounded IDs into **metric labels** (user ids, URLs).
- Logs can carry high-cardinality fields; metrics should not.

```go
level := slog.LevelInfo
if os.Getenv("LOG_LEVEL") == "debug" {
    level = slog.LevelDebug
}
```

OpenTelemetry (When You Need It)

For multi-service traces, adopt OTel exporters rather than inventing headers:

```
incoming HTTP
  --> extract trace context
  --> start span
  --> child client span for outbound HTTP/DB
  --> inject headers
  --> end span
```

Keep slog as the human log stream; attach `trace_id` as a slog attr when a span is present.

Production Checklist

- ☐ JSON logs in production (machine-parseable)
- ☐ `request_id` on every request log line
- ☐ Log level configurable via env
- ☐ Errors logged with `err` attribute once at boundary
- ☐ `/debug/pprof` and `/debug/vars` not public
- ☐ Metrics for rate, errors, latency (RED) on critical paths
- ☐ Timeouts logged with enough context to act
- ☐ PII redaction policy for logs
- ☐ Trace sampling strategy documented

Common Pitfalls

1. **Unstructured printf logs** — hard to query in any backend.
2. **Logging secrets** — tokens in Authorization headers, raw card data.
3. **Metric label explosion** — `path` with raw IDs → millions of series.
4. **Public pprof** — remote attackers profile your heap.
5. **Log-and-return everywhere** — 5 identical errors per failure.
6. **Synchronous remote logging in request path** — prefer local shipper / async agent.
7. **Forgetting `Close` on trace files** — empty or truncated traces.

Exercises

1. **JSON slog** — Configure JSON handler; log `Info` with three attributes; pretty-print with `jq`.
2. **Request middleware** — Inject `X-Request-ID`; assert it appears in a log buffer during a handler test.
3. **expvar counter** — Increment on each hit to `/`; fetch `/debug/vars` and parse JSON in a test.
4. **Error boundary** — Build a small stack of functions that wrap `%w` errors; log only in `main`.
5. **pprof bind** — Serve pprof on `127.0.0.1` only; confirm LAN clients cannot connect.
6. **Level gate** — Read `LOG_LEVEL` and verify Debug lines disappear when set to `info`.

More examples

slog JSON to a buffer (testable logs)

```
mkdir -p /tmp/go-slog-buf && cd /tmp/go-slog-buf
go mod init example.com/slog-buf
```

Save as `main.go`:

```
package main

import (
    "bytes"
    "fmt"
    "log/slog"
    "strings"
)

func main() {
    var buf bytes.Buffer
    log := slog.New(slog.NewJSONHandler(&buf, &slog.HandlerOptions{Level: slog.LevelInfo}))
    log.Debug("hidden")
    log.Info("request", "method", "GET", "path", "/x", "dur_ms", 12)
    out := strings.TrimSpace(buf.String())
    fmt.Println(out)
    fmt.Println("has debug:", strings.Contains(out, "hidden"))
}
```

```
go run .
```

Expected output:

```
{"time":"...","level":"INFO","msg":"request","method":"GET","path":"/x","dur_ms":12}
has debug: false
```

expvar counter + debug vars snapshot

```
mkdir -p /tmp/go-expvar && cd /tmp/go-expvar
go mod init example.com/expvar
```

Save as `main.go`:

```
package main

import (
```

```go
    "expvar"
    "fmt"
    "io"
    "net/http"
    "net/http/httptest"
    "strings"
)

func main() {
    hits := expvar.NewInt("hits")
    mux := http.NewServeMux()
    mux.HandleFunc("GET /", func(w http.ResponseWriter, _ *http.Request) {
        hits.Add(1)
        fmt.Fprintln(w, "ok")
    })
    mux.Handle("GET /debug/vars", expvar.Handler())

    ts := httptest.NewServer(mux)
    defer ts.Close()

    for i := 0; i < 3; i++ {
        r, _ := http.Get(ts.URL + "/")
        r.Body.Close()
    }
    r, _ := http.Get(ts.URL + "/debug/vars")
    defer r.Body.Close()
    body, _ := io.ReadAll(r.Body)
    fmt.Println("hits var:", hits.Value())
    fmt.Println("vars contains hits:", strings.Contains(string(body), "hits"))
}
```

```go
go run .
```

Expected output:

```
hits var: 3
vars contains hits: true
```

Runnable example

Production observability starts with `log/slog` JSON, request IDs, and simple counters. This program logs structured events, correlates a request id, and prints a tiny text exposition counter (Prometheus-shaped, no client library).

```
mkdir -p /tmp/go-observ && cd /tmp/go-observ
go mod init example.com/observ
```

Save as `main.go`:

```
package main

import (
    "bytes"
    "context"
    "fmt"
    "log/slog"
    "net/http"
    "net/http/httptest"
    "sync/atomic"
    "time"
)

type ctxKey string

const requestIDKey ctxKey = "request_id"

func withRequestID(ctx context.Context, id string) context.Context {
    return context.WithValue(ctx, requestIDKey, id)
}

func requestID(ctx context.Context) string {
    if v, ok := ctx.Value(requestIDKey).(string); ok {
        return v
    }
    return ""
}

func main() {
    var buf bytes.Buffer
    log := slog.New(slog.NewJSONHandler(&buf, &slog.HandlerOptions{Level: slog.LevelInfo}))

    var hits atomic.Int64

    mux := http.NewServeMux()
    mux.HandleFunc("GET /work", func(w http.ResponseWriter, r *http.Request) {
        id := r.Header.Get("X-Request-ID")
        if id == "" {
            id = fmt.Sprintf("req-%d", time.Now().UnixNano())
        }
        ctx := withRequestID(r.Context(), id)
        start := time.Now()
```

```go
        hits.Add(1)

        log.Info("handle",
            "msg_event", "work",
            "request_id", requestID(ctx),
            "path", r.URL.Path,
            "dur_ms", time.Since(start).Milliseconds(),
        )
        w.Header().Set("X-Request-ID", id)
        fmt.Fprintln(w, "ok")
    })
    mux.HandleFunc("GET /metrics", func(w http.ResponseWriter, _ *http.Request) {
        // Manual Prometheus text exposition (stdlib only).
        fmt.Fprintf(w, "# HELP work_hits_total Demo counter\n")
        fmt.Fprintf(w, "# TYPE work_hits_total counter\n")
        fmt.Fprintf(w, "work_hits_total %d\n", hits.Load())
    })

    // Drive with httptest (no real listen needed).
    for i := 0; i < 3; i++ {
        req := httptest.NewRequest(http.MethodGet, "/work", nil)
        req.Header.Set("X-Request-ID", fmt.Sprintf("demo-%d", i+1))
        rr := httptest.NewRecorder()
        mux.ServeHTTP(rr, req)
        fmt.Println("status:", rr.Code, "id:", rr.Header().Get("X-Request-ID"))
    }
    rr := httptest.NewRecorder()
    mux.ServeHTTP(rr, httptest.NewRequest(http.MethodGet, "/metrics", nil))
    fmt.Println("--- metrics ---")
    fmt.Print(rr.Body.String())
    fmt.Println("--- logs ---")
    fmt.Print(buf.String())
}
```

```go
go run .
```

Expected output (illustrative):

```
status: 200 id: demo-1
status: 200 id: demo-2
status: 200 id: demo-3
--- metrics ---
# HELP work_hits_total Demo counter
# TYPE work_hits_total counter
work_hits_total 3
--- logs ---
{"time":"...","level":"INFO","msg":"handle","msg_event":"work","request_id":"demo-1",...}
```

...

What to notice

- JSON logs are queryable (`jq`); keep stable field names (`request_id`, `dur_ms`).
- Counters need `_total` and type comments if you want Prometheus-compatible scrapes.
- Correlation ids must ride on context and outbound headers—not only access logs.

Try next

- Redact an `Authorization` header before logging.
- Start a real listener and `curl /metrics`; open pprof on localhost only.

Related Topics

- Structured logging (deep dive)
- Metrics with Prometheus
- Tracing
- Performance / pprof

ADK Go Agents (2.0)

ADK Go Agents (2.0)

Thesis

Google's **Agent Development Kit (ADK)** for Go is a code-first toolkit for building, evaluating, and deploying AI agents — single LLM agents, multi-agent trees, and (in 2.0) **graph-based workflows**. The X post that sparked this chapter announced **ADK Go 1.0** (31 March 2026): OpenTelemetry, plugins, human-in-the-loop tool confirmation, YAML agent configs. By the time you read this, **ADK Go 2.0** is GA (announced **30 June 2026** on adk.dev/2.0) and the import path is `google.golang.org/adk/v2`.

Write new agents against **2.x**. Treat 1.0 material as history and as a migration checklist. This chapter uses **ADK Go v2.6.0** (current on the module proxy as of 2026-10-16) and **Go 1.27.1**.

Mental model

```
ADK Go 1.x                          ADK Go 2.x

google.golang.org/adk               google.golang.org/adk/v2
hierarchical Agent.Run              Workflow Graph: Agents/Tools as nodes
Sequential/Parallel/Loop helpers    graph + dynamic + collaborative workflows
session.NewEvent(id)                session.NewEvent(ctx, id)
custom Run overrides                BeforeAgent / AfterAgent callbacks
```

Fact	Consequence
Module path embeds `/v2`	`go get google.golang.org/adk/v2` — not a tag on the old path
Go **1.25+** required for 2.0	Older toolchains fail module resolution
Agents are graph **nodes**	Bypassing the runner (manual `session.events.append`, old `Run` overrides) breaks routing and retries
HITL / retries are framework-owned	Broad `except`-style swallowing in tools hides failures from `RetryConfig`
X sparked on 1.0	Features below are written for **2.0**; 1.0 notes are called out

Install

```
# Requires Go 1.25+ (this lab: go1.27.1 at /home/box/sdk/go1.27.1)
export PATH=/home/box/sdk/go1.27.1/bin:$PATH
mkdir -p /tmp/adk-hello && cd /tmp/adk-hello
go mod init example.com/adk-hello
go get google.golang.org/adk/v2@v2.6.0
grep adk go.mod
```

Verified on this lab (2026-10-16):

```
go: added google.golang.org/adk/v2 v2.6.0
```

`go.mod` contains a line like `google.golang.org/adk/v2 v2.6.0` (exact pseudo-version / checksum from the proxy). Stay on 1.x only if you must:

```
go get google.golang.org/adk@v1   # legacy line - do not mix with /v2 imports
```

Worked examples

Case 1: Minimal LLM agent (ADK Go 2.0 shape)

The official quickstart (`adk.dev/get-started/go/`) builds a Gemini-backed agent with the Google Search tool. Imports are under `/v2`:

```go
// agent.go - ADK Go 2.0 quickstart shape (from adk.dev; needs GOOGLE_API_KEY to run)
package main

import (
    "context"
    "log"
    "os"

    "google.golang.org/adk/v2/agent"
    "google.golang.org/adk/v2/agent/llmagent"
    "google.golang.org/adk/v2/cmd/launcher"
    "google.golang.org/adk/v2/cmd/launcher/full"
    "google.golang.org/adk/v2/model/gemini"
    "google.golang.org/adk/v2/tool"
    "google.golang.org/adk/v2/tool/geminitool"
    "google.golang.org/genai"
)

func main() {
```

```go
    ctx := context.Background()

    model, err := gemini.NewModel(ctx, "gemini-flash-latest", &genai.ClientConfig{
        APIKey: os.Getenv("GOOGLE_API_KEY"),
    })
    if err != nil {
        log.Fatalf("Failed to create model: %v", err)
    }

    timeAgent, err := llmagent.New(llmagent.Config{
        Name:        "hello_time_agent",
        Model:       model,
        Description: "Tells the current time in a specified city.",
        Instruction: "You are a helpful assistant that tells the current time in a city.",
        Tools: []tool.Tool{
            geminitool.GoogleSearch{},
        },
    })
    if err != nil {
        log.Fatalf("Failed to create agent: %v", err)
    }

    config := &launcher.Config{
        AgentLoader: agent.NewSingleLoader(timeAgent),
    }

    l := full.NewLauncher()
    if err = l.Execute(ctx, config, os.Args[1:]); err != nil {
        log.Fatalf("Run failed: %v\n\n%s", err, l.CommandLineSyntax())
    }
}
```

```go mod tidy
go mod tidy
# Interactive CLI (needs a real API key - not run in this scout sandbox):
# source .env && go run agent.go
# Dev UI:
# go run agent.go web api webui   # http://localhost:8080 - development only
```

Verification without an API key: the module downloaded, `llmagent.Config` fields `Name`, `Model`, `Instruction`, `Tools` match `v2.6.0` source under the module cache, and `gemini.NewModel` / `full.NewLauncher` exist. Live Gemini calls were **not** made (no key in the scout environment).

Case 2: Module path migration (1.x → 2.x)

```
# Before (ADK Go 1.x)
import "google.golang.org/adk/agent/llmagent"

# After (ADK Go 2.x)
import "google.golang.org/adk/v2/agent/llmagent"
```

```
go get google.golang.org/adk/v2
# then rewrite imports, go mod tidy, fix compile breaks from the list below
```

Case 3: `session.NewEvent` needs a context

From the ADK Go 2.0 compatibility notes and `README-v2.md` in the module:

```
// Before (1.x)
ev := session.NewEvent(invocationID)
// or temporary helper:
ev := session.NewEventWithContext(ctx, invocationID)

// After (2.x) - context first; ID/timestamp come from platform providers on ctx
ev := session.NewEvent(ctx, invocationID)
```

Thread the existing agent/tool/callback context. Do not invent `context.Background()` mid-stack unless you are in `main` or test setup. Deterministic replay installs time/UUID providers on `ctx`.

Case 4: What 2.0 adds beyond the 1.0 launch post

1.0 launch theme (Mar 2026)	Where it sits in 2.0
OpenTelemetry tracing	Still first-class; wire a `TraceProvider` through the runner / telemetry helpers
Plugin system (retry-and-reflect, logging)	Prefer framework retries + callbacks; do not hide errors inside tools
`RequireConfirmation` HITL tools	Remains; 2.0 also models pause via event fields such as `RequestedInput`
YAML agent configs	Still supported for persona / tool lists without rebuilds
New in 2.0	Graph workflows, dynamic branching/loops, collaborative coordinator+subagent graphs
New in 2.0	Event schema fields: `NodeInfo`, `Output`, `Routes`, `IsolationScope`, `RequestedInput`

Custom session stores that map events onto rigid SQL columns must grow those fields (or store JSON blobs). Strict API gateways with `additionalProperties: false` must allow them before you write 2.0 sessions into a shared store.

Case 5: Do not bypass the graph

```
// WRONG in 2.0 - mutates session behind the runner's back
// ctx.session.events = append(ctx.session.events, custom)

// RIGHT - yield / return events so the workflow engine persists and routes them
```

Likewise, overrides of old internal `Run` entry points are ignored by the graph scheduler. Move telemetry and policy into `BeforeAgentCallbacks` / `AfterAgentCallbacks` on `llmagent.Config`.

The trap

- **Copy-pasting 1.0 import paths** from the March blog into a 2.0 module — compile errors or accidental dual-module graphs.
- **Catching all errors inside a tool** so the 2.0 retry / HITL machinery never sees a failure.
- **Mixing `adk` and `adk/v2`** in one binary.
- **Shipping ADK Web** (`go run … web api webui`) to production — docs mark it development-only.
- **Assuming the X 1.0 announcement is current API** — upgrade notes on adk.dev/2.0 win.

The boring rule

New Go agents: `google.golang.org/adk/v2`, Go 1.25, pin a minor (`v2.6.0` or current), pass `context` into `session.NewEvent`, let the workflow runtime own events and retries, and keep secrets in the environment — not in the repo.

Try this

1. `go get google.golang.org/adk/v2@latest` in a throwaway module and open `llmagent.Config` in your editor.
2. Port one 1.x agent: rewrite imports, fix `NewEvent`, run tests.
3. Add a `RequireConfirmation: true` tool and exercise the pause path in the dev UI.
4. Read the graph-workflow guide on adk.dev and express a SequentialAgent as an explicit graph.

Sources

- ADK 2.0 overview
- Installation · Go quickstart
- ADK Go 1.0 blog (historical spark)
- github.com/google/adk-go · module `google.golang.org/adk/v2` v2.6.0
- Inspiration: googledevs on X (ADK Go 1.0 thread; chapter written for **2.0**)

Projects

027 Project 27: Kubernetes Event Watcher

027 Build a Kubernetes Event Watcher

Stream cluster Events in real time with `client-go` Watch APIs. Print concise, timestamped lines operators can follow during incident response or CI rollout debugging.

```
kubeconfig -> clientset -> Events.Watch -> ResultChan -> print
```

Problem statement

CLI `kevents`:

- Watch Events in one namespace or all
- Optional field selector (e.g. involved object kind)
- Print: `TIME TYPE REASON OBJECT MESSAGE`
- Handle watch reconnects on error
- Graceful shutdown on SIGINT

Acceptance criteria

- ☐ Opens a Watch and prints events until interrupted
- ☐ Namespace flag works
- ☐ Context cancel stops cleanly (`Stop()` watch)
- ☐ Errors are visible; process does not silent-exit on first blip without log
- ☐ Builds with modules

Setup

```
mkdir kevents && cd kevents
go mod init example.com/kevents
go get k8s.io/client-go@latest
go get k8s.io/apimachinery@latest
go mod tidy
# go 1.27
```

Full `main.go`

```
package main

import (
    "context"
    "flag"
    "fmt"
    "os"
    "os/signal"
    "path/filepath"
    "syscall"
    "time"

    corev1 "k8s.io/api/core/v1"
    metav1 "k8s.io/apimachinery/pkg/apis/meta/v1"
    "k8s.io/apimachinery/pkg/watch"
    "k8s.io/client-go/kubernetes"
    "k8s.io/client-go/tools/clientcmd"
    "k8s.io/client-go/util/homedir"
)

func formatEvent(e *corev1.Event) string {
    obj := e.InvolvedObject.Kind + "/" + e.InvolvedObject.Name
    if e.InvolvedObject.Namespace != "" {
        obj = e.InvolvedObject.Namespace + "/" + obj
    }
    ts := e.LastTimestamp.Time
    if ts.IsZero() {
        ts = e.EventTime.Time
    }
    if ts.IsZero() {
        ts = time.Now()
    }
    return fmt.Sprintf("%s %-12s %-20s %-40s %s",
        ts.Format(time.RFC3339),
        e.Type,
        e.Reason,
        obj,
        e.Message,
    )
}

func main() {
    var kubeconfig string
    if home := homedir.HomeDir(); home != "" {
```

```
        kubeconfig = filepath.Join(home, ".kube", "config")
    }
    kc := flag.String("kubeconfig", kubeconfig, "kubeconfig path")
    ns := flag.String("n", "", "namespace (empty=all)")
    fieldSelector := flag.String("field-selector", "", "field selector")
    flag.Parse()

    cfg, err := clientcmd.BuildConfigFromFlags("", *kc)
    if err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }
    cli, err := kubernetes.NewForConfig(cfg)
    if err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }

    ctx, stop := signal.NotifyContext(context.Background(), os.Interrupt, syscall.SIGTERM)
    defer stop()

    namespace := *ns
    if namespace == "" {
        namespace = metav1.NamespaceAll
    }

    fmt.Fprintln(os.Stderr, "watching events; Ctrl+C to stop")

    for {
        if ctx.Err() != nil {
            return
        }
        w, err := cli.CoreV1().Events(namespace).Watch(ctx, metav1.ListOptions{
            FieldSelector: *fieldSelector,
        })
        if err != nil {
            fmt.Fprintln(os.Stderr, "watch error:", err)
            select {
            case <-ctx.Done():
                return
            case <-time.After(2 * time.Second):
            }
            continue
        }

        func() {
            defer w.Stop()
```

```
            for {
                select {
                case <-ctx.Done():
                    return
                case ev, ok := <-w.ResultChan():
                    if !ok {
                        fmt.Fprintln(os.Stderr, "watch closed; reconnecting")
                        return
                    }
                    if ev.Type == watch.Error {
                        fmt.Fprintf(os.Stderr, "watch event error: %v\n", ev.Object)
                        return
                    }
                    e, ok := ev.Object.(*corev1.Event)
                    if !ok {
                        fmt.Printf("%s %T\n", ev.Type, ev.Object)
                        continue
                    }
                    fmt.Printf("%-8s %s\n", ev.Type, formatEvent(e))
                }
            }
        }()
    }
}
```

Step-by-step build path

1. Client from kubeconfig.
2. `Events(ns).Watch` with ListOptions.
3. Range `ResultChan`; type-assert `*corev1.Event`.
4. On channel close / error, backoff and reconnect.
5. SIGINT via context cancels Watch.

Run and verification

```
go run .
go run . -n default
# generate events:
kubectl run wget --image=busybox --restart=Never -- wget -O- https://kubernetes.default
kubectl delete pod wget --force --grace-period=0
```

Stretch goals

1. Filter `type=Warning` only.
2. JSON lines for log aggregation.
3. Informer/factory instead of raw Watch.
4. Metrics: events/sec counter.

Pitfalls

Pitfall	Fix
Not calling `w.Stop()`	leak; always defer Stop
Ignoring watch close	reconnect loop
Printing only `%T`	format useful fields
No cancel	hang on Ctrl+C

Learning goals

- Watch API lifecycle
- Operator-facing event streams
- Reconnect patterns for long-lived clients

028 Project 28: Kubernetes Rollout Checker

028 Build a Kubernetes Rollout Checker

Check whether a Deployment is fully rolled out: desired replicas match updated and ready counts. Exit 0 when healthy, 1 when pending, 2 on usage/API errors—ideal for CI gates after `kubectl apply`.

```
kubeconfig -> Get Deployment -> compare desired/updated/ready -> exit code
```

Problem statement

```
kroll -name DEPLOY [-n NS] [-wait DURATION] [-interval D]
```

- One-shot check by default
- Optional `-wait` polls until healthy or timeout
- Print a one-line status for logs

Acceptance criteria

- ☐ Reads Deployment status fields correctly
- ☐ Exit 0 only when `ready == desired && updated == desired` (and desired > 0 or allow 0)
- ☐ Exit 1 when still rolling
- ☐ Exit 2 on bad flags / API errors
- ☐ Optional wait loop with timeout

Setup

```
mkdir kroll && cd kroll
go mod init example.com/kroll
go get k8s.io/client-go@latest
go get k8s.io/apimachinery@latest
go mod tidy
# go 1.27
```

Full `main.go`

```
package main

import (
    "context"
    "flag"
    "fmt"
    "os"
    "path/filepath"
    "time"

    metav1 "k8s.io/apimachinery/pkg/apis/meta/v1"
    "k8s.io/client-go/kubernetes"
    "k8s.io/client-go/tools/clientcmd"
    "k8s.io/client-go/util/homedir"
)

type status struct {
    Desired int32
    Updated int32
    Ready   int32
    Avail   int32
}

func (s status) Healthy() bool {
    return s.Ready == s.Desired && s.Updated == s.Desired && s.Avail == s.Desired
}

func getStatus(ctx context.Context, cli *kubernetes.Clientset, ns, name string) (status, err
    d, err := cli.AppsV1().Deployments(ns).Get(ctx, name, metav1.GetOptions{})
    if err != nil {
        return status{}, err
    }
    var desired int32 = 1
    if d.Spec.Replicas != nil {
        desired = *d.Spec.Replicas
    }
    return status{
        Desired: desired,
        Updated: d.Status.UpdatedReplicas,
        Ready:   d.Status.ReadyReplicas,
        Avail:   d.Status.AvailableReplicas,
    }, nil
}
```

```go
func main() {
    var kubeconfig string
    if home := homedir.HomeDir(); home != "" {
        kubeconfig = filepath.Join(home, ".kube", "config")
    }
    kc := flag.String("kubeconfig", kubeconfig, "kubeconfig path")
    name := flag.String("name", "", "deployment name")
    ns := flag.String("n", "default", "namespace")
    wait := flag.Duration("wait", 0, "max wait for healthy (0=check once)")
    interval := flag.Duration("interval", 2*time.Second, "poll interval")
    flag.Parse()

    if *name == "" {
        fmt.Fprintln(os.Stderr, "usage: kroll -name DEPLOY [-n NS] [-wait 2m]")
        os.Exit(2)
    }

    cfg, err := clientcmd.BuildConfigFromFlags("", *kc)
    if err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(2)
    }
    cli, err := kubernetes.NewForConfig(cfg)
    if err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(2)
    }

    deadline := time.Now().Add(*wait)
    for {
        ctx, cancel := context.WithTimeout(context.Background(), 10*time.Second)
        st, err := getStatus(ctx, cli, *ns, *name)
        cancel()
        if err != nil {
            fmt.Fprintln(os.Stderr, "get deployment:", err)
            os.Exit(2)
        }

        fmt.Printf("deployment=%s/%s desired=%d updated=%d ready=%d available=%d\n",
            *ns, *name, st.Desired, st.Updated, st.Ready, st.Avail)

        if st.Healthy() {
            fmt.Println("rollout: healthy")
            os.Exit(0)
        }
        fmt.Println("rollout: pending")
```

```go
        if *wait == 0 || time.Now().After(deadline) {
            if *wait > 0 {
                fmt.Fprintln(os.Stderr, "timeout waiting for rollout")
            }
            os.Exit(1)
        }
        time.Sleep(*interval)
    }
}
```

Run and verification

```bash
kubectl create deploy roll-demo --image=nginx:1.27 --replicas=2
go run . -name roll-demo -n default
go run . -name roll-demo -wait 60s -interval 1s

kubectl set image deploy/roll-demo nginx=nginx:1.27-alpine
go run . -name roll-demo -wait 2m
```

Tests

Unit-test `status.Healthy` without a cluster:

```go
package main

import "testing"

func TestHealthy(t *testing.T) {
    if !(status{2, 2, 2, 2}).Healthy() {
        t.Fatal("expected healthy")
    }
    if (status{2, 1, 1, 1}).Healthy() {
        t.Fatal("expected pending")
    }
}
```

Stretch goals

1. Also check `DeploymentProgressing` condition.
2. Support StatefulSet / DaemonSet.
3. JSON output for CI annotations.
4. Fail if `UnavailableReplicas > 0` after deadline.

Pitfalls

Pitfall	Fix
Ignoring `AvailableReplicas`	include in healthy check
Nil `Spec.Replicas`	default 1 per API conventions
Wait without timeout	always bound wait
Wrong exit codes for CI	document 0/1/2

Learning goals

- Deployment status fields for rollout gates
- Polling with timeout for CI
- Scriptable exit codes

029 Project 29: Terraform Plan Parser

029 Build a Terraform Plan Parser

Parse `terraform show -json` output and summarize create/update/delete/replace actions. This is the foundation for CI policy gates and human-readable plan digests.

```
plan.json -> decode resource_changes -> classify actions -> print + summary
```

Problem statement

```
tf-plan-parse <plan.json>
```

- Read Terraform plan JSON
- For each `resource_changes[]`, print address and actions
- Detect replace (`delete`+`create` in one change)
- Print summary counts
- Exit 2 on usage/IO/JSON errors; 0 otherwise

Acceptance criteria

- ☐ Handles empty plans
- ☐ Counts create/update/delete/replace
- ☐ Replace not double-counted as delete+create in summary (your policy: document)
- ☐ Stable stdout suitable for logs
- ☐ Stdlib only

Setup

```
mkdir tfplanparse && cd tfplanparse
go mod init example.com/tfplanparse
# go 1.27
```

Prerequisite sample:

```
# in a terraform root
terraform plan -out=tfplan
terraform show -json tfplan > plan.json
```

Minimal fixture `testdata/plan.json`:

```
{
  "resource_changes": [
    {
      "address": "aws_instance.web",
      "type": "aws_instance",
      "change": { "actions": ["create"] }
    },
    {
      "address": "aws_instance.old",
      "type": "aws_instance",
      "change": { "actions": ["delete", "create"] }
    }
  ]
}
```

Full `main.go`

```
package main

import (
    "encoding/json"
    "fmt"
    "os"
)

type Plan struct {
    ResourceChanges []ResourceChange `json:"resource_changes"`
}

type ResourceChange struct {
    Address string `json:"address"`
    Type    string `json:"type"`
    Change  struct {
        Actions []string `json:"actions"`
    } `json:"change"`
}

func classify(actions []string) string {
    if len(actions) == 2 && actions[0] == "delete" && actions[1] == "create" {
        return "replace"
    }
    if len(actions) == 1 {
        return actions[0]
```

```
    }
    if len(actions) == 0 {
        return "no-op"
    }
    return "mixed:" + fmt.Sprint(actions)
}

func analyze(p Plan) (counts map[string]int, lines []string) {
    counts = map[string]int{
        "create": 0, "update": 0, "delete": 0, "replace": 0, "no-op": 0, "other": 0,
    }
    for _, rc := range p.ResourceChanges {
        kind := classify(rc.Change.Actions)
        switch kind {
        case "create", "update", "delete", "replace", "no-op":
            counts[kind]++
        default:
            counts["other"]++
        }
        lines = append(lines, fmt.Sprintf("%-8s %s", kind, rc.Address))
    }
    return counts, lines
}

func main() {
    if len(os.Args) != 2 {
        fmt.Fprintln(os.Stderr, "usage: tf-plan-parse <plan.json>")
        os.Exit(2)
    }
    b, err := os.ReadFile(os.Args[1])
    if err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(2)
    }
    var p Plan
    if err := json.Unmarshal(b, &p); err != nil {
        fmt.Fprintln(os.Stderr, "json:", err)
        os.Exit(2)
    }

    counts, lines := analyze(p)
    for _, line := range lines {
        fmt.Println(line)
    }
    fmt.Printf("summary create=%d update=%d delete=%d replace=%d no-op=%d other=%d\n",
        counts["create"], counts["update"], counts["delete"], counts["replace"], counts["no-
}
```

Run and verification

```
go run . testdata/plan.json
# REPLACE aws_instance.old
# create   aws_instance.web
# summary create=1 ...
```

Tests

```go
package main

import "testing"

func TestClassifyReplace(t *testing.T) {
    if classify([]string{"delete", "create"}) != "replace" {
        t.Fatal()
    }
}

func TestAnalyze(t *testing.T) {
    p := Plan{ResourceChanges: []ResourceChange{
        {Address: "a", Change: struct {
            Actions []string `json:"actions"`
        }{Actions: []string{"create"}}},
    }}
    // fix: use proper struct init
    _ = p
}
```

Better test:

```go
func TestAnalyzeCounts(t *testing.T) {
    var p Plan
    p.ResourceChanges = []ResourceChange{
        {Address: "x", Change: struct {
            Actions []string `json:"actions"`
        }{Actions: []string{"create"}}},
    }
    // Simpler: set Actions via temporary
    rc := ResourceChange{Address: "x"}
    rc.Change.Actions = []string{"create"}
    p.ResourceChanges = []ResourceChange{rc}
    rc2 := ResourceChange{Address: "y"}
    rc2.Change.Actions = []string{"delete", "create"}
```

```go
    p.ResourceChanges = append(p.ResourceChanges, rc2)

    c, _ := analyze(p)
    if c["create"] != 1 || c["replace"] != 1 {
        t.Fatalf("%v", c)
    }
}
```

```go
go test ./...
```

Stretch goals

1. Group by provider/type.
2. Filter addresses by regex.
3. Markdown report for PR comments.
4. Read plan from stdin (`-`).

Pitfalls

Pitfall	Fix
Binary `tfplan` not JSON	always `terraform show -json`
Plan format version drift	pin terraform; tolerate missing fields
Counting replace as delete+create	classify first

Learning goals

- Deterministic IaC review tooling
- JSON plan shape for Terraform
- Building blocks for policy-as-code

030 Project 30: Terraform Risk Reporter

030 Build a Terraform Risk Reporter

Use a Terraform plan JSON to **fail CI** when deletes or replacements touch critical resources. This sits on top of the plan parser idea and turns analysis into a gate.

```
plan.json -> find delete/replace -> match risk rules -> exit 1 if risky
```

Problem statement

```
tf-risk [-max-risky N] [-deny-regex RE] <plan.json>
```

- Flag every delete or replace as risky by default
- Optional deny regex on address (e.g. `aws_db_instance`)
- Exit `1` if risky count > max (default 0)
- Exit `0` if clean; `2` on usage/parse errors

Acceptance criteria

- ☐ Detects delete and replace
- ☐ Prints each risky address with actions
- ☐ Exit code suitable for CI
- ☐ Deny regex optional
- ☐ Stdlib only

Setup

```
mkdir tfrisk && cd tfrisk
go mod init example.com/tfrisk
# go 1.27
```

Full main.go

```
package main

import (
    "encoding/json"
    "flag"
    "fmt"
    "os"
    "regexp"
)

type plan struct {
    ResourceChanges []struct {
        Address string `json:"address"`
        Type    string `json:"type"`
        Change  struct {
            Actions []string `json:"actions"`
        } `json:"change"`
    } `json:"resource_changes"`
}

func hasAction(actions []string, x string) bool {
    for _, a := range actions {
        if a == x {
            return true
        }
    }
    return false
}

func isReplace(actions []string) bool {
    return len(actions) == 2 && actions[0] == "delete" && actions[1] == "create"
}

func isRisky(actions []string) bool {
    return hasAction(actions, "delete") || isReplace(actions)
}

func main() {
    maxRisky := flag.Int("max-risky", 0, "allowed risky changes before fail")
    deny := flag.String("deny-regex", "", "if set, only these addresses are risky when delet
    flag.Parse()
    if flag.NArg() != 1 {
        fmt.Fprintln(os.Stderr, "usage: tf-risk [flags] <plan.json>")
        os.Exit(2)
```

```
}

var denyRE *regexp.Regexp
if *deny != "" {
    var err error
    denyRE, err = regexp.Compile(*deny)
    if err != nil {
        fmt.Fprintln(os.Stderr, "deny-regex:", err)
        os.Exit(2)
    }
}

b, err := os.ReadFile(flag.Arg(0))
if err != nil {
    fmt.Fprintln(os.Stderr, err)
    os.Exit(2)
}
var p plan
if err := json.Unmarshal(b, &p); err != nil {
    fmt.Fprintln(os.Stderr, "json:", err)
    os.Exit(2)
}

risky := 0
for _, rc := range p.ResourceChanges {
    a := rc.Change.Actions
    if !isRisky(a) {
        continue
    }
    if denyRE != nil && !denyRE.MatchString(rc.Address) && !denyRE.MatchString(rc.Type)
        // when deny-regex set: only matching resources are "policy risks"
        // still count all deletes as informational? here: skip non-matching
        fmt.Printf("INFO: %s actions=%v (not in deny-regex)\n", rc.Address, a)
        continue
    }
    risky++
    kind := "delete"
    if isReplace(a) {
        kind = "replace"
    }
    fmt.Printf("RISK(%s): %s actions=%v\n", kind, rc.Address, a)
}

fmt.Printf("risky changes=%d max=%d\n", risky, *maxRisky)
if risky > *maxRisky {
    os.Exit(1)
}
```

```
}
```

Policy modes

Mode	Flags	Behavior
Strict	default	any delete/replace fails
Budget	`-max-risky 2`	allow up to 2
Critical only	`-deny-regex 'aws_db|aws_rds'`	only matching addresses gate

Run and verification

```
# fixture from project 29
go run . -max-risky 0 plan.json; echo exit:$?

go run . -deny-regex 'aws_db_instance' plan.json
```

CI sketch:

```
- run: terraform show -json tfplan > plan.json
- run: go run ./tfrisk -max-risky 0 plan.json
```

Tests

```
package main

import "testing"

func TestIsReplace(t *testing.T) {
    if !isReplace([]string{"delete", "create"}) {
        t.Fatal()
    }
    if isReplace([]string{"delete"}) {
        t.Fatal()
    }
}

func TestIsRisky(t *testing.T) {
    if !isRisky([]string{"delete"}) || !isRisky([]string{"delete", "create"}) {
        t.Fatal()
```

```
    }
    if isRisky([]string{"create"}) {
        t.Fatal()
    }
}
```

Stretch goals

1. Severity levels (destroy production tagged resources).
2. Allowlist file of addresses that may be destroyed.
3. GitHub PR comment body markdown.
4. Combine with cost estimator (project 38).

Pitfalls

Pitfall	Fix
Treating update as risk	only delete/replace
Silent success on bad JSON	exit 2
Regex on wrong field	match address and type

Learning goals

- Enforce infrastructure policy as code
- CI exit codes for plan gates
- Reduce risky deploys with automated checks

031 Project 31: Prometheus Exporter

031 Build a Prometheus Exporter

Expose custom application metrics at `/metrics` using the official Prometheus Go client. Instrument a small HTTP handler with counter, histogram, and gauge—the three metric types you will use constantly.

```
/work  -> update metrics
/metrics -> promhttp scrape text
```

Problem statement

Build a demo service on `:9100`:

- `GET /work` simulates work and updates metrics
- `GET /metrics` Prometheus text exposition
- Metrics: `demo_requests_total`, `demo_latency_seconds`, `demo_inflight`

Acceptance criteria

- ☐ `/metrics` returns Prometheus text format
- ☐ Counter increases on `/work`
- ☐ Histogram observes latency
- ☐ Gauge tracks in-flight correctly (inc/dec with defer)
- ☐ Module builds with client_golang

Setup

```
mkdir exporter && cd exporter
go mod init example.com/exporter
go get github.com/prometheus/client_golang/prometheus@latest
go get github.com/prometheus/client_golang/prometheus/promhttp@latest
# go 1.27
go mod tidy
```

Full `main.go`

```go
package main

import (
    "log"
    "math/rand"
    "net/http"
    "time"

    "github.com/prometheus/client_golang/prometheus"
    "github.com/prometheus/client_golang/prometheus/promhttp"
)

func main() {
    reqTotal := prometheus.NewCounter(prometheus.CounterOpts{
        Name: "demo_requests_total",
        Help: "Total completed /work requests",
    })
    latency := prometheus.NewHistogram(prometheus.HistogramOpts{
        Name:    "demo_latency_seconds",
        Help:    "Request latency for /work",
        Buckets: prometheus.DefBuckets,
    })
    inflight := prometheus.NewGauge(prometheus.GaugeOpts{
        Name: "demo_inflight",
        Help: "In-flight /work requests",
    })
    prometheus.MustRegister(reqTotal, latency, inflight)

    mux := http.NewServeMux()
    mux.Handle("/metrics", promhttp.Handler())
    mux.HandleFunc("/work", func(w http.ResponseWriter, r *http.Request) {
        inflight.Inc()
        defer inflight.Dec()
        start := time.Now()
        time.Sleep(time.Duration(50+rand.Intn(200)) * time.Millisecond)
        reqTotal.Inc()
        latency.Observe(time.Since(start).Seconds())
        w.WriteHeader(http.StatusOK)
        _, _ = w.Write([]byte("ok\n"))
    })

    addr := ":9100"
    log.Println("listening", addr, " metrics=/metrics work=/work")
    log.Fatal(http.ListenAndServe(addr, mux))
```

```
}
```

Run and verification

```
go run .
# terminal B
curl -s localhost:9100/work
curl -s localhost:9100/metrics | grep demo_
```

Expected fragments:

```
demo_requests_total ...
demo_inflight ...
demo_latency_seconds_bucket ...
```

Load a bit:

```
for i in $(seq 1 20); do curl -s localhost:9100/work >/dev/null; done
curl -s localhost:9100/metrics | grep demo_requests_total
```

Step-by-step build path

1. Define metric types and Help strings.
2. `MustRegister` with default registry.
3. Instrument handler with defer for gauges.
4. Expose `promhttp.Handler()`.
5. Scrape with curl; later add Prometheus scrape config.

Label cardinality warning

```
// BAD: user_id label → unbounded series
// GOOD: low-cardinality labels: method, code, route template
```

Stretch goals

1. `CounterVec` by HTTP method/status.
2. Custom registry (not global) for tests.
3. `/healthz` separate from metrics.
4. Exemplars / native histograms (advanced).

Pitfalls

Pitfall	Fix
High-cardinality labels	static enum labels only
Forgetting gauge Dec	always defer
Blocking default mux pollution	use dedicated `ServeMux`
Measuring after sleep only	observe full handler duration

Learning goals

- Practical service metrics design
- Counter / histogram / gauge roles
- Observability by default for Go services

032 Project 32: Prometheus Probe Service

032 Build a Prometheus Probe Service

Probe a target URL on demand and export health/latency metrics—similar in spirit to blackbox exporter, but tiny and educational.

```
GET /probe?target=URL -> HTTP GET target -> set probe_up + probe_duration_seconds
GET /metrics -> scrape
```

Problem statement

Service on `:9115`:

- `/probe?target=https://example.com` performs a GET with timeout
- Updates labeled gauges `probe_up` and `probe_duration_seconds`
- Returns `up`/`down` body and HTTP 200/503
- `/metrics` for Prometheus

Acceptance criteria

- ☐ Missing target → 400
- ☐ Successful GET (2xx/3xx policy you document) → `probe_up=1`
- ☐ Error or 4xx/5xx → `probe_up=0`
- ☐ Duration always recorded when attempt finishes
- ☐ HTTP client timeout configured

Setup

```
mkdir probe && cd probe
go mod init example.com/probe
go get github.com/prometheus/client_golang/prometheus@latest
go get github.com/prometheus/client_golang/prometheus/promhttp@latest
go mod tidy
# go 1.27
```

Full `main.go`

```go
package main

import (
    "log"
    "net/http"
    "time"

    "github.com/prometheus/client_golang/prometheus"
    "github.com/prometheus/client_golang/prometheus/promhttp"
)

func main() {
    up := prometheus.NewGaugeVec(prometheus.GaugeOpts{
        Name: "probe_up",
        Help: "1 if last probe succeeded",
    }, []string{"target"})
    dur := prometheus.NewGaugeVec(prometheus.GaugeOpts{
        Name: "probe_duration_seconds",
        Help: "Duration of last probe",
    }, []string{"target"})
    prometheus.MustRegister(up, dur)

    client := &http.Client{Timeout: 2 * time.Second}

    mux := http.NewServeMux()
    mux.Handle("/metrics", promhttp.Handler())
    mux.HandleFunc("/probe", func(w http.ResponseWriter, r *http.Request) {
        target := r.URL.Query().Get("target")
        if target == "" {
            http.Error(w, "missing target", http.StatusBadRequest)
            return
        }
        start := time.Now()
        resp, err := client.Get(target)
        elapsed := time.Since(start).Seconds()
        dur.WithLabelValues(target).Set(elapsed)

        if err != nil {
            up.WithLabelValues(target).Set(0)
            w.WriteHeader(http.StatusServiceUnavailable)
            _, _ = w.Write([]byte("down\n"))
            return
        }
        defer resp.Body.Close()
```

```go
        if resp.StatusCode >= 400 {
            up.WithLabelValues(target).Set(0)
            w.WriteHeader(http.StatusServiceUnavailable)
            _, _ = w.Write([]byte("down\n"))
            return
        }
        up.WithLabelValues(target).Set(1)
        _, _ = w.Write([]byte("up\n"))
    })

    log.Println("probe service on :9115  /probe /metrics")
    log.Fatal(http.ListenAndServe(":9115", mux))
}
```

Run and verification

```bash
go run .
curl -s 'http://localhost:9115/probe?target=https://example.com'
curl -s 'http://localhost:9115/probe?target=http://127.0.0.1:1'
curl -s localhost:9115/metrics | grep probe_
```

Prometheus scrape config sketch:

```yaml
scrape_configs:
  - job_name: blackbox-lite
    metrics_path: /probe
    params:
      target: [https://example.com]
    static_configs:
      - targets: ['localhost:9115']
```

(Real blackbox uses `relabel_configs`; this lab keeps metrics on the probe process itself.)

Stretch goals

1. Histogram of probe durations instead of gauge-only.
2. Support `module=http_2xx|tcp_connect`.
3. Limit label cardinality (hash target or fixed list).
4. Concurrent probe limit with semaphore.

Pitfalls

Pitfall	Fix
Unbounded `target` labels	allowlist or drop after scrape
No timeout	hang worker
Not closing body	connection leak
Following redirects to evil URLs	custom CheckRedirect

Learning goals

- On-demand probes vs push metrics
- Labeled gauges for latest status
- Building blocks for SLO synthetic checks

033 Project 33: eBPF Tracepoint Basics

033 Build eBPF Tracepoint Basics

Capture process exec events from `sys_enter_execve` using eBPF: a minimal kernel program, userspace loader with cilium/ebpf, and a perf/ring buffer reader. **Linux lab only**; requires privileges.

```
kernel tracepoint -> eBPF program -> perf/ring buffer -> userspace Go reader
```

Problem statement

On a Linux VM/lab host:

1. Load a BPF object that attaches to `syscalls/sys_enter_execve`
2. Emit `{pid, comm}` events to userspace
3. Print events until Ctrl+C
4. Detach cleanly on shutdown

Acceptance criteria

- ☐ Documented toolchain install (clang, llvm, bpf2go or prebuilt)
- ☐ Userspace program loads, attaches, reads events
- ☐ Running `ls` / `echo` produces lines
- ☐ SIGINT stops without kernel leftover panic (best-effort detach)
- ☐ Clear notes if hardware/CI cannot run BPF

Setup

```
# Debian/Ubuntu-ish lab
sudo apt-get install -y clang llvm libbpf-dev linux-headers-$(uname -r)

mkdir ebpf-trace && cd ebpf-trace
go mod init example.com/ebpf-trace
go get github.com/cilium/ebpf@latest
go install github.com/cilium/ebpf/cmd/bpf2go@latest
# go 1.27
```

Kernel program `exec.bpf.c` (sketch)

```
//go:build ignore

#include "vmlinux.h"
#include <bpf/bpf_helpers.h>

char LICENSE[] SEC("license") = "Dual BSD/GPL";

struct event {
    __u32 pid;
    char comm[16];
};

struct {
    __uint(type, BPF_MAP_TYPE_PERF_EVENT_ARRAY);
    __uint(key_size, sizeof(__u32));
    __uint(value_size, sizeof(__u32));
} events SEC(".maps");

SEC("tracepoint/syscalls/sys_enter_execve")
int handle_exec(struct trace_event_raw_sys_enter *ctx) {
    struct event e = {};
    e.pid = bpf_get_current_pid_tgid() >> 32;
    bpf_get_current_comm(&e.comm, sizeof(e.comm));
    bpf_perf_event_output(ctx, &events, BPF_F_CURRENT_CPU, &e, sizeof(e));
    return 0;
}
```

Generate bindings (names vary by bpf2go version):

```
go generate ./...
# //go:generate bpf2go -cc clang -cflags "-O2 -g -Wall" bpf exec.bpf.c -- -I/usr/include
```

Userspace `main.go`

```
package main

import (
    "bytes"
    "encoding/binary"
    "fmt"
    "log"
    "os"
```

```
    "os/signal"
    "syscall"

    "github.com/cilium/ebpf/link"
    "github.com/cilium/ebpf/perf"
    "github.com/cilium/ebpf/rlimit"
)

type event struct {
    Pid  uint32
    Comm [16]byte
}

func main() {
    // Allow the current process to lock memory for eBPF maps.
    if err := rlimit.RemoveMemlock(); err != nil {
        log.Fatal(err)
    }

    objs := bpfObjects{}
    if err := loadBpfObjects(&objs, nil); err != nil {
        log.Fatalf("load: %v", err)
    }
    defer objs.Close()

    tp, err := link.Tracepoint("syscalls", "sys_enter_execve", objs.HandleExec, nil)
    if err != nil {
        log.Fatalf("attach: %v", err)
    }
    defer tp.Close()

    rd, err := perf.NewReader(objs.Events, os.Getpagesize())
    if err != nil {
        log.Fatal(err)
    }
    defer rd.Close()

    sig := make(chan os.Signal, 1)
    signal.Notify(sig, os.Interrupt, syscall.SIGTERM)
    go func() {
        <-sig
        _ = rd.Close()
    }()

    fmt.Println("listening for execve; Ctrl+C to stop")
    for {
        rec, err := rd.Read()
```

```go
        if err != nil {
            break
        }
        if rec.LostSamples != 0 {
            fmt.Fprintf(os.Stderr, "lost %d samples\n", rec.LostSamples)
        }
        var e event
        if err := binary.Read(bytes.NewBuffer(rec.RawSample), binary.LittleEndian, &e); err
            continue
        }
        fmt.Printf("pid=%d comm=%s\n", e.Pid, bytes.TrimRight(e.Comm[:], "\x00"))
    }
}
```

> **Note:** `bpfObjects`, `loadBpfObjects`, and field names come from `bpf2go` output. Adjust to your generated API (`objs.HandleExec` vs `objs.bpfPrograms.HandleExec`).

Step-by-step build path

1. Install kernel headers + clang.
2. Write minimal `.bpf.c` with one tracepoint.
3. `bpf2go` generate Go + embed.
4. Load, attach, read perf events.
5. Add filtering (UID, comm prefix) next.

Run and verification

```bash
go generate ./...
go build -o ebpf-trace .
sudo ./ebpf-trace
# other terminal:
ls /tmp; echo hi
# expect comm lines for shell tools
```

Pitfalls

Pitfall	Fix
macOS/Windows host	use Linux VM
Missing CAP_BPF/root	sudo or capabilities
Header/vmlinux mismatch	generate vmlinux.h via bpftool
Lost samples	larger perf buffer; reduce event rate
License section missing	BPF verifier rejects

Pitfall	Fix

Stretch goals

1. Include filename from execve args (careful helper limits).
2. Switch to ringbuf map.
3. Filter by cgroup.
4. Export Prometheus counters of execs/sec.

Learning goals

- Event-driven observability near the kernel
- Userspace loader lifecycle
- Safe iteration practices for low-level tooling

034 Project 34: eBPF XDP Packet Counter

034 Build an eBPF XDP Packet Counter

Attach an XDP program to a network interface, increment a packet counter in a BPF map, and poll the counter from userspace every second. **Linux lab only**—use a VM or spare interface.

```
NIC RX -> XDP program -> map++ -> userspace Lookup every 1s
```

Problem statement

Goals:

- Attach XDP to an interface index (or name)
- Kernel program counts packets (and optionally bytes)
- Userspace prints `packets=N` each second
- Clean detach on exit

Acceptance criteria

- ☐ Document interface selection (`ip link`)
- ☐ Counter increases under traffic (`ping`, `curl`)
- ☐ Userspace reads map without racey panics
- ☐ Defer close/detach on SIGINT
- ☐ Lab safety notes (do not brick remote prod NIC)

Setup

```
mkdir xdp-count && cd xdp-count
go mod init example.com/xdp-count
go get github.com/cilium/ebpf@latest
go install github.com/cilium/ebpf/cmd/bpf2go@latest
# clang/llvm + headers as in project 33
# go 1.27
```

Kernel sketch `count.bpf.c`

```
//go:build ignore

#include "vmlinux.h"
#include <bpf/bpf_helpers.h>

char LICENSE[] SEC("license") = "Dual BSD/GPL";

struct {
    __uint(type, BPF_MAP_TYPE_ARRAY);
    __uint(max_entries, 1);
    __type(key, __u32);
    __type(value, __u64);
} packet_count SEC(".maps");

SEC("xdp")
int xdp_count(struct xdp_md *ctx) {
    __u32 key = 0;
    __u64 *val = bpf_map_lookup_elem(&packet_count, &key);
    if (val) {
        __sync_fetch_and_add(val, 1);
    }
    return XDP_PASS; // do not drop traffic
}
```

Generate:

```
# //go:generate bpf2go -cc clang bpf count.bpf.c -- -I...
go generate ./...
```

Userspace `main.go`

```
package main

import (
    "flag"
    "fmt"
    "log"
    "net"
    "os"
    "os/signal"
    "syscall"
    "time"
```

```go
    "github.com/cilium/ebpf/link"
    "github.com/cilium/ebpf/rlimit"
)

func ifaceIndex(name string) (int, error) {
    iface, err := net.InterfaceByName(name)
    if err != nil {
        return 0, err
    }
    return iface.Index, nil
}

func main() {
    ifname := flag.String("iface", "lo", "interface name")
    flag.Parse()

    if err := rlimit.RemoveMemlock(); err != nil {
        log.Fatal(err)
    }

    idx, err := ifaceIndex(*ifname)
    if err != nil {
        log.Fatal(err)
    }

    objs := bpfObjects{}
    if err := loadBpfObjects(&objs, nil); err != nil {
        log.Fatalf("load: %v", err)
    }
    defer objs.Close()

    l, err := link.AttachXDP(link.XDPOptions{
        Program:   objs.XdpCount, // name from bpf2go
        Interface: idx,
    })
    if err != nil {
        log.Fatalf("attach xdp on %s (%d): %v", *ifname, idx, err)
    }
    defer l.Close()

    fmt.Printf("XDP attached on %s index=%d; Ctrl+C to stop\n", *ifname, idx)

    sig := make(chan os.Signal, 1)
    signal.Notify(sig, os.Interrupt, syscall.SIGTERM)

    ticker := time.NewTicker(time.Second)
```

```go
    defer ticker.Stop()

    for {
        select {
        case <-sig:
            fmt.Println("detaching")
            return
        case <-ticker.C:
            var key uint32 = 0
            var value uint64
            if err := objs.PacketCount.Lookup(&key, &value); err != nil {
                log.Println("lookup:", err)
                continue
            }
            fmt.Printf("packets=%d\n", value)
        }
    }
}
```

Field names (`XdpCount`, `PacketCount`) must match `bpf2go` output—rename to match generated code.

Run and verification

```bash
ip -br link
go generate ./... && go build -o xdp-count .
sudo ./xdp-count -iface lo
# other terminal:
ping -c 20 127.0.0.1
# counter should rise
```

Safety

- Prefer **lab VMs** and `lo` or a throwaway veth pair.
- Always return `XDP_PASS` until you know what you are dropping.
- Remote servers: a bad XDP program can cut SSH—use out-of-band console.

Stretch goals

1. Per-protocol counters (parse Ethernet/IP safely with helpers).
2. Bytes counter as well as packets.
3. `bpftool prog show` / `ip link show` verification notes.
4. Pin maps under `/sys/fs/bpf` for multi-process readers.

Pitfalls

Pitfall	Fix
Hard-coded ifindex	resolve by name
`XDP_DROP` by mistake	start with PASS
No memlock rlimit	`rlimit.RemoveMemlock`
Generated names mismatch	read `*_bpfel.go`

Learning goals

- XDP attach path vs tracepoints
- Array maps for counters
- Operational safety for datapath programs

035 Project 35: Proxmox Multi-Node Scheduler

035 Build a Proxmox Multi-Node Scheduler

Schedule VM placement across Proxmox nodes by free memory and CPU load. Produce an explainable ranking (dry-run) before you ever call migrate/create APIs.

```
fetch node stats -> score nodes -> rank placement order -> print plan
```

Problem statement

CLI that:

- Authenticates with `PVEAPIToken` (env or flags)
- `GET /api2/json/nodes` for CPU/memory stats
- Scores each node: prefer free memory + CPU headroom
- Prints ranked placement list
- Never mutates the cluster in the base project (dry-run only)

Acceptance criteria

- ☐ Fails clearly if env/flags missing
- ☐ Deterministic score function (unit-tested)
- ☐ Sorted highest score first
- ☐ HTTP client timeout + optional TLS skip for lab
- ☐ Dry-run only by default

Setup

```
mkdir pvesched && cd pvesched
go mod init example.com/pvesched
# go 1.27

export PVE_BASE_URL='https://pve.example:8006'
export PVE_TOKEN='user@pam!tokenid=secret'
```

Full `main.go`

```
package main

import (
    "crypto/tls"
    "encoding/json"
    "flag"
    "fmt"
    "net/http"
    "os"
    "sort"
    "time"
)

type NodeStat struct {
    Node   string  `json:"node"`
    CPU    float64 `json:"cpu"`
    Mem    float64 `json:"mem"`
    MaxMem float64 `json:"maxmem"`
    Status string  `json:"status"`
}

type apiResp struct {
    Data []NodeStat `json:"data"`
}

func score(n NodeStat) float64 {
    if n.MaxMem <= 0 {
        return -1
    }
    freeMemRatio := 1 - (n.Mem / n.MaxMem)
    if freeMemRatio < 0 {
        freeMemRatio = 0
    }
    cpuHeadroom := 1 - n.CPU
    if cpuHeadroom < 0 {
        cpuHeadroom = 0
    }
    // weight memory higher for VM placement heuristics
    return freeMemRatio*0.7 + cpuHeadroom*0.3
}

func fetchNodes(base, token string, insecure bool) ([]NodeStat, error) {
    tr := http.DefaultTransport.(*http.Transport).Clone()
    if insecure {
```

```
        tr.TLSClientConfig = &tls.Config{InsecureSkipVerify: true} // lab only
    }
    hc := &http.Client{Timeout: 8 * time.Second, Transport: tr}

    req, err := http.NewRequest(http.MethodGet, base+"/api2/json/nodes", nil)
    if err != nil {
        return nil, err
    }
    req.Header.Set("Authorization", "PVEAPIToken="+token)
    resp, err := hc.Do(req)
    if err != nil {
        return nil, err
    }
    defer resp.Body.Close()
    if resp.StatusCode >= 300 {
        return nil, fmt.Errorf("status %d", resp.StatusCode)
    }
    var out apiResp
    if err := json.NewDecoder(resp.Body).Decode(&out); err != nil {
        return nil, err
    }
    return out.Data, nil
}

func main() {
    base := flag.String("base", os.Getenv("PVE_BASE_URL"), "proxmox base url")
    token := flag.String("token", os.Getenv("PVE_TOKEN"), "api token")
    insecure := flag.Bool("insecure", true, "skip TLS verify (lab)")
    flag.Parse()

    if *base == "" || *token == "" {
        fmt.Fprintln(os.Stderr, "set -base/-token or PVE_BASE_URL/PVE_TOKEN")
        os.Exit(2)
    }

    nodes, err := fetchNodes(*base, *token, *insecure)
    if err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }

    sort.SliceStable(nodes, func(i, j int) bool {
        return score(nodes[i]) > score(nodes[j])
    })

    fmt.Println("placement order (dry-run):")
```

```go
    for i, n := range nodes {
        fmt.Printf("%d) node=%s score=%.3f cpu=%.2f mem=%.0f/%.0f status=%s\n",
            i+1, n.Node, score(n), n.CPU, n.Mem, n.MaxMem, n.Status)
    }
    if len(nodes) > 0 {
        fmt.Printf("recommended: %s\n", nodes[0].Node)
    }
}
```

Run and verification

```go
go run . -base "$PVE_BASE_URL" -token "$PVE_TOKEN"
```

Offline unit test without API:

```go
package main

import "testing"

func TestScorePrefersFreeMem(t *testing.T) {
    a := NodeStat{Mem: 10, MaxMem: 100, CPU: 0.5}
    b := NodeStat{Mem: 90, MaxMem: 100, CPU: 0.1}
    if score(a) <= score(b) {
        t.Fatalf("a=%f b=%f", score(a), score(b))
    }
}
```

Stretch goals

1. Exclude nodes with `status != online`.
2. Factor disk free space into score.
3. Emit JSON plan for a later apply step.
4. Actually create VM on top-ranked node (dangerous—gate with `-apply`).

Pitfalls

Pitfall	Fix
No timeout	hung CLI
Trusting lab TLS skip in prod	proper CA
Non-explainable ML scores	keep weighted formula printed
Mutating by default	dry-run first

Learning goals

- Explainable scheduling scores
- Proxmox API auth patterns
- Safe automation with dry-run defaults

036 Project 36: Proxmox Capacity Balancer

036 Build a Proxmox Capacity Balancer (Dry-Run)

Generate **migration suggestions** from overloaded nodes to healthier nodes. Start with an in-memory model you can unit-test; optionally wire Proxmox APIs later.

```
node CPU map -> find hottest + coolest -> suggest migration path
```

Problem statement

Given per-node CPU utilization (and optionally a list of VMs per node):

1. Identify source = highest CPU above threshold
2. Identify destination = lowest CPU with headroom
3. Print a dry-run migration suggestion
4. Never call migrate unless `-apply` (stretch)

Acceptance criteria

- ☐ Deterministic suggestion from fixed input
- ☐ No suggestion when cluster is balanced
- ☐ Threshold flags (`-hot`, `-cool`)
- ☐ Table output suitable for operators
- ☐ Stdlib only for base version

Setup

```
mkdir pvebalance && cd pvebalance
go mod init example.com/pvebalance
# go 1.27
```

Full `main.go`

```go
package main

import (
    "encoding/json"
    "flag"
    "fmt"
    "os"
    "sort"
)

type Node struct {
    Name string  `json:"name"`
    CPU  float64 `json:"cpu"` // 0..1
}

type Suggestion struct {
    From   string  `json:"from"`
    To     string  `json:"to"`
    FromCPU float64 `json:"from_cpu"`
    ToCPU   float64 `json:"to_cpu"`
    Reason string  `json:"reason"`
}

func suggest(nodes []Node, hot, cool float64) *Suggestion {
    if len(nodes) < 2 {
        return nil
    }
    src := nodes[0]
    dst := nodes[0]
    for _, n := range nodes[1:] {
        if n.CPU > src.CPU {
            src = n
        }
        if n.CPU < dst.CPU {
            dst = n
        }
    }
    if src.Name == dst.Name {
        return nil
    }
    if src.CPU < hot {
        return nil // nothing hot enough
    }
    if dst.CPU > cool {
```

```go
        return nil // no cool target
    }
    return &Suggestion{
        From: src.Name, To: dst.Name,
        FromCPU: src.CPU, ToCPU: dst.CPU,
        Reason: fmt.Sprintf("src cpu %.2f >= hot %.2f; dst cpu %.2f <= cool %.2f",
            src.CPU, hot, dst.CPU, cool),
    }
}

func main() {
    hot := flag.Float64("hot", 0.85, "source CPU threshold")
    cool := flag.Float64("cool", 0.50, "destination max CPU")
    demo := flag.Bool("demo", true, "use built-in demo nodes")
    flag.Parse()

    var nodes []Node
    if *demo {
        nodes = []Node{
            {"pve1", 0.92},
            {"pve2", 0.35},
            {"pve3", 0.41},
        }
    } else {
        // optional: read JSON array from stdin
        if err := json.NewDecoder(os.Stdin).Decode(&nodes); err != nil {
            fmt.Fprintln(os.Stderr, "stdin json:", err)
            os.Exit(2)
        }
    }

    sort.Slice(nodes, func(i, j int) bool { return nodes[i].Name < nodes[j].Name })
    fmt.Println("cluster:")
    for _, n := range nodes {
        fmt.Printf("  %-8s cpu=%.2f\n", n.Name, n.CPU)
    }

    s := suggest(nodes, *hot, *cool)
    if s == nil {
        fmt.Println("suggest: none (balanced or thresholds not met)")
        return
    }
    fmt.Printf("suggest migration: %s -> %s (cpu %.2f -> %.2f)\n", s.From, s.To, s.FromCPU,
    fmt.Println("reason:", s.Reason)
    fmt.Println("mode: dry-run (no API calls)")
}
```

Run and verification

```
go run .
# suggest migration: pve1 -> pve2 ...

go run . -hot 0.99
# none

echo '[{"name":"a","cpu":0.9},{"name":"b","cpu":0.2}]' | go run . -demo=false
```

Tests

```go
package main

import "testing"

func TestSuggest(t *testing.T) {
    nodes := []Node{{"pve1", 0.92}, {"pve2", 0.35}, {"pve3", 0.41}}
    s := suggest(nodes, 0.85, 0.5)
    if s == nil || s.From != "pve1" || s.To != "pve2" {
        t.Fatalf("%+v", s)
    }
}

func TestNoSuggestWhenCool(t *testing.T) {
    nodes := []Node{{"a", 0.9}, {"b", 0.8}}
    if s := suggest(nodes, 0.85, 0.5); s != nil {
        t.Fatal(s)
    }
}
```

```
go test ./...
```

Stretch goals

1. Pick a specific VM on the hot node (largest CPU VM).
2. Fetch live stats from Proxmox API (project 35 client).
3. Simulate after-move CPU estimates.
4. `-apply` with confirmation prompt.

Pitfalls

Pitfall	Fix
Migrating without headroom check	cool threshold
Oscillation (ping-pong)	hysteresis / cooldown timer
Ignoring maintenance nodes	status filter

Learning goals

- Capacity balancing as scored suggestions
- Dry-run first automation
- Testable pure functions for ops tools

037 Project 37: Kubernetes HPA Simulator

037 Build a Kubernetes HPA Simulator

Simulate Horizontal Pod Autoscaler-style replica scaling from a CPU load time series. Learn the control loop (target utilization, min/max replicas, scale-up/down rules) without a cluster.

```
cpu samples -> compare to target -> adjust replicas within [min,max] -> log
```

Problem statement

Model a simplified HPA:

- Target average CPU percent (e.g. 60%)
- Current replicas
- Each tick: observe CPU, scale up if sustained high, scale down if sustained low
- Enforce min/max replicas
- Optional stabilization: require N consecutive ticks before scale down

Acceptance criteria

- ☐ Replicas never leave [min, max]
- ☐ High CPU increases replicas (until max)
- ☐ Low CPU decreases replicas (until min) with optional hysteresis
- ☐ Deterministic mode with fixed seed or fixed series
- ☐ Unit tests for the pure step function

Setup

```
mkdir hpasim && cd hpasim
go mod init example.com/hpasim
# go 1.27
```

Full `main.go`

```go
package main

import (
    "flag"
    "fmt"
    "math/rand"
    "os"
)

type Config struct {
    TargetCPU   float64
    MinReplicas int
    MaxReplicas int
    UpBand      float64 // scale up if cpu > target+UpBand
    DownBand    float64 // scale down if cpu < target-DownBand
    DownStable  int     // consecutive low ticks required
}

type State struct {
    Replicas int
    LowStreak int
}

func step(cfg Config, s State, cpu float64) (State, string) {
    if s.Replicas < cfg.MinReplicas {
        s.Replicas = cfg.MinReplicas
    }
    action := "noop"

    if cpu > cfg.TargetCPU+cfg.UpBand && s.Replicas < cfg.MaxReplicas {
        s.Replicas++
        s.LowStreak = 0
        return s, "scale_up"
    }
    if cpu < cfg.TargetCPU-cfg.DownBand {
        s.LowStreak++
        if s.LowStreak >= cfg.DownStable && s.Replicas > cfg.MinReplicas {
            s.Replicas--
            s.LowStreak = 0
            return s, "scale_down"
        }
        return s, "hold_low"
    }
    s.LowStreak = 0
```

```go
    return s, action
}

func main() {
    ticks := flag.Int("ticks", 30, "simulation ticks")
    target := flag.Float64("target", 60, "target CPU percent")
    minR := flag.Int("min", 1, "min replicas")
    maxR := flag.Int("max", 20, "max replicas")
    seed := flag.Int64("seed", 42, "rng seed")
    demo := flag.Bool("demo-series", false, "use fixed series instead of random")
    flag.Parse()

    if *minR < 1 || *maxR < *minR {
        fmt.Fprintln(os.Stderr, "invalid min/max")
        os.Exit(2)
    }

    cfg := Config{
        TargetCPU:   *target,
        MinReplicas: *minR,
        MaxReplicas: *maxR,
        UpBand:      10,
        DownBand:    20,
        DownStable:  2,
    }
    s := State{Replicas: *minR}
    rng := rand.New(rand.NewSource(*seed))

    fixed := []float64{70, 82, 90, 88, 91, 72, 55, 40, 38, 35, 60, 65}

    for t := 1; t <= *ticks; t++ {
        var cpu float64
        if *demo {
            cpu = fixed[(t-1)%len(fixed)]
        } else {
            cpu = 30 + rng.Float64()*70
        }
        var action string
        s, action = step(cfg, s, cpu)
        fmt.Printf("tick=%02d cpu=%5.1f%% replicas=%d action=%s\n", t, cpu, s.Replicas, acti
    }
}
```

Run and verification

```
go run . -demo-series -ticks 12
go run . -seed 1 -ticks 30
```

Tests

```go
package main

import "testing"

func TestScaleUp(t *testing.T) {
    cfg := Config{TargetCPU: 60, MinReplicas: 1, MaxReplicas: 5, UpBand: 10, DownBand: 20, D
    s := State{Replicas: 1}
    s, a := step(cfg, s, 90)
    if a != "scale_up" || s.Replicas != 2 {
        t.Fatalf("%s %d", a, s.Replicas)
    }
}

func TestRespectMax(t *testing.T) {
    cfg := Config{TargetCPU: 60, MinReplicas: 1, MaxReplicas: 2, UpBand: 0, DownBand: 20, Do
    s := State{Replicas: 2}
    s, a := step(cfg, s, 99)
    if a != "noop" || s.Replicas != 2 {
        t.Fatalf("%s %d", a, s.Replicas)
    }
}

func TestScaleDownNeedsStreak(t *testing.T) {
    cfg := Config{TargetCPU: 60, MinReplicas: 1, MaxReplicas: 10, UpBand: 10, DownBand: 20,
    s := State{Replicas: 3}
    s, _ = step(cfg, s, 20)
    if s.Replicas != 3 {
        t.Fatal("scaled too early")
    }
    s, a := step(cfg, s, 20)
    if a != "scale_down" || s.Replicas != 2 {
        t.Fatalf("%s %d", a, s.Replicas)
    }
}
```

```
go test ./...
```

Stretch goals

1. DesiredReplicas formula: `ceil(current * cpu/target)` like real HPA.
2. Separate scale-up/scale-down cooldown timers.
3. Multi-metric (CPU + RPS).
4. Compare against recorded metrics CSV.

Pitfalls

Pitfall	Fix
Flapping replicas	hysteresis bands + down stable streak
Ignoring min/max	clamp every step
Random-only demos	fixed series for docs/tests

Learning goals

- HPA control-loop intuition
- Stabilization windows
- Pure step functions for simulations

038 Project 38: Terraform Cost Estimator

038 Build a Terraform Cost Estimator (Simple)

Estimate monthly cost **delta** from a Terraform plan JSON by mapping resource types to rough unit prices. Not a substitute for Infracost—but excellent for learning plan analysis and CI annotations.

```
plan.json -> actions × unit price table -> monthly delta $
```

Problem statement

```
tf-cost [-prices prices.json] <plan.json>
```

- `create` adds unit monthly cost
- `delete` subtracts unit monthly cost
- `replace` treat as delete+create (net depends on type change; base: same type → ~0)
- Unknown types contribute 0 (list them as warnings)

Acceptance criteria

- ☐ Parses plan JSON `resource_changes`
- ☐ Prints estimated monthly delta
- ☐ Warns on unknown types that change
- ☐ Exit 2 on parse errors; 0 on success
- ☐ Prices overridable via JSON file

Setup

```
mkdir tfcost && cd tfcost
go mod init example.com/tfcost
# go 1.27
```

`prices.json` example:

```
{
  "aws_instance": 15.0,
  "aws_db_instance": 120.0,
  "aws_lb": 25.0
```

```
}
```

Full `main.go`

```
package main

import (
    "encoding/json"
    "flag"
    "fmt"
    "os"
)

type Plan struct {
    ResourceChanges []struct {
        Address string `json:"address"`
        Type    string `json:"type"`
        Change  struct {
            Actions []string `json:"actions"`
        } `json:"change"`
    } `json:"resource_changes"`
}

func loadPrices(path string) (map[string]float64, error) {
    defaults := map[string]float64{
        "aws_instance":    15.0,
        "aws_db_instance": 120.0,
        "aws_lb":          25.0,
    }
    if path == "" {
        return defaults, nil
    }
    b, err := os.ReadFile(path)
    if err != nil {
        return nil, err
    }
    var m map[string]float64
    if err := json.Unmarshal(b, &m); err != nil {
        return nil, err
    }
    for k, v := range defaults {
        if _, ok := m[k]; !ok {
            m[k] = v
        }
    }
```

```
    return m, nil
}

func estimate(p Plan, cost map[string]float64) (delta float64, unknown []string) {
    seenUnknown := map[string]bool{}
    for _, rc := range p.ResourceChanges {
        unit, ok := cost[rc.Type]
        if !ok {
            if !seenUnknown[rc.Type] && len(rc.Change.Actions) > 0 && rc.Change.Actions[0] !
                // track types that actually change
                for _, a := range rc.Change.Actions {
                    if a == "create" || a == "delete" {
                        seenUnknown[rc.Type] = true
                        unknown = append(unknown, rc.Type)
                        break
                    }
                }
            }
            continue
        }
        actions := rc.Change.Actions
        if len(actions) == 2 && actions[0] == "delete" && actions[1] == "create" {
            // same type replace → approx net 0 for this toy model
            continue
        }
        for _, a := range actions {
            switch a {
            case "create":
                delta += unit
                fmt.Printf("+ $%.2f  %s (%s)\n", unit, rc.Address, rc.Type)
            case "delete":
                delta -= unit
                fmt.Printf("- $%.2f  %s (%s)\n", unit, rc.Address, rc.Type)
            }
        }
    }
    return delta, unknown
}

func main() {
    pricesPath := flag.String("prices", "", "optional prices JSON")
    flag.Parse()
    if flag.NArg() != 1 {
        fmt.Fprintln(os.Stderr, "usage: tf-cost [-prices prices.json] <plan.json>")
        os.Exit(2)
    }
```

```
    cost, err := loadPrices(*pricesPath)
    if err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(2)
    }
    b, err := os.ReadFile(flag.Arg(0))
    if err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(2)
    }
    var p Plan
    if err := json.Unmarshal(b, &p); err != nil {
        fmt.Fprintln(os.Stderr, "json:", err)
        os.Exit(2)
    }

    delta, unknown := estimate(p, cost)
    for _, u := range unknown {
        fmt.Fprintf(os.Stderr, "warn: no price for type %s\n", u)
    }
    fmt.Printf("estimated monthly delta: $%.2f\n", delta)
}
```

Run and verification

```
go run . plan.json
go run . -prices prices.json plan.json
```

Tests

```
package main

import "testing"

func TestEstimateCreate(t *testing.T) {
    var p Plan
    rc := p.ResourceChanges
    _ = rc
    item := struct {
        Address string `json:"address"`
        Type    string `json:"type"`
        Change  struct {
            Actions []string `json:"actions"`
```

```
        } `json:"change"`
    }{Address: "aws_instance.web", Type: "aws_instance"}
    item.Change.Actions = []string{"create"}
    p.ResourceChanges = append(p.ResourceChanges, item)
    delta, _ := estimate(p, map[string]float64{"aws_instance": 15})
    if delta != 15 {
        t.Fatalf("%v", delta)
    }
}
```

Stretch goals

1. Regional price multipliers.
2. Count module expansion addresses.
3. Fail CI if delta > budget (`-max-delta 50`).
4. Integrate with risk reporter output.

Pitfalls

Pitfall	Fix
Treating prices as accurate	label as rough estimate
Ignoring replace	define policy (net 0 vs full)
Double-counting	classify actions carefully

Learning goals

- Plan-driven financial signals
- Extensible price tables
- Honest limits of toy cost models

039 Project 39: Prometheus Alert Rule Tester

039 Build a Prometheus Alert Rule Tester

Evaluate synthetic metric streams against threshold rules with a **for**-style duration (consecutive steps). This teaches alert flapping, hysteresis, and unit-testing rules without a live Prometheus.

```
values[] + threshold + forSteps -> streak -> alert bool per tick
```

Problem statement

Given:

- A series of float samples (CPU%, error rate, …)
- `threshold` and `forSteps` (like `for: 3m` if step=1m)
- Emit per-tick: value, streak, alert firing?

Also support loading samples from a CSV file.

Acceptance criteria

- ☐ Alert fires only after `forSteps` consecutive breaches
- ☐ Streak resets when value drops to/ below threshold
- ☐ Pure function is unit-tested
- ☐ CLI demo with built-in series
- ☐ Stdlib only

Setup

```
mkdir alertrule && cd alertrule
go mod init example.com/alertrule
# go 1.27
```

Full `main.go`

```go
package main

import (
    "encoding/csv"
    "flag"
    "fmt"
    "io"
    "os"
    "strconv"
)

// shouldAlert updates streak based on value and returns whether alert is firing.
// Pass streak by pointer so callers keep state across ticks.
func shouldAlert(v, threshold float64, forSteps int, streak *int) bool {
    if v > threshold {
        *streak++
    } else {
        *streak = 0
    }
    return *streak >= forSteps
}

func evalSeries(vals []float64, threshold float64, forSteps int) []bool {
    out := make([]bool, len(vals))
    streak := 0
    for i, v := range vals {
        out[i] = shouldAlert(v, threshold, forSteps, &streak)
    }
    return out
}

func loadCSV(path string) ([]float64, error) {
    f, err := os.Open(path)
    if err != nil {
        return nil, err
    }
    defer f.Close()
    r := csv.NewReader(f)
    var vals []float64
    for {
        rec, err := r.Read()
        if err == io.EOF {
            break
        }
```

```go
        if err != nil {
            return nil, err
        }
        if len(rec) == 0 {
            continue
        }
        v, err := strconv.ParseFloat(rec[0], 64)
        if err != nil {
            continue // skip header/bad
        }
        vals = append(vals, v)
    }
    return vals, nil
}

func main() {
    threshold := flag.Float64("threshold", 85, "alert if value > threshold")
    forSteps := flag.Int("for", 3, "consecutive steps above threshold")
    csvPath := flag.String("csv", "", "optional CSV of values (first column)")
    flag.Parse()

    vals := []float64{70, 82, 90, 88, 91, 72, 95, 96, 97}
    if *csvPath != "" {
        var err error
        vals, err = loadCSV(*csvPath)
        if err != nil {
            fmt.Fprintln(os.Stderr, err)
            os.Exit(1)
        }
    }

    streak := 0
    fmt.Printf("threshold=%.1f for=%d steps\n", *threshold, *forSteps)
    for i, v := range vals {
        alert := shouldAlert(v, *threshold, *forSteps, &streak)
        fmt.Printf("t=%d val=%5.1f streak=%d alert=%v\n", i, v, streak, alert)
    }
}
```

Run and verification

```go
go run .
# t=0..  alert becomes true once streak hits 3

# CSV:
```

```
printf '10\n90\n91\n92\n50\n' > s.csv
go run . -csv s.csv -threshold 85 -for 3
```

Tests

```go
package main

import "testing"

func TestForDuration(t *testing.T) {
    vals := []float64{90, 90, 90, 10}
    got := evalSeries(vals, 85, 3)
    want := []bool{false, false, true, false}
    for i := range want {
        if got[i] != want[i] {
            t.Fatalf("i=%d got %v want %v", i, got, want)
        }
    }
}

func TestReset(t *testing.T) {
    vals := []float64{90, 10, 90, 90}
    got := evalSeries(vals, 85, 2)
    if got[1] || got[2] || !got[3] {
        t.Fatalf("%v", got)
    }
}
```

```
go test ./...
```

Stretch goals

1. Dual thresholds (warn vs critical).
2. `avg_over_time` style window instead of consecutive.
3. Parse a tiny subset of Prometheus rule YAML.
4. Generate a markdown alert timeline for PR review.

Pitfalls

Pitfall	Fix
Firing on first sample	enforce forSteps

Pitfall	Fix
Off-by-one streak	unit tests with tables
Flapping	longer for + cool-down (stretch)

Learning goals

- Alert `for` duration semantics
- Deterministic rule testing
- Avoiding noisy pages with streaks

040 Project 40: Infra Reconciler Daemon

040 Build an Infra Reconciler Daemon

Model a control-loop daemon that continuously moves **actual** state toward **desired** state. This is the same mental model as Kubernetes controllers and GitOps agents, stripped to a few dozen lines you can reason about.

```
desired config -> diff -> apply one action -> observe -> repeat
```

Problem statement

Daemon-style process:

- Holds `Desired` and `Actual` integers (stand-in for replica counts, firewall rules, …)
- Every tick, apply **at most one** idempotent action
- Desired can change mid-run (config reload simulation)
- Actual can drift (external change)
- Logs every action for auditability
- Stops after max ticks or SIGINT

Acceptance criteria

- ☐ `diff` and `apply` are pure/testable
- ☐ Converges when desired stable
- ☐ Survives desired changes and drift
- ☐ Graceful shutdown on signal
- ☐ Stdlib only

Setup

```
mkdir reconcilerd && cd reconcilerd
go mod init example.com/reconcilerd
# go 1.27
```

Full `main.go`

```go
package main

import (
    "context"
    "flag"
    "fmt"
    "math/rand"
    "os"
    "os/signal"
    "syscall"
    "time"
)

type state struct {
    Desired int
    Actual  int
}

func diff(s state) int { return s.Desired - s.Actual }

func apply(s *state) string {
    d := diff(*s)
    switch {
    case d > 0:
        s.Actual++
        return "scale_up"
    case d < 0:
        s.Actual--
        return "scale_down"
    default:
        return "noop"
    }
}

func main() {
    desired := flag.Int("desired", 5, "initial desired")
    actual := flag.Int("actual", 1, "initial actual")
    ticks := flag.Int("ticks", 40, "max ticks (0=until signal)")
    interval := flag.Duration("interval", 200*time.Millisecond, "tick interval")
    seed := flag.Int64("seed", 1, "drift rng seed")
    driftProb := flag.Float64("drift", 0.1, "per-tick drift probability")
    flag.Parse()

    ctx, stop := signal.NotifyContext(context.Background(), os.Interrupt, syscall.SIGTERM)
```

```
    defer stop()

    s := state{Desired: *desired, Actual: *actual}
    rng := rand.New(rand.NewSource(*seed))
    ticker := time.NewTicker(*interval)
    defer ticker.Stop()

    fmt.Printf("reconciler start desired=%d actual=%d\n", s.Desired, s.Actual)

    for i := 0; *ticks == 0 || i < *ticks; i++ {
        select {
        case <-ctx.Done():
            fmt.Println("shutdown")
            return
        case <-ticker.C:
        }

        // simulate config reload
        if i == 10 {
            s.Desired = 3
            fmt.Printf("tick=%02d CONFIG desired=%d\n", i, s.Desired)
        }

        // simulate external drift
        if rng.Float64() < *driftProb {
            delta := rng.Intn(3) - 1
            s.Actual += delta
            if s.Actual < 0 {
                s.Actual = 0
            }
            fmt.Printf("tick=%02d DRIFT delta=%+d actual=%d\n", i, delta, s.Actual)
        }

        action := apply(&s)
        fmt.Printf("tick=%02d desired=%d actual=%d action=%s diff=%d\n",
            i, s.Desired, s.Actual, action, diff(s))
    }
    fmt.Println("max ticks reached")
}
```

Architecture

```
           watch/load
  desired.yaml              reconciler
                             loop

                                          apply


                                   actual world
```

Run and verification

```
go run . -desired 5 -actual 0 -ticks 25 -drift 0.15 -interval 50ms
# expect scale_up to 5, config change to 3, scale_down, occasional drift recovery
```

Tests

```
package main

import "testing"

func TestApplyConverges(t *testing.T) {
    s := state{Desired: 3, Actual: 0}
    for i := 0; i < 10; i++ {
        apply(&s)
    }
    if s.Actual != 3 || diff(s) != 0 {
        t.Fatalf("%+v", s)
    }
}

func TestNoop(t *testing.T) {
    s := state{Desired: 2, Actual: 2}
    if apply(&s) != "noop" {
        t.Fatal()
    }
}
```

```
go test -race ./...
```

Stretch goals

1. Load desired from JSON file; reload on `SIGHUP`.
2. Workqueue of resource IDs.
3. Per-resource last-error backoff.
4. Metrics: reconciles_total, drift_total.
5. Multi-field state (not just int).

Pitfalls

Pitfall	Fix
Applying huge diffs in one tick	one step; requeue
No drift handling	re-observe actual every tick
Busy loop	ticker / event driven
Non-idempotent apply	same desired → safe noop

Learning goals

- Level-triggered reconciliation
- Drift and config change handling
- Daemon lifecycle with signals

041 Project 41: TinyGo Blink & Toolchain

041 TinyGo: Toolchain, Targets, and Blink

Build your first **TinyGo** firmware loop: install the toolchain, pick a board target, flash (or run in the WASM/simulator path), and understand how TinyGo differs from desktop Go.

Baseline: TinyGo current stable (track tinygo.org release notes); Go 1.22+ host toolchain recommended for building TinyGo itself if you compile from source.

Why TinyGo

Desktop Go	TinyGo
GC, large runtime, OS threads	Smaller runtime, selective GC / freelist modes by target
`net/http`, rich stdlib	Subset of stdlib + `machine` / board packages
`go build` → OS binary	`tinygo build -target=...` → firmware / WASM
Soft realtime only	Soft realtime loops + hardware interrupts (not hard RTOS guarantees)

TinyGo is still **Go syntax and modules**, but the **target** decides available packages and size.

```
Host (dev machine)                    Device / WASM

tinygo build -target=pico   flash   MCU firmware
tinygo run -target=wasip1        wasmtime / browser
```

Step 1: Install TinyGo

```
# macOS (Homebrew example - verify current formula)
brew install tinygo

# or follow https://tinygo.org/getting-started/install/

tinygo version
go version   # host Go still useful for module tools
```

List targets:

```
tinygo targets | head
tinygo info pico          # Raspberry Pi Pico example
tinygo info wasip1        # WASM for host experiments without hardware
```

Step 2: Module + blink (board-agnostic pattern)

```
mkdir tiny-blink && cd tiny-blink
go mod init example.com/tiny-blink
```

`main.go` — **replace pin with your board's LED** (see board docs):

```go
package main

import (
    "machine"
    "time"
)

func main() {
    led := machine.LED // many boards define LED; else machine.GPIO25 etc.
    led.Configure(machine.PinConfig{Mode: machine.PinOutput})

    for {
        led.High()
        time.Sleep(500 * time.Millisecond)
        led.Low()
        time.Sleep(500 * time.Millisecond)
    }
}
```

No hardware? Use WASM smoke build

```
# Compile-only check that TinyGo accepts the package shape
# (blink needs machine.LED - for pure host, use the wasip1 project below)
tinygo build -o /tmp/check.elf -target=pico .
```

Host-only hello (no GPIO):

```go
// main_wasip1.go - build with -target=wasip1
package main
```

```go
import "fmt"

func main() {
    fmt.Println("tinygo wasip1 ok")
}
```

```
tinygo run -target=wasip1 .
# or: tinygo build -o hello.wasm -target=wasip1 . && wasmtime hello.wasm
```

Step 3: Flash (hardware)

Examples (exact flags change by board/probe):

```
# Pico UF2 style (common workflow)
tinygo flash -target=pico .

# Some boards use -port
# tinygo flash -target=arduino-nano33 -port=/dev/ttyACM0 .
```

If flash fails: check udev rules (Linux), cable is data-capable, bootloader mode (Pico BOOTSEL).

Step 4: Size and optimization awareness

```
tinygo build -o firmware.elf -target=pico -size=full .
```

Note code/data/BSS. Optimization flags (`-opt=z` etc.) matter on small MCUs—read current TinyGo docs for supported `-opt` values.

Learning Goals

- Install and verify TinyGo; list targets
- Separate **host Go** vs **device build**
- Produce a continuous loop firmware or WASM hello
- Know where to look for pin definitions (`machine`, board packages)

Extensions

1. Blink two LEDs out of phase.
2. Print `time.Now()` over USB serial if the target supports it (`machine.Serial`).
3. Compare binary size with `-opt` variants.

Common gotchas

Issue	Fix
`machine.LED` undefined	Wrong target or board; set pin explicitly
Flash permission denied	udev / cable / bootloader mode
Import works in `go build` but not TinyGo	Package not in TinyGo stdlib subset
Expect hard realtime	Soft RT only; measure jitter on real hardware

Checkpoint

- ☐ `tinygo version` works
- ☐ Built for at least one target
- ☐ Blink **or** wasip1 hello runs
- ☐ Notes: board name + pin used

042 Project 42: TinyGo Button + Interrupt

042 TinyGo: Button Input and Interrupt-Driven Events

Move from a pure `sleep` loop to **event-driven** MCU code: poll a button, then (where the target supports it) use a pin interrupt so the main loop stays free for other work.

Realtime framing

Style	Latency	CPU use	Typical use
Polling in `for { }`	Bounded by loop rate	Busy or delayed by sleeps	Simple labs
Interrupt on edge	Faster reaction	Idle-friendly	Buttons, encoders, IRQs
Timer IRQ / PWM	Periodic deadlines	Predictable ticks	Soft realtime control

TinyGo is **not** a certified hard-realtime OS. Treat deadlines as **best-effort**: measure, keep ISRs short, defer heavy work to the main loop.

```
Edge on BUTTON pin


  IRQ handler   set flag / push event   main loop reacts (debounce, LED, UART)
```

Step 1: Polling baseline (all targets)

```go
package main

import (
    "machine"
    "time"
)

func main() {
    led := machine.LED
    led.Configure(machine.PinConfig{Mode: machine.PinOutput})
```

```
    btn := machine.BUTTON // or explicit GPIO; active-low common with pull-up
    btn.Configure(machine.PinConfig{Mode: machine.PinInputPullup})

    var last bool = true // pull-up idle high
    for {
        v := btn.Get()
        if v != last {
            // simple debounce
            time.Sleep(20 * time.Millisecond)
            if btn.Get() == v {
                last = v
                if !v { // pressed (active low)
                    led.Set(!led.Get())
                }
            }
        }
        time.Sleep(5 * time.Millisecond)
    }
}
```

```
tinygo flash -target=YOUR_BOARD .
```

Step 2: Interrupt pattern (when supported)

API names vary by TinyGo version/board. Sketch:

```
package main

import (
    "machine"
    "time"
)

var pressed = make(chan struct{}, 1)

func main() {
    led := machine.LED
    led.Configure(machine.PinConfig{Mode: machine.PinOutput})

    btn := machine.BUTTON
    btn.Configure(machine.PinConfig{Mode: machine.PinInputPullup})

    // SetInterrupt API differs by target - consult tinygo.org machine docs for your board.
    // Conceptual:
```

```
    // btn.SetInterrupt(machine.PinFalling, func(p machine.Pin) {
    //  select {
    //  case pressed <- struct{}{}:
    //  default:
    //  }
    // })

    // Fallback if interrupt API unavailable: keep polling in a tight loop
    // and only document the interrupt path for boards that support it.

    for {
        select {
        case <-pressed:
            led.Set(!led.Get())
            time.Sleep(50 * time.Millisecond) // debounce after event
        default:
            // optional idle work / low-power sleep if target supports
            time.Sleep(1 * time.Millisecond)
        }
    }
}
```

Lab requirement: Implement **working** code for **your** board. If interrupts are unsupported, complete polling + debounce and write a short note: *"Interrupt API N/A on target X; measured poll latency …."*

Step 3: Debounce theory

Mechanical buttons bounce for 5–50 ms. Strategies:

1. **Ignore window** after edge (software).
2. **Require stable samples** N times.
3. **Hardware RC** (optional).

Never toggle LEDs raw in an ISR without debounce policy—you will get multi-fires.

Step 4: Measure soft-realtime behavior

Metric	How
Reaction latency	Scope or toggle second pin; or timestamp over UART
Jitter	Stddev of latency over 50 presses

Metric	How
Missed events	Channel full / default branch count

Log 50 presses; record min/avg/max latency if you have a serial console.

Learning Goals

- Poll vs interrupt trade-offs
- Debounce as a correctness requirement
- Keep interrupt work minimal
- Honest soft-realtime expectations

Extensions

1. Long-press vs short-press detection.
2. Two buttons with different actions.
3. Disable interrupts during critical section if API allows.

Checkpoint

☐ Reliable press detection (no multi-toggle spam)

☐ Debounce implemented and explained

☐ Interrupt **or** documented fallback

☐ 50-press notes (even qualitative)

043 Project 43: TinyGo UART Sensor Loop

043 TinyGo: UART Console + Timed Sensor Loop

Build a **periodic control loop**: sample a sensor (or simulated ADC), emit CSV over serial, and keep loop timing steady enough for soft realtime logging.

Architecture

```
  tick every T ms


  read sensor / mock    filter (EMA)    UART CSV line

           deadline miss counter++
```

Step 1: Serial hello

```go
package main

import (
    "fmt"
    "machine"
    "time"
)

func main() {
    machine.Serial.Configure(machine.UARTConfig{BaudRate: 115200})
    for {
        fmt.Fprintf(machine.Serial, "hello tinygo %d\n", time.Now().UnixNano())
        time.Sleep(time.Second)
    }
}
```

Host:

```
# Linux example - adjust device
screen /dev/ttyACM0 115200
```

```
# or: picocom -b 115200 /dev/ttyACM0
```

Step 2: Fixed-rate loop with miss detection

```
package main

import (
    "fmt"
    "machine"
    "time"
)

const period = 50 * time.Millisecond // 20 Hz

func readMockSensor() int {
    // Replace with machine.ADC when available on your board
    return int(time.Now().UnixNano()%1000) / 10 // 0..99 fake
}

func main() {
    machine.Serial.Configure(machine.UARTConfig{BaudRate: 115200})

    var ema float32
    var misses int
    next := time.Now()

    for {
        start := time.Now()
        if start.After(next.Add(period / 2)) {
            misses++
        }

        raw := readMockSensor()
        // EMA: y =  x + (1- )y
        const alpha = 0.2
        ema = alpha*float32(raw) + (1-alpha)*ema

        fmt.Fprintf(machine.Serial, "t_ns=%d raw=%d ema=%.2f misses=%d\n",
            start.UnixNano(), raw, ema, misses)

        next = next.Add(period)
        if d := time.Until(next); d > 0 {
            time.Sleep(d)
        } else {
```

```
            // fell behind - resync
            next = time.Now().Add(period)
            misses++
        }
    }
}
```

Step 3: Real ADC (when target supports it)

Concept:

```
import "machine"

var adc machine.ADC

func initADC() {
    adc = machine.ADC{Pin: machine.ADC0} // board-specific
    adc.Configure(machine.ADCConfig{})
}

func readMockSensor() int {
    return int(adc.Get()) // width depends on target
}
```

Consult your board's pinout. If no ADC: keep mock and still complete timing/miss metrics.

Step 4: Soft realtime checklist

Practice	Why
Fixed `period`	Stable sample rate for filters
Count deadline misses	Visibility when loop overruns
Avoid allocs in hot loop	GC/heap pressure on tiny heaps
Short UART lines	Don't block forever on full buffers

Optional: build with size report and note RAM pressure.

Learning Goals

- UART as primary debug channel
- Timed loops and miss accounting

- Simple digital filter (EMA)
- Path to real ADC sensors

Extensions

1. Change rate via serial command (`r 100` → 100 ms).
2. Threshold alarm: print `ALERT` when ema exceeds N.
3. Dual-rate: fast sample, slow print.

Checkpoint

- ☐ Serial output visible on host
- ☐ Fixed-rate loop with miss counter
- ☐ EMA or similar filter
- ☐ Notes: period chosen and why

044 Project 44: TinyGo Cooperative Tasks (Soft RT)

044 TinyGo: Cooperative Multi-Tasking Without a Full RTOS

Implement a **tiny cooperative scheduler**: multiple tasks share one core by yielding, with per-task periods—useful mental model for soft realtime systems and for understanding why preemptive RTOS exists.

Theory (short)

Model	Description
Cooperative	Task runs until it yields; bad task starves others
Preemptive RTOS	Timer IRQ switches tasks; needs careful shared data
This lab	Cooperative + fixed time slices via `time` checks

```
loop forever:
  for each task:
    if now >= task.next:
       task.run()
       task.next += task.period
```

Step 1: Task table

```go
package main

import (
    "fmt"
    "machine"
    "time"
)

type Task struct {
    Name   string
    Period time.Duration
    Next   time.Time
    Run    func()
}
```

```
func main() {
    machine.Serial.Configure(machine.UARTConfig{BaudRate: 115200})
    led := machine.LED
    led.Configure(machine.PinConfig{Mode: machine.PinOutput})

    var blinkOn bool
    var ticks int

    tasks := []Task{
        {
            Name:   "blink",
            Period: 500 * time.Millisecond,
            Run: func() {
                blinkOn = !blinkOn
                led.Set(blinkOn)
            },
        },
        {
            Name:   "heartbeat",
            Period: 1 * time.Second,
            Run: func() {
                ticks++
                fmt.Fprintf(machine.Serial, "hb ticks=%d\n", ticks)
            },
        },
        {
            Name:   "sensor",
            Period: 100 * time.Millisecond,
            Run: func() {
                // pretend sample
                _ = time.Now().UnixNano() % 100
            },
        },
    }

    now := time.Now()
    for i := range tasks {
        tasks[i].Next = now.Add(tasks[i].Period)
    }

    for {
        now = time.Now()
        for i := range tasks {
            t := &tasks[i]
            if !now.Before(t.Next) {
                t.Run()
```

```
                t.Next = t.Next.Add(t.Period)
                // if still behind, skip ahead
                if now.After(t.Next) {
                    t.Next = now.Add(t.Period)
                }
            }
        }
        time.Sleep(1 * time.Millisecond) // cooperative idle
    }
}
```

Step 2: Inject a "bad" task

Add a task that `time.Sleep(200 * time.Millisecond)` occasionally. Observe heartbeat jitter. **Lesson:** cooperative scheduling fails open when a task hogs the CPU.

Step 3: Metrics

Track:

- `runs` per task
- `late` count when `now.Sub(scheduled) > threshold`
- Print every 5 s over UART

Step 4: Design write-up

In `NOTES.md`:

1. When cooperative is enough (UI blink, slow sensors).
2. When you need interrupts/RTOS (motor control, strict deadlines).
3. How TinyGo's model maps to your board.

Learning Goals

- Multi-rate task tables
- Soft deadline tracking

- Starvation under cooperative scheduling
- Honest boundary vs FreeRTOS-class systems

Extensions

1. Priority: run shortest period first each cycle.
2. Disable a task via serial command.
3. Port one task to pin interrupt (from Project 42).

Checkpoint

- ☐ 3 tasks with different periods
- ☐ UART shows heartbeat
- ☐ Bad-task experiment documented
- ☐ NOTES.md with RT conclusions

045 Project 45: TinyGo I2C Realtime Monitor

045 TinyGo: I2C Device + Realtime Status Monitor

Wire an **I2C sensor or peripheral** (or a mock if no hardware), read it on a fixed schedule, and publish a compact status line over UART—combining bus IO, timing, and error handling for embedded Go.

Hardware options

Path	Hardware	Notes
A	BME280 / SHT30 / MPU6050 etc.	Real I2C; check address + pull-ups
B	I2C EEPROM / expander	Simpler register IO
C	**Mock**	Software sensor if no bus

Many boards: `machine.I2C0` with `SDA`/`SCL` pins—**read your pinout**.

Step 1: Bus init sketch

```go
package main

import (
    "fmt"
    "machine"
    "time"
)

func main() {
    machine.Serial.Configure(machine.UARTConfig{BaudRate: 115200})

    i2c := machine.I2C0
    err := i2c.Configure(machine.I2CConfig{
        Frequency: 100_000, // 100 kHz start; 400 kHz if stable
    })
    if err != nil {
        fmt.Fprintf(machine.Serial, "i2c configure: %v\n", err)
```

```go
        // continue with mock if desired
    }

    const addr uint16 = 0x76 // example - change per device

    ticker := time.NewTicker(200 * time.Millisecond)
    defer ticker.Stop() // if supported; else ignore

    for range ticker.C {
        // Real read example (device-specific registers!):
        // buf := make([]byte, 2)
        // err := i2c.ReadRegister(uint8(addr), 0xD0, buf)
        // Mock:
        v := int(time.Now().UnixNano()%500) / 5
        ok := true
        fmt.Fprintf(machine.Serial, "ok=%v val=%d t=%d\n", ok, v, time.Now().Unix())
        _ = addr
        _ = i2c
    }
}
```

Replace mock with **one** real register read for your chip using the datasheet.

Step 2: Error budget (soft RT)

```go
type Monitor struct {
    Period   time.Duration
    Failures int
    LastOK   time.Time
}

func (m *Monitor) Sample(read func() (int, error)) {
    v, err := read()
    if err != nil {
        m.Failures++
        return
    }
    m.LastOK = time.Now()
    // publish v
    _ = v
}
```

Rules:

- Cap consecutive failures before `SAFE` mode (LED pattern / UART alert).

- Do not block forever on bus errors—timeout if API allows.
- Keep allocations out of the 200 ms path if RAM is tight (`buf` reused).

Step 3: Status LED semantics

LED	Meaning
Slow blink	Healthy samples
Fast blink	Bus errors
Solid	Fatal / halt (optional)

Step 4: Host-side plot (optional)

Log UART to file and plot with any tool:

```
picocom -b 115200 /dev/ttyACM0 | tee samples.log
```

Learning Goals

- I2C configure + address/register mindset
- Timed sampling with failure counts
- Operator-visible status (UART + LED)
- Datasheet-driven integration

Extensions

1. Two sensors on one bus.
2. Change sample rate over UART.
3. Simple threshold alarm relay pin.

Safety

- 3.3 V vs 5 V logic levels—**do not** overvolt pins.
- Shared GND; pull-ups usually 4.7k on SDA/SCL.

Checkpoint

- ☐ Bus init or explicit mock path
- ☐ Fixed-rate samples with failure counter
- ☐ UART status lines
- ☐ Datasheet note or mock justification

000 Projects Index

000 Projects: Build Real Software

This section is hands-on. Most chapters are **build projects** (runnable code + steps). A few are **checklist / drill** projects (harden or verify an existing service against a list)—those are labeled in the chapter title (e.g. *Secure Service Checklist*).

Project tiers

Tier	What you get
Build	Module layout, full or nearly full source, run instructions, extensions
Checklist	Requirements and evidence to collect; you supply or reuse a small service
Index / ideas	Catalogues and workout lists (not a single binary)

How To Use These Projects

1. Read the architecture diagram.
2. Type the code yourself first.
3. Run it.
4. Break it intentionally.
5. Add one extension before moving to the next project.

```
Learning loop

read -> build -> run -> debug -> extend -> repeat
```

Project Path (Beginner -> Advanced)

1. CLI Ping Tool
2. URL Shortener API
3. Concurrent Log Analyzer
4. File Integrity Checker
5. Concurrent Port Scanner
6. Mini Job Runner
7. Linux `ls` Clone
8. Linux `cat` Clone

9. Linux `grep` Clone
10. Linux `tail -f` Clone
11. Linux `du` Clone
12. Linux `find` Clone
13. Linux `wc` Clone
14. HTTP Load Tester
15. TUI System Monitor
16. Proxmox TUI Manager
17. SSH Remote Orchestrator
18. Go Workout: 200 Ten-Minute Exercises
19. Linux ps-lite
20. KV HTTP Store
21. WebSocket Chat
22. Log Shipper CLI
23. Proxmox Batch Operations CLI
24. Controller Pattern Simulator
25. Resource Index + 150 More Ideas
26. Kubernetes Pod Lister
27. Kubernetes Event Watcher
28. Kubernetes Rollout Checker
29. Terraform Plan Parser
30. Terraform Risk Reporter
31. Prometheus Exporter
32. Prometheus Probe Service
33. eBPF Tracepoint Basics
34. eBPF XDP Packet Counter
35. Proxmox Multi-Node Scheduler
36. Proxmox Capacity Balancer
37. Kubernetes HPA Simulator
38. Terraform Cost Estimator
39. Prometheus Alert Rule Tester
40. Infra Reconciler Daemon

TinyGo & soft realtime (embedded)

41. TinyGo Blink & Toolchain
42. TinyGo Button + Interrupt
43. TinyGo UART Sensor Loop
44. TinyGo Cooperative Tasks (Soft RT)
45. TinyGo I2C Realtime Monitor

```
TinyGo track

toolchain/blink -> button/IRQ -> timed sensor UART -> cooperative tasks -> I2C monitor
```

What You Will Practice

- CLI flags and argument parsing
- Networking and HTTP servers
- Goroutines, channels, and worker pools
- File I/O and hashing
- Error handling and retries
- Production-friendly project structure
- **TinyGo** targets, `machine` pins, soft-realtime loops, interrupts, UART/I2C

Step-by-Step Explanation

1. Pick a small set of exercises by track.
2. Timebox each attempt to ten minutes.
3. Record one takeaway and one weakness after each exercise.
4. Revisit chapter references when blocked.
5. Re-solve selected problems from memory weekly.

Learning Goals

- Build consistency, not one-time intensity.
- Improve retrieval and transfer of Go patterns.
- Progress from syntax fluency to engineering fluency.

001 Project 1: CLI Ping Tool

001 Build a CLI Ping Tool (TCP Ping)

We will build a portable ping tool using the standard library. It measures round-trip time by opening a TCP connection.

Why TCP Ping?

Classic ICMP ping needs raw socket permissions. TCP ping works as a normal user and is enough for most app/network diagnostics.

```
CLI flow

input host -> resolve DNS -> connect N times -> record RTT -> print stats
```

Step 1: Create the Module

```
mkdir goping && cd goping
go mod init example.com/goping
```

Step 2: Full `main.go`

```go
package main

import (
    "flag"
    "fmt"
    "math"
    "net"
    "os"
    "time"
)

type result struct {
    ok  bool
    rtt time.Duration
    err error
```

```
}

func tcpPing(host, port string, timeout time.Duration) result {
    start := time.Now()
    conn, err := net.DialTimeout("tcp", net.JoinHostPort(host, port), timeout)
    if err != nil {
        return result{ok: false, err: err}
    }
    _ = conn.Close()
    return result{ok: true, rtt: time.Since(start)}
}

func main() {
    count := flag.Int("c", 4, "number of probes")
    interval := flag.Duration("i", 1*time.Second, "interval between probes")
    timeout := flag.Duration("t", 2*time.Second, "connection timeout")
    port := flag.String("p", "443", "tcp port to probe")
    flag.Parse()

    if flag.NArg() != 1 {
        fmt.Println("usage: goping [flags] <host>")
        flag.PrintDefaults()
        os.Exit(2)
    }

    host := flag.Arg(0)
    ips, err := net.LookupIP(host)
    if err != nil || len(ips) == 0 {
        fmt.Fprintf(os.Stderr, "resolve failed: %v\n", err)
        os.Exit(1)
    }

    fmt.Printf("PING %s (%s) over TCP:%s\n", host, ips[0].String(), *port)

    var sent, recv int
    var minRTT time.Duration = time.Hour
    var maxRTT time.Duration
    var sumRTT time.Duration

    for i := 1; i <= *count; i++ {
        sent++
        r := tcpPing(host, *port, *timeout)
        if r.ok {
            recv++
            if r.rtt < minRTT {
                minRTT = r.rtt
            }
```

```go
            if r.rtt > maxRTT {
                maxRTT = r.rtt
            }
            sumRTT += r.rtt
            fmt.Printf("%d: connected time=%v\n", i, r.rtt)
        } else {
            fmt.Printf("%d: timeout/error: %v\n", i, r.err)
        }

        if i < *count {
            time.Sleep(*interval)
        }
    }

    loss := float64(sent-recv) / float64(sent) * 100
    avg := time.Duration(0)
    if recv > 0 {
        avg = sumRTT / time.Duration(recv)
    } else {
        minRTT = 0
    }

    stdDev := time.Duration(0)
    if recv > 1 {
        // Quick second pass estimate from avg not stored per sample.
        // Keep simple for CLI output consistency.
        _ = math.Sqrt(1)
    }

    fmt.Println("--- stats ---")
    fmt.Printf("%d probes sent, %d received, %.1f%% loss\n", sent, recv, loss)
    fmt.Printf("rtt min/avg/max = %v/%v/%v\n", minRTT, avg, maxRTT)
    _ = stdDev
}
```

Step 3: Run

```go run . google.com
go run . -c 10 -i 500ms -t 1s -p 443 cloudflare.com
```

What to Extend

1. Add per-IP probing when DNS returns multiple addresses.

2. Add JSON output (`-json`) for scripting.
3. Track jitter by storing all RTT samples.

Step-by-Step Explanation

1. Parse probe flags and validate host input.
2. Resolve DNS and print target IP for transparency.
3. Dial TCP with timeout for each probe.
4. Measure latency and aggregate min/avg/max.
5. Print packet loss and summary stats.

Code Anatomy

- `tcpPing` performs one probe and returns a typed result.
- Main loop controls pacing and collects metrics.
- Final summary computes packet loss and latency profile.

Learning Goals

- Network diagnostics with standard library only.
- Timeouts and resilient error reporting.
- Building practical CLI observability tools.

002 Project 2: URL Shortener API

002 Build a URL Shortener API

You will build a small HTTP service with two endpoints:

- `POST /shorten` -> create short URL
- `GET /:code` -> redirect to original URL

```
Request flow

client -> POST /shorten -> store (code -> url) -> return short URL
client -> GET /abc123  -> lookup code -> 302 redirect
```

Full `main.go`

```go
package main

import (
    "crypto/rand"
    "encoding/base64"
    "encoding/json"
    "fmt"
    "log"
    "net/http"
    "net/url"
    "strings"
    "sync"
)

type store struct {
    mu   sync.RWMutex
    data map[string]string
}

type shortenRequest struct {
    URL string `json:"url"`
}

type shortenResponse struct {
    Code string `json:"code"`
```

```
    URL  string `json:"short_url"`
}

func newCode(n int) string {
    b := make([]byte, n)
    if _, err := rand.Read(b); err != nil {
        panic(err)
    }
    return strings.TrimRight(base64.RawURLEncoding.EncodeToString(b), "=")
}

func main() {
    s := &store{data: make(map[string]string)}
    mux := http.NewServeMux()

    mux.HandleFunc("POST /shorten", func(w http.ResponseWriter, r *http.Request) {
        var req shortenRequest
        if err := json.NewDecoder(r.Body).Decode(&req); err != nil {
            http.Error(w, "invalid json", http.StatusBadRequest)
            return
        }

        u, err := url.ParseRequestURI(req.URL)
        if err != nil || u.Scheme == "" || u.Host == "" {
            http.Error(w, "invalid url", http.StatusBadRequest)
            return
        }

        code := newCode(5)
        s.mu.Lock()
        s.data[code] = req.URL
        s.mu.Unlock()

        resp := shortenResponse{
            Code: code,
            URL:  fmt.Sprintf("http://%s/%s", r.Host, code),
        }
        w.Header().Set("Content-Type", "application/json")
        _ = json.NewEncoder(w).Encode(resp)
    })

    mux.HandleFunc("GET /{code}", func(w http.ResponseWriter, r *http.Request) {
        code := r.PathValue("code")
        s.mu.RLock()
        longURL, ok := s.data[code]
        s.mu.RUnlock()
        if !ok {
```

```go
            http.NotFound(w, r)
            return
        }
        http.Redirect(w, r, longURL, http.StatusFound)
    })

    log.Println("listening on :8080")
    log.Fatal(http.ListenAndServe(":8080", mux))
}
```

Run and Test

```bash
go run .

curl -s -X POST localhost:8080/shorten \
  -H 'content-type: application/json' \
  -d '{"url":"https://go.dev/doc/"}'

curl -i localhost:8080/<code-from-response>
```

What to Extend

1. Persistent storage (BoltDB/Postgres).
2. Expiry timestamps.
3. Rate limiting middleware.

Step-by-Step Explanation

1. Define request and response contracts first.
2. Validate inbound input before doing any state changes.
3. Keep handler logic short and move reusable logic into helper functions.
4. Add timeouts and clear error paths.
5. Return consistent responses and status codes.

Code Anatomy

- Handlers parse input, call domain logic, write response.
- Shared state uses synchronization where needed.
- Transport concerns stay separate from business rules.

Learning Goals

- Build reliable service endpoints in Go.
- Understand API ergonomics and operational safety.
- Prepare code structure for tests and persistence later.

003 Project 3: Concurrent Log Analyzer

003 Build a Concurrent Log Analyzer

Read a log file and count log levels using a **worker pool**. Practice fan-out (lines to workers), local aggregation, and fan-in merge without shared mutable maps.

```
file -> producer -> jobs channel -> N workers -> partial counts -> merge -> report
```

Problem statement

```
log-analyzer [-w N] <file>
```

- Stream lines from a file
- Workers count INFO/WARN/ERROR/DEBUG substrings (case-sensitive tokens)
- Merge partial maps in main
- Print a summary table

Acceptance criteria

- ☐ Worker count configurable (`-w`, default `NumCPU`)
- ☐ No shared map writes from workers (each has a local map)
- ☐ Correct totals on a fixture file
- ☐ Terminates cleanly (channels closed, WaitGroup)
- ☐ `go run -race` clean

Setup

```
mkdir logan && cd logan
go mod init example.com/logan
# go 1.27
```

Full `main.go`

```go
package main

import (
    "bufio"
    "flag"
    "fmt"
    "os"
    "runtime"
    "strings"
    "sync"
)

var levels = []string{"INFO", "WARN", "ERROR", "DEBUG"}

func worker(lines <-chan string, out chan<- map[string]int) {
    local := map[string]int{}
    for _, lvl := range levels {
        local[lvl] = 0
    }
    for line := range lines {
        for _, level := range levels {
            if strings.Contains(line, level) {
                local[level]++
            }
        }
    }
    out <- local
}

func analyze(path string, workers int) (map[string]int, error) {
    f, err := os.Open(path)
    if err != nil {
        return nil, err
    }
    defer f.Close()

    if workers < 1 {
        workers = 1
    }

    jobs := make(chan string, 1024)
    partials := make(chan map[string]int, workers)
    var wg sync.WaitGroup
```

```
    for i := 0; i < workers; i++ {
        wg.Go(func() {
            worker(jobs, partials)
        })
    }

    go func() {
        scanner := bufio.NewScanner(f)
        buf := make([]byte, 0, 64*1024)
        scanner.Buffer(buf, 1024*1024)
        for scanner.Scan() {
            jobs <- scanner.Text()
        }
        close(jobs)
        wg.Wait()
        close(partials)
        // surface scan error via second channel in production; simplified here
        _ = scanner.Err()
    }()

    total := map[string]int{}
    for _, lvl := range levels {
        total[lvl] = 0
    }
    for p := range partials {
        for k, v := range p {
            total[k] += v
        }
    }
    return total, nil
}

func main() {
    workers := flag.Int("w", runtime.NumCPU(), "worker count")
    flag.Parse()
    if flag.NArg() != 1 {
        fmt.Fprintln(os.Stderr, "usage: log-analyzer [-w N] <file>")
        os.Exit(2)
    }

    total, err := analyze(flag.Arg(0), *workers)
    if err != nil {
        fmt.Fprintf(os.Stderr, "open failed: %v\n", err)
        os.Exit(1)
    }
```

```go
    fmt.Println("Log Summary")
    for _, lvl := range levels {
        fmt.Printf("%-6s %d\n", lvl+":", total[lvl])
    }
}
```

Step-by-step build path

1. Sequential single-goroutine counter first (correctness baseline).
2. Introduce jobs channel + N workers with **local** maps.
3. Merge partials in one owner goroutine/main.
4. Add `-race` and large fixture stress.
5. Extend matchers (regex, JSON field).

Run and verification

```bash
printf 'INFO start\nERROR boom\nWARN x\nINFO ok\nDEBUG d\nERROR e2\n' > app.log
go run . -w 4 app.log
go run -race . -w 8 app.log
```

Tests

```go
package main

import (
    "os"
    "path/filepath"
    "testing"
)

func TestAnalyze(t *testing.T) {
    dir := t.TempDir()
    path := filepath.Join(dir, "a.log")
    content := "INFO a\nINFO b\nERROR x\n"
    if err := os.WriteFile(path, []byte(content), 0o644); err != nil {
        t.Fatal(err)
    }
    total, err := analyze(path, 2)
    if err != nil {
        t.Fatal(err)
    }
```

```go
    if total["INFO"] != 2 || total["ERROR"] != 1 {
        t.Fatalf("%v", total)
    }
}
```

```go
go test -race ./...
```

Stretch goals

1. JSON output (`-json`).
2. Per-service breakdown if lines contain `service=`.
3. Top-N error messages (heap).
4. Multi-file args.

Pitfalls

Pitfall	Fix
Shared map++ from workers	local maps + merge
Forgetting close(partials)	close after Wait
Scanner default 64K	Buffer increase
Substring false positives	word boundaries / regex

Learning goals

- Leak-free worker pools
- Fan-out / fan-in architecture
- Race-free aggregation patterns

004 Project 4: File Integrity Checker

004 Build a File Integrity Checker

This CLI calculates SHA-256 checksums and verifies files against a manifest.

```
manifest mode: file -> hash -> write "hash path"
verify mode:   read manifest -> re-hash files -> compare -> report
```

Full `main.go`

```go
package main

import (
    "bufio"
    "crypto/sha256"
    "encoding/hex"
    "flag"
    "fmt"
    "io"
    "os"
    "path/filepath"
    "strings"
)

func hashFile(path string) (string, error) {
    f, err := os.Open(path)
    if err != nil {
        return "", err
    }
    defer f.Close()

    h := sha256.New()
    if _, err := io.Copy(h, f); err != nil {
        return "", err
    }
    return hex.EncodeToString(h.Sum(nil)), nil
}

func createManifest(root, out string) error {
    mf, err := os.Create(out)
```

```
    if err != nil {
        return err
    }
    defer mf.Close()

    return filepath.WalkDir(root, func(path string, d os.DirEntry, err error) error {
        if err != nil {
            return err
        }
        if d.IsDir() {
            return nil
        }
        h, err := hashFile(path)
        if err != nil {
            return err
        }
        _, err = fmt.Fprintf(mf, "%s  %s\n", h, path)
        return err
    })
}

func verifyManifest(path string) error {
    f, err := os.Open(path)
    if err != nil {
        return err
    }
    defer f.Close()

    s := bufio.NewScanner(f)
    failed := 0
    for s.Scan() {
        line := s.Text()
        parts := strings.SplitN(line, "  ", 2)
        if len(parts) != 2 {
            return fmt.Errorf("bad line: %q", line)
        }
        expected, filePath := parts[0], parts[1]
        got, err := hashFile(filePath)
        if err != nil || got != expected {
            failed++
            fmt.Printf("FAIL %s\n", filePath)
        } else {
            fmt.Printf("OK   %s\n", filePath)
        }
    }
    if err := s.Err(); err != nil {
        return err
```

```go
    }
    if failed > 0 {
        return fmt.Errorf("%d file(s) failed verification", failed)
    }
    return nil
}

func main() {
    mode := flag.String("mode", "manifest", "manifest|verify")
    root := flag.String("root", ".", "root directory for manifest mode")
    manifest := flag.String("manifest", "checksums.txt", "manifest path")
    flag.Parse()

    var err error
    switch *mode {
    case "manifest":
        err = createManifest(*root, *manifest)
    case "verify":
        err = verifyManifest(*manifest)
    default:
        err = fmt.Errorf("unknown mode: %s", *mode)
    }

    if err != nil {
        fmt.Fprintln(os.Stderr, "error:", err)
        os.Exit(1)
    }
}
```

Run

```bash
go run . -mode manifest -root ./data -manifest checksums.txt
go run . -mode verify -manifest checksums.txt
```

What to Extend

1. Support ignore patterns (`.git`, `node_modules`).
2. Output machine-readable JSON reports.
3. Add optional HMAC signing for manifest authenticity.

Step-by-Step Explanation

1. Parse command-line flags and validate inputs early.
2. Keep the core operation in a small, testable function.
3. Process data as a stream when possible to reduce memory use.
4. Print stable output and meaningful exit codes.
5. Add one extension feature and test edge cases.

Code Anatomy

- `main` handles flags, orchestration, and errors.
- Worker/helper functions hold business logic.
- Output section should be deterministic for scripting and CI usage.

Learning Goals

- Write composable Unix-style Go tools.
- Improve error messages and operator experience.
- Practice iterative improvement over one clear baseline.

005 Project 5: Concurrent Port Scanner

005 Build a Concurrent Port Scanner

Build a CLI that scans a host across a TCP port range using a **bounded worker pool**, collects open ports, and prints them sorted. This is a classic fan-out/fan-in exercise: many dials in flight, controlled concurrency, clean shutdown.

```
ports range -> jobs channel -> workers dial tcp -> open ports channel -> sort -> print
```

Problem statement

Given `-host`, `-from`, `-to`, `-w` (workers), and `-t` (dial timeout), probe every port in range with `net.DialTimeout("tcp", ...)`. Print open ports in ascending order. Exit non-zero on invalid flags.

Acceptance criteria

- ☐ Invalid range (`from > to`, out of 1–65535) exits with code 2 and a clear message
- ☐ Worker count bounds concurrency (not one goroutine per port without limit)
- ☐ Open ports printed sorted; closed/filtered ports silent
- ☐ Program terminates (no goroutine leak after range completes)
- ☐ `go build` succeeds with stdlib only
- ☐ Optional: `-json` or banner grab as stretch (not required for pass)

Setup

```
mkdir portscan && cd portscan
go mod init example.com/portscan
# go.mod
# module example.com/portscan
# go 1.27
```

Full `main.go`

```go
package main

import (
    "flag"
    "fmt"
    "net"
    "os"
    "sort"
    "sync"
    "time"
)

func scan(host string, timeout time.Duration, jobs <-chan int, openPorts chan<- int) {
    for p := range jobs {
        addr := net.JoinHostPort(host, fmt.Sprintf("%d", p))
        conn, err := net.DialTimeout("tcp", addr, timeout)
        if err != nil {
            continue
        }
        _ = conn.Close()
        openPorts <- p
    }
}

func main() {
    host := flag.String("host", "127.0.0.1", "target host")
    from := flag.Int("from", 1, "start port")
    to := flag.Int("to", 1024, "end port")
    workers := flag.Int("w", 200, "worker count")
    timeout := flag.Duration("t", 300*time.Millisecond, "dial timeout")
    flag.Parse()

    if *from < 1 || *to > 65535 || *from > *to {
        fmt.Fprintln(os.Stderr, "invalid port range: need 1 <= from <= to <= 65535")
        os.Exit(2)
    }
    if *workers < 1 {
        fmt.Fprintln(os.Stderr, "workers must be >= 1")
        os.Exit(2)
    }

    jobs := make(chan int, *workers)
    openPorts := make(chan int, 256)
    var wg sync.WaitGroup
```

```go
    for i := 0; i < *workers; i++ {
        wg.Go(func() {
            scan(*host, *timeout, jobs, openPorts)
        })
    }

    go func() {
        for p := *from; p <= *to; p++ {
            jobs <- p
        }
        close(jobs)
        wg.Wait()
        close(openPorts)
    }()

    var open []int
    for p := range openPorts {
        open = append(open, p)
    }
    sort.Ints(open)

    fmt.Printf("open ports on %s (%d-%d):\n", *host, *from, *to)
    for _, p := range open {
        fmt.Println(p)
    }
    fmt.Printf("total open: %d\n", len(open))
}
```

Step-by-step build path

1. **Validate flags** before starting any goroutine.
2. **Create job and result channels**; buffer jobs roughly to worker count to reduce scheduling chatter.
3. **Start N workers** that only dial and optionally send open ports—no shared mutable state except channels.
4. **Producer goroutine** pushes ports, closes jobs, waits for workers, then closes results (fan-in join).
5. **Main aggregates**, sorts, prints. Prefer `sync.WaitGroup.Go` (Go 1.25+) for worker startup.

```
sequence (top → bottom):
  actors: main, producer, workers, openCh
  main --> workers  : start N
  main --> producer : go fill jobs
  producer --> workers : jobs
```

```
  workers --> openCh : open ports
  producer --> openCh : close after Wait
  main --> openCh : range + sort
```

Run and verification

```
# Local: open something you control, e.g. a local HTTP server on :8080
go run . -host 127.0.0.1 -from 1 -to 9000 -w 300 -t 150ms

# Public lab target (be polite; small range)
go run . -host scanme.nmap.org -from 20 -to 100 -w 50 -t 500ms

go build -o portscan .
./portscan -host 127.0.0.1 -from 1 -to 100
```

Quick self-check with a known listener:

```
# terminal A
python3 -m http.server 8765
# terminal B
go run . -host 127.0.0.1 -from 8760 -to 8770 -w 20 -t 200ms
# expect 8765 in the list
```

Race-friendly structure (channels only for shared coordination):

```
go run -race . -host 127.0.0.1 -from 1 -to 200 -w 50 -t 50ms
```

Tests (optional unit)

Extract dial behind an interface for pure tests, or keep a smoke test:

```go
// scan_test.go - integration-style, skip in short mode
package main

import (
    "net"
    "testing"
    "time"
)

func TestDialLocalListener(t *testing.T) {
    ln, err := net.Listen("tcp", "127.0.0.1:0")
    if err != nil {
```

```go
        t.Fatal(err)
    }
    defer ln.Close()
    _, port, _ := net.SplitHostPort(ln.Addr().String())

    conn, err := net.DialTimeout("tcp", net.JoinHostPort("127.0.0.1", port), time.Second)
    if err != nil {
        t.Fatal(err)
    }
    _ = conn.Close()
}
```

```bash
go test -race ./...
```

Stretch goals

1. **CIDR / multi-host**: expand `10.0.0.0/30` into hosts; nest host×port jobs.
2. **Banner grab**: after dial, set read deadline and print first line.
3. **Rate limit**: token bucket so `-w` high does not flood a single target.
4. **UDP probe** (best-effort) or service name map for common ports.
5. **JSON output** for CI: `{"host":"...","open":[22,80]}`.

Pitfalls

Pitfall	Why it hurts	Fix
Unbounded goroutines	file descriptor / memory blowup	worker pool
No timeout	hangs on filtered ports	`DialTimeout`
Shared slice append without sync	data race	send on channel, append in one owner
Forgetting to close `openPorts`	main blocks forever	close after `Wait`
Scanning without permission	legal/ToS issues	only scan hosts you own or have OK for

Code anatomy

- **Producer** pushes port numbers into `jobs`.
- **Workers** consume jobs and emit open ports.
- **Aggregator** (main) merges results and prints a deterministic summary.

Learning goals

- Build leak-free fan-out/fan-in with WaitGroup + channel close.
- Balance throughput (`-w`) against timeout and target load.
- Keep coordination in channels; avoid shared mutable maps without locks.

006 Project 6: Mini Job Runner

006 Build a Mini Job Runner with Retries

This project builds a small in-process job runner with retries and backoff.

```
enqueue jobs -> workers pick jobs -> run handler -> retry on failure -> final status
```

Full `main.go`

```go
package main

import (
    "context"
    "errors"
    "fmt"
    "math/rand"
    "sync"
    "time"
)

type Job struct {
    ID       int
    Payload  string
    Attempts int
}

type Runner struct {
    workers int
    maxTry  int
    jobs    chan Job
}

func NewRunner(workers, maxTry, queueSize int) *Runner {
    return &Runner{
        workers: workers,
        maxTry:  maxTry,
        jobs:    make(chan Job, queueSize),
    }
}
```

```go
func (r *Runner) Enqueue(j Job) {
    r.jobs <- j
}

func process(ctx context.Context, j Job) error {
    select {
    case <-ctx.Done():
        return ctx.Err()
    case <-time.After(100 * time.Millisecond):
        // simulate intermittent failure
        if rand.Intn(3) == 0 {
            return errors.New("temporary failure")
        }
        fmt.Printf("job %d ok payload=%q\n", j.ID, j.Payload)
        return nil
    }
}

func (r *Runner) Start(ctx context.Context) {
    var wg sync.WaitGroup
    for w := 1; w <= r.workers; w++ {
        wg.Add(1)
        go func(workerID int) {
            defer wg.Done()
            for {
                select {
                case <-ctx.Done():
                    return
                case j, ok := <-r.jobs:
                    if !ok {
                        return
                    }

                    err := process(ctx, j)
                    if err == nil {
                        continue
                    }

                    j.Attempts++
                    if j.Attempts < r.maxTry {
                        backoff := time.Duration(j.Attempts*200) * time.Millisecond
                        time.AfterFunc(backoff, func() { r.Enqueue(j) })
                        fmt.Printf("worker=%d retry job=%d attempt=%d err=%v\n", workerID, j
                    } else {
                        fmt.Printf("worker=%d dead-letter job=%d err=%v\n", workerID, j.ID,
                    }
                }
```

```go
            }
        }(w)
    }

    wg.Wait()
}

func (r *Runner) Stop() {
    close(r.jobs)
}

func main() {
    rand.Seed(time.Now().UnixNano())
    ctx, cancel := context.WithTimeout(context.Background(), 5*time.Second)
    defer cancel()

    r := NewRunner(4, 4, 64)
    for i := 1; i <= 20; i++ {
        r.Enqueue(Job{ID: i, Payload: fmt.Sprintf("task-%d", i)})
    }

    go func() {
        <-ctx.Done()
        r.Stop()
    }()

    r.Start(ctx)
}
```

Run

```go
go run .
```

What to Extend

1. Persistent queue storage.
2. Dead-letter queue file.
3. Metrics endpoint (`/metrics`) with processing stats.

Step-by-Step Explanation

1. Model jobs, workers, and outputs explicitly.

2. Bound concurrency using worker pools and buffered channels.
3. Use `sync.WaitGroup` for lifecycle control.
4. Aggregate worker results in one place.
5. Verify behavior under both normal and failure paths.

Code Anatomy

- Producer pushes jobs into a channel.
- Workers consume jobs and emit results.
- Aggregator merges results and prints summary.

Learning Goals

- Build leak-free goroutine patterns.
- Balance throughput and resource limits.
- Understand fan-out/fan-in architecture.

007 Project 7: Build an ls Clone

007 Build an `ls` Clone

Build a practical `ls`-style CLI with flags for hidden files, long format, and sorting. Focus on clean formatting and deterministic ordering for scripts.

```
path -> read entries -> filter -> sort -> format -> print
```

Problem statement

```
gls [-a] [-l] [-sort name|size|time] [path]
```

- Default path `.`
- Hide dotfiles unless `-a`
- `-l` shows mode, size, modtime, name
- Directories get a trailing `/` in long mode

Acceptance criteria

- ☐ Lists directory entries
- ☐ `-a` shows hidden
- ☐ `-l` long format columns
- ☐ Sort by name/size/time
- ☐ Non-zero exit on unreadable path
- ☐ Stdlib only

Setup

```
mkdir gls && cd gls
go mod init example.com/gls
# go 1.27
```

Full `main.go`

```go
package main

import (
    "flag"
    "fmt"
    "io/fs"
    "os"
    "path/filepath"
    "sort"
    "strings"
    "time"
)

type item struct {
    Name string
    Info fs.FileInfo
}

func list(path string, showAll bool, sortBy string) ([]item, error) {
    entries, err := os.ReadDir(path)
    if err != nil {
        return nil, err
    }
    var items []item
    for _, e := range entries {
        name := e.Name()
        if !showAll && strings.HasPrefix(name, ".") {
            continue
        }
        info, err := e.Info()
        if err != nil {
            continue
        }
        items = append(items, item{Name: name, Info: info})
    }
    sort.SliceStable(items, func(i, j int) bool {
        switch sortBy {
        case "size":
            if items[i].Info.Size() == items[j].Info.Size() {
                return strings.ToLower(items[i].Name) < strings.ToLower(items[j].Name)
            }
            return items[i].Info.Size() < items[j].Info.Size()
        case "time":
            ti, tj := items[i].Info.ModTime(), items[j].Info.ModTime()
```

```
            if ti.Equal(tj) {
                return strings.ToLower(items[i].Name) < strings.ToLower(items[j].Name)
            }
            return ti.Before(tj)
        default: // name
            return strings.ToLower(items[i].Name) < strings.ToLower(items[j].Name)
        }
    })
    return items, nil
}

func main() {
    showAll := flag.Bool("a", false, "show hidden files")
    longFmt := flag.Bool("l", false, "long listing format")
    sortBy := flag.String("sort", "name", "name|size|time")
    flag.Parse()

    path := "."
    if flag.NArg() > 0 {
        path = flag.Arg(0)
    }
    switch *sortBy {
    case "name", "size", "time":
    default:
        fmt.Fprintln(os.Stderr, "invalid -sort; want name|size|time")
        os.Exit(2)
    }

    items, err := list(path, *showAll, *sortBy)
    if err != nil {
        fmt.Fprintf(os.Stderr, "read dir failed: %v\n", err)
        os.Exit(1)
    }

    for _, it := range items {
        if *longFmt {
            mode := it.Info.Mode().String()
            sz := it.Info.Size()
            mod := it.Info.ModTime().Format(time.DateTime)
            name := it.Name
            if it.Info.IsDir() {
                name += string(filepath.Separator)
            }
            fmt.Printf("%s %10d %s %s\n", mode, sz, mod, name)
        } else {
            fmt.Println(it.Name)
        }
```

```
    }
}
```

Step-by-step build path

1. `ReadDir` + filter dots.
2. Collect `FileInfo` for size/time sorts.
3. Stable sort with tie-break on name.
4. Long vs short formatters.
5. Validate `-sort` early.

Run and verification

```
go run .
go run . -a -l -sort size /var/log
go run . -sort time -l .
```

Tests

```
package main

import (
    "os"
    "path/filepath"
    "testing"
)

func TestListHidesDot(t *testing.T) {
    dir := t.TempDir()
    _ = os.WriteFile(filepath.Join(dir, "a.txt"), []byte("x"), 0o644)
    _ = os.WriteFile(filepath.Join(dir, ".secret"), []byte("y"), 0o644)
    items, err := list(dir, false, "name")
    if err != nil {
        t.Fatal(err)
    }
    if len(items) != 1 || items[0].Name != "a.txt" {
        t.Fatalf("%v", items)
    }
    items, _ = list(dir, true, "name")
    if len(items) != 2 {
        t.Fatalf("%d", len(items))
    }
```

```
}
```

```
go test ./...
```

Stretch goals

1. Colorize dirs when stdout is a TTY.
2. `-R` recursive.
3. Human sizes (`-h`) in long mode.
4. Multiple path arguments.

Pitfalls

Pitfall	Fix
Locale-dependent sort	document case-fold policy
Following symlinks unexpectedly	`ReadDir` metadata; document
Unstable sort ties	`sort.SliceStable` + name tie-break

Learning goals

- Directory listing APIs
- Formatting operator-friendly tables
- Flag-driven CLI design

008 Project 8: Build a cat Clone

008 Build a `cat` Clone

Create a streaming file printer with line numbers and optional end-of-line markers—stdlib only, Unix-friendly exit codes, and multi-file support like classic `cat`.

```
stdin/files -> scanner -> transform lines -> stdout
```

Problem statement

Implement `gcat` that:

- Reads **stdin** when no file args are given
- Concatenates multiple files in order
- Supports `-n` (number lines) and `-E` (show `$` at end of each line)
- Continues after a missing file (stderr message) like many Unix tools
- Streams line-by-line (does not load whole files)

Acceptance criteria

- ☐ `go run . file.txt` prints file contents
- ☐ No args → read stdin until EOF
- ☐ `-n` numbers lines; numbering continues across files
- ☐ `-E` appends `$` before newline semantics (end of logical line)
- ☐ Missing file: error on stderr, non-zero if any failure (your policy: document it)
- ☐ Works on empty files without hanging

Setup

```
mkdir gcat && cd gcat
go mod init example.com/gcat
# go 1.27
```

Full `main.go`

```go
package main

import (
    "bufio"
    "flag"
    "fmt"
    "io"
    "os"
)

func cat(r io.Reader, number, showEnds bool, start int) (int, error) {
    s := bufio.NewScanner(r)
    // Increase token size for long lines (default 64K may be tight for logs).
    buf := make([]byte, 0, 64*1024)
    s.Buffer(buf, 1024*1024)

    lineNo := start
    for s.Scan() {
        line := s.Text()
        if showEnds {
            line += "$"
        }
        if number {
            fmt.Printf("%6d\t%s\n", lineNo, line)
            lineNo++
        } else {
            fmt.Println(line)
        }
    }
    return lineNo, s.Err()
}

func main() {
    number := flag.Bool("n", false, "number output lines")
    showEnds := flag.Bool("E", false, "show $ at end of line")
    flag.Parse()

    lineNo := 1
    var hadErr bool

    if flag.NArg() == 0 {
        if _, err := cat(os.Stdin, *number, *showEnds, lineNo); err != nil {
            fmt.Fprintln(os.Stderr, "stdin:", err)
            os.Exit(1)
```

```go
        }
        return
    }

    for _, path := range flag.Args() {
        f, err := os.Open(path)
        if err != nil {
            fmt.Fprintf(os.Stderr, "%s: %v\n", path, err)
            hadErr = true
            continue
        }
        n, err := cat(f, *number, *showEnds, lineNo)
        _ = f.Close()
        if err != nil {
            fmt.Fprintf(os.Stderr, "%s: %v\n", path, err)
            hadErr = true
            continue
        }
        lineNo = n
    }

    if hadErr {
        os.Exit(1)
    }
}
```

Step-by-step build path

1. **Flags first** — `-n`, `-E`, then positional paths.
2. **Core `cat(io.Reader, ...)`** — pure stream helper; easy to test with `strings.NewReader`.
3. **Stdin path** when `flag.NArg() == 0`.
4. **Multi-file loop** with continuous line numbers when `-n`.
5. **Scanner buffer** for long log lines; surface `s.Err()`.

Run and verification

```bash
printf 'a\nb\nc\n' > sample.txt
go run . sample.txt
go run . -n -E sample.txt

# multi-file numbering
printf 'x\n' > a.txt; printf 'y\n' > b.txt
go run . -n a.txt b.txt
```

```
# expect line numbers 1 then 2

# stdin
echo hello | go run . -n
```

Tests

```go
package main

import (
    "strings"
    "testing"
)

func TestCatNumbers(t *testing.T) {
    // Capture via redirect in integration tests, or refactor cat to write to io.Writer.
    r := strings.NewReader("one\ntwo\n")
    // Smoke: no error on scan
    _, err := cat(r, false, false, 1)
    if err != nil {
        t.Fatal(err)
    }
}

func TestCatEmpty(t *testing.T) {
    _, err := cat(strings.NewReader(""), true, true, 1)
    if err != nil {
        t.Fatal(err)
    }
}
```

Refactor tip for stronger tests: change signature to `cat(w io.Writer, r io.Reader, ...)`.

```
go test ./...
```

Stretch goals

1. `-b` number non-blank lines only.
2. `-A` show tabs as `^I` and ends as `$`.
3. Binary-safe mode using `io.Copy` when no transforms.
4. Squeeze blank lines (`-s`).
5. Read from `-` as stdin mixed with files.

Pitfalls

Pitfall	Fix
Loading whole file with `ReadFile`	Use scanner / streaming
Default scanner max token 64K	`Scanner.Buffer`
Resetting line numbers per file unintentionally	Pass / return `lineNo`
Silent open errors	Always print to stderr
Treating binary as lines	Document limitation or use raw copy

Learning goals

- Composable Unix-style CLIs with `io.Reader`
- Flag ergonomics and multi-file edge cases
- Streaming over buffering for large inputs

009 Project 9: Build a grep Clone

009 Build a `grep` Clone

Search text with regex, case-insensitive mode, and line numbers. Stream each file line-by-line and print matches in a stable, script-friendly format.

```
files -> scan line by line -> regex match -> print matches
```

Problem statement

Implement `ggrep`:

```
ggrep [flags] <pattern> <file...>
```

Flags:

- `-i` ignore case (prefix `(?i)` or compile with case-folding)
- `-n` show line numbers as `file:line:text`

Exit codes (grep-compatible style is nice-to-have):

- `0` if any match
- `1` if no match
- `2` on usage/regex/IO hard failure (choose and document)

Acceptance criteria

- ☐ Matches printed for each matching line
- ☐ `-i` works for mixed-case patterns
- ☐ `-n` includes line numbers
- ☐ Invalid regex → clear error, exit 2
- ☐ Multiple files supported
- ☐ Streaming (no full-file regex required)

Setup

```
mkdir ggrep && cd ggrep
go mod init example.com/ggrep
# go 1.27
```

Full `main.go`

```go
package main

import (
    "bufio"
    "flag"
    "fmt"
    "os"
    "regexp"
)

func grepFile(path string, re *regexp.Regexp, showLineNo bool) (matched bool, err error) {
    f, err := os.Open(path)
    if err != nil {
        return false, err
    }
    defer f.Close()

    s := bufio.NewScanner(f)
    buf := make([]byte, 0, 64*1024)
    s.Buffer(buf, 1024*1024)

    lineNo := 0
    for s.Scan() {
        lineNo++
        line := s.Text()
        if !re.MatchString(line) {
            continue
        }
        matched = true
        if showLineNo {
            fmt.Printf("%s:%d:%s\n", path, lineNo, line)
        } else {
            fmt.Printf("%s:%s\n", path, line)
        }
    }
    return matched, s.Err()
```

```go
}

func main() {
    ignoreCase := flag.Bool("i", false, "ignore case")
    showLineNo := flag.Bool("n", false, "show line number")
    flag.Parse()

    if flag.NArg() < 2 {
        fmt.Fprintln(os.Stderr, "usage: ggrep [flags] <pattern> <file...>")
        os.Exit(2)
    }

    pattern := flag.Arg(0)
    if *ignoreCase {
        pattern = "(?i)" + pattern
    }
    re, err := regexp.Compile(pattern)
    if err != nil {
        fmt.Fprintln(os.Stderr, "invalid regex:", err)
        os.Exit(2)
    }

    any := false
    hadErr := false
    for _, file := range flag.Args()[1:] {
        matched, err := grepFile(file, re, *showLineNo)
        if err != nil {
            fmt.Fprintf(os.Stderr, "%s: %v\n", file, err)
            hadErr = true
            continue
        }
        if matched {
            any = true
        }
    }

    if hadErr {
        os.Exit(2)
    }
    if !any {
        os.Exit(1)
    }
}
```

Step-by-step build path

1. Parse flags; require pattern + 1 file.
2. Compile regex once (with optional `(?i)`).
3. Per file: open, scan, match, print.
4. Track `any` match and IO errors for exit codes.
5. Raise scanner buffer for long lines.

Run and verification

```
printf 'Error: boom\ninfo ok\nERROR again\n' > app.log
go run . error app.log
go run . -i -n "timeout|error" app.log

# no match → exit 1
go run . zzzzz app.log; echo exit:$?
```

Tests

```go
package main

import (
    "os"
    "path/filepath"
    "regexp"
    "testing"
)

func TestGrepFile(t *testing.T) {
    dir := t.TempDir()
    path := filepath.Join(dir, "t.log")
    if err := os.WriteFile(path, []byte("alpha\nBeta\nalpha2\n"), 0o644); err != nil {
        t.Fatal(err)
    }
    re := regexp.MustCompile(`(?i)alpha`)
    ok, err := grepFile(path, re, true)
    if err != nil || !ok {
        t.Fatalf("matched=%v err=%v", ok, err)
    }
}
```

```
go test -race ./...
```

Stretch goals

1. Recursive directory walk (`-r`) using `filepath.WalkDir`.
2. Invert match (`-v`).
3. Count only (`-c`).
4. Colorize matches on TTY.
5. Read stdin when files omitted.

Pitfalls

Pitfall	Fix
Compiling regex per line	Compile once
Shell globs not expanded by Go	Pass files or implement walk
Catastrophic backtracking	Prefer simpler patterns; timeouts hard in stdlib
Binary files spam	Detect NUL and skip (stretch)

Learning goals

- `regexp` package and RE2 flavor limits
- Streaming text tools
- Grep-compatible exit codes for scripts

010 Project 10: Build a tail -f Clone

010 Build a `tail -f` Clone

Read the last N lines of a file, then optionally follow appended content—core log-watching skill for operators.

```
open file -> ring buffer last N -> print -> poll growth -> print new data
```

Problem statement

```
gtail [-n N] [-f] <file>
```

- Print last `N` lines (default 10)
- `-f` follow: after initial print, stream new bytes as the file grows
- Exit 2 on usage errors; 1 on IO errors

Acceptance criteria

- ☐ Last N lines correct for files with N and <N lines
- ☐ Follow mode prints newly appended lines
- ☐ Does not busy-spin (sleep/poll)
- ☐ Missing file → clear error
- ☐ Stdlib only

Setup

```
mkdir gtail && cd gtail
go mod init example.com/gtail
# go 1.27
```

Full `main.go`

```
package main

import (
    "bufio"
    "context"
    "flag"
    "fmt"
    "io"
    "os"
    "os/signal"
    "syscall"
    "time"
)

func lastNLines(path string, n int) ([]string, error) {
    if n < 0 {
        n = 0
    }
    f, err := os.Open(path)
    if err != nil {
        return nil, err
    }
    defer f.Close()

    buf := make([]string, 0, n)
    s := bufio.NewScanner(f)
    sbuf := make([]byte, 0, 64*1024)
    s.Buffer(sbuf, 1024*1024)
    for s.Scan() {
        if n == 0 {
            continue
        }
        if len(buf) == n {
            copy(buf, buf[1:])
            buf[n-1] = s.Text()
        } else {
            buf = append(buf, s.Text())
        }
    }
    return buf, s.Err()
}

func follow(ctx context.Context, path string, offset int64, poll time.Duration) error {
    f, err := os.Open(path)
```

```
    if err != nil {
        return err
    }
    defer f.Close()

    if _, err := f.Seek(offset, io.SeekStart); err != nil {
        return err
    }

    for {
        select {
        case <-ctx.Done():
            return nil
        default:
        }
        n, err := io.Copy(os.Stdout, f)
        if err != nil {
            return err
        }
        if n == 0 {
            // optional: detect truncation/rotation via Stat size < offset
            select {
            case <-ctx.Done():
                return nil
            case <-time.After(poll):
            }
        }
    }
}

func main() {
    n := flag.Int("n", 10, "show last n lines")
    followMode := flag.Bool("f", false, "follow file")
    poll := flag.Duration("poll", 500*time.Millisecond, "follow poll interval")
    flag.Parse()

    if flag.NArg() != 1 {
        fmt.Fprintln(os.Stderr, "usage: gtail [-n N] [-f] <file>")
        os.Exit(2)
    }

    path := flag.Arg(0)
    lines, err := lastNLines(path, *n)
    if err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }
```

```go
    for _, line := range lines {
        fmt.Println(line)
    }

    if !*followMode {
        return
    }

    st, err := os.Stat(path)
    if err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }

    ctx, stop := signal.NotifyContext(context.Background(), os.Interrupt, syscall.SIGTERM)
    defer stop()

    if err := follow(ctx, path, st.Size(), *poll); err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }
}
```

Step-by-step build path

1. Ring buffer for last N lines (or `container/ring`).
2. Print initial window.
3. Seek to end size; `Copy` loop with poll sleep.
4. SIGINT via context.
5. Stretch: re-open on truncation/rotation.

Run and verification

```bash
printf '1\n2\n3\n4\n5\n' > app.log
go run . -n 3 app.log
# 3 4 5

# follow:
go run . -n 2 -f app.log
# other terminal:
echo new >> app.log
```

Tests

```go
package main

import (
    "os"
    "path/filepath"
    "testing"
)

func TestLastN(t *testing.T) {
    dir := t.TempDir()
    path := filepath.Join(dir, "f")
    _ = os.WriteFile(path, []byte("a\nb\nc\nd\n"), 0o644)
    lines, err := lastNLines(path, 2)
    if err != nil {
        t.Fatal(err)
    }
    if len(lines) != 2 || lines[0] != "c" || lines[1] != "d" {
        t.Fatalf("%v", lines)
    }
}

func TestLastNShortFile(t *testing.T) {
    dir := t.TempDir()
    path := filepath.Join(dir, "f")
    _ = os.WriteFile(path, []byte("only\n"), 0o644)
    lines, err := lastNLines(path, 10)
    if err != nil || len(lines) != 1 {
        t.Fatalf("%v %v", lines, err)
    }
}
```

```go
go test ./...
```

Stretch goals

1. Detect truncate (size < offset) and reopen from start.
2. Inotify/fsnotify instead of poll (Linux).
3. Multiple files (like `tail -f a b`).
4. `-F` forever reopen by name after rotation.

Pitfalls

Pitfall	Fix
Loading whole file for last N	ring buffer while scanning
Busy loop on EOF	sleep/poll or fsnotify
Ignoring rotation	track size/inode
No Ctrl+C in `-f`	signal.NotifyContext

Learning goals

- Ring buffers for trailing windows
- Follow-mode IO
- Operator UX for log tools

011 Project 11: Build a du Clone

011 Build a `du` Clone

Calculate recursive directory size and print human-readable or raw byte output. Use `filepath.WalkDir` and tolerate permission errors without aborting the whole walk.

```
dir walk -> sum file sizes -> print per path + total
```

Problem statement

```
gdu [-h] [path...]
```

- Default path `.`
- Sum sizes of all files under each root
- `-h` human-readable (default true in this lab; use `-h=false` for bytes)
- Multiple roots → print total line

Acceptance criteria

- ☐ Recursive size for directories
- ☐ Human and raw modes
- ☐ Skips unreadable entries without crash
- ☐ Multi-path grand total
- ☐ Stdlib only

Setup

```
mkdir gdu && cd gdu
go mod init example.com/gdu
# go 1.27
```

Full `main.go`

```
package main

import (
    "flag"
    "fmt"
    "io/fs"
    "os"
    "path/filepath"
)

func human(n int64) string {
    units := []string{"B", "KB", "MB", "GB", "TB"}
    v := float64(n)
    i := 0
    for v >= 1024 && i < len(units)-1 {
        v /= 1024
        i++
    }
    return fmt.Sprintf("%.1f%s", v, units[i])
}

func dirSize(root string) (int64, error) {
    var total int64
    err := filepath.WalkDir(root, func(path string, d fs.DirEntry, err error) error {
        if err != nil {
            fmt.Fprintf(os.Stderr, "skip %s: %v\n", path, err)
            return nil
        }
        if d.IsDir() {
            return nil
        }
        info, err := d.Info()
        if err != nil {
            return nil
        }
        total += info.Size()
        return nil
    })
    return total, err
}

func format(sz int64, humanFmt bool) string {
    if humanFmt {
        return fmt.Sprintf("%8s", human(sz))
```

```go
    }
    return fmt.Sprintf("%8d", sz)
}

func main() {
    humanFmt := flag.Bool("h", true, "human readable")
    flag.Parse()

    args := flag.Args()
    if len(args) == 0 {
        args = []string{"."}
    }

    var grand int64
    var hadErr bool
    for _, p := range args {
        sz, err := dirSize(p)
        if err != nil {
            fmt.Fprintf(os.Stderr, "%s: %v\n", p, err)
            hadErr = true
            continue
        }
        grand += sz
        fmt.Printf("%s  %s\n", format(sz, *humanFmt), p)
    }

    if len(args) > 1 {
        fmt.Printf("%s  total\n", format(grand, *humanFmt))
    }
    if hadErr {
        os.Exit(1)
    }
}
```

Step-by-step build path

1. Implement `human` with unit table.
2. `WalkDir` summing file sizes only.
3. Multi-root CLI + total.
4. Log skips to stderr; keep walking.
5. Tests with temp dirs of known size.

Run and verification

```
go run . -h /var/log
go run . -h=false .
go run . -h /tmp /var/tmp
```

Tests

```go
package main

import (
    "os"
    "path/filepath"
    "testing"
)

func TestDirSize(t *testing.T) {
    dir := t.TempDir()
    _ = os.WriteFile(filepath.Join(dir, "a"), []byte("12345"), 0o644)
    _ = os.Mkdir(filepath.Join(dir, "sub"), 0o755)
    _ = os.WriteFile(filepath.Join(dir, "sub", "b"), []byte("xy"), 0o644)
    sz, err := dirSize(dir)
    if err != nil {
        t.Fatal(err)
    }
    if sz != 7 {
        t.Fatalf("got %d", sz)
    }
}

func TestHuman(t *testing.T) {
    if human(1024) != "1.0KB" {
        t.Fatal(human(1024))
    }
}
```

```
go test ./...
```

Stretch goals

1. Apparent size vs allocated blocks (platform-specific).
2. Depth summary (`-d 1` like `du --max-depth`).
3. Exclude globs.

4. Parallel walk with care for FS limits.

Pitfalls

Pitfall	Fix
Aborting walk on permission error	return nil after log
Counting directory entries as files	skip `IsDir`
Binary vs decimal units	document 1024 base

Learning goals

- Recursive FS walks
- Human-readable formatting
- Resilient error handling on real trees

012 Project 12: Build a find Clone

012 Build a `find` Clone

Search a directory tree by name substring, entry type, and minimum size. Use `filepath.WalkDir` for efficient streaming walks without loading the whole tree into memory.

```
walk tree -> filter (name/type/size) -> print matching paths
```

Problem statement

```
gfind [root] [-name SUBSTR] [-type f|d] [-min-size BYTES]
```

Default root is `.`. Filters compose as AND. Name match is case-insensitive substring on the base name (not full path)—document this; classic find uses globs.

Acceptance criteria

- ☐ Walks recursively from root
- ☐ `-name` filters by basename contains (case-insensitive)
- ☐ `-type f` files only; `-type d` dirs only; empty = both
- ☐ `-min-size` applies to files only
- ☐ Permission errors: skip entry (or print to stderr) without aborting whole walk
- ☐ Stdlib only; `go build` clean

Setup

```
mkdir gfind && cd gfind
go mod init example.com/gfind
# go 1.27
```

Full `main.go`

```go
package main

import (
    "flag"
    "fmt"
    "io/fs"
    "os"
    "path/filepath"
    "strings"
)

func match(path string, d fs.DirEntry, name, typeFlag string, minSize int64) (bool, error) {
    base := filepath.Base(path)
    if name != "" && !strings.Contains(strings.ToLower(base), strings.ToLower(name)) {
        return false, nil
    }
    switch typeFlag {
    case "f":
        if d.IsDir() {
            return false, nil
        }
    case "d":
        if !d.IsDir() {
            return false, nil
        }
    case "":
        // ok
    default:
        return false, fmt.Errorf("invalid -type %q (want f, d, or empty)", typeFlag)
    }
    if minSize > 0 && !d.IsDir() {
        info, err := d.Info()
        if err != nil {
            return false, nil // skip unreadable
        }
        if info.Size() < minSize {
            return false, nil
        }
    }
    return true, nil
}

func main() {
    name := flag.String("name", "", "substring to match in filename")
```

```go
    typeFlag := flag.String("type", "", "f=file d=dir")
    minSize := flag.Int64("min-size", 0, "minimum file size in bytes")
    flag.Parse()

    root := "."
    if flag.NArg() > 0 {
        root = flag.Arg(0)
    }

    // Validate type early
    if *typeFlag != "" && *typeFlag != "f" && *typeFlag != "d" {
        fmt.Fprintf(os.Stderr, "invalid -type %q\n", *typeFlag)
        os.Exit(2)
    }

    err := filepath.WalkDir(root, func(path string, d fs.DirEntry, err error) error {
        if err != nil {
            fmt.Fprintf(os.Stderr, "skip %s: %v\n", path, err)
            return nil // keep walking
        }
        ok, mErr := match(path, d, *name, *typeFlag, *minSize)
        if mErr != nil {
            return mErr
        }
        if ok {
            fmt.Println(path)
        }
        return nil
    })
    if err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }
}
```

Step-by-step build path

1. Flags + optional root argument.
2. `WalkDir` callback: handle walk errors without failing the whole run.
3. Centralize filters in `match` for unit tests.
4. Print paths as you go (streaming).
5. Later: add prune (`-maxdepth`) and glob `filepath.Match`.

Run and verification

```
go run . /etc -name conf -type f
go run . . -name go -type f
go run . . -min-size 1048576 -type f   # files   1MiB

# empty name matches everything (careful on huge trees)
go run . /tmp -type d | head
```

Tests

```go
package main

import (
    "os"
    "path/filepath"
    "testing"
)

func TestMatchNameAndType(t *testing.T) {
    dir := t.TempDir()
    f := filepath.Join(dir, "App.LOG")
    if err := os.WriteFile(f, []byte("hi"), 0o644); err != nil {
        t.Fatal(err)
    }
    info, err := os.Stat(f)
    if err != nil {
        t.Fatal(err)
    }
    // DirEntry from WalkDir; use type from os.DirEntry via ReadDir
    entries, _ := os.ReadDir(dir)
    var d os.DirEntry
    for _, e := range entries {
        if e.Name() == "App.LOG" {
            d = e
            break
        }
    }
    ok, err := match(f, d, "log", "f", 0)
    if err != nil || !ok {
        t.Fatalf("ok=%v err=%v size=%d", ok, err, info.Size())
    }
}
```

```go
go test ./...
```

Stretch goals

1. Glob `-name '*.go'` via `filepath.Match`.
2. `-maxdepth N` by counting path separators relative to root.
3. `-exec` style: print null-terminated for `xargs -0`.
4. Concurrent walk with worker pool (careful with FS limits).
5. Skip directories (`.git`, `node_modules`) with a skip list.

Pitfalls

Pitfall	Fix
Returning walk `err` aborts tree	Log and `return nil`
`Info()` on every dir for size	Only when min-size set and file
Symlink loops	`WalkDir` doesn't follow symlinks by default—good
Case-sensitive surprise	Document case-insensitive policy

Learning goals

- `filepath.WalkDir` and `fs.DirEntry`
- Filter composition for CLI tools
- Resilient walks over imperfect trees

013 Project 13: Build a wc Clone

013 Build a `wc` Clone

Count lines, words, and bytes across files—streaming with `bufio.Scanner` and Unicode-aware word boundaries via `unicode.IsSpace`.

```
file -> scanner/read -> count L/W/B -> per file + total
```

Problem statement

```
gwc [file...]
```

- No files → read stdin
- Print `lines words bytes [filename]`
- Multiple files → final `total` row
- Words: maximal non-space sequences (Unicode spaces)

Acceptance criteria

- ☐ Correct counts on simple fixtures
- ☐ Stdin mode works
- ☐ Multi-file total row
- ☐ Missing file: stderr + continue
- ☐ Stdlib only

Setup

```
mkdir gwc && cd gwc
go mod init example.com/gwc
# go 1.27
```

Full main.go

```go
package main

import (
    "bufio"
    "flag"
    "fmt"
    "io"
    "os"
    "unicode"
)

type counts struct {
    lines int
    words int
    bytes int
}

func countReader(r io.Reader) (counts, error) {
    var c counts
    s := bufio.NewScanner(r)
    // Note: Scanner strips newlines; we approximate bytes as len(line)+1 per line.
    // For exact byte counts matching GNU wc, use raw Read and dual-pass logic (stretch).
    buf := make([]byte, 0, 64*1024)
    s.Buffer(buf, 1024*1024)
    for s.Scan() {
        line := s.Text()
        c.lines++
        c.bytes += len(line) + 1 // +1 for '\n' (last line without newline is imperfect)
        inWord := false
        for _, ch := range line {
            if unicode.IsSpace(ch) {
                inWord = false
                continue
            }
            if !inWord {
                c.words++
                inWord = true
            }
        }
    }
    return c, s.Err()
}

func printCounts(c counts, name string) {
```

```
    if name == "" {
        fmt.Printf("%8d %8d %8d\n", c.lines, c.words, c.bytes)
        return
    }
    fmt.Printf("%8d %8d %8d %s\n", c.lines, c.words, c.bytes, name)
}

func main() {
    flag.Parse()
    if flag.NArg() == 0 {
        c, err := countReader(os.Stdin)
        if err != nil {
            fmt.Fprintln(os.Stderr, err)
            os.Exit(1)
        }
        printCounts(c, "")
        return
    }

    var total counts
    var hadErr bool
    for _, path := range flag.Args() {
        f, err := os.Open(path)
        if err != nil {
            fmt.Fprintf(os.Stderr, "%s: %v\n", path, err)
            hadErr = true
            continue
        }
        c, err := countReader(f)
        _ = f.Close()
        if err != nil {
            fmt.Fprintf(os.Stderr, "%s: %v\n", path, err)
            hadErr = true
            continue
        }
        total.lines += c.lines
        total.words += c.words
        total.bytes += c.bytes
        printCounts(c, path)
    }
    if flag.NArg() > 1 {
        printCounts(total, "total")
    }
    if hadErr {
        os.Exit(1)
    }
}
```

Step-by-step build path

1. `countReader(io.Reader)` returns struct.
2. Word state machine with `unicode.IsSpace`.
3. Stdin vs files branching.
4. Aggregate totals.
5. Document newline/byte caveats vs GNU wc.

Run and verification

```
printf 'hello world\n\nfoo\n' > t.txt
go run . t.txt
echo 'a b c' | go run .
go run . t.txt t.txt
```

Tests

```go
package main

import (
    "strings"
    "testing"
)

func TestCountReader(t *testing.T) {
    c, err := countReader(strings.NewReader("hi there\n"))
    if err != nil {
        t.Fatal(err)
    }
    if c.lines != 1 || c.words != 2 {
        t.Fatalf("%+v", c)
    }
}

func TestEmpty(t *testing.T) {
    c, err := countReader(strings.NewReader(""))
    if err != nil || c.lines != 0 || c.words != 0 {
        t.Fatalf("%+v %v", c, err)
    }
}
```

```
go test ./...
```

Stretch goals

1. Exact byte count with `io.Copy` to `byteCounter` writer.
2. Flags `-l`, `-w`, `-c` to select columns.
3. Read from multiple compressed files.
4. Match GNU wc on files without trailing newline.

Pitfalls

Pitfall	Fix
Byte count vs scanner	document or dual-count
ASCII-only words	use unicode
Huge lines	Scanner.Buffer

Learning goals

- Streaming text metrics
- Unicode word splitting
- Multi-file CLI totals

014 Project 14: HTTP Load Tester

014 Build an HTTP Load Tester

Generate concurrent requests and report throughput and latency.

```
N workers -> request loop -> result channel -> aggregate metrics
```

Full `main.go`

```go
package main

import (
    "flag"
    "fmt"
    "net/http"
    "os"
    "sync"
    "time"
)

type result struct {
    ok  bool
    dur time.Duration
}

func main() {
    url := flag.String("url", "http://localhost:8080/health", "target URL")
    concurrency := flag.Int("c", 20, "concurrent workers")
    requests := flag.Int("n", 500, "total requests")
    timeout := flag.Duration("t", 3*time.Second, "request timeout")
    flag.Parse()

    client := &http.Client{Timeout: *timeout}
    results := make(chan result, *requests)

    start := time.Now()
    var wg sync.WaitGroup
    jobs := make(chan struct{}, *requests)
    for i := 0; i < *requests; i++ {
        jobs <- struct{}{}
```

```
    }
    close(jobs)

    for w := 0; w < *concurrency; w++ {
        wg.Add(1)
        go func() {
            defer wg.Done()
            for range jobs {
                t0 := time.Now()
                resp, err := client.Get(*url)
                d := time.Since(t0)
                if err != nil {
                    results <- result{ok: false, dur: d}
                    continue
                }
                _ = resp.Body.Close()
                results <- result{ok: resp.StatusCode < 500, dur: d}
            }
        }()
    }

    wg.Wait()
    close(results)

    var okCount int
    var min, max, sum time.Duration
    min = 1<<63 - 1
    for r := range results {
        if r.ok {
            okCount++
        }
        if r.dur < min {
            min = r.dur
        }
        if r.dur > max {
            max = r.dur
        }
        sum += r.dur
    }

    elapsed := time.Since(start)
    avg := time.Duration(int64(sum) / int64(*requests))
    rps := float64(*requests) / elapsed.Seconds()

    fmt.Printf("target: %s\n", *url)
    fmt.Printf("requests: %d, success: %d, failed: %d\n", *requests, okCount, *requests-okCo
```

```go
    fmt.Printf("latency min/avg/max: %v/%v/%v\n", min, avg, max)
    fmt.Printf("elapsed: %v, throughput: %.2f req/s\n", elapsed, rps)

    if okCount == 0 {
        os.Exit(1)
    }
}
```

Run

```go
go run . -url http://localhost:8080/health -c 50 -n 5000
```

Step-by-Step Explanation

1. Model jobs, workers, and outputs explicitly.
2. Bound concurrency using worker pools and buffered channels.
3. Use `sync.WaitGroup` for lifecycle control.
4. Aggregate worker results in one place.
5. Verify behavior under both normal and failure paths.

Code Anatomy

- Producer pushes jobs into a channel.
- Workers consume jobs and emit results.
- Aggregator merges results and prints summary.

Learning Goals

- Build leak-free goroutine patterns.
- Balance throughput and resource limits.
- Understand fan-out/fan-in architecture.

015 Project 15: TUI System Monitor

015 Build a TUI System Monitor

Create a live terminal UI for CPU, memory, and goroutine stats using Bubble Tea + gopsutil.

```
ticker -> collect metrics -> update model -> render tui frame
```

Setup

```
go mod init example.com/gotop
go get github.com/charmbracelet/bubbletea@latest
go get github.com/charmbracelet/lipgloss@latest
go get github.com/shirou/gopsutil/v4/cpu@latest
go get github.com/shirou/gopsutil/v4/mem@latest
```

Full `main.go`

```go
package main

import (
    "fmt"
    "runtime"
    "time"

    tea "github.com/charmbracelet/bubbletea"
    "github.com/charmbracelet/lipgloss"
    "github.com/shirou/gopsutil/v4/cpu"
    "github.com/shirou/gopsutil/v4/mem"
)

type tickMsg struct{}

type sample struct {
    cpuPct   float64
    memPct   float64
    gorCount int
    ts       time.Time
}
```

```go
type model struct {
    last sample
    err  error
}

func tickCmd() tea.Cmd {
    return tea.Tick(1*time.Second, func(time.Time) tea.Msg { return tickMsg{} })
}

func (m model) Init() tea.Cmd { return tickCmd() }

func (m model) Update(msg tea.Msg) (tea.Model, tea.Cmd) {
    switch msg := msg.(type) {
    case tea.KeyMsg:
        if msg.String() == "q" || msg.String() == "ctrl+c" {
            return m, tea.Quit
        }
    case tickMsg:
        cp, err := cpu.Percent(0, false)
        if err != nil || len(cp) == 0 {
            m.err = err
            return m, tickCmd()
        }
        vm, err := mem.VirtualMemory()
        if err != nil {
            m.err = err
            return m, tickCmd()
        }
        m.last = sample{
            cpuPct:   cp[0],
            memPct:   vm.UsedPercent,
            gorCount: runtime.NumGoroutine(),
            ts:       time.Now(),
        }
        m.err = nil
        return m, tickCmd()
    }
    return m, nil
}

func (m model) View() string {
    title := lipgloss.NewStyle().Bold(true).Foreground(lipgloss.Color("86")).Render("GoTop -
    if m.err != nil {
        return fmt.Sprintf("%s\n\nerror: %v\n\npress q to quit\n", title, m.err)
    }
    return fmt.Sprintf(
```

```
        "%s\n\nCPU:        %6.2f%%\nMemory:     %6.2f%%\nGoroutines: %d\nUpdated:    %s\n\np
        title,
        m.last.cpuPct,
        m.last.memPct,
        m.last.gorCount,
        m.last.ts.Format(time.TimeOnly),
    )
}

func main() {
    p := tea.NewProgram(model{})
    if _, err := p.Run(); err != nil {
        panic(err)
    }
}
```

Run

```
go run .
```

Step-by-Step Explanation

1. Collect node/resource metrics from API sources.
2. Compute deterministic scores per target.
3. Rank candidates by score and policy constraints.
4. Produce dry-run placement/migration decisions.
5. Apply gradually with canary and rollback plan.

Code Anatomy

- Data collection stage fetches capacity and load.
- Scoring stage turns metrics into comparable values.
- Decision stage emits ranked scheduling actions.

Learning Goals

- Build scheduling logic that is explainable and auditable.
- Balance utilization, reliability, and operational safety.
- Prepare for production-grade orchestration workflows.

016 Project 16: Proxmox TUI Manager

016 Build a Proxmox TUI Manager

This project builds an advanced terminal UI that talks to Proxmox API.

Features: - list nodes - list QEMU VMs on selected node - start/stop VM - refresh state

```
TUI keypress -> API client -> Proxmox endpoint -> update model -> redraw UI
```

Setup

```
go mod init example.com/proxmox-tui
go get github.com/charmbracelet/bubbletea@latest
go get github.com/charmbracelet/lipgloss@latest
```

Set env vars:

```
export PVE_BASE_URL="https://proxmox.local:8006"
export PVE_TOKEN="user@pam!token=secret"
```

Full `main.go`

```
package main

import (
    "crypto/tls"
    "encoding/json"
    "fmt"
    "io"
    "net/http"
    "os"
    "strconv"
    "time"

    tea "github.com/charmbracelet/bubbletea"
    "github.com/charmbracelet/lipgloss"
)
```

```go
type node struct {
    Node string `json:"node"`
}

type vm struct {
    VMID   int    `json:"vmid"`
    Name   string `json:"name"`
    Status string `json:"status"`
}

type apiResp[T any] struct {
    Data T `json:"data"`
}

type client struct {
    base string
    hc   *http.Client
}

func newClient(base string) *client {
    return &client{
        base: base,
        hc: &http.Client{Timeout: 8 * time.Second, Transport: &http.Transport{
            TLSClientConfig: &tls.Config{InsecureSkipVerify: true},
        }},
    }
}

func (c *client) do(method, path string, body io.Reader) (*http.Response, error) {
    req, err := http.NewRequest(method, c.base+path, body)
    if err != nil {
        return nil, err
    }
    req.Header.Set("Authorization", "PVEAPIToken="+os.Getenv("PVE_TOKEN"))
    return c.hc.Do(req)
}

func (c *client) listNodes() ([]node, error) {
    resp, err := c.do(http.MethodGet, "/api2/json/nodes", nil)
    if err != nil {
        return nil, err
    }
    defer resp.Body.Close()
    var out apiResp[[]node]
    if err := json.NewDecoder(resp.Body).Decode(&out); err != nil {
        return nil, err
    }
```

```
    return out.Data, nil
}

func (c *client) listVMs(nodeName string) ([]vm, error) {
    resp, err := c.do(http.MethodGet, "/api2/json/nodes/"+nodeName+"/qemu", nil)
    if err != nil {
        return nil, err
    }
    defer resp.Body.Close()
    var out apiResp[[]vm]
    if err := json.NewDecoder(resp.Body).Decode(&out); err != nil {
        return nil, err
    }
    return out.Data, nil
}

func (c *client) vmAction(nodeName string, vmid int, action string) error {
    resp, err := c.do(http.MethodPost, "/api2/json/nodes/"+nodeName+"/qemu/"+strconv.Itoa(vm
    if err != nil {
        return err
    }
    defer resp.Body.Close()
    if resp.StatusCode >= 300 {
        b, _ := io.ReadAll(resp.Body)
        return fmt.Errorf("action failed: %s", string(b))
    }
    return nil
}

type loadedMsg struct {
    nodes []node
    vms   []vm
    err   error
}

type actionMsg struct{ err error }

type model struct {
    cli    *client
    nodes  []node
    vms    []vm
    sel    int
    node   string
    status string
}

func loadCmd(c *client, n string) tea.Cmd {
```

```
    return func() tea.Msg {
        ns, err := c.listNodes()
        if err != nil {
            return loadedMsg{err: err}
        }
        nodeName := n
        if nodeName == "" && len(ns) > 0 {
            nodeName = ns[0].Node
        }
        vs, err := c.listVMs(nodeName)
        return loadedMsg{nodes: ns, vms: vs, err: err}
    }
}

func (m model) Init() tea.Cmd { return loadCmd(m.cli, "") }

func (m model) Update(msg tea.Msg) (tea.Model, tea.Cmd) {
    switch msg := msg.(type) {
    case tea.KeyMsg:
        switch msg.String() {
        case "q", "ctrl+c":
            return m, tea.Quit
        case "r":
            return m, loadCmd(m.cli, m.node)
        case "down", "j":
            if m.sel < len(m.vms)-1 {
                m.sel++
            }
        case "up", "k":
            if m.sel > 0 {
                m.sel--
            }
        case "s": // start
            if len(m.vms) == 0 {
                return m, nil
            }
            v := m.vms[m.sel]
            return m, func() tea.Msg { return actionMsg{err: m.cli.vmAction(m.node, v.VMID,
        case "x": // stop
            if len(m.vms) == 0 {
                return m, nil
            }
            v := m.vms[m.sel]
            return m, func() tea.Msg { return actionMsg{err: m.cli.vmAction(m.node, v.VMID,
        }
    case loadedMsg:
```

```
        if msg.err != nil {
            m.status = "load error: " + msg.err.Error()
            return m, nil
        }
        m.nodes = msg.nodes
        m.vms = msg.vms
        if m.node == "" && len(msg.nodes) > 0 {
            m.node = msg.nodes[0].Node
        }
        if m.sel >= len(m.vms) {
            m.sel = len(m.vms) - 1
            if m.sel < 0 {
                m.sel = 0
            }
        }
        m.status = "loaded"
    case actionMsg:
        if msg.err != nil {
            m.status = "action error: " + msg.err.Error()
            return m, nil
        }
        m.status = "action submitted"
        return m, loadCmd(m.cli, m.node)
    }
    return m, nil
}

func (m model) View() string {
    title := lipgloss.NewStyle().Bold(true).Foreground(lipgloss.Color("39")).Render("Proxmox
    s := title + "\n"
    s += fmt.Sprintf("node: %s | status: %s\n", m.node, m.status)
    s += "keys: j/k move | s start | x stop | r refresh | q quit\n\n"
    for i, v := range m.vms {
        cur := " "
        if i == m.sel {
            cur = ">"
        }
        s += fmt.Sprintf("%s vmid=%d name=%s status=%s\n", cur, v.VMID, v.Name, v.Status)
    }
    return s
}

func main() {
    base := os.Getenv("PVE_BASE_URL")
    if base == "" || os.Getenv("PVE_TOKEN") == "" {
        fmt.Println("set PVE_BASE_URL and PVE_TOKEN")
        os.Exit(2)
```

```go
    }
    m := model{cli: newClient(base), status: "booting"}
    if _, err := tea.NewProgram(m).Run(); err != nil {
        panic(err)
    }
}
```

Run

```go
go run .
```

Notes

- This sample uses `InsecureSkipVerify` for lab environments. Use proper TLS validation in production.
- Use API tokens with least privilege.

Step-by-Step Explanation

1. Collect node/resource metrics from API sources.
2. Compute deterministic scores per target.
3. Rank candidates by score and policy constraints.
4. Produce dry-run placement/migration decisions.
5. Apply gradually with canary and rollback plan.

Code Anatomy

- Data collection stage fetches capacity and load.
- Scoring stage turns metrics into comparable values.
- Decision stage emits ranked scheduling actions.

Learning Goals

- Build scheduling logic that is explainable and auditable.
- Balance utilization, reliability, and operational safety.
- Prepare for production-grade orchestration workflows.

017 Project 17: SSH Remote Orchestrator

017 Build an SSH Remote Orchestrator

Run commands on many hosts concurrently over SSH and aggregate results.

```
host list -> worker pool -> ssh exec per host -> collect stdout/stderr -> summary
```

Setup

```
go mod init example.com/gossh-orchestrator
go get golang.org/x/crypto/ssh@latest
```

Full `main.go`

```go
package main

import (
    "flag"
    "fmt"
    "os"
    "strings"
    "sync"
    "time"

    "golang.org/x/crypto/ssh"
)

type result struct {
    host string
    out  string
    err  error
}

func run(host, user, keyPath, cmd string, timeout time.Duration) result {
    key, err := os.ReadFile(keyPath)
    if err != nil {
        return result{host: host, err: err}
    }
```

```
    signer, err := ssh.ParsePrivateKey(key)
    if err != nil {
        return result{host: host, err: err}
    }

    cfg := &ssh.ClientConfig{
        User:            user,
        Auth:            []ssh.AuthMethod{ssh.PublicKeys(signer)},
        HostKeyCallback: ssh.InsecureIgnoreHostKey(),
        Timeout:         timeout,
    }

    c, err := ssh.Dial("tcp", host+":22", cfg)
    if err != nil {
        return result{host: host, err: err}
    }
    defer c.Close()

    s, err := c.NewSession()
    if err != nil {
        return result{host: host, err: err}
    }
    defer s.Close()

    out, err := s.CombinedOutput(cmd)
    return result{host: host, out: string(out), err: err}
}

func main() {
    hostsArg := flag.String("hosts", "", "comma-separated host list")
    user := flag.String("user", "root", "ssh user")
    key := flag.String("key", os.Getenv("HOME")+"/.ssh/id_rsa", "private key path")
    cmd := flag.String("cmd", "uname -a", "command to run")
    workers := flag.Int("w", 10, "worker count")
    timeout := flag.Duration("t", 5*time.Second, "ssh timeout")
    flag.Parse()

    if *hostsArg == "" {
        fmt.Fprintln(os.Stderr, "provide -hosts")
        os.Exit(2)
    }
    hosts := strings.Split(*hostsArg, ",")

    jobs := make(chan string)
    results := make(chan result)
    var wg sync.WaitGroup
```

```go
    for i := 0; i < *workers; i++ {
        wg.Add(1)
        go func() {
            defer wg.Done()
            for h := range jobs {
                results <- run(strings.TrimSpace(h), *user, *key, *cmd, *timeout)
            }
        }()
    }

    go func() {
        for _, h := range hosts {
            jobs <- h
        }
        close(jobs)
        wg.Wait()
        close(results)
    }()

    ok := 0
    for r := range results {
        if r.err != nil {
            fmt.Printf("[%s] ERROR: %v\n", r.host, r.err)
            continue
        }
        ok++
        fmt.Printf("[%s]\n%s\n", r.host, strings.TrimSpace(r.out))
    }
    fmt.Printf("done: %d/%d succeeded\n", ok, len(hosts))
}
```

Run

```go
go run . -hosts "10.0.0.11,10.0.0.12" -user root -cmd "uptime"
```

Security Notes

- Replace `ssh.InsecureIgnoreHostKey` with known-hosts validation in production.
- Use dedicated restricted SSH keys for automation.

Step-by-Step Explanation

1. Model jobs, workers, and outputs explicitly.
2. Bound concurrency using worker pools and buffered channels.
3. Use `sync.WaitGroup` for lifecycle control.
4. Aggregate worker results in one place.
5. Verify behavior under both normal and failure paths.

Code Anatomy

- Producer pushes jobs into a channel.
- Workers consume jobs and emit results.
- Aggregator merges results and prints summary.

Learning Goals

- Build leak-free goroutine patterns.
- Balance throughput and resource limits.
- Understand fan-out/fan-in architecture.

018 Project 18: Go Workout - 200 Ten-Minute Exercises

018 Go Workout: 200 Ten-Minute Exercises

This chapter uses a **short-exercise format** inspired by workout-style learning books: small tasks, tight scope, immediate practice.

How to Use

- Spend 10 minutes per exercise.
- Do not copy old answers.
- Prefer standard library first.
- After every 10 exercises, write a short reflection.

```
10-minute loop

read task -> code -> run -> test edge case -> note takeaway
```

Foundation Warmup (1-20)

1. Parse two integers from CLI args and print sum.
2. Parse `-name` flag and print greeting.
3. Build a `switch` on day number (1-7).
4. Convert Celsius to Fahrenheit with proper formatting.
5. Reverse a string safely for ASCII.
6. Count vowels in a string.
7. Find min/max in an `[]int`.
8. Remove duplicates from a slice.
9. Implement `contains([]string, string) bool`.
10. Build a frequency map for words in a sentence.
11. Sort a slice of ints ascending.
12. Sort a slice of structs by one field.
13. Write a function returning `(value, error)` for divide.
14. Use `defer` to time function execution.
15. Create a custom error type with context.
16. Marshal struct to JSON with tags.
17. Unmarshal JSON into struct and validate fields.
18. Read env var with fallback default.
19. Parse RFC3339 timestamp.
20. Format duration nicely (`1h2m3s`).

Files and CLI Tools (21-40)

21. Read file and print line count.
22. Read file and print byte count.
23. Read from stdin and uppercase output.
24. Build tiny `head -n` clone.
25. Build tiny `tail -n` clone.
26. Filter lines containing a substring.
27. Replace substring in each line and print.
28. Walk directory and count files.
29. Walk directory and sum file sizes.
30. Print top 10 largest files in a directory tree.
31. Build file extension counter (`.go`, `.md`, etc.).
32. Skip hidden files while walking.
33. Add `-json` output mode to a CLI tool.
34. Add `-quiet` and `-verbose` log levels.
35. Create a checksum (SHA-256) for one file.
36. Compare two files by checksum.
37. Build tiny `wc` (lines/words/bytes).
38. Build tiny `grep` with regex input.
39. Add case-insensitive flag to grep clone.
40. Print first match index and line number.

Structs, Methods, Interfaces (41-60)

41. Define `User` struct and constructor.
42. Add `String()` method to struct.
43. Add pointer receiver mutator method.
44. Add value receiver read-only method.
45. Implement small interface (`Runner`) with two types.
46. Compose two structs via embedding.
47. Resolve method name conflict in embedded structs.
48. Implement `io.Writer` that counts bytes.
49. Implement `io.Reader` over in-memory string.
50. Build a decorator function for timing any function.
51. Build `Option` pattern for server config.
52. Validate config and return combined error.
53. Compare two structs for equality constraints.
54. Parse CSV into typed struct slice.
55. Convert struct slice to map by key field.
56. Implement generic `Map` helper for slices.
57. Implement generic `Filter` helper.
58. Implement generic `Reduce` helper.
59. Create custom type alias with methods.
60. Add unit tests for all methods.

Errors and Reliability (61-80)

61. Wrap errors with `%w` across 3 layers.
62. Detect sentinel error with `errors.Is`.
63. Detect custom type with `errors.As`.
64. Build retry helper with max attempts.
65. Add exponential backoff to retry helper.
66. Add jitter to retry backoff.
67. Build circuit-breaker lite (open/close states).
68. Add timeout via `context.WithTimeout`.
69. Add cancellation via parent context.
70. Build `must` helper for internal tooling only.
71. Build panic recovery middleware for HTTP.
72. Capture stack trace on panic and log.
73. Return structured JSON error response.
74. Add error codes enum for API errors.
75. Validate all CLI input and fail fast.
76. Add dead-letter file for failed jobs.
77. Add graceful shutdown on SIGINT.
78. Add idempotency key check for commands.
79. Add bounded retries for network call.
80. Add fallback path when primary call fails.

Concurrency Basics (81-100)

81. Launch 5 goroutines and wait with `WaitGroup`.
82. Fan-out tasks to worker pool.
83. Fan-in results from workers.
84. Add context cancellation to worker pool.
85. Add timeout to a blocking goroutine.
86. Use buffered channel to smooth bursts.
87. Use unbuffered channel to enforce handoff.
88. Prevent goroutine leak on early return.
89. Build bounded semaphore using channel.
90. Merge two channels into one output.
91. Implement producer-consumer with backpressure.
92. Add `select` with timeout case.
93. Add `select` with default non-blocking case.
94. Protect shared map with mutex.
95. Replace mutex map with `sync.Map` and compare.
96. Use `sync.Once` for singleton init.
97. Use `sync.Cond` for queue notifications.
98. Build rate limiter with ticker.
99. Build token bucket limiter.
100. Profile goroutine count during load.

Networking and HTTP (101-120)

101. Build `/health` endpoint.
102. Add `/ready` endpoint with dependency check.
103. Parse query params safely.
104. Parse JSON request body with size limit.
105. Return JSON response helper function.
106. Add middleware for request logging.
107. Add middleware for panic recovery.
108. Add middleware for request ID.
109. Add middleware for timeout context.
110. Add simple in-memory rate limit middleware.
111. Build URL shortener `POST /shorten`.
112. Build URL shortener `GET /{code}` redirect.
113. Add TTL to short links.
114. Add short-link collision handling.
115. Add ETag support to one endpoint.
116. Add graceful server shutdown with context.
117. Build HTTP client with timeout + retries.
118. Build HTTP client with custom transport.
119. Add gzip compression middleware.
120. Benchmark one endpoint with your load tester.

Data and Persistence (121-140)

121. Use `encoding/csv` to read sample data.
122. Use `encoding/csv` to write report file.
123. Store key-value data in BoltDB (or bbolt).
124. Build repository interface over in-memory store.
125. Add file-backed repository implementation.
126. Add migration version field to data model.
127. Add optimistic locking field (`version`).
128. Add simple cache with TTL map.
129. Add cache invalidation on write.
130. Serialize map to JSON snapshot file.
131. Recover state from snapshot on startup.
132. Add periodic snapshot goroutine.
133. Add checksum to snapshot file.
134. Add snapshot restore validation.
135. Build append-only event log file.
136. Replay event log to rebuild state.
137. Add compaction command for old events.
138. Add corruption detection for events.
139. Add CLI export to CSV.
140. Add CLI import from CSV.

Testing Workout (141-160)

141. Write table-driven tests for parser.
142. Add subtests by category.
143. Add golden file test for CLI output.
144. Add benchmark with `b.Loop`.
145. Add benchmark with `-benchmem` analysis.
146. Add fuzz test for JSON decode path.
147. Add race test and fix discovered issue.
148. Add test helper for temporary fixture files.
149. Add `t.Cleanup` in all integration tests.
150. Use `httptest` for handler test.
151. Use `httptest.Server` for client test.
152. Use `testing/synctest` for flaky timer logic.
153. Add tests for retry backoff boundaries.
154. Add tests for context timeout path.
155. Add tests for panic recovery middleware.
156. Add tests for rate limiter edge case.
157. Add tests for short-link collision behavior.
158. Add tests for file checksum mismatch path.
159. Add integration test with build tag `integration`.
160. Add CI test command matrix in README.

Linux Tooling Track (161-180)

161. Extend `ls` clone with `-R` recursion.
162. Extend `ls` clone with colorized output.
163. Extend `cat` clone with `-T` tab markers.
164. Extend `grep` clone with invert match `-v`.
165. Extend `grep` clone with count-only `-c`.
166. Extend `tail` clone to handle file rotation.
167. Extend `du` clone with max depth flag.
168. Extend `find` clone with regex match mode.
169. Extend `wc` clone with rune count flag.
170. Build mini `xargs` clone.
171. Build mini `cut` clone.
172. Build mini `sort` clone.
173. Build mini `uniq` clone.
174. Build mini `tee` clone.
175. Build mini `which` clone.
176. Build mini `ps` clone using `/proc` (Linux).
177. Build mini `free` clone from `/proc/meminfo`.
178. Build mini `uptime` clone from `/proc/uptime`.
179. Build mini `df` clone using `syscall.Statfs`.
180. Build mini `watch` clone for repeated commands.

Advanced TUI and Infra Track (181-200)

181. Add CPU history sparkline to TUI monitor.
182. Add memory history sparkline to TUI monitor.
183. Add keybinding help modal in TUI.
184. Add theme toggle (light/dark terminal palette).
185. Add status bar with connection state.
186. Add periodic refresh ticker with pause key.
187. Add filtering by VM status in Proxmox TUI.
188. Add node selector screen in Proxmox TUI.
189. Add VM details pane in Proxmox TUI.
190. Add confirmation dialog before VM stop.
191. Add async action queue and progress states.
192. Add retry-on-429 for Proxmox API calls.
193. Add structured logs for every API request.
194. Add audit log file for start/stop actions.
195. Add SSH orchestrator output to JSON report.
196. Add SSH orchestrator parallel limit per subnet.
197. Add host reachability precheck before SSH.
198. Add rollout mode: canary then full batch.
199. Add rollback command set for failed rollout.
200. Build capstone: TUI panel that triggers SSH jobs and shows live results.

Chapter References in This Book

Use these while solving workouts:

- Foundations: `content/01-foundations/`
- Core: `content/02-core/`
- Concurrency: `content/11-concurrency-parallelism/`
- Testing: `content/06-testing/`
- Projects: `content/99-projects/`

Step-by-Step Explanation

1. Pick a small set of exercises by track.
2. Timebox each attempt to ten minutes.
3. Record one takeaway and one weakness after each exercise.
4. Revisit chapter references when blocked.
5. Re-solve selected problems from memory weekly.

Learning Goals

- Build consistency, not one-time intensity.

- Improve retrieval and transfer of Go patterns.
- Progress from syntax fluency to engineering fluency.

019 Project 19: Linux ps-lite

019 Build a Linux `ps`-Lite Tool

Read process info from `/proc` and print a compact process table. Linux-only; detect unsupported platforms cleanly.

```
/proc -> parse /proc/<pid>/stat + comm -> rows -> sort by pid -> table
```

Problem statement

```
gps [-n N]
```

- Scan `/proc` numeric directories
- Parse `stat` for state and RSS (pages → bytes)
- Print PID, state, RSS MiB, comm
- Limit rows with `-n`

Acceptance criteria

- ☐ Works on Linux with `/proc`
- ☐ Clear message on macOS/Windows
- ☐ Sorted by PID
- ☐ Handles processes vanishing mid-scan
- ☐ Stdlib only

Setup

```
mkdir gps && cd gps
go mod init example.com/gps
# go 1.27
```

Full main.go

```go
package main

import (
    "flag"
    "fmt"
    "os"
    "path/filepath"
    "sort"
    "strconv"
    "strings"
)

type proc struct {
    PID   int
    Comm  string
    RSS   int64
    State string
}

func parseProc(pid int) (proc, error) {
    statPath := fmt.Sprintf("/proc/%d/stat", pid)
    b, err := os.ReadFile(statPath)
    if err != nil {
        return proc{}, err
    }
    // comm is in parentheses and may contain spaces: (chrome --type=...)
    s := string(b)
    l := strings.IndexByte(s, '(')
    r := strings.LastIndexByte(s, ')')
    if l < 0 || r < 0 || r <= l {
        return proc{}, fmt.Errorf("bad stat format")
    }
    comm := s[l+1 : r]
    rest := strings.Fields(s[r+2:]) // after ") "
    if len(rest) < 22 {
        return proc{}, fmt.Errorf("short stat")
    }
    state := rest[0]
    // rss is field 24 in /proc/[pid]/stat → index 21 in rest (after state=rest[0])
    rssPages, err := strconv.ParseInt(rest[21], 10, 64)
    if err != nil {
        return proc{}, err
    }
    pageSize := int64(os.Getpagesize())
```

```go
    return proc{PID: pid, Comm: comm, State: state, RSS: rssPages * pageSize}, nil
}

func main() {
    limit := flag.Int("n", 50, "max rows")
    flag.Parse()

    entries, err := os.ReadDir("/proc")
    if err != nil {
        fmt.Fprintln(os.Stderr, "/proc unavailable (Linux only):", err)
        os.Exit(1)
    }

    var rows []proc
    for _, e := range entries {
        if !e.IsDir() {
            continue
        }
        pid, err := strconv.Atoi(e.Name())
        if err != nil {
            continue
        }
        p, err := parseProc(pid)
        if err != nil {
            continue // process exited
        }
        rows = append(rows, p)
    }

    sort.Slice(rows, func(i, j int) bool { return rows[i].PID < rows[j].PID })
    if *limit > len(rows) {
        *limit = len(rows)
    }

    fmt.Printf("%-8s %-4s %-10s %s\n", "PID", "S", "RSS(MB)", "COMM")
    for i := 0; i < *limit; i++ {
        r := rows[i]
        fmt.Printf("%-8d %-4s %-10.1f %s\n",
            r.PID, r.State, float64(r.RSS)/1024.0/1024.0, filepath.Base(r.Comm))
    }
}
```

Step-by-step build path

1. List `/proc` and filter numeric PIDs.
2. Parse `stat` carefully (comm parentheses).

3. Convert RSS pages with `os.Getpagesize()`.
4. Sort and format table.
5. Skip races where PID disappears.

Run and verification

```
go run . -n 40
# compare:
ps -eo pid,stat,rss,comm | head
```

Tests

On non-Linux, skip:

```go
package main

import (
    "os"
    "runtime"
    "testing"
)

func TestParseSelf(t *testing.T) {
    if runtime.GOOS != "linux" {
        t.Skip("linux only")
    }
    p, err := parseProc(os.Getpid())
    if err != nil {
        t.Fatal(err)
    }
    if p.PID == 0 || p.Comm == "" {
        t.Fatalf("%+v", p)
    }
}
```

Stretch goals

1. Sort by RSS descending (`-sort rss`).
2. Filter by comm regex.
3. Read `/proc/pid/cmdline` null-separated.
4. TUI refresh every second.

Pitfalls

Pitfall	Fix
Naive Fields on whole stat	parse comm via parentheses
Hard-coded page size 4096	`os.Getpagesize()`
Crashing when process exits	ignore read errors

Learning goals

- `/proc` as an API
- Robust parsing of kernel text formats
- Linux-specific tooling with clean failure elsewhere

020 Project 20: KV HTTP Store

020 Build a Key-Value HTTP Store

A production-style mini service with CRUD endpoints and periodic snapshot persistence.

```
HTTP API <-> in-memory map + mutex <-> snapshot.json
```

Full `main.go`

```go
package main

import (
    "encoding/json"
    "log"
    "net/http"
    "os"
    "sync"
    "time"
)

type store struct {
    mu sync.RWMutex
    m  map[string]string
}

func (s *store) load(path string) {
    b, err := os.ReadFile(path)
    if err != nil {
        return
    }
    s.mu.Lock()
    defer s.mu.Unlock()
    _ = json.Unmarshal(b, &s.m)
}

func (s *store) save(path string) error {
    s.mu.RLock()
    defer s.mu.RUnlock()
    b, err := json.MarshalIndent(s.m, "", "  ")
    if err != nil {
```

```
        return err
    }
    return os.WriteFile(path, b, 0o644)
}

func main() {
    s := &store{m: map[string]string{}}
    snap := "snapshot.json"
    s.load(snap)

    go func() {
        t := time.NewTicker(10 * time.Second)
        defer t.Stop()
        for range t.C {
            _ = s.save(snap)
        }
    }()

    mux := http.NewServeMux()
    mux.HandleFunc("PUT /kv/{key}", func(w http.ResponseWriter, r *http.Request) {
        key := r.PathValue("key")
        var body struct{ Value string `json:"value"` }
        if err := json.NewDecoder(r.Body).Decode(&body); err != nil {
            http.Error(w, "bad json", http.StatusBadRequest)
            return
        }
        s.mu.Lock()
        s.m[key] = body.Value
        s.mu.Unlock()
        w.WriteHeader(http.StatusNoContent)
    })

    mux.HandleFunc("GET /kv/{key}", func(w http.ResponseWriter, r *http.Request) {
        key := r.PathValue("key")
        s.mu.RLock()
        v, ok := s.m[key]
        s.mu.RUnlock()
        if !ok {
            http.NotFound(w, r)
            return
        }
        _ = json.NewEncoder(w).Encode(map[string]string{"key": key, "value": v})
    })

    mux.HandleFunc("DELETE /kv/{key}", func(w http.ResponseWriter, r *http.Request) {
        key := r.PathValue("key")
        s.mu.Lock()
```

```go
        delete(s.m, key)
        s.mu.Unlock()
        w.WriteHeader(http.StatusNoContent)
    })

    mux.HandleFunc("GET /kv", func(w http.ResponseWriter, r *http.Request) {
        s.mu.RLock()
        defer s.mu.RUnlock()
        _ = json.NewEncoder(w).Encode(s.m)
    })

    log.Println("listening :8081")
    log.Fatal(http.ListenAndServe(":8081", mux))
}
```

Run

```bash
go run .
curl -X PUT localhost:8081/kv/name -d '{"value":"king"}' -H 'content-type: application/json'
curl localhost:8081/kv/name
```

Step-by-Step Explanation

1. Define request and response contracts first.
2. Validate inbound input before doing any state changes.
3. Keep handler logic short and move reusable logic into helper functions.
4. Add timeouts and clear error paths.
5. Return consistent responses and status codes.

Code Anatomy

- Handlers parse input, call domain logic, write response.
- Shared state uses synchronization where needed.
- Transport concerns stay separate from business rules.

Learning Goals

- Build reliable service endpoints in Go.
- Understand API ergonomics and operational safety.
- Prepare code structure for tests and persistence later.

021 Project 21: WebSocket Chat

021 Build a WebSocket Chat Server

A minimal real-time chat backend with a broadcast hub. Clients connect to `/ws`, send text frames, and receive every message fan-out to all peers.

```
clients <-> hub (mutex + map) <-> broadcast WriteMessage
```

Problem statement

- Upgrade HTTP → WebSocket on `/ws`
- Track connected clients
- Broadcast each inbound text message to all clients
- Remove clients on read error/close
- Lab-friendly: `CheckOrigin` allows all (document prod hardening)

Acceptance criteria

- ☐ Two clients can exchange messages
- ☐ Disconnect does not crash the server
- ☐ Mutex protects the client map (race-clean under `-race`)
- ☐ Builds with gorilla/websocket or nhooyr/coder websocket
- ☐ Clear run instructions

Setup

```
mkdir gochat && cd gochat
go mod init example.com/gochat
go get github.com/gorilla/websocket@latest
# go 1.27
go mod tidy
```

Full `main.go`

```go
package main

import (
    "log"
    "net/http"
    "sync"
    "time"

    "github.com/gorilla/websocket"
)

var upgrader = websocket.Upgrader{
    CheckOrigin: func(r *http.Request) bool { return true }, // lab only
}

type hub struct {
    mu      sync.Mutex
    clients map[*websocket.Conn]struct{}
}

func (h *hub) add(c *websocket.Conn) {
    h.mu.Lock()
    defer h.mu.Unlock()
    h.clients[c] = struct{}{}
}

func (h *hub) del(c *websocket.Conn) {
    h.mu.Lock()
    defer h.mu.Unlock()
    delete(h.clients, c)
    _ = c.Close()
}

func (h *hub) broadcast(msg []byte) {
    h.mu.Lock()
    defer h.mu.Unlock()
    for c := range h.clients {
        _ = c.SetWriteDeadline(time.Now().Add(5 * time.Second))
        if err := c.WriteMessage(websocket.TextMessage, msg); err != nil {
            delete(h.clients, c)
            _ = c.Close()
        }
    }
}
```

```
func main() {
    h := &hub{clients: map[*websocket.Conn]struct{}{}}

    http.HandleFunc("/ws", func(w http.ResponseWriter, r *http.Request) {
        c, err := upgrader.Upgrade(w, r, nil)
        if err != nil {
            return
        }
        h.add(c)
        defer h.del(c)

        for {
            _, msg, err := c.ReadMessage()
            if err != nil {
                return
            }
            h.broadcast(msg)
        }
    })

    http.HandleFunc("/", func(w http.ResponseWriter, r *http.Request) {
        w.Header().Set("Content-Type", "text/html; charset=utf-8")
        _, _ = w.Write([]byte(indexHTML))
    })

    log.Println("chat on http://localhost:8090/  ws://localhost:8090/ws")
    log.Fatal(http.ListenAndServe(":8090", nil))
}

const indexHTML = `<!doctype html>
<title>gochat</title>
<pre id="log"></pre>
<input id="m" placeholder="message"/><button onclick="send()">send</button>
<script>
const log = (t) => { logEl.textContent += t + "\n"; };
const logEl = document.getElementById("log");
const ws = new WebSocket("ws://" + location.host + "/ws");
ws.onmessage = (e) => log("< " + e.data);
ws.onopen = () => log("connected");
function send() {
  const v = document.getElementById("m").value;
  ws.send(v);
  log("> " + v);
}
</script>`
```

Run and verification

```
go run .
# open two browser tabs to http://localhost:8090/
# send messages; both tabs should show traffic

go test -race ./...   # if you extract hub tests
```

Manual wscat-style:

```
# if you have websocat or similar
# websocat ws://localhost:8090/ws
```

Tests

```go
package main

import (
    "testing"
)

func TestHubAddDel(t *testing.T) {
    h := &hub{clients: map[*websocket.Conn]struct{}{}}
    // without real conn, just ensure map ops are safe with nil key edge
    // prefer integration test with httptest.Server for deeper coverage
    if len(h.clients) != 0 {
        t.Fatal()
    }
}
```

Better: use `httptest.NewServer` + gorilla dialer (stretch in repo).

Step-by-step build path

1. Upgrade handler.
2. Hub with mutexed set of conns.
3. Read loop → broadcast.
4. Remove on error; write deadlines.
5. Tiny HTML client for demo.

Stretch goals

1. Per-client send goroutine + outbound queue (avoid slow-client blocking hub).

2. Nicknames and join/leave system messages.
3. Rooms via path `/ws/{room}`.
4. Origin allowlist for production.
5. Ping/pong keepalive.

Pitfalls

Pitfall	Fix
Concurrent map without mutex	data race
Broadcast write blocks hub	per-conn writer queues
CheckOrigin true in prod	CSRF-ish WS risks
No write deadline	hung peers stall all

Learning goals

- WebSocket upgrade and framing
- Fan-out hub patterns
- Synchronization around connection sets

022 Project 22: Log Shipper CLI

022 Build a Log Shipper CLI

Tail a log file and POST new lines to an HTTP ingestion endpoint. This project combines follow-mode I/O, JSON payloads, HTTP client timeouts, and optional batching for efficiency.

```
file tail -> line channel -> (optional batch) -> HTTP POST /ingest
```

Problem statement

Build `logship`:

- Seek to end of file (or start with `-from-start`)
- Follow new lines (like `tail -f`)
- POST each line (or batch) as JSON to `-endpoint`
- Survive transient network errors without exiting the process
- Exit cleanly on SIGINT

Acceptance criteria

- ☐ New lines after start are POSTed
- ☐ HTTP client has a timeout
- ☐ Failed POSTs log to stderr and continue following
- ☐ Ctrl+C stops the loop
- ☐ Module builds with stdlib only

Setup

```
mkdir logship && cd logship
go mod init example.com/logship
# go 1.27
```

Full `main.go`

```
package main

import (
    "bufio"
    "bytes"
    "context"
    "encoding/json"
    "flag"
    "fmt"
    "io"
    "net/http"
    "os"
    "os/signal"
    "syscall"
    "time"
)

type payload struct {
    TS   string `json:"ts"`
    Line string `json:"line"`
    File string `json:"file"`
}

func shipLine(ctx context.Context, client *http.Client, endpoint, file, line string) error {
    body, err := json.Marshal(payload{
        TS:   time.Now().Format(time.RFC3339Nano),
        Line: line,
        File: file,
    })
    if err != nil {
        return err
    }
    req, err := http.NewRequestWithContext(ctx, http.MethodPost, endpoint, bytes.NewReader(b
    if err != nil {
        return err
    }
    req.Header.Set("Content-Type", "application/json")
    resp, err := client.Do(req)
    if err != nil {
        return err
    }
    defer resp.Body.Close()
    _, _ = io.Copy(io.Discard, resp.Body)
    if resp.StatusCode >= 300 {
```

```go
        return fmt.Errorf("status %d", resp.StatusCode)
    }
    return nil
}

func main() {
    file := flag.String("file", "app.log", "log file")
    endpoint := flag.String("endpoint", "http://localhost:8080/ingest", "ingest URL")
    fromStart := flag.Bool("from-start", false, "ship existing content first")
    poll := flag.Duration("poll", 300*time.Millisecond, "poll interval on EOF")
    flag.Parse()

    ctx, stop := signal.NotifyContext(context.Background(), os.Interrupt, syscall.SIGTERM)
    defer stop()

    f, err := os.Open(*file)
    if err != nil {
        fmt.Fprintln(os.Stderr, err)
        os.Exit(1)
    }
    defer f.Close()

    if !*fromStart {
        if _, err := f.Seek(0, io.SeekEnd); err != nil {
            fmt.Fprintln(os.Stderr, err)
            os.Exit(1)
        }
    }

    r := bufio.NewReader(f)
    client := &http.Client{Timeout: 3 * time.Second}

    fmt.Fprintf(os.Stderr, "shipping %s -> %s\n", *file, *endpoint)

    for {
        select {
        case <-ctx.Done():
            fmt.Fprintln(os.Stderr, "shutdown")
            return
        default:
        }

        line, err := r.ReadString('\n')
        if err == io.EOF {
            // partial line without newline: keep in buffer via Reader
            select {
            case <-ctx.Done():
```

```go
                return
            case <-time.After(*poll):
            }
            continue
        }
        if err != nil {
            fmt.Fprintln(os.Stderr, "read error:", err)
            select {
            case <-ctx.Done():
                return
            case <-time.After(*poll):
            }
            continue
        }

        if err := shipLine(ctx, client, *endpoint, *file, line); err != nil {
            fmt.Fprintln(os.Stderr, "post failed:", err)
            // continue tailing
        }
    }
}
```

Minimal ingest receiver (for local test)

`ingest.go` (separate program or build tag):

```go
//go:build ignore

package main

import (
    "fmt"
    "io"
    "log"
    "net/http"
)

func main() {
    http.HandleFunc("/ingest", func(w http.ResponseWriter, r *http.Request) {
        b, _ := io.ReadAll(r.Body)
        fmt.Printf("got: %s\n", b)
        w.WriteHeader(http.StatusNoContent)
    })
    log.Println("ingest :8080")
    log.Fatal(http.ListenAndServe(":8080", nil))
```

```
}
```

```
go run ingest.go
# other terminal:
echo 'hello ship' >> app.log
go run . -file app.log -endpoint http://localhost:8080/ingest
# another terminal:
echo 'line two' >> app.log
```

Step-by-step build path

1. Open file; seek end unless `-from-start`.
2. Loop `ReadString('\n')`; on EOF sleep/poll.
3. Marshal JSON; POST with context + timeout.
4. Wire SIGINT via `signal.NotifyContext`.
5. Add batching: accumulate N lines or T ms, one POST.

Batching stretch (sketch)

```
// flush []string every 100 lines or 1s via time.Ticker
type batchPayload struct {
    Lines []payload `json:"lines"`
}
```

Verification

```
go build -o logship .
# with ingest running:
printf 'a\nb\n' >> app.log
# expect two POSTs (or one batch)
```

Pitfalls

Pitfall	Fix
No HTTP timeout	hang forever
Exit on first POST error	use continue + log
Truncate/rotation not handled	detect inode/size shrink; reopen (stretch)
Busy loop on EOF	poll sleep / ticker

Pitfall	Fix
Missing newline at EOF	keep partial buffer (bufio does)

Learning goals

- Follow-mode file IO
- Resilient HTTP clients
- Graceful shutdown with context
- Path to production shippers (batch, backoff, rotation)

023 Project 23: Proxmox Batch Operations CLI

023 Build a Proxmox Batch CLI

Operate many VMs in one command (start/stop/reboot) using Proxmox API token auth and a bounded worker pool. Print a per-VM result summary for automation logs.

```
parse vmid list -> jobs channel -> workers POST status API -> result lines
```

Problem statement

```
pvebatch -node NODE -op start|stop|reboot -ids 101,102,103 [-w 8]
```

- Auth via `-base` / `-token` or `PVE_BASE_URL` / `PVE_TOKEN`
- Concurrent API calls with worker limit
- Non-zero exit if any VM fails (optional policy)

Acceptance criteria

☐ Validates required flags
☐ Worker pool bounds concurrency
☐ Prints ok/fail per VMID
☐ HTTP timeout set
☐ Lab TLS skip documented

Setup

```
mkdir pvebatch && cd pvebatch
go mod init example.com/pvebatch
# go 1.27

export PVE_BASE_URL='https://pve.example:8006'
export PVE_TOKEN='user@pam!tokenid=secret'
```

Full `main.go`

```go
package main

import (
    "crypto/tls"
    "flag"
    "fmt"
    "io"
    "net/http"
    "os"
    "strconv"
    "strings"
    "sync"
    "sync/atomic"
    "time"
)

func action(hc *http.Client, base, token, node string, vmid int, op string) error {
    url := fmt.Sprintf("%s/api2/json/nodes/%s/qemu/%d/status/%s", base, node, vmid, op)
    req, err := http.NewRequest(http.MethodPost, url, nil)
    if err != nil {
        return err
    }
    req.Header.Set("Authorization", "PVEAPIToken="+token)
    resp, err := hc.Do(req)
    if err != nil {
        return err
    }
    defer resp.Body.Close()
    body, _ := io.ReadAll(io.LimitReader(resp.Body, 4096))
    if resp.StatusCode >= 300 {
        return fmt.Errorf("status %d: %s", resp.StatusCode, strings.TrimSpace(string(body)))
    }
    return nil
}

func main() {
    base := flag.String("base", os.Getenv("PVE_BASE_URL"), "proxmox base url")
    token := flag.String("token", os.Getenv("PVE_TOKEN"), "api token")
    node := flag.String("node", "pve", "node name")
    op := flag.String("op", "start", "start|stop|reboot")
    ids := flag.String("ids", "", "comma list vmids")
    workers := flag.Int("w", 8, "parallel workers")
    insecure := flag.Bool("insecure", true, "skip TLS verify (lab)")
    flag.Parse()
```

```
	switch *op {
	case "start", "stop", "reboot":
	default:
		fmt.Fprintln(os.Stderr, "op must be start|stop|reboot")
		os.Exit(2)
	}
	if *base == "" || *token == "" || *ids == "" {
		fmt.Fprintln(os.Stderr, "need -base, -token, -ids (or env PVE_*)")
		os.Exit(2)
	}
	if *workers < 1 {
		os.Exit(2)
	}

	tr := http.DefaultTransport.(*http.Transport).Clone()
	if *insecure {
		tr.TLSClientConfig = &tls.Config{InsecureSkipVerify: true}
	}
	hc := &http.Client{Timeout: 8 * time.Second, Transport: tr}

	jobs := make(chan int)
	var wg sync.WaitGroup
	var fail atomic.Int64

	for i := 0; i < *workers; i++ {
		wg.Go(func() {
			for id := range jobs {
				err := action(hc, *base, *token, *node, id, *op)
				if err != nil {
					fmt.Printf("vm %d fail: %v\n", id, err)
					fail.Add(1)
				} else {
					fmt.Printf("vm %d ok\n", id)
				}
			}
		})
	}

	for _, s := range strings.Split(*ids, ",") {
		id, err := strconv.Atoi(strings.TrimSpace(s))
		if err != nil {
			fmt.Fprintf(os.Stderr, "bad id %q\n", s)
			fail.Add(1)
			continue
		}
		jobs <- id
```

```go
    }
    close(jobs)
    wg.Wait()

    if fail.Load() > 0 {
        os.Exit(1)
    }
}
```

Step-by-step build path

1. Flag/env config and op allowlist.
2. Shared `http.Client` with timeout.
3. Worker pool consuming VMID jobs.
4. Atomic fail counter for exit code.
5. Document token scopes (VM.PowerMgmt).

Run and verification

```go
go run . -node pve -op start -ids 101,102,103 -w 4
go run . -op stop -ids 101
```

Tests

Unit-test ID parsing logic by extracting a helper:

```go
func parseIDs(s string) ([]int, error) {
    var out []int
    for _, p := range strings.Split(s, ",") {
        p = strings.TrimSpace(p)
        if p == "" {
            continue
        }
        id, err := strconv.Atoi(p)
        if err != nil {
            return nil, err
        }
        out = append(out, id)
    }
    return out, nil
}
```

Stretch goals

1. Read VMIDs from file.
2. Status poll until guest agent up.
3. Dry-run mode printing URLs only.
4. JSON summary output.

Pitfalls

Pitfall	Fix
Unbounded goroutines	worker pool
No timeout	hung batch
Insecure TLS in prod	proper CA pool
Ignoring partial failure	exit 1 if any fail

Learning goals

- Concurrent API automation
- WaitGroup + job channel pools
- Operator-friendly batch tooling

024 Project 24: Controller Pattern Simulator

024 Build a Controller/Reconciler Simulator

Practice the Kubernetes-style reconciliation loop **without** Kubernetes dependencies: desired state vs actual state, idempotent steps, external drift, and convergence under a ticker.

```
desired state + actual state -> reconcile loop -> actions -> converge
```

Problem statement

Simulate a Deployment-like controller:

- `DesiredReplicas` is the spec
- `ActualReplicas` is observed status
- Each tick, apply **one** reconcile action: scale up, scale down, or noop
- Occasionally inject **drift** (external delete/create)
- Print a clear timeline so operators can see convergence

Acceptance criteria

- ☐ Reconcile is idempotent: repeated noops when in sync
- ☐ Actual moves toward desired by at most one step per tick (or document multi-step)
- ☐ Drift can move actual away; controller recovers
- ☐ Program ends after N ticks or when stable for M ticks
- ☐ Stdlib only; deterministic mode via fixed RNG seed flag

Setup

```
mkdir reconsim && cd reconsim
go mod init example.com/reconsim
# go 1.27
```

Full `main.go`

```go
package main

import (
    "flag"
    "fmt"
    "math/rand"
    "os"
    "time"
)

type State struct {
    DesiredReplicas int
    ActualReplicas  int
}

func reconcile(s *State) string {
    if s.ActualReplicas < s.DesiredReplicas {
        s.ActualReplicas++
        return "scale_up"
    }
    if s.ActualReplicas > s.DesiredReplicas {
        s.ActualReplicas--
        return "scale_down"
    }
    return "noop"
}

func clampNonNeg(n int) int {
    if n < 0 {
        return 0
    }
    return n
}

func main() {
    desired := flag.Int("desired", 3, "desired replicas")
    actual := flag.Int("actual", 0, "initial actual replicas")
    ticks := flag.Int("ticks", 20, "max ticks")
    driftEvery := flag.Int("drift-every", 6, "inject drift every N ticks (0=off)")
    seed := flag.Int64("seed", 1, "rng seed for drift")
    delay := flag.Duration("delay", 200*time.Millisecond, "tick delay (0 for fast)")
    flag.Parse()

    if *desired < 0 || *actual < 0 {
```

```
        fmt.Fprintln(os.Stderr, "desired/actual must be >= 0")
        os.Exit(2)
    }

    rng := rand.New(rand.NewSource(*seed))
    s := State{DesiredReplicas: *desired, ActualReplicas: *actual}

    stable := 0
    for i := 0; i < *ticks; i++ {
        if *driftEvery > 0 && i > 0 && i%*driftEvery == 0 {
            delta := rng.Intn(3) - 1 // -1,0,1
            s.ActualReplicas = clampNonNeg(s.ActualReplicas + delta)
            fmt.Printf("tick=%02d DRIFT delta=%+d actual=%d\n", i, delta, s.ActualReplicas)
        }

        action := reconcile(&s)
        fmt.Printf("tick=%02d desired=%d actual=%d action=%s\n",
            i, s.DesiredReplicas, s.ActualReplicas, action)

        if action == "noop" {
            stable++
            if stable >= 3 {
                fmt.Println("converged")
                return
            }
        } else {
            stable = 0
        }

        if *delay > 0 {
            time.Sleep(*delay)
        }

        // mid-run desired change demo
        if i == *ticks/2 && *ticks >= 4 {
            s.DesiredReplicas = max(1, s.DesiredReplicas-1)
            fmt.Printf("tick=%02d DESIRED_CHANGED desired=%d\n", i, s.DesiredReplicas)
        }
    }
    fmt.Println("finished ticks")
}

func max(a, b int) int {
    if a > b {
        return a
    }
    return b
```

```
}
```

Step-by-step build path

1. Model `State{Desired, Actual}`.
2. Write pure `reconcile(*State) action` with unit tests.
3. Drive a ticker/loop with logging.
4. Add drift and desired changes.
5. Add convergence detection (stable noops).

Architecture

```
            observe
  Desired                         Actual

               reconcile
            diff  apply one


                action log
```

Run and verification

```
go run . -desired 3 -actual 0 -ticks 25 -seed 42 -delay 0
# expect scale_up until actual=3, noops, maybe drift recovery

go run . -desired 5 -actual 8 -delay 0
# expect scale_down
```

Tests

```go
package main

import "testing"

func TestReconcileScaleUp(t *testing.T) {
    s := State{DesiredReplicas: 2, ActualReplicas: 0}
    if a := reconcile(&s); a != "scale_up" || s.ActualReplicas != 1 {
        t.Fatalf("action=%s actual=%d", a, s.ActualReplicas)
```

```go
    }
}

func TestReconcileNoop(t *testing.T) {
    s := State{DesiredReplicas: 2, ActualReplicas: 2}
    if a := reconcile(&s); a != "noop" {
        t.Fatal(a)
    }
}

func TestReconcileScaleDown(t *testing.T) {
    s := State{DesiredReplicas: 1, ActualReplicas: 3}
    if a := reconcile(&s); a != "scale_down" || s.ActualReplicas != 2 {
        t.Fatalf("action=%s actual=%d", a, s.ActualReplicas)
    }
}
```

```go
go test -race ./...
```

Stretch goals

1. Multiple resources (map of name → state).
2. Exponential backoff after failed apply (simulate errors).
3. Workqueue with rate limiting (client-go style mental model).
4. Level-triggered vs edge-triggered event notes in README.
5. JSON status dump each tick for later graphing.

Pitfalls

Pitfall	Fix
One-shot script instead of loop	Always re-observe actual
Non-idempotent apply	Same desired → noop safely
Unbounded scale steps	Cap max replicas; step by 1
Non-deterministic tests	Fixed seed; pure reconcile

Learning goals

- Controller pattern used in modern platforms
- Convergence over one-shot automation
- Drift handling and observability of actions

025 Project 25: Resource Index + 150 More Ideas

025 Resource Index and Project Idea Backlog

This chapter is a curated idea generator based on well-known Go learning resources and book/tutorial indexes.

Resource Anchors

1. Go by Example: gobyexample.com
2. Exercism Go Track (141 exercises): exercism.org/tracks/go
3. Exercism Go docs/testing: exercism.org/docs/tracks/go
4. Build Your Own X list: github.com/codecrafters-io/build-your-own-x
5. Awesome Go: github.com/uhub/awesome-go
6. Go by Example (book index/public page): simonandschuster.com/…/Go-by-Example

150 Additional Project Ideas

Linux CLI (1-30)

1. `chmod` helper wrapper with safety checks.
2. `chown` dry-run simulator.
3. `tree` command with depth filtering.
4. file deduplicator by checksum.
5. symlink verifier.
6. broken symlink cleaner.
7. directory watcher + action runner.
8. command history analyzer.
9. shell session recorder parser.
10. cron expression validator.
11. mini `top` clone.
12. mini `iotop` style estimator.
13. mini `netstat` parser.
14. mini `ss` parser.
15. process killer by regex.
16. zombie process detector.
17. service health checker.
18. journalctl filter wrapper.
19. colorized log viewer.
20. rotating file logger CLI.
21. backup snapshot tool.

22. rsync-like copy planner.
23. permission drift checker.
24. stale temp-file cleaner.
25. bulk rename utility.
26. grep over compressed files.
27. large-file splitter/merger.
28. file entropy scanner.
29. duplicate line remover.
30. path sanitizer utility.

Networking/Web (31-60)

31. DNS lookup CLI with JSON output.
32. TLS certificate inspector.
33. HTTP header security scanner.
34. simple API gateway with auth key.
35. reverse proxy with rate limit.
36. websocket event broadcaster.
37. webhook receiver with signature check.
38. link checker crawler.
39. HTTP cache proxy.
40. REST pagination helper.
41. OpenAPI schema diff checker.
42. JSON API contract tester.
43. API replay tool from HAR.
44. request signer (HMAC).
45. JWT issue/verify CLI.
46. OAuth device-flow sample app.
47. multipart upload service.
48. resumable upload protocol demo.
49. static site server with live reload.
50. image resize API.
51. URL metadata extractor.
52. API latency SLO dashboard backend.
53. HTTP retry middleware benchmark.
54. gRPC health check service.
55. gRPC proxy to REST bridge.
56. protobuf evolution checker.
57. distributed trace demo app.
58. log correlation ID middleware.
59. CORS policy validator.
60. synthetic uptime checker.

Data/Storage (61-90)

61. sqlite migration runner.

62. seed data loader CLI.
63. query timing profiler.
64. slow-query report generator.
65. CSV-to-SQL importer.
66. SQL-to-CSV exporter.
67. data masking utility.
68. pii scanner for CSV/JSON.
69. key rotation simulation tool.
70. append-only ledger store.
71. LSM-tree toy storage engine.
72. B-tree toy index.
73. bloom filter demo service.
74. write-ahead log simulator.
75. compaction strategy benchmark.
76. cache stampede protector demo.
77. id generator service (ksuid/ulid).
78. event sourcing sample.
79. CQRS read model projector.
80. dead letter queue consumer.
81. stream deduplication worker.
82. retention policy enforcer.
83. object storage sync tool.
84. folder-to-s3 mirror (mock).
85. schema migration drift detector.
86. JSON schema validator CLI.
87. XML-to-JSON converter.
88. parquet writer demo.
89. backup restore verifier.
90. data lineage tracer toy.

Concurrency/Systems (91-120)

91. pipeline backpressure simulator.
92. bounded queue benchmark.
93. lock contention visualizer.
94. goroutine leak detector wrapper.
95. worker auto-scaler simulation.
96. scheduler fairness simulator.
97. token bucket distributed simulation.
98. local message broker toy.
99. pub/sub with retries.
100. exactly-once semantics demo.
101. raft consensus toy model.
102. leader election simulator.
103. heartbeat failure detector.
104. distributed lock prototype.
105. partial failure injection tool.

106. chaos monkey for local services.
107. process supervisor.
108. service dependency graph parser.
109. config hot-reload framework.
110. plugin loader architecture demo.
111. dynamic rule engine toy.
112. policy-as-code evaluator.
113. command bus/event bus demo.
114. saga workflow simulator.
115. batch scheduler prototype.
116. resource quota enforcer.
117. cgroup stats reader (Linux).
118. seccomp profile inspector.
119. ptrace-based syscall tracer (toy).
120. mini container runtime PoC.

TUI/Infra/Platform (121-150)

121. TUI docker image browser.
122. TUI git commit explorer.
123. TUI log tail with filters.
124. TUI database query runner.
125. TUI feature-flag manager.
126. TUI release dashboard.
127. TUI CI pipeline monitor.
128. TUI Kubernetes pod viewer.
129. TUI helm values diff tool.
130. TUI terraform plan viewer.
131. Proxmox node capacity dashboard.
132. Proxmox VM template manager.
133. Proxmox snapshot orchestrator.
134. VM startup order manager.
135. host inventory manager.
136. SSH key rotation orchestrator.
137. parallel command runner with canary.
138. rollout status dashboard.
139. blue/green deployment simulator.
140. incident timeline builder.
141. SRE runbook executor CLI.
142. service dependency outage simulator.
143. alert noise reduction analyzer.
144. SLO burn-rate calculator.
145. policy compliance reporter.
146. artifact promotion workflow.
147. SBOM generator wrapper.
148. CVE triage dashboard.
149. secure secrets sync tool.

150. platform control plane toy app.

Suggested Build Order

1. Pick 10 CLI tasks.
2. Pick 5 network/API tasks.
3. Pick 5 concurrency/system tasks.
4. Pick 3 TUI/infra tasks.
5. Finish with one capstone controller + dashboard.

Step-by-Step Explanation

1. Pick a small set of exercises by track.
2. Timebox each attempt to ten minutes.
3. Record one takeaway and one weakness after each exercise.
4. Revisit chapter references when blocked.
5. Re-solve selected problems from memory weekly.

Learning Goals

- Build consistency, not one-time intensity.
- Improve retrieval and transfer of Go patterns.
- Progress from syntax fluency to engineering fluency.

026 Project 26: Kubernetes Pod Lister

026 Build a Kubernetes Pod Lister

List pods across namespaces using `client-go`. Produce operator-friendly tabular output suitable for scripts and CI smoke checks against a real or kind/minikube cluster.

```
kubeconfig -> clientset -> Pods.List -> print ns/name/phase/node
```

Problem statement

CLI `kpods`:

- Load kubeconfig from `-kubeconfig` (default `~/.kube/config`)
- Optional `-n` namespace (empty = all namespaces)
- Optional label selector `-l`
- Print: `NAMESPACE/NAME  PHASE  NODE  READY`
- Non-zero exit if API call fails

Acceptance criteria

- ☐ Lists pods with `client-go`
- ☐ Namespace filter works
- ☐ Clear errors when kubeconfig/cluster unavailable
- ☐ Context with timeout (no hang forever)
- ☐ Builds with Go modules

Setup

```
mkdir kpods && cd kpods
go mod init example.com/kpods
go get k8s.io/client-go@latest
go get k8s.io/apimachinery@latest
# go 1.27
go mod tidy
```

Requires a reachable cluster (`kubectl get pods` works).

Full `main.go`

```
package main

import (
    "context"
    "flag"
    "fmt"
    "os"
    "path/filepath"
    "time"

    corev1 "k8s.io/api/core/v1"
    metav1 "k8s.io/apimachinery/pkg/apis/meta/v1"
    "k8s.io/client-go/kubernetes"
    "k8s.io/client-go/tools/clientcmd"
    "k8s.io/client-go/util/homedir"
)

func readyCount(p corev1.Pod) (ready, total int) {
    total = len(p.Status.ContainerStatuses)
    for _, cs := range p.Status.ContainerStatuses {
        if cs.Ready {
            ready++
        }
    }
    return ready, total
}

func main() {
    var kubeconfig string
    if home := homedir.HomeDir(); home != "" {
        kubeconfig = filepath.Join(home, ".kube", "config")
    }
    kc := flag.String("kubeconfig", kubeconfig, "kubeconfig path")
    namespace := flag.String("n", "", "namespace (empty=all)")
    label := flag.String("l", "", "label selector")
    timeout := flag.Duration("timeout", 10*time.Second, "API timeout")
    flag.Parse()

    cfg, err := clientcmd.BuildConfigFromFlags("", *kc)
    if err != nil {
        fmt.Fprintln(os.Stderr, "kubeconfig:", err)
        os.Exit(1)
    }
    cli, err := kubernetes.NewForConfig(cfg)
```

```go
    if err != nil {
        fmt.Fprintln(os.Stderr, "client:", err)
        os.Exit(1)
    }

    ctx, cancel := context.WithTimeout(context.Background(), *timeout)
    defer cancel()

    ns := *namespace
    if ns == "" {
        ns = metav1.NamespaceAll
    }

    pods, err := cli.CoreV1().Pods(ns).List(ctx, metav1.ListOptions{
        LabelSelector: *label,
    })
    if err != nil {
        fmt.Fprintln(os.Stderr, "list pods:", err)
        os.Exit(1)
    }

    fmt.Printf("%-48s %-12s %-20s %s\n", "POD", "PHASE", "NODE", "READY")
    for _, p := range pods.Items {
        r, t := readyCount(p)
        fmt.Printf("%-48s %-12s %-20s %d/%d\n",
            p.Namespace+"/"+p.Name,
            string(p.Status.Phase),
            p.Spec.NodeName,
            r, t,
        )
    }
    fmt.Fprintf(os.Stderr, "total: %d\n", len(pods.Items))
}
```

Step-by-step build path

1. Build rest config from kubeconfig.
2. Create `kubernetes.Clientset`.
3. Call `Pods(ns).List` with context timeout.
4. Format rows for humans/scripts.
5. Add label selector and readiness columns.

Run and verification

```go run .
go run . -n kube-system
go run . -l app=myapp
go run . -timeout 5s

# compare with kubectl
kubectl get pods -A --no-headers | wc -l
```

Without a cluster:

```go run . -kubeconfig /nonexistent
# expect non-zero exit and readable error
```

Pitfalls

Pitfall	Fix
No timeout	use `context.WithTimeout`
Hard-coded HOME	`homedir.HomeDir` + flag
Panic on errors	print stderr + exit 1
Huge output	add limit/continue token (stretch)
Version skew	pin client-go close to cluster version

Stretch goals

1. Watch mode (`Watch` + print events).
2. JSON output (`-o json`).
3. Filter by phase (`-field-selector`).
4. In-cluster config path for running as a Pod.
5. Table of restarts from `ContainerStatuses`.

Learning goals

- Use `client-go` safely with context
- Map API objects to operator-facing columns
- Build automation-friendly cluster checks

Appendices

Glossary

Glossary

Short definitions for words this book uses with a fixed meaning. If a chapter taught a default, that default is in the definition.

alignment. The address multiple a type prefers. `int64` wants 8-byte alignment on 64-bit. Padding in a struct exists to satisfy it. See `unsafe.Sizeof` in the unsafe chapter — as a ruler, not as a licence.

benchmark. A function `BenchmarkXxx(*testing.B)` run by `go test -bench`. Reports ns/op. `-benchmem` adds bytes and allocs per op.

build constraint (build tag). A `//go:build` line that includes or excludes a file. `//go:build ignore` keeps a `go generate` helper out of the package.

cgo. The bridge that lets a Go file `import "C"` and call C. Needs a C compiler. Breaks easy cross-compiles. Avoid unless you must.

channel. A typed pipe between goroutines. Send `ch <- v`, receive `v := <-ch`. Buffer size is set at `make`. Close only from the sender side.

compatibility promise. For the standard library: exported signatures stay. For your module: behave the same, or bump a major version (`/v2`).

constructor. An exported function (`Open`, `New`) that returns a valid value. Prefer this over exported struct fields that callers fill in.

context. `context.Context` carries cancellation, a deadline, and optional request values down a call chain. The first argument of an RPC-shaped function.

coverage. The fraction of statements `go test -cover` saw run. A number, not a goal of 100%.

embed (`//go:embed`). Compiler feature that bakes a file into a `string`, `[]byte`, or `embed.FS` at build time. Import `"embed"` or `_ "embed"`.

error. A value that implements `error`. Returned, wrapped with `%w`, inspected with `errors.Is` / `errors.As`. Not an exception.

escape analysis. The compiler's decision that a value must live on the heap (it "escapes") because a pointer to it outlives the stack frame. `go build -gcflags=-m` prints the decisions.

example test. `ExampleXxx` in a `_test.go` file. `// Output:` makes `go test` compare stdout. Shows up in `go doc`.

exported. A name that starts with a capital letter. Visible from other packages. Part of your API.

fuzz. `FuzzXxx(*testing.F)` — the testing tool feeds random inputs that still satisfy your seed corpus.

garbage collection (GC). Automatic reclaim of heap values that nothing points to. You still close files and stop goroutines; GC does not.

generate (`go generate`). Runs `//go:generate` commands. Not part of `go build`. Prefer a `go run` helper you own.

go.mod. The file that names a **module** and its Go version and requirements.

goroutine. A function the runtime can run concurrently. Started with `go f()`. Cheap compared to an OS thread. Not magic: you still need to wait, cancel, or deadlock.

govulncheck. Tool that matches your module graph (and call graph) against the Go vulnerability database.

interface. A set of methods. Satisfied implicitly: no `implements` keyword. Keep it small. Define it where it is used, when you have two real implementations.

iter.Seq / iter.Seq2. Type aliases for push-iterator functions (Go 1.23+). `iter.Seq[V]` is `func(yield func(V) bool)`; `iter.Seq2[K, V]` adds a key. `range` can iterate over them. `slices.Collect` and `maps.Collect` drain them into slices and maps.

ldflags `-X`. Linker flag that overwrites a package-level string at build time. Used to stamp `version`.

maps package. `"maps"` (Go 1.21+) provides generic functions over `map` values: `maps.Clone`, `maps.Copy`, `maps.Keys`, `maps.Values`, `maps.Equal`, `maps.DeleteFunc`. Everything you used to write with a `for range`.

method. A function with a receiver. `func (t Ticket) Label() string`. Value receiver copies; pointer receiver can mutate.

module. A versioned unit of source with a `go.mod` and an import path (`example.com/desk`). Contains one or more packages.

mutex. `sync.Mutex` (or `RWMutex`) for shared memory. Lock, copy of a mutex is a vet finding, unlock. Prefer channels when they fit.

package. A directory of Go files with the same `package` name. The import path is module path plus directory.

panic / recover. Abort the goroutine; `recover` only in `defer`. Not for ordinary errors.

pprof. Profiler shipped as `go tool pprof`. CPU and memory profiles from tests or `net/http/pprof`.

pointer. Address of a value. `*T`. Nil is the zero value. Do not share mutable pointers across goroutines without a lock or a single owner.

race detector. `go test -race`. Finds unsynchronised shared memory. Use it in CI.

reflection (`reflect`). Run-time inspection of types and values. Last resort. `fmt` and `encoding/json` already do the usual jobs.

slice. Header (pointer, length, capacity) plus a backing array. `append` may allocate a new array.

slices package. `"slices"` (Go 1.21+) provides generic functions over slices: `slices.Sort`, `slices.SortFunc`, `slices.Contains`, `slices.Index`, `slices.BinarySearch`, `slices.Compact`, `slices.Reverse`, `slices.Clone`, `slices.Collect`. Prefer over hand-written loops.

range-over-function. Since Go 1.23, `range` accepts a function with signature `func(yield func(V) bool)` or `func(yield func(K, V) bool)`. The runtime calls the function; the function calls `yield` per element; `break` causes `yield` to return `false`. See `iter.Seq` / `iter.Seq2`.

staticcheck. Optional extra static analyser. Not part of the `go` binary.

stringer. A common generator for `String()` methods. This book prefers a hand-written `String` or a tiny `go generate` helper.

table test. A slice of cases in a `TestXxx` function, looped with `t.Run`.

toolchain. The `go` command: run, build, test, vet, fmt, doc, mod.

unsafe. Package that can break the type system. This book uses `unsafe.Sizeof` as a ruler. `unsafe.Pointer` is a contract with the compiler; do not take it.

vet (`go vet`). Static checks that ship with Go. Wrong `Printf` verbs, copied locks, and similar. Run with tests.

workspace (`go.work`). A file that lists modules you are editing together. Local replace without publishing.

zero value. The value of a declared variable with no initialiser: `0`, `""`, `nil`, struct of zeros. Useful when it is a valid empty; dangerous when it is not (then use a constructor).

`CGO_ENABLED`. `0` skips the C compiler. Boring default for pure Go. `1` is required for cgo.

`GOOS` / `GOARCH`. Target OS and architecture for `go build`. Cross-compilation for pure Go is these two variables.

Command Cheat Sheet

Command Cheat Sheet

Commands this book actually uses. Run them from a module directory unless a filename is listed. Substitute `example.com/desk` for your module path.

Everyday

Command	What it does
`go run .`	Compile the package in this directory and run it
`go run file.go`	Compile and run listed files (no `go.mod` needed for a toy)
`go build -o desk .`	Write a binary named `desk`
`go test ./...`	Test every package under this directory
`go test -v ./...`	Same, with names of tests and examples
`gofmt -w .`	Rewrite Go files in place
`go fmt ./...`	Same idea, package-oriented
`go vet ./...`	Compiler-adjacent static checks
`go doc ./ticket`	Print package docs
`go doc ./ticket.Open`	Print one symbol
`go env GOOS GOARCH`	Print this machine's target defaults
`go version`	Print the toolchain version (this book: 1.27)

Modules

Command	What it does
`go mod init example.com/desk`	Create `go.mod`
`go mod tidy`	Add missing modules, drop unused ones
`go get example.com/mod@v1.2.3`	Add or bump a requirement
`go list -m all`	Print the module graph

Tests, benches, profiles

Command	What it does
`go test -cover ./...`	Statement coverage
`go test -race ./...`	Race detector

Command	What it does
`go test -bench=. -benchmem ./join`	Benchmarks plus allocs
`go test -bench=BenchmarkJoinPlus -cpuprofile=cpu.pprof ./join`	CPU profile
`go test -bench=BenchmarkJoinPlus -memprofile=mem.pprof ./join`	Memory profile
`go tool pprof -top cpu.pprof`	Text summary of a profile

Ship

Command	What it does
`CGO_ENABLED=0 go build -o desk .`	Pure-Go binary, no cgo
`go build -ldflags="-X main.version=1.2.3" -o desk .`	Stamp `var version` in `package main`
`GOOS=linux GOARCH=amd64 go build -o desk .`	Cross-compile
`GOOS=linux GOARCH=amd64 CGO_ENABLED=0 go build -ldflags="-X main.version=1.2.3" -o desk .`	Boring release line
`go generate .`	Run `//go:generate` lines
`go run golang.org/x/vuln/cmd/govulncheck@latest ./...`	Vulnerability scan
`go tool dist list`	All `GOOS/GOARCH` pairs

Optional extra

Command	What it does
`staticcheck ./...`	Third-party static analysis (install separately)

Standard library packages (quick reference)

Package	Since	Key items
`slices`	1.21	`Sort`, `SortFunc`, `Contains`, `BinarySearch`, `Compact`, `Reverse`, `Clone`, `Collect`
`maps`	1.21	`Clone`, `Copy`, `Keys`, `Values`, `Equal`, `DeleteFunc`, `Collect`
`cmp`	1.21	`Compare`, `Ordered` constraint
`iter`	1.23	`Seq[V]`, `Seq2[K,V]` — push-iterator types for range-over-function
`log/slog`	1.21	`NewTextHandler`, `NewJSONHandler`, `HandlerOptions`, `With`, `SetDefault`
`sync`	—	`Mutex`, `RWMutex`, `WaitGroup` (`.Go` since 1.25), `Once`, `Map`
`sync/atomic`	—	`Int64`, `Bool`, `Pointer[T]` — typed atomics
`context`	—	`Background`, `WithCancel`, `WithTimeout`, `WithCancelCause` (1.20), `Cause` (1.20)
`testing`	—	`T.Run`, `T.Helper`, `T.TempDir`, `T.Setenv`, `T.Context` (1.21), `B.Loop` (1.24)
`net/http`	—	`ServeMux` with method patterns (1.22), `PathValue` (1.22)

Reminder 1: `go run` is a compile

Save as `hello.go`:

```go
// hello.go
package main

import "fmt"

func main() {
    fmt.Println("desk is open")
}
```

Run:

```
go run hello.go
```

Output:

```
desk is open
```

There is no interpreter. `go build -o hello hello.go` then `./hello` prints the same line.

Reminder 2: `go test` runs examples

Save as `go.mod`:

```
module example.com/desk

go 1.27
```

Save as `greet.go`:

```go
// greet.go
package desk

func Greet() string {
    return "desk is open"
}
```

Save as `greet_test.go`:

```go
// greet_test.go
package desk

import "fmt"

func ExampleGreet() {
    fmt.Println(Greet())
    // Output:
    // desk is open
}
```

Run:

```go
go test -v .
```

Output:

```
=== RUN   ExampleGreet
--- PASS: ExampleGreet (0.00s)
PASS
ok      example.com/desk    0.002s
```

If the printed line drifts from `// Output:`, the test fails. That is documentation you cannot ignore.

The boring rule

- `fmt` / `vet` / `test` on every change.
- `govulncheck` and a stamped `CGO_ENABLED=0` build before you ship.
- When in doubt, `go doc cmd/go` is the long form of this page.

What's New: Go 1.21 – 1.27

What's New: Go 1.21 – 1.27

This appendix is a reference, not a tutorial. Each section lists what changed in that release and shows one complete program that demonstrates two or three of the most useful additions. All programs compile and run with Go 1.27.

The desk domain appears throughout: orders, tickets, prices, shifts.

Go 1.21

Feature	Package / layer	What it does
`min` / `max`	language builtin	Return the smallest or largest of two or more comparable values.
`clear`	language builtin	Zero all elements of a slice or delete all keys of a map.
`slices`	`slices`	Sort, search, compare, and transform slices with generic functions.
`maps`	`maps`	Copy, clone, and compare maps with generic functions.
`cmp`	`cmp`	`cmp.Compare`, `cmp.Or`, and the `cmp.Ordered` constraint.
Structured logging	`log/slog`	`slog.Info`, `slog.Error`, handler configuration, key-value pairs.
One-time helpers	`sync`	`sync.OnceFunc`, `sync.OnceValue`, `sync.OnceValues`.
Test context	`testing`	`t.Context()` returns a context that cancels when the test ends.

```go
// desk121.go
package main

import (
```

```go
    "cmp"
    "fmt"
    "log/slog"
    "os"
    "slices"
)

type Order struct {
    ID    int
    Item  string
    Price float64
}

func main() {
    orders := []Order{
        {1, "keyboard", 89.99},
        {2, "monitor", 349.00},
        {3, "mouse", 29.50},
        {4, "desk pad", 19.95},
    }

    // slices.SortFunc with cmp.Compare for a numeric field.
    slices.SortFunc(orders, func(a, b Order) int {
        return cmp.Compare(a.Price, b.Price)
    })

    logger := slog.New(slog.NewTextHandler(os.Stdout, nil))

    for _, o := range orders {
        logger.Info("order",
            "id", o.ID,
            "item", o.Item,
            "price", o.Price,
        )
    }

    // min / max builtins.
    cheapest := orders[0].Price
    priciest := orders[len(orders)-1].Price
    fmt.Printf("range: %.2f - %.2f\n", cheapest, priciest)

    // clear resets the slice elements to zero values.
    scratch := []int{10, 20, 30}
    clear(scratch)
    fmt.Println("after clear:", scratch)
}
```

Run:

```
go run desk121.go
```

Output:

```
time=... level=INFO msg=order id=4 item="desk pad" price=19.95
time=... level=INFO msg=order id=3 item=mouse price=29.5
time=... level=INFO msg=order id=1 item=keyboard price=89.99
time=... level=INFO msg=order id=2 item=monitor price=349
range: 19.95 - 349.00
after clear: [0 0 0]
```

Go 1.22

Feature	Package / layer	What it does
`for i := range n`	language	Integer range: iterates `i` from `0` to `n-1`. No `int` variable needed.
Per-iteration loop variable	language	Each loop body gets its own copy of `i` and `v`; goroutines no longer share the same address.
Method + path patterns	`net/http`	`mux.HandleFunc("GET /tickets/{id}", h)` matches method and captures `{id}`.
`r.PathValue`	`net/http`	`r.PathValue("id")` extracts a named segment from the matched route.

```go
// desk122.go
package main

import (
    "fmt"
    "net/http"
    "strconv"
)

var tickets = map[int]string{
    1: "Replace keyboard",
    2: "Monitor flicker",
    3: "Desk lamp broken",
```

```go
}

func ticketHandler(w http.ResponseWriter, r *http.Request) {
    raw := r.PathValue("id")
    id, err := strconv.Atoi(raw)
    if err != nil {
        http.Error(w, "bad id", http.StatusBadRequest)
        return
    }
    title, ok := tickets[id]
    if !ok {
        http.Error(w, "not found", http.StatusNotFound)
        return
    }
    fmt.Fprintf(w, "ticket %d: %s\n", id, title)
}

func main() {
    mux := http.NewServeMux()
    mux.HandleFunc("GET /tickets/{id}", ticketHandler)

    // for i := range n - print ticket IDs without a separate counter variable.
    fmt.Println("known ticket IDs:")
    for i := range 3 {
        fmt.Println(" ", i+1)
    }

    fmt.Println("listening on :8122")
    if err := http.ListenAndServe(":8122", mux); err != nil {
        fmt.Println(err)
    }
}
```

Run:

```bash
go run desk122.go &
curl -s http://localhost:8122/tickets/2
```

Output:

```
known ticket IDs:
  1
  2
  3
listening on :8122
ticket 2: Monitor flicker
```

Go 1.23

Feature	Package / layer	What it does
`iter` package	`iter`	Defines `iter.Seq[V]` and `iter.Seq2[K,V]` — the function types a `range` loop can consume.
Range-over-function	language	`for v := range f` works when `f` is `func(yield func(V) bool)`.
`strings.SplitSeq`	`strings`	Like `strings.Split` but yields substrings one at a time via an iterator.
`strings.FieldsSeq`	`strings`	Lazy word-splitting iterator; no intermediate slice allocated.
`maps.Keys` / `maps.Values`	`maps`	Return `iter.Seq` iterators over map keys or values.
`time.Tick` GC fix	`time`	The ticker created by `time.Tick` is now garbage-collected when unreachable.

```go
// desk123.go
package main

import (
    "fmt"
    "maps"
    "strings"
)

func main() {
    // strings.FieldsSeq: iterate words without building a []string.
    shift := "Monday Tuesday Wednesday Thursday Friday"
    fmt.Print("days: ")
    for day := range strings.FieldsSeq(shift) {
        fmt.Printf("%s ", day)
    }
    fmt.Println()

    // maps.Keys iterator - consume keys without an intermediate slice.
    prices := map[string]float64{
        "keyboard": 89.99,
        "monitor":  349.00,
        "mouse":    29.50,
```

```go
    }

    fmt.Println("items in stock:")
    for item := range maps.Keys(prices) {
        fmt.Printf("  %s\n", item)
    }

    // strings.SplitSeq: lazy split on a delimiter.
    csvRow := "order_id,item,qty,price"
    fmt.Print("columns: ")
    for col := range strings.SplitSeq(csvRow, ",") {
        fmt.Printf("[%s]", col)
    }
    fmt.Println()
}
```

Run:

```go
go run desk123.go
```

Output:

```
days: Monday Tuesday Wednesday Thursday Friday
items in stock:
  keyboard
  monitor
  mouse
columns: [order_id][item][qty][price]
```

Go 1.24

Feature	Package / layer	What it does
`b.Loop()`	`testing`	Replaces `for i := 0; i < b.N; i++`; handles timer reset automatically.
`testing/synctest` (experimental)	`testing/synctest`	Runs goroutines in a fake-time bubble to test concurrent code deterministically.
Weak pointers	`weak`	`weak.Pointer[T]` — a reference that does not keep an object alive; useful for caches.

Feature	Package / layer	What it does
`runtime.AddCleanup`	`runtime`	Attach a cleanup function to an object, called on collection (replaces `SetFinalizer`).
`sync/atomic And/Or`	`sync/atomic`	Bitwise AND and OR operations added to integer atomic types.

```go
// desk124_test.go
package main

import (
    "testing"
)

// priceAfterDiscount applies a percentage discount.
func priceAfterDiscount(price, pct float64) float64 {
    return price * (1 - pct/100)
}

// BenchmarkDiscount uses b.Loop() - no manual b.N loop needed.
func BenchmarkDiscount(b *testing.B) {
    price := 349.00
    pct := 15.0
    var result float64
    for b.Loop() {
        result = priceAfterDiscount(price, pct)
    }
    _ = result
}

func TestDiscount(t *testing.T) {
    ctx := t.Context() // cancelled when the test ends
    _ = ctx

    got := priceAfterDiscount(100.0, 10.0)
    if got != 90.0 {
        t.Fatalf("want 90.0, got %f", got)
    }
}
```

Run:

```go
go test -bench=. -benchmem ./desk124_test.go
```

Output:

```
goos: linux
goarch: amd64
BenchmarkDiscount-8   1000000000   0.2990 ns/op   0 B/op   0 allocs/op
PASS
```

Go 1.25

Feature	Package / layer	What it does
`wg.Go(f)`	`sync`	Adds a goroutine to a `sync.WaitGroup` and calls `f()` in it; no separate `Add`/`Done` pair needed.
`testing/synctest` stable	`testing/synctest`	Fake-time goroutine bubble graduates from experimental to stable.
`os.Root`	`os`	A handle to a directory; all path operations are constrained to it, preventing traversal attacks.

```go
// desk125.go
package main

import (
    "fmt"
    "log/slog"
    "os"
    "sync"
)

type Ticket struct {
    ID    int
    Title string
}

func process(t Ticket) error {
    slog.Info("processing", "id", t.ID, "title", t.Title)
    return nil
}

func main() {
    tickets := []Ticket{
```

```go
        {1, "Replace keyboard"},
        {2, "Monitor flicker"},
        {3, "Desk lamp broken"},
        {4, "Chair squeaks"},
    }

    var wg sync.WaitGroup

    // wg.Go launches a goroutine and tracks it - no Add/Done boilerplate.
    for _, tk := range tickets {
        wg.Go(func() {
            if err := process(tk); err != nil {
                slog.Error("failed", "id", tk.ID, "err", err)
            }
        })
    }

    wg.Wait()
    fmt.Fprintln(os.Stdout, "all tickets processed")
}
```

Run:

```go
go run desk125.go
```

Output:

```
time=... level=INFO msg=processing id=1 title="Replace keyboard"
time=... level=INFO msg=processing id=2 title="Monitor flicker"
time=... level=INFO msg=processing id=3 title="Desk lamp broken"
time=... level=INFO msg=processing id=4 title="Chair squeaks"
all tickets processed
```

Go 1.26

Feature	Package / layer	What it does
`errors.AsType[T]`	`errors`	Generic form of `errors.As`; returns `(T, bool)` without declaring a pointer-to-target.
`slices.Chunk`	`slices`	Yields successive sub-slices of length at most n via an iterator.

Feature	Package / layer	What it does
`url.Clone`	`net/url`	Deep-copies a `*url.URL` including its query `Values` map.
`context.AfterFunc` clarification	`context`	Behaviour when the stop function is called after cancellation is now defined by the spec.

```go
// desk126.go
package main

import (
    "errors"
    "fmt"
    "slices"
)

// DeskError carries a numeric code for routing purposes.
type DeskError struct {
    Code    int
    Message string
}

func (e *DeskError) Error() string {
    return fmt.Sprintf("desk error %d: %s", e.Code, e.Message)
}

func processOrder(id int) error {
    if id <= 0 {
        return fmt.Errorf("processOrder: %w", &DeskError{Code: 400, Message: "invalid order
    }
    return nil
}

func main() {
    err := processOrder(-1)

    // errors.AsType[T] - no &target variable needed.
    if de, ok := errors.AsType[*DeskError](err); ok {
        fmt.Printf("caught DeskError code=%d msg=%s\n", de.Code, de.Message)
    }

    // slices.Chunk - batch 5 order IDs into groups of 2.
    orders := []int{101, 102, 103, 104, 105}
    fmt.Println("batches:")
```

```go
    for batch := range slices.Chunk(orders, 2) {
        fmt.Println(" ", batch)
    }
}
```

Run:

```go
go run desk126.go
```

Output:

```
caught DeskError code=400 msg=invalid order id
batches:
  [101 102]
  [103 104]
  [105]
```

Go 1.27

Feature	Package / layer	What it does
`encoding/json/v2`	`encoding/json/v2`	Redesigned JSON package: opt-in, cleaner API, better error messages, strict by default.
`json:",omitzero"`	`encoding/json/v2`	Omits a field when it holds its zero value (works for structs, not just pointers).
`stdlib/uuid`	`uuid`	First-party UUID generation and parsing; no third-party dependency needed.
`strings.CutLast`	`strings`	Like `strings.Cut` but finds the *last* occurrence of the separator.
`bytes.CutLast`	`bytes`	Same as `strings.CutLast` but operates on byte slices.
Promoted field literals	language	Embedded-field names may be omitted in composite literals when unambiguous.
`synctest.Sleep`	`testing/synctest`	Advances fake time inside a synctest bubble, waking goroutines blocked on timers.

```go
// desk127.go
package main

import (
    "fmt"
    "strings"

    jsonv2 "encoding/json/v2"
)

// ShiftEntry records one desk shift.
// HourlyRate is omitted from JSON output when it is zero.
type ShiftEntry struct {
    StaffID    string  `json:"staff_id"`
    Day        string  `json:"day"`
    HourlyRate float64 `json:"hourly_rate,omitzero"`
}

func main() {
    shifts := []ShiftEntry{
        {"emp-001", "Monday", 22.50},
        {"emp-002", "Tuesday", 0},       // HourlyRate omitted in output.
        {"emp-003", "Wednesday", 18.75},
    }

    // encoding/json/v2: Marshal respects omitzero.
    data, err := jsonv2.Marshal(shifts)
    if err != nil {
        panic(err)
    }
    fmt.Println(string(data))

    // strings.CutLast: extract the extension from a versioned filename.
    filename := "report.2025.csv"
    before, after, found := strings.CutLast(filename, ".")
    if found {
        fmt.Printf("base=%q ext=%q\n", before, after)
    }

    // strings.CutLast on a path: find the last path segment.
    path := "/desks/floor2/station7"
    _, segment, _ := strings.CutLast(path, "/")
    fmt.Printf("last segment: %s\n", segment)
}
```

Run:

```
go run desk127.go
```

Output:

```
[{"staff_id":"emp-001","day":"Monday","hourly_rate":22.5},{"staff_id":"emp-002","day":"Tuesday'
base="report.2025" ext="csv"
last segment: station7
```

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